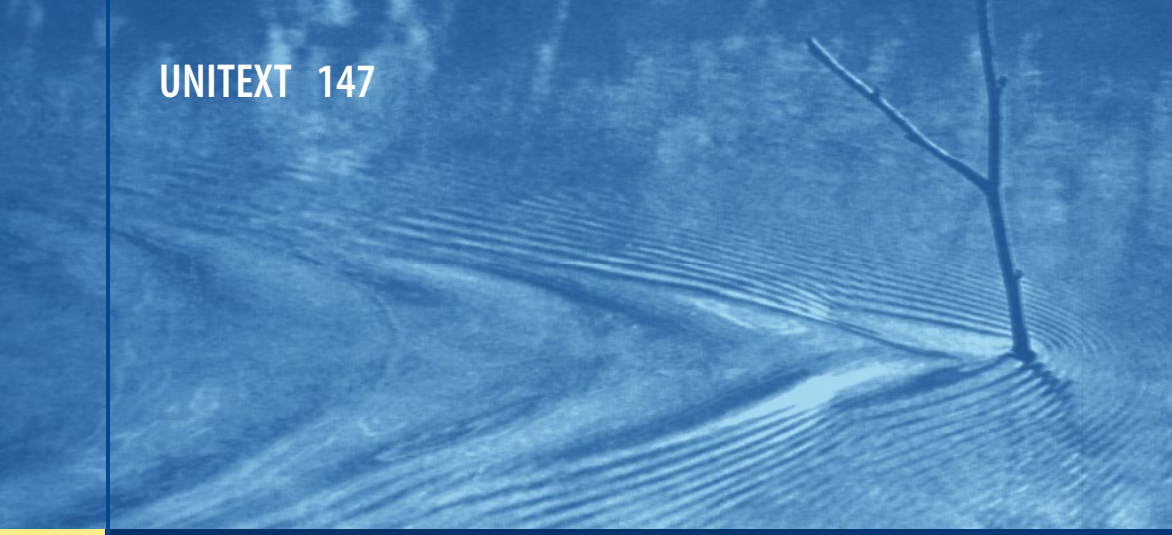

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Sandro Salsa · Gianmaria Verzini

Partial Differential Equations in Action

From Modelling to Theory

Fourth Edition

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To Anna and Giulia

Preface

This book, of which the present one is the fourth edition, is designed as an advanced undergraduate or a first-year graduate course for students from various disciplines like applied mathematics, physics, engineering. It has evolved while teaching courses on partial differential equations (PDEs) during the last few years at the Politecnico of Milan.

The main purpose of these courses was twofold: on the one hand, to train the students to appreciate the interplay between theory and modelling in problems arising in the applied sciences, and on the other hand to give them a solid theoretical background for numerical methods, such as finite elements.

A quick outline of the book is reported below; the most relevant differences with the last edition are written in *italics* .

Chapters 2 to 5 have a rather elementary character with the goal of developing and studying basic problems from the macro-areas of diffusion, propagation and transport, waves and vibrations. The emphasis is, whenever possible, on ideas and connections with concrete aspects, in order to provide intuition and feeling for the subject.

For these chapters, a knowledge of advanced calculus and ordinary differential equations is required. Also, the repeated use of the method of separation of variables assumes some basic results from the theory of Fourier series, which are summarized in the introductory Chapter 1.

Chapter 2 starts with the heat equation and some of its variants in which transport and reaction terms are incorporated. In addition to the classical topics, it is emphasized the connections with simple stochastic processes, such as random walks and Brownian motion. This requires the knowledge of some elements of probability. It is our belief that it is worthwhile presenting this topic as early as possible, even at the price of giving up to a little bit of rigor in the presentation. An application to financial mathematics shows the interaction between probabilistic and deterministic modelling. The last two sections are devoted to two simple non linear models from flow in porous medium and population dynamics. *This last topic comes back from a more general point of view in Section 10.5 of Chapter 10.*

Chapter 3 mainly treats the Laplace/Poisson equation. The main properties of harmonic functions are presented once more emphasizing the probabilistic motivations. The Perron's method of super/sub solutions is included, due to its renewed importance as a solution technique in several contexts, e.g. for fully nonlinear equations. The second part of this chapter deals with representation formulas in terms of potentials. In particular, the basic properties of the Newtonian, single and double layer potentials are presented.

Chapter 4 is devoted to first order equations and in particular to first order scalar conservation laws. The method of characteristics, the notions of shock and rarefaction waves and of entropy solution are introduced through a simple model from traffic dynamics. In the last part, the method of characteristics is extended to quasilinear and fully nonlinear equations in two variables. A subsection on the concepts of *complete integral, general and singular solutions* has been added.

In Chapter 5 the fundamental aspects of waves propagation are examined, leading to the classical formulas of d'Alembert, Kirchhoff and Poisson. Transversal motion of a string and of a membrane, small amplitude of sound waves, radiation from a moving source and a simple model of Thermoacoustic Tomography serve as typical applications of the theory. In the final section, the classical model for surface waves in deep water illustrates the phenomenon of dispersion, with the help of the method of stationary phase.

In the chapters from 6 to 9, the Hilbert spaces methods for the variational formulation and the analysis of linear boundary and initial-boundary value problems are developed. Given the abstract nature of these chapters, an effort is made to provide intuition and motivation about the various concepts and results, running the risk of appearing a bit wordy.

The understanding of these topics requires some basic knowledge of Lebesgue measure and integration, summarized in Appendix A.

Chapter 6 contains the tools from functional analysis in Hilbert spaces, necessary for a correct variational formulation of the most common boundary value problems. The main theme is the solvability of linear abstract variational problems, leading to the Lax-Milgram Theorem and Fredholm's alternative. *A new section treats Nečas theorem and the so called, inf-sup condition.* Emphasis is given to the issues of compactness and weak convergence. Section 6.10 is devoted to the fixed points analysis with the Contraction Mapping Theorem, the Schauder Theorem and *an extended version* of the Leray-Schauder Theorem.

Chapter 7 is divided into two parts. The first one is a brief introduction to the theory of distributions (or generalized functions) of L. Schwartz. In the second one, the most used Sobolev spaces and their basic properties are discussed.

Chapter 8 is devoted to the variational formulation of the most common linear elliptic boundary value problems and their solvability. The development starts with Poisson's equation and continues with general second order equations in divergence form. The last part deals with weak maximum principles and the optimal regularity of the solutions.

The issue in Chapter 9 (*ex Chapter 10*) is the variational formulation of evolution problems, in particular of initial-boundary value problems for second order parabolic

operators in divergence form and for the wave equation. The basic tool is an analogue of the Lax-Milgram Theorem for abstract parabolic problems.

Chapter 10 (*ex Chapter 9*) gathers a number of applications ranging from Elastostatics to Stokes and Navier-Stokes equations. Also a simple control problem is discussed in Section 10.4. *The new section 10.5 on reaction-diffusion equations has been added. In particular, through the use of monotone iterations schemes, a population dynamics model in a heterogeneous environment is examined.*

Chapter 11 presents a brief introduction to the theory of systems of first order conservation laws, in one space dimension. In particular we extend from the scalar case of Chapter 4 the notions of characteristics, shocks, rarefaction waves, contact discontinuities and the entropy condition. The main focus is on the solution of the Riemann problem.

At the end of each chapter, a number of exercises is included. Some of them can be solved by a routine application of the theory or of the methods developed in the text. Other problems are intended as a completion of some arguments or proofs in the text. Also, there are problems in which the student is required to be more autonomous. The most demanding problems are supplied with answers or hints. With respect to the last edition, *new exercises has been added in almost all chapters.*

The order of presentation of the material is clearly a consequence of our ... prejudices. However, the exposition if flexible enough to allow substantial changes without compromising the comprehension and to facilitate a selection of topics for a one or two semester course.

The chapters from 2 to 5 are in practice mutually independent, with the exception of Sections 3.3.1 and 3.5, which presume the knowledge of Sect. 2.6.

In the chapters from 6 to 11, which, in principle, may be presented independently of the previous ones, more attention has to be paid to the order of the arguments. In particular, the material in Chapter 6 and in Sections 7.1-7.4, 7.7-7.10 is necessary for understanding Chapter 8 and Sects. 10.1-10.4 while Chapter 9 and Sect. 10.5 use concepts and results also from Sect. 7.11. Chapter 11 requires the knowledge of Sects 4.4 and 4.5 (and 6.10.1 for Sect. 11.2.2).

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Milan, October 2022 *Sandro Salsa, Gianmaria Verzini*

Contents

1 Introduction 1
1.1 Mathematical Modelling 1
1.2 Partial Differential Equations 2
1.3 Well Posed Problems 5
1.4 Basic Notations and Facts 7
1.5 Fourier Series 12
1.6 The Inverse and the Implicit Function Theorems 16
1.7 Smooth and Lipschitz Domains 17
1.8 Integration by Parts Formulas 19

2 Diffusion 21
2.1 The Diffusion Equation 21
2.1.1 Introduction 21
2.1.2 The conduction of heat 22
2.1.3 Initial boundary value problems ($n = 1$) 24
2.1.4 A solution by separation of variables 27
2.1.5 Problems in dimension $n > 1$ 35
2.2 Uniqueness and Maximum Principles 37
2.2.1 Integral method 37
2.2.2 Maximum principles 39
2.3 The Fundamental Solution 42
2.3.1 Invariant transformations 42
2.3.2 The fundamental solution ($n = 1$) 43
2.3.3 The Dirac distribution 46
2.3.4 The fundamental solution ($n > 1$) 49
2.4 Symmetric Random Walk ($n = 1$) 51
2.4.1 Preliminary computations 51
2.4.2 From random walk to Brownian motion 54
2.4.3 A glance at Brownian motion 56
2.5 Diffusion, Drift and Reaction 59
2.5.1 Random walk with drift. Preliminary calculations 59

2.5.2 The limit equation . . . 60
2.5.3 Pollution in a channel . . . 62
2.5.4 Random walk with drift and reaction . . . 65
2.5.5 Critical dimension in a simple population dynamics . . . 66
2.6 Multidimensional Random Walk . . . 68
2.6.1 The symmetric case . . . 68
2.6.2 Walks with drift and reaction . . . 71
2.7 An Example of Reaction–Diffusion in Dimension $n = 3$ . . . 72
2.8 The Global Cauchy Problem ($n = 1$) . . . 78
2.8.1 The homogeneous case . . . 78
2.8.2 Existence of a solution . . . 79
2.8.3 The nonhomogeneous case. Duhamel's method . . . 80
2.8.4 Global maximum principle and uniqueness . . . 84
2.8.5 Nonnegative solutions . . . 87
2.8.6 The proof of the existence Theorem 2.12 . . . 87
2.9 An Application to Finance . . . 90
2.9.1 European options . . . 90
2.9.2 An evolution model for the price S . . . 91
2.9.3 The Black-Scholes equation . . . 93
2.9.4 The solutions . . . 96
2.9.5 Hedging and self-financing strategy . . . 101
2.10 Some Nonlinear Aspects . . . 103
2.10.1 Nonlinear diffusion. The porous medium equation . . . 103
2.10.2 Nonlinear reaction. Fisher's equation . . . 106
Problems . . . 110

3 The Laplace Equation . . . 115
3.1 Introduction . . . 115
3.2 Boundary value Problems. Uniqueness . . . 116
3.3 Harmonic Functions . . . 118
3.3.1 Discrete harmonic functions . . . 118
3.3.2 Mean value properties . . . 121
3.3.3 Maximum principles . . . 123
3.3.4 The Hopf principle . . . 125
3.3.5 The Dirichlet problem in a disc. Poisson's formula . . . 127
3.3.6 Harnack's inequality and Liouville's theorem . . . 131
3.3.7 Analyticity of harmonic functions . . . 132
3.4 Sub/Superharmonic Functions. The Perron Method . . . 134
3.4.1 Sub/superharmonic functions . . . 134
3.4.2 The method . . . 136
3.4.3 Boundary behavior . . . 138
3.5 A probabilistic solution of the Dirichlet problem . . . 141
3.6 Fundamental Solution and Newtonian Potential . . . 146
3.6.1 The fundamental solution . . . 146
3.6.2 The Newtonian potential . . . 148

3.6.3 A divergence-curl system. Helmholtz decomposition formula 151
3.7 The Green Function . . . 155
3.7.1 An integral identity . . . 155
3.7.2 Green's function . . . 156
3.7.3 Green's representation formula . . . 159
3.7.4 The Neumann function . . . 161
3.8 Uniqueness in Unbounded Domains . . . 163
3.8.1 Exterior problems . . . 163
3.9 Surface Potentials . . . 165
3.9.1 The double and single layer potentials . . . 165
3.9.2 The integral equations of potential theory . . . 170
3.9.3 Proof of Theorem 3.47 . . . 174
Problems . . . 176

4 Scalar Conservation Laws and First Order Equations . . . 181
4.1 Introduction . . . 181
4.2 Linear Transport Equation . . . 182
4.2.1 Pollution in a channel . . . 182
4.2.2 Distributed source . . . 184
4.2.3 Extinction and localized source . . . 185
4.2.4 Inflow and outflow characteristics. A stability estimate . . . 187
4.3 Traffic Dynamics . . . 189
4.3.1 A macroscopic model . . . 189
4.3.2 The method of characteristics . . . 191
4.3.3 The green light problem . . . 193
4.3.4 Traffic jam ahead . . . 198
4.4 Weak (or Integral) Solutions . . . 201
4.4.1 The method of characteristics revisited . . . 201
4.4.2 Weak solutions . . . 204
4.4.3 Piecewise smooth functions and the Rankine-Hugoniot condition . . . 206
4.5 An Entropy Condition . . . 210
4.6 The Riemann problem . . . 213
4.6.1 Convex/concave flux function . . . 213
4.6.2 Vanishing viscosity method. Travelling waves . . . 214
4.6.3 Flux function with inflection points . . . 219
4.7 The Method of Characteristics for Quasilinear Equations . . . 222
4.7.1 Characteristics . . . 222
4.7.2 The Cauchy problem . . . 224
4.7.3 General solutions and Lagrange method of first integrals . . . 231
4.7.4 Underground flow . . . 234
4.8 Fully Nonlinear Equations . . . 236
4.8.1 Characteristic strips . . . 236
4.8.2 The Cauchy Problem . . . 238
4.8.3 Complete integrals, general and singular solutions . . . 243

Problems 246

5 Waves and Vibration 253
5.1 General Concepts 253
5.1.1 Types of waves 253
5.1.2 Group velocity and dispersion relation 255
5.2 Transverse Waves in a String 258
5.2.1 The model 258
5.2.2 Energy 260
5.3 The One-dimensional Wave Equation 262
5.3.1 Initial and boundary conditions 262
5.3.2 Separation of variables 263
5.4 The d'Alembert Formula 269
5.4.1 The homogeneous equation 269
5.4.2 Generalized solutions and propagation of singularities 273
5.4.3 The fundamental solution 276
5.4.4 Nonhomogeneous equation. Duhamel's method 278
5.4.5 Dissipation and dispersion 279
5.5 Second Order Linear Equations 281
5.5.1 Classification 281
5.5.2 Characteristics and canonical form 284
5.6 The Multi-dimensional Wave Equation ($n > 1$) 290
5.6.1 Special solutions 290
5.6.2 Initial boundary value problems. Uniqueness 292
5.7 Two Classical Models 295
5.7.1 Small vibrations of an elastic membrane 295
5.7.2 Small amplitude sound waves 299
5.8 The Global Cauchy Problem 304
5.8.1 Fundamental solution ($n = 3$) and strong Huygens' principle 304
5.8.2 The Kirchhoff formula 307
5.8.3 The Cauchy problem in dimension 2 309
5.9 An Inverse Problem in Thermoacoustic Tomography 311
5.10 The Cauchy Problem with Distributed Sources 315
5.10.1 Retarded potentials ($n = 3$) 315
5.10.2 Radiation from a moving point source 317
5.11 Linear Water Waves 322
5.11.1 A model for surface waves 322
5.11.2 Dimensionless formulation and linearization 326
5.11.3 Deep water waves 327
5.11.4 Interpretation of the solution 330
5.11.5 Asymptotic behavior 331
5.11.6 The method of stationary phase 333
Problems 335

6 Elements of Functional Analysis 339
6.1 Motivations 339
6.2 Norms and Banach Spaces 345
6.3 Hilbert Spaces 350
6.4 Projections and Bases 354
6.4.1 Projections 355
6.4.2 Bases 359
6.5 Linear Operators and Duality 365
6.5.1 Linear operators 365
6.5.2 Functionals and dual space 368
6.5.3 The (Hilbert) adjoint of a bounded operator 371
6.5.4 Ranges and Kernels. The inf-sup condition 373
6.6 Abstract Variational Problems 375
6.6.1 Bilinear forms. The Lax-Milgram Theorem 375
6.6.2 Approximation and Galerkin's method 378
6.6.3 Minimization of quadratic functionals 381
6.6.4 The Nečas Theorem 382
6.7 Compactness and Weak Convergence 384
6.7.1 Compactness 384
6.7.2 Compactness in $C(\overline{\Omega})$ and in $L^p(\Omega)$ 385
6.7.3 Weak convergence and compactness 387
6.7.4 Compact operators 390
6.8 The Fredholm Alternative 392
6.8.1 Hilbert triplets 392
6.8.2 Solvability for abstract variational problems 395
6.8.3 The Fredholm alternative 398
6.9 Spectral Theory for Symmetric Bilinear Forms 400
6.9.1 Spectrum of a matrix 400
6.9.2 Separation of variables revisited 400
6.9.3 Spectrum of a compact self-adjoint operator 402
6.9.4 Application to abstract variational problems 404
6.10 Fixed Points Theorems 409
6.10.1 The Contraction Mapping Theorem 409
6.10.2 The Schauder Theorem 411
6.10.3 The Leray-Schauder Theorem 413
Problems 414

7 Distributions and Sobolev Spaces 419
7.1 Preliminary Ideas 419
7.2 Test Functions and Mollifiers 421
7.3 Distributions 425
7.4 Calculus 429
7.4.1 The derivative in the sense of distributions 429
7.4.2 Gradient, divergence, Laplacian 432
7.5 Operations with Distributions 434

7.5.1 Multiplication. Leibniz rule . . . 434
7.5.2 Composition . . . 436
7.5.3 Division . . . 439
7.5.4 Tensor or direct product . . . 440
7.5.5 Convolution . . . 443
7.6 Tempered Distributions and Fourier Transform . . . 445
7.6.1 Tempered distributions . . . 445
7.6.2 Fourier transform in $\mathcal{S}'(\mathbb{R}^n)$ . . . 447
7.6.3 Fourier transform in $L^2(\mathbb{R}^n)$ . . . 450
7.7 Sobolev Spaces . . . 451
7.7.1 An abstract construction . . . 451
7.7.2 The space $H^1(\Omega)$ . . . 453
7.7.3 The space $H_0^1(\Omega)$ . . . 457
7.7.4 The dual of $H_0^1(\Omega)$ . . . 458
7.7.5 The spaces $H^m(\Omega)$, $m > 1$ . . . 461
7.7.6 Calculus rules . . . 462
7.7.7 Fourier transform and Sobolev spaces . . . 463
7.8 Approximations by Smooth Functions and Extensions . . . 465
7.8.1 Local approximations . . . 465
7.8.2 Extensions and global approximations . . . 465
7.9 Traces . . . 470
7.9.1 Traces of functions in $H^1(\Omega)$ . . . 470
7.9.2 Traces of functions in $H^m(\Omega)$ . . . 473
7.9.3 Trace spaces . . . 474
7.10 Compactness and Embeddings . . . 477
7.10.1 The Rellich Theorem . . . 477
7.10.2 Poincaré's inequalities . . . 479
7.10.3 Sobolev inequality in $\mathbb{R}^n$ . . . 481
7.10.4 Bounded domains . . . 482
7.11 Spaces Involving Time . . . 484
7.11.1 Functions with values into Hilbert spaces . . . 484
7.12 Sobolev spaces involving time . . . 487
Problems . . . 489

8 Variational Formulation of Elliptic Problems . . . 495
8.1 Linear Elliptic Equations . . . 495
8.2 Notions of Solution . . . 497
8.3 Problems for the Poisson Equation . . . 499
8.3.1 Dirichlet problem . . . 499
8.3.2 Neumann, Robin and mixed problems . . . 502
8.3.3 Eigenvalues and eigenfunctions of the Laplace operator . . . 506
8.3.4 An asymptotic stability result . . . 509
8.4 General Equations in Divergence Form . . . 510
8.4.1 Basic assumptions . . . 510
8.4.2 Dirichlet problem . . . 511

8.4.3 Neumann problem . . . 517
8.4.4 Robin and mixed problems . . . 519
8.5 Equilibrium of a Plate . . . 520
8.6 Weak Maximum Principles . . . 522
8.7 Regularity . . . 527
Problems . . . 534

9 Weak Formulation of Evolution Problems . . . 541
9.1 Parabolic Equations . . . 541
9.2 The Cauchy-Dirichlet Problem for the Heat Equation . . . 543
9.3 Abstract Parabolic Problems . . . 546
9.3.1 Formulation . . . 546
9.3.2 Energy estimates. Uniqueness and stability . . . 548
9.3.3 The Faedo-Galerkin approximations . . . 550
9.3.4 Existence . . . 552
9.4 Parabolic PDEs . . . 553
9.4.1 Problems for the heat equation . . . 553
9.4.2 Regularity . . . 555
9.4.3 General Equations . . . 557
9.5 Weak Maximum Principles . . . 559
9.6 The Wave Equation . . . 561
9.6.1 Hyperbolic Equations . . . 561
9.6.2 The Cauchy-Dirichlet problem . . . 562
9.6.3 The method of Faedo-Galerkin . . . 564
9.6.4 Solution of the approximate problem . . . 565
9.6.5 Energy estimates . . . 565
9.6.6 Existence, uniqueness and stability . . . 567
Problems . . . 570

10 More Advanced Topics . . . 573
10.1 The Linear Elastostatic System . . . 573
10.2 The Stokes System . . . 578
10.3 The Stationary Navier Stokes Equations . . . 582
10.3.1 Weak formulation . . . 582
10.3.2 Existence, and uniqueness for small data . . . 582
10.4 A Control Problem . . . 585
10.4.1 Structure of the problem . . . 585
10.4.2 Existence and uniqueness of an optimal pair . . . 587
10.4.3 Lagrange multipliers and optimality conditions . . . 588
10.4.4 An iterative algorithm . . . 589
10.5 Reaction-Diffusion Models . . . 590
10.5.1 Dynamics of a population in a heterogeneous environment . . . 591
10.5.2 A monotone iteration scheme . . . 592
10.5.3 Long time behavior and steady states . . . 596
10.5.4 Regularity . . . 597

10.5.5 Persistence versus extinction . . . 599
Problems . . . 606

11 Systems of Conservation Laws . . . 611
11.1 Introduction . . . 611
11.2 Linear Hyperbolic Systems . . . 616
11.2.1 Characteristics . . . 616
11.2.2 Classical solutions of the Cauchy problem . . . 618
11.2.3 Homogeneous systems with constant coefficients. The Riemann problem . . . 621
11.3 Quasilinear Conservation Laws . . . 625
11.3.1 Characteristics and Riemann invariants . . . 625
11.3.2 Weak (or integral) solutions and the Rankine-Hugoniot condition . . . 628
11.4 The Riemann Problem . . . 629
11.4.1 Rarefaction waves. Genuinely nonlinear systems . . . 631
11.4.2 Solution of the Riemann problem by a single rarefaction wave . . . 633
11.4.3 Lax entropy condition. Shock waves and contact discontinuities . . . 636
11.4.4 Solution of the Riemann problem by a single k-shock . . . 638
11.4.5 The linearly degenerate case . . . 640
11.4.6 Local solution of the Riemann problem . . . 641
11.5 The Riemann Problem for the p-system . . . 641
11.5.1 Shock waves . . . 642
11.5.2 Rarefaction waves . . . 644
11.5.3 The solution in the general case . . . 647
Problems . . . 651

A Measures and Integrals . . . 655
A.1 A Counting Problem . . . 655
A.2 Measures and Measurable Functions . . . 657
A.3 The Lebesgue Integral . . . 659
A.4 Some Fundamental Theorems . . . 660
A.5 Probability Spaces, Random Variables and Their Integrals . . . 662

B Identities and Formulas . . . 665
B.1 Gradient, Divergence, Curl, Laplacian . . . 665
B.2 Formulas . . . 667

References . . . 669

Index . . . 673

Chapter 1
Introduction

1.1 Mathematical Modelling

Mathematical modelling play a big role in the description of a large part of phenomena in the applied sciences and in several aspects of technical and industrial activity.

By a "mathematical model" we mean a set of equations and/or other mathematical relations capable of capturing the essential features of a complex natural or artificial system, in order to describe, forecast and control its evolution. The applied sciences are not confined to the classical ones; in addition to *physics* and *chemistry*, the practice of mathematical modelling heavily affects disciplines like *finance, biology, ecology, medicine, sociology.*

In the industrial activity (e.g. for aerospace or naval projects, nuclear reactors, combustion problems, production and distribution of electricity, traffic control, etc.) the mathematical modeling, involving first the analysis and the numerical simulation and followed by experimental tests, has become a common procedure, necessary for innovation, and also motivated by economic factors. It is clear that all of this is made possible by the enormous computational power now available.

In general, the construction of a mathematical model is based on two main ingredients:

- *general laws*;
- *constitutive relations*.

In this book we shall deal with general laws coming from continuum mechanics and appearing as conservation or balance laws (e.g. of mass, energy, linear momentum, etc.).

The constitutive relations are of an experimental nature and strongly depend on the features of the phenomena under examination. Examples are the Fourier law of heat conduction, the Fick law for the diffusion of a substance or the way the speed of a driver depends on the density of cars ahead.

The outcome of the combination of the two ingredients is usually a *partial differential equation or a system of them.*

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1.2 Partial Differential Equations

A partial differential equation is a relation of the following type:

$$F(x_1, \ldots, x_n, u, u_{x_1}, \ldots, u_{x_n}, u_{x_1x_1}, u_{x_1x_2} \ldots, u_{x_nx_n}, u_{x_1x_1x_1}, \ldots) = 0 \qquad (1.1)$$

where the unknown $u = u(x_1, \ldots x_n)$ is a function of n variables and $u_{x_j}, \ldots$, $u_{x_ix_j}, \ldots$ are its partial derivatives. The highest order of differentiation occurring in the equation is the *order of the equation*.

A first important distinction is between *linear* and *nonlinear* equations.

Equation (1.1) is *linear* if F is linear with respect to u and all its derivatives, otherwise it is *nonlinear*.

A second distinction concerns the types of nonlinearity. We distinguish:

- *Semilinear* equations when F is nonlinear only with respect to u but is linear with respect to all its derivatives, with coefficients depending only on $\mathbf{x} = (x_1, \ldots, x_n)$.
- *Quasi-linear* equations when F is linear with respect to the highest order derivatives of u, with coefficients depending only on $\mathbf{x}$, u and lower order derivatives.
- *Fully nonlinear equations* when F is nonlinear with respect to the highest order derivatives of u.

The theory of linear equations can be considered sufficiently well developed and consolidated, at least for what concerns the most important questions. On the contrary, the nonlinearities present such a rich variety of aspects and complications that a general theory does not appear to be conceivable. The existing results and the new investigations focus on more or less specific cases, especially interesting in the applied sciences.

To give the reader an idea of the wide range of applications we present a series of examples, suggesting one of the possible interpretations. Most of them are considered at various level of deepness in this book. In the examples, $\mathbf{x}$ represents a space variable (usually in dimension $n = 1, 2, 3$) and t is a time variable.

We start with **linear equations**. In particular, equations (1.2)–(1.5) are fundamental and their theory constitutes a starting point for many other equations.

1. *Transport equation* (first order):

$$u_t + \mathbf{v} \cdot \nabla u = 0, \qquad (1.2)$$

where $\nabla = (\partial_{x_1}, \ldots, \partial_{x_n})$. It describes for instance the transport of a solid polluting substance along a channel; here u is the concentration of the substance and $\mathbf{v}$ is the stream speed. We consider the one-dimensional version of (1.2) in Sect. 4.2.

2. *Diffusion* or *heat equation* (second order):

$$u_t - D\Delta u = 0, \qquad (1.3)$$

where $\Delta = \partial_{x_1x_1} + \partial_{x_2x_2} + \ldots + \partial_{x_nx_n}$ is the *Laplace operator*. It describes the conduction of heat through a homogeneous and isotropic medium; u is the temperature

and D encodes the thermal properties of the material. Chapter 2 is devoted to the heat equation and its variants.

3. *Wave equation* (second order):

$$u_{tt} - c^2 \Delta u = 0. \tag{1.4}$$

It describes for instance the propagation of transversal waves of small amplitude in a perfectly flexible and elastic chord (e.g. of a violin) if $n = 1$, or membrane (e.g. of a drum) if $n = 2$. If $n = 3$ it governs the propagation of electromagnetic waves in vacuum or of small amplitude sound waves. Here u may represent the wave amplitude and c is the propagation speed. Chapter 5 is devoted to these equations.

4. *Laplace's or potential equation* (second order):

$$\Delta u = 0, \tag{1.5}$$

where $u = u(\mathbf{x})$. The diffusion and the wave equations model evolution phenomena. The Laplace equation describes the corresponding *steady state*, in which the solution does not depend on time anymore. Together with its nonhomogeneous version

$$\Delta u = f,$$

called *Poisson's* equation, it plays an important role in electrostatics as well. Chapter 3 is devoted to these equations.

5. *Black-Scholes equation* (second order):

$$u_t + \frac{1}{2}\sigma^2 x^2 u_{xx} + rxu_x - ru = 0.$$

Here $u = u(x,t)$, $x \geq 0$, $t \geq 0$. Fundamental in mathematical finance, this equation governs the evolution of the price u of a so called *derivative* (e.g. an *European option*), based on an underlying asset (a stock, a currency, etc.) whose price is x. We meet the Black-Scholes equation in Sect. 2.9.

6. *Vibrating plate* (fourth order):

$$u_{tt} - \Delta^2 u = 0,$$

where $\mathbf{x} = (x_1, x_2)$ and

$$\Delta^2 u = \Delta(\Delta u) = \frac{\partial^4 u}{\partial x_1^4} + 2\frac{\partial^4 u}{\partial x_1^2 \partial x_2^2} + \frac{\partial^4 u}{\partial x_2^4}$$

is the *biharmonic operator*. In the theory of linear elasticity, it models the transversal waves of small amplitude of a homogeneous isotropic plate (see Sect. 8.5 for the stationary version).

7. *Schrödinger equation*(second order):

$$iu_t = \Delta u - V(\mathbf{x})u$$

where i is the complex unit and u is complex valued. This equation is fundamental in quantum mechanics and governs the evolution of a particle subject to a potential V. The function $|u|^2$ represents a *probability density*. We will briefly encounter the Schrödinger equation in Problem 6.6.

Let us list now some examples of **nonlinear equations**.

8. *Burgers equation* (quasilinear, first order):

$$u_t + cuu_x = 0 \qquad (x \in \mathbb{R}).$$

It governs a one dimensional flux of a nonviscous fluid but it is used to model traffic dynamics as well. Its viscous variant

$$u_t + cuu_x = \varepsilon u_{xx} \qquad (\varepsilon > 0)$$

constitutes a basic example of competition between *dissipation* (due to the term εu_{xx}) and *steepening* (shock formation due to the term cuu_x). We will discuss these topics in Sects. 4.4 and 4.6.

9. *Fisher's equation* (semilinear, second order):

$$u_t - D\Delta u = ru(M - u) \qquad (D, r, M \text{ positive constants}).$$

It governs the evolution of a population of density u, subject to diffusion and logistic growth (represented by the right hand side). We will meet Fisher's equation in Sects. 2.10 and 10.5.

10. *Porous medium equation* (quasilinear, second order):

$$u_t = k \operatorname{div}(u^\gamma \nabla u)$$

where $k > 0, \gamma > 1$ are constant. This equation appears in the description of filtration phenomena, e.g. of the motion of water through the ground. We briefly meet the one-dimensional version of the porous medium equation in Sect. 2.10.

11. *Minimal surface equation* (quasilinear, second order):

$$\operatorname{div}\left(\frac{\nabla u}{\sqrt{1 + |\nabla u|^2}}\right) = 0 \qquad (\mathbf{x} \in \mathbb{R}^2).$$

The graph of a solution u minimizes the area among all surfaces $z = v(x_1, x_2)$ whose boundary is a given curve. For instance, soap balls are minimal surfaces. We will not examine this equation (see e.g. *R. McOwen*, [13]).

12. *Eikonal equation* (fully nonlinear, first order):

$$|\nabla u| = c(\mathbf{x}).$$

It appears in geometrical optics: if u is a solution, its level surfaces $u(\mathbf{x}) = t$ describe the position of a light wave front at time t. A bidimensional version is examined at the end of Chap. 4.

Let us now give some examples of **systems**.

13. *Navier's equation of linear elasticity* (three scalar equations of second order):

$$\rho \mathbf{u}_{tt} = \mu \Delta \mathbf{u} + (\mu + \lambda) \operatorname{grad} \operatorname{div} \mathbf{u}$$

where $\mathbf{u} = (u_1(\mathbf{x}, t), u_2(\mathbf{x}, t), u_3(\mathbf{x}, t))$, $\mathbf{x} \in \mathbb{R}^3$. The vector $\mathbf{u}$ represents the displacement from equilibrium of a deformable continuum body of (constant) density ρ. We will examine the stationary version in Sect. 10.1.

14. *Maxwell's equations in vacuum* (eight scalar linear equations of first order):

$$\begin{cases} \mathbf{E}_t - \operatorname{curl} \mathbf{B} = \mathbf{0}, \quad \mathbf{B}_t + \operatorname{curl} \mathbf{E} = \mathbf{0} & \text{(Ampère and Faraday laws)} \\ \operatorname{div} \mathbf{E} = 0, \quad \operatorname{div} \mathbf{B} = 0 & \text{(Gauss laws),} \end{cases}$$

where $\mathbf{E}$ is the electric field and $\mathbf{B}$ is the magnetic induction field. The unit measures are the "natural" ones, i.e. the light speed is $c = 1$ and the magnetic permeability is $\mu_0 = 1$. We will not examine this system (see e.g. *R. Dautray, J.-L. Lions*, [27], vol. 1).

15. *Navier-Stokes equations* (three quasilinear scalar equations of second order and one linear equation of first order):

$$\begin{cases} \mathbf{u}_t + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \Delta \mathbf{u} \\ \operatorname{div} \mathbf{u} = 0, \end{cases}$$

where $\mathbf{u} = (u_1(\mathbf{x}, t), u_2(\mathbf{x}, t), u_3(\mathbf{x}, t))$, $p = p(\mathbf{x}, t)$, $\mathbf{x} \in \mathbb{R}^3$. This equation governs the motion of a viscous, homogeneous and incompressible fluid. Here $\mathbf{u}$ is the fluid speed, p its pressure, ρ its density (constant) and ν is the kinematic viscosity, given by the ratio between the fluid viscosity and its density. The term $(\mathbf{u} \cdot \nabla)\mathbf{u}$ represents the inertial acceleration due to fluid transport. We will meet the Navier-Stokes equations in Sect. 3.6.3 (briefly) and its stationary version in Sect. 10.3.

1.3 Well Posed Problems

Usually, in the construction of a mathematical model, only some of the general laws of continuum mechanics are relevant, while the others are eliminated through the constitutive laws or suitably simplified according to the current situation. In general, additional information are necessary to select or to predict the existence of a unique solution. These information are commonly supplied in the form of *initial and/or boundary conditions*, although other forms are possible. For instance, typical boundary conditions prescribe the value of the solution or of its normal derivative,

or a combination of the two, at the boundary of the relevant domain. A main goal of a theory is to establish suitable conditions on all the data in order to have a problem with the following features:

a) *There exists at least one solution.*
b) *There exists at most one solution.*
c) *The solution depends continuously on the data.*

This last condition requires some explanation. Roughly speaking, property c) states that the correspondence

$$data \to solution \tag{1.6}$$

is *continuous* or, in other words, that a *small error on the data entails a small error on the solution.*

This property is extremely important and may be expressed as a **local stability of the solution with respect to the data**. Think for instance of using a computer to find an approximate solution: the insertion of the data and the computation algorithms entail approximation errors of various type. A significant sensitivity of the solution on small variations of the data would produce an unacceptable result.

The notion of continuity and the error measurements, both in the data and in the solution, are made precise by introducing a suitable notion of *distance.* In dealing with a numerical or a finite dimensional set of data, an appropriate distance may be the usual *euclidean distance*: if $\mathbf{x} = (x_1, x_2, \ldots, x_n)$, $\mathbf{y} = (y_1, y_2, \ldots, y_n)$ then

$$\text{dist}(\mathbf{x}, \mathbf{y}) = |\mathbf{x} - \mathbf{y}| = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}.$$

When dealing for instance with real functions, defined on a set A, common distances are:

$$\text{dist}(f, g) = \max_{\mathbf{x} \in A} |f(\mathbf{x}) - g(\mathbf{x})|,$$

which measures the maximum difference between f and g over A, or

$$\text{dist}(f, g) = \sqrt{\int_A (f - g)^2},$$

related to the so called *root-mean-square distance between* f and g.

Once the notion of distance has been chosen, the continuity of the correspondence (1.6) is easy to understand: *if the distance of the data tends to zero then the distance of the corresponding solutions tends to zero.*

When a problem possesses the properties a), b) c) above it is said to be **well posed**. When using a mathematical model, it is extremely useful, sometimes essential, to deal with well posed problems: the existence of a solution indicates that the model has an intrinsic coherence, while the uniqueness and the stability increase the possibility of providing accurate numerical approximations.

As one can imagine, complex models lead to complicated problems which require rather sophisticated techniques of theoretical analysis. Often, these problems become well posed and efficiently treatable by numerical methods if suitably reformulated in the abstract framework of Functional Analysis, as we will see in Chap. 6.

On the other hand, not only well posed problems are interesting for the applications. There are problems that are intrinsically *ill posed* because of the lack of uniqueness or of stability, but still of great interest for the modern technology. We only mention an important class of ill posed problems, given by the so called **inverse problems**, of which we provide a simple example in Sect. 5.9.

1.4 Basic Notations and Facts

We specify some of the symbols we will constantly use throughout the book and recall some basic notions about sets, topology and functions.

Sets and Topology. We denote by: $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ the sets of natural numbers, (relative) integers, rational, real and complex numbers, respectively. $\mathbb{R}^n$ is the n-dimensional vector space of the n-tuples of real numbers. We denote by $\mathbf{e}^1, \dots, \mathbf{e}^n$ the unit vectors of the canonical base in $\mathbb{R}^n$. In $\mathbb{R}^2$ and $\mathbb{R}^3$ we may denote them by $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$.

The symbols $B_r(\mathbf{x})$ and $\partial B_r(\mathbf{x})$ denote, respectively, the *open* ball and the spherical surface in $\mathbb{R}^n$, with radius r and center at x, that is:

$$B_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{y}| < r\}, \qquad \partial B_r(\mathbf{x}) = \{\mathbf{y} \in \mathbb{R}^n : |\mathbf{x} - \mathbf{y}| = r\}.$$

If there is no need to specify the radius, we write simply $B(\mathbf{x}), \partial B(\mathbf{x})$. The volume of $B_r(\mathbf{x})$ and the area of $\partial B_r(\mathbf{x})$ are given by

$$|B_r(\mathbf{x})| = \frac{\omega_n}{n} r^n \quad \text{and} \quad |\partial B_r(\mathbf{x})| = \omega_n r^{n-1},$$

where ω_n is the surface area of the unit sphere[1] ∂B_1 in $\mathbb{R}^n$; in particular $\omega_2 = 2\pi$ and $\omega_3 = 4\pi$.

Let $A \subseteq \mathbb{R}^n$. A point $\mathbf{x} \in \mathbb{R}^n$ is:

- An *interior point* if there exists a ball $B_r(\mathbf{x}) \subset A$; in particular $\mathbf{x} \in A$.
- A *boundary point* if *any* ball $B_r(\mathbf{x})$ contains points of A and of its complement $\mathbb{R}^n \setminus A$. The set of boundary points of A, the *boundary* of A, is denoted by ∂A; observe that $\partial A = \partial(\mathbb{R}^n \setminus A)$.
- A *cluster point* of A if every ball $B_r(\mathbf{x})$, $r > 0$ contains infinitely many points of A. Note that this is equivalent to asking that there exists a sequence $\{\mathbf{x}_m\} \subset A$, not containing $\mathbf{x}$, such that $\mathbf{x}_m \to \mathbf{x}$ as $m \to +\infty$. If $\mathbf{x} \in A$ and it is not a cluster point for A, we say that $\mathbf{x}$ is an *isolated point* of A.

[1] In general, $\omega_n = n\pi^{n/2}/\Gamma(\frac{1}{2}n + 1)$ where $\Gamma(s) = \int_0^{+\infty} t^{s-1} e^{-t}\, dt$ is the *Euler gamma function.*

A set A is *open* if every point in A is an interior point; a *neighborhood* of a point $\mathbf{x}$ is any open set A such that $\mathbf{x} \in A$.

A set C is *closed* if its complement $\mathbb{R}^n \setminus A$ is open. The set $\overline{A} = A \cup \partial A$ is the *closure* of A; C is closed if and only if $C = \overline{C}$. Also, C is closed if and only if C is *sequentially closed*, that is if, for every sequence $\{\mathbf{x}_m\} \subset C$ such that $\mathbf{x}_m \to \mathbf{x}$, then $\mathbf{x} \in C$.

The unions of any family of open sets is open. The intersection of a *finite number* of open sets is open. The intersection of any family of closed sets is closed. The union of a *finite number* of closed sets is closed.

Since $\mathbb{R}^n$ is simultaneously open and closed, its complement, that is the empty set $\emptyset$, is also open and closed. Only $\mathbb{R}^n$ and $\emptyset$ have this property among the subsets of $\mathbb{R}^n$.

By introducing the above notion of open set, $\mathbb{R}^n$ becomes a *topological space*, equipped with the so called *Euclidean topology*.

An open set A is *connected* if for every pair of points $\mathbf{x}, \mathbf{y} \in A$ there exists a regular curve joining them, entirely contained in A. Equivalently, A, open, is connected if it is not the union of two nonempty open subsets. By a *domain* we mean an *open connected* set. Domains are usually denoted by the letter Ω.

A set A is *convex* if for every $\mathbf{x}, \mathbf{y} \in A$, the segment

$$[\mathbf{x}, \mathbf{y}] = \{\mathbf{x} + s(\mathbf{y} - \mathbf{x}) : 0 \le s \le 1\}$$

is contained in A. Clearly, any convex set is connected.

If $E \subset A$, we say that E is *dense* in A if $\overline{E} = \overline{A}$. This means that any point $\mathbf{x} \in A$ is either an isolated point of E or a cluster point of E. For instance, $\mathbb{Q}$ is dense in $\mathbb{R}$.

A is *bounded* if it is contained in some ball $B_r(\mathbf{0})$. The family of *compact sets* is particularly important. Let $K \subset \mathbb{R}^n$. First, we say that a family F of open sets is an *open covering* of K if

$$K \subset \bigcup_{A \in \mathcal{F}} A.$$

K is *compact* if **every** open covering F of K includes a **finite** covering of K. K is *sequentially compact* if, from every sequence $\{\mathbf{x}_m\} \subset K$, there exists a subsequence $\{\mathbf{x}_{m_k}\}$ such that $\mathbf{x}_{m_k} \to \mathbf{x} \in K$ as $k \to +\infty$.

A subset K of $\mathbb{R}^n$ is compact if and only if it is *closed and bounded*, and if and only if it is *sequentially compact*.

If $\overline{E}$ is compact and contained in A, we write $E \subset\subset A$ and we say that E is *compactly contained* in A.

Relative topology. In some situation it is convenient to consider a subset $E \subset \mathbb{R}^n$ as a topological space in itself, equipped with an *induced* or *relative* Euclidean topology. In this topology we say that a set $A_0 \subset E$ is (*relatively*) *open in* E, if A_0 can be written as the intersection between E and an open set in $\mathbb{R}^n$. That is if $E_0 = E \cap A$ for some A, open in $\mathbb{R}^n$.

Accordingly, a set $E_0 \subseteq E$ is relatively closed in E if $E \setminus E_0$ is relatively open. Clearly, a relatively open/closed set in E could be neither open nor closed in the

whole $\mathbb{R}^n$. For instance, the interval $[-1,0)$ is relatively open in $E=[-1,1]$ and relatively closed in $F=(-2,0)$, since, say,

$$[-1,0)=E\cap(-2,0)=F\cap[-1,1],$$

but it is neither open nor closed in $\mathbb{R}$.

On the other hand, if $E=(-1,0)\cup(0,1)$ then, the two intervals $(-1,0)$ and $(0,1)$, both open in the topology of $\mathbb{R}$, are simultaneously open and closed nonempty subsets in the relative topology of E. Clearly this fact can occur only if E is disconnected. In fact, *if E is connected, E and $\emptyset$ are the only simultaneously relatively open and closed subsets in E.*

Infimum and supremum of a set of real numbers. A (nonempty) set $A\subset\mathbb{R}$ is *bounded from below* if there exists a number l such that $l\le x$ for every $x\in A$. The greatest among the numbers l with such property is called the *infimum* or the *greatest lower bound* of A and denoted by $\inf A$.

More precisely, we say that $\lambda=\inf A$ if:

- $\lambda\le x$ for every $x\in A$ and
- for every $\varepsilon>0$, we can find $\bar{x}\in A$ such that $\bar{x}<\lambda+\varepsilon$.

If $\inf A\in A$, then $\inf A$ is actually called the *minimum* of A, and may be denoted by $\min A$.

Similarly, $A\subset\mathbb{R}$ is *bounded from above* if there exists a number L such that $x\le L$ for every $x\in A$. The smallest among the numbers L with such property is called the *supremum* or the *least upper bound* of A and denoted by $\sup A$.

Precisely, we say that $\Lambda=\sup A$ if:

- $\Lambda\ge x$ for every $x\in A$ and
- for every $\varepsilon>0$, we can find $\bar{x}\in A$ such that $\bar{x}>\Lambda-\varepsilon$.

If $\sup A\in A$, then $\sup A$ is actually called the *maximum* of A, and may be denoted by $\max A$.

If A is unbounded from above or below, we set, respectively,

$$\sup A=+\infty\ \text{ or }\ \inf A=-\infty.$$

Upper and lower limits. Let $\{x_n\}$ be a sequence of real numbers. We say that $l\in\mathbb{R}\cup\{+\infty\}\cup\{-\infty\}$ is a *limit point* of $\{x_n\}$ if there exists a subsequence $\{x_{n_k}\}$ such that $x_{n_k}\to l$ as $k\to+\infty$.

Let E be the set of the limit points of $\{x_n\}$ and put

$$\lambda=\inf E\ \text{ and }\Lambda=\sup E.$$

The extended real numbers λ,Λ are called the *lower* and *upper limits* of $\{x_n\}$. We use the notations

$$\lambda=\liminf_{n\to\infty}x_n\quad\text{and}\quad\Lambda=\limsup_{n\to\infty}x_n.$$

For instance, if $x_n=\cos n$, we have $E=[-1,1]$, so that

$$\lambda = \liminf_{n\to\infty} \cos n = -1 \quad \text{and} \quad \Lambda = \limsup_{n\to\infty} \cos n = 1.$$

Clearly, $\lim_{n\to\infty} x_n = l$ if and only if $E = \{l\}$; then

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = l.$$

Functions. Let $A \subseteq \mathbb{R}^n$ and $u : A \to \mathbb{R}$ be a real valued function, defined in A. We say that u is *bounded from below* (resp. *above*) in A if the image

$$u(A) = \{y \in \mathbb{R}, y = u(\mathbf{x}) \text{ for some } \mathbf{x} \in A\}$$

is *bounded from below* (resp. *above*). The infimum (supremum) of $u(A)$ is called the *infimum* (*supremum*) *of* u and is denoted by

$$\inf_{\mathbf{x}\in A} u(\mathbf{x}) \ \ (\text{resp. } \sup_{\mathbf{x}\in A} u(\mathbf{x})), \qquad \text{or} \qquad \inf_A u \ \ (\text{resp. } \sup_A u).$$

If the infimum (supremum) of $u(A)$ belongs to $u(A)$ then it is *the minimum* (*maximum*) of u.

We say that u is *continuous* at $\mathbf{x} \in A$ if $\mathbf{x}$ is an isolated point of A or if $u(\mathbf{y}) \to u(\mathbf{x})$ as $\mathbf{y} \to \mathbf{x}$. If u is continuous at any point of A we say that u is continuous in A. The set of such functions is denoted by $C(A)$.

If K is compact and $u \in C(K)$ then u attains its maximum and minimum in K (*Weierstrass Theorem*).

The *support* of a function $u : A \to \mathbb{R}$ is the *closure in A of the set where it is different from zero.* In formulas:

$$\operatorname{supp}(u) = \text{closure in } A \text{ of } \{\mathbf{x} \in A : u(\mathbf{x}) \neq 0\}.$$

We will denote by χ_A the *characteristic function of* A: $\chi_A = 1$ in A and $\chi_A = 0$ in $\mathbb{R}^n \setminus A$. The support of χ_A is $\overline{A}$.

A function is *compactly supported* in A if it vanishes outside a compact set contained in A.

We use one of the symbols u_{x_j}, $\partial_{x_j} u$, $\frac{\partial u}{\partial x_j}$ for the first partial derivatives of u, and ∇u or $\operatorname{grad} u$ for the *gradient* of u :

$$\nabla u = \operatorname{grad} u = (u_{x_1}, \ldots, u_{x_n}).$$

Accordingly, we use the notations $u_{x_j x_k}$, $\partial_{x_j x_k} u$, $\frac{\partial^2 u}{\partial x_j \partial x_k}$ and so on for the higher order derivatives.

Given a unit vector $\boldsymbol{\nu}$, we use one the symbols $\nabla u \cdot \boldsymbol{\nu}$, $\partial_{\boldsymbol{\nu}} u$ or $\frac{\partial u}{\partial \boldsymbol{\nu}}$ to denote the derivative of a differentiable function u in the direction $\boldsymbol{\nu}$.

Let Ω be a domain. We say that u is *of class* $C^k(\Omega)$, $k \geq 1$, or that it is a C^k-function in Ω, if u has continuous partial derivatives up to the order k (included) in Ω. The class of continuously differentiable functions of any order in Ω is denoted by

$C^\infty(\Omega)$. The symbol $C^k(\overline{\Omega})$ denotes the set of functions in $C^k(\Omega)$, whose derivatives, up to order k included, have a continuous extension up to $\partial\Omega$.

If $u \in C^1(\Omega)$ then u is differentiable in Ω and we can write, for $\mathbf{x} \in \Omega$ and $\mathbf{h} \in \mathbb{R}^n$, small:

$$u(\mathbf{x}+\mathbf{h}) - u(\mathbf{x}) = \nabla u(\mathbf{x}) \cdot \mathbf{h} + o(\mathbf{h})$$

where the symbol $o(\mathbf{h})$, "*little o of* $\mathbf{h}$", denotes a quantity such that $o(\mathbf{h})/|\mathbf{h}| \to 0$ as $|\mathbf{h}| \to 0$.

Integrals. Up to Chap. 5 included, the integrals can be considered in the Riemann sense (proper or improper). A brief introduction to the Lebesgue measure and integral is provided in Appendix A.

• *Limit, derivative and integrals.* Let $E = K \times (a,b)$, $K \subset \mathbb{R}^n$, compact, and $u : E \to \mathbb{R}$ be continuous. The function

$$\Phi(t) = \int_K u(\mathbf{x},t)d\mathbf{x}$$

is continuous in (a,b) and if $t_0 \in (a,b)$,

$$\lim_{t\to t_0} \Phi(t) = \int_K u(\mathbf{x},t_0)d\mathbf{x}.$$

If u_t exists and it is continuous in E, then Φ is differentiable in (a,b) and

$$\Phi'(t) = \int_K u_t(\mathbf{x},t)d\mathbf{x}.$$

• *Transport in dimension* $n = 1$. Let

$$\varphi(t) = \int_{\alpha(t)}^{\beta(t)} u(x,t)dx.$$

If $\alpha, \beta \in C^1(a,b)$ and $u, u_t \in C(K)$ then $\varphi \in C^1(a,b)$ and

$$\varphi'(t) = \int_{\alpha(t)}^{\beta(t)} u_t(x,t)dx + u(\beta(t))\beta'(t) - u(\alpha(t))\alpha'(t). \tag{1.7}$$

• Let u be a continuous function on an open set $A \subset \mathbb{R}^n$. If

$$\int_B u = 0 \text{ for any ball } B \subset\subset A$$

then $u = 0$ in A.

• *Hölder inequality.* Let $1 < p < \infty$ and $q = p/(p-1)$, the *conjugate exponent of* p. Provided all the integrals are well defined, the following Hölder inequality holds

$$\left|\int_A uv\right| \le \left(\int_A |u|^p\right)^{1/p} \left(\int_A |v|^q\right)^{1/q}. \tag{1.8}$$

The case $p = q = 2$ is known as the *Schwarz inequality*.

Series of functions.

• *Uniform convergence.* A series $\sum_{m=1}^{\infty} u_m$, where $u_m : A \subseteq \mathbb{R}^n \to \mathbb{R}$, is said to be *uniformly convergent in A, with sum u* if, setting $S_N = \sum_{m=1}^{N} u_m$, we have

$$\sup_{\mathbf{x}\in A} |S_N(\mathbf{x}) - u(\mathbf{x})| \to 0 \text{ as } N \to \infty.$$

• *Weierstrass test.* Let $|u_m(\mathbf{x})| \le a_m$, for every $m \ge 1$ and $\mathbf{x} \in A$. If the series $\sum_{m=1}^{\infty} a_m$ is convergent, then $\sum_{m=1}^{\infty} u_m$ converges absolutely and uniformly in A.

• *Limit and series.* Let $\sum_{m=1}^{\infty} u_m$ be uniformly convergent in A. If u_m is continuous at $\mathbf{x}_0$ for every $m \ge 1$, then u is continuous at $\mathbf{x}_0$ and

$$\lim_{\mathbf{x}\to\mathbf{x}_0} \sum_{m=1}^{\infty} u_m(\mathbf{x}) = \sum_{m=1}^{\infty} u_m(\mathbf{x}_0).$$

• *Term by term integration.* Let $\sum_{m=1}^{\infty} u_m$ be uniformly convergent in A. If A is bounded and u_m is integrable in A for every $m \ge 1$, then:

$$\int_A \sum_{m=1}^{\infty} u_m = \sum_{m=1}^{\infty} \int_A u_m.$$

• *Term by term differentiation.* Let A be a bounded open set and $u_m \in C^1(\overline{A})$, for every $m \ge 0$. If:

1. The series $\sum_{m=1}^{\infty} u_m$ is convergent at some $\mathbf{x}_0 \in A$.
2. The series $\sum_{m=1}^{\infty} \partial_{x_j} u_m$ are uniformly convergent in $\overline{A}$, for every $j = 1, \dots, n$.

Then $\sum_{m=1}^{\infty} u_m$ converges uniformly in $\overline{A}$, its sum belongs to $C^1(\overline{A})$ and

$$\partial_{x_j} \sum_{m=1}^{\infty} u_m(\mathbf{x}) = \sum_{m=1}^{\infty} \partial_{x_j} u_m(\mathbf{x}) \qquad (j = 1, \dots, n).$$

1.5 Fourier Series

Fourier coefficients and series. Let u be a $2T$-periodic function in $\mathbb{R}$, integrable on $[-T, T]$. The *Fourier coefficients* of u are given by the formulas

$$a_k = \frac{1}{T} \int_{-T}^{T} u(x) \cos k\omega x \, dx, \qquad b_k = \frac{1}{T} \int_{-T}^{T} u(x) \sin k\omega x \, dx, \tag{1.9}$$

where $k \in \mathbb{N}$ and $\omega = \pi/T$. We can *associate* with u its *Fourier series* by writing

$$u \sim \frac{a_0}{2} + \sum_{k=1}^{\infty} \{a_k \cos k\omega x + b_k \sin k\omega x\}. \tag{1.10}$$

• *Odd and even functions.* If u is an *odd* function, i.e. $u(-x) = -u(x)$, we have $a_k = 0$ for every $k \ge 0$, while

$$b_k = \frac{2}{T}\int_0^T u(x)\sin k\omega x\, dx.$$

Thus, if u is odd, its Fourier series is a *sine* Fourier series:

$$u \sim \textstyle\sum_{k=1}^{\infty} b_k \sin k\omega x.$$

Similarly, if u is *even*, i.e. $u(-x) = u(x)$, we have $b_k = 0$ for every $k \geq 1$, while

$$a_k = \frac{2}{T}\int_0^T u(x)\cos k\omega x\, dx.$$

Thus, if u is even, its Fourier series is a *cosine* Fourier series:

$$u \sim \frac{a_0}{2} + \textstyle\sum_{k=1}^{\infty} a_k \cos k\omega x.$$

• *Fourier coefficients of a derivative*. Let $u \in C^1(\mathbb{R})$ be $2T$-periodic. Then we may compute the Fourier coefficients a'_k and b'_k of u'. We have, integrating by parts, for $k \geq 1$:

$$\begin{aligned} a'_k &= \frac{1}{T}\int_{-T}^{T} u'(x)\cos k\omega x\, dx \\ &= \frac{1}{T}\left[u(x)\cos k\omega x\right]_{-T}^{T} + \frac{k\omega}{T}\int_{-T}^{T} u(x)\sin k\omega x\, dx \\ &= \frac{k\omega}{T}\int_{-T}^{T} u(x)\sin k\omega x\, dx = k\omega b_k \end{aligned}$$

and

$$\begin{aligned} b'_k &= \frac{1}{T}\int_{-T}^{T} u'(x)\sin k\omega x\, dx \\ &= \frac{1}{T}\left[u(x)\sin k\omega x\right]_{-T}^{T} - \frac{k\omega}{T}\int_{-T}^{T} u(x)\cos k\omega x\, dx \\ &= -\frac{k\omega}{T}\int_{-T}^{T} u(x)\cos k\omega x\, dx = -k\omega a_k. \end{aligned}$$

Thus, the Fourier coefficients a'_k and b'_k are related to a_k and b_k by the following formulas:

$$a'_k = k\omega b_k, \qquad b'_k = -k\omega a_k. \tag{1.11}$$

• *Complex form of a Fourier series*. Using the Euler identities

$$e^{\pm ik\omega x} = \cos k\omega x \pm i \sin k\omega x$$

the Fourier series (1.10) can be expressed in the complex form

$$\sum_{k=-\infty}^{\infty} c_k e^{ik\omega x},$$

where the complex Fourier coefficients c_k are given by

$$c_k = \frac{1}{2T}\int_{-T}^{T} u(z)e^{-ik\omega z}dz.$$

The relations among the real and the complex Fourier coefficients are:

$$c_0 = \frac{1}{2}a_0$$

and

$$c_k = \frac{1}{2}(a_k - ib_k),\ c_{-k} = \bar{c}_k \quad \text{for } k > 0.$$

Expansion in Fourier Series. Once a Fourier series is associated to a function u as in (1.10), the following questions arise.

1. Which conditions on u do assure "the convergence" of its Fourier series? Of course there are several notions of convergence (e.g. pointwise, uniform, least squares).
2. If the Fourier series is convergent in some sense, does it always have sum u?

A complete answer to the above questions is not elementary. The convergence of a Fourier series is a rather delicate matter. We indicate some basic results (for the proofs, see e.g. *W. Rudin*, [39] or *R. L. Wheeden, A. Zygmund*, [42].

• *Quadratic mean or L^2-convergence.* This is perhaps the most natural type of convergence for Fourier series (see Sect. 6.4.2). Let

$$S_N(x) = \frac{a_0}{2} + \sum_{k=1}^{N}\{a_k \cos k\omega x + b_k \sin k\omega x\}$$

be the N-partial sum of the Fourier series of u. We have

Theorem 1.1 *Let u be a square integrable function[2] on $(-T, T)$. Then*

$$\lim_{N\to+\infty}\int_{-T}^{T}[S_N(x) - u(x)]^2\,dx = 0.$$

Moreover, the following Parseval relation holds:

$$\frac{1}{T}\int_{-T}^{T}u^2 = \frac{a_0^2}{2} + \sum_{k=1}^{\infty}(a_k^2 + b_k^2). \tag{1.12}$$

Since the numerical series in the right hand side of (1.12) is convergent, we deduce the following important consequence:

Corollary 1.2 (Riemann-Lebesgue) $\lim_{k\to+\infty} a_k = \lim_{k\to+\infty} b_k = 0.$

[2] That is $\int_{-T}^{T} u^2 < \infty$.

• *Pointwise convergence.* We say that u satisfies the *Dirichlet property* in $[-T,T]$ if it is continuous in $[-T,T]$ except possibly at a finite number of points of jump discontinuity and moreover if the interval $[-T,T]$ can be partitioned in a finite number of subintervals such that u is monotone in each one of them.

The following theorem holds.

Theorem 1.3 *If u satisfies the Dirichlet property in $[-T,T]$ then the Fourier series of u converges at each point of $[-T,T]$. Moreover*[3]*:*

$$\frac{a_0}{2} + \sum_{k=1}^{\infty} \{a_k \cos k\omega x + b_k \sin k\omega x\} = \begin{cases} \dfrac{u(x+) + u(x-)}{2} & x \in (-T,T) \\ \dfrac{u(T-) + u(-T+)}{2} & x = \pm T \end{cases}$$

In particular, under the hypotheses of Theorem 1.3, at every point x of continuity of u, the Fourier series converges to $u(x)$.

• *Uniform convergence.* A simple criterion of uniform convergence is provided by the Weierstrass test (see Sect. 1.4). Since

$$|a_k \cos k\omega x + b_k \sin k\omega x| \le |a_k| + |b_k|$$

we deduce: *If the numerical series*

$$\sum_{k=1}^{\infty} |a_k| \quad \textit{and} \quad \sum_{k=1}^{\infty} |b_k|$$

converge, then the Fourier series of u is uniformly convergent in $\mathbb{R}$, with sum u.

This is the case, for instance, if $u \in C^1(\mathbb{R})$ and is $2T$ periodic. In fact, from (1.11) we have for every $k \ge 1$,

$$a_k = -\frac{1}{\omega k} b_k' \quad \text{and} \quad b_k = \frac{1}{\omega k} a_k'.$$

Therefore

$$|a_k| \le \frac{1}{\omega^2 k^2} + (b_k')^2$$

and

$$|b_k| \le \frac{1}{\omega^2 k^2} + (a_k')^2.$$

Now, the series $\sum \frac{1}{k^2}$ is convergent. On the other hand, also the series

$$\sum_{k=1}^{\infty} (a_k')^2 \quad \text{and} \quad \sum_{k=1}^{\infty} (b_k')^2$$

are convergent, by Parseval's relation (1.12) applied to u' in place of u. The conclusion is that *if $u \in C^1(\mathbb{R})$ and $2T$ periodic, its Fourier series is uniformly convergent in $\mathbb{R}$ with sum u.*

Another useful result is a refinement of Theorem 1.3.

[3] We set $f(x\pm) = \lim_{y \to \pm x} f(y)$.

Theorem 1.4 *Assume u satisfies the Dirichlet property in $[-T,T]$. Then:*

a) If u is continuous in $[a,b] \subset (-T,T)$, then its Fourier series converges uniformly in $[a,b]$.
b) If u is continuous in $[-T,T]$ and $u(-T) = u(T)$, then its Fourier series converges uniformly in $[-T,T]$ (and therefore in $\mathbb{R}$).

1.6 The Inverse and the Implicit Function Theorems

• *Inverse Function Theorem.* Consider the following system of n equations

$$\begin{cases} f_1(x_1,..,x_n) = \xi_1 \\ \cdots \\ f_n(x_1,..,x_n) = \xi_n \end{cases} \tag{1.13}$$

or, in concise form,

$$\mathbf{f}(\mathbf{x}) = \boldsymbol{\xi}.$$

System (1.13) can be interpreted as a transformation of coordinates, from $\mathbf{x}$ to $\boldsymbol{\xi}$, and solving it for $\mathbf{x}$ amounts to determine the inverse of $\mathbf{f}$, $\mathbf{x} = \mathbf{f}^{-1}(\boldsymbol{\xi})$.

We denote by $D\mathbf{f}$ the *jacobian matrix* of $\mathbf{f}$, given by

$$D\mathbf{f} = \begin{pmatrix} \nabla f_1 \\ \vdots \\ \nabla f_n \end{pmatrix} = \left(\frac{\partial f_i}{\partial x_j} \right)_{i,j=1,\ldots,n}.$$

The following theorem gives conditions under which an (at least local) inverse exists.

Theorem 1.5 *Let $A \subseteq \mathbb{R}^n$ be open and $\mathbf{f} : A \to \mathbb{R}^n$. Assume that $\mathbf{f} \in C^1(A)$ and that, at $\mathbf{x}_0 \in A$,*

$$\det D\mathbf{f}(\mathbf{x}_0) \neq 0.$$

Then $\mathbf{f}^{-1}$ exists and it is C^1 in a neighborhood V of $\boldsymbol{\xi}_0 = \mathbf{f}(\mathbf{x}_0)$. Moreover, in V,

$$D\mathbf{f}^{-1}(\boldsymbol{\xi}) = (D\mathbf{f}(\mathbf{x}))^{-1}, \quad \boldsymbol{\xi} = \mathbf{f}(\mathbf{x}).$$

Thus, if $U = \mathbf{f}^{-1}(V)$, $\mathbf{f}$ is a one-to-one transformation between U and V, and it is C^1 in both directions. We say that $\mathbf{f}$ is a *local diffeomorphism* between U and V.

• *Implicit Function Theorem.* At times we will need to use functions $u = u(\mathbf{x})$, *implicitly defined* by equations of the form

$$g(\mathbf{x}, u) = 0.$$

More precisely, this means that $g(\mathbf{x}, u(\mathbf{x})) = 0$ for all $\mathbf{x}$ in some set $U \subseteq \mathbb{R}^n$. The following result holds:

Theorem 1.6 *Let $g \in C^1(A)$, $A \subseteq \mathbb{R}^{n+1}$, open. Assume that $g(\mathbf{x}_0, u_0) = 0$ and $g_u(\mathbf{x}_0, u_0) \neq 0$. Then, there exists a unique implicit function $u \in C^1(U)$, where U is a neighborhood of $\mathbf{x}_0$. Moreover, for all $\mathbf{x} \in U$,*

$$u_{x_j}(\mathbf{x}) = -\frac{g_{x_j}(\mathbf{x}, u(\mathbf{x}))}{g_u(\mathbf{x}, u(\mathbf{x}))} \qquad j = 1, ..., n.$$

1.7 Smooth and Lipschitz Domains

We will need, especially in Chaps. 7–10, to distinguish the domains Ω in $\mathbb{R}^n$ according to the degree of smoothness of their boundary.

Definition 1.7 We say that Ω is a C^1-*domain* if for every point $\mathbf{p} \in \partial\Omega$, upon a rotation and translation if necessary, there exists a system of coordinates $(y_1, \ldots, y_{n-1}, y_n) \equiv (\mathbf{y}', y_n)$, with origin at $\mathbf{p}$, a ball $B(\mathbf{p})$ and a function $\varphi_{\mathbf{p}}$ defined in a neighborhood $\mathcal{N}_{\mathbf{p}} \subset \mathbb{R}^{n-1}$ of $\mathbf{y}' = \mathbf{0}'$, such that

$$\varphi_{\mathbf{p}} \in C^1(\mathcal{N}_{\mathbf{p}}), \quad \varphi_{\mathbf{p}}(\mathbf{0}') = 0$$

and

1. $\partial\Omega \cap B(\mathbf{p}) = \{(\mathbf{y}', y_n) : y_n = \varphi_{\mathbf{p}}(\mathbf{y}'), \mathbf{y}' \in \mathcal{N}_{\mathbf{p}}\}$.
2. $\Omega \cap B(\mathbf{p}) = \{(\mathbf{y}', y_n) : y_n > \varphi_{\mathbf{p}}(\mathbf{y}'), \mathbf{y}' \in \mathcal{N}_{\mathbf{p}}\}$.

The first condition expresses the fact that $\partial\Omega$ locally coincides with the graph of a C^1-function. The second one requires that Ω is locally placed on one side of its boundary (see Fig. 1.1).

The boundary of a C^1-domain does not have corners or edges and for every point $\mathbf{p} \in \partial\Omega$, a tangent straight line ($n = 2$) or plane ($n = 3$) or hyperplane ($n > 3$) is well defined, *together* with the *outward* and *inward* normal unit vectors. Moreover these vectors vary continuously on $\partial\Omega$.

The couples $(\varphi_{\mathbf{p}}, \mathcal{N}_{\mathbf{p}})$ appearing in the above definition are called *local charts*. If the functions $\varphi_{\mathbf{p}}$ are all C^k-functions, for some $k \geq 1$, Ω is said to be a C^k-domain. If Ω is a C^k-domain for every $k \geq 1$, it is said to be a C^∞-domain. These are the domains we consider *smooth* domains.

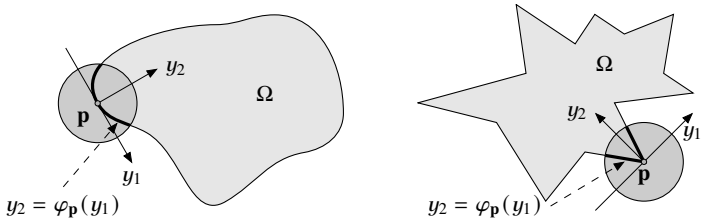

Fig. 1.1 A C^1-domain and a Lipschitz domain

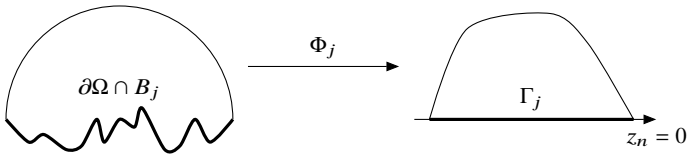

Fig. 1.2 Locally straightening $\partial\Omega$ through the diffeomorphism Φ_j

If Ω is bounded then $\partial\Omega$ is compact and it is possible to cover $\partial\Omega$ by a finite numbers of balls $B_j = B(\mathbf{p}_j)$, $j = 1, \ldots, N$, centered at $\mathbf{p}_j \in \partial\Omega$. Thus, the boundary of Ω can be described by the finite family of local charts $(\varphi_j, \mathcal{N}_j) = (\varphi_{\mathbf{p}_j}, \mathcal{N}_{\mathbf{p}_j})$, $j = 1, \ldots, N$. Observe that the one-to-one transformation (*diffeomorphism*) $\mathbf{z} = \Phi_j(\mathbf{y})$, given by

$$\begin{cases} \mathbf{z}' = \mathbf{y}' \\ z_n = y_n - \varphi_j(\mathbf{y}') \end{cases} \qquad \mathbf{y}' \in \mathcal{N}_j \,, \tag{1.14}$$

maps $\partial\Omega \cap B_j$ into a subset Γ_j of the hyperplane $z_n = 0$, so that $\partial\Omega \cap B_j$ *straightens*, as it is shown in Fig. 1.2.

In a great number of applications the relevant domains are rectangles, prisms, cones, cylinders or unions of them. Very important are polygonal domains obtained by *triangulation procedures* of smooth domains, for numerical approximations. These types of domains belong to the class of *Lipschitz domains*, whose boundary is locally described by the graph of a *Lipschitz* function.

Definition 1.8 We say that $u : \Omega \to \mathbb{R}$ is *Lipschitz* if there exists L such that

$$|u(\mathbf{x}) - u(\mathbf{y})| \leq L\,|\mathbf{x} - \mathbf{y}|$$

for every $\mathbf{x}, \mathbf{y} \in \Omega$. The number L is called the *Lipschitz constant* of u.

Roughly speaking, a function is Lipschitz in Ω if its increment quotients in every direction are bounded. In fact, Lipschitz functions are differentiable at all points of their domain with the exception of a negligible set of points. Precisely, we have (see e.g. *W. P. Ziemer*, [44]):

Theorem 1.9 (Rademacher) *Let u be a Lipschitz function in $\Omega \subseteq \mathbb{R}^n$. Then u is differentiable at every point of Ω, except at a set of points of Lebesgue measure zero.*

Typical real Lipschitz functions in $\mathbb{R}^n$ are $f(\mathbf{x}) = |\mathbf{x}|$ or, more generally, the distance function from a *closed* set, C, defined by

$$f(\mathbf{x}) = \operatorname{dist}(\mathbf{x}, C) = \inf_{\mathbf{y} \in C} |\mathbf{x} - \mathbf{y}| \,.$$

Definition 1.10 We say that a bounded domain Ω is *Lipschitz* if $\partial\Omega$ can be described by a family of local charts $(\varphi_j, \mathcal{N}_j)$, $j = 1, \ldots, N$, where the functions φ_j, $j = 1, \ldots, N$, are Lipschitz or, equivalently, if each map (1.14) is a bi-Lipschitz transformation, that is, both Φ_j and Φ_j^{-1} are Lipschitz.

The so called mixed boundary value problems in dimension $n \geq 2$ require a splitting of the boundary of a domain Ω into subsets having some regularity. Typical examples of regular subsets in dimension $n = 2$ are unions of smooth arcs contained in $\partial\Omega$ and union of faces of a polyhedron for $n = 3$. Most applications deal with sets of this type.

For the sake of completeness, we introduce below a more general notion of regular subsets of $\partial\Omega$. Since the matter could quickly become very technical, we confine ourselves to a common case. Let Γ be a *relatively open* subset of $\partial\Omega$. Γ may have a boundary $\partial\Gamma$ with respect to the relative topology of $\partial\Omega$.

For instance, let Ω be a three dimensional spherical shell, bounded by two concentric spheres Γ_1 and Γ_2. Both Γ_1 and Γ_2 are relatively open[4] subsets of $\partial\Omega$, with no boundary. However, if Γ is a half sphere contained, say, in Γ_1, then its boundary is a circle. We denote by $\overline{\Gamma} = \Gamma \cup \partial\Gamma$ the *closure* of Γ in $\partial\Omega$.

Definition 1.11 Let $\Omega \subset \mathbb{R}^n$, $n \geq 2$ be a bounded domain. We say that a relatively open subset $\Gamma \subset \partial\Omega$ is a *regular subset of* $\partial\Omega$ if:

1. $\overline{\Gamma}$ is locally Lipschitz; that is, for every $\mathbf{p} \in \overline{\Gamma}$ there exists a bi-Lipschitz map $\Psi_{\mathbf{p}}$ that flattens $\overline{\Gamma}$ near p into a subset of the hyperplane $\{z_n = 0\}$ as in Fig. 1.2.
2. If $\partial\Gamma \neq \emptyset$, for every $\mathbf{p} \in \partial\Gamma$, $\Psi_{\mathbf{p}}$ also straightens $\partial\Gamma$, near $\mathbf{p}$, on the hyperplane $\{z_n = 0\}$.

The second conditions implies that Γ lies on one side of $\partial\Gamma$, locally on $\partial\Omega$. The map $\Psi_{\mathbf{p}}$ may be obtained by composing the map $\Phi_{\mathbf{p}}$ that flattens $\partial\Omega$ near $\mathbf{p}$, defined as in (1.14), with another map that straightens $\Phi_{\mathbf{p}}(\partial\Gamma)$ near $\mathbf{z} = \mathbf{0}$, on the hyperplane $\{z_n = 0\}$.

1.8 Integration by Parts Formulas

Let $\Omega \subset \mathbb{R}^n$ be a bounded C^1-domain and

$$\mathbf{F} = (F_1, F_2, \ldots, F_n) : \overline{\Omega} \to \mathbb{R}^n$$

be a vector field with components F_j, $j = 1, \ldots, n$, of class $C^1(\overline{\Omega})$; we write $\mathbf{F} \in C^1(\overline{\Omega}; \mathbb{R}^n)$. The Gauss divergence formula holds:

$$\int_\Omega \operatorname{div} \mathbf{F}\, d\mathbf{x} = \int_{\partial\Omega} \mathbf{F} \cdot \boldsymbol{\nu}\, d\sigma, \tag{1.15}$$

where

$$\operatorname{div} \mathbf{F} = \sum_{j=1}^{n} \partial_{x_j} F_j$$

[4] Also closed, in this case.

denotes the divergence of $\mathbf{F}$, $\boldsymbol{\nu}$ denotes the outward normal unit vector to $\partial\Omega$ and $d\sigma$ is the "surface" measure[5] on $\partial\Omega$, locally given in terms of the local charts by[6]

$$d\sigma = \sqrt{1 + \left|\nabla\varphi_j(\mathbf{y}')\right|^2}\, d\mathbf{y}'.$$

A number of useful identities can be derived from (1.15). Applying (1.15) to $v\mathbf{F}$, with $v \in C^1(\overline{\Omega})$, and recalling the identity

$$\operatorname{div}(v\mathbf{F}) = v \operatorname{div}\mathbf{F} + \nabla v \cdot \mathbf{F},$$

we obtain the following **integration by parts** formula:

$$\int_\Omega v \operatorname{div}\mathbf{F}\, d\mathbf{x} = \int_{\partial\Omega} v\mathbf{F}\cdot\boldsymbol{\nu}\, d\sigma - \int_\Omega \nabla v\cdot\mathbf{F}\, d\mathbf{x}. \tag{1.16}$$

Choosing $\mathbf{F} = \nabla u$, $u \in C^2(\Omega)\cap C^1(\overline{\Omega})$, since

$$\operatorname{div}\nabla u = \Delta u \quad \text{and} \quad \nabla u\cdot\boldsymbol{\nu} = \partial_{\boldsymbol{\nu}} u,$$

the following **Green's identity** follows:

$$\int_\Omega v\Delta u\, d\mathbf{x} = \int_{\partial\Omega} v\partial_{\boldsymbol{\nu}} u\, d\sigma - \int_\Omega \nabla v\cdot\nabla u\, d\mathbf{x}. \tag{1.17}$$

In particular, the choice $v \equiv 1$ yields

$$\int_\Omega \Delta u\, d\mathbf{x} = \int_{\partial\Omega} \partial_{\boldsymbol{\nu}} u\, d\sigma. \tag{1.18}$$

If also $v \in C^2(\Omega)\cap C^1(\overline{\Omega})$, interchanging the roles of u and v in (1.17) and subtracting, we derive a second **Green's identity**:

$$\int_\Omega v\Delta u - u\Delta v\, d\mathbf{x} = \int_{\partial\Omega} (v\partial_{\boldsymbol{\nu}} u - u\partial_{\boldsymbol{\nu}} v)\, d\sigma. \tag{1.19}$$

Remark 1.12 All the above formulas hold for Lipschitz domains as well. In fact, the Rademacher theorem implies that at every point of the boundary of a Lipschitz domain, with the exception of a set of points of surface measure zero, there is a well defined tangent plane. This is enough for extending the formulas (1.16)–(1.19) to Lipschitz domains.

[5] In dimension $n = 2$, $\partial\Omega$ is a curve and $d\sigma$ reduces to the "arc length" measure.

[6] The definition of the integral over the boundary $\partial\Omega$ of a C^1-domain is given in Sect. 7.9.

Chapter 2
Diffusion

2.1 The Diffusion Equation

2.1.1 Introduction

The one-dimensional *diffusion equation* is the linear second order partial differential equation

$$u_t - Du_{xx} = f,$$

in the unknown $u = u(x,t)$, where x is a real space variable, t a time variable, $f = f(x,t)$ a given function and D a positive constant, called *diffusion coefficient.* In space dimension $n > 1$, that is when $\mathbf{x} \in \mathbb{R}^n$, the diffusion equation reads

$$u_t - D\Delta u = f, \tag{2.1}$$

where Δ denotes the *Laplace operator:*

$$\Delta = \sum_{k=1}^{n} \frac{\partial^2}{\partial x_k^2}.$$

When $f \equiv 0$ the equation is said to be *homogeneous* and in this case the *superposition principle* holds: if u and v are solutions of (2.1) and a, b are real (or complex) numbers, $au + bv$ also is a solution of (2.1). More generally, if $u_k(\mathbf{x}, t)$ is a family of solutions of the homogeneous equation depending on the parameter k (integer or real) and $g = g(k)$ is a function rapidly vanishing at infinity, then

$$\sum_{k=1}^{\infty} u_k(\mathbf{x}, t)g(k) \quad \text{and} \quad \int_{-\infty}^{+\infty} u_k(\mathbf{x}, t)g(k)dk$$

are still solutions of the same equation.

A common example of diffusion is given by the *heat conduction* in a solid body. Conduction comes from molecular collision, transferring heat by kinetic en-

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
https://doi.org/10.1007/978-3-031-21853-8_2

ergy, without macroscopic material movement. If the medium is homogeneous and isotropic with respect to the heat propagation, the evolution of the temperature is described by equation (2.1); f represents the intensity of an external distributed source. For this reason eq. (2.1) is also known as the **heat equation.**

On the other hand eq. (2.1) constitutes a much more general diffusion model, where by **diffusion** we mean, for instance, the *transport of a substance due to the molecular motion of the surrounding medium.* In this case, u could represent the concentration of a polluting material or of a solute in a liquid or a gas (dye in a liquid, smoke in the atmosphere) or even a probability density. We may say that the diffusion equation unifies at a macroscopic scale a variety of phenomena, that look quite different when observed at a microscopic scale.

Through equation (2.1) and some variants we will explore the deep connection between probabilistic and deterministic models, according (roughly) to the scheme

$$\text{diffusion processes} \leftrightarrow \text{probability density} \leftrightarrow \text{differential equations.}$$

The *star* in this field is the *Brownian motion,* derived from the name of the botanist Brown, who observed in 1827 the apparently chaotic behavior of certain particles on a water surface, due to the molecular motion.

This irregular motion is now modeled as a *stochastic process* under the terminology of *Wiener process* or *Brownian motion.* The operator $\frac{1}{2}\Delta$ is strictly related to the Brownian motion and indeed it captures and synthesizes the microscopic features of that process. In the theory of stochastic processes, $\frac{1}{2}\Delta$ represents the *infinitesimal generator of the Brownian motion.*

Under equilibrium conditions, that is when there is no time evolution, the solution u depends only on the space variable and satisfies the *stationary* version of the diffusion equation (letting $D = 1$)

$$-\Delta u = f \tag{2.2}$$

($-u_{xx} = f$, in dimension $n = 1$). Equation (2.2) is known as the *Poisson equation*. When $f = 0$, it is called *Laplace's equation* and its solutions are so important in so many fields that they have deserved the special name of **harmonic functions**. These equations will be considered in the next chapter.

2.1.2 The conduction of heat

Heat is a form of energy which it is frequently convenient to consider as separated from other forms. For historical reasons, *calories* instead of Joules are used as units of measurement, each *calorie* corresponding to 4.182 Joules.

We want to derive a mathematical model for the heat conduction in a solid body. We assume that the body is homogeneous and isotropic, with constant *mass density* ρ, and that it can receive energy from an external source (for instance, from an

electrical current or a chemical reaction or from absorption of radiation). Denote by r the time rate per unit mass at which heat is supplied[1] by the external source.

Since heat is a form of energy, it is natural to use the law of conservation of energy, that we can formulate in the following way:

Let V be an arbitrary control volume inside the body. *The time rate of change of thermal energy in V equals the net flux of heat through the boundary ∂V of V, due to the conduction, plus the time rate at which heat is supplied by the external sources.*

If we denote by $e = e(\mathbf{x}, t)$ the thermal energy per unit mass, the total quantity of thermal energy inside V is given by

$$\int_V e\rho \, d\mathbf{x}$$

so that, assuming that the time derivative can be carried inside the integral, its time rate of change is

$$\frac{d}{dt} \int_V e\rho \, d\mathbf{x} = \int_V e_t \rho \, d\mathbf{x}.$$

Denote by $\mathbf{q}$ the *heat flux* vector[2], which specifies the heat flow direction and the magnitude of the rate of flow across a unit area. More precisely, if $d\sigma$ is an area element contained in ∂V with *outer* unit normal $\boldsymbol{\nu}$, then

$$\mathbf{q} \cdot \boldsymbol{\nu} d\sigma$$

is the energy flow rate through $d\sigma$ and therefore the *total inner heat flux* through ∂V is given by

$$-\int_{\partial V} \mathbf{q} \cdot \boldsymbol{\nu} \, d\sigma \underset{\text{(divergence theorem)}}{=} -\int_V \operatorname{div} \mathbf{q} \, d\mathbf{x}.$$

Finally, the contribution due to the external source is given by

$$\int_V r\rho \, d\mathbf{x}.$$

Thus, conservation of energy requires:

$$\int_V e_t \rho \, d\mathbf{x} = -\int_V \operatorname{div} \mathbf{q} \, d\mathbf{x} + \int_V r\rho \, d\mathbf{x}. \tag{2.3}$$

The arbitrariness of V (see Sect. 1.4) allows us to convert the integral equation (2.3) into the pointwise relation

$$e_t \rho = -\operatorname{div} \mathbf{q} + r\rho, \tag{2.4}$$

that constitutes a basic law of heat conduction. However, e and $\mathbf{q}$ are unknown and we need additional information through *constitutive relations* for these quantities. We assume the following:

[1] Dimensions of r: $[r] = [\text{cal}] \times [\text{time}]^{-1} \times [\text{mass}]^{-1}$.

[2] $[\mathbf{q}] = [\text{cal}] \times [\text{length}]^{-2} \times [\text{time}]^{-1}$.

• **Fourier law** of heat conduction. Under "normal" conditions, for many solid materials, the heat flux is a linear function of the temperature gradient, that is:

$$\mathbf{q} = -\kappa \nabla u, \tag{2.5}$$

where u is the absolute temperature and $\kappa > 0$, the *thermal conductivity*[3], depends on the properties of the material. In general, κ may depend on u, $\mathbf{x}$ and t, but often it varies so little in the cases of interest that it is reasonable to neglect its variation. Here we consider κ *constant* so that

$$\operatorname{div} \mathbf{q} = -\kappa \Delta u. \tag{2.6}$$

The minus sign in the law (2.5) reflects the tendency of heat to flow from hotter to cooler regions.

• The thermal energy is a linear function of the absolute temperature:

$$e = c_v u \tag{2.7}$$

where c_v denotes the *specific heat*[4] (at constant volume) of the material. In many cases of interest c_v can be considered constant. The relation (2.7) is reasonably true over not too wide ranges of temperature.

Using (2.6) and (2.7), equation (2.4) becomes

$$u_t = \frac{\kappa}{c_v \rho} \Delta u + \frac{1}{c_v} r \tag{2.8}$$

which is the diffusion equation with $D = \kappa/(c_v \rho)$ and $f = r/c_v$. As we will see, the *thermal diffusivity* D encodes the thermal response time of the material.

2.1.3 Initial boundary value problems ($n = 1$)

As we have mentioned in Ch. 1, the governing equations in a mathematical model must be supplemented by additional information in order to get a *well posed problem*, i.e. a problem that has exactly one solution, depending continuously on the data.

It is not difficult to outline some typical problems for the heat equation that, on physical grounds, appear to be well posed. Consider the evolution of the temperature u inside a cylindrical bar, whose lateral surface is *perfectly insulated* and whose length is much larger than its cross-sectional area A. Although the bar is three dimensional, we may assume that heat moves only down the length of the bar and that the heat transfer intensity is uniformly distributed in each section of the bar. Thus we may assume that $e = e(x,t)$, $r = r(x,t)$, with $0 \le x \le L$. Accordingly, the constitutive relations (2.5) and (2.7) read

[3] $[\kappa] = [\text{cal}] \times [\text{deg}]^{-1} \times [\text{time}]^{-1} \times [\text{length}]^{-1}$ (deg stays for Kelvin degree).

[4] $[c_v] = [\text{cal}] \times [\text{deg}]^{-1} \times [\text{mass}]^{-1}$.

$$e(x,t) = c_v u(x,t), \quad \mathbf{q} = -\kappa u_x \mathbf{i}.$$

By choosing $V = A \times (x, x+\Delta x)$ as the control volume in (2.3), the cross-sectional area A cancels out, and we obtain

$$\int_x^{x+\Delta x} c_v \rho u_t \, dy = \int_x^{x+\Delta x} \kappa u_{xx} \, dy + \int_x^{x+\Delta x} r\rho \, dy$$

that yields for u the one-dimensional heat equation

$$u_t - D u_{xx} = f.$$

We want to study the temperature evolution during an interval of time, say, from $t = 0$ until $t = T$. It is then reasonable to prescribe its initial distribution inside the bar: different initial configurations will correspond to different evolutions of the temperature. Thus we need to prescribe **the initial (or Cauchy) condition**

$$u(x,0) = g(x),$$

where g models the initial temperature profile.

This is not enough to determine a unique evolution; it is necessary to know how the bar interacts with the surroundings. Indeed, starting with a given initial temperature distribution, we can change the evolution of u by controlling the temperature or the heat flux at the two ends of the bar; for instance, we could keep the temperature at a certain fixed level or let it vary in a certain way, depending on time. This amounts to prescribing

$$u(0,t) = h_1(t), \; u(L,t) = h_2(t) \tag{2.9}$$

at any time $t \in (0,T]$. The (2.9) are called **Dirichlet boundary conditions**.

We could also prescribe the heat flux at the end points. Since from Fourier law we have

$$\text{inward heat flow at } x = 0 : -\kappa u_x(0,t),$$

$$\text{inward heat flow at } x = L : \kappa u_x(L,t),$$

the heat flux is assigned through the **Neumann boundary conditions**

$$-u_x(0,t) = h_1(t), \, u_x(L,t) = h_2(t), \qquad t \in (0,T].$$

Another type of boundary condition is the **Robin** or **radiation condition**. Let the surroundings be kept at temperature U and assume that the *inward* heat flux from one end of the bar, say $x = L$, depends linearly on the difference $U - u$, that is

$$\kappa u_x = \gamma(U - u) \qquad (\gamma > 0). \tag{2.10}$$

Letting $\alpha = \gamma/\kappa > 0$ and $h = \gamma U/\kappa$, the Robin condition at $x = L$ reads

$$u_x + \alpha u = h.$$

Formula (2.10) is based on the linear *Newton law of cooling*: the heat loss from the surface of a body is a linear function of the temperature drop $U - u$ from the surroundings to the surface. It represents a good approximation to the radiative loss from a body when $|U - u| / u \ll 1$.

Clearly, it is possible to assign **mixed conditions**: for instance, at one end a Dirichlet condition and at the other one a Neumann condition, or a Robin one.

The problems associated with the above boundary conditions have a corresponding nomenclature. Summarizing, we can state the most common problems for the one dimensional heat equation as follows: *given* $f = f(x,t)$ *(external source) and* $g = g(x)$ *(initial or Cauchy data), determine* $u = u(x,t)$ *such that:*

$$\begin{cases} u_t - Du_{xx} = f & 0 < x < L, 0 < t < T \\ u(x,0) = g(x) & 0 \le x \le L \\ + \text{boundary conditions} & 0 < t \le T \end{cases}$$

where the boundary conditions may be:

- *Dirichlet* $u(0,t) = h_1(t),\ u(L,t) = h_2(t),$
- *Neumann* $-u_x(0,t) = h_1(t),\ u_x(L,t) = h_2(t),$
- *Robin* or *radiation*

$$-u_x(0,t) + \alpha u(0,t) = h_1(t),\ u_x(L,t) + \alpha u(L,t) = h_2(t) \qquad (\alpha > 0),$$

- or *mixed* conditions.

Accordingly, we have the initial-Dirichlet problem, the initial-Neumann problem and so on. When $h_1 = h_2 = 0$, we say that the boundary conditions are **homogeneous**.

Remark 2.1 Observe that only a special part of the boundary of the rectangle

$$Q_T = (0, L) \times (0, T)$$

carries the data, the so-called **parabolic boundary of** Q_T, denoted with $\partial_p Q_T$ (see Fig. 2.1). No final condition (for $t = T, 0 < x < L$) is required.

In important applications, for instance in financial mathematics, x varies over unbounded intervals, typically $(0, \infty)$ or $\mathbb{R}$. In these cases one has to require that the

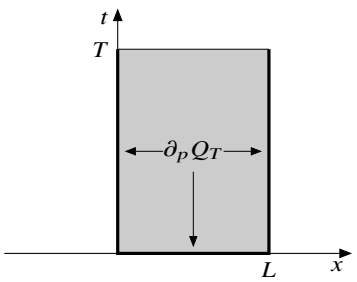

Fig. 2.1 The rectangle Q_T and its parabolic boundary

solution does not grow too much at infinity. We will later consider the *global Cauchy problem*:

$$\begin{cases} u_t - Du_{xx} = f & x \in \mathbb{R}, 0 < t < T \\ u(x,0) = g(x) & x \in \mathbb{R} \\ + \text{conditions as } x \to \pm\infty. \end{cases}$$

2.1.4 A solution by separation of variables

We will prove that, under reasonable hypotheses, the initial Dirichlet, Neumann, Robin and mixed problems are well posed. Sometimes this can be shown using elementary techniques like *the separation of variables method* that we describe below through a simple example of heat conduction. We will come back to this method from a more general point of view in Sect. 6.9.2.

As in the previous section, consider a bar (that we can consider one-dimensional) of length L, initially (at time $t = 0$) at constant temperature u_0. Thereafter, the end point $x = 0$ is kept at the same temperature while the other end $x = L$ is kept at a constant temperature $u_1 > u_0$. We want to know how the temperature evolves inside the bar.

Before making any computations, let us try to conjecture what could happen. Given that $u_1 > u_0$, heat starts flowing from the hotter end, raising the temperature inside the bar and causing a heat outflow into the cold boundary. On the other hand, the interior increase of temperature causes the hot inflow to decrease in time, while the outflow increases. We expect that sooner or later the two fluxes balance each other and that the temperature eventually reaches a steady state distribution. It would also be interesting to know how fast the steady state is reached.

We show that this is exactly the behavior predicted by our mathematical model, given by the heat equation

$$u_t - Du_{xx} = 0, \quad t > 0,\ 0 < x < L$$

with the initial-Dirichlet conditions

$$\begin{aligned} &u(x,0) = u_0, && 0 \le x \le L \\ &u(0,t) = u_0,\ u(L,t) = u_1, && t > 0. \end{aligned}$$

Since we are interested in the long term behavior of our solution, we leave t unlimited. Notice the *jump discontinuity* between the initial and the boundary data at $x = L$; we will take care of this little difficulty later.

• *Dimensionless variables.* First of all we introduce dimensionless variables, that is variables *independent of the units of measurement*. To do that we rescale space, time and temperature with respect to quantities that are characteristic of our problem. For the space variable we can use the length L of the bar as rescaling factor, setting

$$y = \frac{x}{L}$$

which is clearly dimensionless, being a ratio of lengths. Notice that $0 \le y \le 1$. How can we rescale time? Observe that the physical dimensions of the diffusion coefficient D are $[\text{length}]^2 \times [\text{time}]^{-1}$.

Thus the constant $\tau = L^2/D$ gives a characteristic time scale for our diffusion problem. Therefore we introduce the dimensionless time

$$s = \frac{t}{\tau}. \tag{2.11}$$

Finally, we rescale the temperature by setting

$$z(y, s) = \frac{u(Ly, \tau s) - u_0}{u_1 - u_0}.$$

For the dimensionless temperature z we have:

$$z(y, 0) = \frac{u(Ly, 0) - u_0}{u_1 - u_0} = 0, \quad 0 \le y \le 1$$
$$z(0, s) = \frac{u(0, \tau s) - u_0}{u_1 - u_0} = 0, \quad z(1, s) = \frac{u(L, \tau s) - u_0}{u_1 - u_0} = 1.$$

Moreover

$$(u_1 - u_0) z_s = \frac{\partial t}{\partial s} u_t = \tau u_t = \frac{L^2}{D} u_t$$
$$(u_1 - u_0) z_{yy} = \left(\frac{\partial x}{\partial y}\right)^2 u_{xx} = L^2 u_{xx}.$$

Hence, since $u_t = D u_{xx}$,

$$(u_1 - u_0)(z_s - z_{yy}) = \frac{L^2}{D} u_t - L^2 u_{xx} = \frac{L^2}{D} D u_{xx} - L^2 u_{xx} = 0.$$

In conclusion, we find

$$z_s - z_{yy} = 0 \tag{2.12}$$

with the initial condition

$$z(y, 0) = 0 \tag{2.13}$$

and the boundary conditions

$$z(0, s) = 0, \quad z(1, s) = 1. \tag{2.14}$$

We see that, in the dimensionless formulation, the parameters L and D have disappeared, emphasizing the mathematical essence of the problem. On the other hand, we will later show the relevance of the dimensionless variables in test modelling.

• *The steady state solution.* We start solving problem (2.12), (2.13), (2.14) by first determining the steady state solution z^{St}, that satisfies the equation $z_{yy} = 0$ and the boundary conditions (2.14). An elementary computation gives

$$z^{St}(y) = y.$$

In terms of the original variables the steady state solution is

$$u^{St}(x) = u_0 + (u_1 - u_0)\frac{x}{L}$$

corresponding to a uniform heat flux along the bar given by the Fourier law:

$$\text{heat flux} = -\kappa u_x = -\kappa\frac{(u_1 - u_0)}{L}.$$

• *The transient regime.* Knowing the steady state solution, it is convenient to introduce the function

$$U(y, s) = z^{St}(y, s) - z(y, s) = y - z(y, s).$$

Since we expect our solution to eventually reach the steady state, U represents a *transient regime* that should converge to zero as $s \to \infty$. Furthermore, the rate of convergence to zero of U gives information on how fast the temperature reaches its equilibrium distribution. U satisfies (2.12) with initial condition

$$U(y, 0) = y \tag{2.15}$$

and *homogeneous* boundary conditions

$$U(0, s) = 0 \quad \text{and} \quad U(1, s) = 0. \tag{2.16}$$

• *The method of separation of variables.* We are now in a position to find an explicit formula for U using the method of separation of variables. The main idea is to exploit the linear nature of the problem constructing the solution by superposition of simpler solutions of the form $w(s)v(y)$ in which the variables s and y appear in *separated form.* We emphasize that the reduction to **homogeneous boundary conditions** is crucial to carry on the computations.

Step 1. We look for nontrivial solutions to (2.12) of the form

$$U(y, s) = w(s)v(y)$$

with $v(0) = v(1) = 0$. By substitution into (2.12) we find

$$0 = U_s - U_{yy} = w'(s)v(y) - w(s)v''(y)$$

from which, separating the variables,

$$\frac{w'(s)}{w(s)} = \frac{v''(y)}{v(y)}. \tag{2.17}$$

Now, the left hand side in (2.17) is a function of s only, while the right hand side is a function of y only and the equality must hold for every $s > 0$ and every $y \in (0, L)$.

This is possible only when both sides are equal to a common constant λ, say. Hence we have

$$v''(y) - \lambda v(y) = 0 \tag{2.18}$$

with

$$v(0) = v(1) = 0 \tag{2.19}$$

and

$$w'(s) - \lambda w(s) = 0. \tag{2.20}$$

Step 2. We first solve problem (2.18), (2.19). There are three different possibilities for the general solution of (2.18):

a) If $\lambda = 0$,

$$v(y) = A + By \qquad (A, B \text{ arbitrary constants})$$

and the conditions (2.19) imply $A = B = 0$.

b) If λ is a positive real number, say $\lambda = \mu^2 > 0$, then

$$v(y) = Ae^{-\mu y} + Be^{\mu y}$$

and again it is easy to check that the conditions (2.19) imply $A = B = 0$.

c) Finally, if $\lambda = -\mu^2 < 0$, then

$$v(y) = A \sin \mu y + B \cos \mu y.$$

From (2.19) we get

$$\begin{aligned} v(0) &= B = 0 \\ v(1) &= A \sin \mu + B \cos \mu = 0 \end{aligned}$$

from which

$$A \text{ arbitrary}, \qquad B = 0, \qquad \mu_m = m\pi, \ m = 1, 2, \ldots .$$

Thus, only in case c) we find nontrivial solutions, given by

$$v_m(y) = A \sin m\pi y, \qquad \lambda_m = -\mu_m^2 = -m^2\pi^2. \tag{2.21}$$

In this context, (2.18), (2.19) is called an *eigenvalue problem*; the special values λ_m are the *eigenvalues* and the solutions v_m are the corresponding *eigenfunctions*.

With $\lambda = \lambda_m$, the general solution of (2.20) is

$$w_m(s) = Ce^{-m^2\pi^2 s} \qquad (C \text{ arbitrary constant}). \tag{2.22}$$

From (2.21) and (2.22) we obtain damped sinusoidal waves of the form

$$U_m(y, s) = A_m e^{-m^2\pi^2 s} \sin m\pi y.$$

Step 3. Although the solutions U_m satisfy the homogeneous Dirichlet conditions, they do not match the initial condition $U(y,0) = y$. As we already mentioned, we try to construct the correct solution by superposing the U_m, that is, by setting

$$U(y,s) = \sum_{m=1}^{\infty} A_m e^{-m^2\pi^2 s} \sin m\pi y. \tag{2.23}$$

Some questions arise:

Q1. The initial condition requires

$$U(y,0) = \sum_{m=1}^{\infty} A_m \sin m\pi y = y \qquad \text{for } 0 \le y \le 1. \tag{2.24}$$

Is it possible to choose the coefficients A_m in order to satisfy (2.24)? In which sense does U attain the initial data? For instance, is it true, in some sense, that

$$U(z,s) \to y \quad \text{if} \quad (z,s) \to (y,0)?$$

Q2. Any finite linear combination of the U_m is a solution of the heat equation; can we make sure that the same is true for U? The answer is positive if we could differentiate term by term the infinite sum and get

$$(\partial_s - \partial_{yy})U(y,s) = \sum_{m=1}^{\infty} (\partial_s - \partial_{yy})U_m(y,s) = 0. \tag{2.25}$$

What about the boundary conditions?

Q3. Even if we have a positive answer to questions 1 and 2, are we confident that U is the unique solution of our problem and therefore that it describes the correct evolution of the temperature?

Q1. Question 1 is rather general and concerns the *Fourier series expansion* of a function (see Sect. 1.5), in particular of the initial data $g(y) = y$, in the interval $(0,1)$. Due to the homogeneous Dirichlet conditions it is natural to expand $g(y) = y$ in a *sine Fourier series*, that is to expand the *2-periodic* and *odd* function that agrees with y in the interval $(-1,1)$. The Fourier coefficients are given by the formulas

$$A_m = 2\int_0^1 y \sin m\pi y \, dy = -\frac{2}{m\pi}[y\cos m\pi y]_0^1 + \frac{2}{m\pi}\int_0^1 \cos m\pi y \, dy =$$
$$= -2\frac{\cos m\pi}{m\pi} = (-1)^{m+1}\frac{2}{m\pi}.$$

The sine Fourier expansion of $g(y) = y$ on the interval $(0,1)$ is therefore

$$y = \sum_{m=1}^{\infty} (-1)^{m+1}\frac{2}{m\pi}\sin m\pi y. \tag{2.26}$$

Where is the expansion (2.26) valid? It cannot be true at $y = 1$, since $\sin m\pi = 0$ for every m, and we would obtain $1 = 0$. This clearly reflects the presence of the jump discontinuity of the data at $y = 1$.

The theory of Fourier series implies that (2.26) is true at every point $y \in [0, 1)$ and that the series converges uniformly in every interval $[0, a]$, $a < 1$. Moreover, equality (2.26) holds **in the mean quadratic sense** (or $L^2(0, 1)$ sense), that is

$$\int_0^1 \left[y - \sum_{m=1}^{N} (-1)^{m+1} \frac{2}{m\pi} \sin m\pi y \right]^2 dy \to 0 \qquad \text{as } N \to \infty.$$

From (2.23) and the expression of A_m, we obtain the *formal* solution

$$U(y, s) = \sum_{m=1}^{\infty} (-1)^{m+1} \frac{2}{m\pi} e^{-m^2\pi^2 s} \sin m\pi y \tag{2.27}$$

that attains the initial data in the mean quadratic sense, i.e.

$$\lim_{s \to 0^+} \int_0^1 [U(y, s) - y]^2 dy = 0. \tag{2.28}$$

In fact, from Parseval's equality (1.12), we can write

$$\int_0^1 [U(y, s) - y]^2 \, dy = \frac{2}{\pi^2} \sum_{m=1}^{\infty} \frac{(e^{-m^2\pi^2 s} - 1)^2}{m^2}. \tag{2.29}$$

Since for $s \geq 0$

$$\frac{(e^{-m^2\pi^2 s} - 1)^2}{m^2} \leq \frac{1}{m^2}$$

and the series $\sum 1/m^2$ converges, then the series in (2.29) converges uniformly by Weierstrass test (see Sect. 1.4) in $[0, \infty)$ and we can take the limit under the sum, obtaining (2.28).

Q2. The analytical expression of U is rather reassuring: it is a superposition of sinusoids of increasing frequency m and of strongly damped amplitude, because of the negative exponential, at least when $s > 0$. Indeed, for $s > 0$, the rapid convergence to zero of each term and its derivatives in the series (2.27) allows us to differentiate term by term. Precisely, we have

$$\frac{\partial U_m}{\partial s} = \frac{\partial^2 U_m}{\partial y^2} = (-1)^m 2m\pi e^{-m^2\pi^2 s} \sin m\pi y,$$

so that, if $s \geq s_0 > 0$,

$$\left| \frac{\partial U_m}{\partial s} \right|, \left| \frac{\partial^2 U_m}{\partial y^2} \right| \leq 2m\pi e^{-m^2\pi^2 s_0}.$$

Since the numerical series

$$\sum_{m=1}^{\infty} m e^{-m^2\pi^2 s_0}$$

is convergent, we conclude by the Weierstrass test that the series

$$\sum_{m=1}^{\infty} \frac{\partial U_m}{\partial s} \quad \text{and} \quad \sum_{m=1}^{\infty} \frac{\partial^2 U_m}{\partial y^2}$$

converge uniformly in $[0,1] \times [s_0, \infty)$, for every $s_0 > 0$. Thus (2.25) is true and U is a solution of (2.12).

It remains to check the Dirichlet conditions: if $s_1 > 0$,

$$U(z,s) \to 0 \text{ as } (z,s) \to (0,s_1) \text{ or } (z,s) \to (1,s_1).$$

As above, this is true because we can take the two limits under the sum, due to the uniform convergence of the series (2.27) in any region $[0,1] \times [s_0, +\infty)$ with $s_0 > 0$ (choose $s_0 < s_1$). For the same reason, U has continuous derivatives of any order, up to the lateral boundary of the strip $[0,1] \times [s_0, +\infty)$.

Note, in particular, that U *immediately* forgets the initial discontinuity and becomes smooth at any positive time.

Q3. To show that U is indeed the unique solution, we use the so-called *energy method*, that we will develop later in greater generality. Suppose W is another solution of problem (2.12), smooth in any region $[0,1] \times [s_0, \infty)$ with $s_0 > 0$ and satisfying (2.28). Then, by linearity,

$$v = U - W$$

satisfies

$$v_s - v_{yy} = 0 \tag{2.30}$$

and has zero initial-boundary data. Multiplying (2.30) by v, integrating in y over the interval $[0,1]$ and keeping $s > 0$, fixed, we get

$$\int_0^1 v v_s \, dy - \int_0^1 v v_{yy} \, dy = 0. \tag{2.31}$$

Observe that

$$\int_0^1 v v_s \, dy = \frac{1}{2} \int_0^1 \partial_s(v^2) dy = \frac{1}{2} \frac{d}{ds} \int_0^1 v^2 dy. \tag{2.32}$$

Moreover, integrating by parts, we can write

$$\int_0^1 v v_{yy} \, dy = [v(1,s)v_y(1,s) - v(0,s)v_y(0,s)] - \int_0^1 (v_y)^2 dy = -\int_0^1 (v_y)^2 dy$$

since $v(1,s) = v(0,s) = 0$. From (2.31), (2.32) and the previous equation we get

$$\frac{1}{2} \frac{d}{ds} \int_0^1 v^2 dy = -\int_0^1 (v_y)^2 dy \le 0 \tag{2.33}$$

and therefore, the *nonnegative* function

$$E(s) = \int_0^1 v^2(y,s)dy$$

is nonincreasing. On the other hand, using (2.28) for v instead of U, we get

$$E(s) \to 0 \qquad \text{as } s \to 0$$

which forces $E(s) = 0$, for every $s > 0$. But v^2 is nonnegative and continuous in $[0,1]$ if $s > 0$, so that it must be $v(y,s) = 0$ for $y \in [0,1]$ and every $s > 0$. Then, since $v(y,0) = 0$, we deduce $U = W$.

• *Back to the original variables.* In terms of the original variables, our solution is expressed as

$$u(x,t) = u_0 + (u_1 - u_0)\frac{x}{L} - (u_1 - u_0)\sum_{m=1}^{\infty}(-1)^{m+1}\frac{2}{m\pi}e^{-\frac{m^2\pi^2 D}{L^2}t}\sin\frac{m\pi}{L}x.$$

This formula confirms our initial guess about the evolution of the temperature towards the steady state. Indeed, each term of the series converges to zero exponentially as $t \to +\infty$. Since the Weierstrass test works here for $t \geq t_0 > 0$, it is not difficult to show that

$$u(x,t) \to u_0 + (u_1 - u_0)\frac{x}{L} \qquad \text{as } t \to +\infty.$$

Moreover, the first exponential ($m = 1$) in the series decays much more slowly than the others and very soon it determines the main deviation of u from the equilibrium, *independently of the initial condition*. Thus, as $t \to +\infty$, the leading term is the damped sinusoid

$$\frac{2}{\pi}e^{-\frac{\pi^2 D}{L^2}t}\sin\frac{\pi}{L}x.$$

In this mode there is a concentration of heat at $x = L/2$, where the temperature reaches its maximum amplitude $2\exp(-\pi^2 Dt/L^2)/\pi$. At time $t = L^2/D$ the amplitude decays to $2\exp(-\pi^2)/\pi \simeq 3.3 \times 10^{-5}$, about 0.005 per cent of its initial value. This simple calculation shows that to reach the steady state, a time of order L^2/D is required, a fundamental fact in heat diffusion.

Not surprisingly, the scaling factor in (2.11) was exactly $\tau = L^2/D$. The dimensionless formulation is extremely useful in experimental modelling tests. To achieve reliable results, these models must reproduce the same characteristics at different scales. For instance, if our bar were an experimental model of a much bigger beam of length L_0 and diffusion coefficient D_0, to reproduce the same heat diffusion effects, we must choose material (D) and length (L) for our model bar such that

$$\frac{L^2}{D} = \frac{L_0^2}{D_0}.$$

Figure 2.2 shows the solution of the dimensionless problem (2.12), (2.13), (2.14) for

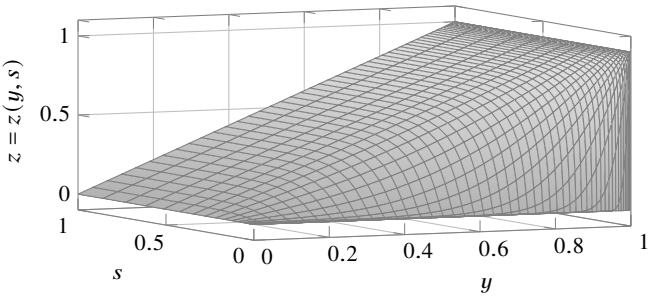

Fig. 2.2 The solution to the dimensionless problem (2.12), (2.13), (2.14)

$0 < s \leq 1$.

2.1.5 Problems in dimension $n > 1$

The formulation of the problems in Sect. 2.1.3 can be easily generalized to any spatial dimension $n > 1$, in particular to $n = 2$ or $n = 3$. Suppose we want to determine the evolution of the temperature in a heat conducting body that occupies a bounded *domain* (*open connected*) $\Omega \subset \mathbb{R}^n$, during an interval of time $[0, T]$. Under the hypotheses of Sect. 2.1.2, the temperature is a function $u = u(\mathbf{x}, t)$ that satisfies the heat equation $u_t - D\Delta u = f$ in the *space-time cylinder* (see Fig. 2.3)

$$Q_T = \Omega \times (0, T).$$

To select a unique solution we have to prescribe, first of all, the *initial distribution*

$$u(\mathbf{x}, 0) = g(\mathbf{x}), \quad \mathbf{x} \in \overline{\Omega},$$

where $\overline{\Omega} = \Omega \cup \partial\Omega$ denotes the *closure* of Ω.

The control of the interaction of the body with the surroundings is modeled through *suitable conditions* on $\partial\Omega$. The most common ones are:

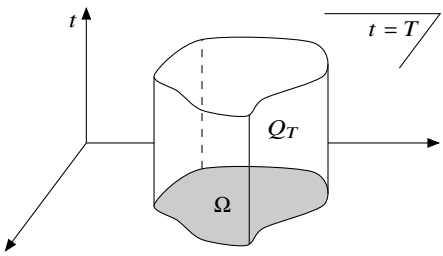

Fig. 2.3 The space-time cylinder Q_T

Dirichlet condition: the temperature is kept at a prescribed level on $\partial\Omega$; this amounts to assigning

$$u(\boldsymbol{\sigma}, t) = h(\boldsymbol{\sigma}, t), \qquad \boldsymbol{\sigma} \in \partial\Omega \text{ and } t \in (0, T].$$

Neumann condition: the heat flux through $\partial\Omega$ is assigned. To model this condition, we assume that the boundary $\partial\Omega$ is a smooth curve or surface, having a tangent line or plane at every point[5] with *outward* unit vector $\boldsymbol{\nu}$. From the Fourier law we have

$$\mathbf{q} = \text{ heat flux} = -\kappa\nabla u$$

so that the *inward heat flux* is

$$-\mathbf{q} \cdot \boldsymbol{\nu} = \kappa\nabla u \cdot \boldsymbol{\nu} = \kappa\partial_{\boldsymbol{\nu}} u.$$

Thus the Neumann condition reads

$$\partial_{\boldsymbol{\nu}} u(\boldsymbol{\sigma}, t) = h(\boldsymbol{\sigma}, t), \qquad \boldsymbol{\sigma} \in \partial\Omega \text{ and } t \in (0, T].$$

Radiation or Robin condition: the *inward* heat flux through $\partial\Omega$ depends linearly on the difference $U - u$ (linear Newton's law of cooling):

$$-\mathbf{q} \cdot \boldsymbol{\nu} = \gamma(U - u), \qquad (\gamma > 0)$$

where U is the surroundings temperature. From the Fourier law we obtain

$$\partial_{\boldsymbol{\nu}} u + \alpha u = h, \qquad \text{on } \partial\Omega \times (0, T]$$

with $\alpha = \gamma/\kappa > 0$, $h = \gamma U/\kappa$.

Mixed conditions: the boundary of Ω is decomposed into various parts where different boundary conditions are prescribed. For instance, a formulation of a mixed Dirichlet-Neumann problem is obtained by writing

$$\partial\Omega = \overline{\Gamma}_D \cup \overline{\Gamma}_N \quad \text{with} \quad \Gamma_D \cap \Gamma_N = \emptyset,$$

where Γ_D and Γ_N are regular open subsets of $\partial\Omega$ in the sense of Definition 1.11, p. 19. Then we assign

$$u = h_1 \text{ on } \overline{\Gamma}_D \times (0, T]$$
$$\partial_{\boldsymbol{\nu}} u = h_2 \text{ on } \Gamma_N \times (0, T].$$

Summarizing, we have the following typical problems: *given* $f = f(\mathbf{x}, t)$ *and* $g = g(\mathbf{x})$*, determine* $u = u(\mathbf{x}, t)$ *such that:*

[5] We can also allow boundaries with corner points, like squares, cones, or edges, like cubes. It is enough that the set of points where the tangent plane does not exist has zero surface measure (zero length in two dimensions). Lipschitz domains have this property (see Sect. 1.7).

$$\begin{cases} u_t - D\Delta u = f & \text{in } Q_T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \overline{\Omega} \\ + \text{ boundary conditions} & \text{on } \partial\Omega \times (0, T] \end{cases}$$

where the boundary conditions could be:

- *Dirichlet*: $u = h$.
- *Neumann*: $\partial_\nu u = h$.
- *Radiation or Robin*: $\partial_\nu u + \alpha u = h \qquad (\alpha > 0)$.
- *Mixed, for instance*:

$$u = h_1 \text{ on } \overline{\Gamma}_D \times (0, T], \qquad \partial_\nu u = h_2 \text{ on } \Gamma_N \times (0, T].$$

Also in dimension $n > 1$, the *global Cauchy problem* is important:

$$\begin{cases} u_t - D\Delta u = f & \mathbf{x} \in \mathbb{R}^n, 0 < t < T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^n \\ + \text{ condition as } |\mathbf{x}| \to \infty. \end{cases}$$

We again emphasize that no final condition (for $t = T, \mathbf{x} \in \Omega$) is required. The data is assigned on the *parabolic boundary* $\partial_p Q_T$ of Q_T, given by the union of the bottom points $\overline{\Omega} \times \{t = 0\}$ and the side points $S_T = \partial\Omega \times (0, T]$:

$$\partial_p Q_T = (\overline{\Omega} \times \{t = 0\}) \cup S_T.$$

2.2 Uniqueness and Maximum Principles

2.2.1 Integral method

Generalizing the energy method used in Sect. 2.1.4, it is easy to show that all the problems we have formulated in the previous section have at most one solution under reasonable conditions on the data. More precisely, suppose u and v are solutions of one of those problems, sharing the same boundary conditions and assuming the initial data in the mean quadratic sense.

Let $w = u - v$; we want to show that $w \equiv 0$. For the time being we do not worry about the precise hypotheses on u and v; we assume they are sufficiently smooth in Q_T up to $\partial_p Q_T$ and observe that w satisfies the homogeneous equation

$$w_t - D\Delta w = 0 \tag{2.34}$$

in $Q_T = \Omega \times (0, T)$, with initial condition

$$w(\mathbf{x}, 0) = 0$$

in $\overline{\Omega}$, and one of the following conditions on $\partial\Omega \times (0,T]$:

$$w = 0 \qquad \text{(Dirichlet)} \tag{2.35}$$

or

$$\partial_\nu w = 0 \qquad \text{(Neumann)} \tag{2.36}$$

or

$$\partial_\nu w + \alpha w = 0 \qquad \alpha > 0, \qquad \text{(Robin)} \tag{2.37}$$

or

$$w = 0 \text{ on } \overline{\Gamma}_D \times (0,T], \quad \partial_\nu w = 0 \text{ on } \Gamma_N \times (0,T] \qquad \text{(mixed)}. \tag{2.38}$$

Multiply equation (2.34) by w and integrate on Ω; we find

$$\int_\Omega w w_t \, d\mathbf{x} = D \int_\Omega w \Delta w \, d\mathbf{x}.$$

Now,

$$\int_\Omega w w_t \, d\mathbf{x} = \frac{1}{2}\frac{d}{dt}\int_\Omega w^2 d\mathbf{x} \tag{2.39}$$

and from Green's identity (1.17) with $u = v = w$,

$$\int_\Omega w \Delta w \, d\mathbf{x} = \int_{\partial\Omega} w \partial_\nu w \, d\sigma - \int_\Omega |\nabla w|^2 \, d\mathbf{x}. \tag{2.40}$$

Then, letting

$$E(t) = \int_\Omega w^2 d\mathbf{x},$$

(2.39) and (2.40) give

$$\frac{1}{2}E'(t) = D\int_{\partial\Omega} w\partial_\nu w \, d\sigma - D\int_\Omega |\nabla w|^2 \, d\mathbf{x}.$$

If the Robin condition (2.37) holds,

$$\int_{\partial\Omega} w\partial_\nu w \, d\sigma = -\alpha \int_{\partial\Omega} w^2 \, d\sigma \le 0.$$

If one of the (2.35), (2.36), (2.38) holds, then

$$\int_{\partial\Omega} w\partial_\nu w \, d\sigma = 0.$$

In any case it follows that $E'(t) \le 0$ and therefore E is a nonincreasing function. Since

$$E(t) \to E(0) = \int_\Omega w^2(\mathbf{x}, 0) \, d\mathbf{x} = 0 \quad \text{as } t \to 0,$$

we must have $E(t) = 0$ for every $t \geq 0$, and this implies $w(\mathbf{x}, t) \equiv 0$ in Ω for every $t > 0$. Thus $u = v$.

The above calculations are completely justified under reasonable conditions, indicated in the following statement. Note that the hypothesis on the initial data is very weak.

Theorem 2.2 *Let Ω be a bounded Lipschitz domain and g be a square integrable function in Ω. The initial Dirichlet, Neumann, Robin and mixed problems have at most one solution, continuous in $\overline{\Omega} \times (0, T)$, together with their first and second spatial derivatives and their first order time derivatives, such that*

$$\int_\Omega (w(\mathbf{x}, t) - g(\mathbf{x}))^2 \, d\mathbf{x} \to 0 \quad \text{as } t \to 0^+.$$

2.2.2 Maximum principles

The fact that heat flows from higher to lower temperature regions implies that a solution of the homogeneous heat equation attains its maximum and minimum values on $\partial_p Q_T$. This result is known as the *maximum principle* and reflects an aspect of the time irreversibility of the phenomena described by the heat equation, in the sense that the future cannot have an influence on the past (*causality principle*). In other words, the value of a solution u at time t is independent of any change of the data after t.

The following simple theorem translates these principles and we state it for functions in the class $C^{2,1}(Q_T) \cap C(\overline{Q}_T)$. These functions are continuous up to the boundary of Q_T, with one time and two spacial derivatives continuous in the interior of Q_T. It is convenient to distinguish between subsolution and supersolution of the heat equation, according to the following definition.

Definition 2.3 A function $w \in C^{2,1}(Q_T)$ such that $w_t - D\Delta w \leq 0$ (≥ 0) in Q_T is called a *subsolution* (*supersolution*) of the diffusion equation.

Theorem 2.4 *Let $w \in C^{2,1}(Q_T) \cap C(\overline{Q}_T)$ such that*

$$w_t - D\Delta w = q(\mathbf{x}, t) \leq 0 \qquad \textit{(resp. } \geq 0\textit{) in } Q_T. \tag{2.41}$$

Then w attains its maximum (resp. minimum) on $\partial_p Q_T$:

$$\max_{\overline{Q}_T} w = \max_{\partial_p Q_T} w \qquad \textit{(resp. } \min_{\overline{Q}_T} w = \min_{\partial_p Q_T} w\textit{)}. \tag{2.42}$$

In particular, if w is negative (resp. positive) on $\partial_p Q_T$, then it is negative (resp. positive) in all Q_T.

Proof We recall that $\partial_p Q_T$ is the union of the base and the lateral boundary of Q_T. Let $q \leq 0$. The case $q \geq 0$ is analogous. We consider two cases.

Case 1. Assume first that $w \in C^{2,1}(\overline{Q}_T)$ and that $q(\mathbf{x},t) < 0$ for all $(\mathbf{x},t)$ in $\overline{Q}_T$. We argue by contradiction. Suppose that w attains its maximum at a point $(\mathbf{x}_0,t_0) \notin \partial_p Q_T$. Then $\mathbf{x}_0 \in \Omega$ and $t_0 \in (0,T]$. From elementary calculus, we have $w_{x_j x_j}(\mathbf{x}_0,t_0) \le 0$ for every $j = 1,\dots,n$, so that

$$\Delta w(\mathbf{x}_0,t_0) \le 0,$$

and

$$\text{either } w_t(\mathbf{x}_0,t_0) = 0, \text{ if } t_0 < T \qquad \text{or } w_t(\mathbf{x}_0,T) \ge 0.$$

In both cases, we obtain the contradiction

$$0 \le w_t(\mathbf{x}_0,t_0) - D\Delta w(\mathbf{x}_0,t_0) = q(\mathbf{x}_0,t_0) < 0.$$

Case 2. Consider now the general case $w \in C^{2,1}(Q_T) \cap C(\overline{Q}_T)$ and $q \le 0$ in $\overline{Q}_T$. We reduce to Case 1. Let $\varepsilon \in (0,T)$ and set $u = w - \varepsilon t$. Then

$$u_t - D\Delta u = q - \varepsilon < 0 \tag{2.43}$$

and $u \in C^{2,1}(\overline{Q}_{T-\varepsilon})$. From Case 1 we deduce that

$$\max_{\overline{Q}_{T-\varepsilon}} u \le \max_{\partial_p Q_{T-\varepsilon}} u \le \max_{\partial_p Q_T} u. \tag{2.44}$$

Since $u \le w \le u + \varepsilon T$, we infer from (2.44),

$$\max_{\overline{Q}_{T-\varepsilon}} w \le \max_{\overline{Q}_{T-\varepsilon}} u + \varepsilon T \le \max_{\partial_p Q_T} u + \varepsilon T \le \max_{\partial_p Q_T} w + \varepsilon T. \tag{2.45}$$

Since w is continuous in $\overline{Q}_T$, we deduce that

$$\max_{\overline{Q}_{T-\varepsilon}} w \to \max_{\overline{Q}_T} w \quad \text{as } \varepsilon \to 0.$$

Hence, letting $\varepsilon \to 0$ in (2.45) we find $\max_{\overline{Q}_T} w \le \max_{\partial_p Q_T} w$, which concludes the proof since the opposite inequality is obvious. □

As an immediate consequence of Theorem 2.4, we have that, if

$$w_t - D\Delta w = 0 \qquad \text{in } Q_T,$$

then w attains its maximum and its minimum on $\partial_p Q_T$:

$$\min_{\partial_p Q_T} w \le w(\mathbf{x},t) \le \max_{\partial_p Q_T} w \qquad \forall (\mathbf{x},t) \in \overline{Q}_T.$$

Moreover (for the proof see Problem 2.6):

Corollary 2.5 (Comparison and stability) *Let $v, w \in C^{2,1}(Q_T) \cap C(\overline{Q}_T)$ be solutions of*

$$v_t - D\Delta v = f_1 \qquad \textit{and} \qquad w_t - D\Delta w = f_2$$

with f_1, f_2 bounded in Q_T. Then:

a) If $v \ge w$ on $\partial_p Q_T$ and $f_1 \ge f_2$ in Q_T then $v \ge w$ in all $\overline{Q_T}$.
b) The following stability estimate holds:

$$\max_{\overline{Q}_T} |v - w| \leq \max_{\partial_p Q_T} |v - w| + T \sup_{Q_T} |f_1 - f_2| \,. \tag{2.46}$$

Corollary 2.5 gives uniqueness of solutions for the initial-Dirichlet problem that are continuous up to $\partial_p Q_T$. Note that it does not require the continuity of any derivatives up to the lateral side.

Inequality (2.46) is a uniform pointwise stability estimate, extremely useful in several applications. In fact if

$$v = g_1, \ w = g_2 \text{ on } \partial_p Q_T$$

and

$$\max_{\partial_p Q_T} |g_1 - g_2| \leq \varepsilon \quad \text{and} \quad \sup_{\overline{Q}_T} |f_1 - f_2| \leq \varepsilon,$$

we deduce

$$\max_{\overline{Q}_T} |v - w| \leq \varepsilon(1 + T).$$

Thus, in finite time, a small uniform distance between the data implies small uniform distance between the corresponding solutions.

Theorem 2.4 is a version of the so called weak maximum principle, weak because this result says nothing about the possibility that a solution achieves its maximum or minimum at an interior point as well. Actually a more precise result is the following, known as *strong maximum principle* (Fig. 2.4). For the proof see *M. H. Protter, H. F. Weinberger*, [16].

Theorem 2.6 *Let $u \in C^{2,1}(Q_T) \cap C(\overline{Q}_T)$ be a subsolution (resp. supersolution) of the heat equation in Q_T. If u attains its maximum (resp. minimum) M at a point $(\mathbf{x}_1, t_1)$ with $x_1 \in \Omega$, $0 < t_1 \leq T$, then*

$$u = M \text{ in } \overline{Q}_{t_1}.$$

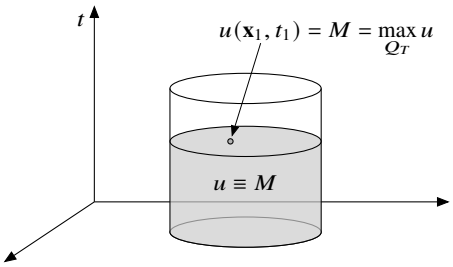

Fig. 2.4 The strong maximum principle

2.3 The Fundamental Solution

There are privileged solutions of the homogeneous diffusion equation that can be used to construct many other ones. In this section we are going to discover one of these special building blocks, the most important one. First we point out some features of the heat equation.

2.3.1 Invariant transformations

The homogeneous diffusion equation has simple but important properties. Let $u = u(\mathbf{x}, t)$ be a solution of

$$u_t - D\Delta u = 0 \qquad \text{in } \mathbb{R}^n \times (0, +\infty). \tag{2.47}$$

- *Time reversal*. The function

$$v(\mathbf{x}, t) = u(\mathbf{x}, -t),$$

obtained by the change of variable $t \longmapsto -t$, is a solution of the **adjoint** or **backward** equation

$$v_t + D\Delta v = 0 \qquad \text{in } \mathbb{R}^n \times (-\infty, 0).$$

Coherently, (2.47) is sometimes called the **forward** equation. The non-invariance of (2.47) with respect to a change of sign in time is another aspect of time irreversibility.

- *Space and time translation invariance*. For $\mathbf{y}, s$ fixed, the function

$$v(\mathbf{x}, t) = u(\mathbf{x} - \mathbf{y}, t - s)$$

is still a solution of (2.47) in $\mathbb{R}^n \times (s, +\infty)$. Clearly, for $\mathbf{x}, t$ fixed the function $u(\mathbf{x} - \mathbf{y}, t - s)$ solves the backward equation with respect to $\mathbf{y}$ and s in $\mathbb{R}^n \times (-\infty, t)$.
- *Parabolic dilations.* The transformation

$$\mathbf{x} \longmapsto a\mathbf{x}, \qquad t \longmapsto bt, \qquad u \longmapsto cu \qquad (a, b, c > 0)$$

represents a dilation (expansion or contraction) of the graph of u. Let us check for which values of a, b, c the function

$$u^*(\mathbf{x}, t) = cu(a\mathbf{x}, bt)$$

is still a solution of (2.47). We have:

$$u^*_t(\mathbf{x}, t) - D\Delta u^*(\mathbf{x}, t) = cbu_t(a\mathbf{x}, bt) - ca^2 D\Delta u(a\mathbf{x}, bt)$$

and so u^* is a solution of (2.47) in $\mathbb{R}^n \times (0, +\infty)$ if $b = a^2$.

This relation suggests the name of *parabolic dilation* for the transformation

$$\mathbf{x} \longmapsto a\mathbf{x} \qquad t \longmapsto a^2 t \qquad (a > 0).$$

Under this transformation the expressions

$$\frac{|\mathbf{x}|^2}{Dt} \qquad \text{or} \qquad \frac{\mathbf{x}}{\sqrt{Dt}}$$

are left unchanged. Moreover, we already observed that they are *dimensionless groups*. Thus it is not surprising that these combinations of the independent variables occur frequently in the study of diffusion phenomena.

• *Dilations and conservation of mass* (or *energy*). Let $u = u(\mathbf{x}, t)$ be a solution of (2.47) in the half-space $\mathbb{R}^n \times (0, +\infty)$. Then we just checked that the function

$$u^*(\mathbf{x}, t) = cu(a\mathbf{x}, a^2 t)$$

is also a solution in the same set. Suppose u satisfies the condition

$$\int_{\mathbb{R}^n} u(\mathbf{x}, t)d\mathbf{x} = q \qquad \text{for every } t > 0. \tag{2.48}$$

If, for instance, u represents the concentration of a substance (density of mass), equation (2.48) states that the total mass is q at every time t. If u is a temperature, (2.48) says that the total internal energy is constant ($= q\rho c_v$). We ask for which a, c the solution u^* still satisfies (2.48). We have

$$\int_{\mathbb{R}^n} u^*(\mathbf{x}, t)d\mathbf{x} = c \int_{\mathbb{R}^n} u(a\mathbf{x}, a^2 t)d\mathbf{x}.$$

Letting $\mathbf{y} = a\mathbf{x}$, so that $d\mathbf{y} = a^n d\mathbf{x}$, we find

$$\int_{\mathbb{R}^n} u^*(\mathbf{x}, t)d\mathbf{x} = ca^{-n} \int_{\mathbb{R}^n} u(\mathbf{y}, a^2 t)d\mathbf{y} = ca^{-n} q$$

and for (2.48) to be satisfied we must have $c = a^n$. In conclusion, if $u = u(\mathbf{x}, t)$ is a solution of (2.47) in the half-space $\mathbb{R}^n \times (0, +\infty)$, satisfying (2.48), the same is true for

$$u^*(\mathbf{x}, t) = a^n u(a\mathbf{x}, a^2 t). \tag{2.49}$$

2.3.2 The fundamental solution ($n = 1$)

We are now in position to construct our special solution, starting with dimension $n = 1$. To help intuition, think for instance of our solution as the concentration of a

substance of total mass q and suppose we want to keep the total mass equal to q at any time.

We have seen that the combination of variables $x/\sqrt{Dt}$ is not only invariant with respect to parabolic dilations but also dimensionless. It is then natural to check if there are solutions of (2.47) involving such dimensionless group. Since $\sqrt{Dt}$ has the dimension of a length, the quantity $q/\sqrt{Dt}$ is a typical order of magnitude for the concentration, so that it makes sense to look for solutions of the form

$$u^*(x,t) = \frac{q}{\sqrt{Dt}} U\left(\frac{x}{\sqrt{Dt}}\right) \tag{2.50}$$

where U is a (dimensionless) function of a single variable.

Here is the main question: is it possible to determine

$$U = U(\xi)$$

such that u^* is a solution of (2.47)? Solutions of the form (2.50) are called *similarity solutions*[6].

Moreover, since we are interpreting u^* as a concentration, we require $U \geq 0$ and the total mass condition yields

$$1 = \frac{1}{\sqrt{Dt}} \int_{\mathbb{R}} U\left(\frac{x}{\sqrt{Dt}}\right) dx \underset{\xi = x/\sqrt{Dt}}{=} \int_{\mathbb{R}} U(\xi) d\xi,$$

so that we require that

$$\int_{\mathbb{R}} U(\xi) d\xi = 1. \tag{2.51}$$

Let us now impose that u^* is a solution to (2.47). We have ($\xi = x/\sqrt{Dt}$):

$$u_t^* = \frac{q}{\sqrt{D}} \left[-\frac{1}{2} t^{-\frac{3}{2}} U(\xi) - \frac{1}{2\sqrt{D}} x t^{-2} U'(\xi) \right] = -\frac{q}{2t\sqrt{Dt}} \left[U(\xi) + \xi U'(\xi) \right]$$
$$u_{xx}^* = \frac{q}{(Dt)^{3/2}} U''(\xi),$$

hence

$$u_t^* - D u_{xx}^* = -\frac{q}{t\sqrt{Dt}} \left\{ U''(\xi) + \frac{1}{2}\xi U'(\xi) + \frac{1}{2} U(\xi) \right\}.$$

We see that for u^* to be a solution of (2.47), U must be a solution in $\mathbb{R}$ of the ordinary differential equation

---

[6] A solution of a particular evolution problem is a *similarity* or *self-similar* solution if its spatial configuration (its graph at a fixed time) remains similar to itself at all times during the evolution. In one space dimension, *self-similar* solutions have the general form

$$u(x,t) = a(t) F(x/b(t))$$

where, preferably, u/a and x/b are dimensionless quantity.

$$U''(\xi) + \frac{1}{2}\xi U'(\xi) + \frac{1}{2}U(\xi) = 0. \tag{2.52}$$

Since $U \geq 0$, (2.51) implies[7]:

$$U(-\infty) = U(+\infty) = 0.$$

On the other hand, (2.52) is invariant with respect to the change of variables

$$\xi \mapsto -\xi$$

and therefore we look for *even solutions*:

$$U(-\xi) = U(\xi).$$

Then we can restrict ourselves to $\xi \geq 0$, asking

$$U'(0) = 0 \quad \text{and} \quad U(+\infty) = 0, \tag{2.53}$$

where $U'(0) = 0$ comes from the smoothness of U as a solution of (2.52). To solve (2.52), observe that it can be written in the form

$$\frac{d}{d\xi}\left\{U'(\xi) + \frac{1}{2}\xi U(\xi)\right\} = 0$$

that yields

$$U'(\xi) + \frac{1}{2}\xi U(\xi) = C \qquad (C \in \mathbb{R}). \tag{2.54}$$

Letting $\xi = 0$ in (2.54) and recalling (2.53), we deduce that $C = 0$ and therefore

$$U'(\xi) + \frac{1}{2}\xi U(\xi) = 0. \tag{2.55}$$

The general solution of (2.55) is

$$U(\xi) = c_0 e^{-\frac{\xi^2}{4}} \qquad (c_0 \in \mathbb{R}).$$

This function is even, integrable and vanishes at infinity. It only remains to choose $c_0 > 0$ in order to ensure (2.51). Since

$$\int_{\mathbb{R}} e^{-\frac{\xi^2}{4}} d\xi \underset{\xi=2z}{=} 2\int_{\mathbb{R}} e^{-z^2} dz = 2\sqrt{\pi}$$

the choice is $c_0 = (4\pi)^{-1/2}$.

Going back to the original variables, we have found the following solution of (2.47)

[7] Rigorously, the precise conditions are: $\liminf_{\xi \to \pm\infty} U(\xi) = 0$.

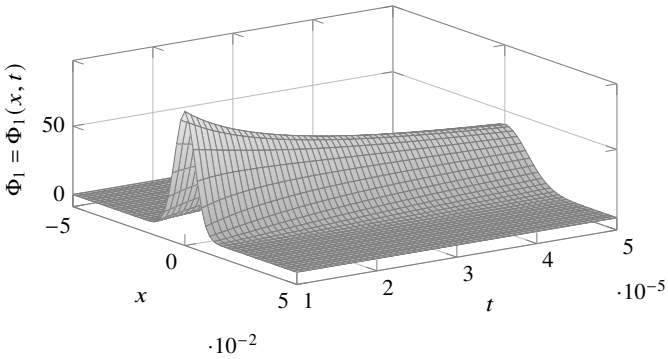

Fig. 2.5 The fundamental solution Φ_1

$$u^*(x,t) = \frac{q}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}, \qquad x \in \mathbb{R},\ t > 0,$$

which is positive, even in x, and satisfies

$$\int_{\mathbb{R}} u^*(x,t)dx = q \qquad \text{for every } t > 0. \tag{2.56}$$

The choice $q = 1$ gives a family of *Gaussians*, parametrized with time, and it is natural to think of a *normal probability density*.

Definition 2.7 The function

$$\Phi_D(x,t) = \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}, \qquad x \in \mathbb{R},\ t > 0, \tag{2.57}$$

is called the *fundamental solution* of eq. (2.47) in dimension $n = 1$ (see Fig. 2.5).

2.3.3 The Dirac distribution

We may ask which kind of initial condition does the fundamental solution satisfy. Thus let us examine the behavior of the fundamental solution Φ_D as $t \to 0^+$. For every fixed $x \neq 0$,

$$\lim_{t\to 0^+} \Phi_D(x,t) = \lim_{t\to 0^+} \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}} = 0 \tag{2.58}$$

while

$$\lim_{t\to 0^+} \Phi_D(0,t) = \lim_{t\to 0^+} \frac{1}{\sqrt{4\pi Dt}} = +\infty. \tag{2.59}$$

If we interpret Φ_D as a concentration or a probability density, eqs. (2.58), (2.59) and (2.56) imply that when $t \to 0^+$ the fundamental solution tends to concentrate mass around the origin; eventually, the whole mass is concentrated at $x = 0$.

The limiting density distribution can be mathematically modeled by the so called *Dirac distribution (or measure) at the origin,* denoted by the symbol δ. The Dirac distribution is not a function in the usual sense of Analysis; if it were, it should have the following properties:

- $\delta(0) = \infty$, $\delta(x) = 0$ for $x \neq 0$,
- $\int_{\mathbb{R}} \delta(x)dx = 1$,

clearly incompatible with any concept of classical function or integral. A rigorous definition of the Dirac measure requires the theory of *generalized functions* or *distributions* of L. Schwartz, that we will consider in Chap. 7. Here we restrict ourselves to some heuristic considerations. Let

$$\mathcal{H}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0, \end{cases}$$

be the characteristic function of the interval $[0, \infty)$, known as the *Heaviside function*. Observe that

$$\frac{\mathcal{H}(x+\varepsilon) - \mathcal{H}(x-\varepsilon)}{2\varepsilon} = \begin{cases} \frac{1}{2\varepsilon} & \text{if } -\varepsilon \leq x < \varepsilon \\ 0 & \text{otherwise.} \end{cases} \tag{2.60}$$

Denote by $I_\varepsilon(x)$ the quotient (2.60); the following properties hold:

a) For every $\varepsilon > 0$,

$$\int_{\mathbb{R}} I_\varepsilon(x)dx = \frac{1}{2\varepsilon} 2\varepsilon = 1.$$

We can interpret I_ε as a *unit impulse of extent* 2ε (Fig. 2.6).

b)

$$\lim_{\varepsilon \downarrow 0} I_\varepsilon(x) = \begin{cases} 0 & \text{if } x \neq 0 \\ \infty & \text{if } x = 0. \end{cases}$$

c) If $\varphi = \varphi(x)$ is a smooth function, vanishing outside a bounded interval, (these kind of functions are called *test functions*), we have

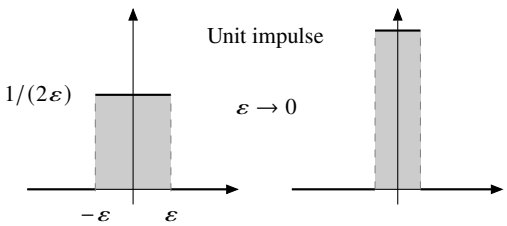

Fig. 2.6 Approximation of the Dirac measure

$$\int_{\mathbb{R}} I_\varepsilon(x)\varphi(x)dx = \frac{1}{2\varepsilon}\int_{-\varepsilon}^{\varepsilon} \varphi(x)dx \underset{\varepsilon\longrightarrow 0}{\longrightarrow} \varphi(0).$$

Properties a) and b) show that I_ε tends to a mathematical object that has precisely the formal features of the Dirac distribution at the origin. In particular c) suggests how to identify this object, that is *through its action on test functions.*

Definition 2.8 We call *Dirac measure at the origin* the generalized function, denoted by δ, that acts on a test function φ as follows:

$$\delta\,[\varphi] = \varphi(0). \tag{2.61}$$

Equation (2.61) is often written in the form $\langle\delta, \varphi\rangle = \varphi(0)$ or even

$$\int \delta(x)\varphi(x)dx = \varphi(0)$$

where the integral symbol is purely formal. Observe that property b) shows that

$$\mathcal{H}' = \delta,$$

whose meaning is given in the following computations, where an integration by parts is used and φ is a test function:

$$\int_{\mathbb{R}} \varphi\, d\mathcal{H} = -\int_{\mathbb{R}} \mathcal{H}(x)\varphi'(x)\,dx = -\int_0^\infty \varphi'(x)\,dx = \varphi(0), \tag{2.62}$$

since φ vanishes for large[8] x.

With the notion of Dirac measure at hand, we can say that Φ_D satisfies the initial conditions

$$\Phi_D(x, 0) = \delta(x).$$

If the unit mass is concentrated at a point $y \neq 0$, we denote by $\delta(x - y)$ the Dirac measure at y, defined through the formula

$$\int \delta(x - y)\varphi(x)dx = \varphi(y).$$

Then, by translation invariance, the fundamental solution $\Phi_D(x - y, t)$ is a solution of the diffusion equation (2.47) in $\mathbb{R} \times (0, +\infty)$, that satisfies the initial condition

$$\Phi_D(x - y, 0) = \delta(x - y).$$

[8] The first integral in (2.62) is a Riemann-Stieltjes integral, that formally can be written as

$$\int \varphi(x)\mathcal{H}'(x)dx$$

and interpreted as *the action of the generalized function* $\mathcal{H}'$ on the test function φ.

Indeed, it is the *unique nonnegative solution*[9] *of (2.47) that satisfies the total mass condition (2.56), with* $q = 1$.

As any solution u of (2.47) has several interpretations (concentration of a substance, probability density, temperature in a bar), also the fundamental solution has several meanings.

We can think of it as a **unit source solution**: $\Phi_D(x,t)$ gives the concentration at the point x, at time t, generated by the diffusion of a **unit mass initially** ($t = 0$) **concentrated at the origin**.

From another perspective, if we imagine a unit mass composed of a large number N of particles, $\Phi_D(x,t)dx$ gives the probability that a single particle is placed between x and $x + dx$ at time t or, equivalently, the percentage of particles inside the interval $(x, x + dx)$ at time t.

Initially Φ_D is zero outside the origin. As soon as $t > 0$, Φ_D becomes positive everywhere: this amounts to saying that the unit mass diffuses instantaneously all over the x-axis and therefore *with infinite speed of propagation*. This could be a problem in using (2.47) as a realistic model, although (see Fig. 2.5) for $t > 0$, small, Φ_D is practically equal to zero outside an interval centered at the origin of length $4D$.

2.3.4 The fundamental solution ($n > 1$)

In space dimension greater than 1, we can more or less repeat the same arguments. We look for positive, radial, self-similar solutions u^* to (2.47), with total mass equal to q at every time, that is

$$\int_{\mathbb{R}^n} u^*(\mathbf{x},t)d\mathbf{x} = q \qquad \text{for every } t > 0. \tag{2.63}$$

Since $q/(Dt)^{n/2}$ is a concentration per unit volume, we set

$$u^*(\mathbf{x},t) = \frac{q}{(Dt)^{n/2}}U(\xi), \qquad \xi = |\mathbf{x}|/\sqrt{Dt}.$$

We have

$$u_t^* = -\frac{q}{2t(Dt)^{n/2}}\left[nU(\xi) + \xi U'(\xi)\right].$$

Set $g(r) = U(r/\sqrt{Dt})$. The expression of the Laplace operator for the radial function g is (see Appendix B, for $n = 3$),

$$\Delta_{\text{polar}} g = g_{rr} + \frac{n-1}{r}g_r = \frac{1}{Dt}U''(r/\sqrt{Dt}) + \frac{n-1}{r\sqrt{Dt}}U'(r/\sqrt{Dt}).$$

[9] Uniqueness is shown in Problem 7.19, p. 492 and it is a consequence of a remarkable theorem, originally due to Widder, stated in Sect. 2.8.5.

Thus, going back to original variables, we get

$$\Delta u^* = \frac{q}{(Dt)^{1+n/2}} \left\{ U''(\xi) + \frac{n-1}{\xi} U'(\xi) \right\}.$$

Therefore, for u^* to be a solution of (2.47), U must be a nonnegative solution in $(0, +\infty)$ of the ordinary differential equation

$$\xi U''(\xi) + (n-1)U'(\xi) + \frac{\xi^2}{2} U'(\xi) + \frac{n}{2} \xi U(\xi) = 0. \tag{2.64}$$

Multiplying by ξ^{n-2}, we can write (2.64) in the form

$$(\xi^{n-1} U')' + \frac{1}{2} (\xi^n U)' = 0$$

that gives

$$\xi^{n-1} U' + \frac{1}{2} \xi^n U = C \qquad (C \in \mathbb{R}). \tag{2.65}$$

Assuming that $\lim_{\xi \to 0^+}$ of U and U' are finite, letting $\xi \to 0^+$ into (2.65), we deduce $C = 0$ and therefore

$$U' + \frac{1}{2} \xi U = 0.$$

Thus we obtain the family of solutions $U(\xi) = c_0 e^{-\frac{\xi^2}{4}}$. The total mass condition requires

$$1 = \frac{1}{(Dt)^{n/2}} \int_{\mathbb{R}^n} U\left(\frac{|\mathbf{x}|}{\sqrt{Dt}} \right) d\mathbf{x} = \frac{c_0}{(Dt)^{n/2}} \int_{\mathbb{R}^n} \exp\left(-\frac{|\mathbf{x}|^2}{4Dt} \right) d\mathbf{x}$$
$$\underset{\mathbf{y}=\mathbf{x}/\sqrt{Dt}}{=} c_0 \int_{\mathbb{R}^n} e^{-|\mathbf{y}|^2} d\mathbf{y} = c_0 \left(\int_{\mathbb{R}} e^{-z^2} dz \right)^n = c_0 (4\pi)^{n/2}$$

and therefore $c_0 = (4\pi)^{-n/2}$. Thus, we have obtained solutions of the form

$$u^*(\mathbf{x}, t) = \frac{q}{(4\pi Dt)^{n/2}} \exp\left(-\frac{|\mathbf{x}|^2}{4Dt} \right), \qquad (t > 0).$$

Once more, the choice $q = 1$ is special.

Definition 2.9 The function

$$\Phi_D(\mathbf{x}, t) = \frac{1}{(4\pi Dt)^{n/2}} \exp\left(-\frac{|\mathbf{x}|^2}{4Dt} \right) \qquad (t > 0)$$

is called the *fundamental solution* of the diffusion equation (2.47).

The remarks after Definition 2.8, p. 48, can be easily generalized to the multidimensional case. In particular, it is possible to define the *n-dimensional Dirac measure* at a point $\mathbf{y}$, denoted by $\delta_n(\mathbf{x} - \mathbf{y})$, through the formula

$$\int \delta_n(\mathbf{x}-\mathbf{y})\varphi(\mathbf{x})d\mathbf{x} = \varphi(\mathbf{y}) \tag{2.66}$$

that expresses the action of $\delta_n(\mathbf{x}-\mathbf{y})$ on the test function φ, smooth in $\mathbb{R}^n$ and vanishing outside a compact set. For $n = 1$ we set $\delta_1 = \delta$.

For fixed $\mathbf{y}$, the fundamental solution $\Phi_D(\mathbf{x}-\mathbf{y}, t)$ is the unique nonnegative solution of the global Cauchy problem

$$\begin{cases} u_t - D\Delta u = 0 & \mathbf{x} \in \mathbb{R}^n,\ t > 0 \\ u(\mathbf{x}, 0) = \delta_n(\mathbf{x}-\mathbf{y}) & \mathbf{x} \in \mathbb{R}^n \end{cases}$$

and it satisfies (2.63) with $q = 1$.

2.4 Symmetric Random Walk ($n = 1$)

In this section we start exploring the connection between probabilistic and deterministic models, in dimension $n = 1$. The main purpose is to construct a Brownian motion, which is a **continuous model** (in both space and time), as a limit of a simple stochastic process, called *random walk*, which is instead a **discrete model** (in both space and time). During the realization of the limiting procedure we shall see how the diffusion equation can be approximated by a *difference equation*. Moreover, this new perspective will better clarify the nature of the diffusion coefficient.

2.4.1 Preliminary computations

Consider a unit mass particle (or a large number of particles of total mass one) that moves randomly along the x axis, according to the following rules: fix

$$h > 0, \text{ space step}, \qquad \tau > 0, \text{ time step}.$$

1. During an interval of time τ, the particle takes one step of h unit length, starting from $x = 0$.
2. The particle moves to the left or to the right with probability $p = \frac{1}{2}$, independently of the previous steps (Fig. 2.7).

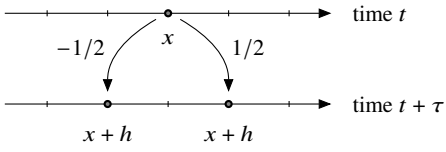

Fig. 2.7 Symmetric random walk

At time $t = N\tau$, that is after N steps, the particle will be located at a point $x = mh$, where $N \geq 0$ and m are integers, with $-N \leq m \leq N$.

Our task is:

Compute the probability $p(x,t)$ to find the particle at x, at time t.

Random walks can occur in a wide variety of situations. To give an example, think of a gambling game in which a *fair coin* is flipped. If heads comes out, the particle moves to the right and the player gains 1 *dollar*; if tails comes out it moves to the left and the player loses 1 *dollar*. Then $p(x,t)$ is the probability to gain m dollars after N throws.

• *Computation of $p(x,t)$.* Let $x = mh$ be the position of the particle after N steps. To reach x, the particle takes some number of steps to the right, say k, and $N - k$ steps to the left. Clearly, $0 \leq k \leq N$ and

$$m = k - (N - k) = 2k - N \tag{2.67}$$

so that N and m are both even or both odd integers and

$$k = \frac{1}{2}(N + m).$$

Thus, $p(x,t) = p_k$ where

$$p_k = \frac{\text{number of walks with } k \text{ steps to the right after } N \text{ steps}}{\text{number of possible walks after } N \text{ steps}}. \tag{2.68}$$

Now, the number of possible walks with k steps to the right and $N - k$ to the left is given by the binomial coefficient[10]

$$C_{N,k} = \binom{N}{k} = \frac{N!}{k!(N-k)!}.$$

On the other hand, the number of possible walks after N steps is 2^N (why?); hence, from (2.68):

$$p_k = \frac{C_{N,k}}{2^N} \qquad x = mh,\ t = N\tau,\ k = \frac{1}{2}(N + m). \tag{2.69}$$

• *Mean displacement and standard deviation of x.* Our ultimate goal is to let h and τ go to zero, keeping x and t fixed, in order to get a continuous walk, which incorporates the main features of the discrete random walk. This is a delicate point, since, if we want to eventually obtain a continuous faithful copy of the random walk,

[10] The set of walks with k steps to the right and $N - k$ to the left is in one to one correspondence with the set of *sequences of* N binary digits, where 1 means *right* and 0 means *left*. There are exactly $C_{N,k}$ of these sequences.

we need to isolate some quantitative parameters, able to capture the essential features of the walk and maintain them unchanged. In our case there are two key parameters[11]

(a) The **mean displacement** of x after N steps $= \langle x \rangle = \langle m \rangle\, h$.

(b) The **variance** of x after N steps $= \left\langle x^2 \right\rangle - \langle x \rangle^2 = \left(\left\langle m^2 \right\rangle - \langle m \rangle^2 \right) h^2$.

First observe that, from (2.67), we have

$$\langle m \rangle = 2 \langle k \rangle - N \tag{2.70}$$

and

$$\left\langle m^2 \right\rangle = 4 \left\langle k^2 \right\rangle - 4 \langle k \rangle N + N^2. \tag{2.71}$$

Since the random variable (whose outcomes are) k follows a binomial distribution of parameters N and $1/2$, we have[12]

$$\text{expected value of } k : \langle k \rangle = \frac{N}{2}, \qquad \text{variance of } k : \left\langle k^2 \right\rangle - \langle k \rangle^2 = \frac{N}{4}$$

whence

$$\text{second moment of } k = \left\langle k^2 \right\rangle = \frac{N(N+1)}{4}.$$

Thus

$$\langle m \rangle = 2 \langle k \rangle - N = 2\frac{N}{2} - N = 0$$

and also $\langle x \rangle = \langle m \rangle\, h = 0$, which is not surprising, given the symmetry of the walk. Furthermore

$$\left\langle m^2 \right\rangle = 4 \left\langle k^2 \right\rangle - 4N \langle k \rangle + N^2 = N^2 + N - 2N^2 + N^2 = N$$

from which variance of $x = \left\langle x^2 \right\rangle = Nh^2$ or

$$\sqrt{\left\langle x^2 \right\rangle} = \sqrt{N} h, \tag{2.72}$$

which is the *standard deviation of* x, since $\langle x \rangle = 0$. Formula (2.72) contains a key information: after N time steps, the average distance from the origin is $\sqrt{N}$ space

[11] If a random variable x takes N possible outcomes $x_1, \ldots, x_N$ with probability $p_1, \ldots, p_N$, its *moments of* (*integer*) *order* $q \geq 1$ are given by

$$E(x^q) = \langle x^q \rangle = \sum_{j=1}^{N} x_j^q\, p_j.$$

The first moment ($q = 1$) is the *mean or expected value of* x, while var $(x) = \left\langle x^2 \right\rangle - \langle x \rangle^2$ is the *variance of* x. The square root of the variance is called *standard deviation*.

[12] If a discrete random variable k follows a *binomial distribution of parameters* N, p, that is $p_k = \binom{N}{k} p^k (1-p)^{N-k}$, one has: expected value of $k = Np$, variance of $k = Np(1-p)$.

steps, that is **the order of the time scale is the square of the space scale**. In other words, if we want to leave the standard deviation unchanged in the limit process, we must rescale the time as the square of the space, that is we must use a *space-time parabolic dilation*!

But let us proceed step by step. The next one is to deduce a *difference equation* for the transition probability $p = p(x,t)$. It is on this equation that we will carry out the limit procedure.

2.4.2 From random walk to Brownian motion

The particle motion has no memory since each move is independent from the previous ones. If the particle location at time $t+\tau$ is x, this means that at time t its location was at $x-h$ or at $x+h$, with equal probability. The total probability formula then gives

$$p(x,t+\tau) = \frac{1}{2}p(x-h,t) + \frac{1}{2}p(x+h,t) \tag{2.73}$$

with the initial conditions

$$p(0,0) = 1 \qquad \text{and} \qquad p(x,0) = 0 \quad \text{if } x \neq 0. \tag{2.74}$$

We now wonder what happens to the difference equation (2.73) when $h \to 0, \tau \to 0$. In the limit, we expect a differential equation in which p plays the role of a density for a continuous probability distribution. If we assume p smooth enough, using Taylor's formula we can write[13]

$$p(x,t+\tau) = p(x,t) + p_t(x,t)\tau + o(\tau),$$
$$p(x \pm h,t) = p(x,t) \pm p_x(x,t)h + \frac{1}{2}p_{xx}(x,t)h^2 + o(h^2).$$

Substituting into (2.73), after some simplifications, we find

$$p_t\tau + o(\tau) = \frac{1}{2}p_{xx}h^2 + (h^2).$$

Dividing by τ, we get

$$p_t + o(1) = \frac{1}{2}\frac{h^2}{\tau}p_{xx} + o\left(\frac{h^2}{\tau}\right). \tag{2.75}$$

This is the crucial point: in the last equation we meet again the combination $\frac{h^2}{\tau}$!!

[13] Recall that the symbol $o(z)$, ("little o of z") denotes a quantity of lower order with respect to z; precisely, $\dfrac{o(z)}{z} \to 0$ when $z \to 0$.

If we want to obtain something nontrivial when $h, \tau \to 0$, **we must require that h^2/τ has a finite and positive limit**; the simplest choice is to keep

$$\frac{h^2}{\tau} = 2D \tag{2.76}$$

for some number $D > 0$ (the number 2 is there for aesthetic reasons only) whose physical dimensions are clearly $[D] = [\text{length}]^2 \times [\text{time}]^{-1}$.

Passing to the limit in (2.75), we get for p the equation

$$p_t = D p_{xx}. \tag{2.77}$$

In the continuous version, the initial condition (2.74) translates into "infinite density at the origin with unit total mass", that is

$$\lim_{t \to 0^+} p(x,t) = \delta(x). \tag{2.78}$$

We have already seen that the unique nonnegative solution of (2.77), (2.78) is

$$p(x,t) = \Phi_D(x,t)$$

and, due to the meaning of p,

$$\int_{\mathbb{R}} p(x,t)dx = 1.$$

Thus, the constant D in (2.76) is precisely the *diffusion coefficient*. Recalling that

$$h^2 = \frac{\langle x^2 \rangle}{N}, \quad \tau = \frac{t}{N},$$

we have

$$\frac{h^2}{\tau} = \frac{\langle x^2 \rangle}{t} = 2D$$

that means: *in unit time, the particle diffuses up to an average distance of* $\sqrt{2D}$. Also, from (2.76) we deduce

$$\frac{h}{\tau} = \frac{2D}{h} \to +\infty. \tag{2.79}$$

This shows that the average speed h/τ of the particle at each step becomes unbounded. Therefore, the fact that the particle diffuses in unit time up to a finite average distance is purely due to the rapid fluctuations of its motion.

What happened in the limit to the random walk? What kind of motion did it become?

We can answer using some more tools from probability theory. Let $x_j = x(j\tau)$ be the position of our particle after j steps and let, for $j \geq 1$,

$$h\xi_j = x_j - x_{j-1}.$$

The ξ_j are *independent, identically distributed random variables*: each one takes on the value 1 or -1 with probability $\frac{1}{2}$. They have expected value $\langle \xi_j \rangle = 0$ and variance $\langle \xi_j^2 \rangle = 1$.

The displacement of the particle after N steps is

$$x_N = h \sum_{j=1}^{N} \xi_j.$$

If we choose

$$h = \sqrt{\frac{2Dt}{N}},$$

that is $\frac{h^2}{\tau} = 2D$, and let $N \to \infty$, the *Central Limit Theorem* insures that x_N converges in law[14] to a random variable $X = X(t)$, normally distributed with mean 0 and variance $2Dt$, whose density is $\Phi_D(x,t)$.

The random walk has become a continuous walk; if $D = 1/2$, it is called (1-dimensional) **Brownian motion** or **Wiener process**, that we will briefly describe in the next section.

2.4.3 A glance at Brownian motion

The random position of a Brownian particle is usually denoted with the symbol $B = B(t)$. The family of random variables $B(t)$, where t plays the role of a parameter, is defined on a common probability space $(\Omega, \mathcal{F}, P)$, where Ω is the set of elementary events, $\mathcal{F}$ a σ-algebra in Ω of measurable events, and P a suitable probability measure in $\mathcal{F}$ (see Appendix A); therefore the right notation should be

$$B = B(t, \omega),$$

with $\omega \in \Omega$, but the dependence on ω is usually omitted and understood (for simplicity or laziness).

The family of random variables $B(t, \omega)$, with time t as a real parameter, is a **continuous stochastic process**. Keeping $\omega \in \Omega$ fixed, we get the real function

$$t \longmapsto B(t, \omega)$$

whose graph describes a Brownian path (see Fig. 2.8).

Keeping t fixed, we get the random variable

$$\omega \longmapsto B(t, \omega).$$

[14] That is, if $N \to +\infty$, for every, $a, b \in \mathbb{R}$, $a < b$, Prob $\{a < x_N < b\} \to \int_a^b \Phi_D(x,t)\, dx$.

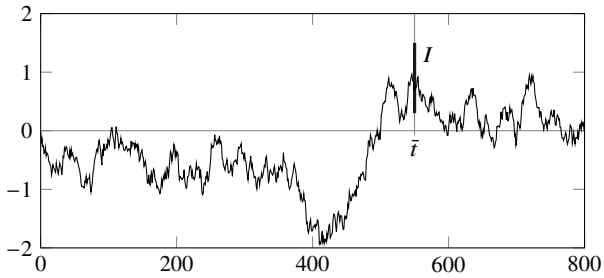

Fig. 2.8 A Brownian path

Without caring too much of what Ω really is, it is important to be able to compute the probability

$$P\{B(t) \in I\}$$

where $I \subseteq \mathbb{R}$ is a reasonable subset of $\mathbb{R}$ (a so called *Borel set*[15]). Figure 2.8 shows the meaning of this computation: fixing t amounts to fixing a vertical straight line, say $t = \bar{t}$. Let I be a subset of this line; in the picture I is an interval. $P\{B(\bar{t}) \in I\}$ is the probability that the particle hits I at time $\bar{t}$.

The main properties of Brownian motion are listed below. To be minimalist we could synthesize everything in the formula[16]

$$dB \sim \sqrt{dt}N(0,1) = N(0,dt) \tag{2.80}$$

where $N(0,1)$ is a normal random variable, with zero mean and variance equal to one.

• *Path continuity.* With probability 1, the possible paths of a Brownian particle are continuous functions

$$t \longmapsto B(t), \quad t \geq 0.$$

Since from (2.79) the instantaneous speed of the particle is infinite, their graphs are nowhere differentiable!

• *Gaussian law for increments.* We can allow the particle to start from a point $x \neq 0$, by considering the process

$$B^x(t) = x + B(t).$$

With every point x is associated a probability P^x, with the following properties (if $x = 0$, $P^0 = P$):

a) $P^x\{B^x(0) = x\} = P\{B(0) = 0\} = 1$.

[15] An interval or a set obtained by countable unions and intersections of intervals, for instance. See Appendix A.

[16] If X is a random variable, we write $X \sim N(\mu, \sigma^2)$ if X has normal distribution with mean μ and variance σ^2.

b) For every $s \geq 0, t \geq 0$, the increment

$$B^x(t+s) - B^x(s) = B(t+s) - B(s) \tag{2.81}$$

has normal law, with *zero mean and variance* t, with density given by

$$\Phi(x,t) \equiv \Phi_{\frac{1}{2}}(x,t) = \frac{1}{\sqrt{2\pi t}} e^{-\frac{x^2}{2t}}.$$

Moreover, the increment (2.81) is independent of any event occurred at a time less or equal than s. For instance, the two events

$$\{B^x(t_2) - B^x(t_1) \in I_2\} \qquad \{B^x(t_1) - B^x(t_0) \in I_1\},$$

with $t_0 < t_1 < t_2$, are independent.

- *Transition probability.* For each Borel set $I \subseteq \mathbb{R}$, a *transition function*

$$P(x,t,I) = P^x\{B^x(t) \in I\}$$

is defined, assigning the probability that the particle, initially at x, belongs to I at time t. We can write:

$$P(x,t,I) = P\{B(t) \in I - x\} = \int_{I-x} \Phi(y,t)dy = \int_I \Phi(y-x,t)dy.$$

- *Invariance.* The motion is invariant with respect to translations.
- *Markov and strong Markov properties.* Let μ be a probability measure[17] on $\mathbb{R}$. If the initial position of the particle is random with a probability distribution μ, we can consider a *Brownian motion with initial distribution* μ, and for it we use the symbol B^μ. With this motion is associated a probability distribution P^μ such that, for every Borel set $F \subseteq \mathbb{R}$,

$$P^\mu\{B^\mu(0) \in F\} = \mu(F).$$

The probability that the particle belongs to I at time t can be computed through the formula

$$P^\mu\{B^\mu(t) \in I\} = \int_{\mathbb{R}} P^x\{B^x(t) \in I\}\,\mu(dx) = \int_{\mathbb{R}} P(x,t,I)\mu(dx).$$

The *Markov property* can be stated as follows: given any condition H, related to the behavior of the particle before time $s \geq 0$, the process

$$Y(t) = B^x(t+s)$$

is a Brownian motion with initial distribution

[17] See Appendix A for the definition of a probability measure μ and of the integral with respect to the measure μ .

$$\mu(I) = P^x \{B^x(s) \in I \,|H\}\,,$$

where $P(A\,|H)$ denotes the conditional probability of A, given H (App. A). This property establishes the independence of the *future* process $B^x(t+s)$ from the *past* (absence of memory) when the *present* $B^x(s)$ is known and reflects the *absence of memory* of the random walk.

In the strong *Markov property*, s is substituted by a random time τ, which depends only on the behavior of the particle in the interval $[0, \tau]$. In other words, to decide whether or not the event $\{\tau \le t\}$ is true, it is enough to know the behavior of the particle up to time t. These kinds of random times are called *stopping times*. An important example is the *first exit time from a domain*, that we will consider in the next chapter. Instead, the random time τ defined by

$$\tau = \inf \{t : B(t) > 10 \text{ and } B(t+1) < 10\}$$

is *not* a stopping time. Indeed (measuring time in *seconds*), τ is "the smallest" among the times t such that the Brownian path is above level 10 at time t, and after one second is below 10. Clearly, to decide whether $\tau \le 3$, say, it is not enough to know the path up to time $t = 3$, since τ involves the behavior of the path up to the *future* time $t = 4$.

2.5 Diffusion, Drift and Reaction

2.5.1 Random walk with drift. Preliminary calculations

The hypothesis of symmetry of our random walk can be removed. Suppose our unit mass particle moves along the x axis with space step $h > 0$, every time interval of duration $\tau > 0$, according to the following rules.

1. The particle starts from $x = 0$.
2. It moves to the right with probability $p_0 \neq \frac{1}{2}$ and to the left with probability $q_0 = 1 - p_0$, independently of the previous steps (Fig. 2.9).

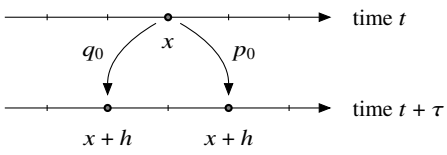

Fig. 2.9 Random walk with drift

Rule 2 breaks the symmetry of the walk and models a particle tendency to move to the right or to the left, according to the sign of $p_0 - q_0$ being positive or negative, respectively. Again we denote by $p = p(x,t)$ the probability that the particle location

is $x = mh$ at time $t = N\tau$ and by k the random variables that counts the number of steps to the right. This time we have

$$p(x,t) = p_k = \binom{N}{k} p_0^k q_0^{N-k} \qquad (m = 2k - N)$$

so that k follows a binomial distribution with parameters N and p_0 and

$$\text{expected value of } k = \langle k \rangle = Np_0, \quad \text{variance of } k = \left\langle k^2 \right\rangle - \langle k \rangle^2 = Np_0q_0.$$

Thus, simple calculations give:

(a)**Mean displacement** x after N steps $= \langle x \rangle = \langle m \rangle\, h = (p_0 - q_0)Nh$.
(b)**Variance** of x after N steps $= \left\langle x^2 \right\rangle - \langle x \rangle^2 = (\left\langle m^2 \right\rangle - \langle m \rangle^2)h^2 = 4p_0q_0Nh^2$.

Observe that, since $t = N\tau$, the ratio

$$\frac{\langle x \rangle}{t} = \frac{(p_0 - q_0)h}{\tau} \tag{2.82}$$

gives the mean displacement of x in unit time: an essential feature of the asymmetric motion. This means that in the limit as $h, \tau \to 0$, we *should keep that ratio under control*.

On the other hand, the *standard deviation* $\sqrt{\mathrm{Var}(x)}$ is still of order $\sqrt{N}h$. Thus, the diffusion features are unchanged with respect to the symmetric case, and codified into the ratio

$$\frac{\sqrt{\mathrm{Var}(x)}}{t} \approx \frac{h^2}{\tau},$$

again to keep under control in the limit process.

2.5.2 The limit equation

This time the total probability formula gives:

$$p(x, t+\tau) = p_0 p(x-h, t) + q_0 p(x+h, t), \tag{2.83}$$

with the usual initial conditions

$$p(0,0) = 1 \quad \text{and} \quad p(x,0) = 0 \quad \text{if } x \neq 0.$$

As in the symmetric case, we want to examine what happens when we pass to the limit for $h \to 0, \tau \to 0$. From Taylor formula, we have

$$p(x, t+\tau) = p(x,t) + p_t(x,t)\tau + o(\tau),$$
$$p(x \pm h, t) = p(x,t) \pm p_x(x,t)h + \frac{1}{2}p_{xx}(x,t)h^2 + o(h^2).$$

Substituting into (2.83), we get

$$p_t\tau + o(\tau) = \frac{1}{2}p_{xx}h^2 + (q_0 - p_0)hp_x + o(h^2). \tag{2.84}$$

A new term appears: $(q_0 - p_0)hp_x$. Dividing by τ, we obtain

$$p_t + o(1) = \frac{1}{2}\frac{h^2}{\tau}p_{xx} + \boxed{\frac{(q_0 - p_0)h}{\tau}}p_x + o\left(\frac{h^2}{\tau}\right). \tag{2.85}$$

Again, here is the crucial point. If we let $h, \tau \to 0$, we realize that the assumption

$$\frac{h^2}{\tau} = 2D \tag{2.86}$$

alone is not sufficient anymore to get something nontrivial from (2.85): indeed, if we keep p_0 and q_0 constant, we have

$$\frac{(q_0 - p_0)h}{\tau} \to \infty$$

and from (2.85) we get a contradiction. Indeed, as anticipated above, we must keep under control the ratio in (2.82). So, what else do we have to require? Writing

$$\frac{(q_0 - p_0)h}{\tau} = \frac{(q_0 - p_0)}{h}\frac{h^2}{\tau}$$

we see we must require, in addition to (2.86), that

$$\frac{q_0 - p_0}{h} \to \beta \tag{2.87}$$

with β finite. Notice that, since $q_0 + p_0 = 1$, (2.87) is equivalent to

$$p_0 = \frac{1}{2} - \frac{\beta}{2}h + o(h) \quad \text{and} \quad q_0 = \frac{1}{2} + \frac{\beta}{2}h + o(h), \tag{2.88}$$

that could be interpreted as a *symmetry of the motion at a microscopic scale.*

With (2.87) at hand, we have

$$\frac{(q_0 - p_0)}{h}\frac{h^2}{\tau} \to 2D\beta \equiv b$$

and (2.85) becomes in the limit,

$$p_t = Dp_{xx} + bp_x. \tag{2.89}$$

We already know that Dp_{xx} models a diffusion phenomenon. Let us *unmask* the term bp_x, by first examining the dimensions of b. Since $q_0 - p_0$ is dimensionless,

being a difference of probabilities, the dimensions of b are those of h/τ, namely of a **velocity.**

Thus the coefficient b codifies the tendency of the limiting continuous motion, to move towards a privileged direction with speed $|b|$: to the right if $b < 0$, to the left if $b > 0$. In other words, there exists a *current of intensity* $|b|$, driving the particle. The random walk has become a **diffusion process with drift**.

The last point of view calls for an analogy with the diffusion of a substance transported along a channel.

2.5.3 Pollution in a channel

In this section we examine a simple convection-diffusion model of a pollutant on the surface of a narrow channel. A water stream of constant speed v transports the pollutant along the positive direction of the x axis. We can neglect the depth of the water (thinking to a floating pollutant) and the transverse dimension (thinking of a very narrow channel).

Our purpose is to derive a mathematical model capable of describing the evolution of the concentration (mass per unit lenght) $c = c(x,t) \geq 0$ of the pollutant. Accordingly, the integral

$$\int_x^{x+\Delta x} c(y,t)dy \tag{2.90}$$

gives the mass inside the interval $(x, x+\Delta x)$ at time t (Fig. 2.10). In the present case there are neither sources nor sinks of pollutant, therefore to construct a model we use the **law of mass conservation**: *the growth rate of the mass contained in an interval* $(x, x+\Delta x)$ *equals the net mass flux into* $(x, x+\Delta x)$ *through the end points.*

From (2.90), assuming we can take the time derivative inside the integral, the growth rate of the mass contained in an interval $(x, x+\Delta x)$ is given by

$$\frac{d}{dt}\int_x^{x+\Delta x} c(y,t)dy = \int_x^{x+\Delta x} c_t(y,t)dy. \tag{2.91}$$

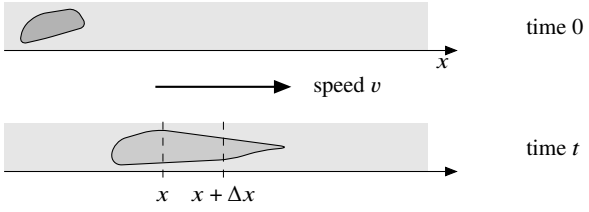

Fig. 2.10 Pollution in a narrow channel

Denote by $q = q(x,t)$ the mass flux (mass per unit time) *entering* the interval $(x, x+\Delta x)$, through the point x at time t. The net mass flux into $(x, x+\Delta x)$ through the end points is

$$q(x,t) - q(x+\Delta x, t). \tag{2.92}$$

Equating (2.91) and (2.92), the law of mass conservation reads

$$\int_x^{x+\Delta x} c_t(y,t)dy = q(x,t) - q(x+\Delta x, t).$$

Dividing by Δx and letting $\Delta x \to 0$, we find the basic law

$$c_t = -q_x. \tag{2.93}$$

At this point we have to decide which kind of mass flux we are dealing with. In other words, we need a *constitutive relation for q*. There are several possibilities, for instance:

a) *Convection.* The flux is determined by the water stream only. This case corresponds to a bulk of pollutant that is driven by the stream, without deformation or expansion. Translating into mathematical terms we find

$$q(x,t) = vc(x,t)$$

where, we recall, v denotes the stream speed.

b) *Diffusion.* The pollutant expands from higher to lower concentration regions. We have seen something like that in heat conduction, where, according to the Fourier law, the heat flux is proportional and opposite to the temperature gradient. Here we can adopt a similar law, that, in this setting, is known as the *Fick's law* and reads

$$q(x,t) = -Dc_x(x,t), \tag{2.94}$$

where the constant D depends on the polluting and has the usual dimensions ($[D] = [\text{length}]^2 \times [\text{time}]^{-1}$).

In our case, convection and diffusion are both present and therefore we superpose the two effects, by writing

$$q(x,t) = vc(x,t) - Dc_x(x,t).$$

From (2.93) we deduce

$$c_t = Dc_{xx} - vc_x \tag{2.95}$$

which constitutes our mathematical model and turns out to be identical to (2.89).

Since D and v are constant, it is easy to determine the evolution of a unit mass of pollutant, initially located at the origin (say). Its concentration is the solution of (2.95) with initial condition

$$c(x,0) = \delta(x)$$

where δ is the Dirac measure at the origin. To find an explicit formula, we can get rid of the drift term $-vc_x$ by setting

$$w(x,t) = c(x,t)e^{hx+kt}$$

with h, k to be chosen suitably. We have:

$$\begin{aligned} w_t &= (c_t + kc)e^{hx+kt} \\ w_x &= (c_x + hc)e^{hx+kt} \\ w_{xx} &= (c_{xx} + 2hc_x + h^2c)e^{hx+kt}. \end{aligned}$$

Using the equation $c_t = Dc_{xx} - vc_x$, we can write

$$\begin{aligned} w_t - Dw_{xx} &= e^{hx+kt}[c_t - Dc_{xx} - 2Dhc_x + (k - Dh^2)c] = \\ &= e^{hx+kt}[(-v - 2Dh)c_x + (k - Dh^2)c]. \end{aligned}$$

Thus if we choose

$$h = -\frac{v}{2D} \qquad \text{and} \qquad k = Dh^2 = \frac{v^2}{4D},$$

w is a solution of the diffusion equation $w_t - Dw_{xx} = 0$, with the initial condition

$$w(x,0) = c(x,0)e^{-\frac{v}{2D}x} = \delta(x)e^{-\frac{v}{2D}x}.$$

In Chap. 7 we show that, in a suitable sense,

$$\delta(x)e^{-\frac{v}{2D}x} = \delta(x),$$

so that, being $w \geq 0$,

$$w(x,t) = \Phi_D(x,t)$$

and finally

$$c(x,t) = e^{\frac{v}{2D}\left(x - \frac{v}{2}t\right)}\Phi_D(x,t) = \frac{1}{\sqrt{4\pi Dt}}e^{-\frac{(x-vt)^2}{4Dt}}. \tag{2.96}$$

The concentration c is thus given by the travelling wave $\Phi_D(x - vt, t)$, in motion to the right with speed v, which coincides with the fundamental solution of equation (2.95).

In realistic situations, the pollutant undergoes some sort of decay, for instance by biological decomposition. The resulting equation for the concentration becomes

$$c_t = Dc_{xx} - vc_x - \gamma c$$

where γ is a rate of decay ($[\gamma] = [\text{time}]^{-1}$). We deal with this case in the next section via a suitable variant of our random walk.

2.5.4 Random walk with drift and reaction

We go back to our 1-dimensional random walk, assuming that the particle loses mass at the constant rate $\gamma > 0$. This means that in an interval of time from t to $t + \tau$ an amount of mass

$$Q(x,t) = \tau\gamma p(x,t)$$

disappears at the position x. The difference equation (2.83) for p becomes

$$p(x,t+\tau) = p_0[p(x-h,t) - Q(x-h,t)] + q_0[p(x+h,t) - Q(x+h,t)].$$

We have:

$$\begin{aligned} p_0 Q(x-h,t) + q_0 Q(x+h,t) &= Q(x,t) + (q_0 - p_0)hQ_x(x,t) + \dots \\ &= \tau\gamma p(x,t) + O(\tau h), \end{aligned}$$

where the symbol "$O(k)$" ("big O of k") denotes a quantity such that $O(k)/k$ remains bounded as $k \to 0$.

Thus, eq. (2.84) modifies into

$$p_t\tau + o(\tau) = \frac{1}{2}p_{xx}h^2 + (q_0 - p_0)hp_x - \tau\gamma p + O(\tau h) + o(h^2).$$

Dividing by τ, letting $h, \tau \to 0$ and assuming

$$\frac{h^2}{\tau} = 2D, \qquad \frac{q_0 - p_0}{h} \to \beta,$$

we get

$$p_t = Dp_{xx} + bp_x - \gamma p \qquad (b = 2D\beta). \tag{2.97}$$

The term $-\gamma p$ appears in (2.97) as a decaying term. On the other hand, as we will see in the next subsection, γ could be *negative*, meaning that this time we have a *creation of mass* at the rate $|\gamma|$. For this reason the last term is generically called a *reaction term* and (2.97) is a *diffusion equation with drift and reaction.* Going back to equation (2.97), it is useful to look separately at the effect of the three terms in its right hand side.

- $p_t = Dp_{xx}$ models pure diffusion. The typical effects are spreading and smoothing, as shown by the typical behavior of the fundamental solution Φ_D.
- $p_t = bp_x$ is a pure transport equation, that we will consider in detail in Chap. 4. The solutions are travelling waves of the form $g(x + bt)$.
- $p_t = -\gamma p$ models pure reaction. The solutions are multiples of $e^{-\gamma t}$, exponentially decaying (increasing) if $\gamma > 0$ ($\gamma < 0$).

So far we have given a probabilistic interpretation for a motion in all $\mathbb{R}$, where no boundary condition is present. Problems 2.11 and 2.12 give a probabilistic interpretation of the Neumann and Dirichlet conditions in terms of *reflecting* and *absorbing* boundaries, respectively.

2.5.5 Critical dimension in a simple population dynamics

When $-\gamma = a > 0$ in (2.97), a competition between reaction and diffusion occurs. We examine this effect on the following simple population dynamics problem:

$$\begin{cases} u_t - Du_{xx} = au & 0 < x < L, t > 0 \\ u(0,t) = u(L,t) = 0 & t > 0 \\ u(x,0) = g(x) & 0 < x < L, \end{cases} \tag{2.98}$$

where u represents the density of a population of individuals. Accordingly, we assume that the initial datum g is nonnegative and nontrivial. In this case, the homogeneous Dirichlet conditions model an hostile external environment[18]. Given this kind of boundary condition, the population decays by diffusion while tends to increase by reaction. Thus the two effects compete and we want to explore which factors determine the overwhelming one.

First of all, since a is constant, we can get rid of the term au by setting

$$u(x,t) = e^{at} w(x,t).$$

We have:

$$u_t = e^{at}(aw + w_t),\ u_x = e^{at} w_x,\ u_{xx} = e^{at} w_{xx}$$

and substituting into the differential equation, after some simple algebra, we find for w the equation

$$w_t - Dw_{xx} = 0,$$

with the same boundary and initial conditions:

$$\begin{aligned} w(0,t) &= w(L,t) = 0 \\ w(x,0) &= g(x). \end{aligned}$$

Then, we can easily exhibit an explicit formula for the solution, using the separation of variables (see Problem 2.1):

$$w(x,t) = \sum_{k=1}^{\infty} b_k \exp\left(-D\frac{k^2\pi^2}{L^2}t\right) \sin\frac{k\pi x}{L} \tag{2.99}$$

with

$$b_k = \frac{2}{L}\int_0^L g(x) \sin\frac{k\pi x}{L}\, dx.$$

If $g \in C^2([0,L])$ and $g(0) = g(L) = 0$, then $|b_k| \le C/k^2$ so that the series (2.99) converges uniformly for $t \ge 0$ and $0 \le x \le L$. Going back to u, we get for the solution to problem (2.98) the following expression:

[18] A homogeneous Neumann condition would represent the evolution of an isolated population, without external exchange.

$$u(x,t) = \sum_{k=1}^{\infty} b_k \exp\left\{\left(a - D\frac{k^2\pi^2}{L^2}\right)t\right\} \sin\frac{k\pi x}{L}. \tag{2.100}$$

Formula (2.100) displays an important difference from the pure diffusion case $a = 0$, as far as the asymptotic behavior for $t \to +\infty$ is concerned. Since $g \geq 0$ is nontrivial, $b_1 > 0$, and the population evolution is determined by the largest exponential in the series (2.100), corresponding to $k = 1$. It is now an easy matter to draw the following conclusions.

1. If
$$a - D\frac{\pi^2}{L^2} < 0, \quad \text{then} \quad \lim_{t\to+\infty} u(x,t) = 0$$
uniformly in $[0, L]$, since $a - D\dfrac{k^2\pi^2}{L^2} < a - D\dfrac{\pi^2}{L^2}$ for every $k > 1$.
2. If
$$a - D\frac{\pi^2}{L^2} > 0, \quad \text{then} \quad \lim_{t\to+\infty} u(x,t) = +\infty$$
for $x \neq 0, L$, since the first exponential blows up exponentially and the other terms are either of lower order or vanish exponentially.
3. If $a - D\dfrac{\pi^2}{L^2} = 0$, (2.100) becomes
$$u(x,t) = b_1 \sin\frac{\pi x}{L} + \sum_{k=2}^{\infty} b_k \exp\left\{\left(a - D\frac{k^2\pi^2}{L^2}\right)t\right\} \sin\frac{k\pi x}{L}.$$
Since $a - Dk^2\pi^2/L^2 < 0$ if $k > 1$, we deduce that
$$u(x,t) \to b_1 \sin\frac{\pi x}{L} \quad \text{as } t \to +\infty, \text{ uniformly in } [0, L].$$

Let us interpret the above results. The coefficients a and D are intrinsic parameters, encoding the features of the population and of the environment. When these parameters are fixed, the habitat size plays a major role. In fact, the value

$$L_0 = \pi\sqrt{\frac{D}{a}}$$

represents a critical value for the population survival. If $L < L_0$ (case 1) the habitat is too small to avoid the extinction of the population; on the contrary, if $L > L_0$ (case 2), one observes exponential growth. If $L = L_0$ (case 3) diffusion and reaction *balance* and the population evolves towards the solution $b_1 \sin(\pi x/L)$ of the stationary problem

$$\begin{cases} -Du_{xx} = au & \text{in } (0, L) \\ u(0) = u(L) = 0, \end{cases}$$

that is the *steady state solution*.

Finally note that if for some $\overline{k} \geq 1$ we have $a - D\overline{k}\pi^2/L^2 > 0$, then for every k, $1 \leq k \leq \overline{k}$, all the values $a - Dk\pi^2/L^2$ are positive and the corresponding terms in the series (2.100) contributes to the exponential growth of the solution. From the model viewpoint, one can say that the *vibration modes*

$$b_k \exp\left\{\left(a - D\frac{k^2\pi^2}{L^2}\right)t\right\} \sin\frac{k\pi x}{L}$$

for $k = 1, 2, \ldots, \overline{k}$ *are activated*.

In this simple example, the habitat is 1-dimensional and the reaction homogeneous (i.e. a is constant). We will deepen and extend this analysis to heterogeneous, n-dimensional models, possibly nonlinear, in Ch. 10, Sect. 10.5 on page 590.

2.6 Multidimensional Random Walk

2.6.1 The symmetric case

What we have done in dimension $n = 1$ can be extended without much effort to dimension $n > 1$, in particular $n = 2, 3$. To define a symmetric random walk, we introduce the *lattice* $\mathbb{Z}^n$ given by the set of points $\mathbf{x} \in \mathbb{R}^n$, whose coordinates are signed integers. Given the *space step* $h > 0$, the symbol $h\mathbb{Z}^n$ denotes the lattice of points whose coordinates are signed integers *multiplied by* h.

Every point $\mathbf{x} \in h\mathbb{Z}^n$, has a "discrete neighborhood" of $2n$ points at distance h, given by

$$\mathbf{x} + h\mathbf{e}_j \qquad \text{and} \qquad \mathbf{x} - h\mathbf{e}_j \qquad (j = 1, \ldots, n),$$

where $\{\mathbf{e}_1, \ldots, \mathbf{e}_n\}$ is the canonical basis in $\mathbb{R}^n$. Our particle moves in $h\mathbb{Z}^n$ according to the following rules (Fig. 2.11).

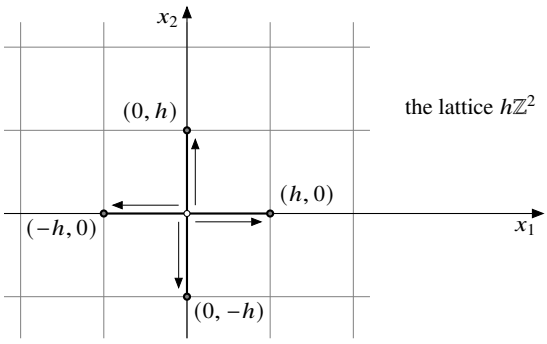

Fig. 2.11 Bidimensional random walk

1. It starts from $\mathbf{x} = \mathbf{0}$.
2. If it is located in $\mathbf{x}$ at time t, at time $t + \tau$ the particle location is at one of the $2n$ points $\mathbf{x} \pm h\mathbf{e}_j$, with probability $p = \frac{1}{2n}$ independently of the previous steps.

As in the 1-dimensional case, our task is to *compute the probability* $p(\mathbf{x}, t)$ *of finding the particle at* $\mathbf{x}$ *at time* t.

Clearly the initial conditions for p are

$$p(\mathbf{0}, 0) = 1 \quad \text{and} \quad p(\mathbf{x}, 0) = 0 \quad \text{if } \mathbf{x} \neq \mathbf{0}.$$

The total probability formula gives

$$p(\mathbf{x}, t + \tau) = \frac{1}{2n} \sum_{j=1}^{n} \left\{ p(\mathbf{x} + h\mathbf{e}_j, t) + p(\mathbf{x} - h\mathbf{e}_j, t) \right\}. \tag{2.101}$$

Indeed, to reach the point $\mathbf{x}$ at time $t + \tau$, at time t the particle must have been located at one of the points in the discrete neighborhood of $\mathbf{x}$ and moved from there towards $\mathbf{x}$ with probability $1/2n$. Keeping $\mathbf{x}$ and t fixed, we want to examine what happens when we let $h \to 0, \tau \to 0$. Assuming that p is defined and smooth in all of $\mathbb{R}^n \times (0, +\infty)$, we use Taylor's formula to write

$$p(\mathbf{x}, t + \tau) = p(\mathbf{x}, t) + p_t(\mathbf{x}, t)\tau + o(\tau)$$
$$p(\mathbf{x} \pm h\mathbf{e}_j, t) = p(\mathbf{x}, t) \pm p_{x_j}(\mathbf{x}, t)h + \frac{1}{2} p_{x_j x_j}(\mathbf{x}, t)h^2 + o(h^2).$$

Substituting into (2.101), after some simplifications, we get

$$p_t \tau + o(\tau) = \frac{h^2}{2n} \Delta p + o(h^2).$$

Dividing by τ we obtain the equation

$$p_t + o(1) = \frac{1}{2n} \frac{h^2}{\tau} \Delta p + o\left(\frac{h^2}{\tau}\right). \tag{2.102}$$

The situation is quite similar to the 1-dimensional case: still, to obtain eventually something nontrivial, we must require that the ratio h^2/τ has a finite and positive limit. The simplest choice is

$$\frac{h^2}{\tau} = 2nD \tag{2.103}$$

with $D > 0$. From (2.103), we deduce that *in unit time, the particle diffuses up to an average distance* $\sqrt{2nD}$. The physical dimensions of D have not changed. Letting $h \to 0, \tau \to 0$ in (2.102), we find for p the diffusion equation

$$p_t = D\Delta p, \tag{2.104}$$

with the initial condition

$$\lim_{t\to 0^+} p(\mathbf{x},t) = \delta_n(\mathbf{x}). \tag{2.105}$$

The unique solution is given by

$$p(\mathbf{x},t) = \Phi_D(\mathbf{x},t) = \frac{1}{(4\pi Dt)^{n/2}} e^{-\frac{|\mathbf{x}|^2}{4Dt}}, \qquad t > 0.$$

The n-dimensional random walk has become a continuous walk; when $D = \frac{1}{2}$, it is called *n-dimensional Brownian motion.* Denote by $\mathbf{B}(t) = \mathbf{B}(t,\omega)$ the random position of a Brownian particle, defined for every $t > 0$ on a probability space $(\Omega, \mathcal{F}, P)$ (see Appendix A).

The family of random variables $\mathbf{B}(t,\omega)$, with time t as a real parameter, is **a vector valued continuous stochastic process**. For $\omega \in \Omega$ fixed, the *vector* function

$$t \longmapsto \mathbf{B}(t,\omega)$$

describes an n-dimensional Brownian path, whose main features are listed below.

- *Path continuity.* With probability 1, the Brownian paths are continuous for $t \geq 0$.
- *Gaussian law for increments.* The process

$$\mathbf{B}^{\mathbf{x}}(t) = \mathbf{x} + \mathbf{B}(t)$$

defines a Brownian motion with start at $\mathbf{x}$. With every point $\mathbf{x}$ is associated a probability $P^{\mathbf{x}}$, with the following properties (if $\mathbf{x} = \mathbf{0}$, $P^{\mathbf{0}} = P$).

a) $P^{\mathbf{x}}\{\mathbf{B}^{\mathbf{x}}(0) = \mathbf{x}\} = P\{\mathbf{B}(0) = \mathbf{0}\} = 1$.
b) For every $s \geq 0, t \geq 0$, the increment

$$\mathbf{B}^{\mathbf{x}}(t+s) - \mathbf{B}^{\mathbf{x}}(s) = \mathbf{B}(t+s) - \mathbf{B}(s) \tag{2.106}$$

follows a *normal law with zero mean value and covariance matrix equal to t* Id_n (Id_n is the $n \times n$ identity matrix), whose density is

$$\Phi(\mathbf{x},t) = \Phi_{\frac{1}{2}}(\mathbf{x},t) = \frac{1}{(2\pi t)^{n/2}} e^{-\frac{|\mathbf{x}|^2}{2t}}.$$

Moreover, (2.106) is independent of any event occurred at any time less than s. For instance, the two events

$$\{\mathbf{B}(t_2) - \mathbf{B}(t_1) \in A_1\} \qquad \{\mathbf{B}(t_1) - \mathbf{B}(t_0) \in A_2\}$$

are independent if $t_0 < t_1 < t_2$.

- *Transition function.* For each Borel set $A \subseteq \mathbb{R}^n$ a *transition function*

$$P(\mathbf{x},t,A) = P^{\mathbf{x}}\{\mathbf{B}^{\mathbf{x}}(t) \in A\}$$

is defined, representing the probability that the particle, initially located at $\mathbf{x}$, belongs to A at time t. We have:

$$P(\mathbf{x}, t, A) = P\{\mathbf{B}(t) \in A - \mathbf{x}\} = \int_{A-\mathbf{x}} \Phi(\mathbf{y}, t) d\mathbf{y} = \int_A \Phi(\mathbf{y} - \mathbf{x}, t) d\mathbf{y}.$$

• *Invariance.* The motion is invariant with respect to rotations and translations.

• *Markov and strong Markov properties.* Let μ be a probability measure on $\mathbb{R}^n$. If the particle has a random initial position with probability distribution μ, we can consider a *Brownian motion with initial distribution* μ, and for it we use the symbol $\mathbf{B}^\mu$. To $\mathbf{B}^\mu$ is associated a probability distribution P^μ such that

$$P^\mu\{\mathbf{B}^\mu(0) \in A\} = \mu(A).$$

The probability that the particle belongs to A at time t can be computed through the formula

$$P^\mu\{\mathbf{B}^\mu(t) \in A\} = \int_{\mathbb{R}^n} P(\mathbf{x}, t, A)\mu(d\mathbf{x}). \tag{2.107}$$

The *Markov property* can be stated as follows: given any condition H, related to the behavior of the particle before time $s \geq 0$, the process $\mathbf{Y}(t) = \mathbf{B}^\mathbf{x}(t+s)$ is a Brownian motion with initial distribution

$$\mu(A) = P^\mathbf{x}\{\mathbf{B}^\mathbf{x}(s) \in A \,|H\}.$$

Again, this property establishes the independence of the *future* process $\mathbf{B}^\mathbf{x}(t+s)$ from the *past*, when the *present* $\mathbf{B}^\mathbf{x}(s)$ is known and encodes the *absence of memory* of the process. In the strong Markov property, a stopping time τ takes the place of s.

• *Expected value.* Given any sufficiently smooth real function $g = g(\mathbf{y})$, $\mathbf{y} \in \mathbb{R}^n$, we can define the real random variable

$$Z(t) = (g \circ \mathbf{B}^\mathbf{x})(t) = g(\mathbf{B}^\mathbf{x}(t)).$$

Its *expectation* is given by the formula

$$E[Z(t)] = \int_{\mathbb{R}^n} g(\mathbf{y}) P(\mathbf{x}, t, d\mathbf{y}) = \int_{\mathbb{R}^n} g(\mathbf{y}) \Phi(\mathbf{y} - \mathbf{x}, t)\, d\mathbf{y}.$$

2.6.2 Walks with drift and reaction

As in the 1-dimensional case, we can construct several variants of the symmetric random walk. For instance, we can allow a different behavior along each direction, by choosing the spatial step h_j depending on $\mathbf{e}_j$. As a consequence the limit process models an anisotropic motion, codified in the matrix

$$\mathbf{D} = \begin{pmatrix} D_1 & 0 & \cdots & 0 \\ 0 & D_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & D_n \end{pmatrix}$$

where $D_j = h_j^2/2n\tau$ is the diffusion coefficient in the direction $\mathbf{e}_j$. The resulting equation for the transition probability $p(\mathbf{x}, t)$ is

$$p_t = \sum_{j=1}^{n} D_j p_{x_j x_j}. \tag{2.108}$$

We may also break the symmetry by asking that along the direction $\mathbf{e}_j$ the probability to go to the left (right) is q_j (resp. p_j) with $\sum_{j=1}^{n}(p_j + q_j) = 1$. If

$$\frac{q_j - p_j}{h_j} \to \beta_j \quad \text{and} \quad b_j = 2D_j\beta_j,$$

the vector $\mathbf{b} = (b_1, \ldots, b_n)$ plays a role of a *drift vector*, reflecting the tendency of motion to move asymmetrically along each coordinate axis. Adding a reaction term of the form cp, the resulting *drift-reaction-diffusion* equation is

$$p_t = \sum_{j=1}^{n} D_j p_{x_j x_j} + \sum_{j=1}^{n} b_j u_{x_j} + cp. \tag{2.109}$$

We let the reader to fill in all the details in the argument leading to equations (2.108) and (2.109). We will deal with general equations of these type in Chap. 9.

2.7 An Example of Reaction–Diffusion in Dimension $n = 3$

In this section we examine a model of reaction–diffusion in a fissionable material. Although we deal with a greatly simplified model, some interesting implications can be drawn.

By shooting neutrons into an uranium nucleus it may happen that the nucleus breaks into two parts, releasing other neutrons already present in the nucleus and causing a chain reaction. Some macroscopic aspects of this phenomenon can be described by means of an elementary model.

Suppose that a cylinder with height h and radius R is made of a fissionable material of constant density ρ, with total mass $M = \pi\rho R^2 h$. At a macroscopic level, the free neutrons diffuse like a chemical in a porous medium, with a flux proportional and opposite to its density gradient. In other terms, if $N = N(x, y, z, t)$ is the *neutron density* and no fission occurs, the *flux of neutrons is equal to* $-\kappa\nabla N$, where κ is a positive constant depending on the material. The mass conservation law then gives $N_t = \kappa\Delta N$. When fission occurs at a constant rate $\gamma > 0$, we get the equation

$$N_t = \kappa \Delta N + \gamma N, \tag{2.110}$$

where reaction and diffusion are competing: diffusion tends to slow down N, while, clearly, the reaction term tends to exponentially increase N. A crucial question is to examine the behavior of N in the long run (i.e. as $t \to +\infty$).

We look for bounded solutions satisfying a homogeneous Dirichlet condition on the boundary of the cylinder, with the idea that the density is higher at the center of the cylinder and very low near the boundary. Then, it is reasonable to assume that N has a radial distribution with respect to the axis of the cylinder. More precisely, using the cylindrical coordinates (r, θ, z), with z along the axis of the cylinder, and

$$x = r\cos\theta, \ y = r\sin\theta,$$

we can write $N = N(r, z, t)$ and the homogeneous Dirichlet condition on the boundary of the cylinder translates into

$$\begin{aligned} &N(R, z, t) = 0 && 0 < z < h \\ &N(r, 0, t) = N(r, h, t) = 0 && 0 < r < R, \end{aligned} \tag{2.111}$$

for every $t > 0$. Accordingly we prescribe an initial condition

$$N(r, z, 0) = N_0(r, z) \tag{2.112}$$

such that

$$N_0(R, z) = 0 \text{ for } 0 < z < h, \text{ and } N_0(r, 0) = N_0(r, h) = 0. \tag{2.113}$$

To solve problem (2.110), (2.111), (2.112), first we get rid of the reaction term by setting

$$N(r, z, t) = \mathcal{N}(r, z, t)e^{\gamma t}. \tag{2.114}$$

Then, writing the Laplace operator in cylindrical coordinates (App. B), $\mathcal{N}$ solves the equation

$$\mathcal{N}_t = \kappa \left[\mathcal{N}_{rr} + \frac{1}{r}\mathcal{N}_r + \mathcal{N}_{zz} \right] \tag{2.115}$$

with the same initial and boundary conditions of N. By maximum principle, we know that there exists only one solution, continuous up to the boundary of the cylinder. To find an explicit formula for the solution, we use the method of separation of variables, first searching for bounded solutions of the form

$$\mathcal{N}(r, z, t) = u(r)v(z)w(t), \tag{2.116}$$

satisfying the homogeneous Dirichlet conditions $u(R) = 0$ and $v(0) = v(h) = 0$.

Substituting (2.116) into (2.115), we find

$$u(r)v(z)w'(t) = \kappa \left[u''(r)v(z)w(t) + \frac{1}{r}u'(r)v(z)w(t) + u(r)v''(z)w(t) \right].$$

Dividing by $\mathcal{N}$ and rearranging the terms, we get,

$$\frac{w'(t)}{Dw(t)} - \left[\frac{u''(r)}{u(r)} + \frac{1}{r}\frac{u'(r)}{u(r)}\right] = \frac{v''(z)}{v(z)}. \tag{2.117}$$

The two sides of (2.117) depend on different variables so that they must be equal to a common constant b. Then for v we have the eigenvalue problem

$$v''(z) - bv(z) = 0$$
$$v(0) = v(h) = 0.$$

The eigenvalues are $b_m \equiv -\eta_m^2 = -\frac{m^2\pi^2}{h^2}$, $m \geq 1$ integer, with corresponding eigenfunctions

$$v_m(z) = \sin \eta_m z.$$

The equation for w and u can be written in the form:

$$\frac{w'(t)}{Dw(t)} + \eta_m^2 = \frac{u''(r)}{u(r)} + \frac{1}{r}\frac{u'(r)}{u(r)} \tag{2.118}$$

where the variables r and t are again separated. This forces the two sides of (2.118) to be equal to a common constant μ. Then the equation for u is

$$u''(r) + \frac{1}{r}u'(r) - \mu u(r) = 0 \tag{2.119}$$

with

$$u(R) = 0 \quad \text{and} \quad u \text{ bounded in } [0, R]\,. \tag{2.120}$$

The (2.119) is a *Bessel equation of order zero with parameter* $-\mu$ (see Sect. 6.4.2); conditions (2.120) force $\mu < 0$. In fact, write Bessel's equation (2.119) in the form $(ru')' - \mu ru = 0$. Multiplying by u and integrating over $(0, R)$, we have

$$\int_0^R (ru')'u dr = \mu \int_0^R ru^2 dr. \tag{2.121}$$

Integrating by parts and using (2.120), we get

$$\int_0^R (ru')'u dr = [(ru')u]_0^R - \int_0^R (u')^2 dr = -\int_0^R (u')^2 dr < 0$$

and from (2.121) we get $\mu < 0$.

Letting $\mu = -\lambda^2$, the only bounded solution of (2.119), (2.120) is $J_0(\lambda r)$, where

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2}\left(\frac{x}{2}\right)^{2k}$$

is the *Bessel function of first kind and order zero* (see Fig. 2.12). To match the

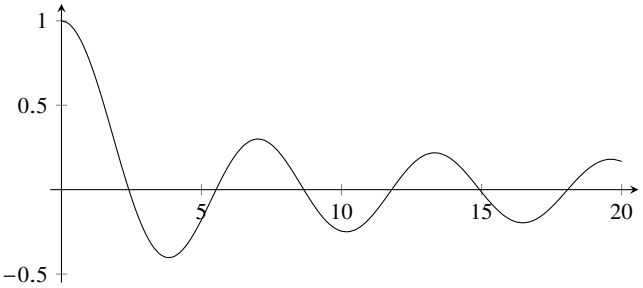

Fig. 2.12 The Bessel function J_0

boundary condition $u(R) = 0$ we require $J_0(\lambda R) = 0$. Now, J_0 has an infinite number of positive simple zeros[19] λ_n, $n \geq 1$:

$$0 < \lambda_1 < \lambda_2 < \ldots < \lambda_n < \ldots .$$

Thus, if $\lambda R = \lambda_n$, we find infinitely many solutions of (2.119), given by

$$u_n(r) = J_0\left(\frac{\lambda_n r}{R}\right)$$

and $\mu = \mu_n = -\lambda_n^2/R^2$. Finally, for w we have the equation

$$w'(t) = D(\mu_n - \eta_m^2)w(t)$$

that gives

$$w_{mn}(t) = \exp\left[D(\mu_n - \eta_m^2)t\right], \qquad c \in \mathbb{R}.$$

To summarize, we have determined so far a countable number of solutions of the differential equation, namely

$$\begin{aligned}\mathcal{N}_{mn}(r,z,t) &= u_n(r)v_m(z)w_{mn}(t) = \\ &= J_0\left(\frac{\lambda_n r}{R}\right)\sin\eta_m z\ \exp\left[-\kappa\left(\eta_m^2 + \frac{\lambda_n^2}{R^2}\right)t\right],\end{aligned}$$

all satisfying the homogeneous Dirichlet conditions. It remains to satisfy the initial condition. Due to the linearity of the problem, we look for a solution obtained by superposition of the $\mathcal{N}_{mn}$, that is

$$\mathcal{N}(r,z,t) = \sum_{n,m=1}^{\infty} c_{mn}\mathcal{N}_{mn}(r,z,t).$$

[19] The zeros of the Bessel functions are known with a considerable degree of accuracy. The first five zeros of J_0 are: 2.4048..., 5.5201..., 8.6537..., 11.7915..., 14.9309....

Then, we choose the coefficients c_{mn} in order to have

$$\sum_{n,m=1}^{\infty} c_{mn}\mathcal{N}_{mn}(r,z,0) = \sum_{n,m=1}^{\infty} c_{mn}J_0\left(\frac{\lambda_n r}{R}\right)\sin\frac{m\pi}{h}z = N_0(r,z). \tag{2.122}$$

Since $N_0(r,0) = N_0(r,h) = 0$, formula (2.122) suggests an expansion of N_0 in sine Fourier series with respect to z. Let

$$c_m(r) = \frac{2}{h}\int_0^h N_0(r,z)\sin\frac{m\pi}{h}z, \qquad m \geq 1,$$

and

$$N_0(r,z) = \sum_{m=1}^{\infty} c_m(r)\sin\frac{m\pi}{h}z.$$

Then (2.122) shows that, for fixed $m \geq 1$, the c_{mn} are the coefficients of the expansion of $c_m(r)$ in the *Fourier-Bessel series*

$$\sum_{n=1}^{\infty} c_{mn}J_0\left(\frac{\lambda_n r}{R}\right) = c_m(r).$$

We are not really interested in the exact formula for the c_{mn}, however we will come back to this point in Remark 2.10 below.

In conclusion, recalling (2.114), the analytic expression of the solution of our original problem is the following:

$$N(r,z,t) = \sum_{n,m=1}^{\infty} c_{mn}J_0\left(\frac{\lambda_n r}{R}\right)\exp\left\{\left(\gamma - \kappa\eta_m^2 - \kappa\frac{\lambda_n^2}{R^2}\right)t\right\}\sin\eta_m z. \tag{2.123}$$

Of course, (2.123) is only a formal solution, since we should check in which sense the boundary and initial condition are attained and that term by term differentiation can be performed. This can be done under reasonable smoothness properties of N_0 and we do not pursue the calculations here.

Rather, we notice that from (2.123) we can draw an interesting conclusion on the long range behavior of N. Consider for instance the value of N at the center of the cylinder, that is at the point $r = 0$ and $z = h/2$; we have, since $J_0(0) = 1$ and $\eta_m^2 = \frac{m^2\pi^2}{h^2}$,

$$N\left(0,\frac{h}{2},t\right) = \sum_{n,m=1}^{\infty} c_{mn}\ \exp\left\{\left(\gamma - \kappa\frac{m^2\pi^2}{h^2} - \kappa\frac{\lambda_n^2}{R^2}\right)t\right\}\sin\frac{m\pi}{2}.$$

The exponential factor is maximized for $m = n = 1$, so, assuming $c_{11} \neq 0$, the leading term in the sum is

$$c_{11} \exp\left\{\left(\gamma - \kappa\frac{\pi^2}{h^2} - \kappa\frac{\lambda_1^2}{R^2}\right)t\right\}.$$

Now we can argue as in Sect. 2.5.5: if $\gamma - \kappa\left(\frac{\pi^2}{h^2} + \frac{\lambda_1^2}{R^2}\right) < 0$, each term in the series goes to zero as $t \to +\infty$ and the reaction dies out. On the opposite, if

$$\gamma - \kappa\left(\frac{\pi^2}{h^2} + \frac{\lambda_1^2}{R^2}\right) > 0,$$

that is

$$\frac{\gamma}{\kappa} > \frac{\pi^2}{h^2} + \frac{\lambda_1^2}{R^2}, \tag{2.124}$$

the leading term increases exponentially with time. To be true, (2.124) requires that the following relations be both satisfied:

$$h^2 > \frac{\kappa\pi^2}{\gamma} \quad \text{and} \quad R^2 > \frac{\kappa\lambda_1^2}{\gamma}. \tag{2.125}$$

Equations (2.125) give a lower bound for the height and the radius of the cylinder. Thus we deduce that *there exists a critical mass of material, below which the reaction cannot be sustained.*

Remark 2.10 A sufficiently smooth function f, for instance of class $C^1([0, R])$, can be expanded in a Fourier-Bessel series, where the Bessel functions $J_0\left(\frac{\lambda_n r}{R}\right)$, $n \geq 1$, play the same role of the trigonometric functions. More precisely (see e.g. *R. Courant, D. Hilbert*, [26], vol. 1), setting $R = 1$ for simplicity, the functions $J_0(\lambda_n r)$ satisfy the following orthogonality relations:

$$\int_0^1 x J_0(\lambda_m x) J_0(\lambda_n x) dx = \begin{cases} 0 & m \neq n \\ \frac{1}{2}c_n^2 & m = n \end{cases}$$

where

$$c_n = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+1)!}\left(\frac{\lambda_n}{2}\right)^{2k+1}.$$

Then

$$f(x) = \sum_{n=0}^{\infty} f_n J_0(\lambda_n x) \tag{2.126}$$

with the coefficients f_n assigned by the formula

$$f_n = \frac{2}{c_n^2}\int_0^1 x f(x) J_0(\lambda_n x) dx.$$

The series (2.126) converges in the following mean square sense: if

$$S_N(x) = \sum_{n=0}^{N} f_n J_0(\lambda_n x)$$

then

$$\lim_{N\to+\infty} \int_0^1 [f(x) - S_N(x)]^2 \, x dx = 0. \tag{2.127}$$

In Chap. 6, we will interpret (2.127) from the point of view of Hilbert space theory.

2.8 The Global Cauchy Problem ($n = 1$)

2.8.1 The homogeneous case

In this section we consider the global Cauchy problem

$$\begin{cases} u_t - Du_{xx} = 0 & \text{in } \mathbb{R} \times (0, \infty) \\ u(x, 0) = g(x) & \text{in } \mathbb{R}, \end{cases} \tag{2.128}$$

where g, the *initial datum*, is given. We confine ourselves to the one dimensional case; techniques, ideas and formulas can be extended without too much effort to the n-dimensional case.

The problem (2.128) models the evolution of the temperature or of the concentration of a substance along a very long (infinite) bar or channel, respectively, given the initial ($t = 0$) distribution.

By heuristic considerations, we can guess what could be a candidate solution. Interpret u as a linear mass concentration. Then, $u(x, t)dx$ gives the mass inside the interval $(x, x + dx)$ at time t.

We want to determine the evolution of $u(x, t)$, due to the diffusion of a mass whose initial concentration is given by g.

The quantity $g(y)dy$ represents the mass concentrated in the interval $(y, y + dy)$ at time $t = 0$. As we have seen, $\Phi(x - y, t)$ is a *unit source solution*, representing the concentration at x at time t, due to the diffusion of a unit mass, initially concentrated in the same interval. Accordingly,

$$\Phi_D(x - y, t)g(y)dy$$

gives the concentration at x at time t, due to the diffusion of the mass $g(y)dy$.

Thanks to the linearity of the diffusion equation, we can use the *superposition principle* and compute the solution as the sum of all contributions. In this way, we get the formula

$$u(x, t) = \int_{\mathbb{R}} \Phi_D(x - y, t)g(y) \, dy = \frac{1}{\sqrt{4\pi Dt}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{4Dt}} g(y) \, dy. \tag{2.129}$$

Clearly, one has to check rigorously that, under reasonable hypotheses on the initial datum g, formula (2.129) really gives the unique solution of the Cauchy problem. This is not a negligible question. First of all, if g grows too much at infinity, more than an exponential of the type e^{ax^2}, $a > 0$, in spite of the rapid convergence to zero of the Gaussian, the integral in (2.129) could be divergent and formula (2.129) loses any meaning.

Even more delicate is the question of the uniqueness of the solution, as we will see later.

Remark 2.11 Formula (2.129) has a probabilistic interpretation. Let $D = \frac{1}{2}$ and let $B^x(t)$ be the position of a Brownian particle, started at x. Let $g(y)$ be the *gain* obtained when the particle crosses y. Then, we can write:

$$u(x,t) = E^x\left[g(B^x(t))\right]$$

where E^x denotes *the expected value* with respect to the probability P^x, with density $\Phi(x-y,t)$, see Appendix A.

In other words: *to compute u at the point (x,t), consider a Brownian particle starting at x, compute its position $B^x(t)$ at time t, and finally compute the expected value of $g(B^x(t))$.*

2.8.2 Existence of a solution

The following theorem states that (2.129) is indeed a solution of the global Cauchy problem under rather general hypotheses on g, satisfied in most of the interesting applications. Although it is a little bit technical, the proof can be found at the end of this section.

Theorem 2.12 *Assume that there exist positive numbers a and c such that*

$$|g(x)| \le ce^{ax^2} \quad \text{for all } x \in \mathbb{R}. \tag{2.130}$$

Let u be given by formula (2.129) and $T < \frac{1}{4aD}$. Then, the following properties hold.

a) There are positive numbers C and A such that

$$|u(x,t)| \le Ce^{Ax^2} \qquad \text{for all } (x,t) \in \mathbb{R} \times (0,T].$$

b) $u \in C^\infty(\mathbb{R} \times (0,T])$ and in the strip $\mathbb{R} \times (0,T]$

$$u_t - Du_{xx} = 0.$$

c) Let $(x,t) \to (x_0, 0^+)$. If g is continuous at x_0 then $u(x,t) \to g(x_0)$.

Remark 2.13 The theorem says that, if we allow an initial data with a controlled exponential growth at infinity expressed by (2.130), then (2.129) is a solution in the strip $\mathbb{R} \times (0,T]$.

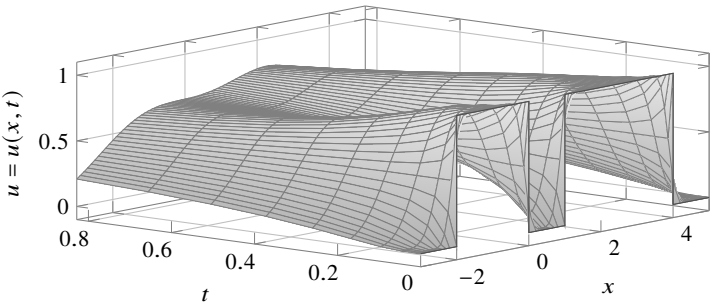

Fig. 2.13 Smoothing effect of the diffusion equation

In some applications (e.g. to Finance), the initial datum grows at infinity no more than $c_1 e^{a_1|x|}$. In this case (2.130) is satisfied by choosing any positive number a and a suitable c. This means that there is really no limitation on T, since

$$T < \frac{1}{4Da}$$

and a can be chosen as small as we like.

Remark 2.14 The property b) shows a typical and important phenomenon connected with the diffusion equation: even if the initial data is discontinuous at some point, immediately after the solution is smooth. The diffusion is therefore a **smoothing process**.

In Fig. 2.13, this phenomenon is shown for the initial data $g(x) = \chi_{(-2,0)}(x) + \chi_{(1,4)}(x)$, where $\chi_{(a,b)}$ denotes the characteristic function of the interval (a, b). By c), if the initial data g is continuous in all of $\mathbb{R}$, then the solution is continuous up to $t = 0$, that is in $\mathbb{R} \times [0, T)$.

2.8.3 The nonhomogeneous case. Duhamel's method

The difference equation (or the total probability formula)

$$p(x, t + \tau) = \frac{1}{2} p(x - h, t) + \frac{1}{2} p(x + h, t),$$

that we found in Sect. 2.4.2 during the analysis of the symmetric random walk, could be considered a probabilistic version of the *mass conservation principle*: the mass located at x *at time* $t + \tau$ *is the sum of the masses diffused from* $x + h$ *and* $x - h$ *at time* t; no mass has been lost or added over the time interval $[t, t + \tau]$. Accordingly, the expression

$$p(x, t+\tau) - \left[\frac{1}{2}p(x-h,t) + \frac{1}{2}p(x+h,t)\right] \tag{2.131}$$

could be considered as a measure of the lost/added mass over the time interval from t to $t+\tau$. Expanding with Taylor's formula as we did in Sect. 2.4.2, keeping $h^2/\tau = 2D$, dividing by τ and letting $h, \tau \to 0$ in (2.131), we find

$$p_t - Dp_{xx}.$$

Thus the differential operator $\partial_t - D\partial_{xx}$ **measures the instantaneous density production rate.**

Suppose now that, from time $t = 0$ until a certain time $t = s > 0$, no mass is present and that at time s a unit mass at the point y (infinite density) appears. We know that we can model this kind of source by means of a Dirac measure at y, that has to be time dependent since the mass appears only at time s. We can write it in the form

$$\delta_2(x-y, t-s).$$

Thus, we are led to the nonhomogeneous equation

$$p_t - Dp_{xx} = \delta_2(x-y, t-s)$$

with $p(x,0) = 0$ as initial condition. What could be the solution? Until $t = s$ nothing happens and *after* s we have $\delta_2(x-y, t-s) = 0$. Therefore it is like starting from time $t = s$ and solving the problem

$$p_t - Dp_{xx} = 0, \qquad x \in \mathbb{R},\ t > s$$

with initial condition (at $t = s$)

$$p(x, s) = \delta(x-y).$$

We have solved this problem when $s = 0$; the solution is $\Phi_D(x-y, t)$. By the time translation invariance of the diffusion equation, we deduce that the solution for any $s > 0$ is given by (see also Problem 7.19, p. 492)

$$p(x,t) = \Phi_D(x-y, t-s), \qquad t > s. \tag{2.132}$$

Consider now a distributed source on the half-plane $t > 0$, capable to produce mass density at the time rate $f(x,t)$. Precisely, $f(x,t)dxdt$ is the mass produced[20] between x and $x+dx$, over the time interval $(t, t+dt)$. If initially no mass is present, we are lead to the *nonhomogeneous Cauchy problem*

$$\begin{cases} v_t - Dv_{xx} = f(x,t) & \text{in } \mathbb{R} \times (0,T) \\ v(x,0) = 0 & \text{in } \mathbb{R}. \end{cases} \tag{2.133}$$

[20] Negative production ($f < 0$) means removal.

As in Sect. 2.8.1, we motivate the form of the solution at the point (x,t) using heuristic considerations.

Let us compute the contribution dv to $v(x,t)$ due to a mass $f(y,s)\,dyds$. It is like having a source term of the form

$$f^*(x,t) = f(x,t)\delta_2(x-y,t-s)$$

and therefore, recalling (2.132), we have

$$dv(x,t) = \Phi_D(x-y,t-s)f(y,s)dyds. \tag{2.134}$$

We obtain the solution $v(x,t)$ by superposition, summing all the contributions (2.134). We split it into the following two steps:

• We sum over y the contributions for fixed s, to get the total density at (x,t), due to the diffusion of mass produced in the time interval $(s, s+ds)$. The result is $w(x,t;s)ds$, where

$$w(x,t;s) = \int_{\mathbb{R}} \Phi_D(x-y,t-s)f(y,s)dy. \tag{2.135}$$

• We sum the above contributions for s ranging from 0 to t:

$$v(x,t) = \int_0^t \int_{\mathbb{R}} \Phi_D(x-y,t-s)f(y,s)dyds.$$

The above construction is an example of application of the *Duhamel's method*, that we state below:

Duhamel's method. The procedure to compute the solution u of problem (2.133) at the point (x,t) consists in the following two steps:

1. Construct a family of solutions of homogeneous Cauchy problems, with variable initial time $s, 0 \le s \le t$, and initial data $f(x,s)$.
2. Integrate the above family with respect to s, over $(0,t)$.

Indeed, let us examine the two steps.

1. Consider the family of homogeneous Cauchy problems

$$\begin{cases} w_t - Dw_{xx} = 0 & x \in \mathbb{R},\ t > s \\ w(x,s;s) = f(x,s) & x \in \mathbb{R}, \end{cases} \tag{2.136}$$

where the initial time s plays the role of a parameter.

The function $\Phi^{y,s}(x,t) = \Phi_D(x-y,t-s)$ is the fundamental solution of the diffusion equation that satisfies for $t = s$ the initial condition

$$\Phi^{y,s}(x,s) = \delta(x-y).$$

Hence, the solution of (2.136) is given by the function (2.135):

$$w(x,t;s) = \int_{\mathbb{R}} \Phi_D(x-y,t-s) f(y,s) dy.$$

Thus, $w(x,t;s)$ is the required family.

2. Integrating w over $(0,t)$ with respect to s, we find

$$v(x,t) = \int_0^t w(x,t;s) ds = \int_0^t \int_{\mathbb{R}} \Phi_D(x-y,t-s) f(y,s) dy ds. \tag{2.137}$$

It is easy to check that v solves (2.133). Indeed, using (2.136) we have

$$v_t - Dv_{xx} = w(x,t;t) + \int_0^t [w_t(x,t;s) - Dw_{xx}(x,t;s)] = f(x,t),$$

and $v(x,0) = 0$.

Everything works under rather mild hypotheses on f, as shown in the following theorem, where the formula for the general Cauchy problem is obtained by superposition of (2.129) and (2.133). For a proof, see *R. McOwen*, [13].

Theorem 2.15 *Let g, T as in Theorem 2.12. If f and its derivatives f_t, f_x, f_{xx} are continuous and bounded in $\mathbb{R} \times [0,T)$, the function*

$$u(x,t) = \int_{\mathbb{R}} \Phi_D(x-y,t) g(y)\, dy + \int_0^t \int_{\mathbb{R}} \Phi(x-y,t-s) f(y,s)\, dy ds \tag{2.138}$$

is continuous with its derivatives u_t, u_x, u_{xx} in $\mathbb{R} \times (0,T]$ and solves the Cauchy problem

$$\begin{cases} u_t - Du_{xx} = f(x,t) & \text{in } \mathbb{R} \times (0,T] \\ u(x,0) = g(x) & \text{in } \mathbb{R}. \end{cases} \tag{2.139}$$

Moreover, $u(x,t) \to g(x_0)$ as $(x,t) \to (x_0,0)$ at any continuity point x_0 of g.

Remark 2.16 Duhamel method is a generalization of the method of arbitrary constants used to solve a Cauchy problem for linear ordinary differential equations. Indeed, for instance, the formula

$$u(t) = \int_0^t e^{a(t-s)} f(s) ds \qquad (a \in \mathbb{R}) \tag{2.140}$$

gives the solution to $u' - au = f(t)$, $u(0) = 0$, and it is obtained by integrating with respect to the parameter s, over $(0,t)$, the family $w(t;s) = e^{a(t-s)} f(s)$, which is the solution to $w' - aw = 0$, $w(s;s) = f(s)$, in perfect analogy with (2.136).

The use of the Duhamel method is not confined to the global Cauchy problem for the heat equation; it works also for all the initial boundary value problems we met in this chapter, see Problem 2.3, and also for other kind of equations, as we shall see in Ch. 5 for the wave equation.

2.8.4 Global maximum principle and uniqueness

The uniqueness of the solution to the global Cauchy problem is still to be discussed. This is not a trivial question, since the following counterexample of Tychonov shows that, in general, there is lack of uniqueness. Define

$$\mathcal{T}(x,t) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \frac{d^k}{dt^k} h(t) \tag{2.141}$$

where $h(t) = \exp(-t^{-2})$. The function $\mathcal{T}$ is a solution of the heat equation ($D = 1$) in $\mathbb{R} \times (0, +\infty)$, continuously vanishing on $t = 0$.

Indeed, it can be checked (see *F. John*, [9]) that there exists $\theta > 0$ such that, for every $t > 0$,

$$\left| \frac{d^k}{dt^k} h(t) \right| \leq \frac{k!}{(\theta t)^k} e^{-t^{-2}/2}$$

and since $k!/(2k)! < 1/k!$, for any $t > 0$ and $x \in \mathbb{R}$, we have

$$|\mathcal{T}(x,t)| \leq e^{-t^{-2}/2} \sum_{k=0}^{\infty} \frac{x^{2k}}{k!(\theta t)^k} = \exp\left(\frac{x^2}{\theta t} - \frac{1}{2t^2} \right). \tag{2.142}$$

Thus, the series (2.141) is convergent for all $x \in \mathbb{R}$ and $t > 0$. Similar estimates hold for the t and x derivatives, allowing to differentiate term by term and to check that

$$\mathcal{T}_t(x,t) = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!} \frac{d^{k+1}}{dt^{k+1}} h(t) = \sum_{k=1}^{\infty} \frac{x^{2k-2}}{(2k-2)!} \frac{d^k}{dt^k} h(t) = \mathcal{T}_{xx}(x,t),$$

so that $\mathcal{T}$ is a solution of the heat equation $u_t - u_{xx} = 0$ in $\mathbb{R} \times (0, +\infty)$. Moreover if x remains bounded, it is clear from (2.142) that $\mathcal{T}(x,t) \to 0$ uniformly in x as $t \to 0$.

Since also the function u identically zero on $\mathbb{R} \times (0, +\infty)$ is a solution of the heat equation, continuously vanishing on $t = 0$, we conclude that, in general, the Cauchy problem is *not well posed*.

What is wrong with $\mathcal{T}$? Ultimately, the problem is that $\mathcal{T}$ grows too much at infinity for small times. For instance, the estimate (2.142) rapidly deteriorates for small $t > 0$ and x large (e.g. $x > 1/t$).

If instead of $1/\theta t$ we had a constant a, as in condition a) of Theorem 2.12, p. 79, then we could assure uniqueness.

In other words, among the class of functions with growth at infinity controlled by an exponential of the type Ce^{Ax^2} for any $t \geq 0$ (the so called *Tychonov class*), the solution to the homogeneous Cauchy problem is unique.

This is a consequence of the following maximum principle.

Theorem 2.17 (Global maximum principle) *Let z be continuous in $\mathbb{R} \times [0,T]$, with derivatives z_x, z_{xx}, z_t continuous in $\mathbb{R} \times (0,T)$, such that, in $\mathbb{R} \times (0,T)$:*

$$z_t - Dz_{xx} \leq 0 \qquad (resp. \geq 0)$$

and

$$z(x,t) \leq Ce^{Ax^2}, \qquad (resp. \geq -Ce^{Ax^2}) \tag{2.143}$$

where $C > 0$. Then

$$\sup_{\mathbb{R}\times[0,T]} z(x,t) \leq \sup_{\mathbb{R}} z(x,0) \qquad (resp. \inf_{\mathbb{R}\times[0,T]} z(x,t) \geq \inf_{\mathbb{R}} z(x,0)).$$

The proof is rather technical (see *F. John*, [9]), but if we assume that z is bounded from above or below ($A = 0$ in (2.143)), then the proof relies on a simple application of the weak maximum principle, Theorem 2.4, p. 39. In Problem 2.14 we ask the reader to fill in the details of the proof.

We now are in position to prove the following uniqueness result.

Corollary 2.18 (Uniqueness I) *Suppose u is a solution of*

$$\begin{cases} u_t - Du_{xx} = 0 & in\ \mathbb{R}\times(0,T) \\ u(x,0) = 0 & in\ \mathbb{R}, \end{cases}$$

continuous in $\mathbb{R}\times[0,T]$, with derivatives u_x, u_{xx}, u_t continuous in $\mathbb{R}\times(0,T)$. If $|u|$ satisfies (2.143) then $u \equiv 0$.

Proof From Theorem 2.17 we have

$$0 = \inf_{\mathbb{R}} u(x,0) \leq \inf_{\mathbb{R}\times[0,T]} u(x,t) \leq \sup_{\mathbb{R}\times[0,T]} u(x,t) \leq \sup_{\mathbb{R}} u(x,0) = 0$$

so that $u \equiv 0$. □

Notice that if

$$|g(x)| \leq ce^{ax^2} \quad \text{for every } x \in \mathbb{R} \quad (c, a \text{ positive}), \tag{2.144}$$

we know from Theorem 2.12 that

$$u(x,t) = \int_{\mathbb{R}} \Phi_D(x-y,t)g(y)dy$$

satisfies the estimate

$$|u(x,t)| \leq Ce^{Ax^2} \quad \text{in } \mathbb{R}\times(0,T] \tag{2.145}$$

and therefore it belongs to the Tychonov class in $\mathbb{R}\times(0,T]$, for $T < 1/(4Da)$. Moreover, if f is as in Theorem 2.15 and

$$v(x,t) = \int_0^t \int_{\mathbb{R}} \Phi_D(x-y,t-s)f(y,s)\,dyds,$$

we easily get the estimate

$$t \inf_{\mathbb{R}\times[0,T)} f \le v(x,t) \le t \sup_{\mathbb{R}\times[0,T)} f, \tag{2.146}$$

for every $x \in \mathbb{R}$, $0 \le t \le T$. In fact:

$$v(x,t) \le \sup_{\mathbb{R}\times[0,T)} f \int_0^t \int_{\mathbb{R}} \Phi_D(x-y,t-s)dyds = t \sup_{\mathbb{R}\times[0,T)} f$$

since

$$\int_{\mathbb{R}} \Phi_D(x-y,t-s)dy = 1$$

for every x, t, s, $t > s$. In the same way it can be shown that $v(x,t) \ge t \inf_{\mathbb{R}\times[0,T)} f$. As a consequence, we have:

Corollary 2.19 (Uniqueness II) *Let g be continuous in $\mathbb{R}$, satisfying (2.145), and let f be as in Theorem 2.15. Then the Cauchy problem (2.139) has a unique solution u in $\mathbb{R}\times[0,T]$, for $T < 1/(4Da)$, belonging to the Tychonov class. This solution is given by (2.138) and moreover*

$$\inf_{\mathbb{R}} g + t \inf_{\mathbb{R}\times[0,T]} f \le u(x,t) \le \sup_{\mathbb{R}} g + t \sup_{\mathbb{R}\times[0,T]} f. \tag{2.147}$$

Proof If u and v are solutions of the same Cauchy problem (2.139), then $w = u - v$ is a solution of (2.139) with $f = g = 0$ and satisfies the hypotheses of Corollary 2.18. It follows that $w(x,t) \equiv 0$.□

• *Stability and comparison.* As in Corollary 2.5, p. 40, the inequality (2.147) is a stability estimate for the correspondence

$$\text{data} \longmapsto \text{solution}.$$

Indeed, let u_1 and u_2 be solutions of (2.139) with data g_1, f_1 and g_2, f_2, respectively. Under the hypotheses of Corollary 2.19, by (2.147) we can write

$$\sup_{\mathbb{R}\times[0,T]} |u_1 - u_2| \le \sup_{\mathbb{R}} |g_1 - g_2| + T \sup_{\mathbb{R}\times[0,T]} |f_1 - f_2|.$$

Therefore if

$$\sup_{\mathbb{R}\times[0,T]} |f_1 - f_2| \le \varepsilon, \quad \sup_{\mathbb{R}} |g_1 - g_2| \le \varepsilon$$

also

$$\sup_{\mathbb{R}\times[0,T]} |u_1 - u_2| \le \varepsilon(1+T)$$

that means *uniform pointwise stability*.

This is not the only consequence of (2.147). We can use it to compare two solutions. For instance, from the left inequality we immediately deduce that if $f \ge 0$ and $g \ge 0$, also $u \ge 0$.

Similarly, if $f_1 \ge f_2$ and $g_1 \ge g_2$, then

$$u_1 \ge u_2.$$

• *Backward equations* arise in several applied contexts, from *control theory* and *dynamic programming* to *probability* and *finance*. An example is the celebrated *Black–Scholes equation,* that we will present in the next section.

Due to the time irreversibility, to have a well posed problem for the backward equation in the time interval $[0, T]$ we must prescribe a *final condition*, that is for $t = T$, rather than an initial one. On the other hand, the change of variable $t \longmapsto T - t$ transforms the backward into the forward equation, so that, from the mathematical point of view, the two equations are equivalent. Up to this change of variable, the theory we have developed so far remains valid.

2.8.5 Nonnegative solutions

We have seen that additional restrictions on a solution u of the global Cauchy problem are needed to ensure uniqueness. A remarkable result, due to Widder, states that even a lower bound on u, e.g. *nonnegativity*, is enough to determine it uniquely. Namely, the following theorem holds (for the proof, see *F. John*, [9]).

Theorem 2.20 *Let u be continuous in $\mathbb{R} \times [0, +\infty)$, with continuous derivatives u_t, u_x, u_{xx} in $\mathbb{R} \times (0, +\infty)$. Assume that u satisfies the following conditions:*

$$\begin{cases} u_t - Du_{xx} = 0 & \text{in } \mathbb{R} \times (0, +\infty) \\ u(x, 0) = g(x) & \text{in } \mathbb{R} \\ u(x, t) \geq 0 & \text{in } \mathbb{R} \times [0, +\infty). \end{cases}$$

Then u is uniquely determined and it is given by

$$u(x, t) = \int_{\mathbb{R}} \Phi_D(x - y, t) g(y)\, dy.$$

2.8.6 The proof of the existence Theorem 2.12

Proof of a). We want to show that u is well defined in the strip $(0, T]$ with $T < \frac{1}{4Da}$. Since $|g(y)| \leq ce^{ay^2}$, we have, for any positive $t \leq T$:

$$|u(x, t)| \leq \frac{c}{\sqrt{4\pi Dt}} \int_{\mathbb{R}} e^{-\frac{(x-y)^2}{4Dt}} e^{ay^2} dy \underset{z=x-y}{=} \frac{c}{\sqrt{4\pi Dt}} \int_{\mathbb{R}} e^{-\frac{z^2}{4Dt} + a(x-z)^2} dz. \tag{2.148}$$

Since $\frac{1}{4Dt} > a$, we can find a small positive ε such that

$$\frac{1 - \varepsilon}{4Dt} > a + \varepsilon, \quad \forall t \in (0, T]. \tag{2.149}$$

Then, adding and subtracting ε,

$$-\frac{1}{4Dt} = -\frac{\varepsilon}{4Dt} - \frac{1 - \varepsilon}{4Dt} \leq -\frac{\varepsilon}{4Dt} - (a + \varepsilon)$$

so that

$$-\frac{z^2}{4Dt}+a(x-z)^2 \le -\frac{\varepsilon z^2}{4Dt}-(a+\varepsilon)z^2+a(x-z)^2 = -\frac{\varepsilon z^2}{4Dt}-\varepsilon z^2-2axz+ax^2. \quad (2.150)$$

Now, let $A = a+\varepsilon^{-1}a^2$; then

$$-\varepsilon z^2-2axz+ax^2 = -\varepsilon z^2-2axz-\varepsilon^{-1}a^2x^2+Ax^2 = -\varepsilon(z+\varepsilon^{-1}ax)^2+Ax^2 \le Ax^2. \quad (2.151)$$

With this choice of A, from (2.148), (2.150) and (2.151), we have

$$\exp\left(a(x-z)^2-\frac{z^2}{4Dt}\right) \le c\exp\left(Ax^2-\frac{\varepsilon z^2}{4Dt}\right) \quad (2.152)$$

and we can write

$$|u(x,t)| \le \frac{ce^{Ax^2}}{\sqrt{4\pi Dt}}\int_{\mathbb{R}} e^{-\frac{\varepsilon z^2}{4Dt}}\,dz \underset{z=\sqrt{4Dt/\varepsilon}\,y}{=} \frac{ce^{Ax^2}}{\sqrt{\varepsilon\pi}}\int_{\mathbb{R}} e^{-y^2}\,dy = \frac{c}{\sqrt{\varepsilon}}e^{Ax^2}.$$

Thus u is well defined in the strip $\mathbb{R}\times(0,T]$, $T<\frac{1}{4Da}$, and moreover a) holds with $C = c/\sqrt{\varepsilon}$.

Proof of b). We need to differentiate under the integral sign. Observe that $\Phi_D \in C^\infty(\mathbb{R}\times(0,\infty))$. Given a point (x,t), we need to bound, in a neighborhood of (x,t), each derivative

$$\partial_t^h\partial_x^k\Phi_D(x-y,t)g(y)$$

by a integrable function $G = G(y)$, in general depending on the derivative itself.

Now, the analytic expression of a derivative $\partial_t^h\partial_x^k\Phi_D(x-y,t)$ is a sum of terms of the form

$$t^{-r}(x-y)^s\exp\left(-\frac{(x-y)^2}{4Dt}\right) \qquad (r,s\ge 0)$$

up to multiplicative constants. Let $x\in[-R,R]$, $0<t_0\le t\le T$. Since for any $b>0$ and any $x,y\in\mathbb{R}$ we have $2xy\le b^{-1}x^2+by^2$, we can write

$$-(x-y)^2 = -x^2+2xy-y^2 \le \frac{R^2}{b}-(1-b)y^2.$$

On the other hand

$$t^{-r}|x-y|^s \le t_0^{-r}(R+|y|)^s.$$

Therefore, we can write, for $x\in[-R,R]$, $t_0\le t\le T$,

$$t^{-r}|x-y|^s\exp\left(-\frac{(x-y)^2}{4Dt}\right)|g(y)| \le$$
$$\le t_0^{-r}(R+|y|)^s\exp\left(\frac{R^2}{bDT}\right)\exp\left(-\left(\frac{1-b}{4DT}-a\right)y^2\right) \equiv G_{R,t_0,s,r,b}(y).$$

If b is chosen so small to have $\frac{1-b}{4DT}-a>0$, then $G_{R,t_0,s,r,b}$ is integrable over $\mathbb{R}$ and therefore differentiation under the integral sign is possible for $x\in[-R,R]$, $t_0\le t\le T$. On the other hand, R is an arbitrary positive number and t_0 is an arbitrary number in $(0,T]$ so that $u\in C^\infty(\mathbb{R}\times(0,T])$ and, in particular,

$$u_t-u_{xx} = \int_{\mathbb{R}}\{\partial_t\Phi_D(x-y,t)-\partial_{xx}\Phi_D(x-y,t)\}\,g(y)dy = 0.$$

Proof of c). Assume g is continuous at x_0. We want to show that, given $\varepsilon > 0$, if $|x - x_0|$ and t are sufficiently small, depending on x_0 and ε, then

$$|u(x,t) - g(x_0)| \leq \varepsilon.$$

Let $\varepsilon > 0$ and $\delta > 0$, $\delta = \delta(\varepsilon, x_0)$, be such that, if $|y - x_0| < \delta$ then

$$|g(y) - g(x_0)| < \varepsilon/2. \tag{2.153}$$

Since $\int_{\mathbb{R}} \Phi_D(x-y)dy = 1$ for every $t > 0$, we can write

$$u(x,t) - g(x_0) = \int_{\mathbb{R}} \Phi_D(x-y,t)[g(y) - g(x_0)]dy =$$

$$= \int_{\{y:|y-x_0|<\delta\}} (\dots)dy + \int_{\{y:|y-x_0|>\delta\}} (\dots)dy \equiv I + II.$$

From (2.153) we have

$$|I| \leq \int_{\{y:|y-x_0|<\delta\}} \Phi_D(x-y,t)\,|g(y) - g(x_0)|\,dy < \varepsilon/2. \tag{2.154}$$

Moreover, since

$$|g(y) - g(x_0)| \leq ce^{ay^2} + ce^{ax_0^2},$$

we get

$$|II| \leq ce^{ax_0^2} \int_{\{y:|y-x_0|>\delta\}} \Phi_D(x-y,t)dy + c\int_{\{y:|y-x_0|>\delta\}} \Phi_D(x-y,t)e^{ay^2}dy. \tag{2.155}$$

Setting $z = x - y$, we find

$$\int_{\{y:|y-x_0|>\delta\}} \Phi_D(x-y,t)dy = \int_{\{z:|z-(x-x_0)|>\delta\}} \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{z^2}{4Dt}}dz.$$

Observe now that, if $|z - (x - x_0)| > \delta$ and $|x - x_0| < \delta/2$, we deduce

$$\delta < |z - (x - x_0)| \leq |z| + |x - x_0| \leq |z| + \frac{\delta}{2}$$

whence

$$\{z : |z - (x - x_0)| > \delta\} \subset \left\{z : |z| > \frac{\delta}{2}\right\}.$$

Therefore, if $|x - x_0| < \delta/2$,

$$\int_{\{z:|z-(x-x_0)|>\delta\}} \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{z^2}{4Dt}}dz \leq \int_{\{z:|z|>\frac{\delta}{2}\}} \frac{1}{\sqrt{4\pi Dt}} e^{-\frac{z^2}{4Dt}}dz = \frac{1}{\sqrt{\pi}} \int_{\left\{s:|s|>\frac{\delta}{4\sqrt{Dt}}\right\}} e^{-s^2}ds$$

and the last integral tends to zero as $t \to 0^+$. Via a similar argument, using (2.152), we can show that also the second integral in (2.155) tends a zero as $t \to 0^+$. Thus, if $|x - x_0| < \delta/2$ and t is sufficiently small, $|II| < \varepsilon/2$ and from (2.154) we get

$$|u(x,t) - g(x_0)| \leq |I| + |II| < \varepsilon$$

from which c) follows. □

2.9 An Application to Finance

2.9.1 European options

In this section we apply the above theory to determine the price of some financial products, in particular of some *derivative* products, called *European options.* A financial product is a *derivative* if its payoff depends on the price behavior of an asset, in jargon *the underlying*, for instance a *stock,* a *currency* or a *commodity.* Among the simplest derivatives are the **European call** and **put options**, that are contracts on a prescribed asset between a *holder* and a *subscriber*, with the following rules.

- At the drawing up time of the contract (say at time $t = 0$) an **exercise** or **strike price** E is fixed.
- At an **expiry date** T, fixed in the future:
  - The holder of a call option **can (but is not obliged to)** exercise the option by **purchasing** the asset at the price E. If the holder decides to buy the asset, the subscriber **must** sell it.
  - The holder of a put option **can (but is not obliged to)** exercise the option by **selling** at the price E. If the holder decides to sell the asset, the subscriber **must** buy it.

Since an option gives to the holder a right without any obligation, the option has a price and the basic question is:

what is the "right" price that must be paid at $t = 0$?

This price certainly depends on the evolution of the price S of the underlying, on the strike price E, on the expiring time T and on the current riskless interest rate $r > 0$.

For instance, for a call, to a lower E corresponds a greater price; the opposite holds for a put. The price fluctuations of the underlying affect in crucial way the value of an option, since they incorporate the amount of risk.

To answer our basic question, we introduce the **value function** $V = V(S, t)$, giving the proper price of the option if at time t the price of the underlying is S. What we need to know is $V(S(0), 0)$. When we like to distinguish between **call** and **put**, we use the notations $C(S, t)$ and $P(S, t)$, respectively.

The problem is then to determine V in agreement with the financial market, where both the underlying and the option are exchanged. We shall use the Black-Scholes method, based on the assumption of a reasonable evolution model for S and on the fundamental principle of *no arbitrage possibilities*.

2.9.2 An evolution model for the price S

Since S depends on more or less foreseeable factors, it is clear that we cannot expect a deterministic model for the evolution of S. To construct it we assume a *market efficiency* in the following sense:

a) The market responds instantaneously to new information on the asset.
b) The price has no memory: its past history is fully stored in the present price, without further information.

Condition a) implies the adoption of a continuous model. Condition b) basically requires that a change dS of the underlying price has the Markov property, like the Brownian motion.

Consider now a time interval from t to $t + dt$, during which S undergoes a change from S to $S + dS$. One of the most common models assumes that the **return** dS/S is given by the sum of two terms.

One is a deterministic term, which gives a contribution μdt due to a constant *drift* μ, representing the average growth rate of S. With this term alone, we would have

$$\frac{dS}{S} = \mu dt$$

and therefore $d \log S = \mu dt$, that gives the exponential growth $S(t) = S(0)e^{\mu t}$.

The other term is stochastic and takes into account the random aspects of the evolution. It gives the contribution

$$\sigma dB$$

where dB is an increment of a Brownian motion and has zero mean and variance dt. The coefficient σ, that we assume to be constant, is called the **volatility** and measures the standard deviation of the return. Summing the contributions we have

$$\frac{dS}{S} = \mu dt + \sigma dB. \tag{2.156}$$

Note the physical dimensions of μ and σ: $[\mu] = [\text{time}]^{-1}$, $[\sigma] = [\text{time}]^{-\frac{1}{2}}$.

Equation (2.156) is a **stochastic differential equation** (*s.d.e.*). To solve it one is tempted to write

$$d \log S = \mu dt + \sigma dB,$$

to integrate between 0 and t, and to obtain

$$\log \frac{S(t)}{S(0)} = \mu t + \sigma(B(t) - B(0)) = \mu t + \sigma B(t)$$

since $B(0) = 0$. However, **this is not correct**. The diffusion term σdB requires the use of the **Itô formula**, a stochastic version of the chain rule. Let us make a few intuitive remarks on this important formula.

Digression on Itô's formula. Let $B = B(t)$ the usual Brownian motion. An Itô process $X = X(t)$ is a solution of a *s.d.e.* of the type

$$dX = a(X,t)\,dt + \sigma(X,t)\,dB \tag{2.157}$$

where a is the *drift term* and σ is the *volatility coefficient*. When $\sigma = 0$, the equation is deterministic and the trajectories can be computed with the usual analytic methods. Moreover, given a smooth function $F = F(x,t)$, we can easily compute the variation of F along those trajectories: it is enough to compute

$$dF = F_t\,dt + F_x\,dX = \{F_t + aF_x\}\,dt.$$

Let now be σ nonzero; the preceding computation would give

$$dF = F_t\,dt + F_x\,dX = \{F_t + aF_x\}\,dt + \sigma F_x\,dB,$$

but **this formula does not give the complete differential of** F. Indeed, using Taylor's formula, one has, letting $X(0) = X_0$:

$$F(X,t) = F(X_0,0) + F_t\,dt + F_x\,dX + \frac{1}{2}\left\{F_{xx}(dX)^2 + 2F_{xt}\,dX\,dt + F_{tt}(dt)^2\right\} + \dots\ .$$

The differential of F along the trajectories of (2.157) is obtained by selecting in the right hand side above the terms which are **linear** with respect to dt or dX. We first find the terms

$$F_t\,dt + F_x\,dX = \{F_t + aF_x\}\,dt + \sigma F_x\,dB.$$

The terms $2F_{xt}\,dX\,dt$ and $F_{tt}(dt)^2$ are of higher order with respect to dt and dX, thus they are not included in the differential. Let us now check the term $(dX)^2$. We have

$$(dX)^2 = (a\,dt + \sigma\,dB)^2 = a^2(dt)^2 + 2a\sigma\,dB\,dt + \boxed{\sigma^2(dB)^2}.$$

While $a^2(dt)^2$ and $2a\sigma\,dB\,dt$ are again of higher order, the framed term **turns out to be exactly**

$$\sigma^2\,dt.$$

Formally, this is a consequence of the basic formula (see (2.80), p. 57)

$$dB \sim \sqrt{dt}\,N(0,1)$$

that assigns $\sqrt{dt}$ for the standard deviation of dB.

Thus the differential of F along the trajectories of (2.157) is given by the following **Itô formula**:

$$dF = \left\{F_t + aF_x + \frac{1}{2}\sigma^2 F_{xx}\right\}dt + \sigma F_x\,dB. \tag{2.158}$$

We are now ready to solve (2.156), that we write in the form

$$dS = \mu S\,dt + \sigma S\,dB.$$

Let $F(S) = \log S$. Since

$$F_t = 0, \qquad F_S = \frac{1}{S}, \qquad F_{SS} = -\frac{1}{S^2}$$

Itô's formula gives, with $X = S$, $a(S,t) = \mu S$, $\sigma(S,t) = \sigma S$,

$$d\log S = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dB.$$

We can now integrate between 0 and t, obtaining

$$\log S(t) = \log S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) t + \sigma B(t). \tag{2.159}$$

Equation (2.159) shows that the random variable $Y = \log S$ has a normal distribution, with mean $\log S_0 + \left(\mu - \frac{1}{2}\sigma^2\right) t$ and variance $\sigma^2 t$. Its probability density is therefore

$$f(y) = \frac{1}{\sqrt{2\pi\sigma^2 t}} \exp\left\{ -\frac{\left(y - \log S_0 - \left(\mu - \frac{1}{2}\sigma^2\right) t\right)^2}{2\sigma^2 t} \right\}$$

and the density of S is given by

$$p(s) = \frac{1}{s} f(\log s) = \frac{1}{s\sqrt{2\pi\sigma^2 t}} \left\{ -\frac{\left(\log s - \log S_0 - \left(\mu - \frac{1}{2}\sigma^2\right) t\right)^2}{2\sigma^2 t} \right\}$$

which is called a **lognormal density**.

2.9.3 The Black-Scholes equation

We now construct a differential equation able to describe the evolution of $V(S,t)$. We work under the following hypotheses:

- S follows a **lognormal law**.
- The volatility σ is constant and known.
- There are no transaction costs or dividends.
- It is possible to buy or sell any number of the underlying asset.
- There is an interest rate $r > 0$, for a riskless investment. This means that 1 dollar in a bank at time $t = 0$ becomes e^{rT} dollars at time $t = T$.
- The market is **arbitrage free**.

The last hypothesis is crucial in the construction of the model and means that *there is no opportunity for instantaneous risk-free profit.* It could be considered as a sort of conservation law for money!

The translation of this principle into mathematical terms is linked with the notion of *hedging* and the existence of *self-financing portfolios*[21] (see Sect. 2.9.5). The basic idea is first to compute the return of V through Itô's formula and then to construct a riskless portfolio Π, consisting of shares of S and the option. By the arbitrage free hypothesis, Π must grow at the current interest rate r, i.e. $d\Pi = r\Pi dt$, which turns out to coincide with the fundamental Black-Scholes equation.

Let us then use Itô's formula to compute the differential of V. Since

$$dS = \mu S\, dt + \sigma S\, dB,$$

we find

$$dV = \left\{ V_t + \mu S V_S + \frac{1}{2}\sigma^2 S^2 V_{SS} \right\} dt + \sigma S V_S dB. \tag{2.160}$$

Now we try to get rid of the risk term $\sigma S V_S dB$ by constructing a portfolio Π, consisting of the option and a quantity[22] $-\Delta$ of underlying:

$$\Pi = V - S\Delta.$$

This is an important financial operation called *hedging*. Consider now the interval of time $(t, t + dt)$ during which Π undergoes a variation $d\Pi$. If we manage to keep Δ equal to its value at t during the interval $(t, t + dt)$, the variation of Π is given by

$$d\Pi = dV - \Delta dS.$$

This is a key point in the whole construction, that needs to be carefully justified. Although we content ourselves with an intuitive level, we will come back to this question in Section 2.9.5.

Using (2.160) we find

$$\begin{aligned} d\Pi &= dV - \Delta dS = \\ &= \left\{ V_t + \mu S V_s + \frac{1}{2}\sigma^2 S^2 V_{SS} - \mu S\Delta \right\} dt + \sigma S(V_S - \Delta) dB. \end{aligned} \tag{2.161}$$

Thus, if we choose

$$\Delta = V_S, \tag{2.162}$$

meaning that Δ is the value of V_S at time t, we eliminate the stochastic component in (2.161). The evolution of the portfolio Π is now entirely deterministic and its dynamics is given by the following equation:

---

[21] A *portfolio* is a collection of *securities* (e.g. stocks) holdings.

[22] We borrow from *finance* the use of the Greek letter Δ in this context. Clearly here it has nothing to do with the Laplace operator.

$$d\Pi = \left\{ V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} \right\} dt. \tag{2.163}$$

The choice (2.162) appears almost .... miraculous, but it is partly justified by the fact that V and S are dependent and the random component in their dynamics is proportional to S. Thus, in a suitable linear combination of V and S such component should disappear.

It is the moment to use the no-arbitrage principle. Investing Π at the riskless rate r, after a time dt we have an increment $r\Pi dt$. Compare $r\Pi dt$ with $d\Pi$ given by (2.163).

- If $d\Pi > r\Pi dt$, we borrow an amount Π to invest in the portfolio. The return $d\Pi$ would be greater of the cost $r\Pi dt$, so that we make an instantaneous riskless profit

$$d\Pi - r\Pi dt.$$

- If $d\Pi < r\Pi dt$, we sell the portfolio Π investing it in a bank at the rate r. This time we would make an instantaneous risk free profit

$$r\Pi dt - d\Pi.$$

Therefore, the arbitrage free hypothesis forces

$$d\Pi = \left\{ V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} \right\} dt = r\Pi dt. \tag{2.164}$$

Substituting

$$\Pi = V - S\Delta = V - V_S S$$

into (2.164), we obtain the celebrated **Black-Scholes equation**:

$$\mathcal{L}V = V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0. \tag{2.165}$$

Note that the coefficient μ, the drift of S, does not appear in (2.165). This fact is apparently counter-intuitive and shows an interesting aspect of the model. The financial meaning of the Black-Scholes equation is emphasized from the following decomposition of its left hand side:

$$\mathcal{L}V = \underbrace{V_t + \frac{1}{2}\sigma^2 S^2 V_{SS}}_{\text{portfolio return}} - \underbrace{r(V - SV_S)}_{\text{bank investment}}.$$

The Black-Scholes equation is a little more general than the equations we have seen so far. Indeed, the diffusion and the drift coefficients are both depending on S. However, as we shall see below, we can transform it into the diffusion equation $u_t = u_{xx}$.

Observe that the coefficient of V_{SS} is positive, so that (2.165) is a **backward equation**. To get a well posed problem, we need a **final condition** (at $t = T$), a side condition at $S = 0$ and one condition for $S \to +\infty$.

• *Final conditions.* We examine what conditions we have to impose at $t = T$.

Call. If at time T we have $S > E$ then we exercise the option, with a profit $S - E$. If $S \le E$, we do not exercise the option with no profit. The *final payoff* of the option is therefore

$$C(S,T) = \max\{S - E, 0\} = (S - E)^+, \qquad S > 0.$$

Put. If at time T we have $S \ge E$, we do not exercise the option, while we exercise the option if $S < E$. The *final payoff* of the option is therefore

$$P(S,T) = \max\{E - S, 0\} = (E - S)^+, \qquad S > 0.$$

• *Boundary conditions.* We now examine the conditions to be imposed at $S = 0$ and for $S \to +\infty$.

Call. If $S = 0$ at a time t, (2.156) implies $S = 0$ thereafter, and the option has no value; therefore

$$C(0,t) = 0, \qquad t \ge 0.$$

As $S \to +\infty$, at time t, the option will be exercised and its value becomes practically equal to S minus the discounted exercise price, that is

$$C(S,t) - (S - e^{-r(T-t)}E) \to 0 \qquad \text{as } S \to \infty.$$

Put. If at a certain time is $S = 0$, so that $S = 0$ thereafter, the final profit is E. Thus, to determine $P(0,t)$ we need to determine the present value of E at time T, that is

$$P(0,t) = Ee^{-r(T-t)}.$$

If $S \to +\infty$, we do not exercise the option, hence

$$P(S,t) = 0 \qquad \text{as } S \to +\infty.$$

2.9.4 The solutions

Let us summarize our model in the two cases.

• *Black-Scholes equation*

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0. \tag{2.166}$$

• *Final payoffs*

$$C(S,T) = (S - E)^+ \qquad \text{(call)}$$
$$P(S,T) = (E - S)^+ \qquad \text{(put)}.$$

- *Boundary conditions*

$$\begin{aligned} &C(0,t) = 0 \ \text{ and } \ C(S,t) - (S - e^{-r(T-t)}E) \to 0 \text{ as } S \to \infty && \text{(call)} \\ &P(0,t) = Ee^{-r(T-t)} \ \text{ and } \ P(S,T) = 0 \text{ as } S \to \infty && \text{(put).} \end{aligned}$$

It turns out that the above problems can be reduced to a global Cauchy problem for the heat equation. In this way it is possible to find explicit formulas for the solutions. First of all we make a change of variables to reduce the Black-Scholes equation to constant coefficients and to pass from backward to forward in time. Also note that $1/\sigma^2$ can be considered an intrinsic reference time while the exercise price E gives a characteristic order of magnitude for S and V. Thus, $1/\sigma^2$ and E can be used as rescaling factors to introduce dimensionless variables.

Let us set

$$x = \log \frac{S}{E}, \quad \tau = \frac{1}{2}\sigma^2(T-t), \quad w(x,\tau) = \frac{1}{E}V\left(Ee^x, T - \frac{2\tau}{\sigma^2}\right).$$

When S goes from 0 to $+\infty$, x varies from $-\infty$ to $+\infty$. When $t = T$ we have $\tau = 0$. Moreover:

$$V_t = -\frac{1}{2}\sigma^2 E w_\tau, \qquad V_S = \frac{E}{S}w_x, \qquad V_{SS} = -\frac{E}{S^2}w_x + \frac{E}{S^2}w_{xx}.$$

Substituting into (2.166), after some simplifications, we get

$$-\frac{1}{2}\sigma^2 w_\tau + \frac{1}{2}\sigma^2(-w_x + w_{xx}) + rw_x - rw = 0$$

or

$$w_\tau = w_{xx} + (k-1)w_x - kw$$

where $k = \frac{2r}{\sigma^2}$ is a dimensionless parameter. By further setting (see Problem 2.16)

$$w(x,\tau) = e^{-\frac{k-1}{2}x - \frac{(k+1)^2}{4}\tau}v(x,\tau)$$

we find that v satisfies

$$v_\tau - v_{xx} = 0, \qquad -\infty < x < +\infty, \ 0 < \tau < T.$$

The final condition for V becomes an initial condition for v. Precisely, after some manipulations, we have

$$v(x,0) = g(x) = \begin{cases} e^{\frac{1}{2}(k+1)x} - e^{\frac{1}{2}(k-1)x} & x > 0 \\ 0 & x \le 0 \end{cases}$$

for the call option, and

$$v(x,0) = g(x) = \begin{cases} e^{\frac{1}{2}(k-1)x} - e^{\frac{1}{2}(k+1)x} & x < 0 \\ 0 & x \ge 0 \end{cases}$$

for the put option.

Finally, the boundary conditions allows to apply the preceding theory and in particular Theorem 2.12, p. 79 and Corollary 2.18, p. 85. As a consequence, the solution is unique and it is given by

$$v(x,\tau) = \frac{1}{\sqrt{4\pi\tau}} \int_{\mathbb{R}} g(y) e^{-\frac{(x-y)^2}{4\tau}} dy.$$

To have a more significant formula, let $y = \sqrt{2\tau}z + x$; then, focusing on the **call** option:

$$v(x,\tau) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} g(\sqrt{2\tau}z + x) e^{-\frac{z^2}{2}} dy =$$

$$= \frac{1}{\sqrt{2\pi}} \left\{ \int_{-x/\sqrt{2\tau}}^{\infty} e^{\frac{1}{2}(k+1)(\sqrt{2\tau}z+x) - \frac{1}{2}z^2} dz - \int_{-x/\sqrt{2\tau}}^{\infty} e^{\frac{1}{2}(k-1)(\sqrt{2\tau}z+x) - \frac{1}{2}z^2} dz \right\}.$$

After some manipulations in the two integrals[23], we obtain

$$v(x,\tau) = e^{\frac{1}{2}(k+1)x + \frac{1}{4}(k+1)^2\tau} N(d_+) - e^{\frac{1}{2}(k-1)x + \frac{1}{4}(k-1)^2\tau} N(d_-)$$

where

$$N(z) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{1}{2}y^2} dy$$

is the distribution of a standard normal random variable and

$$d_\pm = \frac{x}{\sqrt{2\tau}} + \frac{1}{2}(k \pm 1)\sqrt{2\tau}.$$

Going back to the original variables we have, for the **call:**

$$C(S,t) = SN(d_+) - Ee^{-r(T-t)} N(d_-)$$

with

$$d_\pm = \frac{\log(S/E) + \left(r \pm \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}.$$

[23] For instance, to evaluate the first integral, complete the square at the exponent, writing

$$\frac{1}{2}(k+1)(\sqrt{2\tau}z + x) - \frac{1}{2}z^2 = \frac{1}{2}(k+1)x + \frac{1}{4}(k+1)^2\tau - \frac{1}{2}\left[z - \frac{1}{2}(k+1)\sqrt{2\tau}\right]^2.$$

Then, setting $y = z - \frac{1}{2}(k+1)\sqrt{2\tau}$.

$$\int_{-x/\sqrt{2\tau}}^{\infty} e^{\frac{1}{2}(k+1)(\sqrt{2\tau}z+x) - \frac{1}{2}z^2} dz = e^{\frac{1}{2}(k+1)x + \frac{1}{4}(k+1)^2\tau} \int_{-x/\sqrt{2\tau} - (k+1)\sqrt{\tau}/\sqrt{2}}^{\infty} e^{-\frac{1}{2}y^2} dz.$$

The formula for the **put** is

$$P(S,t) = Ee^{-r(T-t)}N(-d_-) - SN(-d_+).$$

It can be shown that

$$\Delta = C_S = N(d_+) > 0 \qquad \text{for the } \textbf{call}$$
$$\Delta = P_S = N(d_+) - 1 < 0 \qquad \text{for the } \textbf{put.}$$

Note that C_S and P_S are strictly increasing with respect to S, since N is a strictly increasing function and d_+ is strictly increasing with S. The functions C, P are therefore *strictly convex functions* of S, for every $t < T$, namely

$$C_{ss} > 0 \quad \text{and} \quad P_{ss} > 0.$$

• *Put-call parity.* Put and call options, with the same exercise price and expiry time, can be connected by forming the following portfolio:

$$\Pi = S + P - C$$

where the minus in front of C shows a so called *short position* (negative holding). For this portfolio, the final payoff is

$$\Pi(S,T) = S + (E - S)^+ - (S - E)^+.$$

If $E \geq S$, we have

$$\Pi(S,T) = S + (E - S) - 0 = E$$

while if $E \leq S$,

$$\Pi(S,T) = S + 0 - (S - E) = E.$$

Thus at expiry the payoff is always equal to E and it constitutes a riskless profit, whose value at t must be equal to the discounted value of E, because of the no arbitrage condition. Hence we find the following relation (*put–call parity*)

$$S + P - C = Ee^{-r(T-t)}. \tag{2.167}$$

Formula (2.167) also shows that, given the value of C (or P), we can find the value of P (or C). From (2.167), since

$$Ee^{-r(T-t)} \leq E$$

and $P \geq 0$, we get

$$C(S,t) = S + P - Ee^{-r(T-t)} \geq S - E$$

and therefore, since $C \geq 0$,

$$C(S,t) \geq (S - E)^+.$$

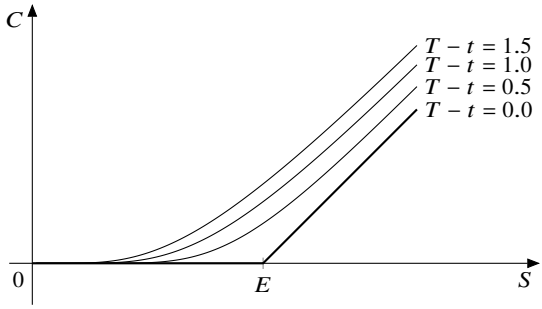

Fig. 2.14 The value function for an European call option

It follows that the value of C *is always greater than the final payoff*. It is not so for a put. In fact

$$P(0,t) = Ee^{-r(T-t)} \leq E$$

so that the value of P is below the final payoff when S is near 0, while it is above just before expiring. Figures 2.14 and 2.15 show the behavior of C and P versus S, for some values of $T - t$ up to expiry.

• *Different volatilities*. The maximum principle arguments in Sect. 2.8 can be used to compare the value of two options with different volatilities σ_1 and σ_2, having the same exercise price E and the same strike time T. Assume that $\sigma_1 > \sigma_2$ and denote by $C^{(1)}$, $C^{(2)}$ the value of the corresponding call options. Diminishing the amount of risk the value of the option should decrease and indeed we want to confirm that

$$C^{(1)} > C^{(2)}, \qquad S > 0,\ 0 \leq t < T.$$

Let $W = C^{(1)} - C^{(2)}$. Then

$$W_t + \frac{1}{2}\sigma_2^2 S^2 W_{SS} + rSW_S - rW = \frac{1}{2}(\sigma_2^2 - \sigma_1^2)S^2 C^{(1)}_{SS} \tag{2.168}$$

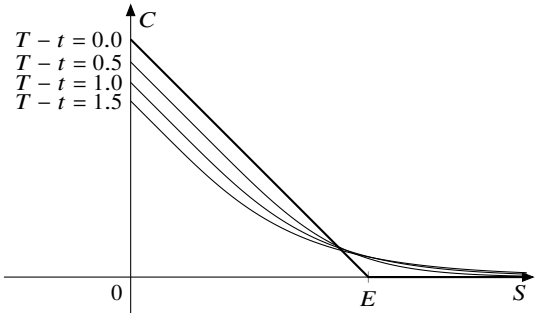

Fig. 2.15 The value function of an European put option

with $W(S,T) = 0$, $W(0,t) = 0$ and $W \to 0$ as $S \to +\infty$.

The (2.168) is a nonhomogeneous equation, whose right hand side is *negative* for $S > 0, 0 < T$, because $C_{SS}^{(1)} > 0$. Since W is continuous in the half strip $[0,+\infty) \times [0,T]$ and vanishes at infinity, it attains its global minimum at a point (S_0, t_0).

We claim that the minimum is zero and cannot be attained at a point in $(0,+\infty) \times [0,T)$. Since the equation is backward, $t_0 = 0$ is excluded. Suppose $W(S_0,t_0) \leq 0$ with $S_0 > 0$ and $0 < t_0 < T$. We have

$$W_t(S_0,t_0) = 0$$

and

$$W_S(S_0,t_0) = 0, \qquad W_{SS}(S_0,t_0) \geq 0.$$

Substituting $S = S_0, t = t_0$ into (2.168) we get a contradiction. Therefore $W = C^{(1)} - C^{(2)} > 0$ for $S > 0, 0 < t < T$.

2.9.5 Hedging and self-financing strategy

The construction of the riskless portfolio Π and, consequently, the use of the *no arbitrage* principle can be made more rigorously, with respect to what we did in Sect. 2.9.3, by introducing the concept of *self-financing* portfolio. The idea is to "duplicate" V by means of a portfolio consisting of a number of shares of S and a bond Z, a free risk investment growing at the rate r, e.g. $Z(t) = e^{rt}$.

To this purpose let us try to determine two processes $\phi = \phi(S,t)$ and $\psi = \psi(S,t)$ such that

$$V = \phi S + \psi Z \qquad (0 \leq t \leq T) \tag{2.169}$$

in order to eliminate any risk factor. In fact, playing the part of the subscriber of a *call* (that has to sell), the risk is that at time T the price $S(T)$ is greater than E, so that the holder will exercise the option. If in the meantime the subscriber has constructed the portfolio (2.169), the profit from it exactly meets the funds necessary to pay the holder. On the other hand, if the option has zero value at time T, the portfolio has no value as well.

For the operation to make sense, it is necessary that the subscriber *does not put extra money in this strategy* (*hedging*). This can be assured by requiring that the portfolio (2.169) be *self-financing* that is, **its changes in value be dependent from variations of S and Z alone**.

In formulas, this amount to requiring

$$dV = \phi dS + \psi dZ \qquad (0 \leq t \leq T). \tag{2.170}$$

What is the financial meaning of (2.170)? We will answer at the end of this section. Actually, we have already met this condition, when we have constructed the portfolio $\Pi = V - S\Delta$ or

$$V = \Pi + S\Delta,$$

by asking that $dV = d\Pi + \Delta dS$. This construction is nothing else that a duplication of V by means of a *self-financing portfolio,* with Π playing the role of Z and choosing $\psi = 1$, $\phi = \Delta$. We have seen that the choice $\Delta = V_S$, together with the no-arbitrage principle, led to the Black-Scholes equation for V.

Changing perspective, we show that: *if V satisfy the Black-Scholes equation*

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0 \tag{2.171}$$

and $\Delta = V_S$, then $V = \Pi + S\Delta$ is self-financing.

Indeed by the Ito formula and (2.171) we get

$$\begin{aligned} dV &= V_S dS + \left\{ V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} \right\} dt = V_S dS + r(V - SV_S)dt \\ &= V_S dS + r\Pi dt = V_S dS + d\Pi \end{aligned}$$

which is the self-financing condition. On the other hand, it is not difficult to check that the choice $\Delta = V_S$ *is the only one* that makes $V = \Pi + S\Delta$ self-financing (see Problem 2.26).

But, what is the real meaning of (2.170)? We see it better in a discrete setting. Consider a sequence of times

$$t_0 < t_1 < \ldots < t_N$$

and suppose that the intervals $(t_j - t_{j-1})$ are very small. Denote by S_j and Z_j the values at t_j of S and Z. Consequently, look for two sequences

$$\phi_j \text{ and } \psi_j$$

corresponding to the quantity of S and Z to be used in the construction of the portfolio (2.169) from t_{j-1} to t_j. Notice that ϕ_j and ψ_j are chosen at time t_{j-1}.

Thus, given the interval (t_{j-1}, t_j),

$$V_j = \phi_j S_j + \psi_j Z_j$$

represents the closing value of the portfolio while

$$\phi_{j+1} S_j + \psi_{j+1} Z_j$$

is the opening value, the amount of money necessary to buy the new one. The **self-financing condition means** that the value V_j of the portfolio at time t_j, determined by the couple (ϕ_j, ψ_j), exactly meets the purchasing cost of the portfolio in the interval (t_j, t_{j+1}), determined by (ϕ_{j+1}, ψ_{j+1}). This means

$$\phi_{j+1} S_j + \psi_{j+1} Z_j = \phi_j S_j + \psi_j Z_j \tag{2.172}$$

or that **the financial gap**

$$D_j = \phi_{j+1} S_j + \psi_{j+1} Z_j - V_j$$

must be zero, otherwise an amount of cash D_j has to be injected to sustain the strategy ($D_j > 0$) or the same amount of money can be drawn from it ($D_j < 0$). From (2.172) we deduce that

$$\begin{aligned} V_{j+1} - V_j &= (\phi_{j+1} S_{j+1} + \psi_{j+1} Z_{j+1}) - (\phi_j S_j + \psi_j Z_j) \\ &= (\phi_{j+1} S_{j+1} + \psi_{j+1} Z_{j+1}) - (\phi_{j+1} S_j + \psi_{j+1} Z_j) \\ &= \phi_{j+1}(S_{j+1} - S_j) + \psi_{j+1}(Z_{j+1} - Z_j) \end{aligned}$$

or

$$\Delta V_j = \phi_{j+1} \Delta S_j + \psi_{j+1} \Delta Z_j$$

whose continuous version is exactly (2.170).

2.10 Some Nonlinear Aspects

All the mathematical models we have examined so far are *linear*. On the other hand, the nature of most real problems is nonlinear. For example, *nonlinear diffusion* has to be taken into account in filtration problems, *nonlinear drift* terms are quite important in fluid dynamics while *nonlinear reaction* terms occur frequently in population dynamics and chemical kinetics.

The presence of a nonlinearity in a mathematical model gives rise to many interesting phenomena that cannot occur in the linear case; typical instances are finite speed of diffusion, finite time blow-up or existence of travelling wave solutions of certain special profiles, each one with its own characteristic velocity.

In this section we try to convey some intuition of what could happen in two typical and important examples from filtration through a porous medium and population dynamics. In Chap. 4, we shall deal with nonlinear transport models.

2.10.1 Nonlinear diffusion. The porous medium equation

Consider a gas of density $\rho = \rho(\mathbf{x}, t)$ flowing through a porous medium. Denote by $\mathbf{v} = \mathbf{v}(\mathbf{x}, t)$ the velocity of the gas and by κ the *porosity* of the medium, representing the volume fraction filled with gas. Conservation of mass reads, in this case:

$$\kappa \rho_t + \operatorname{div}(\rho \mathbf{v}) = 0. \tag{2.173}$$

Besides (2.173), the flow is ruled by the two following constitutive (empirical) laws.

- **Darcy's law:**

$$\mathbf{v} = -\frac{\mu}{\nu}\nabla p \tag{2.174}$$

where $p = p(\mathbf{x}, t)$ is the pressure, μ is the *permeability* of the medium and ν is the *viscosity* of the gas. We assume μ and ν are positive constants.

- **Equation of state:**

$$p = p_0 \rho^a, \qquad p_0 > 0,\ a > 0. \tag{2.175}$$

From (2.174) and (2.175) we have

$$\rho \mathbf{v} = -\frac{\mu p_0 a}{\nu}\rho^a \nabla \rho = -\frac{\mu p_0 a}{\nu(a+1)}\nabla(\rho^{a+1})$$

so that

$$\operatorname{div}(\rho \mathbf{v}) = -\frac{\mu p_0 a}{\nu(a+1)}\Delta(\rho^{a+1}).$$

Setting $m = 1 + a > 1$, from (2.173) we obtain

$$\rho_t = \frac{(m-1)\mu p_0}{\kappa m \nu}\Delta(\rho^m).$$

Rescaling time ($t \mapsto \dfrac{(m-1)\mu p_0}{\kappa m \nu} t$) we finally get the **porous medium equation**

$$\rho_t = \Delta(\rho^m). \tag{2.176}$$

Since

$$\Delta(\rho^m) = \operatorname{div}(m\rho^{m-1}\nabla\rho)$$

we see that the diffusion coefficient is $D(\rho) = m\rho^{m-1}$, showing that the diffusive effect increases with the density.

The porous medium equation can be written in terms of the pressure variable

$$u = p/p_0 = \rho^{m-1}.$$

It is not difficult to check that the equation for u is given by

$$u_t = mu\Delta u + \frac{m}{m-1}|\nabla u|^2 \tag{2.177}$$

showing once more the dependence on u of the diffusion coefficient.

One of the basic questions related to the equation (2.176) or (2.177) is to understand how an initial data ρ_0, confined in a small region Ω, evolves with time. The key object to examine is therefore the unknown *boundary* of the gas $\partial\Omega$, whose speed of expansion we expect to be proportional to $|\nabla u|$ (from (2.174)). This means that we expect a *finite speed of propagation,* in contrast with the classical case $m = 1$.

The porous media equation cannot be treated by elementary means, since at very low density the diffusion has a very low effect and the equation degenerates. However we can get some clue of what happens by examining a sort of fundamental solutions, the so called *Barenblatt solutions*, in spatial dimension 1.

The equation is

$$\rho_t = (\rho^m)_{xx}. \tag{2.178}$$

We look for *nonnegative self-similar* solutions of the form

$$\rho(x,t) = t^{-\alpha}U(xt^{-\beta}) \equiv t^{-\alpha}U(\xi)$$

with U even, $U'(0) = 0$, satisfying

$$\int_{-\infty}^{+\infty} \rho(x,t)dx = 1.$$

This condition requires

$$1 = \int_{-\infty}^{+\infty} t^{-\alpha}U(xt^{-\beta})dx = t^{\beta-\alpha}\int_{-\infty}^{+\infty} U(\xi)d\xi$$

so that we must have $\beta = \alpha$ and $\int_{-\infty}^{+\infty} U(\xi)d\xi = 1$. Substituting into (2.178), we find

$$\alpha t^{-\alpha-1}(-U - \xi U') = t^{-m\alpha-2\alpha}(U^m)''.$$

Thus, if we choose $\alpha = 1/(m+1)$, we get for U the differential equation

$$(m+1)(U^m)'' + \xi U' + U = 0$$

that can be written in the form

$$\frac{d}{d\xi}\left[(m+1)(U^m)' + \xi U\right] = 0.$$

Thus, we have

$$(m+1)(U^m)' + \xi U = \text{constant}.$$

Computing at $\xi = 0$, since $U'(0) = 0$, we deduce that the constant is equal to zero, so that

$$(m+1)(U^m)' = (m+1)mU^{m-1}U' = -\xi U$$

or

$$(m+1)mU^{m-2}U' = -\xi.$$

This in turn is equivalent to

$$\frac{(m+1)m}{m-1}(U^{m-1})' = -\xi$$

whose solution is

$$U(\xi) = \left[A - B_m\xi^2\right]^{1/(m-1)}$$

where A is an arbitrary constant and $B_m = \frac{(m-1)}{2m(m+1)}$. Clearly, to have a physical meaning, we must have

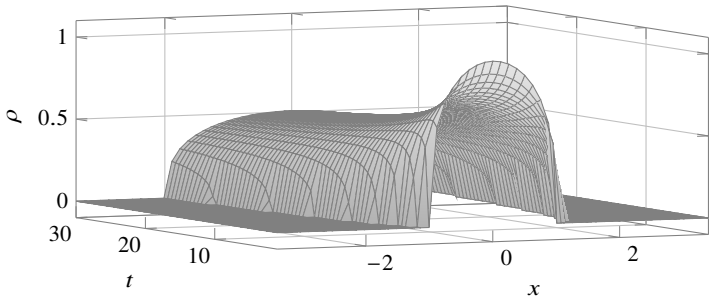

Fig. 2.16 The Barenblatt solution $\rho(x,t) = t^{-1/5}[1 - x^2 t^{-2/5}]_+^{1/3}$ for $1 \le t \le 30$

$$A - B_m \xi^2 \ge 0.$$

In conclusion we have found solutions of the porous medium equation of the form

$$\rho(x,t) = \begin{cases} \dfrac{1}{t^\alpha}\left[A - B_m \dfrac{x^2}{t^{2\alpha}}\right]^{1/(m-1)} & \text{if } x^2 \le At^{2\alpha}/B_m \\ 0 & \text{if } x^2 > At^{2\alpha}/B_m \end{cases} \qquad (\alpha = 1/(m+1)),$$

known as *Barenblatt solutions* (see Fig. 2.16). The points

$$x = \pm\sqrt{A/B_m}\, t^\alpha \equiv \pm r(t)$$

represent the interface between the part filled by gas and the empty part. Its speed of propagation is thus

$$\dot{r}(t) = \alpha\sqrt{A/B_m}\, t^{\alpha-1}.$$

2.10.2 Nonlinear reaction. Fisher's equation

In 1937 Fisher[24] introduced a model for the spatial spread of a so-called *favoured*[25] (or *advantageous*) *gene in a population,* over an infinitely long one dimensional habitat. Denoting by v the gene concentration, Fisher's equation reads

$$v_\tau = Dv_{yy} + rv\left(1 - \frac{v}{M}\right) \qquad \tau > 0,\, y \in \mathbb{R}, \tag{2.179}$$

where D, r, and M are positive parameters. An important question is to determine whether the gene has a typical speed of propagation. Accordingly to the terminology in the introduction, (2.179) is a *semilinear equation* where diffusion is coupled with *logistic growth* through the reaction term

[24] *Fisher, R.A.*, The wave of advance of advantageous gene. Ann. Eugenics, **7**, 355–369, *1937*.

[25] That is a *gene* that has an advantage in the struggle for life.

$$f(v) = rv\left(1 - \frac{v}{M}\right).$$

The parameter r represents a *biological potential* (net birth-death rate, with dimension $[\text{time}]^{-1}$), while M is the *carrying capacity* of the habitat. If we rescale time, space and concentration in the following way

$$t = r\tau, \quad x = \sqrt{r/D}y, \quad u = v/M,$$

(2.179) takes the dimensionless form

$$u_t = u_{xx} + u(1-u), \quad t > 0. \tag{2.180}$$

Note the two equilibria $u \equiv 0$ and $u \equiv 1$. In absence of diffusion, 0 is unstable, and 1 is asymptotically stable. A trajectory with initial data $u(0) = u_0$ between 0 and 1 has the typical behavior shown in Fig. 2.17. Therefore, if

$$u(x,0) = u_0(x), \qquad x \in \mathbb{R}, \tag{2.181}$$

is an initial data for the equation (2.179), with $0 < u_0(x) < 1$, we expect a competitive action between diffusion and reaction, with diffusion trying to spread and lower u_0 against the reaction tendency to increase u towards the equilibrium solution 1. What we intend to show here is the existence of permanent travelling waves solutions connecting the two equilibrium states, that is solutions of the form

$$u(x,t) = U(z), \qquad z = x - ct,$$

with c denoting the propagation speed, satisfying the conditions

$$0 < u < 1, \qquad t > 0, \; x \in \mathbb{R}$$

and

$$\lim_{x \to -\infty} u(x,t) = 1 \quad \text{and} \quad \lim_{x \to +\infty} u(x,t) = 0. \tag{2.182}$$

The first condition in (2.182), states that the gene concentration is saturated at the far left end while the second condition denotes zero concentration at the far right

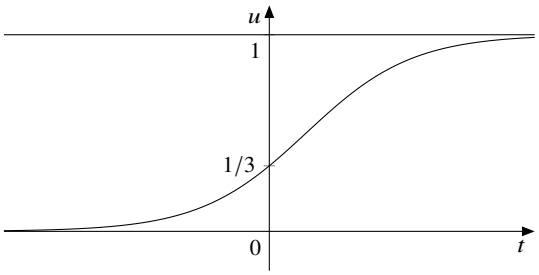

Fig. 2.17 Logistic curve ($u_0 = 1/3$)

end. Clearly, this kind of solutions realizes a balance between diffusion and reaction. Since the equation (2.179) is invariant under the transformation $x \mapsto -x$, it suffices to consider $c > 0$, that is right-moving waves only.

Since

$$u_t = -cU', \quad u_x = U', \quad u_{xx} = U'', \qquad (' = d/dz)$$

substituting $u(x,t) = U(z)$ into (2.180), we find for U the ordinary differential equation

$$U'' + cU' + U - U^2 = 0 \tag{2.183}$$

with

$$\lim_{z\to-\infty} U(z) = 1 \quad \text{and} \quad \lim_{z\to+\infty} U(z) = 0. \tag{2.184}$$

Letting $U' = V$, the equation (2.183) is equivalent to the system

$$\frac{dU}{dz} = V, \quad \frac{dV}{dz} = -cV - U + U^2 \tag{2.185}$$

in the phase plane (U, V). This system has two equilibrium points $(0,0)$ and $(1,0)$, corresponding to two steady states. Our travelling wave solution corresponds to an orbit connecting $(1,0)$ to $(0,0)$, with $0 < U < 1$.

We first examine the local behavior of the orbits near the equilibrium points. The coefficients matrices of the linearized systems at $(0,0)$ and $(1,0)$ are, respectively,

$$J(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & -c \end{pmatrix} \quad \text{and} \quad J(1,0) = \begin{pmatrix} 0 & 1 \\ 1 & -c \end{pmatrix}.$$

The eigenvalues of $J(0,0)$ are

$$\lambda_\pm = \frac{1}{2}\left[-c \pm \sqrt{c^2 - 4}\right],$$

with corresponding eigenvectors

$$\mathbf{h}_\pm = \begin{pmatrix} -c \mp \sqrt{c^2 - 4} \\ 2 \end{pmatrix}.$$

If $c \geq 2$ the eigenvalues are both negative while if $c < 2$ they are complex. Therefore

$$(0,0) \text{ is a } \begin{cases} \text{stable node (sink)} & \text{if } c \geq 2 \\ \text{stable focus (spiral sink)} & \text{if } c < 2. \end{cases}$$

The eigenvalues of $J(1,0)$ are

$$\mu_\pm = \frac{1}{2}\left[-c \pm \sqrt{c^2 + 4}\right],$$

of opposite sign, hence $(1,0)$ is a saddle point. The unstable and stable separatrices leave $(1,0)$ along the directions of the two eigenvectors

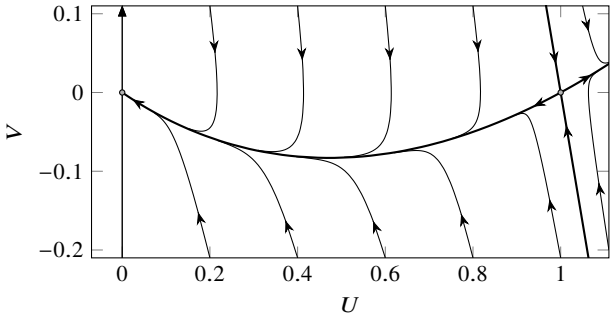

Fig. 2.18 Orbits of the system (2.185) with $c = 3$

$$\mathbf{k}_+ = \begin{pmatrix} c + \sqrt{c^2+4} \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{k}_- = \begin{pmatrix} c - \sqrt{c^2+4} \\ 2 \end{pmatrix},$$

respectively.

Now, the constraint $0 < U < 1$ rules out the case $c < 2$, since in this case U changes sign along any orbit approaching $(0, 0)$. For $c \geq 2$, all orbits in a neighborhood of the origin approach $(0, 0)$ for $z \to +\infty$ asymptotically with slope[26] λ_+. On the other hand, the only orbit going to $(1, 0)$ as $z \to -\infty$ and remaining in the region $0 < U < 1$ is the unstable separatrix γ of the saddle point.

Figure 2.18 shows the orbits configuration in the region of interest (see Problem 2.27). The conclusion is that *for each* $c \geq 2$ *there exists a unique travelling wave solution of equation* (2.179) *with speed* c. *Moreover* U *is strictly decreasing*.

In terms of original variables, there is a unique travelling wave solution for every speed c satisfying the inequality $c \geq c_{\min} = 2\sqrt{rD}$.

Thus, we have a continuous "spectrum" of possible speeds of propagation. It turns out that the minimum speed $c = c_{\min}$ is particularly important.

Indeed, having found a travelling solution is only the beginning of the story. There is a number of questions that arise naturally. Among them, the study of the *stability* of the travelling waves or of the asymptotic behavior (as $t \to +\infty$) of a solution with an initial data u_0 of *transitional* type, that is

$$u_0(x) = \begin{cases} 1 & x \leq a \\ 0 < u_0 < 1 & a < x < b \\ 0 & x \geq b. \end{cases} \tag{2.186}$$

Should we expect that the travelling wave is insensitive to small perturbations? Does the solution with initial condition (2.186) evolve towards one of the travelling waves we have just found?

The interested reader can find the answers in the many specialized texts or papers on the subject, see for instance *J. D. Murray*, [29], vol. 1. Here we only mention that

[26] Except for two orbits tangent to $\mathbf{h}_-$ at $(0, 0)$, in the case $c > 2$.

among the travelling wave solutions we have found, *only the minimum speed one* can be the asymptotic representation of solutions with transitional type initial condition. The biological implication of this result is that $c_{\min}$ *determines the required speed of propagation of an advantageous gene.*

Problems

2.1 Use the method of separation of variables to solve the following initial-Dirichlet problem:

$$\begin{cases} u_t - Du_{xx} = au & 0 < x < L, t > 0 \\ u(0,t) = u(L,t) = 0 & t > 0 \\ u(x,0) = g(x) & 0 \le x \le L. \end{cases}$$

Assume $g(x) \in C^2([0,L])$, $g(0) = g(L) = 0$ and show that the solution is continuous in $[0,L] \times [0,+\infty)$ and of class C^∞ in $(0,L) \times (0,+\infty)$.

2.2 Use the method of separation of variables to solve the following initial-Neumann problem:

$$\begin{cases} u_t - u_{xx} = 0 & 0 < x < L, t > 0 \\ u_x(0,t) = u_x(L,t) = 0 & 0 < x < L \\ u(x,0) = g(x) & t > 0. \end{cases}$$

Show that $u(x,t) \to \frac{1}{L}\int_0^L g(y)dy$ as $t \to +\infty$, for every $x \in (0,L)$.

2.3 Use the Duhamel method to solve the following nonhomogeneous initial-Neumann problem:

$$\begin{cases} u_t - u_{xx} = tx & 0 < x < L, t > 0 \\ u(x,0) = 0 & 0 \le x \le L \\ u_x(0,t) = u_x(L,t) = 0 & t > 0. \end{cases}$$

[*Hint.* Write the solution $w(x,t;s), 0 \le s \le t$, of the homogeneous Neumann problem with initial conditions $w(x,s;s) = sx$. Then $u(x,t) = \int_0^t w(x,t;s)ds$.]

2.4 Use the method of separation of variables to solve (at least formally) the following mixed problem:

$$\begin{cases} u_t - Du_{xx} = 0 & 0 < x < L, t > 0 \\ u(x,0) = g(x) & 0 \le x \le L \\ u_x(0,t) = 0 \\ u_x(L,t) + u(L,t) = U & t > 0. \end{cases}$$

[*Answer.* $u(x,t) = U + \sum_{k\ge1} c_k e^{-D\mu_k^2 t}\cos(\mu_k x)$, where the numbers μ_k are the positive solutions of the equation $\mu\tan(\mu L) = 1$, and $c_k = \frac{1}{\alpha_k}\int_0^L (g(x) - U)\cos(\mu_k x)dx$, $\alpha_k = \int_0^L (\cos\mu_k x)^2 dx$, $k \ge 1$.]

2.5 (*Evolution of a chemical solution*) Consider a tube of length L and constant cross section A, with axis of symmetry along the x axis. The tube contains a fluid with a saline volume concentration of c grams/cm^3. Assume that:

a) $A \ll L$, so that we can assume $c = c(x,t)$.

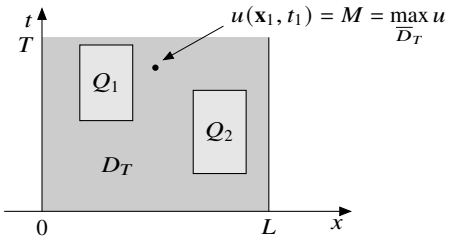

Fig. 2.19 At which points $(\mathbf{x}, t)$, $u(\mathbf{x}, t) = M$?

b) The salt diffuses along the x axis.
c) The velocity of the fluid is negligible.
d) From the left boundary of the pipe, at $x = 0$, a solution of constant concentration C_0 enters the tube at a speed of R_0 cm^3 per second, while at the other hand the solution is removed at the same speed.

1. Using Fick's law (see (2.94), p. 63) to show that c satisfy the diffusion equation with mixed Neumann/Robin at the and of the tube.
2. Solve explicitly the problem and show that $c(x, t)$ tends to an equilibrium concentration $c_\infty(x)$ as $t \to +\infty$.

[*Partial answer.* **1**. Let k be the diffusion coefficient in the Fick's law. The boundary conditions are $c_x(0, t) = -C_0 R_0/kA$, $c_x(L, t) + (R_0/kA)c(L, t) = 0$. **2**. $c_\infty = C_0 + (R_0/kA)(L - x)$.]

2.6 Prove Corollary 2.5, p. 40.

[*Hint.* b) Let $u = v - w$, $M = \sup |f_1 - f_2|$ and apply Theorem 2.4, p. 39, to $z_\pm = \pm u - Mt$.]

2.7 Let $g(t) = M$ for $0 \le t \le 1$ and $g(t) = M - (1 - t)^4$ for $1 < t \le 2$. Let u be the solution of $u_t - u_{xx} = 0$ in $Q_2 = (0, 2) \times (0, 2)$, $u = g$ on $\partial_p Q_2$. Compute $u(1, 1)$ and check that it is the maximum of u. Is this in contrast with the maximum principle of Theorem 2.4, p. 39?

2.8 Suppose $u = u(\mathbf{x}, t)$ is a solution of the heat equation in a plane domain $D_T = Q_T \setminus (\bar{Q}_1 \cup \bar{Q}_2)$ where Q_1 and Q_2 are the rectangles in Fig. 2.19. Assume that u attains its maximum M at the interior point $(\mathbf{x}_1, t_1)$. Where else $u = M$?

2.9 a) Find the self-similar solutions of the equation $u_t - u_{xx} = 0$ of the form $u(x, t) = U(x/\sqrt{t})$ and express the result in term of the *error function*

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-z^2} dz.$$

b) Find the solution of $u_t - u_{xx} = 0$ in $x > 0, t > 0$ satisfying the conditions $u(0, t) = 1$, $u(x, t) \to 0$ as $x \to +\infty$, $t > 0$, and $u(x, 0) = 0$, $x > 0$.

2.10 Determine for which α and β there exist self-similar solutions to $u_t - u_{xx} = f(x)$ of the form $t^\alpha U(x/t^\beta)$ in each one of the following cases:

$$\text{a) } f(x) = 0, \qquad \text{b) } f(x) = 1, \qquad \text{c) } f(x) = x.$$

[*Answer.* a) α arbitrary, $\beta = 1/2$. b) $\alpha = 1, \beta = 1/2$. c) $\alpha = 3/2, \beta = 1/2$.]

2.11 (*Reflecting barriers and Neumann condition*) Consider the symmetric random walk of Sect. 2.4. Suppose that a perfectly *reflecting* barrier is located at the point $L = \overline{m}h + \frac{h}{2} > 0$. By this we mean that if the particle hits the point $L - \frac{h}{2}$ at time t and moves to the right, then it

is reflected and it comes back to $L - \frac{h}{2}$ at time $t + \tau$. Show that when $h, \tau \to 0$ and $h^2/\tau = 2D$, $p = p(x,t)$ is a solution of the problem

$$\begin{cases} p_t - Dp_{xx} = 0 & x < L, t > 0 \\ p(x,0) = \delta(x) & x < L \\ p_x(L,t) = 0 & t > 0 \end{cases}$$

and moreover $\int_{-\infty}^{L} p(x,t)dx = 1$. Compute explicitly the solution.

[*Answer.* $p(x,t) = \Phi_D(x,t) + \Phi_D(x - 2L, t)$.]

2.12 (*Absorbing barriers and Dirichlet condition*) Consider the symmetric random walk of Sect. 2.4. Suppose that a perfectly *absorbing* barrier is located at the point $L = \overline{m}h > 0$. By this we mean that if the particle hits the point $L - h$ at time t and moves to the right then it is absorbed and stops at L. Show that when $h, \tau \to 0$ and $h^2/\tau = 2D$, $p = p(x,t)$ is a solution of the problem

$$\begin{cases} p_t - Dp_{xx} = 0 & x < L, t > 0 \\ p(x,0) = \delta(x) & x < L \\ p(L,t) = 0 & t > 0. \end{cases}$$

Compute explicitly the solution.

[*Answer.* $p(x,t) = \Phi_D(x,t) - \Phi_D(x - 2L, t)$.]

2.13 (*Elastic restoring force*) The one-dimensional motion of a particle follows the following rules, where N is a natural integer and m is an integer:

1. During an interval of time τ the particle takes one step of h unit length, starting from $x = 0$ at time $t = 0$.
2. If at some step the position of the particle is at the point mh, with $-N \le m \le N$, it moves to the right or to the left with probability given, respectively, by:

$$p_0 = \tfrac{1}{2}(1 - \tfrac{m}{N}), \qquad q_0 = \tfrac{1}{2}(1 + \tfrac{m}{N})$$

independently from the previous steps.

Explain why this rules model an elastic restoring force on the particle motion. Show that, if $h^2/\tau = 2D$ and $N\tau = \gamma > 0$, when $h, \tau \to 0$ and $N \to +\infty$, the limit transition probability is a solution of the following equation:

$$p_t = Dp_{xx} + \tfrac{1}{\gamma}(xp_x)_x.$$

2.14 Prove Theorem 2.17, p. 84, under the condition $z(x,t) \le C$, $x \in \mathbb{R}$, $0 \le t \le T$, by filling the details in the following steps.

1. Let $\sup_{\mathbb{R}} z(x,0) = M_0$ and define $w(x,t) = \dfrac{2(C + |M_0|)}{L^2}\left(\dfrac{x^2}{2} + Dt\right) + M_0$.
   Check that $w_t - Dw_{xx} = 0$ and use the maximum principle to show that $w \ge z$ in the rectangle $R_L = [-L, L] \times [0, T]$.
2. Fix an arbitrary point (x_0, t_0) and choose L large enough to have $(x_0, t_0) \in R_L$. Using 1. deduce that $z(x_0, t_0) \le M_0$.

2.15 Show that, if $g(x) = 0$ for $x < 0$ and $g(x) = 1$ for $x > 0$, then

$$\lim_{(x,t)\to(0,0)} \int_{\mathbb{R}} \Phi_D(x - y, t) g(y) dy$$

does not exist.

[*Hint.* Compute the limits along the curves $(x,t) = (0,t)$ and $(x,t) = (\pm t^2, t)$.]

2.16 Find an explicit formula for the solution of the global Cauchy problem

$$\begin{cases} u_t = Du_{xx} + bu_x + cu & x \in \mathbb{R},\ t > 0 \\ u(x,0) = g(x) & x \in \mathbb{R} \end{cases}$$

where D, b, c are constant. Show that, if $c < 0$ and g is bounded, $u(x,t) \to 0$ as $t \to +\infty$.
[*Answer.* $u(x,t) = e^{ct} \int_{\mathbb{R}} \Phi_D(x + bt - y, t) g(y)\, dy$.]

2.17 Find an explicit formula for the solution of the Cauchy problem

$$\begin{cases} u_t = u_{xx} & x > 0, t > 0 \\ u(x,0) = g(x) & x \geq 0 \\ u(0,t) = 0 & t > 0 \end{cases}$$

with g continuous and $g(0) = 0$.

2.18 Let $Q_T = \Omega \times (0,T)$, with Ω bounded domain in $\mathbb{R}^n$. Let $u \in C^{2,1}(Q_T) \cap C(\overline{Q}_T)$ satisfy the equation

$$u_t = D\Delta u + \mathbf{b}(\mathbf{x},t) \cdot \nabla u + c(\mathbf{x},t)u \quad \text{in } Q_T,$$

where $\mathbf{b}$ and c are continuous in $\overline{Q}_T$. Show that if $u \geq 0$ (resp. $u \leq 0$) on $\partial_p Q_T$ then $u \geq 0$ (resp. $u \leq 0$) in Q_T.
[*Hint.* Assume first that $c(\mathbf{x},t) \leq a < 0$. Then reduce to this case by setting $u = ve^{kt}$ with a suitable $k > 0$.]

2.19 Fill in the details in the arguments of Sect. 2.6.2, leading to formulas (2.108) and (2.109).

2.20 (*An invasion problem*) A population of density $P = P(x,y,t)$ and total mass $M(t)$ is initially (at time $t = 0$) concentrated at an isolated point, say, the origin $(0,0)$, it grows at a constant rate $a > 0$ and it diffuses with constant diffusion coefficient D.

1. Write the problem governing the evolution of P and solve it explicitly.
2. Compute the evolution of the mass

$$M(t) = \int_{\mathbb{R}^2} P(x,y,t)\, dxdy.$$

3. Let B_R be the circle centered at $(0,0)$ and radius R. Determine $R = R(t)$ such that

$$\int_{\mathbb{R}^2 \setminus B_{R(t)}} P(x,y,t)\, dxdy = M(0).$$

Call *metropolitan area* the region $B_{R(t)}$ and *rural area* the region $\mathbb{R}^2 \setminus B_{R(t)}$. Compute the speed of the *metropolitan propagation front* $\partial B_{R(t)}$.

[*Hint.* 3. We find: $\int_{\mathbb{R}^2 \setminus B_{R(t)}} P(x,y,t)\, dxdy = M(0) \exp\left\{at - \dfrac{R^2(t)}{4Dt}\right\}$ from which $R(t) = 2t\sqrt{aD}$. The speed of the metropolitan front is therefore equal to $2\sqrt{aD}$.]

2.21 (*Pollution in dimension $n \geq 2$*) Using conservation of mass, write a model of drift–diffusion for the concentration c of a pollutant in dimension $n = 2, 3$, assuming the following constitutive law for the flux vector

$$\mathbf{q}(\mathbf{x},t) = \underbrace{\mathbf{v}(\mathbf{x},t)c(\mathbf{x},t)}_{\text{drift}} - \underbrace{\kappa(\mathbf{x},t)\nabla c(\mathbf{x},t)}_{\text{diffusion}}.$$

[*Answer.* $c_t + \operatorname{div}(\mathbf{v}c - \kappa(\mathbf{x},t)\nabla c) = 0$.]

2.22 Solve the following initial-Dirichlet problem in $B_1 = \{\mathbf{x} \in \mathbb{R}^3 : |\mathbf{x}| < 1\}$:

$$\begin{cases} u_t = \Delta u & \mathbf{x} \in B_1,\ t > 0 \\ u(\mathbf{x}, 0) = 0 & \mathbf{x} \in \overline{B}_1 \\ u(\boldsymbol{\sigma}, t) = 1 & \boldsymbol{\sigma} \in \partial B_1,\ t > 0. \end{cases}$$

Compute $\lim_{t \to +\infty} u$.

[*Hint.* The solution is radial so that $u = u(r, t)$, $r = |\mathbf{x}|$. Observe that $\Delta u = u_{rr} + \frac{2}{r} u_r = \frac{1}{r}(ru)_{rr}$. Let $v = ru$, reduce to homogeneous Dirichlet condition and use separation of variables.]

2.23 Solve the following initial-Neumann problem in $B_1 = \{\mathbf{x} \in \mathbb{R}^3 : |\mathbf{x}| < 1\}$:

$$\begin{cases} u_t = \Delta u & \mathbf{x} \in B_1,\ t > 0 \\ u(\mathbf{x}, 0) = |\mathbf{x}| & \mathbf{x} \in \overline{B}_1 \\ u_{\boldsymbol{\nu}}(\boldsymbol{\sigma}, t) = 1 & \boldsymbol{\sigma} \in \partial B_1,\ t > 0. \end{cases}$$

2.24 Solve the following nonhomogeneous initial-Dirichlet problem in the unit sphere B_1 ($u = u(r, t)$, $r = |\mathbf{x}|$):

$$\begin{cases} u_t - \left(u_{rr} + \dfrac{2}{r} u_r\right) = q e^{-t} & 0 < r < 1,\ t > 0 \\ u(r, 0) = U & 0 \le r \le 1 \\ u(1, t) = 0 & t > 0. \end{cases}$$

[*Answer.* $u(r, t) = \dfrac{2}{r} \displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{\lambda_n} \sin(\lambda_n r) \left\{ \frac{q}{1 - \lambda_n^2} (e^{-t} - e^{-\lambda_n^2 t}) - U e^{-\lambda_n^2 t} \right\}, \quad \lambda_n = n\pi.$]

2.25 Using the maximum principle, compare the values of two call options $C^{(1)}$ and $C^{(2)}$ in the following cases: a) Same exercise price and $T_1 > T_2$; b) Same expiry time and $E_1 > E_2$.

2.26 Referring to Sect. 2.9.5, show that if the portfolio $V = \Pi + S\Delta$ is self-financing, then $\Delta = V_S$.

2.27 Consider system (2.185). Justify carefully the orbit configuration of Fig. 2.18 and in particular that the unstable separatrix γ of the saddle point $(1, 0)$ connects the two equilibrium points $(0, 0)$ and $(1, 0)$, by filling in the details in the following steps:

1. Let $\mathbf{F} = V\mathbf{i} + (-cV + U^2 - U)\mathbf{j}$ and $\mathbf{n}$ be the interior normal to the boundary of the triangle Ω in Fig. 2.20. Show that, if β is suitably chosen, $\mathbf{F} \cdot \mathbf{n} > 0$ along $\partial\Omega$.
2. Deduce that all the orbits of system (2.185) starting at a point in Ω cannot leave Ω (i.e. Ω is a *positively invariant region*) and converge to the origin as $z \to +\infty$.
3. Deduce that the unstable separatrix γ of the saddle point $(1, 0)$ approaches $(0, 0)$ as $z \to +\infty$.

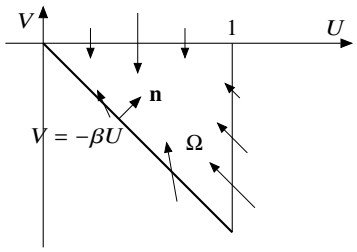

Fig. 2.20 Trapping region for the orbits of the vector field $\mathbf{F} = V\mathbf{i} + (-cV + U^2 - U)\mathbf{j}$

Chapter 3
The Laplace Equation

3.1 Introduction

The Laplace equation $\Delta u = 0$ occurs frequently in the applied sciences, in particular in the study of the *steady state phenomena*. Its solutions are called *harmonic* functions. For instance, the equilibrium position of a homogeneous, perfectly elastic and flexible membrane[1] is a harmonic function, as it is the velocity potential of a homogeneous fluid. Also, the steady state temperature of a homogeneous and isotropic body is a harmonic function and in this case Laplace equation constitutes the stationary counterpart (time independent) of the diffusion equation.

Slightly more generally, Poisson's equation $\Delta u = f$ plays an important role in the theory of *conservative fields* (electrical, magnetic, gravitational, . . .), where the vector field is derived from the gradient of a potential.

For example, let $\mathbf{E}$ be a force field due to a distribution of electric charges in a domain $\Omega \subset \mathbb{R}^3$. Then, in Gaussian (CGS) units, $\operatorname{div} \mathbf{E} = 4\pi\rho$, where ρ represents the density of the charge distribution. When a *potential* u exists such that $\nabla u = -\mathbf{E}$, then $\Delta u = \operatorname{div} \nabla u = -4\pi\rho$, which is Poisson's equation. If the electric field is created by charges located outside Ω, then $\rho = 0$ in Ω and u is harmonic therein. Analogously, the potential of a gravitational field due to a mass distribution is a harmonic function in a region free from mass.

In dimension two, the theories of harmonic and holomorphic functions are strictly connected[2]. Indeed, the real and the imaginary part of a holomorphic function are harmonic. For instance, since the functions

[1] A drum skin is a good approximation.

[2] A complex function $f = f(z)$ is *holomorphic* in an open subset Ω of the complex plane if for every $z_0 \in \Omega$, the following limit exists and it is finite:

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'(z_0).$$

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
https://doi.org/10.1007/978-3-031-21853-8_3

$$z^m = r^m(\cos m\theta + i \sin m\theta), \qquad m \in \mathbb{N},$$

(r, θ polar coordinates) are holomorphic in the whole plane $\mathbb{C}$, the functions

$$u(r,\theta) = r^m \cos m\theta \quad \text{and} \quad v(r,\theta) = r^m \sin m\theta, \qquad m \in \mathbb{N},$$

are harmonic in $\mathbb{R}^2$ (called *elementary harmonics*). In Cartesian coordinates, they are harmonic polynomials; for $m = 1, 2, 3$ we find

$$x, \quad y, \quad xy, \quad x^2 - y^2, \quad x^3 - 3xy^2, \quad 3x^2y - y^3.$$

Other examples are

$$u(x,y) = e^{\alpha x} \cos \alpha y, \quad v(x,y) = e^{\alpha x} \sin \alpha y \qquad (\alpha \in \mathbb{R}),$$

the real and imaginary parts of $f(z) = e^{i\alpha z}$, both harmonic in $\mathbb{R}^2$, and

$$u(r,\theta) = \log r, \quad v(r,\theta) = \theta,$$

the real and imaginary parts of $f(z) = \log_0 z = \log r + i\theta$, harmonic in $\mathbb{R}^2 \setminus (0,0)$ and $\mathbb{R}^2 \setminus \{\theta = 0\}$, respectively.

In this chapter we present the formulation of the most important well posed problems and the classical properties of harmonic functions, focusing mainly on dimensions two and three. As in Chap. 2, we emphasize some probabilistic aspects, exploiting the connection among random walks, Brownian motion and the Laplace operator. A central notion is the concept of *fundamental solution*, that we develop in conjunction with the very basic elements of the so called *potential theory*.

3.2 Boundary value Problems. Uniqueness

Consider the Poisson equation

$$\Delta u = f \quad \text{in } \Omega \tag{3.1}$$

where $\Omega \subset \mathbb{R}^n$ is a **bounded domain**. The boundary value problems associated with equation (3.1) are the stationary counterparts of the corresponding problems for the diffusion equation. Clearly here there is no initial condition. On the boundary $\partial\Omega$ we may assign:

- *Dirichlet data*
$$u = g. \tag{3.2}$$
- *Neumann data*
$$\partial_\nu u = h, \tag{3.3}$$
where ν is the outward normal unit vector to $\partial\Omega$.

- A *Robin* (*radiation*) *condition*

$$\partial_\nu u + \alpha u = h \qquad (\alpha > 0). \tag{3.4}$$

- A *mixed condition*; for instance,

$$\begin{aligned} u &= g \quad \text{on } \overline{\Gamma}_D \\ \partial_\nu u &= h \quad \text{on } \Gamma_N, \end{aligned} \tag{3.5}$$

where $\overline{\Gamma}_D \cup \overline{\Gamma}_N = \partial\Omega$, $\Gamma_D \cap \Gamma_N = \emptyset$, and Γ_D, Γ_N are relatively open subsets of $\partial\Omega$, regular in the sense of Definition 1.11, p. 19.

When $g = h = 0$ we say that the above boundary conditions are *homogeneous.*

We give some interpretations. If u is the position of a homogeneous, perfectly flexible membrane and f is an external distributed load (vertical force per unit surface), then (3.1) models a steady state.

The Dirichlet condition corresponds to fixing the position of the membrane at its boundary. Robin condition describes an elastic attachment at the boundary while a homogeneous Neumann condition corresponds to a free vertical motion of the boundary of the membrane (for more details see Chap. 5, Sect. 5.7.1).

On the other hand, as we have seen in Chap. 2, if u is the steady state concentration of a substance, the Dirichlet condition prescribes the level of u at the boundary, while the Neumann condition prescribes the inward flux of u through the boundary.

Using Green's identity (1.17) we can prove the following uniqueness result.

Theorem 3.1 *Let $\Omega \subset \mathbb{R}^n$ be a smooth, bounded domain. Then there exists at most one solution $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ of (3.1), satisfying on $\partial\Omega$ one of the conditions (3.2), (3.4) or (3.5).*

In the case of the Neumann condition, that is when

$$\partial_\nu u = h \qquad \text{on } \partial\Omega,$$

two solutions differ by a constant.

Proof Let u and v be solutions of the same problem, sharing the same boundary data, and let $w = u - v$. Then w is harmonic and satisfies homogeneous boundary conditions (one among (3.2)-(3.5)). Substituting $u = v = w$ into (1.17) we find

$$\int_\Omega |\nabla w|^2 \, d\mathbf{x} = \int_{\partial\Omega} w \partial_\nu w \, d\sigma.$$

If Dirichlet, Neumann or mixed conditions hold, we have

$$\int_{\partial\Omega} w \partial_\nu w \, d\sigma = 0.$$

When a Robin condition holds,

$$\int_{\partial\Omega} w \partial_\nu w \, d\sigma = -\int_{\partial\Omega} \alpha w^2 d\sigma \leq 0.$$

In any case we obtain that

$$\int_\Omega |\nabla w|^2 \, d\mathbf{x} \le 0. \tag{3.6}$$

From (3.6) we infer $\nabla w = \mathbf{0}$ and therefore $w = u - v =$ constant. This concludes the proof in the case of Neumann condition. In the other cases, the constant must be zero (why?), hence $u = v$. □

Remark 3.2 Consider the Neumann problem

$$\Delta u = f \text{ in } \Omega, \quad \partial_\nu u = h \text{ on } \partial\Omega.$$

Integrating the equation on Ω and using Gauss' formula, we find

$$\int_\Omega f \, d\mathbf{x} = \int_{\partial\Omega} h \, d\sigma. \tag{3.7}$$

The relation (3.7) appears as a *compatibility* condition on the data f and h, that has *necessarily* to be satisfied in order for the Neumann problem to admit a solution. Thus, when having to solve a Neumann problem, the first thing to do is to check the validity of (3.7). If it does not hold, the problem does not have any solution. We will examine later the physical meaning of (3.7).

3.3 Harmonic Functions

3.3.1 Discrete harmonic functions

In Chap. 2 we have examined the connection between Brownian motion and diffusion equation. We now go back to the multidimensional symmetric random walk considered in Sect. 2.6, analyzing its relation with the Laplace operator Δ. For simplicity we will work in dimension $n = 2$ but both arguments and conclusions may be easily extended to any dimension $n > 2$. We fix a time step $\tau > 0$, a space step $h > 0$ and denote by $h\mathbb{Z}^2$ the *lattice* of points $\mathbf{x} = (x_1, x_2)$ whose coordinates are integer multiples of h. Let $p(\mathbf{x}, t) = p(x_1, x_2, t)$ be the transition probability function, giving the probability to find our random particle at $\mathbf{x}$, at time t. From the total probability formula, we found a difference equation for p, that we rewrite in dimension two:

$$p(\mathbf{x}, t+\tau) = \frac{1}{4}\left\{p(\mathbf{x} + h\mathbf{e}_1, t) + p(\mathbf{x} - h\mathbf{e}_1, t) + p(\mathbf{x} + h\mathbf{e}_2, t) + p(\mathbf{x} - h\mathbf{e}_2, t)\right\}.$$

We can write this formula in a more significant way by recognizing at the right hand side the *mean value operator* M_h, whose action on a function $u = u(\mathbf{x})$, gives the average of u over the points of the lattice $h\mathbb{Z}^2$ at distance h from $\mathbf{x}$. We say that these points constitute the *discrete neighborhood of* $\mathbf{x}$ *of radius* h.

Thus, we can write

$$p(\mathbf{x}, t+\tau) = M_h p(\mathbf{x}, t). \tag{3.8}$$

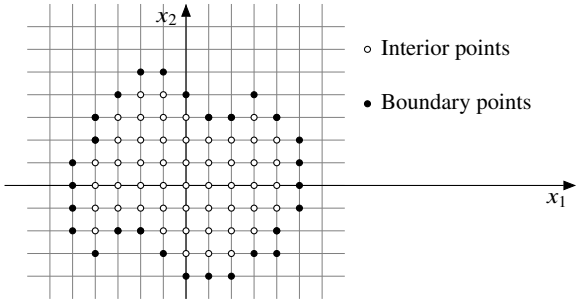


Fig. 3.1 A domain for the discrete Dirichlet problem

In (3.8), the probability p at time $t + \tau$ is determined by the action of M_h over p at the previous time, and then it is natural to interpret the mean value operator as *the generator of the random walk*.

Now we come to the Laplacian. If u is twice continuously differentiable, it is not difficult to show that[3]

$$\lim_{h\to 0} \frac{M_h u(\mathbf{x}) - u(\mathbf{x})}{h^2} = \frac{1}{4}\Delta u(\mathbf{x}). \tag{3.9}$$

The formula (3.9) induces to define, for any fixed $h > 0$, a *discrete Laplace operator* through the formula

$$\Delta_h^* = \frac{4}{h^2}(M_h - I)$$

where I denotes the *identity* operator (i.e. $Iu = u$). Coherently, we say that u is *d-harmonic* (d for *discrete*) if $\Delta_h^* u = 0$. Thus, the value of a *d-harmonic* function at a point $\mathbf{x}$ is given by the average of the values at the points in the discrete neighborhood of $\mathbf{x}$ of radius h.

We can proceed further and define a discrete Dirichlet problem. Let A be a subset of $h\mathbb{Z}^2$. We say that $\mathbf{x} \in A$ is (Fig. 3.1):

- An *interior point* of A if its h-neighborhood is contained in A.
- A *boundary point* if it is not an interior point but its h-neighborhood contains at least an interior point. The set of the boundary points of A, the *boundary* of A, is denoted by ∂A.

The points of A whose h-neighborhood does not contain interior points of A are considered *isolated points.*

[3] Using a second order Taylor's expansion, after some simplifications, we get:

$$M_h u(\mathbf{x}) = u(\mathbf{x}) + \frac{h^2}{4}\left\{u_{x_1x_1}(\mathbf{x}) + u_{x_2x_2}(\mathbf{x})\right\} + o(h^2)$$

from which formula (3.9) comes easily.

We say that A is *connected* if A does not have isolated points and, given any couple of points $\mathbf{x}_0, \mathbf{x}_1$ in A, it is possible to connect them by a walk[4] on $h\mathbb{Z}^2$ entirely contained in A.

Discrete Dirichlet problem. Let A be a *bounded connected* subset of $h\mathbb{Z}^2$ and g be a function defined on the boundary ∂A of A. We want to determine u, defined on A, such that

$$\begin{cases} \Delta_h^* u = 0 \text{ at the interior points of } A \\ u = g \quad \text{on } \partial A. \end{cases} \tag{3.10}$$

We deduce immediately three important properties of a solution u:

1. Maximum principle: *If u attains its maximum or its minimum at an interior point then u is constant*. Indeed, suppose that $\mathbf{x} \in A$ is an interior point and that $u(\mathbf{x}) = M \geq u(\mathbf{y})$, for every $\mathbf{y} \in A$. Since $u(\mathbf{x})$ is the average of the four values of u at the points at distance h from $\mathbf{x}$, at all these points u must be equal to M. Let $\mathbf{x}_1 \neq \mathbf{x}$ be one of these neighboring points. By the same argument, $u(\mathbf{y}) = M$ for every $\mathbf{y}$ in the h-neighborhood of $\mathbf{x}_1$. Since A is connected, proceeding in this way we prove that $u(\mathbf{y}) = M$ at every point of A.
2. *u attains its maximum and its minimum on ∂A*. This is an immediate consequence of 1.
3. *The solution of the discrete Dirichlet problem is unique*. Since Δ_h^* is linear, this is an immediate consequence of 2, applied to the difference of two solutions.

The discrete Dirichlet problem (3.10) has a remarkable probabilistic interpretation that can be used to construct its solution. Let us go back to our random particle. First of all, we want to show that whatever its starting point $\mathbf{x} \in A$ is, the particle hits the boundary *∂A with probability one.*

For every $\Gamma \subseteq \partial A$, we denote by $P(\mathbf{x}, \Gamma)$ the *probability that the particle starting from $\mathbf{x} \in A$ hits ∂A for the first time at a point $\mathbf{y} \in \Gamma$*. We have to prove that $P(\mathbf{x}, \partial A) = 1$ for every $\mathbf{x} \in A$.

For fixed Γ, consider the function

$$\mathbf{x} \longmapsto w_\Gamma(\mathbf{x}) = P(\mathbf{x}, \Gamma)$$

We show that w_Γ is *d-harmonic* in the interior of A, that is $\Delta_h^* w_\Gamma = 0$.

Indeed, to hit Γ starting from $\mathbf{x}$, the particle first hits a point $\mathbf{y}$ in its h-neighborhood with probability $1/4$, and from there it reaches Γ, independently of the first step, with probability $P(\mathbf{y}, \Gamma) = w_\Gamma(\mathbf{y})$. Then, by the total probability formula we can write

$$w_\Gamma(\mathbf{x}) = \frac{1}{4} \sum_{|\mathbf{x}-\mathbf{y}|=h} w_\Gamma(\mathbf{y}) = M_h w_\Gamma(\mathbf{x}),$$

which entails

$$0 = (M_h - I) w_\Gamma = \tfrac{h^2}{4} \Delta_h^* w_\Gamma.$$

[4] Recall that consecutive points in a walk have distance h.

Thus, w_Γ is *d-harmonic* in A. Clearly, if $\mathbf{x} \in \Gamma$ we have $w_\Gamma(\mathbf{x}) = 1$, while if $\mathbf{x} \in \partial A \backslash \Gamma$, $w_\Gamma(\mathbf{x}) = 0$. In particular, $w_{\partial A}(\mathbf{x}) = P(\mathbf{x}, \partial A)$ is *d-harmonic* in A and $w_{\partial A} = 1$ on ∂A. On the other hand, the function $z(\mathbf{x}) \equiv 1$ satisfies the same discrete Dirichlet problem, so that, by the uniqueness property 3 above,

$$w_{\partial A}(\mathbf{x}) \equiv 1 \text{ in } A. \tag{3.11}$$

This means that the particle hits the boundary ∂A *with probability one*.

We now examine the set function

$$\Gamma \mapsto P(\mathbf{x}, \Gamma), \qquad \Gamma \subseteq \partial A. \tag{3.12}$$

From (3.11), $P(\mathbf{x}, \partial A) = 1$. Moreover, by linearity, if $\Gamma_1, \Gamma_2 \subset \partial A$ and $\Gamma_1 \cap \Gamma_2 = \emptyset$, then

$$P(\mathbf{x}, \Gamma_1 \cup \Gamma_2) = P(\mathbf{x}, \Gamma_1) + P(\mathbf{x}, \Gamma_2).$$

Thus, the set function (3.12) defines a probability measure on ∂A, for any fixed $\mathbf{x} \in A$.

We now construct the solution u to (3.10). Interpret the boundary data g as a *payoff*: if the particle starts from $\mathbf{x}$ and hits the boundary for the first time at $\mathbf{y}$, it gains $g(\mathbf{y})$. We have:

Proposition 3.3 *The value $u(\mathbf{x})$ is given by the expected value of the winnings $g(\cdot)$ with respect to the probability $P(\mathbf{x}, \cdot)$. That is*

$$u(\mathbf{x}) = \sum_{\mathbf{y} \in \partial A} g(\mathbf{y}) P(\mathbf{x}, \{\mathbf{y}\}). \tag{3.13}$$

Proof Each term

$$g(\mathbf{y}) P(\mathbf{x}, \{\mathbf{y}\}) = g(\mathbf{y}) w_{\{\mathbf{y}\}}(\mathbf{x})$$

is d-harmonic in A and therefore u is d-harmonic in A as well. Moreover, if $\mathbf{x} \in \partial A$ then $u(\mathbf{x}) = g(\mathbf{x})$, since each term in the sum is equal to $g(\mathbf{x})$ if $\mathbf{y} = \mathbf{x}$ or to zero if $\mathbf{y} \neq \mathbf{x}$. □

As $h \to 0$, formula (3.9) shows that, formally, d-harmonic functions "become" harmonic. Thus, it seems reasonable that appropriate versions of the above properties and results should hold in the continuous case. We start with the mean value properties.

3.3.2 Mean value properties

Guided by their discrete characterization, we want to establish some fundamental properties of the harmonic functions. To be precise, we say that a function u is *harmonic* in a domain $\Omega \subseteq \mathbb{R}^n$ if $u \in C^2(\Omega)$ and $\Delta u = 0$ in Ω.

Since d-harmonic functions are defined through a mean value property, we expect that harmonic functions inherit a mean value property of the following kind: the value

at the center of any ball $B \subset\subset \Omega$, i.e. compactly contained in Ω, equals the average of the values on the boundary ∂B. Actually, something more is true.

Theorem 3.4 *Let u be harmonic in $\Omega \subseteq \mathbb{R}^n$. Then, for any ball $B_R(\mathbf{x}) \subset\subset \Omega$ the following mean value formulas hold:*

$$u(\mathbf{x}) = \frac{n}{\omega_n R^n} \int_{B_R(\mathbf{x})} u(\mathbf{y}) d\mathbf{y} \tag{3.14}$$

$$u(\mathbf{x}) = \frac{1}{\omega_n R^{n-1}} \int_{\partial B_R(\mathbf{x})} u(\boldsymbol{\sigma}) d\sigma \tag{3.15}$$

where ω_n is the surface measure[5] *of ∂B_1.*

Proof Let us start from the second formula. For $r < R$ define

$$g(r) = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r(\mathbf{x})} u(\boldsymbol{\sigma}) d\sigma.$$

Perform the change of variables $\boldsymbol{\sigma} = \mathbf{x} + r\boldsymbol{\sigma}'$. Then

$$\boldsymbol{\sigma}' \in \partial B_1(\mathbf{0}), \quad d\sigma = r^{n-1} d\sigma' \qquad \text{and} \qquad g(r) = \frac{1}{\omega_n} \int_{\partial B_1(\mathbf{0})} u(\mathbf{x} + r\boldsymbol{\sigma}') d\sigma'. \tag{3.16}$$

Thus, carrying the derivatives under the integral and using the divergence theorem ($\boldsymbol{\sigma}' \in \partial B_1(\mathbf{0})$ is both the position and the outer normal direction), we find

$$\begin{aligned} g'(r) &= \frac{1}{\omega_n} \int_{\partial B_1(\mathbf{0})} \frac{d}{dr} u(\mathbf{x} + r\boldsymbol{\sigma}') d\sigma' = \frac{1}{\omega_n} \int_{\partial B_1(\mathbf{0})} \nabla u(\mathbf{x} + r\boldsymbol{\sigma}') \cdot \boldsymbol{\sigma}' d\sigma' \\ &= \frac{r}{\omega_n} \int_{B_1(\mathbf{0})} \Delta u(\mathbf{x} + r\mathbf{y}) d\mathbf{y} = 0. \end{aligned}$$

Thus, g is constant. Since, from (3.16), $g(r) \to u(\mathbf{x})$ for $r \to 0$, we get (3.15).

To obtain (3.14), let $R = r$ in (3.15), multiply by r and integrate both sides between 0 and R. We find

$$\frac{R^n}{n} u(\mathbf{x}) = \frac{1}{\omega_n} \int_0^R dr \int_{\partial B_r(\mathbf{x})} u(\boldsymbol{\sigma}) d\sigma = \frac{1}{\omega_n} \int_{B_R(\mathbf{x})} u(\mathbf{y}) d\mathbf{y}$$

(recall (A.1), Appendix A) from which (3.14) follows. □

Even more significant is a converse of Theorem 3.4. We say that a **bounded** *function u satisfies the mean value property in Ω, if (3.14) or (3.15) holds for any ball $B_R(\mathbf{x}) \subset\subset \Omega$.* Indeed, as we will see below, (3.14) and (3.15) are equivalent. Moreover, it turns out that if u is bounded and possesses the mean value property in a domain Ω, then u is harmonic in Ω. Thus we obtain a characterization of harmonic functions through a mean value property, as in the discrete case. As a byproduct, we deduce that every harmonic function in a domain Ω is continuously differentiable of any order in Ω, that is, it belongs to $C^\infty(\Omega)$. Notice that this is not a trivial fact since it involves derivatives not appearing in the expression of the Laplace operator.

Theorem 3.5 *Let u be bounded in Ω. If u satisfies the mean value property, then $u \in C^\infty(\Omega)$ and it is harmonic in Ω.*

[5] See footnote 1, p. 7.

We postpone the proof to the end of the Sect. 3.3.5. Here we point out the following simple consequence.

Theorem 3.6 *Let $\Omega \subset \mathbb{R}^n$ be a domain and $\{u_k\}_{k\geq 1}$ be a sequence of harmonic functions in Ω. If u_k converges uniformly to u in every compact subset of Ω, then $u \in C^\infty(\Omega)$ and is harmonic in Ω.*

Proof Let $\mathbf{x} \in \Omega$ and $B_r(\mathbf{x}) \subset\subset \Omega$. By Theorem 3.4, for every $k \geq 1$ we can write

$$u_k(\mathbf{x}) = \frac{1}{\omega_n R^n} \int_{B_R(\mathbf{x})} u_k(\mathbf{y}) d\mathbf{y}. \tag{3.17}$$

Since $u_k \to u$ uniformly in $\overline{B_r}(\mathbf{x})$, $\int_{B_R(\mathbf{x})} u_k(\mathbf{y}) d\mathbf{y} \to \int_{B_R(\mathbf{x})} u(\mathbf{y}) d\mathbf{y}$ so that, letting $k \to +\infty$ in (3.17) we get

$$u(\mathbf{x}) = \frac{1}{\omega_n R^n} \int_{B_R(\mathbf{x})} u(\mathbf{y}) d\mathbf{y}.$$

Since $\mathbf{x}$ is an arbitrary point in Ω, we deduce that u has the mean value property and therefore is harmonic. □

3.3.3 Maximum principles

As in the discrete case, a function satisfying the mean value property in a domain[6] Ω cannot attain its maximum or minimum at an *interior point of* Ω, unless it is constant. In case Ω is bounded and u (nonconstant) is continuous up to the boundary of Ω, it follows that u attains both its maximum and minimum **only on** $\partial\Omega$. This result expresses a maximum principle that we state precisely in the following theorem.

Theorem 3.7 *Let $\Omega \subseteq \mathbb{R}^n$ be a domain and $u \in C(\Omega)$. If u has the mean value property and attains its maximum or minimum at $\mathbf{p} \in \Omega$, then u is constant. In particular, if Ω is bounded and $u \in C(\overline{\Omega})$ is not constant, then, for every $\mathbf{x} \in \Omega$,*

$$u(\mathbf{x}) < \max_{\partial\Omega} u \quad \text{and} \quad u(\mathbf{x}) > \min_{\partial\Omega} u.$$

Proof Let $\mathbf{p}$ be a minimum point[7] for u:

$$m = u(\mathbf{p}) \leq u(\mathbf{y}), \qquad \forall \mathbf{y} \in \Omega.$$

We want to show that $u \equiv m$ in Ω. Let $\mathbf{q}$ be another arbitrary point in Ω. Since Ω is connected, it is possible to find a finite sequence of balls $B(\mathbf{x}_j) \subset\subset \Omega$, $j = 0, \ldots, N$, such that (Fig. 3.2):

- $\mathbf{x}_j \in B(\mathbf{x}_{j-1})$, $j = 1, \ldots, N$.
- $\mathbf{x}_0 = \mathbf{p}$, $\mathbf{x}_N = \mathbf{q}$.

The mean value property gives

$$m = u(\mathbf{p}) = \frac{1}{|B(\mathbf{p})|} \int_{B(\mathbf{p})} u(\mathbf{y}) d\mathbf{y}.$$

[6] Recall that a *domain* is an open *connected* set.

[7] The argument for the maximum is the same.

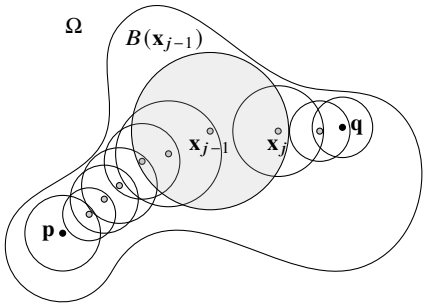


Fig. 3.2 A sequence of overlapping circles connecting the points **p** and **q**

Suppose there exists $\mathbf{z} \in B(\mathbf{p})$ such that $u(\mathbf{z}) > m$. Then, given a circle $B_r(\mathbf{z}) \subset B(\mathbf{p})$, we can write:

$$m = \frac{1}{|B(\mathbf{p})|}\int_{B(\mathbf{p})} u(\mathbf{y})d\mathbf{y} = \frac{1}{|B(\mathbf{p})|}\left\{\int_{B(\mathbf{p})\setminus B_r(\mathbf{z})} u(\mathbf{y})d\mathbf{y} + \int_{B_r(\mathbf{z})} u(\mathbf{y})d\mathbf{y}\right\}. \tag{3.18}$$

By the mean value property,

$$\int_{B_r(\mathbf{z})} u(\mathbf{y})d\mathbf{y} = u(\mathbf{z})\,|B_r(\mathbf{z})| > m\,|B_r(\mathbf{z})|\,,$$

and since $u(\mathbf{y}) \geq m$ for every $\mathbf{y}$, from (3.18) we obtain the contradiction

$$m > \frac{1}{|B(\mathbf{p})|}\left\{m\,|B(\mathbf{p})\setminus B_r(\mathbf{z})| + m\,|B_r(\mathbf{z})|\right\} = m.$$

Thus it must be that $u \equiv m$ in $B(\mathbf{p})$ and in particular $u(\mathbf{x}_1) = m$. We repeat now the same argument with $\mathbf{x}_1$ in place of $\mathbf{p}$ to show that $u \equiv m$ in $B(\mathbf{x}_1)$ and in particular $u(\mathbf{x}_2) = m$. Iterating the procedure we eventually deduce that $u(\mathbf{x}_N) = u(\mathbf{q}) = m$. Since $\mathbf{q}$ is an arbitrary point of Ω, we conclude that $u \equiv m$ in Ω. □

An important consequence of the maximum principle is the following corollary.

Corollary 3.8 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and $g \in C(\partial\Omega)$. The problem*

$$\begin{cases} \Delta u = 0 \text{ in } \Omega \\ u = g \quad \text{on } \partial\Omega \end{cases} \tag{3.19}$$

has at most a solution $u_g \in C^2(\Omega)\cap C(\overline{\Omega})$. Moreover, let u_{g_1} and u_{g_2} be the solutions corresponding to the data $g_1, g_2 \in C(\partial\Omega)$. Then:

a) (Comparison) If $g_1 \geq g_2$ on $\partial\Omega$ then

$$u_{g_1} \geq u_{g_2} \qquad \text{in } \Omega. \tag{3.20}$$

b) (Stability)

$$\left|u_{g_1}(\mathbf{x}) - u_{g_2}(\mathbf{x})\right| \leq \max_{\partial\Omega}|g_1 - g_2| \qquad \text{for every } \mathbf{x} \in \Omega. \tag{3.21}$$

Proof We first show a). Let $w = u_{g_1} - u_{g_2}$. Then w is harmonic and $w = g_1 - g_2 \geq 0$ on $\partial\Omega$. From Theorem 3.7

$$w(\mathbf{x}) \geq \min_{\partial\Omega}(g_1 - g_2) \geq 0 \quad \text{for every } \mathbf{x} \in \Omega.$$

This is (3.20). To prove b), apply Theorem 3.7 to w and $-w$ to find

$$\pm w(\mathbf{x}) \leq \max_{\partial\Omega} |g_1 - g_2| \quad \text{for every } \mathbf{x} \in \Omega$$

which is equivalent to (3.21).

Now if $g_1 = g_2$, (3.21) implies $w = u_{g_1} - u_{g_2} \equiv 0$, so that the Dirichlet problem (3.19) has at most one solution. □

Remark 3.9 Inequality (3.21) is a *stability estimate*. Indeed, suppose g is known within an absolute error less than ε, or, in other words, suppose g_1 is an approximation of g and $\max_{\partial\Omega} |g - g_1| < \varepsilon$; then (3.21) gives

$$\max_{\overline{\Omega}} \left| u_{g_1} - u_g \right| < \varepsilon$$

so that the approximate solution is known within the same absolute error.

Theorem 3.7 states that if $u \in C^2(\Omega) \cap C(\overline{\Omega})$ is a nonconstant harmonic function, then it attains its maximum and minimum *only* at the boundary. On the other hand, one may look for maximum/minimum principles holding for subsolutions/supersolutions of the Laplace equation, as we did for the heat equation (Sect. 2.2.2 on page 39). We provide here a straightforward result, postponing a systematic treatment of this issue to Sect. 3.4.

Theorem 3.10 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and $u \in C^2(\Omega) \cap C(\overline{\Omega})$. Assume that*

$$-\Delta u < 0 \qquad (\textit{resp. } -\Delta u > 0)$$

in Ω. Then, for every $\mathbf{x} \in \Omega$,

$$u(\mathbf{x}) < \max_{\partial\Omega} u \qquad (\textit{resp. } u(\mathbf{x}) > \min_{\partial\Omega} u).$$

Proof Assume by contradiction that $\max_{\overline{\Omega}} u = u(\mathbf{x}_0)$, with $\mathbf{x}_0 \in \Omega$. Then we find the contradiction $\Delta u(\mathbf{x}_0) \leq 0$. Similarly one can treat the other case. □

While in this section we considered only the Laplace operator, maximum principles of various types hold for more general operators, as we shall see in Chap. 8.

3.3.4 The Hopf principle

Another important result concerns the slope of a harmonic function u at a boundary point satisfying a so-called *interior sphere condition*, where u attains an extremum.

Definition 3.11 We say that $\mathbf{x}_0 \in \partial\Omega$ satisfies an *interior sphere condition* if there exists a ball $B_R(\mathbf{p}) \subset \Omega$ such that $\overline{B}_R(\mathbf{p}) \cap \partial\Omega = \{\mathbf{x}_0\}$ (Fig. 3.3).

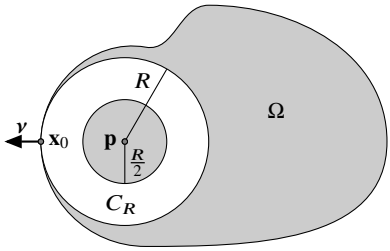


Fig. 3.3 Interior sphere condition at $\mathbf{x}_0$

Theorem 3.12 *Let Ω be a domain in $\mathbb{R}^n$ and $u \in C^2(\Omega) \cap C(\overline{\Omega})$ be harmonic in Ω. Assume that:*

1. *at $\mathbf{x}_0 \in \partial\Omega$ an* interior sphere condition *holds, and denote by $\boldsymbol{\nu}$ the outward unit normal to the sphere at $\mathbf{x}_0$;*
2. *$u(\mathbf{x}_0) = m = \min_{\overline{\Omega}} u$.*

If the derivative of u at $\mathbf{x}_0$ exists, then

$$\partial_{\boldsymbol{\nu}} u(\mathbf{x}_0) < 0$$

unless $u \equiv m$ in Ω.

Proof Let u be nonconstant. Replacing u by $u - m$, we can assume that $m = 0$. Referring to Fig. 3.3, the idea is to compare u with a radial function z, in the spherical shell

$$C_R = B_R(\mathbf{p}) \setminus \overline{B}_{R/2}(\mathbf{p}),$$

with the following properties:

a) $z = m_1 = \min_{\partial B_{R/2}(\mathbf{p})} u$ on $\partial B_{R/2}(\mathbf{p})$, $z = 0$ on $\partial B_R(\mathbf{p})$ and, in particular, $z(\mathbf{x}_0) = 0$.
b) $\partial_{\boldsymbol{\nu}} z(\mathbf{x}_0) < 0$.
c) $\Delta z > 0$ in C_R.

Note that $m_1 > 0$ by Theorem 3.7. The function z is constructed as follows:

$$z(\mathbf{x}) = \frac{m_1}{\gamma}\left\{e^{-\alpha|\mathbf{x}-\mathbf{p}|^2} - e^{-\alpha R^2}\right\}$$

where $\gamma = e^{-\alpha R^2/4} - e^{\alpha R^2}$ and $\alpha > 0$ is to be suitably chosen.

Then a) is satisfied. Observing that the exterior normal to the sphere at $\mathbf{x}_0$ is given by $\boldsymbol{\nu} = (\mathbf{x}_0 - \mathbf{p})/R$, we compute

$$\partial_{\boldsymbol{\nu}} z(\mathbf{x}_0) = \nabla z(\mathbf{x}_0) \cdot \boldsymbol{\nu} = -\frac{2\alpha m_1}{\gamma} e^{-\alpha R^2} (\mathbf{x}_0 - \mathbf{p}) \cdot \frac{(\mathbf{x}_0 - \mathbf{p})}{R} = -\frac{2\alpha R m_1}{\gamma} e^{-\alpha R^2} < 0$$

so that b) holds. Finally, we have, in C_R :

$$\Delta z(\mathbf{x}) = \frac{m_1}{\gamma} e^{-\alpha|\mathbf{x}-\mathbf{p}|^2} \left\{4\alpha^2 |\mathbf{x} - \mathbf{p}|^2 - 2n\alpha\right\} \geq \frac{m_1}{\gamma} e^{-\alpha R^2} \left\{\alpha^2 R^2 - 2n\alpha\right\}$$

and therefore c) is satisfied if $\alpha > 2n/R$.

Once z is constructed, we set $w = u - z$. Then $w \geq 0$ on ∂C_R and $\Delta w < 0$ in C_R. Hence, by Thm. 3.10, $w > 0$ in C_R. Therefore, since $w(\mathbf{x}_0) = 0$, we have

$$\partial_\nu u(\mathbf{x}_0) - \partial_\nu z(\mathbf{x}_0) = \partial_\nu w(\mathbf{x}_0) \le 0$$

and from property b), $\partial_\nu u(\mathbf{x}_0) \le \partial_\nu z(\mathbf{x}_0) < 0$. The proof is complete. □

The key point in Theorem 3.12 is that the normal derivative of u at $\mathbf{x}_0$ cannot be zero. This has a number of consequences, for instance a maximum principle for the Robin problem (see Problem 3.4).

3.3.5 The Dirichlet problem in a disc. Poisson's formula

To prove the existence of a solution to one of the boundary value problems we considered in Sect. 3.2 is not, in general, an elementary task. In Chap. 8, we solve this question in a wider context, using the more advanced tools of Functional Analysis. However, in special cases, elementary methods work, like separation of variables. We use it to compute the solution of the Dirichlet problem in a disc. Precisely, let $B_R = B_R(\mathbf{p}) \subset \mathbb{R}^2$ be the disc of radius R centered at $\mathbf{p} = (p_1, p_2)$ and $g \in C(\partial B_R)$. We want to prove the following theorem.

Theorem 3.13 *The unique solution $u \in C^2(B_R) \cap C(\overline{B}_R)$ of the problem*

$$\begin{cases} \Delta u = 0 & \text{in } B_R \\ u = g & \text{on } \partial B_R \end{cases} \tag{3.22}$$

is given by ***Poisson's formula***

$$u(\mathbf{x}) = \frac{R^2 - |\mathbf{x} - \mathbf{p}|^2}{2\pi R} \int_{\partial B_R(\mathbf{p})} \frac{g(\boldsymbol{\sigma})}{|\mathbf{x} - \boldsymbol{\sigma}|^2} d\sigma. \tag{3.23}$$

In particular, $u \in C^\infty(B_R)$.

Proof The symmetry of the domain suggests the use of polar coordinates

$$x_1 = p_1 + r\cos\theta \qquad x_2 = p_2 + r\sin\theta.$$

Accordingly, let

$$U(r, \theta) = u(p_1 + r\cos\theta, p_2 + r\sin\theta), \; G(\theta) = g(p_1 + R\cos\theta, p_2 + R\sin\theta).$$

The Laplace equation becomes (see Appendix B)

$$U_{rr} + \frac{1}{r}U_r + \frac{1}{r^2}U_{\theta\theta} = 0, \qquad 0 < r < R, 0 \le \theta \le 2\pi, \tag{3.24}$$

with the Dirichlet condition

$$U(R, \theta) = G(\theta), \qquad 0 \le \theta \le 2\pi.$$

Since we ask u to be continuous in $\overline{B}_R$, then U and G have to be continuous in $[0, R] \times [0, 2\pi]$ and $[0, 2\pi]$, respectively; moreover both have to be 2π-periodic with respect to θ, and $U(0, \theta)$ must be independent of θ.

We now use the method of separation of variables, by looking first for solutions of the form

$$U(r,\theta) = v(r)w(\theta),$$

with v, w continuous in $[0,R]$ and $[0,2]$, respectively, and w 2π-periodic. Substitution into (3.24) gives

$$v''(r)w(\theta) + \frac{1}{r}v'(r)w(\theta) + \frac{1}{r^2}v(r)w''(\theta) = 0$$

or, separating the variables,

$$-\frac{r^2v''(r) + rv'(r)}{v(r)} = \frac{w''(\theta)}{w(\theta)}.$$

This identity is possible only when the two quotients have a common constant value λ. Thus we are lead to the ordinary differential equation

$$r^2v''(r) + rv'(r) + \lambda v(r) = 0 \tag{3.25}$$

and to the eigenvalue problem

$$\begin{cases} w''(\theta) - \lambda w(\theta) = 0 \\ w(0) = w(2\pi). \end{cases} \tag{3.26}$$

We leave to the reader to check that problem (3.26) has only the zero solution for $\lambda > 0$. If

$$\lambda = -\mu^2, \quad \mu \geq 0,$$

the differential equation in (3.26) has the general integral

$$w(\theta) = \begin{cases} a + b\theta & \mu = 0 \\ a\cos\mu\theta + b\sin\mu\theta & \mu > 0, \end{cases} \qquad (a, b \in \mathbb{R}).$$

The 2π-periodicity forces $\mu = m$, a nonnegative integer (and $b = 0$ in case $\mu = 0$).

The equation (3.25), with $\lambda = -m^2$, has the general solution[8]

$$v(r) = \begin{cases} d_1\log r + d_2 & m = 0 \\ d_1r^{-m} + d_2r^m & m \geq 1, \end{cases} \qquad (d_1, d_2 \in \mathbb{R}).$$

Since v has to be bounded in $[0,R]$, we exclude $\log r$ and r^{-m}, $m > 0$, and hence $d_1 = 0$.

We have found a countable number of 2π-periodic harmonic functions

$$r^m\{a_m\cos m\theta + b_m\sin m\theta\}, \qquad m = 0, 1, 2, \ldots. \tag{3.27}$$

We now superpose the functions in (3.27) by writing

$$U(r,\theta) = a_0 + \sum_{m=1}^{\infty} r^m\{a_m\cos m\theta + b_m\sin m\theta\}, \tag{3.28}$$

where the coefficients a_m and b_m are still to be chosen in order to satisfy the boundary condition

$$\lim_{(r,\theta)\to(R,\xi)} U(r,\theta) = G(\xi), \qquad \forall\,\xi \in [0, 2\pi]. \tag{3.29}$$

- *Case* $G \in C^1([0,2\pi])$. In this case, G can be expanded in the Fourier series

[8] The (3.25) is an Euler equation. The change of variables $s = \log r$, $z(s) = u(e^s)$, reduces it to the equation $z''(s) - m^2z(s) = 0$.

$$G(\xi) = \frac{\alpha_0}{2} + \sum_{m=1}^{\infty} \{\alpha_m \cos m\xi + \beta_m \sin m\xi\}, \tag{3.30}$$

where

$$\alpha_m = \frac{1}{\pi}\int_0^{2\pi} G(\varphi)\cos m\varphi \, d\varphi, \qquad \beta_m = \frac{1}{\pi}\int_0^{2\pi} G(\varphi)\sin m\varphi \, d\varphi. \tag{3.31}$$

Since G is $C^1([0, 2\pi])$, we have (see Sect. 1.5)

$$\sum_{m=1}^{\infty} |\alpha_m| < \infty, \qquad \sum_{m=1}^{\infty} |\beta_m| < \infty$$

so that, in particular, the series (3.30) is uniformly convergent in $[0, 2\pi]$. Then, the boundary condition (3.29) is satisfied if we choose

$$a_0 = \frac{\alpha_0}{2}, \quad a_m = R^{-m}\alpha_m, \quad b_m = R^{-m}\beta_m.$$

Inserting these values of a_m, b_m into (3.28), we find, for $0 \le r < R$, $0 \le \theta \le 2\pi$,

$$\begin{aligned} U(r,\theta) &= \frac{\alpha_0}{2} + \frac{1}{\pi}\sum_{m=1}^{\infty}\left(\frac{r}{R}\right)^m \int_0^{2\pi} G(\varphi)\{\cos m\varphi \cos m\theta + \sin m\varphi \sin m\theta\}\, d\varphi \\ &= \frac{1}{\pi}\int_0^{2\pi} G(\varphi)\left[\frac{1}{2} + \sum_{m=1}^{\infty}\left(\frac{r}{R}\right)^m \cos m(\varphi-\theta)\right] d\varphi \\ &= \frac{1}{\pi}\int_0^{2\pi} G(\varphi)\left[-\frac{1}{2} + \sum_{m=0}^{\infty}\left(\frac{r}{R}\right)^m \cos m(\varphi-\theta)\right] d\varphi. \end{aligned}$$

Note that in the second equality above, the exchange of sum and integration is possible because of the uniform convergence of the series, since $r < R$. Moreover, for $0 \le r < R$, we can differentiate under the integral sign and then term by term, as many times as we want. Therefore, since for every $m \ge 1$ the functions

$$\left(\frac{r}{R}\right)^m \cos m(\varphi - \theta)$$

are smooth and harmonic, also $U \in C^\infty([0, R) \times [0, 2\pi])$ and is harmonic for $r < R$. To obtain a better formula, observe that

$$\sum_{m=0}^{\infty}\left(\frac{r}{R}\right)^m \cos m(\varphi-\theta) = \mathrm{Re}\left[\sum_{m=0}^{\infty}\left(e^{i(\varphi-\theta)}\frac{r}{R}\right)^m\right].$$

Since

$$\mathrm{Re}\sum_{m=0}^{\infty}\left(e^{i(\varphi-\theta)}\frac{r}{R}\right)^m = \mathrm{Re}\,\frac{1}{1 - e^{i(\varphi-\theta)}\frac{r}{R}} = \frac{R^2 - rR\cos(\varphi-\theta)}{R^2 + r^2 - 2rR\cos(\varphi-\theta)}$$

we find

$$-\frac{1}{2} + \sum_{m=0}^{\infty}\left(\frac{r}{R}\right)^m \cos m(\varphi-\theta) = \frac{1}{2}\,\frac{R^2 - r^2}{R^2 + r^2 - 2rR\cos(\varphi-\theta)}. \tag{3.32}$$

Inserting (3.32) into the formula for U, we get Poisson's formula in polar coordinates:

$$U(r,\theta) = \frac{R^2 - r^2}{2\pi}\int_0^{2\pi} \frac{G(\varphi)}{R^2 + r^2 - 2Rr\cos(\theta-\varphi)}\, d\varphi. \tag{3.33}$$

Going back to Cartesian coordinates[9], we obtain Poisson's formula (3.23). Corollary 3.8, p. 124, assures that (3.33) is indeed the unique solution of the Dirichlet problem (3.22).

• *Case* $G \in C([0, 2\pi])$. Even with G only continuous, formula (3.33) makes perfect sense and defines a harmonic function in B_R. It can be shown that (see Problem 3.22)

$$\lim_{(r,\theta)\to(R,\xi)} U(r,\theta) = G(\xi), \qquad \forall \xi \in [0, 2\pi].$$

Therefore (3.33) is a solution to (3.22); the uniqueness follows once more by Corollary 3.8. □

• *Poisson's formula in dimension* $n > 2$. Theorem 3.13 has an appropriate extension in any number of dimensions. When $B_R = B_R(\mathbf{p})$ is an n-dimensional ball, the solution of the Dirichlet problem (3.22) is given by (see Sect. 3.7.3 for $n = 3$)

$$u(\mathbf{x}) = \frac{R^2 - |\mathbf{x}-\mathbf{p}|^2}{\omega_n R} \int_{\partial B_R(\mathbf{p})} \frac{g(\sigma)}{|\mathbf{x}-\sigma|^n} d\sigma. \tag{3.34}$$

We are now in position to prove Theorem 3.5, p. 122, the converse of the mean value property (*m.v.p.*).

Proof (of Theorem 3.5) First we prove the equivalence of (3.14) and (3.15). The fact that (3.15) implies (3.14) follows as in the end of the proof of Theorem 3.4. To prove the converse, we first show that if u satisfies (3.14) for every $B_r(\mathbf{x}) \subset\subset \Omega$ then u is *continuous*. In fact, let $|u| \le M$ in Ω, $\mathbf{x}, \mathbf{y} \in \Omega$ and let $\overline{B_r}(\mathbf{x})$, $\overline{B_r}(\mathbf{y})$ be contained in Ω. Then by (3.14) we have

$$\begin{aligned} |u(\mathbf{x}) - u(\mathbf{y})| &= \frac{n}{\omega_n r^n} \left| \int_{B_r(\mathbf{x})} u(\mathbf{y}) d\mathbf{y} - \int_{B_r(\mathbf{y})} u(\mathbf{z}) d\mathbf{z} \right| \\ &\le \frac{n}{\omega_n r^n} \int_{B_r(\mathbf{x}) \Delta B_r(\mathbf{y})} |u(\mathbf{z})|\, d\mathbf{z} \le \frac{nM}{\omega_n r^n} |B_r(\mathbf{x}) \Delta B_r(\mathbf{y})|. \end{aligned}$$

where $B_r(\mathbf{x})\Delta B_r(\mathbf{y})$ denotes the *symmetric difference*[10] of $B_r(\mathbf{x})$ and $B_r(\mathbf{y})$. The continuity of u follows, as $|B_r(\mathbf{x})\Delta B_r(\mathbf{y})|/r^n = |B_1(\mathbf{x})\Delta B_1(\mathbf{y})| \to 0$ as $|\mathbf{x}-\mathbf{y}| \to 0$. Now, write (3.14) as

$$\frac{r^n}{n} u(\mathbf{x}) = \frac{1}{\omega_n} \int_{B_r(\mathbf{x})} u(\mathbf{y}) d\mathbf{y} = \frac{1}{\omega_n} \int_0^r ds \int_{\partial B_s(\mathbf{x})} u(\sigma) d\sigma;$$

since u is continuous, we can differentiate both sides with respect to r, obtaining (3.15).

Now consider a ball $B \subset\subset \Omega$. We want to show that u is harmonic and infinitely differentiable in Ω. Denote by v the solution of the Dirichlet problem

$$\begin{cases} \Delta v = 0 & \text{in } B \\ v = u & \text{on } \partial B. \end{cases}$$

We know that

$$v \in C^\infty(B) \cap C(\overline{B})$$

---

[9] With $\sigma = \mathbf{p} + R(\cos\varphi, \sin\varphi)$, $d\sigma = R d\varphi$ and

$$\begin{aligned} |\mathbf{x}-\sigma|^2 &= (r\cos\theta - R\cos\varphi)^2 + (r\sin\theta - R\sin\varphi)^2 \\ &= R^2 + r^2 - 2Rr(\cos\varphi\cos\theta + \sin\varphi\sin\theta) = R^2 + r^2 - 2Rr\cos(\theta-\varphi). \end{aligned}$$

[10] The symmetric difference of two sets is defined by $E\Delta F = (E \setminus F) \cup (F \setminus E)$.

and, being harmonic, it satisfies the m.v.p. in B. Then, also $w = v - u$ satisfies the m.v.p. in B and therefore by Theorem 3.7 it attains its maximum and minimum on ∂B. Since $w = 0$ on ∂B, we conclude that $u = v$ in B. Since B is arbitrary, $u \in C^\infty(\Omega)$ and it is harmonic in Ω. □

3.3.6 Harnack's inequality and Liouville's theorem

From the mean value and Poisson's formulas we deduce another maximum principle, known as Harnack's inequality:

Theorem 3.14 *Let u be harmonic and nonnegative in a domain $\Omega \subset \mathbb{R}^n$. Assume that $B_R(\mathbf{z}) \subset\subset \Omega$. Then, for any $\mathbf{x} \in \overline{B}_R(\mathbf{z})$,*

$$\frac{R^{n-2}(R-r)}{(R+r)^{n-1}}u(\mathbf{z}) \le u(\mathbf{x}) \le \frac{R^{n-2}(R+r)}{(R-r)^{n-1}}u(\mathbf{z}), \tag{3.35}$$

where $r = |\mathbf{x} - \mathbf{z}|$. As a consequence, for every compact set $K \subset \Omega$, there exists a constant C, depending only on n, K and the distance of K from $\partial\Omega$, such that

$$\max_K u \le C \min_K u. \tag{3.36}$$

Proof We can assume that $z = 0$. From Poisson's formula (3.34):

$$u(\mathbf{x}) = \frac{R^2 - |\mathbf{x}|^2}{\omega_n R}\int_{\partial B_R} \frac{u(\boldsymbol{\sigma})}{|\boldsymbol{\sigma} - \mathbf{x}|^n} d\sigma.$$

Observe that $R - |\mathbf{x}| \le |\boldsymbol{\sigma} - \mathbf{x}| \le R + |\mathbf{x}|$ and $R^2 - |\mathbf{x}|^2 = (R - |\mathbf{x}|)(R + |\mathbf{x}|)$. Then, by the mean value property,

$$\begin{aligned} u(\mathbf{x}) &\le \frac{R+|\mathbf{x}|}{(R-|\mathbf{x}|)^{n-1}}\frac{1}{\omega_n R}\int_{\partial B_R} u(\boldsymbol{\sigma})d\sigma \\ &= \frac{R^{n-2}(R+|\mathbf{x}|)}{(R-|\mathbf{x}|)^{n-1}}\frac{1}{\omega_n R^{n-1}}\int_{\partial B_R} u(\boldsymbol{\sigma})d\sigma = \frac{R^{n-2}(R+|\mathbf{x}|)}{(R-|\mathbf{x}|)^{n-1}}u(\mathbf{0}). \end{aligned}$$

Analogously,

$$u(\mathbf{x}) \ge \frac{R^{n-2}(R-|\mathbf{x}|)}{(R+|\mathbf{x}|)^{n-1}}\frac{1}{\omega_n R^{n-1}}\int_{\partial B_R} u(\boldsymbol{\sigma})d\sigma = \frac{R^{n-2}(R-|\mathbf{x}|)}{(R+|\mathbf{x}|)^{n-1}}u(\mathbf{0}).$$

To show (3.36), let $d = \operatorname{dist}(K, \partial\Omega)$ and z_m, z_M be a minimum and a maximum point for u in K, respectively. Then, we can construct a sequence of balls $B_{d/3}(\mathbf{x}_j) \subset\subset \Omega$, $j = 0, \dots, N$, where N depends only on n, K and d, such that (Fig. 3.2, p. 124):

- $x_j \in B(\mathbf{x}_{j-1})$, $j = 1, \dots, N$.
- $x_0 = z_m$, $x_N = z_M$.

A repeated use of inequality (3.35) in each of the balls gives (3.36). We leave the details to the reader. □

Harnack's inequality has important consequences. We present two of them. The first one says that the only harmonic functions in $\mathbb{R}^n$ bounded from below or above are the constant functions.

Corollary 3.15 (Liouville's Theorem) *If u is harmonic in $\mathbb{R}^n$ and $u(\mathbf{x}) \geq M$, then u is constant.*

Proof The function $w = u - M$ is harmonic in $\mathbb{R}^n$ and nonnegative. Fix $\mathbf{x} \in \mathbb{R}^n$ and choose $R > |\mathbf{x}|$; Harnack's inequality gives

$$\frac{R^{n-2}(R-|\mathbf{x}|)}{(R+|\mathbf{x}|)^{n-1}} w(\mathbf{0}) \leq w(\mathbf{x}) \leq \frac{R^{n-2}(R+|\mathbf{x}|)}{(R-|\mathbf{x}|)^{n-1}} w(\mathbf{0}). \tag{3.37}$$

Letting $R \to \infty$ in (3.37) we get $w(\mathbf{0}) \leq w(\mathbf{x}) \leq w(\mathbf{0})$, whence $w(\mathbf{0}) = w(\mathbf{x})$. Since $\mathbf{x}$ is arbitrary we conclude that w, and therefore also u, is constant. □

The second consequence concerns monotone sequences of harmonic functions.

Corollary 3.16 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and $\{u_k\}_{k\geq 1}$ be a sequence of harmonic functions in Ω. Assume that:*

i) for every $k \geq 1$, $u_k \leq u_{k+1}$ in Ω;
ii) $u_k(\mathbf{x})$ converges at some point $\mathbf{x}_0 \in \Omega$.

Then $\{u_k\}_{k\geq 1}$ converges uniformly in every compact set $K \subset \Omega$ to a harmonic function u.

Proof Let $u_k - u_j$, $k > j$. Then $u_k - u_j$ is harmonic and nonnegative in Ω. Let $K \subset \Omega$ be a compact set. We may always assume that $\mathbf{x}_0 \in K$. By (3.36), we can write

$$u_k(\mathbf{x}) - u_j(\mathbf{x}) \leq C(u_k(\mathbf{x}_0) - u_j(\mathbf{x}_0)), \qquad \forall \mathbf{x} \in K.$$

Hence $\{u_k\}$ converges uniformly in K to a function u, which is harmonic, by Theorem 3.6, p. 123.□

3.3.7 Analyticity of harmonic functions

Another important consequence of Poisson's formula is the possibility to control the derivatives of any order of a harmonic function u at a point $\mathbf{x}$, by the maximum of u in a small ball centered at $\mathbf{x}$. Let u be a harmonic function in a domain Ω and $B_r(\mathbf{x}) \subset\subset \Omega$. Since $u \in C^\infty(\Omega)$, we can differentiate the equation $\Delta u = 0$ any number of times and conclude that any derivative of u is still a harmonic function in Ω.

Let us start with an estimate of a first order derivative u_{x_j}. By the Mean Value Theorem and then Gauss formula, we can write

$$u_{x_j}(\mathbf{x}) = \frac{n}{\omega_n r^n} \int_{B_r(\mathbf{x})} u_{x_j}(\mathbf{y}) dy = \frac{n}{\omega_n r^n} \int_{\partial B_r(\mathbf{x})} u(\boldsymbol{\sigma}) \nu_j d\sigma$$

where $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_n) = \frac{\boldsymbol{\sigma} - \mathbf{x}}{r}$, is the exterior normal to $B_r(\mathbf{x})$. Hence

$$\left|u_{x_j}(\mathbf{x})\right| \leq \frac{n}{r} \max_{\partial B_r(\mathbf{x})} |u|. \tag{3.38}$$

To deal with derivatives of any order, it is convenient to introduce a flexible notation. We call *multi-index* an n-tuple $\alpha = (\alpha_1, \dots, \alpha_n)$, where each α_j is a nonnegative integer. Define length and factorial of α by

$$|\alpha| = \alpha_1 + \cdots + \alpha_n \quad \text{and} \quad \alpha! = \alpha_1! \alpha_2! \cdots \alpha_n!$$

respectively. Finally, we set

$$\mathbf{x}^\alpha = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \quad \text{and} \quad D^\alpha = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \frac{\partial^{\alpha_2}}{\partial x_2^{\alpha_2}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}.$$

If $|\alpha| = 0$, we set $D^\alpha u = u$.

Lemma 3.17 *Let u be harmonic in $\Omega \subseteq \mathbb{R}^n$. Then for any ball $B_r(\mathbf{x}) \subset\subset \Omega$ and any multi-index α,*

$$|D^\alpha u(\mathbf{x})| \le \frac{n^k k! e^{k-1}}{r^k} \max_{\partial B_r(\mathbf{x})} |u|, \qquad k = |\alpha|. \tag{3.39}$$

Proof Let $M = \max_{\partial B_r(\mathbf{x})} |u|$. We proceed by induction on k. The step $k = 1$ is provided by (3.38). Assume now that (3.39) holds for $|\alpha| = k$ and any ball $B_\rho(\mathbf{x}) \subset\subset \Omega$. Let $D^\beta u = \partial_{x_j} D^\alpha u$ be a derivative of order $k+1$. Applying step $k = 1$, we write

$$\left|D^\beta u(\mathbf{x})\right| \le \frac{n}{\rho} \max_{\partial B_\rho(\mathbf{x})} |D^\alpha u|, \qquad k = |\alpha|, \tag{3.40}$$

as long as $B_\rho(\mathbf{x}) \subset\subset \Omega$. On the other hand, the induction hypothesis gives, for any point $\mathbf{y} \in \partial B_\rho(\mathbf{x})$,

$$|D^\alpha u(\mathbf{y})| \le \frac{n^k k! e^{k-1}}{\delta^k} \max_{\partial B_\delta(\mathbf{y})} |u|, \qquad k = |\alpha|, \tag{3.41}$$

as long as $\partial B_\delta(\mathbf{y}) \subset\subset \Omega$. To get an estimate in terms of M, we have to choose ρ and δ such that $\partial B_\delta(\mathbf{y}) \subset \overline{B_r}(\mathbf{x})$ for any $\mathbf{y} \in \partial B_\delta(\mathbf{x})$, that is $\delta + \rho \le r$. Choose

$$\rho = \frac{r}{k+1}, \ \delta = r - \frac{r}{k+1} = \frac{kr}{k+1}.$$

Inserting into (3.40) and (3.41) we get, since $\max_{\partial B_\delta(\mathbf{y})} |u| \le M$ and $\left(1 + \frac{1}{k}\right)^k < e$,

$$\begin{aligned} \left|D^\beta u(\mathbf{x})\right| &\le \frac{(k+1)n}{r} \frac{n^k k! e^{k-1}}{r^k \left(\frac{k}{k+1}\right)^k} M \\ &= \frac{n^{k+1}(k+1)! e^{k-1} \left(1 + \frac{1}{k}\right)^k}{r^{k+1}} M < \frac{n^{k+1}(k+1)! e^k}{r^{k+1}} M \end{aligned}$$

which is (3.39) for $k+1$. □

We have seen that if u is harmonic in a domain Ω, then $u \in C^\infty(\Omega)$. On the other hand, in dimension $n = 2$, we know that u is analytic, being the real part of a holomorphic function. Thanks to the estimates (3.39), the same property holds in any dimension. We have:

Theorem 3.18 *Let u be harmonic in Ω. Then u is (real) analytic, that is, for any $\mathbf{x} \in \Omega$ there exists $\rho = \rho(n, dist(\mathbf{x}, \partial\Omega))$ such that*

$$u(\mathbf{z}) = \sum_{k=0}^{\infty} \frac{D^\alpha u(\mathbf{x})}{\alpha!} (\mathbf{z} - \mathbf{x})^\alpha$$

in $\overline{B_\rho}(\mathbf{x})$.

Proof Let $\mathbf{z} - \mathbf{x} = \rho\mathbf{h}$, where $\mathbf{h}$ is a unit vector. Consider the Taylor series for u at $\mathbf{x}$, given by

$$\sum_{|\alpha|=0}^{\infty} \frac{D^\alpha u(\mathbf{x})}{\alpha!} (\mathbf{z} - \mathbf{x})^\alpha = \sum_{k=0}^{\infty} \rho^k \sum_{|\alpha|=k} \frac{D^\alpha u(\mathbf{x})}{\alpha!} \mathbf{h}^\alpha. \tag{3.42}$$

From (3.39) we have, if $|\alpha| = k$,

$$|D^\alpha u(\mathbf{x})| \le k! \left(\frac{ne}{r}\right)^k \max_{\partial B_r(\mathbf{x})} |u|$$

as long as $r < \text{dist}(\mathbf{x}, \partial\Omega)$. Thus, we deduce ($|\mathbf{h}| = 1$) that

$$\rho^k \sum_{|\alpha|=k} \frac{|D^\alpha u(\mathbf{x})|}{\alpha!} \le \max_{\partial B_r(\mathbf{x})} |u| \sum_{|\alpha|=k} \frac{k!}{\alpha!} \left(\frac{ne}{r}\rho\right)^k.$$

Choosing, say, $\rho = r/(2n^2 e)$ and recalling the formula[11]

$$n^k = \sum_{|\alpha|=k} \frac{k!}{\alpha!},$$

we get

$$\sum_{|\alpha|=k} \frac{k!}{\alpha!} \left(\frac{ne}{r}\rho\right)^k = \sum_{|\alpha|=k} \frac{k!}{\alpha!} \frac{1}{(2n)^k} = \frac{1}{2^k}.$$

By the Weierstrass test, the series (3.42) converges uniformly in $\overline{B_\rho}(\mathbf{x})$ and therefore

$$u(\mathbf{x}) = \sum_{|\alpha|=0}^{\infty} \frac{D^\alpha u(\mathbf{x})}{\alpha!} (\mathbf{z} - \mathbf{x})^\alpha \qquad \text{in } \overline{B_\rho}(\mathbf{x}). \qquad \square$$

3.4 Sub/Superharmonic Functions. The Perron Method

3.4.1 Sub/superharmonic functions

In this section we consider the Dirichlet problem

[11] *Multinomial formula*: $(a_1 + \cdots + a_n)^k = \sum_{|\alpha|=k} \frac{k!}{\alpha_1! \ldots \alpha_n!} a_1^{\alpha_1} \ldots a_n^{\alpha_n}$.

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases} \tag{3.43}$$

where $\Omega \subset \mathbb{R}^n$ is a bounded domain and $g \in C(\partial\Omega)$. We say that the *Dirichlet problem* (3.43) *is solvable if, for every* $g \in C(\partial\Omega)$, *there exist a unique solution* $u \in C^\infty(\Omega) \cap C(\overline{\Omega})$.

If Ω is a ball, Poisson's formula (3.34) provides the unique solution of (3.43). In general, while the uniqueness follows from Corollary 3.8, the existence of the solution it not insured, unless $\partial\Omega$ satisfies some particular smoothness conditions (see Sect. 3.4.3).

To prove the existence, a technique due to O. Perron (back in the 20-ties of the last century) consists in taking the supremum u_g of all subharmonic functions smaller than g on $\partial\Omega$, to show that u_g is harmonic and then to examine its boundary behavior. One of the peculiarity of Perron's method lies in its flexibility, and, indeed, it works with much more general operators than the Laplacian and, in particular, with fully nonlinear elliptic operators, of great importance in several applied areas.

To implement the method, we need to introduce the class of sub/super harmonic functions and establish some of their properties. For a function $u \in C(\Omega)$, $\Omega \subseteq \mathbb{R}^n$, denote its volume average over a ball $B_r(\mathbf{x}) \subset\subset \Omega$ by

$$A(u; \mathbf{x}, r) = \frac{n}{\omega_n r^n} \int_{B_r(\mathbf{x})} u(\mathbf{y}) d\mathbf{y}$$

and its spherical average by

$$S(u; \mathbf{x}, r) = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r(\mathbf{x})} u(\boldsymbol{\sigma}) d\sigma.$$

Definition 3.19 A function $u \in C(\Omega)$ is *subharmonic* in Ω if

$$u(\mathbf{x}) \le S(u; \mathbf{x}, r) \tag{3.44}$$

for every $\mathbf{x} \in \Omega$ and every $B_r(\mathbf{x}) \subset\subset \Omega$.

In (3.44), it is equivalent to use volume averages (see Problem 3.8 a)). Superharmonic functions are defined reversing the inequality in (3.44).

Typical subharmonic functions are convex cones or paraboloids (e.g. $u(\mathbf{x}) = |\mathbf{x} - \mathbf{p}|^\alpha$, $\alpha \ge 1$). If $u \in C^2(\Omega)$, then u is subharmonic in Ω if and only if $\Delta u \ge 0$ (see Problem 3.8 d)). The properties we shall need are the following ones.

1. *Maximum principle*. If $u \in C(\overline{\Omega})$ is subharmonic (superharmonic) and nonconstant, then

$$u(\mathbf{x}) < \max_{\partial\Omega} u, \quad (u(\mathbf{x}) > \min_{\partial\Omega} u) \qquad \text{for every } \mathbf{x} \in \Omega.$$

2. If $u_1, u_2, \ldots, u_N$ are subharmonic (superharmonic) in Ω, then

$$u = \max\{u_1, u_2, \ldots, u_N\} \qquad (u = \min\{u_1, u_2, \ldots, u_N\})$$

is subharmonic (superharmonic) in Ω.

3. Let u be subharmonic (superharmonic) in Ω. For a ball $B \subset\subset \Omega$, let $P(u; B)$ be the harmonic function in B, with $P(u; B) = u$ on ∂B. $P(u; B)$ is called the *harmonic lifting* of u in B. Then, the function

$$u^B = \begin{cases} P(u, B) & \text{in } B \\ u & \text{in } \Omega \setminus B \end{cases} \tag{3.45}$$

is subharmonic (superharmonic) in Ω. Notice that $u \le u^B$ ($u \ge u^B$) in Ω by the maximum principle.

The proof of 1. follows repeating verbatim the proof of Theorem 3.7, p. 123.

To prove 2. let $u_1, u_2, \ldots, u_N$ be subharmonic in Ω and $u = \max\{u_1, u_2, \ldots, u_N\}$. For each $j = 1, \ldots, N$, and every ball $B_r(\mathbf{x}) \subset\subset \Omega$, we have

$$u_j(\mathbf{x}) \le S(u_j; \mathbf{x}, r) \le S(u; \mathbf{x}, r).$$

Thus

$$u(\mathbf{x}) \le S(u; \mathbf{x}, r)$$

and u is subharmonic. The other case is similar.

Finally, we show 3. Let u be subharmonic in Ω. By the maximum principle, $u \le P(u, B)$. Let $B_r(\mathbf{x}) \subset\subset \Omega$. If $B_r(\mathbf{x}) \cap B = \emptyset$ or $B_r(\mathbf{x}) \subset B$ there is nothing to prove. In the other case, let w be the harmonic lifting of u^B in $B_r(\mathbf{x})$. Then $u^B \le w$ in Ω, $\Delta w = 0$ in $B_r(\mathbf{x})$ and $w = u_B$ on $\partial B_r(\mathbf{x})$. Thus

$$u^B(\mathbf{x}) \le w(\mathbf{x}) = S(w; \mathbf{x}, r) = S(u^B; \mathbf{x}, r)$$

and therefore u^B is subharmonic. An analogous argument works if u is superharmonic.

3.4.2 The method

We now go back to the Dirichlet problem. As we mentioned, the idea is to construct the solution by looking at the class of subharmonic functions in Ω, smaller than g on the boundary, and to take their supremum. In dimension $n = 1$, the solution of the Dirichlet problem is a line segment l through two fixed endpoints, while the subharmonic functions are convex parabolas. Therefore, Perron's method amounts to construct the segment l by taking the supremum of all convex parabolas below l. Thus, let

$$S_g = \left\{ v \in C(\overline{\Omega}) : v \text{ subharmonic in } \Omega,\ v \le g \text{ on } \partial\Omega \right\}.$$

Our candidate solution is:

$$u_g(\mathbf{x}) = \sup\{v(\mathbf{x}) : v \in S_g\}, \quad \mathbf{x} \in \overline{\Omega}.$$

Note that:

- S_g is nonempty, since
$$v(\mathbf{x}) \equiv \min_{\partial\Omega} g \in S_g.$$
- The function u_g is well defined since, by maximum principle, $u_g \leq \max_{\partial\Omega} g$.

We have to check two things: first that u_g is harmonic in Ω, second that u_g assumes continuously the boundary data. Let us show that u_g is harmonic.

Theorem 3.20 *The function u_g is harmonic in Ω.*

Proof Given $\mathbf{x}_0 \in \Omega$, by the definition of u_g, there exists $\{u_k\}_{k\geq 1} \subset S_g$ such that $u_k(\mathbf{x}_0) \to u_g(\mathbf{x}_0)$ as $k \to \infty$. The functions

$$w_k = \max\{u_1, u_2, \ldots, u_k\}, \, k \geq 1,$$

belong to S_g (by property 2 above) and $w_k \leq w_{k+1}$. Moreover, since $u_k(\mathbf{x}_0) \leq w_k(\mathbf{x}_0) \leq u_g(\mathbf{x}_0)$, we infer that

$$\lim_{k\to+\infty} w_k(\mathbf{x}_0) = u_g(\mathbf{x}_0).$$

Let B be a ball such that $\mathbf{x}_0 \in B$ and $\overline{B} \subset \Omega$. Let w_k^B as in (3.45) with w_k instead of u. Each w_k^B belongs to S_g, too (by property 3 above). For each $k \geq 1$, we have $w_k \leq w_k^B \leq w_{k+1}^B$ and hence,

$$\lim_{k\to+\infty} w_k^B(\mathbf{x}_0) = u_g(\mathbf{x}_0).$$

By Corollary 3.16, p. 132,

$$\lim_{k\to+\infty} w_k^B(\mathbf{x}) = w(\mathbf{x})$$

uniformly in $\overline{B}$, where w is harmonic in B, and clearly $w(\mathbf{x}_0) = u_g(\mathbf{x}_0)$.

We show that $w = u_g$ in B. By the definition of u_g we have $w \leq u_g$. Suppose that there exists $\mathbf{x}_1 \in B$ such that

$$w(\mathbf{x}_1) < u_g(\mathbf{x}_1).$$

Let $\{v_k\} \subset S_g$ such that $v_k(\mathbf{x}_1) \to u_g(\mathbf{x}_1)$. Define, for $k \geq 1$,

$$z_k = \max\{v_1, v_2, \ldots, v_k, w_k\}.$$

Then z_k^B belongs to S_g and $w_k \leq z_k^B \leq u_g$, $v_k \leq z_k^B \leq u_g$ in $\overline{\Omega}$. Thus

$$\lim_{k\to+\infty} z_k^B(\mathbf{x}_0) = u_g(\mathbf{x}_0) \qquad \text{and} \qquad \lim_{k\to+\infty} z_k^B(\mathbf{x}_1) = u_g(\mathbf{x}_1).$$

By Corollary 3.16, the sequence $\left\{z_k^B\right\}$ converges in B to a harmonic function z with $z(\mathbf{x}_1) = u_g(\mathbf{x}_1)$. By construction $w \leq z$ in B and $w(\mathbf{x}_0) = z(\mathbf{x}_0) = u_g(\mathbf{x}_0)$.

Then the function $z - w$ is harmonic, nonnegative with an interior minimum equal to zero at $\mathbf{x}_0$. The maximum principle yields $z - w \equiv 0$ in B that leads to the contradiction

$$w(\mathbf{x}_1) = z(\mathbf{x}_1) = u_g(\mathbf{x}_1) > w(\mathbf{x}_1).$$

Thus, $u_g = w$ in B and u_g is harmonic in B. Since $\mathbf{x}_0$ is arbitrary, we conclude that u_g is harmonic in Ω. □

3.4.3 Boundary behavior

Through Theorem 3.20, we can associate to each $g \in C(\partial\Omega)$ the harmonic function u_g. To prove the solvability of problem (3.43) we have to check if

$$u_g(\mathbf{x}) \to g(\mathbf{p})$$

as $\mathbf{x} \to \mathbf{p}$, for every $\mathbf{p} \in \partial\Omega$. This is not always true, as we shall see later on, and it depends on a particular smoothness condition on $\partial\Omega$. In order to control the behavior of u at a boundary point $\mathbf{p}$, we introduce the key notion of *barrier*.

Definition 3.21 Let $\mathbf{p} \in \partial\Omega$. We say that $h \in C(\overline{\Omega})$ is a *barrier* in Ω at $\mathbf{p}$ if:

i) h is superharmonic in Ω.
ii) $h > 0$ in $\overline{\Omega} \setminus \{\mathbf{p}\}$ and $h(\mathbf{p}) = 0$.

Remark 3.22 The notion of barrier has a local nature. Indeed if there exists a *local* barrier h_{loc} in $B_r(\mathbf{p}) \cap \Omega$ at $\mathbf{p}$, for some ball $B_r(\mathbf{p})$, it is easy to construct a barrier at $\mathbf{p}$ in Ω. Indeed, set $C_r = \Omega \cap (B_r(\mathbf{p}) \setminus B_{r/2}(\mathbf{p}))$ and $m = \min_{\overline{C}_r} h_{\text{loc}}$. Then

$$h(\mathbf{x}) = \begin{cases} \min\{m, h_{\text{loc}}(\mathbf{x})\} & \text{in } \overline{B}_r(\mathbf{p}) \cap \overline{\Omega} \\ m & \text{in } \overline{\Omega} \setminus B_r(\mathbf{p}) \end{cases}$$

is a barrier in Ω at $\mathbf{p}$ (see Problem 3.9)

Definition 3.23 We say that $\mathbf{p} \in \partial\Omega$ is a *regular point* if there exists a barrier at $\mathbf{p}$. If every $\mathbf{p} \in \partial\Omega$ is regular, we say that Ω is *regular* (for the Laplace operator).

The following theorem clarifies the role of a barrier.

Theorem 3.24 *Let $\mathbf{p} \in \partial\Omega$ be a regular point. Then, for every $g \in C(\partial\Omega)$, $u_g(\mathbf{x}) \to g(\mathbf{p})$ as $\mathbf{x} \to \mathbf{p}$.*

Proof Assume that $\mathbf{p} \in \partial\Omega$ is a regular point and that h is a barrier at $\mathbf{p}$. Let $g \in C(\partial\Omega)$. To show that $u_g(\mathbf{x}) \to g(\mathbf{p})$ as $\mathbf{x} \to \mathbf{p}$ it is enough to prove that, for every $\varepsilon > 0$, there exists $k_\varepsilon > 0$ such that

$$g(\mathbf{p}) - \varepsilon - k_\varepsilon h(\mathbf{x}) \le u_g(\mathbf{x}) \le g(\mathbf{p}) + \varepsilon + k_\varepsilon h(\mathbf{x}), \qquad \forall \mathbf{x} \in \overline{\Omega},$$

since $h(\mathbf{x}) \to 0$ as $\mathbf{x} \to \mathbf{p}$. Put

$$z(\mathbf{x}) = g(\mathbf{p}) + \varepsilon + k_\varepsilon h(\mathbf{x}).$$

Then $z \in C(\overline{\Omega})$ and it is superharmonic in Ω. We want to choose $k_\varepsilon > 0$ so that $z \ge g$ on $\partial\Omega$.

By the continuity of g, there exists δ_ε such that, if $\mathbf{y} \in \partial\Omega$ and $|\mathbf{y} - \mathbf{p}| \le \delta_\varepsilon$, then $|g(\mathbf{y}) - g(\mathbf{p})| \le \varepsilon$. For these points $\mathbf{y}$, we have

$$z(\mathbf{y}) = g(\mathbf{p}) - g(\mathbf{y}) + \varepsilon + k_\varepsilon h(\mathbf{y}) + g(\mathbf{y}) \ge g(\mathbf{y})$$

since $g(\mathbf{p}) - g(\mathbf{y}) + \varepsilon \ge 0$ and $h(\mathbf{y}) \ge 0$, from ii) of Definition 3.21. If $\mathbf{y} \in \partial\Omega$ and $|\mathbf{y} - \mathbf{p}| > \delta_\varepsilon$, again from ii) of Definition 3.21, we have $h(\mathbf{y}) \ge m_{\delta_\varepsilon} > 0$. Choosing $k_\varepsilon \ge 2\max|g| / m_{\delta_\varepsilon}$ we get

$$z(\mathbf{y}) = g(\mathbf{p}) + \varepsilon + k_\varepsilon h(\mathbf{y}) \ge g(\mathbf{p}) + \varepsilon + 2\max|g| \ge g(\mathbf{y}).$$

With this choice of k_ε, the maximum principle gives

$$z \geq v, \text{ in } \overline{\Omega}, \quad \forall v \in S_g,$$

from which $z \geq u_g$ in $\overline{\Omega}$. A similar argument gives that $g(\mathbf{p}) - \varepsilon - k_\varepsilon h(\mathbf{x}) \leq u_g(\mathbf{x})$ in $\overline{\Omega}$. □

As a consequence of Theorem 3.24 we have the following corollary.

Corollary 3.25 *The Dirichlet problem (3.43) is solvable if and only if* Ω *is regular.*

Proof The "if" part follows from Theorem 3.24, since the Perron solution u_g gives the unique solution. Assume now that (3.43) is solvable. We must show that there exists a barrier at every $\mathbf{p} \in \partial\Omega$. In fact, let $\tilde{g}(\mathbf{x}) = |\mathbf{x} - \mathbf{p}|^2$. Then $h = u_{\tilde{g}}$ is a barrier at $\mathbf{p}$. Indeed, $h \in C(\overline{\Omega})$ by hypothesis and it is harmonic in Ω by Theorem 3.20.

Moreover, $u_{\tilde{g}}(\mathbf{x}) \geq |\mathbf{x} - \mathbf{p}|^2 > 0$ in $\overline{\Omega} \setminus \{\mathbf{p}\}$, as $\Delta\tilde{g} = 2n > 0$ in $\mathbb{R}^n$, and $h(\mathbf{p}) = \tilde{g}(\mathbf{p}) = 0$. □

We list below some sufficient conditions for a boundary point $\mathbf{p}$ to be regular.

a) Ω *is convex*. At every point $\mathbf{p} \in \partial\Omega$, there exists a so called support hyperplane $\pi(\mathbf{p})$ of equation $(\mathbf{x} - \mathbf{p}) \cdot \boldsymbol{\nu} = 0$, such that Ω is contained in the half space

$$S^+ = \{\mathbf{x} : (\mathbf{x} - \mathbf{p}) \cdot \boldsymbol{\nu} > 0\}$$

and $\pi(\mathbf{p}) \cap \partial\Omega = \{\mathbf{p}\}$. A barrier at $\mathbf{p}$ is given by the affine linear function $h(\mathbf{x}) = (\mathbf{x} - \mathbf{p}) \cdot \boldsymbol{\nu}$.

b) *Exterior sphere condition* (Fig. 3.4). There exists $B_r(\mathbf{x}_0) \subset \mathbb{R}^n \setminus \overline{\Omega}$ such that $\overline{B_r}(\mathbf{x}_0) \cap \partial\Omega = \{\mathbf{p}\}$. Then, a barrier at $\mathbf{p}$ is given by

$$h(\mathbf{x}) = 1 - \frac{r^{n-2}}{|\mathbf{x} - \mathbf{x}_0|^{n-2}} \qquad \text{for } n > 2$$

and

$$h(\mathbf{x}) = \log \frac{|\mathbf{x} - \mathbf{x}_0|}{r} \qquad \text{for } n = 2.$$

c) *Exterior cone condition* (Fig. 3.4). There exists a closed cone $\Gamma_\mathbf{p}$ with vertex at $\mathbf{p}$ and a ball $B_r(\mathbf{p})$ such that $\overline{B_r}(\mathbf{p}) \cap \Gamma_\mathbf{p} \subset \mathbb{R}^n \setminus \Omega$. A barrier at $\mathbf{p}$ in $B_r(\mathbf{p}) \setminus \Gamma_\mathbf{p}$

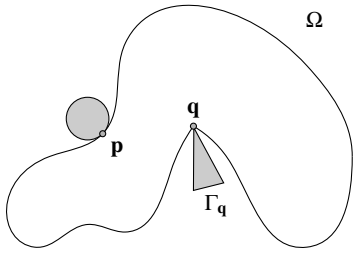
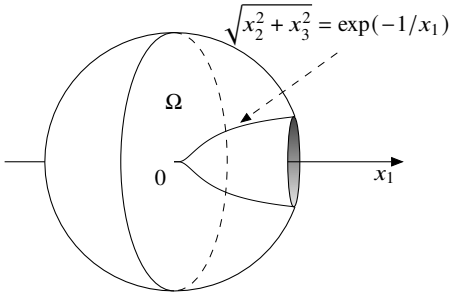


Fig. 3.4 Exterior sphere condition at $\mathbf{p}$ and exterior cone condition at $\mathbf{q}$

Fig. 3.5 The Lebesgue spine



is given by[12] u_g where $g(\mathbf{x}) = |\mathbf{x} - \mathbf{p}|$ and it works also for $B_r(\mathbf{p}) \cap \Omega$, since $B_r(\mathbf{p}) \cap \Omega \subset B_r(\mathbf{p}) \setminus \Gamma_{\mathbf{p}}$. In particular, *a bounded Lipschitz domain is regular*.

Remark 3.26 In the construction of the solution, we could have started with the class

$$S^g = \left\{ v \in C(\overline{\Omega}) : v \text{ superharmonic in } \Omega,\ v \geq g \text{ on } \partial\Omega \right\}$$

and define:

$$u^g(\mathbf{x}) = \inf\{v(\mathbf{x}) : v \in S^g\},\ \mathbf{x} \in \overline{\Omega}.$$

The same proof, with obvious changes, gives that u^g is harmonic in Ω. Clearly $u_g \leq u^g$ and they coincide for every g if and only if Ω is regular for the Dirichlet problem.

A natural question is to ask if nonregular points exist. As we have anticipated, the answer is yes. Here is a celebrated example due to Lebesgue.

Example 3.27 (the Lebesgue spine ($n = 3$)) Let $S_c = \left\{ \rho < e^{-c/x_1},\ x_1 > 0 \right\}$, $0 < c \leq 1$, $\rho^2 = x_2^2 + x_3^2$ and $\Omega = B_1 \setminus \overline{S}_1$ (see Fig. 3.5). Then the origin is an inward exponential cusp and it is not a regular point.

Proof Define

$$w(\mathbf{x}) = \int_0^1 \frac{t}{\sqrt{(x_1 - t)^2 + \rho^2}} dt.$$

We have:

1. w is harmonic[13] in $\mathbb{R}^3 \setminus \{(x_1, 0, 0) : 0 \leq x_1 \leq 1\}$.
2. w is bounded in Ω.

To prove it, consider first $x_1 \leq 0$. Then $t/\sqrt{(x_1 - t)^2 + \rho^2} \leq t/\sqrt{t^2 + x_1^2 + \rho^2} \leq 1$ so that $|w| \leq 1$. If $0 < x_1 < 1$, write

$$w(\mathbf{x}) = \int_0^1 \frac{t - x_1}{\sqrt{(x_1 - t)^2 + \rho^2}} dt + x_1 \int_0^1 \frac{1}{\sqrt{(x_1 - t)^2 + \rho^2}} dt = A(x_1, \rho) + B(x_1, \rho).$$

[12] It is not elementary to show it. This result is known as *Zaremba's Theorem*. We refer to *L. L. Helms*, [8].

[13] We shall see in the next section that w is the Newtonian potential of density t on the segment $l = \{(x_1, 0, 0) : 0 \leq x_1 \leq 1\}$, which is harmonic outside l.

Then, clearly, $|A(x_1,\rho)| \le 1$. Moreover, since $[0,1] \subset [x_1-1, x_1+1]$ and $\rho > e^{-1/x_1}$ in Ω, we have

$$|B(x_1,\rho)| \le x_1 \int_{\rho \le |x_1-t| \le 1} \frac{dt}{|x_1-t|} + \frac{x_1}{\rho} \int_{|x_1-t| \le \rho} dt \le -x_1 2\log\rho + \frac{x_1}{\rho} 2\rho \le 4.$$

3. If $0 < c < 1$, the surface $\partial S_c \cap B_1$ is contained in Ω and we have

$$\lim_{\mathbf{x} \in \partial S_c,\, \mathbf{x} \to 0} w(\mathbf{x}) = 1 + 2c. \tag{3.46}$$

Indeed, the two terms $A(x_1,\rho)$ and $B(x_1,\rho)$ can be explicitly computed. Namely:

$$A(x_1,\rho) = \sqrt{(x_1-1)^2+\rho^2} - \sqrt{x_1^2+\rho^2}$$
$$B(x_1,\rho) = x_1 \log\left[\left(1 - x_1 + \sqrt{(x_1-1)^2+\rho^2}\right)\left(x_1 + \sqrt{x_1^2+\rho^2}\right)\right] - 2x_1\log\rho.$$

Then (3.46) follows easily.

Now, for $c = 1$, the restriction $g = w_{|\partial\Omega}$ is well defined and continuous up to $\mathbf{x} = \mathbf{0}$. Consider u_g. Every point of $\partial\Omega \setminus \{\mathbf{0}\}$ is regular and therefore, if $\mathbf{p} \ne \mathbf{0}$,

$$\lim_{\mathbf{x} \to \mathbf{p}} (w(\mathbf{x}) - u_g(\mathbf{x})) = 0.$$

Being $w - u_g$ bounded, we conclude that $w - u_g \equiv 0$ (see Problem 3.10, b). However, we have seen that along each surface ∂S_c, $0 < c < 1$, w has a different limit as $\mathbf{x} \to \mathbf{0}$. Therefore the limit of u_g as $\mathbf{x} \to \mathbf{0}$ does not exist and, recalling Thm. 3.24, $\mathbf{0}$ can not be regular. □

3.5 A probabilistic solution of the Dirichlet problem

In Sect. 3.3 we solved the discrete Dirichlet problem via a probabilistic method. The key ingredients in the construction of the solution, leading to the representation formula (3.13), were the mean value property and the absence of memory of the random walk (each step is independent of the preceding ones). In the continuous case the appropriate versions of those tools are available, with the Markov property encoding the absence of memory of Brownian motion (see Sect. 2.6). Thus it is reasonable that a continuous counterpart of that technique should provide another way to examine the solvability of the Dirichlet problem (3.43) and to derive a representation formula for the solution. As in Sect. 3.3, we work in dimension $n = 2$, but methods and conclusions can be extended without much effort to any number of dimensions.

Let $\Omega \subset \mathbb{R}^2$ be a bounded domain and $g \in C(\partial\Omega)$. First, we want to derive a representation formula for a candidate solution $u \in C^\infty(\Omega) \cap C(\overline{\Omega})$ of the problem

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{3.47}$$

As in the Perron method in the previous section, the delicate point will be the boundary behavior of u. In the probabilistic framework, this task cannot be treated

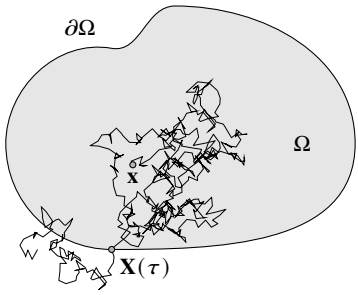


Fig. 3.6 First exit point $\mathbf{X}(\tau)$ from the domain Ω

rigorously in an introductory book. Therefore we will limit ourselves to some heuristic consideration, referring to the specialized literature.

Let $\mathbf{X}(t)$ be the position of a Brownian particle started at $\mathbf{x} \in \Omega$ and define the first exit time from Ω, $\tau = \tau(\mathbf{x})$, as follows (Fig. 3.6):

$$\tau(\mathbf{x}) = \left\{ \inf_{t \geq 0} t : \mathbf{X}(t) \in \mathbb{R}^2 \setminus \Omega \right\}.$$

The time τ is a stopping time: to decide whether the event $\{\tau \leq t\}$ occurs or not, it suffices to observe the process until time t. In fact, for fixed $t \geq 0$, to decide whether or not $\tau \leq t$ is true, it is enough to consider the event

$$E = \{\mathbf{X}(s) \in \Omega, \text{ for all times } s \text{ from 0 until } t, t \text{ included}\}.$$

If this event occurs, then it must be that $\tau > t$. If E does not occur, it means that there are points $X(s)$ outside Ω for some $s \leq t$ and therefore it must be that $\tau \leq t$.

The first thing we have to check is that the particle leaves Ω in finite time, almost surely. Precisely:

Lemma 3.28 *For every* $\mathbf{x} \in \Omega$, $\tau(\mathbf{x})$ *is finite with probability* 1, *that is:*

$$P\{\tau(\mathbf{x}) < \infty\} = 1.$$

Proof It is enough to show that our particle remains inside any circle $B_r = B_r(\mathbf{x})$, with zero probability. If we denote by τ_r the first exit time from B_r, we have to prove that $P\{\tau_r = \infty\} = 0$.

Suppose $\mathbf{X}(t) \in B_r$ until $t = k$ (k integer). Then, for $j = 1, 2, \ldots, k$, it must be that

$$|\mathbf{X}(j) - \mathbf{X}(j-1)| < 2r.$$

Thus (the occurrence of) the event $\{\tau_r > k\}$ implies (the occurrence of) all the events

$$E_j = \{|\mathbf{X}(j) - \mathbf{X}(j-1)| < 2r\}, \qquad j = 1, 2, \ldots, k,$$

and therefore also of their intersection. As a consequence

$$P\{\tau_r > k\} \leq P\left\{\cap_{j=1}^{k} E_j\right\}. \tag{3.48}$$

On the other hand, the increments $\mathbf{X}(j) - \mathbf{X}(j-1)$ are independent and equidistributed according to a standard normal law, hence we can write

$$P\{E_j\} = \frac{1}{2\pi}\int_{\{|\mathbf{z}|<2r\}} \exp\left(-\frac{|\mathbf{z}|^2}{2}\right) d\mathbf{z} \equiv \gamma < 1$$

and

$$P\left\{\cap_{j=1}^{k} E_j\right\} = \prod_{j=1}^{k} P\{E_j\} = \gamma^k. \tag{3.49}$$

Since $\{\tau_r = \infty\}$ implies $\{\tau_r > k\}$, from (3.48) and (3.49) we have

$$P\{\tau_r = \infty\} \le P\{\tau_r > k\} \le \gamma^k.$$

Letting $k \to +\infty$ we get $P\{\tau_r = \infty\} = 0$. □

Lemma 3.28 implies that $\mathbf{X}(t)$ hits the boundary $\partial\Omega$ in finite time $\tau = \tau(\mathbf{x})$, with probability 1. We can therefore introduce on $\partial\Omega$ a probability distribution associated with the random variable $\mathbf{X}(\tau)$ by setting

$$P(\mathbf{x}, \tau, F) = P\{\mathbf{X}(\tau) \in F\} \qquad (\tau = \tau(\mathbf{x})),$$

for every "reasonable" subset $F \subset \partial\Omega$. For fixed $\mathbf{x}$ in Ω, the set function

$$F \longmapsto P(\mathbf{x}, \tau(\mathbf{x}), F)$$

defines a probability measure on $\partial\Omega$, since $P(\mathbf{x}, \tau(\mathbf{x}), \partial\Omega) = 1$, according to Lemma 3.28[14]. $P(\mathbf{x}, \tau(\mathbf{x}), F)$ is called the *escape probability from* Ω, *through* F.

By analogy with formula (3.13), we can now guess the type of formula we expect for the solution u of problem (3.47). To get the value $u(\mathbf{x})$, let a Brownian particle start from $\mathbf{x}$, and let $\mathbf{X}(\tau) \in \partial\Omega$ its first exit point from Ω. Then, compute the random "gain" $g(\mathbf{X}(\tau))$ and take its expected value with respect to the distribution $P(\mathbf{x}, \tau(\mathbf{x}), \cdot)$. This is $u(\mathbf{x})$. Everything works if $\partial\Omega$ is not too bad. Precisely, we have:

Theorem 3.29 *Let Ω be a regular domain in the sense of Definition 3.23 and $g \in C(\partial\Omega)$. The unique solution $u \in C^2(\Omega) \cap C(\overline{\Omega})$ of problem (3.47) is given by*

$$u(\mathbf{x}) = E^{\mathbf{x}}[g(\mathbf{X}(\tau))] = \int_{\partial\Omega} g(\boldsymbol{\sigma}) P(\mathbf{x}, \tau(\mathbf{x}), d\sigma). \tag{3.50}$$

Proof (sketch) For fixed $F \subseteq \partial\Omega$, consider the function

$$u_F : \mathbf{x} \longmapsto P(\mathbf{x}, \tau(\mathbf{x}), F).$$

Claim: u_F is harmonic in Ω. Since u_F is bounded, being a probability, from Theorem 3.5, p. 122, it is enough to show that u_F satisfies the mean value property. Let

$$B_R = B_R(\mathbf{x}) \subset\subset \Omega.$$

If $\tau_R = \tau_R(\mathbf{x})$ is the first exit time from B_R, then $\mathbf{X}(\tau_R)$ has a uniform distribution on ∂B_R, due to the invariance by rotation of the Brownian motion.

This means that, starting from the center, the escape probability from B_R through any arc $K \subset \partial B_R$ is given by

[14] More precisely, $P(\mathbf{x}, \tau(\mathbf{x}), F)$ is well defined on the σ-algebra of the *Borel subsets of* $\partial\Omega$ and it can be shown that it is σ-additive (see Appendix A).

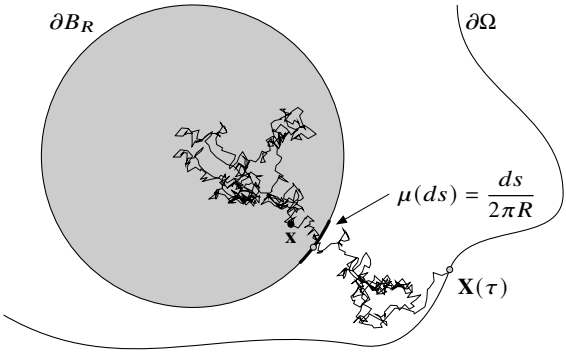


Fig. 3.7 Strong Markov property of a Brownian particle

$$\frac{\text{length of } K}{2\pi R}.$$

Now, before reaching F, the particle must hit ∂B_R. Since τ_R is a stopping time, we may use the strong Markov property. Thus, after τ_R, $\mathbf{X}(t)$ can be considered as a Brownian motion with uniform initial distribution on ∂B_R, expressed by the formula (Fig. 3.7)

$$\mu(ds) = \frac{ds}{2\pi R},$$

where ds is the length element on ∂B_R. Therefore, the particle escapes ∂B_R through some arc of length ds, centered at a point $\mathbf{s}$, with probability $\mu(ds)$ and from there it reaches F with probability $P(\mathbf{s}, \tau(\mathbf{s}), F)$. By the total probability theorem, integrating the product $P(\mathbf{s}, \tau(\mathbf{s}), F)\mu(ds)$ over ∂B_R, we obtain $P(\mathbf{x}, \tau(\mathbf{x}), F)$; namely:

$$P(\mathbf{x}, \tau(\mathbf{x}), F) = \int_{\partial B_R(\mathbf{x})} P(\mathbf{s}, \tau(\mathbf{s}), F)\mu(ds) = \frac{1}{2\pi R}\int_{\partial B_R(\mathbf{x})} P(\mathbf{s}, \tau(\mathbf{s}), F)ds,$$

which is the mean value property for $P(\mathbf{x}, \tau(\mathbf{x}), F)$. Thus u_F is harmonic.

Observe now that if $\boldsymbol{\sigma} \in \partial\Omega$ then $\tau(\boldsymbol{\sigma}) = 0$ and hence

$$P(\boldsymbol{\sigma}, \tau(\boldsymbol{\sigma}), F) = \begin{cases} 1 \text{ if } \boldsymbol{\sigma} \in F \\ 0 \text{ if } \boldsymbol{\sigma} \in \partial\Omega \setminus F. \end{cases}$$

Therefore, if $\sigma \in \partial\Omega$, $P(\boldsymbol{\sigma}, \tau(\boldsymbol{\sigma}), F)$ coincides with the characteristic function of F. Thus, if $d\sigma$ is an arc element on $\partial\Omega$ centered in $\boldsymbol{\sigma}$, the function

$$\mathbf{x} \longmapsto g(\boldsymbol{\sigma})P(\mathbf{x}, \tau(\mathbf{x}), d\sigma) \tag{3.51}$$

is harmonic in Ω, it attains the value $g(\boldsymbol{\sigma})$ on $d\sigma$ and it is zero on $\partial\Omega \setminus d\sigma$. To obtain the harmonic function equal to g on $\partial\Omega$, we integrate over $\partial\Omega$ all the contributions from (3.51). This gives the representation (3.50).

To assert that (3.50) is indeed the required solution, we should check that $u(\mathbf{x}) \to g(\boldsymbol{\sigma})$ when $\mathbf{x} \to \boldsymbol{\sigma}$ for each $\boldsymbol{\sigma} \in \partial\Omega$. It turns out that this is true if and only if Ω is a regular domain in the sense of Definition 3.23, see e.g. *B. Øksendal*, [50]. □

Remark 3.30 The measure

$$F \longmapsto P(\mathbf{x}, \tau(\mathbf{x}), F)$$

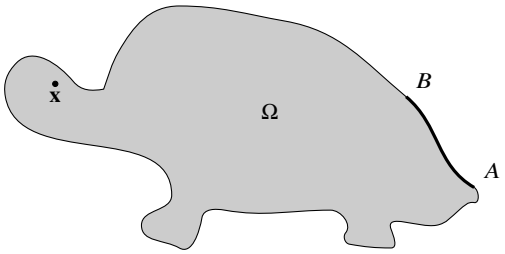


Fig. 3.8 A modification of the Dirichlet data on the arc AB affects the value of the solution at $\mathbf{x}$

is called the *harmonic measure at* $\mathbf{x}$ *of the domain* Ω and in general it can not be expressed by an explicit formula. In the particular case $\Omega = B_R(\mathbf{p})$, Poisson's formula (3.23) indicates that the harmonic measure for the disc $B_R(\mathbf{p})$ is given by

$$P(\mathbf{x}, \tau(\mathbf{x}), d\sigma) = \frac{1}{2\pi R} \frac{R^2 - |\mathbf{x} - \mathbf{p}|^2}{|\boldsymbol{\sigma} - \mathbf{x}|^2} d\sigma.$$

Remark 3.31 Formula (3.50) shows that the value of the solution at a point $\mathbf{x}$ *depends on the boundary data on all* $\partial\Omega$ (except for sets of length zero). In the case of Fig. 3.8, a change of the datum g on the arc AB affects the value of the solution at $\mathbf{x}$, even if this point is far from AB and near $\partial\Omega$.

Recurrence and Brownian motion

We have seen that the solution of a Dirichlet problem can be constructed by using the general properties of a Brownian motion. On the other hand, the deterministic solution of some Dirichlet problems can be used to deduce interesting properties of the Brownian motion.

We examine two simple examples. Recall from Sect. 3.1 that $\log|\mathbf{x}|$ is harmonic in the plane except $\mathbf{x} = \mathbf{0}$.

Let a, R be real numbers, $R > a > 0$. It is easy to check that the function

$$u_R(\mathbf{x}) = \frac{\log R - \log|\mathbf{x}|}{\log R - \log a}$$

is harmonic in the ring $C_{a,R} = \left\{\mathbf{x} \in \mathbb{R}^2 : a < |\mathbf{x}| < R\right\}$ and moreover

$$u_R(\mathbf{x}) = 1 \text{ on } \partial B_a(\mathbf{0}), \qquad u_R(\mathbf{x}) = 0 \text{ on } \partial B_R(\mathbf{0}).$$

Thus $u_R(\mathbf{x})$ represents the escape probability from the ring through $\partial B_a(\mathbf{0})$, starting at $\mathbf{x}$:

$$u_R(\mathbf{x}) = P_R(\mathbf{x}, \tau(\mathbf{x}), \partial B_a(\mathbf{0})).$$

Letting $R \to +\infty$, we get

$$P_R(\mathbf{x}, \tau(\mathbf{x}), \partial B_a(\mathbf{0})) = \frac{\log R - \log|\mathbf{x}|}{\log R - \log a} \to 1 = P_\infty(\mathbf{x}, \tau(\mathbf{x}), \partial B_a(\mathbf{0})).$$

This means that, starting at $\mathbf{x}$, the probability to enter (sooner or later) the disc $B_a(\mathbf{0})$ is 1. Due to the invariance by translations of the Brownian motion, the origin can be replaced by any other point without changing the conclusions. Moreover, since we have proved in Lemma 3.28 that the exit probability from any disc is also 1, we can state the following result: given any point $\mathbf{x}$ and any disc in the plane, a Brownian particle started at $\mathbf{x}$ enters the disc and exits from it an infinite number of times, with probability 1. We say that a bidimensional Brownian motion is *recurrent*.

In three dimensions a Brownian motion is *not recurrent*. In fact (see the next section), the function

$$u_R(\mathbf{x}) = \frac{\frac{1}{|\mathbf{x}|} - \frac{1}{R}}{\frac{1}{a} - \frac{1}{R}}$$

is harmonic in the spherical shell $S_{a,R} = \left\{\mathbf{x} \in \mathbb{R}^3 : a < |\mathbf{x}| < R\right\}$ and

$$u_R(\mathbf{x}) = 1 \text{ on } \partial B_a(\mathbf{0}), \qquad u_R(\mathbf{x}) = 0 \text{ on } \partial B_R(\mathbf{0}).$$

Then $u_R(\mathbf{x})$ represents the escape probability from the shell through $\partial B_a(\mathbf{0})$, starting at $\mathbf{x}$:

$$u_R(\mathbf{x}) = P_R(\mathbf{x}, \tau(\mathbf{x}), \partial B_a(\mathbf{0})).$$

This time, letting $R \to +\infty$, we find

$$P_R(\mathbf{x}, \tau(\mathbf{x}), \partial B_a(\mathbf{0})) = \frac{\frac{1}{|\mathbf{x}|} - \frac{1}{R}}{\frac{1}{a} - \frac{1}{R}} \to \frac{a}{|\mathbf{x}|} = P_\infty(\mathbf{x}, \tau(\mathbf{x}), \partial B_a(\mathbf{0})).$$

Thus, the probability to enter, sooner or later, the sphere $B_a(\mathbf{0})$ is not 1 and it becomes smaller and smaller as the distance of $\mathbf{x}$ from the origin increases.

3.6 Fundamental Solution and Newtonian Potential

3.6.1 The fundamental solution

The Perron method and the probabilistic method leading to formula (3.50), p. 143, are not the only way to construct the solution of the Dirichlet problem. We shall derive deterministic formulas involving various types of potentials, involving a special function, called the *fundamental solution* of the Laplace operator.

As we did for the diffusion equation, let us look at the invariance properties characterizing the operator Δ: the invariances by *translations* and by *rotations*.

Invariance by translations means that the Laplace operator commutes with translations, that is, for each fixed $\mathbf{y}$,

$$\Delta[u(\mathbf{x}-\mathbf{y})] = \Delta u(\mathbf{x}-\mathbf{y}).$$

In particular, if $u = u(\mathbf{x})$ is harmonic in $\Omega \subseteq \mathbb{R}^n$, the function $v(\mathbf{x}) = u(\mathbf{x}-\mathbf{y})$ is harmonic in $\Omega + \mathbf{y} = \{\mathbf{x}+\mathbf{y} : \mathbf{x} \in \Omega\}$.

Invariance by rotations means that the Laplace operator commutes with a rotation in $\mathbb{R}^n$. Since any rotation is represented by an orthogonal matrix $\mathbf{M}$ (i.e. $\mathbf{M}^\top = \mathbf{M}^{-1}$), this means that

$$\Delta[u(\mathbf{M}\mathbf{x})] = \Delta u(\mathbf{M}\mathbf{x}).$$

To check it, observe that, if we denote by D^2u the Hessian of u, we have

$$\Delta u = \operatorname{Tr} D^2 u = \text{trace of the Hessian of } u.$$

Since

$$D^2[u(\mathbf{M}\mathbf{x})] = \mathbf{M}^\top D^2 u(\mathbf{M}\mathbf{x})\mathbf{M}$$

and $\mathbf{M}$ is orthogonal, we have

$$\Delta[u(\mathbf{M}\mathbf{x})] = \operatorname{Tr}[\mathbf{M}^{-1} D^2 u(\mathbf{M}\mathbf{x})\mathbf{M}] = \operatorname{Tr} D^2 u(\mathbf{M}\mathbf{x}) = \Delta u(\mathbf{M}\mathbf{x}).$$

In particular, if $u = u(\mathbf{x})$ is harmonic in $\Omega \subseteq \mathbb{R}^n$, the function $v = u(\mathbf{M}\mathbf{x})$ is harmonic in $\Omega' = \mathbf{M}^{-1}\Omega = \{\mathbf{z} = \mathbf{M}^{-1}\mathbf{x} : \mathbf{x} \in \Omega\}$.

Now, a typical rotationally invariant quantity is *the distance function from a point*, for instance from the origin, that is $r = |\mathbf{x}|$. Thus let us look for *radially symmetric* harmonic functions $u = u(r)$ in $\mathbb{R}^n \setminus \{\mathbf{0}\}$.

Consider first $n = 2$; using polar coordinates and recalling (3.24), we find

$$u_{rr} + \frac{1}{r}u_r = 0 \quad \text{or} \quad \frac{u_{rr}}{u_r} = -\frac{1}{r},$$

from which $u_r = C/r$ and then

$$u(r) = C \log r + C_1, \qquad C, C_1 \in \mathbb{R}.$$

In dimension $n = 3$, using spherical coordinates (r, ψ, θ), $r > 0$, $0 < \psi < \pi$, $0 < \theta < 2\pi$, the operator Δ has the following expression (App. B):

$$\Delta_{\text{polar}} = \underbrace{\frac{\partial^2}{\partial r^2} + \frac{2}{r}\frac{\partial}{\partial r}}_{\text{radial part}} + \underbrace{\frac{1}{r^2}\left\{\frac{1}{(\sin\psi)^2}\frac{\partial^2}{\partial\theta^2} + \frac{\partial^2}{\partial\psi^2} + \cot\psi\frac{\partial}{\partial\psi}\right\}}_{\text{spherical part (Laplace-Beltrami operator)}}.$$

The Laplace equation for $u = u(r)$ becomes

$$u_{rr} + \frac{2}{r}u_r = 0 \quad \text{or} \quad r u_{rr} + 2u_r = (ru)_{rr} = 0,$$

from which

$$u(r) = \frac{C}{r} + C_1, \qquad C, C_1 \in \mathbb{R}.$$

Choose $C_1 = 0$ and $C = \frac{1}{4\pi}$ if $n = 3$, $C = -\frac{1}{2\pi}$ if $n = 2$. The function

$$\Phi(\mathbf{x}) = \begin{cases} -\dfrac{1}{2\pi} \log |\mathbf{x}| & n = 2 \\ \dfrac{1}{4\pi\,|\mathbf{x}|} & n = 3 \end{cases} \tag{3.52}$$

is called the **fundamental solution** for the Laplace operator Δ. As we shall prove in Chap. 7, the above choice of the constant C is made in order to have

$$\Delta\Phi(\mathbf{x}) = -\delta_n(\mathbf{x})$$

where $\delta_n(\mathbf{x})$ denotes *the n-dimensional Dirac measure at* $\mathbf{x} = \mathbf{0}$.

The physical meaning of Φ is remarkable: if $n = 3$, in Gaussian (CGS) units, $4\pi\Phi$ represents the electrostatic potential due to a unitary charge located at the origin and vanishing at infinity[15].

Clearly, if the origin is replaced by a point $\mathbf{y}$, the corresponding potential is $4\pi\Phi(\mathbf{x} - \mathbf{y})$ and

$$\Delta_{\mathbf{x}}\Phi(\mathbf{x} - \mathbf{y}) = -\delta_3(\mathbf{x} - \mathbf{y}).$$

By symmetry, we also have $\Delta_{\mathbf{y}}\Phi(\mathbf{x} - \mathbf{y}) = -\delta_3(\mathbf{x} - \mathbf{y})$.

Remark 3.32 In dimension $n > 3$, the equation for $u(r)$ is (see Problem 3.11)

$$u_{rr} + \frac{n-1}{r} u_r = 0$$

and the fundamental solution of the Laplace operator is

$$\Phi(\mathbf{x}) = \frac{1}{(n-2)\omega_n} \frac{1}{|\mathbf{x}|^{n-2}},$$

where, we recall, ω_n is the surface area of the unit sphere in $\mathbb{R}^n$.

3.6.2 The Newtonian potential

Suppose that $(4\pi)^{-1} f(\mathbf{x})$ gives the density of a charge located inside a compact set in $\mathbb{R}^3$. Then $\Phi(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d\mathbf{y}$ represents the potential at $\mathbf{x}$ due to the charge $(4\pi)^{-1} f(\mathbf{y}) d\mathbf{y}$ inside a small region of volume $d\mathbf{y}$ around $\mathbf{y}$. The full potential is given by the sum of all the contributions; we get

$$u(\mathbf{x}) = \int_{\mathbb{R}^3} \Phi(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d\mathbf{y} = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{f(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d\mathbf{y}, \tag{3.53}$$

[15] In dimension 2,

$$2\pi\Phi(x_1, x_2) = -\log \sqrt{x_1^2 + x_2^2}$$

represents the potential due to a charge of density 1, distributed along the x_3 axis.

which is the *convolution* between Φ and f and it is called the Newtonian potential of f. **Formally**, we have

$$\Delta u(\mathbf{x}) = \int_{\mathbb{R}^3} \Delta_{\mathbf{x}} \Phi(\mathbf{x}-\mathbf{y}) f(\mathbf{y}) d\mathbf{y} = -\int_{\mathbb{R}^3} \delta_3(\mathbf{x}-\mathbf{y}) f(\mathbf{y}) d\mathbf{y} = -f(\mathbf{x}), \tag{3.54}$$

but this calculation is not correct since ($r = |\mathbf{x}|$)

$$\Phi_{x_j x_j}(r) = -\frac{1}{r^3} + \frac{3x_j^2}{r^5},$$

so that $\Phi_{x_j x_j}$ is not integrable in $\mathbb{R}^3$ and it is not possible to differentiate twice under the integral sign.

Nonetheless, under suitable hypotheses on f, it is true that $\Delta u = -f$ (see Theorem 3.33 below). Clearly, the Newtonian potential is not the only solution of the Poisson equation since $u + c$, c constant, is a solution as well. However, it is the only solution vanishing at infinity. All this is stated precisely in the theorem below, where, for simplicity, we assume $f \in C^2(\mathbb{R}^3)$ with compact support[16]. We have:

Theorem 3.33 *Let $f \in C^2(\mathbb{R}^3)$ with **compact** support. Let u be the Newtonian potential of f, defined by (3.53). Then, u is the only solution in $\mathbb{R}^3$ of*

$$\Delta u = -f \tag{3.55}$$

belonging to $C^2(\mathbb{R}^3)$ and vanishing at infinity.

Proof Let $v \in C^2(\mathbb{R}^3)$ be another solution to (3.55), vanishing at infinity. Then $u - v$ is a bounded harmonic function in all $\mathbb{R}^3$ and therefore is constant by Corollary 3.15, p. 132. Since it vanishes at infinity it must be zero; thus $u = v$.

To show that (3.53) belongs to $C^2(\mathbb{R}^3)$ and satisfies (3.55), observe that we can write (3.53) in the alternative form

$$u(\mathbf{x}) = \int_{\mathbb{R}^3} \Phi(\mathbf{y}) f(\mathbf{x}-\mathbf{y}) d\mathbf{y} = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{f(\mathbf{x}-\mathbf{y})}{|\mathbf{y}|} d\mathbf{y}.$$

Since $1/|\mathbf{y}|$ is integrable near the origin and f is zero outside a compact set (see Theorem A.13, p. 661), we can take first and second order derivatives under the integral sign to get

$$u_{x_i x_j}(\mathbf{x}) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \Phi(\mathbf{y}) f_{x_j x_k}(\mathbf{x}-\mathbf{y}) d\mathbf{y}. \tag{3.56}$$

Since $f_{x_j x_k} \in C(\mathbb{R}^3)$, formula (3.56) shows that also $u_{x_i x_j}$ is continuous and therefore $u \in C^2(\mathbb{R}^3)$.

It remains to prove (3.55). Since $\Delta_{\mathbf{x}} f(\mathbf{x}-\mathbf{y}) = \Delta_{\mathbf{y}} f(\mathbf{x}-\mathbf{y})$, from (3.56), we have

$$\Delta u(\mathbf{x}) = \int_{\mathbb{R}^3} \Phi(\mathbf{y}) \Delta_{\mathbf{x}} f(\mathbf{x}-\mathbf{y}) d\mathbf{y} = \int_{\mathbb{R}^3} \Phi(\mathbf{y}) \Delta_{\mathbf{y}} f(\mathbf{x}-\mathbf{y}) d\mathbf{y}.$$

We want to integrate by parts using formula (1.17), but, since Φ has a singularity at $\mathbf{y} = \mathbf{0}$, we have first to isolate the origin, by choosing a small ball $B_r = B_r(\mathbf{0})$ and writing

[16] Recall that the support of a continuous function f is the closure of the set where f is not zero.

$$\Delta u(\mathbf{x}) = \int_{B_r} \cdots \, d\mathbf{y} + \int_{\mathbb{R}^3 \setminus B_r} \cdots \, d\mathbf{y} \equiv I_r + J_r. \tag{3.57}$$

We have, using spherical coordinates,

$$|I_r| \le \frac{\max |\Delta f|}{4\pi} \int_{B_r} \frac{1}{|\mathbf{y}|} \, d\mathbf{y} = \max |\Delta f| \int_0^r \rho \, d\rho = \frac{\max |\Delta f|}{2} r^2$$

so that

$$I_r \to 0 \quad \text{if } r \to 0.$$

Keeping in mind that f vanishes outside a compact set, we can integrate J_r by parts (twice). Denoting by $\boldsymbol{\nu}(\boldsymbol{\sigma})$ the outward[17] unit normal to ∂B_r at $\boldsymbol{\sigma}$, we obtain:

$$\begin{aligned} J_r &= \frac{1}{4\pi r} \int_{\partial B_r} \nabla_{\mathbf{y}} f(\mathbf{x} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) \, d\sigma - \int_{\mathbb{R}^3 \setminus B_r} \nabla \Phi(\mathbf{y}) \cdot \nabla_{\mathbf{y}} f(\mathbf{x} - \mathbf{y}) \, d\mathbf{y} \\ &= \frac{1}{4\pi r} \int_{\partial B_r} \nabla_{\mathbf{y}} f(\mathbf{x} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) \, d\sigma - \int_{\partial B_r} f(\mathbf{x} - \boldsymbol{\sigma}) \nabla \Phi(\boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) \, d\sigma \end{aligned}$$

since $\Delta \Phi = 0$ in $\mathbb{R}^3 \setminus B_r$. We have:

$$\frac{1}{4\pi r} \left| \int_{\partial B_r} \nabla_{\mathbf{y}} f(\mathbf{x} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) \, d\sigma \right| \le r \max |\nabla f| \to 0 \quad \text{as } r \to 0.$$

On the other hand, $\nabla \Phi(\mathbf{y}) = -\mathbf{y} \, |\mathbf{y}|^{-3} / (4\pi)$ and the outward unit normal on ∂B_r is $\boldsymbol{\nu}(\boldsymbol{\sigma}) = -\boldsymbol{\sigma}/r$, so that

$$\int_{\partial B_r} f(\mathbf{x} - \boldsymbol{\sigma}) \nabla \Phi(\boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) \, d\sigma = \frac{1}{4\pi r^2} \int_{\partial B_r} f(\mathbf{x} - \boldsymbol{\sigma}) d\sigma \to f(\mathbf{x}) \quad \text{as } r \to 0.$$

Thus, $J_r \to -f(\mathbf{x})$ as $r \to 0$. Passing to the limit as $r \to 0$ in (3.57) we get

$$\Delta u(\mathbf{x}) = -f(\mathbf{x}).$$

□

Remark 3.34 Theorem 3.33 actually holds under much less restrictive hypotheses on f. For instance, it is enough that $f \in C^1(\mathbb{R}^3)$ and $|f(\mathbf{x})| \le C \, |\mathbf{x}|^{2+\varepsilon}$, for some $\varepsilon > 0$[18].

Remark 3.35 An appropriate version of Theorem 3.33 holds in dimension $n = 2$, with the Newtonian potential replaced by the *logarithmic potential*

$$u(\mathbf{x}) = \int_{\mathbb{R}^2} \Phi(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d\mathbf{y} = -\frac{1}{2\pi} \int_{\mathbb{R}^2} \log |\mathbf{x} - \mathbf{y}| \; f(\mathbf{y}) d\mathbf{y}. \tag{3.58}$$

The logarithmic potential does not vanish at infinity; its asymptotic behavior is (see Problem 3.12)

$$u(\mathbf{x}) = -\frac{M}{2\pi} \log |\mathbf{x}| + O\left(\frac{1}{|\mathbf{x}|}\right) \quad \text{as } |\mathbf{x}| \to +\infty \tag{3.59}$$

[17] With respect to $\mathbb{R}^3 \setminus B_r$.

[18] The proof is not elementary. See e.g. *D. Gilbarg, N. S. Trudinger*, [5].

where

$$M = \int_{\mathbb{R}^2} f(\mathbf{y}) d\mathbf{y}.$$

The logarithmic potential (3.58) is the only solution of $\Delta u = -f$ in $\mathbb{R}^2$ satisfying (3.59).

3.6.3 A divergence-curl system. Helmholtz decomposition formula

Using the properties of the Newtonian potential we can solve the following two problems, that appear in several applications e.g. to linear elasticity, fluid dynamics or electrostatics.

1. *Reconstruction of a vector field* $\mathbf{u} \in \mathbb{R}^3$, *from the knowledge of its curl and divergence*. Precisely, given a scalar f and a vector field $\boldsymbol{\omega}$, we want to find a vector field $\mathbf{u}$ such that

$$\begin{cases} \operatorname{div} \mathbf{u} = f \\ \operatorname{curl} \mathbf{u} = \boldsymbol{\omega} \end{cases} \qquad \text{in } \mathbb{R}^3$$

and vanishing at infinity.

2. *Decomposition of a vector field* $\mathbf{u} \in \mathbb{R}^3$ *into the sum of a curl-free vector field and a divergence-free vector field.* Precisely, given $\mathbf{u}$ vanishing at infinity, we want to find a scalar φ and a vector field $\mathbf{w}$ such that the following *Helmholtz decomposition formula* holds

$$\mathbf{u} = \operatorname{curl} \mathbf{w} + \nabla \varphi. \tag{3.60}$$

Consider **problem 1**. First of all observe that, since $\operatorname{div} \operatorname{curl} \mathbf{u} = 0$, a necessary condition for the existence of a solution is $\operatorname{div} \boldsymbol{\omega} = 0$.

Let us check the uniqueness. If $\mathbf{u}_1$ and $\mathbf{u}_2$ are solutions sharing the same data f and $\boldsymbol{\omega}$, their difference $\mathbf{w} = \mathbf{u}_1 - \mathbf{u}_2$ vanishes at infinity and satisfies

$$\operatorname{div} \mathbf{w} = 0 \quad \text{and} \quad \operatorname{curl} \mathbf{w} = \mathbf{0}, \quad \text{in } \mathbb{R}^3.$$

From $\operatorname{curl} \mathbf{w} = \mathbf{0}$ we infer the existence of a scalar function U such that $\nabla U = \mathbf{w}$. From $\operatorname{div} \mathbf{w} = 0$, we deduce

$$\operatorname{div} \nabla U = \Delta U = 0.$$

Thus U is harmonic. Hence its derivatives, that is the components w_j of $\mathbf{w}$, are bounded harmonic functions in $\mathbb{R}^3$. Liouville's Theorem 3.15, p. 132, implies that each w_j is constant and therefore identically zero, since it vanishes at infinity. We conclude that, under the stated assumptions, *the solution of problem* 1 *is unique*.

To find $\mathbf{u}$, split it into the sum $\mathbf{u} = \mathbf{v} + \mathbf{z}$ and look for $\mathbf{v}$ and $\mathbf{z}$ such that

$$\operatorname{div} \mathbf{v} = f, \qquad \operatorname{curl} \mathbf{v} = \mathbf{0}$$

$$\operatorname{div} \mathbf{z} = 0, \qquad \operatorname{curl} \mathbf{z} = \boldsymbol{\omega}.$$

To find $\mathbf{v}$, let φ be the Newtonian potential of f, that is:

$$\varphi(\mathbf{x}) = -\frac{1}{4\pi}\int_{\mathbb{R}^3}\frac{1}{|\mathbf{x}-\mathbf{y}|}f(\mathbf{y})d\mathbf{y}.$$

Set $\mathbf{v} = \nabla\varphi$; then $\operatorname{curl}\mathbf{v} = 0$ and, under suitable assumptions on f, $\operatorname{div}\mathbf{v} = \Delta\varphi = f$. To find $\mathbf{z}$, recall the identity

$$\operatorname{curl}\operatorname{curl}\mathbf{z} = \nabla(\operatorname{div}\mathbf{z}) - \Delta\mathbf{z}.$$

Assuming $\operatorname{div}\mathbf{z} = 0$, we get

$$\Delta\mathbf{z} = -\operatorname{curl}\operatorname{curl}\mathbf{z} = -\operatorname{curl}\boldsymbol{\omega}$$

so that a candidate solution is

$$\mathbf{z}(\mathbf{x}) = \frac{1}{4\pi}\int_{\mathbb{R}^3}\frac{1}{|\mathbf{x}-\mathbf{y}|}\operatorname{curl}\boldsymbol{\omega}(\mathbf{y})d\mathbf{y}. \tag{3.61}$$

It turns out that $\mathbf{z}$ is the right choice. We summarize the conclusions in the next theorem, also specifying the hypotheses on f and $\boldsymbol{\omega}$ (see Remark 3.34).

Theorem 3.36 *Let $f \in C^1(\mathbb{R}^3)$, $\boldsymbol{\omega} \in C^2(\mathbb{R}^3;\mathbb{R}^3)$ such that $\operatorname{div}\boldsymbol{\omega} = 0$ and, for $|\mathbf{x}|$ large,*

$$|f(\mathbf{x})| \le \frac{M}{|\mathbf{x}|^{2+\varepsilon}}, \qquad |\operatorname{curl}\boldsymbol{\omega}(\mathbf{x})| \le \frac{M}{|\mathbf{x}|^{2+\varepsilon}} \qquad (\varepsilon > 0).$$

Then, the unique solution vanishing at infinity of the system

$$\begin{cases} \operatorname{div}\mathbf{u} = f \\ \operatorname{curl}\mathbf{u} = \boldsymbol{\omega} \end{cases} \qquad \text{in } \mathbb{R}^3$$

is given by the vector field

$$\mathbf{u}(\mathbf{x}) = \int_{\mathbb{R}^3}\frac{1}{4\pi|\mathbf{x}-\mathbf{y}|}\operatorname{curl}\boldsymbol{\omega}(\mathbf{y})d\mathbf{y} - \nabla\int_{\mathbb{R}^3}\frac{1}{4\pi|\mathbf{x}-\mathbf{y}|}f(\mathbf{y})d\mathbf{y}. \tag{3.62}$$

Proof We are left to check that $\mathbf{z}$ given by (3.61) satisfies $\operatorname{div}\mathbf{z} = 0$ and $\operatorname{curl}\mathbf{z} = \boldsymbol{\omega}$. It is not difficult to show that

$$\mathbf{z}(\mathbf{x}) = \frac{1}{4\pi}\operatorname{curl}\int_{\mathbb{R}^3}\frac{\boldsymbol{\omega}(\mathbf{y})}{|\mathbf{x}-\mathbf{y}|}d\mathbf{y}, \tag{3.63}$$

from which $\operatorname{div}\mathbf{z} = 0$ follows. As a consequence $\Delta\mathbf{z} = -\operatorname{curl}\boldsymbol{\omega} = -\operatorname{curl}\operatorname{curl}\mathbf{z}$, or

$$\operatorname{curl}(\operatorname{curl}\mathbf{z} - \boldsymbol{\omega}) = \mathbf{0}.$$

Therefore $\operatorname{curl}\mathbf{z} - \boldsymbol{\omega} = \nabla\psi$ for some scalar ψ. Since $\operatorname{div}\boldsymbol{\omega} = \operatorname{div}\operatorname{curl}\mathbf{z} = 0$, ψ is a bounded harmonic function in $\mathbb{R}^3$. Liouville's Theorem 3.15, p. 132, yields $\psi = 0$ and $\operatorname{curl}\mathbf{z} = \boldsymbol{\omega}$. □

Consider now **problem 2**. Assume that $f = \operatorname{div}\mathbf{u}$ and $\boldsymbol{\omega} = \operatorname{curl}\mathbf{u}$ satisfy the assumptions of Theorem 3.36. Then, recalling (3.63), the representation formula

(3.62) gives

$$\mathbf{u} = \nabla\varphi + \operatorname{curl}\mathbf{w} \tag{3.64}$$

where

$$\varphi(\mathbf{x}) = -\int_{\mathbb{R}^3} \frac{1}{4\pi\,|\mathbf{x}-\mathbf{y}|}\operatorname{div}\mathbf{u}(\mathbf{y})d\mathbf{y} \tag{3.65}$$

and

$$\mathbf{w}(\mathbf{x}) = \int_{\mathbb{R}^3} \frac{1}{4\pi\,|\mathbf{x}-\mathbf{y}|}\operatorname{curl}\mathbf{u}(\mathbf{y})d\mathbf{y}.$$

• *An application to fluid dynamics.* Consider the three dimensional flow of an incompressible fluid, of both constant density ρ and viscosity μ, subject to a conservative external force[19] $\mathbf{F} = \nabla f$.

Let $\mathbf{u} = \mathbf{u}(\mathbf{x}, t)$ denote the velocity field, $p = p(\mathbf{x}, t)$ the pressure and $\mathbf{T} = (T_{ij})_{i,j=1,2,3}$ the stress tensor. The laws of conservation of mass and linear momentum give the following equations:

$$\rho_t + \operatorname{div}(\rho\mathbf{u}) = 0$$

and

$$\rho\frac{D\mathbf{u}}{Dt} = \rho(\mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u}) = \mathbf{F} + \operatorname{div}\mathbf{T}$$

where $\operatorname{div}\mathbf{T}$ is the vector of components $\sum_{j=1,2,3}\partial_{x_j}T_{ij}$, $i = 1, 2, 3$.

The quantity $\frac{D\mathbf{u}}{Dt}$ is called the *material derivative* of $\mathbf{u}$, given by the sum of $\mathbf{u}_t$, the fluid acceleration due to the non-stationary character of the motion, and of $(\mathbf{u}\cdot\nabla)\mathbf{u}$, the inertial acceleration due to fluid transport[20].

The so called Newtonian fluids (e.g. water) are characterized by the constitutive law

$$\mathbf{T} = -p\mathbf{I} + \mu\nabla\mathbf{u}$$

where $\mathbf{I}$ is the identity matrix. Being ρ constant, we get for u and p the celebrated *Navier-Stokes equations*:

$$\operatorname{div}\mathbf{u} = 0 \tag{3.66}$$

and

$$\frac{D\mathbf{u}}{Dt} = \mathbf{u}_t + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\frac{1}{\rho}\nabla p + \nu\Delta\mathbf{u} + \frac{1}{\rho}\nabla f \qquad (\nu = \mu/\rho). \tag{3.67}$$

---

[19] Gravity, for instance.

[20] The i-th component of $(\mathbf{u}\cdot\nabla)\mathbf{u}$ is given by $\sum_{j=1}^{3} u_j\frac{\partial u_i}{\partial x_j}$. Let us compute $\frac{D\mathbf{u}}{Dt}$, for example, for a plane fluid, uniformly rotating with angular speed ω. Then $\mathbf{u}(x, y) = -\omega y\mathbf{i} + \omega x\mathbf{j}$. Since $\mathbf{u}_t = \mathbf{0}$, the motion is stationary and

$$\frac{D\mathbf{u}}{Dt} = (\mathbf{u}\cdot\nabla)\mathbf{u} = \left(-\omega y\frac{\partial}{\partial x} + \omega x\frac{\partial}{\partial y}\right)(-\omega y\mathbf{i} + \omega x\mathbf{j}) = -\omega^2(x\mathbf{i} + y\mathbf{j}),$$

which is the centripetal acceleration.

We look for solution of (3.66), (3.67) subject to a given initial condition

$$\mathbf{u}(\mathbf{x}, 0) = \mathbf{g}(\mathbf{x}) \qquad \mathbf{x} \in \mathbb{R}^3, \tag{3.68}$$

where $\mathbf{g}$ is also divergence-free:

$$\operatorname{div} \mathbf{g} = 0.$$

In general, the system (3.66), (3.67) is extremely difficult to solve. In the case of slow flow, for instance due to high viscosity, the inertial term becomes negligible, compared for instance to $\nu\Delta\mathbf{u}$, and (3.67) simplifies to the linearized equation

$$\mathbf{u}_t = -\frac{1}{\rho}\nabla p + \nu\Delta\mathbf{u} + \frac{1}{\rho}\nabla f. \tag{3.69}$$

It is possible to find an explicit formula for the solution of (3.66), (3.68), (3.69) by writing everything in terms of $\boldsymbol{\omega} = \operatorname{curl} \mathbf{u}$. In fact, taking the curl of (3.69) and (3.68), since

$$\operatorname{curl}\left(-\frac{1}{\rho}\nabla p + \nu\Delta\mathbf{u} + \frac{1}{\rho}\nabla f\right) = \nu\Delta\boldsymbol{\omega},$$

we obtain

$$\begin{cases} \boldsymbol{\omega}_t = \nu\Delta\boldsymbol{\omega} & \mathbf{x} \in \mathbb{R}^3,\ t > 0 \\ \boldsymbol{\omega}(\mathbf{x}, 0) = \operatorname{curl} \mathbf{g}(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^3. \end{cases}$$

This is a global Cauchy problem for the heat equation. If $\mathbf{g} \in C^2(\mathbb{R}^3; \mathbb{R}^3)$ and $\operatorname{curl} \mathbf{g}$ is bounded, we have the unique solution

$$\boldsymbol{\omega}(\mathbf{x}, t) = \frac{1}{(4\pi\nu t)^{3/2}} \int_{\mathbb{R}^3} \exp\left(-\frac{|\mathbf{y}|^2}{4\nu t}\right) \operatorname{curl} \mathbf{g}(\mathbf{x} - \mathbf{y})\, d\mathbf{y}. \tag{3.70}$$

Moreover, for $t > 0$, we can take the divergence operator under the integral in (3.70) and deduce that $\operatorname{div} \boldsymbol{\omega} = 0$. Therefore, if $\operatorname{curl} g(\mathbf{x})$ vanishes rapidly at infinity[21], we can recover $\mathbf{u}$ by solving the system

$$\operatorname{curl} \mathbf{u} = \boldsymbol{\omega}, \quad \operatorname{div} \mathbf{u} = 0,$$

according to formula (3.62) with $f = 0$.

Finally to find the pressure, from (3.69) we have

$$\nabla p = -\rho\mathbf{u}_t + \mu\Delta\mathbf{u} + \nabla f. \tag{3.71}$$

Since

$$\boldsymbol{\omega}_t = \nu\Delta\boldsymbol{\omega},$$

the right hand side is curl-free; hence (3.71) can be solved and determines p up to an additive constant (as it should be).

In conclusion:

[21] $|\operatorname{curl} \mathbf{g}(\mathbf{x})| \le M/|\mathbf{x}|^{2+\varepsilon}$, $\varepsilon > 0$, it is enough.

Proposition 3.37 *Let* $f \in C^1(\mathbb{R}^3)$, $\mathbf{g} \in C^2(\mathbb{R}^3;\mathbb{R}^3)$, *with* $\operatorname{div}\mathbf{g} = 0$ *and* $\operatorname{curl}\mathbf{g}$ *rapidly vanishing at infinity. There exist a unique* $\mathbf{u}$, *with* $\operatorname{curl}\mathbf{u}$ *vanishing at infinity, and* p, *unique up to an additive constant, satisfying the system (3.66), (3.68), (3.69).*

3.7 The Green Function

3.7.1 An integral identity

Formulas (3.53) and (3.58) give a representation of the solution to Poisson's equation in all $\mathbb{R}^n$, $n = 2, 3$. In bounded domains Ω, any representation formula has to take into account the boundary values, as indicated in the following theorem.

To avoid any confusion we clarify some notations. We write $r_{\mathbf{xy}} = |\mathbf{x} - \mathbf{y}|$. If $\mathbf{x}$ is fixed, the symbol $\Phi(\mathbf{x} - \cdot)$ denotes the function

$$\mathbf{y} \longmapsto \Phi(\mathbf{x} - \mathbf{y}).$$

If $\boldsymbol{\sigma} \in \partial\Omega$, $\boldsymbol{\nu} = \boldsymbol{\nu}(\boldsymbol{\sigma})$ denotes the outward unit normal to $\partial\Omega$ at $\boldsymbol{\sigma}$. The (outward) normal derivative of u at $\boldsymbol{\sigma}$ is denoted by one of the symbols

$$\partial_{\boldsymbol{\nu}} u = \frac{\partial u}{\partial \boldsymbol{\nu}} = \nabla u(\boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}).$$

Accordingly,

$$\partial_{\boldsymbol{\nu}} \Phi(\mathbf{x} - \boldsymbol{\sigma}) = \nabla_{\mathbf{y}} \Phi(\mathbf{x} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) = -\nabla_{\mathbf{x}} \Phi(\mathbf{x} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}).$$

Theorem 3.38 *Let* $\Omega \subset \mathbb{R}^n$ *be a bounded, smooth domain and* $u \in C^2(\overline{\Omega})$. *Then, for every* $\mathbf{x} \in \Omega$,

$$\begin{aligned} u(\mathbf{x}) = &- \int_\Omega \Phi(\mathbf{x} - \mathbf{y}) \Delta u(\mathbf{y}) d\mathbf{y} + \\ &+ \int_{\partial\Omega} \Phi(\mathbf{x} - \boldsymbol{\sigma}) \partial_{\boldsymbol{\nu}} u(\boldsymbol{\sigma}) d\sigma - \int_{\partial\Omega} u(\boldsymbol{\sigma}) \partial_{\boldsymbol{\nu}} \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma. \end{aligned} \tag{3.72}$$

The last two terms in the right hand side of (3.72) are called *single* and *double layer potentials* of $\partial_{\boldsymbol{\nu}} u$ and $-u$, respectively. We are going to examine these surface potentials later. The first one is the Newtonian potential of $-\Delta u$ in Ω.

Proof We do it for $n = 3$. The proof for $n = 2$ is similar. For fixed $\mathbf{x} \in \Omega$, we would like to apply Green's identity (1.19), p. 20,

$$\int_\Omega (v\Delta u - u\Delta v) d\mathbf{x} = \int_{\partial\Omega} (v\partial_{\boldsymbol{\nu}} u \ - u\partial_{\boldsymbol{\nu}} v) d\sigma, \tag{3.73}$$

to u and $\Phi(\mathbf{x} - \cdot)$. However, $\Phi(\mathbf{x} - \cdot)$ has a singularity at $\mathbf{x}$, so that it cannot be inserted directly into (3.73). Let us isolate the singularity inside a ball $B_\varepsilon(\mathbf{x})$, with ε small. In the domain $\Omega_\varepsilon = \Omega \setminus \overline{B}_\varepsilon(\mathbf{x})$, $\Phi(\mathbf{x} - \cdot)$ is smooth and harmonic.

Thus, replacing Ω with Ω_ε, we can apply (3.73) to u and $\Phi(\mathbf{x}-\cdot)$. Since

$$\partial\Omega_\varepsilon = \partial\Omega \cup \partial B_\varepsilon(\mathbf{x}),$$

and $\Delta_{\mathbf{y}}\Phi(\mathbf{x}-\mathbf{y}) = 0$, we find:

$$\begin{aligned}\int_{\Omega_\varepsilon} \frac{1}{r_{\mathbf{xy}}}\Delta u\, d\mathbf{y} &= \int_{\partial\Omega_\varepsilon}\left(\frac{1}{r_{\mathbf{x}\sigma}}\frac{\partial u}{\partial \boldsymbol{\nu}} - u\,\frac{\partial}{\partial \boldsymbol{\nu}}\frac{1}{r_{\mathbf{x}\sigma}}\right) d\sigma \\ &= \int_{\partial\Omega}(\cdots)\, d\sigma + \int_{\partial B_\varepsilon(\mathbf{x})}\frac{1}{r_{\mathbf{x}\sigma}}\frac{\partial u}{\partial \boldsymbol{\nu}}\, d\sigma - \int_{\partial B_\varepsilon(\mathbf{x})} u\,\frac{\partial}{\partial \boldsymbol{\nu}}\frac{1}{r_{\mathbf{x}\sigma}}\, d\sigma.\end{aligned} \tag{3.74}$$

We let now $\varepsilon \to 0$ in (3.74). We have:

$$\int_{\Omega_\varepsilon}\frac{1}{r_{\mathbf{x}\sigma}}\Delta u\, d\mathbf{y} = \int_\Omega \chi_{\Omega_\varepsilon}\frac{1}{r_{\mathbf{x}\sigma}}\Delta u\, d\mathbf{y} \to \int_\Omega \frac{1}{r_{\mathbf{x}\sigma}}\Delta u\, d\mathbf{y}, \qquad \text{as } \varepsilon \to 0 \tag{3.75}$$

by the Dominated Convergence Theorem A.7, p. 660, since $\Delta u \in C(\overline{\Omega})$, $r_{\mathbf{x}\sigma}^{-1}$ is positive and integrable in Ω and $\chi_{\Omega_\varepsilon} \to 1$ pointwise in Ω.

On $\partial B_\varepsilon(\mathbf{x})$, we have $r_{\mathbf{x}\sigma} = \varepsilon$ and $|\partial_{\boldsymbol{\nu}} u| \le M$, since $|\nabla u|$ is bounded; then

$$\left|\int_{\partial B_\varepsilon(\mathbf{x})}\frac{1}{r_{\mathbf{x}\sigma}}\frac{\partial u}{\partial \boldsymbol{\nu}}\, d\sigma\right| \le 4\pi\varepsilon M \to 0, \qquad \text{as } \varepsilon \to 0. \tag{3.76}$$

The most delicate term is

$$\int_{\partial B_\varepsilon(\mathbf{x})} u\,\frac{\partial}{\partial \boldsymbol{\nu}}\frac{1}{r_{\mathbf{x}\sigma}}\, d\sigma.$$

On $\partial B_\varepsilon(\mathbf{x})$, the outward (with respect to Ω_ε) unit normal at σ is $\boldsymbol{\nu}(\sigma) = \frac{\mathbf{x}-\sigma}{\varepsilon}$, so that

$$\frac{\partial}{\partial \boldsymbol{\nu}}\frac{1}{r_{\mathbf{x}\sigma}} = \nabla_{\mathbf{y}}\frac{1}{r_{\mathbf{x}\sigma}}\cdot\boldsymbol{\nu}(\sigma) = \frac{\mathbf{x}-\sigma}{\varepsilon^3}\frac{\mathbf{x}-\sigma}{\varepsilon} = \frac{1}{\varepsilon^2}.$$

As a consequence,

$$\int_{\partial B_\varepsilon(\mathbf{x})} u\,\frac{\partial}{\partial \boldsymbol{\nu}}\frac{1}{r_{\mathbf{x}\sigma}}\, d\sigma = \frac{1}{\varepsilon^2}\int_{\partial B_\varepsilon(\mathbf{x})} u\, d\sigma \to 4\pi u(\mathbf{x}) \tag{3.77}$$

as $\varepsilon \to 0$, by the continuity of u.

Letting $\varepsilon \to 0$ in (3.74), from (3.75), (3.76), (3.77) we obtain (3.72). □

3.7.2 Green's function

The function Φ defined in (3.52) is the fundamental solution for the Laplace operator Δ in the whole $\mathbb{R}^n$, for $n = 2, 3$. We can also define a fundamental solution for the Laplace operator in any open set and in particular in any bounded domain $\Omega \subset \mathbb{R}^3$, representing, up to the factor 4π, the potential due to a unit charge, placed at a point $\mathbf{x} \in \Omega$ and equal to zero on $\partial\Omega$.

This function, that we denote by $G(\mathbf{x}, \mathbf{y})$, is called the *Green function in* Ω, *for the operator* Δ; for fixed $x \in \Omega$, G satisfies

$$\Delta_{\mathbf{y}} G(\mathbf{x}, \mathbf{y}) = -\delta_3(\mathbf{x}-\mathbf{y}) \qquad \text{in } \Omega$$

and

$$G(\mathbf{x}, \boldsymbol{\sigma}) = 0, \qquad \text{for } \boldsymbol{\sigma} \in \partial\Omega.$$

More explicitly, the Green function can be written in the form

$$G(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x} - \mathbf{y}) - \varphi(\mathbf{x}, \mathbf{y}) \tag{3.78}$$

where φ, for fixed $\mathbf{x} \in \Omega$, solves the Dirichlet problem

$$\begin{cases} \Delta_{\mathbf{y}} \varphi = 0 & \text{in } \Omega \\ \varphi(\mathbf{x}, \boldsymbol{\sigma}) = \Phi(\mathbf{x} - \boldsymbol{\sigma}) & \text{on } \partial\Omega. \end{cases} \tag{3.79}$$

The Green function has the following important properties (see Problem 3.15):

a) *Positivity*: $G(\mathbf{x}, \mathbf{y}) > 0$ for every $\mathbf{x}, \mathbf{y} \in \Omega$, with $G(\mathbf{x}, \mathbf{y}) \to +\infty$ when $\mathbf{x} - \mathbf{y} \to \mathbf{0}$.
b) *Symmetry*: $G(\mathbf{x}, \mathbf{y}) = G(\mathbf{y}, \mathbf{x})$.

The existence of the Green function for a particular domain depends on the solvability of the Dirichlet problem (3.79). According to Corollary 3.25, p. 139, we know that this is the case if Ω is regular for the Dirichlet problem, for instance, if Ω is a bounded Lipschitz domain.

Even if we know that the Green function exists, explicit formulas are available only for special domains. Sometimes, a technique known as *method of electrostatic images* works. In this method $4\pi\varphi(\mathbf{x}, \cdot)$ is considered as the potential due to an imaginary charge q, placed at a suitable point $\mathbf{x}^*$, the *image* of $\mathbf{x}$, in the complement of Ω. The charge q and the point $\mathbf{x}^*$ have to be chosen so that $4\pi\varphi(\mathbf{x}, \cdot)$, on $\partial\Omega$, is equal to the potential created by the unit charge in $\mathbf{x}$.

The simplest way to illustrate the method is to determine the Green's function for the upper half-space, although this is an unbounded domain. Clearly, we require that G vanishes at infinity.

- *Green's function for the upper half space in* $\mathbb{R}^3$. Let $\mathbb{R}^3_+$ be the upper half space:

$$\mathbb{R}^3_+ = \{(x_1, x_2, x_3) : x_3 > 0\} .$$

Fix $\mathbf{x} = (x_1, x_2, x_3)$ and observe that, if we choose $\mathbf{x}^* = (x_1, x_2, -x_3)$, then, on $y_3 = 0$, we have

$$|\mathbf{x}^* - \mathbf{y}| = |\mathbf{x} - \mathbf{y}| .$$

Thus, if $\mathbf{x} \in \mathbb{R}^3_+$, $\mathbf{x}^*$ belongs to the complement of $\mathbb{R}^3_+$, the function

$$\varphi(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x}^* - \mathbf{y}) = \frac{1}{4\pi\,|\mathbf{x}^* - \mathbf{y}|}$$

is harmonic in $\mathbb{R}^3_+$ and $\varphi(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x} - \mathbf{y})$ on the plane $y_3 = 0$. In conclusion,

$$G(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi\,|\mathbf{x} - \mathbf{y}|} - \frac{1}{4\pi\,|\mathbf{x}^* - \mathbf{y}|} \tag{3.80}$$

is the Green's function for the upper half space.

- *Green's function for the sphere*. Let

$$\Omega = B_R = B_R(\mathbf{0}) \subset \mathbb{R}^3.$$

To determine the Green's function for B_R, set

$$\varphi(\mathbf{x}, \mathbf{y}) = \frac{q}{4\pi\,|\mathbf{x}^* - \mathbf{y}|},$$

$\mathbf{x}$ fixed in B_R, and try to determine $\mathbf{x}^*$, outside B_R, and q, so that

$$\frac{q}{|\mathbf{x}^* - \mathbf{y}|} = \frac{1}{|\mathbf{x} - \mathbf{y}|}, \tag{3.81}$$

when $|\mathbf{y}| = R$. Eq. (3.81) for $|\mathbf{y}| = R$, gives

$$|\mathbf{x}^* - \mathbf{y}|^2 = q^2\,|\mathbf{x} - \mathbf{y}|^2 \tag{3.82}$$

or

$$|\mathbf{x}^*|^2 - 2\mathbf{x}^* \cdot \mathbf{y} + R^2 = q^2(|\mathbf{x}|^2 - 2\mathbf{x} \cdot \mathbf{y} + R^2).$$

Rearranging the terms we have

$$|\mathbf{x}^*|^2 + R^2 - q^2(R^2 + |\mathbf{x}|^2) = 2\mathbf{y} \cdot (\mathbf{x}^* - q^2\mathbf{x}).$$

Since the left hand side does not depend on $\mathbf{y}$, it must be that

$$\mathbf{x}^* = q^2\mathbf{x}$$

and

$$q^4\,|\mathbf{x}|^2 - q^2(R^2 + |\mathbf{x}|^2) + R^2 = 0$$

from which

$$q = R/|\mathbf{x}|\,. \tag{3.83}$$

This works for $\mathbf{x} \neq 0$ and gives (see Fig. 3.9)

$$G(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi}\left[\frac{1}{|\mathbf{x} - \mathbf{y}|} - \frac{R}{|\mathbf{x}|\,|\mathbf{x}^* - \mathbf{y}|}\right], \qquad \mathbf{x}^* = \frac{R^2}{|\mathbf{x}|^2}\mathbf{x},\ \mathbf{x} \neq \mathbf{0}. \tag{3.84}$$

Since

$$|\mathbf{x}^* - \mathbf{y}| = |\mathbf{x}|^{-1}\,(R^4 - 2R^2\mathbf{x} \cdot \mathbf{y} + |\mathbf{y}|^2\,|\mathbf{x}|^2)^{1/2},$$

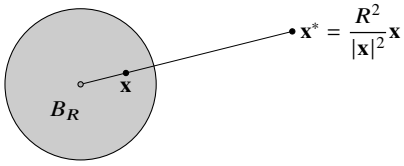


Fig. 3.9 The image $\mathbf{x}^*$ of $\mathbf{x}$ in the construction of the Green's function for the sphere: $|\mathbf{x}^*|\,|\mathbf{x}| = R^2$

when $\mathbf{x} \to \mathbf{0}$ we have

$$\varphi(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi} \frac{R}{|\mathbf{x}|\,|\mathbf{x}^* - \mathbf{y}|} \to \frac{1}{4\pi R}$$

and therefore we can define

$$G(\mathbf{0}, \mathbf{y}) = \frac{1}{4\pi} \left[\frac{1}{|\mathbf{y}|} - \frac{1}{R} \right]. \tag{3.85}$$

Formulas (3.84), (3.85) can be written as

$$G(\mathbf{x}, \mathbf{y}) = \begin{cases} \Phi(\mathbf{x} - \mathbf{y}) - \Phi\left(\frac{R}{|\mathbf{x}|} |\mathbf{x} - \mathbf{y}| \right) & \mathbf{x} \neq \mathbf{0} \\ \Phi(\mathbf{y}) - \Phi(R) & \mathbf{x} = \mathbf{0}. \end{cases} \tag{3.86}$$

Remark 3.39 Formula (3.78) actually defines the Green function for the Laplace operator in a domain $\Omega \subset \mathbb{R}^n$, with $n \geq 2$. See Problem 3.14.

3.7.3 Green's representation formula

From Theorem 3.38 we know that every smooth function u can be written as the sum of the volume (Newtonian) potential of $-\Delta u$, the single layer potential of $\partial_\nu u$ and the double layer potential of u. Suppose u solves the Dirichlet problem

$$\begin{cases} \Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{3.87}$$

Then (3.72) gives, for $x \in \Omega$,

$$\begin{aligned} u(\mathbf{x}) = &- \int_\Omega \Phi(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d\mathbf{y} + \\ &+ \int_{\partial\Omega} \Phi(\mathbf{x} - \boldsymbol{\sigma}) \partial_\nu u(\boldsymbol{\sigma}) d\sigma - \int_{\partial\Omega} g(\boldsymbol{\sigma}) \partial_\nu \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma. \end{aligned} \tag{3.88}$$

This representation formula for u is not satisfactory, since it involves the data f and g but also the normal derivative $\partial_\nu u$, which is unknown. To get rid of $\partial_\nu u$, let $G(\mathbf{x}, \mathbf{y}) = \Phi(\mathbf{x} - \mathbf{y}) - \varphi(\mathbf{x}, \mathbf{y})$ be the Green function in Ω. Since $\varphi(\mathbf{x}, \cdot)$ is harmonic in Ω, we can apply (3.73) to u and $\varphi(\mathbf{x}, \cdot)$; we find

$$\begin{aligned} 0 = &\int_\Omega \varphi(\mathbf{x}, \mathbf{y}) f(\mathbf{y}) d\mathbf{y} \; + \\ &- \int_{\partial\Omega} \varphi(\mathbf{x}, \boldsymbol{\sigma}) \partial_\nu u(\boldsymbol{\sigma}) d\sigma + \int_{\partial\Omega} g(\boldsymbol{\sigma}) \partial_\nu \varphi(\mathbf{x}, \boldsymbol{\sigma}) d\sigma. \end{aligned} \tag{3.89}$$

Adding (3.88), (3.89) and recalling that $\varphi(\mathbf{x}, \boldsymbol{\sigma}) = \Phi(\mathbf{x} - \boldsymbol{\sigma})$ on $\partial\Omega$, we obtain:

Theorem 3.40 *Let Ω be a smooth domain and u be a smooth solution of (3.87). Then:*

$$u(\mathbf{x}) = -\int_\Omega f(\mathbf{y})G(\mathbf{x},\mathbf{y})d\mathbf{y} - \int_{\partial\Omega} g(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}G(\mathbf{x},\boldsymbol{\sigma})d\sigma. \tag{3.90}$$

Thus, the solution of the Dirichlet problem (3.87) can be written as the sum of the two *Green's potentials* in the right hand side of (3.90) and it is known as soon as the Green function in Ω is known. In particular, if u is harmonic, then

$$u(\mathbf{x}) = -\int_{\partial\Omega} g(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}G(\mathbf{x},\boldsymbol{\sigma})d\sigma. \tag{3.91}$$

Comparing with (3.50), we deduce that

$$-\partial_{\boldsymbol{\nu}}G(\mathbf{x},\boldsymbol{\sigma})d\sigma$$

represents the *harmonic measure* in Ω. The function

$$P(\mathbf{x},\boldsymbol{\sigma}) = -\partial_{\boldsymbol{\nu}}G(\mathbf{x},\boldsymbol{\sigma})$$

is called **Poisson's kernel**. Since $G(\mathbf{x},\cdot) > 0$ inside Ω and vanishes on $\partial\Omega$, P is *nonnegative* (actually positive).

On the other hand, the formula

$$u(\mathbf{x}) = -\int_\Omega f(\mathbf{y})G(\mathbf{x},\mathbf{y})d\mathbf{y} \tag{3.92}$$

gives the solution of the Poisson equation $\Delta u = f$ in Ω, vanishing on $\partial\Omega$. From the positivity of G we have that:

$$f \geq 0 \quad \text{in } \Omega \text{ implies } u \leq 0 \text{ in } \Omega,$$

which is another form of the maximum principle.

• *Poisson's kernel and Poisson's formula* ($n = 3$). From (3.84) we can compute Poisson's kernel for the sphere $B_R(\mathbf{0})$ in $\mathbb{R}^3$. If $\mathbf{x} \neq \mathbf{0}$ we have, recalling that $\mathbf{x}^* = R^2\,|\mathbf{x}|^{-2}\,\mathbf{x}$,

$$\nabla_{\mathbf{y}}\left[\frac{1}{|\mathbf{x}-\mathbf{y}|} - \frac{R}{|\mathbf{x}|\,|\mathbf{x}^*-\mathbf{y}|}\right] = \frac{\mathbf{x}-\mathbf{y}}{|\mathbf{x}-\mathbf{y}|^3} - \frac{R}{|\mathbf{x}|}\frac{\mathbf{x}^*-\mathbf{y}}{|\mathbf{x}^*-\mathbf{y}|^3}.$$

If $\boldsymbol{\sigma} \in \partial B_R(\mathbf{0})$, from (3.82) and (3.83), we have $|\mathbf{x}^* - \boldsymbol{\sigma}| = R\,|\mathbf{x}|^{-1}\,|\mathbf{x}-\boldsymbol{\sigma}|$, therefore

$$\nabla_{\mathbf{y}}G(\mathbf{x},\boldsymbol{\sigma}) = \frac{1}{4\pi}\left[\frac{\mathbf{x}-\boldsymbol{\sigma}}{|\mathbf{x}-\boldsymbol{\sigma}|^3} - \frac{|\mathbf{x}|^2}{R^2}\frac{\mathbf{x}^*-\boldsymbol{\sigma}}{|\mathbf{x}-\boldsymbol{\sigma}|^3}\right] = -\frac{\boldsymbol{\sigma}}{4\pi\,|\mathbf{x}-\boldsymbol{\sigma}|^3}\left[1 - \frac{|\mathbf{x}|^2}{R^2}\right].$$

Since on $\partial B_R(\mathbf{0})$ the exterior unit normal is $\boldsymbol{\nu}(\boldsymbol{\sigma}) = \boldsymbol{\sigma}/R$, we have

$$P(\mathbf{x},\boldsymbol{\sigma}) = -\partial_{\boldsymbol{\nu}} G(\mathbf{x},\boldsymbol{\sigma}) = -\nabla_{\mathbf{y}} G(\mathbf{x},\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\boldsymbol{\sigma}) = \frac{R^2-|\mathbf{x}|^2}{4\pi R}\frac{1}{|\mathbf{x}-\boldsymbol{\sigma}|^3}.$$

As a consequence, we obtain Poisson's formula (clearly valid also for $\mathbf{x} = \mathbf{0}$)

$$u(\mathbf{x}) = \frac{R^2-|\mathbf{x}|^2}{4\pi R}\int_{\partial B_R(\mathbf{0})}\frac{g(\boldsymbol{\sigma})}{|\mathbf{x}-\boldsymbol{\sigma}|^3}d\sigma \tag{3.93}$$

for the unique solution of the Dirichlet problem $\Delta u = 0$ in $B_R(\mathbf{0})$ and $u = g$ on $\partial B_R(\mathbf{0})$.

3.7.4 The Neumann function

We can find a representation formula for the solution of a Neumann problem as well. Let u be a smooth solution of the problem

$$\begin{cases}\Delta u = f & \text{in } \Omega\\ \partial_{\boldsymbol{\nu}} u = h & \text{on } \partial\Omega,\end{cases} \tag{3.94}$$

where f and h satisfy the solvability condition

$$\int_{\partial\Omega} h(\boldsymbol{\sigma})d\sigma = \int_{\Omega} f(\mathbf{y})d\mathbf{y}, \tag{3.95}$$

keeping in mind that u is uniquely determined up to an additive constant. From Theorem 3.38 we can write

$$\begin{aligned} u(\mathbf{x}) = &-\int_{\Omega}\Phi(\mathbf{x}-\mathbf{y})f(\mathbf{y})d\mathbf{y} + \\ &+\int_{\partial\Omega} h(\boldsymbol{\sigma})\Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma - \int_{\partial\Omega} u(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma\end{aligned} \tag{3.96}$$

and this time we should get rid of the second integral, containing the unknown datum u on $\partial\Omega$. Mimicking what we have done for the Dirichlet problem, we try to find an analog of the Green function, that is a function $N = N(\mathbf{x},\mathbf{y})$ given by

$$N(\mathbf{x},\mathbf{y}) = \Phi(\mathbf{x}-\mathbf{y}) - \psi(\mathbf{x},\mathbf{y}),$$

where, for $\mathbf{x}$ fixed, ψ is a solution of

$$\begin{cases}\Delta_{\mathbf{y}}\psi = 0 & \text{in } \Omega\\ \partial_{\boldsymbol{\nu}}\psi(\mathbf{x},\boldsymbol{\sigma}) = \partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma}) & \text{on } \partial\Omega,\end{cases}$$

in order to have $\partial_{\boldsymbol{\nu}} N(\mathbf{x},\boldsymbol{\sigma}) = 0$ on $\partial\Omega$. But *this Neumann problem has no solution*, because the solvability condition

$$\int_{\partial\Omega} \partial_\nu \Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma = 0$$

is not satisfied. In fact, letting $u \equiv 1$ in (3.72), we get

$$\int_{\partial\Omega} \partial_\nu \Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma = -1. \tag{3.97}$$

Thus, taking into account (3.97), we require ψ to satisfy

$$\begin{cases} \Delta_{\mathbf{y}}\psi = 0 & \text{in } \Omega \\ \partial_\nu \psi(\mathbf{x},\boldsymbol{\sigma}) = \partial_\nu \Phi(\mathbf{x}-\boldsymbol{\sigma}) + \frac{1}{|\partial\Omega|} & \text{on } \partial\Omega. \end{cases} \tag{3.98}$$

In this way,

$$\int_{\partial\Omega} \left(\partial_\nu \Phi(\mathbf{x}-\boldsymbol{\sigma}) + \frac{1}{|\partial\Omega|} \right) d\sigma = 0$$

and (3.98) is solvable. Note that, with this choice of ψ, we have

$$\partial_\nu N(\mathbf{x},\boldsymbol{\sigma}) = -\frac{1}{|\partial\Omega|} \quad \text{on } \partial\Omega. \tag{3.99}$$

Applying now (3.73) to u and $\psi(\mathbf{x},\cdot)$, we find:

$$0 = \int_\Omega \psi(\mathbf{y}) f(\mathbf{y}) d\mathbf{y} - \int_{\partial\Omega} \psi(\mathbf{x},\boldsymbol{\sigma}) h(\boldsymbol{\sigma}) d\sigma + \int_{\partial\Omega} u(\boldsymbol{\sigma}) \partial_\nu \psi(\boldsymbol{\sigma}) d\sigma. \tag{3.100}$$

Adding (3.100) to (3.96) and using (3.99) we obtain:

Theorem 3.41 *Let Ω be a smooth domain and u be a smooth solution of (3.94). Then:*

$$u(\mathbf{x}) - \frac{1}{|\partial\Omega|} \int_{\partial\Omega} u(\boldsymbol{\sigma}) d\sigma = \int_{\partial\Omega} h(\boldsymbol{\sigma}) N(\mathbf{x},\boldsymbol{\sigma}) d\sigma - \int_\Omega f(\mathbf{y}) N(\mathbf{x},\mathbf{y}) d\mathbf{y}.$$

Thus, the solution of the Neumann problem (3.94) can also be written as the sum of two potentials, up to the additive constant

$$c = \frac{1}{|\partial\Omega|} \int_{\partial\Omega} u(\boldsymbol{\sigma}) d\sigma,$$

the mean value of u on $\partial\Omega$.

The function N is called *Neumann function* (also Green's function for the Neumann problem) and it is defined up to an additive constant.

3.8 Uniqueness in Unbounded Domains

3.8.1 Exterior problems

Boundary value problems in unbounded domains occur in important applications, for instance in the motion of fluids past an obstacle, capacity problems or scattering of acoustic or electromagnetic waves.

As in the case of Poisson's equation in all $\mathbb{R}^n$, a problem in an unbounded domain requires suitable conditions at infinity to be well posed. Consider for example the Dirichlet problem

$$\begin{cases} \Delta u = 0 & \text{in } |\mathbf{x}| > 1 \\ u = 0 & \text{on } |\mathbf{x}| = 1. \end{cases} \tag{3.101}$$

For every real number a, in dimension two and three, respectively,

$$u(\mathbf{x}) = a \log |\mathbf{x}| \quad \text{and} \quad u(\mathbf{x}) = a(1 - 1/|\mathbf{x}|)$$

are solutions to (3.101). Thus there is no uniqueness.

To restore uniqueness, in two dimensions we typically require u to be bounded, while in three dimensions $u(\mathbf{x})$ to have a (finite) limit[22], say u_∞, as $|\mathbf{x}| \to \infty$. Under these conditions, in both cases we select a unique solution.

The problem (3.101) is called *exterior Dirichlet problem*. Given a *bounded* domain Ω, we call *exterior of* Ω the set $\Omega_e = \mathbb{R}^n \setminus \overline{\Omega}$.

Without loss of generality, we will assume that $\mathbf{0} \in \Omega$ and, for simplicity, we will consider only *connected* exterior sets, i.e. **exterior domains**. Note that $\partial\Omega_e = \partial\Omega$.

As we have seen in several occasions, maximum principles are very useful to prove uniqueness. In exterior three dimensional domains we have:

Theorem 3.42 *Let $\Omega_e \subset \mathbb{R}^3$ be an exterior domain and $u \in C^2(\Omega_e) \cap C(\overline{\Omega}_e)$, be harmonic in Ω_e and vanishing as $|\mathbf{x}| \to \infty$. If $u \geq 0$ (resp. $u \leq 0$) on $\partial\Omega_e$ then $u \geq 0$ (resp. $u \leq 0$) in Ω_e.*

Proof Let $u \geq 0$ on $\partial\Omega_e$. Fix $\varepsilon > 0$ and choose r_0 so large that $\Omega \subset \{|\mathbf{x}| < r\}$ and $u \geq -\varepsilon$ on $\{|\mathbf{x}| = r\}$, for every $r \geq r_0$. This is possible since $u(\mathbf{x}) \to 0$ as $|\mathbf{x}| \to +\infty$. Applying Theorem 3.7, p. 123, in the bounded set

$$\Omega_{e,r} = \Omega_e \cap \{|\mathbf{x}| < r\},$$

we get $u \geq -\varepsilon$ in $\Omega_{e,r}$. Since ε is arbitrary and r may be taken as large as we like, we deduce that $u \geq 0$ in Ω_e.

The argument for the case $u \leq 0$ on $\partial\Omega_e$ is similar and we leave the details to the reader. □

An immediate consequence is the following uniqueness result in dimension $n = 3$ (for $n = 2$, see Problem 3.17):

Theorem 3.43 *Let $\Omega_e \subset \mathbb{R}^3$ be an exterior domain. Then there exists at most one solution $u \in C^2(\Omega_e) \cap C(\overline{\Omega}_e)$ of the Dirichlet problem*

[22] That is: $\forall \varepsilon > 0$, $|u(\mathbf{x}) - u_\infty| < \varepsilon$ if $|\mathbf{x}| > N_\varepsilon$.

$$\begin{cases} \Delta u = f & \text{in } \Omega_e \\ u = g & \text{on } \partial\Omega_e \\ u(\mathbf{x}) \to u_\infty & \text{as } |\mathbf{x}| \to \infty. \end{cases}$$

Proof Apply Theorem 3.42 to the difference of two solutions. □

We point out another interesting consequence of Theorem 3.42 and Lemma 3.17, p. 133: a harmonic function vanishing at infinity, for $|\mathbf{x}|$ large, is controlled by the fundamental solution. Actually, more is true:

Theorem 3.44 *Let u be harmonic in $\Omega_e \subset \mathbb{R}^3$ and $u(\mathbf{x}) \to 0$ as $|\mathbf{x}| \to \infty$. There exist $R > 0$, $M > 0$ such that, if $|\mathbf{x}| \geq R$,*

$$|u(\mathbf{x})| \leq \frac{M}{|\mathbf{x}|}, \quad \left|u_{x_j}(\mathbf{x})\right| \leq \frac{M}{|\mathbf{x}|^2}, \quad \left|u_{x_j x_k}(\mathbf{x})\right| \leq \frac{M}{|\mathbf{x}|^3}. \tag{3.102}$$

The proof shows how the constants R, M depend on u.

Proof Choose $a \gg 1$ such that $\Omega \subset \{|\mathbf{x}| \leq a\}$ and $|u(\mathbf{x})| \leq 1$ if $|\mathbf{x}| \geq a$. Let $w(\mathbf{x}) = u(\mathbf{x}) - a/|\mathbf{x}|$. Then w is harmonic for $|\mathbf{x}| \geq a$, $w(\mathbf{x}) \leq 0$ on $|\mathbf{x}| = a$ and vanishes at infinity. Then, by Theorem 3.7, we deduce that

$$w(\mathbf{x}) \leq 0 \qquad \text{for } \{|\mathbf{x}| \geq a\}. \tag{3.103}$$

Setting $v(\mathbf{x}) = u(\mathbf{x}) + a/|\mathbf{x}|$, a similar argument gives $v(\mathbf{x}) \geq 0$ for $\{|\mathbf{x}| \geq a\}$. This and (3.103) implies $|u(\mathbf{x})| \leq a/|\mathbf{x}|$ in $\{|\mathbf{x}| \geq a\}$.

The gradient bound follows from the one on u, using (3.38). In fact, let $|\mathbf{x}| \geq 2a$ and $r = |\mathbf{x}|/2$. Then $B_r(\mathbf{x}) \subset \{|\mathbf{y}| \geq a\}$ and $|u(\mathbf{y})| \leq a/|\mathbf{y}|$ for $\mathbf{y} \in \partial B_r(\mathbf{x})$. From (3.38), we have

$$\left|u_{x_j}(\mathbf{x})\right| \leq \frac{3}{r} \max_{\mathbf{y} \in \partial B_r(\mathbf{x})} |u(\mathbf{y})| \leq \frac{3}{r} \max_{\mathbf{y} \in \partial B_r(\mathbf{x})} \frac{a}{|\mathbf{y}|} = \frac{3a}{r(|\mathbf{x}| - r)} = \frac{12a}{|\mathbf{x}|^2}$$

in $\{|\mathbf{x}| \geq 2a\}$. Similarly we can prove, for $|\mathbf{x}| \geq 3a$,

$$\left|u_{x_j x_k}(\mathbf{x})\right| \leq \frac{243a}{|\mathbf{x}|^3}.$$

□

The estimates (3.102) ensures the validity of the Green identity

$$\int_{\Omega_e} \nabla u \cdot \nabla v \, d\mathbf{x} = \int_{\partial\Omega_e} v \partial_\nu u \, d\sigma \tag{3.104}$$

for any pair $u, v \in C^2(\Omega_e) \cap C^1(\overline{\Omega}_e)$, harmonic in Ω_e and vanishing at infinity. To see this, apply the identity (3.104) in the bounded domain $\Omega_{e,r} = \Omega_e \cap \{|\mathbf{x}| < r\}$. Then, let $r \to \infty$ to get (3.104) in Ω_e.

In turn, via the Green identity (3.104), we can prove an appropriate version of Theorem 3.43 for the exterior Robin/Neumann problem

$$\begin{cases} \Delta u = f & \text{in } \Omega_e \\ \partial_\nu u + ku = g & \text{on } \partial\Omega_e, \ (k \geq 0) \\ u \to u_\infty & \text{as } |\mathbf{x}| \to \infty. \end{cases} \tag{3.105}$$

Observe that the outward unit normal to $\partial\Omega_e$ points inward of Ω and that $k > 0$ corresponds to the Robin problem, while $k = 0$ corresponds to the Neumann problem.

Theorem 3.45 *Let $\Omega_e \subset \mathbb{R}^3$ be an exterior domain. Then there exists at most one solution $u \in C^2(\Omega_e) \cap C^1(\overline{\Omega}_e)$ of problem (3.105).*

Proof Suppose u, v are solutions of (3.105) and let $w = u - v$. Then w is harmonic Ω_e, $\partial_\nu w + kw = 0$ on $\partial\Omega_e$ and $w \to 0$ as $|\mathbf{x}| \to \infty$.

Apply the identity (3.104) with $u = v = w$. Since $\partial_\nu w = -kw$ on $\partial\Omega_e$ we have:

$$\int_{\Omega_e} |\nabla w|^2 \, d\mathbf{x} = \int_{\partial\Omega_e} w\partial_\nu w \, d\sigma = -\int_{\partial\Omega_e} kw^2 d\sigma \leq 0.$$

Thus $\nabla w = 0$, and being Ω_e connected w is constant. Since w vanishes at infinity, $w = 0$. □

3.9 Surface Potentials

In this section we go back to examine the meaning and the main properties of the surface potentials appearing in the identity (3.72). A remarkable consequence is the possibility to convert a boundary value problem into a **boundary integral equation**. This kind of formulation can be obtained for more general operators and more general problems, as soon as a fundamental solution is known. Thus it constitutes a flexible method with important implications. In particular, it constitutes the theoretical basis for the so called *boundary element method*, which may offer several advantages from the point of view of the computational cost in numerical approximations, due to a dimension reduction. Here we present the integral formulations of the main boundary value problems and some basic results.

3.9.1 The double and single layer potentials

The last integral in (3.72) is of the form

$$\mathcal{D}(\mathbf{x}; \mu) = \int_{\partial\Omega} \mu(\sigma)\partial_\nu \Phi(\mathbf{x} - \sigma) d\sigma \tag{3.106}$$

and it is called the **double layer potential of** μ. In three dimensions it may represent the electrostatic potential generated by a *dipole* distribution of moment $\mu/4\pi$ on $\partial\Omega$.

Indeed, for every $\sigma \in \partial\Omega$, let $-q(\sigma)$, $q(\sigma)$ be two charges placed at the points σ, $\sigma + h\nu(\sigma)$, respectively. If $h > 0$ is very small, the pair of charges constitutes a *dipole of axis* $\nu(\sigma)$. The induced potential $u_h(\mathbf{x}, \sigma)$ at $\mathbf{x}$ is given by, setting $4\pi q(\sigma)h = \mu(\sigma)$,

$$\begin{aligned} u_h(\mathbf{x}, \sigma) &= 4\pi q(\sigma)[\Phi(\mathbf{x} - (\sigma + h\nu)) - \Phi(\mathbf{x} - \sigma)] \\ &= \mu(\sigma)\frac{\Phi(\mathbf{x} - (\sigma + h\nu)) - \Phi(\mathbf{x} - \sigma)}{h}. \end{aligned}$$

Since h is very small, we can write, at first order of approximation, $u_h(\mathbf{x}) \simeq \mu(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})$. Integrating on $\partial\Omega$ we obtain $\mathcal{D}(\mathbf{x};\mu)$.

To get a clue of the main features of $\mathcal{D}(\mathbf{x};\mu)$, it is useful to look at the particular case $\mu(\boldsymbol{\sigma}) \equiv 1$, that is

$$\mathcal{D}(\mathbf{x};1) = \int_{\partial\Omega} \partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma. \tag{3.107}$$

Inserting $u \equiv 1$ into (3.72), we get

$$\mathcal{D}(\mathbf{x};1) = -1, \qquad \text{for every } \mathbf{x} \in \Omega. \tag{3.108}$$

On the other hand, if $\mathbf{x} \in \mathbb{R}^n \setminus \overline{\Omega}$ is fixed, $\Phi(\mathbf{x}-\cdot)$ is harmonic in Ω and can be inserted into (3.73), with $u \equiv 1$; the result is

$$\mathcal{D}(\mathbf{x};1) = 0, \qquad \text{for every } \mathbf{x} \in \mathbb{R}^n \setminus \overline{\Omega}. \tag{3.109}$$

What happens for $\mathbf{x} \in \partial\Omega$? First of all we have to check that $\mathcal{D}(\mathbf{x};1)$ is well defined (i.e. finite) on $\partial\Omega$. Indeed the singularity of $\partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})$ becomes critical when $\mathbf{x} \in \partial\Omega$, since as $\boldsymbol{\sigma} \to \mathbf{x}$, the order of infinity equals the topological dimension of $\partial\Omega$. For instance, in the two dimensional case, we have

$$\mathcal{D}(\mathbf{x};1) = -\frac{1}{2\pi}\int_{\partial\Omega} \partial_{\boldsymbol{\nu}} \log|\mathbf{x}-\boldsymbol{\sigma}|\, d\sigma = -\frac{1}{2\pi}\int_{\partial\Omega} \frac{(\mathbf{x}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\boldsymbol{\sigma})}{|\mathbf{x}-\boldsymbol{\sigma}|^2} d\sigma.$$

The order of infinity of the integrand is one and the boundary $\partial\Omega$ is a curve, a one dimensional object. In the three dimensional case we have

$$\mathcal{D}(\mathbf{x};1) = \frac{1}{4\pi}\int_{\partial\Omega} \frac{\partial}{\partial\boldsymbol{\nu}}\frac{1}{|\mathbf{x}-\boldsymbol{\sigma}|} d\sigma = \frac{1}{4\pi}\int_{\partial\Omega} \frac{(\mathbf{x}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\boldsymbol{\sigma})}{|\mathbf{x}-\boldsymbol{\sigma}|^3} d\sigma.$$

The order of infinity of the integrand is two and $\partial\Omega$ is a bidimensional surface.

However, if we assume that Ω is a C^2-*domain*, then (see Gauss Lemma below) $\mathcal{D}(\mathbf{x};1)$ is well defined and equal to $-1/2$ on $\partial\Omega$. To get a clue of what happens, let us examine the geometrical meaning of the integrand in $\mathcal{D}(\mathbf{x};1)$.

• *Geometric interpretation.* To compute the value of $\mathcal{D}(\mathbf{x};1)$ on $\partial\Omega$, we first observe that the formulas (3.108) and (3.109) follow immediately from their geometric interpretation. For simplicity, we assume that Ω is a *convex domain*, but the interpretation holds in any regular C^2-domain.

In dimension two, consider the quantity

$$d\sigma^* = -\frac{(\mathbf{x}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\boldsymbol{\sigma})}{r^2_{\mathbf{x}\boldsymbol{\sigma}}}\, d\sigma = \frac{(\boldsymbol{\sigma}-\mathbf{x})\cdot\boldsymbol{\nu}(\boldsymbol{\sigma})}{r^2_{\mathbf{x}\boldsymbol{\sigma}}}\, d\sigma$$

where we keep the notation $r_{\mathbf{x}\boldsymbol{\sigma}} = |\mathbf{x}-\boldsymbol{\sigma}|$. We have (see Fig. 3.10),

$$\frac{\boldsymbol{\sigma}-\mathbf{x}}{r_{\mathbf{x}\boldsymbol{\sigma}}}\cdot\boldsymbol{\nu}(\boldsymbol{\sigma}) = \cos\varphi$$

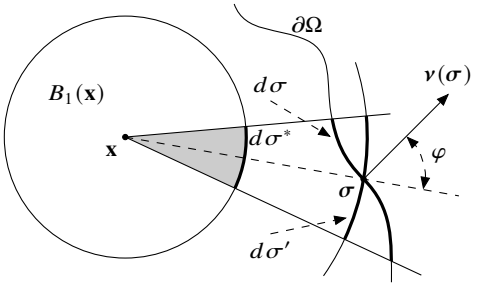


Fig. 3.10 Geometrical interpretation of the integrand in $\mathcal{D}(\mathbf{x}; 1)$, $n = 2$

and therefore

$$d\sigma' = \frac{(\boldsymbol{\sigma} - \mathbf{x}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma})}{r_{\mathbf{x}\boldsymbol{\sigma}}} d\sigma = \cos\varphi \, d\sigma$$

is the projection of the length element $d\sigma$ on the circle $\partial B_{r_{\mathbf{x}\boldsymbol{\sigma}}}(\mathbf{x})$, up to an error of lower order. Then, $d\sigma^* = \frac{d\sigma'}{r_{\mathbf{x}\boldsymbol{\sigma}}}$ is the projection of $d\sigma$ on $\partial B_1(\mathbf{x})$.

Integrating on $\partial\Omega$, the contributions to $d\sigma^*$ sum up to 2π if $\mathbf{x} \in \Omega$ (case a) of Fig. 3.11) and to 0 if $\mathbf{x} \in \mathbb{R}^n \setminus \overline{\Omega}$, due to the sign compensations induced by the orientation of $\boldsymbol{\nu}(\boldsymbol{\sigma})$ (case c) of Fig. 3.11). Thus

$$\int_{\partial\Omega} d\sigma^* = \begin{cases} 2\pi & \text{if } \mathbf{x} \in \Omega \\ 0 & \text{if } \mathbf{x} \in \mathbb{R}^2 \setminus \overline{\Omega}, \end{cases}$$

which are equivalent to (3.108) and (3.109), since

$$\mathcal{D}(\mathbf{x}; 1) = -\frac{1}{2\pi} \int_{\partial\Omega} d\sigma^*.$$

The case b) in Fig. 3.11 corresponds to $\mathbf{x} \in \partial\Omega$. It should be now intuitively clear that the point $\mathbf{x}$ "sees" a total angle of only π radians and therefore $\mathcal{D}(\mathbf{x}; 1) = -1/2$.

The same kind of considerations hold in dimension three; this time, the quantity (see Fig. 3.12)

$$d\sigma^* = -\frac{(\mathbf{x} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma})}{r^3_{\mathbf{x}\boldsymbol{\sigma}}} d\sigma = \frac{(\boldsymbol{\sigma} - \mathbf{x}) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma})}{r^3_{\mathbf{x}\boldsymbol{\sigma}}} d\sigma$$

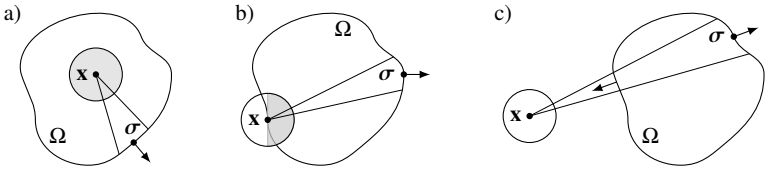


Fig. 3.11 Values of $\int_{\partial\Omega} d\sigma^*$ for $n = 2$. a) 2π; b) π; c) 0

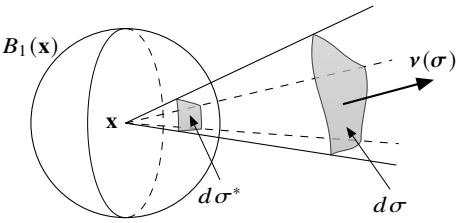


Fig. 3.12 The solid angle $d\sigma^*$, projected from $d\sigma$

is the projection on $\partial B_1(\mathbf{x})$ (solid angle) of the surface element $d\sigma$. Integrating over $\partial\Omega$, the contributions to $d\sigma^*$ sum up to 4π if $x \in \Omega$ and to 0 if $\mathbf{x} \in \mathbb{R}^3 \setminus \overline{\Omega}$.

If $\mathbf{x} \in \partial\Omega$, $\mathbf{x}$ " sees" a total solid angle of measure 2π. Since

$$\mathcal{D}(\mathbf{x}; 1) = -\frac{1}{4\pi} \int_{\partial\Omega} d\sigma^*,$$

we find again the values $-1, 0, -1/2$ in the three cases, respectively.

We gather the above results in the following Gauss Lemma.

Lemma 3.46 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, C^2-domain. Then*

$$\mathcal{D}(\mathbf{x}; 1) = \int_{\partial\Omega} \partial_\nu \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma = \begin{cases} -1 & \mathbf{x} \in \Omega \\ -\frac{1}{2} & \mathbf{x} \in \partial\Omega \\ 0 & \mathbf{x} \in \mathbb{R}^n \setminus \overline{\Omega}. \end{cases}$$

Proof We have to consider only the case $\mathbf{x} \in \partial\Omega$. Consider $n = 3$. The case $n = 2$ is similar. First we prove that $\mathcal{D}(\mathbf{x}; 1)$ is well defined on $\partial\Omega$. Since Ω is a C^2-domain, a neighborhood of $\mathbf{x}$ on $\partial\Omega$, say $\Gamma \subset \partial\Omega$, can be represented in a coordinate system $(\mathbf{y}', y_3) \in \mathbb{R}^2 \times \mathbb{R}$, with origin at $\mathbf{x}$, as a graph $y_3 = \varphi(\mathbf{y}')$ with $\varphi \in C^2$ in $\overline{B_r}(\mathbf{0}')$, $\varphi(\mathbf{0}') = 0$, $\nabla\varphi(\mathbf{0}') = \mathbf{0}'$ and exterior normal at a point $\mathbf{z} = (\mathbf{y}', \varphi(\mathbf{y}'))$ given by $\boldsymbol{\nu} = (\nabla\varphi(\mathbf{y}'), -1)/\sqrt{1 + |\nabla\varphi(\mathbf{y}')|^2}$.

Write

$$\mathcal{D}(\mathbf{x}; 1) = \frac{1}{4\pi} \int_{\partial\Omega\setminus\Gamma} \cdots + \frac{1}{4\pi} \int_{\Gamma} \cdots = I + II.$$

I is well defined since the integrand is continuous on $\partial\Omega \setminus \Gamma$. Consider II. We have $\mathbf{x} = (\mathbf{0}', 0)$. Then

$$II = \frac{1}{4\pi} \int_{B_r(\mathbf{0}')} \frac{\mathbf{y}' \cdot \nabla\varphi(\mathbf{y}') - \varphi(\mathbf{y}')}{[\,|\mathbf{y}'|^2 + \varphi(\mathbf{y}')^2\,]^{3/2}} d\mathbf{y}'.$$

We have $|\varphi(\mathbf{y}') - \mathbf{y}' \cdot \nabla\varphi(\mathbf{y}')| \le C\,|\mathbf{y}'|^2$, and therefore

$$|II| \le \frac{C}{4\pi} \int_{B_r(\mathbf{0}')} \frac{1}{|\mathbf{y}'|} d\mathbf{y}' = \frac{Cr}{2}.$$

Thus $\mathcal{D}(\mathbf{x}; 1)$ is well defined on $\partial\Omega$. To compute its value, apply formula (3.73) to $u \equiv 1$ and $v(\mathbf{y}) = \Phi(\mathbf{x} - \mathbf{y})$ in $\Omega \setminus B_\varepsilon(\mathbf{x})$, to get

$$\int_{\partial\Omega\setminus B_\varepsilon(\mathbf{x})} \partial_\nu \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma = -\int_{\Omega\cap\partial B_\varepsilon(\mathbf{x})} \partial_\nu \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma = -\frac{1}{4\pi\varepsilon^2} \left|\Omega \cap \partial B_\varepsilon(\mathbf{x})\right|.$$

Therefore,

$$\mathcal{D}(\mathbf{x}; 1) = \lim_{\varepsilon\to 0} \int_{\partial\Omega\setminus B_\varepsilon(\mathbf{x})} \partial_\nu \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma = -\frac{1}{2}$$

since $|\Omega \cap \partial B_\varepsilon(\mathbf{x})| - 2\pi\varepsilon^2 \to 0$, as $\varepsilon \to 0$, being Φ of class C^2. □

Thus, when $\mu \equiv 1$, the double layer potential has a jump discontinuity across $\partial\Omega$. Indeed, if $\mathbf{x} \in \partial\Omega$,

$$\lim_{\mathbf{z}\to\mathbf{x},\ \mathbf{z}\in\mathbb{R}^n\setminus\overline{\Omega}} \mathcal{D}(\mathbf{z};1) = \mathcal{D}(\mathbf{x};1) + \frac{1}{2}$$

and

$$\lim_{\mathbf{z}\to\mathbf{x},\ \mathbf{z}\in\Omega} \mathcal{D}(\mathbf{z};1) = \mathcal{D}(\mathbf{x};1) - \frac{1}{2}.$$

These formulas are the key for understanding the general properties of $\mathcal{D}(\mathbf{x};\mu)$, that we state in the following theorem.

Theorem 3.47 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, C^2-domain and μ a continuous function on $\partial\Omega$. Then, $\mathcal{D}(\mathbf{x};\mu)$ is harmonic in $\mathbb{R}^n \setminus \partial\Omega$ it is well defined also on $\partial\Omega$ and the following jump relations hold for every $\mathbf{z} \in \partial\Omega$:*

$$\lim_{\mathbf{x}\to\mathbf{z},\ \mathbf{x}\in\mathbb{R}^n\setminus\overline{\Omega}} \mathcal{D}(\mathbf{x};\mu) = \mathcal{D}(\mathbf{z};\mu) + \frac{1}{2}\mu(\mathbf{z}) \tag{3.110}$$

and

$$\lim_{\mathbf{x}\to\mathbf{z},\ \mathbf{x}\in\Omega} \mathcal{D}(\mathbf{x};\mu) = \mathcal{D}(\mathbf{z};\mu) - \frac{1}{2}\mu(\mathbf{z}). \tag{3.111}$$

We postpone the rather technical proof at the end of the chapter.

Coming back to (3.72), the second integral on the right hand side is of the form

$$\mathcal{S}(\mathbf{x};\psi) = \int_{\partial\Omega} \Phi(\mathbf{x}-\boldsymbol{\sigma})\psi(\boldsymbol{\sigma})d\sigma$$

and it is called the **single layer potential of** ψ.

In three dimensions it represents the electrostatic potential generated by a charge distribution of density $\psi/4\pi$ on $\partial\Omega$. If Ω is a C^2-domain and ψ is continuous on $\partial\Omega$, then $\mathcal{S}$ is continuous across $\partial\Omega$ and

$$\Delta\mathcal{S} = 0 \qquad \text{in } \mathbb{R}^n \setminus \partial\Omega,$$

because there is no problem in differentiating under the integral sign.

Since the flux of an electrostatic potential undergoes a jump discontinuity across a charged surface, we expect a jump discontinuity of the normal derivative of $\mathcal{S}$ across $\partial\Omega$. Precisely

Theorem 3.48 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, C^2-domain and ψ a continuous function on $\partial\Omega$. Then, $\mathcal{S}(\mathbf{x};\psi)$ is harmonic in $\mathbb{R}^n \setminus \partial\Omega$, continuous across $\partial\Omega$ and the following jump relations hold for every $\mathbf{z} \in \partial\Omega$, where $\mathbf{x}_h = \mathbf{z} + h\boldsymbol{\nu}(\mathbf{z})$:*

$$\lim_{h\to 0^+} \nabla\mathcal{S}(\mathbf{x}_h;\psi)\cdot\boldsymbol{\nu}(\mathbf{z}) = \int_{\partial\Omega} \nabla_{\mathbf{z}}\Phi(\mathbf{z}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\mathbf{z})\,\psi(\boldsymbol{\sigma})d\sigma - \frac{1}{2}\psi(\mathbf{z}) \tag{3.112}$$

and

$$\lim_{h\to 0^-} \nabla \mathcal{S}(\mathbf{x}_h;\psi)\cdot\boldsymbol{\nu}(\mathbf{z}) = \int_{\partial\Omega} \nabla_{\mathbf{z}}\Phi(\mathbf{z}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\mathbf{z})\,\psi(\boldsymbol{\sigma})d\sigma + \frac{1}{2}\psi(\mathbf{z}). \tag{3.113}$$

The proof of Theorem 3.48 uses the same techniques of the proof of Theorem 3.47. For the details we refer e.g. to *R. Courant, D. Hilbert*, [26], or *R. B. Guenther, J. W. Lee*, [7].

3.9.2 The integral equations of potential theory

By means of the jump relations (3.111)–(3.113) of the double and single layer potentials, we can reduce the main boundary value problems of potential theory into integral equations of a special form (for the proofs see e.g. *R. B. Guenther, J. W. Lee*, [7]).

Let $\Omega \subset \mathbb{R}^n$ be a smooth domain and $g \in C(\partial\Omega)$. We first show the reduction procedure for the *interior Dirichlet problem*

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{3.114}$$

The starting point is once more the identity (3.72), which gives for the solution u of (3.114) the representation

$$u(\mathbf{x}) = \int_{\partial\Omega} \Phi(\mathbf{x}-\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}u(\boldsymbol{\sigma})d\sigma - \int_{\partial\Omega} g(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma.$$

In Sect. 3.7.3 we used the Green function to get rid of the single layer potential containing the unknown $\partial_{\boldsymbol{\nu}}u$. Here we adopt a different strategy: we forget the single layer potential and try to represent u in the form of a double layer potential, by choosing an appropriate density. In other words, *we seek a continuous function μ on $\partial\Omega$, such that the solution u of* (3.114) *is given by*

$$u(\mathbf{x}) = \int_{\partial\Omega} \mu(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma = \mathcal{D}(\mathbf{x};\mu). \tag{3.115}$$

The function u given by (3.115) is harmonic in Ω, therefore we have only to check the boundary condition

$$\lim_{\mathbf{x}\to\mathbf{z}\in\partial\Omega} u(\mathbf{x}) = g(\mathbf{z}).$$

Letting $\mathbf{x} \to \mathbf{z} \in \partial\Omega$, with $\mathbf{x} \in \Omega$, and taking into account the jump relation (3.111), we obtain for μ the integral equation

$$\int_{\partial\Omega} \mu(\boldsymbol{\sigma})\partial_{\boldsymbol{\nu}}\Phi(\mathbf{z}-\boldsymbol{\sigma})d\sigma - \frac{1}{2}\mu(\mathbf{z}) = g(\mathbf{z}), \qquad \mathbf{z} \in \partial\Omega. \tag{3.116}$$

If $\mu \in C(\partial\Omega)$ solves (3.116), then (3.115) is the solution of (3.114) in $C^2(\Omega) \cap C(\overline{\Omega})$. The following theorem holds.

Theorem 3.49 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, C^2-domain and g a continuous function on $\partial\Omega$. Then, the integral equation (3.116) has a unique solution $\mu \in C(\partial\Omega)$ and the solution $u \in C^2(\Omega) \cap C(\overline{\Omega})$ of the Dirichlet problem (3.114) can be represented as the double layer potential of μ.*

We consider now the *interior Neumann problem*

$$\begin{cases} \Delta u = 0 & \text{in } \Omega \\ \partial_{\boldsymbol{\nu}} u = g & \text{on } \partial\Omega, \end{cases} \tag{3.117}$$

where $g \in C(\partial\Omega)$ satisfies the solvability condition

$$\int_{\partial\Omega} g \, d\sigma = 0. \tag{3.118}$$

This time we seek a continuous function ψ on $\partial\Omega$, such that the solution u of (3.117) is given in the form

$$u(\mathbf{x}) = \int_{\partial\Omega} \psi(\boldsymbol{\sigma}) \Phi(\mathbf{x} - \boldsymbol{\sigma}) d\sigma = \mathcal{S}(\mathbf{x}; \psi). \tag{3.119}$$

The function u given by (3.119) is harmonic in Ω. We check the boundary condition in the form (writing, as in Thm. 3.48, $\mathbf{x}_h = \mathbf{z} + h\boldsymbol{\nu}(\mathbf{z})$)

$$\lim_{h \to 0^-} \nabla u(\mathbf{x}_h) \cdot \boldsymbol{\nu}(\mathbf{z}) = g(\mathbf{z}).$$

Letting $h \to 0^-$ and taking into account the jump relation (3.113), we obtain for ψ the integral equation

$$\int_{\partial\Omega} \psi(\boldsymbol{\sigma}) \nabla_{\mathbf{z}} \Phi(\mathbf{z} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\mathbf{z}) d\sigma + \frac{1}{2}\psi(\mathbf{z}) = g(\mathbf{z}), \qquad \mathbf{z} \in \partial\Omega. \tag{3.120}$$

If $\psi \in C(\partial\Omega)$ solves (3.120), then (3.119) is a solution of (3.117) in the sense specified above.

It turns out that the general solution of (3.120) has the form

$$\psi = \overline{\psi} + C_0 \psi_0 \qquad C_0 \in \mathbb{R},$$

where $\overline{\psi}$ is a particular solution of (3.120) and ψ_0 is a solution of the homogeneous equation

$$\int_{\partial\Omega} \psi_0(\boldsymbol{\sigma}) \nabla_{\mathbf{z}} \Phi(\mathbf{z} - \boldsymbol{\sigma}) \cdot \boldsymbol{\nu}(\mathbf{z}) d\sigma + \frac{1}{2}\psi_0(\mathbf{z}) = 0, \qquad \mathbf{z} \in \partial\Omega. \tag{3.121}$$

As expected, we have infinitely many solutions to the Neumann problem. Observe that

$$\mathcal{S}(\mathbf{x};\psi_0) = \int_{\partial\Omega} \psi_0(\boldsymbol{\sigma})\Phi(\mathbf{x}-\boldsymbol{\sigma})d\sigma$$

is harmonic in Ω, with vanishing normal derivative on $\partial\Omega$, because of (3.113) and (3.121). Consequently, $\mathcal{S}(\mathbf{x};\psi_0)$ is constant and the following theorem holds.

Theorem 3.50 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, C^2-domain and g be a continuous function on $\partial\Omega$ satisfying (3.118). Then, the Neumann problem (3.117) has infinitely many solutions u of the form*

$$u(\mathbf{x}) = \mathcal{S}(\mathbf{x};\overline{\psi}) + C,$$

where $\overline{\psi}$ is a particular solution of (3.120) and C is an arbitrary constant.

Another advantage of the method is that, in principle, exterior problems can be treated as the interior problems, with the same level of difficulty[23]. It is enough to use the exterior jump conditions (3.110), (3.112) and proceed in the same way (see Problem 3.19).

As an example of an elementary application of the method we solve the interior Neumann problem for the disc.

Example 3.51 (the Neumann problem for the disc) Let $B_R = B_R(\mathbf{0}) \subset \mathbb{R}^2$ and consider the Neumann problem

$$\begin{cases} \Delta u = 0 & \text{in } B_R \\ \partial_\nu u = g & \text{on } \partial B_R, \end{cases}$$

where $g \in C(\partial B_R)$ satisfies the solvability condition (3.118). We know that u is unique up to an additive constant. We want to express the solution as a single layer potential:

$$u(\mathbf{x}) = -\frac{1}{2\pi}\int_{\partial B_R} \psi(\boldsymbol{\sigma}) \log|\mathbf{x}-\boldsymbol{\sigma}|\, d\sigma. \tag{3.122}$$

The Neumann condition $\partial_\nu u = g$ on ∂B_R translates into the following integral equation for the density ψ:

$$-\frac{1}{2\pi}\int_{\partial B_R} \frac{(\mathbf{z}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\mathbf{z})}{|\mathbf{z}-\boldsymbol{\sigma}|^2}\psi(\boldsymbol{\sigma})d\sigma + \frac{1}{2}\psi(\mathbf{z}) = g(\mathbf{z}), \qquad \mathbf{z}\in\partial B_R. \tag{3.123}$$

On ∂B_R we have $\boldsymbol{\nu}(\mathbf{z}) = \mathbf{z}/R$, $(\mathbf{z}-\sigma)\cdot\mathbf{z} = R^2 - \mathbf{z}\cdot\boldsymbol{\sigma}$ and moreover, $|\mathbf{z}-\boldsymbol{\sigma}|^2 = 2(R^2 - \mathbf{z}\cdot\boldsymbol{\sigma})$. Hence, (3.123) becomes

$$-\frac{1}{4\pi R}\int_{\partial B_R} \psi(\boldsymbol{\sigma})d\sigma + \frac{1}{2}\psi(\mathbf{z}) = g(\mathbf{z}), \qquad \mathbf{z}\in\partial B_R. \tag{3.124}$$

The solutions of the homogeneous equation ($g = 0$) are constant functions $\psi_0(\mathbf{x}) = C$. A particular solution $\bar{\psi}$, with

[23] Some care is required when solving the exterior Dirichlet problem via a double layer potential, due to the rapid vanishing of its kernel.

$$\int_{\partial B_R} \bar{\psi}(\boldsymbol{\sigma}) d\sigma = 0,$$

is given by $\bar{\psi}(\mathbf{z}) = 2g(\mathbf{z})$.

Thus, the general solution of (3.124) is

$$\psi(\mathbf{z}) = 2g(\mathbf{z}) + C,$$

where C is an arbitrary constant, and the solution of the Neumann problem is given, up to an additive constant, by

$$u(\mathbf{x}) = -\frac{1}{\pi} \int_{\partial B_R} g(\boldsymbol{\sigma}) \log |\mathbf{x} - \boldsymbol{\sigma}| \, d\sigma.$$

The integral equations (3.116) and (3.120) are of the form

$$\int_{\partial\Omega} K(\mathbf{z}, \boldsymbol{\sigma}) \rho(\boldsymbol{\sigma}) d\sigma \pm \frac{1}{2} \rho(\mathbf{z}) = g(\mathbf{z}) \tag{3.125}$$

which are called *Fredholm integral equations of the first kind*. Their solution is based on the following so called **Fredholm alternative**: *either equation (3.125) has exactly one solution for every $g \in C(\partial\Omega)$, or the homogeneous equation*

$$\int_{\partial\Omega} K(\mathbf{z}, \boldsymbol{\sigma}) \phi(\boldsymbol{\sigma}) d\sigma \pm \frac{1}{2} \phi(\mathbf{z}) = 0$$

has a finite number $\phi_1, \ldots, \phi_N$ of nontrivial, linearly independent solutions.

In this last case equation (3.125) is not always solvable and we have:

a) The **adjoint** homogeneous equation

$$\int_{\partial\Omega} K(\boldsymbol{\sigma}, \mathbf{z}) \phi^*(\boldsymbol{\sigma}) d\sigma \pm \frac{1}{2} \phi^*(\mathbf{z}) = 0$$

has N nontrivial linearly independent solutions $\phi_1^*, \ldots, \phi_N^*$.

b) Equation (3.125) has a solution if and only if g satisfies the following N compatibility conditions:

$$\int_{\partial\Omega} \phi_j^*(\boldsymbol{\sigma}) g(\boldsymbol{\sigma}) d\sigma = 0, \quad j = 1, \ldots, N. \tag{3.126}$$

c) If g satisfies (3.126), the general solution of (3.125) is given by

$$\rho = \overline{\rho} + C_1 \phi_1 + \ldots C_N \phi_N$$

where $\overline{\rho}$ is a particular solution of equation (3.125) and $C_1, \ldots, C_N$ are arbitrary real constants.

The analogy with the solution of a system of linear algebraic equations should be evident. We will come back to Fredholm's alternative in Chap. 6.

3.9.3 Proof of Theorem 3.47

If $\mathbf{x} \notin \partial\Omega$ there is no problem in differentiating under the integral sign and, for $\boldsymbol{\sigma}$ fixed on $\partial\Omega$, the function

$$\mathbf{x} \longmapsto \partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma}) = \nabla_{\mathbf{y}}\Phi(\mathbf{x}-\boldsymbol{\sigma})\cdot\boldsymbol{\nu}(\boldsymbol{\sigma})$$

is harmonic. Thus $\mathcal{D}(\mathbf{x};\boldsymbol{\mu})$ is harmonic in $\mathbb{R}^n \setminus \partial\Omega$. With the same proof of Gauss Lemma 3.46, we can show that $\mathcal{D}(\mathbf{x};\boldsymbol{\mu})$ is well defined also for $\mathbf{x} \in \partial\Omega$.

Consider (3.111). The proof of (3.110) is similar. Observe that we cannot take the limit under the integral sign, once more because of the critical singularity of $\partial_{\boldsymbol{\nu}}\Phi(\mathbf{z}-\boldsymbol{\sigma})$ when $\mathbf{z} \in \partial\Omega$.

Let $\mathbf{x} \to \mathbf{z}$ with $\mathbf{x} \in \Omega$. Let $\Gamma_{\mathbf{z}} \subset \partial\Omega$ be a small neighborhood of $\mathbf{z}$ and write

$$\mathcal{D}(\mathbf{x};\boldsymbol{\mu}) = \int_{\partial\Omega\setminus\Gamma_{\mathbf{z}}} \partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})\boldsymbol{\mu}(\boldsymbol{\sigma})d\sigma + \int_{\Gamma_{\mathbf{z}}} \partial_{\boldsymbol{\nu}}\Phi(\mathbf{x}-\boldsymbol{\sigma})\boldsymbol{\mu}(\boldsymbol{\sigma})d\sigma = I + II.$$

The integrand in I does not have any singularity, so

$$I \to \int_{\partial\Omega\setminus\Gamma_{\mathbf{z}}} \partial_{\boldsymbol{\nu}}\Phi(\mathbf{z}-\boldsymbol{\sigma})\boldsymbol{\mu}(\boldsymbol{\sigma})d\sigma \quad \text{as } \mathbf{x} \to \mathbf{z}. \tag{3.127}$$

We can suppose that $\Gamma_{\mathbf{z}}$ is represented in local coordinates by the graph of a C^2-function $\varphi = \varphi(\mathbf{y}')$ with $\mathbf{y}' \in B_\rho(\mathbf{0}') \subset \mathbb{R}^{n-1}$. We choose the reference frame to have

$$\mathbf{z} = (\mathbf{0}', 0),\ \varphi(\mathbf{0}') = 0,\ \nabla\varphi(\mathbf{0}') = \mathbf{0}'$$

and for $\mathbf{x}$ **near** $\mathbf{z}$, $\mathbf{x} \in \Omega$, $\mathbf{x} = (\mathbf{x}', \varphi(\mathbf{x}') + h)$, $h > 0$. Thus, since the unit exterior normal at $(\mathbf{y}', \varphi(\mathbf{y}')) \in \Gamma_{\mathbf{z}}$ is

$$\boldsymbol{\nu}(\mathbf{y}') = (\nabla\varphi(\mathbf{y}'), -1)/\sqrt{1+|\nabla\varphi(\mathbf{y}')|^2},$$

we have

$$II = \frac{1}{4\pi}\int_{B_\rho(\mathbf{0}')} \frac{(\mathbf{x}'-\mathbf{y}')\cdot\nabla\varphi(\mathbf{y}') - [\varphi(\mathbf{x}')+h-\varphi(\mathbf{y}')]}{\left[|\mathbf{x}'-\mathbf{y}'|^2 + |\varphi(\mathbf{x}')+h-\varphi(\mathbf{y}')|^2\right]^{3/2}} \boldsymbol{\mu}(\mathbf{y}', \varphi(\mathbf{y}'))d\mathbf{y}'$$

We split II by writing

$$II = II_1 + II_2$$

where

$$II_1 = \frac{1}{4\pi}\int_{B_\rho(\mathbf{0}')} \frac{(\mathbf{x}'-\mathbf{y}')\cdot\nabla\varphi(\mathbf{y}') - [\varphi(\mathbf{x}')-\varphi(\mathbf{y}')]}{\left[|\mathbf{x}'-\mathbf{y}'|^2 + |\varphi(\mathbf{x}')+h-\varphi(\mathbf{y}')|^2\right]^{3/2}} \boldsymbol{\mu}(\mathbf{y}', \varphi(\mathbf{y}'))d\mathbf{y}'$$

and

$$II_2 = -\frac{h}{4\pi}\int_{B_\rho(\mathbf{0}')} \frac{\boldsymbol{\mu}(\mathbf{y}', \varphi(\mathbf{y}'))}{\left[|\mathbf{x}'-\mathbf{y}'|^2 + |\varphi(\mathbf{x}')+h-\varphi(\mathbf{y}')|^2\right]^{3/2}} d\mathbf{y}'.$$

Since φ is a C^2-function, we have

$$|(\mathbf{x}'-\mathbf{y}')\cdot\nabla\varphi(\mathbf{y}') - [\varphi(\mathbf{x}')-\varphi(\mathbf{y}')]| \le C\,|\mathbf{x}'-\mathbf{y}'|^2$$

whence, if $|\mathbf{x}'| \le \rho$,

$$|II_1| \le \frac{C\max|\mu|}{4\pi}\int_{B_\rho(\mathbf{0}')}\frac{1}{|\mathbf{x}'-\mathbf{y}'|}d\mathbf{y}' \le \frac{C\max|\mu|}{4\pi}\int_{B_{2\rho}(\mathbf{x}')}\frac{1}{|\mathbf{x}'-\mathbf{y}'|}d\mathbf{y}' = \\ = C\max|\mu|\,\rho.$$

Thus, by Lemma 3.52 below, we deduce that

$$II_1 \to \frac{1}{4\pi}\int_{B_\rho(\mathbf{0}')}\frac{-\mathbf{y}'\cdot\nabla\varphi(\mathbf{y}')-\varphi(\mathbf{y}')}{\left[\mathbf{y}'^2+|\varphi(\mathbf{y}')|^2\right]^{3/2}}\mu(\mathbf{y}',\varphi(\mathbf{y}'))d\mathbf{y}' = \int_{\Gamma_\mathbf{z}}\partial_\nu\Phi(\mathbf{z}-\sigma)\mu(\sigma)d\sigma \quad (3.128)$$

as $\mathbf{x}\to\mathbf{0}$. Observe that **(3.127) and (3.128) imply that** $\mathcal{D}(\mathbf{x};\mu)\in C(\partial\Omega)$.

We now show that

$$II_2 \to -\frac{\mu(\mathbf{0})}{2} \qquad \text{as } \mathbf{x}\to\mathbf{z}. \quad (3.129)$$

It is enough to compute the limit of II_2 as $\mathbf{x}\to\mathbf{z}$ along the normal direction at $\mathbf{z}$, that is for $\mathbf{x}'=0$ and $h\to 0^+$, and show that[24]

$$\lim_{h\to0^+} II_2(\mathbf{0}',h) = \lim_{h\to0^+} -\frac{h}{4\pi}\int_{B_\rho(\mathbf{0}')}\frac{\mu(\mathbf{y}',\varphi(\mathbf{y}'))}{\left[|\mathbf{y}'|^2+(h-\varphi(\mathbf{y}'))^2\right]^{3/2}}d\mathbf{y}' = -\frac{\mu(\mathbf{0})}{2}.$$

Set $\mathbf{y}'=h\mathbf{t}$. Then $d\mathbf{y}'=h^2d\mathbf{t}$ and

$$II_2(\mathbf{0}',h) = -\frac{1}{4\pi}\int_{B_{\rho/h}(\mathbf{0}')}\frac{\mu(h\mathbf{t},\varphi(h\mathbf{t}))}{\left[|\mathbf{t}|^2+|1-\varphi(h\mathbf{t})/h|^2\right]^{3/2}}d\mathbf{t}.$$

Using the Mean Value Theorem,

$$\frac{\varphi(h\mathbf{t})}{h} = \nabla\varphi(\widetilde{h}\mathbf{t})\cdot\mathbf{t}$$

with $0<\widetilde{h}<h$. Therefore, since $\nabla\varphi(\mathbf{0}')=\mathbf{0}'$, if h is sufficiently small, we have

$$\left|\nabla\varphi(\widetilde{h}\mathbf{t})\cdot\mathbf{t}\right| \le 1/2 \quad \text{if } |\mathbf{t}|\le 1.$$

Thus we can write

$$\left[|\mathbf{t}|^2+\left|1-\nabla\varphi(\widetilde{h}\mathbf{t})\cdot\mathbf{t}\right|^2\right]^{-3/2} \le f(\mathbf{t}) = \begin{cases} 8 & \text{if } |\mathbf{t}|\le 1 \\ |\mathbf{t}|^{-3} & \text{if } |\mathbf{t}|\ge 1. \end{cases}$$

Therefore the integrand in II_2 is bounded by the integrable function $Mf(\mathbf{t})$, where $M=\max|\mu|$. Then, the Dominated Convergence Theorem gives

[24] Indeed, for $\mathbf{x}$ near $\mathbf{z}$, let $\mathbf{x}_t$ be the projection of $\mathbf{x}$ on $\partial\Omega$. Then write

$$\mathcal{D}(\mathbf{x};\mu)-\mathcal{D}(\mathbf{z};\mu)-\frac{\mu(\mathbf{z})}{2} \\ = \left(\mathcal{D}(\mathbf{x};\mu)-\mathcal{D}(\mathbf{x}_t;\mu)-\frac{\mu(\mathbf{x}_t)}{2}\right)+\left(\frac{\mu(\mathbf{x}_t)}{2}-\frac{\mu(\mathbf{z})}{2}\right)+\left(\mathcal{D}(\mathbf{x}_t;\mu)-\mathcal{D}(\mathbf{z};\mu)\right).$$

By the continuity of μ and $\mathcal{D}(\mathbf{x};\mu)$ along $\partial\Omega$, we can fix $\mathbf{x}_t$ such that the last two terms are arbitrarily small. Then we can move along the normal to $\mathbf{x}_t$, reasoning as in the proof of (3.129), to prove that the first term is small too.

$$II_2(\mathbf{0}', h) \to -\frac{1}{4\pi}\int_{\mathbb{R}^2} \frac{\mu(\mathbf{0})}{\left[|\mathbf{t}|^2+1\right]^{3/2}} d\mathbf{t} = -\frac{\mu(\mathbf{0})}{2} \qquad \text{as } h \to 0^+. \tag{3.130}$$

From (3.127), (3.128), (3.130), we get (3.111). □

Lemma 3.52 *Let $D \subset \mathbb{R}^n$ be a bounded domain and*

$$F(\mathbf{x}) = \int_D g(\mathbf{x}, \mathbf{y}) d\mathbf{y} \qquad (\mathbf{x} \in D),$$

where $g = g(\mathbf{x}, \mathbf{y})$ is continuous in $(\overline{D} \times \overline{D}) \setminus \{\mathbf{x} = \mathbf{y}\}$ and the integral converges absolutely ($g(\mathbf{x}, \cdot) \in L^1(D)$). Assume that:

for every $\varepsilon > 0$, there exists $\rho = \rho(\varepsilon)$ such that, if $\mathbf{x} \in B_\rho \subset D$,

$$\left| \int_{B_\rho} g(\mathbf{x}, \mathbf{y}) d\mathbf{y} \right| \le \varepsilon.$$

Then F is (uniformly) continuous in D.

Proof Fix $\varepsilon > 0$ and $\rho = \rho(\varepsilon)$ as in the hypothesis of the lemma. Let $\mathbf{x}, \mathbf{z} \in B_\rho \subset D$. Write

$$F(\mathbf{x}) - F(\mathbf{z}) = \int_{D \setminus B_\rho} [g(\mathbf{x}, \mathbf{y}) - g(\mathbf{z}, \mathbf{y})] d\mathbf{y} + \int_{B_\rho} [g(\mathbf{x}, \mathbf{y}) - g(\mathbf{z}, \mathbf{y})] d\mathbf{y}.$$

The first integral goes to 0 as $|\mathbf{x} - \mathbf{z}| \to 0$ while the second one is less than or equal to 2ε. □

Problems

3.1 Let B_R be the unit disc centered at $(0, 0)$. Use the method of separation of variables to solve the problem

$$\begin{cases} \Delta u = f & \text{in } B_R \\ u = 1 & \text{on } \partial B_R. \end{cases}$$

Find an explicit formula when $f(x, y) = y$.

[*Hint.* Use polar coordinates; expand $f = f(r, \cdot)$ in sine Fourier series in $[0, 2\pi]$ and derive a set of ordinary differential equations for the Fourier coefficients of $u(r, \cdot)$.]

3.2 Let $C_{1,2} = \left\{(r, \theta) \in \mathbb{R}^2; 1 < r < 2\right\}$. Examine the solvability of the Neumann problem

$$\begin{cases} \Delta u = -1 & \text{in } C_{1,2} \\ \partial_\nu u = \cos\theta & \text{on } r = 1 \\ \partial_\nu u = \lambda(\cos\theta)^2 & \text{on } r = 2 \end{cases} \qquad (\lambda \in \mathbb{R})$$

and write an explicit formula for the solution, when it exists.

3.3 *(Schwarz reflection principle)* a) Let

$$B_1^+ = \left\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 < 1, y > 0\right\}$$

and $u \in C^2(B_1^+) \cap C(\overline{B_1^+})$, harmonic B_1^+, $u(x, 0) = 0$. Show that the function

$$U(x,y)=\begin{cases} u(x,y) & y\geq 0\\ -u(x,-y) & y<0,\end{cases}$$

obtained from u by odd reflection with respect to y, is harmonic in all B_1.
b) State and prove the Schwarz reflection principle in dimension three.

[*Hint.* a) Let v be the solution of $\Delta v=0$ in B_1, $v=U$ on ∂B_1. Define $w(x,y)=v(x,y)+v(x,-y)$ and show that $w\equiv 0$.]

3.4 Let $\Omega\subseteq\mathbb{R}^n$ be a smooth bounded domain. Let $u\in C^2(\Omega)\cap C^1(\overline{\Omega})$ be a solution of the Robin problem

$$\begin{cases}\Delta u=0, & \text{in } \Omega\\ \partial_\nu u+\alpha u=g & \text{on } \partial\Omega,\end{cases}$$

where $\alpha>0$ and $g\in C(\partial\Omega)$. Show that if $g\geq 0(\leq 0)$ on $\partial\Omega$, then $u\geq 0(\leq 0)$ in $\overline{\Omega}$.

3.5 Let $\Omega\subseteq\mathbb{R}^n$ be a domain and u be harmonic in Ω. Use the analyticity Theorem 3.18, p. 134, to show that, if u has a **local** maximum or minimum in Ω, then it is constant.

3.6 Let u be harmonic in $\mathbb{R}^3$ such that

$$\int_{\mathbb{R}^3}|u(\mathbf{x})|^2\,d\mathbf{x}<\infty.$$

Show that $u\equiv 0$.

[*Hint.* Write the mean value formula in a ball $B_R(\mathbf{0})$ for u. Use the Schwarz inequality and let $R\to+\infty$.]

3.7 Let u be harmonic in $\mathbb{R}^n$ and M an orthogonal matrix of order n. Using the mean value property, show that $v(\mathbf{x})=u(\mathbf{Mx})$ is harmonic in $\mathbb{R}^n$.

3.8 Let $u\in C(\Omega)$, $\Omega\subseteq\mathbb{R}^n$. Using the notations of Sect. 3.4.1, prove the following statements:

a) For all $B_r(\mathbf{x})\subset\subset\Omega$, $u(\mathbf{x})\leq S(u;\mathbf{x},r)$ if and only if $u(\mathbf{x})\leq A(u;\mathbf{x},r)$.
b) If $u\in C(\Omega)$ is subharmonic in Ω then $r\longmapsto S(u;\mathbf{x},r)$ and $r\longmapsto A(u;\mathbf{x},r)$ are nondecreasing functions.
c) If u is harmonic in Ω then u^2 is subharmonic in Ω.
d) If $u\in C^2(\Omega)$, u is subharmonic if and only if $\Delta u\geq 0$ in Ω.

Moreover:

e) Let u be subharmonic in Ω and $F:\mathbb{R}\to\mathbb{R}$, smooth. Show that if F is convex and increasing, then $F\circ u$ subharmonic

[*Hint.* a) Let $u(\mathbf{x})\leq S(u;\mathbf{x},s)$. Then $\omega_n s^{n-1}u(\mathbf{x})\leq\int_{\partial B_s(\mathbf{x})}u(\sigma)d\sigma$. Integrating over $(0,r)$ we get $u(\mathbf{x})\leq A(u;\mathbf{x},r)$. Now let $u(\mathbf{x})\leq A(u;\mathbf{x},s)$. Let $v=P(u,B_r(\mathbf{x}))$ be the harmonic lifting of u in $B_r(\mathbf{x})$. Then, by maximum principle, $u(\mathbf{x})\leq v(\mathbf{x})=S(u;\mathbf{x},r)$.
b) Let $s<r$, $B_s(\mathbf{x})\subset B_r(\mathbf{x})\subset\subset\Omega$ and $h=P(u,B_r(\mathbf{x}))$. Then $u\leq h$ in $B_r(\mathbf{x})$ and therefore

$$S(u;\mathbf{x},s)\leq S(h;\mathbf{x},s)=h(\mathbf{x})=S(h;\mathbf{x},r),\quad A(u;\mathbf{x},s)\leq A(h;\mathbf{x},s)=h(\mathbf{x})=A(h;\mathbf{x},r),$$

which is b).]

3.9 Assume that u is superharmonic in Ω and that v is superharmonic in the subdomain $\Omega'\subset\Omega$. Prove that, if $u\leq v$ on $\partial\Omega'$, then

$$w(\mathbf{x})=\begin{cases}\min\{u(\mathbf{x}),v(\mathbf{x})\} & \text{in } \Omega'\\ u(\mathbf{x}) & \text{in } \Omega\setminus\Omega'\end{cases}$$

is superharmonic in Ω.

[*Hint.* Argue as in the proof of property 3, p. 136, using v and ω instead of $P(u,B)$ and B.]

3.10 Let Ω be a bounded domain and $u \in C(\Omega)$.

a) Let u be subharmonic and bounded. Assume that, for every $\mathbf{p} \in \partial\Omega \setminus \{\mathbf{p}_1, \dots, \mathbf{p}_N\}$, $\lim_{\mathbf{x}\to\mathbf{p}} u(\mathbf{x}) \le 0$. Prove that $u \le 0$ in Ω.
b) Deduce that if u is harmonic and bounded in Ω and $\lim_{\mathbf{x}\to\mathbf{p}} u(\mathbf{x}) = 0$ for every $\mathbf{p} \in \partial\Omega \setminus \{\mathbf{p}_1, \dots, \mathbf{p}_N\}$, then $u \equiv 0$ in Ω.

[*Hint.* a) $(n > 2)$ Let $w(\mathbf{x}) = u(\mathbf{x}) - \varepsilon \sum_{j=1}^{N} |\mathbf{x} - \mathbf{p}_j|^{2-n}$, $\varepsilon > 0$. Observe that w is subharmonic in Ω and there exist N balls $B_{\rho_j}(\mathbf{p}_j)$, with ρ_j small, such that $w < 0$ in each $B_{\rho_j}(\mathbf{p}_j) \cap \Omega$. Moreover, on the boundary of $\Omega \setminus \cup_{j=1}^{N} B_{\rho_j}(\mathbf{p}_j)$, $w \le 0$. Conclude by using the maximum principle.]

3.11 a) Let $u : \Omega \subseteq \mathbb{R}^n \to \mathbb{R}$, be radially symmetric, i.e. $u(\mathbf{x}) = u(r)$, $r = |\mathbf{x}|$. Show that

$$\Delta u = u_{rr} + \frac{n-1}{r} u_r .$$

b) Find all radial solutions of $\Delta u = 0$ in $\mathbb{R}^n \setminus \{0\}$, $n \ge 3$.

[*Answer.* b) $u(\mathbf{x}) = A + B|\mathbf{x}|^{2-n}$, with $A, B \in \mathbb{R}$.]

3.12 Let $f \in C^2(\mathbb{R}^2)$ with compact support K and

$$u(\mathbf{x}) = -\frac{1}{2\pi} \int_{\mathbb{R}^2} \log|\mathbf{x} - \mathbf{y}| \; f(\mathbf{y}) d\mathbf{y}.$$

Show that

$$u(\mathbf{x}) = -\frac{M}{2\pi} \log|\mathbf{x}| + O(|\mathbf{x}|^{-1}), \qquad \text{as } |\mathbf{x}| \to +\infty$$

where $M = \int_{\mathbb{R}^2} f(\mathbf{y}) d\mathbf{y}$.

[*Hint.* Write $\log|\mathbf{x} - \mathbf{y}| = \log(|\mathbf{x} - \mathbf{y}| / |\mathbf{x}|) + \log|\mathbf{x}|$ and show that, assuming $\mathbf{y} \in K$, then $|\log(|\mathbf{x} - \mathbf{y}| / |\mathbf{x}|)| \le C/|\mathbf{x}|$.]

3.13 Prove the representation formula (3.72) in dimension two.

3.14 Compute the Green function for the disc of radius R.

[*Answer.*

$$G(\mathbf{x}, \mathbf{y}) = -\frac{1}{2\pi} \left[\log|\mathbf{x} - \mathbf{y}| - \log\left(\frac{|\mathbf{x}|}{R} |\mathbf{x}^* - \mathbf{y}| \right) \right],$$

where $\mathbf{x}^* = R^2 \mathbf{x} |\mathbf{x}|^{-2}$, $\mathbf{x} \neq \mathbf{0}$. Moreover $G(0, \mathbf{y}) = -\frac{1}{2\pi}(\log|\mathbf{y}| - \log R)$.]

3.15 Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain and G be the Green function in Ω. Prove that, for every $\mathbf{x}, \mathbf{y} \in \Omega$, $\mathbf{x} \neq \mathbf{y}$:

a) $G(\mathbf{x}, \mathbf{y}) > 0$.
b) $G(\mathbf{x}, \mathbf{y}) = G(\mathbf{y}, \mathbf{x})$.
c) $G(\mathbf{x}, \mathbf{y}) \le \Phi(\mathbf{x} - \mathbf{y})$.

[*Hint.* a) Let $B_r(\mathbf{x}) \subset \Omega$ and let w be the harmonic function in $\Omega \setminus \overline{B}_r(\mathbf{x})$ such that $w = 0$ on $\partial\Omega$ and $w = 1$ on $\partial B_r(\mathbf{x})$. Show that, for every r small enough, $G(\mathbf{x}, \cdot) > w(\cdot)$ in $\Omega \setminus \overline{B}_r(\mathbf{x})$.
b) For fixed $\mathbf{x} \in \Omega$, define $w_1(\mathbf{y}) = G(\mathbf{x}, \mathbf{y})$ and $w_2(\mathbf{y}) = G(\mathbf{y}, \mathbf{x})$. Apply Green's identity (3.73) in $\Omega \setminus B_r(\mathbf{x})$ to w_1 and w_2. Let $r \to 0$.]

3.16 a) Determine the Green function for the half plane $\mathbb{R}^2_+ = \{(x, y); y > 0\}$ and (formally) derive the Poisson formula

$$u(x, y) = \frac{1}{\pi} \int_{\mathbb{R}} \frac{y}{(x - \xi)^2 + y^2} u(\xi, 0) d\xi$$

for a bounded harmonic function u in $\mathbb{R}^2_+$.

b) Check that if $g(\xi) = u(\xi, 0)$ is continuous and bounded in $\mathbb{R}$, $u(x, y) \to g(x_0)$ if $(x, y) \to (x_0, 0)$.

[*Hint.* Note that $\frac{1}{\pi}\int_{\mathbb{R}} \frac{y}{\xi^2 + y^2} d\xi = 1$, $\forall y > 0$, and write

$$u(x, y) - g(x_0) = \frac{1}{\pi}\int_{\mathbb{R}} \frac{y}{\xi^2 + y^2}(g(x - \xi) - g(x_0))d\xi.$$

Fix $\varepsilon > 0$, and let $|g(x - \xi) - g(x_0)| \le \varepsilon$ if $|\xi| \le \delta$. Set

$$I + II = \frac{1}{\pi}\int_{|\xi| \ge \delta} \frac{y}{\xi^2 + y^2}(g(x - \xi) - g(x_0))d\xi + \frac{1}{\pi}\int_{|\xi| \le \delta} \frac{y}{\xi^2 + y^2}(g(x - \xi) - g(x_0))d\xi.$$

Show that $II \to 0$ as $y \to 0$ and that $|II| \le C\varepsilon$ if $|x - x_0|$ is small.]

3.17 Prove that the exterior Dirichlet problem in the plane has a unique bounded solution $u \in C^2(\Omega_e) \cap C(\overline{\Omega}_e)$, through the following steps. Let w be the difference of two solutions. Then w is harmonic Ω_e, vanishes on $\partial\Omega_e$ and is bounded, say $|w| \le M$.

a) Assume that the $\mathbf{0} \in \Omega$. Let $B_a(\mathbf{0})$ and $B_R(\mathbf{0})$ such that $B_a(\mathbf{0}) \subset \Omega \subset B_R(\mathbf{0})$ and define

$$u_R(\mathbf{x}) = M\frac{\log|\mathbf{x}| - \log|a|}{\log R - \log a}.$$

Use the maximum principle to show that $w \le u_R$ in $C_{a,R} = \{x \in R^2; a < |\mathbf{x}| < R\} \cap \Omega$.

b) Let $R \to \infty$ and deduce that $w \le 0$ in Ω_e.

c) Proceed similarly to show that $w \ge 0$ in Ω_e.

3.18 Find the Poisson formula for the circle B_R, by representing the solution of $\Delta u = 0$ in B_R, $u = g$ on ∂B_R, as a double layer potential.

3.19 Consider the following exterior Neumann-Robin problem in $\mathbb{R}^3$:

$$\begin{cases} \Delta u = 0 & \text{in } \Omega_e \\ \partial_\nu u + ku = g & \text{on } \partial\Omega_e, (k \ge 0) \\ u \to 0 & \text{as } |\mathbf{x}| \to \infty. \end{cases} \tag{3.131}$$

a) Let $k = 0$ and $|u(\mathbf{x})| \le |\mathbf{x}|^{-1-\varepsilon}$ for $|\mathbf{x}|$ large, with $\varepsilon > 0$. Show that the condition $\int_{\partial\Omega} g d\sigma = 0$ is necessary for the solvability of (3.131).

b) Let $k \ge 0$. Represent the solution as a single layer potential and derive the integral equations for the unknown density.

[*Hint.* a) Show that, for R large, $\int_{\partial\Omega} g\, d\sigma = \int_{\{|\mathbf{x}|=R\}} \partial_\nu u\, d\sigma$. Then, use the proof of Theorem 3.44, p. 164, and let $R \to \infty$.]

3.20 *(An ill-posed Robin problem)* Show that $u(x_1, x_2) = x_1$ is a solution of $\Delta u = 0$ in $B = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 < 1\}$ with $\partial_\nu u - u = 0$ on ∂B.

3.21 Solve (formally) the Neumann problem in the half space $\mathbb{R}^3_+$ in the form of a single layer potential

$$u(\mathbf{x}) = \frac{1}{4\pi}\int_{\mathbb{R}^2} \frac{\mu(y_1, y_2)}{[(x_1 - y_1)^2 + (x_2 - y_2)^2 + x_3^2]^{1/2}} dy_1 dy_2.$$

3.22 Let $B = B_1(\mathbf{0}) \subset \mathbb{R}^2$. To complete the proof of Theorem 3.13 we must show that, if $g \in C(\partial B)$ and u is given by formula (3.23), with $R = 1$ and $p = 0$, then

$$\lim_{\mathbf{x} \to \boldsymbol{\xi}} u(\mathbf{x}) = g(\boldsymbol{\xi}) \quad \text{for every } \boldsymbol{\xi} \in \partial B.$$

Fill in the details in the following steps and conclude the proof.

a) First show that

$$\frac{1-|\mathbf{x}|^2}{2\pi}\int_{\partial B}\frac{1}{|\mathbf{x}-\boldsymbol{\sigma}|^2}d\sigma = 1$$

and, therefore, that

$$u(\mathbf{x}) - g(\boldsymbol{\xi}) = \frac{1-|\mathbf{x}|^2}{2\pi}\int_{\partial B}\frac{g(\boldsymbol{\sigma})-g(\boldsymbol{\xi})}{|\mathbf{x}-\boldsymbol{\sigma}|^2}d\sigma.$$

b) For $\delta > 0$, write

$$\begin{aligned} u(\mathbf{x}) - g(\boldsymbol{\xi}) &= \frac{1-|\mathbf{x}|^2}{2\pi}\int_{\partial B\cap\{|\boldsymbol{\sigma}-\boldsymbol{\xi}|<\delta\}}\cdots d\sigma + \frac{1-|\mathbf{x}|^2}{2\pi}\int_{\partial B\cap\{|\boldsymbol{\sigma}-\boldsymbol{\xi}|>\delta\}}\cdots d\sigma \\ &\equiv I + II. \end{aligned}$$

Fix $\varepsilon > 0$ and use the continuity of g to show that, if δ is small enough, then $|I| < \varepsilon$.

c) Show that, if $|\mathbf{x}-\boldsymbol{\xi}| < \delta/2$ and $|\boldsymbol{\sigma}-\boldsymbol{\xi}| > \delta$, then $|\mathbf{x}-\boldsymbol{\sigma}| > \delta/2$ and therefore $\lim_{\mathbf{x}\to\boldsymbol{\xi}} II = 0$.

3.23 Consider the equation

$$Lu \equiv \Delta u + k^2 u = 0 \qquad \text{in } \mathbb{R}^3,$$

known as Helmoltz's or reduced wave equation.

a) Show that the radial solutions $u = u(r)$, $r = |\mathbf{x}|$, satisfying the outgoing Sommerfeld condition

$$u_r + iku = O\left(\frac{1}{r^2}\right) \qquad \text{as } r \to \infty,$$

are of the form

$$\varphi(r;k) = c\frac{e^{-ikr}}{r} \qquad c \in \mathbb{C}.$$

b) For f smooth and compactly supported in $\mathbb{R}^3$ define the potential

$$U(\mathbf{x}) = c_0\int_{\mathbb{R}^3} f(\mathbf{y})\frac{e^{-ik|\mathbf{x}-\mathbf{y}|}}{|\mathbf{x}-\mathbf{y}|}d\mathbf{y}.$$

Select the constant c_0 such that $LU(\mathbf{x}) = -f(\mathbf{x})$.

[*Answer.* b): $c_0 = (4\pi)^{-1}$.]

Chapter 4
Scalar Conservation Laws and First Order Equations

4.1 Introduction

In the main part of this chapter, from Sects. 4.1 to 4.6, we consider equations of the form

$$u_t + q(u)_x = 0, \qquad x \in \mathbb{R}, t > 0. \tag{4.1}$$

In general, $u = u(x,t)$ represents the *density* or the *concentration* of a physical quantity Q (e.g. a mass), $\mathbf{q} = q(u)\mathbf{i}$ is its *flux vector* and $q(u)$ is its *flux function*[1]. Equation (4.1) constitutes a *link* between density and flux and expresses a (**scalar**) **conservation law** for the following reason.

If we consider a control interval (x_1, x_2), the integral

$$\int_{x_1}^{x_2} u(x,t)dx$$

gives the amount of Q contained inside (x_1, x_2) at time t. A *conservation law* states that, without sources or sinks, the rate of change of Q inside (x_1, x_2) is determined by the net flux through the end points of the interval. If the flux is modelled by a vector $\mathbf{q} = q(u)\mathbf{i}$, the law translates into the equation

$$\frac{d}{dt}\int_{x_1}^{x_2} u(x,t)dx = -q(u(x_2,t)) + q(u(x_1,t)), \tag{4.2}$$

since $q > 0$ ($q < 0$) for a flux along the positive (negative) direction of the x axis. If u and q are smooth functions, eq. (4.2) can be rewritten in the form

$$\int_{x_1}^{x_2} [u_t(x,t) + q(u(x,t))_x]\, dx = 0$$

which implies (4.1), due to the arbitrariness of the interval (x_1, x_2).

[1] That is the rate of change of Q. The dimensions of q are $[\text{mass}] \times [\text{time}]^{-1}$.

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At this point we have to decide which type of flux function we are dealing with, or, in other words, we have to establish a *constitutive relation for* q.

In the next section we go back to the model of pollution in a channel, considered in Sect. 2.5.3, neglecting the diffusion and choosing for q a linear function of u, namely:

$$q(u) = vu,$$

where v is constant. The result is a *pure transport* model, in which the vector $v\mathbf{i}$ is the *convection velocity*. In the sequel, we shall use a nonlinear model from traffic dynamics, with speed v depending on u, to introduce and motivate some important concepts and methods, such as the *method of the characteristics.*

The conservation law (4.1) occurs, for instance, in 1-dimensional fluid dynamics where it often describes the formation and propagation of special solution waves, called *shock* and *rarefaction waves*. A shock wave undergoes a *jump discontinuity* and an important question is how to reinterpret the differential equation (4.1) in order to admit discontinuous solutions. This leads to the concept of solution in a generalized or weak sense.

A typical problem associated with equation (4.1) is the *initial value problem:*

$$\begin{cases} u_t + q(u)_x = 0 \\ u(x,0) = g(x) \end{cases} \tag{4.3}$$

where $x \in \mathbb{R}$. Sometimes x varies in a half-line or in a finite interval; as we shall see later, in these cases some other conditions have to be added to obtain a well posed problem.

The conservation law (4.1) is a particular case of first order quasilinear equations of the type ($u = u(x, y)$)

$$a(x,y,u)u_x + b(x,y,u)u_y = c(x,y,u),$$

that we consider in Sect. 4.7. In particular, for this kind of equations, we study the Cauchy problem, extending the method of the characteristics. Finally, in Sect. 4.8, we further generalize to fully nonlinear equations of the form

$$F(u_x, u_y, u, x, y) = 0,$$

concluding with a simple application to geometrical optics.

4.2 Linear Transport Equation

4.2.1 Pollution in a channel

Let us go back to the simple model for the evolution of a pollutant concentration in a narrow channel, considered in Sect. 2.5.3.

When diffusion and transport are both relevant, we have derived the equation

$$c_t = Dc_{xx} - vc_x,$$

where $c = c(x,t)$ is the concentration of the pollutant and $v\mathbf{i}$ is the stream velocity (v constant). We want to discuss here the case of the *pure transport* equation

$$c_t + vc_x = 0 \tag{4.4}$$

that is, when $D = 0$. Introducing the vector $\mathbf{v} = v\mathbf{i} + \mathbf{j}$, equation (4.4) can be written in the form

$$vc_x + c_t = \nabla c \cdot \mathbf{v} = 0,$$

pointing out the orthogonality of ∇c and $\mathbf{v}$. But ∇c is orthogonal to the level lines of c, along which c is constant. Therefore the level lines of c are the straight lines parallel to $\mathbf{v}$, of equation

$$x = vt + x_0.$$

These straight lines are called **characteristics**. Let us compute c along the characteristic $x = vt + x_0$, by letting

$$w(t) = c(x_0 + vt, t).$$

Since[2]

$$\dot{w}(t) = vc_x(x_0 + vt, t) + c_t(x_0 + vt, t),$$

from equation (4.4) we derive the *ordinary differential equation* $\dot{w}(t) = 0$, confirming that c is constant along the characteristic.

We want to determine the evolution of the concentration c, by knowing its initial profile

$$c(x, 0) = g(x). \tag{4.5}$$

The method to compute the solution at a point $(\bar{x}, \bar{t})$, $\bar{t} > 0$, is very simple. Let $x = vt + x_0$ be the equation of the characteristic passing through $(\bar{x}, \bar{t})$ (i.e., $x_0 = \bar{x} - v\bar{t}$).

Then, we move back in time along this characteristic, from $(\bar{x}, \bar{t})$ until the point $(x_0, 0)$ of intersection with the x-axis (see Fig. 4.1).

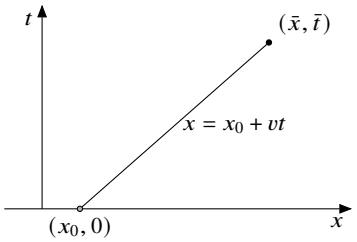

Fig. 4.1 Characteristic line for the linear transport problem

[2] The *dot* denotes derivative with respect to time.

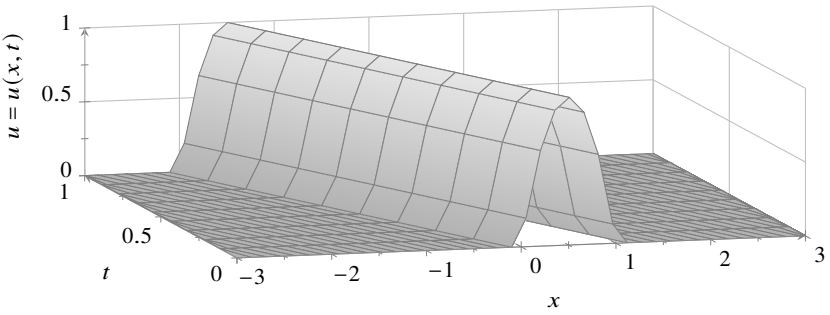

Fig. 4.2 Travelling wave solution of the linear transport equation

Since c is constant along the characteristic and $c(x_0, 0) = g(x_0)$, it must be

$$c(\bar{x}, \bar{t}) = g(x_0) = g(\bar{x} - v\bar{t}).$$

Thus, if $g \in C^1(\mathbb{R})$, the unique solution of the initial value problem (4.4), (4.5) is given by

$$c(x, t) = g(x - vt). \tag{4.6}$$

The solution (4.6) represents a *travelling wave*, moving with speed v in the positive (negative) x-direction if $v > 0$ ($v < 0$).

In Fig. 4.2, an initial profile $g(x) = 4(x - x^2)\chi_{[0,1]}(x)$ is *transported* along the straight lines $x + t =$ constant, i.e. with speed 1 in the negative x-direction.

4.2.2 Distributed source

In presence of an external distributed source along the channel of intensity $f = f(x, t)$ (measured in concentration per unit time), the conservation law (4.2) has to be changed into

$$\frac{d}{dt}\int_{x_1}^{x_2} c(x,t)dx = -q(c(x_2,t)) + q(c(x_1,t)) + \int_{x_1}^{x_2} f(x,t)dx. \tag{4.7}$$

In the pure transport case $q(c) = vc$, (4.7) leads to the nonhomogeneous equation

$$c_t + vc_x = f, \tag{4.8}$$

with the initial condition

$$c(x, 0) = g(x). \tag{4.9}$$

Again, to compute the value of the solution c at a point $(\bar{x}, \bar{t})$ is an easy task. Let $x = x_0 + vt$ be the characteristic passing through $(\bar{x}, \bar{t})$ and compute u along

this characteristic, setting $w(t) = c(x_0 + vt, t)$. From (4.8), w satisfies the ordinary differential equation

$$\dot{w}(t) = vc_x(x_0 + vt, t) + c_t(x_0 + vt, t) = f(x_0 + vt, t)$$

with the initial condition

$$w(0) = g(x_0).$$

Thus

$$w(t) = g(x_0) + \int_0^t f(x_0 + vs, s)ds.$$

Letting $t = \bar{t}$ and recalling that $x_0 = \bar{x} - v\bar{t}$, we get

$$c(\bar{x}, \bar{t}) = w(\bar{t}) = g(\bar{x} - v\bar{t}) + \int_0^t f(\bar{x} - v(\bar{t} - s), s)ds. \tag{4.10}$$

Since $(\bar{x}, \bar{t})$ is arbitrary, if g and f are reasonably smooth functions, (4.10) is the desired solution.

Proposition 4.1 *Let $g \in C^1(\mathbb{R})$ and $f, f_x \in C(\mathbb{R}\times(0, +\infty))$. The unique solution of the initial value problem*

$$\begin{cases} c_t + vc_x = f(x, t) & x \in \mathbb{R},\ t > 0 \\ c(x, 0) = g(x) & x \in \mathbb{R} \end{cases}$$

is given by the formula

$$c(x, t) = g(x - vt) + \int_0^t f(x - v(t - s), s)ds. \tag{4.11}$$

Remark 4.2 Formula (4.11) can be derived using the *Duhamel method*, as in Sect. 2.8.3, p. 80 (see Problem 4.1).

4.2.3 Extinction and localized source

We now consider two other situations, of practical interest in several contexts. Suppose that, due e.g. to *biological decomposition*, the pollutant concentration c decays at the rate

$$r(x, t) = -\gamma c(x, t), \qquad \gamma > 0.$$

Without external sources and diffusion, the mathematical model governing the evolution of c reads

$$c_t + vc_x = -\gamma c,$$

with the initial condition

$$c(x, 0) = g(x).$$

Setting

$$u(x,t) = c(x,t)e^{\frac{\gamma}{v}x}, \tag{4.12}$$

we have

$$u_x = \left(c_x + \frac{\gamma}{v}c\right)e^{\frac{\gamma}{v}x} \quad \text{and} \quad u_t = c_t e^{\frac{\gamma}{v}x}$$

and therefore the equation for u is

$$u_t + vu_x = 0$$

with the initial condition

$$u(x,0) = g(x)e^{\frac{\gamma}{v}x}.$$

From Proposition 4.1, we get

$$u(x,t) = g(x - vt)e^{\frac{\gamma}{v}(x-vt)}$$

and, from (4.12),

$$c(x,t) = g(x - vt)e^{-\gamma t}.$$

Thus, the solution takes the form of a *damped travelling wave.*

We now examine the effect of a source of pollutant placed at a certain point of the channel, e.g. at $x = 0$. Typically, one may think of waste material from industrial machineries. Before the machines start working, for instance before time $t = 0$, we assume that the channel is clean. We recall that, assuming $v > 0$, the stream transports the pollutant along the positive direction of the x axis.

We want to determine the pollutant concentration, supposing that, at the point $x = 0$, c is kept at a constant level $\beta > 0$, for $t > 0$. Then, we have the *boundary condition*

$$c(0,t) = \beta, \qquad \text{for } t > 0 \tag{4.13}$$

and the initial condition

$$c(x,0) = 0, \qquad \text{for } x > 0. \tag{4.14}$$

Thus the relevant domain for our problem is the first quadrant $x > 0, t > 0$.

As before, let $u(x,t) = c(x,t)e^{\frac{\gamma}{v}x}$, which is a solution of $u_t + vu_x = 0$. Then:

$$u(0,t) = c(0,t) = \beta \qquad \text{for } t > 0, \tag{4.15}$$

$$u(x,0) = c(x,0)e^{\frac{\gamma}{v}x} = 0 \qquad \text{for } x > 0. \tag{4.16}$$

Since u is constant along the characteristics u must be of the form

$$u(x,t) = u_0(x - vt) \tag{4.17}$$

where u_0 is to be determined from the boundary condition (4.15) and the initial condition (4.16).

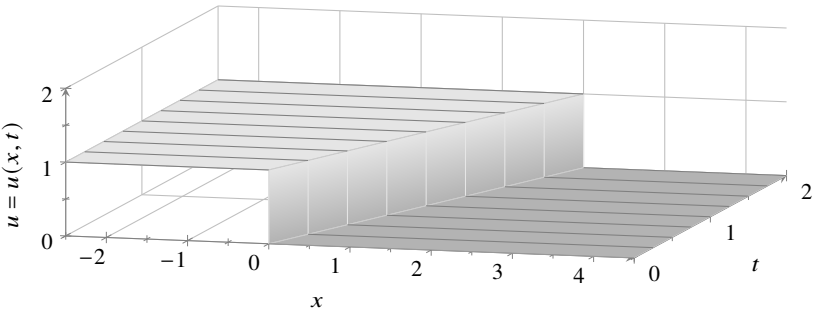

Fig. 4.3 Propagation of a discontinuity

Now, a characteristic $x = vt + x_0$, $x_0 > 0$, leaving from the x-axis, carries *zero data* and hence we deduce that $u = 0$ in all the right sector $x > vt$. This means that the pollutant has not yet reached the point x, at time t, if $x > vt$.

To compute u for $x < vt$, observe that a characteristic leaving the t-axis, i.e. from a point $(0, t_0)$, carries the datum β. Therefore, $u = \beta$ in the upper sector $x < vt$.

Recalling (4.12), we can write our original solution in the compact form

$$c(x,t) = \beta \mathcal{H}(vt - x) e^{-\frac{\gamma}{v}x}$$

where $\mathcal{H}$ is the Heaviside function.

Observe that in $(0, 0)$ there is a jump discontinuity which is *transported along the straight line* $x = vt$. Figure 4.3 shows the solution for $\beta = 3$, $\gamma = 0.7$, $v = 2$.

Thus c is not C^1, and we need to clarify in which sense it is a solution of the equation. We will be back on this question later, from a more general point of view.

4.2.4 Inflow and outflow characteristics. A stability estimate

The domain in the localized source problem is the quadrant $x > 0, t > 0$. To uniquely determine the solution we have used the initial data on the x-axis, $x > 0$, and the boundary data on the t-axis, $t > 0$. The problem is therefore well posed. This is due to the fact that, since $v > 0$, when time increases, *all* the characteristics carry the information (the data) from the boundary of the quadrant *towards its interior* $x > 0$, $t > 0$. We say in this case that the characteristics are **inflow characteristics**.

More generally, consider the equation

$$u_t + a u_x = f(x,t)$$

in the domain $x > 0$, $t > 0$, where a is a constant ($a \neq 0$). The characteristics are the lines

$$x - at = \text{constant}$$

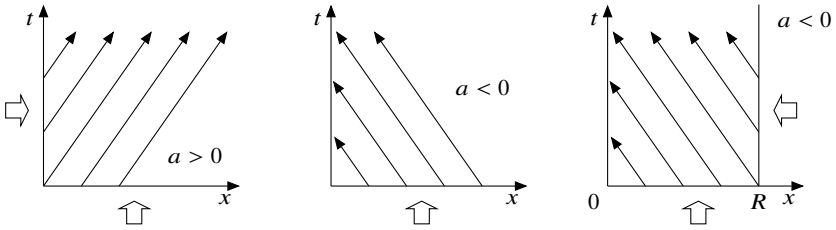

Fig. 4.4 The white arrows indicate where the data should be assigned

as shown in Fig. 4.4. If $a > 0$, we are in the case of the pollutant model: all the characteristics **are inflow** and **the data must be assigned on both semi-axes.**

If $a < 0$, the characteristics leaving the x-axis are **inflow**, while those leaving the t-axis are **outflow**. In this case the initial data alone are sufficient to uniquely determine the solution, while **no data has to be assigned on the semi-axis** $x = 0$, $t > 0$.

Coherently, a problem in the half-strip $0 < x < R$, $t > 0$, besides the initial data, requires a data assignment on the inflow boundary, namely (Fig. 4.4):

$$\begin{cases} u(0,t) = h_0(t) & \text{if } a > 0 \\ u(R,t) = h_R(t) & \text{if } a < 0. \end{cases}$$

The resulting initial-boundary value problem is well posed, since the solution is uniquely determined at every point in the strip by its values along the characteristics. Moreover, a stability estimate can be proved as follows. Consider, for instance, the case $a > 0$ and the problem[3]

$$\begin{cases} u_t + a u_x = 0 & 0 < x < R, t > 0 \\ u(0,t) = h(t) & t > 0 \\ u(x,0) = g(x) & 0 < x < R. \end{cases} \tag{4.18}$$

Multiply the differential equation by u and write

$$u u_t + a u u_x = \frac{1}{2}\frac{\partial}{\partial t}u^2 + \frac{a}{2}\frac{\partial}{\partial x}u^2 = 0.$$

Integrating in x over $(0, R)$, we get:

$$\frac{d}{dt}\int_0^R u^2(x,t)dx + a\left[u^2(R,t) - u^2(0,t)\right] = 0.$$

Now use the data $u(0,t) = h(t)$ and the positivity of a, to obtain

$$\frac{d}{dt}\int_0^R u^2(x,t)dx \le a h^2(t).$$

[3] For the case $u_t + a u_x = f \neq 0$, see Problem 4.2.

Integrating in t we have, using the initial condition $u(x,0) = g(x)$,

$$\int_0^R u^2(x,t)dx \le \int_0^R g^2(x)dx + a\int_0^t h^2(s)ds. \tag{4.19}$$

Now, let u_1 and u_2 be solutions of problem (4.18) with initial data g_1, g_2 and boundary data h_1, h_2 on $x = 0$. Then, by linearity, $w = u_1 - u_2$ is a solution of problem (4.18) with initial data $g_1 - g_2$ and boundary data $h_1 - h_2$ on $x = 0$. Applying the inequality (4.19) to w we have

$$\int_0^R [u_1(x,t) - u_2(x,t)]^2 dx \le \int_0^R [g_1(x) - g_2(x)]^2 dx + a\int_0^t [h_1(s) - h_2(s)]^2 ds.$$

Thus, a quadratic distance of the solutions is controlled by a quadratic distance of the data, at each time. In this sense, the solution of problem (4.18) depends continuously on the initial data and on the boundary data on $x = 0$. We point out that the values of u on $x = R$ do not appear in (4.19).

4.3 Traffic Dynamics

4.3.1 A macroscopic model

Back in 1955, Lighthill and Whitham presented a simple model which explains the most common phenomena observed in heavy traffic. Since human behavior is variable, there is no clear mathematical representation of the behavior of a single driver. Therefore, a (deterministic) traffic model can only capture what happens in a sort of macroscopic average. This makes sense in particular when dealing with heavy traffic where the variability of the driver behaviors is greatly reduced. Thus, our main concern here is to describe phenomena that are observed in a stream of traffic (e.g. on a highway), adopting in this way an *Eulerian point of view*, rather than a *Lagrangian one,* which describes the motion of single cars.

From far away, an intense traffic on a straight section of an highway can be considered as a fluid flow, moving along the positive direction of the x-axis, and described by means of macroscopic variables such as the *density* of cars ρ, their *average speed* V and their *flux function* q:

- The *traffic density* $\rho = \rho(x,t)$, measured in cars per Km, is the (average) number of vehicles per unit road length.
- The *average speed*, or *traffic speed* $V = V(x,t)$, simply corresponds to the average driver's speed.
- The *flux function* or *flow rate* $q = q(x,t)$, measured in cars per hour, gives the number of vehicles that pass a given point x on the road in unit time.

The three (more or less regular) functions ρ, V and q are linked by the simple convection relation

$$q = V\rho. \tag{4.20}$$

Indeed the number of cars passing through the interval $(x, x + dx)$ in the time interval $(t, t + dt)$ is given by $\rho(x,t)dx$ and the corresponding flow rate is $\rho(x,t)dx/dt = \rho(x,t)V(x,t)$, which is (4.20).

To construct a model for the evolution of ρ we assume the following hypotheses.

1. *There is only one lane and overtaking is not allowed.* This is realistic, for instance, for the traffic in a tunnel (see Problem 4.6). Multi-lanes models with overtaking are beyond the scope of this introduction. However, the model we will present is often in agreement with observed dynamics also in this case.

2. *No car "sources" or "sinks".* We consider a road section without exit/entrance gates.

3. *The average speed is not constant and it is a function of the density,* that is

$$V(x,t) = v(\rho(x,t)).$$

This last rather controversial assumption implies that, at a given density, the speed is uniquely determined and that a density change causes an immediate speed variation. Thus drivers immediately adjust to local traffic variations. An advantage of this assumption is that it filters out transients associated to small local adjustements. On the other hand it does not take into account, for instance, accelerations that occur due to sudden changes in the density or bad bahaviors that could cause accidents.

Clearly

$$v'(\rho) = \frac{dv}{d\rho} \leq 0$$

since we expect the speed to decrease as the density increases.

As in Sect. 4.1, from (4.20) and the hypotheses **2** and **3** we derive the conservation law:

$$\rho_t + q(\rho)_x = 0, \tag{4.21}$$

where

$$q(\rho) = v(\rho)\rho.$$

We need a constitutive relation for $v = v(\rho)$. When ρ is small, it is reasonable to assume that the average speed v is more or less equal to the maximal velocity v_m, given by the speed limit. When ρ increases, traffic slows down and stops at the maximum density ρ_m (bumper-to-bumper traffic). We adopt the simplest model consistent with the above considerations, namely

$$v(\rho) = v_m \left(1 - \frac{\rho}{\rho_m}\right), \tag{4.22}$$

so that

$$q(\rho) = v_m \rho \left(1 - \frac{\rho}{\rho_m}\right). \tag{4.23}$$

Since

$$q(\rho)_x = q'(\rho)\rho_x = v_m\left(1 - \frac{2\rho}{\rho_m}\right)\rho_x,$$

equation (4.21) becomes

$$\rho_t + \underbrace{v_m\left(1 - \frac{2\rho}{\rho_m}\right)}_{q'(\rho)}\rho_x = 0. \tag{4.24}$$

According to the terminology in Sect. 1.2, this is a quasilinear equation. We also point out that

$$q''(\rho) = -\frac{2v_m}{\rho_m} < 0,$$

so that q is strictly *concave*. We couple the equation (4.24) with the initial condition

$$\rho(x, 0) = g(x), \qquad \text{with } 0 \le g \le \rho_m. \tag{4.25}$$

4.3.2 The method of characteristics

We want to solve the initial value problem (4.24), (4.25). To compute the density ρ at a point (x, t), we follow the idea we already exploited in the linear transport case without external sources: *to connect the point* (x, t) *with a point* $(x_0, 0)$ *on the x-axis, through a curve along which* ρ *is constant* (Fig. 4.5).

Clearly, if we manage to find such a curve, that we call **characteristic based at** $(x_0, 0)$, the value of ρ at (x, t) is given by $\rho(x_0, 0) = g(x_0)$. Moreover, if this procedure can be repeated for every point (x, t), $x \in \mathbb{R}$, $t > 0$, then we can compute ρ everywhere in the upper halfplane and the problem is completely solved. This is the *method of characteristics*.

Adopting a slightly different point of view, we can implement the above idea as follows: assume that $x = x(t)$ is the equation of the characteristic based at the point $(x_0, 0)$ and that along $x = x(t)$ we *always observe the same initial density* $g(x_0)$. In other words, we have

$$\rho(x(t), t) = g(x_0) \tag{4.26}$$

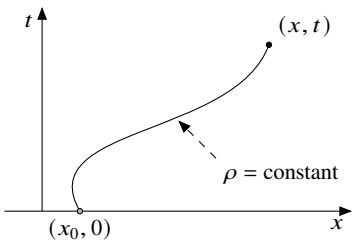

Fig. 4.5 Characteristic curve

for every $t > 0$. If we differentiate the identity (4.26), we get

$$\frac{d}{dt}\rho(x(t),t) = \rho_t(x(t),t) + \rho_x(x(t),t)\dot{x}(t) = 0 \qquad (t > 0). \tag{4.27}$$

On the other hand, (4.24) yields

$$\rho_t(x(t),t) + q'(g(x_0))\rho_x(x(t),t) = 0. \tag{4.28}$$

From the comparison between (4.27) and (4.28) we deduce

$$\dot{x}(t) = q'(g(x_0)).$$

Since $x(0) = x_0$ we find

$$x(t) = q'(g(x_0))t + x_0. \tag{4.29}$$

Thus the characteristics are **straight lines** with slope $q'(g(x_0))$. Different values of x_0 give, in general, different values of the slope (Fig. 4.6).

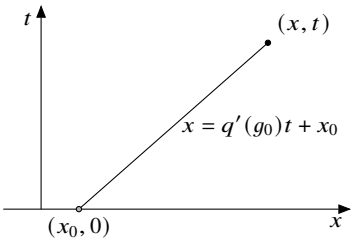

Fig. 4.6 Characteristic straight line ($g_0 = g(x_0)$)

We can now derive a formula for ρ. To compute $\rho(x,t)$, $t > 0$, we move back in time along the characteristic through (x,t) until its base point $(x_0,0)$. Then $\rho(x,t) = g(x_0)$. From (4.29) we have, since $x(t) = x$,

$$x_0 = x - q'(g(x_0))t$$

and finally

$$\rho(x,t) = g(x - q'(g(x_0))t). \tag{4.30}$$

Formula (4.30) represents a **travelling wave** (or a *signal*, a *disturbance*) **propagating with speed** $q'(g(x_0))$ along the positive x-direction.

We emphasize that $q'(g(x_0))$ is the *local wave speed* and it must not be confused with the traffic velocity. In fact, in general,

$$\frac{dq}{d\rho} = \frac{d(\rho v)}{d\rho} = v + \rho\frac{dv}{d\rho} \leq v$$

since $\rho \geq 0$ and $\frac{dv}{d\rho} \leq 0$.

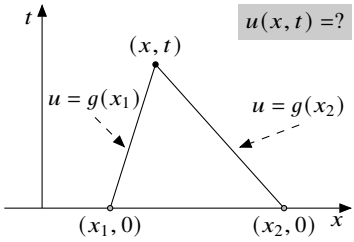

Fig. 4.7 Intersection of characteristics

The different nature of the two speeds becomes more evident if we observe that the wave speed *may be negative* as well. This means that, while the traffic advances along the positive x-direction, the disturbance given by the travelling wave may propagate in the opposite direction. Indeed, in our model (4.23), $\frac{dq}{d\rho} < 0$ when $\rho > \frac{\rho_m}{2}$.

Formula (4.30) seems to be rather satisfactory, since, apparently, it gives the solution of the initial value problem (4.24), (4.25) at every point. Actually, this is true only if x_0 is uniquely determined by (x, t). A more accurate analysis shows that, even if the initial datum g is smooth, the solution may develop a singularity in finite time (e.g. a jump discontinuity). When this occurs, the method of characteristics does not work anymore and formula (4.30) is not effective. A typical case is described in Fig. 4.7: two characteristics based at different points $(x_1, 0)$ and $(x_2, 0)$ intersect at the point (x, t) and the value $u(x, t)$ is not uniquely determined as soon as $g(x_1) \neq g(x_2)$.

In this case we have to weaken the concept of solution and the computation technique. We will come back on these questions later. For the moment, we analyze the method of characteristics in some remarkable cases.

4.3.3 The green light problem

Suppose that bumper-to-bumper traffic is standing at a red light, placed at $x = 0$, while the road ahead is empty. Accordingly, the initial density profile is

$$g(x) = \begin{cases} \rho_m & \text{for } x < 0 \\ 0 & \text{for } x > 0. \end{cases}$$

At time $t = 0$, the traffic light turns green and we want to describe the car flow evolution for $t > 0$. At the beginning, only the cars closer to the light start moving while most of them remain standing.

Since $q'(\rho) = v_m \left(1 - \frac{2\rho}{\rho_m}\right)$, the local wave speed is given by

$$q'(g(x_0)) = \begin{cases} -v_m & \text{for } x_0 < 0 \\ v_m & \text{for } x_0 > 0 \end{cases}$$

and the characteristics are the straight lines

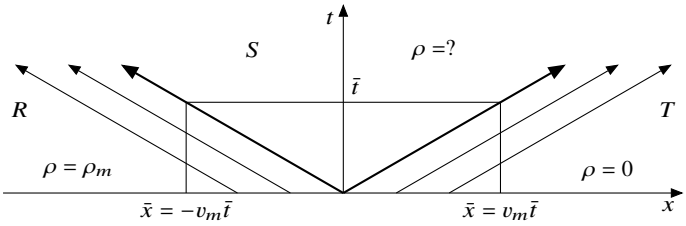

Fig. 4.8 Characteristic for the green light problem

$$\begin{aligned} x &= -v_m t + x_0 \qquad \text{if } x_0 < 0 \\ x &= v_m t + x_0 \qquad \text{if } x_0 > 0. \end{aligned}$$

The lines $x = v_m t$ and $x = -v_m t$ partition the upper half-plane in the three regions R, S and T, shown in Fig. 4.8. Inside R we have $\rho(x,t) = \rho_m$, while inside T we have $\rho(x,t) = 0$. Consider the points on the horizontal line $t = \bar{t}$. At the points $(x,\bar{t}) \in T$ the density is zero: the traffic has not yet arrived in x at time $t = \bar{t}$. We can interpret the point

$$\bar{x} = v_m \bar{t}$$

as the location of the front car, which moves at the maximum speed, since ahead the road is empty.

The cars placed at the points $(x,\bar{t}) \in R$ are still standing. The first car that starts moving at time $t = \bar{t}$ is at the point

$$\bar{x} = -v_m \bar{t}.$$

In particular, it follows that *the green light signal propagates back through the traffic at speed* v_m.

What is the value of the density inside the sector S? No characteristic extends into S, due to the discontinuity of the initial data at the origin, and the method as it stands does not give any information on the value of ρ inside S.

A strategy that may give a reasonable answer is the following:

a) Approximate the initial data by a continuous function g_ε, which converges to g as $\varepsilon \to 0$ at every point x, except 0.
b) Construct the solution ρ_ε of the ε-problem by the method of characteristics.
c) Let $\varepsilon \to 0$ and check that the limit of ρ_ε is a solution of the original problem.

Clearly we run the risk of constructing many solutions, each one depending on the way we regularize the initial data, but for the moment we are satisfied if we construct at least one solution.

a) Let us choose as g_ε the function (Fig. 4.9)

$$g_\varepsilon(x) = \begin{cases} \rho_m & x \le -\varepsilon \\ -\dfrac{\rho_m}{2\varepsilon}(x-\varepsilon) & -\varepsilon \le x \le \varepsilon \\ 0 & x \ge \varepsilon. \end{cases}$$

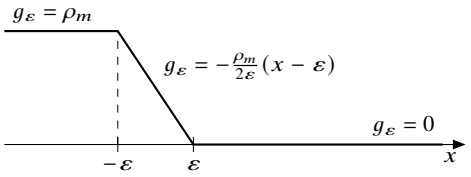

Fig. 4.9 Smoothing of the initial data in the green light problem

When $\varepsilon \to 0$, $g_\varepsilon(x) \to g(x)$ for every $x \neq 0$.

b) The characteristics for the ε-problem are:

$$\begin{aligned} x &= -v_m t + x_0 && \text{if } x_0 \leq -\varepsilon \\ x &= \frac{v_m x_0}{\varepsilon} t + x_0 && \text{if } -\varepsilon \leq x_0 \leq \varepsilon \\ x &= v_m t + x_0 && \text{if } x_0 \geq \varepsilon, \end{aligned}$$

since, for $-\varepsilon \leq x_0 \leq \varepsilon$,

$$q'(g_\varepsilon(x_0)) = v_m \left(1 - \frac{2g_\varepsilon(x_0)}{\rho_m}\right) = \frac{v_m x_0}{\varepsilon}.$$

We say that the characteristics in the region $-v_m t - \varepsilon < x < v_m t + \varepsilon$ form a *rarefaction fan* (Fig. 4.10). Clearly, $\rho_\varepsilon(x,t) = 0$ for $x \geq v_m t + \varepsilon$ and $\rho_\varepsilon(x,t) = \rho_m$ for $x \leq -v_m t - \varepsilon$. Let now (x,t) belong to the region

$$-v_m t - \varepsilon < x < v_m t + \varepsilon.$$

Solving for x_0 in the equation of the characteristic $x = \frac{v_m x_0}{\varepsilon} t + x_0$, we find

$$x_0 = \frac{\varepsilon x}{v_m t + \varepsilon}.$$

Then, along that characteristic,

$$\rho_\varepsilon(x,t) = g_\varepsilon(x_0) = \frac{\rho_m}{2}\left(1 - \frac{x_0}{\varepsilon}\right) = \frac{\rho_m}{2}\left(1 - \frac{x}{v_m t + \varepsilon}\right). \tag{4.31}$$

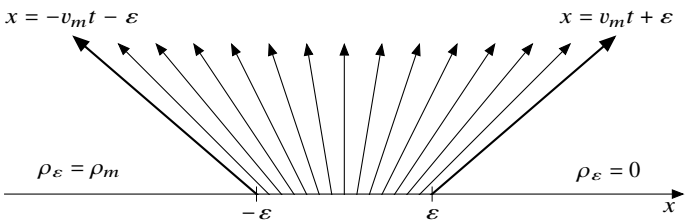

Fig. 4.10 Fanlike characteristics

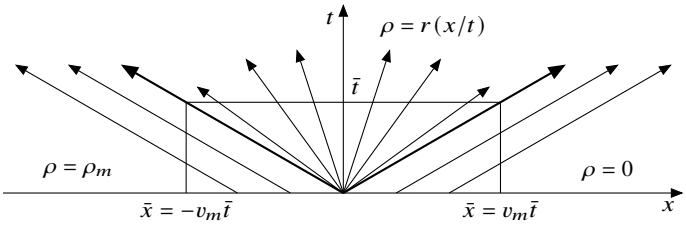

Fig. 4.11 Characteristics in a rarefaction wave

c) Letting $\varepsilon \to 0$ in (4.31) we obtain

$$\rho(x,t) = \begin{cases} \rho_m & \text{for } x \le -v_m t \\ \dfrac{\rho_m}{2}\left(1 - \dfrac{x}{v_m t}\right) & \text{for } -v_m t < x < v_m t \\ 0 & \text{for } x \ge v_m t. \end{cases} \tag{4.32}$$

It is easy to check that ρ is a solution of the equation (4.24) in the regions R, S, T. For fixed t, the function ρ decreases linearly from ρ_m to 0, as x varies from $-v_m t$ to $v_m t$. Moreover, ρ is constant on the fan of straight lines (still called characteristics)

$$x = ht, \qquad -v_m < h < v_m.$$

These type of solutions are called **rarefaction** or **simple waves** (centered at the origin). The characteristics and a typical profile are shown in Figs. 4.11 and 4.12.

Remark 4.3 The formula for $\rho(x,t)$ in the sector S can be obtained, a posteriori, by a formal procedure that emphasizes its self-similar structure. The equation of the characteristics based on $(x_0, 0)$ can be written in the form

$$x = v_m\left(1 - \frac{2g(x_0)}{\rho_m}\right)t + x_0 = v_m\left(1 - \frac{2\rho(x,t)}{\rho_m}\right)t + x_0$$

because $\rho(x,t) = g(x_0)$ along it. Inserting $x_0 = 0$ we obtain

$$x = v_m\left(1 - \frac{2\rho(x,t)}{\rho_m}\right)t.$$

Solving for ρ we find precisely

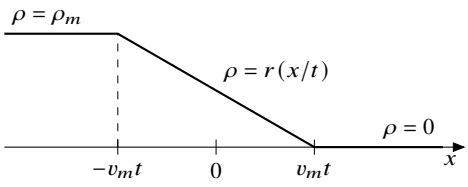

Fig. 4.12 Profile of a rarefaction wave at time t

$$\rho(x,t) = \frac{\rho_m}{2}\left(1 - \frac{x}{v_m t}\right) \qquad (t > 0). \tag{4.33}$$

Since $v_m\left(1 - \frac{2\rho}{\rho_m}\right) = q'(\rho)$, we see that (4.33) is equivalent to

$$\rho(x,t) = r\left(\frac{x}{t}\right)$$

where $r = (q')^{-1}$ is the inverse function of q'.

Indeed the general form of a rarefaction wave, centered at a point (x_0, t_0), is given by

$$\rho(x,t) = r\left(\frac{x - x_0}{t - t_0}\right).$$

Conclusion: we have constructed a continuous solution ρ of the green light problem, *connecting the two constant states* ρ_m and 0 by a rarefaction wave. However, it is not clear in which sense ρ is a solution across the lines $x = \pm v_m t$, since, there, its derivatives undergo a jump discontinuity. Also, it is not clear whether or not (4.32) is the only solution. We will return on these important points.

• *The vehicle paths*. We now examine the path of a vehicle, initially located at $x = -a < 0$. This vehicle, being inside the bumper-to-bumper region, will not move until time $\tau = a/v_m$, at which the green light signal reaches it. From this moment, the vehicle enters the rarefaction region and moves with speed $v = v(\rho)$, where $\rho(x,t)$ is given by (4.33), that is

$$v(\rho(x,t)) = v_m\left(1 - \frac{\rho(x,t)}{\rho_m}\right) = v_m\left(1 - \frac{1}{2}\left(1 - \frac{x}{v_m t}\right)\right) = \frac{v_m}{2} + \frac{1}{2}\frac{x}{t}.$$

Thus, denoting by $x = x(t)$ the vehicle position, we must solve the initial value problem

$$\begin{cases} \dot{x} = \dfrac{v_m}{2} + \dfrac{1}{2}\dfrac{x}{t} \\ x(\tau) = -a. \end{cases}$$

Dividing both sides of the ODE by $\sqrt{t}$, we may write the differential equation in the form

$$\frac{d}{dt}\left(\frac{x}{\sqrt{t}}\right) = \frac{v_m}{2\sqrt{t}}.$$

Integrating over (τ, t) we find, since $x(\tau) = -a$,

$$\frac{x(t)}{\sqrt{t}} + \frac{a}{\sqrt{\tau}} = v_m(\sqrt{t} - \sqrt{\tau}).$$

Inserting $\tau = a/v_m$, we finally get

$$x(t) = v_m t - 2\sqrt{a v_m t}. \tag{4.34}$$

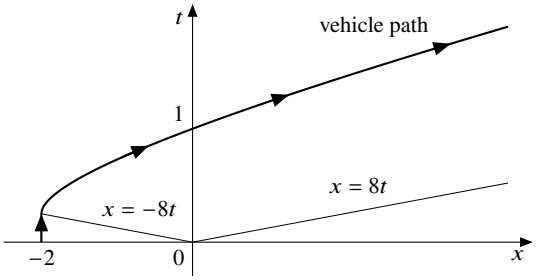

Fig. 4.13 The path of a vehicle starting at $x = -2$

Since $v_m t - 2\sqrt{a v_m t} < v_m t$, the path will never cross the characteristic $x = v_m t$ and hence will never leave the rarefaction region. Thus, after time $\tau = a/v_m$, the path is given by the portion of parabola (4.34), as shown in Fig. 4.13, where $a = 2$ and $v_m = 8$.

Suppose that the light turns on red at time t_{red}. Will our driver be able to go through the traffic light?

We only have to compute how much time it takes to the driver to reach the origin. Since $x(t) = 0$ at time $t_d = 4a/v_m$, our driver will be able to pass through the traffic light if $t_d < t_{\text{red}}$.

4.3.4 Traffic jam ahead

We now assume that the initial density profile is

$$g(x) = \begin{cases} \frac{1}{8}\rho_m & \text{for } x < 0 \\ \rho_m & \text{for } x > 0. \end{cases} \tag{4.35}$$

For $x > 0$, the density is maximal and therefore the traffic is bumper-to-bumper. The cars on the left move with speed $v = \frac{7}{8}v_m$, so that we expect a congestion propagating back into the traffic. We have

$$q'(g(x_0)) = \begin{cases} \frac{3}{4}v_m & \text{if } x_0 < 0 \\ -v_m & \text{if } x_0 > 0 \end{cases}$$

and therefore the characteristics are

$$\begin{aligned} x &= \tfrac{3}{4}v_m t + x_0 \quad \text{if } x_0 < 0 \\ x &= -v_m t + x_0 \quad \text{if } x_0 > 0. \end{aligned}$$

The configuration in Fig. 4.14 shows that the characteristics intersect somewhere in finite time and the theory predicts that ρ becomes a *multivalued* function of position. In other words, ρ should assume two different values at the same point, which clearly makes no sense in our situation. Therefore we have to admit solutions with jump

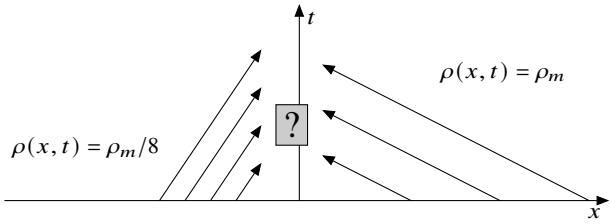

Fig. 4.14 Expecting a discontinuity

discontinuities. But then we have to reexamine the derivation of the conservation law, because the smoothness assumption for ρ does not hold anymore.

Thus let us go back to the conservation of cars in integral form (see (4.2)):

$$\frac{d}{dt}\int_{x_1}^{x_2} \rho(x,t)dx = -q(\rho(x_2,t)) + q(\rho(x_1,t)), \tag{4.36}$$

valid in any control interval (x_1, x_2). Suppose now that ρ is a smooth function up to both sides of a smooth curve having equation

$$x = s(t), \qquad t \in [t_1, t_2]\,,$$

along which ρ undergoes a *jump discontinuity*. For fixed t, let (x_1, x_2) be an interval containing the discontinuity point $x = s(t)$. From (4.36) we have

$$\frac{d}{dt}\left\{\int_{x_1}^{s(t)} \rho(y,t)dy + \int_{s(t)}^{x_2} \rho(y,t)dy\right\} + q(\rho(x_2,t)) - q(\rho(x_1,t)) = 0. \tag{4.37}$$

Applying the transport formula (1.7), p. 11, we find

$$\frac{d}{dt}\int_{x_1}^{s(t)} \rho(y,t)dy = \int_{x_1}^{s(t)} \rho_t(y,t)dy + \rho_-(s(t),t)\dot{s}(t)$$

and

$$\frac{d}{dt}\int_{s(t)}^{x_2} \rho(y,t)dy = \int_{s(t)}^{x_2} \rho_t(y,t)dy - \rho_+(s(t),t)\dot{s}(t),$$

where

$$\rho_+(s(t),t) = \lim_{y\to s(t)^+} \rho(y,t), \qquad \rho_-(s(t),t) = \lim_{y\to s(t)^-} \rho(y,t).$$

Hence, equation (4.37) becomes

$$\int_{x_1}^{x_2} \rho_t(y,t)dy + [\rho_-(s(t),t) - \rho_+(s(t),t)]\,\dot{s}(t) = q(\rho(x_1,t)) - q(\rho(x_2,t)).$$

Letting $x_2 \to s(t)^+$ and $x_1 \to s(t)^-$, we obtain

$$[\rho_-(s(t),t) - \rho_+(s(t),t)]\,\dot{s}(t) = q(\rho_-(s(t),t)) - q(\rho_+(s(t),t)),$$

that is:

$$\dot{s} = \frac{q(\rho_+(s,t)) - q(\rho_-(s,t))}{\rho_+(s,t) - \rho_-(s,t)}. \tag{4.38}$$

The relation (4.38) is an ordinary differential equation for s and it is known as **Rankine-Hugoniot jump condition**. If (4.38) holds, the discontinuity line is called shock curve or simply a shock. Accordingly, the propagating discontinuity (or the solution itself) is called **shock wave**. The jump $\rho_+(s,t) - \rho_-(s,t)$ is called the *strength* of the shock.

The Rankine-Hugoniot condition gives the *shock speed* $\dot{s}$ as the quotient of the flux jump over the density jump. To determine the shock curve we need to know its initial point and the values of ρ from both sides of the curve.

Let us apply the above considerations to the traffic problem with initial condition (4.35). We have

$$\rho_+ = \rho_m, \qquad \rho_- = \frac{1}{8}\rho_m,$$

while

$$q(\rho_+) = 0, \qquad q(\rho_-) = \frac{7}{64} v_m \rho_m$$

and (4.38) gives

$$\dot{s}(t) = \frac{q(\rho_+) - q(\rho_-)}{\rho_+ - \rho_-} = -\frac{1}{8} v_m.$$

Since clearly $s(0) = 0$, the shock curve is the straight line $x = -\frac{1}{8} v_m t$.

Note that *the slope is negative: the shock propagates back with speed* $-\frac{1}{8} v_m$, as it is revealed by the brake lights of the cars, slowing down because of a traffic jam ahead.

As a consequence, the solution of our problem is given by the following formula (Fig. 4.15)

$$\rho(x,t) = \begin{cases} \frac{1}{8}\rho_m & x < -\frac{1}{8} v_m t \\ \rho_m & x > -\frac{1}{8} v_m t. \end{cases}$$

This time we say that the two constant states $\frac{1}{8}\rho_m$ and ρ_m are connected by a **shock wave**.

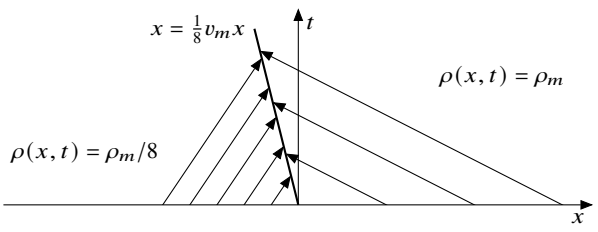

Fig. 4.15 Shock wave

4.4 Weak (or Integral) Solutions

4.4.1 The method of characteristics revisited

The method of characteristics applied to the problem

$$\begin{cases} u_t + q(u)_x = 0 \\ u(x,0) = g(x) \end{cases} \tag{4.39}$$

gives the travelling wave (see (4.30) with $x_0 = \xi$)

$$u(x,t) = g(x - q'(g(\xi))t) \qquad (q' = \frac{dq}{du}) \tag{4.40}$$

with local speed $q'(g(\xi))$, in the positive x-direction. Since $u(x,t) \equiv g(\xi)$ along the characteristic

$$x = q'(g(\xi))t + \xi, \tag{4.41}$$

based at $(\xi, 0)$, from (4.40) we obtain that u is implicitly defined by the equation

$$G(x,t,u) \equiv u - g(x - q'(u)t) = 0. \tag{4.42}$$

If g and q' are smooth functions, the Implicit Function Theorem 1.6, p. 17, implies that equation (4.42) defines u as a function of (x,t), as long as the condition

$$G_u(x,t,u) = 1 + tq''(u)g'(x - q'(u)t) \neq 0 \tag{4.43}$$

holds. Since along the characteristic (4.41) we have $g'(x - q'(u)t) = g'(\xi)$ and $q''(u) = q''(g(\xi))$, an immediate consequence is that if $q'' \circ g$ and g' have the same sign, the solution given by the method of characteristics is defined and smooth for all times $t \geq 0$. Precisely, we have:

Proposition 4.4 *Assume that $q \in C^2(\mathbb{R})$, $g \in C^1(\mathbb{R})$ and $g'(\xi)q''(g(\xi)) \geq 0$ in $\mathbb{R}$. Then formula (4.42) defines the unique solution u of problem (4.39) in the half-plane $t \geq 0$. Moreover, $u \in C^1(\mathbb{R} \times [0,\infty))$ and*

$$u_x(x,t) = -\frac{G_x(x,t,u)}{G_u(x,t,u)}, \qquad u_t(x,t) = -\frac{G_t(x,t,u)}{G_u(x,t,u)}. \tag{4.44}$$

Thus, if $q'' \circ g$ and g' have the same sign, the characteristics cannot intersect. This is not surprising since

$$g'(\xi)q''(g(\xi)) = \frac{d}{d\xi}q'(g(\xi)),$$

so that the condition $g'(\xi)q''(g(\xi)) \geq 0$ means the the characteristics have increasing slope, preventing any intersection. Note that in the ε-approximation of the green light problem, q is concave and g_ε is decreasing. Although g_ε is not smooth, the

characteristics do not intersect and ρ_ε is well defined for all times $t > 0$. In the limit as $\varepsilon \to 0$, the discontinuity of g reappears and the fan of characteristics produces a rarefaction wave.

What happens if $q'' \circ g$ and g' have a different sign over an interval $[a, b]$? Proposition 4.4 still holds for small times, since $G_u \sim 1$ if $t \sim 0$, but when time goes on, we expect the formation of a shock. Indeed, suppose, for instance, that q is concave and g is increasing. The family of characteristics based on a point in the interval $[a, b]$ is

$$x = q'(g(\xi))t + \xi, \qquad \xi \in [a, b]\,. \tag{4.45}$$

Since $q'(g(\xi))$ decreases in $[a, b]$, we expect an intersection of characteristics along a shock curve. The main question is to find the positive time t_s (*breaking time*) and the location x_s **of first appearance of the shock.**

According to the above discussion, the breaking time must coincide with the first time t at which the expression

$$G_u(x, t, u) = 1 + tq''(u)g'(x - q'(u)t)$$

becomes zero[4]. Computing G_u along the characteristic (4.45), we have $u = g(\xi)$ and

$$G_u(x, t, u) = 1 + tq''(g(\xi))g'(\xi).$$

Assume that the positive function

$$z(\xi) = -q''(g(\xi))g'(\xi)$$

attains its maximum only at the point $\xi_M \in [a, b]$. Then $z(\xi_M) > 0$ and

$$t_s = \min_{\xi \in [a,b]} \frac{1}{z(\xi)} = \frac{1}{z(\xi_M)}. \tag{4.46}$$

Since x_s belongs to the characteristics $x = q'(g(\xi_M))t + \xi_M$, we find

$$x_s = \frac{q'(g(\xi_M))}{z(\xi_M)} + \xi_M. \tag{4.47}$$

Example 4.5 Consider the initial value problem

$$\begin{cases} u_t + (1 - 2u)u_x = 0 \\ u(x, 0) = \frac{1}{2}\arctan(\pi x). \end{cases} \tag{4.48}$$

We have $q(u) = u - u^2$, $q'(u) = 1 - 2u$, $q''(u) = -2$, and $g(\xi) = \frac{1}{2}\arctan(\pi\xi)$, $g'(\xi) = \pi/(2 + 2\pi^2\xi^2)$. Therefore, the function

$$z(\xi) = -q''(g(\xi))g'(\xi) = \frac{\pi}{1 + \pi^2\xi^2}$$

[4] Indeed, by formulas (4.44), u_x and u_t become unbounded as (x, t) approaches a zero of G_u.

has a maximum at $\xi_M = 0$ and $z(0) = \pi$. The breaking-time is $t_s = 1/\pi$ and

$$x_s = q'(g(\xi_M))t_s + \xi_M = 1/\pi.$$

Thus, the shock curve starts from $(1/\pi, 1/\pi)$. For $0 \le t < 1/\pi$ the solution u is smooth and implicitly defined by the equation

$$G(x,t,u) = u - \frac{1}{2}\arctan\left[\pi x - \pi(1-2u)t\right] = 0. \tag{4.49}$$

After $t = 1/\pi$, equation (4.49) defines u as a multivalued function of (x,t) and does not define a solution anymore. Figure 4.16 shows what happens for the times

$$t = 0,\ \frac{1}{\pi},\ 0.5,\ 0.7,\ 0.18.$$

The thick line corresponds to graph of u at the breaking time $t = 1/\pi$.

How does the solution of Example 4.5 evolve after $t = 1/\pi$? We have to insert a shock into the multivalued graph in Fig. 4.16, in a way that the conservation law is preserved. We will see that the correct insertion point is prescribed by the Rankine-Hugoniot condition. It turns out that this corresponds to cutting off from the multivalued profile the two equal area lobes A and B as described in Fig. 4.17, at time $t = 0.7$ (Whitham equal area rule[5]).

Remark 4.6 The breaking point (x_s, t_s) has an interesting geometrical meaning. In fact, it turns out that if $g'(\xi)q''(g(\xi)) < 0$ in $[a,b]$, the family of characteristics (4.45) admits an *envelope* and (x_s, t_s) is the point on the envelope with minimum time coordinate (see Problem 4.7).

Recall that the *envelope* of a family of curves $\phi(x,t,\xi) = 0$, depending on the parameter ξ, is a curve $\psi(x,t) = 0$ *tangent at each one of its points to a curve of the family*. If the family of curves $\phi(x,t,\xi) = 0$ has an envelope, its parametric equations are obtained by solving the system

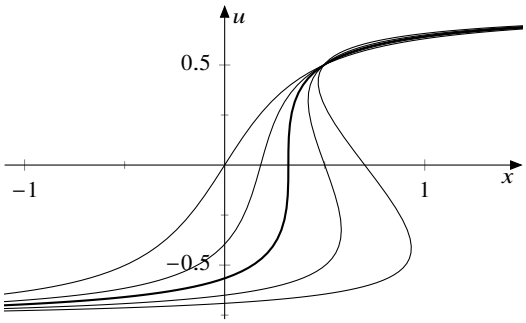

Fig. 4.16 The graphs implicitly defined by equation (4.49) for $t = 0, 0.18, \mathbf{1/\pi}, 0.5, 0.7$

[5] See *G. B. Whitham*, [33].

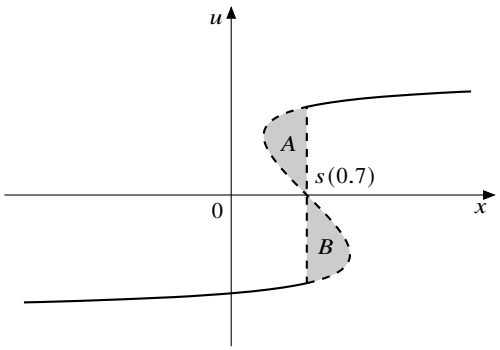

Fig. 4.17 Inserting a shock by the Whitham *equal-area rule* in Example 4.5

$$\begin{cases} \phi(x,t,\xi) = 0 \\ \phi_\xi(x,t,\xi) = 0 \end{cases}$$

with respect to x and t. On the other hand, a solution of this system may produce a locus of irregular points, see Problem 4.8.

4.4.2 Weak solutions

We have seen that the method of characteristics is not sufficient, in general, to determine the solution of an initial value problem for all times $t > 0$. In the green light problem a rarefaction wave was used to construct the solution in a region not covered by characteristics. In the traffic jam case the solution undergoes a shock, propagating according to the Rankine-Hugoniot condition.

Some questions naturally arise.

Q1. *In which sense is the differential equation satisfied across a shock or, more generally, across a separation curve where the constructed solution is not differentiable?*

A way to "solve" the problem would be simply not to care about those points. However, in this case it would be possible to construct solutions that do not have any connection with the physical meaning of the conservation law.

Q2. *Is the solution unique?*

Q3. *If there is no uniqueness, is there a criterion to select the "physically correct" solution?*

To answer, we need first of all to introduce a more flexible notion of solution, in which the derivatives of the solution are not directly involved. Let us go back to the problem

$$\begin{cases} u_t + q(u)_x = 0 & x \in \mathbb{R}, t > 0 \\ u(x,0) = g(x) & x \in \mathbb{R} \end{cases} \tag{4.50}$$

where q is a smooth function and g is bounded. Assume for the moment that u is a smooth solution, at least of class C^1 in $\mathbb{R} \times [0, \infty)$. We say that u is a **classical solution.**

Let v be a smooth function, with compact support in $\mathbb{R} \times [0, \infty)$. Thus v vanishes outside a compact set contained in $\mathbb{R} \times [0, \infty)$. We call v a *test function*. Multiplying the differential equation by v and integrating on $\mathbb{R} \times (0, \infty)$, we get

$$\int_0^\infty \int_{\mathbb{R}} [u_t + q(u)_x]\, v \, dxdt = 0. \tag{4.51}$$

The idea is to carry the derivatives onto the test function v via an integration by parts. If we integrate by parts the first term with respect to t and we use the initial condition, we obtain[6]:

$$\begin{aligned}\int_0^\infty \int_{\mathbb{R}} u_t v \, dxdt &= -\int_0^\infty \int_{\mathbb{R}} u v_t \, dxdt - \int_{\mathbb{R}} u(x,0)v(x,0)dx \\ &= -\int_0^\infty \int_{\mathbb{R}} u v_t \, dxdt - \int_{\mathbb{R}} g(x)v(x,0)dx.\end{aligned}$$

Integrating by parts the second term in (4.51) with respect to x, we have:

$$\int_0^\infty \int_{\mathbb{R}} q(u)_x v \, dxdt = -\int_0^\infty \int_{\mathbb{R}} q(u) v_x \, dxdt.$$

Then, equation (4.51) becomes

$$\int_0^\infty \int_{\mathbb{R}} [u v_t + q(u) v_x] \, dxdt + \int_{\mathbb{R}} g(x)v(x,0)dx = 0. \tag{4.52}$$

We have obtained an integral equation, valid **for every test function** v. Observe that no derivative of u appears in (4.52).

On the other hand, suppose that a smooth function u satisfies (4.52) for every test function v. Integrating by parts in the reverse order, we arrive to the equation

$$\int_0^\infty \int_{\mathbb{R}} [u_t + q(u)_x]\, v \, dxdt + \int_{\mathbb{R}} [g(x) - u(x,0)]\, v(x,0)dx = 0, \tag{4.53}$$

which is true for every test function v.

If we choose v vanishing for $t = 0$, then the second integral is zero and the arbitrariness of v implies

$$u_t + q(u)_x = 0 \text{ in } \mathbb{R} \times (0, +\infty). \tag{4.54}$$

Choosing now v not vanishing for $t = 0$, from (4.53) and (4.54), we get

[6] Since v is compactly supported and u, v are smooth, there is no problem in exchanging the order of integration.

$$\int_{\mathbb{R}} [g(x) - u(x,0)]\, v(x,0)dx = 0.$$

Once more, the arbitrariness of v implies

$$u(x,0) = g(x) \text{ in } \mathbb{R}.$$

Therefore u is a classical solution of problem (4.50).

Conclusion: *a function* $u \in C^1(\mathbb{R} \times [0 \times \infty))$ *is a classical solution of problem* (4.50) *if and only if the equation* (4.52) *holds for every test function* v.

Now, (4.52) makes perfect sense for u and g merely locally integrable[7], and therefore it constitutes an alternative **weak** or **integral** formulation of problem (4.50). This motivates the following definition.

Definition 4.7 Let g be locally integrable in $\mathbb{R}$. A function u, locally integrable in $\mathbb{R} \times [0, \infty)$, is called a *weak* (or *integral*) *solution* of problem (4.50) if equation (4.52) holds for every test function v in $\mathbb{R} \times [0, \infty)$.

We point out that a weak solution may be discontinuous, since the definition requires only that it is locally integrable.

4.4.3 Piecewise smooth functions and the Rankine-Hugoniot condition

Definition 4.7 looks rather satisfactory, because of its flexibility. However we have to understand which information is hidden in the integral formulation about the behavior of weak solutions across a discontinuity curve. First we introduce the class of piecewise-C^1 functions:

Definition 4.8 We say that a locally bounded function $u : \mathbb{R} \times [0, \infty) \to \mathbb{R}$ is a *piecewise-C^1 function* if there exist a finite number of C^1 curves Γ_j, $j = 1, \ldots, N$, of equation $x = s_j(t)$, defined in some interval $I_j \subseteq [0, +\infty)$, such that:

a) The curves Γ_j can intersect only at their endpoints and u is C^1 outside $\bigcup_{j=1}^{N} \Gamma_j$.
b) Across each Γ_j, either u is continuous or it undergoes a jump discontinuity.

Moreover, the jumps

$$u_+(s_j(t), t) - u_-(s_j(t), t)$$

are continuous on I_j, $j = 1, \ldots, N$.

We now show that, in the class of piecewise-C^1 solutions, the only admissible discontinuities in the half plane $t > 0$ are those prescribed by the Rankine-Hugoniot condition, as it is stated in the following theorem. Here the initial datum plays no role, so we examine weak solutions of the differential equation only, by considering test functions supported in $\mathbb{R} \times (0, \infty)$.

[7] u and g locally integrable means integrable on any compact subset of $\mathbb{R} \times [0, +\infty)$, $\mathbb{R}$ respectively.

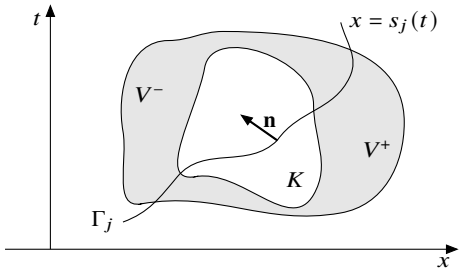

Fig. 4.18 Domain splitted by a discontinuity curve

Theorem 4.9 *Let $u : \mathbb{R} \times [0, \infty) \to \mathbb{R}$ be a piecewise-C^1 function. Then u is a weak solution of the equation*

$$u_t + q(u)_x = 0 \tag{4.55}$$

in the half plane $t > 0$ if and only if:

1. *u is a classical solution of (4.55) in the region where u is C^1;*
2. *the Rankine-Hugoniot jump condition*

$$\dot{s}_j(t) = \frac{q(u_+(s_j(t), t)) - q(u_-(s_j(t), t))}{u_+(s_j(t), t) - u_-(s_j(t), t)}$$

holds along each discontinuity line Γ_j, in the interior points of I_j.

Proof Assume u is a weak solution of (4.55). We have already seen in the previous subsection that u is a classical solution of

$$u_t + q(u)_x = 0$$

in any open set where it is C^1. To prove 2, let Γ_j be one of the discontinuity curves, of equation $x = s_j(t)$ and let V be an open set, contained in the half-plane $t > 0$. Since we consider only interior points of I_j, we can assume that no other curve Γ_h $h \neq j$, intersects V. Then V is partitioned into two disjoint open domains V^+ and V^- by Γ_j as in Fig. 4.18. In both V^+ and V^-, u is a classical solution. Choose now a test function v, supported in a compact set $K \subset V$, such that $K \cap \Gamma_j$ is nonempty. Since $v(x, 0) = 0$, we can write:

$$\begin{aligned} 0 &= \int_0^\infty \int_{\mathbb{R}} [uv_t + q(u)v_x]\, dxdt \\ &= \int_{V^+} [uv_t + q(u)v_x]\, dxdt + \int_{V^-} [uv_t + q(u)v_x]\, dxdt. \end{aligned}$$

Integrating by parts and observing that $v = 0$ on $\partial V^+ \setminus \Gamma_j$, we have:

$$\begin{aligned} \int_{V^+} [uv_t + q(u)v_x]\, dxdt &= -\int_{V^+} [u_t + q(u)_x]\, v\, dxdt + \int_{\Gamma_j} [u_+ n_2 + q(u_+)n_1]\, v\, dl \\ &= \int_{\Gamma_j} [u_+ n_2 + q(u_+)n_1]\, v\, dl \end{aligned}$$

where u_+ denotes the value of u on Γ_j, from the V^+ side, $\mathbf{n} = (n_1, n_2)$ is the outward unit normal vector on ∂V^+ and dl denotes the arc length on Γ_j. Similarly, since $\mathbf{n}$ is inward with respect to V^-:

$$\int_{V^-} [uv_t + q(u)v_x]\, dxdt = -\int_{\Gamma_j} [u_- n_2 + q(u_-)n_1]\, v\, dl$$

where u_- denotes the value of u on Γ_j, from the V^- side. Therefore we deduce that

$$\int_{\Gamma_j} \{[q(u_+) - q(u_-)]\, n_1 + [u_+ - u_-]\, n_2\}\, v dl = 0.$$

Since u is a piecewise-C^1 function, the jumps $[q(u_+) - q(u_-)]$ and $[u_+ - u_-]$ are continuous along Γ_j, and hence the arbitrariness of v yields

$$[q(u_+) - q(u_-)]\, n_1 + [u_+ - u_-]\, n_2 = 0 \tag{4.56}$$

on Γ_j. If u is continuous across Γ_j, (4.56) is automatically satisfied. If $u_+ \neq u_-$ we write the relation (4.56) more explicitly. Since $x = s_j(t)$ on Γ_j, we have

$$\mathbf{n} = (n_1, n_2) = \frac{1}{\sqrt{1 + (\dot{s}_j(t))^2}}(-1, \dot{s}_j(t)).$$

Hence (4.56) becomes, after simple calculations,

$$\dot{s}_j(t) = \frac{q(u_+(s_j(t), t)) - q(u_-(s_j(t), t))}{u_+(s_j(t), t) - u_-(s_j(t), t)} \tag{4.57}$$

which is the *Rankine-Hugoniot* condition along Γ_j.

On the other hand, if u is piecewise continuous and satisfies conditions 1 and 2, it is easy to check that u is a weak solution of (4.55). □

By Theorem 4.9, any function constructed by connecting classical solutions and rarefaction waves in a continuous way is a weak solutions. The same is true for shock waves, since they satisfy the Rankine-Hugoniot condition. Thus, *the solutions of the green light and of the traffic jam problems are precisely weak solutions*.

Definition 4.7 gives a satisfactory answer to question Q1. The other two questions require a deeper analysis, as the following example shows.

Example 4.10 (Non-uniqueness) Imagine a flux of particles along the x-axis, each one moving with constant speed. Suppose that $u = u(x, t)$ represents *the velocity field*, which gives the speed of the particle located at x at time t. If $x = x(t)$ is the *path* of a particle, its velocity at time t is given by

$$\dot{x}(t) = u(x(t), t) \equiv \text{constant}.$$

Thus, we have

$$\begin{aligned} 0 &= \frac{d}{dt} u(x(t), t) = u_t(x(t), t) + u_x(x(t), t)\dot{x}(t) \\ &= u_t(x(t), t) + u_x(x(t), t)\, u(x(t), t). \end{aligned}$$

Therefore $u = u(x, t)$ satisfies the *Burgers equation*

$$u_t + u u_x = u_t + \left(\frac{u^2}{2}\right)_x = 0 \tag{4.58}$$

which is a conservation law with $q(u) = u^2/2$. Note that q is strictly convex: $q'(u) = u$ and $q''(u) = 1$. We couple (4.58) with the initial condition $u(x, 0) = g(x)$, where

$$g(x) = \begin{cases} 0 & x < 0 \\ 1 & x > 0. \end{cases}$$

The characteristics are the straight lines

$$x = g(x_0)t + x_0. \tag{4.59}$$

Therefore, $u = 0$ if $x < 0$ and $u = 1$ if $x > t$. The region

$$S = \{(x, t) : 0 < x < t\}$$

is not covered by characteristics. As in the green light problem, we connect the states 0 and 1 through a *rarefaction wave* (see Remark 4.3). Since $q'(u) = u$, we have

$$r(s) = (q')^{-1}(s) = s,$$

so that we construct the weak solution (Fig. 4.19)

$$u(x, t) = \begin{cases} 0 & x \le 0 \\ x/t & 0 < x < t \\ 1 & x \ge t. \end{cases} \tag{4.60}$$

However, u **is not the unique weak solution**! There exists also a *shock wave* solution. In fact, since

$$u_- = 0, u_+ = 1, q(u_-) = 0, q(u_+) = \frac{1}{2},$$

the Rankine-Hugoniot condition yields

$$\dot{s}(t) = \frac{q(u_+) - q(u_-)}{u_+ - u_-} = \frac{1}{2}.$$

Given the discontinuity at $x = 0$ of the initial data, the shock curve starts at $s(0) = 0$ and it is the straight line $x = \frac{t}{2}$. Hence, the function

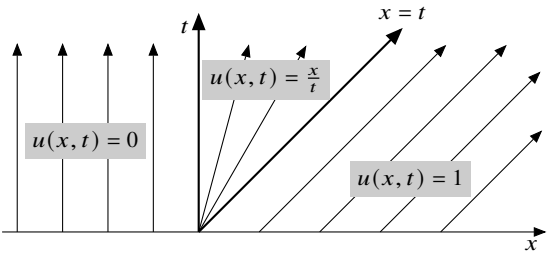

Fig. 4.19 The rarefaction wave in Example 4.10

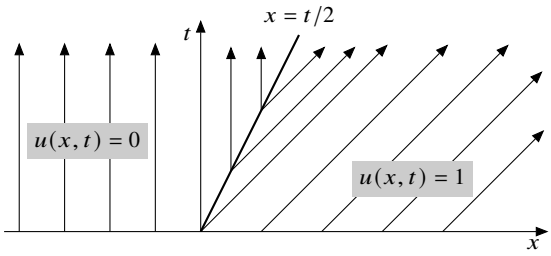

Fig. 4.20 A *non-physical* shock

$$w(x,t) = \begin{cases} 0 & x < \frac{t}{2} \\ 1 & x > \frac{t}{2} \end{cases}$$

is another weak solution (Fig. 4.20).

As we shall see, this shock wave has to be considered as a *not physically acceptable solution*.

4.5 An Entropy Condition

The previous example shows that the answer to question Q2 is *negative* and question Q3 becomes relevant. We need a criterion to establish which is the physically correct solution.

Let us go back to classical solutions. From Proposition 4.4, p. 201, we have seen that the equation

$$G(x,t,u) \equiv u - g(x - q'(u)t) = 0$$

defines the unique classical solution u of problem (4.50), at least for small times. The Implicit Function Theorem gives

$$u_x(x,t) = -\frac{G_x(x,t,u)}{G_u(x,t,u)} = \frac{g'(x - q'(u)t)}{1 + tg'(x - q'(u)t)q''(u)}.$$

If we assume

$$g'(\xi) > 0, q''(u) \geq q''_{\min} > 0, \qquad \text{for all } \xi \in \mathbb{R}, u \in \mathbb{R},$$

then

$$1 + tg'(x - q'(u)t)q''(u) \geq tg'(x - q'(u)t)q''_{\min}$$

and we get

$$u_x(x,t) \leq \frac{E}{t}, \qquad \text{where } E = \frac{1}{q''_{\min}}.$$

Using the mean value theorem we deduce the following condition[8]:
there exists $E \geq 0$ *such that, for every* $x, z \in \mathbb{R}$, $z > 0$, *and every* $t > 0$,

$$u(x+z,t) - u(x,t) \leq \frac{E}{t} z. \tag{4.61}$$

The inequality (4.61) does not involve any derivative of u and makes perfect sense for discontinuous solutions as well. It turns out (see Theorem 4.12 below) that it constitutes an effective criterion to select physically admissible shocks. By analogy with gas dynamics, where a condition like (4.61) implies that the entropy increases across a shock, it is called **entropy condition**[9]. Actually, several types of entropy conditions have been formulated in the literature; another one similar to equation (4.65) below, is introduced in Sect. 4.6.3.

A weak solution satisfying (4.61) is said to be an **entropic solution** and a number of consequences follows directly from it.

- The function
$$x \longmapsto u(x,t) - \frac{E}{t} x$$
is *decreasing*. In fact, let $x + z = x_2$, $x = x_1$ and $z > 0$. Then $x_2 > x_1$ and (4.61) is equivalent to
$$u(x_2,t) - \frac{E}{t} x_2 \leq u(x_1,t) - \frac{E}{t} x_1. \tag{4.62}$$
A remarkable consequence is that, for every $t > 0$, either $u(\cdot,t)$ is continuous or it has at most a countable number of jump discontinuities.
- If x is a jump discontinuity point for $u(\cdot,t)$, then
$$u_+(x,t) < u_-(x,t), \tag{4.63}$$
where
$$u_\pm(x,t) = \lim_{y \to x^\pm} u(y,t).$$
To show it, choose $x_1 < x < x_2$ and let x_1 and x_2 both go to x in (4.62).
- Since q is **strictly convex**, (4.63) is equivalent to
$$q'(u_+) < \sigma(u_-,u_+) < q'(u_-), \tag{4.64}$$
where we have set
$$\sigma(u_-,u_+) = \frac{q(u_+) - q(u_-)}{u_+ - u_-},$$
and also to
$$\sigma(u_-,u_+) < \sigma(u_-,u_*), \tag{4.65}$$
for every state u_* between u_- and u_+.

[8] For a suitable z^* between 0 e z, one has: $u(x+z,t) - u(x,t) = u_x(x+z^*,t)z$.

[9] The denomination is due to *Courant and Friedrichs, 1948*.

- Finally, if $x = s(t)$ is a shock curve, by the Rankine-Hugoniot jump condition, (4.64) reads

$$q'(u_+(s,t)) < \dot{s} < q'(u_-(s,t)) \tag{4.66}$$

along the curve.

Remark 4.11 If

$$g' < 0, \qquad q'' < q''_{\max} < 0,$$

the inequality (4.61) changes into

$$u(x+z,t) - u(x,t) \geq -\frac{E}{t}z$$

with $E = \frac{1}{|q''_{\max}|}$. Thus, at a jump discontinuity (x,t) we have

$$u_+(x,t) > u_-(x,t)$$

while (4.64), (4.65) and (4.66) remain unchanged.

The condition (4.66) is called **entropy inequality** and it has a remarkable meaning: *the states on the left (right) of the shock curve tends to propagate faster (slower) than the shock, giving rise to a phenomenon called self-sharpening*, since no state can cross the shock curve (see Example 4.5, p. 202). Thus the characteristics **hit forward in time** the shock line. In other words, it is not possible to go back in time along a characteristic and hit a shock line, expressing a sort of *irreversible phenomenon*.

Indeed, if the characteristics hit a shock curve *backward in time*, the discontinuity wave is to be considered *non-physical*.

The geometrical meaning of conditions (4.64) and (4.65) is illustrated in Fig. 4.21 in the concave case. In both cases (concave or convex), (4.65) implies that going along the chord AB from A to B one "sees" the graph of q on the left. Equivalently, we have $u_+ > u_-$ in the concave case and $u_+ < u_-$ in the convex case.

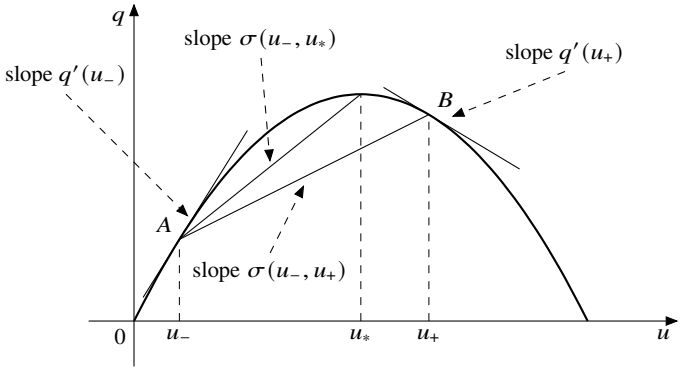

Fig. 4.21 Geometrical meaning of the entropy inequality

The above considerations lead us to select the entropy solutions as the only physically meaningful ones.

Thus, in Example 4.10, p. 208, the solution w represents a non-physical shock since it does not satisfy the entropy condition. The correct solution is therefore the simple wave (4.60).

The following result holds[10].

Theorem 4.12 *Let $q \in C^2(\mathbb{R})$ be strictly convex or concave and g locally integrable and bounded in $\mathbb{R}$. There exists a unique entropic solution of the problem*

$$\begin{cases} u_t + q(u)_x = 0 & x \in \mathbb{R},\ t > 0 \\ u(x,0) = g(x) & x \in \mathbb{R} \end{cases} \tag{4.67}$$

and $|u(x,t)| \le \sup_{\mathbb{R}} |g| = M$. Moreover, let v be another entropic solution of (4.67) with initial data h such that $\sup_{\mathbb{R}} |h| \le M$. Then the following stability estimates holds, for every $x_1, x_2 \in \mathbb{R}$, $t > 0$:

$$\int_{x_1}^{x_2} |u(x,t) - v(x,t)|\, dx \le \int_{x_1 - At}^{x_2 + At} |g(x) - h(x)|\, dx \tag{4.68}$$

where $A = \max |q'|$ over the interval $[-M, M]$.

4.6 The Riemann problem

4.6.1 Convex/concave flux function

We now apply Theorem 4.12 to solve explicitly problem (4.50) with initial data

$$g(x) = \begin{cases} u_R & x > 0 \\ u_L & x < 0, \end{cases} \tag{4.69}$$

where u_R and u_L are constant states, $u_R \neq u_L$. This problem is known as **Riemann problem**, and it is particularly important also in the construction of numerical approximation methods. The traffic problems in Sects. 4.3.3 and 4.3.4 are precisely examples of Riemann problems. We have:

Theorem 4.13 *Let $q \in C^2(\mathbb{R})$ be strictly convex and $q'' \ge h > 0$ or strictly concave and $q'' \le -h < 0$.*

a) If $q'' \ge h$ and $u_L > u_R$ or $q'' \le -h$ and $u_L < u_R$, the unique entropy solution is given by the shock wave

$$u(x,t) = \begin{cases} u_R & x > \sigma(u_L, u_R)t \\ u_L & x < \sigma(u_L, u_R)t \end{cases} \tag{4.70}$$

[10] For the proof, see e.g. *J. Smoller*, [20].

where

$$\sigma(u_L, u_R) = \frac{q(u_R) - q(u_L)}{u_R - u_L}.$$

b) If $q'' \geq h$ and $u_L \leq u_R$ or $q'' \leq -h$ and $u_L > u_R$, the unique entropy solution is given by the rarefaction wave

$$u(x,t) = \begin{cases} u_L & x < q'(u_L)t \\ r\left(\frac{x}{t}\right) & q'(u_L)t < x < q'(u_R)t \\ u_R & x > q'(u_R)t \end{cases}$$

where $r = (q')^{-1}$ is the inverse function of q'.

Proof We consider only the convex case; the concave case is perfectly similar.

a) Cleary $u_+ = u_R$ and $u_- = u_L$, so that the discontinuous solution (4.70) satisfies the Rankine Hugoniot condition and therefore it is a weak solution. Moreover, since $u_R < u_L$ the entropy condition (4.61) holds as well, so that u is the unique entropic solution of problem (4.69) by Theorem 4.12.

b) Since

$$r(q'(u_R)) = u_R \quad \text{and} \quad r(q'(u_L)) = u_L,$$

u is continuous in the half-plane $t > 0$. We now check that u satisfies the equation $u_t + q(u)_x = 0$ in the region

$$S = \left\{(x,t) : q'(u_L) < \frac{x}{t} < q'(u_R)\right\}.$$

Let $u(x,t) = r\left(\frac{x}{t}\right)$.We have:

$$u_t + q(u)_x = -r'\left(\frac{x}{t}\right)\frac{x}{t^2} + q'(r)r'\left(\frac{x}{t}\right)\frac{1}{t} = r'\left(\frac{x}{t}\right)\frac{1}{t}\left[q'(r) - \frac{x}{t}\right] \equiv 0.$$

Thus, u is a weak solution in the upper half-plane.

Let us check the entropy condition. We consider only the case

$$q'(u_L)t \leq x < x + z \leq q'(u_R)t$$

leaving the other ones to the reader. Since $q'' \geq h > 0$, we have

$$r'(s) = \frac{1}{q''(r)} \leq \frac{1}{h} \qquad (s = q'(r)),$$

so that, for a suitable z^*, $0 < z^* < z$,

$$u(x+z) - u(x,t) = r\left(\frac{x+z}{t}\right) - r\left(\frac{x}{t}\right) = r'\left(\frac{x+z^*}{t}\right)\frac{z}{t} \leq \frac{1}{h}\frac{z}{t}$$

which is the entropy condition with $E = 1/h$. □

4.6.2 Vanishing viscosity method. Travelling waves

There is another instructive and perhaps more natural way to construct entropic solutions of the conservation law

$$u_t + q(u)_x = 0 \quad x \in \mathbb{R}, t > 0 \tag{4.71}$$

the so called *vanishing viscosity method*. This method consists in viewing (4.71) as the limit for $\varepsilon \to 0^+$ of the equation

$$u_t + q(u)_x = \varepsilon u_{xx} \quad x \in \mathbb{R}, t > 0 \tag{4.72}$$

which is a second order conservation law with flux function

$$\tilde{q}(u, u_x) = q(u) - \varepsilon u_x, \tag{4.73}$$

where ε is a *small positive* number. Although we recognize εu_{xx} as a diffusion term, this kind of model arises mostly in fluid dynamics where u is the fluid velocity and ε is its *kinematic viscosity*, from which comes the name of the method.

There are several good reasons in favor of this approach. First of all, a small amount of diffusion or viscosity makes the mathematical model more realistic in most applications. Note that εu_{xx} becomes relevant only when u_{xx} is large, that is in a region where u_x changes rapidly and a shock occurs. For instance in our model of traffic dynamics, it is natural to assume that drivers would slow down when they see increased (relative) density ahead. Thus, an appropriate model for their velocity is

$$\tilde{v}(\rho) = v(\rho) - \varepsilon \frac{\rho_x}{\rho}$$

which corresponds to the flux function $\tilde{q}(\rho) = \rho v(\rho) - \varepsilon \rho_x$ for the flow-rate of cars.

Another reason comes from the fact that shocks constructed by the vanishing viscosity method are *physical shocks,* since they satisfy the entropy inequality. This is due, in principle, to the time irreversibility present in a diffusion equation that, in the limit, selects the physical admissible solution.

As for the heat equation, we expect to obtain smooth solutions of (4.72) even with discontinuous initial data. On the other hand, the competition between the steepening effect due to the nonlinear term and the diffusion due to εu_{xx} may force the evolution towards a shock wave. We are precisely interested in examining this kind of situations.

To this aim we consider solutions of (4.72) connecting two constant states u_L and u_R, that is satisfying the conditions

$$\lim_{x \to -\infty} u(x, t) = u_L, \qquad \lim_{x \to +\infty} u(x, t) = u_R. \tag{4.74}$$

Since we are looking for shock waves, it is reasonable to seek a solution depending only on a coordinate $\xi = x - vt$ moving with the (unknown) shock speed v. Thus, let us look for *travelling wave solutions* of (4.72) of the form

$$u(x, t) = U(x - vt) \equiv U(\xi),$$

with

$$U(-\infty) = u_L \quad \text{and} \quad U(+\infty) = u_R \tag{4.75}$$

and $u_L \neq u_R$. We have

$$u_t = -v\frac{dU}{d\xi}, \quad u_x = \frac{dU}{d\xi}, \quad u_{xx} = \frac{d^2U}{d\xi^2},$$

so that we obtain for U the ordinary differential equation

$$\frac{dq(U)}{d\xi} - v\frac{dU}{d\xi} = \varepsilon\frac{d^2U}{d\xi^2}$$

which can be integrated to yield

$$q(U) - vU + A = \varepsilon\frac{dU}{d\xi}$$

where A is an arbitrary constant. Assuming that $\dfrac{dU}{d\xi} \to 0$ as $\xi \to \pm\infty$ and using (4.75) we get

$$q(u_L) - vu_L + A = 0 \quad \text{and} \quad q(u_R) - vu_R + A = 0. \tag{4.76}$$

Subtracting these two equations we find

$$v = \frac{q(u_R) - q(u_L)}{u_R - u_L} \equiv \bar{v} \tag{4.77}$$

and then

$$A = \frac{-q(u_R)u_L + q(u_L)u_R}{u_R - u_L} \equiv \bar{A}. \tag{4.78}$$

Thus, if there exists a travelling wave solution satisfying conditions (4.74), it moves with a speed $\bar{v}$ predicted by the Rankine-Hugoniot formula. Still it is not clear whether such travelling wave solution exists. To verify this, examine the equation

$$\varepsilon\frac{dU}{d\xi} = q(U) - \bar{v}U + \bar{A}. \tag{4.79}$$

From (4.76), equation (4.79) has the two equilibria $U = u_R$ and $U = u_L$. Moreover, conditions (4.75) require u_R to be *asymptotically stable* and u_L *unstable*. At this point, we need to have information on the shape of q.

Assume $q'' < 0$. Then the phase diagram for eq. (4.79) is described in Fig. 4.22, for the two cases $u_L > u_R$ and $u_L < u_R$. Between u_L and u_R, $q(U) - \bar{v}U + \bar{A} > 0$ and, as the arrows indicate, U is *increasing*. We see that only the case $u_L < u_R$ is compatible with conditions (4.75) and this corresponds precisely to a shock formation for the non-diffusive conservation law. Thus,

$$q'(u_L) - \bar{v} > 0 \quad \text{and} \quad q'(u_R) - \bar{v} < 0$$

or

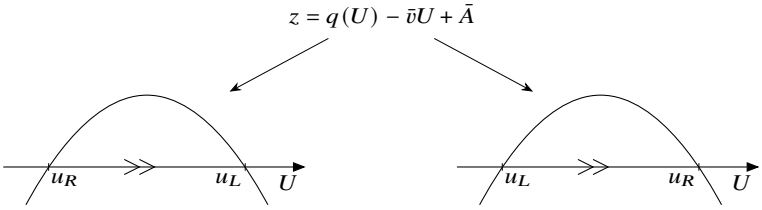

Fig. 4.22 Case (b) only is compatible with conditions (4.74)

$$q'(u_R) < \bar{v} < q'(u_L) \tag{4.80}$$

which is *the entropy inequality*.

Similarly, if $q'' > 0$, a travelling wave solution connecting the two states u_R and u_L exists only if $u_L > u_R$, U is *decreasing* and (4.80) holds.

Let us see what happens when $\varepsilon \to 0^+$. Assume, for instance, $q'' < 0$. As we have noticed, for ε small, the diffusion term becomes relevant where u_x changes rapidly and therefore we expect that our travelling wave increases abruptly from a value $U(\xi_1)$ close to u_L to a value $U(\xi_2)$ close to u_R, within a narrow region called the *transition layer*. For instance, we may choose ξ_1 and ξ_2 such that

$$U(\xi_2) - U(\xi_1) \geq (1 - \beta)(u_R - u_L)$$

with a positive β, very close to 0. We call the number $\varkappa = \xi_2 - \xi_1$ *thickness* of the transition layer. To compute it, we separate the variables U and ξ in (4.79) and integrate over (ξ_1, ξ_2); this yields

$$\xi_2 - \xi_1 = \varepsilon \int_{U(\xi_1)}^{U(\xi_2)} \frac{ds}{q(s) - \bar{v}s + \bar{A}}.$$

Thus, the thickness of the transition layer is proportional to ε. As $\varepsilon \to 0^+$, the transition region becomes narrower and narrower, and eventually a shock wave that satisfies the entropy inequality is obtained.

In the next example, we examine this phenomenon for the important case of the *Burgers, viscous* equation.

Example 4.14 (Burgers shock solution) In this example we construct the entropic solution of the Riemann problem for the Burgers equation with initial data u_L and u_R, $u_L > u_R$, as a limit for $\varepsilon \to 0^+$ of travelling wave solutions to the viscous Burgers equation

$$u_t + uu_x = \varepsilon u_{xx} \tag{4.81}$$

connecting the states u_L and u_R.

First we find an explicit formula for the travelling wave solutions. Note that

$$q(u) = u^2/2$$

is convex, so we will find $u(x,t) = U(\xi)$, $\xi = x - \bar{v}t$, with U *decreasing*. The formulas (4.77) and (4.78) give

$$\bar{v} = \frac{u_L + u_R}{2} \quad \text{and} \quad \bar{A} = -\frac{1}{2}u_L u_R.$$

The equation (4.79) becomes

$$2\varepsilon \frac{dU}{d\xi} = U^2 - 2\bar{v}U - 2\bar{A} = (U - u_L)(U - u_R)$$

and can be easily integrated to give (recall that $u_R < U < u_L$),

$$\frac{1}{u_L - u_R} \log \frac{u_L - U}{U - u_R} = \frac{\xi}{2\varepsilon} + k \qquad (k \in \mathbb{R})$$

or

$$U(\xi) = u_R + \frac{u_L - u_R}{1 + c \exp\left\{\dfrac{u_L - u_R}{2\varepsilon}\xi\right\}} \qquad (c = e^{k(u_L - u_R)} > 0).$$

Thus we have found a family of travelling waves given by

$$u(x,t) = u_R + \frac{u_L - u_R}{1 + c \exp\left\{\dfrac{u_L - u_R}{2\varepsilon}(x - \bar{v}t)\right\}} \qquad (c = e^{k(u_L - u_R)} > 0).$$

(see Fig. 4.23). When $\varepsilon \to 0^+$,

$$u(x,t) \to w(x,t) = \begin{cases} u_L & x < \bar{v}t \\ u_R & x > \bar{v}t \end{cases}$$

which is the entropic shock solution for the nonviscous Burgers equation with initial data u_L if $x < 0$ and u_R if $x > 0$.

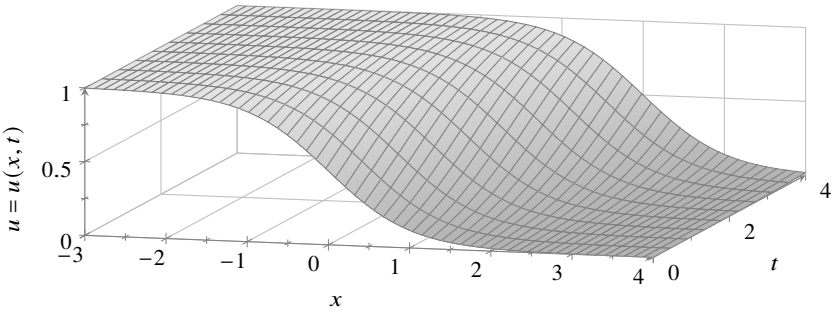

Fig. 4.23 The travelling wave in Example 4.14, with $c = 1$

Remark 4.15 Remarkably, there is an explicit representation formula for the solution of the initial value problem for the viscous Burgers equation (see Problem 4.13).

4.6.3 Flux function with inflection points

When the flux curve exhibits a finite number inflection points we have to find other criteria to select physically admissible discontinuities. In these cases, limiting ourselves to the class of piecewise-C^1 weak solutions, a slight generalization of (4.65), that we call *condition E*, acts as a proper entropy condition. It reads:

$$\sigma(u_-, u_+) \leq \sigma(u_-, u_*) \tag{4.82}$$

along a discontinuity curve, for every state u_* between u_- and u_+. Observe that now we do not know *a priori* which is the greater value between u_- and u_+. Also note that the equality sign in (4.82) includes also the linear case $q(u) = vu$, v constant. In this case, the jump discontinuity propagates with the same speed of the states on both sides: the corresponding wave is called *contact discontinuity* (see Fig. 4.3, p. 187). The following uniqueness theorem holds[11]:

Theorem 4.16 *There exists at most a piecewise-C^1 weak solution of problem* (4.50)*, satisfying condition* (4.82)*.*

Consider now a Riemann problem with states u_L for $x < 0$ and u_R for $x > 0$. According to (4.82), the states u_L, u_R can be connected by an admissible shock as long as the chord joining the two points $(u_L, q(u_L))$ and $(u_R, q(u_R))$ lies above (resp. below) the graph of q if $u_R < u_L$ (resp. $u_R > u_L$).

On the other hand, we know that the states u_L, u_R can be connected by a rarefaction wave as long as the arc joining the two points $(u_L, q(u_L))$ and $(u_R, q(u_R))$ is strictly concave (resp. convex) if $u_R < u_L$ (resp. $u_R > u_L$).

An immediate consequence is that, *if q has inflection points, in general the Riemann problem cannot be solved by a single wave.*

We first examine the **case** $u_R < u_L$. Referring, for instance, to Figure 4.24, the states u_L and u_R cannot be directly connected by a single shock since the chord joining $(u_R, q(u_R))$, $(u_L, q(u_L))$ would cross the flux curve, violating the entropy condition (4.82).

Also we cannot use a single rarefaction wave since q' is not invertible between u_R and u_L. Thus, to construct the entropic solution, the right (and only!) way to proceed[12] is to introduce suitable intermediate states that can be connected either by an admissible discontinuity (entropic shock or contact discontinuity) or by a rarefaction wave.

---

[11] See *O.A. Oleinik,* Uniqueness and stability of the generalized solution of the Cauchy Problem for a quasilinear equation, Amer. Math. Soc. Transl., Ser 2, 33, 285–290, *1963*.

[12] See Problem 4.15.

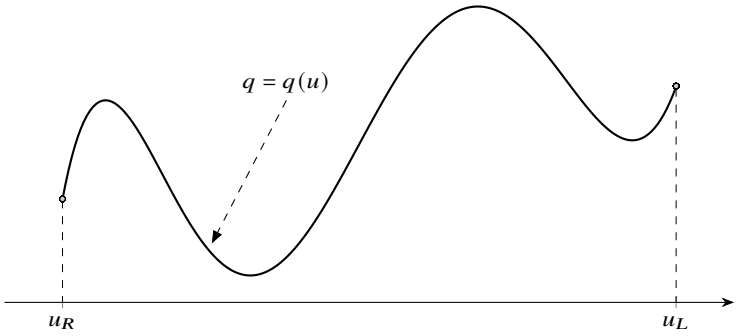

Fig. 4.24 Flux function with three inflexion points

To this purpose we introduce the *upper concave envelope* q^c of q in $[u_R, u_L]$, that is the lowest concave function that lives over q in the interval $[u_R, u_L]$. Figure 4.25 shows q^c for the flux function in Fig. 4.24, which has 3 inflection points.

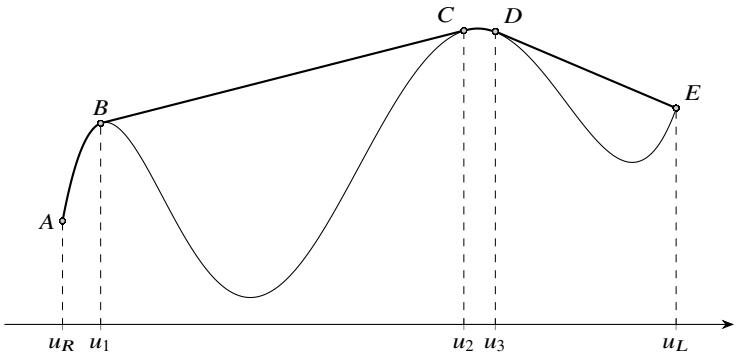

Fig. 4.25 Upper concave envelope of the function in Fig. 4.24

We use this example to describe the construction of the entropic solution. We see that the interval $[u_R, u_L]$ is decomposed into the subintervals $[u_R, u_1]$ and $\left[u_2, u_3\right]$, in which q^c is strictly concave (*rarefaction intervals*), alternating with the subintervals $[u_1, u_2]$ and $\left[u_3, u_L\right]$ where the graph of q^c is a straight line (*shock intervals*). Then our entropic solution has the following structure, where r denotes the inverse of q' on each rarefaction interval:

- the characteristics $x = q'(u_L)t + \xi$, $\xi < 0$, carry the value $u = u_L$ into the sector $t > 0$, $x < q'(u_3)t = \sigma(u_L, u_3)t$;
- across the shock line $x = \sigma(u_L, u_3)t$, u jumps down from $u = u_L$ to $u = u_3$. Note that only the characteristics of equation $x = q'(u_L)t + \xi$, $\xi < 0$, coming from the left, impinge on the shock line. For this reason this kind of discontinuity is more properly called a *semishock*;

- the states u_3 and u_2 are connected by the rarefaction wave $w_{u_3,u_2}(x,t) = r(x/t)$, defined in the sector $\sigma(u_L,u_3)t < x < \sigma(u_2,u_1)t$;
- across the shock line $x = \sigma(u_2,u_1)t$, u jumps down from $u = u_2$ to $u = u_1$. Since

$$\sigma(u_2,u_1) = q'(u_2) = q'(u_1),$$

the jump discontinuity moves at the same speed of the states at both sides and therefore it is a *contact discontinuity*;
- the states u_1 and u_R are connected by the rarefaction wave $w_{u_1,u_R}(x,t) = r(x/t)$, defined in the sector $\sigma(u_2,u_1)t < x < q'(u_R)t$;
- finally, $u = u_R$ in the sector $t > 0$, $q'(u_R)t < x$.

The characteristic configuration and the profile of u at time t are described in Fig. 4.26 and 4.27, respectively.

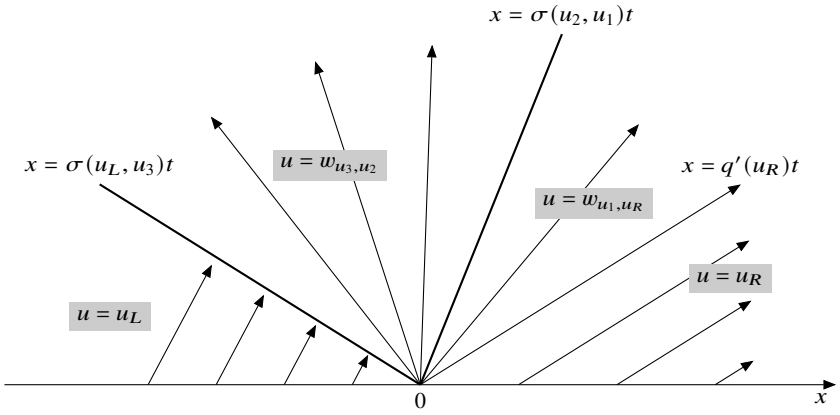

Fig. 4.26 Back semishock followed by a rarefaction wave, a contact discontinuity and then by a front rarefaction wave. Characteristics configuration

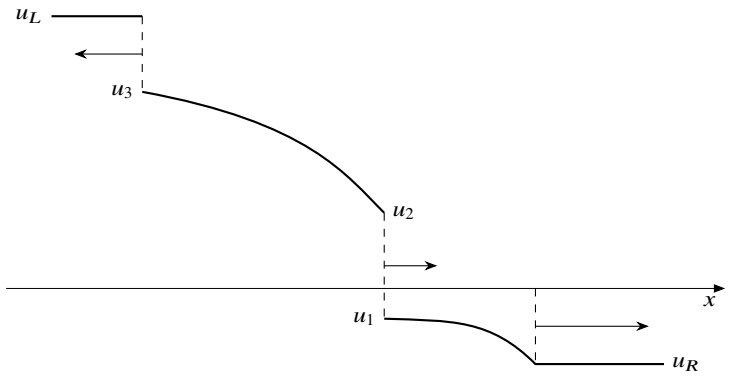

Fig. 4.27 Profile at a fixed time of the solution in Fig. 4.26

The **case** $u_L < u_R$ can be treated similarly. This time one needs to introduce the lower convex envelope q_c of q. The interval $[u_L, u_R]$ is decomposed into subintervals in which q_c is strictly convex (*rarefaction intervals*), alternating with subintervals where the graph of q_c is a straight line (*shock intervals*). Then the construction of the entropic solution of the Riemann problem proceed as in the previous case (see Problem 4.14).

4.7 The Method of Characteristics for Quasilinear Equations

In this section we apply the method of characteristics to general quasilinear equations. We consider the case of two independent variables, where the intuition is supported by the geometric interpretation.

4.7.1 Characteristics

We consider equations of the form

$$a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u), \tag{4.83}$$

where

$$u = u(x, y)$$

and a, b, c are *continuously differentiable functions*.

The solutions of (4.83) can be constructed via geometric arguments. The tangent plane to the graph of a solution u at a point (x_0, y_0, z_0) has equation

$$u_x(x_0, y_0)(x - x_0) + u_y(x_0, y_0)(y - y_0) - (z - z_0) = 0$$

and the vector

$$\mathbf{n}_0 = (u_x(x_0, y_0), u_y(x_0, y_0), -1)$$

is *normal* to that plane. Introducing the vector

$$\mathbf{v}_0 = (a(x_0, y_0, z_0), b(x_0, y_0, z_0), c(x_0, y_0, z_0)),$$

equation (4.83) implies that

$$\mathbf{n}_0 \cdot \mathbf{v}_0 = 0.$$

Thus, $\mathbf{v}_0$ is tangent to the graph of u (Fig. 4.28). In other words, (4.83) says that, at every point (x, y, z), the graph of any solution is tangent to the vector field

$$\mathbf{v}(x, y, z) = (a(x, y, z), b(x, y, z), c(x, y, z)).$$

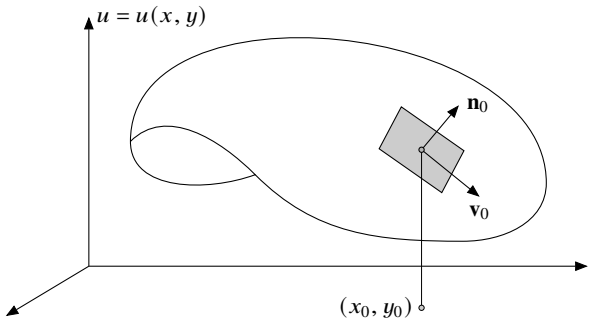

Fig. 4.28 Integral surface

In this case we say that the graph of a solution is an **integral surface** of the vector field **v**.

Now, we can construct an integral surfaces of **v** as union of **integral curves** of **v**, that is curves tangent to **v** at every point. These curves are solutions of the system

$$\frac{dx}{dt} = a(x, y, z), \qquad \frac{dy}{dt} = b(x, y, z), \qquad \frac{dz}{dt} = c(x, y, z) \tag{4.84}$$

and are called **characteristics**. Note that $z = z(t)$ gives the values u along the *projected characteristic* $(x(t), y(t))$, that is

$$z(t) = u(x(t), y(t)). \tag{4.85}$$

In fact, differentiating (4.85) and using (4.84) and (4.83), we have

$$\begin{aligned}\frac{dz}{dt} &= u_x(x(t), y(t))\frac{dx}{dt} + u_y(x(t), y(t))\frac{dy}{dt} \\ &= a(x(t), y(t), z(t))u_x(x(t), y(t)) + b((x(t), y(t), z(t))u_y(x(t), y(t)) \\ &= c(x(t), y(t), z(t)).\end{aligned}$$

Thus, along a characteristic the partial differential equation (4.83) degenerates into an ordinary differential equation.

In the case of a conservation law (with $t = y$)

$$u_y + q'(u)u_x = 0 \qquad\qquad (q'(u) = \frac{dq}{du}),$$

we have introduced the notion of characteristic in a slightly different way, but we shall see later that there is no contradiction.

The following proposition is a consequence of the above geometric reasoning and of the existence and uniqueness theorem for systems of ordinary differential equations (see *R. Courant, D. Hilbert*, [26], vol. 2).

Proposition 4.17 *Under the above notation:*

a) Let the surface S be the graph of a C^1-function $u = u(x, y)$. If S is union of characteristics then u is a solution of the equation (4.83).
b) Every integral surface S of the vector field $\mathbf{v}$ *is union of characteristics. Namely: every point of S belongs exactly to one characteristic, entirely contained in S.*
c) Two integral surfaces intersecting at one point intersect along the whole characteristic passing through that point.

4.7.2 The Cauchy problem

Proposition 4.17 gives a characterization of the integral surfaces as a union of characteristics. The problem is how to construct such unions to get a smooth surface. One way to proceed is to look for solutions *u whose values are prescribed on a curve γ_0, contained in the x, y plane*.

In other words, suppose that

$$x(s) = f(s), \qquad y(s) = g(s), \qquad s \in I \subseteq \mathbb{R},$$

is a parametrization of γ_0. We look for a solution u of (4.83) such that

$$u(f(s), g(s)) = h(s), \qquad s \in I, \tag{4.86}$$

where $h = h(s)$ is a given function. We assume that I is a neighborhood of $s = 0$, and that f, g, h are *continuously differentiable* in I.

The system (4.83), (4.86) is called **Cauchy problem**. If we consider the three-dimensional curve Γ_0 given by the parametrization

$$x(s) = f(s), \qquad y(s) = g(s), \qquad z(s) = h(s),$$

then, solving the Cauchy problem (4.83), (4.86) amounts to *determining an integral surface containing* Γ_0.

The data are often assigned in the form of *initial values*

$$u(x, 0) = h(x),$$

with y playing the role of "time". In this case, γ_0 is the axis $y = 0$ and x plays the role of the parameter s. Then a parametrization of Γ_0 is given by

$$x = x, \qquad y = 0, \qquad z(x) = h(x).$$

By analogy, we often refer to Γ_0 as to the *initial curve*. The strategy to solve a Cauchy problem comes from its geometric meaning: since the graph of a solution $u = u(x, y)$ is a union of characteristics, we determine those flowing out from Γ_0 by solving the system

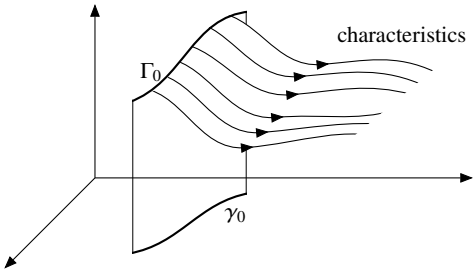

Fig. 4.29 Characteristics flowing out of Γ_0

$$\frac{dx}{dt} = a(x, y, z), \qquad \frac{dy}{dt} = b(x, y, z), \qquad \frac{dz}{dt} = c(x, y, z), \tag{4.87}$$

with the family of initial conditions

$$x(0) = f(s), \quad y(0) = g(s), \quad z(0) = h(s), \tag{4.88}$$

parametrized by $s \in I$. The union of the characteristics of this family should give the graph of u, see Fig. 4.29. Why only *should*? We will come back later to this question.

Under our hypotheses, the Cauchy problem (4.87), (4.88) has exactly one solution

$$x = X(s, t), \quad y = Y(s, t), \quad z = Z(s, t) \tag{4.89}$$

in a neighborhood of $t = 0$, for every $s \in I$.

A couple of questions arise:

a) Does the system of the three equations (4.89) define a function $z = u(x, y)$?
b) Even if the answer to the question **a**) is positive, is $z = u(x, y)$ the unique solution of the Cauchy problem (4.83), (4.86)?

Let us consider a neighborhood of $s = t = 0$, setting

$$X(0, 0) = f(0) = x_0, \quad Y(0, 0) = g(0) = y_0, \quad Z(0, 0) = h(0) = z_0.$$

The answer to the question **a**) is positive if we can solve for s and t the first two equations in (4.89), and find $s = S(x, y)$ and $t = T(x, y)$ of class C^1 in a neighborhood of (x_0, y_0), such that

$$S(x_0, y_0) = 0, \qquad T(x_0, y_0) = 0.$$

Then, from the third equation $z = Z(s, t)$, we get

$$z = Z(S(x, y), T(x, y)) = u(x, y). \tag{4.90}$$

From the Inverse Function Theorem 1.5, p. 16, the system

$$\begin{cases} X(s, t) = x \\ Y(s, t) = y \end{cases}$$

defines

$$s = S(x, y) \quad \text{and} \quad t = T(x, y)$$

in a neighborhood of (x_0, y_0) if the Jacobian

$$J(0,0) = \begin{vmatrix} X_s(0,0) & Y_s(0,0) \\ X_t(0,0) & Y_t(0,0) \end{vmatrix} \neq 0. \tag{4.91}$$

From (4.87) and (4.88), we have

$$X_s(0,0) = f'(0), \quad Y_s(0,0) = g'(0)$$

and

$$X_t(0,0) = a(x_0, y_0, z_0), \quad Y_t(0,0) = b(x_0, y_0, z_0),$$

so that (4.91) becomes

$$J(0,0) = \begin{vmatrix} f'(0) & g'(0) \\ a(x_0, y_0, z_0) & b(x_0, y_0, z_0) \end{vmatrix} \neq 0 \tag{4.92}$$

or

$$b(x_0, y_0, z_0) f'(0) \neq a(x_0, y_0, z_0) g'(0). \tag{4.93}$$

Condition (4.93) means that *the vectors*

$$(a(x_0, y_0, z_0), b(x_0, y_0, z_0)) \quad \text{and} \quad (f'(0), g'(0))$$

are not parallel.

In conclusion: *if condition* (4.92) *holds, then* (4.90) *is a well defined* C^1*-function.*

Now consider question **b**). The above construction of u implies that the surface $z = u(x, y)$ contains Γ_0 and all the characteristics flowing out from Γ_0, so that u is a solution of the Cauchy problem (4.83), (4.86). Moreover, by Proposition 4.17 c), two integral surfaces containing Γ_0 must contain the same characteristics and therefore coincide.

We summarize everything in the following theorem, recalling that

$$(x_0, y_0, z_0) = (f(0), g(0), h(0)).$$

Theorem 4.18 *Let* a, b, c *be* C^1*-functions in a neighborhood of* (x_0, y_0, z_0) *and* f, g, h *be* C^1*-functions in* I*. If* $J(0,0) \neq 0$*, then, in a neighborhood of* (x_0, y_0)*, there exists a unique* C^1*-solution* $u = u(x, y)$ *of the Cauchy problem*

$$\begin{cases} a(x, y, u) u_x + b(x, y, u) u_y = c(x, y, u) \\ u(f(s), g(s)) = h(s). \end{cases} \tag{4.94}$$

Moreover, u *is defined by the parametric equations* (4.99).

Remark 4.19 If a, b, c and f, g, h are C^k-functions, $k \geq 2$, then u is a C^k-function as well.

It remains to examine what happens when $J(0,0) = 0$, that is when the vectors $(a(x_0, y_0, z_0), b(x_0, y_0, z_0))$ and $(f'(0), g'(0))$ are parallel.

Suppose that there exists a C^1-solution u of the Cauchy problem (4.94). Differentiating the second equation in (4.94) we get

$$h'(s) = u_x(f(s), g(s))f'(s) + u_y(f(s), g(s))g'(s). \tag{4.95}$$

Computing at $x = x_0$, $y = y_0$, $z = z_0$ and $s = 0$, we obtain

$$\begin{cases} a(x_0, y_0, z_0)u_x(x_0, y_0) + b(x_0, y_0, z_0)u_y(x_0, y_0) = c(x_0, y_0, z_0) \\ f'(0)u_x(x_0, y_0) + g'(0)u_y(x_0, y_0) = h'(0). \end{cases} \tag{4.96}$$

Since u is a solution of the Cauchy problem, the vector

$$(u_x(x_0, y_0), u_y(x_0, y_0))$$

is a solution of the algebraic system (4.96). But then, from Linear Algebra, we know that the condition

$$\operatorname{rank} \begin{pmatrix} a(x_0, y_0, z_0) & b(x_0, y_0, z_0) & c(x_0, y_0, z_0) \\ f'(0) & g'(0) & h'(0) \end{pmatrix} = 1 \tag{4.97}$$

must hold and therefore the two vectors

$$(a(x_0, y_0, z_0), b(x_0, y_0, z_0), c(x_0, y_0, z_0)) \quad \text{and} \quad (f'(0), g'(0), h'(0)) \tag{4.98}$$

must be parallel. This is equivalent to saying that Γ_0 is tangent to the characteristic curve at (x_0, y_0, z_0). When this occurs, we say that (x_0, y_0, z_0) is a **characteristic point for** Γ_0.

Conclusion: *If $J(0,0) = 0$, a necessary condition for the existence of a C^1-solution $u = u(x, y)$ of the Cauchy problem in a neighborhood of (x_0, y_0) is that (x_0, y_0, z_0) be a characteristic point for Γ_0.*

Now, assume Γ_0 *itself is a characteristic* and let $P_0 = (x_0, y_0, z_0) \in \Gamma_0$. If we choose a curve Γ^* transversal[13] to Γ_0 at P_0, by Theorem 4.18 there exists a unique integral surface containing Γ^* and, by Proposition 4.17 c), p. 223, this surface contains Γ_0. In this way we can construct *infinitely many smooth solutions*.

We have seen that if $J(0,0) = 0$ and a C^1-solution exists in a neighborhood of (x_0, y_0) then, necessarily, P_0 is characteristic for Γ_0. On the other hand, it may occur that $J(0,0) = 0$, that P_0 is non-characteristic Γ_0 and that solutions of the Cauchy problem exist anyway; clearly, these solutions **cannot** be of class C^1 (see Example 4.20 below).

Let us summarize the steps to solve the Cauchy problem (4.94):

Step 1. Determine the solution

[13] That is, with non-collinear tangent vectors.

$$x = X(s,t), \quad y = Y(s,t), \quad z = Z(s,t) \tag{4.99}$$

of the characteristic system

$$\frac{dx}{dt} = a(x,y,z), \quad \frac{dy}{dt} = b(x,y,z), \quad \frac{dz}{dt} = c(x,y,z), \tag{4.100}$$

with initial conditions

$$x(s,0) = f(s), \quad y(s,0) = g(s), \quad z(s,0) = h(s), \qquad s \in I.$$

Step 2. Compute $J(s,t)$ on the initial curve Γ_0, that is:

$$J(s,0) = \begin{vmatrix} f'(s) & g'(s) \\ X_t(0,s) & Y_t(0,s). \end{vmatrix}.$$

We examine the following cases:

2a. $J(s,0) \neq 0$, for every $s \in I$. This means that Γ_0 does not have characteristic points. Then, in a neighborhood of Γ_0, there exists a unique solution $u = u(x,y)$ of the Cauchy problem, defined by the parametric equations (4.99).

2b. $J(s_0,0) = 0$ for some $s_0 \in I$, and $P_0 = (f(s_0), g(s_0), h(s_0))$ is a characteristic point for Γ_0. A C^1-solution may exist in a neighborhood of P_0 only if the rank condition (4.97) holds at P_0.

2c. $J(s_0,0) = 0$ for some $s_0 \in I$ and P_0 is not characteristic for Γ_0. There are no C^1-solutions in a neighborhood of P_0. There may exist less regular solutions.

2d. Γ_0 is a characteristic. Then there exist infinitely many C^1-solutions in a neighborhood of Γ_0.

Example 4.20 Consider the nonhomogeneous Burgers equation

$$uu_x + u_y = 1. \tag{4.101}$$

As in Example 4.10, if y is the time variable y, $u = u(x,y)$ represents *a velocity field* of a flux of particles along the x-axis. Equation (4.101) states that the acceleration of each particle is equal to 1. Assume that

$$u(x,0) = h(x), \qquad x \in \mathbb{R}.$$

The characteristics are solutions of the system

$$\frac{dx}{dt} = z, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = 1$$

and the initial curve Γ_0 has the parametrization

$$x = f(s) = s, \quad y = g(s) = 0, \quad z = h(s), \qquad s \in \mathbb{R}.$$

The characteristics flowing out from Γ_0 are

$$X(s,t) = s + \frac{t^2}{2} + th(s), \qquad Y(s,t) = t, \qquad Z(s,t) = h(s) + t.$$

Since

$$J(s,t) = \begin{vmatrix} 1 + th'(s) & 0 \\ t + h(s) & 1 \end{vmatrix} = 1 + th'(s),$$

we have $J(s,0) = 1$ and we are in the case **2a**: in a neighborhood of Γ_0 there exists a unique C^1-solution. If, for instance, $h(s) = s$, we find the solution

$$u = \frac{2x - y^2}{2 + 2y} + y, \qquad (x \in \mathbb{R},\ y > -1).$$

Now consider the same equation with initial condition

$$u\left(\frac{y^2}{4}, y\right) = \frac{y}{2},$$

equivalent to assigning the values of u on the parabola $x = \frac{y^2}{4}$. A parametrization of Γ_0 is given by

$$x = s^2, \quad y = 2s, \quad z = s, \qquad s \in \mathbb{R}.$$

Solving the characteristic system with these initial conditions, we find

$$X(s,t) = s^2 + ts + \frac{t^2}{2}, \qquad Y(s,t) = 2s + t, \qquad Z(s,t) = s + t. \tag{4.102}$$

Observe that Γ_0 does not have any characteristic point, since its tangent vector $(2s, 2, 1)$ is never parallel to the characteristic direction $(s, 1, 1)$. However

$$J(s,t) = \begin{vmatrix} 2s + t & 2 \\ s + t & 1 \end{vmatrix} = -t$$

which vanishes for $t = 0$, i.e. exactly on Γ_0. We are in the case **2c**. Solving for s and t, $t \neq 0$, in the first two equations (4.102), and substituting into the third one, we find

$$u(x,y) = \frac{y}{2} \pm \sqrt{x - \frac{y^2}{4}}.$$

We have found two solutions of the Cauchy problem, satisfying the differential equation in the region $x > \frac{y^2}{4}$. However, these solutions are not smooth in a neighborhood of Γ_0, since on Γ_0 they are not differentiable.

• *Conservation laws*. According to the new definition, the characteristics of the equation

$$u_y + q'(u)u_x = 0, \qquad (q'(u) = \frac{dq}{du}),$$

with initial conditions

$$u(x,0) = g(x),$$

are the three-dimensional solution curves of the system

$$\frac{dx}{dt} = q'(z), \qquad \frac{dy}{dt} = 1, \qquad \frac{dz}{dt} = 0$$

with initial conditions

$$x(s,0) = s, \quad y(s,0) = 0, \quad z(s,0) = g(s), \qquad s \in \mathbb{R}.$$

Integrating, we find

$$z = g(s), \qquad x = q'(g(s))t + s, \qquad y = t.$$

The projections of these straight-lines on the x, y plane are

$$x = q'(g(s))y + s,$$

which coincide with the "old characteristics", called *projected characteristics* in the general quasilinear context.

- *Linear equations*. Consider a linear equation

$$a(x,y)u_x + b(x,y)u_y = 0. \tag{4.103}$$

Introducing the vector $\mathbf{w}(x,y) = (a(x,y), b(x,y))$, we may write equation (4.103) in the form

$$D_{\mathbf{w}}u = \nabla u \cdot \mathbf{w} = 0.$$

Thus, every solution of the (4.103) is constant along the integral lines of the vector field w, i.e. along the projected characteristics, solutions of the reduced characteristic system

$$\frac{dx}{dt} = a(x,y), \qquad \frac{dy}{dt} = b(x,y), \tag{4.104}$$

locally equivalent to the ordinary differential equation

$$b(x,y)dx - a(x,y)dy = 0.$$

If a *first integral*[14] $\psi = \psi(x,y)$ of the system (4.104) is known, then the family of the projected characteristics is given in implicit form by

$$\psi(x,y) = k, \qquad k \in \mathbb{R}$$

and the formula

$$u(x,y) = G(\psi(x,y)),$$

[14] We recall that a *first integral* (also called *constant of motion*) for a system of ODE $\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x})$, is a nonconstant C^1-function $\varphi = \varphi(\mathbf{x})$, which is constant along the trajectories of the system, i.e. such that $\nabla\varphi \cdot \mathbf{f} \equiv 0$.

defines a family of solutions depending on an arbitrary C^1-function $G = G(r)$, called *general solution*[15] of (4.103), as we shall see in the next section. G can then be determined by the Cauchy data.

Example 4.21 Let us solve the problem

$$\begin{cases} yu_x + xu_y = 0 \\ u(x,0) = x^4. \end{cases} \tag{4.105}$$

Here $\mathbf{w} = (y, x)$ and the reduced characteristic system is

$$\frac{dx}{dt} = y, \qquad \frac{dy}{dt} = x,$$

locally equivalent to

$$xdx - ydy = 0.$$

Integrating, we find that the projected characteristics are the hyperbolas

$$\psi(x,y) = x^2 - y^2 = k$$

and therefore $\psi(x,y) = x^2 - y^2$ is a first integral. Then, the general solution of the equation is

$$u(x,y) = G(x^2 - y^2).$$

Imposing the Cauchy condition, we have

$$G(x^2) = x^4$$

from which $G(r) = r^2$. The solution of the Cauchy problem is

$$u(x,y) = (x^2 - y^2)^2.$$

Remark 4.22 The PDE in (4.105) has the form (4.103) with $\operatorname{div}\mathbf{w} = \operatorname{div}(a,b) = 0$. Then $\operatorname{curl}(b,-a) = \mathbf{0}$ and there exists a scalar potential ϕ such that $\nabla\phi = (b,-a)$. Therefore $\nabla\phi \cdot (a,b) = 0$ and ϕ is a first integral. Thus a general solution of (4.103) is, in this case, $u(x,y) = G(\phi(x,y))$, with G arbitrary.

4.7.3 General solutions and Lagrange method of first integrals

We have seen that, in the linear case, we can construct a *general solution*, depending on an arbitrary function, from the knowledge of a first integral for the reduced characteristic system. The method can be extended to equations of the form

[15] The notion of general solution of a first order PDE parallels that of general integral of an ODE; this does not mean that it contains *all* the possible solutions, see Problem 4.20.

$$a(x,y,u)u_x + b(x,y,u)u_y = c(x,y,u). \tag{4.106}$$

We say that two *first integrals* $\varphi = \varphi(x,y,u)$ and $\psi = \psi(x,y,u)$ are *independent* if $\nabla\varphi$ and $\nabla\psi$ are nowhere colinear. Then:

Theorem 4.23 *Let $\varphi = \varphi(x,y,u)$, $\psi = \psi(x,y,u)$ be two independent first integrals of the characteristic system associated to* (4.106)*. The general solution of* (4.106) *is defined in an implicit way as*

$$F(\varphi(x,y,u), \psi(x,y,u)) = 0$$

where $F = F(h,k)$ is any C^1-function such that

$$F_h\varphi_u + F_k\psi_u \neq 0.$$

Proof It is based on the following two observations. First, the function

$$w = F(\varphi(x,y,u), \psi(x,y,u)) \tag{4.107}$$

is a first integral. In fact,

$$\nabla w = F_h\nabla\varphi + F_k\nabla\psi$$

so that

$$\nabla w \cdot (a,b,c) = F_h\nabla\varphi \cdot (a,b,c) + F_k\nabla\psi \cdot (a,b,c) \equiv 0$$

since φ and ψ are first integrals. Moreover, by hypothesis,

$$w_u = F_h\varphi_u + F_k\psi_u \neq 0.$$

Second, if w is a first integral and $w_u \neq 0$, then, the equation

$$w(x,y,u) = 0 \tag{4.108}$$

defines implicitly an integral surface $u = u(x,y)$ of (4.106). In fact, since w is a first integral, it satisfies the equation

$$a(x,y,u)w_x + b(x,y,u)w_y + c(x,y,u)w_u = 0. \tag{4.109}$$

Moreover, from the Implicit Function Theorem, we have

$$u_x = -\frac{w_x}{w_u}, \qquad u_y = -\frac{w_y}{w_u},$$

and from (4.109) we get easily (4.106). □

Remark 4.24 As a by-product of the proof, we have that the general solution $w = w(x,y,u)$ of the *linear* three-dimensional homogeneous equation (4.109) is given by (4.107).

Remark 4.25 The search for first integrals is sometimes simplified by writing the characteristic system in the form

$$\frac{dx}{a(x,y,u)} = \frac{dy}{b(x,y,u)} = \frac{du}{c(x,y,u)}.$$

Example 4.26 Consider again the nonhomogeneous Burgers equation

$$uu_x + u_y = 1$$

with initial condition

$$u\left(\frac{1}{2}y^2, y\right) = y.$$

A parametrization of the initial curve Γ_0 is

$$x = \frac{1}{2}s^2, \quad y = s, \quad u = s$$

and therefore Γ_0 is a characteristic. We are in the case **2d**. Let us use the Lagrange method. To find two independent first integrals, we write the characteristic system in the form

$$\frac{dx}{u} = dy = du$$

or

$$dx = udu, \qquad dy = du.$$

Integrating these two equations, we get

$$x - \frac{1}{2}u^2 = c_1, \qquad y - u = c_2$$

and therefore

$$\varphi(x, y, u) = x - \frac{1}{2}u^2$$
$$\psi(x, y, u) = y - u$$

are two first integral. Since

$$\nabla\varphi(x, y, u) = (1, 0, -u)$$

and

$$\nabla\psi(x, y, u) = (0, 1, -1)$$

we see that they are also independent. Thus, the general solution of Burgers equation is given by

$$F\left(x - \frac{1}{2}u^2, y - u\right) = 0$$

where F is an arbitrary C^1-function.

Finally, to satisfy the initial condition, it is enough to choose F such that

$$F(0, 0) = 0.$$

As expected, there exist infinitely many solutions of the Cauchy problem.

4.7.4 Underground flow

We apply the methodology presented in the previous sections to a model for the underground flow of a fluid (like water). In the wet region, only a fraction of any control volume is filled with fluid. This fraction, denoted by ϕ, is called *porosity* and, in general, it depends on position, temperature and pressure. Here, we assume that ϕ depends on position only: $\phi = \phi(x, y, z)$.

If ρ is the fluid density and $\mathbf{q} = (q_1, q_2, q_3)$ is the flux vector (the volumetric flow rate of the fluid), the conservation of mass yields, in this case,

$$\phi\rho_t + \operatorname{div}(\rho\mathbf{q}) = 0.$$

For $\mathbf{q}$ the following modified *Darcy's law* is commonly adopted:

$$\mathbf{q} = -\frac{k}{\mu}(\nabla p + \rho\mathbf{g})$$

where p is the pressure and $\mathbf{g}$ is the gravity acceleration; $k > 0$ is the *permeability* of the medium and μ is the fluid *viscosity*. Thus, we have:

$$\phi\rho_t - \operatorname{div}\left[\frac{\rho k}{\mu}(\nabla p + \rho\mathbf{g})\right] = 0. \tag{4.110}$$

Now, suppose that two *immiscible* fluids, of density ρ_1 and ρ_2, flow underground. Immiscible means that the two fluids cannot dissolve one into the other or chemically interact. In particular, the conservation law holds for the mass of each fluid. Thus, if we denote by S_1 and S_2 the fractions (*saturations*) of the available space filled by the two fluids, respectively, we can write

$$\phi(S_1\rho_1)_t + \operatorname{div}(\rho_1\mathbf{q}_1) = 0 \tag{4.111}$$

$$\phi(S_2\rho_2)_t + \operatorname{div}(\rho_2\mathbf{q}_2) = 0. \tag{4.112}$$

We assume that $S_1 + S_2 = 1$, i.e. that the medium is completely saturated and that capillarity effects are negligible. We set $S_1 = S$ and $S_2 = 1 - S$. The Darcy law for the two fluids becomes

$$\mathbf{q}_1 = -k\frac{k_1}{\mu_1}(\nabla p + \rho_1\mathbf{g})$$

$$\mathbf{q}_2 = -k\frac{k_2}{\mu_2}(\nabla p + \rho_2\mathbf{g})$$

where k_1, k_2 are the *relative permeability coefficients*, in general depending on S.

We make now some other simplifying assumptions:

a) Gravitational effects are negligible.
b) $k, \phi, \rho_1, \rho_2, \mu_1, \mu_2$ are constant.

c) $k_1 = k_1(S)$ and $k_2 = k_2(S)$ are known functions.

Equations (4.111) and (4.112) become:

$$\phi S_t + \operatorname{div} \mathbf{q}_1 = 0, \qquad -\phi S_t + \operatorname{div} \mathbf{q}_2 = 0, \tag{4.113}$$

while the Darcy laws take the form

$$\mathbf{q}_1 = -k\frac{k_1}{\mu_1}\nabla p, \qquad \mathbf{q}_2 = -k\frac{k_2}{\mu_2}\nabla p. \tag{4.114}$$

Letting $\mathbf{q} = \mathbf{q}_1 + \mathbf{q}_2$ and adding the two equations in (4.113) we have

$$\operatorname{div} \mathbf{q} = 0.$$

Adding the two equations in (4.114) yields, setting $K(S) = \left(\frac{k_1(S)}{\mu_1} + \frac{k_2(S)}{\mu_2}\right)^{-1}$,

$$\nabla p = -\frac{1}{k}K(S)\mathbf{q}$$

from which

$$\Delta p = \operatorname{div} \nabla p = -\frac{1}{k}\mathbf{q} \cdot \nabla K(S).$$

From the first equations in (4.113) and (4.114) we get

$$\begin{aligned}\phi S_t &= -\operatorname{div} \mathbf{q}_1 = \frac{k}{\mu_1}[\nabla k_1(S) \cdot \nabla p + k_1(S)\Delta p] \\ &= -\frac{1}{\mu_1}[K(S)\mathbf{q} \cdot \nabla k_1(S) + k_1(S)\mathbf{q} \cdot \nabla K(S)] \\ &= \mathbf{q} \cdot \nabla H(S) = H'(S)\mathbf{q} \cdot \nabla S\end{aligned}$$

where

$$H(S) = -\frac{1}{\mu_1}k_1(S)K(S).$$

When $\mathbf{q}$ is known, the resulting *quasilinear* equation for the saturation S is

$$\phi S_t = H'(S)\mathbf{q} \cdot \nabla S,$$

known as the *Bukley-Leverett* equation.

In particular, if $\mathbf{q}$ can be considered one-dimensional and constant, i.e. $\mathbf{q} = q\mathbf{i}$, we have

$$qH'(S)S_x - \phi S_t = 0$$

which is of the form (4.106), with $u = S$ and $y = t$. The characteristic system is (see Remark 4.25, p. 232)

$$\frac{dx}{qH'(S)S} = -\frac{dt}{\phi} = \frac{dS}{0}.$$

Two first integrals are

$$w_1 = \phi x + qH'(S)St \quad \text{and} \quad w_2 = S.$$

Thus, the general solution is given by

$$F(\phi x + qH'(S)St, S) = 0.$$

The choice

$$F(w_1, w_2) = w_2 - f(w_1),$$

yields $S = f(\phi x + qH'(S)t)$ that satisfies the initial condition $S(x, 0) = f(\phi x)$.

4.8 Fully Nonlinear Equations

4.8.1 Characteristic strips

We extend the characteristic method to *fully nonlinear* equations of the form

$$F(x, y, u, u_x, u_y) = 0. \tag{4.115}$$

We assume that $F = F(x, y, u, p, q)$ is a smooth function of its arguments and, to avoid trivial cases, that the nondegeneracy condition $F_p^2 + F_q^2 \neq 0$ holds. In the quasilinear case,

$$F(x, y, u, p, q) = a(x, y, u)p + b(x, y, u)q - c(x, y, u)$$

and

$$F_p = a(x, y, u), \quad F_q = b(x, y, u), \tag{4.116}$$

so that $F_p^2 + F_q^2 \neq 0$ says that a and b do not vanish simultaneously.

Equation (4.115) has a geometrical interpretation as well. Let $u = u(x, y)$ be a smooth solution and consider a point (x_0, y_0, z_0) on its graph. Equation (4.115) constitutes a link between the components u_x and u_y of the normal vector

$$\mathbf{n}_0 = (-u_x(x_0, y_0), -u_y(x_0, y_0), 1)$$

but it is a little more complicated than in the quasilinear case[16] and is not a priori clear what a characteristic system for equation (4.115) should be. Reasoning by analogy

[16] If, for instance $F_q \neq 0$, by the Implicit Function Theorem, the equation $F(x_0, y_0, z_0, p, q) = 0$ defines $q = q(p)$ so that

$$F(x_0, y_0, z_0, p, q(p)) = 0.$$

Therefore, the possible tangent plane to u at (x_0, y_0, z_0) form a one parameter family of planes, given by

$$p(x - x_0) + q(p)(y - y_0) - (z - z_0) = 0.$$

with the quasilinear case, from (4.116) we are led to the equations

$$\begin{aligned} \frac{dx}{dt} &= F_p(x, y, z, p, q) \\ \frac{dy}{dt} &= F_q(x, y, z, p, q), \end{aligned} \tag{4.117}$$

where $z(t) = u(x(t), y(t))$ and

$$p = p(t) = u_x(x(t), y(t)), \qquad q = q(t) = u_y(x(t), y(t)). \tag{4.118}$$

Thus, taking account of (4.117), the equation for z is:

$$\frac{dz}{dt} = u_x \frac{dx}{dt} + u_y \frac{dy}{dt} = pF_p + qF_q. \tag{4.119}$$

Equations (4.119) and (4.117) correspond to the characteristic system (4.84), but with two more unknowns: $p(t)$ and $q(t)$. We need two more equations. Proceeding formally, assuming that u is *twice continuously differentiable*, from (4.117) and (4.118) we can write

$$\begin{aligned} \frac{dp}{dt} &= u_{xx}(x(t), y(t)) \frac{dx}{dt} + u_{xy}(x(t), y(t)) \frac{dy}{dt} \\ &= u_{xx}(x(t), y(t)) F_p + u_{xy}(x(t), y(t)) F_q. \end{aligned}$$

We have to get rid of the second order derivatives. Since u is a solution of (4.115), the identity

$$F(x, y, u(x, y), u_x(x, y), u_y(x, y)) \equiv 0$$

holds. Partial differentiation with respect to x yields, since $u_{xy} = u_{yx}$:

$$F_x + F_u u_x + F_p u_{xx} + F_q u_{xy} \equiv 0.$$

Computing along $x = x(t)$, $y = y(t)$, we get

$$u_{xx}(x(t), y(t)) F_p + u_{xy}(x(t), y(t)) F_q = -F_x - p(t) F_u. \tag{4.120}$$

Thus, we deduce for p the following differential equation:

$$\frac{dp}{dt} = -F_x(x, y, z, p, q) - pF_u(x, y, z, p, q).$$

Similarly, we find

$$\frac{dq}{dt} = -F_y(x, y, z, p, q) - qF_u(x, y, z, p, q).$$

This family, in general, envelopes a cone with vertex at (x_0, y_0, z_0), called *Monge cone*. Each possible tangent plane touches the Monge cone along a generatrix.

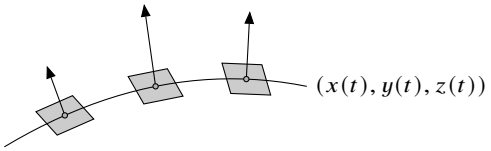

Fig. 4.30 Characteristic strip

In conclusion, we are led to the following **characteristic system** of five autonomous equations:

$$\frac{dx}{dt} = F_p, \qquad \frac{dy}{dt} = F_q, \qquad \frac{dz}{dt} = pF_p + qF_q \tag{4.121}$$

and

$$\frac{dp}{dt} = -F_x - pF_u, \qquad \frac{dq}{dt} = -F_y - qF_u. \tag{4.122}$$

Observe that $F = F(x, y, u, p, q)$ is a **first integral** of (4.121), (4.122). In fact

$$\begin{aligned}\frac{d}{dt}F(x(t), y(t), z(t), p(t), q(t)) &= F_x\frac{dx}{dt} + F_y\frac{dy}{dt} + F_u\frac{dz}{dt} + F_p\frac{dp}{dt} + F_q\frac{dq}{dt}\\ &= F_xF_p + F_yF_q + F_u(pF_p + qF_q) + F_p(-F_x - pF_u) + F_q(-F_y - qF_u) \equiv 0\end{aligned}$$

and therefore, if $F(x(t_0), y(t_0), z(t_0), p(t_0), q(t_0)) = 0$ at some t_0, then

$$F(x(t), y(t), z(t), p(t), q(t)) \equiv 0. \tag{4.123}$$

Thus, the curve

$$x = x(t), \quad y = y(t), \quad z = z(t),$$

still called a *characteristic curve*, lives on an integral surface, while

$$p = p(t), \quad q = q(t)$$

give the normal vector at each point, and it can be related to a portion of the tangent plane, as shown in Fig. 4.30. For this reason, a solution $(x(t), y(t), z(t), p(t), q(t))$ of (4.121), (4.122) is called *characteristic strip.*

4.8.2 The Cauchy Problem

As usual, the *Cauchy problem* consists in looking for a solution u of (4.115), assuming prescribed values on a given curve γ_0 in the xy plane. If γ_0 has the parametrization

$$x = f(s), \; y = g(s), \qquad s \in I \subseteq \mathbb{R}$$

we want that

$$u(f(s), g(s)) = h(s), \qquad s \in I, \tag{4.124}$$

where $h = h(s)$ is a given function. We assume that $0 \in I$ and that f, g, h are smooth functions in I.

Thus, let Γ_0 be the *initial curve*, given by the parametrization

$$x = f(s), \qquad y = g(s), \qquad z = h(s). \tag{4.125}$$

Equations (4.125) only specify the *initial condition* (at $t = 0$) for x, y and z. To solve the characteristic system, we have first to complete Γ_0 into a *strip*

$$(f(s), g(s), h(s), \varphi(s), \psi(s)),$$

with the two functions $\varphi(s)$ and $\psi(s)$ representing the initial values for p and q, that is

$$\varphi(s) = u_x(f(s), g(s)) \quad \text{and} \quad \psi(s) = u_y(f(s), g(s)).$$

The functions φ and ψ cannot be chosen arbitrarily. In fact, a first condition comes from differentiating $h(s) = u(f(s), g(s))$. The result is the so called *strip condition*

$$h'(s) = \varphi(s) f'(s) + \psi(s) g'(s). \tag{4.126}$$

On the other hand, $\varphi(s)$ and $\psi(s)$ determine the normal to the solution surface at any point $(f(s), g(s), h(s))$ and must also satisfy the condition

$$F(f(s), g(s), h(s), \varphi(s), \psi(s)) = 0. \tag{4.127}$$

Now we are in position to give a (formal) procedure to construct a solution of our **Cauchy problem**: *Determine a solution $u = u(x, y)$ of*

$$F(x, y, u, u_x, u_y) = 0,$$

such that $u(f(s), g(s)) = h(s)$.

1. Solve for $\varphi(s)$ and $\psi(s)$ the (nonlinear) system

$$\begin{cases} F(f(s), g(s), h(s), \varphi(s), \psi(s)) = 0 \\ \varphi(s) f'(s) + \psi(s) g'(s) = h'(s). \end{cases} \tag{4.128}$$

2. Solve the characteristic system (4.121), (4.122) with initial conditions

$$x(0) = f(s), y(0) = g(s), z(0) = h(s), p(0) = \varphi(s), q(0) = \psi(s).$$

   Suppose we find the solution

$$x = X(s, t), \ y = Y(s, t), \ z = Z(s, t), \ p = P(s, t), \ q = Q(s, t).$$

3. Solve $x = X(s, t), y = Y(s, t)$ for s, t in terms of x, y. Substitute $s = S(x, y)$ and $t = T(x, y)$ into $z = Z(s, t)$ to yield a solution $z = u(x, y)$.

Example 4.27 We want to solve the equation

$$u = u_x^2 - 3u_y^2$$

with initial condition $u(x,0) = x^2$. We have $F(x,y,u,p,q) = p^2 - 3q^2 - u$ and the characteristic system is

$$\frac{dx}{dt} = 2p, \quad \frac{dy}{dt} = -6q, \quad \frac{dz}{dt} = 2p^2 - 6q^2 = 2z \tag{4.129}$$

$$\frac{dp}{dt} = p, \qquad \frac{dq}{dt} = q. \tag{4.130}$$

A parametrization of the initial line Γ_0 is

$$f(s) = s,\ g(s) = 0,\ h(s) = s^2.$$

To complete the initial strip we solve the system

$$\begin{cases} \varphi^2 - 3\psi^2 = s^2 \\ \varphi = 2s. \end{cases}$$

There are two solutions:

$$\varphi(s) = 2s, \qquad \psi(s) = \pm s.$$

The choice of $\psi(s) = s$ yields, integrating the equations (4.130),

$$P(s,t) = 2se^t, \qquad Q(s,t) = se^t$$

whence, from (4.129),

$$X(s,t) = 4s(e^t - 1) + s, \qquad Y(s,t) = -6s(e^t - 1), \qquad Z(s,t) = s^2 e^{2t}.$$

Solving the first two equations for s, t and substituting into the third one, we get

$$u(x,y) = \left(x + \frac{y}{2}\right)^2.$$

The choice of $\psi(s) = -s$ yields

$$u(x,y) = \left(x - \frac{y}{2}\right)^2.$$

As the example shows, in general there is *no uniqueness*, unless system (4.128) has a unique solution. On the other hand, if this system has no (real) solution, then equation (4.133) has no solution as well.

Furthermore, observe that if $(x_0, y_0, z_0) = (f(0), g(0), h(0))$ and (p_0, q_0) is a solution of the system

$$\begin{cases} F(x_0, y_0, z_0, p_0, q_0) = 0 \\ p_0 f'(0) + q_0 g'(0) = h'(0), \end{cases} \tag{4.131}$$

by the Inverse Function Theorem 1.5, the condition

$$\begin{vmatrix} f'(0) & g'(0) \\ F_p(x_0, y_0, z_0, p_0, q_0) & F_q(x_0, y_0, z_0, p_0, q_0) \end{vmatrix} \neq 0 \tag{4.132}$$

ensures the existence of a solution $\varphi(s)$ and $\psi(s)$ of (4.128), in a neighborhood of $s = 0$. Condition (4.132) corresponds to (4.92) in the quasilinear case. It implies that *the initial curve is not tangent at* (x_0, y_0, z_0) *to a characteristic direction.*

The following theorem summarizes the above discussion on the Cauchy problem

$$F(x, y, u, u_x, u_y) = 0 \tag{4.133}$$

with initial curve Γ_0 given by

$$x = f(s), \ y = g(s), \ z = h(s). \tag{4.134}$$

Theorem 4.28 *Assume that:*

1. *F is a smooth function defined in a domain $D \subseteq \mathbb{R}^5$ and $F_p^2 + F_q^2 \neq 0$.*
2. *f, g, h are twice continuously differentiable in a neighborhood of $s = 0$.*
3. *(p_0, q_0) is a solution of the system (4.131) where $(x_0, y_0, z_0) = (f(0), g(0), h(0))$, and condition (4.132) holds.*

Then, in a neighborhood of (x_0, y_0), there exists a C^2 solution $z = u(x, y)$ of the Cauchy problem (4.133), (4.134).

- *Geometrical optics.* The equation

$$c^2(u_x^2 + u_y^2) = 1 \qquad (c > 0) \tag{4.135}$$

is called *eikonal equation* and arises in (two dimensional) geometrical optics. Indeed, assume that the level lines γ_t of equation

$$u(x, y) = t \tag{4.136}$$

represent the "wave fronts" of a wave perturbation (i.e. light) moving with time t and *normal* propagation speed c, that we assume to be constant. An orthogonal trajectory to the wave fronts coincides with a *light ray*. Therefore, a point $(x(t), y(t))$ on a ray satisfies the identity

$$u(x(t), y(t)) = t \tag{4.137}$$

and its velocity vector $\mathbf{v} = (\dot{x}, \dot{y})$ is colinear to ∇u, with $|\mathbf{v}| = c$. Therefore

$$\nabla u \cdot \mathbf{v} = |\nabla u| \, |\mathbf{v}| = c \, |\nabla u| \, .$$

On the other hand, differentiating (4.137) we get

$$\nabla u \cdot \mathbf{v} = u_x \dot{x} + u_y \dot{y} = 1,$$

from which the eikonal equation $c^2 |\nabla u|^2 = 1$.

Geometrically, if we fix a point (x_0, y_0, z_0), the equation $c^2(p^2 + q^2) = 1$ states that the family of planes

$$z - z_0 = p(x - x_0) + q(y - y_0),$$

tangent to an integral surface at (x_0, y_0, z_0), all make a fixed angle $\theta = \arctan |\nabla u|^{-1} = \arctan c$ with the z-axis. This family envelopes the circular cone

$$(x - x_0)^2 + (y - y_0)^2 = c^2 (z - z_0)^2$$

called the light cone, with opening angle 2θ.

The eikonal equation is of the form (4.115), with[17]

$$F(x, y, u, p, q) = \frac{1}{2} \left[c^2(p^2 + q^2) - 1 \right].$$

The characteristic system is[18]:

$$\frac{dx}{d\tau} = c^2 p, \quad \frac{dy}{d\tau} = c^2 q, \quad \frac{dz}{d\tau} = c^2 p^2 + c^2 q^2 = 1 \tag{4.138}$$

and

$$\frac{dp}{d\tau} = 0, \quad \frac{dq}{d\tau} = 0. \tag{4.139}$$

An initial curve Γ_0

$$x = f(s), \qquad y = g(s), \qquad z = h(s),$$

can be completed into an initial strip, by solving for ϕ and ψ the system

$$\begin{cases} \phi(s)^2 + \psi(s)^2 = c^{-2} \\ \phi(s) f'(s) + \psi(s) g'(s) = h'(s). \end{cases} \tag{4.140}$$

This system has *two real and distinct solutions* if

$$f'(s)^2 + g'(s)^2 > c^2 h'(s)^2 \tag{4.141}$$

while it has *no real solution* if[19]

[17] The factor $\frac{1}{2}$ is there for esthetic reasons.

[18] Using τ as a parameter along the characteristics.

[19] System (4.140) is equivalent to finding the intersection between the circle $\xi^2 + \eta^2 = c^{-2}$ and the straight line $f'\xi + g'\eta = h'$. The distance of the center $(0, 0)$ from the line is given by

$$d = \frac{|h'|}{\sqrt{(f')^2 + (g')^2}}$$

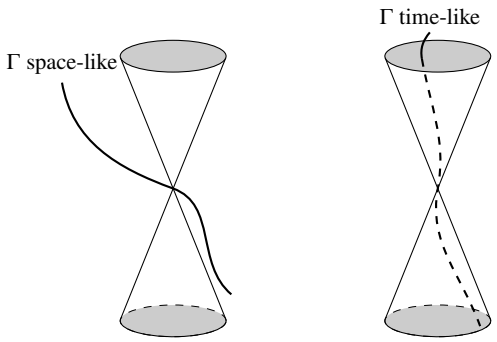

Fig. 4.31 Space-like and time-like initial curves

$$f'(s)^2 + g'(s)^2 < c^2 h'(s)^2. \tag{4.142}$$

If (4.141) holds, Γ_0 forms an angle greater than θ with the z-axis and therefore it is exterior to the light cone (Fig. 4.31). In this case we say that Γ_0 is *space-like* and we can find two different solutions of the Cauchy problem. If (4.142) holds, Γ_0 is contained in the light cone and we say it is *time-like*. The Cauchy problem does not have any solution.

Given a space-like curve Γ_0 and ϕ, ψ, solutions of the system (4.140), the corresponding characteristic strip is, for s fixed,

$$x(\tau) = f(s) + c^2\phi(s)\tau, \qquad y(\tau) = g(s) + c^2\psi(s)\tau, \qquad z(\tau) = h(s) + \tau$$
$$p(\tau) = \phi(s), \qquad q(\tau) = \psi(s).$$

Observe that the point $(x(\tau), y(\tau))$ moves along the characteristic with speed

$$\sqrt{\dot{x}^2(\tau) + \dot{y}^2(\tau)} = c^2\sqrt{\phi^2(s) + \psi^2(s)} = c$$

and direction $(\phi(s), \psi(s)) = (p(\tau), q(\tau))$. Therefore, the projected characteristics are coincident with the light rays. Moreover, we see that the fronts γ_τ can be constructed from Γ_0 by shifting any point on Γ_0 along a ray at a distance $c\tau$. Thus, the wave fronts constitute a family of *parallel* curves.

4.8.3 Complete integrals, general and singular solutions

Consider the two parameters family of spheres of equation

$$(x - a)^2 + (y - b)^2 + z^2 = 1. \tag{4.143}$$

By thinking $z = u(x, y)$ and differentiating with respect to x and y we get,

---

so that there are 2 real intersections if $d < c^{-1}$, while there is no real intersection if $d > c^{-1}$.

$$(x-a)+u\,u_x=0 \qquad \text{and} \qquad (y-b)+u\,u_y=0.$$

Eliminating the two constants a and b we find the PDE

$$u^2(u_x^2+u_y^2)=1-u^2. \tag{4.144}$$

Thus (4.143) defines a two-parameter family of solutions of (4.144). This is an example of complete integral. Precisely, by a *complete integral* of an equation $F(x,y,u,u_x,u_y)=0$ we mean a solution $u=u(x,y;a,b)$, depending on two *independent* arbitrary constants a,b. *Independent* means that u cannot be written as $u=u(x,y;g(a,b))$, for some real function g; for instance $u=u(x,y;a-b)$ would actually depend only on the single parameter $c=g(a,b)=a-b$. A sufficient (but not necessary) condition for independence is provided by

$$\det\begin{pmatrix} u_{xa} & u_{xb} \\ u_{ya} & u_{yb} \end{pmatrix} \neq 0 \tag{4.145}$$

as it is easy to check.

To find a complete integral, it could be convenient to write (formally) the characteristic system (4.121), (4.122), in the form ($p=u_x, q=u_y$)

$$\frac{dx}{F_p}=\frac{dy}{F_q}=\frac{du}{pF_p+qF_q}=-\frac{dp}{F_x+pF_u}=-\frac{dq}{F_y+qF_u}. \tag{4.146}$$

Example 4.29 (The Clairaut equation) A complete integral for equations of the form

$$u=xu_x+yu_y+f(u_x,u_y) \tag{4.147}$$

is given by

$$u(x,y;a,b)=ax+by+f(a,b) \tag{4.148}$$

representing a two-parameters family of planes.

Indeed, the last equation in (4.146) gives

$$\frac{dp}{0}=\frac{dq}{0} \quad \text{or} \quad p=a,\ q=b, \qquad a,b \text{ constants}.$$

From (4.147) we get easily (4.149).

Example 4.30 A complete integral for the geometrical optics equation

$$c^2(u_x^2+u_y^2)=1, \qquad c>0,$$

is

$$u(x,y;\varphi,C)=c^{-1}(x\sin\varphi+y\cos\varphi)+C, \tag{4.149}$$

a family of planes making an angle $\theta=\arctan c$ with the z-axis, where φ and C are clearly independent (even though (4.145) does not hold).

Indeed, the last two equations in (4.146) gives

$$\frac{dp}{0} = \frac{dq}{0} \quad \text{or} \quad p = C_1,\ q = C_2, \qquad C_1, C_2 \text{ constants.}$$

Thus $c^2(C_1^2 + C_2^2) = 1$ from which $C_1 = c^{-1}\sin\varphi$, $C_2 = c^{-1}\cos\varphi$. Since $du = pdx + qdy$ we get easily (4.149).

Remark 4.31 The equations of the form $F(u_x, u_y) = 0$, with e.g. $F_q \neq 0$, admit complete integrals of the form $u = ax + f(a)y + b$ where $q = f(a)$ is implicitly defined by $F(a, q) = 0$.

Remark 4.32 From a complete integral $u = u(x, y; a, b)$ we may generate a *general solution* by choosing an arbitrary C^1 function w relating the two parmeters a and b, say $b = w(a)$, and then construct the envelope of the one-parameter family of surfaces given by the graphs of $u = u(x, y; a, w(a))$. If this envelope exists and is C^1, then it is a *general solution* depending on w. In principle, this envelope can be determined by determining $a = a(x, y)$ from the equation

$$\partial_a \left[u(x, y; a, w(a))\right] = 0 \tag{4.150}$$

and computing $u = u(x, y; a(x, y))$[20]. Particular solutions, not included in the complete integral, may be generated from particular choices of w.

Example 4.33 (The Clairaut equation — continued) Consider the Clairaut equation. Choosing for instance $f(a, b) = a^2 + b$ and $b = w(a) = a$ we have $u = ax + ay + a^2 + a$ and

$$\partial_a u(x, y; a, w(a)) = x + y + 2a + 1 = 0.$$

We find $a = -(1 + x + y)/2$ and $u = -\frac{(1+x+y)^2}{4}$, which is clearly not included in the complete integral (4.148).

Other kinds of solutions can be constructed directly as *envelopes* of complete integrals. As for curves, the envelope of a family S of surfaces is a surface tangent at each point to a member of S. Precisely, assume that $u = u(x, y; a, b)$ is a complete integral for $F(x, y, u, u_x, u_y) = 0$ and that we can solve the equations

$$\partial_a u(x, y; a, b) = 0, \quad \partial_b u(x, y; a, b) = 0$$

for the parameters a, b as C^1-functions of x, y: $a = a(x, y)$, $b = b(x, y)$. Then $v(x, y) = u(x, y; a(x, y), b(x, y))$ is the envelope of the two-parameter family given by the graphs of $u = u(x, y; a, b)$ and it is a solution of $F(x, y, u, u_x, u_y) = 0$ as well, called *singular solution*.

Indeed, at each point (x, y), v is tangent to the graph of $u = u(x, y; a, b)$ for $a = a(x, y), b = b(x, y)$; therefore $\nabla u = \nabla v$ at (x, y), for $a = a(x, y), b = b(x, y)$.

[20] We must always check that by solving the system (4.150) we really get the required envelope. Indeed the solution to (4.150) could be a locus of irregular points that do not satisfy the equation (see Remark 4.6, p. 203).

Example 4.34 (The Clairaut equation — continued) Choosing for instance $f(a,b) = a^2 + b^2$ we have $u = ax + by + a^2 + b^2$ and

$$\partial_a u(x,y;a,b) = x + 2a = 0.$$
$$\partial_b u(x,y;a,b) = y + 2b = 0.$$

We find the singular solution

$$v(x,y) = -\frac{x^2}{2} - \frac{y^2}{2} + \frac{x^2}{4} + \frac{y^2}{4} = -\frac{1}{4}(x^2 + y^2).$$

Problems

4.1 Using *Duhamel's method* (see Sect. 2.8.3, p. 80), solve the problem

$$\begin{cases} c_t + vc_x = f(x,t) & x \in \mathbb{R}, t > 0 \\ c(x,0) = 0 & x \in \mathbb{R}. \end{cases}$$

Find an explicit formula when $f(x,t) = e^{-t}\sin x$.
[*Hint.* For fixed $s \geq 0$ and $t > s$, solve

$$\begin{cases} w_t + vw_x = 0 \\ w(x,s;s) = f(x,s). \end{cases}$$

Then, integrate w with respect to s over $(0,t)$.]

4.2 Consider the following problem $(a > 0)$:

$$\begin{cases} u_t + au_x = f(x,t) & 0 < x < R, t > 0 \\ u(0,t) = 0 & t > 0 \\ u(x,0) = 0 & 0 < x < R. \end{cases}$$

Prove the stability estimate

$$\int_0^R u^2(x,t)dx \leq e^t \int_0^t \int_0^R f^2(x,s)dxds, \qquad t > 0.$$

[*Hint.* Multiply by u the equation. Use $a > 0$ and the inequality $2fu \leq f^2 + u^2$ to obtain

$$\frac{d}{dt}\int_0^R u^2(x,t)dx \leq \int_0^R f^2(x,t)dx + \int_0^R u^2(x,t)dx.$$

Prove that if $E(t)$ satisfies $E'(t) \leq G(t) + E(t),\ E(0) = 0$, then $E(t) \leq e^t \int_0^t G(s)ds$.]

4.3 Solve the Burgers equation $u_t + uu_x = 0$ with initial data

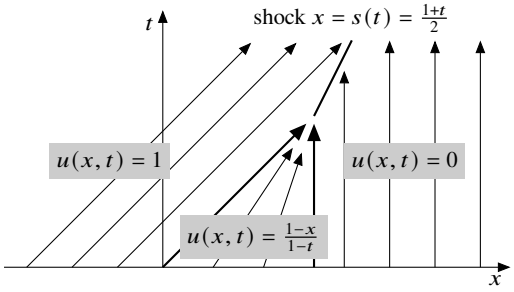

Fig. 4.32 The solution of Problem 4.3

$$g(x) = \begin{cases} 1 & x \le 0 \\ 1 - x & 0 < x < 1 \\ 0 & x \ge 1. \end{cases}$$

[*Answer.* See Fig. 4.32.]

4.4 Solve the problem

$$\begin{cases} u_t + uu_x = 0 & x \in \mathbb{R}, t > 0 \\ u(x,0) = g(x) & x \in \mathbb{R} \end{cases}$$

where

$$g(x) = \begin{cases} 0 & x < 0 \\ 1 & 0 < x < 1 \\ 0 & x > 1. \end{cases}$$

4.5 *A rather frequent traffic situation.* Consider the model of Sect. 4.3.1 and suppose that two traffic lights are placed at distance L, at the points $x = -\beta L$, $x = \alpha L$, with $\beta > 0$, $0 < \alpha < 1/2$, $\alpha + \beta = 1$. At time $t = 0$, the car distribution is given by

$$\begin{cases} \rho_m & x < -\beta L \\ 0 & -\beta L < x < 0 \\ \rho_m & 0 < x < \alpha L \\ 0 & x > \alpha L. \end{cases}$$

Thus, there is bumper-to-bumper congestion before the first light, a piece of clear road between $-\beta L$ and 0, a congested queue of length αL before the second light, beyond which the road is clear.

a) Introduce dimensionless variables

$$u = \frac{\rho}{\rho_m}, z = \frac{x}{L}, \tau = \frac{v_m t}{L},$$

to get the equation $u_\tau + \tilde{q}(u)_z = 0$, where $\tilde{q}(u) = u(1-u)$, with initial condition

$$\begin{cases} 1 & z < -\beta \\ 0 & -\beta < z < 0 \\ 1 & 0 < z < \alpha \\ 0 & z > \alpha. \end{cases}$$

b) Assuming that both lights turn on green at $\tau = 0$, determine the characteristic configuration and the shock curve.

c) Compute the time τ_α at which the shock curve intersects the vertical line $z = \alpha$. Determine the path of the vehicles started from the queue in the road section between 0 and α. Deduce how long the second light should stay on green, in order to let all the cars in the queue to pass the light at $z = \alpha$.

[*Partial answer.* b) There is a shock starting $(0, 0)$ and two rarefaction waves centered at $(-\beta, 0)$ and $(\alpha, 0)$. The shock curve has equation

$$z = \begin{cases} 0 & 0 < \tau < \alpha \\ \alpha + \tau - 2\sqrt{\alpha t} & \alpha \le \tau < 1/4\alpha \\ 2(1-2\alpha)\tau - (1-2\alpha) & 1/4\alpha \le \tau. \end{cases}$$

The characteristic fans of the two rarefaction waves are delimited by the lines (from left to right) $z = -\beta - \tau$, $z = -\beta + \tau$ and $z = \alpha - \tau$, $z = \alpha + \tau$ respectively.
c) $\tau_\alpha = 1/(2 - 4\alpha)$. The green time must be greater then τ_α.]

4.6 *Traffic in a tunnel.* A rather realistic model for the car speed in a very long tunnel is the following:

$$v(\rho) = \begin{cases} v_m & 0 \le \rho \le \rho_c \\ \lambda \log(\frac{\rho_m}{\rho}) & \rho_c \le \rho \le \rho_m \end{cases}$$

where $\lambda = \frac{v_m}{\log(\rho_m/\rho_c)}$.

Observe that v is continuous also at

$$\rho_c = \rho_m e^{-v_m/\lambda},$$

which represents a *critical density*: if $\rho \le \rho_c$ the drivers are free to reach the speed limit. Typical values are: $\rho_c = 7$ car/Km, $v_m = 90$ Km/h, $\rho_m = 110$ car/Km, $v_m/\lambda = 2.75$.

Assume the entrance is placed at $x = 0$ and that the cars are waiting (with maximum density) the tunnel to open to the traffic at time $t = 0$. Thus, the initial density is

$$\rho = \begin{cases} \rho_m & x < 0 \\ 0 & x > 0. \end{cases}$$

a) Determine density and car speed; draw their graphs.
b) Determine and draw in the x, t plane the trajectory of a car initially at $x = x_0 < 0$, and compute the time it takes to enter the tunnel.

4.7 Consider the equation $u_t + q'(u)u_x = 0$, with initial condition $u(x, 0) = g(x)$. Assume that $g, q' \in C^1([a, b])$ and

$$g'(\xi)q''(g(\xi)) < 0$$

in $[a, b]$. Show that the family of characteristics

$$x = q'(g(\xi))t + \xi, \qquad \xi \in [a, b] \tag{4.151}$$

admits an *envelope* and that the point (x_s, t_s) of shock formation, given by formulas (4.46) and (4.47), is the point on this envelope with minimum time coordinate (Fig. 4.33).

4.8 Check that, for the family of curves $(x - a)^2 - y^3 = 0$, $a \in \mathbb{R}$, the system (4.6) produces the x-axis, which is the locus of cusps of the family.

4.9 Find the solutions of the problems

$$\begin{cases} u_t \pm uu_x = 0 & t > 0, x \in \mathbb{R} \\ u(x, 0) = x & x \in \mathbb{R}. \end{cases}$$

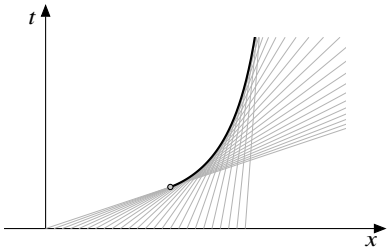

Fig. 4.33 Envelope of characteristics and point of shock formation

4.10 Draw the characteristics and describe the evolution for $t \to +\infty$ of the solution of the problem

$$\begin{cases} u_t + uu_x = 0 & t > 0, x \in \mathbb{R} \\ u(x,0) = \begin{cases} \sin x & 0 < x < \pi \\ 0 & x \le 0 \text{ or } x \ge \pi. \end{cases} \end{cases}$$

4.11 Show that, for every $\alpha > 1$, the function

$$u_\alpha(x,t) = \begin{cases} -1 & 2x \le -(1+\alpha)t \\ -\alpha & -(1+\alpha)t < 2x < 0 \\ \alpha & 0 < 2x < (\alpha+1)t \\ 1 & (\alpha+1)t \le 2x \end{cases}$$

is a weak solution of the problem

$$\begin{cases} u_t + uu_x = 0 & t > 0, x \in \mathbb{R} \\ u(x,0) = \begin{cases} -1 & x < 0 \\ 1 & x > 0. \end{cases} \end{cases}$$

Is it also an entropic solution, at least for some α?

4.12 *Time irreversibility.* Let u_1 and u_2 be the entropic solutions of the Burgers equation $u_t + (u^2/2)_x = 0$ with initial data

$$g_1(x) = \begin{cases} 1 & x < 1/2 \\ 0 & x > 1/2 \end{cases} \text{ and } g_2(x) = \begin{cases} 1 & x < 0 \\ 1 - x & 0 \le x \le 1 \\ 0 & x > 1. \end{cases}$$

Show that at time $t = 1$,

$$u_1(x,1) = u_2(x,2) = \begin{cases} 1 & x < 1 \\ 0 & x > 1. \end{cases}$$

Thus we cannot reconstruct the initial data of a weak solution u from the knowledge of u at a later time.

4.13 *Hopf-Cole transformation.* Consider the following problem for the viscous Burgers equation

$$\begin{cases} u_t + uu_x = \varepsilon u_{xx} & t > 0, x \in \mathbb{R} \\ u(0,x) = g(x) & x \in \mathbb{R} \end{cases} \tag{4.152}$$

where g is smooth and bounded, and $\varepsilon > 0$.

a) Show that the transformation

$$u = -2\varepsilon \frac{\varphi_x}{\varphi}$$

changes problem (4.152) into the following global Cauchy problem for the heat equation:

$$\begin{cases} \varphi_t + \varepsilon \varphi_{xx} = 0 & t > 0, x \in \mathbb{R} \\ \varphi(x,0) = \varphi_0(x) & x \in \mathbb{R} \end{cases} \tag{4.153}$$

where $\varphi_0(x) = \exp\left\{-\frac{1}{2\varepsilon}\int_a^x g(s)ds\right\}$, $a \in \mathbb{R}$.

b) Solve the problem (4.153) and show that the solution to problem (4.152) is independent on a and given by

$$u(x,t) = \frac{\int_{\mathbb{R}} \varphi_0(y) \frac{x-y}{t} \exp(-\frac{(x-y)^2}{4\varepsilon t}) dy}{\int_{\mathbb{R}} \varphi_0(y) \exp(-\frac{(x-y)^2}{4\varepsilon t}) dy.}$$

Remark: It can be proved that as $\varepsilon \to 0^+$, $u(x,t)$ tends pointwise to the solution of problem (4.152) for the nonviscous Burgers equation.

4.14 Consider the conservation law $u_t + q(u)_x = 0$, where the graph of q is shown in Fig. 4.24, p. 220. Exchange the roles of u_R and u_L, so that $u_L < u_R$, and construct the entropic solution of the Riemann problem.

[*Hint.* Since $u_L < u_R$, replace the graph of q by its lower convex envelope, shown in Fig. 4.34, and proceed as in Sect. 4.6.3.]

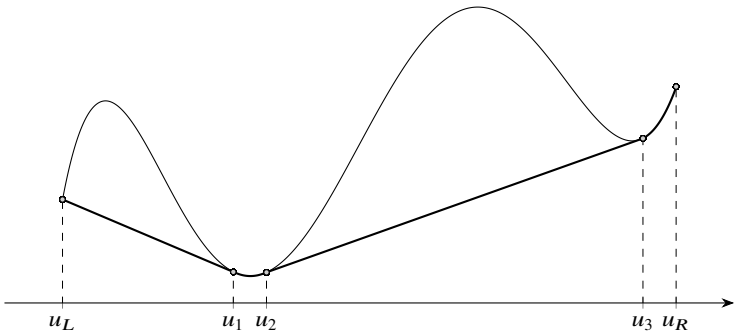

Fig. 4.34 Lower convex envelope of the flux function in Fig. 4.24, p. 220

4.15 With reference to Problem 4.14, explain why it is *not possible* to construct an entropic solution of the Riemann problem by using the upper concave envelope of q.

4.16 Find the solution of the linear equation

$$u_x + xu_y = y$$

satisfying the initial condition $u(0,y) = g(y)$, $y \in \mathbb{R}$, with

$$\text{a) } g(y) = \cos y \qquad \text{and} \qquad \text{b) } g(y) = y^2.$$

[*Answer of a).* $u = xy - \frac{x^3}{3} + \cos(y - \frac{x^2}{2})$.]

4.17 Consider the linear equation

$$au_x + bu_y = c(x, y),$$

with the initial condition $u(x, 0) = h(x)$. Assume that a, b are constants $(b \neq 0)$, and c, h are smooth and bounded.

a) Show that

$$u(x, y) = h(x - \gamma y) + \int_0^{y/b} c(a\tau + x - \gamma y, b\tau)d\tau, \qquad \gamma = a/b.$$

b) Deduce that a jump discontinuity of h at x_0 propagates into a jump of the same size for u, along the characteristic of equation $x - \gamma y = x_0$.

4.18 Let

$$D = \{(x, y) : y > x^2\}$$

and $a = a(x, y)$ be a continuous function in $\overline{D}$.

a) Given $g \in C(\mathbb{R})$, check the solvability of the linear problem

$$\begin{cases} a(x, y)u_x - u_y = -u & (x, y) \in D \\ u(x, x^2) = g(x) & x \in \mathbb{R}. \end{cases}$$

b) Examine the cases

$$a(x, y) = y/2 \quad \text{and} \quad g(x) = \exp(-\gamma x^2),$$

where γ is a real parameter.

4.19 Solve the Cauchy problem

$$\begin{cases} xu_x - yu_y = u - y & x > 0, y > 0 \\ u(y^2, y) = y & y > 0. \end{cases}$$

May a solution exist in a neighborhood of the origin?

[*Answer.* $u(x, y) = (y + x^{2/3}y^{-1/3})/2$. No solution can exist in neighborhood of $(0, 0)$.]

4.20 Check that a general solution of the equation

$$u_x + u_y = u^{1/3},$$

is given by

$$G(x - y, 3u^{2/3} - 2y) = 0, \tag{4.154}$$

with G arbitrary C^1 function. Note that the solution $u = 0$ is not included in (4.154).

4.21 Consider a cylindrical pipe with axis along the x-axis, filled with a fluid moving along the positive direction. Let $\rho = \rho(x, t)$ and $q = \frac{1}{2}\rho^2$ be the fluid density and the flux function. Assume the walls of the pipe are composed by porous material, from which the fluid leaks at the rate $H = k\rho^2$, $k > 0$.

a) Following the derivation of the conservation law given in the introduction, show that ρ satisfies the equation

$$\rho_t + \rho\rho_x = -k\rho^2.$$

b) Solve the Cauchy problem with $\rho(x, 0) = 1$.

[*Answer.* b) $\rho(x, t) = 1/(1 + kt)$.]

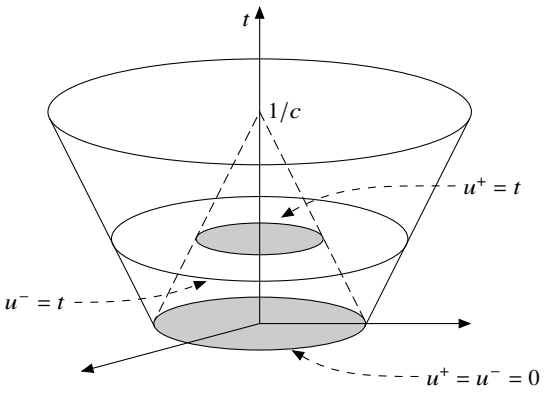

Fig. 4.35 Solutions of Problem 4.24

4.22 Solve the Cauchy problem

$$\begin{cases} u_x = -(u_y)^2 & x > 0, y \in \mathbb{R} \\ u(0, y) = 3y & y \in \mathbb{R}. \end{cases}$$

4.23 Solve the Cauchy problem

$$\begin{cases} u_x^2 + u_y^2 = 4u & (x, y) \in \mathbb{R}^2 \\ u(x, -1) = x^2 & x \in \mathbb{R}. \end{cases}$$

[*Answer.* $u(x, y) = x^2 + (y + 1)^2$.]

4.24 Solve the Cauchy problem

$$\begin{cases} c^2(u_x^2 + u_y^2) = 1 & (x, y) \in \mathbb{R}^2 \\ u(\cos s, \sin s) = 0 & s \in \mathbb{R}. \end{cases}$$

[*Answer.* There are two solutions $u^{\pm}(x, y) = \frac{\pm 1}{c}\left\{1 - \sqrt{x^2 + y^2}\right\}$ whose wave fronts are shown in Fig. 4.35.]

4.25 Check that $u = \frac{1}{4}(y + b)^2 + ax$ is a complete integral of the equation

$$xu_x + u_y^2 - u = 0.$$

Note that the solution $u = 0$ is not included. What kind of solution is it?

4.26 Find the complete integral of the equation $u_x u_y - u^2 = 0$.
[*Hint.* Write $u(x, y) = f(x)g(y)$.]

Chapter 5
Waves and Vibration

5.1 General Concepts

5.1.1 Types of waves

Our daily experience deals with sound waves, electromagnetic waves (as radio or light waves), deep or surface water waves, elastic waves in solid materials. Oscillatory phenomena manifest themselves also in less macroscopic and known contexts and ways. This is the case, for instance, of rarefaction and shock waves in traffic dynamics or of electrochemical waves in human nervous system and in the regulation of the heart beat. In quantum physics, everything can be described in terms of wave functions, at a sufficiently small scale.

Although the above phenomena share many similarities, they show several differences as well. For example, progressive water waves propagate a disturbance, while standing waves do not. Sound waves need a supporting medium, while electromagnetic waves do not. Electrochemical waves interact with the supporting medium, in general modifying it, while water waves do not.

Thus, it seems too hard to give a general definition of *wave*, capable of covering all the above cases, so that we limit ourselves to introduce some terminology and general concepts, related to specific types of waves. We start with one-dimensional waves.

a. Progressive or **travelling** waves are disturbances described by a function of the following form:

$$u(x,t) = g(x - vt).$$

For $t = 0$, we have $u(x,0) = g(x)$, which is the "initial" profile of the perturbation. This profile propagates without change of shape with speed $|v|$, in the positive (negative) x-direction if $v > 0$ ($v < 0$). We have already met this kind of waves in Chaps. 2 and 4.

b. Harmonic waves are particular progressive waves of the form

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
https://doi.org/10.1007/978-3-031-21853-8_5

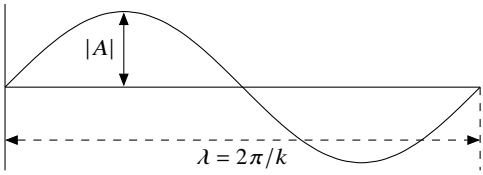

Fig. 5.1 Sinusoidal wave

$$u(x,t) = A\exp\{i(kx-\omega t)\}, \qquad A,k,\omega \in \mathbb{R}. \tag{5.1}$$

It is understood that only the *real part* or the *imaginary part*, that is

$$A\cos(kx-\omega t) \qquad \text{or} \qquad A\sin(kx-\omega t),$$

is of interest, but the complex notation may often simplify the computations. In (5.1) we distinguish, considering for simplicity ω and k positive (Fig. 5.1):

- The wave *amplitude* $|A|$.
- The *wave number* k, which is the number of complete oscillations in the space interval $[0, 2\pi]$, and the *wavelength*

$$\lambda = \frac{2\pi}{k},$$

  which is the distance between successive maxima (*crest*) or minima (*troughs*) of the waveform.
- The *angular frequency* ω, and the *frequency*

$$f = \frac{\omega}{2\pi}$$

  which is the number of complete oscillations in one second (Hertz) at a fixed space position.
- The *wave or phase speed*

$$c_p = \frac{\omega}{k},$$

  which is the crests (or troughs) speed.

c. Standing waves are of the form

$$u(x,t) = B\cos kx\cos\omega t.$$

In these disturbances, the basic sinusoidal wave, $\cos kx$, is modulated by the time dependent oscillation $B\cos\omega t$. A standing wave may be generated, for instance, by superposing two harmonic waves with the same amplitude, propagating in opposite directions:

$$A\cos(kx-\omega t) + A\cos(kx+\omega t) = 2A\cos kx\cos\omega t. \tag{5.2}$$

Consider now waves in dimension $n > 1$.

d. Plane waves. Scalar plane waves are of the form

$$u(\mathbf{x}, t) = f(\mathbf{k} \cdot \mathbf{x} - \omega t).$$

The disturbance propagates in the direction of $\mathbf{k}$ with speed $c_p = \omega/|\mathbf{k}|$. The planes of equation

$$\theta(\mathbf{x}, t) = \mathbf{k} \cdot \mathbf{x} - \omega t = \text{constant}$$

constitute the wave-fronts. *Harmonic or monochromatic plane waves* have the form

$$u(\mathbf{x}, t) = A \exp\{i(\mathbf{k} \cdot \mathbf{x} - \omega t)\}.$$

Here $\mathbf{k}$ is the *wave number vector* and ω is the *angular frequency*. The vector $\mathbf{k}$ is orthogonal to the wave front and $|\mathbf{k}|/2\pi$ gives the number of waves per unit length. The scalar $\omega/2\pi$ still gives the number of complete oscillations in one second (Hertz) at a fixed space position.

e. Spherical waves are of the form

$$u(\mathbf{x}, t) = v(r, t)$$

where $r = |\mathbf{x} - \mathbf{x}_0|$ and $\mathbf{x}_0 \in \mathbb{R}^n$ is a fixed point. In particular $u(\mathbf{x}, t) = e^{i\omega t} v(r)$ represents a stationary spherical wave, while $u(\mathbf{x}, t) = v(r - ct)$ is a progressive wave whose wavefronts are the spheres $r - ct = \text{constant}$, moving with speed $|c|$ (outgoing if $c > 0$, incoming if $c < 0$).

5.1.2 Group velocity and dispersion relation

Many oscillatory phenomena can be modelled by linear equations whose solutions are superpositions of harmonic waves with angular frequency depending on the wave number:

$$\omega = \omega(k). \tag{5.3}$$

A typical example is the wave system produced by dropping a stone in a pond.

If ω is linear, e.g. $\omega(k) = ck$, $c > 0$, the crests move with speed c, independent of the wave number. However, if $\omega(k)$ is not proportional to k, the crests move with speed $c_p = \omega(k)/k$, that depends on the wave number. In other words, the crests move at different speeds for different wavelengths. As a consequence, the various components in a wave packet, given by the superposition of harmonic waves of different wavelengths, will eventually separate or disperse. For this reason, (5.3) is called **dispersion relation.**

In the theory of dispersive waves, the **group velocity**, given by

$$c_g = \omega'(k),$$

is a central notion, mainly for the following three reasons.

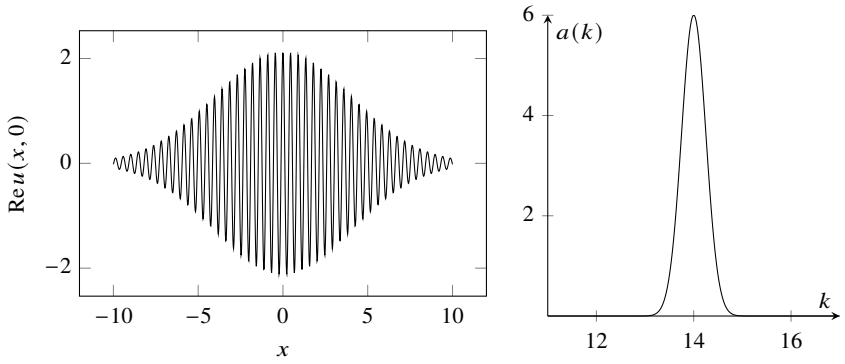

Fig. 5.2 Wave packet and its Fourier transform

1. *The group velocity is the speed at which an isolated wave packet moves as a whole*. A wave packet may be obtained by the superposition of dispersive harmonic waves, for instance through a Fourier integral of the form

$$u(x,t) = \int_{-\infty}^{+\infty} a(k) e^{i[kx - \omega(k)t]} dk \tag{5.4}$$

where only the real part has a physical meaning. Consider a localized wave packet, with wave number $k \approx k_0$, almost constant, and with amplitude slowly varying with x. Then, the packet contains a large number of crests and the amplitudes $|a(k)|$ of the various Fourier components are negligible except that in a small neighborhood of k_0, say, $(k_0 - \delta, k_0 + \delta)$.

Figure 5.2 shows the initial profile of a Gaussian packet,

$$\operatorname{Re} u(x,0) = \frac{3}{\sqrt{2}} \exp\left\{-\frac{x^2}{32}\right\} \cos 14x,$$

slowly varying with x, with $k_0 = 14$, and its Fourier transform:

$$a(k) = 6 \exp\{-8(k-14)^2\}.$$

As we can see, the amplitudes $|a(k)|$ of the various Fourier components are negligible except when k is near k_0.
Then we may write

$$\omega(k) \approx \omega(k_0) + \omega'(k_0)(k - k_0) = \omega(k_0) + c_g(k - k_0)$$

and

$$u(x,t) \approx e^{i\{k_0 x - \omega(k_0)t\}} \int_{k_0-\delta}^{k_0+\delta} a(k) e^{i(k-k_0)(x - c_g t)} dk. \tag{5.5}$$

Thus, u turns out to be well approximated by the product of two waves. The first one is a pure harmonic wave (*carrier*) with relatively short wavelength $2\pi/k_0$ and phase speed $\omega(k_0)/k_0$. The second one depends on x, t through the combination $x - c_g t$, and is a superposition of waves of very small wavenumbers $k - k_0$, which correspond to very large wavelengths. We may interpret the second factor as a sort of envelope of the short waves of the packet (*modulation wave*), that is the packet as a whole, which therefore moves with the group speed.

2. *An observer traveling at the group velocity sees constantly waves of the same wavelength* $2\pi/k$, after the transitory effects due to a localized initial perturbation (e.g. a stone thrown into a pond). In other words, c_g is the propagation speed of the wave numbers.

Imagine dropping a stone into a pond. The water perturbation looks complicated, at the beginning. After a sufficiently long time, the various Fourier components will be quite dispersed and the perturbation will appear as a slowly modulated wave train, almost sinusoidal near every point, with a local wave number $k(x,t)$ and a *local frequency* $\omega(x,t)$. If the water is deep enough, we expect that, at each fixed time t, the wavelength increases with the distance from the stone (longer waves move faster, see Sect. 5.11.4) and that, at each fixed point x, the wavelength tends to decrease with time.

Thus, the essential features of the wave system can be observed at a relatively long distance from the location of the initial disturbance and after some time has elapsed.

We may assume that the free surface displacement u is given by a Fourier integral of the form (5.4) and we are interested on the behavior of u for $t \gg 1$. An important tool comes from the *method of stationary phase* (see Sect. 5.11.6) which gives an asymptotic formula for integrals of the form

$$I(t) = \int_{-\infty}^{+\infty} f(k) e^{it\varphi(k)} dk \tag{5.6}$$

as $t \to +\infty$. We can put u into the form (5.6) by writing

$$u(x,t) = \int_{-\infty}^{+\infty} a(k) e^{it[k\frac{x}{t} - \omega(k)]} dk,$$

by moving from the origin at a fixed speed V (thus $x = Vt$) and then defining

$$\varphi(k) = kV - \omega(k).$$

Assume for simplicity that φ has only one stationary point k_0, that is

$$\omega'(k_0) = V,$$

and that $\omega''(k_0) \neq 0$. Then, according to the method of stationary phase, we can write

$$u(Vt,t) = \sqrt{\frac{\pi}{|\omega''(k_0)|}}\frac{a(k_0)}{\sqrt{t}}\exp\{it\,[k_0V - \omega(k_0)]\} + O(t^{-1}). \tag{5.7}$$

Thus, if we allow errors of order t^{-1}, by moving with speed $V = \omega'(k_0) = c_g$, *the same wave number* k_0 *always appears* at the position $x = c_g t$. Note that the amplitude decreases like $t^{-1/2}$ as $t \to +\infty$. This is a typical attenuation effect of dispersion.

3. *Energy is transported at the group velocity by waves of wavelength* $2\pi/k$. In a wave packet like (5.5), the energy is proportional to

$$\int_{k_0-\delta}^{k_0+\delta} |a(k)|^2\,dk \simeq 2\delta\,|a(k_0)|^2$$

so that it moves at the same speed of k_0, that is c_g (see e.g. *L. A. Segel*, [32]).

Since the energy travels at the group velocity, there are significant differences in the wave system according to the sign of $c_g - c_p$, as we will see in Sect. 5.11.

5.2 Transverse Waves in a String

5.2.1 The model

We now derive a classical model for the small transverse vibration of a **tightly stretched** horizontal string (e.g. the string of a guitar). We assume the following hypotheses:

1. *Each point of the string undergoes vertical displacements only*. Horizontal displacements can be neglected.
2. *The vertical displacement of a point depends on time and on its position on the string.* If we denote by u the vertical displacement of a point located at x when the string is at rest, then we have $u = u(x,t)$.
3. *The vibrations of the string have small amplitude compared to its length.* This entails that the changes in the slope of the string from the horizontal equilibrium position are very small so that, $|u_x(x,t)| \ll 1$. Note that this makes sense, since the slope is dimensionless.
4. *The string is perfectly flexible*. This means that it offers no resistance to bending. In particular, by the absence of distributed moments along the string, the stress at any point on the string can be modelled by a tangential force $\mathbf{T}$ of magnitude τ, called tension. Fig. 5.3 shows how the forces due to the *tension* acts at the end points of a small segment of the string.
5. *Friction is negligible.*

Under the above assumptions, the equation of motion of the string can be derived from *conservation of mass* and *Newton's law.*

Let $\rho_0 = \rho_0(x)$ be the linear density of the string at rest and $\rho = \rho(x,t)$ be its density at time t. Consider an arbitrary part of the string between x and $x + \Delta x$ and

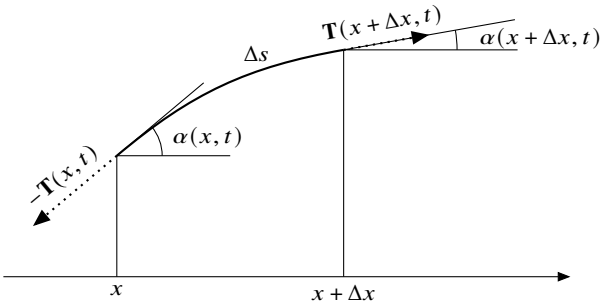

Fig. 5.3 Tension at the end points of a small segment of a string

denote by Δs the corresponding length element at time t. Then, the conservation of mass yields

$$\rho_0(x)\Delta x = \rho(x,t)\Delta s. \tag{5.8}$$

To write Newton's law of motion we have to determine the forces acting on our small piece of string, namely: the force exerted by the right part of the string at $x+\Delta x$ (i.e. $\mathbf{T}(x+\Delta x,t)$), the force exerted by the left part of the string at x (i.e. $-\mathbf{T}(x,t)$), and the body forces such as gravity and external loads.

Since the motion is vertical, the horizontal forces have to balance. On the other hand these forces come from the tension only, so that if $\tau(x,t)$ denotes the magnitude of the tension at x at time t, we can write (Fig. 5.3):

$$\tau(x+\Delta x,t)\cos\alpha(x+\Delta x,t) - \tau(x,t)\cos\alpha(x,t) = 0.$$

Dividing by Δx and letting $\Delta x \to 0$, we obtain

$$\frac{\partial}{\partial x}\left[\tau(x,t)\cos\alpha(x,t)\right] = 0,$$

from which

$$\tau(x,t)\cos\alpha(x,t) = \tau_0(t), \tag{5.9}$$

where $\tau_0(t)$ is *positive*.

The vertical forces are given by the vertical component of the tension and by the body forces. Using (5.9), the scalar vertical component of the tension at x, at time t, is given by:

$$\tau_{\text{vert}}(x,t) = \tau(x,t)\sin\alpha(x,t) = \tau_0(t)\tan\alpha(x,t) = \tau_0(t)u_x(x,t).$$

Therefore, the (scalar) vertical component of the force acting on our small piece of string, due to the tension, is

$$\tau_{\text{vert}}(x+\Delta x,t) - \tau_{\text{vert}}(x,t) = \tau_0(t)\left[u_x(x+\Delta x,t) - u_x(x,t)\right].$$

Denote by $f(x,t)$ the magnitude of the (vertical) body forces per unit mass. Then, using (5.8), the magnitude of the body forces acting on the string segment is given by:

$$\int_x^{x+\Delta x} \rho(y,t) f(y,t) ds = \int_x^{x+\Delta x} \rho_0(y) f(y,t) dy.$$

Thus, using (5.8) again and observing that u_{tt} is the (scalar) vertical acceleration, Newton's law gives:

$$\int_x^{x+\Delta x} \rho_0(y) u_{tt}(y,t) dy = \tau_0(t)[u_x(x+\Delta x,t) - u_x(x,t)] + \int_x^{x+\Delta x} \rho_0(y) f(y,t) dy.$$

Dividing by Δx and letting $\Delta x \to 0$, we obtain the equation

$$u_{tt} - c^2(x,t) u_{xx} = f(x,t) \tag{5.10}$$

where

$$c(x,t) = \sqrt{\frac{\tau_0(t)}{\rho_0(x)}}.$$

If the string is homogeneous, then ρ_0 is constant. If moreover the string is **perfectly elastic**, then τ_0 is constant as well, since the horizontal tension is nearly the same as for the string at rest, in the horizontal position. For instance, guitar and violin strings are nearly homogeneous, perfectly flexible and elastic. We shall come back to eq. (5.10) shortly.

5.2.2 Energy

Suppose that our *perfectly flexible and elastic* string has length L at rest, in the horizontal position. We may identify its initial position with the segment $[0, L]$ on the x axis. Since $u_t(x,t)$ is the vertical velocity of the point located at x, the expression

$$E_{\text{cin}}(t) = \frac{1}{2} \int_0^L \rho_0 u_t^2 \, dx \tag{5.11}$$

represents the total **kinetic energy during the vibrations**. The string stores **potential energy** too, due to the work of elastic forces. These forces stretch an element of string of length Δx at rest, by

$$\Delta s - \Delta x = \int_x^{x+\Delta x} \sqrt{1+u_x^2}\, dy - \Delta x = \int_x^{x+\Delta x} (\sqrt{1+u_x^2} - 1) dy \approx \frac{1}{2} u_x^2 \Delta x,$$

since[1] $|u_x| \ll 1$. Thus, the work done by the elastic forces on that string element is

[1] Recall that, at first order, if $\varepsilon \ll 1$, $\sqrt{1+\varepsilon} - 1 \simeq \varepsilon/2$.

$$dW = \frac{1}{2}\tau_0 u_x^2 \Delta x.$$

Summing all the contributions, the total elastic internal energy is given by

$$E_{\text{pot}}(t) = \frac{1}{2}\int_0^L \tau_0 u_x^2 \, dx. \tag{5.12}$$

From (5.11) and (5.12) we find

$$E(t) = E_{\text{cin}}(t) + E_{\text{pot}}(t) = \frac{1}{2}\int_0^L [\rho_0 u_t^2 + \tau_0 u_x^2] \, dx. \tag{5.13}$$

Let us compute the variation of E. Taking the time derivative under the integral, we find (remember that $\rho_0 = \rho_0(x)$ and τ_0 is constant),

$$\dot{E}(t) = \int_0^L [\rho_0 u_t u_{tt} + \tau_0 u_x u_{xt}] \, dx.$$

Integrating by parts we get

$$\int_0^L \tau_0 u_x u_{xt} \, dx = \tau_0[u_x(L,t)u_t(L,t) - u_x(0,t)u_t(0,t)] - \tau_0 \int_0^L u_t u_{xx} dx,$$

whence

$$\dot{E}(t) = \int_0^L [\rho_0 u_{tt} - \tau_0 u_{xx}] u_t dx + \tau_0[u_x(L,t)u_t(L,t) - u_x(0,t)u_t(0,t)].$$

Using (5.10), we find:

$$\dot{E}(t) = \int_0^L \rho_0 f u_t \, dx + \tau_0[u_x(L,t)u_t(L,t) - u_x(0,t)u_t(0,t)]. \tag{5.14}$$

In (5.14), the first term takes into account the energy due to external distributed forces, while the other two terms describe the energy exchange at the endpoints.

In particular, if

$$f = 0$$

and, at the endpoints,

$$\text{either } u_x = 0 \text{ or } u \text{ is constant (hence } u_t = 0),$$

we deduce

$$\dot{E}(t) = 0.$$

This implies

$$E(t) = E(0)$$

which *expresses the conservation of energy*.

5.3 The One-dimensional Wave Equation

5.3.1 Initial and boundary conditions

Equation (5.10) is called the *vibrating string equation*, or *one-dimensional wave equation*. The coefficient c has the dimensions of a velocity and in fact, we will shortly see that it represents the wave propagation speed along the string. When $f \equiv 0$, the equation is *homogeneous* and the *superposition principle holds*: if u_1 and u_2 are solutions of

$$u_{tt} - c^2 u_{xx} = 0 \tag{5.15}$$

and a, b are (real or complex) scalars, then $au_1 + bu_2$ is a solution as well. More generally, if $u_k(x,t)$ is a family of solutions depending on the parameter k (integer or real) and $g = g(k)$ is a function rapidly vanishing at infinity, then

$$\sum_{k=1}^{\infty} u_k(x,t)g(k) \quad \text{and} \quad \int_{-\infty}^{+\infty} u_k(x,t)g(k)dk$$

are still solutions of (5.15).

Suppose we are considering the space-time region $0 < x < L$, $0 < t < T$. In a reasonable well posed problem for the (one-dimensional) heat equation it is appropriate to assign the initial profile of the temperature, because of the presence of a first order time derivative, and a boundary condition at both ends $x = 0$ and $x = L$, because of the second order spatial derivative.

By analogy with the Cauchy problem for second order ordinary differential equations, the presence of the second order time derivative in (5.10) suggests that not only the initial profile of the string has to be assigned, but the initial velocity has to be prescribed as well.

Thus, our initial (or Cauchy) data are

$$u(x,0) = g(x), \qquad u_t(x,0) = h(x), \qquad x \in [0, L]\,.$$

The boundary data are formally similar to those for the heat equation:

- *Dirichlet data* describe the displacement at the end points of the string:

$$u(0,t) = a(t), \qquad u(L,t) = b(t), \qquad t > 0.$$

  If $a(t) = b(t) \equiv 0$ (homogeneous data), both ends are fixed, with zero displacement.
- *Neumann data* describe an applied (scalar) vertical tension at the end points. As in the derivation of the wave equation, we may model this tension by $-\tau_0 u_x(0,t)$ and $\tau_0 u_x(L,t)$, respectively, so that the Neumann conditions take the form

$$-\tau_0 u_x(0,t) = a(t), \qquad \tau_0 u_x(L,t) = b(t), \qquad t > 0.$$

In the special case of homogeneous data, $a(t) = b(t) \equiv 0$, both ends of the string are attached to a frictionless sleeve and are free to move vertically.

- *Robin data* describe a linear elastic attachment at the end points. One way to realize this type of boundary condition is to attach an end point to a linear spring[2] whose other end is fixed. Then, if $k > 0$ is the elastic constant of the spring, the tension forces at the end points are equal to $-ku(0,t)$ and $-ku(L,t)$, respectively. This translates into assigning

$$\tau_0 u_x(0,t) = ku(0,t), \qquad \tau_0 u_x(L,t) = -ku(L,t), \qquad t > 0.$$

In several concrete situations, *mixed conditions* have to be assigned. For instance, Robin data at $x = 0$ and Dirichlet data at $x = L$.

- *Global Cauchy problem*. We may think of a string of infinite length and assign only the initial data

$$u(x,0) = g(x), \qquad u_t(x,0) = h(x), \qquad x \in \mathbb{R}.$$

Although physically unrealistic, it turns out that the solution of the global Cauchy problem is of fundamental importance. We shall solve it in Sect. 5.4.

Under reasonable assumptions on the data, the above problems are well posed. In the next section we use the method of separation of variables to show the well posedness for a Cauchy-Dirichlet problem.

Remark 5.1 Other kinds of problems for the wave equation are the so called *Goursat problem* and the *characteristic Cauchy problem*. Two examples are given in Problems 5.11, 5.12.

5.3.2 Separation of variables

Suppose that the vibration of a violin chord is modelled by the following Cauchy-Dirichlet problem

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & 0 < x < L, t > 0 \\ u(0,t) = u(L,t) = 0 & t > 0 \\ u(x,0) = g(x),\ u_t(x,0) = h(x) & 0 \le x \le L \end{cases} \tag{5.16}$$

where $c^2 = \tau_0/\rho_0$ is constant.

We want to check whether this problem is *well posed*, that is, whether a solution exists, is unique and it is stable (i.e. it depends "continuously" on the data g and h). For the time being we proceed formally, without worrying too much about the correct hypotheses on g and h and the regularity of u.

[2] That is a spring which obeys Hooke's law: the strain is a linear function of the stress.

• *Existence*. Since the boundary conditions are homogeneous[3], we try to construct a solution by separation of variables.

Step 1. We start looking for nontrivial solutions of the form

$$U(x,t) = w(t)v(x)$$

with $v(0) = v(L) = 0$. Inserting U into the wave equation we find

$$0 = U_{tt} - c^2 U_{xx} = w''(t)v(x) - c^2 w(t)v''(x)$$

that is, separating the variables,

$$\frac{1}{c^2}\frac{w''(t)}{w(t)} = \frac{v''(x)}{v(x)}. \tag{5.17}$$

We have reached a familiar situation: (5.17) is an identity between two functions, one depending on t only and the other one depending on x only. Therefore the two sides of (5.17) must be both equal to the same constant, say λ. Thus, we are led to the equation

$$w''(t) - \lambda c^2 w(t) = 0 \tag{5.18}$$

and to the *eigenvalue problem*

$$v''(x) - \lambda v(x) = 0 \tag{5.19}$$

$$v(0) = v(L) = 0. \tag{5.20}$$

Step 2 (solution of the eigenvalue problem). There are three possibilities for the general integral of (5.19).

a) If $\lambda = 0$, then $v(x) = A + Bx$ and the (5.20) imply $A = B = 0$.
b) If $\lambda = \mu^2 > 0$, then $v(x) = Ae^{-\mu x} + Be^{\mu x}$ and again the (5.20) imply $A = B = 0$.
c) If $\lambda = -\mu^2 < 0$, then $v(x) = A\sin\mu x + B\cos\mu x$. From (5.20) we get

$$\begin{aligned} v(0) &= B = 0 \\ v(1) &= A\sin\mu L + B\cos\mu L = 0, \end{aligned}$$

whence

$$A \text{ arbitrary}, \ B = 0, \ \mu L = m\pi, \ m = 1, 2, \ldots .$$

Thus, only in case c) we find nontrivial solutions, given by

$$v_m(x) = A\sin\mu_m x, \qquad \mu_m = \frac{m\pi}{L}. \tag{5.21}$$

Step 3. Insert $\lambda = -\mu_m^2 = -m^2\pi^2/L^2$ into (5.18). Then, the general integral is

$$w_m(t) = C\cos(\mu_m ct) + D\sin(\mu_m ct) \qquad (C, D \in \mathbb{R}). \tag{5.22}$$

[3] Remember that this assumption is essential for using the separation of variables method.

From (5.21) and (5.22) we construct the family of solutions

$$U_m(x,t) = [a_m \cos(\mu_m ct) + b_m \sin(\mu_m ct)] \sin \mu_m x, \qquad m = 1, 2, \ldots$$

where a_m and b_m are arbitrary constants.

U_m is called the m^{th}-**normal mode** of vibration or m^{th}-*harmonic*, and it represents a *standing wave* with frequency $mc/2L$. The first harmonic and its frequency $c/2L$, the lowest possible, are said to be *fundamental*. All the other frequencies are *integral multiples* of the fundamental one. It seems that a violin chord produces good quality tones, pleasant to the ear, because of this reason. This is not the case, for instance, for a vibrating membrane like a drum, as we will see shortly.

Step 4. Superposition of the normal modes. If the initial conditions are

$$u(x,0) = A \sin \mu_m x \qquad u_t(x,0) = B \sin \mu_m x$$

then the solution of our problem is U_m, with $a_m = A$ and $b_m = B/(\mu_m c)$, and the string vibrates at its m^{th}-mode. In general, the solution is constructed by superposing the harmonics U_m through the formula

$$u(x,t) = \sum_{m=1}^{\infty} [a_m \cos(\mu_m ct) + b_m \sin(\mu_m ct)] \sin \mu_m x, \tag{5.23}$$

where the coefficients a_m and b_m have to be chosen in order to satisfy the initial conditions

$$u(x,0) = \sum_{m=1}^{\infty} a_m \sin \mu_m x = g(x) \tag{5.24}$$

and

$$u_t(x,0) = \sum_{m=1}^{\infty} c\mu_m b_m \sin \mu_m x = h(x), \tag{5.25}$$

for $0 \le x \le L$.

Looking at (5.24) and (5.25), it is natural to assume that both g and h have an expansion in Fourier sine series in the interval $[0, L]$. Let

$$\hat{g}_m = \frac{2}{L} \int_0^L g(x) \sin \mu_m x \, dx \quad \text{and} \quad \hat{h}_m = \frac{2}{L} \int_0^L h(x) \sin \mu_m x \, dx$$

be the Fourier sine coefficients of g and h. If we choose

$$a_m = \hat{g}_m, \qquad b_m = \frac{\hat{h}_m}{\mu_m c}, \tag{5.26}$$

then (5.23) becomes

$$u(x,t) = \sum_{m=1}^{\infty} \left[\hat{g}_m \cos(\mu_m ct) + \frac{\hat{h}_m}{\mu_m c} \sin(\mu_m ct) \right] \sin \mu_m x \tag{5.27}$$

and it satisfies (5.24) and (5.25).

Although every U_m is a smooth solution of the wave equation, in principle (5.27) is only a formal solution, unless we may differentiate term by term twice with respect to both x and t, obtaining

$$(\partial_{tt} - c^2\partial_{xx}^2)u(x,t) = \sum_{m=1}^{\infty}(\partial_{tt} - c^2\partial_{xx}^2)U_m(x,t) = 0. \tag{5.28}$$

This is possible if $\hat{g}_m$ and $\hat{h}_m$ vanish sufficiently fast as $m \to +\infty$. In fact, differentiating term by term twice, we have

$$u_{xx}(x,t) = -\sum_{m=1}^{\infty}\left[\mu_m^2\hat{g}_m\cos(\mu_m ct) + \frac{\mu_m\hat{h}_m}{c}\sin(\mu_m ct)\right]\sin\mu_m x \tag{5.29}$$

and

$$u_{tt}(x,t) = -\sum_{m=1}^{\infty}\left[\mu_m^2\hat{g}_m c^2\cos(\mu_m ct) + \mu_m\hat{h}_m c\sin(\mu_m ct)\right]\sin\mu_m x. \tag{5.30}$$

Thus, if, for instance,

$$|\hat{g}_m| \le \frac{C}{m^4} \quad \text{and} \quad |\hat{h}_m| \le \frac{C}{m^3}, \tag{5.31}$$

then

$$|\mu_m^2\hat{g}_m| \le \frac{C\pi^2}{L^2m^2}, \quad \text{and} \quad |\mu_m\hat{h}_m| \le \frac{C\pi}{Lm^2}$$

so that, by the Weierstrass test, the series in (5.29), (5.30) converge uniformly in $[0,L]\times[0,+\infty)$. Since also the series (5.27) is uniformly convergent in $[0,L]\times[0,+\infty)$, differentiation term by term is allowed and u is a C^2 solution of the wave equation.

Under which assumptions on g and h does formula (5.31) hold?
Let $g \in C^4([0,L])$, $h \in C^3([0,L])$ and assume the following compatibility conditions:

$$\begin{aligned} &g(0) = g(L) = g''(0) = g''(L) = 0\\ &h(0) = h(L) = 0. \end{aligned} \tag{5.32}$$

Then (5.31) hold[4].

Moreover, under the same assumptions, u attains the initial and boundary data in the pointwise sense. Indeed it is not difficult to check that $u(y,s) \to 0$ as $(y,s) \to$

[4] It is an exercise on integration by parts. For instance, if $f \in C^4([0,L])$ and $f(0) = f(L) = f''(0) = f''(L) = 0$, then, integrating by parts four times, we have

$$\hat{f}_m = \int_0^L f(x)\sin\left(\frac{m\pi}{L}\right)dx = \frac{L^4}{\pi^4m^4}\int_0^L f^{(4)}(x)\sin\left(\frac{m\pi}{L}\right)dx$$

and $|\hat{f}_m| \le \max|f^{(4)}|\,\frac{L^5}{\pi^4m^4}$.

$(0,t)$ or $(y,s) \to (L,t)$, for every $t \geq 0$, and

$$u(y,t) \to g(x),\ u_t(y,t) \to h(x), \qquad \text{as } (y,t) \to (x,0), \tag{5.33}$$

for every $x \in [0,L]$. We conclude that u is a solution of problem (5.16).

• *Uniqueness*. To show that (5.27) is the unique solution of problem (5.16), we use the conservation of energy. Let u and v be solutions of (5.16). Then $w = u - v$ is a solution of the same problem with zero initial and boundary data. We want to show that $w \equiv 0$.

Formula (5.13) gives, for the total mechanical energy,

$$E(t) = E_{\text{cin}}(t) + E_{\text{pot}}(t) = \frac{1}{2}\int_0^L [\rho_0 w_t^2 + \tau_0 w_x^2]\, dx.$$

In our case, by (5.14) we have

$$\dot{E}(t) = 0,$$

since $f = 0$ and $w_t(L,t) = w_t(0,t) = 0$, whence

$$E(t) = E(0)$$

for every $t \geq 0$. Since, in particular, $w_t(x,0) = w_x(x,0) = 0$, we have

$$E(t) = E(0) = 0$$

for every $t \geq 0$. On the other hand, $E_{\text{cin}}(t) \geq 0$, $E_{\text{pot}}(t) \geq 0$, so that we deduce

$$E_{\text{cin}}(t) = 0,\ E_{\text{pot}}(t) = 0,$$

which force $w_t = w_x = 0$ for every x, t. Therefore w is constant and since $w(x,0) = 0$, we conclude that $w(x,t) = 0$ for every $t \geq 0$.

• *Stability*. We want to show that small perturbations on the data yield small perturbations on the solution. Clearly, we need to establish how we intend to measure the distance for the data and for the corresponding solutions. For the initial data, we use the *quadratic or L^2-distance*, given by[5]

$$\|g_1 - g_2\|_0 = \left(\int_0^L |g_1(x) - g_2(x)|^2\, dx\right)^{1/2}.$$

For functions depending also on time, we define

$$\|u - v\|_{0,\infty} = \sup_{t>0}\left(\int_0^L |u(x,t) - v(x,t)|^2\, dx\right)^{1/2}$$

which measures the maximum in time of the quadratic distance in space.

[5] The symbol $\|g\|$ denotes a *norm* of g. See Chap. 6.

Now, let u_1 and u_2 be solutions of problem (5.16), corresponding to the data g_1, h_1 and g_2, h_2, respectively. Their difference $w = u_1 - u_2$ is a solution of the same problem with Cauchy data $g = g_1 - g_2$ and $h = h_1 - h_2$. From (5.27) we know that

$$w(x,t) = \sum_{m=1}^{\infty} \left[\hat{g}_m \cos(\mu_m c t) + \frac{\hat{h}_m}{\mu_m c} \sin(\mu_m c t) \right] \sin \mu_m x.$$

From Parseval's identity (1.12) and the elementary inequality $(a+b)^2 \le 2(a^2+b^2)$, $a, b \in \mathbb{R}$, we can write

$$\begin{aligned} \int_0^L |w(x,t)|^2 \, dx &= \frac{L}{2} \sum_{m=1}^{\infty} \left[\hat{g}_m \cos(\mu_m c t) + \frac{\hat{h}_m}{\mu_m c} \sin(\mu_m c t) \right]^2 \\ &\le L \sum_{m=1}^{\infty} \left[\hat{g}_m^2 + \left(\frac{\hat{h}_m}{\mu_m c} \right)^2 \right]. \end{aligned}$$

Since $\mu_m \ge \pi/L$, using Parseval's identity for g and h, we obtain

$$\begin{aligned} \int_0^L |w(x,t)|^2 \, dx &\le L \max \left\{ 1, \left(\frac{L}{\pi c} \right)^2 \right\} \sum_{m=1}^{\infty} \left[\hat{g}_m^2 + \hat{h}_m^2 \right] \\ &= 2 \max \left\{ 1, \left(\frac{L}{\pi c} \right)^2 \right\} \left[\|g\|_0^2 + \|h\|_0^2 \right] \end{aligned}$$

whence the stability estimate

$$\|u_1 - u_2\|_{0,\infty}^2 \le 2 \max \left\{ 1, \left(\frac{L}{\pi c} \right)^2 \right\} \left[\|g_1 - g_2\|_0^2 + \|h_1 - h_2\|_0^2 \right]. \tag{5.34}$$

Thus, close data produce close solutions.

Summarizing, we have proved the following result:

Theorem 5.2 *Let $g \in C^4([0,L])$ and $h \in C^3([0,L])$ satisfy conditions (5.32). Then, problem (5.16) has a unique solution $u \in C^2([0,L] \times (0,+\infty))$ given by (5.27). Moreover u satisfies the stability estimate*

$$\|u\|_{0,\infty}^2 \le 2 \max \left\{ 1, \left(\frac{L}{\pi c} \right)^2 \right\} \left[\|g\|_0^2 + \|h\|_0^2 \right].$$

Remark 5.3 From (5.27), the chord vibration is given by the superposition of harmonics corresponding to the nonzero Fourier coefficients of the initial data. The complex of such harmonics determines a particular feature of the emitted sound, known as the *timbre*, a sort of signature of the musical instrument!

Remark 5.4 The hypotheses we have made on g and h are unnaturally restrictive. For example, if we pluck a violin chord at a point, the initial profile is continuous but

has a corner at that point and cannot be even C^1. A physically realistic assumption for the initial profile g is *continuity*.

Similarly, if we are willing to model the vibration of a chord set into motion by a strike of a little hammer, we should allow discontinuity in the initial velocity. Thus it is realistic to assume h *bounded*.

Under these weaker hypotheses, the separation of variables method does not work. On the other hand, we have already faced a similar situation in Chap. 4, where the necessity to admit discontinuous solutions of a conservation law has led to a more general and flexible formulation of the initial value problem. Also for the wave equation it is possible to introduce suitable *weak* formulations of the various initial-boundary value problems, in order to include realistic initial data and solutions with a low degree of regularity. A first attempt is shown in Sect. 5.4.2. A weak formulation, more suitable also for numerical methods, is treated in Chap. 9.

5.4 The d'Alembert Formula

5.4.1 The homogeneous equation

In this section we establish the celebrated formula of d'Alembert for the solution of the following global Cauchy problem:

$$\begin{cases} u_{tt} - c^2 u_{xx} = 0 & x \in \mathbb{R},\, t > 0 \\ u(x,0) = g(x), \quad u_t(x,0) = h(x) & x \in \mathbb{R}. \end{cases} \tag{5.35}$$

To find the solution of (5.35), we first factorize the wave equation in the following way:

$$(\partial_t - c\partial_x)(\partial_t + c\partial_x)u = 0. \tag{5.36}$$

Now, let

$$v = u_t + cu_x. \tag{5.37}$$

Then v solves the linear transport equation (see Sect. 4.2.1)

$$v_t - cv_x = 0,$$

whence

$$v(x,t) = \psi(x + ct),$$

where ψ is an arbitrary differentiable function. From (5.37) we have

$$u_t + cu_x = \psi(x + ct)$$

and formula (4.11), p. 185, yields

$$u(x,t) = \int_0^t \psi(x - c(t-s) + cs)\, ds + \varphi(x - ct),$$

where φ is another arbitrary differentiable function.

Letting $x - ct + 2cs = y$, we find

$$u(x,t) = \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(y)dy + \varphi(x - ct). \tag{5.38}$$

To determine ψ and φ, we impose the initial conditions

$$u(x,0) = \varphi(x) = g(x) \tag{5.39}$$

and

$$u_t(x,0) = \psi(x) - c\varphi'(x) = h(x),$$

whence

$$\psi(x) = h(x) + cg'(x). \tag{5.40}$$

Inserting (5.40) and (5.39) into (5.38) we get:

$$\begin{aligned} u(x,t) &= \frac{1}{2c} \int_{x-ct}^{x+ct} [h(y) + cg'(y)]\ dy + g(x - ct) \\ &= \frac{1}{2c} \int_{x-ct}^{x+ct} h(y)dy + \frac{1}{2} [g(x + ct) - g(x - ct)] + g(x - ct) \end{aligned}$$

and, finally, the **d'Alembert formula**

$$u(x,t) = \frac{1}{2} [g(x + ct) + g(x - ct)] + \frac{1}{2c} \int_{x-ct}^{x+ct} h(y)dy. \tag{5.41}$$

• *Uniqueness and stability*. If $g \in C^2(\mathbb{R})$ and $h \in C^1(\mathbb{R})$, formula (5.41) defines a C^2-solution in the half-plane $\mathbb{R} \times [0, +\infty)$. On the other hand, a C^2-solution u in $\mathbb{R} \times [0, +\infty)$ has to be given by (5.41), just because we can apply to u the procedure we have used to solve the Cauchy problem. Thus the solution is unique. However, observe that no regularizing effect takes place here: the solution u remains no more than C^2 for any $t > 0$. Thus, there is a striking difference with diffusion phenomena, governed by the heat equation.

Furthermore, let u_1 and u_2 be the solutions corresponding to the data g_1, h_1 and g_2, h_2, respectively. Then, if all the data are bounded, the d'Alembert formula for $u_1 - u_2$ yields, for every $x \in \mathbb{R}$ and $t \in [0, T]$,

$$|u_1(x,t) - u_2(x,t)| \le \|g_1 - g_2\|_\infty + T \|h_1 - h_2\|_\infty$$

where

$$\|g_1 - g_2\|_\infty = \sup_{x \in \mathbb{R}} |g_1(x) - g_2(x)|, \qquad \|h_1 - h_2\|_\infty = \sup_{x \in \mathbb{R}} |h_1(x) - h_2(x)|.$$

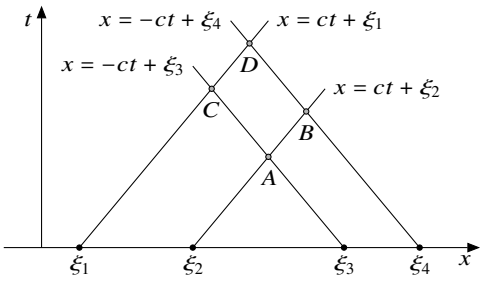

Fig. 5.4 Characteristic parallelogram

Therefore, we have stability in the pointwise uniform sense, at least for finite time.

• *Progressive waves*. Let

$$F(x) = \frac{1}{2}g(x) + \frac{1}{2c}\int_0^x h(y)dy, \quad G(x) = \frac{1}{2}g(x) - \frac{1}{2c}\int_0^x h(y)dy.$$

Then we may write (5.41) in the form

$$u(x,t) = F(x+ct) + G(x-ct) \tag{5.42}$$

which gives u as a *superposition of two progressive waves moving at constant speed c in the negative and positive x-direction*, respectively. Thus, these waves are not dispersive.

The two terms in (5.42) are respectively constant along the two families of straight lines γ^+ and γ^- given by

$$x + ct = \text{constant}, \qquad x - ct = \text{constant}.$$

These lines are called *characteristics*[6] and carry important information, as we will see in the next subsection.

• *Characteristic parallelogram*. An interesting consequence of (5.42) comes from looking at Fig. 5.4. Consider the *characteristic parallelogram* with vertices at the points A, B, C, D. We have, with a little abuse of notation,

$$\begin{aligned} F(A) = F(C), \quad G(A) &= G(B) \\ F(D) = F(B), \quad G(D) &= G(C). \end{aligned}$$

Summing these relations we get

$$[F(A) + G(A)] + [F(D) + G(D)] = [F(C) + G(C)] + [F(B) + G(B)]$$

which is equivalent to

$$u(A) + u(D) = u(C) + u(B). \tag{5.43}$$

[6] In fact they are the *characteristics* for the two first order factors in the factorization (5.36). See Sect. 4.2.

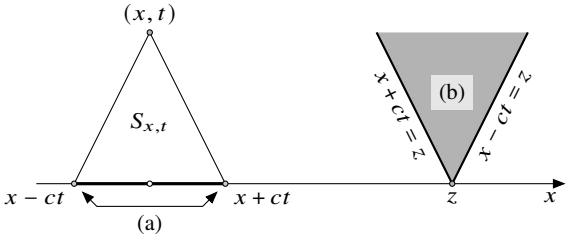

Fig. 5.5 (a) Domain of dependence of (x, t); (b) range of influence of z

Thus, knowing u at three points of a characteristic parallelogram, we can compute u at the fourth one.

Actually, eq. (5.43) leads to a generalized formulation of the wave equation, as it is shown in Problem 5.7.

• *Domains of dependence and of influence*. From d'Alembert formula it follows that the value of u at the point (x, t) depends on the values of g at the points $x - ct$ and $x + ct$ and on the values of h over the whole interval

$$[x - ct, x + ct].$$

This interval is called **domain of dependence of** (x, t) (Fig. 5.5).

From a different perspective, the values of g and h at a point z affect the value of u at the points (x, t) in the sector

$$z - ct \le x \le z + ct,$$

which is called **range of influence** of z (Fig. 5.5). This entails that a disturbance initially localized at z is not felt at a point x until time

$$t = \frac{|x - z|}{c}.$$

Remark 5.5 Differentiating the last term in (5.41) with respect to time we get:

$$\begin{aligned} \frac{\partial}{\partial t}\frac{1}{2c}\int_{x-ct}^{x+ct} h(y)dy &= \frac{1}{2c}\left[ch(x + ct) - (-c)h(x - ct)\right] \\ &= \frac{1}{2}\left[h(x + ct) + h(x - ct)\right] \end{aligned}$$

which has the form of the first term with g replaced by h. It follows that if w_h denotes the solution of the problem

$$\begin{cases} w_{tt} - c^2 w_{xx} = 0 & x \in \mathbb{R}, t > 0 \\ w(x, 0) = 0, w_t(x, 0) = h(x) & x \in \mathbb{R}, \end{cases} \tag{5.44}$$

then d'Alembert formula can be written in the form

$$u(x,t) = \frac{\partial}{\partial t} w_g(x,t) + w_h(x,t). \tag{5.45}$$

Actually, (5.45) can be established without relying on d'Alembert formula, as we will see later.

5.4.2 Generalized solutions and propagation of singularities

In Remark 5.4, p. 268, we have emphasized the necessity of a weak formulation of problem (5.35) to include more physically realistic data. On the other hand, the d'Alembert formula makes perfect sense even for *g continuous* and *h locally integrable*. The question is in which sense the resulting function satisfies the wave equation, since, in principle, it is not even differentiable, only continuous. There are several ways to weaken the notion of solution to include this case; here, for instance, we mimic what we did for conservation laws.

Assuming for the moment that u is a smooth solution of the global Cauchy problem, we multiply the wave equation by a C^2*-test function* v, defined in $\mathbb{R}\times[0,+\infty)$ and compactly supported. Integrating over $\mathbb{R}\times[0,+\infty)$ we obtain

$$\int_0^\infty \int_{\mathbb{R}} [u_{tt} - c^2 u_{xx}] v \, dxdt = 0.$$

Let us now integrate twice by parts both terms, in order to carry all the derivatives from u to v. This yields

$$\int_0^\infty \int_{\mathbb{R}} c^2 u_{xx} v \, dxdt = \int_0^\infty \int_{\mathbb{R}} c^2 u v_{xx} \, dxdt,$$

since v is zero outside a compact subset of $\mathbb{R}\times[0,+\infty)$, and

$$\begin{aligned}\int_0^\infty \int_{\mathbb{R}} u_{tt} v \, dxdt &= -\int_{\mathbb{R}} u_t(x,0)v(x,0)dx - \int_0^\infty \int_{\mathbb{R}} u_t v_t \, dxdt \\ &= -\int_{\mathbb{R}} [u_t(x,0)v(x,0) - u(x,0)v_t(x,0)] \, dx + \int_0^\infty \int_{\mathbb{R}} u v_{tt} dxdt.\end{aligned}$$

Using the Cauchy data $u(x,0) = g(x)$ and $u_t(x,0) = h(x)$, we obtain the integral equation

$$\int_0^\infty \int_{\mathbb{R}} u[v_{tt} - c^2 v_{xx}] \, dxdt + \int_{\mathbb{R}} [g(x)v_t(x,0) - h(x)v(x,0)] \, dx = 0. \tag{5.46}$$

Note that (5.46) makes sense for e.g. u continuous, g continuous and h locally integrable. Conversely, if u is a C^2 function that satisfies (5.46) **for every test function** v, then one can check that u is a solution of problem (5.35).

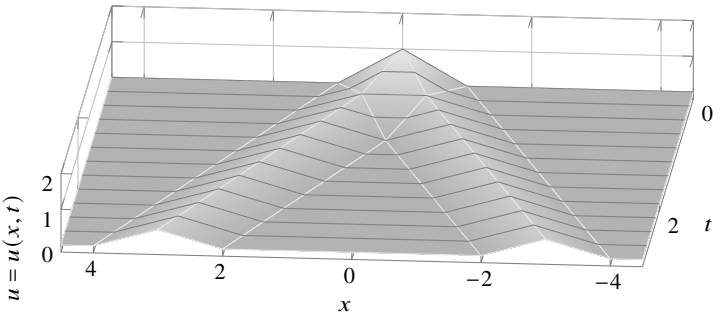

Fig. 5.6 Chord plucked at the origin ($c = 1$)

Thus we may adopt the following definition.

Definition 5.6 Let $g \in C(\mathbb{R})$ and h be locally integrable in $\mathbb{R}$. We say that $u \in C(\mathbb{R} \times [0, +\infty))$ is a *generalized solution* of problem (5.35) if (5.46) holds **for every** test function v.

If g is continuous and h is locally integrable, it can be shown that formula (5.41) represents precisely a generalized solution.

Figure 5.6 shows the wave propagation along a chord of infinite length, plucked at the origin and originally at rest, modelled by the solution of the problem

$$\begin{cases} u_{tt} - u_{xx} = 0 & x \in \mathbb{R}, t > 0 \\ u(x, 0) = g(x), u_t(x, 0) = 0 & x \in \mathbb{R} \end{cases}$$

where g has a triangular profile. As we see, this generalized solution displays lines of discontinuities of the first derivatives, while outside these lines it is smooth.

We want to show that these lines are characteristics. More generally, consider a region $G \subset \mathbb{R} \times (0, +\infty)$, divided into two domains G^+ and G^- by a smooth curve Γ of equation $x = s(t)$, as in Fig. 5.7. Let

$$\mathbf{n} = n_1\mathbf{i} + n_2\mathbf{j} = \frac{1}{\sqrt{1 + (\dot{s}(t))^2}}(-\mathbf{i} + \dot{s}(t)\mathbf{j}) \tag{5.47}$$

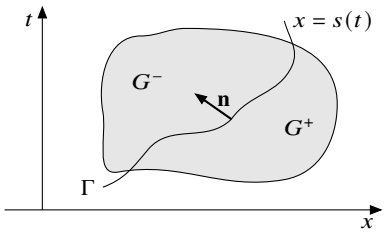

Fig. 5.7 Line of discontinuity of first derivatives

be the unit normal to Γ, pointing outward to G^+.

Let f be a function defined in G, having a jump discontinuity across Γ. To denote this jump we use the symbol

$$[f(s(t),t)] = f^+(s(t),t) - f^-(s(t),t)$$

or simply $[f]$ when there is no risk of confusion, where

$$f^\pm(s(t),t) = \lim_{x \to s(t)^\pm} f(x,t).$$

Now, let u be a generalized solution of Cauchy problem (5.35), of class C^2 in G^+ and G^-, with bounded second derivatives, and whose first derivatives undergo a jump discontinuity on Γ. We want to prove that:

Proposition 5.7 Γ *is a characteristic.*

Proof First of all observe that, from our hypotheses, we have $[u] = 0$ and $[u_x], [u_t] \neq 0$. Moreover, the jumps $[u_x]$ and $[u_t]$ are continuous along Γ.

By analogy with conservation laws, we expect that the integral formulation (5.46) should imply a sort of Rankine-Hugoniot condition, relating the jumps of the derivatives with the slope of Γ, and expressing the balance of linear momentum across Γ.

In fact, let v be a test function with compact support in G. Inserting v into (5.46), we can write

$$0 = \int_G (c^2 u v_{xx} - u v_{tt})\, dxdt = \int_{G^+} (\dots) dxdt + \int_{G^-} (\dots)\, dxdt. \tag{5.48}$$

From Gauss formula, since $v = 0$ on ∂G (dl denotes the arc length on Γ),

$$\begin{aligned}\int_{G^+} (c^2 u v_{xx} - u v_{tt}) dxdt &= \int_\Gamma (n_1 c^2 u^+ v_x - n_2 u^+ v_t)\, dl - \int_{G^+} (c^2 u_x v_x - u_t v_t)\, dxdt \\ &= \int_\Gamma (n_1 c^2 v_x - n_2 v_t) u^+\, dl - \int_\Gamma (n_1 c^2 u_x^+ - n_2 u_t^+) v\, dl,\end{aligned}$$

since $\int_{G^+} (c^2 u_{xx}^+ - u_{tt}^+) v dxdt = 0$. Similarly,

$$\int_{G^-} (c^2 u^- v_{xx} - u^- v_{tt})\, dxdt = -\int_\Gamma (n_1 c^2 v_x - n_2 v_t) u^-\, dl + \int_\Gamma (n_1 c^2 u_x^- - n_2 u_t^-) v\, dl,$$

since $\int_{G^{(2)}} (c^2 u_{xx}^- - u_{tt}^-) v dxdt = 0$ as well.

Thus, since $[u] = 0$ on Γ, or more explicitly $[u(s(t),t)] \equiv 0$, (5.48) yields

$$\int_\Gamma (c^2 [u_x] n_1 - [u_t] n_2) v\, dl = 0.$$

Due to the arbitrariness of v and the continuity of $[u_x]$ and $[u_t]$ on Γ, we deduce

$$c^2 [u_x] n_1 - [u_t] n_2 = 0, \qquad \text{on } \Gamma,$$

or, recalling (5.47),

$$\dot{s} = -c^2 \frac{[u_x]}{[u_t]} \qquad \text{on } \Gamma, \tag{5.49}$$

which is the analogue of the Rankine-Hugoniot condition for conservation laws.

On the other hand, differentiating $[u(s(t),t)] \equiv 0$ we obtain

$$\frac{d}{dt}\,[u(s(t),t)] = [u_x(s(t),t)]\dot{s}(t) + [u_t(s(t),t)] \equiv 0$$

or

$$\dot{s} = -\frac{[u_t]}{[u_x]} \qquad \text{on } \Gamma. \tag{5.50}$$

Equations (5.49) and (5.50) entail

$$\dot{s}(t) = \pm c$$

which yields

$$s(t) = \pm ct + \text{constant}$$

showing that Γ is a characteristic. □

5.4.3 The fundamental solution

It is rather instructive to solve the global Cauchy problem with $g \equiv 0$ and a special h: *the Dirac delta at a point* ξ, that is $h(x) = \delta(x-\xi)$. This problem models, for instance, the vibrations of a violin string generated by a *unit impulse localized at* ξ (a strike of a sharp hammer). The corresponding solution is called the **fundamental solution** and plays the same role of the fundamental solution for the diffusion equation.

Certainly, the *Dirac delta* is a quite unusual datum, out of reach of the theory we have developed so far. Therefore, we proceed formally. Thus, let $K = K(x,\xi,t)$ denote our fundamental solution and apply d'Alembert formula; we find

$$K(x,\xi,t) = \frac{1}{2c}\int_{x-ct}^{x+ct} \delta(y-\xi)dy, \tag{5.51}$$

which at first glance looks like a math-*UFO*. To get a more explicit formula, we first compute $\int_{-\infty}^{x} \delta(y)dy$. To do it, recall that (see Sect. 2.3.3), if $\mathcal{H}$ is the *Heaviside* function and

$$I_\varepsilon(y) = \frac{\mathcal{H}(y+\varepsilon) - \mathcal{H}(y-\varepsilon)}{2\varepsilon} = \begin{cases} \frac{1}{2\varepsilon} & -\varepsilon \le y < \varepsilon \\ 0 & \text{everywhere else} \end{cases} \tag{5.52}$$

is the unit impulse of extent ε, then $\lim_{\varepsilon\downarrow 0} I_\varepsilon(y) = \delta(y)$. Then it seems appropriate to compute $\int_{-\infty}^{x} \delta(y)dy$ by means of the formula

$$\int_{-\infty}^{x} \delta(y)dy = \lim_{\varepsilon\downarrow 0}\int_{-\infty}^{x} I_\varepsilon(y)dy.$$

Now, we have:

$$\int_{-\infty}^{x} I_\varepsilon(y)dy = \begin{cases} 0 & x \le -\varepsilon \\ (x+\varepsilon)/2\varepsilon & -\varepsilon < x < \varepsilon \\ 1 & x \ge \varepsilon. \end{cases}$$

Letting $\varepsilon \to 0$ we deduce that (the value at zero is irrelevant)

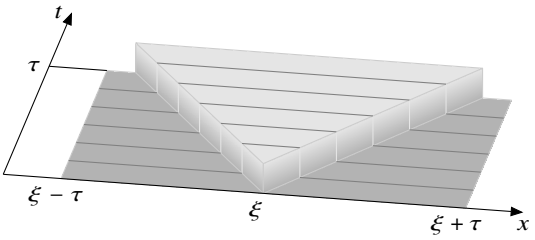

Fig. 5.8 The fundamental solution $K(x, \xi, t)$, $0 < t \leq \tau$

$$\int_{-\infty}^{x} \delta(y)dy = \mathcal{H}(x), \tag{5.53}$$

which actually is not surprising, if we remember that $\mathcal{H}' = \delta$. Everything works nicely. Let us go back to our math-*UFO*, by now . . . identified; we write

$$\int_{x-ct}^{x+ct} \delta(y-\xi)dy = \lim_{\varepsilon \downarrow 0} \int_{-\infty}^{x+ct} I_\varepsilon(y-\xi)dy - \lim_{\varepsilon \downarrow 0} \int_{-\infty}^{x-ct} I_\varepsilon(y-\xi)dy.$$

Then, using (5.51), (5.52) and (5.53), we conclude:

$$K(x,\xi,t) = \frac{1}{2c}\left\{\mathcal{H}(x-\xi+ct) - \mathcal{H}(x-\xi-ct)\right\}. \tag{5.54}$$

Figure 5.8 shows the graph of $K(x, \xi, t)$, with $c = 1$. Note how the initial discontinuity at $x = \xi$ propagates along the characteristics $x = \xi \pm t$.

We have found the fundamental solution (5.54) through d'Alembert formula. Conversely, using the fundamental solution we may derive d'Alembert formula. Namely, consider the solution w_h of the Cauchy problem (5.44), with data (see Remark 5.5, p. 272)

$$w(x,0) = 0, \quad w_t(x,0) = h(x), \quad x \in \mathbb{R}.$$

We may write

$$h(x) = \int_{-\infty}^{+\infty} \delta(x-\xi)h(\xi)d\xi$$

looking at $h(x)$ as a superposition of impulses $\delta(x-\xi)h(\xi)$, concentrated at ξ. Then, we may construct w_h by superposing the solutions of the same problem with data $\delta(x-\xi)h(\xi)$ instead of h. But these solutions are given by

$$K(x,\xi,t)h(\xi)$$

and therefore we obtain

$$w_h(x,t) = \int_{-\infty}^{+\infty} K(x,\xi,t)h(\xi)d\xi.$$

More explicitly, from (5.54):

$$\begin{aligned} w_h(x,t) &= \frac{1}{2c}\int_{-\infty}^{+\infty} \{H(x-\xi+ct) - H(x-\xi-ct)\}\, h(\xi)d\xi \\ &= \frac{1}{2c}\int_{-\infty}^{x+ct} h(\xi)d\xi - \frac{1}{2c}\int_{-\infty}^{x-ct} h(\xi)d\xi \\ &= \frac{1}{2c}\int_{x-ct}^{x+ct} h(y)dy. \end{aligned}$$

At this point, (5.45) yields d'Alembert formula.

We shall use this method to construct the solution of the global Cauchy problem in dimension 3.

5.4.4 Nonhomogeneous equation. Duhamel's method

To solve the nonhomogeneous problem

$$\begin{cases} u_{tt} - c^2 u_{xx} = f(x,t) & x \in \mathbb{R}, t > 0 \\ u(x,0) = 0, u_t(x,0) = 0 & x \in \mathbb{R}, \end{cases} \tag{5.55}$$

we use the Duhamel's method (see Sect. 2.8.3, p. 80). For $s \geq 0$ fixed, let $w = w(x,t;s)$ be the solution of problem

$$\begin{cases} w_{tt} - c^2 w_{xx} = 0 & x \in \mathbb{R}, t \geq s \\ w(x,s;s) = 0, w_t(x,s;s) = f(x,s) & x \in \mathbb{R}. \end{cases} \tag{5.56}$$

Since the wave equation is invariant under (time) translations, from (5.41) we get

$$w(x,t;s) = \frac{1}{2c}\int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dy.$$

Then, the solution of (5.55) is given by

$$u(x,t) = \int_0^t w(x,t;s)ds = \frac{1}{2c}\int_0^t ds \int_{x-c(t-s)}^{x+c(t-s)} f(y,s)dy = \frac{1}{2c}\int_{S_{x,t}} f(y,s)dyds. \tag{5.57}$$

where $S_{x,t}$ is the *triangular sector* in Fig. 5.5. In fact, $u(x,0) = 0$ and

$$u_t(x,t) = w(x,t;t) + \int_0^t w_t(x,t;s)ds = \int_0^t w_t(x,t;s)ds$$

since $w(x,t;t) = 0$. Thus $u_t(x,0) = 0$. Moreover,

$$u_{tt}(x,t) = w_t(x,t;t) + \int_0^t w_{tt}(x,t;s)ds = f(x,t) + \int_0^t w_{tt}(x,t;s)ds$$

and

$$u_{xx}(x,t) = \int_0^t w_{xx}(x,t;s)ds.$$

Therefore, since $w_{tt} - c^2 w_{xx} = 0$,

$$u_{tt}(x,t) - c^2 u_{xx}(x,t) = f(x,t) + \int_0^t \left\{ w_{tt}(x,t;s)ds - c^2 w_{xx}(x,t;s) \right\} ds = f(x,t).$$

Everything works and gives the unique solution in $C^2(\mathbb{R} \times [0,+\infty))$, under rather natural hypotheses on f: we require f and f_x to be continuous in $\mathbb{R} \times [0,+\infty)$.

Note from (5.57) that the value of u at the point (x,t) depends on the values of the forcing term f *in the whole triangular sector* $S_{x,t}$.

As in the case of the heat equation, Duhamel's method can be used to solve any initial boundary value problem (see Problem 5.3).

5.4.5 Dissipation and dispersion

Dissipation and dispersion effects are quite important in wave propagation phenomena. Let us go back to our model for the vibrating string, with *fixed end points*, assuming that its weight is negligible and that there are *no external loads*.

• *External damping*. External factors of dissipation like friction due to the medium may be included into the model through some empirical constitutive law. We may assume, for instance, a *linear law* of friction expressing a force per unit mass proportional to the speed of vibration. Then a vertical force of intensity

$$-\int_x^{x+\Delta x} k\rho u_t ds = -\int_x^{x+\Delta x} k\rho_0 u_t dx$$

acts on the segment of string between x and $x + \Delta x$, where $k > 0$ is a damping constant. The final equation takes the form

$$\rho_0 u_{tt} - \tau_0 u_{xx} + k\rho_0 u_t = 0. \tag{5.58}$$

The same calculations in Sect. 5.2.2 yield

$$\dot{E}(t) = -\int_0^L k\rho_0 u_t^2 \, dx = -2kE_{\text{cin}}(t) \leq 0 \tag{5.59}$$

which shows a rate of energy dissipation proportional to the kinetic energy.

For equation (5.58), the usual initial-boundary value problems are still well posed under reasonable assumptions on the data. In particular, the uniqueness of the solution follows from (5.59), since $E(0) = 0$ implies $E(t) = 0$ for all $t > 0$.

• *Internal damping.* The derivation of the wave equation in Sect. 5.2.1 leads to

$$\rho_0 u_{tt} = (\tau_{\text{vert}})_x$$

where τ_{vert} is the (scalar) vertical component of the tension. The hypothesis of vibrations of small amplitude corresponds to taking

$$\tau_{\text{vert}} \simeq \tau_0 u_x, \tag{5.60}$$

where τ_0 is the intensity of the horizontal component of the tension. In other words, we assume that the vertical forces due to the tension at two end points of a string element are proportional to the relative displacement of these points. On the other hand, the string vibrations convert kinetic energy into heat, because of the friction among the string particles. The amount of heat increases with the speed of vibration while, at the same time, the intensity of the vertical tension decreases. Thus, the vertical tension depends not only on the relative displacements u_x, but also on how fast these displacements change with time[7]. Hence, we modify (5.60) by inserting a term proportional to u_{xt}:

$$\tau_{\text{vert}} = \tau_0 u_x + \gamma u_{xt} \tag{5.61}$$

where γ is a *positive* constant. The positivity of γ follows from the fact that energy dissipation lowers the intensity of the vertical tension, so that the slope u_x decreases if $u_x > 0$ and therefore $u_{xt} = (u_x)_t < 0$, while it increases if $u_x < 0$ and $u_{xt} > 0$. Using the law (5.61), we derive the third order equation

$$\rho_0 u_{tt} - \tau_0 u_{xx} - \gamma u_{xxt} = 0. \tag{5.62}$$

In spite of the presence of the term u_{xxt}, the usual initial-boundary value problems are again well posed under reasonable assumptions on the data. In particular, uniqueness of the solution follows once again from dissipation of energy, since, in this case,

$$\dot{E}(t) = -\gamma \int_0^L u_{xt}^2 \, dx \leq 0.$$

• *Dispersion.* When the string is under the action of a vertical elastic restoring force proportional to u, the equation of motion becomes

$$u_{tt} - c^2 u_{xx} + \lambda u = 0 \qquad (\lambda > 0) \tag{5.63}$$

known as the *linearized Klein-Gordon equation.* To emphasize the effect of the zero order term λu, let us seek for *harmonic waves solutions* of the form

$$u(x,t) = A e^{i(kx - \omega t)}.$$

[7] In the movie *The Legend of 1900* there is a spectacular demo of this phenomenon.

Inserting u into (5.63) we find the *dispersion relation*

$$\omega^2 - c^2k^2 = \lambda \qquad \Longrightarrow \qquad \omega(k) = \pm\sqrt{c^2k^2 + \lambda}.$$

Thus, these waves are dispersive with phase and group velocities given respectively by

$$c_p(k) = \frac{\sqrt{c^2k^2 + \lambda}}{|k|}, \quad c_g = \frac{d\omega}{dk} = \frac{c^2\,|k|}{\sqrt{c^2k^2 + \lambda}}.$$

Observe that $c_g < c_p$.

A wave packet solution can be obtained by an integration over all possible wave numbers k :

$$u(x,t) = \int_{-\infty}^{+\infty} A(k)e^{i[kx-\omega(k)t]}dk \tag{5.64}$$

where $A(k)$ is the Fourier transform of the initial condition:

$$A(k) = \int_{-\infty}^{+\infty} u(x,0)e^{-ikx}dx.$$

This entails that, even if the initial condition is *localized* inside a small interval, *all* the wavelengths contribute to the value of u. Although we observe a decaying in amplitude of order $t^{-1/2}$ (see formula (5.7), p. 258), these dispersive waves do not dissipate energy. For example, since the ends of the string are fixed, the total mechanical energy is given by

$$E(t) = \frac{\rho_0}{2}\int_0^L (u_t^2 + c^2u_x^2 + \lambda u^2)dx$$

and one may check that $\dot{E}(t) = 0, t > 0$.

5.5 Second Order Linear Equations

5.5.1 Classification

To derive formula (5.42), p. 271, we may use the characteristics in the following way. We change variables by setting

$$\xi = x + ct, \qquad \eta = x - ct \tag{5.65}$$

or

$$x = \frac{\xi + \eta}{2}, \qquad t = \frac{\xi - \eta}{2c} \tag{5.66}$$

and define

$$U(\xi,\eta) = u\left(\frac{\xi+\eta}{2}, \frac{\xi-\eta}{2c}\right).$$

Then

$$U_\xi = \frac{1}{2}u_x + \frac{1}{2c}u_t$$

and, since $u_{tt} = c^2 u_{xx}$, we obtain

$$U_{\xi\eta} = \frac{1}{4}u_{xx} - \frac{1}{4c}u_{xt} + \frac{1}{4c}u_{xt} - \frac{1}{4c^2}u_{tt} = 0.$$

The equation

$$U_{\xi\eta} = 0 \tag{5.67}$$

is called the *canonical form* of the wave equation; its solution is immediate:

$$U(\xi,\eta) = F(\xi) + G(\eta)$$

and going back to the original variables, (5.42) follows.

Consider now a general equation of the form:

$$au_{tt} + 2bu_{xt} + cu_{xx} + du_t + eu_x + hu = f \tag{5.68}$$

with (x,t) varying, in general, in a domain Ω. We assume that the coefficients a, b, c, d, e, h, f are C^2 functions in Ω. The sum of second order terms

$$a(x,t)u_{tt} + 2b(x,t)u_{xt} + c(x,t)u_{xx} \tag{5.69}$$

is called **principal part** of eq. (5.68) and determines the type of equation according to the following classification. Consider the algebraic equation

$$H(p,q) = ap^2 + 2bpq + cq^2 = 1 \qquad (a > 0) \tag{5.70}$$

in the plane pq. If $b^2 - ac > 0$, (5.70) defines a hyperbola, if $b^2 - ac = 0$ a parabola and if $b^2 - ac < 0$ an ellipse. Accordingly, equation (5.68) is called:

a) **Hyperbolic** when $b^2 - ac > 0$.
b) **Parabolic** when $b^2 - ac = 0$.
c) **Elliptic** when $b^2 - ac < 0$.

Note that the quadratic form $H(p,q)$ is, in the three cases, *indefinite, nonnegative, positive*, respectively. The above classification extends to equations in any number of variables, as we shall see later on.

It may happen that a single equation is of different type in different subdomains. In this case the equation is said of *mixed type*. For instance, the *Tricomi* equation $xu_{tt} - u_{xx} = 0$ is hyperbolic in the half plane $x > 0$, and elliptic in the half plane $x < 0$. We will refer to domains in which the equation (5.68) is of a *single* type.

Basically, all the second order equations in two variables we have met so far are particular cases of (5.68). Specifically,

- The *wave* equation
$$u_{tt} - c^2 u_{xx} = 0$$
is *hyperbolic*: $a(x,t) = 1$, $c(x,t) = -c^2$, and the other coefficients are zero.
- The *diffusion* equation
$$u_t - D u_{xx} = 0$$
is *parabolic*: $c(x,t) = -D$, $d(x,t) = 1$, and the other coefficients are zero.
- *Laplace* equation (using y instead of t)
$$u_{xx} + u_{yy} = 0$$
is *elliptic*: $a(x,y) = 1$, $c(x,y) = 1$, and the other coefficients are zero.

May we reduce to a canonical form, similar to (5.67), the diffusion and the Laplace equations? Let us briefly examine why the change of variables (5.65) works for the wave equation.

Decompose the wave operator as follows

$$\partial_{tt} - c^2 \partial_{xx} = (\partial_t + c\partial_x)(\partial_t - c\partial_x). \tag{5.71}$$

If we introduce the vectors $\mathbf{v} = (c, 1)$ and $\mathbf{w} = (-c, 1)$, then (5.71) can be written in the form

$$\partial_{tt} - c^2 \partial_{xx} = \partial_{\mathbf{v}} \partial_{\mathbf{w}}.$$

On the other hand, the characteristics

$$x + ct = \text{constant}, \quad x - ct = \text{constant}$$

of the two first order equations

$$\phi_t - c\phi_x = 0 \quad \text{and} \quad \psi_t + c\psi_t = 0,$$

corresponding to the two factors in (5.71), are straight lines in the direction of $\mathbf{w}$ and $\mathbf{v}$, respectively. The change of variables

$$\xi = \phi(x,t) = x + ct \qquad \eta = \psi(x,t) = x - ct$$

maps the straight lines $x + ct = 0$ and $x - ct = 0$ into $\xi = 0$ and $\eta = 0$, respectively. Moreover, recalling (5.66), we have

$$\partial_\xi = \frac{1}{2c}(\partial_t + c\partial_x) = \frac{1}{2c}\partial_{\mathbf{v}}, \qquad \partial_\eta = \frac{1}{2c}(\partial_t - c\partial_x) = \frac{1}{2c}\partial_{\mathbf{w}}.$$

Thus, the wave operator is converted into a multiple of its canonical form:

$$\partial_{tt} - c^2 \partial_{xx} = \partial_{\mathbf{v}} \partial_{\mathbf{w}} = 4c^2 \partial_{\xi\eta}.$$

Once the characteristics are known, the change of variables (5.65) reduces the wave equation to the form (5.67).

Proceeding in the same way, for the diffusion operator we would have

$$\partial_{xx} = \partial_x \partial_x .$$

Therefore we find only one family of characteristics, given by

$$t = \text{constant}.$$

Thus, no change of variables is necessary and the diffusion equation is already in its canonical form.

For the Laplace operator we find

$$\partial_{xx} + \partial_{yy} = (\partial_y + i\partial_x)(\partial_y - i\partial_x)$$

and there are two families of *complex* characteristics given by

$$\phi(x, y) = x + iy = \text{constant}, \qquad \psi(x, y) = x - iy = \text{constant}.$$

The change of variables

$$z = x + iy, \qquad \overline{z} = x - iy$$

leads to the equation

$$\partial_{z\overline{z}} U = 0$$

whose general solution is

$$U(z, \overline{z}) = F(z) + G(\overline{z}).$$

This formula may be considered as a characterization of the harmonic functions in the complex plane.

It should be clear, however, that the characteristics for the diffusion and the Laplace equations do not play the same relevant role as they do for the wave equation.

5.5.2 Characteristics and canonical form

Let us go back to the equation in general form (5.68). Can we reduce its principal part to a canonical form? There are at least two substantial reasons to examine the question.

The first one is tied to the type of well posed problems associated with (5.68): which kind of data have to be assigned and where, in order to find a unique and stable solution? It turns out that hyperbolic, parabolic and elliptic equations share their well posed problems with their main prototypes: the wave, diffusion and Laplace equations, respectively. Also the choice of numerical methods depends very much on the type of problem to be solved.

The second reason comes from the different features the three types of equation exhibit. Hyperbolic equations model oscillatory phenomena *with finite speed of propagation of the disturbances*, while for parabolic equation, "information" travels with infinite speed. Finally, elliptic equations model stationary situations, with no evolution in time.

To obtain the canonical form of the principal part we try to apply the ideas at the end of the previous subsection, performing a change of independent variables of the type

$$\xi = \phi(x,t), \qquad \eta = \psi(x,t) \tag{5.72}$$

with ϕ and ψ twice continuously differentiable. We have to make sure that the transformation (5.72) is *invertible*, at least locally. According to the Inverse Function Theorem 1.5, it is enough that the Jacobian matrix of the transformation is not singular, that is

$$J = \det \begin{pmatrix} \phi_x & \phi_t \\ \psi_x & \psi_t \end{pmatrix} = \phi_x \psi_t - \phi_t \psi_x \neq 0. \tag{5.73}$$

In this way, at least locally, the inverse transformation

$$x = \Phi(\xi,\eta), \qquad t = \Psi(\xi,\eta)$$

is well defined. Let

$$U(\xi,\eta) = u(\Phi(\xi,\eta),\Psi(\xi,\eta)) \qquad \text{or} \qquad u(x,t) = U(\phi(x,t),\psi(x,t)).$$

Then

$$u_x = \phi_x U_\xi + \psi_x U_\eta, \qquad u_t = \phi_t U_\xi + \psi_t U_\eta$$

and moreover:

$$u_{xx} = \phi_x^2 U_{\xi\xi} + 2\phi_x\psi_x U_{\xi\eta} + \psi_x^2 U_{\eta\eta} + \phi_{xx} U_\xi + \psi_{xx} U_\eta,$$

$$u_{tt} = \phi_t^2 U_{\xi\xi} + 2\phi_t\psi_t U_{\xi\eta} + \psi_t^2 U_{\eta\eta} + \phi_{tt} U_\xi + \psi_{tt} U_\eta,$$

$$u_{xt} = \phi_t\phi_x U_{\xi\xi} + (\phi_x\psi_t + \phi_t\psi_x) U_{\xi\eta} + \psi_t\psi_x U_{\eta\eta} + \phi_{xt} U_\xi + \psi_{xt} U_\eta.$$

Therefore the equation (5.68) assumes the form

$$AU_{\xi\xi} + 2BU_{\xi\eta} + CU_{\eta\eta} + DU_\xi + EU_\eta + hU = F \tag{5.74}$$

where (all the functions are evaluated at $x = \Phi(\xi,\eta)$ and $t = \Psi(\xi,\eta)$)

$$A = a\phi_t^2 + 2b\phi_t\phi_x + c\phi_x^2, \qquad C = a\psi_t^2 + 2b\psi_t\psi_x + c\psi_x^2,$$
$$B = a\phi_t\psi_t + b(\phi_x\psi_t + \phi_t\psi_x) + c\phi_x\psi_x,$$
$$D = a\phi_{tt} + 2b\phi_{xt} + c\phi_{xx} + d\phi_t + e\phi_x, \qquad E = a\psi_{tt} + 2b\psi_{xt} + c\psi_{xx} + d\psi_t + e\psi_x$$
$$F = f.$$

Observe that the transformed equation (5.74) has the same form of the original equation and **does not change type** since it can be easily checked that

$$B^2 - AC = J^2(b^2 - ac). \tag{5.75}$$

We now choose ϕ and ψ according to the type of equation. First of all, note that, if $a = c = 0$, the principal part is already in the form (5.67), so that we may assume $a > 0$ (or equivalently $c > 0$). Then we decompose the differential operator in (5.69) into the product of two first order factors, as follows[8]:

$$a\partial_{tt} + 2b\partial_{xt} + c\partial_{xx} = a(\partial_t - \Lambda^+\partial_x)(\partial_t - \Lambda^-\partial_x) \tag{5.76}$$

where

$$\Lambda^{\pm} = \frac{-b \pm \sqrt{b^2 - ac}}{a}.$$

Case 1: $b^2 - ac > 0$, the equation is **hyperbolic.** The two factors in (5.76) represent the derivatives along the direction fields

$$\mathbf{v}(x,t) = (-\Lambda^+(x,t), 1) \quad \text{and} \quad \mathbf{w}(x,t) = (-\Lambda^-(x,t), 1)$$

respectively, so that we may write

$$a\partial_{tt} + 2b\partial_{xt} + c\partial_{xx} = a\partial_{\mathbf{v}}\partial_{\mathbf{w}}.$$

The vector fields $\mathbf{v}$ and $\mathbf{w}$ are tangent at any point to the characteristics

$$\phi(x,t) = k_1 \quad \text{and} \quad \psi(x,t) = k_2 \qquad (k_1, k_2 \in \mathbb{R}) \tag{5.77}$$

of the following *first-order* equations

$$\phi_t - \Lambda^+\phi_x = 0 \qquad \text{and} \qquad \psi_t - \Lambda^-\psi_x = 0 \tag{5.78}$$

that may be written in the compact form

$$av_t^2 + 2bv_xv_t + cv_x^2 = 0 \tag{5.79}$$

where v stands for either ϕ or ψ. The curves (5.77) are called *characteristics* also for the equation (5.68) and they are the solution curves of the ordinary differential equations

$$\frac{dx}{dt} = -\Lambda^+, \ \frac{dx}{dt} = -\Lambda^-, \tag{5.80}$$

respectively. Note that the two equations (5.80) can be put into the compact form

$$a\left(\frac{dx}{dt}\right)^2 - 2b\frac{dx}{dt} + c = 0. \tag{5.81}$$

[8] Remember that $ax^2 + 2bxy + cy^2 = a(x - x_1y)(x - x_2y)$, where $x_{1,2} = [-b \pm \sqrt{b^2 - ac}]/a$.

By analogy with the case of the wave equation, we expect that the change of variables

$$\xi = \phi(x,t), \qquad \eta = \psi(x,t) \tag{5.82}$$

should straighten the characteristics, at least locally, converting $\partial_{\mathbf{v}}\partial_{\mathbf{w}}$ into a multiple of $\partial_{\xi\eta}$.

Observe that the vectors $\nabla\phi$ and $\nabla\psi$ are orthogonal to $\mathbf{v}$ and $\mathbf{w}$, respectively, and that $\mathbf{v}$, $\mathbf{w}$ are nowhere collinear (since $b^2 - ac > 0$). Therefore the Jacobian J of the transformation does not vanish.

Now, we have

$$A = C = 0$$

since ϕ and ψ both satisfy (5.79). Moreover, from (5.75), we have

$$B = J\sqrt{b^2 - ac} \neq 0.$$

Therefore the equation (5.74) assumes the form

$$U_{\xi\eta} = \frac{1}{2B}(F - DU_\xi - EU_\eta - hU),$$

which is the *canonical form of* (5.68).

Example 5.8 Consider the equation

$$xu_{tt} - (1+x^2)u_{xt} = 0, \tag{5.83}$$

Since $b^2 - ac = (1+x^2)/4 > 0$, (5.83) is hyperbolic. Equation (5.81) is

$$x\left(\frac{dx}{dt}\right)^2 + (1+x^2)\frac{dx}{dt} = 0$$

which yields,

$$\frac{x}{1+x^2}\frac{dx}{dt} = -1 \quad \text{and} \quad \frac{dx}{dt} = 0.$$

Thus, the characteristics curves are:

$$\phi(x,t) = e^{2t}(1+x^2) = k_1 \quad \text{and} \quad \psi(x,t) = x = k_2.$$

We set

$$\xi = e^{2t}(1+x^2) \quad \text{and} \quad \eta = x.$$

After routine calculations, we find $D = E = 0$, so that the canonical form is

$$U_{\xi\eta} = 0.$$

The solution of (5.83) is therefore

$$u(x,t) = F(e^{2t}(1+x^2)) + G(x)$$

with F and G arbitrary C^2 functions.

Case 2: $b^2 - ac \equiv 0$, the equation is **parabolic**. There exists *only one* family of characteristics, given by $\phi(x,t) = k$, where ϕ is a solution of the first order equation

$$a\phi_t + b\phi_x = 0,$$

since $\Lambda^+ = \Lambda^- = -b/a$. If ϕ is known, choose any smooth function ψ such that $\nabla\phi$ and $\nabla\psi$ are *linearly independent* and

$$a\psi_t^2 + 2b\psi_t\psi_x + c\psi_x^2 = C \neq 0.$$

Set

$$\xi = \phi(x,t), \qquad \eta = \psi(x,t)$$

and

$$U(\xi,\eta) = u(\Phi(\xi,\eta), \Psi(\xi,\eta)).$$

For the derivatives of U we may use the computations done in **case 1**. However, observe that, since $b^2 - ac = 0$ and $a\phi_t + b\phi_x = 0$, we have

$$\begin{aligned} B &= a\phi_t\psi_t + b(\phi_t\psi_x + \phi_x\psi_t) + c\phi_x\psi_x = \psi_t(a\phi_t + b\phi_x) + \psi_x(b\phi_t + c\phi_x) \\ &= b\psi_x\left(\phi_t + \frac{c}{b}\phi_x\right) = b\psi_x\left(\phi_t + \frac{b}{a}\phi_x\right) = \frac{b}{a}\psi_x(a\phi_t + b\phi_x) = 0. \end{aligned}$$

Thus, the equation for U becomes

$$U_{\eta\eta} = \frac{1}{C}(F - DU_\xi - EU_\eta - hU)$$

which is the *canonical form* of the equation (5.68).

Example 5.9 The equation

$$u_{tt} - 6u_{xt} + 9u_{xx} = u$$

is parabolic everywhere. The family of characteristics is

$$\phi(x,t) = 3t + x = k.$$

Choose $\psi(x,t) = x$ and set

$$\xi = 3t + x, \qquad \eta = x.$$

Since $\nabla\phi = (3,1)$ and $\nabla\psi = (1,0)$, the gradients are independent and we set

$$U(\xi,\eta) = u\left(\frac{\xi - \eta}{3}, \eta\right).$$

We have $C = 9$ and $D = E = 0$, so that the equation for U is

$$9U_{\eta\eta} - U = 0$$

whose solution is

$$U(\xi, \eta) = F(\xi)e^{-\eta/3} + G(\xi)e^{\eta/3}$$

with F and G arbitrary C^2 functions. Finally, we find

$$u(x,t) = F(3t+x)e^{-x/3} + G(3t+x)e^{x/3}.$$

Case 3: $b^2 - ac < 0$, the equation is **elliptic**. In this case there are no real characteristics. If the coefficients a, b, c are analytic functions, i.e. they can be locally expanded in Taylor series, we can consider equation (5.81) for complex t and x and proceed as in **Case 1**, to find two families of complex conjugate characteristics

$$\Phi(x,y) = \phi(x,y) + i\psi(x,y) = z_1, \quad \bar{\Phi}(x,y) = \phi(x,y) - i\psi(x,y) = z_2 \qquad (z_1, z_2 \in \mathbb{C})$$

with $\nabla\phi$ and $\nabla\psi$ linearly independent. To obtain a real canonical form, let

$$\xi = \phi(x,y), \eta = \psi(x,y).$$

Substituting $v = \Phi$ into the equation (5.79) and separating the real and the imaginary parts we obtain

$$a\phi_t^2 + 2b\phi_t\phi_x + c\phi_x^2 = a\psi_t^2 + 2b\psi_t\psi_x + c\psi_x^2$$

and

$$a\phi_t\psi_t + b(\phi_x\psi_t + \phi_t\psi_x) + c\phi_x\psi_x = 0.$$

Therefore

$$A = C \text{ and } B = 0.$$

Moreover, from (5.75), we have $B^2 - AC = -AC < 0$ so that $A = C$ does not vanish. The equation for $U = u(\phi(x,y), \psi(x,y))$ is

$$U_{\xi\xi} + U_{\eta\eta} = \frac{1}{A}(F - DU_\xi - EU_\eta - hU)$$

which is the canonical form of equation (5.68).

Example 5.10 The equation

$$2u_{tt} + 2u_{xt} + u_{xx} - u = 0$$

is elliptic everywhere with constant coefficients. The complex characteristic curves solve the differential equations

$$\frac{dx}{dt} = \frac{1 \pm i}{2}.$$

We find the curves

$$\phi(x,t) \pm i\psi(x,t) = 2x - t \pm it = \text{constant}.$$

Letting

$$\xi = 2x - t, \ \eta = t,$$

we find the canonical form

$$U_{\xi\xi} + U_{\eta\eta} - \frac{1}{2}U = 0.$$

5.6 The Multi-dimensional Wave Equation ($n > 1$)

5.6.1 Special solutions

The wave equation

$$u_{tt} - c^2 \Delta u = f \tag{5.84}$$

constitutes a basic model for describing a remarkable number of oscillatory phenomena in dimension $n > 1$. Here $u = u(\mathbf{x}, t)$, $\mathbf{x} \in \mathbb{R}^n$ and, as in the one-dimensional case, c is the *speed of propagation*. If $f \equiv 0$, the equation is *homogeneous* and the *superposition principle* holds. Let us examine some relevant solutions of (5.84).

- *Plane waves*. If $\mathbf{k} \in \mathbb{R}^n$ and $\omega^2 = c^2 |\mathbf{k}|^2$, the function

$$u(\mathbf{x}, t) = w(\mathbf{x} \cdot \mathbf{k} - \omega t)$$

is a solution of the homogeneous (5.84). Indeed,

$$u_{tt}(\mathbf{x}, t) - c^2 \Delta u(\mathbf{x}, t) = \omega^2 w''(\mathbf{x} \cdot \mathbf{k} - \omega t) - c^2 |\mathbf{k}|^2 w''(\mathbf{x} \cdot \mathbf{k} - \omega t) = 0.$$

We have already seen in Sect. 5.1.1 that the planes

$$\mathbf{x} \cdot \mathbf{k} - \omega t = \text{constant}$$

constitute the wave fronts, moving at speed $c_p = \omega/|\mathbf{k}|$ in the $\mathbf{k}$ direction. The scalar $\lambda = 2\pi/|\mathbf{k}|$ is the wavelength. If $w(z) = Ae^{iz}$, the wave is said *monochromatic or harmonic*.

- *Cylindrical waves* ($n = 3$) are of the form

$$u(\mathbf{x}, t) = w(r, t)$$

where $\mathbf{x} = (x_1, x_2, x_3)$, $r = \sqrt{x_1^2 + x_2^2}$. In particular, solutions of the form $u(\mathbf{x}, t) = e^{i\omega t} w(r)$ represent stationary cylindrical waves, that can be found by solving the homogeneous equation (5.84), using the separation of variables method, in axially symmetric domains.

If the axis of symmetry is the x_3 axis, it is appropriate to use the cylindrical coordinates $x_1 = r \cos\theta, x_2 = r \sin\theta, x_3$. Then, the wave equation becomes (App. B)

$$u_{tt} - c^2 \left(u_{rr} + \frac{1}{r} u_r + \frac{1}{r^2} u_{\theta\theta} + u_{x_3 x_3} \right) = 0.$$

Looking for standing waves of the form $u(r,t) = e^{i\lambda ct} w(r)$, $\lambda \geq 0$, we find, after dividing by $c^2 e^{i\lambda ct}$,

$$w''(r) + \frac{1}{r} w' + \lambda^2 w = 0.$$

This is a Bessel equation of zero order. We know from Sect. 2.7 that the only solutions bounded at $r = 0$ are

$$w(r) = aJ_0(\lambda r), \qquad a \in \mathbb{R},$$

where, we recall,

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2} \right)^{2k}$$

is the Bessel function of first kind of zero order. In this way we get waves of the form

$$u(r,t) = aJ_0(\lambda r) e^{i\lambda ct}.$$

- *Spherical waves* ($n = 3$) are of the form

$$u(\mathbf{x}, t) = w(r,t)$$

where $\mathbf{x} = (x_1, x_2, x_3)$, $r = |\mathbf{x}| = \sqrt{x_1^2 + x_2^2 + x_3^2}$. In particular

$$u(\mathbf{x}, t) = e^{i\omega t} w(r)$$

represents a standing spherical wave and can be determined by solving the homogeneous equation (5.84) via the method of separation of variables in spherically symmetric domains. In this case, spherical coordinates

$$x_1 = r\cos\theta \sin\psi, \; x_2 = r \sin\theta \sin\psi, \; x_3 = r\cos\psi,$$

are appropriate and the wave equation becomes (App. B)

$$\frac{1}{c^2} u_{tt} - u_{rr} - \frac{2}{r} u_r - \frac{1}{r^2} \left\{ \frac{1}{(\sin\psi)^2} u_{\theta\theta} + u_{\psi\psi} + \frac{\cos\psi}{\sin\psi} u_\psi \right\} = 0. \tag{5.85}$$

Let us look for solution of the form

$$u(r,t) = e^{i\lambda ct} w(r), \quad \lambda \geq 0.$$

We find, after simplifying out $e^{i\lambda ct}$,

$$w'' + \frac{2}{r} w' + \lambda^2 w = 0$$

which can be written[9]

$$(rw)'' + \lambda^2 rw = 0.$$

Thus, $v = rw$ is solution of

$$v'' + \lambda^2 v = 0$$

which gives $v(r) = a\cos(\lambda r) + b\sin(\lambda r)$ and hence the attenuated spherical waves

$$w(r,t) = ae^{i\lambda ct}\frac{\cos(\lambda r)}{r}, \qquad w(r,t) = be^{i\lambda ct}\frac{\sin(\lambda r)}{r}. \tag{5.86}$$

Let us now determine *the general form of a spherical wave* in $\mathbb{R}^3$. Inserting $u(\mathbf{x},t) = w(r,t)$ into (5.85) we obtain

$$w_{tt} - c^2\left\{w_{rr} + \frac{2}{r}w_r\right\} = 0$$

which can be written in the form

$$(rw)_{tt} - c^2(rw)_{rr} = 0. \tag{5.87}$$

Then, formula (5.42) gives

$$w(r,t) = \frac{F(r+ct)}{r} + \frac{G(r-ct)}{r} \equiv w_{\text{in}}(r,t) + w_{\text{out}}(r,t), \tag{5.88}$$

which represents the superposition of two attenuated progressive spherical waves. The wave fronts of w_{out} are the spheres $r - ct = k$, expanding as time goes on. Hence, w_{out} represents an outgoing wave. On the contrary, the wave w_{in} is incoming, since its wave fronts are the contracting spheres $r + ct = k$.

5.6.2 Initial boundary value problems. Uniqueness

Let

$$Q_T = \Omega \times (0,T)$$

be a *space-time cylinder*, where Ω is a bounded C^1 or Lipschitz domain in $\mathbb{R}^n$. A typical initial boundary value problems for the wave equation in Q_T consists in assigning an initial data (at time $t = 0$) and some appropriate conditions on the boundary $\partial\Omega$ of Ω for $t > 0$.

More specifically, we may pose the following problems: Determine $u = u(\mathbf{x},t)$ such that:

$$\begin{cases} u_{tt} - c^2\Delta u = f(\mathbf{x},t) & \text{in } Q_T \\ u(\mathbf{x},0) = g(\mathbf{x}),\ u_t(\mathbf{x},0) = h(\mathbf{x}) & \text{in } \Omega \\ + \text{ boundary conditions} & \text{on } \partial\Omega \times [0,T) \end{cases} \tag{5.89}$$

[9] Thanks to the providential presence of the factor 2 in the coefficient of w'.

where the boundary conditions can be:

1. $u = h$ (Dirichlet).
2. $\partial_\nu u = h$ (Neumann).
3. $\partial_\nu u + \alpha u = h$, $\alpha = \alpha(\boldsymbol{\sigma}) > 0$ (Robin).
4. $u = h_1$ on $\Gamma_D \times [0, T)$ and $\partial_\nu u = h_2$ on $\Gamma_N \times [0, T)$ (mixed problem) with Γ_N a relatively open subset of $\partial\Omega$ and $\Gamma_D = \partial\Omega \setminus \Gamma_N$.

The *global Cauchy problem*

$$\begin{cases} u_{tt} - c^2 \Delta u = f & \mathbf{x} \in \mathbb{R}^n, t > 0 \\ u(\mathbf{x}, 0) = g(\mathbf{x}),\ u_t(\mathbf{x}, 0) = h(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^n, \end{cases} \tag{5.90}$$

is quite important also in dimension $n > 1$. We will examine it with some details later on. Particularly relevant are the different features that the solutions exhibit for $n = 2$ and $n = 3$.

Under rather natural hypotheses on the data, problem (5.89) has at most one solution. To see it, we may use once again the conservation of energy, which is proportional to:

$$E(t) = \frac{1}{2} \int_\Omega \left\{ u_t^2 + c^2 \left| \nabla u \right|^2 \right\} d\mathbf{x}.$$

The growth rate is:

$$\dot{E}(t) = \int_\Omega \left\{ u_t u_{tt} + c^2 \nabla u_t \cdot \nabla u \right\} d\mathbf{x}.$$

Integrating by parts, we have

$$\int_\Omega c^2 \nabla u_t \cdot \nabla u \, d\mathbf{x} = c^2 \int_{\partial\Omega} u_\nu u_t \, d\sigma - \int_\Omega c^2 u_t \Delta u \, d\mathbf{x}$$

whence, since $u_{tt} - c^2 \Delta u = f$,

$$\dot{E}(t) = \int_\Omega \left\{ u_{tt} - c^2 \Delta u \right\} u_t \, d\mathbf{x} + c^2 \int_{\partial\Omega} u_\nu u_t \, d\sigma = \int_\Omega f u_t \, d\mathbf{x} + c^2 \int_{\partial\Omega} u_\nu u_t \, d\sigma.$$

It is now easy to prove the following result.

Theorem 5.11 *Problem (5.89), coupled with one of the boundary conditions (a)–(d) above, has at most one solution in $C^2(Q_T) \cap C^1(\overline{Q}_T)$.*

Proof Let u_1 and u_2 be solutions of the same problem, sharing the same data. Their difference $w = u_1 - u_2$ is a solution of the homogeneous equation, with zero initial/boundary data. We show that $w(\mathbf{x}, t) \equiv 0$.

In the case of Dirichlet, Neumann and mixed conditions, since either $w_\nu = 0$ or $w_t = 0$ on $\partial\Omega \times [0, T)$, we have $\dot{E}(t) = 0$. Thus, since $E(0) = 0$, we infer:

$$E(t) = \frac{1}{2} \int_\Omega \left\{ w_t^2 + c^2 \left| \nabla w \right|^2 \right\} d\mathbf{x} = 0, \qquad \forall t > 0.$$

Therefore, for each $t > 0$, both w_t and $|\nabla w(\mathbf{x}, t)|$ vanish and hence $w(\mathbf{x}, t)$ is constant. Then $w(\mathbf{x}, t) \equiv 0$, since $w(\mathbf{x}, 0) = 0$. For the Robin problem, since α is time independent, we have

$$\dot{E}(t) = -c^2 \int_{\partial\Omega} \alpha w w_t \, d\sigma = -\frac{c^2}{2} \frac{d}{dt} \int_{\partial\Omega} \alpha w^2 \, d\sigma$$

that is

$$\frac{d}{dt} \left\{ E(t) + \frac{c^2}{2} \int_{\partial\Omega} \alpha w^2 \, d\sigma \right\} = 0.$$

Hence,

$$E(t) + \frac{c^2}{2} \int_{\partial\Omega} \alpha w^2 d\sigma = k, \text{ constant.}$$

Since $E(0) = 0$ and $w(\mathbf{x}, 0) = 0$ it follows that $k = 0$. Since $\alpha > 0$, we easily conclude that $w \equiv 0$. □

Uniqueness for the global Cauchy problem follows from another energy inequality. *First a remark*. For better clarity, let us consider the case $n = 2$. Suppose that a disturbance governed by the homogeneous wave equation ($f = 0$) is felt at $\mathbf{x}_0$ at time t_0. Since the disturbances travel with speed c, $u(\mathbf{x}_0, t_0)$ is only affected by the values of the initial data in the circle $B_{ct_0}(\mathbf{x}_0)$. More generally, at time $t_0 - t$, $u(\mathbf{x}_0, t_0)$ is determined by the values of u in the circle $B_{c(t_0-t)}(\mathbf{x}_0)$. As t varies from 0 to t_0, the union of the circles $B_{c(t_0-t)}(\mathbf{x}_0)$ in the $\mathbf{x}t$ space coincides with the so called *backward or retrograde cone* with vertex at $(\mathbf{x}_0, t_0)$ and opening $\theta = \arctan c$, given by (see Fig. 5.9):

$$C_{\mathbf{x}_0,t_0} = \{(\mathbf{x}, t) : |\mathbf{x} - \mathbf{x}_0| \leq c(t_0 - t), 0 \leq t \leq t_0\}.$$

Thus, given a point $\mathbf{x}_0$, it is natural to introduce an energy associated with its backward cone by the formula

$$e(t) = \frac{1}{2} \int_{B_{c(t_0-t)}(\mathbf{x}_0)} (u_t^2 + c^2 |\nabla u|^2) d\mathbf{x}, \qquad 0 \leq t < t_0.$$

It turns out that $e(t)$ is a decreasing function. Namely, going back to dimension n:

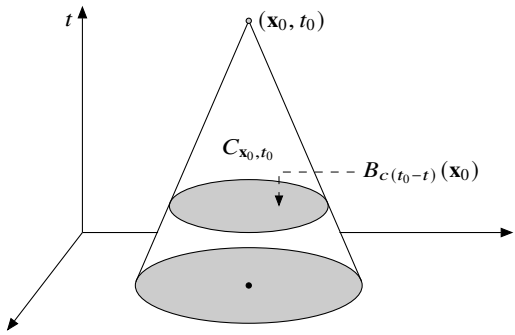

Fig. 5.9 Retrograde cone

Lemma 5.12 *Let u be a C^2-solution of the homogeneous wave equation in $\mathbb{R}^n \times [0, +\infty)$. Then*

$$\dot{e}(t) \leq 0.$$

Proof We may write

$$e(t) = \frac{1}{2} \int_0^{c(t_0-t)} dr \int_{\partial B_r(\mathbf{x}_0)} (u_t^2 + c^2 |\nabla u|^2) d\sigma$$

so that

$$\dot{e}(t) = -\frac{c}{2} \int_{\partial B_{c(t_0-t)}(\mathbf{x}_0)} (u_t^2 + c^2 |\nabla u|^2) d\sigma + \int_{B_{c(t_0-t)}(\mathbf{x}_0)} (u_t u_{tt} + c^2 \nabla u \cdot \nabla u_t) d\mathbf{x}.$$

An integration by parts yields

$$\int_{B_{c(t_0-t)}(\mathbf{x}_0)} \nabla u \cdot \nabla u_t \, d\mathbf{x} = \int_{\partial B_{c(t_0-t)}(\mathbf{x}_0)} u_t u_{\boldsymbol{\nu}} \, d\sigma - \int_{B_{c(t_0-t)}(\mathbf{x}_0)} u_t \Delta u \, d\mathbf{x}$$

whence

$$\begin{aligned} \dot{e}(t) &= \int_{B_{c(t_0-t)}(\mathbf{x}_0)} u_t (u_{tt} - c^2 \Delta u) d\mathbf{x} + \frac{c}{2} \int_{\partial B_{c(t_0-t)}(\mathbf{x}_0)} (2c u_t u_{\boldsymbol{\nu}} - u_t^2 - c^2 |\nabla u|^2) d\sigma \\ &= \frac{c}{2} \int_{\partial B_{c(t_0-t)}(\mathbf{x}_0)} (2c u_t u_{\boldsymbol{\nu}} - u_t^2 - c^2 |\nabla u|^2) d\sigma. \end{aligned}$$

Now, we have

$$|u_t u_{\boldsymbol{\nu}}| \leq |u_t| \, |\nabla u| \,,$$

so that

$$2c u_t u_{\boldsymbol{\nu}} - u_t^2 - c^2 |\nabla u|^2 \leq 2c |u_t| \, |\nabla u| - u_t^2 - c^2 |\nabla u|^2 = -(u_t - c |\nabla u|)^2 \leq 0$$

and therefore $\dot{e}(t) \leq 0$. □

Two almost immediate consequences are stated in the following theorem:

Theorem 5.13 *Let $u \in C^2(\mathbb{R}^n \times [0, +\infty))$ be a solution of the Cauchy problem (5.90). If $g \equiv h \equiv 0$ in $B_{ct_0}(\mathbf{x}_0)$ and $f \equiv 0$ in $C_{\mathbf{x}_0,t_0}$ then $u \equiv 0$ in $C_{\mathbf{x}_0,t_0}$. Consequently, problem (5.90) has at most one solution in $C^2(\mathbb{R}^n \times [0, +\infty))$.*

5.7 Two Classical Models

5.7.1 Small vibrations of an elastic membrane

In Sect. 5.2.1 we have derived a model for the small transverse vibrations of a string. Similarly, we may derive the equation governing the small transverse vibrations of a highly stretched membrane (think e.g. of a drum), at rest in the horizontal position. We briefly sketch the derivation leaving it to the reader to fill in the details. Assume the following hypotheses.

1. *The vibrations of the membrane are small and vertical*. This means that the changes from the plane horizontal shape are very small compared to the membrane dimension and horizontal displacements are negligible.
2. *The vertical displacement of a point of the membrane depends on time and on its position at rest only*. Thus, if u denotes the vertical displacement of a point located at rest at (x, y), we have

$$u = u(x, y, t).$$

3. *The membrane is perfectly flexible and elastic.* In other words, there is no resistance to bending. In particular, the stress in the membrane can be modelled by a tangential force $\mathbf{T}$ of magnitude τ, called tension[10]. Perfect elasticity means that τ is a *constant*.
4. *Friction is negligible.*

Under the above assumptions, the equation of motion of the membrane can be derived from conservation of mass and Newton's law.

Let $\rho_0 = \rho_0(x, y)$ be the surface mass density of the membrane at rest and consider a small *rectangular* piece of membrane, with vertices at the points A, B, C, D of coordinates (x, y), $(x + \Delta x, y)$, $(x, y + \Delta y)$ and $(x + \Delta x, y + \Delta y)$, respectively. Denote by ΔS the corresponding area at time t. Then, conservation of mass yields

$$\rho(x, y, t)\Delta S = \rho_0(x, y)\Delta x \Delta y. \tag{5.91}$$

To write Newton's law of motion we have to determine the forces acting on our small piece of membrane. Since the motion is vertical, the horizontal forces have to balance.

The vertical forces are given by body forces (e.g. gravity and external loads) and the vertical component of the tension.

Denote by $f(x, y, t)\mathbf{k}$ the resultant of the body forces per unit mass. Then, using (5.91), the body forces acting on the membrane element are given by:

$$\rho(x, y, t) f(x, y, t)\Delta S\, \mathbf{k} = \rho_0(x, y) f(x, y, t)\Delta x \Delta y\, \mathbf{k}.$$

Along the edges AB and CD, the tension is perpendicular to the x-axis and almost parallel to the y-axis (see footnote 10). Its (scalar) vertical components are respectively given by

$$\tau_{\text{vert}}(x, y, t) \simeq \tau u_y(x, y, t)\Delta x, \quad \tau_{\text{vert}}(x, y + \Delta y, t) \simeq \tau u_y(x, y + \Delta y, t)\Delta x.$$

[10] The tension $\mathbf{T}$ has the following meaning. Consider a small region on the membrane, delimited by a closed curve γ. The material on one side of γ exerts on the material on the other side a *force per unit length* $\mathbf{T}$ *(pulling)* along γ. A constitutive law for $\mathbf{T}$ is

$$\mathbf{T}(x, y, t) = \tau(x, y, t)\mathbf{N}(x, y, t), \quad (x, y) \in \gamma,$$

where $\mathbf{N}$ is the outward unit normal vector to γ, tangent to the membrane.
Again, the tangentiality of the tension force is due to the absence of distributed moments over the membrane.

Similarly, along the edges AC and BD, the tension is perpendicular to the y-axis and almost parallel to the x-axis. Its (scalar) vertical components are respectively given by

$$\tau_{\text{vert}}(x,y,t) \simeq \tau u_x(x,y,t)\Delta y, \quad \tau_{\text{vert}}(x+\Delta x,y,t) \simeq \tau u_x(x+\Delta x,y,t)\Delta y.$$

Thus, using again (5.91) and observing that u_{tt} is the (scalar) vertical acceleration, Newton's law gives:

$$\begin{aligned}&\rho_0(x,y)\Delta x\Delta y\, u_{tt} = \\ &= \tau[u_y(x,y+\Delta y,t) - u_y(x,y,t)]\Delta x + \tau[u_x(x+\Delta x,y,t) - u_x(x,y,t)]\Delta y + \\ &\quad +\rho_0(x,y) f(x,y,t)\Delta x\Delta y.\end{aligned}$$

Dividing for $\Delta x\Delta y$ and letting $\Delta x, \Delta y \to 0$, we obtain the equation

$$u_{tt} - c^2(u_{yy} + u_{xx}) = f(x,y,t) \tag{5.92}$$

where $c(x,y,t) = \sqrt{\tau/\rho_0(x,y)}$.

• *Square membrane.* Consider a membrane occupying at rest the square Q_a of side a, pinned at the boundary. We want to study its vibrations when the membrane initially takes a horizontal position, with speed $h = h(x,y)$. If there is no external load and the weight of the membrane is negligible, the vibrations are governed by the following initial-boundary value problem:

$$\begin{cases} u_{tt} - c^2\Delta u = 0 & 0 < x < a,\ 0 < y < a,\ t > 0 \\ u(x,y,0) = 0,\ u_t(x,y,0) = h(x,y) & 0 < x < a,\ 0 < y < a \\ u(0,y,t) = u(a,y,t) = 0 & 0 \le y \le a,\ t \ge 0 \\ u(x,0,t) = u(x,a,t) = 0 & 0 \le x \le a,\ t \ge 0. \end{cases}$$

The square shape of the membrane and the homogeneous boundary conditions suggest the use of the method of separation of variables. Let us seek for a solution under the form

$$u(x,y,t) = v(x,y)q(t)$$

with $v = 0$ at the boundary. Substituting into the wave equation, we find

$$q''(t)v(x,y) - c^2 q(t)\Delta v(x,y) = 0$$

and, separating the variables,

$$\frac{q''(t)}{c^2 q(t)} = \frac{\Delta v(x,y)}{v(x,y)} = \eta, \qquad (\eta \text{ constant}),$$

whence the equation

$$q''(t) + c^2\eta q(t) = 0 \tag{5.93}$$

and the *eigenvalue problem*

$$\begin{cases} \Delta v - \eta v = 0 & \text{in } Q_a \\ v(0,y) = v(a,y) = v(x,0) = v(x,a) = 0, & 0 \le x, y \le a. \end{cases} \tag{5.94}$$

Multiplying the equation in (5.94) with v, integrating by parts on Q_a and using the boundary conditions we obtain

$$0 = \int_{Q_a} (\Delta v - \eta v) v = -\int_{Q_a} \left(|\nabla v|^2 + \eta v^2 \right).$$

Therefore, if $\eta \ge 0$, the problem (5.94) has only the trivial solution $v = 0$. Thus we can write $\eta = -\lambda^2 < 0$. We now solve the eigenvalue problem, using once more time the method of separation of variables and setting $v(x,y) = X(x)Y(y)$, with the conditions

$$X(0) = X(a) = 0, \qquad Y(0) = Y(a) = 0.$$

Substituting into (5.94), we obtain

$$\frac{Y''(y)}{Y(y)} + \lambda^2 = -\frac{X''(x)}{X(x)} = \mu^2$$

where μ is a new constant.

Letting $\gamma^2 = \lambda^2 - \mu^2$, we have to solve the following two one-dimensional eigenvalue problems, in $0 < x < a$ and $0 < y < a$, respectively:

$$\begin{cases} X''(x) + \mu^2 X(x) = 0 \\ X(0) = X(a) = 0 \end{cases} \qquad \begin{cases} Y''(y) + \gamma^2 Y(y) = 0 \\ Y(0) = Y(a) = 0. \end{cases}$$

The solutions are:

$$X(x) = A \sin(\mu_m x), \quad \mu_m = \frac{m\pi}{a}, \qquad Y(y) = B \sin(\gamma_n y), \quad \gamma_n = \frac{n\pi}{a}$$

where A, B are arbitrary constants and $m, n = 1, 2, \ldots$. Since $\lambda^2 = \gamma^2 + \mu^2$, we have

$$\lambda_{mn}^2 = \frac{\pi^2}{a^2}(m^2 + n^2), \qquad m, n = 1, 2, \ldots \tag{5.95}$$

corresponding to the eigenfunctions $v_{mn}(x,y) = \sin(\mu_m x) \sin(\gamma_n y)$. For $\lambda = \lambda_{mn}$, the general integral of (5.93) is

$$q_{mn}(t) = a \cos(c\lambda_{mn} t) + b \sin(c\lambda_{mn} t) \qquad (a, b \in \mathbb{R}).$$

Thus we have found infinitely many special solutions to the wave equations, of the form

$$u_{mn} = [a \cos(c\lambda_{mn} t) + b \sin(c\lambda_{mn} t)] \sin(\mu_m x) \sin(\gamma_n y),$$

which, moreover, vanish on the boundary.

Every u_{mn} is a standing wave corresponding to a particular mode of vibration of the membrane. The *fundamental frequency* is $f_{11} = c\sqrt{2}/2a$, while the other frequencies are $f_{mn} = c\sqrt{m^2+n^2}/2a$. [11]

Going back to our problem, to find a solution which satisfies the initial conditions, we superpose the modes u_{mn} by defining

$$u(x,y,t) = \sum_{m,n=1}^{\infty} [a_{mn}\cos(c\lambda_{mn}t) + b_{mn}\sin(c\lambda_{mn}t)]\sin(\mu_m x)\sin(\gamma_n y).$$

Since $u(x,y,0) = 0$, we choose $a_{mn} = 0$ for every $m,n \geq 1$. From $u_t(x,y,0) = h(x,y)$ we find the condition

$$\sum_{m,n=1}^{\infty} cb_{mn}\lambda_{mn}\sin(\mu_m x)\sin(\gamma_n y) = h(x,y). \tag{5.96}$$

Therefore, we assume that h can be expanded in a double Fourier sine series as follows:

$$h(x,y) = \sum_{m,n=1}^{\infty} h_{mn}\sin(\mu_m x)\sin(\gamma_n y),$$

where the coefficients h_{mn} are given by

$$h_{mn} = \frac{4}{a^2}\int_Q h(x,y)\sin\left(\frac{m\pi}{a}x\right)\sin\left(\frac{n\pi}{a}y\right)\,dxdy.$$

Then, if we choose $b_{mm} = h_{mm}/c\lambda_{mn}$, (5.96) is satisfied. Thus, we have constructed the formal solution

$$u(x,y,t) = \sum_{m,n=1}^{\infty} \frac{h_{mn}}{c\lambda_{mn}}\sin(c\lambda_{mn}t)\sin(\mu_m x)\sin(\gamma_n y). \tag{5.97}$$

If the coefficients $h_{mm}/c\lambda_{mn}$ vanish fast enough as $m,n \to +\infty$, it can be shown that (5.97) gives the unique solution[12].

5.7.2 Small amplitude sound waves

Sound waves are small disturbances in the density and pressure of a compressible gas. In an isotropic gas, their propagation can be described in terms of a single

[11] Note that the frequencies f_{mn} are **not** integer multiple of the fundamental one, as in the case of the vibrating string. This seems to be the cause of the low quality of tones produced, e.g., by a drum.

[12] We leave it to the reader to find appropriate smoothness hypotheses on h, in order to ensure that (5.97) is the unique solution.

scalar quantity. Moreover, due to the small amplitudes involved, it is possible to *linearize* the equations of motion, within a reasonable range of validity. Three are the relevant equations: two of them express *conservation of mass* and *balance of linear momentum*, the other one is a *constitutive* relation between density and pressure.

Conservation of mass provides the relation between the gas density $\rho = \rho(\mathbf{x}, t)$ and its velocity $\mathbf{v} = \mathbf{v}(\mathbf{x}, t)$:

$$\rho_t + \operatorname{div}(\rho \mathbf{v}) = 0. \tag{5.98}$$

The balance of linear momentum describes how the volume of gas occupying a region V reacts to the pressure exerted by the rest of the gas. Assuming that the *viscosity* of the gas is *negligible*, this force is given by the *normal pressure* $-p\boldsymbol{\nu}$ on the boundary of V ($\boldsymbol{\nu}$ is the exterior normal to ∂V).

Thus, if there are no significant external forces, in particular the gravitational force is negligible, the linear momentum equation is

$$\frac{D\mathbf{v}}{Dt} \equiv \mathbf{v}_t + (\mathbf{v} \cdot \nabla)\mathbf{v} = -\frac{1}{\rho}\nabla p. \tag{5.99}$$

The last equation is an empirical relation between p and ρ. Since the pressure fluctuations are very rapid, the compressions/expansions of the gas are *adiabatic, without any loss of heat.*

In these conditions, if $\gamma = c_p/c_v$ is the (adimensional) ratio of the specific heats of the gas ($\gamma \approx 1.4$ in air) then p/ρ^γ is constant, so that we can write

$$p = f(\rho) = C\rho^\gamma \tag{5.100}$$

with C constant.

The system of equations (5.98), (5.99), (5.100) is quite complicated and extremely difficult to solve in its general form. Here we consider sound waves that are only small perturbations of static atmospheric conditions and this allows a major simplification. In fact, if we consider a static atmosphere, where ρ_0 and p_0 are constant density and pressure, with zero velocity field, we may write

$$\rho = (1 + s)\rho_0,$$

where s is a small dimensionless quantity, called *condensation* and representing the relative variation of the density from equilibrium. Then, from (5.100), we have

$$p - p_0 \approx f'(\rho_0)(\rho - \rho_0) = s\rho_0 f'(\rho_0) \tag{5.101}$$

and

$$\nabla p \approx \rho_0 f'(\rho_0)\nabla s.$$

Now, if v is also small, we may keep in (5.99) only first order terms in s and v. Thus, we may neglect the convective acceleration $(\mathbf{v} \cdot \nabla)\mathbf{v}$ and approximate (5.99) and (5.98) with the linear equations

$$\mathbf{v}_t = -c_0^2 \nabla s \tag{5.102}$$

and

$$s_t + \operatorname{div} \mathbf{v} = 0, \tag{5.103}$$

respectively, where we have set $c_0 = \sqrt{f'(\rho_0)} = \sqrt{\gamma p_0/\rho_0}$. Note that c_0 has the dimension of a velocity: $[c_0] = [\text{length}] \times [\text{time}]^{-1}$.

Remark 5.14 Let us pause to examine the range of validity of the above linearization. First let us pass to adimensional variables. Suppose that V is an average value of $|\mathbf{v}|$. Moreover, let λ and T be typical orders of magnitude for space and time in the wave propagation, such as wavelength and period. Rescale $\mathbf{v}$, s, $\mathbf{x}$ and t as follows:

$$\boldsymbol{\xi} = \frac{\mathbf{x}}{\lambda}, \ \tau = \frac{t}{T}, \quad \mathbf{U}(\boldsymbol{\xi}, \tau) = \frac{\mathbf{v}(\lambda\boldsymbol{\xi}, T\tau)}{V}, \quad \sigma(\boldsymbol{\xi}, \tau) = s(\lambda\boldsymbol{\xi}, T\tau). \tag{5.104}$$

Then

$$\mathbf{v}_t = \frac{V}{T}\mathbf{U}_\tau, \ \operatorname{div} \mathbf{v} = \frac{V}{\lambda} \operatorname{div}_{\boldsymbol{\xi}} \mathbf{U}, \ s_t = \frac{1}{T}\sigma_\tau, \ \nabla s = \frac{1}{\lambda}\nabla_{\boldsymbol{\xi}}\sigma.$$

Substituting into (5.102) and (5.103), we obtain

$$\frac{V}{T}\mathbf{U}_\tau + \frac{c_0^2}{\lambda}\nabla_{\boldsymbol{\xi}}\sigma = \mathbf{0} \quad \text{and} \quad \frac{1}{T}\sigma_\tau + \frac{V}{\lambda} \operatorname{div}_{\boldsymbol{\xi}} \mathbf{U} = 0.$$

In these equations for the adimensional quantities $\mathbf{U}$ and σ, the coefficients must be of the same order of magnitude, therefore

$$\frac{V}{T} \approx \frac{c_0^2}{\lambda} \quad \text{and} \quad \frac{1}{T} \approx \frac{V}{\lambda}$$

which implies

$$\frac{\lambda}{T} \approx c_0.$$

Hence, c_0 is a typical propagation speed, namely it is the **sound speed**.

In the linearization process, we neglected convective acceleration, viscosity and gravity. Now, the convective acceleration is negligible with respect to (say) $\mathbf{v}_t$, if

$$\frac{V^2}{\lambda}\mathbf{U} \cdot \nabla_{\boldsymbol{\xi}}\mathbf{U} \ll \frac{V}{T}\mathbf{U}_\tau \quad \text{or} \quad V \ll c_0.$$

Thus, if the gas speed is much smaller than the sound speed, our linearization gives a good approximation. The ratio $M = V/c_0$ is called **Mach number**.

Viscosity is modeled by the term $\mu\Delta\mathbf{v}$ in the left hand side of (5.99), where μ is the kinematic viscosity. In order to neglect this term compared to $\mathbf{v}_t$ one needs that

$$\mu\frac{V}{\lambda^2} \ll \frac{V}{T}$$

that is $\lambda \gg \mu/c_0$. In air $\mu/c_0 \approx 3 \times 10^{-7}$m so that we require that the wavelength is not too short:

$$\lambda \gg 3 \times 10^{-7}\text{m}.$$

In terms of frequency, by $f = c_0/\lambda$, this translates into $f \ll 10^9$ Hertz, well above the audible range.

Finally, it is possible to show that gravity is negligible if $f \gg 0.03$ Hertz, well below the audible range.

The following theorem, in which we assume that both s and $\mathbf{v}$ are smooth functions, follows from (5.102) and (5.103).

Theorem 5.15 *a) The condensation s is a solution of the wave equation*

$$s_{tt} - c_0^2 \Delta s = 0 \tag{5.105}$$

where $c_0 = \sqrt{f'(\rho_0)} = \sqrt{\gamma p_0/\rho_0}$ is the speed of sound.

b) If $\mathbf{v}(\mathbf{x}, 0) = \mathbf{0}$, there exists an acoustic potential ϕ such that $\mathbf{v} = \nabla\phi$. Moreover ϕ satisfies (5.105) as well.

Proof a) Taking the divergence on both sides of (5.102) and the t-derivative on both sides of (5.103) we get, respectively:

$$\operatorname{div} \mathbf{v}_t = -c_0^2 \Delta s$$

and

$$s_{tt} + (\operatorname{div} \mathbf{v})_t = 0.$$

Since $(\operatorname{div} \mathbf{v})_t = \operatorname{div} \mathbf{v}_t$, equation (5.105) follows.

b) We have (see (5.102))

$$\mathbf{v}_t = -c_0^2 \nabla s.$$

Then, since $\mathbf{v}(\mathbf{x}, 0) = 0$,

$$\mathbf{v}(\mathbf{x}, t) = -c_0^2 \int_0^t \nabla s(\mathbf{x}, z)\, dz = \nabla \left(-c_0^2 \int_0^t s(\mathbf{x}, z)\, dz \right)$$

so that

$$\phi(\mathbf{x}, t) = -c_0^2 \int_0^t s(\mathbf{x}, z)\, dz$$

is an acoustic potential: $\mathbf{v} = \nabla\phi$. Finally, from (5.103),

$$\phi_{tt} = -c_0^2 s_t = c_0^2 \operatorname{div} \mathbf{v} = c_0^2 \Delta \phi$$

which is (5.105) for ϕ. □

Once the potential ϕ is known, the velocity field $\mathbf{v}$, the condensation s and the pressure fluctuation $p - p_0$ can be recovered from the following formulas:

$$\mathbf{v} = \nabla\phi, \qquad s = -\frac{1}{c_0^2}\phi_t, \qquad p - p_0 = -\rho_0 \phi_t.$$

Consider, for instance, a plane wave represented by the following potential:

$$\phi(\mathbf{x}, t) = w(\mathbf{x} \cdot \mathbf{k} - \omega t).$$

We know that if

$$c_0^2 |\mathbf{k}|^2 = \omega^2,$$

ϕ is a solution of (5.105). In this case, we have:

$$\mathbf{v} = w'\mathbf{k}, \qquad s = \frac{\omega}{c_0^2} w', \qquad p - p_0 = \rho_0 \omega w'.$$

Example 5.16 (Motion of a gas in a tube) Consider a straight cylindrical tube with axis along the x_1-axis, filled with gas in the region $x_1 > 0$ of density ρ_0, at pressure p_0. A flat piston, whose face moves according to the relation $x_1 = h(t)$, sets the gas into motion. We assume that $h(t) = 0$ for $t \leq 0$, $|h(t)| \ll 1$ and $|h'(t)| \ll c_0$. Under these conditions, the motion of the piston generates sound waves of small amplitude and the acoustic potential ϕ is a solution of the homogeneous wave equation. To compute ϕ we need suitable boundary conditions. The continuity of the normal velocity of the gas at the contact surface with the piston gives

$$\phi_{x_1}(h(t), x_2, x_3, t) = h'(t).$$

Since $h(t) \sim 0$, we may approximate this condition by

$$\phi_{x_1}(0, x_2, x_3, t) = h'(t). \tag{5.106}$$

At the tube walls the normal velocity of the gas is zero, so that, if $\boldsymbol{\nu}$ denotes the outward unit normal vector on the tube wall, we have

$$\nabla\phi \cdot \boldsymbol{\nu} = 0. \tag{5.107}$$

Finally, since the waves are generated by the piston movement, we may look for a solution of the form:

$$\phi(\mathbf{x}, t) = w(x_1 - c_0 t).$$

This is an *outgoing plane wave*[13]

$$\phi(\mathbf{x}, t) = w(\mathbf{x} \cdot \mathbf{n} - c_0 t),$$

where $\mathbf{n} = (1, 0, 0)$, and we have

$$\nabla\phi \cdot \boldsymbol{\nu} = w'(\mathbf{x} \cdot \mathbf{n} - c_0 t)\mathbf{n} \cdot \boldsymbol{\nu} = 0$$

for every $\boldsymbol{\nu}$ orthogonal to the wall tube.

From (5.106) we get

$$w'(-c_0 t) = h'(t)$$

so that ($r = -c_0 t$)

$$w(r) = -c_0 h\left(-\frac{r}{c_0}\right).$$

Hence, the acoustic potential is given by

[13] We do not expect *incoming* waves, which should be generated by sources placed far from the piston.

$$\phi(\mathbf{x},t) = -c_0 h\left(t - \frac{x_1}{c_0}\right),$$

which represents a *progressive wave* propagating along the tube. In this case:

$$\mathbf{v} = h'\left(t - \frac{x_1}{c_0}\right)\mathbf{i}, \quad s = \frac{1}{c_0}h'\left(t - \frac{x_1}{c_0}\right), \quad p = c_0\rho_0 h'\left(t - \frac{x_1}{c_0}\right) + p_0.$$

5.8 The Global Cauchy Problem

5.8.1 Fundamental solution ($n = 3$) and strong Huygens' principle

In this section we consider the global Cauchy problem for the three-dimensional homogeneous wave equation:

$$\begin{cases} u_{tt} - c^2\Delta u = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ u(\mathbf{x},0) = g(\mathbf{x}), \quad u_t(\mathbf{x},0) = h(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^3. \end{cases} \tag{5.108}$$

We know from Theorem 5.13 that problem (5.108) has at most one solution $u \in C^2(\mathbb{R}^3 \times [0,+\infty))$. Our purpose here is to show that the solution u exists and to express it in terms of the data g and h, through an explicit formula. Our derivation is rather heuristic so that, for the time being, we do not worry too much about the correct hypotheses on h and g, which we assume as smooth as we need to carry out the calculations.

First we need a lemma that reduces the problem to the case $g = 0$ (and which actually holds in any dimension). Denote by w_h the solution of the problem

$$\begin{cases} w_{tt} - c^2\Delta w = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ w(\mathbf{x},0) = 0, \quad w_t(\mathbf{x},0) = h(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^3. \end{cases} \tag{5.109}$$

Lemma 5.17 *If $w_g \in C^3(\mathbb{R}^3 \times [0,+\infty))$, then $v = \partial_t w_g$ solves the problem*

$$\begin{cases} v_{tt} - c^2\Delta v = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ v(\mathbf{x},0) = g(\mathbf{x}), \quad v_t(\mathbf{x},0) = 0 & \mathbf{x} \in \mathbb{R}^3. \end{cases} \tag{5.110}$$

Therefore the solution of (5.108) *is given by*

$$u = \partial_t w_g + w_h. \tag{5.111}$$

Proof Let $v = \partial_t w_g$. Differentiating the wave equation with respect to t, we have

$$0 = \partial_t(\partial_{tt} w_g - c^2\Delta w_g) = (\partial_{tt} - c^2\Delta)\partial_t w_g = v_{tt} - c^2\Delta v.$$

Moreover,

$$v(\mathbf{x},0) = \partial_t w_g(\mathbf{x},0) = g(\mathbf{x}), \qquad v_t(\mathbf{x},0) = \partial_{tt} w_g(\mathbf{x},0) = c^2 \Delta w_g(\mathbf{x},0) = 0.$$

Thus, v is a solution of (5.110) and $u = v + w_h$ is the solution of (5.108). □

The lemma shows that, once the solution of (5.109) is determined, the solution of the complete problem (5.108) is given by (5.111).

Therefore, we focus on the solution of (5.109), first with a special choice of h, given by the three-dimensional Dirac measure at $\mathbf{0}$, $\delta_3(\mathbf{x})$. For example, in the case of sound waves, this initial datum models a sudden change of the air density, concentrated at the origin. If w represents the relative density variation (condensation) with respect to a static atmosphere, then w solves the problem

$$\begin{cases} w_{tt} - c^2 \Delta w = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ w(\mathbf{x},0) = 0, \quad w_t(\mathbf{x},0) = \delta_3(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^3. \end{cases} \tag{5.112}$$

The solution of (5.112), which we denote by $K(\mathbf{x},t)$, is called the **fundamental solution** of the three-dimensional wave equation. To solve (5.112) we approximate the Dirac measure by the *fundamental solution* of the three-dimensional diffusion equation. Indeed, from Sect. 2.3.4 (choosing $t = \varepsilon$, $D = 1$, $n = 3$), we know that

$$\Phi(\mathbf{x},\varepsilon) = \frac{1}{(4\pi\varepsilon)^{3/2}} \exp\left\{-\frac{|\mathbf{x}|^2}{4\varepsilon}\right\} \to \delta_3(\mathbf{x}) \quad \text{as } \varepsilon \to 0^+.$$

By Theorem 5.13, the problem (5.112) with $\delta_3(\mathbf{x})$ replaced by $\Phi(\mathbf{x},\varepsilon)$ has a unique solution w_ε. Since the Laplace operator is rotationally invariant and $\Phi(\mathbf{x},\varepsilon)$ is radially symmetric, we deduce that w_ε shares the same type of symmetry and it is a spherical wave of the form

$$w_\varepsilon = w_\varepsilon(r,t), \quad r = |\mathbf{x}|.$$

Thus, from (5.88) we may write

$$w_\varepsilon(r,t) = \frac{F(r+ct)}{r} + \frac{G(r-ct)}{r}. \tag{5.113}$$

The initial conditions require

$$F(r) + G(r) = 0 \qquad \text{and} \qquad c(F'(r) - G'(r)) = r\Phi(r,\varepsilon)$$

or

$$F = -G \qquad \text{and} \qquad G'(r) = -r\Phi(r,\varepsilon)/2c.$$

Integrating the second relation yields

$$G(r) = -\frac{1}{2c(4\pi\varepsilon)^{3/2}} \int_0^r s \exp\left\{-\frac{s^2}{4\varepsilon}\right\} ds = \frac{1}{4\pi c} \frac{1}{\sqrt{4\pi\varepsilon}} \left(\exp\left\{-\frac{r^2}{4\varepsilon}\right\} - 1\right)$$

and finally

$$w_\varepsilon(r,t) = \frac{1}{4\pi c r}\left\{\frac{1}{\sqrt{4\pi\varepsilon}}\exp\left\{-\frac{(r-ct)^2}{4\varepsilon}\right\} - \frac{1}{\sqrt{4\pi\varepsilon}}\exp\left\{-\frac{(r+ct)^2}{4\varepsilon}\right\}\right\}.$$

Now observe that the function

$$\widetilde{\Phi}(r,\varepsilon) = \frac{1}{\sqrt{4\pi\varepsilon}}\exp\left\{-\frac{r^2}{4\varepsilon}\right\}$$

is the fundamental solution of the *one-dimensional* diffusion equation with $x = r$ and $t = \varepsilon$. We let now $\varepsilon \to 0^+$. Since $r + ct > 0$ for every $t > 0$,

$$\frac{1}{\sqrt{4\pi\varepsilon}}\exp\left\{-\frac{(r+ct)^2}{4\varepsilon}\right\} \to 0$$

while

$$\frac{1}{\sqrt{4\pi\varepsilon}}\exp\left\{-\frac{(r-ct)^2}{4\varepsilon}\right\} \to \delta(r-ct).$$

Therefore we conclude that, for $t > 0$,

$$K(\mathbf{x},t) = \frac{\delta(|\mathbf{x}| - ct)}{4\pi c\,|\mathbf{x}|}. \tag{5.114}$$

Due to the presence of the Dirac delta, both in problem (5.112) and in formula (5.114), the fundamental solution K satisfies (5.112) in a suitable sense. A rigorous justification uses the theory of Schwartz distributions of Chap. 7[14].

Let now the initial datum be concentrated at a point $\mathbf{y}$: $w_t(\mathbf{x},0) = \delta_3(\mathbf{x}-\mathbf{y})$. Due to the invariance under translation in space of the physical system, the corresponding fundamental solution is given by

$$K(\mathbf{x}-\mathbf{y},t) = \frac{\delta(|\mathbf{x}-\mathbf{y}| - ct)}{4\pi c\,|\mathbf{x}-\mathbf{y}|} \qquad (t>0),$$

which represents an *outgoing travelling wave*, initially supported at $\mathbf{y}$ and thereafter on the surface

$$\partial B_{ct}(\mathbf{y}) = \{\mathbf{x} : |\mathbf{x}-\mathbf{y}| = ct\}.$$

The union of the surfaces $\partial B_{ct}(\mathbf{y})$ is called the **support** of $K(\mathbf{x}-\mathbf{y},t)$ and coincides with the **boundary** of the *forward space-time cone, with vertex at* $(\mathbf{y},0)$ and opening $\theta = \arctan c$, given by

$$C^*_{\mathbf{y},0} = \{(\mathbf{x},t) : |\mathbf{x}-\mathbf{y}| \le ct,\, t>0\}.$$

Following the terminology of Sect. 5.4, $\partial C^*_{\mathbf{y},0}$ constitutes the **range of influence of the point y**.

The fact that the range of influence of the point $\mathbf{y}$ is *only the boundary* of the forward cone and not the full cone has important consequences on the nature of the

[14] See Example 7.27, p. 438 and Problems 7.17, 7.18.

disturbances governed by the three-dimensional wave equation. The most striking phenomenon is that a perturbation generated at time $t = 0$ by a point source placed at $\mathbf{y}$ is felt at the point $\mathbf{x}_0$ **only at time** $t_0 = |\mathbf{x}_0 - \mathbf{y}|/c$ (Fig. 5.10). This is known as *strong Huygens' principle* and explains why *sharp signals* are propagated from a point source.

We will shortly see that this is not the case in two dimensions.

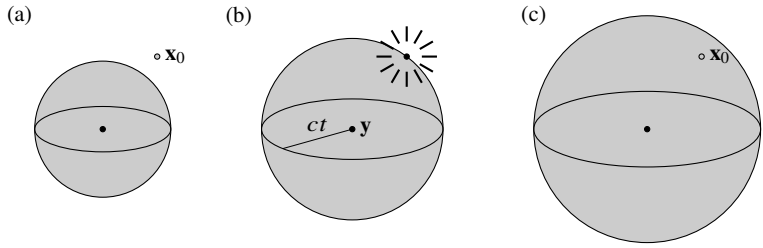

Fig. 5.10 Huygens principle. (a) $ct < |\mathbf{x}_0 - \mathbf{y}|$; (b) $ct = |\mathbf{x}_0 - \mathbf{y}|$; (c) $ct > |\mathbf{x}_0 - \mathbf{y}|$

5.8.2 The Kirchhoff formula

Using the fundamental solution as in Sect. 5.4.3, we may derive a formula for the solution of (5.109) with a general datum h. Since

$$h(\mathbf{x}) = \int_{\mathbb{R}^3} \delta_3(\mathbf{x} - \mathbf{y}) h(\mathbf{y}) d\mathbf{y},$$

we may see h as a superposition of the impulses $\delta_3(\mathbf{x} - \mathbf{y})h(\mathbf{y})$, localized at $\mathbf{y}$, of strength $h(\mathbf{y})$. Accordingly, the solution of (5.109) is given by the superposition of the corresponding solutions $K(\mathbf{x} - \mathbf{y}, t)h(\mathbf{y})$, that is, for $t > 0$,

$$w_h(\mathbf{x}, t) = \int_{\mathbb{R}^3} K(\mathbf{x} - \mathbf{y}, t) h(\mathbf{y}) d\mathbf{y} = \int_{\mathbb{R}^3} \frac{\delta(|\mathbf{x} - \mathbf{y}| - ct)}{4\pi c\, |\mathbf{x} - \mathbf{y}|} h(\mathbf{y}) d\mathbf{y}. \tag{5.115}$$

Thus w_h is the **x-convolution of** h **with the fundamental solution**. Using the formula

$$\int_0^\infty \delta(r - ct) f(r) dr = f(ct)$$

we can write

$$w_h(\mathbf{x}, t) = \int_0^\infty \frac{\delta(r - ct)}{4\pi c r} dr \int_{\partial B_r(\mathbf{x})} h(\boldsymbol{\sigma}) d\sigma = \frac{1}{4\pi c^2 t} \int_{\partial B_{ct}(\mathbf{x})} h(\boldsymbol{\sigma}) d\sigma.$$

Lemma 5.17, p. 304, and the above intuitive argument lead to the following theorem:

Theorem 5.18 (Kirchhoff's formula) *Let $g \in C^3(\mathbb{R}^3)$ and $h \in C^2(\mathbb{R}^3)$. Then,*

$$u(\mathbf{x},t) = \frac{\partial}{\partial t}\left[\frac{1}{4\pi c^2 t}\int_{\partial B_{ct}(\mathbf{x})} g(\boldsymbol{\sigma})d\sigma\right] + \frac{1}{4\pi c^2 t}\int_{\partial B_{ct}(\mathbf{x})} h(\boldsymbol{\sigma})d\sigma \tag{5.116}$$

is the unique solution $u \in C^2(\mathbb{R}^3 \times [0,+\infty))$ of problem (5.108).

Proof Letting $\boldsymbol{\sigma} = \mathbf{x} + ct\boldsymbol{\omega}$, where $\boldsymbol{\omega} \in \partial B_1(\mathbf{0})$, we have $d\sigma = c^2 t^2 d\omega$ and we may write

$$w_g(\mathbf{x},t) = \frac{1}{4\pi c^2 t}\int_{\partial B_{ct}(\mathbf{x})} g(\boldsymbol{\sigma})d\sigma = \frac{t}{4\pi}\int_{\partial B_1(\mathbf{0})} g(\mathbf{x}+ct\boldsymbol{\omega})d\omega.$$

Since $g \in C^3(\mathbb{R}^3)$, this formula shows that w_g satisfies the hypotheses of Lemma 5.17. Since $h \in C^2(\mathbb{R}^3)$, it follows that

$$w_h(\mathbf{x},t) = \frac{1}{4\pi c^2 t}\int_{\partial B_{ct}(\mathbf{x})} h(\boldsymbol{\sigma})d\sigma = \frac{t}{4\pi}\int_{\partial B_1(\mathbf{0})} h(\mathbf{x}+ct\boldsymbol{\omega})d\omega$$

belongs to $C^2(\mathbb{R}^3 \times [0,+\infty))$ and therefore, to conclude the proof, it is enough to check that w_h solves problem (5.109). We have:

$$\partial_t w_h(\mathbf{x},t) = \frac{1}{4\pi}\int_{\partial B_1(\mathbf{0})} h(\mathbf{x}+ct\boldsymbol{\omega})d\omega + \frac{ct}{4\pi}\int_{\partial B_1(\mathbf{0})} \nabla h(\mathbf{x}+ct\boldsymbol{\omega})\cdot\boldsymbol{\omega}\, d\omega. \tag{5.117}$$

Thus,

$$w_h(\mathbf{x},0) = 0 \quad \text{and} \quad \partial_t w_h(\mathbf{x},0) = h(\mathbf{x}).$$

Moreover, by the Gauss formula, we may write

$$\begin{aligned}\frac{ct}{4\pi}\int_{\partial B_1(\mathbf{0})} \nabla h(\mathbf{x}+ct\boldsymbol{\omega})\cdot\boldsymbol{\omega}\, d\omega &= \frac{1}{4\pi ct}\int_{\partial B_{ct}(\mathbf{x})} \partial_{\boldsymbol{\nu}} h(\boldsymbol{\sigma})d\sigma \\ &= \frac{1}{4\pi ct}\int_{B_{ct}(\mathbf{x})} \Delta h(\mathbf{y})d\mathbf{y} = \frac{1}{4\pi ct}\int_0^{ct} dr \int_{\partial B_r(\mathbf{x})} \Delta h(\boldsymbol{\sigma})d\sigma,\end{aligned}$$

whence, from (5.117),

$$\begin{aligned}\partial_{tt} w_h(\mathbf{x},t) = \frac{c}{4\pi}\int_{\partial B_1(\mathbf{0})} \nabla h(\mathbf{x}+ct\boldsymbol{\omega})\cdot\boldsymbol{\omega}\, d\omega - \frac{1}{4\pi c t^2}\int_{B_{ct}(\mathbf{x})} \Delta h(\mathbf{y})d\mathbf{y} \\ + \frac{1}{4\pi t}\int_{\partial B_{ct}(\mathbf{x})} \Delta h(\boldsymbol{\sigma})d\sigma = \frac{1}{4\pi t}\int_{\partial B_{ct}(\mathbf{x})} \Delta h(\boldsymbol{\sigma})d\sigma.\end{aligned}$$

On the other hand,

$$\Delta w_h(\mathbf{x},t) = \frac{t}{4\pi}\int_{\partial B_1(\mathbf{0})} \Delta h(\mathbf{x}+ct\boldsymbol{\omega})d\omega = \frac{1}{4\pi c^2 t}\int_{\partial B_{ct}(\mathbf{x})} \Delta h(\boldsymbol{\sigma})d\sigma$$

and therefore

$$\partial_{tt} w_h - c^2 \Delta w_h = 0. \qquad \square$$

Using formula (5.117) with g instead of h, we may write the Kirchhoff formula in the following form:

$$u(\mathbf{x},t)=\frac{1}{4\pi c^2t^2}\int_{\partial B_{ct}(\mathbf{x})}\{g(\boldsymbol{\sigma})+\nabla g(\boldsymbol{\sigma})\cdot(\boldsymbol{\sigma}-\mathbf{x})+th(\boldsymbol{\sigma})\}\,d\sigma. \tag{5.118}$$

The presence of the gradient of g in (5.118) suggests that, unlike in the one-dimensional case, the solution u *may be more irregular than the data*. Indeed, if $g\in C^k(\mathbb{R}^3)$ and $h\in C^{k-1}(\mathbb{R}^3)$, $k\geq 2$, we can only guarantee that u is C^{k-1} and u_t is C^{k-2} at a later time.

Formula (5.118) makes perfect sense also for $g\in C^1(\mathbb{R}^3)$ and h bounded. Clearly, under these weaker hypotheses, (5.118) satisfies the wave equation in an appropriate generalized sense, as in Sect. 5.4.2, for instance. In this case, scattered singularities in the initial data h may concentrate at later time on smaller sets, giving rise to stronger singularities (*focusing effect*, see Problem 5.17).

According to (5.118), $u(\mathbf{x},t)$ depends upon the data g and h only on the surface $\partial B_{ct}(\mathbf{x})$, which therefore coincides with the **domain of dependence** for $(\mathbf{x},t)$.

Assume that the support of g and h is a compact set D. Then $u(\mathbf{x},t)$ is different from zero only for $t_{\min}<t<t_{\max}$ where $t_{\min}$ and $t_{\max}$ are the first and the last time t such that

$$D\cap\partial B_{ct}(\mathbf{x})\neq\emptyset.$$

In other words, a disturbance, initially localized inside D, starts affecting the point $\mathbf{x}$ at time $t_{\min}$ and ceases to affect it after time $t_{\max}$. This is another way to express the *strong Huygens' principle*.

Fix t and consider the union of all the spheres $\partial B_{ct}(\boldsymbol{\xi})$ as $\boldsymbol{\xi}$ varies on ∂D. The envelope of these surfaces constitutes the *wave front* and bounds the support of u, which spreads at speed c (see Problem 5.16).

5.8.3 The Cauchy problem in dimension 2

The solution of the Cauchy problem in two dimensions can be obtained from Kirchhoff formula, using the so called *Hadamard's method of descent*. Consider first the problem

$$\begin{cases} w_{tt}-c^2\Delta w=0 & \mathbf{x}\in\mathbb{R}^2,\ t>0\\ w(\mathbf{x},0)=0,\quad w_t(\mathbf{x},0)=h(\mathbf{x}) & \mathbf{x}\in\mathbb{R}^2. \end{cases} \tag{5.119}$$

The key idea is to "embed" the two-dimensional problem (5.119) into a three-dimensional setting. More precisely, write the points in $\mathbb{R}^3$ as $(\mathbf{x},x_3)$ and set $h(\mathbf{x},x_3)=h(\mathbf{x})$, where now $\mathbf{x}\in\mathbb{R}^2$. The solution U of the three-dimensional problem is given by Kirchhoff's formula:

$$U(\mathbf{x},x_3,t)=\frac{1}{4\pi c^2t}\int_{\partial B_{ct}(\mathbf{x},x_3)}h\,d\sigma. \tag{5.120}$$

We claim that, since h does not depend on x_3, U is independent of x_3 as well, and therefore the solution of (5.119) is given by (5.120) with, say, $x_3=0$.

To prove the claim, note that the spherical surface $\partial B_{ct}(\mathbf{x}, x_3)$ is the union of two hemispheres, whose equations are

$$y_3 = F_\pm(\mathbf{y}) = x_3 \pm \sqrt{c^2t^2 - r^2},$$

where $r = |\mathbf{y} - \mathbf{x}|$. On both hemispheres we have:

$$d\sigma = \sqrt{1 + |\nabla F_\pm|^2}\, d\mathbf{y} = \sqrt{1 + \frac{r^2}{c^2t^2 - r^2}}\, d\mathbf{y} = \frac{ct}{\sqrt{c^2t^2 - r^2}}\, d\mathbf{y}$$

and therefore may write

$$U(\mathbf{x}, x_3, t) = U(\mathbf{x}, t) = \frac{1}{2\pi c} \int_{B_{ct}(\mathbf{x})} \frac{h(\mathbf{y})}{\sqrt{c^2t^2 - |\mathbf{x} - \mathbf{y}|^2}} d\mathbf{y}.$$

Thus, U is independent of x_3 as claimed. From the above calculations, and recalling Lemma 5.17, p. 304, we deduce the following theorem.

Theorem 5.19 (Poisson's formula) *Let $g \in C^3(\mathbb{R}^2)$ and $h \in C^2(\mathbb{R}^2)$. Then,*

$$u(\mathbf{x}, t) = \frac{1}{2\pi c} \left\{ \frac{\partial}{\partial t} \int_{B_{ct}(\mathbf{x})} \frac{g(\mathbf{y}) d\mathbf{y}}{\sqrt{c^2t^2 - |\mathbf{x} - \mathbf{y}|^2}} + \int_{B_{ct}(\mathbf{x})} \frac{h(\mathbf{y}) d\mathbf{y}}{\sqrt{c^2t^2 - |\mathbf{x} - \mathbf{y}|^2}} \right\}$$

is the unique solution $u \in C^2(\mathbb{R}^2 \times [0, +\infty))$ of the problem

$$\begin{cases} u_{tt} - c^2 \Delta u = 0 & \mathbf{x} \in \mathbb{R}^2,\ t > 0 \\ u(\mathbf{x}, 0) = g(\mathbf{x}), \quad u_t(\mathbf{x}, 0) = h(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^2. \end{cases}$$

Also Poisson's formula can be written in a somewhat more explicit form. Indeed, letting $\mathbf{y} - \mathbf{x} = ct\mathbf{z}$, we have

$$d\mathbf{y} = c^2t^2 d\mathbf{z}, \quad |\mathbf{x} - \mathbf{y}|^2 = c^2t^2\, |\mathbf{z}|^2$$

whence

$$\int_{B_{ct}(\mathbf{x})} \frac{g(\mathbf{y})}{\sqrt{c^2t^2 - |\mathbf{x} - \mathbf{y}|^2}} d\mathbf{y} = ct \int_{B_1(\mathbf{0})} \frac{g(\mathbf{x} + ct\mathbf{z})}{\sqrt{1 - |\mathbf{z}|^2}} d\mathbf{z}.$$

Then

$$\begin{aligned} &\frac{\partial}{\partial t} \int_{B_{ct}(\mathbf{x})} \frac{g(\mathbf{y})}{\sqrt{c^2t^2 - |\mathbf{x} - \mathbf{y}|^2}} d\mathbf{y} \\ &= c \int_{B_1(\mathbf{0})} \frac{g(\mathbf{x} + ct\mathbf{z})}{\sqrt{1 - |\mathbf{z}|^2}} d\mathbf{z} + c^2t \int_{B_1(\mathbf{0})} \frac{\nabla g(\mathbf{x} + ct\mathbf{z}) \cdot \mathbf{z}}{\sqrt{1 - |\mathbf{z}|^2}} d\mathbf{z} \end{aligned}$$

and, going back to the original variables, we obtain

$$u(\mathbf{x},t)=\frac{1}{2\pi ct}\int_{B_{ct}(\mathbf{x})}\frac{g(\mathbf{y})+\nabla g(\mathbf{y})\cdot(\mathbf{y}-\mathbf{x})+th(\mathbf{y})}{\sqrt{c^2t^2-|\mathbf{x}-\mathbf{y}|^2}}d\mathbf{y}. \tag{5.121}$$

Poisson's formula displays an important difference with respect to its three-dimensional analogue, Kirchhoff's formula. In fact the *domain of dependence* of the point $(\mathbf{x},t)$ is given by **the full circle**

$$B_{ct}(\mathbf{x})=\{\mathbf{y}:\ |\mathbf{x}-\mathbf{y}|<ct\}\,.$$

This entails that a disturbance, initially localized at $\boldsymbol{\xi}$, starts affecting the point $\mathbf{x}$ at time $t_{\min}=|\mathbf{x}-\boldsymbol{\xi}|/c$. However, this effect does not vanish for $t>t_{\min}$, since $\boldsymbol{\xi}$ still belongs to the circle $B_{ct}(\mathbf{x})$ after $t_{\min}$.

It is the type of phenomenon one may observe by placing a cork on still water and dropping a stone not too far away. The cork remains undisturbed until it is reached by the wave front, but its oscillations persist thereafter.

Thus, sharp signals do not exist in dimension two and *the strong Huygens principle does not hold*. It could be observed that if the initial data are compactly supported, then (5.121) shows a decay in time of order $1/t$ for any fixed x, as $t\to\infty$. This decay becomes of order $1/t^2$ if $h\equiv 0$.

Remark 5.20 An examination of Poisson's formula reveals that the fundamental solution for the two dimensional wave equation is given by

$$K(\mathbf{x},t)=\frac{1}{2\pi c}\frac{1}{\sqrt{c^2t^2-|\mathbf{x}|^2}}\chi_{B_{ct}(\mathbf{x})}, \tag{5.122}$$

where $\chi_{B_{ct}(\mathbf{x})}$ is the characteristic function of $B_{ct}(\mathbf{x})$. Its support is the **full** forward space-time cone, with vertex at $(\mathbf{0},0)$ and opening $\theta=\arctan c$, given by

$$C^*_{\mathbf{0},0}=\{(\mathbf{x},t):|\mathbf{x}|\le ct,\,t\ge 0\}\,.$$

5.9 An Inverse Problem in Thermoacoustic Tomography

Thermoacoustic (or photoacoustic) tomography (TAT) is a hybrid imaging technique used to visualize the capability of a medium to absorb non-ionizing radiations (of electromagnetic nature) and it is largely exploited in several areas of applied sciences, notably medical diagnostics.

A short electromagnetic impulse, such as a laser pulse of duration of about 20 picoseconds ($=20\cdot 10^{-12}$ seconds), is sent through a biological object. Part of the energy is absorbed throughout the body and the amount of absorption in a given part of it depends on the biological properties of that part. For instance, many tumoral cells absorb much more electromagnetic energy (in a proper energy band) than

healthy cells. In this case, the knowledge of the *absorption map* of the body is of invaluable importance in cancer detection.

The optical pulse radiation causes tissues heating (in the range of millikelvin) with thermoelastic expansion/contraction, which in turn results in the generation of propagating ultrasound (pressure) waves (the photoacoustic effect). The pressure waves can be measured by means of transducers placed on a surface S surrounding the body. The goal is *to reconstruct the absorption map, or something related to, from these measurements.*

From a mathematical point of view, the reconstruction problem can be formulated as a so called *inverse problem*, typically *ill posed*. Several techniques have been employed to solve it, under suitable hypotheses[15]. Here we briefly sketch one of the most simple mathematical models, and describe the *time reversal method*, based, in dimension $n = 3$, on the *strong Huygens principle* and on the invariance of the wave equation under a time reversal. Indeed, throughout this section we shall work in dimension $n = 3$.

The mathematical model for the pressure

As usual we make use of general conservation laws and of suitable constitutive relations. The biological object is assimilated to an inviscid fluid, homogeneous and isotropic with respect to acoustic waves propagation (see *O. Scherzer et al.*, [31]). We denote by $\rho = \rho(\mathbf{x}, t)$ and $p = p(\mathbf{x}, t)$ the density and the acoustic pressure of the medium, respectively.

We adopt the following assumptions:

- External forces are negligible.
- The velocity v of the "fluid" is very small (negligible).
- The (relative) variations of pressure and density are very small:

$$p = p_0(1 + \delta^* p)$$
$$\rho = \rho_0(1 + \delta^* \rho)$$

with p_0, ρ_0 constant and $|\delta^* p| \ll 1$ and $|\delta^* \rho| \ll 1$.

The general laws are: the *linearized mass conservation*

$$\rho_t + \operatorname{div}(\rho \mathbf{v}) \simeq \rho_t + \rho_0 \operatorname{div} \mathbf{v} = 0$$

and the *linearized Euler equation*

$$\rho \mathbf{v}_t + (\mathbf{v} \cdot \nabla)\mathbf{v} \simeq \rho_0 \mathbf{v}_t = -\nabla p.$$

Combining the two laws, we get

[15] We refer for instance to *P. Kuchment and L Kunyansky*, Mathematics of thermoacoustic tomography, European J. Appl. Math. 19, 191–224, *2008*.

$$\rho_{tt} - \Delta p = 0. \tag{5.123}$$

We now need to choose how to model:

1. The absorption of the electromagnetic power.
2. The relation between absorption and temperature variation.
3. The effect of temperature variations on the thermal expansion of the absorbing tissue.

1. The absorption r is modelled by the heat source function (energy per unit time)

$$r(\mathbf{x}, t) = I_{em}(\mathbf{x}, t)\mu_{abs}(\mathbf{x})$$

where μ_{abs} is the *absorption map* and I_{em} is the *radiation intensity*. Given its impulsive nature, we may write

$$I_{em}(\mathbf{x}, t) \simeq J(\mathbf{x})\delta(t) \tag{5.124a}$$

for a proper spatial intensity distribution $J(\mathbf{x})$. Thus we think that before time $t = 0$ nothing happens ($p \equiv 0$) and that the pulse is concentrated at time $t = 0$.

2. Heat conduction effects are negligible. Hence, we have

$$c_v(\mathbf{x})T_t = r(\mathbf{x}, t),$$

where $c_v(\mathbf{x})$ is the specific heat at constant volume and T is the temperature.

3. We adopt the *linearized law of thermal expansion*, given by:

$$\beta(\mathbf{x})T_t = \frac{1}{v_s^2}p_t - \rho_t$$

where β is the coefficient of *thermal expansion* and v_s is the sound speed (which we assume to be constant).

From 1, 2 and 3 we have:

$$\frac{1}{v_s^2}p_t - \rho_t = \frac{J(\mathbf{x})\mu_{abs}(\mathbf{x})\beta(\mathbf{x})}{c_v(\mathbf{x})}\delta(t) \equiv f(\mathbf{x})\delta(t) \tag{5.125}$$

where f is called *energy deposition function.*

Taking the t-derivative and using (5.123), we formally obtain the equation

$$p_{tt} - v_s^2\Delta p = v_s^2 f(\mathbf{x})\delta'(t), \qquad \mathbf{x} \in \mathbb{R}^3, t \in \mathbb{R} \tag{5.126}$$

where we think $p \equiv 0$ for $t < 0$ and $f \equiv 0$ outside the body. However, the meaning of $\delta'(t)$ has to be clarified.

As in Sect. 2.3.3, we can define δ' *through its action on test functions*. Thus, take a smooth function φ vanishing outside a compact interval and write, after a formal integration by parts:

$$\int \delta'(t)\varphi(t)dt = -\int \delta(t)\varphi'(t)dt = -\varphi'(0). \tag{5.127}$$

Indeed, as we will see in Chap. 7, equation (5.127) gives the true meaning of δ'.

Still it is not clear how to solve (5.126). Fortunately, equation (5.126) has an equivalent formulation in more standard terms. Indeed it is possible to prove (see Problem 7.18) that p is a solution of (5.126) if and only if p solves the problem

$$\begin{cases} p_{tt} - v_s^2 \Delta p = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ p(\mathbf{x}, 0) = v_s^2 f(\mathbf{x}),\ p_t(\mathbf{x}, 0) = 0 & \mathbf{x} \in \mathbb{R}^3 \end{cases}. \tag{5.128}$$

This is our model for the ultrasound waves.

Remark 5.21 As we shall see in Chap. 7, Sect. 7.5.4, the "product" $J(\mathbf{x})\delta(t)$ in formula (5.124a), takes the name of **direct** or **tensor product** between $J(\mathbf{x})$ and $\delta(t)$, and it should be more properly written as

$$J(\mathbf{x}) \otimes \delta(t).$$

Similarly, the two direct products on the right of (5.125) and (5.126) should be written as $f(\mathbf{x}) \otimes \delta(t)$ and $f(\mathbf{x}) \otimes \delta'(t)$, respectively. The symbol $\otimes$ emphasizes that the two factors in the tensor product act on a test function $\psi(\mathbf{x}, t)$ separately, each one with respect to its own variable.

The inverse problem. The time reversal method

Assume now we are able to measure the pressure on a closed (nice) surface S surrounding the body, for $0 \le t \le T$. This means that we know p on the surface $S \times [0, T]$. Thus, supposing that the measured datum is g, we can add this information to our problem and write:

$$\begin{cases} p_{tt} - v_s^2 \Delta p = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ p(\mathbf{x}, 0) = v_s^2 f(\mathbf{x}),\ p_t(\mathbf{x}, 0) = 0 & \mathbf{x} \in \mathbb{R}^3 \\ p(\boldsymbol{\sigma}, t) = g(\boldsymbol{\sigma}, t) & \boldsymbol{\sigma} \in S,\ 0 \le t \le T. \end{cases} \tag{5.129}$$

Our problem is *to reconstruct f from the knowledge of g.* In fact, from the knowledge of f we can extract information on the absorption map through formula (5.125).[16]

At first glance, it seems that our reconstruction problem is unsolvable, since many different initial data give rise to solutions with the same lateral data. Nevertheless, there are two elements that make the problem solvable. First, the wave equation in (5.129) is valid in the whole space $\mathbb{R}^3$, not only in the region inside S, and second, f is compactly supported inside the body, hence inside S.

Now, the idea of the time reversal method is simple. Since the Huygens principle holds and f is compactly supported, after a long enough time, the pressure waves

[16] See, for instance, L.V. Wang, H. Hu, *Biomedical Optics, Principles and Imaging*. Wiley-Interscience (2007).

leave S. This means that, if T is large enough,

$$p(\mathbf{x}, T) = p_t(\mathbf{x}, T) = 0 \text{ inside } S.$$

But then p is a solution of the final problem

$$\begin{cases} p_{tt} - v_s^2 \Delta p = 0 & \text{inside } S \times (0, T) \\ p(\mathbf{x}, T) = p_t(\mathbf{x}, T) = 0 & \text{inside } S \\ p(\boldsymbol{\sigma}, t) = g(\boldsymbol{\sigma}, t) & \boldsymbol{\sigma} \in S,\ 0 \le t \le T. \end{cases} \tag{5.130}$$

Since the wave equation is invariant by time reversal, we know that, under reasonable assumptions on g, problem (5.130) has at most one solution. The existence and, above all, the stability estimates on the solution, follow by the methods we will develop in Chap. 9. Then, via standard numerical methods, one is able to compute $p(\mathbf{x}, 0) = f(\mathbf{x})$ and to solve the inverse problem.

5.10 The Cauchy Problem with Distributed Sources

5.10.1 Retarded potentials ($n = 3$)

The solution of the nonhomogeneous Cauchy problem can be obtained via Duhamel's method. We give the details for $n = 3$ only (for $n = 2$, see Problem 5.20). By linearity it is enough to derive a formula for the solution of the problem with zero initial data:

$$\begin{cases} u_{tt} - c^2 \Delta u = f(\mathbf{x}, t) & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ u(\mathbf{x}, 0) = 0, \quad u_t(\mathbf{x}, 0) = 0 & \mathbf{x} \in \mathbb{R}^3. \end{cases} \tag{5.131}$$

Assume that $f \in C^2(\mathbb{R}^3 \times [0, +\infty))$. For $s \ge 0$ fixed, let $w = w(\mathbf{x}, t; s)$ be the solution of the problem

$$\begin{cases} w_{tt} - c^2 \Delta w = 0 & \mathbf{x} \in \mathbb{R}^3, t \ge s \\ w(\mathbf{x}, s; s) = 0, \quad w_t(\mathbf{x}, s; s) = f(\mathbf{x}, s) & \mathbf{x} \in \mathbb{R}^3. \end{cases}$$

Since the wave equation is invariant under space-time translations, w is given by Kirchhoff's formula with t replaced by $t - s$:

$$w(\mathbf{x}, t; s) = \frac{1}{4\pi c^2 (t - s)} \int_{\partial B_{c(t-s)}(\mathbf{x})} f(\boldsymbol{\sigma}, s) d\sigma.$$

Then, as one can check mimicking the proof in dimension one (Sect. 5.4.4),

$$u(\mathbf{x}, t) = \int_0^t w(\mathbf{x}, t; s)\, ds = \frac{1}{4\pi c^2} \int_0^t \frac{ds}{(t - s)} \int_{\partial B_{c(t-s)}(\mathbf{x})} f(\boldsymbol{\sigma}, s) d\sigma \tag{5.132}$$

is the unique solution $u \in C^2(\mathbb{R}^3 \times [0, +\infty))$ of (5.131).

Formula (5.132) shows that $u(\mathbf{x}, t)$ depends on the values of f in the **full backward cone**

$$C_{\mathbf{x},t} = \{(\mathbf{z}, s)\colon\ |\mathbf{z} - \mathbf{x}| \le c(t - s),\, 0 \le s \le t\}\,.$$

Note that (5.132) may also be written in the form

$$u(\mathbf{x}, t) = \frac{1}{4\pi c^2} \int_{B_{ct}(\mathbf{x})} \frac{1}{|\mathbf{x} - \mathbf{y}|} f\left(\mathbf{y}, t - \frac{|\mathbf{x} - \mathbf{y}|}{c}\right) d\mathbf{y} \tag{5.133}$$

which is a so called *retarded potential*. In fact, the value of u at $\mathbf{x}$ at time t does not depend on the source at the same time, but rather at the earlier time

$$t_{\text{ret}} = t - \frac{|\mathbf{x} - \mathbf{y}|}{c}.$$

The time lag $t - t_{\text{ret}}$ is precisely the time required by the source effect to propagate from $\mathbf{y}$ to $\mathbf{x}$. This is due to the finite speed c of propagation of the disturbances.

We can also write u in terms of the fundamental solution

$$K(\mathbf{x} - \mathbf{y}, t - s) = \frac{\delta(|\mathbf{x} - \mathbf{y}| - c(t - s))}{4\pi c\,|\mathbf{x} - \mathbf{y}|}.$$

Indeed, recalling formula (5.115), u can be expressed by the space-time convolution

$$u(\mathbf{x}, t) = \int_0^t \int_{\mathbb{R}^3} K(\mathbf{x} - \mathbf{y}, t - s) f(\mathbf{y}, s) d\mathbf{y} ds. \tag{5.134}$$

Important applications to Acoustics or Electromagnetism require more general f, modeling for instance various types of point sources. The method still works, as we shall see in the next example and, more significantly, in the next section.

- *Point source*. Let

$$f(\mathbf{x}, t) = \delta_3(\mathbf{x}) q(t).$$

f models a source concentrated at $\mathbf{x} = \mathbf{0}$, with strength $q = q(t)$, that we assume smoothly varying with time for $t \ge 0$, and *null* for $t < 0$. Formally, substituting this source into (5.133) we obtain, since $\delta_3(\mathbf{y})$ requires to set $\mathbf{y} = \mathbf{0}$ inside the integral,

$$u(\mathbf{x}, t) = \frac{1}{4\pi c^2} \frac{1}{|\mathbf{x}|} q\left(t - \frac{|\mathbf{x}|}{c}\right). \tag{5.135}$$

Note that $u = 0$ for $|\mathbf{x}| > ct$. To justify the formula (5.135), we approximate δ_3 by a smooth function $\psi_\varepsilon = \psi_\varepsilon(\mathbf{x})$ supported in the ball $B_\varepsilon = B_\varepsilon(\mathbf{0})$, with unit strength, that is

$$\int_{B_\varepsilon} \psi_\varepsilon(\mathbf{x}) d\mathbf{x} = 1, \tag{5.136}$$

and such that

$$\psi_\varepsilon \to \delta_3 \ \text{ as } \varepsilon \to 0. \tag{5.137}$$

A function like ψ_ε is called an *approximation of the identity* and can be constructed in several ways (see Problem 7.1). Recall that the limit in (5.137) means

$$\int_{\mathbb{R}^n} \psi_\varepsilon(\mathbf{x})\varphi(\mathbf{x})d\mathbf{x} \to \varphi(\mathbf{0}) \qquad \text{as} \quad \varepsilon \to 0,$$

for every smooth function φ, compactly supported in $\mathbb{R}^3$.

Let $f_\varepsilon(\mathbf{x}, t) = \psi_\varepsilon(\mathbf{x})q(t)$. Assume that $\varepsilon < |\mathbf{x}| < ct$ and $\varepsilon < ct - |\mathbf{x}|$. Then $B_\varepsilon \subset B_{ct}(\mathbf{x})$, $\mathbf{x} \notin B_\varepsilon$ and the integration in (5.133) can be restricted to B_ε. We find

$$u_\varepsilon(\mathbf{x}, t) = \frac{1}{4\pi c^2} \int_{B_\epsilon} q\left(t - \frac{|\mathbf{x} - \mathbf{y}|}{c}\right) \frac{\psi_\varepsilon(\mathbf{y})}{|\mathbf{x} - \mathbf{y}|} d\mathbf{y}. \tag{5.138}$$

Letting $\varepsilon \to 0$ we get

$$u(\mathbf{x}, t) = \frac{1}{4\pi c^2} \frac{1}{|\mathbf{x}|} q\left(t - \frac{|\mathbf{x}|}{c}\right)$$

for all $\mathbf{x} \neq \mathbf{0}$, which is (5.135).

5.10.2 Radiation from a moving point source

A more interesting case occurs when a source (of acoustic waves, say) is moving. For simplicity we consider a point source of constant strength S, moving along the x_3-axis with constant speed $v > 0$. We can model this kind of source by the formula

$$f(\mathbf{x}, t) = S\delta_3(\mathbf{x} - vt\mathbf{k}) = S\delta_3(x_1, x_2, x_3 - vt)$$

where $\mathbf{k}$ is the unit vector along the x_3-axis. The idea is that the contributions of f come only from the points where the argument of δ_3 vanishes, that is from the points $\mathbf{x} = vt\mathbf{k}$. Here $\mathbf{x} \in \mathbb{R}^3$ and t goes (ideally) from $-\infty$ to $+\infty$.

Thus, we want to find a potential u such that

$$u_{tt} - c^2\Delta u = S\delta_3(\mathbf{x} - vt\mathbf{k}) \qquad \text{in } \mathbb{R}^3 \times \mathbb{R}. \tag{5.139}$$

This problem is rather anomalous, not only due to the presence of a travelling delta function as a source, but also because it is formulated in the entire space-time and there are not initial conditions. To find u we first proceed in a formal[17] way, expressing u as a retarded potential, obtained by substituting

$$f(\mathbf{y}, t) = S\delta_3(\mathbf{y} - vt\mathbf{k})$$

into (5.133); we find

[17] But, we believe, instructive.

$$u(\mathbf{x},t)=\frac{S}{4\pi c^2}\int\frac{1}{|\mathbf{x}-\mathbf{y}|}\delta_3\left(\mathbf{y}-v\left(t-\frac{|\mathbf{x}-\mathbf{y}|}{c}\right)\mathbf{k}\right)d\mathbf{y} \tag{5.140}$$

where the "integration" is extended over the set where the argument of δ_3 vanishes. It is actually rather delicate to interpret rigorously formula (5.140), since it involves the composition of the δ_3 measure with a nonlinear function of $\mathbf{y}$, smooth outside $\mathbf{x}$. Let us write (5.140) in a simpler form. To do it, we use the following two properties of the δ_n measure, whose proof can be found in Chap. 7.

The first property (Example 7.34, p. 441) states that

$$\delta_n(\mathbf{x})=\delta(x_1)\delta(x_2)\cdots\delta(x_n)$$

where the right hand side denotes a *tensor product of delta's* (see Remark 5.21, p. 314). This formula simply means that the action of the n-dimensional delta measure can be decomposed into the actions of the 1-dimensional delta measures $\delta(x_1),\delta(x_2),\ldots,\delta(x_n)$. In formulas, this means, taking $n=3$ for simplicity:

$$\int\delta_3(\mathbf{x})\varphi(\mathbf{x})d\mathbf{x}=\int\delta_1(x_1)\left(\int\delta(x_2)\left(\int\delta(x_3)\varphi(x_1,x_2,x_3)dx_3\right)dx_2\right)dx_1$$

for every smooth function φ, compactly supported.

The second one (Example 7.24, p. 437) shows how δ_n behaves under a space dilation in $\mathbb{R}^n$. We need it in dimension $n=1$:

$$\delta(ax)=\frac{1}{|a|}\delta(x),\qquad\forall a\neq 0. \tag{5.141}$$

In particular, choosing $a=-1$, we deduce that δ_n is an even distribution, i.e.

$$\delta_n(-\mathbf{x})=\delta_n(\mathbf{x}). \tag{5.142}$$

Using the above properties, we can write our moving source in the following way:

$$S\delta_3(\mathbf{x}-vt\mathbf{k})=S\delta_3(x_1,x_2,x_3-vt)=\frac{S}{v}\delta(x_1)\delta(x_2)\delta\left(t-\frac{x_3}{v}\right).$$

Let us now go back to (5.140) and evaluate the action of $\delta(y_1)\delta(y_2)$, which demands to set $y_1=y_2=0$ everywhere inside the integral. We get:

$$u(\mathbf{x},t)=\frac{S}{4\pi c^2 v}\int\frac{1}{|\mathbf{x}-y_3\mathbf{k}|}\delta\left(t-\frac{|\mathbf{x}-y_3\mathbf{k}|}{c}-\frac{y_3}{v}\right)dy_3. \tag{5.143}$$

The "integration" in (5.143) is extended along the y_3-axis, over the set where the argument of δ vanishes. As we shall see shortly, (5.143) can be handled by means of formula (7.24), in Proposition 7.29, p. 439, and allows us to derive a simple analytical expression for the moving source potential.

However, before computing explicitly u, we draw some interesting information from (5.143). In particular we may ask: *which points* $\mathbf{x}$ *can be affected by the source*

radiation within time t? To answer, observe that the argument of δ vanishes at the points satisfying the equation

$$vt - y_3 = \frac{v}{c} |\mathbf{x} - y_3\mathbf{k}| . \tag{5.144}$$

From (5.143), $u(\mathbf{x}, t)$ at a given point $\mathbf{x}$ and at a given time t is obtained by the superposition of the potentials generated by the moving source when this was located at the points $(0, 0, y_3)$, where y_3 is a solution of (5.144). Clearly we must have $y_3 \le vt$. For instance, the perturbation originated at $y_3 = 0$ has reached, at time t, the sphere $\partial B_{ct}(\mathbf{0})$. More generally, formula (5.144) shows that the perturbation originated at any point $\mathbf{p} = (0, 0, y_3)$, with $y_3 < vt$, has reached, at time t, the points $\mathbf{x}$ on the sphere $\partial B_{\rho_t}(\mathbf{p})$ with

$$\rho_t = \frac{c}{v}(vt - y_3).$$

We now distinguish two cases: the *subsonic case* $v < c$ and the *supersonic case* $v > c$. Set $m = v/c$.

The subsonic case $m < 1$. Let us evaluate explicitly the "integral" in (5.143). The first thing to do is to compute the roots of (5.144), for $(\mathbf{x}, t)$ fixed. Squaring both sides, we get

$$v^2t^2 - 2vty_3 + y_3^2 = \frac{v^2}{c^2}(|\mathbf{x}|^2 - 2x_3y_3 + y_3^2)$$

or

$$(1 - m^2)y_3^2 - 2(vt - m^2x_3)y_3 + v^2t^2 - m^2 |\mathbf{x}|^2 = 0.$$

Setting $\mathbf{x}' = (x_1, x_2)$, after some algebra we find the roots

$$y_3^{\pm} = \frac{1}{1 - m^2}\left[vt - m^2x_3 \pm m\sqrt{|\mathbf{x}'|^2 (1 - m^2) + (vt - x_3)^2}\right]. \tag{5.145}$$

Since (5.144) must hold, we must check if $vt - y_3^{\pm} \ge 0$. Thus we compute

$$vt - y_3^{\pm} = \frac{1}{1 - m^2}\left[m^2(x_3 - vt) \mp m\sqrt{|\mathbf{x}'|^2 (1 - m^2) + (vt - x_3)^2}\right].$$

Consider y_3^+. Then the denominator is positive and the numerator is negative since it is less than

$$m^2(x_3 - vt) - m |vt - x_3| \le (m^2 - m) |vt - x_3| \le 0.$$

Thus $vt - y_3^+ \le 0$. On the contrary, $vt - y_3^- \ge 0$ since $m^2(x_3 - vt) + m |vt - x_3| \ge 0$ and therefore the only admissible root is y_3^-.

Now we use formula (7.24), p. 439, with

$$g(y_3) = t - \frac{|\mathbf{x} - y_3\mathbf{k}|}{c} - \frac{y_3}{v}.$$

Accordingly, we can write

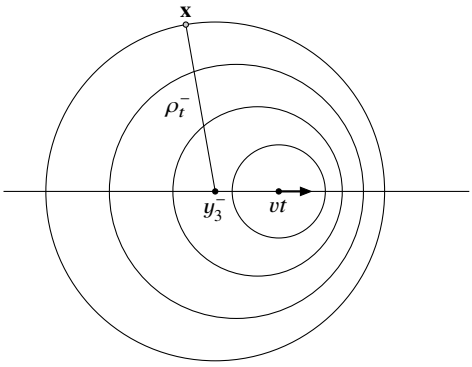

Fig. 5.11 The subsonic case: $\rho_t^- = \frac{c}{v}\left|vt - y_3^-\right|$

$$\delta(g(y_3)) = \frac{\delta(y_3 - y_3^-)}{\left|g'(y_3^-)\right|}$$

and (5.143) yields

$$u(\mathbf{x}, t) = \frac{S}{4\pi c^2 v} \frac{1}{\left|\mathbf{x} - y_3^- \mathbf{k}\right| \left|g'(y_3^-)\right|}.$$

Now, $g'(y_3) = -\frac{y_3 - x_3}{c|\mathbf{x} - y_3\mathbf{k}|} - \frac{1}{v}$ and, recalling (5.144) and (5.145),

$$\left|\mathbf{x} - y_3^- \mathbf{k}\right| \left|g'(y_3^-)\right| = \frac{\left|c\left|\mathbf{x} - y_3^- \mathbf{k}\right| + v(y_3^- - x_3)\right|}{cv} = \frac{\left|(vt - y_3^-)c^2/v + v(y_3^- - x_3)\right|}{cv}$$
$$= \frac{c}{v^2}\left|vt - m^2 x_3 - (1 - m^2) y_3^-\right| = \frac{1}{v}\sqrt{|\mathbf{x}'|^2 (1 - m^2) + (vt - x_3)^2}.$$

In conclusion, we obtain

$$u(\mathbf{x}, t) = \frac{S}{4\pi c^2} \frac{1}{\sqrt{|\mathbf{x}'|^2 (1 - m^2) + (vt - x_3)^2}} \tag{5.146}$$

which is defined in $\mathbb{R}^3 \setminus (0, 0, vt)$.

The radiation effect is described in Fig. 5.11, where the **k**-direction is horizontal. Any point **x** can be reached at time t by the perturbation originated by the source when it was located at a suitable point $\mathbf{p} = (0, 0, y_3^-)$. A similar picture occurs when we hear the siren of an ambulance moving along a straight road at constant speed.

The supersonic case $m > 1$. In this case, the family of spheres $\partial B_{\rho_t}(\mathbf{p})$, centered at $\mathbf{p} = (0, 0, y_3)$, with radius $\rho_t = (c/v)(vt - y_3)$, $y_3 < vt$, has an envelope, given by the circular cone Γ around the **k** axis, with vertex at $(0, 0, vt)$ and opening $\theta = \arcsin(c/v)$. The equation of Γ, known as the **Mach cone** (Fig. 5.12), is

$$vt - x_3 = \cot\theta \ |\mathbf{x}'| = \sqrt{(m^2 - 1)}\,|\mathbf{x}'|\,. \tag{5.147}$$

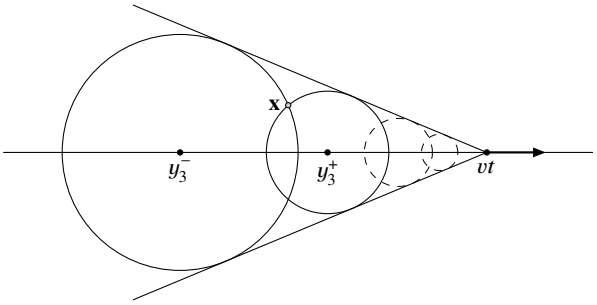

Fig. 5.12 The Mach cone in the supersonic case

Only the points inside Γ can be reached by the perturbation. Hence $u = 0$ outside Γ. To compute the "integral" in (5.143), we observe that both the roots $y_3^{\pm}$ given in (5.145) satisfy the condition $vt - y_3^{\pm} \geq 0$, since both the numerator and the denominator are negative (recall that $x_3 - vt \leq 0$ from (5.147)). Thus both radiations from $(0, 0, y_3^+)$ and $(0, 0, y_3^-)$ contributes to the potential u and formula (7.24) gives

$$\delta(g(y_3)) = \frac{\delta(y_3 - y_3^-)}{|g'(y_3^-)|} + \frac{\delta(y_3 - y_3^+)}{|g'(y_3^+)|}.$$

However, since

$$|\mathbf{x} - y_3^- \mathbf{k}| \, |g'(y_3^-)| = |\mathbf{x} - y_3^+ \mathbf{k}| \, |g'(y_3^+)| = \frac{1}{v} \sqrt{(vt - x_3)^2 - |\mathbf{x}'|^2 \, (m^2 - 1)},$$

we finally obtain

$$u(\mathbf{x}, t) = \begin{cases} \dfrac{S}{2\pi c^2} \dfrac{1}{\sqrt{(vt - x_3)^2 - |\mathbf{x}'|^2 \, (m^2 - 1)}} & \text{inside } \Gamma, \\ 0 & \text{outside } \Gamma. \end{cases} \tag{5.148}$$

The radiation effect is described in Fig. 5.12. A similar configuration arises when a jet is flying at a supersonic speed.

Remark 5.22 On Γ the potential u becomes infinite. Thus, if we interpret u as a sound potential ϕ or a condensation s, as in Sect. 5.7.2, near Γ, the small amplitude assumption leading to the wave equation is not anymore consistent: thus the formula (5.148) is not the proper one, for points near Γ.

What still remains to do is to check rigorously that the potentials u, given in (5.146) **and** (5.148) **satisfies** (5.139) **in a suitable sense.** This is done in Example 7.36, p. 442, in Chap. 7.

5.11 Linear Water Waves

A great variety of interesting phenomena occurs in the analysis of water waves. Here we briefly analyze surface water waves, that is, disturbances of the free surface of an incompressible fluid, resulting from the balance between a restoring force, due to gravity and/or surface tension, and fluid inertia, due to an external action (such as wind, passage of a ship, sub-sea earthquakes). We will focus on the special case of *linear waves*, whose amplitude is small compared to wavelength, analyzing the dispersive relation in the approximation of deep water.

5.11.1 A model for surface waves

We start by deriving a basic model for surface water waves, assuming the following hypotheses.

1. The fluid has *constant density* ρ and *negligible viscosity*. In particular, the force exerted on a control fluid volume V by the rest of the fluid is given by the normal pressure $-p\boldsymbol{\nu}$ on ∂V, where $\boldsymbol{\nu}$ is the exterior normal unit vector to ∂V.
2. The *atmospheric pressure* above the fluid surface is *constant* (e.g. no wind) and we neglect the small air pressure variations, due to the fluid motion. In this case the fluid surface is called a **free surface**.
3. The motion is *laminar* (no breaking waves or turbulence) and *two dimensional*. This means that in a suitable coordinate system x, z, where the coordinate x measures horizontal distance and z is a vertical coordinate (Fig. 5.13), we can describe the free surface by a function $z = h(x, t)$, while the velocity vector has the form $\mathbf{w} = u(x, z, t)\mathbf{i} + v(x, z, t)\mathbf{k}$.
4. The motion is *irrotational*, so that there exists a (smooth) scalar potential $\phi = \phi(x, z, t)$ such that:
$$\mathbf{w} = \nabla\phi = \phi_x\mathbf{i} + \phi_z\mathbf{k}.$$

The mathematical model governing the waves motion involves the unknowns h and ϕ, together with initial conditions and suitable conditions at the boundary of our relevant domain, composed by the free surface, the lower boundary and the lateral sides.

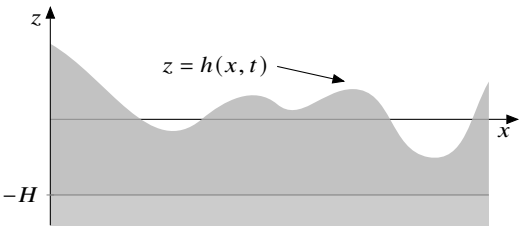

Fig. 5.13 Vertical section of the fluid region

We assume that the side boundaries are so far away that their influence can be neglected. Therefore x varies all along the real axis.

Furthermore, we assume, for simplicity, that the lower boundary is flat, at the level $z = -H$.

Two equations for h and ϕ come from conservation of mass and balance of linear momentum, taking into account hypotheses 1 to 4 above.

The *mass conservation* gives:

$$\operatorname{div} \mathbf{w} = \Delta\phi = 0, \qquad x \in \mathbb{R}, \ -H < z < h(x,t). \tag{5.149}$$

Thus, ϕ is a harmonic function (at any time t).

The *balance of linear momentum* yields:

$$\mathbf{w}_t + (\mathbf{w} \cdot \nabla)\mathbf{w} = \mathbf{g} - \frac{1}{\rho}\nabla p \tag{5.150}$$

where $\mathbf{g}$ is the gravitational acceleration.

Let us rewrite (5.150) in terms of the potential ϕ. From the identity

$$\mathbf{w} \times \operatorname{curl} \mathbf{w} = \frac{1}{2}\nabla(|\mathbf{w}|^2) - (\mathbf{w} \cdot \nabla)\mathbf{w},$$

since $\operatorname{curl} \mathbf{w} = \mathbf{0}$, we obtain

$$(\mathbf{w} \cdot \nabla)\mathbf{w} = \frac{1}{2}\nabla(|\nabla\phi|^2).$$

Moreover, writing $\mathbf{g} = \nabla(-gz)$, (5.150) becomes

$$\frac{\partial}{\partial t}(\nabla\phi) + \frac{1}{2}\nabla(|\nabla\phi|^2) = -\frac{1}{\rho}\nabla p + \nabla(-gz)$$

or

$$\nabla\left\{\phi_t + \frac{1}{2}|\nabla\phi|^2 + \frac{p}{\rho} + gz\right\} = 0.$$

As a consequence

$$\phi_t + \frac{1}{2}|\nabla\phi|^2 + \frac{p}{\rho} + gz = C(t)$$

where $C = C(t)$ is an arbitrary function. Since ϕ is uniquely defined up to an additive function of time, we can choose $C(t) \equiv 0$ without affecting the velocity field.

In this case, we obtain **Bernoulli's equation**

$$\phi_t + \frac{1}{2}|\nabla\phi|^2 + \frac{p}{\rho} + gz = 0. \tag{5.151}$$

We consider now the boundary conditions. We impose the so called **bed condition**, according to which the normal component of the velocity vanishes on the bottom; therefore

$$\phi_z(x,-H,t)=0, \qquad x\in\mathbb{R}. \tag{5.152}$$

More delicate is the condition to be prescribed on the free surface $z=h(x,t)$. In fact, since this surface is itself an unknown of the problem, we actually need *two conditions* on it.

The first one comes from Bernoulli's equation. Namely, the total pressure on the free surface is given by

$$p=p_f-p_{at}=-\sigma h_{xx}\left\{1+h_x^2\right\}^{-3/2}. \tag{5.153}$$

In (5.153) the term p_f is the pressure of the fluid below the free surface, p_{at} is the atmospheric pressure, while the last term is due to the *surface tension*, as we will shortly see.

Thus, inserting $z=h(x,t)$ and (5.153) into (5.151), we obtain the following **dynamic condition at the free surface:**

$$\phi_t+\frac{1}{2}|\nabla\phi|^2-\frac{\sigma h_{xx}}{\rho\left\{1+h_x^2\right\}^{3/2}}+gh=0, \qquad x\in\mathbb{R},\, z=h(x,t). \tag{5.154}$$

A second condition follows imposing that the fluid particles on the free surface always remain there. If the particle path is described by the equations $x=x(t)$, $z=z(t)$, this amounts to requiring that

$$z(t)-h(x(t),t)\equiv 0.$$

Differentiating this equation yields

$$\dot z(t)-h_x(x(t),t)\dot x(t)-h_t(x(t),t)=0$$

that is, since $\dot x(t)=\phi_x(x(t),z(t),t)$ and $\dot z(t)=\phi_z(x(t),z(t),t)$,

$$\phi_z-h_t-\phi_x h_x=0, \qquad x\in\mathbb{R},\, z=h(x,t), \tag{5.155}$$

which is known as the **kinematic condition at the free surface**.

Finally, we require a reasonable behavior of ϕ and h as $x\to\pm\infty$, for instance

$$\int_{\mathbb{R}}|\phi|\,dx<\infty, \quad \int_{\mathbb{R}}|h|\,dx<\infty \quad \text{and} \quad \phi,\,h\to 0 \ \text{ as } x\to\pm\infty. \tag{5.156}$$

Equation (5.149) and the boundary conditions (5.152), (5.154), (5.155) constitute our model for water waves. After a brief justification of formula (5.153), in the next subsection we go back to the above model, deriving a dimensionless formulation and a linearized version of it.

• *Effect of surface tension.* In a water molecule the two hydrogen atoms take an asymmetric position with respect to the oxygen atom. This asymmetric structure generates an electric dipole moment. Inside a bulk of water these moments balance,

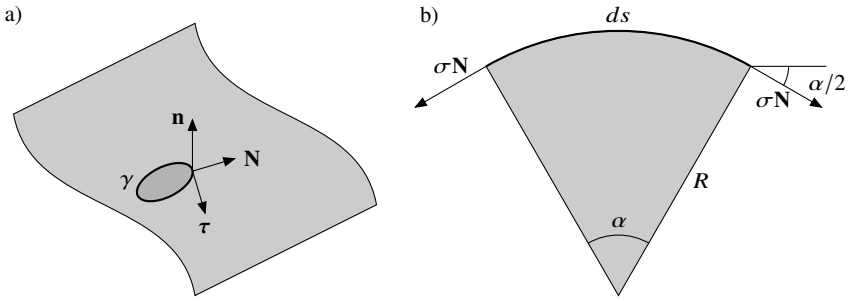

Fig. 5.14 Surface tension $\sigma\mathbf{N}$

but on the surface they tend to be parallel and create a macroscopic inter-molecular force per unit length, confined to the surface, called *surface tension*.

The way this force manifests itself is similar to the action exerted on a small portion of an elastic material by the surrounding material and described by a *stress vector*, representing a force per unit area, on the boundary of the portion. Similarly, consider a small region on the water surface, delimited by a closed curve γ. The surface water on one side of γ exerts on the water on the other side a (*pulling*) *force per unit length* $\mathbf{f}$ along γ.

Let $\mathbf{n}$ be a unit vector normal to the water surface and $\boldsymbol{\tau}$ be a unit tangent vector to γ (Fig. 5.14a) such that $\mathbf{N} = \boldsymbol{\tau} \times \mathbf{n}$ points outwards the region bounded by γ. A simple constitutive law for $\mathbf{f}$ is

$$\mathbf{f}(\mathbf{x}, t) = \sigma(\mathbf{x}, t)\mathbf{N}(\mathbf{x}, t), \qquad \mathbf{x} \in \gamma.$$

Thus, $\mathbf{f}$ acts in the direction of $\mathbf{N}$; its magnitude σ, independent of $\mathbf{N}$, is called **surface tension**.

Formula (5.153) is obtained by balancing the net vertical component of the force produced by surface tension with the difference of the pressure forces across the surface.

Consider the section ds of a small surface element shown in Fig. 5.14b. A surface tension of magnitude σ acts tangentially at both ends. Up to higher order terms, the downward vertical component is given by $2\sigma \sin(\alpha/2)$. On the other hand, this force is equal to $(p_{at} - p_f)ds$ where p_f is the fluid pressure beneath the surface. Thus,

$$(p_{at} - p_f)ds = \pm 2\sigma \sin(\alpha/2),$$

where the "+" sign ("−" sign) corresponds to a *convex* (*concave*) surface element. Since we have $ds = R\alpha$ and, for small α, $2\sin(\alpha/2) \approx \alpha$, we may write

$$p_{at} - p_f \approx \pm\frac{\sigma}{R} = \sigma\kappa \tag{5.157}$$

where κ is the **curvature of the surface**. If the atmospheric pressure prevails, the curvature is positive and the surface is convex, otherwise the curvature is negative

and the surface is concave, as in Fig. 5.14b. If the surface is described by $z = h(x,t)$, we have

$$\kappa = \frac{h_{xx}}{\left\{1 + h_x^2\right\}^{3/2}}$$

which, inserted into (5.157), gives (5.153).

5.11.2 Dimensionless formulation and linearization

The nonlinearities in (5.154), (5.155) and the fact that the free surface itself is an unknown of the problem make the above model quite difficult to analyze by elementary means. However, if we restrict our considerations to waves *whose amplitude is much smaller than their wavelength*, then both difficulties disappear. In spite of this simplification, the resulting theory has a rather wide range of applications, since it is not rare to observe waves with amplitude from 1 to 2 meters and a wavelength of up to a kilometer or more.

To perform a correct linearization procedure, we first introduce *dimensionless* variables. Denote by λ, A and T, an average *wavelength, amplitude and period*[18], respectively. Set

$$\tau = \frac{t}{T}, \qquad \xi = \frac{x}{\lambda}, \qquad \eta = \frac{z}{\lambda}.$$

Since the dimensions of h and ϕ are, respectively [length] and $[\text{length}]^2 \times [\text{time}]^{-1}$, we may rescale ϕ and h by setting:

$$\Phi(\xi,\eta,\tau) = \frac{T}{\lambda A}\phi(\lambda\xi,\lambda\eta,T\tau),$$

$$\Gamma(\xi,\tau) = \frac{1}{A}h(\lambda\xi,\lambda\eta,T\tau).$$

In terms of these dimensionless variables, our model becomes, after elementary calculations:

$$\begin{array}{ll}
\Delta\Phi = 0, & -H_0 < \eta < \varepsilon\Gamma(\xi,\tau),\ \xi \in \mathbb{R} \\
\Phi_\tau + \dfrac{\varepsilon}{2}|\nabla\Phi|^2 + \mathcal{F}\left\{\Gamma - \mathcal{B}\Gamma_{\xi\xi}\left\{1+\varepsilon^2\Gamma_\xi^2\right\}^{3/2}\right\} = 0, & \eta = \varepsilon\Gamma(\xi,\tau), \xi \in \mathbb{R} \\
\Phi_\eta - \Gamma_\tau - \varepsilon\Phi_\xi\Gamma_\xi = 0 & \eta = \varepsilon\Gamma(\xi,\tau), \xi \in \mathbb{R} \\
\Phi_\eta(\xi,-H_0,\tau) = 0, & \xi \in \mathbb{R},
\end{array}$$

where we have emphasized the four dimensionless combinations[19]

$$\varepsilon = \frac{A}{\lambda}, \qquad H_0 = \frac{H}{\lambda}, \qquad \mathcal{F} = \frac{gT^2}{\lambda}, \qquad \mathcal{B} = \frac{\sigma}{\rho g\lambda^2}. \tag{5.158}$$

[18] That is, the time a crest takes to travel a distance of order λ.

[19] Note that the relevant parameters are reduced from seven $(A, \lambda, T, H, g, \sigma, \rho)$ to four.

The parameter $\mathcal{B}$, called *Bond number*, measures the relevance of surface tension while $\mathcal{F}$, the *Froude number*, measures the relevance of gravity.

At this point, the assumption of *small amplitude compared to the wavelength*, translates simply into

$$\varepsilon = \frac{A}{\lambda} \ll 1$$

and the linearization of the above system is achieved by letting $\varepsilon = 0$:

$$\begin{array}{ll} \Delta\Phi = 0, & -H_0 < \eta < 0,\ \xi \in \mathbb{R} \\ \Phi_\tau + \mathcal{F}\left\{\Gamma - \mathcal{B}\Gamma_{\xi\xi}\right\} = 0, & \eta = 0, \xi \in \mathbb{R} \\ \Phi_\eta - \Gamma_\tau = 0, & \eta = 0, \xi \in \mathbb{R} \\ \Phi_\eta(\xi, -H_0, \tau) = 0, & \xi \in \mathbb{R}. \end{array}$$

Going back to the original variables, we finally obtain the linearized system

$$\begin{cases} \Delta\phi = 0, & -H < z < 0,\ x \in \mathbb{R} & \text{(Laplace)} \\ \phi_t + gh - \frac{\sigma}{\rho} h_{xx} = 0, & z = 0, x \in \mathbb{R} & \text{(dynamic)} \\ \phi_z - h_t = 0, & z = 0, x \in \mathbb{R} & \text{(kinematic)} \\ \phi_z(x, -H, t) = 0, & x \in \mathbb{R} & \text{(bed condition).} \end{cases} \tag{5.159}$$

It is possible to obtain an equation for ϕ only. Differentiate twice with respect to x the kinematic equation and use $\phi_{xx} = -\phi_{zz}$; this yields

$$h_{txx} = \phi_{zxx} = -\phi_{zzz}. \tag{5.160}$$

Differentiate the dynamic condition with respect to t , then use $h_t = \phi_z$ and (5.160). The result is:

$$\phi_{tt} + g\phi_z + \frac{\sigma}{\rho}\phi_{zzz} = 0, \qquad z = 0, x \in \mathbb{R}. \tag{5.161}$$

5.11.3 Deep water waves

We solve now system (5.159) with the following initial conditions:

$$\phi(x, z, 0) = 0, \quad h(x, 0) = h_0(x), \quad h_t(x, 0) = 0. \tag{5.162}$$

Thus, initially ($t = 0$) the fluid velocity is zero and the free surface has been perturbed into a non-horizontal profile h_0, that we assume (for simplicity) smooth, even (i.e. $h_0(-x) = h_0(x)$) and *compactly supported*. In addition we consider the case of *deep water* ($H \ggg 1$) so that the bed condition can be replaced by[20]

$$\phi_z(x, z, t) \to 0 \quad \text{as } z \to -\infty. \tag{5.163}$$

[20] For the case of finite depth see Problem 5.21.

The resulting initial-boundary value problem is not of the type we considered so far, but we are reasonably confident that it is well posed. Since x varies over the whole real axis, we may use the Fourier transform with respect to x, setting

$$\widehat{\phi}(k,z,t) = \int_{\mathbb{R}} e^{-ikx}\phi(x,z,t)dx, \qquad \widehat{h}(k,t) = \int_{\mathbb{R}} e^{-ikx}h(x,t)dx.$$

Note that, the assumptions on h_0 implies that $\widehat{h}_0(k) = \widehat{h}(k,0)$ rapidly vanishes as $|k| \to \infty$ and that $\widehat{h}_0(-k) = \widehat{h}_0(k)$. Moreover, since $\widehat{\phi}_{xx} = -k^2\widehat{\phi}$, the Laplace equation transforms into the ordinary differential equation

$$\widehat{\phi}_{zz} - k^2\widehat{\phi} = 0,$$

whose general integral is

$$\widehat{\phi}(k,z,t) = A(k,t)e^{|k|z} + B(k,t)e^{-|k|z}. \tag{5.164}$$

From (5.163) we deduce $B(k,t) = 0$, so that

$$\widehat{\phi}(k,z,t) = A(k,t)e^{|k|z}. \tag{5.165}$$

Transforming (5.161) we get

$$\widehat{\phi}_{tt} + g\widehat{\phi}_z + \frac{\sigma}{\rho}\widehat{\phi}_{zzz} = 0, \qquad z = 0,\ k \in \mathbb{R}$$

and (5.165) yields for A the equation

$$A_{tt} + (g|k| + \frac{\sigma}{\rho}|k|^3)A = 0.$$

Thus, we obtain

$$A(k,t) = a(k)e^{i\omega(k)t} + b(k)e^{-i\omega(k)t}$$

where

$$\omega(k) = \sqrt{g|k| + \frac{\sigma}{\rho}|k|^3},$$

is a **dispersion relation** and

$$\widehat{\phi}(k,z,t) = \left\{a(k)e^{i\omega(k)t} + b(k)e^{-i\omega(k)t}\right\}e^{|k|z}.$$

To determine $a(k)$ and $b(k)$, observe that the *dynamic condition* gives

$$\widehat{\phi}_t(k,0,t) + \left\{g + \frac{\sigma}{\rho}k^2\right\}\widehat{h}(k,t) = 0, \qquad k \in \mathbb{R} \tag{5.166}$$

from which

$$i\omega(k)\left\{a(k)e^{i\omega(k)t} - b(k)e^{-i\omega(k)t}\right\} + \left(g + \frac{\sigma}{\rho}k^2\right)\widehat{h}(k,t) = 0, \qquad k \in \mathbb{R}$$

and for $t = 0$

$$i\omega(k)\left\{a(k) - b(k)\right\} + \left(g + \frac{\sigma}{\rho}k^2\right)\widehat{h}_0(k) = 0. \tag{5.167}$$

Similarly, the *kinematic condition* gives

$$\widehat{\phi}_z(k,0,t) + \widehat{h}_t(k,t) = 0, \qquad k \in \mathbb{R}. \tag{5.168}$$

From (5.164) we have

$$\widehat{\phi}_z(k,0,t) = |k|\left\{a(k)e^{i\omega(k)t} + b(k)e^{-i\omega(k)t}\right\}$$

and since $\widehat{h}_t(k,0) = 0$, we get, from (5.168) for $t = 0$ and $k \neq 0$,

$$a(k) + b(k) = 0. \tag{5.169}$$

Then from (5.167) and (5.169) we have ($k \neq 0$)

$$a(k) = -b(k) = \frac{i\left(g + \frac{\sigma}{\rho}k^2\right)}{2\omega(k)}\widehat{h}_0(k)$$

and therefore

$$\widehat{\phi}(k,z,t) = \frac{i\left(g + \frac{\sigma}{\rho}k^2\right)}{2\omega(k)}\left\{e^{i\omega(k)t} - e^{-i\omega(k)t}\right\}e^{|k|z}\widehat{h}_0(k).$$

From (5.166) we deduce:

$$\widehat{h}(k,t) = \left(g + \frac{\sigma}{\rho}k^2\right)^{-1}\widehat{\phi}_t(k,0,t) = \frac{1}{2}\left\{e^{i\omega(k)t} + e^{-i\omega(k)t}\right\}\widehat{h}_0(k)$$

and finally, transforming back[21]

$$h(x,t) = \frac{1}{4\pi}\int_{\mathbb{R}}\left\{e^{i(kx-\omega(k)t)} + e^{i(kx+\omega(k)t)}\right\}\widehat{h}_0(k)\,dk. \tag{5.170}$$

[21] Note that, since also $\omega(k)$ is even, we may write

$$h(x,t) = \frac{1}{2\pi}\int_{\mathbb{R}}\cos\left[kx - \omega(k)t\right]\hat{h}_0(k)dk.$$

5.11.4 Interpretation of the solution

We now examine some remarkable features of the solution (5.170). The surface displacement appears in **wave packet** form and the dispersion relation

$$\omega(k) = \sqrt{g\,|k| + \frac{\sigma}{\rho}\,|k|^3}$$

shows that each Fourier component of the initial free surface propagates both in the positive and negative x-directions. The phase and group velocities are (considering only $k > 0$, for simplicity)

$$c_p(k) = \frac{\omega(k)}{k} = \sqrt{\frac{g}{k} + \frac{\sigma k}{\rho}} \quad \text{and} \quad c_g(k) = \omega'(k) = \frac{g + 3\sigma k^2/\rho}{2\sqrt{gk + \sigma k^3/\rho}}.$$

Thus, we see that the speed c_p of a wave of wavelength $\lambda = 2\pi/k$ **depends on its wavelength**. A fundamental parameter is

$$B^* = 4\pi^2 \mathcal{B} = \frac{\sigma k^2}{\rho g}$$

where $B = \frac{\sigma}{\rho g \lambda^2}$ is the Bond number. For water, under "normal" conditions,

$$\rho = 1 \text{ gr/cm}^3,\ \sigma = 72 \text{ gr/sec}^2,\ g = 980 \text{ cm/sec}^2 \tag{5.171}$$

so that $B^* = 1$ for wavelengths $\lambda \simeq 1.7$ cm. When $\lambda \gg 1.7$ cm, then $B^* \ll 1$, $k = \frac{2\pi}{\lambda} \ll 1$ and surface tension becomes negligible. This is the case of **gravity waves** (generated e.g. by dropping a stone into a pond) whose phase speed and group velocity are well approximated by

$$c_p \simeq \sqrt{\frac{g}{k}} = \sqrt{\frac{g\lambda}{2\pi}} \quad \text{and} \quad c_g \simeq \frac{1}{2}\sqrt{\frac{g}{k}} \simeq \frac{1}{2}c_p.$$

Thus, **longer waves move faster** and **energy is slower than the crests** (half speed).

On the other hand, if $\lambda \ll 1.7$ cm, then $B^* \gg 1$, $k = \frac{2\pi}{\lambda} \gg 1$ and this time surface tension prevails over gravity. In fact, short wavelengths are associated with relative high curvature of the free surface and high curvature is concomitant with large surface tension effects. This is the case of **capillarity waves** (generated e.g. by raindrops in a pond) and their speed and group velocity are well approximated by

$$c_p \simeq \sqrt{\frac{\sigma k}{\rho}} = \sqrt{\frac{2\pi\sigma}{\lambda\rho}} \quad \text{and} \quad c_g \simeq \frac{3}{2}\sqrt{\frac{\sigma k}{\rho}} \simeq \frac{3}{2}c_p.$$

Thus **shorter waves move faster** and **energy is faster than the crests**.

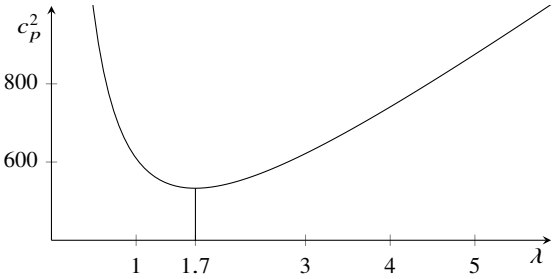

Fig. 5.15 c_p^2 versus λ

When both gravity and surface tension are relevant (**capillarity-gravity waves**), Fig. 5.15 shows the graph of c_p^2 versus λ, for water, with the values (5.171):

$$c_p^2 = 156.97\,\lambda + \frac{452.39}{\lambda}.$$

The main feature of this graph is the presence of the minimum

$$c_{\min} = 23 \text{ cm/sec}$$

corresponding just to the value $\lambda = 1.7$ cm. The consequence is curious: linear gravity and capillarity deep water waves can appear simultaneously only when the speed is greater than 23 cm/sec. A typical situation occurs when a small obstacle (e.g. a twig) moves at speed v in still water. The motion of the twig results in the formation of a wave system that moves along with it, with gravity waves behind and capillarity waves ahead. In fact, the result above shows that this wave system can actually appear only if $v > 23$ cm/sec.

5.11.5 Asymptotic behavior

As we have already observed, the behavior of a wave packet is dominated for short times by the initial conditions and only after a relatively long time it is possible to observe the intrinsic features of the perturbation. For this reason, information about the asymptotic behavior of the packet as $t \to +\infty$ are important. Thus, we need a good asymptotic formula for the integral in (5.170) when $t \gg 1$.

For simplicity, we consider gravity waves only, for which $\omega(k) \simeq \sqrt{g\,|k|}$.

Let us follow a particle $x = x(t)$ moving along the positive $x-$direction with constant speed $v > 0$, so that $x = vt$. Inserting $x = vt$ into (5.170) we find

$$\begin{aligned} h(vt,t) &= \frac{1}{4\pi}\int_{\mathbb{R}} e^{it(kv-\omega(k))}\widehat{h}_0(k)\,dk + \frac{1}{4\pi}\int_{\mathbb{R}} e^{it(kv+\omega(k))}\widehat{h}_0(k)\,dk \\ &\equiv h_1(vt,t) + h_2(vt,t). \end{aligned}$$

According to Theorem 5.23 in the next subsection (see also Remark 5.25, p. 335), with

$$\varphi(k) = kv - \omega(k),$$

if there exists exactly one stationary point for φ, i.e. only one point k_0 such that

$$\omega'(k_0) = v \quad \text{and} \quad \varphi''(k_0) = -\omega''(k_0) \neq 0,$$

we may estimate h_1, for $t \gg 1$, by the following formula:

$$h_1(vt, t) = \frac{A(k_0)}{\sqrt{t}} \exp\{it[k_0 v - \omega(k_0)]\} + O(t^{-1}) \tag{5.172}$$

where

$$A(k_0) = \widehat{h}_0(k_0)\sqrt{\frac{1}{8\pi\,|\omega''(k_0)|}}\exp i\left\{-\frac{\pi}{4}\operatorname{sign}\omega''(k_0)\right\}.$$

We have ($k \neq 0$),

$$\omega'(k) = \frac{1}{2}\sqrt{g}\,|k|^{-1/2}\operatorname{sign}(k) \qquad \text{and} \qquad \omega''(k) = -\frac{\sqrt{g}}{4}\,|k|^{-3/2}.$$

Since $v > 0$, the equation $\omega'(k_0) = v$ gives the unique *point of stationary phase*

$$k_0 = \frac{g}{4v^2} = \frac{gt^2}{4x^2}.$$

Moreover,

$$k_0 v - \omega(k_0) = -\frac{g}{4v} = -\frac{gt}{4x}$$

and

$$\omega''(k_0) = -\frac{2v^3}{g} = -\frac{2x^3}{gt^3} < 0.$$

Hence, from (5.172), we find

$$h_1(vt, t) = \frac{1}{4}\widehat{h}_0\left(\frac{g}{4v^2}\right)\sqrt{\frac{g}{\pi t v^3}}\exp i\left\{-\frac{gt}{4v} + \frac{\pi}{4}\right\} + O(t^{-1}).$$

Similarly, since $\widehat{h}_0(k_0) = \widehat{h}_0(-k_0$, we find

$$h_2(vt, t) = \frac{1}{4}\widehat{h}_0\left(\frac{g}{4v^2}\right)\sqrt{\frac{g}{\pi t v^3}}\exp i\left\{\frac{gt}{4v} - \frac{\pi}{4}\right\} + O(t^{-1}).$$

Finally,

$$h(vt, t) = h_1(vt, t) + h_2(vt, t) = \widehat{h}_0\left(\frac{g}{4v^2}\right)\sqrt{\frac{g}{4\pi v^3 t}}\cos\left\{\frac{gt}{4v} - \frac{\pi}{4}\right\} + O(t^{-1}).$$

This formula shows that, for large x and t, with $x/t = v$, constant, the wave packet is locally sinusoidal with wave number

$$k(x,t) = \frac{gt}{4vx} = \frac{gt^2}{4x^2}.$$

In other words, an observer moving at the constant speed $v = x/t$ sees a dominant wavelength $2\pi/k_0$, where k_0 is the solution of $\omega'(k_0) = x/t$. The amplitude decreases as $t^{-1/2}$. This is due to the dispersion of the various Fourier components of the initial configuration, after a sufficiently long time.

5.11.6 The method of stationary phase

The *method of stationary phase*, essentially due to Laplace, gives an asymptotic formula for integrals of the form

$$I(t) = \int_a^b f(k)e^{it\varphi(k)}dk \qquad (-\infty \le a < b \le \infty)$$

as $t \to +\infty$. Actually, only the real part of $I(t)$, in which the factor $\cos[t\varphi(k)]$ appears, is of interest. Now, as t is very large and $\varphi(k)$ varies, we expect that $\cos[t\varphi(k)]$ oscillates eventually much more than f. For this reason, the contributions of the intervals where $\cos[t\varphi(k)] > 0$ will balance those in which $\cos[t\varphi(k)] < 0$, so that we expect that $I(t) \to 0$ as $t \to +\infty$, just as the Fourier coefficients of an integrable function tend to zero as the frequency goes to infinity.

To obtain information on the vanishing speed, assume φ is constant on a certain interval J. On this interval $\cos[t\varphi(k)]$ is constant as well and hence there are neither oscillations nor cancellations. Thus, it is reasonable that, for $t \gg 1$, the relevant contributions to $I(t)$ come from intervals where φ is constant or at least almost constant. The same argument suggests that eventually, a however small interval, containing a stationary point k_0 for φ, will contribute to the integral much more than any other interval without stationary points.

The method of stationary phase makes the above argument precise through the following theorem.

Theorem 5.23 *Let f and φ belong to $C^2([a,b])$. Assume that*

$$\varphi'(k_0) = 0, \varphi''(k_0) \neq 0 \quad \text{and} \quad \varphi'(k) \neq 0 \text{ for } k \neq k_0.$$

Then, as $t \to +\infty$

$$\int_a^b f(k)e^{it\varphi(k)}dk = \sqrt{\frac{2\pi}{|\varphi''(k_0)|}}\frac{f(k_0)}{\sqrt{t}}\exp\left\{i\left[t\varphi(k_0) + \frac{\pi}{4}\operatorname{sign}\varphi''(k_0)\right]\right\} + O(t^{-1}).$$

To prove the theorem we need first a lemma.

Lemma 5.24 *Let f, φ as in Theorem 5.23. Let $[c,d] \subseteq [a,b]$ and assume that $|\varphi'(k)| \geq C > 0$ in $[c,d]$. Then*

$$\int_c^d f(k) e^{it\varphi(k)} dk = O(t^{-1}) \qquad t \to +\infty. \tag{5.173}$$

Proof Integrating by parts we get (multiplying and dividing by φ'):

$$\int_c^d \frac{f}{\varphi'} \varphi' e^{it\varphi} dk = \frac{1}{it} \left\{ \frac{f(d) e^{it\varphi(d)}}{\varphi'(d)} - \frac{f(c) e^{it\varphi(c)}}{\varphi'(c)} - \int_c^d \frac{f'\varphi' - f\varphi''}{(\varphi')^2} e^{it\varphi} dk \right\}.$$

Thus, from $\left|e^{it\varphi(k)}\right| \leq 1$ and our hypotheses, we have

$$\left| \int_c^d f\, e^{it\varphi} dk \right| \leq \frac{1}{Ct} \left\{ |f(d)| + |f(c)| + \frac{1}{C} \int_c^d |f'\varphi' - f\varphi''|\, dk \right\} \leq \frac{K}{t}$$

which gives (5.173). □

Proof (of Theorem 5.23) Without loss of generality, we may assume $k_0 = 0$, so that $\varphi'(0) = 0$, $\varphi''(0) \neq 0$. From Lemma 5.24, it is enough to consider the integral

$$\int_{-\varepsilon}^{\varepsilon} f(k) e^{it\varphi(k)} dk$$

where $\varepsilon > 0$ is as small as we wish. For simplicity, we give the proof in the case φ is a quadratic polynomial, that is

$$\varphi(k) = \varphi(0) + Ak^2, \; A = \frac{1}{2}\varphi''(0).$$

Write

$$f(k) = f(0) + \frac{f(k) - f(0)}{k} k \equiv f(0) + q(k)k,$$

and observe that, since $f \in C^2([-\varepsilon, \varepsilon])$, $q'(k)$ is bounded in $[-\varepsilon, \varepsilon]$. Then, we have:

$$\int_{-\varepsilon}^{\varepsilon} f(k) e^{it\varphi(k)} dk = 2f(0) e^{it\varphi(0)} \int_0^{\varepsilon} e^{itAk^2} dk + e^{it\varphi(0)} \int_{-\varepsilon}^{\varepsilon} q(k) k e^{itAk^2} dk.$$

Now, an integration by parts shows that the second integral is $O(1/t)$ as $t \to \infty$ (the reader should check the details). In the first integral, if $A > 0$, let $tAk^2 = y^2$. Then

$$\int_0^{\varepsilon} e^{itAk^2} dk = \frac{1}{\sqrt{tA}} \int_0^{\varepsilon\sqrt{tA}} e^{iy^2} dy.$$

Since[22]

$$\int_0^{\varepsilon\sqrt{tA}} e^{iy^2} dy = \frac{\sqrt{\pi}}{2} e^{i\frac{\pi}{4}} + O\left(\frac{1}{\varepsilon\sqrt{tA}}\right),$$

we get

[22] Recall that $e^{i\pi/4} = (\sqrt{2} + i\sqrt{2})/2$. Moreover, the following formulas hold:

$$\left| \frac{\sqrt{\pi}}{2\sqrt{2}} - \int_0^{\lambda} \cos(y^2) dy \right| \leq \frac{1}{\lambda}, \quad \left| \frac{\sqrt{\pi}}{2\sqrt{2}} - \int_0^{\lambda} \sin(y^2) dy \right| \leq \frac{1}{\lambda}.$$

$$\int_0^{\varepsilon} f(k)e^{it\varphi(k)}dk = \sqrt{\frac{2\pi}{|\varphi''(0)|}}\frac{f(0)}{\sqrt{t}}\exp\left\{i\left[\varphi(0)t+\frac{\pi}{4}\right]\right\}+O\left(\frac{1}{t}\right),$$

which proves the theorem when $A > 0$. The proof is similar if $A < 0$. □

Remark 5.25 Theorem 5.23 holds for integrals extended over the whole real axis as well (actually this is the most interesting case) as long as, in addition, f is bounded, $|\varphi'(\pm\infty)| \geq C > 0$, and $\int_{\mathbb{R}} |f'\varphi' - f\varphi''|\,(\varphi')^{-2}dk < \infty$. Indeed, it is easy to check that Lemma 5.24 is true under these hypotheses and then the proof of Theorem 5.23 is exactly the same.

Problems

5.1 The chord of a guitar of length L is plucked at its middle point and then released. Write the mathematical model which governs the vibrations and solve it. Compute the energy $E(t)$.

5.2 Solve the problem

$$\begin{cases} u_{tt} - u_{xx} = 0 & 0 < x < 1, t > 0 \\ u(x,0) = u_t(x,0) = 0 & 0 \leq x \leq 1 \\ u_x(0,t) = 1, u(1,t) = 0 & t \geq 0. \end{cases}$$

5.3 (*Forced vibration*) Using Duhamel method, solve the problem

$$\begin{cases} u_{tt} - u_{xx} = g(t)\sin x & 0 < x < \pi, t > 0 \\ u(x,0) = u_t(x,0) = 0 & 0 \leq x \leq \pi \\ u(0,t) = u(\pi,t) = 0 & t \geq 0. \end{cases}$$

[*Hint.* Find the solution $w(x,t;s)$ of $w_{tt} - w_{xx} = 0$ in $(0,\pi)\times(s,+\infty)$, with $w(x,s;s) = 0, w_t(x,s;s) = g(s)\sin x, 0 \leq x \leq \pi$, and $w(0,t;s) = w(\pi,t;s) = 0, t \geq s$. Then $u(x,t) = \int_0^t w(x,t;s)ds = \cdots$.]

5.4 (*Equipartition of energy*) Let $u = u(x,t)$ be the solution of the global Cauchy problem for the equation $u_{tt} - c^2u_{xx} = 0$, with initial data $u(x,0) = g(x)$, $u_t(x,0) = h(x)$. Assume that g and h are smooth functions with compact support contained in the interval (a,b). Show that there exists T such that, for $t \geq T$, $E_{\text{cin}}(t) = E_{\text{pot}}(t)$.

5.5 Solve the global Cauchy problem for the equation $u_{tt} - c^2u_{xx} = 0$, with the following initial data:

a) $u(x,0) = 1$ if $|x| < a$, $u(x,0) = 0$ if $|x| > a$; $u_t(x,0) = 0$.
b) $u(x,0) = 0$; $u_t(x,0) = 1$ if $|x| < a$, $u_t(x,0) = 0$ if $|x| > a$.

5.6 Consider the global Cauchy problem for the equation $u_{tt} - c^2u_{xx} = 0$. Can we choose the initial data g and h in order to have a solution of the form $u(x,t) = F(x-ct)$ or $u(x,t) = G(x+ct)$?

5.7 Check that formula (5.43) may be written in the following form:

$$u(x+c\tau-c\eta, t+\tau+\eta) - u(x+c\tau, t+\tau) - u(x-c\eta, t+\eta) + u(x,t) = 0 \qquad (5.174)$$

for nonnegative τ, η. Show that if u is a C^2 function and satisfies (5.174) for all $(x,t) \in \mathbb{R}\times(0,+\infty)$ and all $\tau \geq 0$, $\eta \geq 0$, then $u_{tt} - c^2 u_{xx} = 0$. Thus, (5.174) can be considered as a weak formulation of the wave equation.

5.8 The small longitudinal free *vibrations of an elastic bar* are governed by the following equation

$$\rho(x)\sigma(x)\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x}\left[E(x)\sigma(x)\frac{\partial u}{\partial x}\right] \tag{5.175}$$

where u is the longitudinal displacement, ρ is the linear density of the material, σ is the cross section of the bar and E is its *Young's modulus*[23].

Assume the bar has constant cross section but it is constructed by welding together two bars, of different (constant) Young's modulus E_1, E_2 and density ρ_1, ρ_2, respectively.

Since the two bars are welded together, the displacement u is continuous across the junction, which we locate at $x = 0$. In this case:

a) Give a weak formulation of the global initial value problem for equation (5.175).
b) Deduce that the following jump condition must hold at $x = 0$:

$$E_1 u_x(0-,t) = E_2 u_x(0+,t) \qquad t > 0. \tag{5.176}$$

c) Let $c_j^2 = E_j/\rho_j$, $j = 1, 2$. A left incoming wave $u_{\text{inc}}(x,t) = \exp[i(x - c_1 t)]$ produces at the junction a reflected wave $u_{\text{ref}}(x,t) = a\exp[i(x + c_1 t)]$ and a transmitted wave $u_{\text{tr}}(x,t) = b\exp[i(x - c_2 t)]$. Determine a, b and interpret the result.

[*Hint.* c) Look for a solution of the form $u = u_{\text{inc}} + u_{\text{ref}}$ for $x < 0$ and $u = u_{\text{tr}}$ for $x > 0$. Use the continuity of u and the jump condition (5.176).]

5.9 Determine the characteristics of Tricomi's equation $u_{tt} - t u_{xx} = 0$. [*Answer.* $3x \pm 2t^{3/2} = k$, for $t > 0$.]

5.10 Classify the equation $t^2 u_{tt} + 2t u_{xt} + u_{xx} - u_x = 0$ and find the characteristics. After a reduction to canonical form, find the solution.

[*Answer.* $U(x,t) = F(te^{-x}) + G(te^{-x})e^x$, with F, G arbitrary.]

5.11 Consider the following *characteristic Cauchy problem* for the wave equation in the half-plane $x > t$ (note that the data are the values of u and $u_{\boldsymbol{\nu}}$ on the characteristic $x = t$):

$$\begin{cases} u_{tt} - u_{xx} = 0 & x > 0 \\ u(x,x) = f(x) & x \in \mathbb{R} \\ u_{\boldsymbol{\nu}}(x,x) = g(x) & x \in \mathbb{R} \end{cases}$$

where $\boldsymbol{\nu} = (1,-1)/\sqrt{2}$. Establish whether or not this problem is well posed.

5.12 Consider the following so called *Goursat problem* for the wave equation in the sector $-t < x < t$ (note that the data are the values of u on the characteristics $x \pm t = 0$, $x > 0$):

$$\begin{cases} u_{tt} - u_{xx} = 0 & -t < x < t \\ u(x,x) = f(x), u(x,-x) = g(x) & x > 0 \\ f(0) = g(0). \end{cases}$$

Establish whether or not this problem is well posed.

[23] E is the proportionality factor in the *strain-stress* relation given by Hooke's law: T (strain) $= E\,\varepsilon$ (stress). Here $\varepsilon \simeq u_x$. For steel, $E = 2\times 10^{11}$ dine/cm^2, for aluminium, $E = 7\times 10^{12}$ dine/cm^2.

5.13 (*Ill posed non-characteristic Cauchy problem for the heat equation*) Check that for every integer k, the function

$$u_k(x,t) = \frac{1}{k}(\cosh kx \cos kx \cos 2k^2 t - \sinh kx \sin kx \sin 2k^2 t)$$

solves $u_t = u_{xx}$ and the (non-characteristic) initial conditions:

$$u(0,t) = \frac{1}{k}\cos 2k^2 t,\ u_x(0,t) = 0.$$

Deduce that the corresponding Cauchy problem in the half-plane $x > 0$ is **ill posed**.

5.14 (*Circular membrane*) A perfectly flexible and elastic membrane has at rest the shape of the circle $B_1 = \{(x,y) : x^2 + y^2 \le 1\}$. If the boundary is fixed and there are no external loads, the vibrations of the membrane are governed by the following system:

$$\begin{cases} u_{tt} - c^2(u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}) = 0 & 0 < r < 1, 0 \le \theta \le 2\pi, t > 0 \\ u(r,\theta,0) = g(r,\theta), u_t(r,\theta,0) = h(r,\theta) & 0 < r < 1, 0 \le \theta \le 2\pi \\ u(1,\theta,t) = 0 & 0 \le \theta \le 2\pi, t \ge 0. \end{cases}$$

In the case $h = 0$ and $g = g(r)$, use the method of separation of variables to find the solution

$$u(r,t) = \sum_{n=1}^{\infty} a_n J_0(\lambda_n r)\cos\lambda_n t$$

where J_0 is the Bessel function of order zero, $\lambda_1, \lambda_2, \ldots$ are the zeros of J_0 and the coefficients a_n are given by

$$a_n = \frac{2}{c_n^2}\int_0^1 g(s)J_0(\lambda_n s)s\,ds$$

where (see Remark 2.10, p. 77)

$$c_n = \sum_{n=1}^{\infty} \frac{(-1)^k}{k!(k+1)!}\left(\frac{\lambda_n}{2}\right)^{2k+1}.$$

5.15 (*Circular waveguide*) Consider the equation $u_{tt} - c^2\Delta u = 0$ in the cylinder

$$C_R = \{(r,\theta,z) : 0 \le r \le R, 0 \le \theta \le 2\pi,\ -\infty < z < +\infty\}.$$

Determine the axially symmetric solutions of the form $u(r,z,t) = v(r)w(z)h(t)$ satisfying the Neumann condition $u_r = 0$ on $r = R$.

[*Answer.* $u_n(r,z,t) = \exp\{-i(\omega t - kz)\}J_0(\mu_n r/R)$, $n \in \mathbb{N}$, where J_0 is the Bessel function, μ_n are its stationary points ($J_0'(\mu_n) = 0$) and $\frac{\omega^2}{c^2} = k^2 + \frac{\mu_n^2}{R^2}$.]

5.16 Let u be the solution of $u_{tt} - c^2\Delta u = 0$ in $\mathbb{R}^3 \times (0, +\infty)$ with data

$$u(\mathbf{x},0) = g(\mathbf{x}) \quad \text{and} \quad u_t(\mathbf{x},0) = h(\mathbf{x}),$$

both supported in the sphere $\bar{B}_\rho(\mathbf{0})$. Describe the support of $u(\cdot,t)$ for $t > 0$.

[*Answer.* For $\rho \ge ct$, the support of $u(\cdot,t)$ is $\bar{B}_{\rho+ct}(\mathbf{0})$. For $\rho < ct$, the support of $u(\cdot,t)$ is the spherical shell $\bar{B}_{\rho+ct}(\mathbf{0}) \setminus B_{ct-\rho}(\mathbf{0})$, of width 2ρ, which expands at speed c.]

5.17 (*Focusing effect*) Solve the problem

$$\begin{cases} w_{tt} - c^2\Delta w = 0 & \mathbf{x} \in \mathbb{R}^3, t > 0 \\ w(\mathbf{x},0) = 0, \quad w_t(\mathbf{x},0) = h(|\mathbf{x}|) & \mathbf{x} \in \mathbb{R}^3 \end{cases}$$

where ($r = |\mathbf{x}|$)

$$h(r) = \begin{cases} 1 & 0 \le r \le 1 \\ 0 & r > 1. \end{cases}$$

Check that $w(r,t)$ displays a discontinuity at the origin at time $t = 1/c$.

5.18 Let u be a regular and bounded solution of $u_{tt} - c^2\Delta u = 0$ in $\mathbb{R}^n \times (0,+\infty)$ with $u(\mathbf{x},0) = g(\mathbf{x})$, $u_t(\mathbf{x},0) = 0$ in $\mathbb{R}^n$ ($n = 1,2,3$). Define

$$w(\mathbf{x},t) = \frac{c}{\sqrt{4D\pi t}} \int_{\mathbb{R}} u(\mathbf{x},s) \exp\left(-\frac{c^2 s^2}{4Dt}\right) ds.$$

Show (formally) that w solves the heat equation $w_t - D\Delta w = 0$ in $\mathbb{R}^n \times (0,+\infty)$ with $w(\mathbf{x},0) = g(\mathbf{x})$ in $\mathbb{R}^n$.

5.19 (*Dissipative term*) Consider the problem

$$\begin{cases} u_{tt}(\mathbf{x},t) + k u_t(\mathbf{x},t) = c^2 \Delta u(\mathbf{x},t) & \mathbf{x} \in \mathbb{R}^2,\ t > 0 \\ u(\mathbf{x},0) = 0,\ u_t(\mathbf{x},0) = g(\mathbf{x}) & \mathbf{x} \in \mathbb{R}^2. \end{cases}$$

a) Determine $\alpha \in \mathbb{R}$ so that $v(\mathbf{x},t) = e^{\alpha t} u(\mathbf{x},t)$ solves an equation without first-order term (but with a zero-order one) on $\mathbb{R}^2 \times (0,+\infty)$.
b) Find $\beta \in \mathbb{R}$ so that $w(x_1,x_2,x_3,t) = w(\mathbf{x},x_3,t) = e^{\beta x_3} v(\mathbf{x},t)$ solves an equation with second-order terms only, on $\mathbb{R}^3 \times (0,+\infty)$.
c) Determine the solution u of the original problem.

5.20 *Retarded potential* ($n = 2$)

a) Show that the solution of the two-dimensional nonhomogeneous Cauchy problem with zero initial data is given by

$$u(\mathbf{x},t) = \frac{1}{2\pi c} \int_0^t \int_{B_{c(t-s)}(\mathbf{x})} \frac{1}{\sqrt{c^2(t-s)^2 - |\mathbf{x}-\mathbf{y}|^2}} f(\mathbf{y},s)\, d\mathbf{y} ds.$$

b) *Point source.* Show that if $f(\mathbf{x},t) = \delta_2(\mathbf{x}) q(t)$, q smooth for $t \ge 0$ and zero for < 0 then

$$u(\mathbf{x},t) = \begin{cases} \frac{1}{2\pi c} \int_0^{t-\frac{|\mathbf{x}|}{c}} \frac{q(s)}{\sqrt{c^2(t-s)^2 - |\mathbf{x}|^2}} ds & \text{for } |\mathbf{x}| < ct, \\ 0 & \text{for } |\mathbf{x}| > ct. \end{cases}$$

5.21 For *linear gravity waves* ($\sigma = 0$), examine the case of uniform finite depth, replacing condition (5.163) by $\phi_z(x,-H,t) = 0$ under the initial conditions (5.162).

a) Write the dispersion relation.

Deduce that:

b) The phase and group velocity have a finite upper bound.
c) The square of the phase velocity in deep water ($H \gg \lambda$) is proportional to the wavelength.
d) Linear shallow water waves ($H \ll \lambda$) are not dispersive.

[*Answer.* a) $\omega^2 = gk\tanh(kH)$, b) $c_{p\max} = \sqrt{kH}$, c) $c_p^2 \sim g\lambda/2\pi$, d) $c_p^2 \sim gH$.]

5.22 For linear gravity waves ($\sigma = 0$), determine the travelling wave solutions of the linearized system (5.159) with potential of the form

$$\phi(x,z,t) = F(x-ct)G(z).$$

Rediscover the dispersion relation found in Problem 5.21, a).

[*Answer.* $\phi(x,z,t) = \cosh k(z+H)\,\{A\cos k(x-ct) + B\sin k(x-ct)\}$, A, B arbitrary constants and $c^2 = g\tanh(kH)/k$.]

Chapter 6
Elements of Functional Analysis

6.1 Motivations

The main purpose in the previous chapters has been to introduce part of the basic and classical theory of some important equations of mathematical physics. The emphasis on phenomenological aspects and the connection with a probabilistic point of view should have conveyed to the reader some intuition and feeling about the interpretation and the limits of those models.

The few rigorous theorems and proofs we have presented had the role of bringing to light the main results on the qualitative properties of the solutions and justifying, partially at least, the well-posedness of the relevant boundary and initial/boundary value problems we have considered.

However, these purposes are somehow in competition with one of the most important role of modern mathematics, which is to reach a unifying vision of large classes of problems under a common structure, capable not only of increasing theoretical understanding, but also of providing the necessary flexibility to guide the numerical methods which will be used to compute approximate solutions.

This conceptual jump requires a change of perspective, based on the introduction of abstract methods, historically originating from the vain attempts to solve basic problems (e.g. in electrostatics) at the end of the 19th century. It turns out that the new level of knowledge opens the door to the solution of complex problems in modern technology.

These abstract methods, in which analytical and geometrical aspects fuse, are the core of the branch of Mathematics called Functional Analysis.

It could be useful for understanding the subsequent development of the theory, to examine in an informal way how the main ideas come out, working on a couple of specific examples.

Let us go back to the derivation of the diffusion equation, in Sect. 2.1.2. If the body is heterogeneous or anisotropic, maybe with discontinuities in its thermal parameters (e.g. due to the mixture of two different materials), the Fourier law of heat conduction for the flux function $\mathbf{q}$ takes the form

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
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$$\mathbf{q} = -\mathbf{A}\nabla u,$$

where the matrix $\mathbf{A} = \mathbf{A}(\mathbf{x})$ satisfies the condition

$$\mathbf{q} \cdot \nabla u = -\mathbf{A}\nabla u \cdot \nabla u \leq 0 \qquad (\textit{ellipticity condition}),$$

reflecting the tendency of heat to flow from hotter to cooler regions. If $\rho = \rho(\mathbf{x})$ and $c_v = c_v(\mathbf{x})$ are the density and the specific heat of the material, and $f = f(\mathbf{x})$ is the rate of external heat supply per unit volume, we are led to the diffusion equation

$$\rho c_v u_t - \operatorname{div}(\mathbf{A}\nabla u) = f.$$

In stationary conditions, $u(\mathbf{x}, t) = u(\mathbf{x})$, and we are reduced to

$$-\operatorname{div}(\mathbf{A}\nabla u) = f. \tag{6.1}$$

Since the matrix $\mathbf{A}$ encodes the conductivity properties of the medium, we expect a low degree of regularity of $\mathbf{A}$, but then a natural question arises, since we cannot compute the divergence of $\mathbf{A}\nabla u$:

$$\textit{what is the meaning of equation (6.1)}? \tag{6.2}$$

We have already faced similar situations in Sect. 4.4.2, where we have introduced discontinuous solutions of a conservation law, and in Sect. 5.4.2, where we have considered solutions of the wave equation with irregular initial data. Let us follow the same ideas.

Suppose we want to solve equation (6.1) in a bounded domain Ω, with *zero* boundary data (Dirichlet problem). Formally, we multiply the differential equation by a smooth test function v *vanishing on* $\partial\Omega$, and we integrate over Ω:

$$\int_\Omega -\operatorname{div}(\mathbf{A}\nabla u) v \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}.$$

Using Gauss formula we obtain the equation

$$\int_\Omega \mathbf{A}\nabla u \cdot \nabla v \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x} \quad \forall v \in \mathring{C}^1(\overline{\Omega}), \tag{6.3}$$

where $\mathring{C}^1(\overline{\Omega})$ is the set of functions in $C^1(\overline{\Omega})$, vanishing on $\partial\Omega$.

We call equation (6.3) the *weak* or *variational* formulation of our Dirichlet problem and we may say that $u \in \mathring{C}^1(\overline{\Omega})$ is a *weak* or *variational* solution of our Dirichlet problem if (6.3) holds *for every* $v \in \mathring{C}^1(\overline{\Omega})$.

Note that (6.3) makes perfect sense for $\mathbf{A}$ and f bounded (possibly discontinuous). Fine, but:

$$\textit{Is the variational problem (6.3) well posed}?$$

Things are not so straightforward, as we have experienced in Sect. 4.4.3 and, actually, it turns out that the space $\mathring{C}^1(\overline{\Omega})$ is not the proper choice, although it seems

to be the natural one. To see why, let us consider another example, somewhat more revealing.

Consider the equilibrium position of a stretched membrane having the shape of a square Ω, subject to an external load f (force per unit mass) and kept at level zero on $\partial\Omega$.

Since there is no time evolution, the position of the membrane may be described by a function $u = u(\mathbf{x})$, solution of the Dirichlet problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases} \tag{6.4}$$

(assuming for simplicity $c = 1$ in (5.92)). For problem (6.4), eq. (6.3) becomes

$$\int_\Omega \nabla u \cdot \nabla v \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x} \qquad \forall v \in \mathring{C}^1(\overline{\Omega}). \tag{6.5}$$

Now, this equation has an interesting physical interpretation. The integral in the left hand side represents the work done by the internal elastic forces, due to a *virtual displacement* v. On the other hand $\int_\Omega fv$ expresses the work done by the external forces to produce that virtual displacement.

Thus, the variational formulation (6.5) states that these two works balance, which constitutes a version of the *principle of virtual work*.

There is more, if we bring into play the energy. In fact, the *total potential energy* is proportional to

$$E(v) = \underbrace{\frac{1}{2}\int_\Omega |\nabla v|^2 \, d\mathbf{x}}_{\text{internal elastic energy}} - \underbrace{\int_\Omega f v \, d\mathbf{x}}_{\text{external potential energy}}. \tag{6.6}$$

Since nature likes to save energy, the equilibrium position u corresponds to the minimizer of (6.6) among all the *admissible* configurations $v \in \mathring{C}^1(\overline{\Omega})$. This fact is closely connected with the principle of virtual work and, actually, it is equivalent to it (see Sect. 8.3.1).

Thus, changing point of view, instead of looking for a variational solution of (6.5) we may, equivalently, look for a minimizer of $E(v)$ over $\mathring{C}^1(\overline{\Omega})$.

However there is a drawback. It turns out that the minimum problem *does not have a solution*, in general. The reason is that we are looking in the wrong set of admissible functions.

Why $\mathring{C}^1(\overline{\Omega})$ is a wrong choice? To be minimalist, it is like looking for the minimizer of the function

$$f(x) = (x - \pi)^2$$

over the rational numbers $\mathbb{Q}$. Clearly $\inf_{x\in\mathbb{Q}} f(x) = 0$, but it is not the *minimum!* If we enlarge the numerical set from $\mathbb{Q}$ to $\mathbb{R}$ we clearly have $\min_{x\in\mathbb{R}} f(x) = f(\pi) = 0$.

Similarly, the space $\mathring{C}^1(\overline{\Omega})$ is too narrow to have any hope of finding the minimizer there and we are forced to enlarge the set of admissible functions. To this purpose

observe that $\mathring{C}^1(\overline{\Omega})$ is not naturally tied to the physical meaning of $E(v)$, which is an energy and only requires *the gradient of u to be square integrable,* that is $|\nabla u| \in L^2(\Omega)$. There is no need of *a priori* continuity of the derivatives, actually neither of u. Thus, the correct functional setting turns out to be the so called *Sobolev space* $H_0^1(\Omega)$, whose elements are exactly the functions belonging to $L^2(\Omega)$, together with their first derivatives, vanishing on $\partial\Omega$. We could call them functions of finite energy!

Although we feel we are on the right track, there is a price to pay, to put everything in a rigorous perspective and avoid risks of contradiction or nonsenses. In fact many questions arise immediately.

The first one is analogous to question (6.2): what do we mean by the *gradient* of a function which is only in $L^2(\Omega)$, maybe with a lot of discontinuities? Second: a function in $L^2(\Omega)$ is, in principle, well defined except on sets of measure zero. But, then, since $\partial\Omega$ is precisely a set of measure zero,

what does it mean "vanishing on $\partial\Omega$"?

We shall answer these questions in Chap. 7. We may anticipate that, for the first one, the idea is the same we used to define the *Dirac delta* as a derivative of the Heaviside function, resorting to a weaker notion of derivative (we shall say *in the sense of distributions*), based on the formula of integration by parts and on the introduction of a suitable set of test function.

For the second question, there is a way to introduce in a suitable coherent way a so called *trace operator* which associates to a function $u \in L^2(\Omega)$, with gradient in $L^2(\Omega)$, a function $u|_{\partial\Omega}$ representing its values on $\partial\Omega$ (see Sect. 7.9). The elements of $H_0^1(\Omega)$ vanish on $\partial\Omega$ in the sense that they have zero trace.

Another question is:

what makes the space $H_0^1(\Omega)$ *so special?*

Here the conjunction between geometrical and analytical aspects comes into play. First of all, although it is an infinite-dimensional vector space, we may endow $H_0^1(\Omega)$ with a structure which reflects as much as possible the structure of a finite dimensional vector space like $\mathbb{R}^n$, where the life is obviously easier.

Indeed, in this vector space (thinking of $\mathbb{R}$ as the scalar field) we may introduce an *inner product* given by

$$(u, v)_{H_0^1(\Omega)} = \int_\Omega \nabla u \cdot \nabla v \, d\mathbf{x}$$

with the same properties of an inner product in $\mathbb{R}^n$. Then, it makes sense to talk about *orthogonality* between two functions u and v *in* $H^1(\Omega)$, expressed by the vanishing of their inner product:

$$(u, v)_{H_0^1(\Omega)} = 0.$$

Having defined the inner product $(\cdot, \cdot)_{H_0^1(\Omega)}$, we may define the *size* (*norm*) of u by

$$\|u\|_{H_0^1(\Omega)} = \sqrt{(u,u)_{H_0^1(\Omega)}}$$

and the distance between u and v by

$$\mathrm{dist}(u,v) = \|u - v\|_{H_0^1(\Omega)}\,.$$

Thus, we may say that a sequence $\{u_n\} \subset H_0^1(\Omega)$ converges to u in $H_0^1(\Omega)$ if $\mathrm{dist}(u_n, u) \to 0$ as $n \to \infty$. It may be observed that all of this can be done, even more comfortably, in the space $\mathring{C}^1(\overline{\Omega})$. This is true, but with a key difference.

Let us use once more an analogy with an elementary fact. The minimizer of the function $f(x) = (x - \pi)^2$ does not exist among the rational numbers $\mathbb{Q}$, although it can be approximated as much as one likes by these numbers. If from a very practical point of view, rational numbers could be considered satisfactory enough, certainly it is not so from the point of view of the development of science and technology, since, for instance, no one could even conceive the achievements of *integral and differential calculus* without the real number system.

As $\mathbb{R}$ is the *completion of* $\mathbb{Q}$, in the sense that $\mathbb{R}$ contains all the limits of sequences in $\mathbb{Q}$ that converge somewhere, the same is true for $H_0^1(\Omega)$ with respect to $\mathring{C}^1(\overline{\Omega})$. This makes $H_0^1(\Omega)$ a so called *Hilbert space* and it gives a big advantage with respect to $\mathring{C}^1(\overline{\Omega})$, which we illustrate going back to our membrane problem and precisely to equation (6.5). This time we use a geometrical interpretation.

In fact, (6.5) means that we are searching for an element u, whose inner product with any element v of $H_0^1(\Omega)$ reproduces "the action of f on v", given by the linear map

$$v \longmapsto \int_\Omega f v \, d\mathbf{x}.$$

This is a familiar situation in Linear Algebra. Any function $F : \mathbb{R}^n \to \mathbb{R}$, which is *linear*, that is such that

$$F(a\mathbf{x} + b\mathbf{y}) = aF(\mathbf{x}) + bF(\mathbf{y}) \qquad \forall a, b \in \mathbb{R}, \forall \mathbf{x}, \mathbf{y} \in \mathbb{R}^n,$$

can be *expressed as the inner product with a unique representative vector* $\mathbf{z}_F \in \mathbb{R}^n$ (Representation Theorem). This amounts to saying that there is exactly one solution $\mathbf{z}_F$ of the equation

$$\mathbf{z} \cdot \mathbf{y} = F(\mathbf{y}), \qquad \text{for every } \mathbf{y} \in \mathbb{R}^n. \tag{6.7}$$

The structure of the two equations (6.5), (6.7) is the same: on the left hand side there is an *inner product* and on the other one a *linear map*.

Thus another natural question arises:

Is there any analogue of the Representation Theorem in $H_0^1(\Omega)$?

The answer is yes (see the Riesz Theorem 6.31, p. 370) and holds in any Hilbert space, not only in $H_0^1(\Omega)$. In infinite dimensional spaces it is required the study of *linear, continuous functionals* and the related key concept of *dual space*. Then,

an abstract result of geometrical nature, implies the well-posedness of a concrete boundary value problem.

What about equation (6.3)? If the matrix $\mathbf{A}$ is symmetric and strictly positive, the left hand side of (6.3) *still defines an inner product* in $H_0^1(\Omega)$ and again the Riesz Theorem yields the well-posedness of the Dirichlet problem.

If $\mathbf{A}$ is not symmetric, things change only a little. Various generalizations of the Riesz Theorem (e.g. the Lax-Milgram Theorem 6.42, p. 376) allow the unified treatment of more general problems, through their *weak* or *variational formulation.* Actually, as we have experienced with equation (6.3), the variational formulation is often the only way of formulating and solving a problem, without losing its original features.

The above arguments should have convinced the reader of the existence of a general Hilbert space structure, underlying a large class of problems arising in the applications. In this chapter we develop the tools of Functional Analysis, essential for a correct variational formulation of a wide variety of boundary value problems. The results we present constitute the theoretical basis for numerical methods such as *finite elements*, or more generally *Galerkin's methods*, and this makes the theory even more attractive and important.

More advanced results, related to general solvability questions and the so called spectral properties of elliptic operators (formalizing the notions of eigenvalues and eigenfunctions) are included in the final part of this chapter.

A final comment is in order. Look again at the minimization problem above. We have enlarged the class of admissible configurations from a class of quite smooth functions to a rather wide class of functions. What kind of solutions are we finding with these abstract methods? If the data (e.g. Ω and f, for the membrane) are regular, could the corresponding solutions be irregular? If yes, this does not sound too good! In fact, although we are working in a setting of possibly irregular configurations, it turns out that the solution actually possesses its natural degree of regularity, once more confirming the intrinsic coherence of the method.

It also turns out that the knowledge of the optimal regularity of the solution plays an important role in the error control for numerical methods. However, this part of the theory is rather technical and we do not have much space to treat it in detail. We shall only state some of the most common results.

The power of abstract methods is not restricted to stationary problems. As we shall see, Sobolev spaces depending on time can be introduced for the treatment of evolution problems, both of diffusive or wave propagation type (see Chap. 9).

In this introductory book, the emphasis is mainly to *linear* problems. However, in the last section, we present some of the most important tools of nonlinear analysis, the *Contractions principle,* the *Schauder* and the *Leray-Schauder* theorems. In Chaps. 10 and 11, we will have occasion to apply these theorems to a variety of situations, in particular, to the stationary Navier-Stokes equation (Sect. 10.3).

6.2 Norms and Banach Spaces

It may be useful for future developments, to introduce *norm* and *distance* independently of an *inner product*, to emphasize better their axiomatic properties.

Let X be a linear space over the scalar field $\mathbb{R}$ or $\mathbb{C}$. A *norm* in X, is a real function

$$\|\cdot\| : X \to \mathbb{R} \tag{6.8}$$

such that, for each scalar λ and every $x, y \in X$, the following properties hold:

$$\begin{array}{ll} \mathcal{N}_1 : \|x\| \geq 0; \|x\| = 0 \text{ if and only if } x = 0 & (\textit{positivity}); \\ \mathcal{N}_2 : \|\lambda x\| = |\lambda| \, \|x\| & (\textit{homogeneity}); \\ \mathcal{N}_3 : \|x + y\| \leq \|x\| + \|y\| & (\textit{triangular inequality}). \end{array}$$

A norm is introduced to measure the size (or the "length") of each vector $x \in X$, so that properties $\mathcal{N}_1$, $\mathcal{N}_2$, $\mathcal{N}_3$ should appear as natural requirements, describing its interaction with the linear structure of the space.

A *normed space* is a linear space X endowed with a norm $\|\cdot\|$. A norm induces a *distance* between two vectors given by

$$d(x, y) = \|x - y\| \,, \tag{6.9}$$

which makes X into a metric space.

More generally, a *metric space* is a set M, endowed with a distance $d : M \times M \to \mathbb{R}$, satisfying the following three properties; for every $x, y, z \in M$:

$$\begin{array}{ll} \mathcal{D}_1 : d(x, y) \geq 0, d(x, y) = 0, \text{ if and only if } x = y & (\textit{positivity}). \\ \mathcal{D}_2 : d(x, y) = d(y, x) & (\textit{symmetry}). \\ \mathcal{D}_3 : d(x, z) \leq d(x, y) + d(y, z) & (\textit{triangular inequality}). \end{array}$$

Notice that a metric space *does not need to be a linear* space. We say that a sequence $\{x_n\} \subset M$ *converges* to x in M, and we write $x_m \to x$ in M, if

$$d(x_m, x) \to 0 \qquad \text{as } m \to \infty.$$

The limit is unique: if $d(x_m, x) \to 0$ and $d(x_m, y) \to 0$ as $m \to \infty$, then, from the triangular inequality, we have

$$d(x, y) \leq d(x_m, x) + d(x_m, y) \to 0 \text{ as } m \to \infty$$

which implies $x = y$. We call

$$B_r(x) = \{y \in M : d(x, y) < r\}$$

the (open) ball of radius r, centered at x. As in Sect. 1.4, we can define a *topology in* M, induced by the metric, repeating verbatim all the definitions in that section and maintaining also the same properties. In particular, *a set is closed (compact) if and only if it is sequentially closed (compact)*.

An important distinction is between convergent and Cauchy sequences. A sequence $\{x_m\} \subset M$ is a *Cauchy* (or *fundamental*) sequence if

$$d(x_m, x_k) \to 0 \qquad \text{as } m, k \to \infty.$$

If $x_m \to x$ in M, from the triangular inequality, we may write

$$d(x_m, x_k) \le d(x_m, x) + d(x_k, x) \to 0 \quad \text{as } m, k \to \infty,$$

and therefore

$$\{x_m\} \text{ convergent } \textbf{implies} \text{ that } \{x_m\} \text{ is a Cauchy sequence.} \tag{6.10}$$

The converse in not true, in general. Take $M = \mathbb{Q}$, with the usual norm given by $|x|$. The sequence of rational numbers

$$x_m = \left(1 + \frac{1}{m}\right)^m$$

is a Cauchy sequence but it is *not* convergent in $\mathbb{Q}$, since its limit is the irrational number e.

A metric space in which every Cauchy sequence converges is called **complete**. The normed spaces which are complete with respect to the induced distance (6.9) deserves a special name.

Definition 6.1 A complete, normed linear space is called *Banach space*.

The notion of convergence (or of limit) can be extended to functions from a metric space into another, always reducing it to the convergence of distances, that are real functions.

Let (M_1, d_1), (M_2, d_2) be two metric spaces and let $F : M_1 \to M_2$. We say that F is continuous at $x \in M_1$ if

$$d_2(F(y), F(x)) \to 0 \qquad \text{when } d_1(x, y) \to 0$$

or, equivalently, if, for every sequence $\{x_m\} \subset M_1$,

$$d_1(x_m, x) \to 0 \quad \text{implies} \quad d_2(F(x_m), F(x)) \to 0.$$

F is *continuous in* M if it is *continuous at every* $x \in M$. In particular:

Proposition 6.2 *Every norm in a linear space X is continuous in X.*

Proof Let $\|\cdot\|$ be a norm in X. From the triangular inequality, we may write

$$\|y\| \le \|y - x\| + \|x\| \quad \text{and} \quad \|x\| \le \|y - x\| + \|y\|$$

whence

$$|\,\|y\| - \|x\|\,| \le \|y - x\|\,.$$

Thus, if $\|y - x\| \to 0$ then

$$|\|y\| - \|x\|| \to 0,$$

which is the continuity of the norm. □

Definition 6.3 Two norms $\|\cdot\|_1$ and $\|\cdot\|_2$, defined in the same space X, are *equivalent* if there exist two positive numbers c_1, c_2 such that

$$c_1 \|x\|_2 \le \|x\|_1 \le c_2 \|x\|_2 \quad \text{for every } x \in X.$$

Two equivalent norms induce the same topology. Some examples are in order.

Spaces of continuous functions. Let $\Omega \subset \mathbb{R}^n$ be a bounded *open set*.

• *The space* $C(\overline{\Omega})$. The symbol $C(\overline{\Omega})$ denotes the set of (real or complex) continuous functions on $\overline{\Omega}$, endowed with the norm (called the *maximum norm*)

$$\|f\|_{C(\overline{\Omega})} = \max_{\overline{\Omega}} |f| .$$

A sequence $\{f_m\}$ converges to f in $C(\overline{\Omega})$ if

$$\max_{\overline{\Omega}} |f_m - f| \to 0,$$

that is, if f_m *converges uniformly to* f in $\overline{\Omega}$. Since a uniform limit of continuous functions is continuous, $C(\overline{\Omega})$ is a Banach space.

Note that other norms may be introduced in $C(\overline{\Omega})$, for instance the $L^2(\Omega)$ *norm*

$$\|f\|_{L^2(\Omega)} = \left(\int_\Omega |f|^2 \right)^{1/2} .$$

Equipped with this norm $C(\overline{\Omega})$ is *not complete*. Let, for example $\Omega = (-1, 1) \subset \mathbb{R}$. The sequence

$$f_m(t) = \begin{cases} 0 & -1 \le t \le 0 \\ mt & 0 \le t \le \frac{1}{m} \\ 1 & \frac{1}{m} \le t \le 1 \end{cases} \qquad (m \ge 1),$$

contained in $C([-1, 1])$, is a Cauchy sequence with respect to the L^2 norm. In fact (letting $m > k$),

$$\|f_m - f_k\|^2_{L^2(0,1)} = \int_0^1 |f_m(t) - f_k(t)|^2 \, dt = (m-k)^2 \int_0^{\frac{1}{m}} t^2 dt + \int_{\frac{1}{m}}^{\frac{1}{k}} (1 - kt)^2 dt$$
$$= \frac{(m-k)^2}{3m^3} + \frac{(m-k)^3}{3km^3} < \frac{1}{3}\left(\frac{1}{m} + \frac{1}{k}\right) \to 0 \qquad \text{as } m, k \to \infty.$$

However, f_m converges in $L^2(-1, 1)$-norm to the Heaviside function

$$\mathcal{H}(t) = \begin{cases} 1 & t \ge 0 \\ 0 & t < 0, \end{cases}$$

which is discontinuous at $t = 0$ and therefore does not belong to $C([-1, 1])$.

• *The spaces* $C^k(\overline{\Omega})$, $k \geq 0$ *integer*. The symbol $C^k(\overline{\Omega})$ denotes the set of functions whose derivatives, up to the order k included, are continuous in Ω and can be extended continuously up to $\partial\Omega$.

To denote a derivative of order m, it is convenient to use the symbol (see Sect. 3.3.7)

$$D^\alpha = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}},$$

where $\alpha = (\alpha_1, \ldots, \alpha_n)$ is an n-tuple of nonnegative integers (*multi-index*), of length

$$|\alpha| = \alpha_1 + \ldots + \alpha_n = m.$$

We endow $C^k(\overline{\Omega})$ with the norm (*maximum norm of order* k)

$$\|f\|_{C^k(\overline{\Omega})} = \|f\|_{C(\overline{\Omega})} + \sum_{|\alpha|=1}^{k} \|D^\alpha f\|_{C(\overline{\Omega})}.$$

If $\{f_m\}$ is a Cauchy sequence in $C^k(\overline{\Omega})$, all the sequences $\{D^\alpha f_m\}$ with $0 \leq |\alpha| \leq k$ are Cauchy sequences in $C(\overline{\Omega})$. From the theorems on term by term differentiation of sequences (see Sect. 1.4), it follows that the resulting space is a Banach space.

• *The spaces* $C^{0,\alpha}(\overline{\Omega})$, $0 \leq \alpha \leq 1$. We say that a function f is *Hölder continuous in* Ω, *with exponent* α, if

$$\sup_{\substack{\mathbf{x},\mathbf{y}\in\overline{\Omega} \\ \mathbf{x}\neq\mathbf{y}}} \frac{|f(\mathbf{x}) - f(\mathbf{y})|}{|\mathbf{x} - \mathbf{y}|^\alpha} \equiv C_H(f;\Omega) < \infty. \tag{6.11}$$

The quotient in the left hand side of (6.11) represents an "incremental quotient of order α". The number $C_H(f;\Omega)$ is called the *Hölder constant of* f in Ω. If $\alpha = 1$, f *is Lipschitz continuous*, while if $\alpha = 0$ it is merely continuous. Typical examples of Hölder continuous functions in $\mathbb{R}^n$ with exponent α are the powers $f(\mathbf{x}) = |\mathbf{x}|^\alpha$.

We use the symbol $C^{0,\alpha}(\overline{\Omega})$ to denote the set of functions of $C(\overline{\Omega})$, Hölder continuous in Ω with exponent α. Endowed with the norm

$$\|f\|_{C^{0,\alpha}(\overline{\Omega})} = \|f\|_{C(\overline{\Omega})} + C_H(f;\Omega),$$

$C^{0,\alpha}(\overline{\Omega})$ becomes a Banach space.

• *The spaces* $C^{k,\alpha}(\overline{\Omega})$, $k \geq 1$ *integer*, $0 \leq \alpha \leq 1$. $C^{k,\alpha}(\overline{\Omega})$ denotes the set of functions of $C^k(\overline{\Omega})$, whose derivatives up to order k included, are Hölder continuous in Ω with exponent α. Endowed with the norm

$$\|f\|_{C^{k,\alpha}(\overline{\Omega})} = \|f\|_{C^k(\overline{\Omega})} + \sum_{|\beta|=k} C_H(D^\beta f;\Omega)$$

$C^{k,\alpha}(\overline{\Omega})$ becomes a Banach space.

We shall see in Chaps. 7 and 8 the usefulness of these spaces.

Remark 6.4 With the introduction of *function spaces* we are actually making a step towards abstraction, regarding a function from a different perspective. In calculus we see it as a point map while here we have to consider it as *a single element* (or a point or a vector) *of a vector space.*

Summable and bounded functions. Let Ω be an open set in $\mathbb{R}^n$ and $p \geq 1$ a real number. We denote by $L^p(\Omega)$ the set of functions $f : \Omega \to \mathbb{R}$ such that $|f|^p$ is Lebesgue integrable in Ω. Identifying two functions f and g when they are equal a.e.[1] in Ω, $L^p(\Omega)$ becomes a Banach space (see e.g. *K. Yosida*, [43]) when equipped with the norm (*integral norm of order p*)

$$\|f\|_{L^p(\Omega)} = \left(\int_\Omega |f|^p\right)^{1/p}.$$

The identification of two functions equal a.e. amounts to saying that an element of $L^p(\Omega)$ is not a single function but, actually, an equivalence class of functions, different from one another only on subsets of measure zero. At first glance, this fact could be annoying, but after all, the situation is perfectly analogous to some familiar ones. Take for instance the *rational numbers*. A rational number is rigorously defined as an equivalent class of fractions: the fractions 2/3, 4/6, 8/12 . . . . represent the same number, but in practice one uses the more convenient representative, namely 2/3. Similarly, when dealing with a "function" in $L^p(\Omega)$, for most practical purposes, one may refer to the more convenient representative of the class. However, as we shall see later, some care is due, especially when pointwise properties are involved.

The symbol $L^\infty(\Omega)$ denotes the set of essentially bounded functions in Ω. Recall that $f : \Omega \to \mathbb{R}$ is essentially bounded if there exists M such that

$$|f(\mathbf{x})| \leq M \qquad \text{a.e. in } \Omega \tag{6.12}$$

(see Appendix A). The infimum of all numbers M with the property (6.12) is called essential supremum of $|f|$, and denoted by

$$\|f\|_{L^\infty(\Omega)} = \operatorname*{ess\,sup}_\Omega |f|.$$

Note that the essential supremum may differ from the supremum. As an example take the characteristic function of the rational numbers $\chi_\mathbb{Q}$, equal to 1 on $\mathbb{Q}$ and 0 on $\mathbb{R} \setminus \mathbb{Q}$. Then $\sup_\mathbb{R} \chi_\mathbb{Q} = 1$, but $\left\|\chi_\mathbb{Q}\right\|_{L^\infty(\mathbb{R})} = 0$, since $\mathbb{Q}$ has zero Lebesgue measure.

If we identify two functions when they are equal a.e., $\|f\|_{L^\infty(\Omega)}$ is a norm in $L^\infty(\Omega)$, and $L^\infty(\Omega)$ becomes a Banach space. Hölder inequality (1.8), p. 11, for real functions, may be now rewritten in terms of norms as follows:

[1] A property is valid *almost everywhere* in a set Ω, *a.e.* in short, if it is true at all points in Ω, but for a subset of measure zero (see Appendix A).

$$\left| \int_\Omega fg \right| \le \|f\|_{L^p(\Omega)} \, \|g\|_{L^q(\Omega)} \,, \tag{6.13}$$

where $q = p/(p-1)$ is the conjugate exponent of p, allowing also the case $p = 1$, $q = \infty$.

Note that, if Ω has finite measure and $1 \le p_1 < p_2 \le \infty$, from (6.13) we have, choosing $g \equiv 1$, $p = p_2/p_1$ and $q = p_2/(p_2 - p_1)$:

$$\int_\Omega |f|^{p_1} \le |\Omega|^{1/q} \, \|f\|^{p_1}_{L^{p_2}(\Omega)}$$

and therefore $L^{p_2}(\Omega) \subset L^{p_1}(\Omega)$. If the measure of Ω is infinite, this inclusion is not true, in general; for instance, $f \equiv 1$ belongs to $L^\infty(\mathbb{R})$ but is not in $L^p(\mathbb{R})$ for $1 \le p < \infty$.

Remark 6.5 If Ω is bounded, $C(\overline{\Omega}) \subset L^p(\Omega)$ for every $1 \le p \le \infty$. If $p < \infty$, from Theorem A.11, p. 661, we deduce that $C(\overline{\Omega})$ is dense in $L^p(\Omega)$. This means that if $f \in L^p(\Omega)$, $1 \le p < \infty$, there exists a sequence $\{f_k\} \subset C(\overline{\Omega})$ such that $f_k \to f$ in $L^p(\Omega)$. Thus $C(\overline{\Omega})$ is a **non-closed** subspace of $L^p(\Omega)$, $1 \le p < \infty$.

Analogous considerations can be done for functions $f : \Omega \to \mathbb{C}$. When necessary, we will denote the complex L^p spaces by the symbols $L^p_{\mathbb{C}}(\Omega)$ or $L^p(\Omega; \mathbb{C})$.

6.3 Hilbert Spaces

Let X be a linear space over $\mathbb{R}$. An inner or scalar product in X is a function

$$(\cdot, \cdot) : X \times X \to \mathbb{R}$$

with the following three properties. For every $x, y, z \in X$ and every scalar $\lambda, \mu \in \mathbb{R}$:

$$\begin{array}{ll} \mathcal{H}_1 : (x,x) \ge 0 \text{ and } (x,x) = 0 \text{ if and only if } x = 0 & (\textit{positivity}); \\ \mathcal{H}_2 : (x,y) = (y,x) & (\textit{symmetry}); \\ \mathcal{H}_3 : (\mu x + \lambda y, z) = \mu(x,z) + \lambda(y,z) & (\textit{bilinearity}). \end{array}$$

A linear space endowed with an inner product is called an *inner product space*. Property $\mathcal{H}_3$ shows that the inner product is linear with respect to its first argument. From $\mathcal{H}_2$, the same is true for the second argument as well. Then, we say that $(\cdot, \cdot)$ constitutes a *symmetric bilinear form* in X. When different inner product spaces are involved it may be necessary the use of notations like $(\cdot, \cdot)_X$, to avoid confusion.

If the scalar field is $\mathbb{C}$, then $(\cdot, \cdot) : X \times X \to \mathbb{C}$, and property $\mathcal{H}_2$ has to be extended by

$$\mathcal{H}_2^* : \ (x,y) = \overline{(y,x)} \quad (\textit{antisymmetry});$$

where the bar denotes complex conjugation[2]. As a consequence, we have

[2] Notice that, if (x, y) is real, $\mathcal{H}_2^*$ and $\mathcal{H}_2$ coincide.

$$(z, \mu x + \lambda y) = \overline{\mu}(z, x) + \overline{\lambda}(z, y)$$

and we say that $(\cdot, \cdot)$ is *antilinear* with respect to its second argument or that it is a *hermitian form* in X.

An inner product induces a norm, given by

$$\|x\| = \sqrt{(x, x)}. \tag{6.14}$$

In fact, properties $\mathcal{N}_1$ and $\mathcal{N}_2$ in the definition of norm are immediate, while the triangular inequality $\mathcal{N}_3$ is a consequence of the following important theorem.

Theorem 6.6 *Let X be an inner product space, real or complex, and $x, y \in X$. Then:*

1. ***Schwarz's inequality**:*

$$|(x, y)| \leq \|x\| \, \|y\| \, . \tag{6.15}$$

Moreover equality holds in (6.15) if and only if x and y are linearly dependent.

2. ***Parallelogram law**:*

$$\|x + y\|^2 + \|x - y\|^2 = 2 \, \|x\|^2 + 2 \, \|y\|^2 \, .$$

Proof 1. We mimic the finite dimensional proof for complex vector spaces. Let $z \in \mathbb{C}$ and $x, y \in X$. Using the properties of the inner product and (6.14), we may write:

$$\forall z \in \mathbb{C} : 0 \leq (zx - y, zx - y) = |z|^2 \, \|x\|^2 - 2 \, \mathrm{Re}[z(x, y)] + \|y\|^2 \, .$$

Choosing[3] $z = \overline{(x, y)} \, \|x\|^{-2}$ we find

$$0 \leq |(x, y)|^2 \, \|x\|^{-2} - 2 \, |(x, y)|^2 \, \|x\|^{-2} + \|y\|^2 = - \, |(x, y)|^2 \, \|x\|^{-2} + \|y\|^2$$

from which the (6.15). Equality is possible only if $zx - y = 0$, i.e. if x and y are linearly dependent.
2. Just observe that

$$\|x \pm y\|^2 = (x \pm y, y \pm y) = \|x\|^2 \pm 2 \, \mathrm{Re}(x, y) + \|y\|^2 \, . \qquad \square$$

The parallelogram law generalizes an elementary result in euclidean plane geometry: *in a parallelogram, the sum of the squares of the sides length equals the sum of the squares of the diagonals length.*

The Schwarz inequality implies that the inner product is continuous; in fact, writing

$$(w, z) - (x, y) = (w - x, z) + (x, z - y),$$

we have

$$|(w, z) - (x, y)| \leq \|w - x\| \, \|z\| + \|x\| \, \|z - y\|$$

so that, if $w \longrightarrow x$ and $z \longrightarrow y$, then $(w, z) \longrightarrow (x, y)$.

[3] If $z = s + it$, then the value $\overline{(x, y)} \, \|x\|^{-2}$ minimizes the (real) convex polynomial

$$\varphi(s, t) = (s^2 + t^2) \, \|x\|^2 + 2 \, \mathrm{Re} \, [(s + it)(x, y)] + \|y\|^2 \, .$$

Definition 6.7 Let H be an inner product space. We say that H is a *Hilbert space* if it is complete with respect to the norm (6.14), induced by the inner product.

Two Hilbert spaces H_1 and H_2 are *isometric* if *there exists a one-to-one and onto linear map* $L : H_1 \to H_2$, called *isometry*, which preserves the norm, that is:

$$\|x\|_{H_1} = \|Lx\|_{H_2}, \qquad \forall x \in H_1. \tag{6.16}$$

An isometry preserves also the inner product since from

$$\|x - y\|^2_{H_1} = \|L(x - y)\|^2_{H_2} = \|Lx - Ly\|^2_{H_2},$$

we get

$$\|x\|^2_{H_1} - 2\,\mathrm{Re}(x, y)_{H_1} + \|y\|^2_{H_1} = \|Lx\|^2_{H_2} - 2\,\mathrm{Re}(Lx, Ly)_{H_2} + \|Ly\|^2_{H_2}$$

and from (6.16) we infer

$$\mathrm{Re}(x, y)_{H_1} = \mathrm{Re}(Lx, Ly)_{H_2}, \qquad \forall x, y \in H_1.$$

Replacing y by iy we also get

$$\mathrm{Im}(x, y)_{H_1} = \mathrm{Im}(Lx, Ly)_{H_2}, \qquad \forall x, y \in H_1.$$

Example 6.8 $\mathbb{R}^n$ is a Hilbert space with respect to the usual inner product

$$(\mathbf{x}, \mathbf{y})_{\mathbb{R}^n} = \mathbf{x} \cdot \mathbf{y} = \sum_{j=1}^{n} x_j y_j, \qquad \mathbf{x} = (x_1, \ldots, x_n), \mathbf{y} = (y_1, \ldots, y_n).$$

The induced norm is

$$|\mathbf{x}| = \sqrt{\mathbf{x} \cdot \mathbf{x}} = \sqrt{\sum_{j=1}^{n} x_j^2}.$$

More generally, if $\mathbf{A} = (a_{ij})_{i,j=1,\ldots,n}$ is a square matrix of order n, *symmetric* and *positive*,

$$(\mathbf{x}, \mathbf{y})_{\mathbf{A}} = \mathbf{x} \cdot \mathbf{A}\mathbf{y} = \mathbf{A}\mathbf{x} \cdot \mathbf{y} = \sum_{i=1}^{n} a_{ij} x_i y_j \tag{6.17}$$

defines another scalar product in $\mathbb{R}^n$. Actually, *every* inner product in $\mathbb{R}^n$ may be written in the form (6.17), with a suitable matrix $\mathbf{A}$.

$\mathbb{C}^n$ is a Hilbert space with respect to the inner product

$$(\mathbf{x}, \mathbf{y})_{\mathbb{C}^n} = \sum_{j=1}^{n} x_j \overline{y}_j \qquad \mathbf{x} = (x_1, \ldots, x_n), \mathbf{y} = (y_1, \ldots, y_n).$$

It is easy to show that every real (resp. complex) linear space of dimension n is isometric to $\mathbb{R}^n$ (resp. $\mathbb{C}^n$).

Example 6.9 $L^2(\Omega)$ and $L^2_{\mathbb{C}}(\Omega)$ are Hilbert spaces with respect to the inner products

$$(u,v)_{L^2(\Omega)} = \int_\Omega uv \quad \text{and} \quad (u,v)_{L^2_{\mathbb{C}}(\Omega)} = \int_\Omega u\overline{v}.$$

Example 6.10 Let $l^2_{\mathbb{C}}$ be the set of complex sequences $\mathbf{x} = \{x_m\}$ such that

$$\sum_{m\in\mathbb{Z}} |x_m|^2 < \infty.$$

For $\mathbf{x} = \{x_m\}$ and $\mathbf{y} = \{y_m\}$, define

$$(\mathbf{x},\mathbf{y})_{l^2_{\mathbb{C}}} = \sum_{m\in\mathbb{Z}} x_i\overline{y}_j.$$

Then $(\mathbf{x},\mathbf{y})_{l^2_{\mathbb{C}}}$ is an inner product which makes $l^2_{\mathbb{C}}$ into a Hilbert space over $\mathbb{C}$. This space constitutes the discrete analogue of $L^2(0,2\pi)$. Indeed, each $u \in L^2(0,2\pi)$ has an expansion in Fourier series (Sec. 1.5)

$$u(x) = \sum_{m\in\mathbb{Z}} \widehat{u}_m e^{imx},$$

where

$$\widehat{u}_m = \frac{1}{2\pi}\int_0^{2\pi} u(x)e^{-imx}dx.$$

Note that $\overline{\widehat{u}}_m = \widehat{u}_{-m}$, since u is a real function. From Parseval's identity, we have

$$(u,v)_{L^2(0,2\pi)} = \int_0^{2\pi} uv = 2\pi \sum_{m\in\mathbb{Z}} \widehat{u}_m\widehat{v}_{-m}$$

and (Bessel's equation)

$$\|u\|^2_{L^2(0,2\pi)} = \int_0^{2\pi} u^2 = 2\pi \sum_{m\in\mathbb{Z}} |\widehat{u}_m|^2 .$$

Example 6.11 (a Sobolev space) It is possible to use the frequency space $l^2_{\mathbb{C}}$ introduced in the previous example to define the derivatives of a function in $L^2(0,2\pi)$ in a weak or generalized sense. Let $u \in C^1(\mathbb{R})$, 2π-periodic. The Fourier coefficients of u' are given by

$$\widehat{u'}_m = im\widehat{u}_m$$

and we may write

$$\|u'\|^2_{L^2(0,2\pi)} = \int_0^{2\pi} (u')^2 = 2\pi \sum_{m\in\mathbb{Z}} m^2 |\widehat{u}_m|^2 . \tag{6.18}$$

Thus, both sequences $\{\widehat{u}_m\}$ and $\{m\widehat{u}_m\}$ belong to $l^2_{\mathbb{C}}$. But the right hand side in (6.18) does not involve u' directly, so that it makes perfect sense to define

$$H^1_{\text{per}}(0,2\pi) = \left\{u \in L^2(0,2\pi) : \{\widehat{u}_m\}\,,\ \{m\widehat{u}_m\} \in l^2_{\mathbb{C}}\right\}$$

and introduce the inner product

$$(u,v)_{H^1_{\text{per}}(0,2\pi)} = 2\pi \sum_{m\in\mathbb{Z}} (1+m^2)\widehat{u}_m\widehat{v}_{-m}$$

which makes $H^1_{\text{per}}(0,2\pi)$ into a Hilbert space. Since

$$\{m\widehat{u}_m\} \in l^2_{\mathbb{C}},$$

with each $u \in H^1_{\text{per}}(0,2\pi)$ is associated the (real) function $v \in L^2(0,2\pi)$ given by

$$v(x) = \sum_{m\in\mathbb{Z}} im\widehat{u}_m e^{imx}.$$

We see that v may be considered as *a generalized derivative of* u. Moreover, since

$$u(x) = \sum_{m\in\mathbb{Z}} \widehat{u}_m e^{imx}$$

and

$$\left|\widehat{u}_m e^{imx}\right| = \frac{1}{m} m\,|\widehat{u}_m| \le \frac{1}{2}\left(\frac{1}{m^2} + m^2\,|\widehat{u}_m|^2\right),$$

the Weierstrass test entails that the Fourier series of u converges uniformly in $\mathbb{R}$. Thus u has a continuous, 2π-periodic extension to all $\mathbb{R}$. Therefore $H^1_{\text{per}}(0,2\pi)$ coincides with the space of 2π-periodic functions in $L^2(0,2\pi)$ together with their generalized first derivatives. Finally observe that, if we use the symbol u' also for the generalized derivative of u, the inner product in $H^1_{\text{per}}(0,2\pi)$ can be written in the form

$$(u,v)_{H^1_{\text{per}}(0,2\pi)} = \int_0^{2\pi} (u'v' + uv).$$

6.4 Projections and Bases

Unless stated explicitly, **from now on we consider Hilbert spaces over** $\mathbb{R}$ (real Hilbert spaces), endowed with inner product $(\cdot,\cdot)$ and induced norm $\|\cdot\|$.

6.4.1 Projections

Hilbert spaces are the ideal setting to solve problems in infinitely many dimensions. They unify through the inner product and the induced norm, both an analytical and a geometric structure. As we shall shortly see, we may coherently introduce the concepts of orthogonality, projection and basis, prove a infinite-dimensional Pythagoras' Theorem (an example is just Bessel's equality) and introduce other operations, extremely useful from both a theoretical and practical point of view.

As in the finite-dimensional case, two elements x, y belonging to an inner product space are called *orthogonal* or *normal* if $(x, y) = 0$, and we write $x \perp y$.

Now, if we consider a subspace V of $\mathbb{R}^n$, e.g. a hyperplane through the origin, every $\mathbf{x} \in \mathbb{R}^n$ has a unique orthogonal projection onto V. In fact, if $\dim V = k$ and the unit vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k$ constitute an *orthonormal basis* in V, we may always find an orthonormal basis in $\mathbb{R}^n$, given by

$$\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_k, \mathbf{w}_{k+1}, \ldots, \mathbf{w}_n,$$

where $\mathbf{w}_{k+1}, \ldots, \mathbf{w}_n$ are suitable unit vectors. Thus, if

$$\mathbf{x} = \sum_{j=1}^{k} x_j \mathbf{v}_j + \sum_{j=k+1}^{n} x_j \mathbf{w}_j,$$

the projection of $\mathbf{x}$ onto V is given by

$$P_V \mathbf{x} = \sum_{j=1}^{k} x_j \mathbf{v}_j.$$

On the other hand, the projection $P_V \mathbf{x}$ can be characterized through the following property, which does not involve a basis in $\mathbb{R}^n$: *$P_V \mathbf{x}$ is the point in V that minimizes the distance from* $\mathbf{x}$, that is

$$|P_V \mathbf{x} - \mathbf{x}| = \inf_{\mathbf{y} \in V} |\mathbf{y} - \mathbf{x}| \,. \tag{6.19}$$

In fact, if $\mathbf{y} = \sum_{j=1}^{k} y_j \mathbf{v}_j$, we have

$$|\mathbf{y} - \mathbf{x}|^2 = \sum_{j=1}^{k} (y_j - x_j)^2 + \sum_{j=k+1}^{n} x_j^2 \geq \sum_{j=k+1}^{N} x_j^2 = |P_V \mathbf{x} - \mathbf{x}|^2 \,. \tag{6.20}$$

In this case, the "infimum" in (6.19) is actually a "minimum".

The uniqueness of $P_V \mathbf{x}$ follows from the fact that, if $\mathbf{y}^* \in V$ and

$$|\mathbf{y}^* - \mathbf{x}| = |P_V \mathbf{x} - \mathbf{x}| \,,$$

then, from (6.20) with $\mathbf{y} = \mathbf{y}^*$, we must have

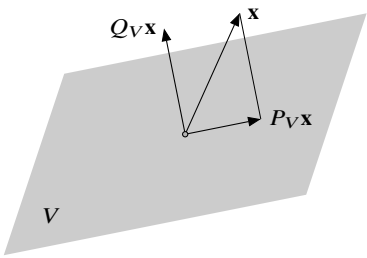

Fig. 6.1 Projection Theorem: $Q_V\mathbf{x} = \mathbf{x} - P_V\mathbf{x}$

$$\sum_{j=1}^{k}(y_j^* - x_j)^2 = 0,$$

whence $y_j^* = x_j$ for $j = 1, \ldots, k$, and therefore $\mathbf{y}^* = P_V\mathbf{x}$. Since

$$(\mathbf{x} - P_V\mathbf{x}) \perp \mathbf{v}, \qquad \forall \mathbf{v} \in V,$$

every $\mathbf{x} \in \mathbb{R}^n$ may be written in a unique way in the form $\mathbf{x} = \mathbf{y} + \mathbf{z}$, with $\mathbf{y} \in V$ and $\mathbf{z} \in V^\perp$, where $V^\perp$ denotes the subspace of the vectors orthogonal to V (see Fig. 6.1).

Then, we say that $\mathbb{R}^n$ is *direct sum* of the subspaces V and $V^\perp$ and we write

$$\mathbb{R}^n = V \oplus V^\perp.$$

Finally,

$$|\mathbf{x}|^2 = |\mathbf{y}|^2 + |\mathbf{z}|^2$$

which is Pythagoras' Theorem in $\mathbb{R}^n$.

We may extend all the above consideration to infinite-dimensional Hilbert spaces H, if we consider **closed subspaces** V of H. Here closed means with respect to the convergence induced by the norm in H. Recall that a subset $U \subset H$ is closed in H if it contains all the limit points of sequences in U. Observe that if V has finite dimension k, it is automatically closed, since it is isometric to $\mathbb{R}^k$. Also, a closed subspace of H is a Hilbert space as well, with respect to the inner product in H.

Theorem 6.12 (Projection Theorem) *Let V be a closed subspace of a Hilbert space H. Then, for every $x \in H$, there exists a unique element $P_V x \in V$ such that*

$$\|P_V x - x\| = \inf_{v \in V} \|v - x\| . \tag{6.21}$$

Moreover, the following properties hold:

1. *$P_V x = x$ if and only if $x \in V$.*
2. *Let $Q_V x = x - P_V x$. Then $Q_V x \in V^\perp$ and*

$$\|x\|^2 = \|P_V x\|^2 + \|Q_V x\|^2 .$$

Proof Let

$$d = \inf_{v \in V} \|v - x\| .$$

By the definition of greatest lower bound, we may select a sequence $\{v_m\} \subset V$, such that $\|v_m - x\| \to d$ as $m \to \infty$. Indeed, for every integer $m \geq 1$ there exists $v_m \in V$ such that

$$d \leq \|v_m - x\| < d + \frac{1}{m}. \tag{6.22}$$

Letting $m \to \infty$ in (6.22), we get $\|v_m - x\| \to d$.

We now show that $\{v_m\}$ is a Cauchy sequence. In fact, using the parallelogram law for the vectors $v_k - x$ and $v_m - x$, we obtain

$$\|v_k + v_m - 2x\|^2 + \|v_k - v_m\|^2 = 2\|v_k - x\|^2 + 2\|v_m - x\|^2. \tag{6.23}$$

Since $\frac{v_k+v_m}{2} \in V$, we may write

$$\|v_k + v_m - 2x\|^2 = 4\left\|\frac{v_k + v_m}{2} - x\right\|^2 \geq 4d^2$$

whence, from (6.23):

$$\begin{aligned}\|v_k - v_m\|^2 &= 2\|v_k - x\|^2 + 2\|v_m - x\|^2 - \|v_k + v_m - 2x\|^2 \\ &\leq 2\|v_k - x\|^2 + 2\|v_m - x\|^2 - 4d^2.\end{aligned}$$

Letting $k, m \to \infty$, the right hand side goes to zero and therefore

$$\|v_k - v_m\| \to 0$$

as well. This proves that $\{v_m\}$ is a Cauchy sequence.

Since H is complete, v_m converges to an element $w \in H$ which belongs to V, because V *is closed*. Using the norm continuity (Proposition 6.2) we deduce

$$\|v_m - x\| \to \|w - x\| = d$$

so that w realizes the minimum distance from x among the elements in V.

We have to prove the uniqueness of w. Suppose $\bar{w} \in V$ is another element such that $\|\bar{w} - x\| = d$. The parallelogram law, applied to the vectors $w - x$ and $\bar{w} - x$, yields

$$\begin{aligned}\|w - \bar{w}\|^2 &= 2\|w - x\|^2 + 2\|\bar{w} - x\|^2 - 4\left\|\frac{w + \bar{w}}{2} - x\right\|^2 \\ &\leq 2d^2 + 2d^2 - 4d^2 = 0\end{aligned}$$

whence $w = \bar{w}$. We have proved that there exists a unique element $w = P_V x \in V$ such that

$$\|x - P_V x\| = d.$$

To prove 1, note that, since V is closed, $x \in V$ if and only if $d = 0$, which means $x = P_V x$.

To show 2, let $Q_V x = x - P_V x$, $v \in V$ and $t \in \mathbb{R}$. Since $P_V x + tv \in V$ for every t, we have:

$$\begin{aligned}d^2 &\leq \|x - (P_V x + tv)\|^2 = \|Q_V x - tv\|^2 = \|Q_V x\|^2 - 2t(Q_V x, v) + t^2\|v\|^2 \\ &= d^2 - 2t(Q_V x, v) + t^2\|v\|^2.\end{aligned}$$

Erasing d^2 and dividing by $t > 0$, we get that the second degree polynomial

$$P(t) \equiv t^2\|v\|^2 - 2t(Q_V x, v)$$

is always nonnegative, whence $(Q_V x, v)^2 \leq 0$. Thus $(Q_V x, v) = 0$, which means $Q_V x \in V^\perp$ and implies that

$$\|x\|^2 = \|P_V x + Q_V x\|^2 = \|P_V x\|^2 + \|Q_V x\|^2 ,$$

concluding the proof. □

The elements $P_V x$, $Q_V x$ are called **orthogonal projections** of x onto V and $V^\perp$, respectively. The greatest lower bound (6.21) is actually a minimum. Moreover, thanks to properties 1, 2, we say that H is *direct sum* of V and $V^\perp$, that is:

$$H = V \oplus V^\perp .$$

Note that

$$V^\perp = \{0\} \qquad \text{if and only if} \qquad V = H .$$

Remark 6.13 Another characterization of $P_V x$ is the following (see Problem 6.4): $u = P_V x$ *if and only if*

$$\begin{cases} u \in V & \text{and} \\ (x-u, v) = 0, \ \forall v \in V . \end{cases}$$

Remark 6.14 It is useful to point out that, even if V is *not a closed* subspace of H, the subspace $V^\perp$ is *always closed.* In fact, if $y_n \to y$ and $\{y_n\} \subset V^\perp$, we have, for every $x \in V$,

$$(y, x) = \lim (y_n, x) = 0$$

whence $y \in V^\perp$.

Example 6.15 Let $\Omega \subset \mathbb{R}^n$ be a set of finite measure. Consider in $L^2(\Omega)$ the 1-dimensional subspace V of the constant functions (a basis is given by $f \equiv 1$, for instance). Since it is finite-dimensional, V is closed in $L^2(\Omega)$. Given $f \in L^2(\Omega)$, to find the projection $P_V f$, we solve the minimization problem

$$\min_{\lambda \in \mathbb{R}} \int_\Omega (f - \lambda)^2 .$$

Since

$$\int_\Omega (f-\lambda)^2 = \int_\Omega f^2 - 2\lambda \int_\Omega f + \lambda^2 \, |\Omega| ,$$

we see that the minimizer is

$$\lambda = \frac{1}{|\Omega|} \int_\Omega f .$$

Therefore

$$P_V f = \frac{1}{|\Omega|} \int_\Omega f \quad \text{and} \quad Q_V f = f - \frac{1}{|\Omega|} \int_\Omega f .$$

Thus, the subspace $V^\perp$ is given by the functions $g \in L^2(\Omega)$ with *zero mean value.* In fact these functions are orthogonal to $f \equiv 1$:

$$(g, 1)_{L^2(\Omega)} = \int_\Omega g = 0 .$$

6.4.2 Bases

A Hilbert space H is called **separable** if there exists a *countable dense subset*of H.

Definition 6.16 An *orthonormal basis* in a separable Hilbert space H is a sequence $\{w_k\}_{k\geq 1} \subset H$ such that[4]

$$\begin{cases} (w_k, w_j) = \delta_{kj} & k, j \geq 1, \\ \|w_k\| = 1 & k \geq 1 \end{cases}$$

and every $x \in H$ may be expanded in the form

$$x = \sum_{k=1}^{\infty} (x, w_k) w_k. \tag{6.24}$$

The series (6.24) is called **generalized Fourier series** and the numbers

$$c_k = (x, w_k)$$

are the *Fourier coefficients* of x with respect to the basis $\{w_k\}_{k\geq 1}$. Moreover (Pythagoras again!):

$$\|x\|^2 = \sum_{k=1}^{\infty} (x, w_k)^2.$$

If $\{w_k\}_{k\geq 1}$ is an orthonormal sequence, the projection of $x \in H$ onto the subspace V spanned by, say, $w_1, \ldots, w_N$, is given by

$$P_V x = \sum_{k=1}^{N} (x, w_k) w_k.$$

Thus, for any other linear combination

$$S_N = \sum_{k=1}^{N} a_j w_j$$

we have

$$\|x - P_V x\| \leq \|x - S_N\|. \tag{6.25}$$

An example of separable Hilbert space is $L^2(\Omega)$, $\Omega \subseteq \mathbb{R}^n$. In particular, the set of functions

$$\frac{1}{\sqrt{2\pi}}, \frac{\cos x}{\sqrt{\pi}}, \frac{\sin x}{\sqrt{\pi}}, \frac{\cos 2x}{\sqrt{\pi}}, \frac{\sin 2x}{\sqrt{\pi}}, \ldots, \frac{\cos mx}{\sqrt{\pi}}, \frac{\sin mx}{\sqrt{\pi}}, \ldots$$

[4] δ_{kj} is the Kronecker symbol: $\delta_{ij} = 0$ if $j \neq i$, $\delta_{ii} = 1$.

constitutes an orthonormal basis in $L^2(0, 2\pi)$ (see e.g. *R. L. Wheeden, A. Zygmund*, [42]).

The following proposition is useful to check that an orthonormal sequence $\{w_k\}_{k\geq 1}$ is a basis for H.

Proposition 6.17 *An orthonormal sequence $\{w_k\}_{k\geq 1} \subset H$ is a basis for H if and only if one of the following conditions is satisfied.*

a) If $x \in H$ is orthogonal to w_k for every $k \geq 1$, then $x = 0$.
b) The set of all finite linear combination of the $w_k's$ is dense in H.

Proof If $\{w_k\}_{k\geq 1} \subset H$ is a basis for H and $x \in H$ is orthogonal to w_k for every $k \geq 1$, from (6.24) we get $x = 0$. Viceversa, if $\{w_k\}_{k\geq 1} \subset H$ is not a basis, then for some x we have $x \neq \sum_k (x, w_k) w_k \equiv \tilde{x}$; then $x - \tilde{x} \neq 0$ is orthogonal to all the w_k's.

We prove the second part. If $\{w_k\}_{k\geq 1} \subset H$ is a basis for H, every x can be approximated by the partial sums of the Fourier series (6.24) and therefore b) holds. Viceversa, if b) holds, given $x \in H$ and $\varepsilon > 0$, we can find $S_N = \sum_{k=1}^N a_j w_j$ such that $\|x - S_N\| \leq \varepsilon$. From (6.25) we infer that

$$\left\| x - \sum_{k=1}^{N} (x, w_k) w_k \right\| \leq \|x - S_N\| \leq \varepsilon.$$

Since ε was arbitrary, (6.24) holds for x and we conclude that $\{w_k\}_{k\geq 1}$ is a basis for H. □

It turns out that:

Proposition 6.18 *Every separable Hilbert space H admits a countable orthonormal basis.*

Proof Let $\{z_k\}_{k\geq 1}$ be dense in H. Disregarding, if necessary, those elements which are spanned by other elements in the sequence, we may assume that $\{z_k\}_{k\geq 1}$ constitutes an *independent set*, i.e. every finite subset of $\{z_k\}_{k\geq 1}$ is composed by independent elements.

Then, an orthonormal basis $\{w_k\}_{k\geq 1}$ is obtained by applying to $\{z_k\}_{k\geq 1}$ the following so called *Gram-Schmidt process*. First, we construct by induction a sequence $\{\tilde{w}\}_{k\geq 1}$ as follows. Let $\tilde{w}_1 = z_1$. Once $\tilde{w}_{k-1}$ is known, we construct $\tilde{w}_k$, $k \geq 2$, by subtracting from z_k its components with respect to $\tilde{w}_1, \ldots, \tilde{w}_{k-1}$:

$$\tilde{w}_k = z_k - \frac{(z_k, \tilde{w}_{k-1})}{\|\tilde{w}_{k-1}\|^2} \tilde{w}_{k-1} - \cdots - \frac{(z_k, \tilde{w}_1)}{\|\tilde{w}_1\|^2} \tilde{w}_1.$$

In this way, $\tilde{w}_k$ is orthogonal to $\tilde{w}_1, \ldots, \tilde{w}_{k-1}$. Finally, set

$$w_k = \frac{\tilde{w}_k}{\|\tilde{w}_k\|}.$$

Note that z_k is a linear combination of $w_1, w_2, \ldots w_k$. Since $\{z_k\}_{k\geq 1}$ is dense in H, then the set of all finite linear combination of the $w_k's$ is dense in H as well. By Proposition 6.17, $\{w_k\}_{k\geq 1}$ is a countable orthonormal basis in H. □

In the applications, orthonormal bases arise from solving particular boundary value problems, often in relation to the separation of variables method. Typical examples come from the vibrations of a nonhomogeneous string or from heat conduction in a rod with nonconstant thermal properties c_v, ρ, κ. The first example leads to the wave equation

$$\rho(x)u_{tt} - \tau u_{xx} = 0.$$

Setting $u(x,t) = v(x)z(t)$, we find for the spatial factor the equation

$$\tau v'' + \lambda \rho v = 0.$$

The second example leads to the heat equation

$$c_v \rho u_t - (\kappa u_x)_x = 0$$

and, separating the variables, we find for $v = v(x)$ the equation

$$(\kappa v')' + \lambda c_v \rho v = 0.$$

These equations are particular cases of a general class of ordinary differential equations of the form

$$(pu')' + qu + \lambda wu = 0, \tag{6.26}$$

called *Sturm-Liouville* equations. Usually one looks for solutions of (6.26) in an interval (a,b), $-\infty \le a < b \le +\infty$, satisfying suitable conditions at the end points. The natural assumptions on p and q are $p \neq 0$ in (a,b) and $p, q, 1/p$ locally integrable in (a,b). The function w plays the role of a *weight function*, continuous and *positive* in (a,b).

In general, the resulting boundary value problem has nontrivial solutions only for particular values of λ, called *eigenvalues*. The corresponding solutions are called *eigenfunctions* and it turns out that, when suitably normalized, they constitute an orthonormal basis in the Hilbert space $L^2_w(a,b)$, the set of Lebesgue measurable functions in (a,b), such that

$$\|u\|^2_{L^2_w} = \int_a^b u^2(x)w(x)dx < \infty,$$

endowed with the inner product

$$(u,v)_{L^2_w} = \int_a^b u(x)v(x)w(x)dx.$$

We list below some examples (for the proofs see *R. Courant, D. Hilbert*, [26], vol. 1).

- Consider the problem

$$\begin{cases} (1-x^2)u'' - xu' + \lambda u = 0 & \text{in } (-1,1) \\ u(-1) < \infty, \quad u(1) < \infty. \end{cases}$$

The differential equation is known as *Chebyshev* equation and may be written in the form (6.26):

$$((1-x^2)^{1/2}u')' + \lambda(1-x^2)^{-1/2}u = 0,$$

which shows the proper weight function $w(x) = (1 - x^2)^{-1/2}$. The eigenvalues are $\lambda_n = n^2$, $n = 0, 1, 2, \ldots$. The corresponding eigenfunctions are the *Chebyshev polynomials* T_n, recursively defined by $T_0(x) = 1$, $T_1(x) = x$ and

$$T_{n+1} = 2xT_n - T_{n-1} \qquad (n > 1).$$

For instance:

$$T_2(x) = 2x^2 - 1, \ T_3(x) = 4x^3 - 3x, \ T_4(x) = 8x^4 - 8x^2 + 1.$$

The normalized polynomials

$$\sqrt{1/\pi}\, T_0, \sqrt{2/\pi}\, T_1, \ldots, \sqrt{2/\pi}\, T_n, \ldots$$

constitute an orthonormal basis in $L^2_w(-1, 1)$.

- Consider the problem (see also Problem 8.4)

$$((1 - x^2)u')' + \lambda u = 0 \qquad \text{in } (-1, 1)$$

with weighted Neumann conditions

$$(1 - x^2)u'(x) \to 0 \quad \text{as } x \to \pm 1.$$

The differential equation is known as *Legendre* equation. The eigenvalues are $\lambda_n = n(n + 1)$, $n = 0, 1, 2, \ldots$. The corresponding eigenfunctions are the *Legendre polynomials*, defined by $L_0(x) = 1$, $L_1(x) = x$,

$$(n + 1)L_{n+1} = (2n + 1)xL_n - nL_{n-1} \qquad (n > 1)$$

or by *Rodrigues formula*

$$L_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n}(x^2 - 1)^n \qquad (n \geq 0).$$

For instance, $L_2(x) = (3x^2 - 1)/2$, $L_3(x) = (5x^3 - 3x)/2$. The normalized polynomials

$$\sqrt{\frac{2n + 1}{2}} L_n$$

constitute an orthonormal basis in $L^2(-1, 1)$ (here $w(x) \equiv 1$). Every function $f \in L^2(-1, 1)$ has an expansion

$$f(x) = \sum_{n=0}^{\infty} f_n L_n(x)$$

where $f_n = \frac{2n+1}{2} \int_{-1}^{1} f(x)L_n(x)dx$, with convergence in $L^2(-1, 1)$.

- Consider the problem

$$\begin{cases} u'' - 2xu' + 2\lambda u = 0 & \text{in } (-\infty, +\infty) \\ e^{-x^2/2}u(x) \to 0 & \text{as } x \to \pm\infty. \end{cases}$$

The differential equation is known as *Hermite* equation (see Problem 6.6) and may be written in the form (6.26):

$$(e^{-x^2}u')' + 2\lambda e^{-x^2}u = 0$$

which shows the proper weight function $w(x) = e^{-x^2}$. The eigenvalues are $\lambda_n = n$, $n = 0, 1, 2, \ldots$. The corresponding eigenfunctions are the *Hermite polynomials*, defined by Rodrigues formula

$$H_n(x) = (-1)^n e^{x^2} \frac{d^n}{dx^n} e^{-x^2} \qquad (n \geq 0).$$

For instance

$$H_0(x) = 1, \; H_1(x) = 2x, \;\; H_2(x) = 4x^2 - 2, \; H_3(x) = 8x^3 - 12x.$$

The normalized polynomials $(\sqrt{\pi}2^n n!)^{-1/2} H_n$ constitute an orthonormal basis in $L^2_w(\mathbb{R})$, with $w(x) = e^{-x^2}$. Every $f \in L^2_w(\mathbb{R})$ has an expansion

$$f(x) = \sum_{n=0}^{\infty} f_n H_n(x)$$

where $f_n = (\sqrt{\pi}2^n n!)^{-1} \int_{\mathbb{R}} f(x) H_n(x) e^{-x^2} dx$, with convergence in $L^2_w(\mathbb{R})$.

• After separating variables in the model for the vibration of a circular membrane, the following parametric *Bessel's equation of order p* arises (see Problem 6.8):

$$x^2u'' + xu' + (\lambda x^2 - p^2)u = 0 \qquad x \in (0, 1) \tag{6.27}$$

where $p \geq 0$, $\lambda \geq 0$, with the boundary conditions

$$u(0) \text{ finite}, \quad u(1) = 0. \tag{6.28}$$

Equation (6.27) may be written in Sturm-Liouville form as

$$(xu')' + \left(\lambda x - \frac{p^2}{x}\right) u = 0$$

which shows the proper weight function $w(x) = x$. The simple rescaling $z = \sqrt{\lambda}x$ reduces (6.27) to the Bessel's equation of order p

$$z^2 \frac{d^2u}{dz^2} + z\frac{du}{dz} + (z^2 - p^2)u = 0, \tag{6.29}$$

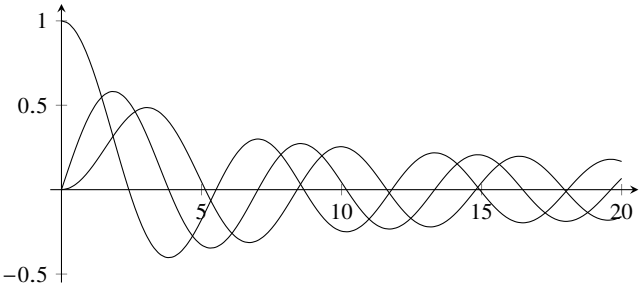

Fig. 6.2 Graphs of J_0, J_1 and J_2

where the dependence on the parameter λ is removed. The only bounded solutions of (6.29) are the *Bessel's functions of first kind and order* p, given by (see Fig. 6.2)

$$J_p(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{\Gamma(k+1)\Gamma(k+p+1)} \left(\frac{z}{2}\right)^{p+2k} \tag{6.30}$$

where

$$\Gamma(s) = \int_0^{\infty} e^{-t} t^{s-1} dt \tag{6.31}$$

is the Euler gamma function. In particular, if $p = n \geq 0$, integer:

$$J_n(z) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k!(k+n)!} \left(\frac{z}{2}\right)^{n+2k}.$$

For every $p \geq 0$, there exists an infinite, increasing sequence $\{\alpha_{pj}\}_{j\geq 1}$ of positive zeroes of J_p:

$$J_p(\alpha_{pj}) = 0 \qquad (j = 1, 2, \ldots).$$

Then, the eigenvalues of problem (6.27), (6.28) are given by $\lambda_{pj} = \left(\frac{\alpha_{pj}}{a}\right)^2$, with corresponding eigenfunctions $u_{pj}(x) = J_p\left(\frac{\alpha_{pj}}{a}x\right)$. The normalized eigenfunctions

$$\frac{\sqrt{2}}{aJ_{p+1}(\alpha_{pj})} J_p\left(\frac{\alpha_{pj}}{a}x\right)$$

constitute an orthonormal basis in $L^2_w(0,a)$, with $w(x) = x$. Every function $f \in L^2_w(0,a)$ has an expansion in *Fourier-Bessel series*

$$f(x) = \sum_{j=1}^{\infty} f_j J_p\left(\frac{\alpha_{pj}}{a}x\right),$$

where

$$f_j = \frac{2}{a^2 J_{p+1}^2(\alpha_{pj})} \int_0^a x f(x) J_p(\frac{\alpha_{pj}}{a} x) dx,$$

convergent in $L_w^2(0, a)$.

6.5 Linear Operators and Duality

6.5.1 Linear operators

Let H_1 and H_2 be Hilbert spaces. A linear operator from H_1 into H_2 is a function

$$L : H_1 \to H_2$$

such that[5] $\forall \alpha, \beta \in \mathbb{R}$ and $\forall x, y \in H_1$

$$L(\alpha x + \beta y) = \alpha Lx + \beta Ly.$$

For every linear operator we define its *kernel*, $\mathcal{N}(L)$ and *range,* $\mathcal{R}(L)$, as follows:

Definition 6.19 The *kernel*, or *null space*, of L is the preimage of the null vector in H_2:

$$\mathcal{N}(L) = \{x \in H_1 : Lx = 0\}.$$

The *range*, or *image*, of L is the set of all outputs from points in H_1:

$$\mathcal{R}(L) = \{y \in H_2 : \exists x \in H_1, Lx = y\}.$$

$\mathcal{N}(L)$ and $\mathcal{R}(L)$ are linear subspaces of H_1 and H_2, respectively. If $\mathcal{N}(L) = \{0\}$ then we say that L is *injective* or a *one-to-one* operator. In this case there exists an inverse operator $L^{-1} : \mathcal{R}(L) \to H_1$ such that $L \circ L^{-1} = I_{H_1}$, the identity operator in H_1, and $L^{-1} \circ L = I_{\mathcal{R}(L)}$, the identity operator in $\mathcal{R}(L)$.

If $\mathcal{R}(L) = H_2$ we say that L is *surjective* or *onto*. If $L : H_1 \to H_2$ is one-to-one and onto, the equation

$$Lx = y \tag{6.32}$$

has one and only one solution $x \in H_1$ for every $y \in H_2$.

Our main objects will be the linear bounded operators.

Definition 6.20 A linear operator $L : H_1 \to H_2$ is *bounded* if there exists a number C such that

$$\|Lx\|_{H_2} \leq C \|x\|_{H_1}, \qquad \forall x \in H_1. \tag{6.33}$$

The number C controls the norm expansion operated by L on the elements of H_1. In particular, if $C < 1$, L contracts the size of the vectors in H_1.

[5] Notation: if L is linear, when no confusion arises, we may write Lx instead of $L(x)$.

If $x \neq 0$, using the linearity of L, we may write (6.33) in the form

$$\left\| L\left(\frac{x}{\|x\|_{H_1}} \right) \right\|_{H_2} \leq C,$$

which is equivalent to

$$\sup_{\|x\|_{H_1}=1} \|Lx\|_{H_2} = K(L) < \infty, \tag{6.34}$$

since $x/\|x\|_{H_1}$ is a unit vector in H_1. Clearly $K(L) \leq C$.

Proposition 6.21 *A linear operator $L : H_1 \to H_2$ is bounded if and only if it is continuous.*

Proof Let L be bounded. From (6.33) we have, $\forall x, x_0 \in H_1$,

$$\|L(x - x_0)\|_{H_2} \leq C \, \|x - x_0\|_{H_1}$$

so that, if $\|x - x_0\|_{H_1} \to 0$, also $\|Lx - Lx_0\|_{H_2} = \|L(x - x_0)\|_{H_2} \to 0$. This shows the continuity of L.

Let L be continuous. In particular, L is continuous at $x = 0$ so that there exists δ such that

$$\|Lx\|_{H_2} \leq 1 \quad \text{if } \|x\|_{H_1} \leq \delta.$$

Choose now $y \in H_1$, with $\|y\|_{H_1} = 1$, and let $z = \delta y$. We have $\|z\|_{H_1} = \delta$ which implies

$$\delta \, \|Ly\|_{H_2} = \|Lz\|_{H_2} \leq 1$$

or

$$\|Ly\|_{H_2} \leq \frac{1}{\delta}$$

and (6.34) holds with $K \leq C = \frac{1}{\delta}$. □

If $L : H_1 \to H_2$ is a bounded operator, $\mathcal{N}(L)$ is a *closed subspace* of H_1. In fact, if $\{x_n\} \subset \mathcal{N}(L)$ and $x_n \to x$, then $0 = Lx_n \to Lx$ so that $x \in \mathcal{N}(L)$; thus $\mathcal{N}(L)$ contains all its limit points and therefore is closed. However the subspace $\mathcal{R}(L)$ could not be closed in H_2 as the following example shows.

Example 6.22 (A bounded operator whose range is not closed) Let $H_1 = H_2 = l^2$. Define $L : l^2 \to l^2$ as follows. If $x = \{x_n\}_{n\geq 1} \in l^2$, set

$$Lx = y, \text{ where } y = \{x_n/\sqrt{n}\}_{n\geq 1}.$$

Then $y \in l^2$ and

$$\|y\|_{l^2}^2 = \textstyle\sum_{n=1}^{\infty} \frac{1}{n} x_n^2 \leq \sum_{n=1}^{\infty} x_n^2 = \|x\|_{l^2}^2$$

so that L is bounded. Now note that the element $y = \{1/n\}_{n\geq 1}$ belongs to l^2 but not to $\mathcal{R}(L)$, as it is easy to check. On the other hand $\mathcal{R}(L)$ contains all the elements $y^k = \{y_n^k\}$, where

$$y_n^k = \begin{cases} 1/n & 1 \leq n \leq k \\ 0 & n \geq k. \end{cases} \qquad k = 1, 2, \ldots$$

Since $y^k \to y$ in l^2 it follows that $\mathcal{R}(L)$ is not closed.

Given two Hilbert spaces H_1 and H_2, we denote by

$$\mathcal{L}(H_1, H_2)$$

the family of all linear bounded operators from H_1 into H_2. If $H_1 = H_2 = H$, we simply write $\mathcal{L}(H)$. $\mathcal{L}(H_1, H_2)$ becomes a linear space if we define, for $x \in H_1$ and $\lambda \in \mathbb{R}$,

$$(G + L)(x) = Gx + Lx$$
$$(\lambda L)x = \lambda Lx.$$

Also, it is easy to check that the number $K(L)$ in (6.34) satisfies the properties of a norm and therefore we may choose it as a norm in $\mathcal{L}(H_1, H_2)$:

$$\|L\|_{\mathcal{L}(H_1,H_2)} = \sup_{\|x\|_{H_1}=1} \|Lx\|_{H_2}. \tag{6.35}$$

Thus, for every $L \in \mathcal{L}(H_1, H_2)$, we have

$$\|Lx\|_{H_2} \le \|L\|_{\mathcal{L}(H_1,H_2)} \|x\|_{H_1}.$$

The resulting space is complete, so that:

Proposition 6.23 *Endowed with the norm* (6.35)*, $\mathcal{L}(H_1, H_2)$ is a Banach space.*

Example 6.24 Let $\mathbf{A}$ be an $m \times n$ real matrix. The map

$$L : \mathbf{x} \longmapsto \mathbf{Ax}$$

is a linear operator from $\mathbb{R}^n$ into $\mathbb{R}^m$. To compute $\|L\|_{\mathcal{L}(\mathbb{R}^n,\mathbb{R}^m)}$, note that

$$\|\mathbf{Ax}\|^2 = \mathbf{Ax} \cdot \mathbf{Ax} = \mathbf{A}^\top \mathbf{Ax} \cdot \mathbf{x}.$$

The matrix $\mathbf{A}^\top \mathbf{A}$ is symmetric and nonnegative and therefore, from Linear Algebra,

$$\sup_{\|\mathbf{x}\|=1} \mathbf{A}^\top \mathbf{Ax} \cdot \mathbf{x} = \Lambda_M$$

where Λ_M is the maximum eigenvalue of $\mathbf{A}^\top \mathbf{A}$. Thus, $\|L\|_{\mathcal{L}(\mathbb{R}^n,\mathbb{R}^m)} = \sqrt{\Lambda_M}$.

Example 6.25 Let V be a *closed subspace* of a Hilbert space H. The projections

$$x \longmapsto P_V x, \qquad x \longmapsto Q_V x,$$

defined in Theorem 6.12, page 356, are bounded linear operators from H into H. In fact, they are linear (see Problem 6.4), and from

$$\|x\|^2 = \|P_V x\|^2 + \|Q_V x\|^2,$$

it follows immediately that

$$\|P_V x\| \leq \|x\|, \qquad \|Q_V x\| \leq \|x\|$$

so that (6.33) holds with $C = 1$. Since $P_V x = x$ when $x \in V$ and $Q_V x = x$ when $x \in V^\perp$, it follows that

$$\|P_V\|_{\mathcal{L}(H)} = \|Q_V\|_{\mathcal{L}(H)} = 1.$$

Finally, observe that

$$\mathcal{N}(P_V) = \mathcal{R}(Q_V) = V^\perp \quad \text{and} \quad \mathcal{N}(Q_V) = \mathcal{R}(P_V) = V.$$

Example 6.26 Let V and H be Hilbert spaces, with[6] $V \subset H$. Considering an element in V as an element of H, we define the operator $I_{V\to H} : V \to H$,

$$I_{V\to H}(u) = u,$$

which is called *embedding of V into H*. $I_{V\to H}$ is clearly a linear operator and it is also bounded if there exists a constant C such that

$$\|u\|_H \leq C \|u\|_V, \qquad \text{for every } u \in V.$$

In this case, we say that V *is continuously embedded in* H and we write

$$V \hookrightarrow H.$$

For instance, $H^1_{\text{per}}(0, 2\pi) \hookrightarrow L^2(0, 2\pi)$.

An important theorem in Functional Analysis states that (for the proof, see e.g. *K. Yosida*, [43]):

Theorem 6.27 *Let $L \in \mathcal{L}(H_1, H_2)$ be one-to-one and onto. Then the inverse operator $L^{-1} : H_2 \to H_1$ is bounded.*

In other words, Theorem 6.27 states that, if $L \in \mathcal{L}(H_1, H_2)$ is one-to-one and onto, and $x = L^{-1}y$ is the solution of equation $Lx = y$, the following estimate holds

$$\|x\|_{H_1} \leq \left\|L^{-1}\right\|_{\mathcal{L}(H_2,H_1)} \|y\|_{H_2}$$

for every $y \in H_2$.

We say that L is a **continuous isomorphism** between H_1 and H_2.

6.5.2 Functionals and dual space

A linear operator $L : H \to \mathbb{R}$ takes the name of (real) **functional**.

[6] The inner products in V and H may be different.

Definition 6.28 The collection of all bounded linear functionals on a Hilbert space H is called the *dual space* of H and it is denoted by H^* (instead of $\mathcal{L}(H, \mathbb{R})$).

Example 6.29 Let $H = L^2(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ and fix $g \in L^2(\Omega)$. The functional defined by

$$L_g : f \longmapsto \int_\Omega fg$$

is linear and bounded. In fact, Schwarz inequality yields

$$\left|L_g f\right| = \left|\int_\Omega fg\right| \leq \left(\int_\Omega |f|^2\right)^{1/2} \left(\int_\Omega |g|^2\right)^{1/2} = \|g\|_{L^2(\Omega)} \, \|f\|_{L^2(\Omega)}$$

so that $L_g \in L^2(\Omega)^*$ and $\left\|L_g\right\|_{L^2(\Omega)^*} \leq \|g\|_{L^2(\Omega)}$. Actually $\left\|L_g\right\|_{L^2(\Omega)^*} = \|g\|_{L^2(\Omega)}$ since, choosing $f = g$, we have

$$\|g\|^2_{L^2(\Omega)} = L_g(g) \leq \left\|L_g\right\|_{L^2(\Omega)^*} \|g\|_{L^2(\Omega)}$$

whence also $\left\|L_g\right\|_{L^2(\Omega)^*} \geq \|g\|_{L^2(\Omega)}$.

Example 6.30 The functional in Example 6.29 is induced by the inner product with a fixed element in $L^2(\Omega)$. More generally, let H be a Hilbert space. For fixed $y \in H$, the functional

$$L_1 : x \longmapsto (x, y)$$

is continuous. In fact the Schwarz inequality yields

$$|(x, y)| \leq \|x\| \, \|y\| \,,$$

whence $L_1 \in H^*$ and $\|L_1\|_{H^*} \leq \|y\|$. Actually $\|L_1\|_{H^*} = \|y\|$ since, choosing $x = y$, we have

$$\|y\|^2 = |L_1 y| \leq \|L_1\|_{H^*} \, \|y\| \,,$$

or $\|L_1\|_{H^*} \geq \|y\|$. Observe that this argument provides the following alternative definition of the norm of an element $y \in H$:

$$\|y\| = \sup_{\|x\|=1} |(x, y)| \,. \tag{6.36}$$

To identify the dual space of a Hilbert space H is crucial in many instances. Example 6.30 shows that the inner product with a fixed element y in H defines an element of H^*, whose norm is exactly $\|y\|$. From Linear Algebra it is well known that all linear functionals in a finite-dimensional space can be represented in that way. Precisely, if $L : \mathbb{R}^n \to \mathbb{R}$ is linear there exists a vector $\mathbf{a} \in \mathbb{R}^n$ such that, for every $\mathbf{h} \in \mathbb{R}^n$,

$$L\mathbf{h} = \mathbf{a} \cdot \mathbf{h}$$

and $\|L\|_{(\mathbb{R}^n)^*} = |\mathbf{a}|$. The following theorem, known as **Riesz representation Theorem**, says that an analogous result holds in Hilbert spaces.

Theorem 6.31 *Let H be a Hilbert space. For every $L \in H^*$ there exists a unique $u_L \in H$ such that:*

$$Lx = (x, u_L), \qquad \forall x \in H.$$

Moreover, $\|u_L\| = \|L\|_{H^*}$.

Proof Let $\mathcal{N} = \mathcal{N}(L)$ be the kernel of L. As we already noticed, $\mathcal{N}$ is a *closed* subspace of H. If $\mathcal{N} = H$, then L is the *null operator* and $u_L = 0$. If $\mathcal{N} \subset H$, by the Projection Theorem, there exists $z \in \mathcal{N}^\perp$, $z \neq 0$. Thus $Lz \neq 0$ and, given any $x \in H$, the element

$$w = x - \frac{Lx}{Lz} z$$

belongs to $\mathcal{N}$. In fact

$$Lw = L\left(x - \frac{Lx}{Lz} z\right) = Lx - \frac{Lx}{Lz} Lz = 0.$$

Since $z \in \mathcal{N}^\perp$, we have

$$0 = (w, z) = (x, z) - \frac{Lx}{Lz} \|z\|^2$$

which entails

$$Lx = \frac{L(z)}{\|z\|^2} (x, z).$$

Therefore if $u_L = L(z) \|z\|^{-2} z$, then $Lx = (x, u_L)$.

For the uniqueness, observe that, if $v \in H$ and

$$Lx = (x, v), \quad \text{for every } x \in H,$$

subtracting this equation from $Lx = (x, u_L)$, we infer

$$(x, u_L - v) = 0, \qquad \text{for every } x \in H$$

which forces $v = u_L$ (by choosing $x = u_L - v$).

To show that $\|u_L\| = \|L\|_{H^*}$, use the Schwarz inequality

$$|(x, u_L)| \leq \|x\| \, \|u_L\|$$

to get

$$\|L\|_{H^*} = \sup_{\|x\|=1} |Lx| = \sup_{\|x\|=1} |(x, u_L)| \leq \|u_L\| .$$

On the other hand,

$$\|u_L\|^2 = (u_L, u_L) = Lu_L \leq \|L\|_{H^*} \, \|u_L\|$$

whence

$$\|u_L\| \leq \|L\|_{H^*} .$$

Thus $\|u_L\| = \|L\|_{H^*}$. □

The Riesz map $\mathcal{R}_H : H^* \to H$, given by

$$L \longmapsto u_L = \mathcal{R}_H L,$$

is called *canonical isometry* and we say that u_L is the *Riesz element associated with* L, with respect to the scalar product $(\cdot, \cdot)$. Moreover, H^* endowed with the inner product

$$(L_1, L_2)_{H^*} = (u_{L_1}, u_{L_2}) \tag{6.37}$$

is clearly a Hilbert space. Thus, in the end, the Representation Theorem 6.31, allows the **identification of a Hilbert space with its dual**.

Remark 6.32 A few words about **notations**. The symbol $(\cdot,\cdot)$ or $(\cdot,\cdot)_H$ denotes the inner product in a Hilbert space H. Let now $L \in H^*$. For the *action* of the functional L on an element $x \in H$ we used the symbol Lx. Sometimes, when it is useful or necessary to emphasize the *duality (or pairing)* between H and H^*, we shall use the notation $\langle L, x\rangle_*$ or even $\langle L, x\rangle_{H^*,H}$.

6.5.3 The (Hilbert) adjoint of a bounded operator

The concept of *adjoint operator* extends the notion of transpose of an $m \times n$ matrix $\mathbf{A}$ and plays a crucial role in several situations, for instance, in determining compatibility conditions for the solvability of several problems. The transpose $\mathbf{A}^\top$ is characterized by the identity

$$(\mathbf{Ax}, \mathbf{y})_{\mathbb{R}^m} = (\mathbf{x}, \mathbf{A}^\top \mathbf{y})_{\mathbb{R}^n}, \qquad \forall \mathbf{x} \in \mathbb{R}^n, \forall \mathbf{y} \in \mathbb{R}^m.$$

We extend precisely this relation to define the adjoint of a bounded linear operator. Let $L \in \mathcal{L}(H_1, H_2)$. If $y \in H_2$ is fixed, the real map

$$T_y : x \longmapsto (Lx, y)_{H_2}$$

defines an element of H_1^*. In fact

$$\left|T_y x\right| = \left|(Lx, y)_{H_2}\right| \le \|Lx\|_{H_2} \|y\|_{H_2} \le \|L\|_{\mathcal{L}(H_1,H_2)} \|y\|_{H_2} \|x\|_{H_1}$$

so that $\left\|T_y\right\|_{H_1^*} \le \|L\|_{\mathcal{L}(H_1,H_2)} \|y\|_{H_2}$.

From Riesz's Theorem, there exists a unique $w \in H_1$ depending on y, which we denote by $w = L^* y$, such that

$$T_y x = (x, L^* y)_{H_1}, \qquad \forall x \in H_1, \forall y \in H_2. \tag{6.38}$$

The relation (6.38) defines L^* as a (linear) operator from H_2 into H_1, which is called the *(Hilbert) adjoint* of L. Precisely:

Definition 6.33 The operator $L^* : H_2 \to H_1$ defined by the identity

$$(Lx, y)_{H_2} = (x, L^* y)_{H_1}, \qquad \forall x \in H_1, \forall y \in H_2 \tag{6.39}$$

is called the *adjoint* of L.

Example 6.34 Let $L : H_1 \to H_2$ be an isometry. Then $L^* = L^{-1}$. In fact, since L preserves the inner product, we can write

$$(x, L^* y)_{H_1} = (Lx, y)_{H_2} = (x, L^{-1} y)_{H_1}, \qquad \forall x \in H_1, \forall y \in H_2.$$

In particular, if $\mathcal{R}_H : H^* \to H$ is the Riesz map, then $\mathcal{R}_H^* = \mathcal{R}_H^{-1} : H \to H^*$.

Example 6.35 Let $T : L^2(0,1) \to L^2(0,1)$ be the linear map

$$(Tu)(x) = \int_0^x u(t)dt.$$

The Schwarz inequality gives

$$\left| \int_0^x u \right|^2 \leq x \int_0^x u^2,$$

whence

$$\begin{aligned} \|Tu\|^2_{L^2(0,1)} = \int_0^1 |Tu|^2 = \int_0^1 \left| \int_0^x u \right|^2 dx \leq \int_0^1 \left(x \int_0^x u^2 \right) dx \\ \leq \frac{1}{2} \int_0^1 u^2 = \frac{1}{2} \|u\|^2_{L^2(0,1)} \end{aligned}$$

and therefore T is bounded. To compute T^*, observe that

$$\begin{aligned} (Tu, v)_{L^2(0,1)} = \int_0^1 \left[v(x) \int_0^x u(y)dy \right] dx = \text{[exchanging the order of integration]} \\ = \int_0^1 \left[u(y) \int_y^1 v(x)dx \right] dy = (u, T^*v)_{L^2(0,1)}. \end{aligned}$$

Thus,

$$(T^*v)(x) = \int_x^1 v(t)dt.$$

Symmetric matrices correspond to selfadjoint operators. We say that L is **selfadjoint** if $H_1 = H_2$ and $L^* = L$. Then, (6.39) reduces to

$$(Lx, y) = (x, Ly).$$

Example 6.36 An example of selfadjoint operator in a Hilbert space H is the projection P_V on a closed subspace V of H; in fact, recalling the Projection Theorem:

$$(P_V x, y) = (P_V x, P_V y + Q_V y) = (P_V x, P_V y) = (P_V x + Q_V x, P_V y) = (x, P_V y).$$

Important self-adjoint operators are associated with *inverses of differential operators*, as we will see in Chap. 8.

The following properties are immediate consequences of the definition of adjoint.

Proposition 6.37 *Let* $L, L_1 \in \mathcal{L}(H_1, H_2)$ *and* $L_2 \in \mathcal{L}(H_2, H_3)$. *Then:*

a) $L^* \in \mathcal{L}(H_2, H_1)$. *Moreover* $(L^*)^* = L$ *and*

$$\|L^*\|_{\mathcal{L}(H_2,H_1)} = \|L\|_{\mathcal{L}(H_1,H_2)}.$$

b) $(L_2L_1)^* = L_1^*L_2^*$.

c) *L is a continuous isomorphism between H_1 and H_2 if and only if L^* is a continuous isomorphism between H_2 and H_1. Moreover*

$$(L^{-1})^* = (L^*)^{-1}.$$

Proof For a) and b) see Problem 6.10.

c) Since from a) $(L^*)^* = L$, it is enough to show that, if L is a continuous isomorphism between H_1 and H_2, then L^* is the same between H_2 and H_1. Therefore we show that the equation

$$L^*y = x \tag{6.40}$$

has a unique solution and

$$\exists c > 0 : \|y\|_{H_2} \le c\,\|x\|_{H_1}. \tag{6.41}$$

Equation (6.40) is equivalent to $(L^*y, z)_{H_1} = (x, z)_{H_1}\ \forall z \in H_1$. Set $z = L^{-1}w$ to get the equivalent equation

$$(L^*y, L^{-1}w)_{H_1} = (y, w)_{H_1} = (x, L^{-1}w)_{H_1} \quad \forall w \in H_2.$$

Since L^{-1} is continuous,

$$|(x, L^{-1}w)_{H_1}| \le \left\|L^{-1}\right\|_{\mathcal{L}(H_2,H_1)} \|x\|_{H_1}\,\|w\|_{H_2}$$

so that the functional $G : w \mapsto (x, L^{-1}w)_{H_1}$ belongs to H_2^* with

$$\|G\|_{H_2^*} \le \left\|L^{-1}\right\|_{\mathcal{L}(H_2,H_1)} \|x\|_{H_1}.$$

By the Riesz Representation Theorem, there is a unique $y \in H_2$ such that

$$(y, w)_{H_2} = (x, L^{-1}w)_{H_1} \quad \forall w \in H_2$$

and

$$\|y\|_{H_2} = \|G\|_{H_2^*} \le \left\|L^{-1}\right\|_{\mathcal{L}(H_2,H_1)} \|x\|_{H_1},$$

which is (6.41) with $\left\|L^{-1}\right\|_{\mathcal{L}(H_2,H_1)} = c$.

Finally, since from b) $I_{H_1} = (L^{-1}L)^* = L^*(L^{-1})^*$, it follows that $(L^{-1})^* = (L^*)^{-1}$. □

6.5.4 Ranges and Kernels. The inf-sup condition

The next theorem extends some relations, well known in the finite-dimensional case.

Theorem 6.38 *Let $L \in \mathcal{L}(H_1, H_2)$. Then*

a) $\mathcal{N}(L^*) = \mathcal{R}(L)^\perp$ *or* $\overline{\mathcal{R}(L)} = \mathcal{N}(L^*)^\perp$

b) $\mathcal{N}(L) = \mathcal{R}(L^*)^\perp$ *or* $\overline{\mathcal{R}(L^*)} = \mathcal{N}(L)^\perp$.

Proof a) We show the first relation. The second one is obtained by passing to the orthogonal, remembering that $\mathcal{R}(L)$ could not be closed. We have:

$$\begin{aligned} y \in \mathcal{N}(L^*) \iff L^*y = 0 &\iff (L^*y, x)_{H_1} = 0,\ \forall x \in H_1 \\ &\iff (y, Lx)_{H_2} = 0,\ \forall x \in H_1 \iff y \in \mathcal{R}(L)^\perp. \end{aligned}$$

b) Let $L = L^*$ in a) and use $(L^*)^* = L$. □

We stress that in the relations a) and b) above, $\mathcal{R}(L)^\perp$ and $\mathcal{R}(L^*)^\perp$ are *always* closed subsets of H_2 and H_1 respectively. For $\mathcal{R}(L)$ and $\mathcal{R}(L^*)$ this is not always the case, as Example 6.22 shows, so that it is necessary to consider their closure when passing to the orthogonal. In particular Theorem 6.38 indicates that a key information for the solvability of the equation $Lx = y$ would be to know that $\mathcal{R}(L)$ is *closed*.

Indeed, assume that $\mathcal{R}(L)$ is closed. Then we deduce that

$$y \in \mathcal{R}(L) \iff y \text{ is orthogonal to every element of the adjoint kernel } \mathcal{N}(L^*)$$

or, in other words, that the equation

$$Lx = y \text{ is solvable} \iff (y, w)_{H_2} = 0, \forall w : L^* w = 0.$$

In particular,

$$L \textit{ is onto if and only if } L^* \textit{ is one-to-one.} \tag{6.42}$$

Equivalently, *either* the equation $Lx = y$ has a solution for all $y \in H_2$ *or* the homogeneous adjoint equation $L^* w = 0$ has non trivial solutions (a version of the so called *Alternative Theorem* we will meet later on).

When can we insure that $\mathcal{R}(L)$ is closed? Of course this is true if $\mathcal{R}(L)$ has finite dimension. The next theorem gives a *necessary and sufficient condition* for L to have *both closed range and to be one-to-one*. Then, since $\mathcal{R}(L)$ is closed, it is itself a Hilbert subspace of H_2 and, by Theorem 6.27, $L^{-1} : \mathcal{R}(T) \to H_1$ *is continuous*.

Theorem 6.39 *Let $L \in \mathcal{L}(H_1, H_2)$. Then L is one-to-one and $\mathcal{R}(L) \subseteq H_2$ is closed if and only if*

$$\exists \alpha > 0 : \forall x \in H_1, \quad \|Lx\|_{H_2} \geq \alpha \|x\|_{H_1}. \tag{6.43}$$

Proof Let (6.43) hold. If $Lx = 0$ then $x = 0$ and L is injective. Let now $\{y_n\} \subset \mathcal{R}(L)$, $y_n \to y$. Then $y_n = Lx_n$ for some $x_n \in H_1$, and (6.43) gives

$$\|x_n - x_m\|_{H_1} \leq \alpha^{-1} \|y_n - y_m\|_{H_2}.$$

Therefore, since $\{y_n\}$ is a Cauchy sequence in H_2, $\{x_n\}$ is also a Cauchy sequence in H_1 and there exists $x \in H_1$ such that $x_n \to x$. Since L is continuous, $y_n = Lx_n \to Lx = y$. Thus $y \in \mathcal{R}(L)$ and $\mathcal{R}(L)$ is closed.

Viceversa, let L be injective with closed range. Then L has an inverse $L^{-1} : \mathcal{R}(L) \to H_1$ and (since $\mathcal{R}(L)$ is closed) Theorem 6.27 implies that L^{-1} is continuous; therefore $\exists c > 0$ such that

$$\forall y \in \mathcal{R}(L) : \left\|L^{-1} y\right\|_{H_1} \leq c \|y\|_{H_2}$$

which is (6.43) with $\alpha = c^{-1}$, $y = Lx$. □

Using (6.42), the following corollary is now immediate.

Corollary 6.40 *Let $L \in \mathcal{L}(H_1, H_2)$. Then L is a continuous isomorphism* ***if and only if*** *the following two conditions are fulfilled:*

1. *(6.43) holds;*
2. $\mathcal{N}(L^*) = \{0\}$.

Moreover,

$$\left\|L^{-1}\right\|_{\mathcal{L}(H_2,H_1)} \leq \alpha^{-1}. \tag{6.44}$$

The conditions 1 and 2 above can be expressed in equivalent ways, sometimes more useful. Indeed, by recalling that (see (6.36), p. 369)

$$\|Lx\|_{H_2} = \sup_{y\in H_2} \frac{(Lx,y)_{H_2}}{\|y\|_{H_2}},$$

we see that (6.43) is equivalent to the following so called **inf-sup condition**:

$$\exists \alpha > 0 : \inf_{x\in H_1} \sup_{y\in H_2} \frac{(Lx,y)_{H_2}}{\|x\|_{H_1}\,\|y\|_{H_2}} \geq \alpha. \tag{6.45}$$

Moreover, since $L^*y = 0$ if and only if $(x, L^*y)_{H_1} = (Lx, y)_{H_2} = 0$ for every $x \in H_1$, the condition 2 can be written in terms of L, in the following form,

$$\left[(Lx,y)_{H_2} = 0,\ \forall x \in H_1\right] \implies y = 0. \tag{6.46}$$

We close this section with another important theorem. For the proof see e.g. *K. Yosida*, [43].

Theorem 6.41 *Let $L \in \mathcal{L}(H_1, H_2)$. Then $\mathcal{R}(L)$ is closed if and only if $\mathcal{R}(L^*)$ is closed.*

6.6 Abstract Variational Problems

6.6.1 Bilinear forms. The Lax-Milgram Theorem

In the variational formulation of several boundary value problems a key role is played by bilinear forms. Given two linear spaces V, W, a (real) **bilinear form** in $V \times W$ is a function

$$a : V \times W \to \mathbb{R}$$

satisfying the following properties:

a) For every $v \in W$, the function $u \longmapsto a(u, v)$ is linear in V.
b) For every $u \in V$, the function $v \longmapsto a(u, v)$ is linear in W.

When $V = W$, we simply say that a is a *bilinear form in V*. Here are some examples, in which Ω is a bounded domain in $\mathbb{R}^n$.

- A typical example of bilinear form in a Hilbert space is its inner product.
- The formula

$$a(u,v) = \int_a^b (p(x)u'v' + q(x)u'v + r(x)uv)dx$$

where p, q, r are bounded functions, defines a bilinear form in $C^1([a,b])$. More generally,

$$a(u,v) = \int_\Omega (\alpha(\mathbf{x})\ \nabla u \cdot \nabla v + u\mathbf{b}(\mathbf{x}) \cdot \nabla v + a_0(\mathbf{x})uv)d\mathbf{x},$$

or

$$a(u,v) = \int_\Omega \alpha(\mathbf{x})\ \nabla u \cdot \nabla v d\mathbf{x} + \int_{\partial\Omega} huv\ d\sigma,$$

(α, $\mathbf{b}$, a_0, h bounded) are bilinear forms in $C^1(\overline{\Omega})$.

- A bilinear form in $C^2(\overline{\Omega})$, involving higher order derivatives, is

$$a(u,v) = \int_\Omega \Delta u\ \Delta v\ d\mathbf{x}.$$

- The formula

$$b(\mathbf{u},q) = \int_\Omega \operatorname{div}\mathbf{u}\ q\ d\mathbf{x}$$

defines a bilinear form in $C^1(\overline{\Omega};\mathbb{R}^n) \times L^2(\Omega)$.

Let V, W be Hilbert spaces, a be a bilinear form in $V \times W$ and $F \in W^*$. Our purpose is to examine the well-posedness of the following problem, called *abstract variational problem*:

$$\begin{cases} \text{Find } u \in V \text{ such that} \\ a(u,v) = \langle F, v\rangle_*, \quad \forall v \in W \end{cases} \tag{6.47}$$

As we shall see, many boundary values problems can be recast in this form. In the case $V = W$, sufficient conditions are provided by the following fundamental result known as the **Lax-Milgram Theorem**:

Theorem 6.42 *Let V be a (real) Hilbert space endowed with inner product $(\cdot,\cdot)$ and norm $\|\cdot\|$. Let $a = a(u,v)$ be a bilinear form in V. If:*

*1. a is **continuous**, i.e. there exists a constant M such that*

$$|a(u,v)| \le M\ \|u\|\ \|v\|, \qquad \forall u, v \in V.$$

*2. a is V-**coercive**, i.e. there exists a constant $\alpha > 0$ such that*

$$a(v,v) \ge \alpha\ \|v\|^2, \qquad \forall v \in V, \tag{6.48}$$

then there exists a unique solution $\bar{u} \in V$ of problem (6.47). Moreover, the following stability estimate holds:

$$\|\bar{u}\| \le \frac{1}{\alpha}\ \|F\|_{V^*}. \tag{6.49}$$

Remark 6.43 The coercivity inequality (6.48) may be considered as an abstract version of the *energy* or *integral estimates* we met in the previous chapters. Usually, it is the key estimate to prove in order to apply Theorem 6.42.

Proof (of Theorem 6.42) We split it into several steps.

1. *Reformulation of problem* (6.47). For every fixed $u \in V$, by the continuity of a, the linear map

$$v \mapsto a(u, v)$$

is bounded in V and therefore it defines an element of V^*. From the Riesz Representation Theorem and the linearity of a with respect to u, there exists a unique linear operator $A : V \to V$ such that

$$a(u, v) = (Au, v), \qquad \forall v \in V. \tag{6.50}$$

Since $F \in V^*$ as well, there exists a unique $z_F \in V$ such that

$$Fv = (z_F, v), \qquad \forall v \in V$$

and moreover $\|F\|_{V^*} = \|z_F\|$. Then, problem (6.47) can be recast in the following way:

$$\begin{cases} \text{Find } u \in V \text{ such that} \\ (Au, v) = (z_F, v), \quad \forall v \in V \end{cases}$$

which, in turn, is equivalent to **finding u such that**

$$Au = z_F. \tag{6.51}$$

We want to show that (6.51) has exactly one solution which means that $A : V \to V$ is a *continuous isomorphism*.

2. *Continuity of* A. Observe that

$$\|Au\|^2 = (Au, Au) = a(Au, u) \le M \|Au\| \|u\|$$

whence

$$\|Au\| \le M \|u\| .$$

3. Now we use Corollary 6.40. The coercivity of a yields

$$\alpha \|u\|^2 \le a(u, u) = (Au, u) \le \|Au\| \|u\|$$

whence

$$\|u\| \le \frac{1}{\alpha} \|Au\|_V \tag{6.52}$$

which is condition (6.43). On the other hand, if

$$(Au, v) = 0 \qquad \forall u \in V,$$

choosing $u = v$ we get, again from the coercivity of a, $0 = (Av, v) = a(v, v) \ge \alpha \|v\|^2$, whence $v = 0$ and also condition (6.46) is satisfied. Therefore A is a continuous isomorphism.

4. *Solution of problem* (6.47). Since A is one-to-one and $\mathcal{R}(A) = V$, there exists exactly one solution $\bar{u} \in V$ of equation

$$Au = z_F.$$

From point **1**, $\bar{u}$ is the unique solution of problem (6.47) as well.

5. *Stability estimate.* From (6.52) with $u = \bar{u}$, we obtain

$$\|\bar{u}\| \le \frac{1}{\alpha} \|A\bar{u}\| = \frac{1}{\alpha} \|z_F\| = \frac{1}{\alpha} \|F\|_{V^*}$$

Remark 6.44 Since a is a continuous bilinear form, we can define an operator $L \in \mathcal{L}(V, V^*)$ that associates to every $u \in V$ the functional $Lu \in V^*$ given by

$$v \mapsto a(u, v).$$

In other terms, L is uniquely defined by the relation

$$\langle Lu, v\rangle_* = a(u, v), \quad \forall u, v \in V.$$

Then the abstract variational problem is equivalent to the equation

$$Lu = F \quad \text{in } V^*$$

and, by the Lax-Milgram Theorem, L is a *continuous isomorphism between* V and V^*. The inverse operator L^{-1} acts as the solution map $F \longmapsto u(F)$ and inequality (6.49) yields

$$\left\|L^{-1}\right\|_{\mathcal{L}(V^*,V)} \leq \frac{1}{\alpha}.$$

This inequality (or (6.49)) is called *stability estimate* for the following reason. The functional F, an element of V^*, encodes the "data" of the problem (6.47). If $F_1, F_2 \in V^*$ and u_1, u_2 are the corresponding solutions, inequality (6.49) gives

$$\|u_1 - u_2\| \leq \frac{1}{\alpha} \|F_1 - F_2\|_{V^*}.$$

Thus, close data imply close solutions. The stability constant $1/\alpha$ plays an important role, since it controls the norm variation of the solutions in terms of the variations on the data, measured by $\|F_1 - F_2\|_{V^*}$. This entails, in particular, that the more the coerciveness constant α is large, the more "stable" is the solution.

6.6.2 Approximation and Galerkin's method

The solution u of the abstract variational problem (6.47), satisfies the equation

$$a(u, v) = Fv \tag{6.53}$$

for *every* v in the Hilbert space V. In concrete applications, it is important to compute approximate solutions with a given degree of accuracy and the infinite dimension of V is the main obstacle. Often, however, V is separable and may be written as a *union of finite-dimensional subspaces*, so that, in principle, it could be reasonable to obtain approximate solutions by "projecting" equation (6.53) on those subspaces. This is the idea of **Galerkin's method**. In principle, the higher the dimension of the subspace the better should be the degree of approximation. More precisely, the idea is to construct a sequence $\{V_k\}$ of subspaces of V with the following properties:

a) Every V_k is *finite-dimensional*: $\dim V_k = k$,

b) $V_k \subset V_{k+1}$ (actually, not strictly necessary),
c) $\overline{\cup V_k} = V$.

To realize the projection, assume that the vectors $\psi_1, \psi_2, \ldots, \psi_k$ span V_k. Then, we look for an approximation of the solution u in the form

$$u_k = \sum_{j=1}^{k} c_j \psi_j, \tag{6.54}$$

by solving the problem

$$a(u_k, v) = Fv \qquad \forall v \in V_k. \tag{6.55}$$

Since $\{\psi_1, \psi_2, ..., \psi_k\}$ constitutes a basis in V_k, (6.55) amounts to requiring

$$a(u_k, \psi_r) = F\psi_r \qquad r = 1, ..., k. \tag{6.56}$$

Substituting (6.54) into (6.56), we obtain the k linear algebraic equations

$$\sum_{j=1}^{k} a(\psi_j, \psi_r) c_j = F\psi_r \qquad r = 1, 2, ..., k, \tag{6.57}$$

for the unknown coefficients $c_1, c_2, ..., c_k$. Introducing the vectors

$$\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_k \end{pmatrix}, \quad \mathbf{F} = \begin{pmatrix} F\psi_1 \\ F\psi_2 \\ \vdots \\ F\psi_k \end{pmatrix}$$

and the matrix $\mathbf{A} = (a_{rj})$, with entries $a_{rj} = a(\psi_j, \psi_r)$, $j, r = 1, ..., k$, we may write (6.57) in the compact form

$$\mathbf{Ac} = \mathbf{F}. \tag{6.58}$$

The matrix $\mathbf{A}$ is called *stiffness matrix* and clearly plays a key role in the numerical analysis of the problem.

If the bilinear form a is coercive, $\mathbf{A}$ is *strictly positive*. In fact, let $\boldsymbol{\xi} \in \mathbb{R}^k$. Then, by linearity and coercivity:

$$\begin{aligned} \mathbf{A}\boldsymbol{\xi} \cdot \boldsymbol{\xi} &= \sum_{r,j=1}^{k} a_{rj} \xi_r \xi_j = \sum_{r,j=1}^{k} a(\psi_j, \psi_r) \xi_r \xi_j \\ &= \sum_{r,j=1}^{k} a(\xi_j \psi_j, \xi_r \psi_r) = a\left(\sum_{i=1}^{k} \xi_j \psi_j, \sum_{j=1}^{k} \xi_r \psi_r \right) \geq \alpha \|\mathbf{v}\|^2 \end{aligned}$$

where

$$\mathbf{v} = \sum_{j=1}^{k} \xi_j \psi_j \in V_k.$$

Since $\{\psi_1, \psi_2, ..., \psi_k\}$ is a basis in V_k, we have $\mathbf{v} = \mathbf{0}$ if and only if $\boldsymbol{\xi} = \mathbf{0}$. Therefore $\mathbf{A}$ is strictly positive and, in particular, non singular.

Thus, for each $k \geq 1$, there exists a unique solution $u_k \in V_k$ of (6.58). We want to show that $u_k \to u$, as $k \to \infty$, i.e. the convergence of the method, and give a control of the approximation error.

For this purpose, we prove the following lemma, known as *Céa's Lemma*, which also emphasizes the role of the continuity and the coercivity constants (M and α, respectively) of the bilinear form a.

Lemma 6.45 *Assume that the hypotheses of the Lax-Milgram Theorem hold and let u be the solution of problem (6.47). If u_k is the solution of problem (6.56), then*

$$\|u - u_k\| \leq \frac{M}{\alpha} \inf_{v \in V_k} \|u - v\| . \tag{6.59}$$

Proof We have

$$a(u_k, v) = Fv, \qquad \forall v \in V_k$$

and

$$a(u, v) = Fv, \qquad \forall v \in V_k.$$

Subtracting the two equations we obtain

$$a(u - u_k, v) = 0, \qquad \forall v \in V_k.$$

In particular, since $v - u_k \in V_k$, we have

$$a(u - u_k, v - u_k) = 0, \qquad \forall v \in V_k,$$

which implies

$$\begin{aligned} a(u - u_k, u - u_k) &= a(u - u_k, u - v) + a(u - u_k, v - u_k) \\ &= a(u - u_k, u - v). \end{aligned}$$

Then, by the continuity and the coercivity of a,

$$\alpha \|u - u_k\|^2 \leq a(u - u_k, u - u_k) \leq M \|u - u_k\| \|u - v\| ,$$

whence

$$\|u - u_k\| \leq \frac{M}{\alpha} \|u - v\| . \tag{6.60}$$

This inequality holds for every $v \in V_k$, with $\frac{M}{\alpha}$ independent of k. Therefore (6.60) still holds if we take in the right hand side the infimum over all $v \in V_k$. □

- **Convergence of Galerkin's method**. Since we have assumed that

$$\overline{\cup V_k} = V,$$

there exists a sequence $\{w_k\} \subset V_k$ such that $w_k \to u$ as $k \to \infty$. Lemma 6.45 gives, for every k:

$$\|u - u_k\| \leq \frac{M}{\alpha} \inf_{v \in V_k} \|u - v\| \leq \frac{M}{\alpha} \|u - w_k\|,$$

whence $\|u - u_k\| \to 0$.

6.6.3 Minimization of quadratic functionals

When a is *symmetric*, i.e. if

$$a(u, v) = a(v, u), \qquad \forall u, v \in V,$$

the abstract variational problem (6.47) is equivalent to a *minimization problem*. In fact, consider the quadratic functional

$$E(v) = \frac{1}{2} a(v, v) - Fv.$$

We have:

Theorem 6.46 *Let a be symmetric. Then $\bar{u}$ is solution of problem (6.47) if and only if $\bar{u}$ is a minimizer of E, that is*

$$E(\bar{u}) = \min_{v \in V} E(v).$$

Proof For every $\varepsilon \in \mathbb{R}$ and every "variation" $v \in V$ we have

$$\begin{aligned} E(\bar{u} + \varepsilon v) - E(\bar{u}) &= \left\{ \frac{1}{2} a(\bar{u} + \varepsilon v, \bar{u} + \varepsilon v) - F(\bar{u} + \varepsilon v) \right\} - \left\{ \frac{1}{2} a(\bar{u}, \bar{u}) - F\bar{u} \right\} \\ &= \varepsilon \{ a(\bar{u}, v) - Fv \} + \frac{1}{2} \varepsilon^2 a(v, v). \end{aligned}$$

Now, if $\bar{u}$ is the solution of problem (6.47), then $a(\bar{u}, v) - Fv = 0$. Therefore

$$E(\bar{u} + \varepsilon v) - E(\bar{u}) = \frac{1}{2} \varepsilon^2 a(v, v) \geq 0$$

so that $\bar{u}$ minimizes E. On the other hand, if $\bar{u}$ is a minimizer of E, then

$$E(\bar{u} + \varepsilon v) - E(\bar{u}) \geq 0,$$

which entails

$$\varepsilon \{ a(\bar{u}, v) - Fv \} + \frac{1}{2} \varepsilon^2 a(v, v) \geq 0$$

for every $\varepsilon \in \mathbb{R}$. Dividing by $\varepsilon \neq 0$ and letting $\varepsilon \to 0$ we get

$$a(\bar{u}, v) - Fv = 0, \qquad \forall v \in V,$$

and $\bar{u}$ is a solution of problem (6.47). □

Letting $\varphi(\varepsilon) = E(\bar{u} + \varepsilon v)$, from the above calculations we have

$$\varphi'(0) = a(\bar{u}, v) - Fv.$$

Thus, the linear functional

$$v \longmapsto a(\bar{u}, v) - Fv$$

appears as the **derivative of** E at $\bar{u}$ along the direction v and we write

$$E'(\bar{u})v = a(\bar{u}, v) - Fv. \tag{6.61}$$

In Calculus of Variations E' is called **first variation** and denoted by δE. The variational equation

$$E'(u)v = a(u, v) - Fv = 0, \qquad \forall v \in V \tag{6.62}$$

is called the **Euler equation** for the functional E.

Remark 6.47 A bilinear form a, symmetric and coercive, induces in V the inner product

$$(u, v)_a = a(u, v).$$

In this case, existence, uniqueness and stability for problem (6.47) follow directly from Riesz's Representation Theorem. In particular, *there exists a unique minimizer* $\bar{u}$ *of* E.

6.6.4 The Nečas Theorem

In this subsection we go back to our variational problem

$$\begin{cases} \text{Find } u \in V \text{ such that} \\ a(u, v) = \langle F, v \rangle_*, \quad \forall v \in W. \end{cases} \tag{6.63}$$

Where a is a continuous bilinear form in $V \times W$. We have seen that, in the case $V = W$, if a is coercive, then problem (6.63) is well posed (Lax-Milgram). However, the *coercivity condition*

$$\exists \alpha > 0 : a(u, u) \geq \alpha \|u\|^2 \quad \forall u \in V \tag{6.64}$$

is by no means *necessary* to have a well posed problem as it is is already evident in the finite dimensional case $V = W = \mathbb{R}^n$. Indeed, consider the (continuous) bilinear form $a(\mathbf{u}, \mathbf{v}) = (\mathbf{Mu}, \mathbf{v})_{\mathbb{R}^n}$, for some square matrix $\mathbf{M}$ of order n. If $\mathbf{b} \in \mathbb{R}^n$, the problem

$$a(\mathbf{u}, \mathbf{v}) = (\mathbf{b}, \mathbf{v})_{\mathbb{R}^n} \qquad \forall \mathbf{v} \in \mathbb{R}^n \tag{6.65}$$

is equivalent to the equation

$$\mathbf{Mu} = \mathbf{b} \tag{6.66}$$

which is well posed *if and only if* $\mathbf{M}$ is invertible. On the other hand, the coercivity assumption reads:

$$\exists \alpha > 0 : (\mathbf{Mu}, \mathbf{u})_{\mathbb{R}^n} = a(\mathbf{u}, \mathbf{u}) \geq \alpha \, \|\mathbf{u}\|^2_{\mathbb{R}^n} . \tag{6.67}$$

Since $(\mathbf{Mu}, \mathbf{u})_{\mathbb{R}^n} = \frac{1}{2}((\mathbf{M} + \mathbf{M}^\top)\mathbf{u}, \mathbf{u})_{\mathbb{R}^n}$, (6.67) means that *the eigenvalues of the symmetric matrix* $\mathbf{M}+\mathbf{M}^\top$ *are all positive*. However this is only a sufficient condition for a matrix to be invertible. In fact if, for instance, $\mathbf{M}$ is symmetric, $\mathbf{M}$ is invertible if and only if its eigenvalues *do not vanish*, but they do not have to be positive.

Here comes into play the *inf-sup condition* (6.45) of Corollary 6.40, p. 374. In the present case this condition reads

$$\exists \alpha > 0 : \forall \mathbf{u} \in \mathbb{R}^n, \ \sup_{\mathbf{0} \neq \mathbf{v} \in \mathbb{R}^n} \frac{(\mathbf{Mu}, \mathbf{v})_{\mathbb{R}^n}}{\|\mathbf{v}\|_{\mathbb{R}^n}} \geq \alpha \, \|\mathbf{u}\|_{\mathbb{R}^n} .$$

The supremum is attained for $\mathbf{v}$ parallel to $\mathbf{Mu}$, hence the inf-sup condition becomes

$$\forall \mathbf{u} \in \mathbb{R}^n, \ \sqrt{(\mathbf{Mu}, \mathbf{Mu})_{\mathbb{R}^n}} = \sqrt{(\mathbf{M}^\top \mathbf{Mu}, \mathbf{u})_{\mathbb{R}^n}} \geq \alpha \, \|\mathbf{u}\|_{\mathbb{R}^n} . \tag{6.68}$$

Denote by $\lambda_j^2 \geq 0$, the eigenvalues of the symmetric matrix $\mathbf{M}^\top \mathbf{M}$. Then, condition (6.68) means that

$$\forall j = 1, ..., n, \quad \left|\lambda_j\right| \geq \alpha,$$

precisely a necessary and sufficient condition for M to be invertible.

Conclusion: if $V = W = \mathbb{R}^n$, *the inf-sup condition alone is necessary and sufficient for the well-posedness of problem (6.63).*

In infinitely many dimension (or even if $V = \mathbb{R}^n, W = \mathbb{R}^m$, $m \neq n$) to achieve well-posedness *the inf-sup condition* alone is not enough. The following theorem (Nečas, 1962) is a rephrasing of Corollary 6.40 in terms of bilinear forms.

Theorem 6.48 *Let a be a continuous bilinear form:*

$$\exists M : |a(u, v)| \leq M \, \|u\|_V \, \|v\|_W \quad \forall u \in V, \forall v \in W. \tag{6.69}$$

Then, the problem (6.63) is well-posed for all $F \in W^*$ ***if and only if*** *a satisfies the following conditions:*

1. there exists $\alpha > 0$ *such that*

$$\forall u \in V, \quad \sup_{0 \neq v \in V} \frac{a(u, v)}{\|v\|_W} \geq \alpha \, \|u\|_V , \quad \text{(inf-sup condition)} \tag{6.70}$$

2.

$$[a(u, v) = 0, \ \forall u \in V] \Rightarrow v = 0. \tag{6.71}$$

Moreover, the solution u_F *to (6.63) satisfies the stability estimate*

$$\|u_F\|_V \leq \frac{1}{\alpha} \, \|F\|_{W^*} . \tag{6.72}$$

Proof As we did in the proof of the Lax-Milgram Theorem, we first convert the problem (6.63) into the operator form

$$Bu = z_F \quad \text{in } W. \tag{6.73}$$

where $B \in \mathcal{L}(V, W)$,

$$a(u, v) = (Bu, v)_W \quad \forall v \in W, \tag{6.74}$$

and z_F is the Riesz element associated to F. The proof or Theorem 6.48 reduces to prove that the conditions (6.70) and (6.71) are necessary and sufficient for B to be a continuous isomorphism with

$$\left\|B^{-1}\right\|_{\mathcal{L}(W,V)} \leq \frac{1}{\alpha}.$$

Now, (6.70) is equivalent to

$$\forall u \in V : \|Bu\|_W = \sup_{0 \neq v \in W} \frac{(Bu, v)_W}{\|v\|_W} = \sup_{0 \neq v \in W} \frac{a(u, v)}{\|v\|_W} \geq \alpha \|u\|_V .$$

On the other hand, (6.71) is equivalent to,

$$[a(u, v) = (Bu, v)_W = 0, \ \ \forall u \in X] \Longrightarrow v = 0.$$

By Corollary 6.40 and (6.46) we conclude the proof. □

It is important to examine the solvability of problem (6.63) when the Nečas Theorem does not hold. We shall come back to this question in Section 6.8, for simplicity only in the case $V = W$. But first we need the notion of **compact operator**.

6.7 Compactness and Weak Convergence

6.7.1 Compactness

The solvability of boundary value problems and the analysis of numerical methods involve several questions of convergence. In typical situations one is able to construct a sequence of approximations and the main task is to prove that this sequence converges to a solution of the problem in a suitable sense. It is often the case that, through *energy type estimates*, one is able to show that these sequences of approximations are *bounded* in some Hilbert space. How can we use this information? Although we cannot expect these sequences to converge, we may reasonably look for *convergent subsequences*, which is already quite satisfactory. In technical words, we are asking to our sequences to have a *compactness property*. Let us spend a few words on this important topological concept (for the proofs, see e.g. *K. Yosida*, [43]). Once more, the difference between finite and infinite dimension plays a big role.

Let X be a normed space. The general definition of compact set involves open coverings: an *open covering* of $E \subseteq X$ is a family of *open* sets whose union contains E.

Definition 6.49 We say that $E \subseteq X$ is *compact* if from every open covering of E it is possible to extract a finite subcovering of E.

It is somewhat more convenient to work with *precompact* sets as well, that is sets *whose closure is compact*. In finite dimensional spaces the characterization of precompact sets is well known: $E \subset \mathbb{R}^n$ is precompact if and only if E is bounded. What about infinitely many dimensions? Let us introduce a characterization of precompact sets in normed spaces in terms of convergent sequences, much more comfortable to use. First, let us agree that a subset E of a normed space X is *sequentially precompact* (resp. *compact*), if for every sequence $\{x_k\} \subset E$ there exists a subsequence $\{x_{k_s}\}$, convergent in X (resp. in E). We have:

Theorem 6.50 *Let X be a normed space and $E \subset X$. Then E is precompact (compact) if and only if it is sequentially precompact (compact).*

While a compact set is always *closed and bounded* (see Problem 6.13), the following example exhibits a closed and bounded set which is *not* compact.

Example 6.51 Consider the real Hilbert space (see Problem 6.3)

$$l^2 = \left\{ \mathbf{x} = \{x_k\}_{k\geq 1} : \sum_{k=1}^{\infty} x_k^2 < \infty,\ x_k \in \mathbb{R} \right\}$$

endowed with

$$(\mathbf{x}, \mathbf{y}) = \sum_{k=1}^{\infty} x_k y_k \qquad \text{and} \qquad \|\mathbf{x}\|^2 = \sum_{k=1}^{\infty} x_k^2.$$

Let $E = \{\mathbf{e}^k\}_{k\geq 1}$, where $\mathbf{e}^1 = \{1, 0, 0, \ldots\}$, $\mathbf{e}^2 = \{0, 1, 0, \ldots\}$, etc.. Observe that E constitutes an orthonormal basis in l^2. Then, E is closed and bounded in l^2. However, *E is not sequentially compact.* Indeed, if $j \neq k$,

$$\left\|\mathbf{e}^j - \mathbf{e}^k\right\| = \sqrt{2}$$

and therefore no subsequence of $\{\mathbf{e}^k\}_{k\geq 1}$ can be convergent.

Thus, in infinite dimensions, closed and bounded does not imply compact. Actually, this can only happen in finite-dimensional spaces. In fact:

Theorem 6.52 *Let X be a Banach space. The closed unit ball $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\}$ is compact if and only if X is finite-dimensional.*

6.7.2 Compactness in $C(\overline{\Omega})$ and in $L^p(\Omega)$

To recognize that a subset of a Banach or Hilbert space is compact is usually a hard task. A characterization of the compact subsets in $C(\overline{\Omega})$, where Ω is a bounded domain is given by the following important theorem, known as the *Ascoli-Arzelà Theorem*[7]:

[7] Recall that the norm in $C(\overline{\Omega})$ is $\|f\|_{C(\overline{\Omega})} = \max_{\overline{\Omega}} |f|$.

Theorem 6.53 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and $E \subset C(\overline{\Omega})$. Then E is precompact in $C(\overline{\Omega})$ if and only if the following conditions hold.*

1. *E is bounded, that is there exists $M > 0$ such that*

$$\|u\|_{C(\overline{\Omega})} \leq M, \qquad \forall u \in E. \tag{6.75}$$

2. *E is equicontinuous, that is, for every $\varepsilon > 0$ there exists $\delta = \delta(\varepsilon) > 0$ such that if $\mathbf{x}, \mathbf{x}+\mathbf{h} \in \overline{\Omega}$ and $|\mathbf{h}| < \delta$ then*

$$|u(\mathbf{x}+\mathbf{h}) - u(\mathbf{x})| < \varepsilon, \qquad \forall u \in E. \tag{6.76}$$

In general, the condition 1 is rather easy to prove, while the equicontinuity condition 2 could be difficult to check. A particularly significant case occurs when E is bounded in $C^{0,\alpha}(\overline{\Omega})$, $0 < \alpha \leq 1$ or in $C^1(\overline{\Omega})$. Then:

Corollary 6.54 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and $E \subset C^{0,\alpha}(\overline{\Omega})$, $0 < \alpha \leq 1$. If there exists $M > 0$ such that*

$$\|u\|_{C^{0,\alpha}(\overline{\Omega})} \leq M, \qquad \forall u \in E \tag{6.77}$$

then E is precompact in $C(\overline{\Omega})$.

Proof Condition (6.77) means that

$$\|u\|_{C(\overline{\Omega})} + \sup_{\substack{\mathbf{x},\mathbf{y}\in\overline{\Omega} \\ \mathbf{x}\neq\mathbf{y}}} \frac{|u(\mathbf{x}) - u(\mathbf{y})|}{|\mathbf{x}-\mathbf{y}|^{\alpha}} \leq M, \qquad \forall u \in E.$$

Thus, in particular, $\|u\|_{C(\overline{\Omega})} \leq M$, which is (6.75). Moreover,

$$|u(\mathbf{x}+\mathbf{h}) - u(\mathbf{x})| \leq M\,|\mathbf{h}|^{\alpha}, \qquad \forall \mathbf{x}, \mathbf{x}+\mathbf{h} \in \overline{\Omega},\ \forall u \in E \tag{6.78}$$

which gives the equicontinuity of E. □

In $L^p(\Omega)$, $1 \leq p < \infty$, there exists an analogue of Corollary 6.54, due to M. Riesz, Fréchet and Kolmogorov.

Theorem 6.55 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain and $S \subset L^p(\Omega)$, $1 \leq p < \infty$. If:*

1. *S is bounded, i.e. there exists K such that*

$$\|u\|_{L^p(\Omega)} \leq K, \quad \forall u \in S.$$

2. *There exist α and L, positive, such that, if u is extended by zero outside Ω,*

$$\|u(\cdot+\mathbf{h}) - u(\cdot)\|_{L^p(\Omega)} \leq L\,|\mathbf{h}|^{\alpha}, \quad \textit{for every } \mathbf{h} \in \mathbb{R}^n \textit{ and } u \in S,$$

then S is pre-compact.

The second condition expresses an *equicontinuity in norm* of all the elements in S. We shall meet this condition in Sect. 7.10.1.

6.7.3 Weak convergence and compactness

We have seen that compactness in a normed space is equivalent to sequential compactness. In the applications, this translates into a very strong requirement for an approximating sequence.

Fortunately, in normed spaces, and in particular in Hilbert spaces, there is another notion of convergence, much more flexible, which turns out to be perfectly adapted to the variational formulation of boundary value problems.

Let H be a Hilbert space with inner product $(\cdot,\cdot)$ and norm $\|\cdot\|$. If $F \in H^*$ and $\|x_k - x\| \to 0$, then $Fx_k \to Fx$. In fact

$$|Fx_k - Fx| = |F(x_k - x)| \le \|F\|_{H^*} \|x_k - x\| .$$

However, see Example 6.57 below, it could be that

$$Fx_k \to Fx$$

for every $F \in H^*$, even if $\|x_k - x\| \not\to 0$. Then, we say that x_k *weakly converges* to x. Precisely:

Definition 6.56 A sequence $\{x_k\} \subset H$ *weakly converges* to $x \in H$, and we write

$$x_k \rightharpoonup x$$

(with an "half arrow"), if

$$Fx_k \to Fx, \qquad \forall F \in H^*.$$

To emphasize the difference, the convergence in norm is sometime called *strong convergence*. From the Riesz Representation Theorem, it follows that $\{x_k\} \subset H$ weakly converges to $x \in H$ if and only if

$$(x_k, y) \to (x, y), \qquad \forall y \in H.$$

The weak limit is unique, since $x_k \rightharpoonup x$ and $x_k \rightharpoonup z$ imply

$$(x - z, y) = 0 \qquad \forall y \in H,$$

whence $x = z$. Moreover, the Schwarz inequality gives

$$|(x_k - x, y)| \le \|x_k - x\| \, \|y\|$$

so that *strong convergence* implies *weak convergence*, which should not be surprising. The two notions of convergence are equivalent in finite-dimensional spaces. It is not so in infinite dimensions, as the following example shows.

Example 6.57 Let $H = L^2(0, 2\pi)$. The sequence $v_k(x) = \cos kx$, $k \ge 1$, is *weakly convergent to zero.* In fact, for every $f \in L^2(0, 2\pi)$, Corollary 1.2, p. 14, on the

Fourier coefficients of f, implies that

$$(f, v_k)_{L^2(0,2\pi)} = \int_0^{2\pi} f(x) \cos kx \, dx \to 0$$

as $k \to \infty$. However

$$\|v_k\|_{L^2(0,2\pi)} = \sqrt{\pi}$$

and therefore $\{v_k\}_{k\geq 1}$ does not converge strongly to zero.

Remark 6.58 If $L \in \mathcal{L}(H_1, H_2)$ and $x_k \rightharpoonup x$ in H_1 we cannot say that $Lx_k \to Lx$ in H_2. However, if $F \in H_2^*$, then $F \circ L \in H_1^*$ and therefore

$$F(Lx_k) = (F \circ L)x_k \to (F \circ L)x = F(Lx).$$

Thus, if L is (strongly) continuous then it is *weakly sequentially continuous* as well.

Warning: *Not always strong implies weak*! Take a strongly closed set $E \subset H$. Thus E contains all the limits of strongly convergent sequences $\{x_k\} \subset E$. Can we deduce that E is *weakly sequentially closed* as well? The answer is no: if $x_k \rightharpoonup x$ (only weakly), since the convergence is not strong, we can not affirm that $x \in E$. Indeed, E is not weakly sequentially closed, in general (see however Problem 6.16).

Take for instance

$$E = \{v \in L^2(0, 2\pi) : \|v\|_{L^2(0,2\pi)} = \sqrt{\pi}\}.$$

Then E is a strongly closed (and bounded) subset of $L^2(0, 2\pi)$. However, E contains the sequence $\{v_k\}$, where $v_k(x) = \cos kx$, and we have seen in Example 6.57 that $v_k \rightharpoonup 0 \notin E$. Hence E is *not* weakly sequentially closed.

We have observed that the norm in a Hilbert space is strongly continuous. With respect to weak convergence, the norm is only (sequentially) lower semicontinuous, as property 2 in the following theorem shows.

Theorem 6.59 *Let $\{x_k\} \subset H$ such that $x_k \rightharpoonup x$. Then*

1. *$\{x_k\}$ is bounded.*
2. $\|x\| \leq \liminf_{k\to\infty} \|x_k\|$.

We omit the proof[8] of 1. For the second point, it is enough to observe that

$$\|x\|^2 = \lim_{k\to\infty} (x_k, x) \leq \|x\| \liminf_{k\to\infty} \|x_k\|$$

and divide by $\|x\|$.

The usefulness of weak convergence is revealed by the following compactness result. Basically, it says that if we substitute *strong* with *weak* convergence, any bounded sequence in a Hilbert space is weakly pre-compact. Precisely:

[8] It is a consequence of the Banach-Steinhaus Theorem. See e.g. *K. Yosida*, [43].

Theorem 6.60 *Every bounded sequence in a Hilbert space H contains a subsequence which is weakly convergent to an element $x \in H$.*

Proof We give it under the additional hypothesis that H is separable[9]. Thus, there exists a sequence $\{z_k\}$ dense in H. Let now $\{x_j\} \subset H$ be a bounded sequence: $\|x_j\| \leq M$, $\forall j \geq 1$. We split the proof into three steps.

1. We use a "diagonal" process, to construct a subsequence $\{x_s^{(s)}\}$, such that the real sequence $(x_s^{(s)}, z_k)$ is convergent for every fixed z_k. To do this, observe that the sequence $\{(x_j, z_1)\}$ is bounded in $\mathbb{R}$ and therefore there exists $\{x_j^{(1)}\} \subset \{x_j\}$ such that $\{(x_j^{(1)}, z_1)\}$ is convergent. For the same reason, from $\{x_j^{(1)}\}$ we may extract a subsequence $\{x_j^{(2)}\}$ such that $\{(x_j^{(2)}, z_2)\}$ is convergent. By induction, we construct $\{x_j^{(k)}\}$ such that

$$\left\{(x_j^{(k)}, z_k)\right\}$$

converges. Consider the diagonal sequence $\{x_s^{(s)}\}$, obtained by selecting $x_1^{(1)}$ from $\{x_j^{(1)}\}$, $x_2^{(2)}$ from $\{x_j^{(2)}\}$ and so on. Then,

$$\left\{(x_s^{(s)}, z_k)\right\}$$

is convergent for every fixed $k \geq 1$.

2. We now use the density of $\{z_k\}$ in H, to show that $(x_s^{(s)}, z)$ converges for every $z \in H$. In fact, for fixed $\varepsilon > 0$ and $z \in H$, we may find z_k such that $\|z - z_k\| < \varepsilon$. Write

$$(x_s^{(s)} - x_m^{(m)}, z) = (x_s^{(s)} - x_m^{(m)}, z - z_k) + (x_s^{(s)} - x_m^{(m)}, z_k).$$

If j and m are large enough, we have

$$\left|(x_s^{(s)} - x_m^{(m)}, z_k)\right| < \varepsilon$$

since $(x_s^{(s)}, z_k)$ is convergent. Moreover, from the Schwarz inequality,

$$\left|(x_s^{(s)} - x_m^{(m)}, z - z_k)\right| \leq \left\|x_s^{(s)} - x_m^{(m)}\right\| \|z - z_k\| \leq 2M\varepsilon.$$

Thus, if j and m are large enough, we have

$$\left|(x_s^{(s)} - x_m^{(m)}, z)\right| \leq (2M + 1)\varepsilon,$$

hence the sequence $(x_s^{(s)} - x_m^{(m)}, z)$ is a Cauchy sequence in $\mathbb{R}$ and therefore convergent.

3. From **2**, we may define a linear functional T in H by setting

$$Tz = \lim_{s \to \infty} (x_s^{(s)}, z).$$

Since $\left\|x_s^{(s)}\right\| \leq M$, we have $|Tz| \leq M\|z\|$, whence $T \in H^*$. From the Riesz Representation theorem, there exists a unique $x_\infty \in H$ such that

$$Tz = (x_\infty, z), \qquad \forall z \in H.$$

---

[9] There is actually no loss of generality since we can always replace H by the closure H_0 of the subspace generated by the finite linear combinations of elements of the sequence. Clearly H_0 is separable.

Thus

$$(x_s^{(s)}, z) \to (x_\infty, z), \ \forall z \in H,$$

which means $x_s^{(s)} \rightharpoonup x_\infty$. □

Example 6.61 Let $H = L^2(\Omega)$, $\Omega \subseteq \mathbb{R}^n$ and consider a sequence $\{u_k\}_{k\geq 1} \subset L^2(\Omega)$. To say that $\{u_k\}$ is bounded means that

$$\|u_k\|_{L^2(\Omega)} \leq M, \qquad \text{for every } k \geq 1.$$

Theorem 6.60 implies the existence of a subsequence $\{u_{k_m}\}_{m\geq 1}$ and of $u \in L^2(\Omega)$ such that, as $m \to +\infty$,

$$\int_\Omega u_{k_m} v \to \int_\Omega uv, \qquad \text{for every } v \in L^2(\Omega).$$

6.7.4 Compact operators

By definition, every operator in $\mathcal{L}(H_1, H_2)$ transforms bounded sets in H_1 into bounded sets in H_2. The subclass of operators that transform *bounded sets* into *precompact sets* is particularly important.

Definition 6.62 Let H_1 and H_2 be Hilbert spaces and $L \in \mathcal{L}(H_1, H_2)$. We say that L is *compact* if, for every bounded $E \subset H_1$, the image $L(E)$ is *precompact* in H_2.

An equivalent characterization of compact operators may be given in terms of weak convergence. Indeed, a linear operator is compact if and only if "it converts weak convergence into strong convergence". Precisely:

Proposition 6.63 *Let $L \in \mathcal{L}(H_1, H_2)$. L is compact if and only if, for every sequence $\{x_k\} \subset H_1$,*

$$x_k \rightharpoonup 0 \quad \text{in } H_1 \qquad \text{implies} \qquad Lx_k \to 0 \quad \text{in } H_2. \tag{6.79}$$

Proof Assume that (6.79) holds. Let $E \subset H_1$ be bounded, and $\{z_k\} \subset L(E)$. Then $z_k = Lx_k$ for some $x_k \in E$. From Theorem 6.60, there exists a subsequence $\{x_{k_s}\}$ weakly convergent to $x \in H_1$. Then

$$y_s = x_{k_s} - x \rightharpoonup 0$$

in H_1 and, from (6.79), $Ly_s \to 0$ in H_2, that is

$$z_{k_s} = Lx_{k_s} \to Lx \equiv z$$

in H_2. Thus, $L(E)$ is sequentially precompact, and therefore precompact in H_2. Viceversa, let L be compact and $x_k \rightharpoonup 0$ in H_1. Suppose $Lx_k \not\to 0$. Then, for some $\bar{\varepsilon} > 0$ and infinitely many indexes k_j, we have $\|Lx_{k_j}\| > \bar{\varepsilon}$. Since

$$x_{k_j} \rightharpoonup 0,$$

by Theorem 6.59, $\{x_{k_j}\}$ is bounded in H_1, so that $\{Lx_{k_j}\}$ contains a subsequence (that we still call) $\{Lx_{k_j}\}$ strongly (and therefore weakly) convergent to some $y \in H_2$. On the other hand, we

have $Lx_{k_j} \rightharpoonup 0$ as well, which entails $y = 0$. Thus

$$\|Lx_{k_j}\| \to 0.$$

Contradiction. □

Example 6.64 Let $H^1_{\text{per}}(0, 2\pi)$ be the Hilbert space introduced in Example 6.11, p. 353. The embedding of $H^1_{\text{per}}(0, 2\pi)$ into $L^2(0, 2\pi)$ is compact (see Problem 6.18).

Example 6.65 From Theorem 6.52, the identity operator $I : H \to H$ is compact if and only if $\dim H < \infty$. Also, any bounded operator with finite dimensional range is compact.

Example 6.66 Let $Q = (0, 1) \times (0, 1)$ and $g \in C(\overline{Q})$. Consider the integral operator

$$Tv(x) = \int_0^1 g(x, y)v(y)dy. \tag{6.80}$$

We want to show that T is compact from $L^2(0, 1)$ into $L^2(0, 1)$. In fact, for every $x \in (0, 1)$, the Schwarz inequality gives

$$|Tv(x)| \le \int_0^1 |g(x, y)v(y)|\, dy \le \|g(x, \cdot)\|_{L^2(0,1)}\, \|v\|_{L^2(0,1)}\,, \tag{6.81}$$

whence

$$\int_0^1 |Tv(x)|^2\, dx \le \|g\|^2_{L^2(Q)}\, \|v\|^2_{L^2(0,1)}$$

which implies that $Tv \in L^2(0, 1)$ and that T is bounded.

To check compactness, we use Proposition 6.63. Let $\{v_k\} \subset L^2(0, 1)$ such that $v_k \rightharpoonup 0$, that is

$$\int_0^1 v_k w \to 0, \qquad \text{for every } w \in L^2(0, 1). \tag{6.82}$$

We have to show that $Tv_k \to 0$ in $L^2(0, 1)$. Being weakly convergent, $\{v_k\}$ is bounded so that

$$\|v_k\|_{L^2(0,1)} \le M, \tag{6.83}$$

for some M and every k. From (6.81) we have

$$|Tv_k(x)| \le M\, \|g(x, \cdot)\|_{L^2(0,1)}\,.$$

Moreover, inserting $w(\cdot) = g(x, \cdot)$ into (6.82), we infer that

$$Tv_k(x) = \int_0^1 g(x, y)v_k(y)dy \to 0, \qquad \text{for every } x \in (0, 1).$$

From the Dominated Convergence Theorem (Appendix A) we infer that $Tv_k \to 0$ in $L^2(0, 1)$. Therefore T is compact.

The following proposition is useful.

Proposition 6.67 *Let $L : H_1 \to H_2$ be compact. Then:*

a) $L^* : H_2 \to H_1$ is compact.
b) If $G \in \mathcal{L}(H_2, H_3)$ or $G \in \mathcal{L}(H_0, H_1)$, the operator $G \circ L$ or $L \circ G$ is compact.

Proof a) We use Proposition 6.63. Let $\{x_k\} \subset H_2$ and $x_k \rightharpoonup 0$. Let us show that $\|L^* x_k\|_{H_1} \to 0$. We have:

$$\|L^* x_k\|_{H_1}^2 = (L^* x_k, L^* x_k)_{H_1} = (x_k, LL^* x_k)_{H_2}.$$

Since $L^* \in \mathcal{L}(H_2, H_1)$, we have

$$L^* x_k \rightharpoonup 0$$

in H_1 and the compactness of L entails

$$LL^* x_n \to 0$$

in H_2. Since $\|x_k\| \le M$, we finally have

$$\|L^* x_k\|_{H_1}^2 = (x_k, LL^* x_k)_{H_2} \le M \, \|LL^* x_k\|_{H_2} \to 0.$$

b) We leave it as an exercise. □

6.8 The Fredholm Alternative

6.8.1 Hilbert triplets

In this Section we examine the solvability of the abstract variational problem

$$\begin{cases} \text{Find } u \in V \text{ such that} \\ a(u, v) = \langle F, v \rangle_*, \quad \forall v \in V \end{cases} \tag{6.84}$$

when the Lax-Milgram Theorem 6.42, p. 376, or, more generally, the Nečas Theorem cannot be applied. In this situation it may happen that the problem does not have a solution, unless certain compatibility conditions on F are satisfied. A typical example is given by the Neumann problem

$$\begin{cases} -\Delta u = f \text{ in } \Omega \\ \partial_\nu u = g \text{ on } \partial\Omega. \end{cases} \tag{6.85}$$

We will see in Chapter 8 that problem (6.85) can be suitably recast in the form (6.84). It turns out that a necessary and sufficient solvability condition is given by

$$\int_\Omega f + \int_{\partial\Omega} g = 0 \tag{6.86}$$

(also recall Remark 3.2, p. 118). Moreover, if (6.86) holds, there are infinitely many solutions, differing among each other by an additive constant. Condition (6.86) has

both a precise physical interpretation in terms of a resultant of forces at equilibrium and a deep mathematical meaning, with roots in Linear Algebra!

Indeed, the results we are going to present are extensions of well known facts concerning the solvability of linear algebraic systems of the form

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{6.87}$$

where $\mathbf{A}$ is an $n \times n$ matrix and $\mathbf{b} \in \mathbb{R}^n$. The following dichotomy holds: *either (6.87) has a unique solution for every* $\mathbf{b}$ *or the homogeneous equation* $\mathbf{A}\mathbf{x} = \mathbf{0}$ *has nontrivial solutions*.

More precisely, system (6.87) is solvable if and only if $\mathbf{b}$ *belongs to the column space of* $\mathbf{A}$, which is the orthogonal complement of $\mathcal{N}(\mathbf{A}^\top)$. If $\mathbf{w}_1, \ldots, \mathbf{w}_s$ span $\mathcal{N}(\mathbf{A}^\top)$, this is equivalent to the s compatibility conditions, $0 \le s \le n$,

$$\mathbf{b} \cdot \mathbf{w}_j = \mathbf{0} \qquad j = 1, \ldots, s.$$

Finally, the kernels $\mathcal{N}(\mathbf{A})$ and $\mathcal{N}(\mathbf{A}^\top)$ have the same dimension and if $\mathbf{v}_1, \ldots, \mathbf{v}_s$ span $\mathcal{N}(\mathbf{A})$, the general solution of (6.87) is given by

$$\mathbf{x} = \bar{\mathbf{x}} + \sum_{j=1}^{s} c_j \mathbf{v}_j$$

where $\bar{\mathbf{x}}$ is a particular solution of (6.87) and $c_1, \ldots, c_s$ are arbitrary constants.

The extension to infinite-dimensional spaces requires some care. In particular, in order to state an analogous dichotomy theorem for the variational problem (6.84), we need to clarify the general setting, to avoid confusion.

The problem involves two Hilbert spaces: V, the space where we seek the solution, and V^*, which the data F belongs to. Let us introduce a third space H, "intermediate" between V and V^*.

Remember that we use the symbols $\langle \cdot, \cdot \rangle_*$ or Fv to denote the duality between V^* and V or, if strictly necessary, the symbol $\langle \cdot, \cdot \rangle_{V^*,V}$.

In the applications to boundary value problems, usually $H = L^2(\Omega)$, with Ω bounded domain in $\mathbb{R}^n$, while V is a Sobolev space. In practice, we often meet a pair of Hilbert spaces V, H with the following properties:

1. $V \hookrightarrow H$, i.e. V *is continuously embedded in* H. Recall that this simply means that the embedding operator $I_{V \to H}$, from V into H, introduced in Example 6.26, p. 368, is continuous or, equivalently, that there exists a constant C such that

$$\|u\|_H \le C \|u\|_V , \qquad \forall u \in V. \tag{6.88}$$

2. V *is dense in* H.

Using Riesz's Representation Theorem we may identify H with H^*. Also, we may *continuously embed* H into V^*, so that any element in H can be thought as an element of V^*. To see it, observe that, for any fixed $u \in H$, the linear functional $T_u : V \to \mathbb{R}$, defined by

$$\langle T_u, v\rangle_* = (u, v)_H, \qquad v \in V, \tag{6.89}$$

is continuous in V. In fact, the Schwarz inequality and (6.88) give

$$|(u, v)_H| \le \|u\|_H \, \|v\|_H \le C \, \|u\|_H \, \|v\|_V \, . \tag{6.90}$$

Thus, the map $u \longmapsto T_u$ is continuous from H into V^*, with $\|T_u\|_{V^*} \le C \, \|u\|_H$. Moreover, if $T_u = 0$, then

$$0 = \langle T_u, v\rangle_* = (u, v)_H, \quad \forall v \in V$$

which forces $u = 0$, by the density of V in H.

Therefore, the map $u \longmapsto T_u$ is one to one and defines the continuous embedding $I_{H \hookrightarrow V^*}$ of H into V^*. This allows the *identification* of $u \in H$ with $T_u \in V^*$. In particular, instead of (6.89), we can write

$$\langle T_u, v\rangle_* = \langle u, v\rangle_* = (u, v)_H, \qquad \forall v \in V, \tag{6.91}$$

regarding u on the left in the last equality as an element of V^* and on the right as an element of H.

Finally, V (and therefore also H) *is dense*[10] *in* V^*. Summarizing, we have

$$V \hookrightarrow H \hookrightarrow V^*$$

with *dense embeddings*. We call $\{V, H, V^*\}$ a **Hilbert triplet**.

Warning: Given the identification of H with H^*, the scalar product in H achieves a privileged role. As a consequence, a further identification of V with V^* through the inverse Riesz map $\mathcal{R}_V^{-1}$ is forbidden, since it would give rise to a nonsense, unless $H = V$. Indeed, in the Hilbert triplet setting, we have the embedding $J : V \hookrightarrow V^*$, defined by

$$I_{H \hookrightarrow V^*} \circ I_{V \hookrightarrow H}. \tag{6.92}$$

Accordingly, for $u \in V$ we have from (6.91),

$$\langle Ju, v\rangle_* = \langle u, v\rangle_* = (u, v)_H, \qquad \forall v \in V. \tag{6.93}$$

If now we identify $u \in V$ with an element of V^* through $\mathcal{R}_V^{-1} : V \to V^*$, we would have also

$$\left\langle \mathcal{R}_V^{-1} u, v\right\rangle_* = \langle u, v\rangle_* = (u, v)_V, \qquad \forall v \in V, \tag{6.94}$$

---

[10] It is enough to show that the only element of V^*, orthogonal (in V^*) to T_v for every $v \in V$, is $F = 0$. Thus, let $F \in V^*$ and $(T_v, F)_{V^*} = 0$ for every v. Let $\mathcal{R}_V : V^* \to V$ be the Riesz's operator. Then we can write

$$0 = (T_v, F)_{V^*} = (\mathcal{R}_V T_v, \mathcal{R}_V F)_V = \langle T_v, \mathcal{R}_V F\rangle_* = (v, \mathcal{R}_V F)_H, \quad \forall v \in V$$

where the second equality comes from the definition (6.37) of the scalar product in V^*, the third one from the definition of $\mathcal{R}_V$ and the fourth one from (6.91). Choosing in particular $v = \mathcal{R}_V F$, we conclude that $\|\mathcal{R}_V F\|_H = 0$ and therefore that $F = 0$, since $\mathcal{R}_V$ is an isometry.

which is compatible with (6.93) only if $(u,v)_H = (u,v)_V$, for every $v \in V$. Since V is complete and dense in H, this forces $H = V$.

6.8.2 Solvability for abstract variational problems

To state the main result of this section we need to introduce weakly coercive forms and their adjoints.

Definition 6.68 We say that the bilinear form $a = a(u,v)$ is *weakly coercive with respect to the pair* (V,H) if there exist $\lambda_0 \in \mathbb{R}$ and $\alpha > 0$ such that

$$a(v,v) + \lambda_0 \|v\|_H^2 \geq \alpha \|v\|_V^2 \qquad \forall v \in V.$$

We say that the number α is the *weak coercivity constant* of a.

The *adjoint* form a^* of a is given by $a^*(u,v) = a(v,u)$, obtained by interchanging the arguments in the analytical expression of a. In the applications to boundary value problems, a^* is associated with the so called *formal adjoint* of a differential operator (see Sect. 8.4.2).

We shall denote by $\mathcal{N}(a)$ and $\mathcal{N}(a^*)$, the set of solutions u and w, respectively, of the variational problems

$$a(u,v) = 0, \quad \forall v \in V \qquad \text{and} \qquad a^*(w,v) = 0, \quad \forall v \in V.$$

Observe that $\mathcal{N}(a)$ and $\mathcal{N}(a^*)$ are both subspaces of V, playing the role of kernels for a and a^*.

Theorem 6.69 *Let $\{V,H,V^*\}$ be a Hilbert triplet, with V compactly embedded in H. Let a be a bilinear form in V, continuous and weakly coercive with respect to (V,H). Then:*

a) Either equation

$$a(u,v) = \langle F, v\rangle_*, \qquad \forall v \in V, \tag{6.95}$$

has a unique solution u_F for every $F \in V^*$ and there exist a constant C, independent of u_F and F, such that

$$\|u_F\|_V \leq C \|F\|_{V^*}, \tag{6.96}$$

b) or

$$0 < \dim \mathcal{N}(a) = \dim \mathcal{N}(a_*) = d < \infty$$

and (6.95) is solvable if and only if $\langle F, w\rangle_* = 0$ for every $w \in \mathcal{N}(a^*)$.

The proof of Theorem 6.69 relies on a more general result, known as Fredholm's Alternative, stated in the next section.

Some comments are in order. The following dichotomy holds: either (6.95) has a unique solution for every $F \in V^*$ or the homogeneous equation $a(u,v) = 0$ has

nontrivial solutions. In other words, existence of a solution to equation (6.95) for every $F \in V^*$ implies uniqueness; viceversa, uniqueness for equation (6.95) implies existence for every $F \in V^*$. The same conclusions hold for the adjoint equation

$$a^*(u, v) = \langle F, v \rangle_*, \qquad \forall v \in V.$$

If $w_1, w_2, \ldots, w_d$ span $\mathcal{N}(a^*)$, (6.95) is solvable if and only if the *d compatibility conditions*

$$\langle F, w_j \rangle_* = 0, \quad j = 1, \ldots, d,$$

hold. In this case, equation (6.95) has infinitely many solutions given by

$$u = u_F + \sum_{j=1}^{d} c_j z_j,$$

where u_F is a particular solution of (6.95), $\{z_1, \ldots, z_d\}$ span $\mathcal{N}(a)$ and $c_1, \ldots, c_d$ are arbitrary constants.

Remark 6.70 If alternative a) in Theorem 6.69 holds, the abstract variational problem is well posed whence, by Nečas Theorem, a satisfies the inf-sup condition (6.70).

We shall apply Theorem 6.69 to boundary value problems in Chap. 8. Here is however a preliminary example.

Example 6.71 Let $V = H^1_{\text{per}}(0, 2\pi)$, $H = L^2(0, 2\pi)$ and assume that $w = w(t)$ is a *positive,* continuous function in $[0, 2\pi]$. We know (Example 6.64, p. 391) that V is compactly embedded in H. Moreover, V is dense in H. In fact, as we shall see in Chap. 7, $C^1_0(0, 2\pi)$, the set of continuously differentiable functions, compactly supported in $(0, 2\pi)$, which is clearly contained in V, is dense in H. Thus $\{V, H, V^*\}$ is a Hilbert triplet.

Given $f \in H$, consider the variational problem

$$\int_0^{2\pi} u'v'w dt = \int_0^{2\pi} fv \, dt, \qquad \forall v \in V. \tag{6.97}$$

The bilinear form $a(u, v) = \int_0^{2\pi} u'v'w dt$ is continuous in V but it is not V-coercive. In fact

$$|a(u, v)| \le w_{\max} \|u'\|_{L^2(0,2\pi)} \|v'\|_{L^2(0,2\pi)} \le w_{\max} \|u\|_{H^1(0,2\pi)} \|v\|_{H^1(0,2\pi)},$$

but $a(u, u) = 0$ if u is constant. However it is weakly coercive with respect to (V, H), since

$$a(u, u) + \|u\|^2_{L^2(0,2\pi)} = \int_0^{2\pi} (u')^2 w dt + \int_0^{2\pi} u^2 dt \ge \min\{w_{\min}, 1\} \left\|u^2\right\|_{H^1(0,2\pi)}.$$

Moreover,

$$\left| \int_0^{2\pi} fv\, dt \right| \leq \|f\|_{L^2(0,2\pi)} \|v\|_{L^2(0,2\pi)} \leq \|f\|_{L^2(0,2\pi)} \|v\|_{H^1(0,2\pi)}$$

hence the functional $F : v \mapsto \int_0^{2\pi} fv\, dt$ defines an element of V^*.

We are under the hypotheses of Theorem 6.69. The bilinear form is symmetric, so that $\mathcal{N}(a) = \mathcal{N}(a_*)$. The solutions of the homogeneous equation

$$a(u, v) = \int_0^{2\pi} u'v'w dt = 0, \qquad \forall v \in V, \tag{6.98}$$

are the constant functions. In fact, letting $v = u$ in (6.98) we obtain

$$\int_0^{2\pi} (u')^2 w dt = 0$$

which forces $u(t) \equiv c$, constant, since $w > 0$. Then, $\dim \mathcal{N}(a) = 1$. Thus, from Theorem 6.69 we can draw the following conclusions: equation (6.97) is solvable if and only if

$$\langle F, 1 \rangle_* = \int_0^{2\pi} f dt = 0.$$

Moreover, in this case, (6.97) has infinitely many solutions of the form $u = \bar{u} + c$.

The variational problem has a simple interpretation as a boundary value problem. By a (formal) integration by parts, recalling that $v(0) = v(2\pi)$, we may rewrite (6.97) as

$$\int_0^{2\pi} [(-wu')' - f]v dt + v(0)[w(2\pi)u'(2\pi) - w(0)u'(0)] = 0, \qquad \forall v \in V.$$

Choosing v vanishing at 0, we are left with

$$\int_0^{2\pi} [-(wu')' - f]v dt = 0, \qquad \forall v \in V, v(0) = v(2\pi) = 0,$$

which forces

$$(u'w)' = -f.$$

Then

$$v(0)[w(2\pi)u'(2\pi) - w(0)u'(0)] = 0, \qquad \forall v \in V,$$

which, in turn, forces

$$w(2\pi)u'(2\pi) = w(0)u'(0).$$

Thus, problem (6.97) constitutes the variational formulation of the following boundary value problem:

$$\begin{cases} (wu')' = -f & \text{in } (0, 2\pi) \\ u(0) = u(2\pi) \\ w(2\pi)u'(2\pi) = w(0)u'(0). \end{cases}$$

It is important to point out that the periodicity condition $u(0) = u(2\pi)$ is forced by the choice of the space V while the Neumann type periodicity condition is *encoded* in the variational equation (6.97).

6.8.3 The Fredholm alternative

We introduce some terminology. Let V_1, V_2 Hilbert spaces and $\Phi : V_1 \to V_2$. We say that Φ is a *Fredholm operator* if $\mathcal{N}(\Phi)$ and $\mathcal{R}(\Phi)^\perp$ have *finite dimension*. The index of Φ is the integer

$$\operatorname{ind}(\Phi) = \dim \mathcal{N}(\Phi) - \dim \mathcal{R}(\Phi)^\perp = \dim \mathcal{N}(\Phi) - \dim \mathcal{N}(\Phi^*).$$

We have the following result, known as the *Fredholm Alternative Theorem*[11]:

Theorem 6.72 *Let V be a Hilbert space and $K \in \mathcal{L}(V)$ be a compact operator. Let I be the identity operator in V. Then*

$$\Phi = I - K$$

is a Fredholm operator with zero index. Moreover, $\Phi^* = I - K^*$,

$$\mathcal{R}(\Phi) = \mathcal{N}(\Phi^*)^\perp \tag{6.99}$$

and

$$\mathcal{N}(\Phi) = \{0\} \iff \mathcal{R}(\Phi) = V. \tag{6.100}$$

Formula (6.99) implies that $\mathcal{R}(\Phi)$ is closed. Formula (6.100) shows that Φ is one-to-one *if and only if* it is onto. In this case, Theorem 6.27, p. 368, implies that both Φ^{-1} and $(\Phi^*)^{-1}$ belong to $\mathcal{L}(V)$. In other words, uniqueness for the equation

$$u - Ku = f \tag{6.101}$$

with $f = 0$, is equivalent to existence for every $f \in V$ and viceversa. Moreover, the solution of (6.101) satisfies the estimate

$$\|u\|_V \leq \left\|\Phi^{-1}\right\|_{\mathcal{L}(V)} \|f\|_V .$$

If Φ is not one-to-one, let $d = \dim \mathcal{R}(\Phi)^\perp = \dim \mathcal{N}(\Phi^*) > 0$. Then, (6.99) says that equation (6.101) is solvable if and only if $f \perp \mathcal{N}(\Phi^*)$, that is, if and only if $(f, w)_V = 0$ for every solution w of

$$w - K^* w = 0. \tag{6.102}$$

If $\{w_1, w_2, \ldots, w_d\}$ span $\mathcal{N}(\Phi^*)$, this is equivalent to saying that the d *compatibility relations*

[11] Compare with Thm. 6.38. For the proof, see e.g. *H. Brezis*, [35].

$$(f, w_j)_V = 0, \quad j = 1, \dots, d,$$

are the necessary and sufficient conditions for the solvability of (6.101).

Analogous considerations hold for the adjoint $\Phi^* = I - K^*$ and the associated equation

$$v - K^* v = g.$$

Clearly, Theorem 6.72 holds for operators $K - \lambda I$ with $\lambda \neq 0$. The case $\lambda = 0$ *cannot be included*. Trivially, for the operator $K = 0$ (which is compact), we have $\mathcal{N}(K) = V$, hence, if $\dim V = \infty$, Theorem 6.72 does not hold. A more significant example is the one-dimensional range operator

$$Kx = L(x)x_0,$$

where $L \in V^*$ and x_0 is fixed in V. Assume $\dim V = \infty$. From the Riesz Representation Theorem, there exists $z \in V$ such that $Lx = (z, x)_V$, for every $x \in V$. Thus, $\mathcal{N}(K)$ is given by the subspace of the elements in V orthogonal to z, which has infinitely many dimensions.

Proof (of Theorem 6.69, sketch) The strategy is to write equation

$$a(u, v) = \langle F, v \rangle_* \tag{6.103}$$

in the form

$$(I_V - K)u = g$$

where I_V is the identity operator in V, and $K : V \to V$ is compact.

Let $J : V \to V^*$ be the embedding of V into V^* defined by (6.92), p. 394. Recall that J is the composition of the embeddings $I_{V \hookrightarrow H}$ and $I_{H \hookrightarrow V^*}$. Since $I_{V \hookrightarrow H}$ is compact and $I_{H \hookrightarrow V^*}$ is continuous, by Proposition 6.67, p. 392, we infer that J is compact. We write (6.103) in the form

$$a_{\lambda_0}(u, v) \equiv a(u, v) + \lambda_0 (u, v)_H = \langle \lambda_0 Ju + F, v \rangle_*$$

where $\lambda_0 > 0$ is such that $a_{\lambda_0}(u, v)$ is V-coercive (recall that $\langle Ju, v \rangle_* = (u, v)_H$, from (6.93)).

Since, for fixed $u \in V$, the linear map $v \mapsto a_{\lambda_0}(u, v)$ is continuous in V, there exists $L \in \mathcal{L}(V, V^*)$ such that

$$\langle Lu, v \rangle_* = a_{\lambda_0}(u, v), \qquad \forall u, v \in V.$$

Thus, equation $a(u, v) = \langle F, v \rangle_*$ is equivalent to

$$\langle Lu, v \rangle_* = \langle \lambda_0 Ju + F, v \rangle_*,$$

for every $v \in V$, and therefore to

$$Lu = \lambda_0 Ju + F. \tag{6.104}$$

Since a_{λ_0} is V-coercive, by the Lax-Milgram Theorem, the operator L is a continuous isomorphism between V and V^* and (6.104) can be written in the form

$$u - \lambda_0 L^{-1} Ju = L^{-1} F.$$

Letting

$$g = L^{-1} F \in V \qquad \text{and} \qquad K = \lambda_0 L^{-1} J,$$

equation (6.104) becomes

$$(I_V - K)u = g$$

where $K : V \to V$.

Since J is compact and L^{-1} is continuous, K is compact. Applying the Fredholm Alternative Theorem and rephrasing the conclusions in terms of bilinear forms, we conclude the proof (we omit the rather long and technical details). □

6.9 Spectral Theory for Symmetric Bilinear Forms

6.9.1 Spectrum of a matrix

Let $\mathbf{A}$ be an $n \times n$ matrix and $\lambda \in \mathbb{C}$. Then, either the equation

$$\mathbf{Ax} - \lambda \mathbf{x} = \mathbf{b}$$

has a unique solution for every b or there exists $u \neq 0$ such that

$$\mathbf{Au} = \lambda \mathbf{u}.$$

In the last case we say that λ, u constitutes an *eigenvalue-eigenvector pair*. The set of eigenvalues of $\mathbf{A}$ is called the *spectrum* of A, denoted by $\sigma_P(\mathbf{A})$. If $\lambda \notin \sigma_P(\mathbf{A})$ the resolvent matrix $(\mathbf{A} - \lambda \mathbf{I})^{-1}$ is well defined. The set

$$\rho(\mathbf{A}) = \mathbb{C} \setminus \sigma_P(\mathbf{A})$$

is called the *resolvent set* of A. If $\lambda \in \sigma_P(\mathbf{A})$, the kernel $\mathcal{N}(\mathbf{A} - \lambda \mathbf{I})$ is the subspace spanned by the eigenvectors corresponding to λ and it is called the *eigenspace* of λ. Note that $\sigma_P(\mathbf{A}) = \sigma_P(\mathbf{A}^\intercal)$.

The symmetric matrices are particularly important: all the eigenvalues $\lambda_1, \ldots, \lambda_n$ are real (possibly of multiplicity greater than 1) and there exists in $\mathbb{R}^n$ an orthonormal basis of eigenvectors $u_1, \ldots, u_n$. The action of $\mathbf{A}$ can be written as the sum of the projections into its eigenspaces, according to he following formula[12], called *spectral decomposition* of A:

$$\mathbf{A} = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top + \ldots + \lambda_n \mathbf{u}_n \mathbf{u}_n^\top.$$

We are going to extend these concepts in the Hilbert space setting. A motivation is ... the method of separation of variables.

6.9.2 Separation of variables revisited

Using the method of separation of variables, in the first chapters we have constructed solutions of boundary value problems by superposition of special solutions. However,

[12] Here $\mathbf{u}_j$ are column vectors; $\mathbf{u}_j \mathbf{u}_j^\top$ is an $n \times n$ matrix, sometimes denoted by $\mathbf{u}_j \otimes \mathbf{u}_j$ ($\mathbf{u}_j$ tensor $\mathbf{u}_j$).

explicit computations can be performed only when the geometry of the relevant domain is quite particular. What may we say in general? Let us consider an example from diffusion.

Suppose we have to solve the problem

$$\begin{cases} u_t = \Delta u & (x,y) \in \Omega,\ t > 0 \\ u(x,y,0) = g(x,y) & (x,y) \in \Omega \\ u(x,y,t) = 0 & (x,y) \in \partial\Omega,\ t > 0 \end{cases}$$

where Ω is a bounded bi-dimensional domain. Let us look for solutions of the form

$$u(x,y,t) = v(x,y)w(t).$$

Substituting into the differential equation, with some elementary manipulations, we obtain

$$\frac{w'(t)}{w(t)} = \frac{\Delta v(x,y)}{v(x,y)} = -\lambda,$$

where λ is a constant, which leads to the two problems

$$w' + \lambda w = 0 \qquad t > 0 \tag{6.105}$$

and

$$\begin{cases} -\Delta v = \lambda v & \text{in } \Omega \\ v = 0 & \text{on } \partial\Omega. \end{cases} \tag{6.106}$$

A number λ such that there exists a nontrivial solution v of (6.106) is called a *Dirichlet eigenvalue* of the operator $-\Delta$ in Ω and v is a corresponding eigenfunction. Now, the original problem can be solved if the following two properties hold:

a) There exists a sequence of (real) eigenvalues λ_k with corresponding eigenvectors u_k. Solving (6.105) for $\lambda = \lambda_k$ yields

$$w_k(t) = ce^{-\lambda_k t} \qquad c \in \mathbb{R}.$$

b) The initial data g can be expanded in series of eigenfunctions:

$$u(x,y) = \sum g_k u_k(x,y).$$

Then, the solution is given by

$$u(x,y,t) = \sum g_k u_k(x,y) e^{-\lambda_k t}$$

where the series converges in some suitable sense.

Condition b) requires that the set of Dirichlet eigenfunctions of $-\Delta$ constitutes a basis in the space of initial data. This leads to the problem of determining the *spectrum* of a linear operator in a Hilbert space and, in particular, of self-adjoint compact operators. Indeed, it turns out that the solution map of a symmetric varia-

tional boundary value problem is often a self-adjoint compact operator. We will go back to the above problem in Sect. 8.3.4.

6.9.3 Spectrum of a compact self-adjoint operator

We define resolvent and spectrum for a bounded linear operator. Although the natural setting is the complex field $\mathbb{C}$, we limit ourselves to $\mathbb{R}$, mainly for simplicity but also because this is the interesting case for us.

Definition 6.73 Let H be a (real) Hilbert space, $L \in \mathcal{L}(H)$, and I be the identity in H.

a) The *resolvent set* $\rho(L)$ of L is the set of real numbers λ such that $L - \lambda I$ is one-to-one and onto:

$$\rho(L) = \{\lambda \in \mathbb{R} : L - \lambda I \text{ is one-to-one and onto}\} .$$

b) The (real) *spectrum* $\sigma(L)$ of L is

$$\sigma(L) = \mathbb{R} \setminus \rho(L).$$

Remark 6.74 If $\lambda \in \rho(L)$, the *resolvent* $(L - \lambda I)^{-1}$ is bounded (see Theorem 6.27, p. 368).

If H is finite dimensional, any linear operator in $\mathcal{L}(H)$ is represented by a matrix, so that its spectrum is given by the set of its eigenvalues. In infinitely many dimensions the spectrum may be divided in three subsets. In fact, if $\lambda \in \sigma(L)$, different things can go wrong with $(L-\lambda I)^{-1}$. First of all, it may happen that $L-\lambda I$ is not one-to-one so that $(L - \lambda I)^{-1}$ does not even exist.

This means that

$$\mathcal{N}(L - \lambda I) \neq \{0\}$$

or, in other words, that the equation

$$Lu = \lambda u \tag{6.107}$$

has nontrivial solutions. Then, we say that λ is an *eigenvalue* of L and that the nonzero solutions of (6.107) are the *eigenvectors* corresponding to λ. The linear space $\mathcal{N}(L - \lambda I)$ spanned by these eigenvectors is called the *eigenspace* of λ. The dimension of $\mathcal{N}(L - \lambda I)$ is called the (geometric) *multiplicity* of λ.

Definition 6.75 The set $\sigma_P(L)$ of the eigenvalues of L is called the *point spectrum* of L.

Other things can occur. $L - \lambda I$ is one-to-one, $\mathcal{R}(L - \lambda I)$ is dense in H, but $(L - \lambda I)^{-1}$ is unbounded. Then, we say that λ belongs to the *continuous spectrum* of L, denoted by $\sigma_C(L)$.

Finally, $L - \lambda I$ is one-to-one but $\mathcal{R}(L - \lambda I)$ is not dense in H. This defines the *residual spectrum* of L.

Example 6.76 Let $H = l^2$ and $L : l^2 \to l^2$ be the *shift* operator which maps $\mathbf{x} = \{x_1, x_2, \ldots\} \in l^2$ into $\mathbf{y} = \{0, x_1, x_2, \ldots\}$. We have

$$(L - \lambda I)x = \{-\lambda x_1, x_1 - \lambda x_2, x_2 - \lambda x_3, \ldots\}.$$

If $\lambda \neq 0$, then $\lambda \in \rho(L)$. In fact for every $\mathbf{z} = \{z_1, z_2, \ldots\} \in l^2$,

$$(L - \lambda I)^{-1}\mathbf{z} = \left\{-\frac{z_1}{\lambda}, -\frac{z_2}{\lambda} - \frac{z_1}{\lambda^2}, \ldots\right\}.$$

Since $\mathcal{R}(L)$ contains only sequences whose first element is zero, $\mathcal{R}(L)$ is *not dense in* l^2, therefore $0 \in \sigma_R(L) = \sigma(L)$.

We are mainly interested in the spectrum of a compact self-adjoint operator. The following theorem is fundamental (see *H. Brezis*, [35] for the proof).

Theorem 6.77 *Let K be a compact, selfadjoint operator on a separable, infinite dimensional Hilbert space H. Then:*

a) $0 \in \sigma(K)$ *and* $\sigma(K) \setminus \{0\} = \sigma_P(K) \setminus \{0\}$. *Moreover, every nonzero eigenvalue has finite multiplicity.*
b) $\sigma_P(K) \setminus \{0\}$ *is a finite set or it can be ordered in a sequence*

$$|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_m| \geq \cdots$$

with $\lambda_m \to 0$ *as* $m \to \infty$, *and where each eigenvalue appears a finite number of times, according to its multiplicity.*
c) H has an orthonormal basis $\{u_m\}$ *consisting of eigenvectors of K.*

A few comments are in order.

• If $\dim H = \infty$, the spectrum of a compact selfadjoint operator contains always $\lambda = 0$, which is not necessarily an eigenvalue[13]. This follows immediately from Theorem 6.52, p. 385. Indeed if $0 \in \rho(K)$, $I_H = K^{-1}K$, would be a compact operator, contradicting Theorem 6.52. The other elements in $\sigma(K)$ are all eigenvalues, arranged in an infinite sequence $\{\lambda_m\}_{m\geq 1}$ converging to zero.

• Let u_j and u_k be eigenvectors corresponding to two *different* eigenvalues λ_j, λ_k, including possibly 0. Then $(u_j, u_k) = 0$. Indeed, from

$$Ku_j = \lambda_j u_j \quad \text{and} \quad Ku_k = \lambda_k u_k$$

we get

$$\lambda_j(u_j, u_k) = (Ku_j, u_k) = (u_j, Ku_k) = \lambda_k(u_j, u_k)$$

and hence $(u_j, u_k) = 0$, since $\lambda_j \neq \lambda_k$.

[13] Actually, properties a) and b) hold for any compact operator.

• If $\lambda \neq 0$ is an eigenvalue, the Fredholm Alternative Theorem 6.72 implies that the eigenspace $\mathcal{N}(K - \lambda I)$ has finite dimension and therefore every nonzero eigenvalue appears in the sequence $\{\lambda_m\}_{m\geq 1}$ only a finite number of times. For instance, if $\dim \mathcal{N}(K - \lambda_1 I) = d_1$, in the sequence $\{\lambda_m\}_{m\geq 1}$ we have $\lambda_1 = \lambda_2 = \cdots = \lambda_{d_1}$; if $\dim \mathcal{N}(K - \lambda_{d_1+1} I) = d_2$, we have

$$\lambda_{d_1+1} = \lambda_{d_1+2} = \cdots = \lambda_{d_1+d_2}$$

and so on.

• The separability of H comes into play only when $0 \in \sigma_P(K)$, to make sure that $\mathcal{N}(K)$ has a countable orthonormal basis. The union of the bases of $\mathcal{N}(K)$ and all the $\mathcal{N}(K - \lambda_m I)$, $m \geq 1$, forms a basis in H.

• A consequence of Theorem 6.77 is the *spectral decomposition* formula for K. If $u \in H$ and $\{u_m\}$ is an orthonormal set of eigenvectors corresponding to all nonzero eigenvalues $\{\lambda_m\}_{m\geq 1}$, we can describe the action of K as follows:

$$Ku = \sum_{m\geq 1} (Ku, u_m)u_m = \sum_{m\geq 1} \lambda_m (u, u_m)u_m, \quad \forall u \in H. \tag{6.108}$$

6.9.4 Application to abstract variational problems

We now apply Theorems 6.69 and 6.77 to our abstract variational problems. The setting is the same of Theorem 6.69, given by a Hilbert triplet $\{V, H, V^*\}$, with *compact embedding* of V into H. We also assume that H is *separable*. Let a be a bilinear form in V, continuous and (V, H)-weakly coercive; in particular:

$$a_{\lambda_0}(v, v) \equiv a(v, v) + \lambda_0 \|v\|_H^2 \geq \alpha \|v\|_V^2, \qquad \forall v \in V \quad (\alpha > 0). \tag{6.109}$$

The notion of resolvent and spectrum can be easily defined. Consider the problem

$$a(u, v) = \lambda (u, v)_H + \langle F, v\rangle_*, \qquad \forall v \in V. \tag{6.110}$$

The *resolvent set* $\rho(a)$ is the set of real numbers λ such that (6.110) has a unique solution $u(F) \in V$, for every $F \in V^*$ and the solution map

$$S(a, \lambda) : F \longmapsto u(F) \tag{6.111}$$

is a continuous isomorphism between V^* and V.

The (real) *spectrum* is $\sigma(a) = \mathbb{R} \setminus \rho(a)$, while the *point spectrum* $\sigma_P(a)$ is the subset of the spectrum given by the eigenvalues, that is the numbers λ such that the homogeneous problem

$$a(u, v) = \lambda (u, v)_H, \qquad \forall v \in V \tag{6.112}$$

has nontrivial solutions $u \in V$, called *eigenvectors* corresponding to λ. We call *eigenspace* of λ the space spanned by the corresponding eigenfunctions and we denote it by $\mathcal{N}(a, \lambda)$. The dimension of $\mathcal{N}(a, \lambda)$ is called *multiplicity* of λ. Note that by the Alternative Theorem 6.69, $\sigma(a) = \sigma_P(a)$.

Theorem 6.78 *Let $\{V, H, V^*\}$ be a Hilbert triplet, with H separable and V compactly embedded in H. Assume that H is infinite dimensional. Let a be a symmetric bilinear form in V, continuous and weakly coercive ((6.109) holds). We have:*

a) $\sigma(a) = \sigma_P(a) \subset (-\lambda_0, +\infty)$. *The set of the eigenvalues is infinite and can be ordered in a nondecreasing sequence $\{\lambda_m\}_{m\geq 1}$, where each λ_m appears a (finite) number of times according to its multiplicity. Moreover, $\lambda_m \to +\infty$.*

b) If u_k, u_m are eigenvectors corresponding to two different eigenvalues λ_k and λ_m, respectively, then

$$a(u_k, u_m) = 0 = (u_k, u_m)_H.$$

Moreover, H has an orthonormal basis $\{w_m\}$, with w_m corresponding to λ_m, for all $m \geq 1$.

c) $\left\{w_m/\sqrt{\lambda_m + \lambda_0}\right\}$ *constitutes an orthonormal basis in V, with respect to the scalar product*

$$((u, v)) = a(u, v) + \lambda_0 (u, v)_H. \tag{6.113}$$

Before proving Theorem 6.78 we make some preliminary considerations.

Since $H \hookrightarrow V^*$, we can consider the map $S_{\lambda_0} \in \mathcal{L}(H)$, given by the restriction to H of the solution map $S(a_{\lambda_0}, 0)$. Let us explore the relation between $\sigma_P(S_{\lambda_0})$ and $\sigma_P(a_{\lambda_0})$.

First of all, note that $0 \notin \sigma_P(S_{\lambda_0})$, otherwise we should have $S_{\lambda_0} f = 0$ for some $f \in H$, $f \neq 0$, which leads to the contradiction

$$0 = a_{\lambda_0}(0, v) = (f, v)_H, \quad \forall v \in V.$$

Also, it is clear that $0 \in \rho(a_{\lambda_0})$, whence

$$\sigma_P(a_{\lambda_0}) \subset (0, +\infty). \tag{6.114}$$

Lemma 6.79 *$\lambda \in \sigma_P(a_{\lambda_0})$ if and only if $\mu = 1/\lambda \in \sigma_P(S_{\lambda_0})$. Moreover:*

$$\mathcal{N}(a_{\lambda_0}, \lambda) = \mathcal{N}(S_{\lambda_0} - \mu I)$$

and from Theorem 6.69, p. 395, $\mathcal{N}(a_{\lambda_0}, \lambda)$ has dimension $d < \infty$.

Proof Let λ be an eigenvalue of a_{λ_0} and f be a corresponding eigenvector, that is,

$$a_{\lambda_0}(f, v) = \lambda (f, v)_H, \quad \forall v \in V. \tag{6.115}$$

Then $f \in V \subset H$ and setting $\mu = 1/\lambda$, equation (6.115) is equivalent to

$$a_{\lambda_0}(\mu f, v) = (f, v)_H$$

that is

$$S_{\lambda_0} f = \mu f. \tag{6.116}$$

Since (6.115) and (6.116) are equivalent, the lemma follows. □

We are now in position to prove the Theorem 6.78.

Proof From (6.114) if follows that $\sigma(a) = \sigma_P(a) \subset (-\lambda_0, +\infty)$. Observe now that, since $S_{\lambda_0}(H) \subset V$ and V is compactly embedded into H, S_{λ_0} is compact as an operator from H into H. Also, by the symmetry of a, S_{λ_0} is selfadjoint, that is

$$(S_{\lambda_0} f, g)_H = (f, S_{\lambda_0} g)_H, \qquad \text{for all } f, g \in H.$$

In fact, let $u = S_{\lambda_0} f$ and $w = S_{\lambda_0} g$. Then, for every $v \in V$,

$$a_{\lambda_0}(u, v) = (f, v)_H \quad \text{and} \quad a_{\lambda_0}(w, v) = (g, v)_H.$$

In particular,

$$a_{\lambda_0}(u, w) = (f, w)_H \quad \text{and} \quad a_{\lambda_0}(w, u) = (g, u)_H$$

so that, since $a_{\lambda_0}(u, w) = a_{\lambda_0}(w, u)$ and $(g, u)_H = (u, g)_H$, we can write

$$(S_{\lambda_0} f, g)_H = (u, g)_H = (f, w)_H = (f, S_{\lambda_0} g)_H.$$

Since we observed that $0 \notin \sigma_P(S_{\lambda_0})$, from Theorem 6.77 it follows that $\sigma(S_{\lambda_0}) \setminus \{0\} = \sigma_P(S_{\lambda_0})$ and the eigenvalues of S_{λ_0} form an infinite nonincreasing sequence $\{\mu_m\}$, with $\mu_m \to 0$. Recalling Lemma 6.79, $\sigma_P(a_{\lambda_0})$ consists of a sequence $\{\lambda_m^0\}$ that we can list in nondecreasing order, according to the multiplicity of each eigenvalue, with $\lambda_m^0 \to +\infty$. This proves a), with $\lambda_m = \lambda_m^0 - \lambda_0$. To prove b), let u_k and u_m be eigenvectors corresponding to λ_k and λ_m, respectively, $\lambda_k \neq \lambda_m$. Then

$$a(u_k, u_m) = \lambda_k (u_k, u_m)_H \quad \text{and} \quad a(u_k, u_m) = \lambda_m (u_k, u_m)_H.$$

Subtracting the two equations we get

$$(\lambda_k - \lambda_m)(u_k, u_m)_H = 0$$

from which $(u_k, u_m)_H = 0 = a(u_k, u_m)$. Moreover, from c) in Theorem 6.77, H has an orthonormal basis $\{w_m\}_{m \geq 1}$ consisting of eigenvectors corresponding to $\{\lambda_m\}_{m \geq 1}$. Finally, if δ_{mk} denotes the Kronecker symbol, we have

$$a(w_m, w_k) = \lambda_m (w_m, w_k)_H = \lambda_m \delta_{mk}$$

so that

$$((w_m, w_k)) = a_{\lambda_0}(w_m, w_k) = (\lambda_m + \lambda_0)\, \delta_{mk}.$$

This shows that $\left\{w_m / \sqrt{\lambda_m + \lambda_0}\right\}$ constitutes an orthonormal system in V, endowed with the scalar product (6.113). To check that it is indeed a basis, we show that the only element in V, orthogonal to every w_m, is $v = 0$. Thus, let $v \in V$ and assume that

$$((v, w_m)) \equiv a(v, w_m) + \lambda_0 (v, w_m) 0, \qquad \forall m \geq 1.$$

Since $a(v, w_m) = \lambda_m (v, w_m)_H$, we deduce that

$$(\lambda_m + \lambda_0)(v, w_m)_H = 0, \qquad \forall m \geq 1.$$

Since $\{w_m\}$ is a basis in H, it follows that $v = 0$. The proof is complete. □

Under the hypotheses of Theorem 6.78, every element $u \in H$ has the Fourier expansion

$$\sum_{m=1}^{\infty} c_m w_m \tag{6.117}$$

where $c_m = (u, w_m)_H$ are the Fourier coefficients of u and

$$\|u\|_H^2 = \sum_{m=1}^{\infty} c_m^2 < \infty.$$

It turns out that also the elements of V can be characterized in terms of their Fourier coefficients with respect to the basis $\{w_m\}_{m\geq 1}$. In fact the following theorem holds, where *we adopt in V the inner product* $((u, v)) = a_{\lambda_0}(u, v)$ and the induced norm $\sqrt{((v, v))}$.

Theorem 6.80 *Let $u \in H$. Then u belongs to V if and only if*

$$((u, u)) = \sum_{m=1}^{\infty} (\lambda_m + \lambda_0) c_m^2 < \infty. \tag{6.118}$$

Proof Let $\tilde{w}_m = w_m/\sqrt{\lambda_0 + \lambda_m}$. Since $\{\tilde{w}\}_{m\geq 1}$ is an orthonormal basis in V with respect to the scalar product (6.113), u belongs to V if and only if

$$\sum_{m=1}^{\infty} ((u, \tilde{w}_m))^2 < \infty.$$

Now,

$$((u, \tilde{w}_m)) = \frac{1}{\sqrt{\lambda_m + \lambda_0}}\, a_{\lambda_0}(u, w_m) = \sqrt{\lambda_m + \lambda_0}\, c_m$$

and (6.118) follows. □

Using the above considerations, we can easily prove an important *variational* characterizations of the eigenvalues of a, in terms of the so called *Rayleigh quotient* associated to a, defined by

$$R(u) = \frac{a(u, u)}{\|u\|_H^2} \qquad (u \neq 0). \tag{6.119}$$

Observe that minimizing $R(u)$ over $V \setminus \{0\}$ is equivalent to minimize $a(u, u)$ over V with the constraint $\|u\|_H^2 = 1$. We start with a variational principle for the first eigenvalue.

Theorem 6.81 *Assume a, V and H are as in Theorem 6.78.*

1. *Let λ_1 be the first eigenvalue of a and u_1 be a corresponding eigenvector. Then*

$$\lambda_1 = R(u_1) = \min_{v\in V, v\neq 0} R(v) = \min_{v\in V, \|v\|_H = 1} a(v, v). \tag{6.120}$$

2. *If $R(w) = \lambda_1 = \min_{v\in V, v\neq 0} R(v)$, then w is an eigenvector of a, corresponding to λ_1, that is w is a solution of the equation*

$$a(w,v) = \lambda_1 (w,v)_H \qquad \forall v \in V.$$

Proof 1. Let $\{\lambda_m\}_{m\geq 1}$ be the nondecreasing sequence of all the eigenvalues of a, each one counted according its multiplicity, and $\{w_m\}_{m\geq 1}$ be a corresponding orthonormal basis in H of eigenvectors. For every $v \in V$ we can write

$$v = \sum_{m=1}^{\infty} (v, w_m)_H w_m$$

where the series converges in V (see Thm. 6.80). Since $\lambda_m \geq \lambda_1$ for every $m > 1$, and $a(v, w_m) = \lambda_m (v, w_m)$, we can write

$$a(v,v) = \sum_{m=1}^{\infty} (v,w_m)_H a(v,w_m) = \sum_{m=1}^{\infty} \lambda_m (v,w_m)_H^2 \geq \lambda_1 \sum_{m=1}^{\infty} (v,w_m)_H^2 = \lambda_1 \|v\|_H^2 .$$

Moreover,

$$a(w_1, w_1) = \lambda_1.$$

We infer that $\lambda_1 \leq R(v)$ for every $v \in V$ and that $\lambda_1 = R(w_1)$. Since $R(w_1) = R(u_1)$, (6.120) follows.

2. Let d be the dimension of the eigenspace corresponding to λ_1, spanned by $w_1, \ldots, w_{d_1}$. From $R(w) = \lambda_1$ and $\|w\|_H^2 = \sum (w, w_m)_H^2$ we have

$$\lambda_1 \sum_{m=1}^{\infty} (w,w_m)_H^2 = a(w,w) = \lambda_1 \sum_{m=1}^{d_1} (w,w_m)_H^2 + \sum_{m=d_1+1}^{\infty} \lambda_m (w,w_m)_H^2.$$

This forces

$$(\lambda_m - \lambda_1)(w,w_m)_H^2 = 0$$

for $m \geq d_1 + 1$, since $\lambda_m > \lambda_1$ when $m \geq d_1 + 1$. Then $w = \sum_{m=1}^{d_1} (w, w_m)_H w_m$ and therefore w is an eigenvector of a corresponding to λ_1. □

Also the other eigenvalues have a similar variational characterization. Assume a, V and H are as in Theorem 6.78. Let $W_k = \text{span}\{w_1, \ldots, w_k\}$. We have:

Theorem 6.82 *The following characterization holds for the k-th eigenvalue, $k \geq 2$:*

$$\lambda_k = \min \left\{ R(v) : v \neq 0,\ v \in V \cap W_{k-1}^{\perp} \right\}. \tag{6.121}$$

Moreover, the minimum is attained at any eigenvector corresponding to λ_k.

Proof For every $v \in V \cap W_{k-1}^{\perp}$, we can write:

$$v = \sum_{m=1}^{\infty} (v,w_m)_H w_m = \sum_{m=k}^{\infty} (v,w_m)_H w_m.$$

Since $\lambda_m \geq \lambda_k$ for every $m \geq k$, we get

$$a(v,v) = \sum_{m=k}^{\infty} \lambda_m (v,w_m)_H^2 \geq \lambda_k \sum_{m=k}^{\infty} (v,w_m)_H^2 = \lambda_k \|v\|_H^2 . \tag{6.122}$$

Moreover, if w is an eigenvector corresponding to λ_k, we have

$$w = \sum_{w_m \in N(a,\lambda_k)} (w, w_m) w_m$$

and therefore

$$a(w,w) = \lambda_k \sum_{w_m \in N(a,\lambda_k)} (w,w_m)_H^2 = \lambda_k \|w\|_H^2$$

so that $\lambda_k = R(w)$. The second part of the statement follows as in Theorem 6.81. □

There is another variational principle characterizing the eigenvalues in our functional setting, known as the *minimax property*[14], that avoids the use of any orthonormal basis of H. We refer to Problem 6.23 for both statement and proof.

6.10 Fixed Points Theorems

Fixed point theorems constitute a fundamental tool for solving *nonlinear* problems which can be reformulated by an equation of the form $F(x) = x$. A solution of this equation is a point which is mapped by F into itself and for this reason it is called *fixed point* of F.

In general, F is an operator defined in some metric or normed space X into itself. Here we consider two types of fixed point theorems:

1. Fixed points for maps that contract distances (so called *contractions*).
2. Fixed points involving *compactness and/or convexity.*

The most important theorem in the first class is the *Contraction Mapping Theorem* whose natural setting is a complete metric space. This theorem gives a sufficient condition for the existence, uniqueness and stability of a fixed point, constructed through an iterative procedure, known as the *method of successive approximation*.

Among those in the second class, we consider the theorems of *Schauder* and of *Leray-Schauder*, stated in Banach spaces. These theorems do not give any information on the uniqueness of the fixed point.

We shall use a fixed point technique based on the above theorems in Sects. 10.3 and 11.2.

6.10.1 The Contraction Mapping Theorem

Let (M,d) denote a metric space with distance d and

$$F : M \to M.$$

We say that F is a strict contraction if there exists a number ρ, $0 < \rho < 1$, such that

$$d(F(x),F(y)) \le \rho\, d(x,y), \quad \forall x,y \in M. \tag{6.123}$$

[14] Due to *Lord Rayleigh*, Theory of Sound, London, *1894*.

Clearly, every strict contraction is continuous. For instance, a function $F : \mathbb{R}^n \to \mathbb{R}^n$ is a strict contraction if it is differentiable and $\sup_{\mathbb{R}^n} \left|\nabla F_j(\mathbf{x})\right| \leq \rho < 1/\sqrt{n}, j = 1, \ldots, n$.

The inequality (6.123) expresses the fact that F contracts the distance between any two points, by at least a percentage $\rho < 1$. This fact confers to a recursive sequence $x_{m+1} = F(x_m)$, $m \geq 0$, starting from an arbitrary point $x_0 \in M$, a remarkable asymptotic stability, particularly useful in numerical approximation methods. Precisely, the following result holds.

Theorem 6.83 *Let (M, d) be a complete metric space and $F : M \to M$ be a strict contraction:*

$$d(F(x), F(y)) \leq \rho\, d(x, y), \qquad \forall x, y \in M$$

for some $\rho < 1$. Then there exists a unique fixed point $x^* \in M$ of F. Moreover, for any choice of $x_0 \in M$, the recursive sequence

$$x_{m+1} = F(x_m), \quad m \geq 0 \tag{6.124}$$

converges to x^*.

Proof Fix $x_0 \in M$. We show that the sequence (6.124) is a Cauchy sequence. In fact we have, for $j \geq 1$,

$$x_{j+1} = F(x_j), \quad x_j = F(x_{j-1})$$

so that

$$d(x_{j+1}, x_j) = d(F(x_j), F(x_{j-1})) \leq \rho d(x_j, x_{j-1}).$$

Iterating from $j - 1$ to $j = 1$, we get

$$d(x_{j+1}, x_j) \leq \rho^j d(x_1, x_0).$$

From the triangular inequality, if $m > k$,

$$d(x_m, x_k) \leq \sum_{j=k}^{m-1} d(x_{j+1}, x_j) \leq d(x_1, x_0) \sum_{j=k}^{m-1} \rho^j \leq d(x_1, x_0) \frac{\rho^k}{1-\rho}.$$

Thus, if $k \to \infty$, $d(x_m, x_k) \to 0$ and $\{x_m\}$ is a Cauchy sequence. By the completeness of (M, d), there exists $x^* \in M$ such that $x_m \to x^*$. Letting $m \to \infty$ in the recursive relation (6.124), we obtain $x^* = F(x^*)$, i.e. x^* is a fixed point for F. If y^* is another fixed point for F, we have

$$d(x^*, y^*) = d(F(x^*), F(y^*)) \leq \rho d(x^*, y^*)$$

or

$$(1 - \rho) d(x^*, y^*) \leq 0$$

which implies $d(x^*, y^*) = 0$. Therefore $x^* = y^*$, i.e. the fixed point is unique. □

We emphasize that the initial point x_0 is arbitrarily chosen in M. From a different perspective, Theorem 6.83 affirms that the dynamical system defined by the recursive sequence (6.124) has a unique equilibrium point x^*. Moreover, x^* is asymptotically stable, with attraction basin coincident with M.

6.10.2 The Schauder Theorem

The Schauder Theorem extends to a Banach space setting the following Brouwer Fixed Point Theorem, valid in $\mathbb{R}^n$:

Theorem 6.84 *Let $S \subset \mathbb{R}^n$ be a closed sphere and $T : S \to S$ be a continuous map. Then T has a fixed point $\mathbf{x}^*$.*

In dimension $n = 1$, Theorem 6.84 amounts to say that the graph of a continuous function $f : [0, 1] \to [0, 1]$ intersects the straight line $y = x$ at least one time. In dimension $n > 1$, there are several proofs, all rather nontrivial[15].

The Brouwer Theorem still holds if the sphere S is substituted by a set S' *homeomorphic* (i.e. *topologically equivalent*) to S. This means that there exists a one-to-one and onto function $\varphi : S \leftrightarrow S'$ (*a homeomorphism*) such that φ and φ^{-1} are continuous. A homeomorphism is a mathematical realization of a continuous deformation. Thus, any continuous deformation of S is homeomorphic to S.

The difficulty in extending Brouwer's Theorem to infinitely many dimensions relies on the fact that a closed sphere in an infinite dimensional Banach or Hilbert space is *never compact*. A way of overcoming the difficulty is to consider *compact and convex* sets. Recall that E is *convex* if, for every $x, y \in E$, the line segment of endpoints x, y is contained in E.

Let X be a Banach space and $E \subset X$. The *convex envelope* of E, denoted by $\text{co}\{E\}$, is the smallest convex set containing E. In other words:

$$\text{co}\{E\} = \{\cap F : F \text{ convex}, E \subset F\}.$$

The symbol $\overline{\text{co}}\{E\}$ denotes the closure of $\text{co}\{E\}$, and is called the *closed convex envelope* of E. An important property of the convex envelopes is expressed in the following proposition (for the proof, see e.g. *A. E. Taylor, D. C. Lay*, [41]).

Proposition 6.85 *If E is compact, then $\overline{\text{co}}\{E\}$ is compact.*

An example of compact and convex set is the closed convex envelope of a finite number of points:

$$\overline{\text{co}}\{x_1, x_2, \ldots, x_N\} = \left\{ \sum_{i=1}^{N} \lambda_i x_i : 0 \le \lambda_i \le 1, \sum_{i=1}^{N} \lambda_i = 1 \right\}.$$

Theorem 6.86 (Schauder) *Let X be a Banach space. We are given:*

a) $A \subset X$, compact and convex.
b) $T : A \to A$, continuous.

Then T has a fixed point $x^* \in A$.

[15] A particularly elegant proof can be found in *P. Lax,* The American Mathematical Monthly, vol. 106, No. 6.

Proof The idea is to approximate T by means of operators acting on finite dimensional spaces in order to apply Brouwer's Theorem.

Since A is compact, for every $\varepsilon > 0$, we can find a covering of A consisting of a finite number n_ε of open balls $B_1 = B_\varepsilon(x_1), \dots, B_{n_\varepsilon} = B_\varepsilon(x_{n_\varepsilon})$ of radius ε and centers $x_j \in A$, $j = 1, \dots, n_\varepsilon$. Let now

$$A_\varepsilon = \overline{\text{co}}\left\{x_1, x_2, \dots, x_{n_\varepsilon}\right\}.$$

Being A closed and convex, $A_\varepsilon \subseteq A$. Moreover, A_ε is homeomorphic to the closed unit ball in $\mathbb{R}^{M_\varepsilon}$, for a suitable M_ε, $M_\varepsilon \leq n_\varepsilon$. Define $P_\varepsilon : A \to A_\varepsilon$ by

$$P_\varepsilon(x) = \sum_{i=1}^{n_\varepsilon} \alpha_i x_i, \quad \alpha_i = \frac{\text{dist}(x, A - B_i)x_i}{\sum_{i=1}^{n_\varepsilon} \text{dist}(x, A - B_i)}, x \in A.$$

Note that $\text{dist}(x, A - B_i) \neq 0$ if $x \in B_i$, so that the denominator does not vanish, for every $x \in A$. Moreover, $0 \leq \alpha_i \leq 1$ and $\sum_{i=1}^{n_\varepsilon} \alpha_i = 1$, so that $P_\varepsilon(x)$ is given by a convex combination of the points x_i and $P_\varepsilon(x) \in A_\varepsilon$. P_ε is continuous, being a quotient of finite linear combination of distances and, for every $x \in A$, we have

$$\|P_\varepsilon(x) - x\| \leq \sum_{i=1}^{n_\varepsilon} \alpha_i \|x_i - x\| < \varepsilon \tag{6.125}$$

since $\text{dist}(x, A - B_i) = 0$, if $x \notin B_i$.

Now, the operator

$$P_\varepsilon \circ T : A_\varepsilon \to A_\varepsilon$$

is continuous and, by Brouwer's Theorem, there exists a fixed point x_ε:

$$(P_\varepsilon \circ T)(x_\varepsilon) = x_\varepsilon.$$

Using the compactness of A, there exists a subsequence $\left\{x_{\varepsilon_j}\right\}$ and a point $x^* \in A$ such that $x_{\varepsilon_j} \to x^*$ as $\varepsilon_j \to 0$. From (6.125) with $x = T(x_{\varepsilon_j})$, we get

$$\left\|x_{\varepsilon_j} - T(x_{\varepsilon_j})\right\| = \left\|(P_{\varepsilon_j} \circ T)(x_{\varepsilon_j}) - T(x_{\varepsilon_j})\right\| < \varepsilon_j.$$

Letting $\varepsilon_j \to 0$, by the continuity of the norm, we obtain $\|x^* - T(x^*)\| = 0$, i.e. $x^* = T(x^*)$. □

The application of Schauder's Theorem requires the compactness of A, which is a strong requirement in infinitely many dimensions. In the applications to boundary value problems, it is much more convenient to formulate some variants in which the compactness is required to the image $T(A)$ or to the operator T itself, rather than to A. A first variant is the following.

Theorem 6.87 *Let X be a Banach space. Assume that:*

a) $A \subseteq X$ is closed and convex.
b) $T : A \to A$ is continuous.
c) $\overline{T(A)}$ is compact in X.

Then T has a fixed point $x^* \in A$.

Proof Let $K = \overline{\text{co}}\{\overline{T(A)}\}$, i.e. the closed convex envelope of $\overline{T(A)}$. Since $\overline{T(A)}$ is compact, K is compact by Proposition 6.85 and $K \subseteq A$. Moreover $T(K) \subseteq T(A) \subseteq K$. Thus, the restriction $T : K \to K$ falls under the hypotheses of Theorem 6.86 and hence T has a fixed point $x^* \in K$. □

A second variant uses the *compactness* of T. We recall that T is compact if the image of a bounded set has compact closure. We emphasize that if T is linear and compact then is clearly continuous, but this is not the case if T is nonlinear[16].

Theorem 6.88 *Let X be a Banach space. Assume that:*

a) $A \subset X$ is closed, bounded and convex.
b) $T : A \to A$ is a continuous and compact operator.

Then T has a fixed point $x^* \in A$.

Proof Observe that $\overline{T(A)}$ is compact in A and use Theorem 6.87. □

6.10.3 The Leray-Schauder Theorem

A further variant of Schauder's Theorem requires the compactness of the operator T and the existence of a family of operators T_s, $0 \leq s \leq 1$, where $T_1 = T$ and T_0 is a compact operator which has a fixed point.

The simplest example is provided by the family $T_s = sT$, with $T_0 = 0$, which has the obvious fixed point $x = 0$. The main hypothesis in the following theorem asserts that the family of operators T_s, $0 \leq s \leq 1$, *has no fixed point on the boundary of a suitable bounded set*. In this way, any family of fixed points, moving continuously with s, can neither exit nor enter such bounded set. Under this assumption, T has a fixed point.

Theorem 6.89 (Leray-Schauder) *Let X be a Banach space and $T : X \to X$ such that:*

a) T is continuous and compact.
b) There exists M such that

$$\|x\| \neq M \tag{6.126}$$

for every solution (x, s) of the equation $x = sT(x)$ with $0 \leq s \leq 1$.

Then T has a fixed point.

Proof Let

$$B_M = \{x \in X : \|x\| \leq M\}$$

and define $P : B_M \to B_M$ by

$$P(x) = \begin{cases} T(x) & \text{if } \|T(x)\| \leq M \\ M\dfrac{T(x)}{\|T(x)\|} & \text{if } \|T(x)\| > M. \end{cases}$$

B_M is closed, convex and bounded and P is continuous. Moreover, since $T(B_M)$ is precompact, $P(B_M)$ also is precompact. By Theorem 6.87, there exists $x^* \in B_M$ such that $P(x^*) = x^*$.

We claim that x^* is a fixed point for T. If not, we should have $P(x^*) \neq T(x^*)$. Therefore, it must be $\|T(x^*)\| > M$ and

[16] Convince yourself using functions of real variables.

$$x^* = P(x^*) = \frac{M}{\|T(x^*)\|} T(x^*) \tag{6.127}$$

or

$$x^* = sT(x^*), \quad \text{with } s = \frac{M}{\|T(x^*)\|} < 1.$$

From (6.126) we infer $\|x^*\| \neq M$ while (6.127) implies $\|x^*\| = M$. Contradiction. □

Remark 6.90 An important particular way to check assumption b) above is to show that, if the family of operators $T_s = sT$, $0 \leq s \leq 1$, has fixed points, then these points have norm *less than* M. This is a so called *a priori estimate* for the fixed points of the family T_s; "*a priori*" because we do not know whether or not they exist.

Problems

6.1 (*Heisenberg Uncertainty Principle*) Let $\psi \in C^1(\mathbb{R})$ such that $x\,[\psi(x)]^2 \to 0$ as $|x| \to \infty$ and $\int_{\mathbb{R}} |\psi(x)|^2 dx = 1$. Show that

$$1 \leq 4 \int_{\mathbb{R}} x^2 \, |\psi(x)|^2 \, dx \int_{\mathbb{R}} |\psi'(x)|^2 \, dx.$$

(If ψ is a Schrödinger wave function, the first factor in the right hand side measures the spread of the position of a particle, while the second one measures the spread of its momentum.)

6.2 Let H be a Hilbert space and $a(u, v)$ be a symmetric and nonnegative bilinear form in H:

$$a(u, v) = a(v, u) \quad \text{and} \quad a(u, v) \geq 0, \quad \forall u, v \in H.$$

Show that

$$|a(u, v)| \leq \sqrt{a(u, u)}\sqrt{a(v, v)}.$$

[*Hint.* Mimic the proof of the Schwarz inequality, see Theorem 6.6, p. 351.]

6.3 Show the completeness of l^2 (see Example 6.51, p. 385).

[*Hint.* Take a Cauchy sequence $\{\mathbf{x}^k\}$ where $\mathbf{x}^k = \{x_m^k\}_{m \geq 1}$. In particular, $|x_m^k - x_m^h| \to 0$ as $h, k \to \infty$ and therefore $x_m^h \to x_m$ for every m. Define $\mathbf{x} = \{x_m\}_{m \geq 1}$ and show that $\mathbf{x}^k \to \mathbf{x}$ in l^2.]

6.4 a) Let H be a Hilbert space and V be a closed subspace of H. Show that $u = P_V x$ if and only if

$$\begin{cases} u \in V, \\ (x - u, v) = 0, \ \forall v \in V. \end{cases}$$

b) Using a), show that the projection operator P_V is linear.

6.5 Let $f \in L^2(-1, 1)$. Find the polynomial of degree not greater than n that gives the best approximation of f in the mean quadratic sense, that is, the polynomial p that minimizes

$$\int_{-1}^{1} (f - q)^2$$

among all polynomials q with degree not greater than n.

[*Answer.*

$$p(x) = a_0 L_0(x) + a_1 L_1(x) + \ldots + a_n L_n(x),$$

where L_n is the n-th Legendre polynomials and $a_j = (n + 1/2)(f, L_n)_{L^2(-1,1)}$.]

6.6 (*Hermite's equation and the quantum mechanics harmonic oscillator*) Consider the equation

$$w'' + (2\lambda + 1 - x^2)w = 0\,, \qquad x \in \mathbb{R} \tag{6.128}$$

with $w(x) \to 0$ as $x \to \pm\infty$.

a) Show that the change of variables $z = we^{x^2/2}$ transforms (6.128) into Hermite's equation for z :

$$z'' - 2xz' + 2\lambda z = 0$$

with $e^{-x^2/2} z(x) \to 0$ as $x \to \pm\infty$.

b) Consider the Schrödinger wave equation for the harmonic oscillator

$$\psi'' + \frac{8\pi^2 m}{h^2}(E - 2\pi^2 m\nu^2 x^2)\psi = 0 \qquad x \in \mathbb{R}$$

where m is the mass of the particle, E is the total energy, h is the Plank constant and ν is the vibrational frequency. The physically admissible solutions are those satisfying the following conditions:

$$\psi \to 0 \ \text{ as } x \to \pm\infty \quad \text{ and } \quad \|\psi\|_{L^2(\mathbb{R})} = 1.$$

Show that there is a solution if and only if

$$E = h\nu(n + \frac{1}{2}) \qquad n = 0, 1, 2 \ldots .$$

and, for each n, the corresponding solution is given by

$$\psi_n(x) = k_n H_n(2\pi\sqrt{\nu m/h}\,x) \exp(-\frac{2\pi^2 \nu m}{h} x^2)$$

where $k_n = (\frac{4\pi\nu m}{2^{2n}(n!)^2 h})^{1/2}$ and H_n is the n-th Hermite polynomial.

6.7 Using separation of variables, solve the following steady state diffusion problem in three dimensions (r, θ, φ spherical coordinates, $0 \le \theta \le 2\pi$, $0 \le \varphi \le \pi$):

$$\begin{cases} \Delta u = 0 & r < 1, 0 < \varphi < \pi \\ u(1, \varphi) = g(\varphi) & 0 \le \varphi \le \pi. \end{cases}$$

[*Answer.* $u(r, \varphi) = \sum_{n=0}^{\infty} a_n r^n L_n(\cos\varphi)$,

where L_n is the n-th Legendre polynomial and $a_n = \frac{2n+1}{2} \int_{-1}^{1} g(\cos^{-1} x) L_n(x) dx$. At a certain point, the change of variable $x = \cos\varphi$ is required.]

6.8 The vertical displacement u of a circular membrane of radius 1 satisfies the bidimensional wave equation $u_{tt} = \Delta u$, with boundary condition $u(1, \theta, t) = 0$. Supposing the membrane initially at rest, write a formal solution of the problem.

[*Answer.* $u(r, \theta, t) = \sum_{p,j=0}^{\infty} J_p(\alpha_{pj} r) \left\{A_{pj} \cos p\theta + B_{pj} \sin p\theta\right\} \cos(\sqrt{\alpha_{pj}}\, t)$,

where the coefficients A_{pj} and B_{pj} are determined by the expansion of the initial condition $u(r, \theta, 0) = g(r, \theta)$ and $\left\{\alpha_{pj}\right\}_{j \ge 1}$ is the sequence of positive zeros of the Bessel's functions of first kind and order J_p defined in (6.30), p. 364.]

6.9 In calculus, we say that $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0$ if there exists a linear mapping $L : \mathbb{R}^n \to \mathbb{R}$ such that $f(\mathbf{x}_0 + \mathbf{h}) - f(\mathbf{x}_0) = L\mathbf{h} + o(\|\mathbf{h}\|), \quad \text{as } \mathbf{h} \to \mathbf{0}$.
Determine the Riesz elements associated with L, with respect to the following inner products:

a) $(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{y} = \sum_{j=1}^n x_j y_j$,
b) $(\mathbf{x}, \mathbf{y})_{\mathbf{A}} = \mathbf{A}\mathbf{x} \cdot \mathbf{y} = \sum_{i,j=1}^n a_{ij} x_i y_j$, where $\mathbf{A} = (a_{ij})$ is a positive and symmetric matrix.

6.10 Prove a) and b) in Proposition 6.37, p. 372.

[*Hint.* First show that $\|L^*\|_{\mathcal{L}(H_2,H_1)} \le \|L\|_{\mathcal{L}(H_1,H_2)}$ and then that $(L^*)^* = L$. Reverse the role of L and L^* to show that $\|L^*\|_{\mathcal{L}(H_2,H_1)} \ge \|L\|_{\mathcal{L}(H_1,H_2)}$.]

6.11 Let $\mathbf{a} = \{a_n\}_{n\ge1}$ be a sequence of positive real numbers, such that $a_n \to +\infty$ as $n \to \infty$. Define

$$l^2_a = \left\{\mathbf{x} = \{x_n\}_{n\ge1} : \{a_n x_n\}_{n\ge1} \in l^2\right\}.$$

a) Show that l^2_a is a Hilbert space with the inner product

$$(\mathbf{x}, \mathbf{y})_{l^2_a} = \sum a_n^2 x_n y_n.$$

b) Show that the dual space of l^2_a can be identified with

$$(l^2_a)^* = \left\{\mathbf{z} = \{z_n\}_{n\ge1} : \left\{a_n^{-1} z_n\right\}_{n\ge1} \in l^2\right\},$$

with the duality given by $\langle \mathbf{z}, \mathbf{x}\rangle = (\mathbf{z}, \mathbf{x})_{l^2}$.

6.12 Show that the condition (6.71), p. 383, can be stated in other two equivalent ways:

$$\forall v \in W,\ v \ne 0 : \ \sup_{u\in V} a(u, v) > 0,$$

or

$$\forall v \ne 0,\ \exists u \in V : a(u, v) \ne 0.$$

6.13 Let $E \subset X$, where X is a Banach space. Prove the following facts:

a) If E is compact, then it is closed and bounded.
b) Let $E \subset F$ and F be compact; if E is closed then E is compact.

6.14 (*Projection on a closed convex set*) Let H be a Hilbert space and $E \subset H$, closed and convex.

a) Show that, for every $x \in H$, there is a unique element $P_E x \in E$ (called the *projection of x on E*) such that

$$\|P_E x - x\| = \inf_{v\in E} \|v - x\|. \tag{6.129}$$

b) Show that $x^* = P_E x$ if and only if $x^* \in E$ and the following **variational inequality** holds:

$$(x^* - x, v - x^*) \ge 0, \qquad \text{for every } v \in E. \tag{6.130}$$

c) Give a geometrical interpretation of (6.130).

[*Hint.* a) Follow the proof of the Projection Theorem 6.12, p. 356. b) Let $0 \le t \le 1$ and define

$$\varphi(t) = \|x^* + t(v - x^*) - x\|^2 \qquad v \in E.$$

Show that $x^* = P_E x$ if and only if $\varphi'(0) \ge 0$. Check that $\varphi'(0) \ge 0$ is equivalent to (6.130).]

6.15 Let $\Omega \subseteq \mathbb{R}^n$ be an open set, $a, b \in \mathbb{R}$ and

$$E = \{u \in L^2(\Omega) : a \le u(\mathbf{x}) \le b \quad \text{a.e. in } \Omega\}.$$

1. Show that E is a closed and convex subset of $L^2(\Omega)$.
2. Let $z \in L^2(\Omega)$ and $z^* = P_E z$ (see Problem 6.14). Show that the variational inequality (6.130) is equivalent, in this case, to the pointwise inequality

$$\forall v \in E : (z^*(\mathbf{x}) - z(\mathbf{x}))(v(\mathbf{x}) - z^*(\mathbf{x})) \geq 0, \quad \text{a.e. in } \Omega. \tag{6.131}$$

3. Derive the formulas

$$z^*(\mathbf{x}) = \max\{a, \min\{z(\mathbf{x}), b\}\} = \min\{b, \max\{a, z(\mathbf{x})\}\}. \tag{6.132}$$

[*Hint.* 1. Recall that if $u_k \to u$ in $L^2(\Omega)$, there exists a subsequence $\{u_{k_j}\}$, such that $u_{k_j} \to u$ a.e. in Ω.
2. The implication (6.131)$\Longrightarrow$(6.130) is obvious. To prove the opposite one, argue by contradiction, assuming that (6.130) is true and there exists $\bar{v} \in E$ such that

$$(z^*(\mathbf{x}) - z(\mathbf{x}))(\bar{v}(\mathbf{x}) - z^*(\mathbf{x})) < 0$$

in $\Omega' \subset \Omega$, with $|\Omega'| > 0$. Let $v = \bar{v}$ in Ω' and $v = z^*$ in $\Omega \setminus \Omega'$ and deduce a contradiction with (6.130).
3. Show that (6.131) implies, a.e. in Ω, $z^*(\mathbf{x}) = z(\mathbf{x})$ if $a \leq z(\mathbf{x}) \leq b$, $z^*(\mathbf{x}) = a$ if $z(\mathbf{x}) < a$ and $z^*(\mathbf{x}) = b$, if $z(\mathbf{x}) > b$.]

6.16 Let H be a Hilbert space and $E \subset H$, be closed and convex. Show that E is *weakly sequentially closed*.

[*Hint.* Let $\{x_k\} \subset E$ such that $x_k \rightharpoonup x$. Use (6.130) to show that $P_E x = x$, so that $x \in E$.]

6.17 Let H be a Hilbert space and $\{x_k\} \subset E$ such that $x_k \rightharpoonup x$. Show that if:

$$\text{i) } \|x_k\| \to \|x\| \text{ as } k \to \infty \qquad \text{or} \qquad \text{ii) } \|x_k\| \leq \|x\| \text{ for every } k \geq 1,$$

then $x_k \to x$.

6.18 Show that the embedding of $H^1_{\text{per}}(0, 2\pi)$ into $L^2(0, 2\pi)$ is compact.

[*Hint.* Let $\{u_k\} \subset H^1_{\text{per}}(0, 2\pi)$ with

$$\|u_k\|^2 = \sum_{m \in \mathbb{Z}} (1 + m^2)\, |\widehat{u_k}_m|^2 < M.$$

Show that, by a diagonal process, it is possible to select indexes k_j such that, for each m, $\widehat{u_{k_j}}_m$ converges to some number U_m. Let

$$u(x) = \sum_{m \in \mathbb{Z}} U_m e^{imx}$$

and show that $u_{k_j} \to u$ in $L^2(0, 2\pi)$.]

6.19 Let $\{e_k\}_{k \geq 1}$ be an orthonormal basis in an infinite dimensional Hilbert space H. Show that $e_k \rightharpoonup 0$ as $k \to \infty$.

[*Hint.* Every $x \in H$ has the expansion $x = \sum_{k=1}^{\infty} (x, e_k) e_k$ with $\|x\|^2 = \cdots$.]

6.20 Let V and W be two closed subspaces of a Hilbert space H, with inner product $(\cdot, \cdot)$. Let $x_0 \in H$ and define the following sequence of projections (see Fig. 6.3):

$$x_{2n+1} = P_V(x_{2n}), \quad x_{2n+2} = P_W(x_{2n+1}), \qquad n \geq 0.$$

Prove that:

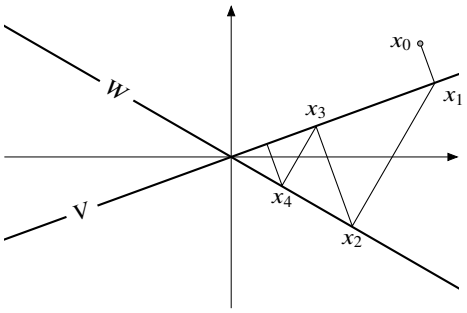

Fig. 6.3 The sequence of projections in Problem 6.20

a) If $V \cap W = \{0\}$ then $x_n \to 0$,
b) If $V \cap W \neq \{0\}$, then $x_n \to P_{V\cap W}(x_0)$,

by filling in the details in the following steps.

1. Check that $\|x_{n+1}\|^2 = (x_{n+1}, x_n)$. By computing $\|x_{n+1} - x_n\|^2$, show that $\|x_n\|$ is decreasing (hence $\|x_n\| \to l \geq 0$) and $\|x_{n+1} - x_n\| \to 0$.
2. If $V \cap W = \{0\}$, show that if a subsequence $x_{2n_k} \rightharpoonup x$, then $x_{2n_k+1} \rightharpoonup x$ as well. Deduce that $x = 0$ (so that the entire sequence converges weakly to 0).
3. Show that $\|x_n\|^2 = (x_{n+1}, x_{n-2}) = (x_{n+2}, x_{n-3}) = \cdots = (x_{2n-1}, x_0)$ and deduce that $x_n \to 0$.
4. If $V \cap W \neq \{0\}$, let $z_n = x_n - P_{V\cap W}(x_0)$ and reduce this case to the case a).

6.21 Let $L : L^2(\mathbb{R}) \to L^2(\mathbb{R})$ be defined by $Lv(x) = v(-x)$ a.e.. Show that $\sigma(L) = \sigma_P(L) = \{1, -1\}$. Find the corresponding eigenfunctions.

6.22 Let $S_{\lambda_0} : H \to H$ be the operator defined in Section 6.9.4. Show that $0 \in \sigma_C(S_{\lambda_0})$.

6.23 (*Minimax property of the eigenvalues*) Let V, H and a as in Theorem 6.78, p. 405. Denote by $R(v)$ the Rayleigh quotient (6.119) and by $\{\lambda_m\}_{m\geq 1}$ the nondecreasing sequence of eigenvalue of the bilinear form a.

1. Let $v_1, \ldots, v_{k-1}$ be $k-1$ linearly independent elements in V, $k \geq 2$. Set
$$\lambda_k^*(v_1, \ldots, v_{k-1}) = \inf\left\{R(v) : (v, v_j)_H = 0,\ j = 1, \ldots, k-1\right\}.$$
Show that $\lambda_k^* \leq \lambda_k$.
2. Show that
$$\max_{S_k} \lambda_k^*(v_1, \ldots, v_{k-1}) = \lambda_k,$$
where $S_k = \left\{(v_1, \ldots, v_{k-1}),\ v_j \in V,\ j = 1, \ldots, k-1, \text{linearly independent}\right\}$.

[*Hint.* 1. Show that it is possible to construct a linear combination of the first k eigenfunctions $\bar{v} = \sum_1^k c_j w_j$, which is orthogonal to each v_j, $j = 1, \ldots, k-1$. Prove that $R(\bar{v}) \leq \lambda_k$. 2. Observe that $\lambda_k^*(w_1, \ldots, w_{k-1}) = \lambda_k$.]

6.24 Let (M, d) be a complete metric space and $F(\cdot, \cdot) : M \times \mathbb{R} \to M$ with the following properties:

1. There exists ρ, $0 < \rho < 1$, such that $d(F(x, b), F(y, b)) \leq \rho d(x, y)$, $\forall x, y \in M$, $\forall b \in \mathbb{R}$.
2. There exists a number $L > 0$ such that $d(F(x, b_1), F(x, b_1)) \leq L\,|b_1 - b_2|$, $\forall x \in M$, $\forall b_1, b_2 \in \mathbb{R}$.

Show that the equation $F(x, b) = x$ has a unique fixed point $x^* = x^*(b)$ for every $b \in \mathbb{R}$ and

$$d(x^*(b_1), x^*(b_2)) \leq \frac{L}{1-\rho}\,|b_1 - b_2|, \quad \forall b_1, b_2 \in \mathbb{R}.$$

Chapter 7
Distributions and Sobolev Spaces

7.1 Preliminary Ideas

We have seen the concept of *Dirac measure* arising in connection with the fundamental solutions of the diffusion and the wave equations. Another interesting situation is the following, where the Dirac measure models a mechanical impulse.

Consider a mass m moving along the x-axis with constant speed $v\mathbf{i}$ (see Fig. 7.1). At time $t = t_0$ an *elastic* collision with a vertical wall occurs. After the collision, the mass moves with opposite speed $-v\mathbf{i}$. If v_2, v_1 denote the scalar speeds at times t_1, t_2, $t_1 < t_2$, by the laws of mechanics we should have

$$m(v_2 - v_1) = \int_{t_1}^{t_2} F(t)dt,$$

where F denotes the intensity of the force acting on m. When $t_1 < t_2 < t_0$ or $t_0 < t_1 < t_2$, then $v_2 = v_1 = v$ or $v_2 = v_1 = -v$, respectively, and therefore $F = 0$: no force is acting on m before and after the collision. However, if $t_1 < t_0 < t_2$, the left hand side is equal to $2mv \neq 0$. If we insist to model the intensity of the force by a function F, the integral in the right hand side is zero and we obtain a contradiction.

Indeed, in this case, F is a force concentrated at time t_0, of intensity $2mv$, that is

$$F(t) = 2mv\,\delta(t - t_0).$$

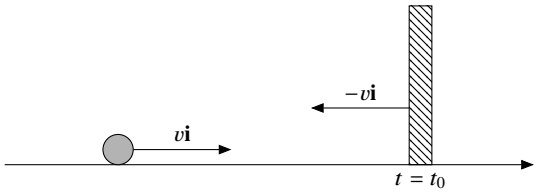

Fig. 7.1 Elastic collision at time $t = t_0$

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
https://doi.org/10.1007/978-3-031-21853-8_7

In this chapter we see how the Dirac delta is perfectly included in the *theory of distributions or Schwartz generalized functions*. We already mentioned in Sect. 2.3.3 that the key idea in this theory is to describe a mathematical object through its action on smooth test functions φ, with compact support. In the case of the Dirac δ, such action is expressed by the formula (see Definition 2.8, p. 48)

$$\int \delta(x)\varphi(x)dx = \varphi(0)$$

where, we recall, the integral symbol is purely formal. As we shall shortly see, the appropriate notation is $\langle \delta, \varphi \rangle = \varphi(0)$.

Of course, by a principle of coherence, among the generalized functions we should be able to recover the *usual* functions in Analysis. This fact implies that the choice of the test functions cannot be arbitrary. In fact, let $\Omega \subseteq \mathbb{R}^n$ be a domain and take for instance a function $u \in L^2(\Omega)$. A natural way to define the action of u on a test φ is

$$\langle u, \varphi \rangle = \int_\Omega u\varphi \, d\mathbf{x}.$$

If we let φ be varying over all $L^2(\Omega)$, we know from the last chapter, that $\langle u, \varphi \rangle$ identifies uniquely u. Indeed, if $v \in L^2(\Omega)$ is such that $\langle u, \varphi \rangle = \langle v, \varphi \rangle$ for every $\varphi \in L^2(\Omega)$, we have

$$0 = \langle u - v, \varphi \rangle = \int_\Omega (u - v)\varphi \, d\mathbf{x} \qquad \forall \varphi \in L^2(\Omega) \tag{7.1}$$

which forces (why?) $u = v$ a.e. in Ω.

On the other hand, we cannot use L^2-functions as test functions since, for instance, $\langle \delta, \varphi \rangle = \varphi(0)$ does not have any meaning.

We ask: is it possible to reconstruct u from the knowledge of $(u, \varphi)_{L^2(\Omega)}$, when φ varies on a set of *nice* functions?

Certainly this is impossible if we use only a restricted set of *test* functions. However, it is possible to recover u from the value of $(u, \varphi)_{L^2(\Omega)}$, when φ varies in a **dense** set in $L^2(\Omega)$. In fact, let $(u, \varphi)_{L^2(\Omega)} = (v, \varphi)_{L^2(\Omega)}$ for every test function. Given $\psi \in L^2(\Omega)$, there exists a sequence of test functions $\{\varphi_k\}$ such that $\|\varphi_k - \psi\|_{L^2(\Omega)} \to 0$. Then[1],

$$0 = \int_\Omega (u - v)\varphi_k \, d\mathbf{x} \to \int_\Omega (u - v)\psi \, d\mathbf{x}$$

so that (7.1) still holds *for every* $\psi \in L^2(\Omega)$ and $(u, \varphi)_{L^2(\Omega)}$ identifies a unique element in $L^2(\Omega)$.

Thus, the set of test functions must be *dense* in $L^2(\Omega)$ if we want L^2-functions to be seen as distributions. In the next section we construct an appropriate set of test functions.

[1] From $\left| \int_\Omega (u - v)(\varphi_k - \psi) d\mathbf{x} \right| \le \|u - v\|_{L^2(\Omega)} \, \|\varphi_k - \psi\|_{L^2(\Omega)}$.

However, the main purpose of introducing the Schwartz distributions is not restricted to a mere extension of the notion of function but it relies on the possibility of broadening the domain of *calculus* in a significant way, opening the door to an enormous amount of new applications. Here the key idea is to use integration by parts to carry the derivatives onto the test functions and therefore the test functions have to be smooth. Actually, this is not a new procedure. For instance, we have used it in Sect. 2.3.3, when we have interpreted the Dirac delta at $x = 0$ as the derivative of the Heaviside function $\mathcal{H}$, (see formula (2.62) and footnote 8, p. 48).

Also, the weakening of the notion of solution of conservation laws (Sect. 4.4.2) or of the wave equation (Sect. 5.4.2) follows more or less the same pattern.

In the first part of this chapter we give the basic concepts of the theory of Schwartz distributions, mainly finalized to the introduction of Sobolev spaces. The basic references are the books [40] by *L. Schwartz*, 1966, and [37] by *I. M. Gel'fand, G. E. Shilov*, 1964, to which we refer for the proofs we do not present here. With the exception of Sec. 7.6 where the Fourier transform is introduced, we can think to real-valued functions and distributions. In any case, since the domain is always a subset of $\mathbb{R}^n$, everything extends to complex valued functions and distributions, by considering their real and complex parts.

7.2 Test Functions and Mollifiers

Recall that, given a continuous function v, defined in a domain $\Omega \subseteq \mathbb{R}^n$, the support of v is given by the closure, in the *relative topology* of Ω (see Sec. 1.4), of the set of points where v is different from zero:

$$\text{supp}(v) = \Omega \cap \text{ closure of } \{\mathbf{x} \in \Omega : v(\mathbf{x}) \neq 0\}\,.$$

Actually, the support or, better, the essential support, is defined also for measurable functions, not necessarily continuous in Ω. Namely, let Z be the union of the open sets on which $v = 0$ a.e. Then, $\Omega \setminus Z$ is called the *essential support* of v and we use the same symbol $\text{supp}(v)$ to denote it.
We say that v is *compactly supported* in Ω, if $\text{supp}(v)$ is a *compact* subset of Ω.

Definition 7.1 Denote by $C_0^\infty(\Omega)$ the set of functions belonging to $C^\infty(\Omega)$, compactly supported in Ω. We call *test functions* the elements of $C_0^\infty(\Omega)$.

The reader can easily check that the function given by

$$\eta(\mathbf{x}) = \begin{cases} c \exp\left(\frac{1}{|\mathbf{x}|^2 - 1}\right) & 0 \leq |\mathbf{x}| < 1 \\ 0 & |\mathbf{x}| \geq 1 \end{cases} \qquad (c \in \mathbb{R}) \tag{7.2}$$

belongs to $C_0^\infty(\mathbb{R}^n)$.

The function (7.2) is a typical and important example of test function. Indeed, we will see below that many other test functions can be generated by convolution with (7.2).

Let us briefly recall the definition and the main properties of the convolution of two functions. Given two functions u and v defined in $\mathbb{R}^n$, the *convolution* $u * v$ of u and v is given by the formula:

$$(u * v)(\mathbf{x}) = \int_{\mathbb{R}^n} u(\mathbf{x}-\mathbf{y})v(\mathbf{y})d\mathbf{y} = \int_{\mathbb{R}^n} u(\mathbf{y})v(\mathbf{x}-\mathbf{y})d\mathbf{y}.$$

It can be proved that (*Young's Theorem*): *if* $u \in L^p(\mathbb{R}^n)$ *and* $v \in L^q(\mathbb{R}^n)$, $p, q \in [1, \infty]$, *then* $u * v \in L^r(\mathbb{R}^n)$ *where* $\frac{1}{r} = \frac{1}{p} + \frac{1}{q} - 1$ *and*

$$\|u * v\|_{L^r(\mathbb{R}^n)} \leq \|u\|_{L^p(\mathbb{R}^n)} \|v\|_{L^q(\mathbb{R}^n)}.$$

The convolution is a very useful *regularization* device. Indeed, consider the function η defined in (7.2). We have:

$$\eta \geq 0 \quad \text{and} \quad \text{supp}(\eta) = \overline{B}_1(\mathbf{0}),$$

where, we recall, $B_R(\mathbf{0}) = \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x}| < R\}$. Choose

$$c = \left(\int_{B_1(\mathbf{0})} \exp\left(\frac{1}{|\mathbf{x}|^2 - 1} \right) d\mathbf{x} \right)^{-1}$$

so that $\int_{\mathbb{R}^n} \eta = 1$. Set, for $\varepsilon > 0$,

$$\eta_\varepsilon(\mathbf{x}) = \frac{1}{\varepsilon^n} \eta\left(\frac{\mathbf{x}}{\varepsilon} \right). \tag{7.3}$$

This function belongs to $C_0^\infty(\mathbb{R}^n)$ (and therefore to all $L^p(\mathbb{R}^n)$), with support equal to $\overline{B}_\varepsilon(\mathbf{0})$, and still $\int_{\mathbb{R}^n} \eta_\varepsilon = 1$.

Let now $f \in L^p(\Omega)$, $1 \leq p \leq \infty$. If we set $f \equiv 0$ outside Ω, we obtain a function in $L^p(\mathbb{R}^n)$, denoted by $\tilde{f}$, for which the convolution $\eta_\varepsilon * \tilde{f}$ is well defined in all $\mathbb{R}^n$:

$$f_\varepsilon(\mathbf{x}) = (\eta_\varepsilon * \tilde{f})(\mathbf{x}) = \int_\Omega \eta_\varepsilon(\mathbf{x}-\mathbf{y}) f(\mathbf{y}) d\mathbf{y} = \int_{B_\varepsilon(\mathbf{0})} \eta_\varepsilon(\mathbf{z}) \tilde{f}(\mathbf{x}-\mathbf{z}) d\mathbf{z}.$$

Observe that, since $\int_{\mathbb{R}^n} \eta_\varepsilon = 1$, $\tilde{f} * \eta_\varepsilon$ may be considered as a *convex weighted average of* $\tilde{f}$ and, as such, we expect a smoothing effect on $\tilde{f}$. Indeed, even if f is very irregular, f_ε is a C^∞-function. For this reason η_ε is called a *mollifier*. Moreover, as $\varepsilon \to 0$, f_ε is an approximation of f in the sense explained in the following important lemma.

Lemma 7.2 *Let* $f \in L^p(\Omega)$, $1 \leq p \leq \infty$*; then* f_ε *has the following properties:*

a) The support of f_ε *is contained in a* ε*-neighborhood of the support of* f *(Fig. 7.2):*

$$\text{supp}(f_\epsilon) \subseteq \{\mathbf{x} \in \mathbb{R}^n : \text{dist}(\mathbf{x}, \text{supp}(f)) \leq \varepsilon\}.$$

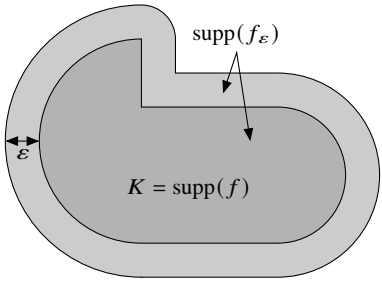

Fig. 7.2 Support of the convolution with a ε-mollifier

b) $f_\varepsilon \in C^\infty(\mathbb{R}^n)$ *and, if* $\mathrm{supp}(f)$ *is a compact set* $K \subset \Omega$, *then* $f_\varepsilon \in C_0^\infty(\Omega)$, *for* $\varepsilon < \mathrm{dist}(K, \partial\Omega)$.
c) If $f \in C(\Omega)$, $f_\varepsilon \to f$ *uniformly in every compact* $K \subset \Omega$, *as* $\varepsilon \to 0$.
d) If $1 \le p < \infty$, *then*

$$\|f_\varepsilon\|_{L^p(\Omega)} \le \|f\|_{L^p(\Omega)} \quad \textit{and} \quad \|f_\varepsilon - f\|_{L^p(\Omega)} \to 0 \ \textit{as}\ \varepsilon \to 0.$$

Proof a) Let $K = \mathrm{supp}(f)$. If $|\mathbf{z}| \le \varepsilon$ and $\mathrm{dist}(\mathbf{x}, K) > \varepsilon$, then $\tilde{f}(\mathbf{x} - \mathbf{z}) = 0$ so that $f_\varepsilon(\mathbf{x}) = 0$.

b) Since $\mathbf{x} \mapsto \eta_\varepsilon(\mathbf{x} - \mathbf{y}) \in C_0^\infty(\mathbb{R}^n)$ for every $\mathbf{y}$, f_ε is continuous and there is no problem in differentiating under the integral sign, obtaining all the time a continuous function. Thus $f_\varepsilon \in C^\infty(\mathbb{R}^n)$. From a), if K is compact, the support of f_ε is compact as well and contained in Ω, if $\varepsilon < \mathrm{dist}(K, \partial\Omega)$. Therefore $f_\varepsilon \in C_0^\infty(\Omega)$.

c) Since $\int_{\mathbb{R}^n} \eta_\varepsilon = 1$, We can write

$$f_\varepsilon(\mathbf{x}) - f(\mathbf{x}) = \int_{\{|\mathbf{z}| \le \varepsilon\}} \eta(\mathbf{z}) \left[\tilde{f}(\mathbf{x} - \mathbf{z}) - f(\mathbf{x})\right] d\mathbf{z}.$$

If $\mathbf{x} \in K \subset \Omega$, K compact, and $\varepsilon < \frac{1}{2}\mathrm{dist}(K, \partial\Omega)$, then $\tilde{f}(\mathbf{x} - \mathbf{z}) = f(\mathbf{x} - \mathbf{z})$ so that

$$|f_\varepsilon(\mathbf{x}) - f(\mathbf{x})| \le \sup_{|\mathbf{z}| \le \varepsilon} |f(\mathbf{x} - \mathbf{z}) - f(\mathbf{x})|\,.$$

Since f is uniformly continuous in every compact subset of Ω, we have that

$$\sup_{|\mathbf{z}| \le \varepsilon} |f(\mathbf{x} - \mathbf{z}) - f(\mathbf{x})| \to 0,$$

uniformly in $\mathbf{x}$, as $\varepsilon \to 0$. Thus $f_\varepsilon \to f$ uniformly in K.

d) From the Hölder inequality, we have, for $q = p/(p-1)$,

$$\begin{aligned} f_\varepsilon(\mathbf{x}) &= \int_\Omega \eta_\varepsilon(\mathbf{x} - \mathbf{y}) f(\mathbf{y}) d\mathbf{y} = \int_\Omega (\eta_\varepsilon(\mathbf{x} - \mathbf{y}))^{1/q} (\eta_\varepsilon(\mathbf{x} - \mathbf{y}))^{1/p} f(\mathbf{y}) d\mathbf{y} \\ &\le \left(\int_\Omega \eta_\varepsilon(\mathbf{x} - \mathbf{y}) \, |f(\mathbf{y})|^p \, d\mathbf{y} \right)^{1/p}. \end{aligned}$$

This inequality and the Fubini Theorem A.14, p. 662, yield

$$\|f_\varepsilon\|_{L^p(\Omega)} \le \|f\|_{L^p(\Omega)}\,. \tag{7.4}$$

In fact:

$$\|f_\varepsilon\|^p_{L^p(\Omega)} = \int_\Omega |f_\varepsilon(\mathbf{x})|^p \, d\mathbf{x} \le \int_\Omega \left(\int_\Omega \eta_\varepsilon(\mathbf{x}-\mathbf{y}) \, |f(\mathbf{y})|^p \, d\mathbf{y} \right) d\mathbf{x}$$
$$= \int_\Omega |f(\mathbf{y})|^p \left(\int_\Omega \eta_\varepsilon(\mathbf{x}-\mathbf{y}) d\mathbf{x} \right) d\mathbf{y} \le \int_\Omega |f(\mathbf{y})|^p \, d\mathbf{y} = \|f\|^p_{L^p(\Omega)} .$$

From Theorem A.11, p. 661, given any $\delta > 0$, there exists $g \in C_0(\Omega)$ such that $\|g - f\|_{L^p(\Omega)} < \delta$. Then, (7.4) implies

$$\|g_\varepsilon - f_\varepsilon\|_{L^p(\Omega)} \le \|g - f\|_{L^p(\Omega)} < \delta.$$

Moreover, since the support of g is compact, $g_\varepsilon \to g$ uniformly in Ω, by c), so that, we have, in particular, $\|g_\varepsilon - g\|_{L^p(\Omega)} < \delta$, for ε small. Thus,

$$\|f - f_\varepsilon\|_{L^p(\Omega)} \le \|f - g\|_{L^p(\Omega)} + \|g - g_\varepsilon\|_{L^p(\Omega)} + \|g_\varepsilon - f_\varepsilon\|_{L^p(\Omega)} \le 3\delta.$$

This shows that $\|f - f_\varepsilon\|_{L^p(\Omega)} \to 0$ as $\varepsilon \to 0$. □

Remark 7.3 Let $f \in L^1_{\text{loc}}(\Omega)$, i.e. $f \in L^1(\Omega')$ for every[2] $\Omega' \subset\subset \Omega$. The convolution $f_\varepsilon(\mathbf{x})$ is well defined if $\mathbf{x}$ stays ε-away from $\partial\Omega$, that is if $\mathbf{x}$ belongs to the set

$$\Omega_\varepsilon = \{\mathbf{x} \in \Omega : \text{dist}(\mathbf{x}, \partial\Omega) > \varepsilon\} .$$

Moreover, $f_\varepsilon \in C^\infty(\Omega_\varepsilon)$.

Remark 7.4 In general $\|f - f_\varepsilon\|_{L^\infty(\Omega)} \not\to 0$ as $\varepsilon \to 0$. However $\|f_\varepsilon\|_{L^\infty(\Omega)} \le \|f\|_{L^\infty(\Omega)}$ is clearly true.

Example 7.5 Let $\Omega' \subset\subset \Omega$ and $f = \chi_{\Omega'}$ be the characteristic function of Ω'. Then, $f_\varepsilon = \chi_{\Omega'} * \eta_\varepsilon \in C^\infty_0(\Omega)$ as long as $\varepsilon < \text{dist}(\Omega', \partial\Omega)$. Note that $0 \le f_\varepsilon \le 1$. In fact

$$f_\varepsilon(\mathbf{x}) = \int_\Omega \eta_\varepsilon(\mathbf{x}-\mathbf{y}) \chi_{\Omega'}(\mathbf{y}) d\mathbf{y} = \int_{\Omega' \cap B_\varepsilon(\mathbf{x})} \eta_\varepsilon(\mathbf{x}-\mathbf{y}) d\mathbf{y} \le \int_{B_\varepsilon(\mathbf{0})} \eta_\varepsilon(\mathbf{y}) d\mathbf{y} = 1.$$

Moreover, $f \equiv 1$ in Ω'_ε . In fact, if $\mathbf{x} \in \Omega'_\varepsilon$, the ball $B_\varepsilon(\mathbf{x})$ is contained in Ω' and therefore

$$\int_{\Omega' \cap B_\varepsilon(\mathbf{x})} \eta_\varepsilon(\mathbf{x}-\mathbf{y}) d\mathbf{y} = \int_{B_\varepsilon(\mathbf{0})} \eta_\varepsilon(\mathbf{y}) d\mathbf{y} = 1.$$

A consequence of Lemma 7.2 is the following approximation theorem.

Theorem 7.6 *$C^\infty_0(\Omega)$ is dense in $L^p(\Omega)$ for every $1 \le p < \infty$.*

Proof Denote by $L^p_c(\Omega)$ the space of functions in $L^p(\Omega)$, with (essential) support compactly contained in Ω. Let $f \in L^p_c(\Omega)$ and $K = \text{supp}(f)$. From Lemma 7.2.a, we know that $\text{supp}(f_\varepsilon)$ is contained in an ε-neighborhood of K, which is still a compact subset of Ω, for ε small.

Since by Lemma 7.2.d, $f_\varepsilon \to f$ in $L^p(\Omega)$, we deduce that $C^\infty_0(\Omega)$ is dense in $L^p_c(\Omega)$, if $1 \le p < \infty$. On the other hand, $L^p_c(\Omega)$ is dense in $L^p(\Omega)$; in fact, let $\{K_m\}$ be a sequence of compact subsets of Ω such that

$$K_m \subset K_{m+1} \qquad \text{and} \qquad \cup K_m = \Omega.$$

Denote by χ_{K_m} the characteristic function of K_m. Then, we have

[2] $\Omega' \subset\subset \Omega$ means that the closure of Ω' is a compact subset of Ω.

$$\{\chi_{K_m} f\} \subset L_c^p(\Omega) \quad \text{and} \quad \|\chi_{K_m} f - f\|_{L^p} \to 0 \quad \text{as } m \to +\infty$$

by the Dominated Convergence Theorem A.7, p. 660, since $|\chi_{K_m} f| \le |f|$. □

7.3 Distributions

We now endow $C_0^\infty(\Omega)$ with a suitable notion of convergence. Recall that, for every multi-index α, the symbol

$$D^\alpha = \frac{\partial^{\alpha_1}}{\partial x_1^{\alpha_1}} \cdots \frac{\partial^{\alpha_n}}{\partial x_n^{\alpha_n}}, \qquad \alpha = (\alpha_1, \ldots, \alpha_n),$$

denotes a derivative of order $|\alpha| = \alpha_1 + \ldots + \alpha_n$. If $|\alpha| = 0$ we set $D^\alpha \varphi = \varphi$.

Definition 7.7 Let $\{\varphi_k\} \subset C_0^\infty(\Omega)$ and $\varphi \in C_0^\infty(\Omega)$. We say that

$$\varphi_k \to \varphi \qquad \text{in } C_0^\infty(\Omega), \quad \text{as } k \to +\infty$$

if the following two properties hold:

1. There exists a compact set $K \subset \Omega$ containing the support of every φ_k.
2. $\forall \alpha = (\alpha_1, \ldots, \alpha_n)$, $|\alpha| \ge 0$, $D^\alpha \varphi_k \to D^\alpha \varphi$ uniformly in Ω.

It is not difficult to show that the limit so defined is *unique*. When endowed with the above notion of convergence the space $C_0^\infty(\Omega)$ becomes a *topological vector space,* denoted by $\mathcal{D}(\Omega)$.

Following the discussion in the first section, we focus on the linear functionals in $\mathcal{D}(\Omega)$. If L is one of those, we shall use the *bracket* (or *pairing*) $\langle L, \varphi \rangle$ or $\langle L, \varphi \rangle_{\mathcal{D}'(\Omega)}$, if there is any risk of confusion, to denote the action of L on a test function φ. We say that a linear functional

$$L : \mathcal{D}(\Omega) \to \mathbb{R}$$

is *continuous* in $\mathcal{D}(\Omega)$ if

$$\langle L, \varphi_k \rangle \to \langle L, \varphi \rangle, \qquad \text{whenever } \varphi_k \to \varphi \text{ in } \mathcal{D}(\Omega). \tag{7.5}$$

Note that, given the linearity of L, it is enough to check (7.5) in the case $\varphi = 0$.

Definition 7.8 A *distribution* in Ω is a linear continuous functional in $\mathcal{D}(\Omega)$. The set of distributions is denoted by $\mathcal{D}'(\Omega)$.

Two distributions L_1 and L_2 coincide when their action on every test function is the same, i.e. if

$$\langle L_1, \varphi \rangle = \langle L_2, \varphi \rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

To every real-valued function $u \in L^2(\Omega)$ corresponds the functional I_u whose action on φ is

$$\langle I_u, \varphi \rangle = \int_\Omega u\varphi \, d\mathbf{x},$$

which is certainly continuous in $\mathcal{D}(\Omega)$. Therefore I_u is a distribution in $\mathcal{D}'(\Omega)$ and we have seen at the end of Sect. 7.1 that I_u may be identified with u.

Thus, the notion of distribution generalizes the notion of function (in $L^2(\Omega)$) and, for real functions, the pairing $\langle \cdot, \cdot \rangle$ between $\mathcal{D}(\Omega)$ and $\mathcal{D}'(\Omega)$ generalizes the inner product in $L^2(\Omega)$.

The functional I_u is actually well defined for any function $u \in L^1_{\text{loc}}(\Omega)$, that, therefore, can be identified with it. On the other hand, if u is a function and $u \notin L^1_{\text{loc}}$, u **cannot** represent a distribution. A typical example is $u(x) = 1/x$ which does not belong to $L^1_{\text{loc}}(\mathbb{R})$. However, there is a distribution closely related to $1/x$ as we show in Example 7.12, p. 427.

• *Measures*. Special classes of distributions are obtained by relaxing the condition 2 in Definition 7.7, requiring that it holds only for $|\alpha| = 0$. Precisely, let $L \in \mathcal{D}'(\Omega)$. If $\langle L, \varphi_k \rangle \to 0$ whenever the supports of the φ_k are contained in the same compact K and (only) $\varphi_k \to 0$ uniformly, L is called a *measure*. Equivalently, L is a measure if and only if, for every φ there exists a constant C, *depending on L and the support of φ*, such that

$$|\langle L, \varphi \rangle| \leq C \max |\varphi| . \tag{7.6}$$

Every function $u \in L^1_{\text{loc}}(\Omega)$ is a measure since

$$\left| \int_\Omega u\varphi \right| \leq \|u\|_{L^1(\text{supp}(\varphi))} \max |\varphi| .$$

Another important measure is the Dirac delta, defined below.

Definition 7.9 The n-dimensional *Dirac delta* at the origin, denoted by δ_n, is the distribution in $\mathcal{D}'(\mathbb{R}^n)$, whose action is

$$\langle \delta_n, \varphi \rangle = \varphi(\mathbf{0}). \tag{7.7}$$

If $n = 1$ we simply write δ instead of δ_n.

Quite often, especially in the applied sciences, the Dirac delta at $\mathbf{0}$ is denoted by $\delta_n(\mathbf{x})$ and the formula (7.7) is written in the symbolic form

$$\int \delta_n(\mathbf{x})\varphi(\mathbf{x}) = \varphi(\mathbf{0}).$$

Similarly, the Dirac delta at $\mathbf{y}$ is denoted by $\delta_n(\mathbf{x} - \mathbf{y})$ and defined by the relation

$$\int \delta_n(\mathbf{x} - \mathbf{y})\varphi(\mathbf{x}) = \varphi(\mathbf{y}).$$

These are the (abuse of) notations that we already used in the previous chapters.

$\mathcal{D}'(\Omega)$ is a *linear space*. Indeed if α, β are real scalars, $\varphi \in \mathcal{D}(\Omega)$ and $L_1, L_2 \in \mathcal{D}'(\Omega)$, we define $\alpha L_1 + \beta L_2 \in \mathcal{D}'(\Omega)$ by means of the formula

$$\langle \alpha L_1 + \beta L_2, \varphi \rangle = \alpha \langle L_1, \varphi \rangle + \beta \langle L_2, \varphi \rangle.$$

In $\mathcal{D}'(\Omega)$ we may introduce a notion of (weak) convergence: $\{L_k\}$ *converges to* L *in* $\mathcal{D}'(\Omega)$ *if*

$$\langle L_k, \varphi \rangle \to \langle L, \varphi \rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

If $1 \le p \le \infty$, we have the **continuous embeddings**:

$$L^p(\Omega) \hookrightarrow \mathcal{D}'(\Omega).$$

This means that, if $u_k \to u$ in $L^p(\Omega)$, then[3] $u_k \to u$ in $\mathcal{D}'(\Omega)$ as well.

With respect to this type of convergence, $\mathcal{D}'(\Omega)$ possesses a *completeness property* that may be used to construct a distribution or to recognize that some linear functional in $\mathcal{D}(\Omega)$ is a distribution. Precisely, one can prove the following result.

Proposition 7.10 *Let* $\{F_k\} \subset \mathcal{D}'(\Omega)$ *such that* $\lim_{k\to\infty}\langle F_k, \varphi\rangle$ *exists and is finite for all* $\varphi \in \mathcal{D}(\Omega)$. *Call* $F(\varphi)$ *this limit. Then,* $F \in \mathcal{D}'(\Omega)$ *and* $F_k \to F$ *in* $\mathcal{D}'(\Omega)$.

In particular, if the numerical series

$$\sum_{k=1}^{\infty} \langle F_k, \varphi \rangle$$

converges for all $\varphi \in \mathcal{D}(\Omega)$, then $\sum_{k=1}^{\infty} F_k = F \in \mathcal{D}'(\Omega)$.

Example 7.11 For every $\varphi \in \mathcal{D}(\mathbb{R})$, the numerical series

$$\sum_{k=-\infty}^{\infty} \langle \delta(x-k), \varphi \rangle = \sum_{k=-\infty}^{\infty} \varphi(k)$$

is convergent, since only a finite number of terms is different from zero (those with $k \in \mathbb{Z}$ belonging to the support of φ). From Proposition 7.10, we get that the series

$$\operatorname{comb}(x) = \sum_{k=-\infty}^{\infty} \delta(x-k) \tag{7.8}$$

is convergent in $\mathcal{D}'(\mathbb{R})$ and its sum is a distribution called the **Dirac comb**. This name is due to the fact it models a train of impulses concentrated at the integers (see Fig. 7.3, using some ... fantasy).

Example 7.12 Let $h_r(x) = 1 - \chi_{[-r,r]}(x)$ be the characteristic function of the set $\mathbb{R} \setminus [-r, r]$. Define

[3] For instance, let $\varphi \in \mathcal{D}(\Omega)$. We have, by Hölder's inequality:

$$\left| \int_\Omega (u_k - u)\varphi d\mathbf{x} \right| \le \|u_k - u\|_{L^p(\Omega)} \, \|\varphi\|_{L^q(\Omega)}$$

where $q = p/(p-1)$. Then, if $\|u_k - u\|_{L^p(\Omega)} \to 0$, also $\int_\Omega (u_k - u)\varphi d\mathbf{x} \to 0$, showing the convergence of $\{u_k\}$ in $\mathcal{D}'(\Omega)$.

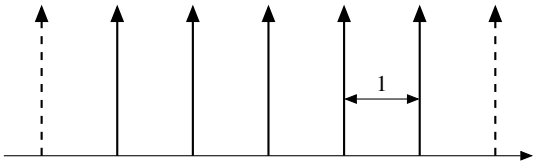

Fig. 7.3 A train of impulses

$$\text{p. v.}\,\frac{1}{x} = \lim_{r\to 0}\frac{1}{x}h_r(x).$$

We want to show that p. v. $\frac{1}{x}$ defines a distribution in $\mathcal{D}'(\mathbb{R})$, called *principal value of* $\frac{1}{x}$. By Proposition 7.10 it is enough to check that, for all $\varphi \in \mathcal{D}(\mathbb{R})$, the limit

$$\lim_{r\to 0}\int_{\mathbb{R}}\frac{1}{x}h_r(x)\varphi(x)dx$$

is finite. Indeed, assume that $\text{supp}(\varphi) \subset [-a, a]$. Then, for $0 < r < a$,

$$\int_{\mathbb{R}}\frac{1}{x}h_r(x)\varphi(x)dx = \int_{\{r<|x|<a\}}\frac{\varphi(x)}{x}dx = \int_{\{r<|x|<a\}}\frac{\varphi(x)-\varphi(0)}{x}dx$$

since

$$\int_{\{r<|x|<a\}}\frac{\varphi(0)}{x}dx = 0,$$

due to the odd symmetry of $1/x$. Now, we have

$$\varphi(x) - \varphi(0) = \varphi'(0)x + o(x), \qquad \text{as } x \to 0,$$

so that

$$\frac{\varphi(x)-\varphi(0)}{x} = \varphi'(0) + o(1), \qquad \text{as } x \to 0.$$

This implies that $[\varphi(x) - \varphi(0)]/x$ is summable in $[-a, a]$ and therefore

$$\lim_{r\to 0}\int_{\{r<|x|<a\}}\frac{\varphi(x)-\varphi(0)}{x}dx = \int_{\{|x|<a\}}\frac{\varphi(x)-\varphi(0)}{x}dx$$

is a finite number. Thus, p. v. $\frac{1}{x} \in \mathcal{D}'(\mathbb{R})$ and the above computations yield

$$\langle \text{p. v.}\,\frac{1}{x}, \varphi\rangle = \lim_{r\to 0}\int_{\{r<|x|\}}\frac{\varphi(x)}{x}dx \equiv \text{p. v.}\int_{\mathbb{R}}\frac{\varphi(x)}{x}dx$$

where p. v. stays for *principal value*.

• *Support of a distribution*. The Dirac δ_n *is concentrated at the origin*. More precisely, we say that its support coincides with the origin. The support of a general distribution F may be defined in the following way. We want to characterize the smallest closed set outside of which F vanishes. However, we cannot proceed as in

the case of a function, since a distribution is defined on the elements of $\mathcal{D}(\Omega)$, not on subsets of $\mathbb{R}^n$.

Thus, let us start saying that $F \in \mathcal{D}'(\Omega)$ *vanishes in an open set* $A \subset \Omega$ if

$$\langle F, \varphi \rangle = 0$$

for every $\varphi \in \mathcal{D}(\Omega)$ whose support is contained in A. Let $\mathcal{A}$ be *the union of all open sets where* F *vanishes*. A is open. Then, we define:

$$\operatorname{supp}(F) = \Omega \setminus \mathcal{A}.$$

For example, $\operatorname{supp}(\text{comb}) = \mathbb{Z}$.

Remark 7.13 Let $F \in \mathcal{D}'(\Omega)$ with compact support K. Then the bracket $\langle F, v \rangle$ is well defined for all $v \in C^\infty(\Omega)$, **not necessarily with compact support**. In fact, let $\varphi \in \mathcal{D}(\Omega)$, $0 \le \varphi \le 1$, such that $\varphi \equiv 1$ in an open neighborhood of K (see Example 7.5, p. 424). Then $v\varphi \in \mathcal{D}(\Omega)$ and we can define

$$\langle F, v \rangle = \langle F, v\varphi \rangle .$$

Note that $\langle F, v\varphi \rangle$ is independent of the choice of φ. Indeed if ψ has the same property of φ, then

$$\langle F, v\varphi \rangle - \langle F, v\psi \rangle = \langle F, v(\varphi - \psi) \rangle = 0$$

since $\varphi - \psi = 0$ in an open neighborhood of K.

7.4 Calculus

7.4.1 The derivative in the sense of distributions

A central concept in the theory of the Schwartz distributions is the notion of *weak* or *distributional derivative*. Clearly we have to abandon the classical definition, since, for instance, we are going to define the derivative for a function $u \in L^1_{\text{loc}}$, which may be quite irregular.

The idea is to carry the derivative onto the test functions, as if we were using the integration by parts formula.

Let us start from a function $u \in C^1(\Omega)$. If $\varphi \in \mathcal{D}(\Omega)$, denoting by $\boldsymbol{\nu} = (\nu_1, \dots, \nu_n)$ the outward normal unit vector to $\partial\Omega$, we have

$$\int_\Omega \varphi \partial_{x_i} u \, d\mathbf{x} = \int_{\partial\Omega} \varphi u \, \nu_i \, d\mathbf{x} - \int_\Omega u \partial_{x_i} \varphi \, d\mathbf{x} = - \int_\Omega u \partial_{x_i} \varphi \, d\mathbf{x}$$

since $\varphi = 0$ in a neighborhood[4] of $\partial\Omega$. The equation

[4] This allows the integration by parts without regularity assumptions on $\partial\Omega$, since the integral can be restricted to a smooth domain contained in Ω and containing $\operatorname{supp}(\varphi)$.

$$\int_\Omega \varphi\, \partial_{x_i} u\, d\mathbf{x} = -\int_\Omega u\, \partial_{x_i}\varphi\, d\mathbf{x},$$

interpreted in $\mathcal{D}'(\Omega)$, becomes

$$\langle \partial_{x_i} u, \varphi\rangle = -\langle u, \partial_{x_i}\varphi\rangle. \tag{7.9}$$

Formula (7.9) shows that the action of $\partial_{x_i} u$ on the test function φ equals the action of u on the test function $-\partial_{x_i}\varphi$. On the other hand, formula (7.9) makes perfect sense if we replace u by any $F \in \mathcal{D}'(\Omega)$ and it is not difficult to check that it defines a continuous linear functional in $\mathcal{D}(\Omega)$. This leads to the following fundamental notion:

Definition 7.14 Let $F \in \mathcal{D}'(\Omega)$. The *derivative* $\partial_{x_i} F$ is the distribution defined by the formula

$$\langle \partial_{x_i} F, \varphi\rangle = -\langle F, \partial_{x_i}\varphi\rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

From (7.9), if $u \in C^1(\Omega)$, its derivatives in the sense of distributions coincide with the classical ones. This is the reason we keep the same notations in the two cases.

Note that the derivative of a distribution is **always defined**! Moreover, since any derivative of a distribution is a distribution, we deduce that **every distribution possesses derivatives of any order** (in $\mathcal{D}'(\Omega)$):

$$\langle D^\alpha F, \varphi\rangle = (-1)^{|\alpha|}\langle F, D^\alpha \varphi\rangle.$$

For example, the second order derivative

$$\partial_{x_i x_k} F = \partial_{x_i}(\partial_{x_k} F)$$

is defined by

$$\langle \partial_{x_i x_k} F, \varphi\rangle = \langle F, \partial_{x_i x_k}\varphi\rangle. \tag{7.10}$$

Not only. Since φ is smooth, then $\partial_{x_i x_k}\varphi = \partial_{x_k x_i}\varphi$ so that (7.10) yields

$$\partial_{x_i x_k} F = \partial_{x_k x_i} F.$$

Thus, for **all** $F \in \mathcal{D}'(\Omega)$ we may always reverse the order of differentiation *without any restriction*.

Example 7.15 Let $u(x) = \mathcal{H}(x)$, the Heaviside function. In $\mathcal{D}'(\mathbb{R})$ we have $\mathcal{H}' = \delta$. In fact, let $\varphi \in \mathcal{D}(\mathbb{R})$. By definition,

$$\langle \mathcal{H}', \varphi\rangle = -\langle \mathcal{H}, \varphi'\rangle.$$

On the other hand, $\mathcal{H} \in L^1_{\text{loc}}(\mathbb{R})$, hence

$$\langle \mathcal{H}, \varphi'\rangle = \int_{\mathbb{R}} \mathcal{H}(x)\varphi'(x)dx = \int_0^\infty \varphi'(x)dx = -\varphi(0)$$

whence

$$\langle \mathcal{H}', \varphi \rangle = \varphi(0) = \langle \delta, \varphi \rangle$$

or $\mathcal{H}' = \delta$. Note that $\mathcal{H}' = 0$ a.e. in the pointwise sense. Thus, the derivative in the a.e. sense *does not* coincide with the derivative in the distributional sense.

Example 7.16 Let $F = \delta_n$. Then $\partial_{x_j} \delta_n$ is defined in $\mathcal{D}'(\mathbb{R}^n)$ by

$$\langle \partial_{x_j} \delta_n, \varphi \rangle = -\langle \delta_n, \partial_{x_j} \varphi \rangle = -\partial_{x_j} \varphi(\mathbf{0}), \quad \forall \varphi \in \mathcal{D}'(\mathbb{R}^n).$$

Note that $\partial_{x_j} \delta_n$ is not a measure and has support at the origin.

Another aspect of the idyllic relationship between calculus and distributions is given by the following theorem, which expresses the continuity in $\mathcal{D}'(\Omega)$ of every derivative D^α.

Proposition 7.17 *If $F_k \to F$ in $\mathcal{D}'(\Omega)$ then $D^\alpha F_k \to D^\alpha F$ in $\mathcal{D}'(\Omega)$ for any multi-index α.*

Proof $F_k \to F$ in $\mathcal{D}'(\Omega)$ means that $\langle F_k, \varphi \rangle \to \langle F, \varphi \rangle$, $\forall \varphi \in \mathcal{D}(\Omega)$. In particular, since $D^\alpha \varphi \in \mathcal{D}(\Omega)$,

$$\langle D^\alpha F_k, \varphi \rangle = (-1)^{|\alpha|} \langle F_k, D^\alpha \varphi \rangle \to (-1)^{|\alpha|} \langle F, D^\alpha \varphi \rangle = \langle D^\alpha F, \varphi \rangle. \qquad \square$$

As a consequence, if $\sum_{k=1}^{\infty} F_k = F$ in $\mathcal{D}'(\Omega)$, then

$$\sum_{k=1}^{\infty} D^\alpha F_k = D^\alpha F \qquad \text{in } \mathcal{D}'(\Omega).$$

Thus, *term by term differentiation is always permitted* in $\mathcal{D}'(\Omega)$.

More difficult is the proof of the following theorem, which expresses a well known fact for functions.

Proposition 7.18 *Let Ω be a domain in $\mathbb{R}^n$. If $F \in \mathcal{D}'(\Omega)$ and $\partial_{x_j} F = 0$ for every $j = 1, \ldots, n$, then F is a constant function.*

Proof We do it in dimension $n = 1$, with $\Omega = (a, b)$. We have $\langle F', \eta \rangle = -\langle F, \eta' \rangle = 0$ for all $\eta \in \mathcal{D}(a, b)$. Let $\varphi, \varphi_0 \in \mathcal{D}(a, b)$ with $\int_a^b \varphi_0 = 1$, and define

$$\eta(x) = \int_a^x \left(\varphi - \varphi_0 \int_a^b \varphi \right).$$

Then $\eta \in \mathcal{D}(a, b)$ (check it!) and

$$0 = \langle F, \eta' \rangle = \left\langle F, \varphi - \varphi_0 \int_a^b \varphi \right\rangle$$

or

$$\langle F, \varphi \rangle = \langle c, \varphi \rangle$$

where $c = \langle F, \varphi_0 \rangle$. Since φ is arbitrary, it follows that $F = c$ in $\mathcal{D}'(a, b)$. $\square$

7.4.2 Gradient, divergence, Laplacian

There is no problem to define *vector valued distributions*. The space of test functions is $\mathcal{D}(\Omega;\mathbb{R}^n)$, i.e. the set of vectors $\boldsymbol{\varphi} = (\varphi_1, \dots, \varphi_n)$ whose components belong to $\mathcal{D}(\Omega)$.

A distribution $\mathbf{F} \in \mathcal{D}'(\Omega;\mathbb{R}^n)$ is given by $\mathbf{F} = (F_1, \dots, F_n)$ with $F_j \in \mathcal{D}'(\Omega)$, $j = 1, \dots, n$. The pairing between $\mathcal{D}(\Omega;\mathbb{R}^n)$ and $\mathcal{D}'(\Omega;\mathbb{R}^n)$ is defined by

$$\langle \mathbf{F}, \boldsymbol{\varphi} \rangle = \sum_{i=1}^{n} \langle F_i, \varphi_i \rangle. \tag{7.11}$$

- The *gradient* of the scalar distribution $F \in \mathcal{D}'(\Omega)$, $\Omega \subset \mathbb{R}^n$, is simply

$$\nabla F = (\partial_{x_1} F, \partial_{x_2} F, \dots, \partial_{x_n} F).$$

Clearly $\nabla F \in \mathcal{D}'(\Omega;\mathbb{R}^n)$. If $\boldsymbol{\varphi} \in \mathcal{D}(\Omega;\mathbb{R}^n)$, we have

$$\langle \nabla F, \boldsymbol{\varphi} \rangle = \sum_{i=1}^{n} \langle \partial_{x_i} F, \varphi_i \rangle = -\sum_{i=1}^{n} \langle F, \partial_{x_i} \varphi_i \rangle = -\langle F, \operatorname{div} \boldsymbol{\varphi} \rangle$$

whence

$$\langle \nabla F, \boldsymbol{\varphi} \rangle = -\langle F, \operatorname{div} \boldsymbol{\varphi} \rangle \tag{7.12}$$

which shows the action of ∇F on $\boldsymbol{\varphi}$.

- For $\mathbf{F} \in \mathcal{D}'(\Omega;\mathbb{R}^n)$, we set

$$\operatorname{div} \mathbf{F} = \sum_{i=1}^{n} \partial_{x_i} F_i.$$

Clearly $\operatorname{div} \mathbf{F} \in \mathcal{D}'(\Omega)$. If $\varphi \in \mathcal{D}(\Omega)$, then

$$\langle \operatorname{div} \mathbf{F}, \varphi \rangle = \left\langle \sum_{i=1}^{n} \partial_{x_i} F_i, \varphi \right\rangle = -\sum_{i=1}^{n} \langle F_i, \partial_{x_i} \varphi \rangle = -\langle \mathbf{F}, \nabla \varphi \rangle$$

whence

$$\langle \operatorname{div} \mathbf{F}, \varphi \rangle = -\langle \mathbf{F}, \nabla \varphi \rangle. \tag{7.13}$$

- The Laplace operator is defined in $\mathcal{D}'(\Omega)$ by

$$\Delta F = \sum_{i=1}^{n} \partial_{x_i x_i} F.$$

If $\varphi \in \mathcal{D}(\Omega)$, then

$$\langle \Delta F, \varphi \rangle = \langle F, \Delta \varphi \rangle .$$

Using (7.12), (7.13) we get

$$\langle \Delta F, \varphi \rangle = \langle F, \operatorname{div} \nabla \varphi \rangle = - \langle \nabla F, \nabla \varphi \rangle = \langle \operatorname{div} \nabla F, \varphi \rangle$$

whence $\Delta = \operatorname{div} \nabla$ also in $\mathcal{D}'(\Omega)$.

Example 7.19 Consider the **fundamental solution** for the Laplace operator in $\mathbb{R}^3$ (Sect. 3.6.1, p. 146)

$$u(\mathbf{x}) = \frac{1}{4\pi} \frac{1}{|\mathbf{x}|}.$$

Observe that $u \in L^1_{\text{loc}}(\mathbb{R}^3)$ so that $u \in \mathcal{D}'(\Omega)$. We want to show that, in $\mathcal{D}'(\mathbb{R}^3)$,

$$-\Delta u = \delta_3. \tag{7.14}$$

First of all, if $\Omega \subset \mathbb{R}^3$ and $\mathbf{0} \notin \Omega$, we know that u is *harmonic in* Ω, that is

$$\Delta u = 0 \qquad \text{in } \Omega,$$

in the classical sense and therefore also in $\mathcal{D}'(\mathbb{R}^3)$. Thus, let $\varphi \in \mathcal{D}(\mathbb{R}^3)$ with $\mathbf{0} \in \operatorname{supp}(\varphi)$. We have, since $u \in L^1_{\text{loc}}(\mathbb{R}^3)$:

$$\langle \Delta u, \varphi \rangle = \langle u, \Delta \varphi \rangle = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{1}{|\mathbf{x}|} \Delta \varphi(\mathbf{x}) d\mathbf{x}. \tag{7.15}$$

We would like to carry the Laplacian onto $1/|\mathbf{x}|$. However, this cannot be done directly, since the integrand is not continuous at $\mathbf{0}$. Therefore we exclude a small sphere $B_r = B_r(\mathbf{0})$ from our integration region and write

$$\int_{\mathbb{R}^3} \frac{1}{|\mathbf{x}|} \Delta \varphi(\mathbf{x}) d\mathbf{x} = \lim_{r \to 0} \int_{B_R \setminus B_r} \frac{1}{|\mathbf{x}|} \Delta \varphi(\mathbf{x}) d\mathbf{x} \tag{7.16}$$

where $B_R = B_R(\mathbf{0})$ is a sphere containing the support of φ. Recalling that $\varphi = 0$ and $\nabla \varphi = \mathbf{0}$ on ∂B_R, an integration by parts in the spherical shell $C_{R,r} = B_R \setminus B_r$ yields

$$\int_{C_{R,r}} \frac{1}{|\mathbf{x}|} \Delta \varphi(\mathbf{x}) d\mathbf{x} = \int_{\partial B_r} \frac{1}{r} \partial_\nu \varphi(\mathbf{x}) d\sigma - \int_{C_{R,r}} \nabla \left(\frac{1}{|\mathbf{x}|} \right) \cdot \nabla \varphi(\mathbf{x}) d\mathbf{x}$$

where $\boldsymbol{\nu} = -\frac{\mathbf{x}}{|\mathbf{x}|}$ is the *outward* normal unit vector on ∂B_r. Integrating once more by parts the last integral, we obtain:

$$\begin{aligned} \int_{C_{R,r}} \nabla \left(\frac{1}{|\mathbf{x}|} \right) \cdot \nabla \varphi(\mathbf{x}) d\mathbf{x} &= \int_{\partial B_r} \partial_\nu \left(\frac{1}{|\mathbf{x}|} \right) \varphi(\mathbf{x}) d\sigma - \int_{C_{R,r}} \Delta \left(\frac{1}{|\mathbf{x}|} \right) \varphi(\mathbf{x}) d\mathbf{x} \\ &= \int_{\partial B_r} \partial_\nu \left(\frac{1}{|\mathbf{x}|} \right) \varphi(\mathbf{x}) d\sigma, \end{aligned}$$

since $\Delta(\frac{1}{|\mathbf{x}|}) = 0$ inside $C_{R,r}$. From the above computations we infer

$$\int_{C_{R,r}} \frac{1}{|\mathbf{x}|} \Delta \varphi(\mathbf{x}) d\mathbf{x} = \int_{\partial B_r} \frac{1}{r} \partial_\nu \varphi(\mathbf{x}) d\sigma - \int_{\partial B_r} \partial_\nu \left(\frac{1}{|\mathbf{x}|} \right) \varphi(\mathbf{x}) d\sigma. \tag{7.17}$$

We have:

$$\frac{1}{r}\left|\int_{\partial B_r}\partial_\nu\varphi(\mathbf{x})d\sigma\right| \le \frac{1}{r}\int_{\partial B_r}|\partial_\nu\varphi(\mathbf{x})|\,d\sigma \le 4\pi r\max_{\mathbb{R}^3}|\nabla\varphi|$$

and therefore

$$\lim_{r\to 0}\int_{\partial B_r}\frac{1}{r}\partial_\nu\varphi(\mathbf{x})d\sigma = 0.$$

Moreover, since

$$\partial_\nu\left(\frac{1}{|\mathbf{x}|}\right) = \nabla\left(\frac{1}{|\mathbf{x}|}\right)\cdot\left(-\frac{\mathbf{x}}{|\mathbf{x}|}\right) = \left(-\frac{\mathbf{x}}{|\mathbf{x}|^3}\right)\cdot\left(-\frac{\mathbf{x}}{|\mathbf{x}|}\right) = \frac{1}{|\mathbf{x}|^2},$$

we may write

$$\int_{\partial B_r}\partial_\nu\left(\frac{1}{|\mathbf{x}|}\right)\varphi(\mathbf{x})d\sigma = 4\pi\frac{1}{4\pi r^2}\int_{\partial B_r}\varphi(\mathbf{x})d\sigma \to 4\pi\varphi(\mathbf{0})$$

as $r\to 0$. Thus, from (7.17) we get

$$\lim_{r\to 0}\int_{B_R\setminus B_r}\frac{1}{|\mathbf{x}|}\Delta\varphi(\mathbf{x})d\mathbf{x} = -4\pi\varphi(\mathbf{0})$$

and finally (7.15) yields

$$\langle\Delta u,\varphi\rangle = -\varphi(\mathbf{0}) = -\langle\delta_3,\varphi\rangle$$

whence $-\Delta u = \delta_3$ in $\mathcal{D}'(\mathbb{R}^3)$.

7.5 Operations with Distributions

7.5.1 Multiplication. Leibniz rule

Let us analyze the multiplication between two distributions. Does it make any sense to define, for instance, the product $\delta\cdot\delta = \delta^2$ as a distribution in $\mathcal{D}'(\mathbb{R})$?

Things are not so smooth. An idea for defining δ^2 may be the following: take a sequence $\{u_k\}$ of functions in $L^1_{\text{loc}}(\mathbb{R})$ such that $u_k\to\delta$ in $\mathcal{D}'(\mathbb{R})$, compute u_k^2 and set

$$\delta^2 = \lim_{k\to\infty}u_k^2 \qquad \text{in } \mathcal{D}'(\mathbb{R}).$$

Since we may approximate δ in $\mathcal{D}'(R)$ in many ways (see Problem 7.2), it is necessary that the definition does not depend on the approximating sequence. In other words, to compute δ^2 we must be free to choose any approximating sequence of δ. However, this is illusory. Indeed choose

$$u_k = k\chi_{[0,1/k]}.$$

We have $u_k \to \delta$ in $\mathcal{D}'(\mathbb{R})$ but, if $\varphi \in \mathcal{D}(\mathbb{R})$, by the Mean Value Theorem we have

$$\int_{\mathbb{R}} u_k^2 \varphi = k^2 \int_0^{1/k} \varphi = k\varphi(x_k)$$

for some $x_k \in [0, 1/k]$. Now, if $\varphi(0) > 0$, say, we deduce that

$$\int_{\mathbb{R}} u_k^2 \varphi \to +\infty, \qquad k \to +\infty$$

so that $\left\{u_k^2\right\}$ does not converge in $\mathcal{D}'(\mathbb{R})$.

The method does not work and it seems that there is no other reasonable way to define δ^2 in $\mathcal{D}'(\Omega)$. Thus, we simply give up defining δ^2 or, in general, the product of two distributions, as a distribution. However, if $F \in \mathcal{D}'(\Omega)$ and $u \in C^\infty(\Omega)$, we may define the product uF by the formula

$$\langle uF, \varphi \rangle = \langle F, u\varphi \rangle, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

First of all, this makes sense since $u\varphi \in \mathcal{D}(\Omega)$. Also, if $\varphi_k \to \varphi$ in $\mathcal{D}(\Omega)$, then $u\varphi_k \to u\varphi$ in $\mathcal{D}(\Omega)$ and

$$\langle uF, \varphi_k \rangle = \langle F, u\varphi_k \rangle \to \langle F, u\varphi \rangle = \langle uF, \varphi \rangle,$$

so that uF is a well defined element of $\mathcal{D}'(\Omega)$.

Example 7.20 Let $u \in C^\infty(\mathbb{R})$. We have

$$u\delta = u(0)\delta.$$

Indeed, if $\varphi \in \mathcal{D}(\mathbb{R})$,

$$\langle u\delta, \varphi \rangle = \langle \delta, u\varphi \rangle = u(0)\varphi(0) = \langle u(0)\delta, \varphi \rangle.$$

In particular

$$x\delta = 0.$$

Note that the product $u\delta$ makes sense even if u is *only continuous*.

The *Leibniz rule* holds: let $F \in \mathcal{D}'(\Omega)$ and $u \in C^\infty(\Omega)$; then

$$\partial_{x_i}(uF) = u\, \partial_{x_i} F + \partial_{x_i} u\, F\,. \tag{7.18}$$

In fact, let $\varphi \in \mathcal{D}(\Omega)$; we have:

$$\left\langle \partial_{x_i}(uF), \varphi \right\rangle = -\left\langle uF, \partial_{x_i}\varphi \right\rangle = -\left\langle F, u\partial_{x_i}\varphi \right\rangle$$

while

$$\begin{aligned}\left\langle u\,\partial_{x_i}F+\partial_{x_i}u\,F,\varphi\right\rangle &= \left\langle \partial_{x_i}F,u\varphi\right\rangle+\left\langle F,\varphi\partial_{x_i}u\right\rangle\\ &= -\left\langle F,\partial_{x_i}(u\varphi)\right\rangle+\left\langle F,\varphi\partial_{x_i}u\right\rangle=-\left\langle F,u\partial_{x_i}\varphi\right\rangle\end{aligned}$$

and (7.18) follows.

Example 7.21 From $x\delta=0$ and Leibniz rule we obtain

$$\delta+x\delta'=0.$$

More generally,

$$x^m\delta^{(k)}=0\quad\text{ in }\mathcal{D}'(\mathbb{R}),\quad\text{ if } 0\le k<m.$$

7.5.2 Composition

Composition in $\mathcal{D}'(\mathbb{R})$ requires caution as well. For instance, if $F=\delta$ and $v(x)=x^3$, is there a natural way to define $F\circ v$ as a distribution in $\mathcal{D}'(\mathbb{R})$?

As above, consider the sequence $u_k=k\chi_{[0,1/k]}$ and compute $w_k=u_k\circ v$. If $\varphi\in\mathcal{D}(\mathbb{R})$, we have

$$\int_{\mathbb{R}}w_k\varphi=k\int_{\mathbb{R}}\chi_{[0,1/k]}(x^3)\varphi(x)dx=k\int_0^{k^{-1/3}}\varphi(x)dx=k^{2/3}\varphi(x_k)$$

for some $x_k\in[0,1/k]$. Then, if $\varphi(0)>0$, $\int_{\mathbb{R}}w_k\varphi\to+\infty$ and $F\circ v$ does not make any sense. Thus, it seems hard to define the composition between two general distributions. To see what can be done, let us start analyzing the case of two functions.

Let $\Omega',\Omega\subseteq\mathbb{R}^n$ and $\psi:\Omega'\to\Omega$ be **one to one**, with ψ and ψ^{-1} **of class** C^∞. If $F:\Omega\to\mathbb{R}$ is a continuous function we may consider the composition

$$w(\mathbf{x})=(F\circ\psi)(\mathbf{x})=F(\psi(\mathbf{x})).$$

For $\varphi\in\mathcal{D}(\Omega')$, we have, using the change of variables $\mathbf{y}=\psi(\mathbf{x})$:

$$\int_{\Omega'}F(\psi(\mathbf{x}))\varphi(\mathbf{x})d\mathbf{x}=\int_{\Omega}F(\mathbf{y})\varphi(\psi^{-1}(\mathbf{y}))\left|\det J_{\psi^{-1}}(\mathbf{y})\right|d\mathbf{y}\tag{7.19}$$

where $J_{\psi^{-1}}$ denotes the Jacobian of the transformation ψ^{-1}. Observe that

$$\varphi\circ\psi^{-1}\cdot\left|\det J_{\psi^{-1}}\right|\in\mathcal{D}(\Omega'),$$

with support given by $K=\psi(K')$, where K' is the support of φ. Thus, in terms of distributions, (7.19) writes

$$\langle F\circ\psi,\varphi\rangle_{\mathcal{D}'(\Omega')}=\langle F,\varphi\circ\psi^{-1}\cdot\left|\det J_{\psi^{-1}}\right|\rangle_{\mathcal{D}'(\Omega)}.\tag{7.20}$$

This formula makes sense also if $F \in \mathcal{D}'(\Omega)$ and it is not difficult to check that if $\varphi_k \to 0$ in $\mathcal{D}(\Omega')$ then $\varphi_k \circ \psi^{-1} \cdot \left|\det J_{\psi^{-1}}\right| \to 0$ in $\mathcal{D}(\Omega)$. Thus, (7.20) defines a distribution in $\mathcal{D}'(\Omega')$. Precisely:

Definition 7.22 If $F \in \mathcal{D}'(\Omega)$ and $\psi : \Omega' \to \Omega$ is one to one, with ψ and ψ^{-1} of class C^∞, then formula (7.20) defines the *composition* $F \circ \psi$ as an element of $\mathcal{D}'(\Omega')$.

Let us check that (7.20) behaves well with respect to convergence in $\mathcal{D}'(\Omega')$.

Proposition 7.23 *Let ψ be as in Definition 7.22. If $F_k \to F$ in $\mathcal{D}'(\Omega)$, then $F_k \circ \psi \to F \circ \psi$ in $\mathcal{D}'(\Omega')$.*

Proof Let $\varphi \in \mathcal{D}(\Omega')$. Then

$$\begin{aligned}\lim_{k\to\infty} \langle F_k \circ \psi, \varphi\rangle_{\mathcal{D}'(\Omega')} &= \lim_{k\to\infty} \langle F_k, \varphi \circ \psi^{-1} \cdot \left|\det J_{\psi^{-1}}\right|\rangle_{\mathcal{D}'(\Omega)} \\ &= \langle F, \varphi \circ \psi^{-1} \cdot \left|\det J_{\psi^{-1}}\right|\rangle_{\mathcal{D}'(\Omega)} = \langle F \circ \psi, \varphi\rangle_{\mathcal{D}'(\Omega')} . \qquad \square\end{aligned}$$

Abuses of notation are quite common, like $F(\psi(\mathbf{x}))$ to denote $F \circ \psi$. For instance, we have repeatedly used the (comfortable and incorrect) notation $\delta(\mathbf{x} - \mathbf{a})$ instead of the (uncomfortable and correct) notation $\delta \circ \tau_{\mathbf{a}}$, with $\tau_{\mathbf{a}}(\mathbf{x}) = \mathbf{x} - \mathbf{a}$.

Example 7.24 In $\mathcal{D}'(\mathbb{R}^n)$, we have

$$\delta_n(a\mathbf{x}) = \frac{1}{|a|^n}\delta_n(\mathbf{x}) \qquad (a \in \mathbb{R}, a \neq 0).$$

Using formula (7.20) we may extend to distributions some properties, typical of functions. We list some of them.

We say that $F \in \mathcal{D}'(\mathbb{R}^n)$ is:

- *Radial*, if

$$F(\mathbf{Ax}) = F(\mathbf{x}), \qquad \textit{for every orthogonal matrix } \mathbf{A}.$$

- *Homogeneous of degree λ*, if

$$F(t\mathbf{x}) = t^\lambda F(\mathbf{x}), \qquad \forall t > 0.$$

- *Even, odd* if, respectively,

$$F(-\mathbf{x}) = F(\mathbf{x}),\ F(-\mathbf{x}) = -F(\mathbf{x}).$$

- *Periodic with period* $\mathbf{P}$, if

$$F(\mathbf{x} + \mathbf{P}) = F(\mathbf{x}).$$

Example 7.25 a) $\delta_n \in \mathcal{D}'(\mathbb{R}^n)$ is radial, even and homogeneous of degree $\lambda = -n$.
b) p. v. $\frac{1}{x} \in \mathcal{D}'(\mathbb{R})$ is odd and homogeneous of degree $\lambda = -1$.
c) In $\mathcal{D}'(\mathbb{R})$, comb is periodic with period 1.

Example 7.26 Let $f : \mathbb{R} \to \mathbb{R}$ be one-to-one, with f and f^{-1} of class C^∞ and $f(x_0) = 0$. Then necessarily $f'(x_0) \neq 0$ and the following formula holds in $\mathcal{D}'(\mathbb{R})$:

$$\delta(f(x)) = \frac{\delta(x - x_0)}{|f'(x_0)|}. \tag{7.21}$$

In fact, let $\varphi \in \mathcal{D}(\mathbb{R})$. Setting $g = f^{-1}$, we have, since $\det J_g(y) = g'(y) = [f'(g(y))]^{-1}$:

$$\langle \delta \circ f, \varphi \rangle = \langle \delta, \varphi \circ g \cdot \left| \det J_g \right| \rangle = \left\langle \delta, \frac{\varphi \circ g}{|f' \circ g|} \right\rangle = \frac{\varphi(g(0))}{|f'(g(0))|} = \frac{\varphi(x_0)}{|f'(x_0)|}$$

and

$$\left\langle \frac{\delta(x - x_0)}{|f'(x_0)|}, \varphi \right\rangle = \frac{1}{|f'(x_0)|} \langle \delta, \varphi(x + x_0) \rangle = \frac{\varphi(x_0)}{|f'(x_0)|}.$$

Comparing the two formulas we get (7.21).

Example 7.27 Let $r > 0$. In $\mathcal{D}'(\mathbb{R}^3)$, we have:

$$\langle \delta(|\mathbf{x}| - r), \varphi(\mathbf{x}) \rangle = \int_{\partial B_r} \varphi(\boldsymbol{\sigma}) d\sigma = r^2 \int_{\partial B_1} \varphi(r\boldsymbol{\omega}) d\omega,$$

for every $\varphi \in \mathcal{D}'(\mathbb{R}^3)$. Indeed, the formula corresponds to the following formal change to spherical coordinates (recall formula (A.1), p. 662):

$$\int_{\mathbb{R}^3} \delta(|\mathbf{x}| - r) \varphi(\mathbf{x}) d\mathbf{x} = \int_0^{+\infty} \delta(\rho - r) \int_{\partial B_\rho} \varphi(\boldsymbol{\sigma}) d\sigma d\rho = \int_{\partial B_r} \varphi(\boldsymbol{\sigma}) d\sigma.$$

• *Restriction of a distribution*. It is possible to define the restriction of a distribution $F \in \mathcal{D}'(\Omega)$ to an open set $\Omega' \subset \Omega$ by simply defining

$$\langle F_{|\Omega'}, \varphi \rangle_{\mathcal{D}'(\Omega')} = \langle F, \tilde{\varphi} \rangle_{\mathcal{D}'(\Omega)} \tag{7.22}$$

where $\varphi \in \mathcal{D}(\Omega')$ and $\tilde{\varphi} \in \mathcal{D}(\Omega)$ is the extension to zero of φ outside Ω'. Clearly, (7.22) defines $F_{|\Omega'}$ as a distribution in $\mathcal{D}'(\Omega')$, since if $\varphi_k \to 0$ in $\mathcal{D}(\Omega')$ then $\tilde{\varphi}_k \to 0$ in $\mathcal{D}(\Omega)$. Also:

$$F_k \to F \text{ in } \mathcal{D}'(\Omega) \text{ implies } F_{k\,|\Omega'} \to F_{|\Omega'} \text{ in } \mathcal{D}'(\Omega')$$

as it is easy to check.

Example 7.28 Let $\Omega \subset \mathbb{R}^n$ be an open set. If $\mathbf{0} \in \Omega$ then $\langle \delta_{|\Omega}, \varphi \rangle = \varphi(\mathbf{0})$ for all $\varphi \in \mathcal{D}(\Omega)$. If $\mathbf{0} \notin \Omega$ then $\langle \delta_{|\Omega}, \varphi \rangle = 0$ for all $\varphi \in \mathcal{D}'(\Omega)$ and therefore $\delta_{|\Omega}$ is the zero function.

We can generalize the formula (7.21) to smooth functions f having (finite or infinite) simple zeros without cluster points. First, let $f \in C^\infty(\mathbb{R})$ and define $\delta \circ f \in \mathcal{D}'(\mathbb{R})$ by the formula

$$\langle \delta \circ f, \varphi \rangle = \lim_{\varepsilon \to 0} \langle I_\varepsilon(f), \varphi \rangle \tag{7.23}$$

where $I_\varepsilon(t) = 1/\varepsilon$ for $|t| < \varepsilon/2$ and $I_\varepsilon(t) = 0$ otherwise, provided the limit exists for every test function $\varphi \in \mathcal{D}(\mathbb{R})$.

Proposition 7.29 *Assume that f vanishes only at the points x_j, with $f'(x_j) \neq 0$, $j = 1, 2, \ldots$. Then the limit in (7.23) exists and*

$$\delta(f(x)) = \sum_{j \geq 1} \frac{\delta(x - x_j)}{|f'(x_j)|} \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{7.24}$$

Proof Let $\varphi \in \mathcal{D}(\mathbb{R})$ with $\text{supp}(\varphi) = [a, b]$. Only a finite number of zeros of f belong to $[a, b]$, say $x_1, \ldots, x_N$. Let ε so small that $|f(x)| < \varepsilon/2$ and $f'(x) \neq 0$ on the union of N disjoint intervals $I_j = (x_j - \eta, x_j + \eta)$, $j = 1, \ldots, N$. Denote by f_j the restriction of f to I_j and let $g_j = f_j^{-1}$. We have :

$$\langle I_\varepsilon(f(x)), \varphi \rangle = \sum_{j=1}^{N} \int_{I_j} I_\varepsilon(f_j(x)) \varphi(x) dx \underset{y = f_j(x)}{=} \sum_{j=1}^{N} \frac{1}{\varepsilon} \int_{-\varepsilon/2}^{\varepsilon/2} \varphi(g_j(y)) \left| f_j'(g_j(y)) \right|^{-1} dy.$$

Letting $\varepsilon \to 0$, we find, since $g_j(0) = x_j$:

$$\lim_{\varepsilon \to 0} \langle I_\varepsilon(f(x)), \varphi \rangle = \sum_{j=1}^{N} \varphi(g_j(0)) \left| f_j'(g_j(0)) \right|^{-1} = \sum_{j=1}^{N} \frac{\varphi(x_j)}{\left| f_j'(x_j) \right|}$$

which means (7.24) in terms of distributions. □

Example 7.30 Let $f(x) = x^2 - a^2$ $(a > 0)$. Then

$$\delta(f(x)) = \frac{\delta(x + a) + \delta(x - a)}{2a}.$$

7.5.3 Division

The division in $\mathcal{D}'(\Omega)$ is rather delicate, even restricting to $F \in \mathcal{D}'(\Omega)$ and $u \in C^\infty(\Omega)$. To divide F by u means to find $G \in \mathcal{D}'(\Omega)$ such that $uG = F$. If u never vanishes there is no problem, since in this case $1/u \in C^\infty(\Omega)$ and the answer is simply

$$G = \frac{1}{u} F.$$

If u vanishes somewhere, things get complicated. We only consider a particular case in one dimension.

Let $I \subseteq \mathbb{R}$ be an open interval and $u \in C^\infty(I)$. If u vanishes at z, we say that z is *a zero of order $m(z)$* if the derivatives of u up to order $m(z) - 1$, included, vanish at z, while the derivative of order $m(z)$ does not vanish at z. For instance, $z = 0$ is a zero of order 3 for $u(x) = \sin x - x$.

One can prove the following proposition.

Proposition 7.31 *Assume that u vanishes at isolated points $z_1, z_2, \ldots$ with order $m(z_1), m(z_2), \ldots$. Then, the equation*

$$uG = 0$$

has infinitely many solutions in $\mathcal{D}'(I)$, given by the following formula:

$$G = \sum_j \sum_{k=0}^{m(z_j)-1} c_{jk}\delta^{(k)}(x - z_j) \tag{7.25}$$

where c_{jk} are arbitrary constants and $\delta^{(k)}$ is the derivative of δ of order k.

Example 7.32 The solutions in $\mathcal{D}'(\mathbb{R})$ of the equation

$$xG = 0$$

are the distributions of the form $G = c\delta$, with $c \in \mathbb{R}$. To solve the equation

$$xG = 1, \tag{7.26}$$

we add to the solutions of the homogeneous equation $xG = 0$ a particular solution of (7.26). It turns out that one of these is

$$G_1 = \text{p. v.}\,\frac{1}{x}.$$

In fact, if $\varphi \in \mathcal{D}(\mathbb{R})$, from Example 7.12, p. 427, we get

$$\left\langle x \cdot \text{p. v.}\,\frac{1}{x}, \varphi \right\rangle = \left\langle \text{p. v.}\,\frac{1}{x}, x\varphi \right\rangle =$$
$$= \text{p. v.} \int_{\mathbb{R}} \frac{x\varphi(x)}{x} dx = \int_{\mathbb{R}} \varphi(x)dx = \langle 1, \varphi \rangle$$

whence

$$x \cdot \text{p. v.}\,\frac{1}{x} = 1.$$

Therefore, the general solution of (7.26) is

$$G = \text{p. v.}\,\frac{1}{x} + c\delta, \qquad c \in \mathbb{R}. \tag{7.27}$$

7.5.4 Tensor or direct product

The *tensor* or *direct product* of two distributions F, G, denoted by $F \otimes G$, is an (important) operation that enables us to construct a distribution in $\mathcal{D}'(\mathbb{R}^m \times \mathbb{R}^n)$ starting from two distributions in $\mathcal{D}'(\mathbb{R}^m)$ and $\mathcal{D}'(\mathbb{R}^n)$, respectively. To avoid confusions,

we call $\mathbf{x}$ the independent variable in $\mathbb{R}^m$ and $\mathbf{y}$ the one in $\mathbb{R}^n$. If we were dealing with two functions $u \in L^1_{\text{loc}}(\mathbb{R}^m)$ and $v \in L^1_{\text{loc}}(\mathbb{R}^n)$, the tensor product is defined simply by the formula

$$w(\mathbf{x}, \mathbf{y}) = (u \otimes v)(\mathbf{x}, \mathbf{y}) = u(\mathbf{x})v(\mathbf{y}).$$

As a distribution, the action of w on a test function $\varphi \in \mathcal{D}(\mathbb{R}^m \times \mathbb{R}^n)$ is given by (using Fubini's Theorem)

$$\int_{\mathbb{R}^m \times \mathbb{R}^n} w(\mathbf{x}, \mathbf{y})\varphi(\mathbf{x}, \mathbf{y}) d\mathbf{x} d\mathbf{y} =$$
$$\int_{\mathbb{R}^m} u(\mathbf{x}) \left(\int_{\mathbb{R}^n} v(\mathbf{y})\varphi(\mathbf{x}, \mathbf{y}) d\mathbf{y} \right) d\mathbf{x} = \int_{\mathbb{R}^n} v(\mathbf{y}) \left(\int_{\mathbb{R}^m} u(\mathbf{x})\varphi(\mathbf{x}, \mathbf{y}) d\mathbf{x} \right) d\mathbf{y}.$$

This formula can be extended to distributions. Indeed, let $F \in \mathcal{D}'(\mathbb{R}^m)$, $G \in \mathcal{D}'(\mathbb{R}^n)$ and $\varphi \in \mathcal{D}(\mathbb{R}^m \times \mathbb{R}^n)$. Then the two functions

$$\mathbf{x} \longmapsto \langle G(\mathbf{y}), \varphi(\mathbf{x}, \mathbf{y}) \rangle \quad \text{and} \quad \mathbf{y} \longmapsto \langle F(\mathbf{x}), \varphi(\mathbf{x}, \mathbf{y}) \rangle$$

are test functions in $\mathcal{D}(\mathbb{R}^m)$ and $\mathcal{D}(\mathbb{R}^n)$, respectively. The following Theorem holds (see *K. Yosida*, [43] for a proof).

Theorem 7.33 *Given $F \in \mathcal{D}'(\mathbb{R}^m)$ and $G \in \mathcal{D}'(\mathbb{R}^n)$, there exists a unique distribution $W \in \mathcal{D}'(\mathbb{R}^m \times \mathbb{R}^n)$ such that, $\forall \varphi \in \mathcal{D}(\mathbb{R}^m \times \mathbb{R}^n)$:*

$$\langle W, \varphi \rangle = \langle F(\mathbf{x}), \langle G(\mathbf{y}), \varphi(\mathbf{x}, \mathbf{y}) \rangle \rangle = \langle G(\mathbf{y}), \langle F(\mathbf{x}), \varphi(\mathbf{x}, \mathbf{y}) \rangle \rangle . \tag{7.28}$$

In particular, $\forall \varphi_1 \in \mathcal{D}(\mathbb{R}^m)$, $\forall \varphi_2 \in \mathcal{D}(\mathbb{R}^n)$,

$$\langle W, \varphi_1 \varphi_2 \rangle = \langle F, \varphi_1 \rangle \langle G, \varphi_2 \rangle .$$

The distribution W defined in Theorem 7.33 is called the *tensor or direct product of the distributions F and G*, denoted by $F \otimes G$, as we have already mentioned. It turns out that the tensor product is *associative* and there is no difficulty in defining the tensor product of any number k of distributions, so that we can write without ambiguity

$$F_1 \otimes F_2 \otimes \cdots \otimes F_k .$$

Example 7.34 Let us check that $\delta_3(\mathbf{x}) = \delta(x_1) \otimes \delta(x_2) \otimes \delta(x_3)$. In fact, let $\varphi \in \mathcal{D}(\mathbb{R}^3)$. We have, on one side, $\langle \delta_3, \varphi \rangle = \varphi(\mathbf{0})$, on the other side

$$\begin{aligned} \langle \delta(x_1) \otimes \delta(x_2) \otimes \delta(x_3), \varphi(x_1, x_2, x_3) \rangle &= \langle \delta(x_1) \otimes \delta(x_2), \varphi(x_1, x_2, 0) \rangle \\ &= \langle \delta(x_1), \varphi(x_1, 0, 0) \rangle = \varphi(\mathbf{0}). \end{aligned}$$

Example 7.35 Let $g \in L^1_{\text{loc}}(\mathbb{R})$. Let us compute $g(x) \otimes \delta'(y)$. Let $\varphi \in \mathcal{D}(\mathbb{R} \times \mathbb{R})$. We have:

$$\langle g(x)\otimes\delta'(y),\varphi(x,y)\rangle = \langle g(x),\langle\delta'(y),\varphi(x,y)\rangle\rangle = -\big\langle g(x),\big\langle\delta(y),\varphi_y(x,y)\big\rangle\big\rangle$$
$$= -\big\langle g(x),\varphi_y(x,0)\big\rangle = -\int_{\mathbb{R}} g(x)\varphi_y(x,0)dx.$$

Very often, in applied contexts, the symbol $\otimes$ is omitted, as we actually did in Chap. 5, in Sects. 5.9 and 5.10.

Example 7.36 In this example we go back to Sect. 5.10.2 and prove that the potentials in formulas (5.146) and (5.148), pp. 320, 321, are solutions in the sense of distributions of the moving source equation, written with the proper symbology:

$$u_{tt} - c^2\Delta u = S\delta(x_1)\otimes\delta(x_2)\otimes\delta(x_3 - vt) \quad \text{in } \mathcal{D}'(\mathbb{R}^3\times\mathbb{R}), \tag{7.29}$$

in the subsonic and supersonic case, respectively. We put $\mathbf{x}' = (x_1, x_2)$.

- The **subsonic case** $m = v/c < 1$. The corresponding potential is given by

$$u(\mathbf{x}', x_3, t) = \frac{S}{4\pi c^2}\frac{1}{\sqrt{|\mathbf{x}'|^2(1-m^2) + (vt - x_3)^2}}. \tag{7.30}$$

Since u is a travelling wave along the x_3-axis, it is convenient to introduce the new coordinates $\boldsymbol{\xi}' = \mathbf{x}'\sqrt{1-m^2}$, $\xi_3 = vt - x_3$ and look at the function

$$U(\boldsymbol{\xi}', \xi_3) = \frac{S}{4\pi c^2}\frac{1}{\sqrt{|\boldsymbol{\xi}'|^2 + \xi_3^2}} = \frac{S}{4\pi c^2}\frac{1}{|\boldsymbol{\xi}|}.$$

Then $u(\mathbf{x}', x_3, t) = U(\mathbf{x}'\sqrt{1-m^2}, vt - x_3)$ and

$$u_{tt} = v^2 U_{\xi_3\xi_3}, \quad u_{x_jx_j} = (1-m^2)U_{\xi_j\xi_j},\ j = 1,2, \quad u_{x_3x_3} = U_{\xi_3\xi_3}.$$

Inserting into (7.29), we get

$$(v^2 - c^2)U_{\xi_3\xi_3} - c^2(1-m^2)\Delta_2 U = S\,\delta_2(\boldsymbol{\xi}'/\sqrt{1-m^2})\otimes\delta(\xi_3), \quad \text{in } \mathcal{D}'(\mathbb{R}^3).$$

Dividing by $v^2 - c^2 = -c^2(1-m^2)$ and recalling that

$$\delta_2(\boldsymbol{\xi}'/\sqrt{1-m^2}) = (1-m^2)\delta_2(\boldsymbol{\xi}'),$$

we conclude that u is a solution of (7.29) if and only if U satisfies the equation:

$$\Delta_3 U = -\frac{S}{c^2}\delta_2(\boldsymbol{\xi}')\otimes\delta(\xi_3) = -\frac{S}{c^2}\delta_3(\boldsymbol{\xi}), \quad \text{in } \mathcal{D}'(\mathbb{R}^3). \tag{7.31}$$

From Example 7.19, p. 433, we know that $-\frac{1}{4\pi|\boldsymbol{\xi}|}$ is the fundamental solution of Δ_3. Thus U satisfies (7.31) and the subsonic case is completed.

- The **supersonic case** $m = v/c > 1$. The corresponding potential is given by

$$u(\mathbf{x},t)=\begin{cases}\dfrac{S}{2\pi c^2}\dfrac{1}{\sqrt{(vt-x_3)^2-|\mathbf{x}'|^2\,(m^2-1)}} & vt-x_3>|\mathbf{x}'|\sqrt{(m^2-1)}\\ 0 & vt-x_3<|\mathbf{x}'|\sqrt{(m^2-1)}.\end{cases}$$

In this case, we introduce the coordinates $\boldsymbol{\xi}'=\mathbf{x}'\sqrt{m^2-1}$, $\xi_3=vt-x_3$ and look at the function

$$U(\boldsymbol{\xi}',\xi_3)=\frac{S}{2\pi c^2}\frac{1}{\sqrt{\xi_3^2-|\boldsymbol{\xi}'|^2}},\quad \text{for } \xi_3>|\boldsymbol{\xi}'|\,,$$

and $U=0$ for $\xi_3<|\boldsymbol{\xi}'|$. Then $u(\mathbf{x}',x_3,t)=U(\mathbf{x}'\sqrt{m^2-1},vt-x_3)$ and

$$u_{tt}=v^2U_{\xi_3\xi_3},\quad u_{x_jx_j}=(m^2-1)U_{\xi_j\xi_j},\ j=1,2,\quad u_{x_3x_3}=U_{\xi_3\xi_3}.$$

Inserting into (7.29), we get

$$(v^2-c^2)U_{\xi_3\xi_3}-c^2(m^2-1)\Delta_2U=S\,(m^2-1)\delta_2(\boldsymbol{\xi}')\otimes\delta(\xi_3)\quad \text{in } \mathcal{D}'(\mathbb{R}^3).$$

Dividing by $v^2-c^2=c^2(m^2-1)$, we conclude that u is a solution of (7.29) if and only if U satisfies the equation:

$$U_{\xi_3\xi_3}-\Delta_2U=\frac{S}{c^2}\,\delta_2(\boldsymbol{\xi}')\otimes\delta(\xi_3)=\frac{S}{c^2}\delta_3(\boldsymbol{\xi})\qquad \text{in } \mathcal{D}'(\mathbb{R}^3).\tag{7.32}$$

Recalling Remark 5.20, p. 311, we recognize that $(c^2/S)U$ is the fundamental solution of the two-dimensional wave equation (with $t=\xi_3, c=1$) (see Problem 7.18) and therefore (7.32) holds true. This completes the supersonic case.

7.5.5 Convolution

The convolution of two distributions may be defined with some restrictions as well. Let us see why. If $u, w\in L^1(\mathbb{R}^n)$ and $\varphi\in\mathcal{D}(\mathbb{R}^n)$ we may write:

$$\begin{aligned}\langle u*w,\varphi\rangle&=\int_{\mathbb{R}^n}\left[\int_{\mathbb{R}^n}u(\mathbf{x}-\mathbf{y})w(\mathbf{y})d\mathbf{y}\right]\varphi(\mathbf{x})\,d\mathbf{x}\\&=\int_{\mathbb{R}^n}\int_{\mathbb{R}^n}u(\mathbf{x})w(\mathbf{y})\varphi(\mathbf{x}+\mathbf{y})\,d\mathbf{y}d\mathbf{x}=\langle u(\mathbf{x})\otimes w(\mathbf{y}),\varphi(\mathbf{x}+\mathbf{y})\rangle\,.\end{aligned}$$

Now, the question is: may we give any meaning to this formula when u and v are two generic distributions T and V in $\mathcal{D}'(\mathbb{R}^n)$? The answer is negative, simply because the function $(\mathbf{x},\mathbf{y})\longmapsto\varphi(\mathbf{x}+\mathbf{y})$ does not have compact support[5] in $\mathbb{R}^n\times\mathbb{R}^n$ (unless $\varphi\equiv 0$).

[5] For instance: if $\varphi\in\mathcal{D}'(\mathbb{R})$ and $\operatorname{supp}(\varphi)=[a,b]$, then the support of $\varphi(x+y)$ in $\mathbb{R}^2$ is the unbounded strip $a\le x+y\le b$.

However, there are special cases in which the brackets

$$\begin{aligned}\langle T * V, \varphi\rangle &= \langle T(\mathbf{x}) \otimes V(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle \\ &= \langle T(\mathbf{x}), \langle V(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle\rangle = \langle V(\mathbf{y}), \langle T(\mathbf{x}), \varphi(\mathbf{x}+\mathbf{y})\rangle\rangle\,.\end{aligned} \tag{7.33}$$

make sense. Indeed it is enough that *the support of* $\varphi(\mathbf{x}+\mathbf{y})$ *has a compact intersection with the support of* $T(\mathbf{x}) \otimes V(\mathbf{y})$. To see why, simply note that in this case we could replace $\varphi(\mathbf{x}+\mathbf{y})$ by any test function in $\mathcal{D}(\mathbb{R}^{2n})$ that agrees with $\varphi(\mathbf{x}+\mathbf{y})$ on the support of $T(\mathbf{x}) \otimes V(\mathbf{y})$, mimicking Remark 7.13, and nothing changes.

An important case occurs when *at least one of the two distributions has compact support,* as in the following theorem.

Theorem 7.37 *Let* $T, V \in \mathcal{D}'(\mathbb{R}^n)$*, with either* T *or* V *compactly supported. Then the formula*

$$\langle T * V, \varphi\rangle = \langle T(\mathbf{x}) \otimes V(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle\,, \qquad \varphi \in \mathcal{D}(\mathbb{R}^n)$$

defines an element of $\mathcal{D}'(\mathbb{R}^n)$*, called the* ***convolution*** *of* T *and* V*. The following properties hold.*

a) The convolution is commutative: $T * V = V * T$.
b) It commutes with derivatives:

$$D^\alpha(T * V) = D^\alpha T * V = T * D^\alpha V$$

for every multi-index $\alpha = (\alpha_1, \ldots, \alpha_n)$.
c) The Dirac delta at the origin acts as the identity: $V * \delta_n = \delta_n * V = V$*. In particular,*

$$\delta_n * \delta_n = \delta_n.$$

d) If T *has compact support and* $u \in C^\infty(\mathbb{R}^n)$*, then* $T * u$ *is well defined and it coincides with* C^∞*-function.*

Proof a) follows from the definition of tensor product, eq. (7.28). For b) we have:

$$\begin{aligned}\langle D^\alpha(T * V), \varphi\rangle &= (-1)^{|\alpha|}\,\langle T(\mathbf{x}) \otimes V(\mathbf{y}), D^\alpha\varphi(\mathbf{x}+\mathbf{y})\rangle \\ &= (-1)^{|\alpha|}\,\langle V(\mathbf{y}), \langle T(\mathbf{x}), D^\alpha\varphi(\mathbf{x}+\mathbf{y})\rangle\rangle \\ &= \langle V(\mathbf{y}), \langle D^\alpha T(\mathbf{x}), \varphi(\mathbf{x}+\mathbf{y})\rangle\rangle \\ &= \langle V * D^\alpha T, \varphi\rangle = \langle D^\alpha T * V, \varphi\rangle\,.\end{aligned}$$

Using a), $D^\alpha(T * V) = D^\alpha(V * T) = D^\alpha V * T = T * D^\alpha V$. c) follows from

$$\begin{aligned}\langle V * \delta_n, \varphi\rangle &= \langle V(\mathbf{x}) \otimes \delta_n(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle = \langle V(\mathbf{x}), \langle\, \delta_n(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle\rangle \\ &= \langle V, \varphi\rangle\,.\end{aligned}$$

Finally, for d) first note that $\mathbf{x} \mapsto \langle u(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle$ is C^∞-function on $\mathbb{R}^n$. Recalling Remark 7.13, p. 429, the bracket

$$\langle T * u, \varphi\rangle = \langle T(\mathbf{x}), \langle u(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle\rangle$$

makes sense since T has compact support. We have:

$$\begin{aligned}\langle T * u, \varphi\rangle &= \langle T(\mathbf{x}), \langle u(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})\rangle\rangle \\ &= \left\langle T(\mathbf{x}), \int_{\mathbb{R}^n} u(\mathbf{y}), \varphi(\mathbf{x}+\mathbf{y})dy\right\rangle = \left\langle T(\mathbf{x}), \int_{\mathbb{R}^n} u(\mathbf{x}-\mathbf{y}), \varphi(\mathbf{y})dy\right\rangle \\ &= \int_{\mathbb{R}^n} \langle T(\mathbf{x}), u(\mathbf{x}-\mathbf{y})\rangle\, \varphi(\mathbf{y})dy.\end{aligned}$$

Hence $T * u$ coincide with the C^∞- function $\mathbf{y} \longmapsto \langle T(\mathbf{x}), u(\mathbf{x}-\mathbf{y})\rangle$. □

Warning: The convolution of functions is associative. For distributions, the convolution is in general, *not associative*. In fact, consider the three distributions $1, \delta', H$; we have:

$$\delta' * 1 = (\delta * 1)' = 1' = 0$$

whence

$$\mathcal{H} * (\delta' * 1) = \mathcal{H} * 0 = 0.$$

However,

$$(\mathcal{H} * \delta') * 1 = (\mathcal{H}' * \delta) * 1 = (\delta * \delta) * 1 = 1.$$

The problem is that two out of three factors (1 and H) have noncompact support. *If at least two factors have compact support one can show that the convolution is associative.*

7.6 Tempered Distributions and Fourier Transform

7.6.1 Tempered distributions

We now introduce the Fourier transform $\hat{F}$ of a distribution. As usual, the idea is to define the action of $\hat{F}$ by carrying the transform onto the test functions. However a problem immediately arises: unless $\varphi \in \mathcal{D}(\mathbb{R}^n)$ is not identically zero, its Fourier transform

$$\widehat{\varphi}(\boldsymbol{\xi}) = \int_{\mathbb{R}^n} e^{-i\mathbf{x}\cdot\boldsymbol{\xi}}\varphi(\mathbf{x})d\mathbf{x}$$

cannot belong[6] to $\mathcal{D}(\mathbb{R}^n)$. Thus, it is necessary to choose a larger space of test functions. It turns out that the correct one consists in the set of functions *rapidly vanishing at* ∞, which obviously contains $\mathcal{D}(\mathbb{R}^n)$. Even if φ is real valued, $\widehat{\varphi}$ could be complex-valued, hence it is natural to consider **complex-valued functions and distributions**.

[6] Let $n = 1$ and $\varphi \in \mathcal{D}(\mathbb{R})$. Assume that $\mathrm{supp}(\varphi) \subset (-a, a)$. We may write

$$\widehat{\varphi}(\xi) = \int_{-a}^{a} e^{-ix\xi}\varphi(x)dx = \int_{-a}^{a}\sum_{n=0}^{\infty}\frac{(-ix\xi)^n}{n!}\varphi(x)dx = \sum_{n=0}^{\infty}\frac{(-i\xi)^n}{n!}\int_{-a}^{a} x^n\varphi(x)dx.$$

Since $\left|\int_{-a}^{a} x^n\varphi(x)dx\right| \le \max|\varphi|\, a^n$, it follows that $\widehat{\varphi}$ is an *analytic* function in all $\mathbb{C}$. Therefore $\widehat{\varphi}$ cannot vanish outside a compact interval, unless $\widehat{\varphi} \equiv 0$. But then $\varphi \equiv 0$ as well.

Definition 7.38 We denote by $\mathcal{S}(\mathbb{R}^n)$ the space of functions $v \in C^\infty(\mathbb{R}^n)$ *rapidly vanishing at infinity*, i.e. such that

$$D^\alpha v(\mathbf{x}) = o(|\mathbf{x}|^{-m}), \qquad |\mathbf{x}| \to \infty,$$

for all $m \in \mathbb{N}$ and every multi-index α.

We endow $\mathcal{S}(\mathbb{R}^n)$ with an "ad hoc" notion of convergence. If $\beta = (\beta_1, \dots, \beta_n)$ is a multi-index, we set

$$\mathbf{x}^\beta = x_1^{\beta_1} \cdots x_n^{\beta_n}.$$

Definition 7.39 Let $\{v_k\} \subset \mathcal{S}(\mathbb{R}^n)$ and $v \in \mathcal{S}(\mathbb{R}^n)$. We say that

$$v_k \to v \qquad \text{in } \mathcal{S}(\mathbb{R}^n)$$

if for every pair of multi-indexes α, β,

$$\mathbf{x}^\beta D^\alpha v_k \to \mathbf{x}^\beta D^\alpha v, \quad \text{uniformly in } \mathbb{R}^n.$$

Clearly $\mathcal{D}(\mathbb{R}^n)$ is a subspace of $\mathcal{S}(\mathbb{R}^n)$ and if $\{v_k\} \subset \mathcal{D}(\mathbb{R}^n)$, $v_k \to v$ in $\mathcal{D}(\mathbb{R}^n)$, then

$$v_k \to v \quad \text{in } \mathcal{S}(\mathbb{R}^n)$$

as well, since each v_k vanishes outside a common compact set so that the multiplication by $\mathbf{x}^\beta$ does not have any influence.

Example 7.40 The function $v(\mathbf{x}) = e^{-|\mathbf{x}|^2}$ belongs to $\mathcal{S}(\mathbb{R}^n)$. On the other hand, the function $v(\mathbf{x}) = e^{-|\mathbf{x}|^2} \sin(e^{|\mathbf{x}|^2})$ does not (why?).

$\mathcal{D}(\mathbb{R}^n)$ *is dense in* $\mathcal{S}(\mathbb{R}^n)$. In fact, given $v \in \mathcal{S}(\mathbb{R}^n)$, let

$$v_k(\mathbf{x}) = v(\mathbf{x})\rho(|\mathbf{x}|/k)$$

where $\rho = \rho(s)$, $s \geq 0$, is a nonnegative C^∞-function, equal to 1 in $[0, 1]$ and zero for $s \geq 2$ (see Fig. 7.4). It can be checked that $\{v_k\} \subset \mathcal{D}(\mathbb{R}^n)$ and $v_k \to v$ in $\mathcal{S}(\mathbb{R}^n)$, since $\rho(|\mathbf{x}|/k)$ is equal to 1 for $\{|\mathbf{x}| < k\}$ and zero for $\{|\mathbf{x}| > 2k\}$.

The Fourier transform will be defined for complex-valued linear continuous functional over $\mathcal{S}(\mathbb{R}^n)$. These are the so called *tempered distributions*. Precisely:

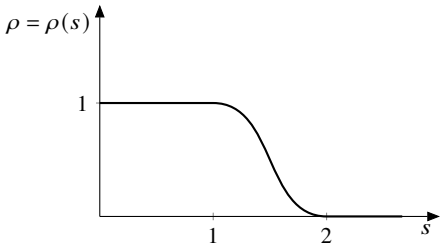

Fig. 7.4 A smooth decreasing and nonnegative function, equal to 1 in $[0, 1]$ and vanishing for $s \geq 2$

Definition 7.41 A *tempered distribution* in $\mathbb{R}^n$ is a complex-valued linear continuous functional in $\mathcal{S}(\mathbb{R}^n)$. The set of tempered distributions is denoted by $\mathcal{S}'(\mathbb{R}^n)$.

Clearly, the restriction to $\mathcal{D}(\mathbb{R}^n)$ of every tempered distribution T defines an element of $\mathcal{D}'(\mathbb{R}^n)$. Moreover, since $\mathcal{D}(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$, if $T_1, T_2 \in \mathcal{S}'(\mathbb{R}^n)$ and $T_1 = T_2$ as elements in $\mathcal{D}'(\mathbb{R}^n)$, then $T_1 = T_2$ as elements in $\mathcal{S}'(\mathbb{R}^n)$.

Like $\mathcal{D}'(\Omega)$, $\mathcal{S}'(\mathbb{R}^n)$ possesses a completeness property that may be used to construct a tempered distribution or to recognize that some linear functional in $\mathcal{D}'(\mathbb{R}^n)$ is a tempered distribution. First, we say that a sequence $\{T_k\} \subset \mathcal{S}'(\mathbb{R}^n)$ converges to T in $\mathcal{S}'(\mathbb{R}^n)$ if

$$\langle T_k, v\rangle \to \langle T, v\rangle\,, \quad \forall v \in \mathcal{S}(\mathbb{R}^n).$$

We have:

Proposition 7.42 *Let $\{T_k\} \subset \mathcal{S}'(\mathbb{R}^n)$ such that*

$$\lim_{k\to\infty} \langle T_k, v\rangle \text{ exists and is finite,} \quad \forall v \in \mathcal{S}(\mathbb{R}^n).$$

Then, this limit defines $T \in \mathcal{S}'(\mathbb{R}^n)$ and T_k converges to T in $\mathcal{S}'(\mathbb{R}^n)$.

Example 7.43 The Dirac comb is a tempered distribution. In fact, if $v \in \mathcal{S}(\mathbb{R})$, we have

$$\langle \text{comb}, v\rangle = \sum_{k=-\infty}^{\infty} v(k)$$

and the series is convergent since $v(k) \to 0$ more rapidly than $|k|^{-m}$ for every $m > 0$. From Proposition 7.42, comb $\in \mathcal{S}'(\mathbb{R})$.

Remark 7.44 (Convolution) If $T \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}(\mathbb{R}^n)$, the convolution is well defined by formula

$$(T * v)(\mathbf{x}) = \langle T, v(\mathbf{x} - \cdot)\rangle\,.$$

Then, $T * v \in \mathcal{S}'(\mathbb{R}^n)$ and coincides with a function in $C^\infty(\mathbb{R}^n)$.

7.6.2 Fourier transform in $\mathcal{S}'(\mathbb{R}^n)$

For a complex-valued function $u \in L^1(\mathbb{R}^n)$, its Fourier transform is given by

$$\widehat{u}(\boldsymbol{\xi}) = \mathcal{F}\,[u]\,(\boldsymbol{\xi}) = \int_{\mathbb{R}^n} e^{-i\mathbf{x}\cdot\boldsymbol{\xi}} u(\mathbf{x}) d\mathbf{x}.$$

It could be that, even if u is compactly supported, $\widehat{u} \notin L^1(\mathbb{R}^n)$. For instance, if $p_a(x) = \frac{1}{2}\chi_{[-a,a]}(x)$ then

$$\widehat{p}_a(\xi) = \frac{\sin(a\xi)}{\xi}$$

which is not in $L^1(\mathbb{R})$ (see Appendix A). When also $\widehat{u} \in L^1(\mathbb{R}^n)$, u can be reconstructed from $\widehat{u}$ through the following inversion formula:

Theorem 7.45 *Let $u \in L^1(\mathbb{R}^n)$, $\widehat{u} \in L^1(\mathbb{R}^n)$. Then*

$$u(\mathbf{x}) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^n} e^{i\mathbf{x}\cdot\boldsymbol{\xi}} \widehat{u}(\boldsymbol{\xi}) d\boldsymbol{\xi} \equiv \mathcal{F}^{-1}\left[\widehat{u}\right](\mathbf{x}). \tag{7.34}$$

In particular, the inversion formula (7.34) holds for $u \in \mathcal{S}(\mathbb{R}^n)$, since (exercise) $\widehat{u} \in \mathcal{S}(\mathbb{R}^n)$ as well. Moreover, it can be proved that

$$u_k \to u \qquad \text{in } \mathcal{S}(\mathbb{R}^n)$$

if and only if

$$\widehat{u}_k \to \widehat{u} \qquad \text{in } \mathcal{S}(\mathbb{R}^n),$$

which means that

$$\mathcal{F}, \mathcal{F}^{-1} : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$$

are *continuous, one-to-one and onto operators in* $\mathcal{S}(\mathbb{R}^n)$.

Now observe that, if $u, v \in \mathcal{S}(\mathbb{R}^n)$,

$$\begin{aligned}\langle \hat{u}, v\rangle &= \int_{\mathbb{R}^n} \left(\int_{\mathbb{R}^n} e^{-i\mathbf{x}\cdot\boldsymbol{\xi}} u(\mathbf{x}) d\mathbf{x} \right) v(\boldsymbol{\xi}) d\boldsymbol{\xi} \\ &= \int_{\mathbb{R}^n} u(\mathbf{x}) \left(\int_{\mathbb{R}^n} e^{-i\mathbf{x}\cdot\boldsymbol{\xi}} v(\boldsymbol{\xi}) d\boldsymbol{\xi} \right) d\mathbf{x} = \langle u, \hat{v}\rangle .\end{aligned}$$

so that the following *weak Parseval identity* holds:

$$\langle \hat{u}, v\rangle = \langle u, \hat{v}\rangle . \tag{7.35}$$

The key point is that the last bracket makes sense for $u = T \in \mathcal{S}'(\mathbb{R}^n)$ as well, and defines a tempered distribution. In fact:

Lemma 7.46 *Let $T \in \mathcal{S}'(\mathbb{R}^n)$. The linear functional*

$$v \mapsto \langle T, \widehat{v}\rangle , \qquad \forall v \in \mathcal{S}(\mathbb{R}^n)$$

is a tempered distribution.

Proof If $v_k \to 0$ in $\mathcal{S}(\mathbb{R}^n)$ then $\hat{v}_k \to 0$ in $\mathcal{S}(\mathbb{R}^n)$ and $\langle T, \hat{v}_k\rangle \to 0$. Thus, $v \mapsto \langle T, \hat{v}\rangle$ is a tempered distribution. □

We are now in position to define the Fourier transform of $T \in \mathcal{S}'(\mathbb{R}^n)$.

Definition 7.47 Let $T \in \mathcal{S}'(\mathbb{R}^n)$. The *Fourier transform* $\widehat{T} = \mathcal{F}[T]$ is the tempered distribution defined by

$$\langle \mathcal{F}[T], v\rangle = \langle u, \mathcal{F}[v]\rangle , \qquad \forall v \in \mathcal{S}(\mathbb{R}^n).$$

We see that the transform has been carried onto the test function $v \in \mathcal{S}(\mathbb{R}^n)$. As a consequence, all the properties valid for functions, continue to hold for tempered distributions. We list some of them. Let $T \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}(\mathbb{R}^n)$.

1. *Translation*. Let $\tau_{\mathbf{a}}(\mathbf{x}) = (\mathbf{x} - \mathbf{a})$, $\mathbf{a} \in \mathbb{R}^n$. Then,

$$\mathcal{F}\left[T(\mathbf{x}-\mathbf{a})\right](\boldsymbol{\xi}) = e^{-i\mathbf{a}\cdot\boldsymbol{\xi}}\hat{T}(\boldsymbol{\xi}) \quad \text{and} \quad \mathcal{F}\left[e^{i\mathbf{a}\cdot\mathbf{x}}T\right](\boldsymbol{\xi}) = \hat{T}(\boldsymbol{\xi}-\mathbf{a}). \tag{7.36}$$

In fact, for instance ($v = v(\boldsymbol{\xi}), \hat{v} = \hat{v}(\mathbf{x})$):

$$\begin{aligned}\langle \mathcal{F}\left[T(\mathbf{x}-\mathbf{a})\right], v\rangle &= \langle T(\mathbf{x}-\mathbf{a}), \hat{v}\rangle = \langle T, \hat{v}(\mathbf{x}+\mathbf{a})\rangle \\ &= \left\langle T, \mathcal{F}\left[e^{-i\mathbf{a}\cdot\boldsymbol{\xi}}v\right]\right\rangle = \langle \hat{T}, e^{-i\mathbf{a}\cdot\boldsymbol{\xi}}v\rangle = \langle e^{-i\mathbf{a}\cdot\boldsymbol{\xi}}\hat{T}, v\rangle.\end{aligned}$$

2. *Rescaling*. Let $h \in \mathbb{R}$, $h \neq 0$, and $\psi(\mathbf{x}) = h\mathbf{x}$, $\psi^{-1}(\mathbf{x}) = h^{-1}\mathbf{x}$. Then

$$\mathcal{F}\left[T \circ \psi\right] = \frac{1}{|h|^n}\mathcal{F}\left[T\right] \circ \psi^{-1}. \tag{7.37}$$

With abuse of notation, we may write (7.37) in the more explicit form,

$$\mathcal{F}\left[T(h\mathbf{x})\right](\boldsymbol{\xi}) = \frac{1}{|h|^n}\hat{T}\left(\frac{\boldsymbol{\xi}}{h}\right).$$

In fact, using (7.20)[7], since $\det J_{\psi^{-1}} = h^{-n}$ and the formula (7.37) holds for v, we can write,

$$\begin{aligned}\langle \mathcal{F}\left[T \circ \psi\right], v\rangle &= \langle T \circ \psi, \hat{v}\rangle = \left\langle T, \frac{1}{|h|^n}\hat{v}\left(\frac{\mathbf{x}}{h}\right)\right\rangle \\ &= \langle T, \mathcal{F}\left[v(h\boldsymbol{\xi})\right]\rangle = \langle \hat{T}, v(h\boldsymbol{\xi})\rangle = \left\langle \frac{1}{|h|^n}\hat{T}\left(\frac{\boldsymbol{\xi}}{h}\right), v\right\rangle.\end{aligned}$$

In particular, choosing $h = -1$, it follows that if T is *even (odd)* then $\hat{T}$ is *even (odd)*.

3. *Derivatives*:

$$a)\ \mathcal{F}\left[\partial_{x_j}T\right] = i\xi_j\hat{T} \quad \text{and} \quad b)\ \mathcal{F}\left[x_jT\right] = i\partial_{\xi_j}\hat{T}.$$

Namely:

$$\left\langle \mathcal{F}\left[\partial_{x_j}T\right], v\right\rangle = \left\langle \partial_{x_j}T, \hat{v}\right\rangle = -\left\langle T, \partial_{x_j}\hat{v}\right\rangle = \left\langle T, \mathcal{F}\left[i\xi_j v\right]\right\rangle = \langle i\xi_j\hat{T}, v\rangle.$$

For the second formula, we have:

$$\begin{aligned}\left\langle \mathcal{F}\left[x_jT\right], v\right\rangle &= \left\langle x_jT, \hat{v}\right\rangle = \left\langle T, x_j\hat{v}\right\rangle \\ &= \left\langle T, -i\mathcal{F}\left[\partial_{\xi_j}v\right]\right\rangle = \langle \hat{T}, -i\partial_{\xi_j}v\rangle = \langle i\partial_{\xi_j}\hat{T}, v\rangle.\end{aligned}$$

[7] In view of the density of $\mathcal{D}'(\mathbb{R}^n)$ in $\mathcal{S}'(\mathbb{R}^n)$.

4. *Convolution*[8]. If $T \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}(\mathbb{R}^n)$, then $T * v \in \mathcal{S}'(\mathbb{R}^n)$ and

$$\mathcal{F}[T * v] = \hat{T} \cdot \hat{v}.$$

This formula holds also in the case of $T, F \in \mathcal{S}'(\mathbb{R}^n)$ and F or T has compact support.

Example 7.48 We know that $\delta_n \in \mathcal{S}'(\mathbb{R}^n)$. We have:

$$\widehat{\delta_n} = 1, \qquad \widehat{1} = (2\pi)^n \delta_n.$$

In fact:

$$\langle \widehat{\delta_n}, v \rangle = \langle \delta_n, \hat{v} \rangle = \int_{\mathbb{R}^n} v(\mathbf{x}) d\mathbf{x} = \langle 1, v \rangle .$$

For the second formula, using (7.34) we have:

$$\begin{aligned} \langle \mathcal{F}[1], v \rangle = \langle 1, \hat{v} \rangle &= \int_{\mathbb{R}^n} \hat{v} \, d\mathbf{x} = (2\pi)^n v(\mathbf{0}) \\ &= \langle (2\pi)^n \delta_n, v \rangle . \end{aligned}$$

Example 7.49 Transform of x_j :

$$\widehat{x_j} = i(2\pi)^n \partial_{\xi_j} \delta_n.$$

Indeed, from 3, b) and Example 7.48, we may write

$$\mathcal{F}[x_j] = \mathcal{F}\left[x_j \cdot 1\right] = i\partial_{\xi_j} \mathcal{F}[1] = i(2\pi)^n \partial_{\xi_j} \delta_n.$$

7.6.3 Fourier transform in $L^2(\mathbb{R}^n)$

Since $L^2(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$, the Fourier transform is well defined for all functions in $L^2(\mathbb{R}^n)$. The following theorem holds, where $\overline{v}$ denotes the complex conjugate of v.

Theorem 7.50 *$u \in L^2(\mathbb{R}^n)$ if and only if $\hat{u} \in L^2(\mathbb{R}^n)$. Moreover, if $u, v \in L^2(\mathbb{R}^n)$, the following strong Parseval identity holds:*

$$(\hat{u}, \hat{v})_{L^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} \hat{u} \, \overline{\hat{v}} = (2\pi)^n \int_{\mathbb{R}^n} u \, \overline{v} = (2\pi)^n (u, v)_{L^2(\mathbb{R}^n)}. \tag{7.38}$$

In particular

$$\|\widehat{u}\|^2_{L^2(\mathbb{R}^n)} = (2\pi)^n \|u\|^2_{L^2(\mathbb{R}^n)} . \tag{7.39}$$

Formula (7.39) shows that the *Fourier transform is an isometry in* $L^2(\mathbb{R}^n)$ (but for the factor $(2\pi)^{-n}$).

[8] We omit the proof.

Proof Since $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$, it is enough to prove (7.38) for $u, v \in \mathcal{S}(\mathbb{R}^n)$. We have:

$$\begin{aligned}(\hat{u}, \hat{v})_{L^2(\mathbb{R}^n)} &= \int_{\mathbb{R}^n} \hat{u}(\boldsymbol{\xi})\, \overline{\hat{v}(\boldsymbol{\xi})} = \int_{\mathbb{R}^n} \widehat{u}(\boldsymbol{\xi}) \int_{\mathbb{R}^n} \overline{v(\mathbf{x}) e^{-i\mathbf{x}\cdot\boldsymbol{\xi}}}\, d\mathbf{x}\, d\boldsymbol{\xi} \\ &= \int_{\mathbb{R}^n} \overline{v(\mathbf{x})} \int_{\mathbb{R}^n} \widehat{u}(\boldsymbol{\xi}) e^{i\mathbf{x}\cdot\boldsymbol{\xi}}\, d\boldsymbol{\xi}\, d\mathbf{x} \\ &= (2\pi)^n \int_{\mathbb{R}^n} u(\mathbf{x})\overline{v(\mathbf{x})}\, d\mathbf{x} = (2\pi)^n (u, v)_{L^2(\mathbb{R}^n)}.\end{aligned}$$

□

Example 7.51 Let us compute

$$\int_{\mathbb{R}} \left(\frac{\sin x}{x}\right)^2 dx.$$

We know that the Fourier transform of $p_1 = \frac{1}{2}\chi_{[-1,1]}$ is

$$\hat{p}_1(\xi) = 2\frac{\sin \xi}{\xi},$$

which belongs to $L^2(\mathbb{R})$. Thus, (7.39) yields

$$4\int_{\mathbb{R}} \left(\frac{\sin \xi}{\xi}\right)^2 d\xi = 2\pi \int_{\mathbb{R}} (\chi_{[-1,1]}(x))^2 dx = 4\pi$$

whence

$$\int_{\mathbb{R}} \left(\frac{\sin x}{x}\right)^2 dx = \pi.$$

7.7 Sobolev Spaces

7.7.1 An abstract construction

Sobolev spaces constitute one of the most relevant functional settings for the treatment of boundary value problems. Here, we will be mainly concerned with (*real*) Sobolev spaces based on $L^2(\Omega)$, developing only the theoretical elements we will need in the sequel[9]. In general, Ω is a domain in $\mathbb{R}^n$. In particular, for $n = 1$, Ω reduces to an interval (a, b), $-\infty \le a < b \le \infty$.

The following abstract theorem is a flexible tool for generating Sobolev Spaces. The ingredients of the construction are:

- The space $\mathcal{D}'(\Omega; \mathbb{R}^m)$, for some $m \ge 1$; in particular, for $m = 1$, $\mathcal{D}'(\Omega)$.

[9] We omit the most technical proofs, that can be found, for instance, in the classical books *R. A. Adams, J. J. F. Fournier*, [34], *V. G. Maz'ja*, [38] or *W. P. Ziemer*, [44].

- Two Hilbert spaces H and Z with $Z \hookrightarrow D'(\Omega;\mathbb{R}^m)$. In particular

$$v_k \to v \text{ in } Z \quad \textbf{implies} \quad v_k \to v \text{ in } \mathcal{D}'(\Omega;\mathbb{R}^m). \tag{7.40}$$

- A linear continuous operator $L\colon H \to \mathcal{D}'(\Omega;\mathbb{R}^m)$ (such as a gradient, with $m = n$, or a divergence, with $m = 1$).

We have:

Theorem 7.52 *Define*

$$W = \{v \in H : Lv \in Z\}$$

and

$$(u,v)_W = (u,v)_H + (Lu,Lv)_Z. \tag{7.41}$$

Then W is a Hilbert space with inner product given by (7.41). The embedding of W in H is continuous and the restriction of L to W is continuous from W into Z.

Proof It is easy to check that (7.41) has all the properties of an inner product, with induced norm

$$\|u\|_W = \sqrt{\|u\|_H^2 + \|Lu\|_Z^2}.$$

Thus W is an inner-product space. It remains to check its completeness. Let $\{v_k\}$ be a Cauchy sequence in W. We must show that there exists $v \in H$ such that

$$v_k \to v \text{ in } H$$

and

$$Lv_k \to Lv \text{ in } Z.$$

Observe that $\{v_k\}$ and $\{Lv_k\}$ are Cauchy sequences in H and Z, respectively. Thus, there exist $v \in H$ and $z \in Z$ such that

$$v_k \to v \quad \text{in } H \qquad \text{and} \quad Lv_k \to z \quad \text{in } Z.$$

The continuity of L and (7.40) yield

$$Lv_k \to Lv \text{ in } \mathcal{D}'(\Omega;\mathbb{R}^n) \quad \text{and} \quad Lv_k \to z \text{ in } \mathcal{D}'(\Omega;\mathbb{R}^n).$$

Since the limit of a sequence in $\mathcal{D}'(\Omega;\mathbb{R}^n)$ is unique, we infer that $Lv = z$. Therefore

$$Lv_k \to Lv \text{ in } Z$$

and W is a Hilbert space.

The continuity of the embedding $W \subset H$ follows from

$$\|u\|_H \le \|u\|_W$$

while the continuity of $L_{|W} : W \to Z$ follows from

$$\|Lu\|_Z \le \|u\|_W.$$

The proof is complete. □

7.7.2 The space $H^1(\Omega)$

Let $\Omega \subseteq \mathbb{R}^n$ be a domain. Choose in Theorem 7.52:

$$H = L^2(\Omega), \qquad Z = L^2(\Omega;\mathbb{R}^n) \hookrightarrow \mathcal{D}'(\Omega;\mathbb{R}^n)$$

and $L : H \to \mathcal{D}'(\Omega;\mathbb{R}^n)$ given by

$$L = \nabla,$$

where the gradient is considered in the sense of distributions. Then, W is the Sobolev space of the functions in $L^2(\Omega)$, whose *first derivatives in the sense of distributions (also called **weak** or **distributional** derivatives) are functions in* $L^2(\Omega)$. For this space we use the symbol[10] $H^1(\Omega)$. Thus:

$$H^1(\Omega) = \left\{v \in L^2(\Omega) : \nabla v \in L^2(\Omega;\mathbb{R}^n)\right\}.$$

In other words, if $v \in H^1(\Omega)$, every partial derivative $\partial_{x_i} v$ is a function $v_i \in L^2(\Omega)$. This means that

$$\left\langle \partial_{x_i} v, \varphi \right\rangle = -(v, \partial_{x_i}\varphi)_{L^2(\Omega)} = (v_i, \varphi)_{L^2(\Omega)}, \qquad \forall \varphi \in \mathcal{D}(\Omega)$$

or, more explicitly,

$$\int_\Omega v(\mathbf{x})\partial_{x_i}\varphi(\mathbf{x})d\mathbf{x} = -\int_\Omega v_i(\mathbf{x})\varphi(\mathbf{x})d\mathbf{x}, \qquad \forall \varphi \in \mathcal{D}(\Omega). \tag{7.42}$$

In many applied situations, the Dirichlet integral

$$\int_\Omega |\nabla v|^2$$

represents an energy. The functions in $H^1(\Omega)$ are therefore associated with *configurations having finite energy*. From Theorem 7.52 and the separability of $L^2(\Omega)$, we have[11]:

Proposition 7.53 *$H^1(\Omega)$ is a separable Hilbert space, continuously embedded in $L^2(\Omega)$. The gradient operator is continuous from $H^1(\Omega)$ into $L^2(\Omega;\mathbb{R}^n)$.*

The inner product and the norm in $H^1(\Omega)$ are given, respectively, by

[10] Also $H^{1,2}(\Omega)$ or $W^{1,2}(\Omega)$ are used (see also Problem 7.23).

[11] If we associate with each element u of $H^1(\Omega)$ the vector $u, u_{x_1}, \ldots, u_{x_n}$, we see that $H^1(\Omega)$ is isometric to a subspace of

$$L^2(\Omega) \times L^2(\Omega) \times \ldots \times L^2(\Omega) = L^2(\Omega;\mathbb{R}^{n+1})$$

which is separable because $L^2(\Omega)$ is separable.

$$(u,v)_{H^1(\Omega)} = \int_\Omega uv\, d\mathbf{x} + \int_\Omega \nabla u \cdot \nabla v\, d\mathbf{x}$$

and

$$\|u\|^2_{H^1(\Omega)} = \int_\Omega u^2 d\mathbf{x} + \int_\Omega |\nabla u|^2\, d\mathbf{x}.$$

Example 7.54 Let $\Omega = B_{1/2}(\mathbf{0}) = \left\{\mathbf{x} \in \mathbb{R}^2 : |\mathbf{x}| < 1/2\right\}$ and

$$u(\mathbf{x}) = (-\ln|\mathbf{x}|)^a, \qquad \mathbf{x} \neq 0.$$

We have, using polar coordinates,

$$\int_{B_{1/2}(\mathbf{0})} u^2 d\mathbf{x} = 2\pi \int_0^{1/2} (-\ln r)^{2a} r dr < \infty, \qquad \text{for every } a \in \mathbb{R},$$

so that $u \in L^2(B_{1/2}(\mathbf{0}))$ for every $a \in \mathbb{R}$. Moreover, away from the origin, u is a smooth function and therefore its distributional derivative coincides with its classical one, given by:

$$u_{x_i} = -a x_i\, |\mathbf{x}|^{-2}\, (-\ln|\mathbf{x}|)^{a-1},\ i = 1, 2.$$

Now, u_{x_i} belongs to $L^1_{\text{loc}}(B_{1/2})$ for every a, and to affirm that $v_i = u_{x_i}$ represents the distributional derivative of u we have to check that (7.42) holds with $v_i = u_{x_i}$. Let $\varphi \in \mathcal{D}(\Omega)$. We have:

$$\int_{B_{1/2}} u_{x_i}\varphi = \lim_{\varepsilon \to 0} \int_{B_{1/2}\setminus B_\varepsilon} u_{x_i}\varphi.$$

Integrating by parts we obtain, denoting by $\boldsymbol{\nu}$ the interior normal to ∂B_ε,

$$\int_{B_{1/2}\setminus B_\varepsilon} u_{x_i}\varphi = \int_{\partial B_\varepsilon} u\varphi\nu_i d\sigma - \int_{B_{1/2}\setminus B_\varepsilon} u\varphi_{x_i} d\mathbf{x}.$$

Since

$$\left|\int_{\partial B_\varepsilon} u\varphi\nu_i d\sigma\right| \leq 2\pi\varepsilon(-\ln\varepsilon)^a \max|\varphi| \to 0 \text{ and } \int_{B_{1/2}\setminus B_\varepsilon} u\varphi_{x_i} \to \int_{B_{1/2}} u\varphi_{x_i}$$

as $\varepsilon \to 0$, we see that u_{x_i} satisfies (7.42) and it is the distributional derivative of u in $B_{1/2}$. We have

$$|\nabla u| = \left|a(-\ln|\mathbf{x}|)^{a-1}\right| |\mathbf{x}|^{-1}.$$

Using polar coordinates, we get

$$\int_{B_{1/2}(\mathbf{0})} |\nabla u|^2\, d\mathbf{x} = 2\pi a^2 \int_0^{1/2} |\ln r|^{2a-2}\, r^{-1} dr.$$

This integral is finite only if $2-2a>1$ or $a<1/2$. Since ∇u represents the gradient of u in the sense of distributions, we conclude that $u\in H^1(B_{1/2}(\mathbf{0}))$ only if $a<1/2$. We point out that when $a>0$, u is **unbounded** near $\mathbf{0}$.

We have affirmed that the Sobolev spaces constitute an adequate functional setting to solve boundary value problems. This point requires that we go more deeply into the arguments in Sect. 6.1 and that we make some necessary observations. When we write $f\in L^2(\Omega)$, we may think of a single function

$$f:\Omega\to\mathbb{R}\ (\text{or }\mathbb{C}),$$

square summable in the Lebesgue sense. However, if we want to exploit the Hilbert space structure of $L^2(\Omega)$, we need to identify two functions when they are equal a.e. in Ω. Adopting this point of view, each element in $L^2(\Omega)$ is actually an *equivalence class* of which f is a *representative*. The drawback here is that *it does not make sense anymore to compute the value of f at a single point*, since a point is a set with measure zero!

The same considerations hold for "functions" in $H^1(\Omega)$, since

$$H^1(\Omega)\subset L^2(\Omega).$$

On the other hand, if we deal with a boundary value problem, it is clear that *we would like to compute the solution at any point in Ω*!

Even more important is the question of *the trace of a function on the boundary of a domain*. By *trace* of f on $\partial\Omega$ we mean the restriction of f to $\partial\Omega$. In a Dirichlet or Neumann problem we assign precisely the trace of the solution or of its normal derivative on $\partial\Omega$, which is a set with measure zero. Does this make any sense if $u\in H^1(\Omega)$?

It could be objected that, after all, one always works with a single representative and that the numerical approximation of the solution only involves a finite number of points, making meaningless the distinction between functions in $L^2(\Omega)$ or in $H^1(\Omega)$ or continuous. Then, why do we have to struggle to give a precise meaning to the trace of a function in $H^1(\Omega)$?

One reason comes from numerical analysis itself, in particular from the need to keep under control the approximation errors and to provide stability estimates.

Let us ask, for instance: if a Dirichlet data is known within an error of order ε in L^2-norm on $\partial\Omega$, can we estimate in terms of ε the corresponding error in the solution?

If we are satisfied with an L^2 or an L^∞ norm *in the interior* of the domain, this kind of estimate may be available. But often, an *energy estimate* is required, involving the norm in $L^2(\Omega)$ of the gradient of the solution. In this case, the L^2-norm of the boundary data is not sufficient and it turns out that the exact information on the data, necessary to restore an energy estimate, is encoded in the trace characterization of Sect. 7.9.

We shall introduce the notion of *trace on $\partial\Omega$* for a function in $H^1(\Omega)$, using an approximation procedure with smooth functions. However, there are two cases, in

which the trace problem may be solved quite simply: the one-dimensional case and the case of functions with zero trace. We start with the first case.

• *Characterization of* $H^1(a,b)$, $a,b \in \mathbb{R}$. As the Example 7.54 shows, a function in $H^1(\Omega)$ may be unbounded. In dimension $n = 1$ this cannot occur. In fact, the elements in $H^1(a,b)$ are continuous functions[12] in $[a,b]$.

Proposition 7.55 *Let $u \in L^2(a,b)$. Then $u \in H^1(a,b)$ if and only if u is continuous in $[a,b]$ and there exists $w \in L^2(a,b)$ such that*

$$u(y) = u(x) + \int_x^y w(s)ds, \qquad \forall x, y \in [a,b]. \tag{7.43}$$

Moreover $u' = w$ (both a.e. and in the sense of distributions).

Proof Assume that u is continuous in $[a,b]$ and that (7.43) holds with $w \in L^2(a,b)$. Choose $x = a$. Replacing, if necessary, u by $u - u(a)$, we may assume $u(a) = 0$, so that

$$u(y) = \int_a^y w(s)ds, \qquad \forall x, y \in [a,b].$$

Let $\varphi \in \mathcal{D}(a,b)$. We have:

$$\langle u', \varphi \rangle = -\langle u, \varphi' \rangle = -\int_a^b u(s)\varphi'(s)ds = -\int_a^b \left[\int_a^s w(t)dt\right] \varphi'(s)ds =$$

(exchanging the order of integration)

$$= -\int_a^b \left[\int_t^b \varphi'(s)ds\right] w(t)dt = \int_a^b \varphi(t)w(t)dt = \langle w, \varphi \rangle .$$

Thus $u' = w$ in $\mathcal{D}'(a,b)$ and therefore $u \in H^1(a,b)$. From the Lebesgue Differentiation Theorem (Appendix A) we deduce that $u' = w$ a.e. as well.

Viceversa, let $u \in H^1(a,b)$. Define

$$v(x) = \int_a^x u'(s)ds, \qquad x \in [a,b]. \tag{7.44}$$

The function v is continuous in $[a,b]$ and the above proof shows that $v' = u'$ in $\mathcal{D}'(a,b)$. Then, by Proposition 7.18, p. 431,

$$u = v + C, \quad C \in \mathbb{R},$$

and therefore u is continuous in $[a,b]$ as well. Moreover, (7.44) yields

$$u(y) - u(x) = v(y) - v(x) = \int_x^y u'(s)ds$$

which is (7.43). □

Since a function $u \in H^1(a,b)$ is continuous in $[a,b]$, the value $u(x_0)$ at every point $x_0 \in [a,b]$ makes perfect sense. In particular *the trace of u* at the end points of the interval is given by the values $u(a)$ and $u(b)$.

[12] Rigorously: every *equivalence class in $H^1(a,b)$ has a representative which is continuous in $[a,b]$*.

7.7.3 The space $H_0^1(\Omega)$

Let $\Omega \subseteq \mathbb{R}^n$ be a domain. We introduce an important subspace of $H^1(\Omega)$.

Definition 7.56 We denote by $H_0^1(\Omega)$ the closure of $\mathcal{D}(\Omega)$ in $H^1(\Omega)$.

Thus $u \in H_0^1(\Omega)$ if and only if there exists a sequence $\{\varphi_k\} \subset \mathcal{D}(\Omega)$ such that $\varphi_k \to u$ in $H^1(\Omega)$, that is, such that both $\|\varphi_k - u\|_{L^2(\Omega)} \to 0$ and $\|\nabla\varphi_k - \nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \to 0$ as $k \to \infty$.

Since the test functions in $\mathcal{D}(\Omega)$ have zero trace on $\partial\Omega$, every $u \in H_0^1(\Omega)$ "inherits" this property and it is reasonable to consider the elements $H_0^1(\Omega)$ as the functions in $H^1(\Omega)$ with zero trace on $\partial\Omega$. Clearly, $H_0^1(\Omega)$ is a Hilbert subspace of $H^1(\Omega)$.

An important property that holds in $H_0^1(\Omega)$, particularly useful in the solution of boundary value problems, is expressed by the following inequality of Poincaré. Recall that the diameter of a set Ω is given by

$$\operatorname{diam}(\Omega) = \sup_{\mathbf{x},\mathbf{y}\in\Omega} |\mathbf{x} - \mathbf{y}| .$$

Theorem 7.57 *Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. There exists a positive constant C_P (a Poincaré's constant) depending only on n and $\operatorname{diam}(\Omega)$, such that, for every $u \in H_0^1(\Omega)$,*

$$\|u\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} . \tag{7.45}$$

Proof We use a strategy which is rather common for proving formulas in $H_0^1(\Omega)$. First, we prove (7.45) for $v \in \mathcal{D}(\Omega)$; then, if $u \in H_0^1(\Omega)$, we select a sequence $\{v_k\} \subset \mathcal{D}(\Omega)$ converging to u in the $H^1(\Omega)$ norm as $k \to \infty$, that is

$$\|v_k - u\|_{L^2(\Omega)} \to 0, \qquad \|\nabla v_k - \nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \to 0.$$

In particular

$$\|v_k\|_{L^2(\Omega)} \to \|u\|_{L^2(\Omega)} , \qquad \|\nabla v_k\|_{L^2(\Omega;\mathbb{R}^n)} \to \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} .$$

Since (7.45) holds for every v_k, we have

$$\|v_k\|_{L^2(\Omega)} \leq C_P \|\nabla v_k\|_{L^2(\Omega;\mathbb{R}^n)} .$$

Letting $k \to \infty$ we obtain (7.45) for u. Thus, it is enough to prove (7.45) for $v \in \mathcal{D}(\Omega)$. Assume without loss of generality that $\mathbf{0} \in \Omega$, and set $\max_{\mathbf{x}\in\Omega} |\mathbf{x}| \leq M = \operatorname{diam}(\Omega) < \infty$. Applying the Gauss Divergence Theorem and recalling that $v = 0$ on $\partial\Omega$ we can write

$$\int_\Omega \operatorname{div}(v^2\mathbf{x}) d\mathbf{x} = 0. \tag{7.46}$$

Now,

$$\operatorname{div}(v^2\mathbf{x}) = 2v\nabla v \cdot \mathbf{x} + nv^2$$

so that (7.46) yields

$$\int_\Omega v^2 d\mathbf{x} = -\frac{2}{n}\int_\Omega v\nabla v \cdot \mathbf{x}\, d\mathbf{x}.$$

Since Ω is bounded, using the Schwarz inequality, we get

$$\int_\Omega v^2 d\mathbf{x} = \frac{2}{n}\left|\int_\Omega v\nabla v\cdot\mathbf{x}\,d\mathbf{x}\right| \le \frac{2M}{n}\left(\int_\Omega v^2 d\mathbf{x}\right)^{1/2}\left(\int_\Omega |\nabla v|^2\,d\mathbf{x}\right)^{1/2}.$$

Simplifying, it follows that

$$\|v\|_{L^2(\Omega)} \le C_P\,\|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)}$$

with $C_P = 2M/n$. □

Inequality (7.45) implies that in $H_0^1(\Omega)$ the norm $\|u\|_{H^1(\Omega)}$ is equivalent to $\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}$. Indeed

$$\|u\|_{H^1(\Omega)} = \sqrt{\|u\|^2_{L^2(\Omega)} + \|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)}}$$

and from (7.45),

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \le \|u\|_{H^1(\Omega)} \le \sqrt{C_P^2+1}\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}.$$

When Ω is bounded, unless explicitly stated, **we will choose in $H_0^1(\Omega)$**

$$(u,v)_{H_0^1(\Omega)} = (\nabla u,\nabla v)_{L^2(\Omega;\mathbb{R}^n)} \quad \text{and} \quad \|u\|_{H_0^1(\Omega)} = \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}$$

as **inner product** and **norm**, respectively.

7.7.4 The dual of $H_0^1(\Omega)$

In the applications of the Lax-Milgram theorem to boundary value problems, the dual of $H_0^1(\Omega)$ plays an important role. In fact it deserves a special symbol.

Definition 7.58 We denote by $H^{-1}(\Omega)$ the dual of $H_0^1(\Omega)$ with the norm

$$\|F\|_{H^{-1}(\Omega)} = \sup\left\{|Fv| : v\in H_0^1(\Omega),\ \|v\|_{H_0^1(\Omega)}\le 1\right\}.$$

The first thing to observe is that, since $\mathcal{D}(\Omega)$ is dense (by definition) and continuously embedded in $H_0^1(\Omega)$, $H^{-1}(\Omega)$ *is a space of distributions*. This means two things:

a) If $F\in H^{-1}(\Omega)$, its restriction to $\mathcal{D}(\Omega)$ is a distribution.
b) If $F,G\in H^{-1}(\Omega)$ and $F\varphi = G\varphi$ for every $\varphi\in\mathcal{D}(\Omega)$, then $F=G$.

To prove a) it is enough to note that if $\varphi_k\to\varphi$ in $\mathcal{D}(\Omega)$, then $\varphi_k\to\varphi$ in $H_0^1(\Omega)$ as well, and therefore $F\varphi_k\to F\varphi$. Thus $F\in\mathcal{D}'(\Omega)$.

To prove b), let $u\in H_0^1(\Omega)$ and $\varphi_k\to u$ in $H_0^1(\Omega)$, with $\varphi_k\in\mathcal{D}(\Omega)$. Then, since $F\varphi_k = G\varphi_k$, we may write

$$Fu = \lim_{k\to+\infty} F\varphi_k = \lim_{k\to+\infty} G\varphi_k = Gu, \qquad \forall u\in H_0^1(\Omega),$$

whence $F = G$.

Thus, $H^{-1}(\Omega)$ is in one-to-one correspondence with a subspace of $\mathcal{D}'(\Omega)$ and in this sense we will write

$$H^{-1}(\Omega) \subset \mathcal{D}'(\Omega).$$

The following theorem gives a characterization of the elements of $H^{-1}(\Omega)$.

Theorem 7.59 *$H^{-1}(\Omega)$ is the set of distributions of the form*

$$F = f_0 + \operatorname{div} \mathbf{f} \tag{7.47}$$

where $f_0 \in L^2(\Omega)$ and $\mathbf{f} = (f_1, \ldots, f_n) \in L^2(\Omega; \mathbb{R}^n)$. Moreover:

$$\|F\|_{H^{-1}(\Omega)} \leq \|f_0\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}. \tag{7.48}$$

Proof Let $F \in H^{-1}(\Omega)$. From Riesz's Representation Theorem, there exists a unique $u \in H_0^1(\Omega)$ such that

$$(u, v)_{H_0^1(\Omega)} = Fv, \qquad \forall v \in H_0^1(\Omega).$$

Since Ω can be unbounded we use in $H_0^1(\Omega)$ the full inner product in $H^1(\Omega)$ and we get:

$$(u, v)_{H_0^1(\Omega)} = (\nabla u, \nabla v)_{L^2(\Omega;\mathbb{R}^n)} + (u, v)_{L^2(\Omega)} = -\langle \operatorname{div} \nabla u, v\rangle + (u, v)_{L^2(\Omega)}$$

in $\mathcal{D}'(\Omega)$; it follows that (7.47) holds with $f_0 = u$ and $\mathbf{f} = -\nabla u$. Moreover,

$$\|F\|_{H^{-1}(\Omega)} = \|u\|_{H_0^1(\Omega)} = (\|f_0\|^2_{L^2(\Omega)} + \|\mathbf{f}\|^2_{L^2(\Omega;\mathbb{R}^n)})^{1/2} \leq \|f_0\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}.$$

Viceversa, let $F = f_0 + \operatorname{div} \mathbf{f}$, with $f_0 \in L^2(\Omega)$ and $\mathbf{f} = L^2(\Omega; \mathbb{R}^n)$. Then $F \in \mathcal{D}'(\Omega)$ and, letting $\langle F, v\rangle = Fv$, we have;

$$Fv = \int_\Omega f_0 v \, d\mathbf{x} - \int_\Omega \mathbf{f} \cdot \nabla v \, d\mathbf{x}, \qquad \forall v \in \mathcal{D}(\Omega).$$

From the Schwarz inequality we have

$$|Fv| \leq \left\{\|f_0\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}\right\} \|v\|_{H_0^1(\Omega)}. \tag{7.49}$$

Thus, F is continuous in the H_0^1-norm. It remains to show that F has a unique continuous extension to all $H_0^1(\Omega)$. Take $u \in H_0^1(\Omega)$ and $\{v_k\} \subset \mathcal{D}(\Omega)$ such that $\|v_k - u\|_{H_0^1(\Omega)} \to 0$. Then, (7.49) yields

$$|Fv_k - Fv_h| \leq \left\{\|f_0\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}\right\} \|v_k - v_h\|_{H_0^1(\Omega)}.$$

Therefore $\{Fv_k\}$ is a Cauchy sequence in $\mathbb{R}$ and converges to a limit we may denote by Fu, which is independent of the sequence approximating u, as it is not difficult to check. Finally, since

$$|Fu| = \lim_{k\to\infty} |Fv_k| \quad \text{and} \quad \|u\|_{H_0^1(\Omega)} = \lim_{k\to\infty} \|v_k\|_{H_0^1(\Omega)},$$

from (7.49) we get:

$$|Fu| \leq \left\{\|f_0\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}\right\} \|u\|_{H_0^1(\Omega)}$$

showing that $F \in H^{-1}(\Omega)$ and that (7.48) holds. □

Theorem 7.59 says that the elements of $H^{-1}(\Omega)$ are represented by a linear combination of functions in $L^2(\Omega)$ and their first derivatives (in the sense of distributions). In particular, $L^2(\Omega) \hookrightarrow H^{-1}(\Omega)$ and $\left\{H_0^1(\Omega), L^2(\Omega), H^{-1}(\Omega)\right\}$ is a Hilbert triplet.

Example 7.60 If $n = 1$, the (restriction of the) Dirac δ belongs to $H^{-1}(-a, a)$. Indeed, we have $\delta = \mathcal{H}'$ in $\mathcal{D}'(-a, a)$, where $\mathcal{H}$ is the Heaviside function, and $\mathcal{H} \in L^2(-a, a)$.
However, if $n \geq 2$ and $\mathbf{0} \in \Omega$, $\delta_n \notin H^{-1}(\Omega)$. For instance, let $n = 2$ and $\Omega = B_{1/2}(\mathbf{0})$. Suppose $\delta_2 \in H^{-1}(\Omega)$. Then we could write

$$\delta_2 = f_0 + \operatorname{div} \mathbf{f}$$

for some $f_0 \in L^2(\Omega)$ and $\mathbf{f} \in L^2(\Omega; \mathbb{R}^2)$. Thus, for every $\varphi \in \mathcal{D}(\Omega)$,

$$\varphi(\mathbf{0}) = \langle \delta_2, \varphi \rangle = \langle f_0 + \operatorname{div} \mathbf{f}, \varphi \rangle = \int_\Omega [f_0 \varphi - \mathbf{f} \cdot \nabla \varphi] \, d\mathbf{x}.$$

From the Schwarz inequality, it follows that

$$|\varphi(\mathbf{0})| \leq \left\{ \|f_0\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)} \right\} \|\varphi\|_{H_0^1(\Omega)}$$

and, using the density of $\mathcal{D}(\Omega)$ in $H_0^1(\Omega)$, this estimate should hold for any $u \in H_0^1(\Omega)$ as well. But this is impossible, since in $H_0^1(\Omega)$ there are functions which are unbounded near the origin, as we have seen in Example 7.54, p. 454.

Example 7.61 Let Ω be a smooth, bounded domain in $\mathbb{R}^n$. Let $u = \chi_\Omega$ be its characteristic function. Since $\chi_\Omega \in L^2(\mathbb{R}^n)$, the distribution $\mathbf{F} = \nabla \chi_\Omega$ belongs to $H^{-1}(\mathbb{R}^n; \mathbb{R}^n)$. The support of $\mathbf{F} = \nabla \chi_\Omega$ coincides with $\partial\Omega$ and its action on a test $\boldsymbol{\varphi} \in \mathcal{D}(\mathbb{R}^n; \mathbb{R}^n)$ is described by the following formula:

$$\langle \nabla \chi_\Omega, \boldsymbol{\varphi} \rangle = -\int_{\mathbb{R}^n} \chi_\Omega \operatorname{div} \boldsymbol{\varphi} \, d\mathbf{x} = -\int_{\partial\Omega} \boldsymbol{\varphi} \cdot \boldsymbol{\nu} \, d\sigma.$$

Thus $\langle \nabla \chi_\Omega, \boldsymbol{\varphi} \rangle$ gives the inward flux of the vector field $\boldsymbol{\varphi}$ through $\partial\Omega$.

It is important to avoid confusion between $H^{-1}(\Omega)$ and $H^1(\Omega)^*$, the dual of $H^1(\Omega)$. Since, in general, $\mathcal{D}(\Omega)$ is **not dense** in $H^1(\Omega)$, the space $H^1(\Omega)^*$ is **not** a space of distributions. Indeed, although the restriction to $\mathcal{D}(\Omega)$ of every $T \in H^1(\Omega)^*$ is a distribution, this restriction does not identify T. As a simple example, take a constant $\mathbf{f} \in \mathbb{R}^n$ and define

$$Tu = \int_\Omega \mathbf{f} \cdot \nabla u \, d\mathbf{x}.$$

Since

$$|Tu| \leq \sqrt{|\Omega|} \, |\mathbf{f}| \, \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)},$$

we infer that $T \in H^1(\Omega)^*$. However, the restriction of T to $\mathcal{D}(\Omega)$ is the zero distribution, since in $\mathcal{D}'(\Omega)$ we have

$$\langle T, \varphi \rangle = -\langle \operatorname{div} \mathbf{f}, \varphi \rangle = 0, \qquad \forall \varphi \in \mathcal{D}(\Omega).$$

7.7.5 The spaces $H^m(\Omega)$, $m > 1$

By involving higher order derivatives, we may construct new Sobolev spaces. Let N be the number of multi-indexes $\alpha = (\alpha_1, \ldots, \alpha_n)$ such that $|\alpha| = \sum_{i=1}^n \alpha_i \leq m$. Choose in Theorem 7.52

$$H = L^2(\Omega), \qquad Z = L^2(\Omega; \mathbb{R}^N) \subset \mathcal{D}'(\Omega; \mathbb{R}^N),$$

and $L : L^2(\Omega) \to \mathcal{D}'(\Omega; \mathbb{R}^N)$ given by

$$Lv = \{D^\alpha v\}_{|\alpha| \leq m}.$$

Then W is the Sobolev space of the functions in $L^2(\Omega)$, *whose derivatives* (in the sense of distributions) *up to order m included, are functions in* $L^2(\Omega)$. For this space we use the symbol $H^m(\Omega)$. Thus:

$$H^m(\Omega) = \left\{v \in L^2(\Omega) : D^\alpha v \in L^2(\Omega), \quad \forall \alpha : |\alpha| \leq m\right\}.$$

From Theorem 7.52 and the separability of $L^2(\Omega)$, we deduce:

Proposition 7.62 *$H^m(\Omega)$ is a separable Hilbert space, continuously embedded in $L^2(\Omega)$. The operators D^α, $|\alpha| \leq m$, are continuous from $H^m(\Omega)$ into $L^2(\Omega)$.*

The inner product and the norm in $H^m(\Omega)$ are given, respectively, by

$$(u, v)_{H^m(\Omega)} = \sum_{|\alpha| \leq m} \int_\Omega D^\alpha u D^\alpha v \, d\mathbf{x}$$

and

$$\|u\|^2_{H^m(\Omega)} = \sum_{|\alpha| \leq m} \int_\Omega |D^\alpha u|^2 \, d\mathbf{x}.$$

If $u \in H^m(\Omega)$, any derivative of u of order $k \leq m$ belongs to $H^{m-k}(\Omega)$ and $H^m(\Omega) \hookrightarrow H^{m-k}(\Omega)$, $k \geq 1$.

Example 7.63 Let $B_{1/2}(\mathbf{0}) \subset \mathbb{R}^3$ and consider $u(\mathbf{x}) = |\mathbf{x}|^{-a}$. It is easy to check (see Problem 7.24) that $u \in H^1(B_{1/2}(\mathbf{0}))$ if $a < 1/2$. The second order derivatives of u are given by:

$$u_{x_i x_j} = a(a+2) x_i x_j \, |\mathbf{x}|^{-a-4} - a\delta_{ij} \, |\mathbf{x}|^{-a-2}.$$

Then

$$\left|u_{x_i x_j}\right| \leq |a(a+2)| \, |\mathbf{x}|^{-a-2}$$

so that $u_{x_i x_j} \in L^2(B_{1/2}(0))$ if $2a + 4 < 3$, or $a < -\frac{1}{2}$. As in Example 7.54, p. 454, one can show that the $u_{x_i x_j}$ also are the distributional second derivatives of u. Thus $u \in H^2(B_{1/2}(\mathbf{0}))$ if $a < -1/2$.

7.7.6 Calculus rules

Most calculus rules in $H^m(\Omega)$ are formally similar to the classical ones, although some of their proofs are not so trivial. We list here a few of them.

• *Derivative of a product.* Let $u \in H^1(\Omega)$ and $v \in \mathcal{D}(\Omega)$. Then $uv \in H^1(\Omega)$ and

$$\nabla(uv) = u\nabla v + v\nabla u. \tag{7.50}$$

Formula (7.50) holds if both $u, v \in H^1(\Omega)$ as well. In this case, however,

$$uv \in L^1(\Omega) \quad \text{and} \quad \nabla(uv) \in L^1(\Omega; \mathbb{R}^n).$$

• *Change of variables*. Let $u \in H^1(\Omega)$ and $g : \Omega' \to \Omega$ be onto, one-to-one and Lipschitz, with g^{-1} Lipschitz. Then, the composition $u \circ g : \Omega' \to \mathbb{R}$ belongs to $H^1(\Omega')$ and

$$\partial_{x_i} [u \circ g](\mathbf{x}) = \sum_{k=1}^{n} \partial_{x_k} u(g(\mathbf{x})) \partial_{x_i} g_k(\mathbf{x}) \tag{7.51}$$

both a.e. in Ω and in $\mathcal{D}'(\Omega)$. In particular, the Lipschitz change of variables $y = g(\mathbf{x})$ transforms $H^1(\Omega)$ into $H^1(\Omega')$.

• *Chain rule* (see Problem 7.30). Let $u \in H^1(\Omega)$ and $f : \mathbb{R} \to \mathbb{R}$ be Lipschitz. Assume that $f(0) = 0$ if Ω is unbounded. Then, the composition $f \circ u : \Omega \to \mathbb{R}$ belongs to $H^1(\Omega)$ and

$$\nabla [f \circ u] = f'(u)\nabla u \tag{7.52}$$

both a.e. in Ω and in $\mathcal{D}'(\Omega)$. If $f(0) = 0$ and $u \in H_0^1(\Omega)$ then $f \circ u \in H_0^1(\Omega)$.

In particular, choosing respectively

$$f(t) = |t|, \quad f(t) = \max\{t, 0\} \quad \text{and} \quad f(t) = -\min\{t, 0\},$$

we deduce the following result:

Proposition 7.64 *Let $u \in H^1(\Omega)$ (resp $H_0^1(\Omega)$). Then*

$$|u|, \ u^+ = \max\{u, 0\} \ \text{ and } \ u^- = -\min\{u, 0\}$$

all belong to $H^1(\Omega)$ (resp. to $H_0^1(\Omega)$). Moreover, the following formulas hold both a.e. in Ω and in $\mathcal{D}'(\Omega)$:

$$\nabla u^+ = \nabla u \, \chi_{\{u>0\}}, \ \nabla u^- = \nabla u \, \chi_{\{u<0\}}$$

and

$$\nabla u = \nabla u^+ - \nabla u^-, \ \ \nabla(|u|) = \text{sign}(u)\nabla u.$$

Finally, let Ω be a bounded Lipschitz domain. If $u_k \rightharpoonup u$ (resp. $u_k \to u$) in $H^1(\Omega)$ then

$$u_k^{\pm} \rightharpoonup u^{\pm}, \ |u_k| \rightharpoonup |u| \quad (\text{resp. } u_k^{\pm} \to u^{\pm}, \ |u_k| \to |u|) \qquad \text{in } H^1(\Omega).$$

Proof The first part of the proposition is a direct consequence of (7.52). We first prove the final assertions for $|u|$. Let $u_k \rightharpoonup u$ in $H^1(\Omega)$. As we shall see later, by Rellich's Theorem 7.86, p. 478, the embedding of $H^1(\Omega)$ into $L^2(\Omega)$ is compact. Then, by Proposition 6.63, p. 390, $u_k \to u$ strongly in $L^2(\Omega)$. Since

$$\| |u_k| - |u| \|_{L^2(\Omega)} \le \|u_k - u\|_{L^2(\Omega)}$$

we see that $|u_k| \to |u|$ in $L^2(\Omega)$. On the other hand, $\{|u_k|\}$ is bounded in $H^1(\Omega)$, thus there exists a subsequence $\{|u_{k_j}|\}$ weakly convergent in $H^1(\Omega)$. But then the limit must be $|u|$ and we conclude that the whole sequence $\{|u_k|\}$ converges weakly to $|u|$ in $H^1(\Omega)$.

Let now $u_k \to u$ in $H^1(\Omega)$. We have just seen that

$$\nabla(|u_k|) \rightharpoonup \nabla(|u|)$$

in $L^2(\Omega)$. Moreover

$$|\nabla(|u_k|)| = |\nabla u_k| \to |\nabla u| = |\nabla(|u|)|$$

in $L^2(\Omega)$. We conclude (see Problem 6.17) that

$$\nabla(|u_k|) \longrightarrow \nabla(|u|)$$

in $L^2(\Omega)$ and therefore $|u_k| \longrightarrow |u|$ in $H^1(\Omega)$. The statements for u^+ and u^- follows from those for $|u|$ and the formulas

$$u^+ = \frac{u + |u|}{2} \qquad \text{and} \qquad u^- = \frac{u - |u|}{2}. \qquad \square$$

Remark 7.65 As a consequence of Proposition 7.64, if $u \in H^1(\Omega)$ is constant in a measurable set $K \subseteq \Omega$, then $\nabla u = \mathbf{0}$ a.e. in K.

7.7.7 Fourier transform and Sobolev spaces

The spaces $H^m(\mathbb{R}^n)$, $m \ge 1$, may be defined in terms of the Fourier transform. In fact, by Theorem 7.50, p. 450,

$$u \in L^2(\mathbb{R}^n) \qquad \text{if and only if} \qquad \widehat{u} \in L^2(\mathbb{R}^n)$$

and

$$\|u\|^2_{L^2(\mathbb{R}^n)} = (2\pi)^{-n} \|\widehat{u}\|^2_{L^2(\mathbb{R}^n)}.$$

It follows that, for every multi-index α with $|\alpha| \le m$,

$$D^\alpha u \in L^2(\mathbb{R}^n) \qquad \text{if and only if} \qquad \boldsymbol{\xi}^\alpha \widehat{u} \in L^2(\mathbb{R}^n)$$

and

$$\|D^\alpha u\|^2_{L^2(\mathbb{R}^n)} = (2\pi)^{-n} \|\boldsymbol{\xi}^\alpha \widehat{u}\|^2_{L^2(\mathbb{R}^n)}.$$

Observe now that for all α, $0 \le |\alpha| \le m$, there exists a positive constant C such that:

$$|\boldsymbol{\xi}^\alpha|^2 \le |\boldsymbol{\xi}|^{2|\alpha|} \le (1 + |\boldsymbol{\xi}|^2)^m \le C(1 + \sum_{0 \le |\alpha| \le m} |\boldsymbol{\xi}^\alpha|^2), \quad \forall \boldsymbol{\xi} \in \mathbb{R}^n.$$

The first two inequalities follow from $\alpha_j \le |\alpha| \le m$, while the third one from the binomial formula. Thus we obtain the following result.

Proposition 7.66 *Let $u \in L^2(\mathbb{R}^n)$. Then:*

a) $u \in H^m(\mathbb{R}^n)$ if and only if

$$(1 + |\boldsymbol{\xi}|^2)^{m/2} \widehat{u} \in L^2(\mathbb{R}^n).$$

b) The norms

$$\|u\|_{H^m(\mathbb{R}^n)} \qquad \text{and} \qquad \left\|(1 + |\boldsymbol{\xi}|^2)^{m/2} \widehat{u}\right\|_{L^2(\mathbb{R}^n)}$$

are equivalent.

- *Sobolev spaces of real order*. The norm

$$\|u\|_{H^m(\mathbb{R}^n)} = \left\|(1 + |\boldsymbol{\xi}|^2)^{m/2} \widehat{u}\right\|_{L^2(\mathbb{R}^n)} \tag{7.53}$$

makes perfect sense even if m is not an integer and we are led to the following definition.

Definition 7.67 Let $s \in \mathbb{R}$, $0 < s < \infty$. We denote by $H^s(\mathbb{R}^n)$ the space of functions $u \in L^2(\mathbb{R}^n)$ such that

$$|\boldsymbol{\xi}|^s \, \widehat{u} \in L^2(\mathbb{R}^n).$$

Formally, saying that

$$\left|\xi_j\right|^s \hat{u} \in L^2(\mathbb{R}^n)$$

amounts to saying that the "derivative of order s with respect to x_j" of u belongs to $L^2(\mathbb{R}^n)$. Then

$$u \in H^s(\mathbb{R}^n)$$

if all the "derivatives of order s" of u belong to $L^2(\mathbb{R}^n)$. We have:

Proposition 7.68 *$H^s(\mathbb{R}^n)$ is a Hilbert space with inner product and norm given by*

$$(u, v)_{H^s(\mathbb{R}^n)} = \int_{\mathbb{R}^n} (1 + |\boldsymbol{\xi}|^2)^s \widehat{u}\, \overline{\widehat{v}}\, d\xi$$

and

$$\|u\|_{H^s(\mathbb{R}^n)} = \left\|(1 + |\boldsymbol{\xi}|^2)^{s/2} \widehat{u}\right\|_{L^2(\mathbb{R}^n)}.$$

The space $H^{1/2}(\mathbb{R}^n)$ of the L^2-functions possessing "half derivatives" in $L^2(\mathbb{R}^n)$ plays an important role in Sect. 7.9.

7.8 Approximations by Smooth Functions and Extensions

7.8.1 Local approximations

The functions in $H^1(\Omega)$ may be quite irregular. However, using mollifiers, any $u \in H^1(\Omega)$ may be approximated *locally* by smooth functions, in the sense that the approximation holds in every compact subset of Ω.

Denote by $\eta_\varepsilon = \frac{1}{\varepsilon^n}\eta(\frac{\mathbf{x}}{\varepsilon})$ the mollifier introduced in Sect. 7.2. For u defined in Ω, denote by $\tilde{u}$ the zero extension of u outside Ω. We have:

Theorem 7.69 *Let $u \in H^1(\Omega)$ and, for $\varepsilon > 0$, small, define*

$$u_\varepsilon = \eta_\varepsilon * \tilde{u}.$$

Then, if $\varepsilon \to 0$,

1. *$u_\varepsilon \in C^\infty(\mathbb{R}^n)$ and $u_\varepsilon \to u$ in $L^2(\Omega)$.*
2. *$\nabla u_\varepsilon \to \nabla u$ in $L^2(\Omega';\mathbb{R}^n)$, for every $\Omega' \subset\subset \Omega$.*

Proof Property 1 follows from Lemma 7.2, b and d, p. 422. To prove 2, note first that, if $\varepsilon < \text{dist}(\Omega', \partial\Omega)$ and $x \in \Omega'$, the function $y \longmapsto \eta_\varepsilon(\mathbf{x} - \mathbf{y})$ belongs to $\mathcal{D}(\Omega)$. Then, for every $j = 1, 2, \ldots, n$, we have

$$\partial_{x_j} u_\varepsilon(\mathbf{x}) = \int_\Omega \partial_{x_j}\eta_\varepsilon(\mathbf{x} - \mathbf{y})u(\mathbf{y})d\mathbf{y} = -\int_\Omega \partial_{y_j}\eta_\varepsilon(\mathbf{x} - \mathbf{y})u(\mathbf{y})d\mathbf{y}$$

and finally, integrating by parts,

$$\partial_{x_j} u_\varepsilon(\mathbf{x}) = \int_\Omega \eta_\varepsilon(\mathbf{x} - \mathbf{y})\partial_{y_j}u(\mathbf{y})d\mathbf{y} = (\partial_{y_j}u)_\varepsilon(\mathbf{x}). \tag{7.54}$$

Then, 2 follows from property d of Lemma 7.2, applied to Ω'. □

7.8.2 Extensions and global approximations

By Theorem 7.69, we may approximate a function in $H^1(\Omega)$ by smooth functions, as long as we stay at positive distance from $\partial\Omega$. We wonder whether an approximation is possible in all $\overline{\Omega}$. First we give the following definition.

Definition 7.70 We denote by $\mathcal{D}(\overline{\Omega})$ the set of restrictions to $\overline{\Omega}$ of functions in $\mathcal{D}(\mathbb{R}^n)$.

Thus, $\varphi \in \mathcal{D}(\overline{\Omega})$ if there is $\psi \in \mathcal{D}(\mathbb{R}^n)$ such that $\varphi = \psi$ in $\overline{\Omega}$. Clearly, $\mathcal{D}(\overline{\Omega}) \subset C^\infty(\overline{\Omega})$. We want to establish whether

$$\mathcal{D}(\overline{\Omega}) \text{ **is dense in** } H^1(\Omega). \tag{7.55}$$

The case $\Omega = \mathbb{R}^n$ is special, since $\mathcal{D}(\Omega)$ coincides with $\mathcal{D}(\overline{\Omega})$. We have:

Theorem 7.71 *$\mathcal{D}(\mathbb{R}^n)$ is dense in $H^1(\mathbb{R}^n)$. In particular $H^1(\mathbb{R}^n) = H_0^1(\mathbb{R}^n)$.*

Proof First observe that $H_c^1(\mathbb{R}^n)$, the subspace of functions with compact (essential) support in $\mathbb{R}^n$, is dense in $H^1(\mathbb{R}^n)$. In fact, let $u \in H^1(\mathbb{R}^n)$ and $v \in \mathcal{D}(\mathbb{R}^n)$, such that $0 \le v \le 1$ and $v \equiv 1$ if $|\mathbf{x}| \le 1$. Define

$$u_s(\mathbf{x}) = v\left(\frac{\mathbf{x}}{s}\right) u(\mathbf{x}).$$

Then $u_s \in H_c^1(\mathbb{R}^n)$ and

$$\nabla u_s(\mathbf{x}) = v\left(\frac{\mathbf{x}}{s}\right) \nabla u(\mathbf{x}) + \frac{1}{s} u(\mathbf{x}) \nabla v\left(\frac{\mathbf{x}}{s}\right).$$

From the Dominated Convergence Theorem, it follows that[13]

$$u_s \to u \quad \text{in } H^1(\mathbb{R}^n), \quad \text{as } s \to \infty.$$

On the other hand $\mathcal{D}(\mathbb{R}^n)$ is dense in $H_c^1(\mathbb{R}^n)$. In fact, if $u \in H_c^1(\mathbb{R}^n)$, we have

$$u_\varepsilon = u * \eta_\varepsilon \in \mathcal{D}(\mathbb{R}^n)$$

and $u_\varepsilon \to u$ in $H^1(\mathbb{R}^n)$. □

However, for general Ω, (7.55) is not true, as the following example shows.

Example 7.72 Consider, for instance, the planar domain in polar coordinates

$$\Omega = \{(\rho, \theta) : 0 < \rho < 1, 0 < \theta < 2\pi\}.$$

The domain Ω coincides with the open unit circle $B_1(0,0)$, centered at the origin, without the radius

$$\{(\rho, \theta) : 0 < \rho < 1, \theta = 0\}.$$

The closure $\overline{\Omega}$ is given by the full closed circle. Let

$$u(x_1, x_2) = \tilde{u}(\rho, \theta) = \rho^{1/2} \cos(\theta/2), \qquad (x_1 = \rho \cos\theta,\ x_2 = \rho \sin\theta).$$

Then $u \in L^2(\Omega)$, since u is bounded. Moreover (see Appendix B),

$$|\nabla u|^2 = \tilde{u}_\rho^2 + \frac{1}{\rho^2}\tilde{u}_\theta^2 = \frac{1}{4\rho} \qquad \text{in } \Omega,$$

so that $u \in H^1(\Omega)$. However, $\tilde{u}(\rho, 0^+) = \rho^{1/2}$ while $\tilde{u}(\rho, 2\pi^-) = -\rho^{1/2}$. Thus, u has a jump discontinuity across $\theta = 0$ and therefore no sequence of smooth functions in $\overline{\Omega}$ can converge to u in $H^1(\Omega)$.

Indeed, assume that $\{u_k\} \subset C^\infty(\overline{B_1})$ converges to u in $H^1(\Omega)$ as $k \to +\infty$. Then also

$$\int_{B_1} \nabla u_k \cdot \boldsymbol{\varphi} = \int_\Omega \nabla u_k \cdot \boldsymbol{\varphi} \to \int_\Omega \nabla u \cdot \boldsymbol{\varphi}, \qquad \forall \boldsymbol{\varphi} \in \mathcal{D}(B_1).$$

However,

$$\int_\Omega \nabla u_k \cdot \boldsymbol{\varphi} = -\int_\Omega u_k \operatorname{div} \boldsymbol{\varphi} \to -\int_\Omega u \operatorname{div} \boldsymbol{\varphi}$$

[13] Observe that $|u_s| \le |u|$ and, for $s \ge 1$, $|\nabla u_s| \le |\nabla u| + M\,|u|$, where $M = \max |\nabla v|$.

while, writing $B_1^{\pm} = B_1 \cap \{\mathbf{x} : \pm x_2 > 0\}$,

$$\int_\Omega \nabla u \cdot \boldsymbol{\varphi} = \int_{B_1^+} \nabla u \cdot \boldsymbol{\varphi} + \int_{B_1^-} \nabla u \cdot \boldsymbol{\varphi} = -\int_\Omega u \operatorname{div} \boldsymbol{\varphi} - 2\int_0^1 \sqrt{x_1}\varphi_2(x_1, 0)dx_1.$$

Contradiction.

The difficulty in Example 7.72 is that the domain Ω lies on both sides of part of its boundary (the radius $0 < \rho < 1, \theta = 0$). Hence, to hope that (7.55) is true, we have to avoid domains with this anomaly and consider domains with some degree of regularity, for $n \geq 2$.

Thus, assume that the domain Ω, for $n \geq 2$, is a C^1 or even a Lipschitz domain. Then Theorem 7.71 suggests a strategy to prove (7.55): given $u \in H^1(\Omega)$, extend the definition of u to all $\mathbb{R}^n$ in order to obtain a function in $H^1(\mathbb{R}^n)$ and then apply Theorem 7.71. The first thing to do is to introduce an extension operator:

Definition 7.73 We say that a linear operator $E : H^1(\Omega) \to H^1(\mathbb{R}^n)$ is an *extension operator* if, for all $u \in H^1(\Omega)$:

1. $E(u) = u$ in Ω;
2. If Ω is bounded, there exists a relatively compact set Ω' such that supp $E(u) \subset \Omega'$, for every u;
3. E is continuous:
$$\|Eu\|_{H^1(\mathbb{R}^n)} \leq c(n, \Omega) \|u\|_{H^1(\Omega)}.$$

How do we construct E? The first thing that comes into mind is to define $Eu = 0$ outside Ω (trivial extension). This certainly works if $u \in H_0^1(\Omega)$. In fact, it can be proved that $u \in H_0^1(\Omega)$ *if and only if its trivial extension belongs to* $H^1(\mathbb{R}^n)$.

However, the trivial extension works in this case only. For instance, let $u \in H^1(a, b)$ with, say, $u(a) = A \neq 0, u(b) = B \neq 0$. Let Eu be the trivial extension of u. Then, in $\mathcal{D}'(\mathbb{R})$, $(Eu)' = \chi_{(a,b)}u' + A\delta(x - a) - B\delta(x - b)$ which is not even a function.

Thus, we have to use another method. If Ω is a half space, that is

$$\Omega = \mathbb{R}^n_+ = \{(x_1, \ldots, x_n) : x_n > 0\} \qquad (n \geq 1)$$

an extension operator can be defined by a reflection method as follows:

• *Reflection method.* Let $u \in H^1(\mathbb{R}^n_+)$. Write $\mathbf{x} = (\mathbf{x}', x_n)$, $\mathbf{x}' \in \mathbb{R}^{n-1}$. We reflect in an even way with respect to the hyperplane $x_n = 0$, by setting $Eu = \tilde{u}$ where

$$\tilde{u}(\mathbf{x}) = u(\mathbf{x}', |x_n|).$$

Then, it is possible to prove that, in $\mathcal{D}'(\mathbb{R}^n)$:

$$\tilde{u}_{x_j}(\mathbf{x}) = \begin{cases} u_{x_j}(\mathbf{x}', |x_n|) & j < n \\ u_{x_n}(\mathbf{x}', |x_n|)\operatorname{sign} x_n & j = n. \end{cases} \tag{7.56}$$

It is now easy to check that E has the properties 1,2,3 listed above. In particular,

$$\|Eu\|^2_{H^1(\mathbb{R}^n)} = 2\,\|u\|^2_{H^1(\mathbb{R}^n_+)}\,.$$

• *Extension operator*[14], $n \geq 2$. Suppose now that Ω is a bounded *Lipschitz* domain. To construct an extension operator we use two rather general ideas, which may be applied in several different contexts: *localization* and *reduction to a half space*.

Localization. It is based on the following lemma. Given a set K, by *open covering* of K we mean a collection $\mathcal{O}$ of open sets, such that $K \subset \cup_{O\in\mathcal{O}} O$.

Lemma 7.74 (Partition of unity) *Let $K \subset \mathbb{R}^n$ be a compact set and $O_1, \ldots, O_N$ be an open covering of K. There exist functions $\psi_1, \ldots, \psi_N$ with the following properties:*

1. *For every $j = 1, \ldots, N$, $\psi_j \in C_0^\infty(O_j)$ and $0 \leq \psi_j \leq 1$.*
2. *For every $\mathbf{x} \in K$, $\sum_{j=1}^N \psi_j(\mathbf{x}) = 1$.*

Proof (sketch) Since $K \subset \cup_{j=1}^N O_j$ and each O_j is open, we can find open sets $K_j \subset\subset O_j$ such that

$$K \subset \cup_{j=1}^N K_j\,.$$

Let χ_{K_j} be the characteristic function of K_j and η_ε the mollifier (7.3), p. 422. Define $\varphi_{j,\varepsilon} = \eta_\varepsilon * \chi_{K_j}$. According to Example 7.3, p. 424, we may fix ε so small in order to have $\varphi_{j,\varepsilon} \in C_0^\infty(O_j)$ and $\varphi_{j,\varepsilon} > 0$ on K_j. Then the functions

$$\psi_j = \frac{\varphi_{j,\varepsilon}}{\sum_{s=1}^N \varphi_{s,\varepsilon}}$$

satisfy conditions 1 and 2. □

The set of functions $\psi_1, \ldots, \psi_N$ is called a *partition of unity for K, associated with the covering $O_1, \ldots, O_N$*. Now, if $u : K \to \mathbb{R}$, the localization procedure consists in writing

$$u = \sum_{j=1}^N \psi_j u \tag{7.57}$$

i.e. as a sum of functions $u_j = \psi_j u$ supported in O_j.

Reduction to a half space. Take an open covering of $\partial\Omega$ by N balls $B_j = B(\mathbf{p}_j)$, $j = 1, \ldots, N$, centered at $\mathbf{p}_j = (\mathbf{p}'_j, p_{jn}) \in \partial\Omega$ and such that $\partial\Omega \cap B_j$, up to a rotation R_j of coordinates if necessary, is given by a graph of a Lipschitz function

$$x_n = \varphi_j(\mathbf{x}'), \qquad \mathbf{x}' \in \mathcal{N}_j \subset \mathbb{R}^{N-1},$$

where $\mathcal{N}_j$ is a neighborhood of $\mathbf{p}'_j \in \mathbb{R}^{n-1}$ and $p_{jn} = \varphi_j(\mathbf{p}'_j)$. This is possible, since $\partial\Omega$ is compact. Moreover, let $A_0 \subset \Omega$ be an open set containing $\Omega \setminus \cup_{j=1}^N B_j$ (Fig. 7.5).

Then, $A_0, B_1, \ldots, B_N$ is an open covering of $\overline{\Omega}$. Let $\psi_0, \psi_1, \ldots, \psi_N$ be a partition of unity for $\overline{\Omega}$, associated with $A_0, B_1, \ldots, B_N$.

[14] For $n = 1$, see Problem 7.28.

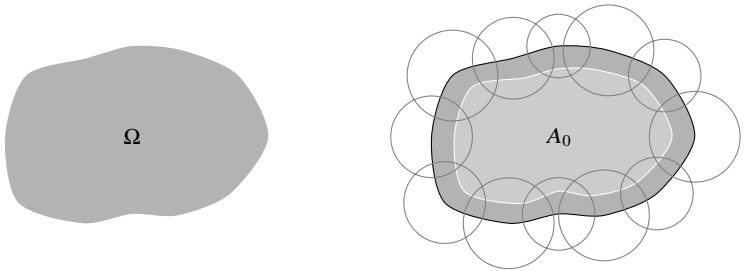

Fig. 7.5 A set Ω and an open covering of its closure

For each B_j, $1 \le j \le N$, introduce the bi-Lipschitz transformation $\mathbf{z} = \Phi_j(\mathbf{x})$, given by

$$\begin{cases} \mathbf{z}' = \mathbf{x}' \\ z_n = x_n - \varphi_j(\mathbf{x}'). \end{cases} \tag{7.58}$$

Set (Fig. 7.6)

$$\Phi_j(B_j \cap \Omega) \equiv U_j \subset \mathbb{R}^n_+$$

and

$$\Phi_j(B_j \cap \partial\Omega) \equiv \Gamma_j \subset \partial\mathbb{R}^n_+ = \{z_n = 0\}\,.$$

Let $u \in H^1(\Omega)$ and $u_j = \psi_j u$. Then,

$$w_j = u_j \circ R_J^{-1} \circ \Phi_j^{-1}$$

is supported in $U_j \cup \Gamma_j$, so that, by extending it to zero in $\mathbb{R}^n_+ \setminus U_j$, we have $w_j \in H^1(\mathbb{R}^n_+)$.

The function $Ew_j = \tilde{w}_j$, obtained by the reflection method, belongs to $H^1(\mathbb{R}^n)$. Now we go back to the original coordinates defining

$$Eu_j = \tilde{w}_j \circ \Phi_j, \qquad 1 \le j \le N,$$

in B_j and $Eu_j = 0$ outside B_j. Finally, let $u_0 = \psi_0 u_0$ and let Eu_0 be the trivial extension of u_0. Set

$$Eu = Eu_0 + \sum_{j=1}^{N} Eu_j \circ R_j.$$

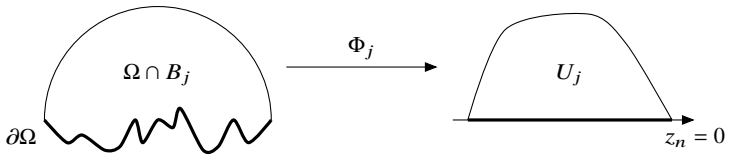

Fig. 7.6 The bi-Lipschitz transformation Φ_j *flattens* $B_j \cap \partial\Omega$

At this point, it is not difficult to show that E satisfies the requirements 1, 2, 3 of Definition 7.73. We have proved the following

Theorem 7.75 *Let Ω be either $\mathbb{R}^n_+$ or a bounded, Lipschitz domain. Then, there exists an extension operator $E : H^1(\Omega) \to H^1(\mathbb{R}^n)$.*

An immediate consequence of Theorems 7.75 and 7.71, is the following global approximation result:

Theorem 7.76 *Let Ω be either $\mathbb{R}^n_+$ or a bounded, Lipschitz domain. Then $\mathcal{D}(\overline{\Omega})$ **is dense in** $H^1(\Omega)$. In other words, if $u \in H^1(\Omega)$, there exists a sequence $\{u_m\} \subset \mathcal{D}(\overline{\Omega})$ such that*

$$\|u_m - u\|_{H^1(\Omega)} \to 0 \quad \text{as } m \to +\infty.$$

7.9 Traces

7.9.1 Traces of functions in $H^1(\Omega)$

The possibility of approximating any element $u \in H^1(\Omega)$, $\Omega \subseteq \mathbb{R}^n$, $n \geq 2$, by smooth functions in $\overline{\Omega}$ represents a key tool for introducing the notion of *restriction of u on $\Gamma = \partial\Omega$*. Such restriction is called the **trace of** u on Γ and it will be an element of $L^2(\Gamma)$. Recall that in one dimension, $H^1(a,b) \subset C([a,b])$ hence the trace of u at the end points, given by the values $u(a)$ and $u(b)$, makes perfect sense.

Observe that if $\Omega = \mathbb{R}^n_+$, then $\Gamma = \partial\mathbb{R}^n_+ = \mathbb{R}^{n-1}$ and $L^2(\Gamma)$ is well defined. If Ω is a Lipschitz domain, we define $L^2(\Gamma)$ by localization. More precisely, let $B_1, \ldots, B_N$ be an open covering of Γ by balls centered at points on Γ, as in Sect. 7.8.2. If $g : \Gamma \to \mathbb{R}$, write

$$g = \sum_{j=1}^{N} \psi_j g$$

where $\psi_1, \ldots, \psi_N$ is a partition of unity for Γ, associated with $B_1, \ldots, B_N$. Since $\Gamma \cap B_j$ is the graph of a Lipschitz function $x_n = \varphi_j(\mathbf{x}')$, $\mathbf{x}' \in \mathcal{N}_j$, on $\Gamma \cap B_j$ there is a natural notion of "surface element", given by

$$d\sigma = \sqrt{1 + \left|\nabla\varphi_j\right|^2}\, d\mathbf{x}'.$$

Thus, for $1 \leq p < \infty$, we may define

$$\int_{\Gamma\cap B_j} \psi_j\, |g|^p\, d\sigma = \int_{\mathcal{N}_j} \psi_j(\mathbf{x}', \varphi_j(\mathbf{x}'))\, \left|g(\mathbf{x}', \varphi_j(\mathbf{x}'))\right|^p \sqrt{1 + \left|\nabla\varphi_j(\mathbf{x}')\right|^2}\, d\mathbf{x}'.$$

We say that $g \in L^p(\Gamma)$ if[15]

[15] Observe that the norm (7.59) depends on the particular covering and partition of unity. However, it can be shown that norms corresponding to different coverings and partitions of unity are all equivalent and induce the same topology on $L^p(\Gamma)$.

$$\|g\|^p_{L^p(\Gamma)} = \int_\Gamma |g|^p \, d\sigma = \sum_{j=1}^N \int_{\Gamma\cap B_j} \psi_j \, |g|^p \, d\sigma < \infty. \tag{7.59}$$

With the norm (7.59) and the usual identification of functions if they are equal a.e. with respect to the surface measure $d\sigma$, $L^p(\Gamma)$ is a Banach space. $L^2(\Gamma)$ is a Hilbert space with respect to the inner product

$$(g,h)_{L^2(\Gamma)} = \sum_{j=1}^N \int_{\Gamma\cap B_j} \psi_j \, gh \, d\sigma.$$

Let us go back to our trace problem. The strategy consists in the following two steps.

Let

$$\tau_0 : \mathcal{D}(\overline{\Omega}) \to L^2(\Gamma)$$

be the operator that associates to every function v its restriction $v_{|\Gamma}$ to Γ: $\tau_0 v = v_{|\Gamma}$. This makes perfect sense, since each $v \in \mathcal{D}(\overline{\Omega})$ is continuous up to Γ.

First step. Show that $\|\tau_0 v\|_{L^2(\Gamma)} \le c(\Omega, n)\, \|v\|_{H^1(\Omega)}$. Thus, τ_0 is continuous from $\mathcal{D}(\overline{\Omega}) \subset H^1(\Omega)$ into $L^2(\Gamma)$.

Second step. Extend τ_0 to all $H^1(\Omega)$ by using the density of $\mathcal{D}(\overline{\Omega})$ in $H^1(\Omega)$.

An elementary analogy may be useful. Suppose we have a function $f : \mathbb{Q} \to \mathbb{R}$ and we want to define the value of f at an irrational point x. What do we do? Since $\mathbb{Q}$ is dense in $\mathbb{R}$, we select a sequence $\{r_k\} \subset \mathbb{Q}$ such that $r_k \to x$. Then we compute $f(r_k)$ and show that $\{f(r_n)\}$ is a Cauchy sequence (*First step*). Then set $f(x) = \lim_{k\to\infty} f(r_k)$ and prove that the limit does not depend on the approximating sequence $\{r_n\}$ (*second step*).

Theorem 7.77 *Let Ω be either $\mathbb{R}^n_+$ or a bounded, Lipschitz domain. Then there exists a linear operator (trace operator) $\tau_0 : H^1(\Omega) \to L^2(\Gamma)$ such that:*

1. $\tau_0 u = u|_\Gamma$ *if* $u \in \mathcal{D}(\overline{\Omega})$.
2. $\|\tau_0 u\|_{L^2(\Gamma)} \le c(\Omega, n)\, \|u\|_{H^1(\Omega)}$.

Proof Let $\Omega = \mathbb{R}^n_+$. First, we prove inequality 2 for $u \in \mathcal{D}(\overline{\Omega})$. In this case $\tau_0 u = u(\mathbf{x}', 0)$ and we show that

$$\int_{\mathbb{R}^{n-1}} |u(\mathbf{x}', 0)|^2 \, d\mathbf{x}' \le 2\, \|u\|^2_{H^1(\mathbb{R}^n_+)}, \qquad \forall u \in \mathcal{D}(\overline{\Omega}) \tag{7.60}$$

(a more precise estimate is given in Problem 7.31). For every $x_n \in (0,1)$ we may write:

$$u^2(\mathbf{x}', 0) = u^2(\mathbf{x}', x_n) - \int_0^{x_n} \frac{d}{dt} u^2(\mathbf{x}', t)\, dt = u^2(\mathbf{x}', x_n) - 2\int_0^{x_n} u(\mathbf{x}', t) u_{x_n}(\mathbf{x}', t)\, dt.$$

Since by the Schwarz inequality

$$\left(\int_0^1 \left| u(\mathbf{x}', t) u_{x_n}(\mathbf{x}', t) \right| dt \right)^2 \le \int_0^1 u^2(\mathbf{x}', t)\, dt \int_0^1 u^2_{x_n}(\mathbf{x}', t)\, dt,$$

we deduce that (using the elementary inequality $2ab \le a^2 + b^2$)

$$u^2(\mathbf{x}',0) \le u^2(\mathbf{x}',x_n) + \int_0^1 u^2(\mathbf{x}',t)\,dt + \int_0^1 u_{x_n}^2(\mathbf{x}',t)\,dt$$
$$\le u^2(\mathbf{x}',x_n) + \int_0^1 u^2(\mathbf{x}',t)\,dt + \int_0^1 |\nabla u(\mathbf{x}',t)|^2\,dt.$$

Integrating both sides on $\mathbb{R}^{n-1}$ with respect to $\mathbf{x}'$ and on $(0,1)$ with respect to x_n, we easily obtain (7.60).

Assume now $u \in H^1(\mathbb{R}^n_+)$. Since $\mathcal{D}(\overline{\Omega})$ is dense in $H^1(\mathbb{R}^n_+)$, we can select $\{u_k\} \subset \mathcal{D}(\overline{\Omega})$ such that $u_k \to u$ in $H^1(\mathbb{R}^n_+)$.

The linearity of τ_0 and estimate (7.60) yield

$$\|\tau_0 u_h - \tau_0 u_k\|_{L^2(\mathbb{R}^{n-1})} \le \sqrt{2}\,\|u_h - u_k\|_{H^1(\mathbb{R}^n_+)}.$$

Since $\{u_k\}$ is a Cauchy sequence in $H^1(\mathbb{R}^n_+)$, we infer that $\{\tau_0 u_k\}$ is a Cauchy sequence in $L^2(\mathbb{R}^{n-1})$. Therefore, there exists $u_0 \in L^2(\mathbb{R}^{n-1})$ such that

$$\tau_0 u_k \to u_0 \qquad \text{in } L^2(\mathbb{R}^{n-1}).$$

The limiting element u_0 does not depend on the approximating sequence $\{u_k\}$. In fact, if $\{v_k\} \subset \mathcal{D}(\overline{\Omega})$ and $v_k \to u$ in $H^1(\mathbb{R}^n_+)$, then

$$\|v_k - u_k\|_{H^1(\mathbb{R}^n_+)} \to 0.$$

From

$$\|\tau_0 v_k - \tau_0 u_k\|_{L^2(\mathbb{R}^{n-1})} \le \sqrt{2}\,\|v_k - u_k\|_{H^1(\mathbb{R}^n_+)}$$

it follows that $\tau_0 v_k \to u_0$ in $L^2(\mathbb{R}^{n-1})$ as well.

Thus, if $u \in H^1(\mathbb{R}^n_+)$, it makes sense to define $\tau_0 u = u_0$. It should be clear that τ_0 has the properties 1, 2.

If Ω is a bounded Lipschitz domain, the theorem can be proved once more by localization and reduction to a half space. We omit the details. □

Definition 7.78 The function $\tau_0 u$ is called the *trace* of u on Γ.

The following integration by parts formula for functions in $H^1(\Omega)$ is a consequence of the trace Theorem 7.77.

Corollary 7.79 *Assume Ω is either $\mathbb{R}^n_+$ or a bounded, Lipschitz domain. Let $u \in H^1(\Omega)$ and $\mathbf{v} \in H^1(\Omega;\mathbb{R}^n)$. Then*

$$\int_\Omega \nabla u \cdot \mathbf{v}\,d\mathbf{x} = -\int_\Omega u\ \operatorname{div}\mathbf{v}\,d\mathbf{x} + \int_\Gamma (\tau_0 u)(\tau_0 \mathbf{v})\cdot \boldsymbol{\nu}\,d\sigma, \tag{7.61}$$

where $\boldsymbol{\nu}$ is the outward unit normal to Γ and $\tau_0\mathbf{v} = (\tau_0 v_1, \ldots, \tau_0 v_n)$.

Proof Formula (7.61) holds if $u \in \mathcal{D}(\overline{\Omega})$ and $\mathbf{v} \in \mathcal{D}(\overline{\Omega};\mathbb{R}^n)$. Let $u \in H^1(\Omega)$ and $\mathbf{v} \in H^1(\Omega;\mathbb{R}^n)$. Select $\{u_k\} \subset \mathcal{D}(\overline{\Omega})$, $\{\mathbf{v}_k\} \subset \mathcal{D}(\overline{\Omega};\mathbb{R}^n)$ such that $u_k \to u$ in $H^1(\Omega)$ and $\mathbf{v}_k \to \mathbf{v}$ in $H^1(\Omega;\mathbb{R}^n)$. Then:

$$\int_\Omega \nabla u_k \cdot \mathbf{v}_k\,d\mathbf{x} = -\int_\Omega u_k \operatorname{div}\mathbf{v}_k\,d\mathbf{x} + \int_\Gamma (\tau_0 u_k)\ (\tau_0 \mathbf{v}_k)\cdot \boldsymbol{\nu}\,d\sigma.$$

Letting $k \to \infty$, by the continuity of τ_0, we obtain (7.61). □

It is not surprising that the kernel of τ_0 is precisely $H^1_0(\Omega)$:

$$\tau_0 u = 0 \iff u \in H_0^1(\Omega).$$

However, only the proof of the " $\Longleftarrow$ " part is trivial. The proof of the "$\Longrightarrow$" part is rather technical and we omit it.

For the trace $\tau_0 u$ we also use the notation $u_{|\Gamma}$ and, if there is no risk of confusion, inside a boundary integral, we will write simply

$$\int_{\partial\Omega} u \, d\sigma.$$

In a similar way, we may define the trace of $u \in H^1(\Omega)$ on a relatively open subset $\Gamma_0 \subset \Gamma$, regular in the sense of Definition 1.11, p. 19.

Theorem 7.80 *Assume Ω is either $\mathbb{R}^n_+$ or a bounded, Lipschitz domain. Let Γ_0 be an open subset of Γ. Then there exists a trace operator $\tau_{\Gamma_0} : H^1(\Omega) \to L^2(\Gamma_0)$ such that:*

1. $\tau_{\Gamma_0} u = u|_{\Gamma_0}$ *if* $u \in \mathcal{D}(\overline{\Omega})$.
2. $\left\|\tau_{\Gamma_0} u\right\|_{L^2(\Gamma_0)} \leq c(\Omega, n) \, \|u\|_{H^1(\Omega)}$.

The function $\tau_{\Gamma_0} u$ is called *the trace of* u on Γ_0, often denoted by $u_{|\Gamma_0}$. The kernel of τ_{Γ_0} is denoted by $H^1_{0,\Gamma_0}(\Omega)$:

$$\tau_{\Gamma_0} u = 0 \iff u \in H^1_{0,\Gamma_0}(\Omega).$$

This space can be characterized in another way. Let V_{0,Γ_0} be the set of functions in $\mathcal{D}(\overline{\Omega})$ vanishing in a neighborhood of $\overline{\Gamma}_0$. Then:

Proposition 7.81 $H^1_{0,\Gamma_0}(\Omega)$ *is the closure of* V_{0,Γ_0} *in* $H^1(\Omega)$.

7.9.2 Traces of functions in $H^m(\Omega)$

We have seen that a function $u \in H^m(\mathbb{R}^n_+)$, $m \geq 1$, has a trace on $\Gamma = \partial\mathbb{R}^n_+$. However, if $m = 2$, every derivative of u belongs to $H^1(\mathbb{R}^n_+)$, so that it has a trace on Γ. In particular, we may define the trace of $\partial_{x_n} u$ on Γ. Let

$$\tau_1 u = (\partial_{x_n} u)_{|\Gamma}.$$

In general, for $m \geq 2$, we may define the trace on Γ of the derivatives $\partial^j_{x_n} u = \frac{\partial^j u}{\partial x_n^j}$ for $j = 0, 1, \ldots, m-1$ and set

$$\tau_j u = (\partial^j_{x_n} u)_{|\Gamma}.$$

In this way, we construct a linear operator $\tau : H^m(\mathbb{R}^n_+) \to L^2(\Gamma; \mathbb{R}^m)$, given by

$$\tau u = (\tau_0 u, \ldots, \tau_{m-1} u).$$

From Theorem 7.77, τ satisfies the following conditions:

1. $\tau u = (u_{|\Gamma}, (\partial_{x_n} u)_{|\Gamma}, \ldots, (\partial_{x_n}^{m-1} u)_{|\Gamma})$, if $u \in \mathcal{D}(\overline{\mathbb{R}^n_+})$.
2. $\|\tau u\|_{L^2(\Gamma;\mathbb{R}^m)} \leq c \, \|u\|_{H^m(\mathbb{R}^n_+)}$.

The operator τ associates to $u \in H^m(\mathbb{R}^n_+)$ the trace on Γ of u and its derivatives up to the order $m-1$, in the positive direction of the x_n axis. This direction corresponds to the *interior normal* to $\Gamma = \partial\mathbb{R}^n_+$.

Analogously, for a bounded domain Ω we may define the trace on Γ of the (interior or exterior) normal derivatives of u, up to order $m-1$. This requires Ω to be at least a C^m-domain, because of the flattering procedure. The following theorem holds, where $\boldsymbol{\nu}$ denotes the *exterior unit normal* to $\partial\Omega$.

Theorem 7.82 *Assume Ω is either $\mathbb{R}^n_+$ or a bounded, C^m-domain, $m \geq 2$. Then there exists a trace operator $\tau : H^m(\Omega) \to L^2(\Gamma;\mathbb{R}^m)$ such that:*

1. $\tau u = (u_{|\Gamma}, \frac{\partial u}{\partial \boldsymbol{\nu}}_{|\Gamma}, \ldots, \frac{\partial^{m-1} u}{\partial \boldsymbol{\nu}^{m-1}}_{|\Gamma})$ *if* $u \in \mathcal{D}(\overline{\Omega})$.
2. $\|\tau u\|_{L^2(\Gamma;\mathbb{R}^m)} \leq c(\Omega, n) \, \|u\|_{H^m(\Omega)}$.

Similarly, we may define a trace of the (interior or exterior) normal derivatives of u, up to order $m-1$, on an open regular subset $\Gamma_0 \subset \Gamma$.

It turns out that the kernel of the operator τ is given by the *closure of* $\mathcal{D}(\Omega)$ in $H^m(\Omega)$, denoted by $H^m_0(\Omega)$. Precisely:

$$\tau u = (0, \ldots, 0) \quad \Longleftrightarrow \quad u \in H^m_0(\Omega).$$

Clearly, $H^m_0(\Omega)$ is a Hilbert subspace of $H^m(\Omega)$. If $u \in H^m_0(\Omega)$, u and its normal derivatives up to order $m-1$ have zero trace on Γ.

7.9.3 Trace spaces

The operator $\tau_0 : H^1(\Omega) \to L^2(\Gamma)$ is not surjective. In fact the image of τ_0 is *strictly contained* in $L^2(\Gamma)$. In other words, there are functions in $L^2(\Gamma)$ which are not traces of functions in $H^1(\Omega)$. So, the natural question is: which functions $g \in L^2(\Gamma)$ are traces of functions in $H^1(\Omega)$? The answer is not elementary: roughly speaking, we could characterize them as *functions possessing half derivatives in* $L^2(\Gamma)$ (see Sect. 7.7.7, p. 463). It is as if in the restriction to the boundary, a function of $H^1(\Omega)$ loses "one half of each derivative". To give an idea of what this means, let us consider the case $\Omega = \mathbb{R}^n_+$, $n \geq 2$. First a density lemma. For the proof, see Problem 7.34.

Lemma 7.83 *$\mathcal{D}(\mathbb{R}^n)$ is dense into $H^{1/2}(\mathbb{R}^n)$.*

Theorem 7.84 *The linear operator $\tau_0 : H^1(\mathbb{R}^n_+) \to H^{1/2}(\mathbb{R}^{n-1})$ is continuous and*

$$\|\tau_0 u\|_{H^{1/2}(\Gamma)} \leq \frac{1}{2\pi} \, \|u\|_{H^1(\mathbb{R}^n_+)} \, . \tag{7.62}$$

*Moreover, there exists a linear and continuous operator (**lifting operator**) $\mathcal{P}$: $H^{1/2}(\mathbb{R}^{n-1}) \to H^1(\mathbb{R}^n_+)$ such that*

$$\tau_0(\mathcal{P}g) = g, \qquad \forall g \in H^{1/2}(\mathbb{R}^{n-1}).$$

In particular, $\mathcal{R}(\tau_0) = H^{1/2}(\mathbb{R}^{n-1})$.

Proof Let $u \in H^1(\mathbb{R}^n_+)$ and $g = \tau_0 u$. First we show the estimate (7.62). This implies that $\mathcal{R}(\tau_0) \subseteq H^{1/2}(\mathbb{R}^{n-1})$ and that $\tau_0 : H^1(\mathbb{R}^n_+) \to H^{1/2}(\mathbb{R}^{n-1})$ is continuous.

By the density of $\mathcal{D}(\overline{\mathbb{R}^n_+})$ into $H^1(\mathbb{R}^n_+)$ it is enough to show (7.62) for $u \in \mathcal{D}(\overline{\mathbb{R}^n_+})$. Thus let $u \in \mathcal{D}(\overline{\mathbb{R}^n_+})$ and extend it to all $\mathbb{R}^n$ by even reflection with respect to the plane $x_n = 0$. We write the points in $\mathbb{R}^n$ as $x = (\mathbf{x}', x_n)$, with $\mathbf{x}' = (x_1, \dots, x_{n-1})$. Now we express $\widehat{g}$ in terms of $\widehat{u}$. By the Fourier inversion formula, we can write

$$u(\mathbf{x}', x_n) = \frac{1}{(2\pi)^n} \int_{\mathbb{R}^{n-1}} e^{i\mathbf{x}'\cdot\boldsymbol{\xi}'} \left(\int_{\mathbb{R}} \widehat{u}(\boldsymbol{\xi}', \xi_n) e^{ix_n\xi_n} d\xi_n \right) d\boldsymbol{\xi}'$$

so that

$$g(\mathbf{x}') = u(\mathbf{x}', 0) = \frac{1}{(2\pi)^{n-1}} \int_{\mathbb{R}^{n-1}} e^{i\mathbf{x}'\cdot\boldsymbol{\xi}'} \left(\frac{1}{2\pi} \int_{\mathbb{R}} \widehat{u}(\boldsymbol{\xi}', \xi_n) d\xi_n \right) d\boldsymbol{\xi}'.$$

This shows that

$$\widehat{g}(\boldsymbol{\xi}') = \frac{1}{2\pi} \int_{\mathbb{R}} \widehat{u}(\boldsymbol{\xi}', \xi_n) d\xi_n.$$

Thus:

$$\|g\|^2_{H^{1/2}(\mathbb{R}^{n-1})} = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^{n-1}} (1 + |\boldsymbol{\xi}'|^2)^{1/2} \left| \int_{\mathbb{R}} \widehat{u}(\boldsymbol{\xi}', \xi_n) d\xi_n \right|^2 d\boldsymbol{\xi}'.$$

Now note the following two facts. First, from the Schwarz inequality, we can write

$$\begin{aligned} \left| \int_{\mathbb{R}} \widehat{u}(\boldsymbol{\xi}', \xi_n) d\xi_n \right| &\le \int_{\mathbb{R}} (1 + |\boldsymbol{\xi}|^2)^{-1/2} (1 + |\boldsymbol{\xi}|^2)^{1/2} \, |\widehat{u}(\boldsymbol{\xi}', \xi_n)| \, d\xi_n \\ &\le \left(\int_{\mathbb{R}} (1 + |\boldsymbol{\xi}|^2) \, |\widehat{u}(\boldsymbol{\xi}', \xi_n)|^2 \, d\xi_n \right)^{1/2} \left(\int_{\mathbb{R}} (1 + |\boldsymbol{\xi}|^2)^{-1} d\xi_n \right)^{1/2}. \end{aligned}$$

Second, since $\int_{\mathbb{R}} (a^2 + t^2)^{-1} dt = \left[\frac{1}{a} \arctan(\frac{t}{a}) \right]_{-\infty}^{+\infty} = \frac{\pi}{|a|}$,

$$\int_{\mathbb{R}} (1 + |\boldsymbol{\xi}|^2)^{-1} d\xi_n = \int_{\mathbb{R}} (1 + |\boldsymbol{\xi}'|^2 + \xi_n^2)^{-1} d\xi_n = \frac{\pi}{(1 + |\boldsymbol{\xi}'|^2)^{1/2}}.$$

Thus,

$$\|g\|^2_{H^{1/2}(\mathbb{R}^{n-1})} \le \frac{1}{4\pi} \int_{\mathbb{R}^n} (1 + |\boldsymbol{\xi}|^2) \, |\widehat{u}(\boldsymbol{\xi})|^2 \, d\boldsymbol{\xi} = \frac{1}{4\pi} \|u\|^2_{H^1(\mathbb{R}^n)} = \frac{1}{2\pi} \|u\|^2_{H^1(\mathbb{R}^n_+)}$$

which is (7.62).

It remains to construct a lifting operator $\mathcal{P}$. Take any $g \in \mathcal{D}(\mathbb{R}^{n-1})$ and define

$$\mathcal{P}g(\mathbf{x}', x_n) = u(\mathbf{x}', x_n) = \frac{1}{(2\pi)^{n-1}} \int_{\mathbb{R}^{n-1}} e^{-(1+|\boldsymbol{\xi}'|^2)^{1/2} x_n} \widehat{g}(\boldsymbol{\xi}') e^{i\mathbf{x}'\cdot\boldsymbol{\xi}'} d\boldsymbol{\xi}' \qquad x_n \ge 0.$$

Then, clearly $\tau_0(\mathcal{P}g)(\mathbf{x}') = u(\mathbf{x}', 0) = g(\mathbf{x}')$ and the partial Fourier transform of u with respect to $\mathbf{x}'$ is

$$\hat{u}(\boldsymbol{\xi}', x_n) = e^{-(1+|\boldsymbol{\xi}'|^2)^{1/2} x_n} \widehat{g}(\boldsymbol{\xi}').$$

From (7.39),

$$\|\mathcal{P}g\|^2_{H^1(\mathbb{R}^n_+)} = \frac{1}{(2\pi)^{n-1}} \int_0^{+\infty} \int_{\mathbb{R}^{n-1}} (1+|\xi'|^2) e^{-2(1+|\xi'|^2)^{1/2} x_n} \, |\widehat{g}(\xi')|^2 \, d\xi' dx_n$$
$$= \frac{1}{2^n \pi^{n-1}} \int_{\mathbb{R}^{n-1}} (1+|\xi'|^2)^{1/2} \, |\widehat{g}(\xi')|^2 \, d\xi' = \frac{1}{2^n \pi^{n-1}} \|g\|^2_{H^{1/2}(\mathbb{R}^{n-1})}.$$

Thus, $\mathcal{P} : \mathcal{D}(\mathbb{R}^{n-1}) \to H^1(\mathbb{R}^n_+)$ is linear and continuous. By the density of $\mathcal{D}(\mathbb{R}^{n-1})$ into $H^{1/2}(\mathbb{R}^{n-1})$, $\mathcal{P}$ extends to a linear, continuous operator from $H^{1/2}(\mathbb{R}^{n-1})$ into $H^1(\mathbb{R}^n_+)$. In particular, $H^{1/2}(\mathbb{R}^{n-1}) \subseteq \mathcal{R}(\tau_0)$. □

If Ω is a bounded, Lipschitz domain, it is possible to define $H^{1/2}(\Gamma)$ by localization and reduction to the half space, as we did for $L^2(\Gamma)$. In this way we can endow $H^{1/2}(\Gamma)$ with an inner product that makes it a Hilbert space, continuously embedded in $L^2(\Gamma)$. It turns out that $H^{1/2}(\Gamma)$ coincides with $\mathcal{R}(\tau_0)$:

$$H^{1/2}(\Gamma) = \left\{ \tau_0 u : u \in H^1(\Omega) \right\}. \tag{7.63}$$

Actually, slightly changing our point of view, we take (7.63) as a definition of $H^{1/2}(\Gamma)$ and endow $H^{1/2}(\Gamma)$ with the norm

$$\|g\|_{H^{1/2}(\Gamma)} = \inf \left\{ \|w\|_{H^1(\Omega)} : w \in H^1(\Omega), \ \tau_0 w = g \right\}. \tag{7.64}$$

This norm is equal to the smallest among the norms of all elements in $H^1(\Omega)$ sharing the same trace g on Γ and takes into account that the trace operator τ_0 is not injective, since we know that the kernel of τ_0 is $H^1_0(\Omega)$. In particular, the following *trace inequality* holds:

$$\|\tau_0 u\|_{H^{1/2}(\Gamma)} \leq \|u\|_{H^1(\Omega)}, \tag{7.65}$$

which means that the trace operator τ_0 is continuous from $H^1(\Omega)$ onto $H^{1/2}(\Gamma)$.

In a similar way, if Γ_0 is a relatively open and regular subset of Γ, we may define

$$H^{1/2}(\Gamma_0) = \left\{ \tau_{\Gamma_0} u : u \in H^1(\Omega) \right\}, \tag{7.66}$$

endowed with the norm

$$\|g\|_{H^{1/2}(\Gamma_0)} = \inf \left\{ \|w\|_{H^1(\Omega)} : w \in H^1(\Omega), \ \tau_{\Gamma_0} w = g \right\}. \tag{7.67}$$

In particular, the following trace inequality holds:

$$\left\| \tau_{\Gamma_0} u \right\|_{H^{1/2}(\Gamma_0)} \leq \|u\|_{H^1(\Omega)}, \tag{7.68}$$

which means that the trace operator τ_{Γ_0} is continuous from $H^1(\Omega)$ *onto* $H^{1/2}(\Gamma_0)$.

We summarize and complete the above discussion in the following theorem.

Theorem 7.85 *The spaces $H^{1/2}(\Gamma)$ and $H^{1/2}(\Gamma_0)$, defined respectively by (7.63) and (7.66), are Hilbert spaces continuously embedded into $L^2(\Gamma)$ and $L^2(\Gamma_0)$. The trace operators*

$$\tau_0 : H^1(\Omega) \to H^{1/2}(\Gamma) \qquad \text{and} \qquad \tau_{\Gamma_0} : H^1(\Omega) \to H^{1/2}(\Gamma_0)$$

are surjective and continuous. In particular, there exist two linear isometric ***lifting*** *operators*

$$\mathcal{P}_0 : H^{1/2}(\Gamma) \to H^1(\Omega) \ \textit{and}\ \mathcal{P}_{\Gamma_0} : H^{1/2}(\Gamma_0) \to H^1(\Omega).$$

That is: $\tau_0(\mathcal{P}_0 g) = g$, $\forall g \in H^{1/2}(\Gamma)$, $\tau_{\Gamma_0}(\mathcal{P}_{\Gamma_0} g) = g$, $\forall g \in H^{1/2}(\Gamma_0)$ *and*

$$\|\mathcal{P}_0 g\|_{H^1(\Omega)} = \|g\|_{H^{1/2}(\Gamma)}, \quad \left\|\mathcal{P}_{\Gamma_0} g\right\|_{H^1(\Omega)} = \|g\|_{H^{1/2}(\Gamma_0)}.$$

Proof Consider $H^{1/2}(\Gamma)$. Let $H^*(\Omega)$ be the orthogonal complement to $H_0^1(\Omega)$ in $H^1(\Omega)$, that is

$$H^1(\Omega) = H_0^1(\Omega) \oplus H^*(\Omega).$$

Then $H^*(\Omega)$ is a Hilbert space. Take the restriction τ_0^* of τ_0 to $H^*(\Omega)$. Clearly τ_0^* is an isomorphism between $H^*(\Omega)$ and $H^{1/2}(\Gamma)$. Moreover τ_0^* is norm-preserving. To see it note that if $u \in H^*(\Omega)$ and $\tau_0^* u = g$, the counter-images $\tau_0^{-1} g$ of g are of the form $w = u + v$ with $v \in H_0^1(\Omega)$. Since

$$\|u + v\|^2_{H^1(\Omega)} = \|u\|^2_{H^1(\Omega)} + \|v\|^2_{H^1(\Omega)}$$

we have

$$\|g\|_{H^{1/2}(\Gamma)} = \inf\left\{\|w\|_{H^1(\Omega)} : w \in H^1(\Omega),\ \tau_0 w = g\right\} = \|u\|^2_{H^1(\Omega)}.$$

Thus τ_0^* is a linear isometry between $H^*(\Omega)$ and $H^{1/2}(\Gamma)$. This implies that $H^{1/2}(\Gamma)$ is a Hilbert space with inner product

$$(g, h)_{H^{1/2}(\Gamma)} = ((\tau_0^*)^{-1} g, (\tau_0^*)^{-1} h)_{H^1(\Omega)}$$

and that $\mathcal{P}_0 = (\tau_0^*)^{-1}$ is an isometric lifting operator. The same arguments work for $H^{1/2}(\Gamma_0)$. □

Finally, if Ω is $\mathbb{R}^n_+$ or a bounded C^m domain, $m \geq 2$, the space of traces of functions in $H^m(\Omega)$ is the fractional order Sobolev space $H^{m-1/2}(\Gamma)$, still showing a loss of "half derivative". Coherently, the trace of a normal derivative undergoes a loss of one more derivative and belongs to $H^{m-3/2}(\Gamma)$; the derivatives of order $m-1$ have traces in $H^{1/2}(\Gamma)$. Thus we obtain:

$$\tau : H^m(\Omega) \to (H^{m-1/2}(\Gamma), H^{m-3/2}(\Gamma), \ldots, H^{1/2}(\Gamma)).$$

The kernel of τ is $H_0^m(\Omega)$.

7.10 Compactness and Embeddings

7.10.1 The Rellich Theorem

Since

$$\|u\|_{L^2(\Omega)} \leq \|u\|_{H^1(\Omega)},$$

$H^1(\Omega)$ is *continuously embedded* in $L^2(\Omega)$ that is, if a sequence $\{u_k\}$ converges to u in $H^1(\Omega)$ it converges to u in $L^2(\Omega)$ as well.

If we assume that Ω is a **bounded, Lipschitz** domain (a *bounded interval* (a,b) for $n=1$), then the embedding of $H^1(\Omega)$ in $L^2(\Omega)$ is also **compact** (see also Problem 7.27). Thus, a bounded sequence $\{u_k\} \subset H^1(\Omega)$ has the following important properties:

There exist a subsequence $\{u_{k_j}\}$ *and* $u \in H^1(\Omega)$ *such that*

a) $u_{k_j} \to u$ *in* $L^2(\Omega)$.
b) $u_{k_j} \rightharpoonup u$ *in* $H^1(\Omega)$ *(i.e.* u_{k_j} *converges weakly*[16] *to* u *in* $H^1(\Omega)$*).*

Actually, only property a) follows from the compactness of the embedding. Property b) expresses a general fact in every Hilbert space H: every bounded subset $E \subset H$ is *sequentially weakly compact* (Theorem 6.60, p. 389).

Theorem 7.86 *Let* Ω *be a bounded, Lipschitz domain. Then* $H^1(\Omega)$ *is compactly embedded in* $L^2(\Omega)$.

Proof We use the compactness criterion expressed in Theorem 6.55, p. 386. First, observe that, for every $v \in \mathcal{D}(\mathbb{R}^n)$ we may write

$$v(\mathbf{x}+\mathbf{h}) - v(\mathbf{x}) = \int_0^1 \frac{d}{dt} v(\mathbf{x}+t\mathbf{h})\,dt = \int_0^1 \nabla v(\mathbf{x}+t\mathbf{h}) \cdot \mathbf{h}\,dt$$

whence

$$|v(\mathbf{x}+\mathbf{h}) - v(\mathbf{x})|^2 = \left| \int_0^1 \nabla v(\mathbf{x}+t\mathbf{h}) \cdot \mathbf{h}\,dt \right|^2 \le |\mathbf{h}|^2 \int_0^1 |\nabla v(\mathbf{x}+t\mathbf{h})|^2\,dt.$$

Integrating on $\mathbb{R}^n$ we find

$$\int_{\mathbb{R}^n} |v(\mathbf{x}+\mathbf{h}) - v(\mathbf{x})|^2\,d\mathbf{x} \le |\mathbf{h}|^2 \int_{\mathbb{R}^n} d\mathbf{x} \int_0^1 |\nabla v(\mathbf{x}+t\mathbf{h})|^2\,dt \le |\mathbf{h}|^2 \,\|\nabla v\|^2_{L^2(\mathbb{R}^n;\mathbb{R}^n)}$$

so that

$$\int_{\mathbb{R}^n} |v(\mathbf{x}+\mathbf{h}) - v(\mathbf{x})|^2\,d\mathbf{x} \le |\mathbf{h}|^2 \,\|\nabla v\|^2_{L^2(\mathbb{R}^n,\mathbb{R}^n)}\,. \tag{7.69}$$

Since $\mathcal{D}(\mathbb{R}^n)$ is dense in $H^1(\mathbb{R}^n)$, we infer that (7.69) holds for every $u \in H^1(\mathbb{R}^n)$ as well. Let now $S \subset H^1(\Omega)$ be bounded, i.e. there exists a number M such that:

$$\|u\|_{H^1(\Omega)} \le M, \qquad \forall u \in S.$$

By Theorem 7.75, p. 470, every $u \in S$ has an extension $\widetilde{u} \in H^1(\mathbb{R}^n)$, with support contained in a common open set Ω', with $\Omega \subset\subset \Omega' \subset\subset \mathbb{R}^n$. Thus, $\widetilde{u} \in H^1_0(\Omega')$ and moreover,

$$\|\nabla \widetilde{u}\|_{L^2(\Omega';\mathbb{R}^n)} \le c\,\|u\|_{H^1(\Omega)} \le cM.$$

Denote by $\widetilde{S}$ the set of such extensions. Then (7.69) holds for every $\widetilde{u} \in \widetilde{S}$:

$$\int_{\Omega'} |\widetilde{u}(\mathbf{x}+\mathbf{h}) - \widetilde{u}(\mathbf{x})|^2\,d\mathbf{x} \le |\mathbf{h}|^2 \,\|\nabla \widetilde{u}\|^2_{L^2(\mathbb{R}^n;\mathbb{R}^n)} \le c^2 M^2\,|\mathbf{h}|^2\,.$$

Theorem 6.55, p. 386, implies that $\widetilde{S}$ is precompact in $L^2(\Omega')$, which implies that S is precompact in $L^2(\Omega)$. □

[16] See Sect. 6.7.

7.10.2 Poincaré's inequalities

Under suitable hypotheses, the norm $\|u\|_{H^1(\Omega)}$ is equivalent to $\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}$. This means that there exists a constant C_P, independent of u, such that

$$\|u\|_{L^2(\Omega)} \leq C_P \, \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \, . \tag{7.70}$$

Inequalities like (7.70) are called *inequalities of Poincaré* and play a big role in the variational treatment of boundary value problems, as we shall realize in the next chapter. We have already proved in Theorem 7.57, p. 457, that (7.70) holds if $u \in H_0^1(\Omega)$, that is for functions vanishing on $\Gamma = \partial\Omega$. On the other hand, (7.70) cannot hold if $u =$ constant $\neq 0$. Roughly speaking, (7.70) is valid on any subspace of $H^1(\Omega)$ which *does not contain nontrivial constants*. This is true whenever u vanishes in some "nontrivial set". For instance, if u belongs to one of the following closed subspaces of $H^1(\Omega)$, then (7.70) holds:

1. $\Omega = (a, b)$ and $u(c) = 0$ where c is a (fixed) point in $[a, b]$.
2. $u \in H^1_{0,\Gamma_0}(\Omega)$ (u vanishes on a nonempty relatively open subset $\Gamma_0 \subset \Gamma$) or, more generally, $\int_{\Gamma_0} u \, d\sigma = 0$, $n \geq 2$.
3. $u \in H^1(\Omega)$ and $\int_E u = 0$ where $E \subseteq \Omega$ has positive measure, that is, $|E| = a > 0$. Note that, if $E = \Omega$, then u has mean value zero in Ω, $n \geq 1$.

Theorem 7.87 *Let Ω be a bounded Lipschitz domain ($\Omega = (a, b)$ if $n = 1$), and V be a closed subspace of $H^1(\Omega)$ such that $1 \notin V$. Then there exists a constant C_P such that*

$$\|u\|_{L^2(\Omega)} \leq C_P \, \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \, , \qquad \textit{for every } u \in V. \tag{7.71}$$

Proof By contradiction suppose (7.71) is not true. This means that for every integer $j \geq 1$, there exists $u_j \in V$ such that

$$\|u_j\|_{L^2(\Omega)} > j \, \|\nabla u_j\|_{L^2(\Omega;\mathbb{R}^n)} \, . \tag{7.72}$$

Normalize u_j in $L^2(\Omega)$ by setting

$$w_j = \frac{u_j}{\|u_j\|_{L^2(\Omega)}}.$$

Then, from (7.72),

$$\|w_j\|_{L^2(\Omega)} = 1 \quad \text{and} \quad \|\nabla w_j\|_{L^2(\Omega;\mathbb{R}^n)} < \frac{1}{j} \leq 1.$$

Thus $\{w_j\}$ is bounded in $H^1(\Omega)$. By Rellich's Theorem and since V is closed, there exists a subsequence $\{w_{j_k}\}$ and $w \in V$ such that

- $w_{j_k} \to w$ in $L^2(\Omega)$.
- $\nabla w_{j_k} \rightharpoonup \nabla w$ in $L^2(\Omega;\mathbb{R}^n)$.

The continuity of the norm gives

$$\|w\|_{L^2(\Omega)} = \lim_{j_k \to +\infty} \|w_{j_k}\|_{L^2(\Omega)} = 1.$$

On the other hand, the weak sequential lower semicontinuity of the norm (Theorem 6.59, p. 388) yields

$$\|\nabla w\|_{L^2(\Omega;\mathbb{R}^n)} \leq \liminf_{j_k\to\infty} \left\|\nabla w_{j_k}\right\|_{L^2(\Omega;\mathbb{R}^n)} = 0$$

so that $\nabla w = \mathbf{0}$ a.e. and in $\mathcal{D}'(\Omega;\mathbb{R}^n)$. Since Ω is connected, w is constant by Proposition 7.18, p. 431, and since $w \in V$, we infer $w = 0$, in contradiction with $\|w\|_{L^2(\Omega)} = 1$. □

Remark 7.88 Let Ω be a bounded Lipschitz domain and $E \subset \Omega$, with $|E| = a > 0$. If $u \in H^1(\Omega)$, set

$$u_E = \frac{1}{|E|}\int_E u \tag{7.73}$$

and $w = u - u_E$. Then $\int_E w = 0$ and (7.71) holds for w. Thus, we get a Poincaré's inequality of the form:

$$\|u - u_E\|_{L^2(\Omega)} \leq C_P \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,. \tag{7.74}$$

A similar inequality holds if E is replaced by nonempty relatively open subset $\Gamma_0 \subseteq \Gamma$. As a consequence, the following norms are *all equivalent* to the standard norm in $H^1(\Omega)$:

$$\sqrt{\int_\Omega |\nabla u|^2 + (u_E)^2}, \quad \sqrt{\int_\Omega |\nabla u|^2 + (u^2)_E} \tag{7.75}$$

or

$$\sqrt{\int_\Omega |\nabla u|^2 + (u_{\Gamma_0})^2}, \quad \sqrt{\int_\Omega |\nabla u|^2 + (u^2)_{\Gamma_0}}. \tag{7.76}$$

Let us prove, for instance, that

$$c\left\{\int_\Omega |\nabla u|^2 + (u_E)^2\right\} \leq \|u\|^2_{H^1(\Omega)} \leq C\left\{\int_\Omega |\nabla u|^2 + (u_E)^2\right\} \tag{7.77}$$

where the constants c, C depends only on n, Ω and E. We have, from (7.74), using the inequality $2ab \leq \varepsilon^{-1}a^2 + \varepsilon b^2$,

$$\begin{aligned}\int_\Omega u^2 &\leq C_P^2 \int_\Omega |\nabla u|^2 + 2u_E \int_\Omega u \;\leq C_P^2 \int_\Omega |\nabla u|^2 + 2u_E\,|\Omega|^{1/2}\left(\int_\Omega u^2\right)^{1/2}\\ &\leq C_P^2 \int_\Omega |\nabla u|^2 + \frac{|\Omega|}{\varepsilon}u_E^2 + \varepsilon\int_\Omega u^2.\end{aligned}$$

Choosing $\varepsilon = \frac{1}{2}$ we have

$$\int_\Omega u^2 \leq 2C_P^2 \int_\Omega |\nabla u|^2 + 4\,|\Omega|\,u_E^2$$

from which the inequality in the right hand side of (7.77) follows with $C(n,\Omega,E) = \max\left\{2C_P^2 + 1, 4\,|\Omega|\right\}$. The inequality on the left of (7.77) is easier and we leave it as an exercise.

7.10.3 Sobolev inequality in $\mathbb{R}^n$

From Proposition 7.55, p. 456, we know that the elements of $H^1(\mathbb{R})$ are continuous and (Problem 7.25) vanish as $x \to \pm\infty$. Moreover, if $u \in \mathcal{D}(\mathbb{R})$, we may write

$$u^2(x) = \int_{-\infty}^{x} \frac{d}{ds} u^2(s) ds = 2 \int_{-\infty}^{x} u(s) u'(s) ds.$$

Using the Schwarz inequality and $2ab \le a^2 + b^2$, we get

$$u(x)^2 \le 2 \|u\|_{L^2(\mathbb{R})} \|u'\|_{L^2(\mathbb{R})} \le \|u\|^2_{L^2(\mathbb{R})} + \|u'\|^2_{L^2(\mathbb{R})} = \|u\|^2_{H^1(\mathbb{R})} .$$

Since $\mathcal{D}(\mathbb{R})$ is dense in $H^1(\mathbb{R})$, this inequality holds if $u \in H^1(\mathbb{R})$ as well. We have proved:

Proposition 7.89 *Let $u \in H^1(\mathbb{R})$. Then $u \in L^\infty(\mathbb{R})$ and*

$$\|u\|_{L^\infty(\mathbb{R})} \le \|u\|_{H^1(\mathbb{R})} .$$

On the other hand, Example 7.54, p. 454, implies that, when $\Omega \subseteq \mathbb{R}^n$, $n \ge 2$,

$$u \in H^1(\Omega) \not\Rightarrow u \in L^\infty(\Omega).$$

However, it is possible to prove that u is actually p-summable with a suitable $p > 2$. Moreover, the L^p-norm of u can be estimated by the H^1-norm of u. To guess which exponent p is the correct one, assume that the inequality

$$\|u\|_{L^p(\mathbb{R}^n)} \le c \|\nabla u\|_{L^2(\mathbb{R}^n)} \tag{7.78}$$

is valid **for every** $u \in \mathcal{D}(\mathbb{R}^n)$, *where c may depend on p* and n **but not on** u. We now use a typical "dimensional analysis" argument. Inequality (7.78) must be *invariant under dilations* in the following sense. Let $u \in \mathcal{D}(\mathbb{R}^n)$ and for $\lambda > 0$ set

$$u_\lambda(\mathbf{x}) = u(\lambda \mathbf{x}).$$

Then $u_\lambda \in \mathcal{D}(\mathbb{R}^n)$ so that inequality (7.78) must be true for u_λ, with c **independent of** λ:

$$\|u_\lambda\|_{L^p(\mathbb{R}^n)} \le c \|\nabla u_\lambda\|_{L^2(\mathbb{R}^n;\mathbb{R}^n)} . \tag{7.79}$$

Now,

$$\int_{\mathbb{R}^n} |u_\lambda|^p \, d\mathbf{x} = \int_{\mathbb{R}^n} |u(\lambda \mathbf{x})|^p \, d\mathbf{x} = \frac{1}{\lambda^n} \int_{\mathbb{R}^n} |u(\mathbf{y})|^p \, d\mathbf{y}$$

while

$$\int_{\mathbb{R}^n} |\nabla u_\lambda(\mathbf{x})|^2 \, d\mathbf{x} = \int_{\mathbb{R}^n} |\lambda \nabla u(\lambda \mathbf{x})|^2 \, d\mathbf{x} = \frac{1}{\lambda^{n-2}} \int_{\mathbb{R}^n} |\nabla u(\mathbf{y})|^2 \, d\mathbf{y}.$$

Therefore, (7.79) becomes

$$\frac{1}{\lambda^{n/p}}\left(\int_{\mathbb{R}^n}|u|^p\,d\mathbf{y}\right)^{1/p}\leq c(n,p)\frac{1}{\lambda^{(n-2)/2}}\left(\int_{\mathbb{R}^n}|\nabla u|^2\,d\mathbf{y}\right)^{1/2}$$

or

$$\|u\|_{L^p(\mathbb{R}^n)}\leq c\lambda^{1-\frac{n}{2}+\frac{n}{p}}\,\|\nabla u\|_{L^2(\mathbb{R}^n;\mathbb{R}^n)}\,.$$

The only way to get a constant independent of λ is to choose p such that

$$1-\frac{n}{2}+\frac{n}{p}=0.$$

Hence, if $n>2$, the correct p is given by

$$2^*=\frac{2n}{n-2}$$

which is called the *Sobolev exponent* for $H^1(\mathbb{R}^n)$. The following theorem of Sobolev, Gagliardo, Nirenberg holds:

Theorem 7.90 *Let $u\in H^1(\mathbb{R}^n), n\geq 3$. Then $u\in L^{2^*}(\mathbb{R}^n)$ with $2^*=\frac{2n}{n-2}$, and the following inequality holds.*

$$\|u\|_{L^{2^*}(\mathbb{R}^n)}\leq c\,\|\nabla u\|_{L^2(\mathbb{R}^n,\,\mathbb{R}^n)}\tag{7.80}$$

where$c=c(n)$.

In the case $n=2$, the correct statement is:

Proposition 7.91 *Let $u\in H^1(\mathbb{R}^2)$. Then $u\in L^p(\mathbb{R}), 2\leq p<\infty$, and*

$$\|u\|_{L^p(\mathbb{R}^2)}\leq c(p)\,\|u\|_{H^1(\mathbb{R}^2)}\,.$$

7.10.4 Bounded domains

We now consider bounded domains. In dimension $n=1$, the elements of $H^1(a,b)$ are continuous in $[a,b]$ and therefore bounded as well. Furthermore, the following inequality holds:

$$\|v\|_{L^\infty(a,b)}\leq C^*\,\|v\|_{H^1(a,b)}\tag{7.81}$$

with

$$C^*=\sqrt{2}\max\left\{(b-a)^{-1/2},(b-a)^{1/2}\right\}.$$

Indeed, by the Schwarz inequality we have, for any $x,y\in[a,b]$, $y>x$:

$$u(y)=u(x)+\int_x^y u'(s)ds\leq u(x)+\sqrt{b-a}\,\|u'\|_{L^2(a,b)}$$

whence, using the elementary inequality $(A+B)^2\leq 2A^2+2B^2$,

$$u(y)^2 \le 2u(x)^2 + 2(b-a)\,\|u'\|^2_{L^2(a,b)}\,.$$

Integrating over (a,b) with respect to x we get

$$(b-a)u(y)^2 \le 2\,\|u\|^2_{L^2(a,b)} + 2(b-a)^2\,\|u'\|^2_{L^2(a,b)}$$

from which (7.81) follows easily.

In dimension $n \ge 2$, the improvement in summability is indicated in the following theorem.

Theorem 7.92 *Let Ω be a bounded, Lipschitz domain. Then:*

1. *If $n > 2$, $H^1(\Omega) \hookrightarrow L^p(\Omega)$ for $2 \le p \le \frac{2n}{n-2}$. Moreover, if $2 \le p < \frac{2n}{n-2}$, the embedding of $H^1(\Omega)$ in $L^p(\Omega)$ is compact.*
2. *If $n = 2$, $H^1(\Omega) \hookrightarrow L^p(\Omega)$ for $2 \le p < \infty$, with compact embedding.*

In the above cases

$$\|u\|_{L^p(\Omega)} \le c(n,p,\Omega)\,\|u\|_{H^1(\Omega)}\,.$$

For instance, in the important case $n = 3$, we have

$$2^* = \frac{2n}{n-2} = 6.$$

Hence $H^1(\Omega) \hookrightarrow L^6(\Omega)$ and

$$\|u\|_{L^6(\Omega)} \le c(\Omega)\,\|u\|_{H^1(\Omega)}\,.$$

When the embedding of $H^1(\Omega)$ in $L^p(\Omega)$ is compact, the Poincaré inequality in Theorem 7.87, p. 479 may be replaced by (see Problem 7.26)

$$\|u\|_{L^p(\Omega)} \le c(n,p,\Omega)\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,. \tag{7.82}$$

Theorem 7.92 shows what we can conclude about a H^1-function with regards to further regularity. It is natural to expect something more for H^m-functions, with $m > 1$. In fact, the following result holds.

Theorem 7.93 *Let Ω be a bounded, Lipschitz domain, and $m > n/2$. Then*

$$H^m(\Omega) \hookrightarrow C^{k,\alpha}(\overline{\Omega}), \tag{7.83}$$

for k integer, $0 \le k < m - \frac{n}{2}$ and any $\alpha \in (0,1)$ if n is even, $0 < \alpha \le 1/2$ if n is odd. The embedding in (7.83) is compact for any $\alpha \in (0,1)$ if n is even and for $\alpha \ne 1/2$ if n is odd, and

$$\|u\|_{C^{k,\alpha}(\overline{\Omega})} \le c(n,m,\alpha,\Omega)\,\|u\|_{H^m(\Omega)}\,.$$

Theorem 7.93 implies that, in dimension $n = 2$, two derivatives occurs to get (Hölder) continuity:

$$H^2(\Omega) \hookrightarrow C^{0,\alpha}(\overline{\Omega}), \quad 0 < \alpha < 1.$$

In fact, n is even, $m = 2$, $m - \frac{n}{2} = 1$. Similarly

$$H^3(\Omega) \hookrightarrow C^{1,\alpha}(\overline{\Omega}),\ 0 < \alpha < 1,$$

since $m - \frac{n}{2} = 2$. In dimension $n = 3$ we have

$$H^2(\Omega) \hookrightarrow C^{0,1/2}(\overline{\Omega})$$

and

$$H^3(\Omega) \hookrightarrow C^{1,1/2}(\overline{\Omega}),$$

since $m - \frac{n}{2} = \frac{1}{2}$ in the first case and $m - \frac{n}{2} = \frac{3}{2}$ in the second.

Remark 7.94 If $u \in H^m(\Omega)$ for every $m \geq 1$, then $u \in C^\infty(\overline{\Omega})$. This kind of results is very useful in the regularity theory for boundary value problems.

7.11 Spaces Involving Time

7.11.1 Functions with values into Hilbert spaces

The natural functional setting for evolution problems requires spaces which involve time. Given a function $u = u(\mathbf{x}, t)$, it is often convenient to separate the roles of space and time adopting the following point of view. Assume that $t \in [0, T]$ and that, for every t, or at least for a.e. t, the function $u(\cdot, t)$ belongs to a *separable* Hilbert space V (e.g. $L^2(\Omega)$ or $H^1(\Omega)$).

Then, we may consider u as a function of the real variable t with values into V:

$$u : [0, T] \to V.$$

When we adopt this convention, we write $u(t)$ and $\dot{u}(t)$ instead of $u(\mathbf{x}, t)$ and $u_t(\mathbf{x}, t)$.

• *Spaces of continuous functions.* We start by introducing the set $C([0, T]; V)$ of the continuous functions $u : [0, T] \to V$, endowed with the norm

$$\|u\|_{C([0,T];V)} = \max_{0 \leq t \leq T} \|u(t)\|_V .$$

$C([0, T]; V)$ is Banach space. We say that $v : [0, T] \to V$ is the (strong) derivative of u if

$$\left\| \frac{u(t + h) - u(t)}{h} - v(t) \right\|_V \to 0$$

for each $t \in [0, T]$. We write as usual u' or $\dot{u}$ for the derivative of u.

The symbol $C^1([0, T]; V)$ denotes the Banach space of functions whose derivative exists and belongs to $C([0, T]; V)$, endowed with the norm

$$\|u\|_{C^1([0,T];V)} = \|u\|_{C([0,T];V)} + \|\dot{u}\|_{C([0,T];V)}$$

Similarly we can define the spaces $C^k([0,T];V)$ and $C^\infty([0,T];V)$, while $\mathcal{D}(0,T;V)$ denotes the subspace of $C^\infty([0,T];V)$ of the functions compactly supported in $(0,T)$.

• *Integrals and spaces of summable or integrable functions.* We can extend to these types of functions the notions of measurability and integral, without too much effort, following more or less the procedure outlined in Appendix A. First, we introduce the set of functions $s : [0,T] \to V$ which assume only a finite number of values. These functions are called *simple* and are of the form

$$s(t) = \sum_{j=1}^{N} \chi_{E_j}(t) u_j \qquad (0 \le t \le T), \tag{7.84}$$

where, $u_1, \dots, u_N \in V$ and $E_1, \dots, E_N$ are Lebesgue measurable, mutually disjoint subsets of $[0,T]$.

We say that $f : [0,T] \to V$ is *measurable* if there exists a sequence of simple functions $s_k : [0,T] \to V$ such that, as $k \to \infty$,

$$\|s_k(t) - f(t)\|_V \to 0, \qquad \text{a.e. in } [0,T].$$

It is not difficult to prove that, if f is measurable and $v \in V$, the (real) function

$$t \longmapsto (f(t), v)_V$$

is Lebesgue measurable in $[0,T]$. The notion of integral is defined first for simple functions. If s is given by (7.84), we define

$$\int_0^T s(t)dt = \sum_{j=1}^{N} |E_j| u_j.$$

Then:

Definition 7.95 We say that $f : [0,T] \to V$ is *summable* in $[0,T]$ if it is measurable and there exists a sequence $s_k : [0,T] \to V$ of simple functions such that

$$\int_0^T \|s_k(t) - f(t)\|_V \, dt \to 0, \qquad \text{as } k \to +\infty. \tag{7.85}$$

If f is summable in $[0,T]$, we define the integral of f as follows:

$$\int_0^T f(t)dt = \lim_{k \to +\infty} \int_0^T s_k(t)dt, \qquad \text{as } k \to +\infty. \tag{7.86}$$

Since for simple functions it is easy to check that

$$\left\| \int_0^T [s_h(t) - s_k(t)]\, dt \right\|_V \leq \int_0^T \|s_h(t) - s_k(t)\|_V\, dt$$
$$\leq \int_0^T \|s_h(t) - f(t)\|_V\, dt + \int_0^T \|s_k(t) - f(t)\|_V\, dt,$$

it follows from (7.85) that the sequence $\left\{\int_0^T s_k(t)dt\right\}$ is a Cauchy sequence in V. Thus, the limit (7.86) is well defined and does not depend on the choice of the approximating sequence $\{s_k\}$. Moreover, the following important theorem holds, due to *Bochner*:

Theorem 7.96 *A measurable function $f : [0,T] \to V$ is summable in $[0,T]$ if and only if the real function $t \to \|f(t)\|_V$ is summable in $[0,T]$. Moreover*

$$\left\| \int_0^T f(t)dt \right\|_V \leq \int_0^T \|f(t)\|_V\, dt \tag{7.87}$$

and

$$\left(u, \int_0^T f(t)\, dt\right)_V = \int_0^T (u, f(t))_V dt \qquad \forall u \in V. \tag{7.88}$$

The inequality (7.87) is well known in the case of real or complex functions. By Riesz's Representation Theorem, (7.88) shows that the action of any element of V^* commutes with the integral.

Once the definition of integral has been given, we can introduce the spaces $L^p(0,T;V)$, $1 \leq p \leq \infty$.

We define $L^p(0,T;V)$ as the set of measurable functions $u : [0,T] \to V$ such that: if $1 \leq p < \infty$

$$\|u\|_{L^p(0,T;V)} = \left(\int_0^T \|u(t)\|_V^p\, dt \right)^{1/p} < \infty \tag{7.89}$$

while if $p = \infty$

$$\|u\|_{L^\infty(0,T;V)} = \operatorname*{ess\,sup}_{0 \leq t \leq T} \|u(t)\|_V < \infty.$$

Endowed with the above norms, $L^p(0,T;V)$ becomes a Banach space for $1 \leq p \leq \infty$. If $p = 2$, the norm (7.89) is induced by the inner product

$$(u,v)_{L^2(0,T;V)} = \int_0^T (u(t), v(t))_V dt$$

that makes $L^2(0,T;V)$ into a Hilbert space.

If $u \in L^1(0,T;V)$, the function $t \longmapsto \int_0^t u(s)ds$ is continuous and

$$\frac{d}{dt} \int_0^t u(s)ds = u(t) \quad \text{a.e. in } (0,T).$$

Proposition 7.97 *Let $1 \le p < \infty$. Then:*

a) $\mathcal{D}(0,T;V)$ is dense in $L^p(0,T;V)$.
b) $(L^p(0,T;V))^*$ is isometric to and can be identified with $L^q(0,T;V^*)$, where $q = p/(p-1)$.

We conclude this subsection with a useful result (see Problem 7.32): the weak convergence in $L^2(0,T;V)$ preserves boundedness in $L^\infty(0,T;V)$.

Proposition 7.98 *Let $\{u_k\} \subset L^2(0,T;V)$, weakly convergent to u. Assume that*

$$\operatorname*{ess\,sup}_{t\in[0,T]} \|u_k(t)\|_V \le C$$

with C independent of k. Then, also,

$$\operatorname*{ess\,sup}_{t\in[0,T]} \|u(t)\|_V \le C.$$

7.12 Sobolev spaces involving time

To define Sobolev spaces, we need the notion of derivative in the sense of distributions for functions $u \in L^1_{\text{loc}}(0,T;V)$. In general, the weak derivative $\dot{u}$ is defined by the linear operator

$$\varphi \longmapsto \langle \dot{u}, \varphi \rangle = -\int_0^T u(t)\dot{\varphi}(t)dt \tag{7.90}$$

from $\mathcal{D}(0,T)$ into V. As usual, the crochet $\langle \cdot, \cdot \rangle$ denotes the *action* of the distribution $\dot{u}$ on the test function φ. Indeed, we have

$$\|\langle \dot{u}, \varphi \rangle\|_V = \left\| \int_0^T u(t)\dot{\varphi}(t)dt \right\|_V \le \int_0^T |\dot{\varphi}(t)|\, \|u(t)\|_V\, dt \le K \max |\dot{\varphi}(t)|$$

where K is the integral of $\|u(t)\|_V$ over the support of $\dot{\varphi}$. Therefore $\dot{u}$ is continuous with respect to the convergence in $\mathcal{D}(0,T)$ and defines an element of $\mathcal{D}'(0,T;V)$.

Thus, we say that $\dot{u} \in L^1_{\text{loc}}(0,T;V)$ is the derivative in the sense of distribution of u if

$$\langle \dot{u}, \varphi \rangle = \int_0^T \varphi(t)\dot{u}(t)dt = -\int_0^T \dot{\varphi}(t)u(t)dt \tag{7.91}$$

for every $\varphi \in \mathcal{D}(0,T)$ or, equivalently, if

$$\int_0^T \varphi(t)(\dot{u}(t), v)_V dt = -\int_0^T \dot{\varphi}(t)(u(t), v)_V dt, \quad \forall v \in V. \tag{7.92}$$

We denote by $H^1(0,T;V)$ the Sobolev space of the functions $u \in L^2(0,T;V)$ such that $\dot{u} \in L^2(0,T;V)$. This is a Hilbert space with inner product

$$(u,v)_{H^1(0,T;V)} = \int_0^T \{(u(t),v(t))_V + (\dot{u}(t),\dot{v}(t))_V\}\,dt.$$

Since functions in $H^1(a,b)$ are continuous in $[a,b]$, it makes sense to consider the value of u at any point of $[a,b]$. In a certain way, the functions in $H^1(0,T;V)$ depends only on the real variable t, so that the following theorem is not surprising.

Theorem 7.99 *Let $u \in H^1(0,T;V)$. Then, $u \in C([0,T];V)$ and*

$$\max_{0\le t\le T} \|u(t)\|_V \le C(T)\,\|u\|_{H^1(0,T;V)}\,.$$

Moreover, the fundamental theorem of calculus holds:

$$u(t) = u(s) + \int_s^t \dot{u}(r)dr, \qquad 0 \le s \le t \le T.$$

If V and W are separable Hilbert spaces with

$$V \hookrightarrow W$$

it makes sense to define the Sobolev space

$$H^1(0,T;V,W) = \left\{u \in L^2(0,T;V) : \dot{u} \in L^2(0,T;W)\right\},$$

where $\dot{u}$ is intended in the sense of distribution in $\mathcal{D}'(0,T;W)$, endowed with the norm

$$\|u\|^2_{H^1(0,T;V,W)} = \|u\|^2_{L^2(0,T;V)} + \|\dot{u}\|^2_{L^2(0,T;W)}\,.$$

Indeed, consider the linear functional

$$\varphi \longmapsto \langle \dot{u},\varphi\rangle = \int_0^T \varphi(t)\dot{u}(t)dt = -\int_0^T \dot{\varphi}(t)u(t)dt$$

and observe that, since $V \hookrightarrow W$, we have, for some constant C,

$$\|\langle \dot{u},\varphi\rangle\|_W \le \max|\dot{\varphi}(t)| \int_0^T \|u(t)\|_W\,dt \le C \max|\dot{\varphi}(t)|\,\sqrt{T}\left(\int_0^T \|u(t)\|^2_V\,dt\right)^{1/2}.$$

Therefore $\dot{u}$ defines a distribution in $\mathcal{D}'(0,T;W)$.

This situation typically occurs in the applications to initial-boundary value problems. In fact, given a Hilbert triplet $\{V,H,V^*\}$, with V and H separable, we shall see in Chap. 9 that the natural functional setting for those problems is precisely the space $H^1(0,T;V,V^*)$. The following result is fundamental (for the proof, see *R. Dautray, J.-L. Lions*, [27], vol. 5).

Theorem 7.100 *Let $\{V,H,V^*\}$ be a Hilbert triplet, with V and H separable. Then:*

a) $C^\infty([0,T];V)$ is dense in $H^1(0,T;V,V^*)$.

b) $H^1(0,T;V,V^*) \hookrightarrow C([0,T];H)$, *that is*

$$\|u\|_{C([0,T];H)} \le C(T)\,\|u\|_{H^1(0,T;V,V^*)}\,.$$

c) If $u, v \in H^1(0,T;V,V^*)$, *the following integration by parts formula holds:*

$$\int_s^t \{\langle \dot u(r), v(r)\rangle_* + \langle \dot v(r), u(r)\rangle_*\}\,dr = (u(t),v(t))_H - (u(s),v(s))_H$$

for all $s,t \in [0,T]$.

Remark 7.101 From the integration by parts formula we infer,

$$\frac{d}{dt}(u(t),v(t))_H = \langle \dot u(t), v(t)\rangle_* + \langle \dot v(t), u(t)\rangle_*$$

a.e. $t \in [0,T]$ and (letting $u = v$)

$$\int_s^t \frac{d}{dt}\,\|u(r)\|_H^2\,dt = \|u(t)\|_H^2 - \|u(s)\|_H^2\,. \tag{7.93}$$

Problems

7.1 A distribution $L \in \mathcal{D}'(\Omega)$ is said to be positive if $\langle L, \varphi\rangle \ge 0$ for every $\varphi \in \mathcal{D}(\Omega)$, $\varphi \ge 0$ in Ω. Show that if L is positive, then it is a measure.

[*Hint.* Let $\varphi \in \mathcal{D}(\Omega)$, $\varphi \ge 0$, with support$(\varphi) = K$. Let $\psi \in \mathcal{D}(\Omega)$ such that $\psi \ge 0$ and $\psi \equiv 1$ on K. Observe that $\psi\,\|\varphi\|_{L^\infty(\Omega)} \pm \varphi \ge 0$ and prove that (7.6), p. 426, holds]

7.2 (*approximations of* δ_n)

a) Let $B_r = B_r(\mathbf{0}) \subset \mathbb{R}^n$. Show that, if χ_{B_r} is the characteristic function of B_r,

$$\lim_{r\to 0} \frac{1}{|B_r|}\chi_{B_r} = \delta_n, \qquad \text{in } \mathcal{D}'(\mathbb{R}^n).$$

b) Let η_ε be the mollifier in (7.3). Show that $\lim_{\varepsilon\to 0}\eta_\varepsilon = \delta_n$, in $\mathcal{D}'(\mathbb{R}^n)$.
c) Let $\Phi_D(\mathbf{x},t)$ be the fundamental solution of the heat equation $u_t = D\Delta u$. Show that, as $t \to 0^+$,

$$\Phi_D(\cdot,t) \to \delta_n, \ \text{ in } \ \mathcal{D}'(\mathbb{R}^n).$$

d) Let $f \in L^1(\mathbb{R}^n)$ with the following properties:

$$f \ge 0,\ f(-\mathbf{x}) = f(\mathbf{x}), \int_{\mathbb{R}^n} f = 1.$$

Define $f_k(\mathbf{x}) = k^n f(k\mathbf{x})$. Show that

$$\lim_{k\to+\infty} f_k = \delta_n, \qquad \text{in } \mathcal{D}'(\mathbb{R}^n).$$

7.3 Show that the series

$$\sum_{k=1}^{\infty} c_k \sin kx$$

converges in $\mathcal{D}'(\mathbb{R})$ if the numerical sequence $\{c_k\}$ is *slowly increasing*, i.e. if there exists $p \in \mathbb{R}$ such that $|c_k| \leq k^p$ as $k \to \infty$.

7.4 Show that if $F \in \mathcal{D}'(\mathbb{R}^n)$, $v \in \mathcal{D}(\mathbb{R}^n)$ and v vanishes in an open set containing the support of F, then $\langle F, v\rangle = 0$. Is it true that $\langle F, v\rangle = 0$ if v vanishes *only* on the support of F?

7.5 Let $u(x) = |x|$ and $S(x) = \operatorname{sign}(x)$. Prove that $u' = S$ in $\mathcal{D}'(\mathbb{R})$.

7.6 Prove that $x^m \delta^{(k)} = 0$ in $\mathcal{D}'(\mathbb{R})$, if $0 \leq k < m$.

7.7 Let $u(x) = \ln|x|$. Then, $u' = \text{p. v. } \frac{1}{x}$, in $\mathcal{D}'(\mathbb{R})$.
[*Hint.* Write $\langle u', \varphi\rangle = -\langle u, \varphi'\rangle = -\int_{\mathbb{R}} \ln|x|\, \varphi'(x)dx = -\lim_{\varepsilon\to 0} \int_{\{|x|>\varepsilon\}} \ln|x|\, \varphi'(x)dx$, and integrate by parts.]

7.8 Let $n = 3$ and $\mathbf{F} \in \mathcal{D}'(\Omega; \mathbb{R}^3)$. Define $\operatorname{curl} \mathbf{F} \in \mathcal{D}'(\Omega; \mathbb{R}^3)$ by the formula

$$\operatorname{curl} \mathbf{F} = (\partial_{x_2} F_3 - \partial_{x_3} F_2, \partial_{x_3} F_1 - \partial_{x_1} F_3, \partial_{x_1} F_2 - \partial_{x_2} F_1).$$

Check that, for all $\boldsymbol{\varphi} \in \mathcal{D}(\Omega; \mathbb{R}^3)$,

$$\langle \operatorname{curl} \mathbf{F}, \boldsymbol{\varphi}\rangle = \langle \mathbf{F}, \operatorname{curl} \boldsymbol{\varphi}\rangle .$$

7.9 Show that if $u(x_1, x_2) = \frac{1}{2\pi} \ln(x_1^2 + x_2^2)$, then

$$-\Delta u = \delta_2, \qquad \text{in } \mathcal{D}'(\mathbb{R}^2).$$

7.10 Show that if $u_k \to u$ in $L^p(\mathbb{R}^n)$ then $u_k \to u$ in $\mathcal{S}'(\mathbb{R}^n)$.

7.11 Solve the equation $x^2 G = 0$ in $\mathcal{D}'(\mathbb{R})$.

7.12 Let $u \in C^\infty(\mathbb{R})$ with compact support in $[0, 1]$. Compute $\operatorname{comb} * u$.
[*Answer.* $\sum_{k=-\infty}^{+\infty} u(x - k)$, the periodic extension of u over $\mathbb{R}$.]

7.13 Let $\mathcal{H} = \mathcal{H}(x)$ be the Heaviside function. Prove that

$$a)\ \mathcal{F}[\operatorname{sign}(x)] = \frac{2}{i}\ \text{p. v. } \frac{1}{\xi}, \qquad b)\ \mathcal{F}[\mathcal{H}] = \pi\delta + \frac{1}{i}\ \text{p. v. } \frac{1}{\xi}.$$

[*Hint.* a) Let $u(x) = \operatorname{sign}(x)$. Note that $u' = 2\delta$. Transform this equation to obtain $\xi \widehat{u}(\xi) = -2i$. Solve this equation using formula (7.27), and recall that $\widehat{u}$ is odd while δ is even. b) Write $\mathcal{H}(x) = \frac{1}{2} + \frac{1}{2}\operatorname{sign}(x)$ and use a).]

7.14 Show that $\operatorname{comb}_{\mathcal{P}}(x) = \sum_{k=-\infty}^{+\infty} \delta(x - k\mathcal{P})$ is periodic of period $\mathcal{P}$. Compute the Fourier transform of $\operatorname{comb}_{\mathcal{P}}$.

7.15 Compute $\delta \circ f$ where $f(x) = \sin x$.
[*Answer.* $\sum_{k=-\infty}^{+\infty} \delta(x - k\pi) \equiv \operatorname{comb}_{\pi}(x)$.]

7.16 Let $v \in S(\mathbb{R})$ and define $\varphi(x) = \sum_{k=-\infty}^{+\infty} v(x + 2\pi k)$.

a) Show that φ is well defined in $\mathbb{R}$, 2π-periodic and has the Fourier expansion

$$\varphi(x) = \frac{1}{2\pi} \sum_{n=-\infty}^{+\infty} \widehat{v}(n) e^{inx}.$$

b) Deduce the **Poisson's formula**

$$\sum_{k=-\infty}^{+\infty} \widehat{v}(k) = 2\pi \sum_{k=-\infty}^{+\infty} v(2\pi k). \tag{7.94}$$

c) Show that $\mathcal{F}[\mathrm{comb}] = \frac{1}{2\pi}\,\mathrm{comb}_{2\pi}$.

7.17 a) Use Proposition 7.10, p. 427, to show that

$$F_r(\mathbf{x}) = \frac{\delta(|\mathbf{x}| - r)}{|\mathbf{x}|} \quad \text{and} \quad G_r(\mathbf{x}) = \frac{\partial_r \delta(|\mathbf{x}| - r)}{|\mathbf{x}|}$$

are well defined as distributions in $\mathcal{D}'(\mathbb{R}^3)$.

b) Use Example 7.27, p. 438, to show that $F_r \to 0$ and $G_r \to 4\pi\delta_3$ in $\mathcal{D}'(\mathbb{R}^3)$, as $r \to 0$.

7.18 Let $g, h \in C_0^\infty(\mathbb{R}^n)$.

a) Show that $u \in C^2(\mathbb{R}^n \times [0, +\infty))$ is a solution of the global Cauchy problem

$$\begin{cases} u_{tt} - c^2\Delta u = 0, & \mathbf{x} \in \mathbb{R}^n,\ t > 0 \\ u(\mathbf{x}, 0) = g(\mathbf{x}),\ u_t(\mathbf{x}, 0) = h(\mathbf{x}), & \mathbf{x} \in \mathbb{R}^n \end{cases} \qquad (n = 1, 2, 3) \tag{7.95}$$

if and only if $u^0(\mathbf{x}, t)$, the extension of u to zero for $t < 0$, is a solution in $\mathcal{D}'(\mathbb{R}^n \times \mathbb{R})$ of the equation

$$u^0_{tt} - c^2\Delta u^0 = g(\mathbf{x}) \otimes \delta'(t) + h(\mathbf{x}) \otimes \delta(t). \tag{7.96}$$

b) Let $K = K(\mathbf{x}, t)$ be the fundamental solution of the wave equation in one of the dimensions $n = 1, 2, 3$. Deduce from a) that $K^0(\mathbf{x}, t)$ satisfies the equation

$$K^0_{tt} - c^2\Delta K^0 = \delta_n(\mathbf{x}) \otimes \delta(t), \quad \text{in } \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}).$$

[*Hint.* a) Let u satisfy (7.96). This means that, for every $\varphi \in \mathcal{D}(\mathbb{R}^n \times \mathbb{R})$,

$$\int_{\mathbb{R}} \int_{\mathbb{R}^n} u^0(\varphi_{tt} - c^2\Delta\varphi)\,d\mathbf{x}\,dt = -\int_{\mathbb{R}^n} g(\mathbf{x})\varphi_t(\mathbf{x}, 0)\,d\mathbf{x} + \int_{\mathbb{R}^n} h(\mathbf{x})\varphi(\mathbf{x}, 0)\,d\mathbf{x}. \tag{7.97}$$

Integrate twice by parts the first integral, taking into account that $u^0 = u^0(\mathbf{x}, t) = \mathcal{H}(t)u(\mathbf{x}, t)$, to get,

$$\begin{aligned} &\int_0^{+\infty} \int_{\mathbb{R}^n} (u_{tt} - c^2\Delta u)\varphi\,d\mathbf{x}\,dt \\ &= \int_{\mathbb{R}^n} [u(\mathbf{x}, 0) - g(\mathbf{x})]\,\varphi_t(\mathbf{x}, 0)\,d\mathbf{x} - \int_{\mathbb{R}^n} [u_t(\mathbf{x}, 0) - h(\mathbf{x})]\,\varphi(\mathbf{x}, 0)\,d\mathbf{x}. \end{aligned} \tag{7.98}$$

Choose arbitrarily $\varphi \in \mathcal{D}(\mathbb{R}^n \times (0, +\infty))$ to recover the wave equation. Then choose $\psi_0, \psi_1 \in \mathcal{D}(\mathbb{R}^n)$ and $b(t) \in C_0^\infty(\mathbb{R})$ such that $b(0) = 1$ and $b'(0) = 0$. Observe that the function

$$\psi(\mathbf{x}, t) = (\psi_0(\mathbf{x}) + t\psi_1(\mathbf{x}))b(t)$$

belongs to $\mathcal{D}(\mathbb{R}^n \times \mathbb{R})$. Insert ψ into (7.98) and use the arbitrariness of ψ_0, ψ_1 to deduce that u satisfy also the initial conditions. Viceversa, let u satisfy (7.95). A double integration by parts gives, using the initial conditions,

$$0 = \int_0^{+\infty} \int_{\mathbb{R}^n} u(\varphi_{tt} - c^2\Delta\varphi)\,d\mathbf{x}\,dt + \int_{\mathbb{R}^n} g(\mathbf{x})\varphi_t(\mathbf{x}, 0)\,d\mathbf{x} - \int_{\mathbb{R}^n} h(\mathbf{x})\varphi(\mathbf{x}, 0)\,d\mathbf{x}$$

which is equivalent to (7.97).

b) Recall that $K(\mathbf{x},0)=0$, $K_t(\mathbf{x},0)=\delta_n(\mathbf{x})$.]

7.19 (*Uniqueness of the fundamental solution of the heat equation*) Let u be a *nonnegative* solution of the equation

$$u_t - D\Delta u = 0 \quad \text{in } \mathbb{R}^n \times (0,+\infty)$$

such that $u(\mathbf{x},0)=\delta_n$, that is

$$\langle u(\mathbf{x},t), \varphi(\mathbf{x})\rangle \to \varphi(\mathbf{0}) \quad \text{in } \mathcal{D}'(\mathbb{R}^n) \quad \text{as } t \to 0^+.$$

Show that $u(\mathbf{x},t)=\Phi_D(\mathbf{x},t)$, for $t>0$.

[*Answer.* Let η_ε be the mollifier in (7.3); define

$$\delta_\varepsilon = \eta_\varepsilon * \delta \quad \text{and} \quad u_\varepsilon = \eta_\varepsilon * u.$$

Then δ_ε is smooth and $\delta_\varepsilon \to \delta$ in $\mathcal{D}'(\mathbb{R}^n)$ and $u_\varepsilon \geq 0$ is a smooth solution of the heat equation with initial condition $u_\varepsilon(\mathbf{x},0)=\delta_\varepsilon$. Since u_ε is nonnegative and smooth, the Widder Theorem (see Thm. 2.20, p. 87) implies that

$$u_\varepsilon(\mathbf{x},t) = \int_{\mathbb{R}^n} \Phi_D(\mathbf{x}-\mathbf{y},t)\,\delta_\varepsilon(\mathbf{y})\,d\mathbf{y}.$$

Now, as $\varepsilon \to 0$, for each $t>0$, we have both $u_\varepsilon \to \Phi_D$ and $u_\varepsilon = \eta_\varepsilon * u \to u$ in $\mathcal{D}'(\mathbb{R}^n)$. Therefore $u(\mathbf{x},t)=\Phi_D(\mathbf{x},t)$.]

7.20 Let $g \in C_0^\infty(\mathbb{R}^n)$, $f \in C_0^\infty(\mathbb{R}^n \times (0,+\infty))$.

a) Show that $u \in C^{2,1}(\mathbb{R}^n \times [0,+\infty))$ is a solution of the global Cauchy problem

$$\begin{cases} u_t - \Delta u = f(\mathbf{x},t), & \mathbf{x}\in\mathbb{R}^n, t>0 \\ u(\mathbf{x},0)=g(\mathbf{x}), & \mathbf{x}\in\mathbb{R}^n \end{cases} \tag{7.99}$$

if and only if $u^0(\mathbf{x},t)$, the extension of u to zero for $t<0$, is a solution of the equation

$$u^0_t - \Delta u^0 = f^0(\mathbf{x},t) + g(\mathbf{x}) \otimes \delta(t) \quad \text{in } \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}). \tag{7.100}$$

b) Let $\Phi = \Phi(\mathbf{x},t)$ be the fundamental solution of the heat equation in dimension $n=1,2,3$. Deduce from a) that $\Phi^0(\mathbf{x},t)$ satisfies

$$\Phi^0_t - \Delta\Phi^0 = \delta_n(\mathbf{x}) \otimes \delta(t) \quad \text{in } \mathcal{D}'(\mathbb{R}^n \times \mathbb{R}).$$

7.21 Choose in Theorem 7.52, p. 452,

$$H = L^2(\Omega;\mathbb{R}^n), \quad Z = L^2(\Omega) \subset \mathcal{D}'(\Omega)$$

and $L : H \to \mathcal{D}'(\Omega)$ given by $L = \operatorname{div}$. Identify the resulting space W.

7.22 Let X and Z be Banach spaces with $Z \hookrightarrow \mathcal{D}'(\Omega;\mathbb{R}^m)$ (e.g. $L^p(\Omega)$ or $L^p(\Omega;\mathbb{R}^n)$). Let $L : X \to \mathcal{D}'(\Omega;\mathbb{R}^m)$ be a linear continuous operator (e.g. a gradient or a divergence). Define

$$W = \{v \in X : Lv \in Z\}$$

with norm

$$\|u\|^2_W = \|u\|^2_X + \|Lu\|^2_Z .$$

Prove that W is a Banach space, continuously embedded in X.

7.23 (*The Sobolev spaces $W^{1,p}$*) Let $\Omega \subseteq \mathbb{R}^n$ be a domain. For $p \geq 1$, define

$$W^{1,p}(\Omega) = \{v \in L^p(\Omega) : \nabla v \in L^p(\Omega; \mathbb{R}^n)\}\,.$$

Using the result of Problem 7.22, show that $W^{1,p}(\Omega)$ is a Banach space.

7.24 Let $\Omega = B_{1/2}(\mathbf{0}) \subset \mathbb{R}^n$, $n > 2$, and $u(\mathbf{x}) = |\mathbf{x}|^{-a}$, $\mathbf{x} \neq \mathbf{0}$. Determine for which values of a, $u \in H^2(\Omega)$.

7.25 Let $u \in H^s(\mathbb{R})$. Prove that, if $s > 1/2$, $u \in C(\mathbb{R})$ and $u(x) \to 0$ as $x \to \pm\infty$.

[*Hint.* Show that $\widehat{u} \in L^1(\mathbb{R})$.]

7.26 Let u and Ω satisfy the hypotheses of Theorem 7.87, p. 479. Prove that, if the embedding $W^{1,p}(\Omega) \hookrightarrow L^p(\Omega)$ is compact, $1 < p < +\infty$ then

$$\|u\|_{L^p(\Omega)} \leq c(n, p, \Omega)\, \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,.$$

7.27 Let Ω be a bounded domain (not necessarily Lipschitz). Show that $H_0^1(\Omega)$ is compactly embedded in $L^2(\Omega)$.

[*Hint.* Extend u by zero outside Ω.]

7.28 Let $a' < a < b < b'$ and define the operator $E : H^1(a, b) \to H^1(\mathbb{R}^n)$ by $Eu = \psi u^*$, where

$$u^*(x) = \begin{cases} u(a) & x \leq a \\ u(x) & a \leq x \leq b \\ u(b) & x \geq b \end{cases}$$

and $\psi \in C_0^\infty(a', b')$, $0 \leq \psi \leq 1$, $\psi = 1$ for $a \leq x \leq b$. Show that E defines an extension operator, according to Definition 7.73, 467.

7.29 Let $n > 1$ and

$$\Omega = \left\{(\mathbf{x}', x_n) : \mathbf{x}' \in \mathbb{R}^{n-1}, 0 < x_n < d\right\}.$$

Show that the Poincaré's inequality holds in $H_0^1(\Omega)$.

7.30 Prove the chain rule in Sect. 7.7.6 under the further assumption that $f \in C^1(\mathbb{R})$ and that Ω is a bounded Lipschitz domain.

[*Hint.* There exists $\{u_m\} \subset C^1(\overline{\Omega})$, $u_m \to u$ in $H^1(\Omega)$ and a.e. in Ω, as $m \to \infty$. From $|f(u_m) - f(u)| \leq \max|f'||u_m - u|$ deduce that $f(u_m) \to f(u)$ in $L^2(\Omega)$ and a.e. in Ω, as $m \to \infty$. Using the dominated convergence Theorem, show that

$$f'(u_m)\partial_{x_j} u_m - f'(u)\partial_{x_j} u \to 0$$

in $L^2(\Omega)$, as $m \to \infty$.]

7.31 Modify slightly the proof of Theorem 7.77, p. 471, to prove that for every $\varepsilon > 0$

$$\|\tau_0 u\|^2_{L^2(\mathbb{R}^{n-1})} \leq \frac{1+\varepsilon}{\varepsilon}\, \|u\|^2_{L^2(\mathbb{R}^n_+)} + \varepsilon\, \|\nabla u\|^2_{L^2(\mathbb{R}^n_+;\mathbb{R}^n)}\,. \tag{7.101}$$

7.32 Let Ω be a bounded, Lipschitz domain and let Γ_0 a relatively open subset of $\partial\Omega$.

a) Show that

$$(u, v)_{\tilde{H}^1(\Omega)} = \int_{\Gamma_0} u_{|\Gamma_0}\, v_{|\Gamma_0}\, d\sigma + \int_\Omega \nabla u \cdot \nabla v\, d\mathbf{x}$$

is an inner product in $H^1(\Omega)$.

b) Show that the induced norm is equivalent to $\|u\|_{H^1(\Omega)}$.

[*Hint.* See Remark 7.88, p. 480.]

7.33 Let Ω be a bounded Lipschitz domain and $\tau_0 : H^1(\Omega) \to L^2(\Gamma)$ be the trace operator. Show that $\tau_0(\mathcal{D}(\overline{\Omega}))$ is dense both in $L^2(\Gamma)$ and in $H^{1/2}(\Gamma)$, $\Gamma = \partial\Omega$.

7.34 Show that $\mathcal{D}(\mathbb{R}^n)$ is dense in $H^{1/2}(\mathbb{R}^n)$.

[*Hint.* a) First check that $\|u\|_{H^{1/2}(\mathbb{R}^n)} \leq \|u\|_{H^1(\mathbb{R}^n)}$ and, since $\mathcal{D}(\mathbb{R}^n)$ is dense in $H^1(\mathbb{R}^n)$, it is enough to show that $H^1(\mathbb{R}^n)$ is dense in $H^{1/2}(\mathbb{R}^n)$.

b) Let $u \in H^{1/2}(\mathbb{R}^n)$ and define $u_m = \mathcal{F}^{-1}(\hat{u}_m)$ where $\hat{u}_m(\boldsymbol{\xi}) = e^{-|\boldsymbol{\xi}|^2/m}\hat{u}(\boldsymbol{\xi})$. Check that $u_m \in H^1(\mathbb{R}^n)$ and $u_m \to u$ in $H^{1/2}(\mathbb{R}^n)$.]

7.35 Let Ω be a bounded Lipschitz domain, $\Gamma = \partial\Omega$ and $g : \Gamma \to \mathbb{R}$ be Lipschitz continuous with Lipschitz constant L, that is

$$|g(\boldsymbol{\sigma}_1) - g(\boldsymbol{\sigma}_2)| \leq L|\boldsymbol{\sigma}_1 - \boldsymbol{\sigma}_2|$$

for every $\boldsymbol{\sigma}_1, \boldsymbol{\sigma}_2 \in \Gamma$.

a) Show that the function $\tilde{g} : \mathbb{R}^n \to \mathbb{R}$, defined by

$$\tilde{g}(\mathbf{x}) = \min_{\boldsymbol{\sigma}\in\Gamma}\{g(\boldsymbol{\sigma}) + L|\mathbf{x} - \boldsymbol{\sigma}|\}$$

is Lipschitz continuous in $\mathbb{R}^n$ with Lipschitz constant L and $\tilde{g} = g$ on Γ. Thus $\tilde{g}$ is a Lipschitz extension of g in $\mathbb{R}^n$.

b) Deduce that $g \in H^{1/2}(\Gamma)$.

[*Hint.* a) Let $\tilde{g}(\mathbf{x}) = g(\boldsymbol{\sigma}_\mathbf{x}) + L|\mathbf{x} - \boldsymbol{\sigma}_\mathbf{x}|$. Observe that
$g(\mathbf{x}) - g(\mathbf{y}) = g(\boldsymbol{\sigma}_\mathbf{x}) + L|\mathbf{x} - \boldsymbol{\sigma}_\mathbf{x}| - g(\boldsymbol{\sigma}_\mathbf{y}) - L|\mathbf{y} - \boldsymbol{\sigma}_\mathbf{y}| \leq L(|\mathbf{x} - \boldsymbol{\sigma}_\mathbf{x}| - |\mathbf{y} - \boldsymbol{\sigma}_\mathbf{y}|) \leq \cdots.$]

7.36 Prove Proposition 7.98, p. 487.

[*Hint.* Prove that, for all $s, t \in [0, T]$,

$$\int_s^t (u_k(r), v)_V dr \leq C\,\|v\|_V\,(t - s)$$

and use the Lebesgue Differentiation Theorem A.12, p. 661.]

Chapter 8
Variational Formulation of Elliptic Problems

8.1 Linear Elliptic Equations

Poisson's equation $\Delta u = f$ is the simplest among the *elliptic equations*, according to the classification in Sect. 5.5, at least in dimension two. This type of equations plays an important role in the modelling of a large variety of phenomena, often of stationary nature. Typically, in drift, diffusion and reaction models like those considered in Chap. 2, a stationary solution corresponds to a steady state, with no more dependence on time.

Elliptic equations appear in the theory of electrostatic and electromagnetic potentials or in the search of the vibration modes of elastic structures as well (e.g. through the method of separation of variables for the wave equation). Let us define precisely what we mean by *(linear) elliptic equation* in dimension n.

Let $\Omega \subseteq \mathbb{R}^n$ be a domain, $\mathbf{A}(\mathbf{x}) = (a_{ij}(\mathbf{x}))$ a square matrix of order n, $\mathbf{b}(\mathbf{x}) = (b_1(\mathbf{x}), \dots, b_n(\mathbf{x}))$, $\mathbf{c}(\mathbf{x}) = (c_1(\mathbf{x}), \dots, c_n(\mathbf{x}))$ vector fields in $\mathbb{R}^n$, $a = a(\mathbf{x})$ and $f = f(\mathbf{x})$ real functions. An equation of the form

$$-\sum_{i,j=1}^{n} \partial_{x_i}(a_{ij}(\mathbf{x})u_{x_j}) + \sum_{i=1}^{n} \partial_{x_i}(b_i(\mathbf{x})u) + \sum_{i=1}^{n} c_i(\mathbf{x})u_{x_i} + a(\mathbf{x})u = f(\mathbf{x}) \tag{8.1}$$

or

$$-\sum_{i,j=1}^{n} a_{ij}(\mathbf{x})u_{x_i x_j} + \sum_{i=1}^{n} b_i(\mathbf{x})u_{x_i} + a(\mathbf{x})u = f(\mathbf{x}) \tag{8.2}$$

is said to be **elliptic in** Ω if $\mathbf{A}$ is **positive** in Ω, i.e. if the following *ellipticity condition* holds:

$$\sum_{i,j=1}^{n} a_{ij}(\mathbf{x})\xi_i\xi_j > 0, \qquad \forall \mathbf{x} \in \Omega, \forall \boldsymbol{\xi} \in \mathbb{R}^n, \boldsymbol{\xi} \neq \mathbf{0}.$$

We say that (8.1) is in **divergence form** since it may be written as

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$$\underbrace{-\operatorname{div}(\mathbf{A}(\mathbf{x})\nabla u)}_{\text{diffusion}} + \underbrace{\operatorname{div}(\mathbf{b}(\mathbf{x})u) + \mathbf{c}(\mathbf{x})\cdot\nabla u}_{\text{transport}} + \underbrace{a(\mathbf{x})u}_{\text{reaction}} = \underbrace{f(\mathbf{x})}_{\text{external source}} \tag{8.3}$$

which emphasizes the particular structure of the highest order terms. Usually, the first term models the diffusion in heterogeneous or anisotropic media, when the constitutive law for the flux function $\mathbf{q}$ is given by the Fourier or the Fick law:

$$\mathbf{q} = -\mathbf{A}\nabla u.$$

Here u may represent, for instance, a temperature or the concentration of a substance. Thus, the term $-\operatorname{div}(\mathbf{A}\nabla u)$ is associated with thermal or molecular diffusion. The matrix $\mathbf{A}$ is called *diffusion matrix*; the dependence of $\mathbf{A}$ on $\mathbf{x}$ denotes *heterogeneous/anisotropic* diffusion.

The examples in Chap. 2 explain the meaning of the other terms in equation (8.3). In particular, $\operatorname{div}(\mathbf{b}\,u)$ models *convection* or *transport* and corresponds to a flux function given by

$$\mathbf{q} = \mathbf{b}\,u.$$

The vector $\mathbf{b}$ has the dimensions of a velocity. Think, for instance, of the fumes emitted by a factory installations, which diffuse and are transported by the wind. In this case $\mathbf{b}$ is the wind velocity. Note that, if $\operatorname{div}\mathbf{b} = 0$, then $\operatorname{div}(\mathbf{b}\,u)$ reduces to $\mathbf{b}\cdot\nabla u$ which is of the same form of the third term $\mathbf{c}\cdot\nabla u$.

The term $a\,u$ models (linear) *reaction*. If u is the concentration of a substance, a represents the rate of decomposition ($a > 0$) or growth ($a < 0$).

Finally, f represents an external action, distributed in Ω, e.g. the rate of heat per unit mass supplied by an external source.

If the entries a_{ij} of the matrix $\mathbf{A}$ and the components b_j of $\mathbf{b}$ are all differentiable, we may compute the divergence of both $\mathbf{A}\nabla u$ and $\mathbf{b}u$, and reduce (8.1) to the *non-divergence* form

$$-\sum_{i,j=1}^{n} a_{ij}(\mathbf{x})u_{x_i x_j} + \sum_{k=1}^{n} \tilde{b}_k(\mathbf{x})u_{x_k} + \tilde{c}(\mathbf{x})u = f(\mathbf{x})$$

where

$$\tilde{b}_k(\mathbf{x}) = -\sum_{i=1}^{n} \partial_{x_i} a_{ik}(\mathbf{x}) + b_k(\mathbf{x}) + c_k(\mathbf{x}) \quad \text{and} \quad \tilde{c}(\mathbf{x}) = \operatorname{div}\mathbf{b}(\mathbf{x}) + a(\mathbf{x}).$$

However, when a_{ij} or b_j are *not differentiable*, we must keep the divergence form and interpret the differential equation (8.3) in a suitable weak sense.

A *non-divergence form equation* is also associated with diffusion phenomena through stochastic processes which generalize the Brownian motion, called *diffusion processes* (see e.g. *B. Øksendal*, [50]).

In the next section we give a brief account of the various notions of solution available for these kinds of equations. Since the Poisson equation (where $\mathbf{A}$ is the

identity and the other coefficients vanish) is both in divergence and non-divergence form, we use the Dirichlet problem for this equation as a model problem.

8.2 Notions of Solution

Assume we are given a domain $\Omega \subset \mathbb{R}^n$ and two real functions $f : \Omega \to \mathbb{R}$, $g : \partial\Omega \to \mathbb{R}$. We want to determine a function u satisfying the equation

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega. \end{cases} \tag{8.4}$$

Let us examine what we mean by *solving* the above problem. The obvious part is the final goal: we would like to show *existence, uniqueness and stability of the solution*; then, based on these results, we want to *compute* the solution by Numerical Analysis methods.

Less obvious is the *meaning of solution*. In fact, in principle, every problem may be formulated in several ways and a different notion of solution is associated with each way. What is important in the applications is to select the "most efficient notion" for the problem under examination, where "efficiency" may stand for the best compromise between *simplicity of both formulation and theoretical treatment*, *sufficient flexibility and generality, adaptability to numerical methods.*

Here is a (nonexhaustive!) list of various notions of solution for problem (8.4).

• **Classical** solutions are twice continuously differentiable functions; the differential equation and the boundary condition are satisfied in the usual pointwise sense.

• **Strong** solutions belong to the Sobolev space $H^2(\Omega)$. Thus, they possess derivatives in $L^2(\Omega)$ up to the second order, in the sense of distributions.

The differential equation is satisfied in the pointwise sense, a.e. with respect to the Lebesgue measure in Ω, while the boundary condition is satisfied in the sense of traces.

• **Distributional** solutions belong to $L^1_{\text{loc}}(\Omega)$ and the equation holds in the sense of distributions, i.e.:

$$\int_\Omega -u\Delta\varphi d\mathbf{x} = \int_\Omega f\varphi d\mathbf{x}, \quad \forall \varphi \in \mathcal{D}(\Omega).$$

The boundary condition is satisfied in a very weak sense.

• **Variational** (or **weak**) solutions belong to the Sobolev space $H^1(\Omega)$. The boundary value problem is recast within the framework of the abstract variational theory developed in Sect. 6.6. Often the new formulation represents a version of the *principle of virtual work*.

• **Viscosity** solutions. In the simplest case, a viscosity solution is a **continuous function** in $\overline{\Omega}$, satisfying the following conditions: for every $\mathbf{x}_0 \in \Omega$, and every smooth function φ such that $\varphi(\mathbf{x}_0) = u(\mathbf{x}_0)$ and

$$\varphi(\mathbf{x}) \geq u(\mathbf{x}) \quad \text{or} \quad \varphi(\mathbf{x}) \leq u(\mathbf{x}))$$

in a neighborhood of $\mathbf{x}_0$, it holds

$$-\Delta\varphi(\mathbf{x}_0) \leq f(\mathbf{x}_0) \text{ in the first case} \quad \text{and} \quad \Delta\varphi(\mathbf{x}_0) \leq f(\mathbf{x}_0) \text{ in the second one.}$$

As we see, the action of the differential operator is carried onto the test function, in a proper pointwise fashion.[1]

Clearly, all these notions of solution must be connected by a *coherence principle*, which may be stated as follows: if all the data (domain, coefficients, boundary data, forcing terms) and the solution are smooth, *all the above notions* must be equivalent. Thus, the *non-classical* notions constitute a generalization of the classical one.

An important task, with consequences in the error control in numerical methods, is to establish the optimal degree of regularity of a non-classical solution.

More precisely, let u be a non-classical solution of the Poisson problem (8.4). The question is:
how much does the regularity of f, g *and of the domain* Ω *affect the regularity of the solution?*

An exhaustive answer requires rather complicated tools. In the sequel we shall indicate only the most relevant results.

The theory for classical and strong solutions is well established and can be found, e.g. in the book [5] by *D. Gilbarg, N. S. Trudinger*. From the numerical point of view, *the method of finite differences* best fits the differential structure of the problem and aims at approximating classical solutions.

The distributional theory is well developed, is quite general, but is not the most appropriate framework for solving boundary value problems. Indeed, the sense in which the boundary values are attained is one of the most delicate points, when one is willing to widen the notion of solution.

The notion of viscosity solution is designed to deal with fully nonlinear operators, when no integration by parts is available. Important examples come from Stochastic Optimal Control and Dynamic Programming. For the basic theory, we refer to the by now classical paper of *M. Crandall, H. Ishii, P.L. Lions,* User's Guide to Viscosity Solutions of Second Order Partial Differential equations, Bull. A.M.S., 1983.

For our purposes, the most convenient notion of solution is the variational (or weak) one: it leads to a quite flexible formulation, with a sufficiently high degree of generality, and a basic theory relying on the Lax-Milgram Theorem and its variants (Sect. 6.6). Moreover, the analogy (and often the coincidence) with the principle of virtual work indicates a direct connection with the physical interpretation.

Finally, the variational formulation is the most natural to implement the *Galerkin method* (*finite elements, spectral elements, etc. . .*), widely used in the numerical approximation of the solutions of boundary value problems.

[1] To get a clue of the meaning of viscosity solution, look at the problem $u'' = 0$, in the interval $(0, 1)$, $u(0) = 0$, $u(1) = 1$, whose solution is clearly $u(x) = x$. The conditions in the notion of viscosity solution prescribe that, if a parabola $\varphi = \varphi(x)$ touches from above (below) the graph of $u(x) = x$ at a point x_0, then $\varphi''(x_0) \geq 0$ (≤ 0).

Thus, we shall develop the basic theory of elliptic equations in divergence form, recasting the most common boundary value problems within the functional framework of the abstract variational problems of Sect. 6.6. To present the main ideas behind the variational formulation, we start from the Poisson equation (with zero order term). All the results we present hold in any dimension $n \geq 1$. In dimension $n = 1$, Ω is an interval $(a, b) \subset \mathbb{R}$ and $\partial_\nu u$ at the boundary means $-u_x(a)$ and $u_x(b)$.

8.3 Problems for the Poisson Equation

In this section we examine the weak formulation of the most common boundary value problems for the equation $-\Delta u + a(\mathbf{x})u = f$. In general one can proceed along the following steps.

1. Select a space of *smooth test functions, compatible with the boundary conditions.* Multiply the differential equation by a *test* function and integrate over the domain Ω.

2. Assume all the data of the problem are smooth. Integrate by parts the diffusion term, using the boundary conditions and obtaining an *integral equation* (*the variational formulation*).

3. Check that the choice of the test functions and the consequent variational formulation are the correct ones; that is, recover the original formulation by integrating by parts in reverse order. This means that for classical solutions the two formulations are equivalent (coherence principle).

4. Interpret the integral equation as an abstract variational problem (Sect. 6.6) in a suitable Hilbert space. In general, this is a Sobolev space, given by the closure of the space of test functions in a suitable norm.

At this point, the main tools for solving the *abstract variational problem* are the Lax-Milgram Theorem 6.42, p. 376 (or the Riesz Representation Theorem 6.31, p. 370, if the bilinear form is symmetric) and the Fredholm Alternative Theorem 6.69, p. 395.

8.3.1 Dirichlet problem

Let $\Omega \subset \mathbb{R}^n$ be a bounded domain. We examine the following problem:

$$\begin{cases} -\Delta u + a(\mathbf{x})u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{8.5}$$

To achieve a weak formulation, let us follow the steps 1–4 above.

1. We first assume that a and f are smooth and that $u \in C^2(\Omega) \cap C(\overline{\Omega})$ is a classical solution of (8.5). We select $\mathcal{D}(\Omega)$ as the space of test functions, having continuous derivatives of any order and compact support in Ω. In particular, *they vanish in a*

neighborhood of $\partial\Omega$. Let $v \in \mathcal{D}(\Omega)$ and multiply the differential equation by v. We get

$$\int_\Omega \{-\Delta u + a(\mathbf{x})u - f\}\, v \, d\mathbf{x} = 0. \tag{8.6}$$

2. We Integrate by parts and use the boundary condition. We obtain

$$\int_\Omega \{\nabla u \cdot \nabla v + a(\mathbf{x})uv\} \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}, \qquad \forall v \in \mathcal{D}(\Omega). \tag{8.7}$$

Thus (8.5) implies (8.7).

3. Now assume that (8.7) is true. Integrating by parts in the reverse order we return to (8.6), which entails $-\Delta u + a(\mathbf{x})u - f = 0$ in Ω, due to the arbitrariness of v (recall that $\mathcal{D}(\Omega)$ is dense in $L^2(\Omega)$). Thus, *for classical solutions, the two formulations (8.5) and (8.7) are equivalent.*

4. Observe that (8.7) only involves first order derivatives of the solution and of the test function. Then, enlarging the space[2] of test functions to $H_0^1(\Omega)$, the closure of $\mathcal{D}(\Omega)$ in the norm $\|u\|_{H_0^1(\Omega)} = \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}$, we may state the weak formulation of problem (8.5) as follows:

Determine $u \in H_0^1(\Omega)$ such that

$$\int_\Omega \{\nabla u \cdot \nabla v + a(\mathbf{x})uv\} \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}, \qquad \forall v \in H_0^1(\Omega). \tag{8.8}$$

Introducing the bilinear form

$$B(u, v) = \int_\Omega \{\nabla u \cdot \nabla v + a(\mathbf{x})uv\} \, d\mathbf{x}$$

and the linear functional

$$Lv = \int_\Omega f v \, d\mathbf{x},$$

equation (8.8) corresponds to the abstract variational problem

$$B(u, v) = Lv, \quad \forall v \in H_0^1(\Omega).$$

The well-posedness of this problem follows from the Lax-Milgram Theorem under the hypothesis $a(\mathbf{x}) \geq 0$ a.e. in Ω. Recall that a Poincaré inequality holds in $H_0^1(\Omega)$, (see Theorem 7.57, p. 457):

$$\|u\|_{L^2(\Omega)} \leq C_P \, \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \, .$$

We have:

[2] By density, in step 1 we may equivalently have chosen a larger space of enough regular test functions, e.g. the space of $C^1(\overline{\Omega})$ functions vanishing at the boundary.

Theorem 8.1 *Assume that* $f \in L^2(\Omega)$ *and that* $0 \le a(\mathbf{x}) \le \alpha_0$ *a.e. in* Ω. *Then, problem* (8.8) *has a unique solution* $u \in H_0^1(\Omega)$. *Moreover*

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \le C_P \, \|f\|_{L^2(\Omega)} \, .$$

Proof We check that the hypotheses of the Lax-Milgram Theorem hold, with $V = H_0^1(\Omega)$.

Continuity of the bilinear form B. The Schwarz and Poincaré inequalities yield:

$$\begin{aligned} |B(u,v)| &\le \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \, \|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)} + \alpha_0 \, \|u\|_{L^2(\Omega)} \, \|v\|_{L^2(\Omega)} \\ &\le (1 + C_P^2 \alpha_0) \, \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \, \|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)} \end{aligned}$$

so that B is continuous in $H_0^1(\Omega)$.

Coercivity of B. It follows from

$$B(u,u) = \int_\Omega |\nabla u|^2 \, d\mathbf{x} + \int_\Omega a(\mathbf{x}) u^2 d\mathbf{x} \ge \|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)}$$

since $a \ge 0$.

Continuity of L. The Schwarz and Poincaré inequalities give

$$\begin{aligned} |Lv| = \left| \int_\Omega f v \, d\mathbf{x} \right| &\le \|f\|_{L^2(\Omega)} \, \|v\|_{L^2(\Omega)} \\ &\le C_P \, \|f\|_{L^2(\Omega)} \, \|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)} \, . \end{aligned}$$

Hence $L \in H^{-1}(\Omega)$ and $\|L\|_{H^{-1}(\Omega)} \le C_P \, \|f\|_{L^2(\Omega)}$. The conclusions follow from the Lax-Milgram Theorem. □

Remark 8.2 Suppose that $a = 0$ and that u represents the equilibrium position of an elastic membrane. Then $B(u,v)$ represents the work done by the elastic internal forces, due to a *virtual displacement* v. On the other hand Lv expresses the work done by the external forces. The weak formulation (8.8) states that these two works balance, and it constitutes a version of the *principle of virtual work*.

Furthermore, due to the symmetry of B, the solution u of the problem **minimizes in** $H_0^1(\Omega)$ **the Dirichlet functional**

$$E(u) = \underbrace{\frac{1}{2} \int_\Omega |\nabla u|^2 \, d\mathbf{x}}_{\text{internal elastic energy}} - \underbrace{\int_\Omega f u \, d\mathbf{x}}_{\text{external potential energy}}$$

which represents the **total potential energy**. Equation (8.8) constitutes the Euler equation for E (recall Sect. 6.6.3, p. 381).

Thus, in agreement with the principle of virtual work, *u minimizes the potential energy among all the admissible configurations.*

Similar observations can be made for the other types of boundary conditions.

8.3.2 Neumann, Robin and mixed problems

Let $\Omega \subset \mathbb{R}^n$ be a bounded, *Lipschitz* domain. We examine the following Neumann problem:

$$\begin{cases} -\Delta u + a(\mathbf{x})u = f & \text{in } \Omega \\ \partial_\nu u = g & \text{on } \partial\Omega \end{cases} \tag{8.9}$$

where $\boldsymbol{\nu}$ denotes the outward normal unit vector to $\partial\Omega$. As for the Dirichlet problem, to derive a variational formulation, we follows the steps 1–4.

1. Assume that a, f and g are smooth and that $u \in C^2(\Omega) \cap C^1(\overline{\Omega})$ is a classical solution of (8.9). Since there is no particularly convenient way to choose the space of test functions in order to incorporate the Neumann boundary condition, we choose $\mathcal{D}(\overline{\Omega})$ as the space of test functions, having continuous derivatives of any order up to $\partial\Omega$. Let $v \in \mathcal{D}(\overline{\Omega})$, arbitrary, and multiply the Poisson equation by v. Integrating over Ω, we get

$$\int_\Omega \{-\Delta u + au\}\, v\, d\mathbf{x} = \int_\Omega fv\, d\mathbf{x}. \tag{8.10}$$

2. An integration by parts gives

$$-\int_{\partial\Omega} \partial_\nu u\, v d\sigma + \int_\Omega \{\nabla u \cdot \nabla v\ + auv\}\, d\mathbf{x} = \int_\Omega fv\, d\mathbf{x}, \quad \forall v \in \mathcal{D}(\overline{\Omega}). \tag{8.11}$$

Using the Neumann condition we may write

$$\int_\Omega \{\nabla u \cdot \nabla v + auv\}\, d\mathbf{x} = \int_\Omega fv\, d\mathbf{x} + \int_{\partial\Omega} gv\, d\sigma, \qquad \forall v \in \mathcal{D}(\overline{\Omega}). \tag{8.12}$$

Thus (8.9) implies (8.12).

3. On the other hand, suppose that (8.12) is true. Integrating by parts in the reverse order, we find

$$\int_\Omega \{-\Delta u + au - f\}\, v\, d\mathbf{x} + \int_{\partial\Omega} \partial_\nu u\, v d\sigma = \int_{\partial\Omega} gv\, d\sigma, \tag{8.13}$$

for every $\forall v \in \mathcal{D}(\overline{\Omega})$. Since $\mathcal{D}(\Omega) \subset \mathcal{D}(\overline{\Omega})$ we may insert any $v \in \mathcal{D}(\Omega)$ into (8.13), to get

$$\int_\Omega \{-\Delta u + \ au - f\ \}\, v\, d\mathbf{x} = 0.$$

The arbitrariness of $v \in \mathcal{D}(\Omega)$ entails $-\Delta u + au - f = 0$ in Ω. Therefore (8.13) becomes

$$\int_{\partial\Omega} \partial_\nu u\, v d\sigma = \int_{\partial\Omega} gv\, d\sigma, \qquad \forall v \in \mathcal{D}(\overline{\Omega})$$

and the arbitrariness of $v \in \mathcal{D}(\overline{\Omega})$ forces $\partial_\nu u = g$, recovering the Neumann condition as well (if Ω is smooth, the set of the restrictions to $\partial\Omega$ of functions in $\mathcal{D}(\overline{\Omega})$ is

dense in $L^2(\partial\Omega)$, see Problem 7.33, p. 494). *Thus, for classical solutions, the two formulations (8.9) and (8.12) are equivalent.*

4. Recall now that, by Theorem 7.76, p. 470, $\mathcal{D}(\overline{\Omega})$ is dense in $H^1(\Omega)$, which therefore constitutes the natural Sobolev space for the Neumann problem. Then, enlarging the space of test functions to $H^1(\Omega)$, we may give the **weak** formulation of problem (8.9) as follows:

Determine $u \in H^1(\Omega)$ such that

$$\int_\Omega \{\nabla u \cdot \nabla v + auv\}\, d\mathbf{x} = \int_\Omega fv d\mathbf{x} + \int_{\partial\Omega} gv d\sigma, \qquad \forall v \in H^1(\Omega). \tag{8.14}$$

We point out that the Neumann condition is *encoded in (8.14)* and not forced as for the Dirichlet boundary conditions. Since we used the density of $\mathcal{D}(\overline{\Omega})$ in $H^1(\Omega)$ and the trace of v on $\partial\Omega$, some regularity of the domain is needed, even in the variational formulation (Lipschitz is enough). Introducing the bilinear form

$$B(u,v) = \int_\Omega \{\nabla u \cdot \nabla v + auv\}\ d\mathbf{x} \tag{8.15}$$

and the linear functional

$$Lv = \int_\Omega fv\, d\mathbf{x} + \int_{\partial\Omega} gv d\sigma, \tag{8.16}$$

(8.14) may be formulated as the abstract variational problem

$$B(u,v) = Lv, \quad \forall v \in H^1(\Omega).$$

The following theorem states the well-posedness of this problem under reasonable hypotheses on the data. Recall from Theorem 7.77, p. 471, the *trace inequality*

$$\|v\|_{L^2(\partial\Omega)} \le \overline{C}(n,\Omega)\, \|v\|_{H^1(\Omega)}\,. \tag{8.17}$$

Theorem 8.3 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, Lipschitz domain, $f \in L^2(\Omega)$, $g \in L^2(\partial\Omega)$ and $0 < c_0 \le a(\mathbf{x}) \le \alpha_0$ a.e. in Ω. Then, problem (8.14) has a unique solution $u \in H^1(\Omega)$. Moreover,*

$$\|u\|_{H^1(\Omega)} \le \frac{1}{\min\{1, c_0\}} \left\{ \|f\|_{L^2(\Omega)} + \overline{C}\, \|g\|_{L^2(\partial\Omega)} \right\}.$$

Proof We check that the hypotheses of the Lax-Milgram Theorem hold, with $V = H^1(\Omega)$.

Continuity of the bilinear form B. The Schwarz inequality yields:

$$\begin{aligned} |B(u,v)| &\le \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\, \|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)} + \alpha_0\, \|u\|_{L^2(\Omega)}\, \|v\|_{L^2(\Omega)} \\ &\le (1+\alpha_0)\, \|u\|_{H^1(\Omega)}\, \|v\|_{H^1(\Omega)} \end{aligned}$$

so that B is continuous in $H^1(\Omega)$.

Coercivity of B. It follows from

$$B(u,u) = \int_\Omega |\nabla u|^2 \, d\mathbf{x} + \int_\Omega a u^2 d\mathbf{x} \geq \min\{1, c_0\} \, \|u\|^2_{H^1(\Omega)}$$

since

$$a(\mathbf{x}) \geq c_0 > 0 \quad \text{a.e. in } \Omega.$$

Continuity of L. From the Schwarz inequality and (8.17) we get:

$$\begin{aligned} |Lv| &\leq \left| \int_\Omega f v \, d\mathbf{x} \right| + \left| \int_{\partial\Omega} g v \, d\sigma \right| \leq \|f\|_{L^2(\Omega)} \, \|v\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)} \, \|v\|_{L^2(\partial\Omega)} \\ &\leq \left\{ \|f\|_{L^2(\Omega)} + \overline{C} \, \|g\|_{L^2(\partial\Omega)} \right\} \|v\|_{H^1(\Omega)} \, . \end{aligned}$$

Therefore L is continuous in $H^1(\Omega)$ with

$$\|L\|_{H^1(\Omega)^*} \leq \|f\|_{L^2(\Omega)} + \overline{C} \, \|g\|_{L^2(\partial\Omega)} \, .$$

The conclusion follows from the Lax-Milgram Theorem. □

Remark 8.4 The condition $a(\mathbf{x}) \geq c_0 > 0$ for a.e. $\mathbf{x} \in \Omega$, in Theorem 8.3, may be relaxed by asking that[3]

$$a(\mathbf{x}) \geq 0 \quad \text{and} \quad \int_\Omega a(\mathbf{x}) d\mathbf{x} = I_0 > 0.$$

Indeed, in this case,

$$a(\mathbf{x}) \geq c_1 > 0$$

on some set E of positive measure. By Remark 7.88, p. 480, we can write

$$B(u,u) \geq \min\{1, c_1\} \left(\int_\Omega |\nabla u|^2 \, d\mathbf{x} + \int_E u^2 d\mathbf{x} \right) \geq C \, \|u\|^2_{H^1(\Omega)}$$

where C depends on c_1, $|\Omega|$ and $|E|$, recovering the coercivity of B.

If $a \equiv 0$, *neither the existence nor the uniqueness* of a solution is guaranteed. Two solutions with the same Neumann data differ by a constant. A way to restore uniqueness is to select a solution with, e.g., zero mean value, that is

$$\int_\Omega u(\mathbf{x}) d\mathbf{x} = 0.$$

The existence of a solution requires the following **compatibility condition** on the data f and g:

$$\int_\Omega f \, d\mathbf{x} + \int_{\partial\Omega} g \, d\sigma = 0, \tag{8.18}$$

obtained by substituting $v = 1$ into the equation

$$\int_\Omega \nabla u \cdot \nabla v \, dx = \int_\Omega f v \, d\mathbf{x} + \int_{\partial\Omega} g v \, d\sigma.$$

[3] Useful in some control problems.

Note that, since Ω is bounded, the function $v = 1$ belongs to $H^1(\Omega)$.

If $a \equiv 0$ and (8.18) does not hold, problem (8.9) has no solution. Viceversa, we shall see later that, if this condition is fulfilled, a solution exists.

If $g = 0$, (8.18) has a simple interpretation. Indeed problem (8.9) is a model for the equilibrium configuration of a membrane whose boundary is free to slide along a vertical guide. The compatibility condition $\int_\Omega f d\mathbf{x} = 0$ expresses the obvious fact that, at equilibrium, the resultant of the external loads must vanish.

Robin problem. The same arguments leading to the weak formulation of the Neumann problem (8.9) may be used for the problem

$$\begin{cases} -\Delta u + a(\mathbf{x})u = f & \text{in } \Omega \\ \partial_\nu u + h(\mathbf{x})u = g & \text{on } \partial\Omega. \end{cases} \tag{8.19}$$

Inserting into (8.11) the Robin condition

$$\partial_\nu u = -hu + g \quad \text{on } \partial\Omega$$

we obtain the following **variational formulation**:

Determine $u \in H^1(\Omega)$ such that

$$\int_\Omega \{\nabla u \cdot \nabla v + auv\}\, d\mathbf{x} + \int_{\partial\Omega} huv\, d\sigma = \int_\Omega fv\, d\mathbf{x} + \int_{\partial\Omega} gv\, d\sigma, \quad \forall v \in H^1(\Omega).$$

We have:

Theorem 8.5 *Let Ω, f, g and a be as in Theorem 8.3, and $0 \le h(\mathbf{x}) \le h_0$ a.e. on $\partial\Omega$. Then, problem (8.19) has a unique weak solution $u \in H^1(\Omega)$. Moreover*

$$\|u\|_{H^1(\Omega)} \le \frac{1}{\min\{1, c_0\}} \left\{ \|f\|_{L^2(\Omega)} + \overline{C}\, \|g\|_{L^2(\partial\Omega)} \right\}.$$

Proof Introducing the bilinear form

$$\tilde{B}(u, v) = B(u, v) + \int_{\partial\Omega} huv\, d\sigma,$$

the variational formulation becomes

$$\tilde{B}(u, v) = Lv, \qquad \forall v \in H^1(\Omega)$$

where B and L are defined in (8.15) and (8.16), respectively.

From the Schwarz inequality and (8.17), we infer

$$\left| \int_{\partial\Omega} huv\, d\sigma \right| \le h_0 \|u\|_{L^2(\partial\Omega)} \|v\|_{L^2(\partial\Omega)} \le \overline{C}^2 h_0 \|u\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)}.$$

On the other hand, the positivity of α, a_0 and h entails that

$$\tilde{B}(u, u) \ge B(u, u) \ge \min\{1, c_0\} \|u\|^2_{H^1(\Omega)}.$$

The conclusions follow easily. □

Remark 8.6 As in Theorem 8.3, we may relax the condition $a_0(\mathbf{x}) \geq c_0 > 0$ for a.e. $\mathbf{x} \in \Omega$. See Problem 8.17.

Mixed Dirichlet-Neumann problem. Let Γ_D and Γ_N be nonempty relatively open subsets of $\partial\Omega$, with $\Gamma_N = \partial\Omega \setminus \overline{\Gamma}_D$, regular in the sense of Definition 1.11, p. 19. Consider the problem

$$\begin{cases} -\Delta u + a(\mathbf{x})u = f & \text{in } \Omega \\ u = 0 & \text{on } \Gamma_D \\ \partial_\nu u = g & \text{on } \Gamma_N . \end{cases}$$

The correct functional setting is $H^1_{0,\Gamma_D}(\Omega)$, i.e. the set of functions in $H^1(\Omega)$ with zero trace on Γ_D. From Theorem 7.87, p. 479, the Poincaré inequality holds in $H^1_{0,\Gamma_D}(\Omega)$ and therefore we may choose the norm

$$\|u\|_{H^1_{0,\Gamma_D}(\Omega)} = \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} .$$

From (8.10) and the Gauss formula, we obtain, since $u = 0$ on Γ_D,

$$-\int_{\Gamma_N} \partial_\nu u \, v d\sigma + \int_\Omega \{\nabla u \cdot \nabla v \, + auv\} \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}, \quad \forall v \in \mathcal{D}(\overline{\Omega}).$$

Using the Neumann condition on Γ_N yields the following **variational formulation**:

Determine $u \in H^1_{0,\Gamma_D}(\Omega)$ such that, $\forall v \in H^1_{0,\Gamma_D}(\Omega)$,

$$\int_\Omega \nabla u \cdot \nabla v \, d\mathbf{x} + \int_\Omega auv \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x} + \int_{\Gamma_N} gv \, d\sigma.$$

Using the trace inequality (Theorem 7.80, p. 473)

$$\|v\|_{L^2(\Gamma_N)} \leq \widetilde{C} \, \|v\|_{H^1(\Omega)} , \tag{8.20}$$

the proof of the next theorem follows the usual pattern.

Theorem 8.7 *Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain. Assume $f \in L^2(\Omega)$, $g \in L^2(\Gamma_N)$ and $0 \leq a(\mathbf{x}) \leq \alpha_0$ a.e. in Ω. Then the mixed problem has a unique solution $u \in H^1_{0,\Gamma_D}(\Omega)$. Moreover:*

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \leq C_P \, \|f\|_{L^2(\Omega)} + \widetilde{C} \, \|g\|_{L^2(\Gamma_N)} .$$

8.3.3 Eigenvalues and eigenfunctions of the Laplace operator

In Sect. 6.9.2 we have seen how the efficacy of the separation of variables method for a given problem relies on the existence of a basis of eigenfunctions associated with that problem. The abstract results in Sect. 6.9.4, concerning the spectrum of

a weakly coercive bilinear form, constitute the appropriate tools for analyzing the spectral properties of uniformly elliptic operators and in particular of the Laplace operator. It is important to point out that the spectrum of a differential operator *must be associated with specific homogeneous boundary conditions.*

Thus, for instance, we may consider the *Dirichlet eigenfunctions* for the Laplace operator in a bounded domain Ω, i.e. the nontrivial solutions of the problem

$$\begin{cases} -\Delta u = \lambda u & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{8.21}$$

A weak solution of problem (8.21) is a function $u \in H_0^1(\Omega)$ such that

$$(\nabla u, \nabla v)_{L^2(\Omega;\mathbb{R}^n)} = \lambda (u, v)_{L^2(\Omega)}, \qquad \forall v \in H_0^1(\Omega).$$

Since Ω is bounded, the bilinear form is $H_0^1(\Omega)$-coercive so that Theorem 6.78, p. 405, gives:

Theorem 8.8 *Let Ω be a bounded domain. Then:*

a) There exists in $L^2(\Omega)$ an orthonormal basis $\{u_k\}_{k\geq 1}$ consisting of Dirichlet eigenfunctions for the Laplace operator.

b) The corresponding eigenvalues $\{\lambda_k\}_{k\geq 1}$ are all positive and may be arranged in a nondecreasing sequence

$$0 < \lambda_1 < \lambda_2 \leq \cdots \leq \lambda_k \leq \cdots \tag{8.22}$$

with $\lambda_k \to +\infty$. Every eigenvalue appears in the sequence (8.22) a finite number of times, according to its multiplicity.

c) The sequence $\left\{u_k/\sqrt{\lambda_k}\right\}_{k\geq 1}$ constitutes an orthonormal basis in $H_0^1(\Omega)$, with respect to the scalar product $(u, v)_{H_0^1(\Omega)} = (\nabla u, \nabla v)_{L^2(\Omega;\mathbb{R}^n)}$.

Actually, the strict inequality $\lambda_1 < \lambda_2$ in (8.22) is a consequence of Theorem 8.9 below, which states that λ_1 is a simple eigenvalue, that is of multiplicity 1.

From Theorem 6.81, p. 407, we deduce the following **variational principle for λ_1**, the **principal Dirichlet eigenvalue**:

$$\lambda_1 = \min\left\{R(u) : u \in H_0^1(\Omega), u \neq 0\right\} \tag{8.23}$$

where $R(u)$ is the Rayleigh quotient:

$$R(u) = \frac{\int_\Omega |\nabla u|^2 \, d\mathbf{x}}{\int_\Omega u^2 d\mathbf{x}}. \tag{8.24}$$

Moreover, the minimum is attained at any *eigenvector* corresponding to λ_1. Thus,

$$\|u\|_{L^2(\Omega)} \leq \frac{1}{\sqrt{\lambda_1}} \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \tag{8.25}$$

and equality holds when u is an eigenvector corresponding to λ_1. An interesting consequence is that $1/\sqrt{\lambda_1}$ is the best (i.e. *the smallest*) *Poincaré constant for the domain* Ω.

More information on λ_1 and its eigenspace are contained in the following theorem.

Theorem 8.9 *Let Ω be a bounded, Lipschitz domain. Then the principal eigenvalue λ_1 is simple, that is the corresponding eigenspace is 1-dimensional. Moreover, every eigenfunction corresponding to λ_1 has constant sign in Ω.*

Proof Let u_1 be an eigenvector corresponding to λ_1. First we prove that u_1 has constant sign in Ω. Let $w = |u_1|$. By Proposition 7.64, p. 462, we have $\nabla w = \operatorname{sign}(u_1)\nabla u_1$ and, consequently, $R(w) = R(u_1)$. From Theorem 6.81, p. 407, we deduce that w also is an eigenvector corresponding to λ_1, that is it satisfies the equation

$$(\nabla w, \nabla v)_{L^2(\Omega;\mathbb{R}^n)} = \lambda_1 (w, v)_{L^2(\Omega)}, \qquad \forall v \in H^1(\Omega).$$

Theorems 8.35, p. 533, and 8.29, p. 529, give $w \in C^\infty(\Omega) \cap C(\overline{\Omega})$, so that w is a classical solution of the equation

$$-\Delta w = \lambda_1 w, \quad \text{in } \Omega$$

and $w = 0$ on $\partial\Omega$. Since $w \geq 0$ and $\lambda_1 > 0$, w is superharmonic in Ω and the maximum principle (see Sect. 3.4) gives either $w \equiv 0$ or $w > 0$ in Ω. Thus u_1 never vanishes in Ω and therefore, being continuous, it has constant sign in Ω.

Suppose now by contradiction that the λ_1-eigenspace has dimension > 1. Then there should exist another eigenfunction corresponding to λ_1, orthogonal to u_1 in $L^2(\Omega)$:

$$\int_\Omega u_1 u_2 \, d\mathbf{x} = 0.$$

But this is impossible because every λ_1-eigenfunction has constant sign in Ω. □

Also the other eigenvalues can be variationally characterized. For instance, denoting by V_1 the eigenspace corresponding to λ_1, we have (see Problem 8.18):

$$\lambda_2 = \min \left\{ R(v) : v \neq 0, v \in H_0^1(\Omega) \cap V_1^\perp \right\}. \tag{8.26}$$

Similar results hold for the other types of boundary value problems as well. For instance, the *Neumann eigenfunctions* for the Laplace operator in Ω are the nontrivial solutions of the problem

$$\begin{cases} -\Delta u = \mu u & \text{in } \Omega \\ \partial_\nu u = 0 & \text{on } \partial\Omega. \end{cases} \tag{8.27}$$

A weak solution of (8.27) is a function $u \in H^1(\Omega)$ such that

$$B(u, v) \equiv (\nabla u, \nabla v)_{L^2(\Omega;\mathbb{R}^n)} = \mu (u, v)_{L^2(\Omega)}, \quad \forall v \in H^1(\Omega).$$

If Ω is a bounded Lipschitz domain, $H^1(\Omega)$ is compactly embedded into $L^2(\Omega)$ and the bilinear form B is clearly weakly coercive with constant, say, $\lambda_0 = 1$. Applying Theorem 6.78, we find:

Theorem 8.10 *Let Ω be a bounded Lipschitz domain.*

a) There exists in $L^2(\Omega)$ an orthonormal basis $\{u_k\}_{k\geq 1}$ consisting of Neumann eigenfunctions for the Laplace operator.
b) The corresponding eigenvalues form a nondecreasing sequence

$$0 = \mu_1 < \mu_2 \leq \cdots \leq \mu_k \leq \cdots \tag{8.28}$$

with $\mu_k \to +\infty$. Each eigenvalue appears in the sequence (8.28) a finite number of times, according to its multiplicity. In particular, the principal eigenvalue $\mu_1 = 0$ is simple (corresponding to constant eigenfunctions).
c) The sequence $\{u_k/\sqrt{\mu_k + 1}\}_{k\geq 1}$ constitutes an orthonormal basis in $H^1(\Omega)$, with respect to the scalar product $(u, v)_{H^1(\Omega)} = (u, v)_{L^2(\Omega)} + (\nabla u, \nabla v)_{L^2(\Omega;\mathbb{R}^n)}$.

8.3.4 An asymptotic stability result

The results in the last subsection may be used sometimes to prove the asymptotic stability of a steady state solution of an evolution equation as time $t \to +\infty$.

As an example, consider the following problem for the heat equation. Suppose that u is the (unique and smooth) solution of

$$\begin{cases} u_t - \Delta u = f(\mathbf{x}) & \mathbf{x} \in \Omega, t > 0 \\ u(\mathbf{x}, 0) = u_0(\mathbf{x}) & \mathbf{x} \in \Omega \\ u(\boldsymbol{\sigma}, t) = 0 & \boldsymbol{\sigma} \in \partial\Omega, t > 0, \end{cases}$$

where Ω is a smooth, bounded domain. Denote by $u_\infty = u_\infty(\mathbf{x})$ the solution of the stationary problem

$$\begin{cases} -\Delta u_\infty = f & \text{in } \Omega \\ u_\infty = 0 & \text{on } \partial\Omega. \end{cases}$$

Proposition 8.11 *For $t \geq 0$, we have*

$$\|u(\cdot, t) - u_\infty\|_{L^2(\Omega)} \leq e^{-\lambda_1 t} \left\{ \|u_0\|_{L^2(\Omega)} + {\lambda_1}^{-1} \|f\|_{L^2(\Omega)} \right\} \tag{8.29}$$

where λ_1 is the first Dirichlet eigenvalue for the Laplace operator in Ω.

Proof Set $g(\mathbf{x}) = u_0(\mathbf{x}) - u_\infty(\mathbf{x})$. The function $w(\mathbf{x}, t) = u(\mathbf{x}, t) - u_\infty(\mathbf{x})$ solves the problem

$$\begin{cases} w_t - \Delta w = 0 & \mathbf{x} \in \Omega, t > 0 \\ w(\mathbf{x}, 0) = g(\mathbf{x}) & \mathbf{x} \in \Omega \\ w(\boldsymbol{\sigma}, t) = 0 & \boldsymbol{\sigma} \in \partial\Omega, t > 0. \end{cases} \tag{8.30}$$

Let us use the method of separation of variables and look for solutions of the form $w(\mathbf{x}, t) = v(\mathbf{x})z(t)$. We find

$$\frac{z'(t)}{z(t)} = \frac{\Delta v(\mathbf{x})}{v(\mathbf{x})} = -\lambda$$

with λ constant. Thus we are led to the eigenvalue problem

$$\begin{cases} -\Delta v = \lambda v & \text{in } \Omega \\ v = 0 & \text{on } \partial\Omega. \end{cases}$$

From Theorem 8.8, there exists in $L^2(\Omega)$ an orthonormal basis $\{u_k\}_{k\geq 1}$ consisting of eigenvectors, corresponding to a sequence of nondecreasing eigenvalues $\{\lambda_k\}$, with $\lambda_1 > 0$ and $\lambda_k \to +\infty$. Then, if $g_k = (g, u_k)_{L^2(\Omega)}$, we can write

$$g = \sum_1^\infty g_k u_k \qquad \text{and} \qquad \|g\|^2_{L^2(\Omega)} = \sum_{k=1}^\infty g_k^2.$$

As a consequence, we find $z_k(t) = e^{-\lambda_k t}$ and finally

$$w(\mathbf{x}, t) = \sum_1^\infty e^{-\lambda_k t} g_k u_k(\mathbf{x}).$$

Thus,

$$\|u(\cdot, t) - u_\infty\|^2_{L^2(\Omega)} = \|w(\cdot, t)\|^2_{L^2(\Omega)} = \sum_{k=1}^\infty e^{-2\lambda_k t} g_k^2$$

and, since $\lambda_k > \lambda_1$ for every k, we deduce that

$$\|u(\cdot, t) - u_\infty\|^2_{L^2(\Omega)} \leq \sum_{k=1}^\infty e^{-2\lambda_1 t} g_k^2 = e^{-2\lambda_1 t} \|g\|^2_{L^2(\Omega)}.$$

Recalling (8.25) and Theorem 8.1, p. 500 (with $C_P = \frac{1}{\sqrt{\lambda_1}}$), we get

$$\|u_\infty\|_{L^2(\Omega)} \leq \frac{1}{\sqrt{\lambda_1}} \|\nabla u_\infty\|_{L^2(\Omega)} \leq \frac{1}{\lambda_1} \|f\|_{L^2(\Omega)},$$

and hence

$$\|g\|_{L^2(\Omega)} \leq \|u_0\|_{L^2(\Omega)} + \|u_\infty\|_{L^2(\Omega)} \leq \|u_0\|_{L^2(\Omega)} + \frac{1}{\lambda_1} \|f\|_{L^2(\Omega)},$$

giving (8.29). □

Proposition 8.11 implies that the steady state u_∞ is *asymptotically stable in the $L^2(\Omega)$-norm* as $t \to +\infty$. The speed of convergence is exponential and it is determined by the first eigenvalue λ_1 (compare with Sect. 2.1.4).

8.4 General Equations in Divergence Form

8.4.1 Basic assumptions

In this section we consider boundary value problems for elliptic operators with general diffusion and transport terms. Let $\Omega \subset \mathbb{R}^n$ be a **bounded domain** and set

$$\mathcal{E}u = -\operatorname{div}(\mathbf{A}(\mathbf{x})\nabla u) + \operatorname{div}(\mathbf{b}(\mathbf{x})u) + \mathbf{c}(\mathbf{x}) \cdot \nabla u + a(\mathbf{x})u \tag{8.31}$$

where

$$\mathbf{A} = (a_{ij})_{i,j=1,\ldots,n}, \quad \mathbf{b} = (b_1, \ldots, b_n), \quad \mathbf{c} = (c_1, \ldots, c_n) \quad \text{and} \quad a$$

are all real functions.

Throughout this section, we will work under the following assumptions.

1. The differential operator $\mathcal{E}$ is **uniformly elliptic**, i.e. there exist **positive numbers** α and M such that:

$$\alpha\,|\boldsymbol{\xi}|^2 \le \sum_{i,j=1}^{n} a_{ij}(\mathbf{x})\xi_i\xi_j \quad \text{and} \quad \left|a_{ij}(\mathbf{x})\right| \le M, \quad \forall \boldsymbol{\xi} \in \mathbb{R}^n, \text{ for a.e. } \mathbf{x} \in \Omega. \tag{8.32}$$

2. The coefficients $\mathbf{b}$, $\mathbf{c}$ and a are all **bounded**:

$$|\mathbf{b}(\mathbf{x})| \le \beta_0, \quad |\mathbf{c}(\mathbf{x})| \le \gamma_0, \quad |a(\mathbf{x})| \le \alpha_0, \quad \text{a.e. in } \Omega. \tag{8.33}$$

The uniform ellipticity condition (8.32) states that $\mathbf{A}$ is positive[4] in Ω. If $\mathbf{A}$ is symmetric its minimum eigenvalue is bounded from below by α, called *ellipticity constant*. We point out that at this level of generality, we allow discontinuities also of the diffusion matrix $\mathbf{A}$, of the transport coefficients $\mathbf{b}$ and $\mathbf{c}$, in addition to the reaction coefficient a.

We want to extend to these type of operators the theory developed so far. The uniform ellipticity is a key property. In this section, we first indicate some sufficient conditions assuring the well-posedness of the usual boundary value problems, based on the use of the Lax-Milgram Theorem.

On the other hand, these conditions may be sometimes rather restrictive. When they are not satisfied, precise information on solvability and well-posedness can be obtained from the Fredholm Alternative Theorem 6.69, p. 395.

As in the preceding sections, we start from the Dirichlet problem.

8.4.2 Dirichlet problem

Consider the problem

$$\begin{cases} \mathcal{E}u = f + \operatorname{div}\mathbf{f} & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{8.34}$$

where $f \in L^2(\Omega)$ and $\mathbf{f} \in L^2(\Omega;\mathbb{R}^n)$.

A comment on the right hand side of (8.34) is in order. We have denoted by $H^{-1}(\Omega)$ the dual of $H_0^1(\Omega)$. We know (Theorem 7.59, p. 459) that every element $F \in H^{-1}(\Omega)$ can be identified with an element in $\mathcal{D}'(\Omega)$ of the form

$$F = f + \operatorname{div}\mathbf{f}.$$

Thus, the right hand side of (8.34) represents an element of $H^{-1}(\Omega)$. Since we are using in $H_0^1(\Omega)$ the norm $\|\nabla u\|_{L^2(\Omega)}$ we have

[4] If $\mathbf{A}$ is only *nonnegative*, the equation is *degenerate elliptic* and things get too complicated for this introductory book.

$$\|F\|_{H^{-1}(\Omega)} \leq C_P \, \|f\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)} \, . \tag{8.35}$$

As in Sect. 8.3, to derive a variational formulation of (8.34), we first assume that all the coefficients and the data f, $\mathbf{f}$ are smooth. Then, we multiply the equation by a test function $v \in \mathcal{D}(\Omega)$ and integrate over Ω:

$$\int_\Omega \{-\operatorname{div}(\mathbf{A}\nabla u - \mathbf{b}u)\, v\}\, d\mathbf{x} + \int_\Omega (\mathbf{c}\cdot\nabla u + au)v \, d\mathbf{x} = \int_\Omega (f + \operatorname{div}\mathbf{f})v \, d\mathbf{x}.$$

Integrating by parts, we find, since $v = 0$ on $\partial\Omega$:

$$\int_\Omega \{-\operatorname{div}(\mathbf{A}\nabla u - \mathbf{b}u)\, v\}\, d\mathbf{x} = \int_\Omega \{\mathbf{A}\nabla u \cdot \nabla v \ - \mathbf{b}u \cdot \nabla v\}\, d\mathbf{x}$$

and

$$\int_\Omega v \operatorname{div}\mathbf{f} \, d\mathbf{x} = -\int_\Omega \mathbf{f}\cdot\nabla v \, d\mathbf{x}.$$

Thus, the resulting equation is:

$$\int_\Omega \{\mathbf{A}\nabla u \cdot \nabla v - \mathbf{b}u\cdot\nabla v + \mathbf{c}v\cdot\nabla u \ + auv\}\, d\mathbf{x} = \int_\Omega \{fv - \mathbf{f}\cdot\nabla v\ \}\, d\mathbf{x} \tag{8.36}$$

for every $v \in \mathcal{D}(\Omega)$.

It is not difficult to check that if the domain and the other data are smooth, for classical solutions, the two formulations (8.34) and (8.36) are equivalent.

We now enlarge the space of test functions to $H_0^1(\Omega)$ and introduce the bilinear form

$$B(u,v) = \int_\Omega \{\mathbf{A}\nabla u \cdot \nabla v - \mathbf{b}u\cdot\nabla v + \mathbf{c}v\cdot\nabla u \ + auv\}\, d\mathbf{x}$$

and the linear functional

$$Fv = \int_\Omega \{fv - \mathbf{f}\cdot\nabla v\ \}\, d\mathbf{x}.$$

Then, the weak formulation of problem (8.34) is the following:

Determine $u \in H_0^1(\Omega)$ such that

$$B(u,v) = Fv, \qquad \forall v \in H_0^1(\Omega). \tag{8.37}$$

A set of hypotheses that ensure the well-posedness of the problem is indicated in the following proposition.

Proposition 8.12 *Assume that hypotheses* (8.32) *and* (8.33) *hold and that $f \in L^2(\Omega)$, $\mathbf{f} \in L^2(\Omega;\mathbb{R}^n)$. Then, if* $\mathbf{b}$ *and* $\mathbf{c}$ *have Lipschitz components and*

$$\frac{1}{2}\operatorname{div}(\mathbf{b}-\mathbf{c}) + a \geq 0, \ \textit{a.e. in } \Omega, \tag{8.38}$$

problem (8.37) *has a unique solution. Moreover, the following stability estimate holds:*

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \le \frac{1}{\alpha}\left\{C_P\,\|f\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}\right\}. \tag{8.39}$$

Proof We apply the Lax-Milgram Theorem with $V = H_0^1(\Omega)$. The continuity of B in V follows easily. In fact, the Schwarz inequality and the bounds in (8.32), (8.33) give:

$$\left|\int_\Omega \mathbf{A}\nabla u\cdot\nabla v\,d\mathbf{x}\right| \le \int_\Omega \sum_{i,j=1}^n \left|a_{ij}u_{x_i}v_{x_j}\right| d\mathbf{x}$$
$$\le n^2 M\int_\Omega |\nabla u|\,|\nabla v|\,d\mathbf{x} \le n^2 M\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,\|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)}.$$

Moreover, using Poincaré's inequality,

$$\left|\int_\Omega [\mathbf{b}u\cdot\nabla v - \mathbf{c}v\cdot\nabla u]\;d\mathbf{x}\right| \le (\beta_0+\gamma_0)C_P\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,\|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)}$$

and

$$\left|\int_\Omega auv\,d\mathbf{x}\right| \le \alpha_0\int_\Omega |u|\,|v|\,d\mathbf{x} \le \alpha_0 C_P^2\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,\|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)}.$$

Thus, we can write

$$|B(u,v)| \le (n^2M + (\beta_0+\gamma_0)C_P + \alpha_0 C_P^2)\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,\|\nabla v\|_{L^2(\Omega;\mathbb{R}^n)}$$

which shows the continuity of B. Let us analyze the coercivity of B. We have:

$$B(u,u) = \int_\Omega \left\{\mathbf{A}\nabla u\cdot\nabla u - (\mathbf{b}-\mathbf{c})u\cdot\nabla u + au^2\right\} d\mathbf{x}.$$

Observe that, since $u = 0$ on $\partial\Omega$, integrating by parts we obtain

$$\int_\Omega (\mathbf{b}-\mathbf{c})u\cdot\nabla u\,d\mathbf{x} = \frac{1}{2}\int_\Omega (\mathbf{b}-\mathbf{c})\cdot\nabla u^2 d\mathbf{x} = -\frac{1}{2}\int_\Omega \operatorname{div}(\mathbf{b}-\mathbf{c})\,u^2 d\mathbf{x}.$$

Therefore, from (8.32) and (8.38), it follows that

$$B(u,u) \ge \alpha\int_\Omega |\nabla u|^2\,d\mathbf{x} + \int_\Omega \left[\frac{1}{2}\operatorname{div}(\mathbf{b}-\mathbf{c}) + a\right] u^2 d\mathbf{x} \ge \alpha\,\|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)}$$

so that B is V-coercive. Since we already know that $F \in H^{-1}(\Omega)$, the Lax-Milgram Theorem and (8.35) give existence, uniqueness and the stability estimate (8.39). □

Remark 8.13 If $\mathbf{A}$ is symmetric and $\mathbf{b} = \mathbf{c} = \mathbf{0}$, the solution u is a minimizer in $H_0^1(\Omega)$ for the "energy" functional

$$E(u) = \int_\Omega \left\{\mathbf{A}\nabla u\cdot\nabla u + au^2 - 2fu + 2\mathbf{f}\cdot\nabla u\right\} d\mathbf{x}.$$

As in Remark 8.2, p. 501, the equation (8.37) constitutes the Euler equation for E.

• *Nonhomogeneous Dirichlet conditions*. If Ω is a Lipschitz domain, we can consider a nonhomogeneous Dirichlet condition

$$u = g \quad \text{on } \partial\Omega,$$

with $g \in H^{1/2}(\partial\Omega)$. First we reduce the problem to a homogeneous boundary condition by considering a lifting $\tilde{g}$ of g in $H^1(\Omega)$ (see Thm. 7.84, p. 474) and setting $w = u - \tilde{g}$. Then $w \in H_0^1(\Omega)$ and the equation for w takes the form

$$\mathcal{E}w = f + \operatorname{div}(\mathbf{f} + \mathbf{A}\nabla\tilde{g} - \mathbf{b}\tilde{g}) - \mathbf{c} \cdot \nabla\tilde{g} - a\tilde{g}.$$

From (8.32) and (8.33) we have

$$\mathbf{c} \cdot \nabla\tilde{g} + a\tilde{g} \in L^2(\Omega) \quad \text{and} \quad \mathbf{A}\nabla\tilde{g} - \mathbf{b}\tilde{g} \in L^2(\Omega; \mathbb{R}^n).$$

Therefore, the Lax-Milgram Theorem yields existence, uniqueness and the estimate

$$\|\nabla w\|_{L^2(\Omega;\mathbb{R}^n)} \leq C \left\{ \|f\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)} + \|\tilde{g}\|_{H^1(\Omega)} \right\}, \tag{8.40}$$

valid for any extension $\tilde{g}$ of g, with $C = C(\alpha, n, M, \beta_0, \gamma_0, \alpha_0, \Omega)$. Since

$$\|u\|_{H^1(\Omega)} \leq (C_P + 1) \|\nabla w\|_{L^2(\Omega;\mathbb{R}^n)} + \|\tilde{g}\|_{H^1(\Omega)}$$

and (by Sect. 7.9.3)

$$\|g\|_{H^{1/2}(\partial\Omega)} = \inf \left\{ \|\tilde{g}\|_{H^1(\Omega)} : \tilde{g} \in H^1(\Omega), \ \ \tilde{g}|_{\partial\Omega} = g \right\},$$

by taking the lowest upper bound with respect to $\tilde{g}$, from (8.40) we can write (with a slightly different C)

$$\|u\|_{H^1(\Omega)} \leq C \left\{ \|f\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)} + \|g\|_{H^{1/2}(\partial\Omega)} \right\}.$$

Example 8.14 Figure 8.1 shows the solution of the following Dirichlet problem in the upper half circle $B_1^+(0,0) \subset \mathbb{R}^2$:

$$\begin{cases} -u_{\rho\rho} - \frac{1}{\rho} u_\rho - \frac{1}{\rho^2} u_{\theta\theta} - \rho u_\theta = 0 & \rho < 1, 0 < \theta < \pi \\ u(1,\theta) = \sin(\theta/2) & 0 \leq \theta \leq \pi \\ u(\rho,0) = 0, \ u(\rho,\pi) = \rho & \rho \leq 1 \end{cases} \tag{8.41}$$

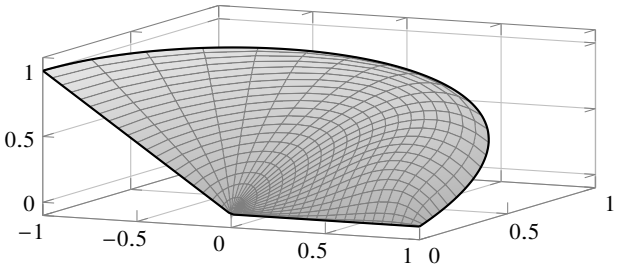

Fig. 8.1 The solution of problem (8.41)

where (ρ, θ) denotes polar coordinates. Note that, in rectangular coordinates, the differential equation writes

$$-\Delta u + y u_x - x u_y = 0$$

so that $-\rho u_\theta$ represents a transport term of the type $\mathbf{c} \cdot \nabla u$ with $\mathbf{c} = (y, -x)$. Since $\operatorname{div} \mathbf{c} = 0$ and $a = 0$, Proposition 8.12 ensures the well posedness of the problem.

Alternative for the Dirichlet problem. We will see later that problem (8.34) is actually well posed under the condition

$$\operatorname{div} \mathbf{b} + a \geq 0 \quad \text{a.e. in } \Omega,$$

which does not involve the coefficient $\mathbf{c}$. In particular, this condition is fulfilled if $a(\mathbf{x}) \geq 0$ and $\mathbf{b}(\mathbf{x}) = 0$ a.e. in Ω. In general however, we cannot prove that the bilinear form B is coercive. What we may affirm is that B is **weakly coercive**, i.e. *there exists* $\lambda_0 \in \mathbb{R}$ *such that:*

$$\tilde{B}(u, v) = B(u, v) + \lambda_0 (u, v)_{L^2(\Omega)}$$

is coercive. In fact, from the elementary inequality $|ab| \leq \varepsilon a^2 + \frac{1}{4\varepsilon} b^2$, $\varepsilon > 0$, we get

$$\begin{aligned} \left| \int_\Omega (\mathbf{b} - \mathbf{c}) u \cdot \nabla u \, d\mathbf{x} \right| &\leq (\beta_0 + \gamma_0) \int_\Omega |u \cdot \nabla u| \, d\mathbf{x} \\ &\leq \varepsilon \, \|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)} + \frac{(\beta_0 + \gamma_0)^2}{4\varepsilon} \, \|u\|^2_{L^2(\Omega)} . \end{aligned}$$

Therefore:

$$\tilde{B}(u, u) \geq (\alpha - \varepsilon) \, \|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)} + (\lambda_0 - \frac{(\beta_0 + \gamma_0)^2}{4\varepsilon} - \alpha_0) \, \|u\|^2_{L^2(\Omega)} . \tag{8.42}$$

If we choose $\varepsilon = \alpha/2$ and $\lambda_0 = (\beta_0 + \gamma_0)^2/4\varepsilon + \alpha_0$, we obtain

$$\tilde{B}(u, u) \geq \frac{\alpha}{2} \, \|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)}$$

which shows the coercivity of $\tilde{B}$. Introduce now the Hilbert triplet

$$V = H^1_0(\Omega), H = L^2(\Omega), V^* = H^{-1}(\Omega)$$

and recall that, since Ω is a bounded domain, $H^1_0(\Omega)$ is dense and compactly embedded in $L^2(\Omega)$. Finally, define the adjoint bilinear form of B by

$$B^*(u, v) = \int_\Omega \left\{ (\mathbf{A}^\top \nabla u + \mathbf{c} u) \cdot \nabla v - \mathbf{b} v \cdot \nabla u + a u v \right\} \, d\mathbf{x} \equiv B(v, u),$$

associated with the formal adjoint $\mathcal{E}^*$ of $\mathcal{E}$, given by

$$\mathcal{E}^* u = -\operatorname{div}(\mathbf{A}^\top \nabla u) - \operatorname{div}(\mathbf{c}u) - \mathbf{b}\cdot\nabla u + au.$$

We are now in position to apply Theorem 6.69, p. 395, to our variational problem. The conclusions are:

1. *The kernels $\mathcal{N}(B)$* and *$\mathcal{N}(B^*)$, given respectively by the solutions of the homogeneous problems*

$$B(u,v) = 0, \quad \forall v \in H_0^1(\Omega) \qquad \text{and} \qquad B^*(w,v) = 0, \quad \forall v \in H_0^1(\Omega),$$

   share the same dimension d, $0 \le d < \infty$.
2. *The problem*

$$B(u,v) = Fv, \qquad \forall v \in H_0^1(\Omega)$$

   has a solution if and only if $Fw = 0$ for every $w \in \mathcal{N}(B^*)$.

Let us translate the statements 1 and 2 into a less abstract language:

Theorem 8.15 *Let Ω be a bounded, Lipschitz domain, $f \in L^2(\Omega)$ and $\mathbf{f} \in L^2(\Omega;\mathbb{R}^n)$. Assume (8.32) and (8.33) hold. Then, we have the following alternative:*

a) Either $\mathcal{E}$ is a continuous isomorphism between $H_0^1(\Omega)$ and $H^{-1}(\Omega)$, that is the problem (8.34) has a unique weak solution for every $f, \mathbf{f}$ and there exists $C = C(\alpha, n, M, \beta_0, \gamma_0, \alpha_0, \Omega)$ such that

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \le C\left\{\|f\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}\right\},$$

   or the homogeneous and the adjoint homogeneous problems, given, respectively, by

$$\begin{cases} \mathcal{E}u = 0 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases} \qquad \begin{cases} \mathcal{E}^* w = 0 & \text{in } \Omega \\ w = 0 & \text{on } \partial\Omega \end{cases} \tag{8.43}$$

   have each d linearly independent solution, $0 < d < \infty$.

b) Moreover, problem (8.34) has a solution if and only if

$$\int_\Omega \{fw - \mathbf{f}\cdot\nabla w\}\, d\mathbf{x} = 0 \tag{8.44}$$

   for every solution w of the adjoint homogeneous problem.

The Alternative Theorem 8.15 implies that if we can show the uniqueness of the solution of problem (8.34), then automatically we infer both the existence and the stability estimate.

To show uniqueness, the weak maximum principles in Sect. 8.6 are quite useful. We will be back to this argument there.

The conditions (8.44) constitute *d compatibility conditions* that the data have to satisfy in order for a solution to exist.

8.4.3 Neumann problem

Let Ω be a bounded, *Lipschitz* domain. The Neumann condition for an operator in the divergence form (8.31) assigns on $\partial\Omega$ the flux naturally associated with the operator. This flux is composed by two terms: $\mathbf{A}\nabla u \cdot \boldsymbol{\nu}$, due to the diffusion term $-\operatorname{div}\mathbf{A}\nabla u$ (internal flux), and $-\mathbf{b}u \cdot \boldsymbol{\nu}$, due to the convective term $\operatorname{div}(\mathbf{b}u)$ (internal flux of $\mathbf{b}$), where $\boldsymbol{\nu}$ is the outward unit normal on $\partial\Omega$. We set

$$\partial_{\nu}^{\mathcal{E}} u \equiv (\mathbf{A}\nabla u - \mathbf{b}u) \cdot \boldsymbol{\nu} = \sum_{i,j=1}^{n} a_{ij} u_{x_j} \nu_i - u \sum_j b_j \nu_j.$$

We call $\partial_{\nu}^{\mathcal{E}} u$ the *conormal derivative* of u. Thus, the Neumann problem is:

$$\begin{cases} \mathcal{E}u = f & \text{in } \Omega \\ \partial_{\nu}^{\mathcal{E}} u = g & \text{on } \partial\Omega. \end{cases} \tag{8.45}$$

The variational formulation of problem (8.45) may be obtained by the usual integration by parts technique. It is enough to note, that, multiplying the differential equation $\mathcal{E}u = f$ by a test function $v \in H^1(\Omega)$ and using the Neumann condition, we get, formally:

$$\int_{\Omega} \{(\mathbf{A}\nabla u - \mathbf{b}u)\nabla v + (\mathbf{c} \cdot \nabla u)v + auv\}\, d\mathbf{x} = \int_{\Omega} fv\, d\mathbf{x} + \int_{\partial\Omega} gv\, d\sigma.$$

Introducing the bilinear form

$$B(u, v) = \int_{\Omega} \{(\mathbf{A}\nabla u - \mathbf{b}u)\nabla v + (\mathbf{c} \cdot \nabla u)v + auv\}\, d\mathbf{x} \tag{8.46}$$

and the linear functional

$$Fv = \int_{\Omega} fv\, d\mathbf{x} + \int_{\partial\Omega} gv\, d\sigma,$$

we are led to the following **variational formulation**, that can be easily checked to be equivalent to the original problem, when all the data are smooth:

Determine $u \in H^1(\Omega)$ *such that*

$$B(u, v) = Fv, \quad \forall v \in H^1(\Omega). \tag{8.47}$$

If the size of $\mathbf{b} - \mathbf{c}$ is small compared to the diffusive and the reaction terms, the problem (8.47) is well-posed, as the following proposition shows.

Proposition 8.16 *Assume that hypotheses* (8.32) *and* (8.33) *hold and that* $f \in L^2(\Omega)$, $g \in L^2(\partial\Omega)$. *If* $a(\mathbf{x}) \geq c_0 > 0$ *a.e. in* Ω *and*

$$m \equiv \min\{\alpha - (\beta_0 + \gamma_0)/2, c_0 - (\beta_0 + \gamma_0)/2\} > 0, \tag{8.48}$$

then, problem (8.47) *has a unique solution. Moreover, the following stability estimate holds:*

$$\|u\|_{H^1(\Omega)} \le \frac{1}{m}\left\{\|f\|_{L^2(\Omega)} + \overline{C}(n,\Omega)\,\|g\|_{L^2(\partial\Omega)}\right\}.$$

Proof We check that the assumptions of Lax-Milgram theorem hold. Since

$$|B(u,v)| \le (n^2 M + \beta_0 + \gamma_0 + \alpha_0)\,\|u\|_{H^1(\Omega)}\,\|v\|_{H^1(\Omega)},$$

B is continuous in $H^1(\Omega)$. Moreover, we may write

$$B(u,u) \ge \alpha\int_\Omega |\nabla u|^2\,d\mathbf{x} - \left|\int_\Omega [(\mathbf{b}-\mathbf{c})\cdot\nabla u]\;u\,d\mathbf{x}\right| + \int_\Omega a u^2 d\mathbf{x}.$$

From Schwarz's inequality and the inequality $2ab \le a^2 + b^2$, we obtain

$$\left|\int_\Omega [(\mathbf{b}-\mathbf{c})\cdot\nabla u]\;u\,d\mathbf{x}\right| \le (\beta_0+\gamma_0)\,\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}\,\|u\|_{L^2(\Omega)} \le \frac{(\beta_0+\gamma_0)}{2}\,\|u\|^2_{H^1(\Omega)}.$$

Thus, if (8.48) holds, we get $B(u,u) \ge m\,\|u\|^2_{H^1(\Omega)}$ and therefore B is coercive. Finally, using (8.17), it is not difficult to check that $F \in H^1(\Omega)^*$, with

$$\|F\|_{H^1(\Omega)^*} \le \|f\|_{L^2(\Omega)} + \overline{C}(n,\Omega)\,\|g\|_{L^2(\partial\Omega)}.$$

□

Alternative for the Neumann problem. The bilinear form B is coercive also under the condition

$$(\mathbf{b}-\mathbf{c})\cdot\boldsymbol{\nu} \le 0 \;\text{ a.e. on }\; \partial\Omega \quad\text{and}\quad \frac{1}{2}\operatorname{div}(\mathbf{b}-\mathbf{c}) + a \ge c_0 > 0 \;\text{ a.e in } \Omega,$$

as it can be checked following the proof of Proposition 8.12, p. 512.

However, in general the bilinear form B is only **weakly coercive**. In fact, choosing in (8.42) $\varepsilon = \alpha/2$ and $\lambda_0 = (\beta_0+\gamma_0)^2/2\varepsilon + 2\alpha_0$, we easily get

$$\tilde{B}(u,u) = B(u,u) + \lambda_0\,\|u\|^2_{L^2(\Omega)} \ge \frac{\alpha}{2}\,\|\nabla u\|^2_{L^2(\Omega)} + \left(\frac{(\beta_0+\gamma_0)^2}{2\alpha} + \alpha_0\right)\|u\|^2_{L^2(\Omega)}$$

and therefore $\tilde{B}$ is coercive. Applying Theorem 6.69, p. 395, we obtain the following alternative:

Theorem 8.17 *Let Ω be a bounded, Lipschitz domain. Assume that* (8.32) *and* (8.33) *hold. Then, if $f \in L^2(\Omega)$ and $g \in L^2(\partial\Omega)$:*

a) Either problem (8.45) *has a unique solution $u \in H^1(\Omega)$ and*

$$\|u\|_{H^1(\Omega)} \le C(n,\alpha,M,\beta_0,\gamma_0,\alpha_0,\Omega)\left\{\|f\|_{L^2(\Omega)} + \|g\|_{L^2(\partial\Omega)}\right\}$$

or the homogeneous and the adjoint homogeneous problems

$$\begin{cases}\mathcal{E}u = 0 & \text{in } \Omega\\ (\mathbf{A}\nabla u - \mathbf{b}u)\cdot\boldsymbol{\nu} = 0 & \text{on } \partial\Omega\end{cases} \quad \text{and} \quad \begin{cases}\mathcal{E}^* w = 0 & \text{in } \Omega\\ (\mathbf{A}^\top\nabla w + \mathbf{c}w)\cdot\boldsymbol{\nu} = 0 & \text{on } \partial\Omega\end{cases}$$

have each d linearly independent solutions, $0 < d < \infty$.
b) Moreover, problem (8.47) has a solution if and only if

$$Fw = \int_\Omega fw \, d\mathbf{x} + \int_{\partial\Omega} gw \, d\sigma = 0 \tag{8.49}$$

for every solution w of the adjoint homogeneous problem.

Remark 8.18 Again, uniqueness implies existence. Note that if $\mathbf{b} = \mathbf{c} = \mathbf{0}$ and $a = 0$, then the solutions of the adjoint homogeneous problem are the constant functions. Therefore $d = 1$ and the compatibility condition (8.49) reduces to the well known equation

$$\int_\Omega f \, d\mathbf{x} + \int_{\partial\Omega} g \, d\sigma = 0.$$

8.4.4 Robin and mixed problems

The variational formulation of the Robin problem

$$\begin{cases} \mathcal{E}u = f & \text{in } \Omega \\ \partial_\nu^{\mathcal{E}} u + hu = g & \text{on } \partial\Omega \end{cases} \tag{8.50}$$

is obtained by replacing the bilinear form B in problem (8.47), by

$$\tilde{B}(u, v) = B(u, v) + \int_{\partial\Omega} huv \, d\sigma.$$

If $0 \le h(\mathbf{x}) \le h_0$ a.e. on $\partial\Omega$, Proposition 8.16 still holds for problem (8.50).

As for the Neumann problem, in general the bilinear form B is only *weakly coercive* and a theorem perfectly analogous to Theorem 8.17 holds. We leave the details as an exercise.

Let now Γ_D, Γ_N be nonempty, relatively open and regular subsets of $\partial\Omega$ with $\Gamma_N = \partial\Omega \setminus \overline{\Gamma}_D$. Consider the mixed Dirichlet-Neumann problem

$$\begin{cases} \mathcal{E}u = f & \text{in } \Omega \\ u = 0 & \text{on } \Gamma_D \\ \partial_\nu^{\mathcal{E}} u = g & \text{on } \Gamma_N. \end{cases}$$

As in Sect. 8.3.2, the correct functional setting is $H^1_{0,\Gamma_D}(\Omega)$ in which we can use the norm $\|u\|_{H^1_{0,\Gamma_D}(\Omega)} = \|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)}$. Introducing the linear functional

$$Fv = \int_\Omega fv \, d\mathbf{x} + \int_{\Gamma_N} gv \, d\sigma,$$

the **variational formulation** is the following:

Determine $u \in H^1_{0,\Gamma_D}(\Omega)$ *such that*

$$B(u,v) = Fv, \quad \forall v \in H^1_{0,\Gamma_D}(\Omega). \tag{8.51}$$

Proceeding as in Proposition 8.12, p. 512, we may prove the following result:

Proposition 8.19 *Assume that hypotheses* (8.32) *and* (8.33) *hold and that* $f \in L^2(\Omega)$, $g \in L^2(\Gamma_N)$. *If* $\mathbf{b}$ *and* $\mathbf{c}$ *have Lipschitz components in* $\overline{\Omega}$ *and*

$$(\mathbf{b}-\mathbf{c})\cdot\boldsymbol{\nu} \le 0 \text{ a.e. on } \Gamma_N, \quad \frac{1}{2}\operatorname{div}(\mathbf{b}-\mathbf{c}) + a \ge 0, \quad \text{a.e. in } \Omega,$$

then problem (8.51) *has a unique solution* $u \in H^1_{0,\Gamma_D}(\Omega)$. *Moreover, the following stability estimate holds:*

$$\|\nabla u\|_{L^2(\Omega;\mathbb{R}^n)} \le \frac{1}{\alpha}\left\{C_P \|f\|_{L^2(\Omega)} + \overline{C}\,\|g\|_{L^2(\partial\Omega)}\right\}.$$

For the mixed problem as well, in general the bilinear form is only *weakly coercive* and we may resort to the Alternative Theorem, achieving a result similar to Theorem 8.17. Only note that the compatibility conditions (8.49) take now the form

$$Fw = \int_\Omega f w \, d\mathbf{x} + \int_{\Gamma_N} g w \, d\sigma = 0$$

for every solution w of the adjoint problem

$$\begin{cases} \mathcal{E}^* w = 0 & \text{in } \Omega \\ w = 0 & \text{on } \Gamma_D \\ (\mathbf{A}^\top \nabla w + \mathbf{c} w)\cdot\boldsymbol{\nu} = 0 & \text{on } \Gamma_N. \end{cases}$$

8.5 Equilibrium of a Plate

The range of application of the variational theory is not confined to second order equations. In particular, we can deal with a fourth order equation (so called *biharmonic equation*), modeling the equilibrium of a plate. More precisely, in this section we consider the vertical deflection $u = u(x,y)$ of a bent plate of small thickness (compared with the other dimensions) under the action of a normal load. If $\Omega \subset \mathbb{R}^2$ is a domain representing the transversal section of the plate, it can be shown that u is governed by the fourth order elliptic equation[5]

[5] It is possible to give the definition of *ellipticity* for an operator of order higher than two. See *M. Renardy, R. C. Rogers*, [18]. For instance, consider the linear operator with constant coefficients $\mathcal{L} = \sum_{|\alpha|=m} a_\alpha D^\alpha$, $m \ge 2$, where $\alpha = (\alpha_1, \ldots, \alpha_n)$ is a multi-index. Associate with $\mathcal{L}$ its **symbol**, given by

$$S_{\mathcal{L}}(\boldsymbol{\xi}) = \sum_{|\alpha|=m} a_\alpha (i\boldsymbol{\xi})^\alpha.$$

$$\Delta\Delta u = \Delta^2 u = \frac{q}{D} \equiv f \qquad \text{in } \Omega,$$

where q is the external distributed load and D encodes the elastic properties of the material. The operator Δ^2 is called **biharmonic** or **bi-laplacian** and the solutions of $\Delta^2 u = 0$ are called *biharmonic functions*. In two dimensions, the explicit expression of Δ^2 is given by

$$\Delta^2 = \frac{\partial^4}{\partial x^4} + 2\frac{\partial^4}{\partial x^2 \partial y^2} + \frac{\partial^4}{\partial y^4}.$$

If the plate is rigidly fixed along its boundary (*clamped plate*), then u and its normal derivative must vanish on $\partial\Omega$. Thus, we are led to the following boundary value problem:

$$\begin{cases} \Delta^2 u = f & \text{in } \Omega \\ u = \partial_\nu u = 0 & \text{on } \partial\Omega. \end{cases} \tag{8.52a}$$

To derive a variational formulation, we choose $\mathcal{D}(\Omega)$ as space of test functions. This choice takes into account the boundary conditions. Now, we multiply the biharmonic equation by a function $v \in \mathcal{D}(\Omega)$ and integrate over Ω:

$$\int_\Omega \Delta^2 u \, v \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}. \tag{8.53}$$

Integrating by parts twice and using the conditions $v = \partial_\nu v = 0$ on $\partial\Omega$, we get:

$$\begin{aligned} \int_\Omega \Delta^2 u \, v \, d\mathbf{x} &= \int_\Omega (\mathrm{div}\nabla\Delta u) v \, d\mathbf{x} = \int_{\partial\Omega} \partial_\nu(\Delta u) v \, d\sigma - \int_\Omega \nabla\Delta u \cdot \nabla v \, d\mathbf{x} \\ &= -\int_{\partial\Omega} \Delta u \partial_\nu v \, d\sigma + \int_\Omega \Delta u \Delta v \, d\mathbf{x} = \int_\Omega \Delta u \Delta v \, d\mathbf{x}. \end{aligned}$$

Thus, (8.53) becomes

$$\int_\Omega \Delta u \Delta v \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}. \tag{8.54}$$

Now we enlarge the space of test functions by taking the closure of $\mathcal{D}(\Omega)$ in $H^2(\Omega)$, which is $H_0^2(\Omega)$. Note that (see Sect. 7.9.2, p. 473) this is precisely the space of functions u such that u and $\partial_\nu u$ have zero trace on $\partial\Omega$. We are led to the following variational formulation:

Determine $u \in H_0^2(\Omega)$ such that

$$\int_\Omega \Delta u \Delta v \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}, \qquad \forall v \in H_0^2(\Omega). \tag{8.55}$$

To analyze the well-posedness of problem (8.55) we need the following lemma (for its proof, see Problem 8.22).

Then $\mathcal{L}$ is said to be **elliptic** if $S_{\mathcal{L}}(\boldsymbol{\xi}) \neq 0$ for every $\boldsymbol{\xi} \in \mathbb{R}^n$, $\boldsymbol{\xi} \neq \mathbf{0}$. The symbol of $\mathcal{L} = \Delta^2$ in 2 dimensions is $-\xi_1^4 - 2\xi_1^2\xi_2^2 - \xi_2^4$, which is negative if $(\xi_1, \xi_2) \neq (0, 0)$. Thus Δ^2 is elliptic. Note that, for $m = 2$, we recover the usual definition of ellipticity.

Lemma 8.20 *If Ω is a bounded C^2-domain, we have*

$$\|\Delta u\|_{L^2(\Omega)} \le \|u\|_{H_0^2(\Omega)} \le C(n,\Omega)\,\|\Delta u\|_{L^2(\Omega)} \quad \forall u \in H^2(\Omega)\cap H_0^1(\Omega).$$

Thus we may choose

$$\int_\Omega \Delta u\cdot\Delta v\,d\mathbf{x} \quad \text{and} \quad \left(\int_\Omega |\Delta u|^2\,d\mathbf{x}\right)^{1/2}$$

as inner product and norm in the Hilbert space $H^2(\Omega)\cap H_0^1(\Omega)$.

The following result holds:

Proposition 8.21 *Let Ω be a bounded, C^2-domain. If $f \in L^2(\Omega)$, there exists a unique solution $u \in H_0^2(\Omega)$ of (8.55). Moreover,*

$$\|\Delta u\|_{L^2(\Omega)} \le C(n,\Omega)\,\|f\|_{L^2(\Omega)}\,.$$

Proof Note that $H_0^2(\Omega)$ is a closed subspace of $H^2(\Omega)\cap H_0^1(\Omega)$; then, by Lemma 8.20, we may choose the bilinear form

$$B(u,v) = \int_\Omega \Delta u\cdot\Delta v\,d\mathbf{x}$$

as the inner product in $H_0^2(\Omega)$. On the other hand, setting,

$$Lv = \int_\Omega fv\,d\mathbf{x},$$

from Lemma 8.20, we have:

$$|L(v)| = \int_\Omega |fv|\;d\mathbf{x} \le \|f\|_{L^2(\Omega)}\,\|v\|_{L^2(\Omega)} \le C(n,\Omega)\,\|f\|_{L^2(\Omega)}\,\|\Delta v\|_{L^2(\Omega)}$$

so that $L \in H_0^2(\Omega)^*$. We conclude the proof directly from the Riesz Representation Theorem. □

8.6 Weak Maximum Principles

In Chap. 3 we have given a version of the maximum principle for the Laplace equation. This principle has an extension valid for general divergence form operators. First some remark on the inequalities at the boundary for Sobolev functions, in dimension $n > 1$.

Let Ω be a bounded Lipschitz domain and $u \in H^1(\Omega)$. We know that $u = 0$ on $\partial\Omega$ in the sense of the traces if and only if $u \in H_0^1(\Omega)$. Analogously (recall Sect. 7.7.6), we say that $u \ge 0$ on $\partial\Omega$ *in the sense of the traces*, if the *negative* part of u has zero trace on $\partial\Omega$, that is $u^- = \max\{-u,0\} \in H_0^1(\Omega)$. Similarly, $u \le 0$ on $\partial\Omega$ *in the sense of the traces* if its *positive part* has zero trace on $\partial\Omega$, that is $u^+ = \max\{u,0\} \in H_0^1(\Omega)$.

Of course, if $u \in H^1(\Omega) \cap C(\overline{\Omega})$, the above inequalities reduce to the classical ones for $u_{|\partial\Omega}$. In the sequel, we will omit to specify *in the sense of the traces*, unless there is a risk of confusion.

Other inequalities follow in a natural way. For instance, $u \le v$ on $\partial\Omega$ means $u - v \le 0$ on $\partial\Omega$. Thus, we may define:

$$\sup_{\partial\Omega} u = \inf\{k \in \mathbb{R} : u \le k \text{ on } \partial\Omega\}, \quad \inf_{\partial\Omega} u = \sup\{k \in \mathbb{R} : u \ge k \text{ on } \partial\Omega\}$$

which coincide with the usual *greatest lower bound* and *lowest upper bound* when the trace of u belongs to $C(\partial\Omega)$. Moreover, consistently, if $u \in H^1(\Omega)$ then

$$\sup_{\partial\Omega} u \le \operatorname*{ess\,sup}_{\Omega} u, \qquad \inf_{\partial\Omega} u \ge \operatorname*{ess\,inf}_{\Omega} u. \tag{8.56}$$

Indeed, for instance, if $\operatorname{ess\,sup}_\Omega u = \Lambda \in \mathbb{R}$, then $u - \Lambda \le 0$ a.e. in Ω, so that $(u-\Lambda)^+$ belongs to $H_0^1(\Omega)$. Therefore $\sup_{\partial\Omega} u \le \Lambda$.

Assume now that $u \in H^1(\Omega)$ satisfies the inequality

$$\mathcal{E}u = -\operatorname{div}(\mathbf{A}(\mathbf{x})\nabla u) + \operatorname{div}(\mathbf{b}(\mathbf{x})u) + \mathbf{c}(\mathbf{x}) \cdot \nabla u + a(\mathbf{x})u \le 0 \qquad (\ge 0)$$

in a weak sense, that is, for every $v \in H_0^1(\Omega)$, $v \ge 0$ a.e. in Ω,

$$B(u,v) = \int_\Omega \{(\mathbf{A}\nabla u - \mathbf{b}u) \cdot \nabla v + \mathbf{c}v \cdot \nabla u + auv\}\, dx \le 0 \qquad (\ge 0). \tag{8.57}$$

The following weak maximum principle holds

Theorem 8.22 *Let Ω be a bounded Lipschitz domain*[6]*. Assume that (8.32) and (8.33) hold. Let $u \in H^1(\Omega)$ and $\mathcal{E}u \le 0$ ($\mathcal{E}u \ge 0$) in a weak sense.*

a) If $\mathbf{b}$ is Lipschitz continuous and $\operatorname{div}\mathbf{b} + a \ge 0$ a.e. in Ω, then[7]

$$\operatorname*{ess\,sup}_{\Omega} u \le \sup_{\partial\Omega} u^+ \qquad (\operatorname*{ess\,inf}_{\Omega} u \ge \inf_{\partial\Omega}(-u^-)). \tag{8.58}$$

b) If $\operatorname{div}\mathbf{b} + a = 0$ a.e. in Ω, then

$$\operatorname*{ess\,sup}_{\Omega} u = \sup_{\partial\Omega} u \qquad (\operatorname*{ess\,inf}_{\Omega} u = \inf_{\partial\Omega} u). \tag{8.59}$$

In particular, in both cases, if $u \le 0$ ($u \ge 0$) on $\partial\Omega$, then $u \le 0$ ($u \ge 0$) a.e. in Ω.

Proof We prove only the first of the two inequalities in (8.58) and (8.59). The proof of the others is perfectly analogous.

---

[6] As usual Ω is a bounded interval in dimension $n = 1$.

[7] With a little more effort, condition a) can be replaced by $\mathbf{b} \in L^\infty(\Omega;\mathbb{R}^n)$ and $\operatorname{div}\mathbf{b} + a \ge 0$ in $\mathcal{D}'(\Omega)$, that is

$$\int_\Omega \{-\mathbf{b} \cdot \nabla\varphi + a\varphi\} \ge 0 \qquad \forall \varphi \in \mathcal{D}(\Omega),\ \varphi \ge 0 \text{ in } \Omega.$$

a) Let $l^+ = \sup_{\partial\Omega} u^+$. We may assume that $l^+ < \infty$, otherwise there is nothing to prove. Suppose that

$$l^+ < \Lambda = \operatorname*{ess\,sup}_{\Omega} u.$$

We want to reach a contradiction. Let λ be such that $l^+ < \lambda < \Lambda$ and select as a test function

$$v_\lambda = \max\{u - \lambda, 0\} = (u - \lambda)^+,$$

which clearly belongs to $H_0^1(\Omega)$. Note that $v_\lambda > 0$ on a set of positive measure. Let

$$G_\lambda = \operatorname{supp} \nabla u \cap \{u > \lambda\}.$$

Then $\nabla v_\lambda = \nabla u$ a.e. on G_λ while $\nabla v_\lambda = \mathbf{0}$ a.e. outside G_λ. Choosing $v = v_\lambda$ as a test function in (8.57), we find

$$\int_\Omega \{(\mathbf{A}\nabla u - \mathbf{b}u) \cdot \nabla v_\lambda + \mathbf{c}v_\lambda \cdot \nabla u + auv_\lambda\}\, d\mathbf{x} \le 0.$$

Integrating by parts the term in $\mathbf{b}$ (possible since it is Lipschitz), after a simple rearrangement of terms, we obtain

$$\begin{aligned}\int_{G_\lambda} \mathbf{A}\nabla v_\lambda \cdot \nabla v_\lambda\, d\mathbf{x} &\le -\int_{G_\lambda} (\mathbf{c} + \mathbf{b})v_\lambda \cdot \nabla v_\lambda\, d\mathbf{x} - \int_{\{u>\lambda\}} (\operatorname{div}\mathbf{b} + a)uv_\lambda\, d\mathbf{x} \\ &\le -\int_{G_\lambda} (\mathbf{c} + \mathbf{b})v_\lambda \cdot \nabla v_\lambda\, d\mathbf{x},\end{aligned}$$

since $\operatorname{div}\mathbf{b} + a \ge 0$ a.e. in Ω and $uv_\lambda > 0$ a.e. on $\{u > \lambda\}$.

Using the uniform ellipticity condition (8.32) and the bounds (8.33) for $\mathbf{b}$ and $\mathbf{c}$, we obtain

$$\begin{aligned}\alpha \|\nabla v_\lambda\|^2_{L^2(G_\lambda;\mathbb{R}^n)} &\le (\beta_0 + \gamma_0) \int_{G_\lambda} |v_\lambda \nabla v_\lambda|\, d\mathbf{x} \\ \text{(by the Schwarz inequality)} &\le (\beta_0 + \gamma_0) \|v_\lambda\|_{L^2(G_\lambda)} \|\nabla v_\lambda\|_{L^2(G_\lambda;\mathbb{R}^n)}.\end{aligned}$$

Simplifying by $\|\nabla v_\lambda\|_{L^2(G_\lambda;\mathbb{R}^n)}$, we have

$$\|\nabla v_\lambda\|_{L^2(G_\lambda;\mathbb{R}^n)} \le \frac{\gamma_0}{\alpha} \|v_\lambda\|_{L^2(G_\lambda)}. \tag{8.60}$$

Consider now $n > 2$. Using the Sobolev Embedding Theorem 7.92, p. 483, Hölder's inequality and (8.60), we get

$$\|v_\lambda\|_{L^{2n/(n-2)}(\Omega)} \le C_s \|\nabla v_\lambda\|_{L^2(\Omega;\mathbb{R}^n)} = C_s \|\nabla v_\lambda\|_{L^2(G_\lambda;\mathbb{R}^n)} \le \frac{C_s\gamma_0}{\alpha} |G_\lambda|^{1/n} \|v_\lambda\|_{L^{2n/(n-2)}(\Omega)}$$

or

$$|G_\lambda| \ge \alpha^n \gamma_0^{-n} C_s^{-n} \equiv \xi > 0. \tag{8.61}$$

Since ξ is independent of λ, the inequality (8.61) remains valid if we let $\lambda \to \Lambda$. We infer that $\operatorname{supp} \nabla u \cap \{u = \Lambda\}$ is a set of positive measure. But, by Remark 7.65, $\nabla u = 0$ a.e. on every measurable set where u is constant and we have reached a contradiction.

If $n = 1, 2$ an inequality similar to (8.61) follows by using the Sobolev Embedding Theorem with any $p > 2$. We leave the details to the reader.

b) Let $l = \sup_{\partial\Omega} u$, $l < \lambda < \Lambda$ and $v_\lambda = (u - \lambda)^+$. Then proceed as in a), and conclude by (8.56).□

Remark 8.23 It is not possible to substitute $\sup_{\partial\Omega} u^+$ by $\sup_{\partial\Omega} u$ or $\inf_{\partial\Omega}(-u^-)$ with $\inf_{\partial\Omega} u$ in (8.58), and such inequalities may be strict. A counterexample in dimension *one* is shown in Fig. 8.2. The solution of $-u'' + u = 0$ in $(0, 1)$, $u(0) = u(1) = -1$, has a *negative maximum* which is greater than -1 and strictly negative.

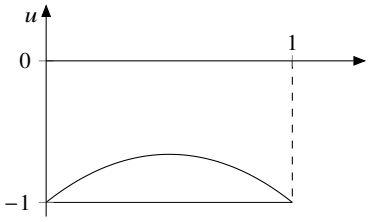

Fig. 8.2 The solution of $-u'' + u = 0$ in $(0,1)$, $u(0) = u(1) = -1$

Theorem 8.22, in conjunction with the Alternative Theorem 8.15, p. 516, gives the following well-posedness result for the Dirichlet problem.

Corollary 8.24 *Under the hypotheses of Theorem 8.22, for every* $f \in L^2(\Omega)$, $\mathbf{f} \in L^2(\Omega;\mathbb{R}^n)$, $g \in H^{1/2}(\partial\Omega)$, *the Dirichlet problem*

$$\begin{cases} \mathcal{L}u = f + \operatorname{div}\mathbf{f} & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases}$$

has a unique solution $u \in H^1(\Omega)$ *and*

$$\|u\|_{H^1(\Omega)} \le C(n,\alpha,M,\beta_0,\gamma_0,\alpha_0,\Omega)\left\{\|f\|_{L^2(\Omega)} + \|\mathbf{f}\|_{L^2(\Omega)} + \|g\|_{H^{1/2}(\partial\Omega)}\right\}.$$

Proof We use Fredholm's Alternative. The bilinear form B in (8.57) is weakly coercive in $H_0^1(\Omega)$. Therefore, to prove the Corollary it is enough to show uniqueness for the Dirichlet problem. In fact, let u, w be two solutions and set

$$z = u - w \in H_0^1(\Omega).$$

Then z solves the homogeneous problem, that is $B(z,v) = 0$, for every $v \in H_0^1(\Omega)$. From Theorem 8.22 we deduce that

$$0 = \inf_{\partial\Omega}(-z^-) \le \operatorname*{ess\,inf}_{\Omega} z \le \operatorname*{ess\,sup}_{\Omega} z \le \sup_{\partial\Omega} z^+ = 0.$$

Therefore $z = 0$ a.e. in Ω. □

There are similar maximum principles yielding uniqueness and therefore well-posedness, for problems involving mixed conditions.

To deal with mixed Dirichlet-Neumann problem we consider two regular and relatively open set $\Gamma_D, \Gamma_N \subset \partial\Omega$, with $\Gamma_D \neq \emptyset$ and $\Gamma_N = \partial\Omega \setminus \overline{\Gamma}_D$. Here we say that $u \ge 0$ (resp. $u \le 0$) on Γ_D if and only if $u^- \in H^1_{0,\Gamma_D}(\Omega)$ (resp. $u^+ \in H^1_{0,\Gamma_D}(\Omega)$).

With the same technique used for Theorem 8.22, we can prove the following result (compare with Example 8.32, p. 530).

Theorem 8.25 *Assume that (8.32) and (8.33) hold. Let* $u \in H^1(\Omega)$ *satisfy*

$$\int_\Omega \{(\mathbf{A}\nabla u - \mathbf{b}u)\cdot\nabla v + \mathbf{c}v\cdot\nabla u + auv\}\,d\mathbf{x} \le 0 \quad (\ge 0) \tag{8.62}$$

for every $v \in H^1_{0,\Gamma_D}(\Omega)$, $v \ge 0$ *a.e. in* Ω, *with* $\Gamma_D \neq \emptyset$, *regular and relatively open in* $\partial\Omega$. *Let* $\Gamma_N = \partial\Omega \setminus \overline{\Gamma}_D$.

a) If $\mathbf{b}$ *is Lipschitz continuous and*

$$\mathbf{b}\cdot\boldsymbol{\nu}\le 0 \ \text{ a.e. on } \Gamma_N, \quad \operatorname{div}\mathbf{b}+a\ge 0, \quad \text{a.e. in } \Omega, \tag{8.63}$$

then

$$\operatorname*{ess\,sup}_{\Omega} u \le \sup_{\Gamma_D} u^+ \qquad (\operatorname*{ess\,inf}_{\Omega} u \ge \inf_{\Gamma_D}(-u^-)). \tag{8.64}$$

b) If

$$\mathbf{b}\cdot\boldsymbol{\nu}\le 0 \ \text{ a.e. on } \Gamma_N, \quad \operatorname{div}\mathbf{b}+a= 0, \quad \text{a.e. in } \Omega, \tag{8.65}$$

then

$$\operatorname*{ess\,sup}_{\Omega} u = \sup_{\Gamma_D} u \qquad (\operatorname*{ess\,inf}_{\Omega} u = \inf_{\Gamma_D} u). \tag{8.66}$$

In particular, in both cases, if $u\le 0$ *(resp.* $u\ge 0$*) on* Γ_D*, then* $u\le 0$ *(resp.* $u\ge 0$*) a.e. in* Ω*.*

Proof We prove only the first of the two inequalities in (8.64) and (8.66). The proof of the others is perfectly analogous.

a) Let $l_D^+ = \sup_{\Gamma_D} u^+ < \infty$ and $\Lambda = \sup_\Omega u$. Assume by contradiction that $l_D^+ < \lambda < \Lambda$ and set $v_\lambda = (u-\lambda)^+$, $l_D^+ < \lambda < \Lambda$. Note that v_λ belongs to $H^1_{0,\Gamma_D}(\Omega)$. Since $\mathbf{b}$ is Lipschitz continuous, for every $v\in H^1_{0,\Gamma_D}(\Omega)$, we can integrate by parts to write

$$\begin{aligned}\int_\Omega -\mathbf{b}u\cdot\nabla v\,d\mathbf{x} &= \int_\Omega \operatorname{div}(\mathbf{b}u)v\,d\mathbf{x} - \int_{\Gamma_N} uv\,\mathbf{b}\cdot\boldsymbol{\nu}\,d\sigma\\ &= \int_\Omega \left[\operatorname{div}\mathbf{b}\,uv + \mathbf{b}v\cdot\nabla u\right]\,d\mathbf{x} - \int_{\Gamma_N} uv\,\mathbf{b}\cdot\boldsymbol{\nu}\,d\sigma.\end{aligned}$$

Inserting into (8.62) we get, after a simple rearrangements of terms:

$$\int_\Omega \mathbf{A}\nabla u\cdot\nabla v\,d\mathbf{x} \le -\int_\Omega \{(\mathbf{c}+\mathbf{b})v\cdot\nabla u + (\operatorname{div}\mathbf{b}+a)uv\}\,d\mathbf{x} + \int_{\Gamma_N} uv\,\mathbf{b}\cdot\boldsymbol{\nu}\,d\sigma.$$

Choosing $v=v_\lambda$ as a test function, we find (keeping the notations in the proof of Theorem 8.22)

$$\int_{G_\lambda} \mathbf{A}\nabla v_\lambda\cdot\nabla v_\lambda\,d\mathbf{x} \le -\int_{G_\lambda}(\mathbf{c}+\mathbf{b})v_\lambda\cdot\nabla v_\lambda\,d\mathbf{x}$$

since $\operatorname{div}\mathbf{b}+a\ge 0$ a.e. in Ω, $\mathbf{b}\cdot\boldsymbol{\nu}\le 0$ a.e. on Γ_N and $u, v_\lambda > 0$ a.e. on $\{u>\lambda\}$.

The rest of the proof follows as in Theorem 8.22.

b) Let $l_D = \sup_{\Gamma_D} u < \infty$. Assume $l<\Lambda$ and set $v_\lambda = (u-\lambda)^+$, with $l<\lambda<\Lambda$. Note that v_λ belongs to $H^1_{0,\Gamma_D}(\Omega)$. Then proceed as in a), and conclude using (8.56). □

Using Theorem 6.69, p. 395, we have:

Theorem 8.26 *Under the hypotheses of Theorem 8.25, for every* $f\in L^2(\Omega)$, $g\in L^2(\Gamma_N)$*, the mixed Dirichlet-Neumann problem*

$$\begin{cases}\mathcal{E}u = f & \text{in } \Omega\\ u=0 & \text{on } \Gamma_D\\ \partial_{\boldsymbol{\nu}}^{\mathcal{E}}u = g & \text{on } \Gamma_N\end{cases}$$

has a unique solution $u\in H^1(\Omega)$ *and*

$$\|u\|_{H^1(\Omega)} \le C(n, \alpha, M, \beta_0, \gamma_0, \alpha_0, \Omega) \left\{ \|f\|_{L^2(\Omega)} + \|g\|_{L^2(\Gamma_N)} \right\}.$$

Proof Let u, w be two solutions and set $z = u - w$. Then $z \in H^1_{0,\Gamma_D}(\Omega)$ and solves the homogeneous problem, that is $B(z, v) = 0$, for every $v \in H^1_{0,\Gamma_D}(\Omega)$. From Theorem 8.25 we deduce that

$$0 = \inf_{\Gamma_D}(-z^-) \le \operatorname*{ess\,inf}_{\Omega} z \le \operatorname*{ess\,sup}_{\Omega} z \le \sup_{\Gamma_D} z^+ = 0.$$

Therefore $z = 0$ a.e. in Ω. The conclusions follow from Fredholm's Alternative. □

8.7 Regularity

An important task, in general rather technically complicated, is to establish the optimal regularity of a weak solution in relation to the degree of smoothness of the data, namely, *the domain Ω, the boundary data, the coefficients of the operator and the forcing term.* To get a clue of what happens, consider for example the following problem:

$$\begin{cases} -\Delta u + u = F & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

where $F \in H^{-1}(\Omega)$, Ω bounded. Under this hypothesis, the Lax-Milgram Theorem yields a (unique) solution $u \in H^1_0(\Omega)$ and we cannot get much more, in terms of smoothness. Indeed, from Sobolev inequalities (see Sect. 7.10.4) it follows that $u \in L^p(\Omega)$ with $p = \frac{2n}{n-2}$, if $n \ge 3$, or $u \in L^p(\Omega)$, with $2 \le p < \infty$, if $n = 2$. However, this gain in integrability does not seriously increase the smoothness of u.

Reversing our point of view, we may say that, starting from a function in $H^1_0(\Omega)$ and applying to it a second order operator "two orders of differentiability are lost": the loss of one order drives from $H^1_0(\Omega)$ into $L^2(\Omega)$ while a further loss leads to $H^{-1}(\Omega)$. It is as if the upper index -1 indicates a "lack" of one order of differentiability.

Nevertheless, consider the case in which $u \in H^1(\mathbb{R}^n)$ is a solution of the equation

$$-\Delta u + u = f \quad \text{in } \mathbb{R}^n. \tag{8.67}$$

We ask: if $f \in L^2(\mathbb{R}^n)$ *what is the optimal regularity of* u?

Following the above argument, our conclusions would be: we start from $u \in H^1(\mathbb{R}^n)$, and applying the second order operator $-\Delta + I$, where I denotes the identity operator, we end up with $f \in L^2(\mathbb{R})$ instead of $f \in H^{-1}(\Omega)$.

Thus we conclude that the starting function should actually be in $H^2(\mathbb{R}^n)$ rather than $H^1(\mathbb{R}^n)$. Indeed this is true and can be easily proved using the Fourier transform, as follows: since

$$\widehat{\partial_{x_i} u}(\boldsymbol{\xi}) = i\xi_i \widehat{u}(\boldsymbol{\xi}), \qquad \widehat{\partial_{x_i x_j} u}(\boldsymbol{\xi}) = -\xi_i \xi_j \widehat{u}(\boldsymbol{\xi}), \qquad -\widehat{\Delta u}(\xi) = |\boldsymbol{\xi}|^2 \, \widehat{u}(\boldsymbol{\xi}),$$

equation (8.67) becomes

$$(1 + |\boldsymbol{\xi}|^2)\widehat{u}(\boldsymbol{\xi}) = \widehat{f}(\boldsymbol{\xi}).$$

Using the H^2 norm in (7.53), p. 464, we deduce that

$$\|u\|^2_{H^2(\mathbb{R}^n)} = \int_{\mathbb{R}^n} (1+|\boldsymbol{\xi}|^2)^2\, |\widehat{u}(\boldsymbol{\xi})|^2\, d\boldsymbol{\xi} = \int_{\mathbb{R}^n} \left|\widehat{f}(\boldsymbol{\xi})\right|^2 d\boldsymbol{\xi} = (2\pi)^n \|f\|^2_{L^2(\mathbb{R}^n)}, \tag{8.68}$$

where the last equality follows from formula (7.39), p. 450.

Thus, $u \in H^2(\mathbb{R}^n)$ and moreover, we have obtained the stability estimate (8.68). We may go further. If $f \in H^1(\mathbb{R}^n)$, that is if f has first partial derivatives in $L^2(\mathbb{R}^n)$, a similar computation yields $u \in H^3(\mathbb{R}^n)$. Iterating this argument, we conclude that for every $m \geq 0$,

$$\text{if } f \in H^m(\mathbb{R}^n) \qquad \text{then} \qquad u \in H^{m+2}(\mathbb{R}^n).$$

Using the Sobolev embedding theorems of Sect. 7.10.4 in (say) any ball of $\mathbb{R}^n$, we infer that, if m is sufficiently large, u is a *classical solution*. In fact, if $u \in H^{m+2}(\mathbb{R}^n)$ then

$$u \in C^k(\mathbb{R}^n) \text{ for } k < m+2-\frac{n}{2},$$

and therefore it is enough that $m > \frac{n}{2}$ to have u at least in $C^2(\mathbb{R}^n)$. An immediate consequence is:

$$\text{if } f \in C^\infty(\mathbb{R}^n) \qquad \text{then} \qquad u \in C^\infty(\mathbb{R}^n).$$

This kind of results can be extended to any *uniformly elliptic* operators $\mathcal{E}$ in divergence form and to the solutions of the Dirichlet, Neumann and Robin problems. The regularity for mixed problems is more complicated and requires delicate conditions along the border between Γ_D and Γ_N. We will not insist on this subject, providing only an example[8].

There are two kinds of regularity results, concerning interior regularity and global (up to the boundary) regularity, respectively. Since the proofs are quite technical we only state the main results (for the proofs, see e.g. *D. Gilbarg, N. S. Trudinger*, [5]).

In all the theorems below u is a weak solution of

$$\mathcal{E}u = -\operatorname{div}(\mathbf{A}(\mathbf{x})\nabla u - \mathbf{b}(\mathbf{x})u) + \mathbf{c}(\mathbf{x})\cdot\nabla u + a(\mathbf{x})u = f \quad \text{in } \Omega$$

where Ω is a bounded domain. We keep the hypotheses (8.32) and (8.33).

Interior regularity

The next theorem is a H^2-interior regularity result. Note that the boundary of the domain does not play any role. We have:

Theorem 8.27 *Let the coefficients a_{ij} and b_j, $i,j = 1,\dots,n$, be Lipschitz in $\overline{\Omega}$, $f \in L^2(\Omega)$. Then $u \in H^2_{loc}(\Omega)$ and it satisfies the differential equation in the*

[8] See *Haller-Dintelmann, Meyer, Reberg, Schiela*, Holder continuity and optimal control for nonsmooth elliptic domains, Appl. Math. Optim. 60: 397–428, *2009*.

pointwise a.e. sense in Ω*. Moreover, if* $\Omega' \subset\subset \Omega$*,*

$$\|u\|_{H^2(\Omega')} \leq C_2 \left\{ \|f\|_{L^2(\Omega)} + \|u\|_{L^2(\Omega)} \right\}. \tag{8.69}$$

Thus, u is a strong solution (see Sect. 8.2) in Ω. The constant C_2 depends only on all the relevant parameters $\alpha, \beta_0, \gamma_0, \alpha_0, M$ and also on the distance of Ω' from $\partial\Omega$ and the Lipschitz constant of a_{ij} and b_j, $i, j = 1, \ldots, n$.

Remark 8.28 The presence of the norm $\|u\|_{L^2(\Omega)}$ in the right hand side of (8.69) is necessary[9] and due to the fact that the bilinear form B associated to $\mathcal{E}$ is only weakly coercive.

If we increase the regularity of the coefficients, the smoothness of u increases according to the following theorem:

Theorem 8.29 *Let* $f \in H^m(\Omega)$*,* $a_{ij}, b_j \in C^{m+1}(\overline{\Omega})$ *and* $c_j, a_0 \in C^m(\overline{\Omega})$*,* $m \geq 1$*,* $i, j = 1, \ldots, n$*. Then* $u \in H^{m+2}_{loc}(\Omega)$ *and if* $\Omega' \subset\subset \Omega$*,*

$$\|u\|_{H^{m+2}(\Omega')} \leq C_m \left\{ \|f\|_{H^m(\Omega)} + \|u\|_{L^2(\Omega)} \right\}.$$

As a consequence, if $a_{ij}, b_j, c_j, a_0, f \in C^\infty(\Omega)$*, then* $u \in C^\infty(\Omega)$ *as well.*

The constant C_m depends only on all the relevant parameters α, β_0, γ_0, α_0, M, on the distance of Ω' from $\partial\Omega$ and on the norms of a_{ij}, b_j in $C^{m+1}(\overline{\Omega})$ and of c_j, a in $C^m(\overline{\Omega})$, $i, j = 1, \ldots, n$.

Global regularity

We now focus on the optimal regularity of a solution (nonnecessarily unique!) of the boundary value problems we have considered in the previous sections.

Consider first H^2-regularity. If $u \in H^2(\Omega)$, its trace on $\partial\Omega$ belongs to $H^{3/2}(\partial\Omega)$ so that a Dirichlet data g_D has to be taken in this space. On the other hand, the trace of the normal derivative belongs to $H^{1/2}(\partial\Omega)$ and hence we have to assign a Neumann or a Robin data g_N in this space. Also, the domain has to be smooth enough, say C^2, in order to define the traces of u and $\partial_\nu u$.

Thus, assume that u is a solution of $\mathcal{E}u = f$ in Ω, subject to one of the following boundary conditions:

$$u = g_D \in H^{3/2}(\partial\Omega)$$

or

$$\partial^{\mathcal{E}}_\nu u + hu = g_N \in H^{1/2}(\partial\Omega),$$

with

$$0 \leq h(\sigma) \leq h_0 \quad \text{a.e. on } \partial\Omega.$$

We have:

[9] For instance, $u(x) = \sin x$ is a solution of the equation $u'' + u = 0$. Clearly we cannot control any norm of u with the norm of the right hand side alone!

Theorem 8.30 *Let Ω be a bounded, C^2-domain. Assume that $f \in L^2(\Omega)$, a_{ij}, b_j, $i, j = 1, \dots, n$, and h are Lipschitz in $\overline{\Omega}$ and on $\partial\Omega$, respectively. Then $u \in H^2(\Omega)$ and*

$$\|u\|_{H^2(\Omega)} \le C_{2,D} \left\{ \|u\|_{L^2(\Omega)} + \|f\|_{L^2(\Omega)} + \|g_D\|_{H^{3/2}(\partial\Omega)} \right\} \text{ (Dirichlet)},$$
$$\|u\|_{H^2(\Omega)} \le C_{2,N} \left\{ \|u\|_{L^2(\Omega)} + \|f\|_{L^2(\Omega)} + \|g_N\|_{H^{1/2}(\partial\Omega)} \right\} \text{ (Neumann/Robin)}.$$

If we increase the regularity of the domain, the coefficients and the data, the smoothness of u increases accordingly to the following theorem.

Theorem 8.31 *Let Ω be a bounded C^{m+2}-domain. Assume that $a_{ij}, b_j \in C^{m+1}(\overline{\Omega})$, $c_j, a \in C^m(\overline{\Omega}), i, j = 1, \dots, n$, $f \in H^m(\Omega)$. If $g_D \in H^{m+3/2}(\partial\Omega)$ or $g_N \in H^{m+1/2}(\partial\Omega)$ and $h \in C^{m+1}(\partial\Omega)$, then $u \in H^{m+2}(\Omega)$ and moreover,*

$$\|u\|_{H^{m+2}(\Omega)} \le C_{m,D} \left\{ \|u\|_{L^2(\Omega)} + \|f\|_{H^m(\Omega)} + \|g_D\|_{H^{m+3/2}(\partial\Omega)} \right\}$$

for the Dirichlet problem,

$$\|u\|_{H^{m+2}(\Omega)} \le C_{m,N} \left\{ \|u\|_{L^2(\Omega)} + \|f\|_{H^m(\Omega)} + \|g_N\|_{H^{m+1/2}(\partial\Omega)} \right\}$$

for the Neumann/Robin problem. In particular, if Ω is a C^∞-domain, all the coefficients are in $C^\infty(\overline{\Omega})$ and the boundary data are in $C^\infty(\partial\Omega)$, then $u \in C^\infty(\overline{\Omega})$.

The constants $C_{m,D}$ and $C_{m,N}$ in the above theorems depend on α, on the norms of the coefficients and the data in their respective spaces, and on the smoothness character of Ω.

Mixed problems

As we have already mentioned, the regularity for mixed problems is more complicated and requires delicate conditions along the border between Γ_D and Γ_N. However, the following example gives an idea of what can happen.

Example 8.32 The function $u(r, \theta) = r^{\frac{1}{2}} \sin \frac{\theta}{2}$ is a weak solution in the half circle

$$S_\pi = \{(r, \theta) : 0 < r < 1, \quad 0 < \theta < \pi\}$$

of the mixed problem[10]

$$\begin{cases} \Delta u = 0 & \text{in } S_\pi \\ u(1, \theta) = \sin \frac{\theta}{2} & 0 < \theta < \pi \\ u(r, 0) = 0 \text{ and } \partial_{x_2} u(r, \pi) = 0 & 0 \le r < 1. \end{cases} \tag{8.70}$$

Namely, $|\nabla u|^2 = u_r^2 + \frac{1}{r^2} u_\theta^2 = \frac{1}{4r}$, so that

[10] With a little abuse of notation we set $u(x_1, x_2) = u(r, \theta)$, with $x_1 = r \cos\theta$, $x_2 = r \sin\theta$.

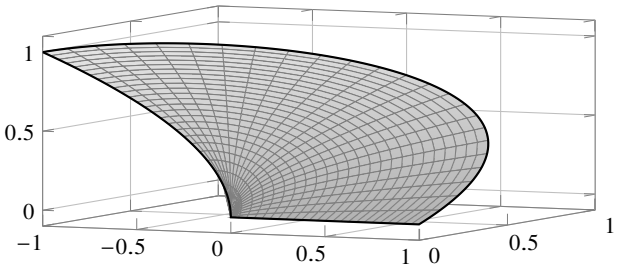

Fig. 8.3 The solution of the mixed problem (8.70)

$$\int_{S_\pi} |\nabla u|^2 \, dx_1 dx_2 = \frac{\pi}{4}$$

whence $u \in H^1(S_\pi)$. Moreover, $u(r, 0) = 0$ and

$$\partial_{x_2} u = u_r \sin\theta + \frac{1}{r} u_\theta \cos\theta = \frac{1}{2\sqrt{r}} \cos\frac{\theta}{2}$$

whence $\partial_{x_2} u(r, \pi) = 0$. However, along the half-line $\theta = \pi/2$, for example, we have $\left|\partial_{x_i x_j} u\right| \sim r^{-\frac{3}{2}}$ for $r \sim 0$, so that

$$\int_{S_\alpha} \left|\partial_{x_i x_j} u\right|^2 dx_1 dx_2 \sim \int_0^1 r^{-2} dr = \infty$$

and therefore $u \notin H^2(S_\pi)$. Thus, the solution has a low order of regularity near the origin, even though the boundary of S_π is flat there. Note that the origin separates the Dirichlet and Neumann regions (see Fig. 8.3).

Conclusion: *in general, the optimal regularity of the solution of a mixed problem is less than H^2 near the boundary between the Dirichlet and Neumann regions.*

Regularity in Lipschitz domains

The above regularity results hold for C^2 or smoother domains. However, in several applied situations, Lipschitz domains are the relevant ones. Let us examine the following situation.

Example 8.33 Consider the plane sector:

$$S_\alpha = \{(r, \theta) : 0 < r < 1, -\alpha/2 < \theta < \alpha/2\} \qquad (0 < \alpha < 2\pi).$$

The function (see Fig. 8.4)

$$u(r, \theta) = r^{\frac{\pi}{\alpha}} \cos\frac{\pi}{\alpha}\theta$$

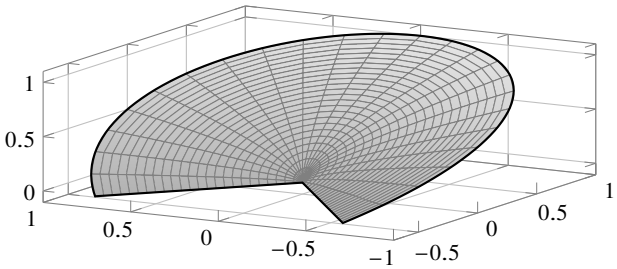

Fig. 8.4 The case $\alpha = \frac{3}{2}\pi$ in Example 8.33

is harmonic in S_α, since it is the real part of $f(z) = z^{\frac{\pi}{\alpha}}$, which is holomorphic in S_α. Furthermore,

$$u(r, -\alpha/2) = u(r, \alpha/2) = 0, \qquad 0 \le r \le 1 \tag{8.71}$$

and

$$u(1, \theta) = \cos \frac{\pi}{\alpha}\theta, \qquad -\alpha/2 \le \theta \le \alpha/2. \tag{8.72}$$

We focus on a neighborhood of the origin. If $\alpha = \pi$, S_α is a semicircle and

$$u(r, \theta) = \operatorname{Re} z = x \in C^\infty(\overline{S}_\alpha).$$

Suppose $\alpha \neq \pi$. Since

$$|\nabla u|^2 = u_r^2 + \frac{1}{r^2}u_\theta^2 = \frac{\pi^2}{\alpha^2} r^{2(\frac{\pi}{\alpha}-1)},$$

we have

$$\int_{S_\alpha} |\nabla u|^2 \, dx_1 dx_2 = \frac{\pi^2}{\alpha} \int_0^1 r^{2\frac{\pi}{\alpha}-1} dr = \frac{\pi}{2}$$

so that $u \in H^1(S_\alpha)$ and is the unique weak solution of $\Delta u = 0$ in S_α with the boundary conditions (8.71), (8.72). It is easy to check that for every $i, j = 1, 2$, $\left|\partial_{x_i x_j} u\right| \sim r^{\frac{\pi}{\alpha}-2}$, as $r \to 0$, whence

$$\int_{S_\alpha} \left|\partial_{x_i x_j} u\right|^2 dx_1 dx_2 \simeq \int_0^1 r^{2\frac{\pi}{\alpha}-3} dr.$$

This integral is convergent only for $2\frac{\pi}{\alpha} - 3 > -1$, i. e. $\alpha < \pi$. The conclusion is that $u \in H^2(S_\alpha)$ *if and only if* $\alpha \le \pi$, i.e. if *the sector is convex.* If $\alpha > \pi$, $u \notin H^2(S_\alpha)$.

Conclusion: *in a neighborhood of a non-convex angle, we expect a low degree of regularity of the solution (less than H^2).*

Thus, we may expect H^2-regularity for bounded, convex domains. Indeed the following theorem holds, where we consider for simplicity only the Laplace operator

(for the proof, see *P. Grisvard*, [6]). Note that a convex domain has a Lipschitz boundary.

Proposition 8.34 *Let Ω be a bounded, convex domain. Then, if $f \in L^2(\Omega)$ and $\lambda > 0$, the weak solution of the problems*

$$-\Delta u = f \text{ in } \Omega, \qquad u = 0 \text{ on } \partial\Omega$$

and

$$-\Delta u + \lambda u = f \text{ in } \Omega, \qquad \partial_\nu u = 0 \text{ on } \partial\Omega$$

belongs to $H^2(\Omega)$.

Increasing the summability of f we can obtain at least continuity up to the boundary of a Lipschitz domain. Precisely, the following slightly more general result holds for the Dirichlet problem. Note that no regularity is required to the coefficients of the operator $\mathcal{E}$, only uniform ellipticity and boundedness (recall (8.31), (8.32) and (8.33)).

Theorem 8.35 *Let Ω be a bounded, Lipschitz domain. Assume that $u \in H_0^1(\Omega)$ is a solution of $\mathcal{E}u = f + \operatorname{div} \mathbf{F}$ in Ω. If $f \in L^p(\Omega)$ and $\mathbf{F} \in L^q(\Omega; \mathbb{R}^n)$ with $p > n/2$ and $q > n$, $n \geq 2$, then $u \in C^{0,\sigma}(\overline{\Omega})$ for some $\sigma \in (0, 1]$, and*

$$\|u\|_{H^1(\Omega)} + \|u\|_{C^{0,\sigma}(\overline{\Omega})} \leq c \left\{ \|u\|_{L^2(\Omega)} + \|f\|_{L^p(\Omega)} + \|\mathbf{F}\|_{L^q(\Omega;\mathbb{R}^n)} \right\}. \tag{8.73}$$

For the proof of this theorem, see *D. Gilbarg, N. S. Trudinger*, [5]. The numbers σ and c depend only on all the relevant parameters α, β_0, γ_0, α_0, M and on the diameter and the Lipschitz constant of Ω.

A similar result hold for the Robin/Neumann conditions[11].

Theorem 8.36 *Let Ω be a bounded, Lipschitz domain. Assume that $u \in H^1(\Omega)$ is a solution of*

$$\begin{cases} \mathcal{E}u = f & \text{in } \Omega \\ \partial_\nu^{\mathcal{E}} u + hu = g & \text{on } \partial\Omega \,. \end{cases}$$

If $f \in L^p(\Omega)$, $g \in L^q(\partial\Omega)$, with $p > n/2$, $q > n - 1$, $n \geq 2$, then $u \in C^{0,\sigma}(\overline{\Omega})$ for some $\sigma \in (0, 1]$, and

$$\|u\|_{H^1(\Omega)} + \|u\|_{C^{0,\sigma}(\overline{\Omega})} \leq c \left\{ \|u\|_{L^2(\Omega)} + \|f\|_{L^p(\Omega)} + \|g\|_{L^q(\partial\Omega)} \right\}.$$

Again, σ and c depend only on all the relevant parameters $\alpha, \beta_0, \gamma_0, \alpha_0, c_0, h_0$, M and on the diameter and the Lipschitz constant of Ω.

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[11] see *R. Nittka*, Regularity of solutions of linear second order elliptic and parabolic boundary value problems on Lipschitz domains, J. Differential Equations 251, 860–880, 2011.

Problems

8.1 Write the weak formulation of the following problem:

$$\begin{cases} (x^2+1)u'' - xu' = \sin 2\pi x & 0 < x < 1 \\ u(0) = u(1) = 0. \end{cases}$$

Show that there exists a unique solution $u \in H_0^1(0,1)$ and find an estimate for $\|u'\|_{L^2(0,1)}$.

8.2 Write the weak formulation for the equation

$$(p(x)u')' + b(x)u' + a(x)u = 0 \qquad \text{in } (0,1),$$

with Robin and mixed conditions. Assume p, a, b bounded and $p(x) \geq \alpha > 0$ in $(0,1)$. Give sufficient conditions on p, b, a which assure existence and uniqueness of the solution and give a stability estimate.

8.3 Write the weak formulation of the following problem:

$$\begin{cases} \cos x\, u'' - \sin x\, u' - xu = 1 & 0 < x < \pi/6 \\ u'(0) = -u(0),\ u(\pi/6) = 0. \end{cases}$$

Discuss existence and uniqueness and derive a stability estimates.

8.4 (*The Legendre equation*) Let

$$X = \left\{ v \in L^2(-1,1) \colon (1-x^2)^{1/2} v' \in L^2(-1,1) \right\}$$

with inner product

$$(u,v)_X = \int_{-1}^{1} \left[uv + (1-x^2)u'v' \right] dx.$$

a) Check that $(u,v)_X$ is indeed an inner product and that X is a Hilbert space.
b) Given $f \in L^2(-1,1)$, study the variational problem

$$(u,v)_X = \int_{-1}^{1} fv\, dx, \qquad \text{for every } v \in X. \tag{8.74}$$

c) Determine the boundary value problem whose variational formulation is (8.74).

[*Hint.* a) Use Theorem 7.52, p. 452, with

$$V = L^2(-1,1) \ \text{ and } \ Z = L^2_w(-1,1),\ w(x) = (1-x^2)^{1/2}.$$

Check that $Z \hookrightarrow \mathcal{D}'(-1,1)$. b) Use the Lax-Milgram Theorem.
Answer. c) The boundary value problem is

$$\begin{cases} -\left[(1-x^2)u' \right]' + u = f & -1 < x < 1 \\ (1-x^2)u'(x) \to 0 & \text{as } x \to \pm 1. \end{cases}$$

This is a Legendre equation with the natural Neumann conditions at both end points.]

8.5 Let

$$V = H^1_{per}(0,2\pi) = \left\{ u \in H^1(0,2\pi) : u(0) = u(2\pi) \right\}$$

and F be the linear functional

$$F : v \longmapsto \int_0^{2\pi} tv(t)\, dt.$$

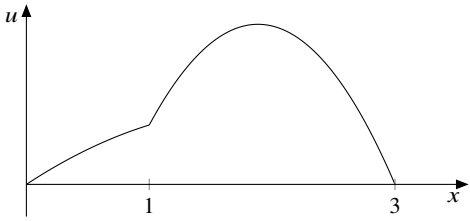

Fig. 8.5 The solution of the transmission problem $(p(x)u')' = -1$, $u(0) = u(3) = 0$, with $p(x) = 3$ in $(0, 1)$ and $p(x) = 1/2$ in $(1, 3)$

a) Check that $F \in V^*$.
b) According to Riesz's Theorem 6.31, p. 370, there is a unique element $u \in V$ such that

$$(u, v)_{H^1(0,2\pi)} = Fv, \text{ for every } v \in V.$$

Determine explicitly u.

8.6 (*Transmission conditions* (I)) Consider the problem

$$\begin{cases} (p(x)u')' = f & \text{in } (a, b) \\ u(a) = u(b) = 0 \end{cases} \tag{8.75}$$

where $f \in L^2(a, b)$, $p(x) = p_1 > 0$ in (a, c) and $p(x) = p_2 > 0$ in (c, b).
Show that problem (8.75) has a unique weak solution in $H^1(a, b)$, satisfying the conditions:

$$\begin{cases} p_1 u'' = f & \text{in } (a, c) \\ p_2 u'' = f & \text{in } (c, b) \\ p_1 u'(c-) = p_2 u'(c+). \end{cases}$$

Observe the jump of the derivative of u at $x = c$ (Fig. 8.5).

8.7 Let $\Omega = (0, 1) \times (0, 1) \subset \mathbb{R}^2$. Prove that the functional

$$E(v) = \frac{1}{2} \int_\Omega \left\{ |\nabla v|^2 - 2xv \right\} dxdy$$

has a unique minimizer $u \in H_0^1(\Omega)$. Write the Euler equation and find an explicit formula for u.

8.8 Consider the following subspace of $H^1(\Omega)$:

$$V = \left\{ u \in H^1(\Omega) : \frac{1}{|\partial\Omega|} \int_{\partial\Omega} u \, d\sigma = 0 \right\}.$$

a) Show that V is a Hilbert space with inner product $(\cdot, \cdot)_{H^1(\Omega)}$ and find which boundary value problem has the following weak formulation:

$$\int_\Omega \{ \nabla u \cdot \nabla v + uv \} \, d\mathbf{x} = \int_\Omega f v \, d\mathbf{x}, \qquad \forall v \in V.$$

b) Show that if $f \in L^2(\Omega)$ there exists a unique solution.

[*Answer.* a) $-\Delta u + u = f$ in Ω, $\partial_\nu u =$ constant on $\partial\Omega$. b) Use the Riesz Representation Theorem or the Lax-Milgram Theorem.]

8.9 Let $\Omega \subset \mathbb{R}^n$ and $g \in H^{1/2}(\partial\Omega)$. Define

$$H^1_g(\Omega) = \left\{ v \in H^1(\Omega) \colon v = g \quad \text{on } \partial\Omega \right\}.$$

Prove the following theorem, known as **The Dirichlet principle**: *Among all the functions* $v \in H^1_g(\Omega)$*, the harmonic one minimizes the Dirichlet integral*

$$D(v) = \int_\Omega |\nabla v|^2 \, d\mathbf{x}.$$

[*Hint.* In $H^1(\Omega)$ use the inner product

$$(u, v)_{1,\partial} = \int_{\partial\Omega} uv \, d\sigma + \int_\Omega \nabla u \cdot \nabla v \, d\mathbf{x}$$

and the corresponding norm $\|u\|_{1,\partial}$ (see Remark 7.88, p. 480). Then, minimizing $D(v)$ over $H^1_g(\Omega)$ is equivalent to minimizing $\|v\|^2_{1,\partial}$. Let $u \in H^1_g(\Omega)$ be harmonic in Ω. If $v \in H^1_g(\Omega)$, write $v = u + w$, with $w \in H^1_0(\Omega)$. Show that $(u, w)_{1,\partial} = 0$ and conclude that $\|u\|^2_{1,\partial} \le \|v\|^2_{1,\partial}$.]

8.10 (*A simple system*) Let $\Omega \subset \mathbb{R}^n$ be a bounded Lipschitz domain. Consider the Neumann problem for the following system:

$$\begin{cases} -\Delta u_1 + u_1 - u_2 = f_1 & \text{in } \Omega \\ -\Delta u_2 + u_1 + u_2 = f_2 & \text{in } \Omega \\ \partial_\nu u_1 = \partial_\nu u_2 = 0 & \text{on } \partial\Omega. \end{cases}$$

Derive a variational formulation and establish a well-posedness theorem.

[*Hint.* Variational formulation:

$$\int_\Omega \{\nabla u_1 \cdot \nabla v_1 + \nabla u_2 \cdot \nabla v_2 + u_1 v_1 - u_2 v_1 + u_1 v_2 + u_2 v_2\} = \int_\Omega (f_1 v_1 + f_2 v_2)$$

for every $(v_1, v_2) \in H^1(\Omega) \times H^1(\Omega)$.]

8.11 (*Transmission conditions* (II)) Let Ω_1 and Ω be bounded, Lipschitz domains in $\mathbb{R}^n$ such that $\Omega_1 \subset\subset \Omega$. Let $\Omega_2 = \Omega \setminus \overline{\Omega}_1$. In Ω_1 and Ω_2 consider the following bilinear forms

$$a_k(u, v) = \int_{\Omega_k} \mathbf{A}^k(\mathbf{x}) \nabla u \cdot \nabla v \, d\mathbf{x} \qquad (k = 1, 2)$$

with A^k *uniformly elliptic*. Assume that the entries of A^k are **continuous in** $\overline{\Omega}_k$, but that the matrix

$$\mathbf{A}(\mathbf{x}) = \begin{cases} \mathbf{A}^1(\mathbf{x}) & \text{in } \overline{\Omega}_1 \\ \mathbf{A}^2(\mathbf{x}) & \text{in } \Omega_2 \end{cases}$$

may have a jump across $\Gamma = \partial\Omega_1$. Let $u \in H^1_0(\Omega)$ be the weak solution of the equation

$$a(u, v) = a_1(u, v) + a_2(u, v) = (f, v)_{L^2(\Omega)} \qquad \forall v \in H^1_0(\Omega),$$

where $f \in L^2(\Omega)$.

a) Which boundary value problem does u satisfy?
b) Which conditions on Γ do express the coupling between u_1 and u_2?

[*Answer.* b) $u_1|_\Gamma = u_2|_\Gamma$ and $\mathbf{A}^1 \nabla u_1 \cdot \boldsymbol{\nu} = \mathbf{A}^2 \nabla u_2 \cdot \boldsymbol{\nu}$, where $\boldsymbol{\nu}$ points outward with respect to Ω_1.]

8.12 Find the mistake in the following argument. Consider the Neumann problem

$$\begin{cases} -\Delta u + \mathbf{c} \cdot \nabla u = f & \text{in } \Omega \\ \partial_{\boldsymbol{\nu}} u = 0 & \text{on } \partial\Omega \end{cases} \tag{8.76}$$

with Ω smooth, $\mathbf{c} \in C^1(\overline{\Omega})$ and $f \in L^2(\Omega)$. Let $V = H^1(\Omega)$ and

$$B(u, v) = \int_\Omega \{\nabla u \cdot \nabla v + (\mathbf{c} \cdot \nabla u)v\} \, d\mathbf{x}.$$

If $\operatorname{div} \mathbf{c} = 0$, we may write

$$\int_\Omega (\mathbf{c} \cdot \nabla u)u \, d\mathbf{x} = \frac{1}{2} \int_\Omega \mathbf{c} \cdot \nabla(u^2) \, d\mathbf{x} = \frac{1}{2} \int_{\partial\Omega} u^2 \mathbf{c} \cdot \boldsymbol{\nu} \, d\sigma.$$

Thus, if $\mathbf{c} \cdot \boldsymbol{\nu} \geq c_0 > 0$ then, recalling Remark 7.88, p. 480,

$$B(u, u) \geq \|\nabla u\|^2_{L^2(\Omega)} + \frac{c_0}{2} \|u\|^2_{L^2(\partial\Omega)} \geq C \|u\|^2_{H^1(\Omega)}$$

so that B is V-coercive and problem (8.76) has a unique solution!!

8.13 Let $Q = (0, \pi) \times (0, \pi)$. Study the solvability of the Dirichlet problem

$$\begin{cases} \Delta u + 2u = f & \text{in } Q \\ u = 0 & \text{on } \partial Q. \end{cases}$$

In particular, examine the cases $f(x, y) = 1$ and $f(x, y) = x - \pi/2$.

8.14 Let

$$B_1^+ = \left\{(x, y) \in \mathbb{R}^2 \colon x^2 + y^2 < 1, \, y > 0\right\}.$$

Examine the solvability of the Robin problem

$$\begin{cases} -\Delta u = f & \text{in } B_1^+ \\ \partial_{\boldsymbol{\nu}} u + yu = 0 & \text{on } \partial B_1^+. \end{cases}$$

8.15 Let $\Omega = (0, 1) \times (0, 1)$, $a \in R$. Examine the solvability of the mixed problem

$$\begin{cases} \Delta u + au = 1 & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \setminus \{y = 0\} \\ \partial_{\boldsymbol{\nu}} u = x & \text{on } \{y = 0\}. \end{cases}$$

8.16 Let $\Omega \subset \mathbb{R}^n$ be a bounded, Lipschitz domain and $a, b \in L^\infty(\Omega)$, with $\int_\Omega b(\mathbf{x}) d\mathbf{x} \neq 0$. Consider the following (nonlocal) Neumann problem:

$$\begin{cases} -\Delta u + a(\mathbf{x}) \int_\Omega b(\mathbf{z}) u(\mathbf{z}) d\mathbf{z} = f(\mathbf{x}) & \text{in } \Omega \\ \partial_{\boldsymbol{\nu}} u = 0 & \text{on } \partial\Omega \end{cases} \tag{8.77}$$

where $f \in L^2(\Omega)$.

a) Give a weak formulation of the problem.
b) Analyze its solvability.

8.17 Let $u \in H^1(\Omega)$ satisfy the Robin problem

$$\int_\Omega \{\mathbf{A}\nabla u \cdot \nabla v + auv\} \, d\mathbf{x} + \int_{\partial\Omega} huv \, d\sigma = \int_\Omega v \, d\mathbf{x}, \tag{8.78}$$

for every $v \in H^1(\Omega)$. Assume that (8.32) and (8.33), p. 511, hold. Moreover, assume that $a \geq 0$ a.e. in Ω, $h \in L^\infty(\partial\Omega)$, $h \geq 0$ a.e. on $\partial\Omega$, and

$$\int_{\partial\Omega} h \, d\sigma + \int_{\Omega} a \, d\mathbf{x} = q > 0. \tag{8.79}$$

Prove that there exists a unique solution of (8.78) and derive a stability estimate.

8.18 Let λ_1 be the principal Dirichlet eigenvalue for the Laplace operator and V_1 be the corresponding eigenspace.

a) Show that the second Dirichlet eigenvalue λ_2 satisfy the following variational principle, where $R(v)$ is the Rayleigh quotient (8.24), p. 507:

$$\lambda_2 = \min \left\{ R(v) : v \neq 0, v \in H_0^1(\Omega) \cap V_1^{\perp}, \right\}.$$

b) State and prove a similar variational principle for the n-th Dirichlet eigenvalue.

8.19 Let $\Omega \subset \mathbb{R}^n$ be a bounded, regular domain and $a \in C^{\infty}(\overline{\Omega})$, $a \geq 0$ in Ω. Consider the following eigenvalue problem:

$$\begin{cases} -\Delta u + a(\mathbf{x})u = \lambda u & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

a) Check the existence of an orthonormal basis of eigenfunctions $\{w_k\}_{k\geq 1}$ in $L^2(\Omega)$, and of a corresponding nondecreasing sequence of eigenvalues $\{\lambda_k\}_{k\geq 1}$ with $\lambda_k \to +\infty$. Suitably normalized, $\{w_k\}_{k\geq 1}$ is orthonormal in $H_0^1(\Omega)$ with respect to an appropriate inner product. Which one is it?

b) Show that the principal eigenvalue $\lambda_1 = \lambda_1(\Omega, a)$ is positive. Which Rayleigh quotient does λ_1 minimize?

c) Infer the following monotonicity and stability properties of λ_1 with respect to the coefficient a:

1. If $a_1 \leq a_2$ in Ω then $\lambda_1(\Omega, a_1) \leq \lambda_1(\Omega, a_2)$.
2. $|\lambda_1(\Omega, a_1) - \lambda_1(\Omega, a_2)| \leq \|a_1 - a_2\|_{L^{\infty}(\Omega)}$.
3. If $\Omega_1 \subset \Omega_2$ then $\lambda_1(\Omega_1, a) \geq \lambda_1(\Omega_2, a)$.

8.20 Let

$$Eu = -\operatorname{div}(\mathbf{A}(\mathbf{x})\nabla u) + a(\mathbf{x})u,$$

with $\mathbf{A} = (a_{ij})_{ij=1,\dots,n}$ *symmetric*. Assume that Ω is a regular bounded domain and that a_{ij} and a belong to $C^{\infty}(\overline{\Omega})$. State and prove the analogue of Theorem 8.8.

8.21 Let $\Omega \subset \mathbb{R}^n$ be a bounded, Lipschitz domain. Show that for every $u \in H^2(\Omega)$ such that $\partial_{\nu} u = 0$ on $\partial\Omega$, the following inequality holds

$$\int_{\Omega} (\Delta u)^2 \geq \mu_2 \int_{\Omega} |\nabla u|^2 \tag{8.80}$$

where μ_2 is the first positive Neumann eigenvalue for the operator $-\Delta$ in Ω. When does the equality sign hold in (8.80)?

[*Hint.* Let $\{\mu_j\}_{j\geq 1}$, $(\mu_1 = 0)$ and $\{u_j\}_{j\geq 1}$ be, respectively, the sequences of Neumann eigenvalues and of the corresponding eigenfunctions. Check that

$$\|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)} = \sum_{j=2}^{\infty} \mu_j (u, u_j)^2_{L^2(\Omega)}$$

and, since $\{u_j\}_{j\geq 1}$ is nondecreasing, that

$$\|\Delta u\|^2_{L^2(\Omega)} = \sum_{j=2}^{\infty} \mu_j^2 (u, u_j)^2_{L^2(\Omega)} \geq \mu_2 \sum_{j=2}^{\infty} \mu_j (u, u_j)^2_{L^2(\Omega)} = \mu_2 \|\nabla u\|^2_{L^2(\Omega;\mathbb{R}^n)}. \quad]$$

8.22 Prove Lemma 8.20.

[*Hint.* Let $-\Delta u = f$ in Ω. The Lax-Milgram Theorem and Theorem 8.30, p. 530, imply that the solution of the Dirichlet problem

$$\begin{cases} -\Delta u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

belongs to $H^2(\Omega) \cap H_0^1(\Omega)$ and that

$$\|u\|_{H^2(\Omega)} \leq C(n,\Omega)\, \|f\|_{L^2(\Omega)} = C(n,\Omega)\, \|\Delta u\|_{L^2(\Omega)}\,. \tag{8.81}$$

Moreover, clearly we have $\|\Delta u\|_{L^2(\Omega)} \leq \|u\|_{H^2(\Omega)}$.]

8.23 Derive a variational formulation of the following problem

$$\begin{cases} \Delta^2 u = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \\ \Delta u + \rho \partial_{\boldsymbol{\nu}} u = 0 & \text{on } \partial\Omega \end{cases}$$

where ρ is a positive constant and Ω is a bounded, C^2-domain. Show that the right functional setting is $H^2(\Omega) \cap H_0^1(\Omega)$, i.e. the space of functions in $H^2(\Omega)$ with zero trace. Prove the well-posedness of the resulting problem.

[*Hint.* The variational formulation is

$$\int_\Omega \Delta u \cdot \Delta v \, d\mathbf{x} + \int_{\partial\Omega} \rho \partial_{\boldsymbol{\nu}} u \cdot \partial_{\boldsymbol{\nu}} v \, d\sigma = \int_\Omega f v \, d\mathbf{x}, \ \forall v \in H^2(\Omega) \cap H_0^1(\Omega).$$

To show the well-posedness, use Lemma 8.20 and the inequality

$$\|\partial_{\boldsymbol{\nu}} v\|_{L^2(\partial\Omega)} \leq C(n,\Omega)\, \|\Delta v\|_{L^2(\Omega)}\,,$$

$\forall v \in H^2(\Omega) \cap H_0^1(\Omega)$.]

8.24 (*Mean value formulas for bi-harmonic functions*) Let $u \in C^4(\Omega)$, $\Omega \subset \mathbb{R}^n$, be a bi-harmonic function: $\Delta^2 u = \Delta(\Delta u) = 0$ in Ω. Prove the following mean value formulas: *for every ball* $B_r(x) \subset\subset \Omega$,

$$u(\mathbf{x}) = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r(\mathbf{x})} u(\sigma) d\sigma - \frac{r^2}{2n} \Delta u(\mathbf{x}). \tag{8.82}$$

and

$$u(\mathbf{x}) = \frac{n}{\omega_n r^n} \int_{B_r(\mathbf{x})} u(\mathbf{y}) d\mathbf{y} - \frac{r^2}{2(n+2)} \Delta u(\mathbf{x}). \tag{8.83}$$

[*Hint.* As in Sect. 3.3.2, p. 121, let $g(r) = \frac{1}{\omega_n r^{n-1}} \int_{\partial B_r(\mathbf{x})} u(\sigma) d\sigma$ and, noting that Δu is harmonic, check that

$$g'(r) = \frac{1}{\omega_n r^{n-1}} \int_{B_r(\mathbf{x})} \Delta u(\mathbf{y}) d\mathbf{y} = \frac{r}{n} \Delta u(\mathbf{x}).$$

Integrate over $(0, r)$ to get (8.82). To prove (8.83), write

$$\frac{n}{\omega_n r^n} \int_{B_r(\mathbf{x})} u(\mathbf{y}) d\mathbf{y} = \frac{n}{\omega_n r^n} \int_0^r ds \int_{\partial B_s(\mathbf{x})} u(\sigma) d\sigma$$

and use (8.82) for the ball $B_s(\mathbf{x})$.]

8.25 A) Let V, H be Hilbert spaces with $V \hookrightarrow H$. Denote by $a(u,v)$, $a_1(u,v)$, $a_2(u,v)$ three nonnegative bilinear forms, on V, such that

$$a(u,v) = a_1(u,v) + a_2(u,v), \quad \forall u, v \in V.$$

Define

$$\Lambda = \inf_{u\in V, u\neq 0} \frac{a(u,u)}{\|u\|_H^2} \quad \text{and} \quad \lambda_j = \inf_{u\in V, u\neq 0} \frac{a_j(u,u)}{\|u\|_H^2}, \quad j = 1, 2.$$

Which one of the following inequality is true?

$$\text{a) } \Lambda \geq \lambda_1 + \lambda_2 \qquad \text{b) } \Lambda \leq \lambda_1 + \lambda_2.$$

B) Let $V_{r,s} = H_0^1(r,s)$, $0 < r < s$. Compute

$$\lambda(r,s) = \inf_{u\in V_{r,s}} \frac{\int_r^s (u')^2}{\int_r^s u^2}.$$

C) Let Ω be the sector of elliptic ring defined by $(p, q > 0)$:

$$\Omega = \left\{(x,y) \in \mathbb{R}^2 : \frac{1}{4} < \frac{x^2}{p^2} + \frac{y^2}{q^2} < 1, \ \ x > 0, \ \ y > 0\right\}.$$

Using the results in A) and B), show that, if Λ is the first Dirichlet eigenvalue for the operator $-\Delta$ in Ω, then:

$$\Lambda \geq \frac{4\pi^2}{3}\left(\frac{1}{p^2} + \frac{1}{q^2}\right).$$

8.26 Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain, $f \in L^2(\Omega)$. Assume that f has the following expansion:

$$f = \sum_{k=1}^{\infty} f_k u_k$$

where $f_k = (f, u_k)_{L^2(\Omega)}$ and $\{u_k\}_{k\geq 1}$ is a sequence of orthonormal (in $L^2(\Omega)$) Dirichlet eigenvectors for the Laplace operator with corresponding eigenvalues $\{\lambda_k\}_{k\geq 1}$.

a) Show that the solution $u \in H_0^1(\Omega)$ of the Dirichlet problem $-\Delta u = f$ in Ω, $u = 0$ on $\partial\Omega$ has the following expansion:

$$u = -\sum_{k=1}^{\infty} \frac{f_k}{\lambda_k} u_k, \tag{8.84}$$

with convergence in $H_0^1(\Omega)$.

b) Let $G = G(\mathbf{x}, \mathbf{y})$ be the Green function for the Laplace operator in Ω. Show that $G(\mathbf{x}, \mathbf{y})$ has the following expansion:

$$G(\mathbf{x}, \mathbf{y}) = -\sum_{k=1}^{\infty} \frac{1}{\lambda_k} u_k(\mathbf{x}) u_k(\mathbf{y})$$

with convergence in $L^2(\Omega \times \Omega)$ and that

$$\int_\Omega \int_\Omega (G(\mathbf{x}, \mathbf{y}))^2 d\mathbf{x} d\mathbf{y} = \sum_{k=1}^{\infty} \frac{1}{\lambda_k^2}.$$

[*Hint.* b) Fix $\mathbf{x}$ and write $G(\mathbf{x}, \mathbf{y}) = -\sum_{k=1}^{\infty} g_k(\mathbf{x}) u_k(\mathbf{y})$, where $g_k(\mathbf{x}) = \int_\Omega G(\mathbf{x}, \mathbf{z}) u_k(\mathbf{z}) d\mathbf{z}$. Recall from formula (3.92), p. 160, that the solution u of the Dirichlet problem in a) has the representation

$$u(\mathbf{x}) = \int_\Omega G(\mathbf{x}, \mathbf{y}) f(\mathbf{y}) d\mathbf{y}.$$

Insert the expansion of G and f, use the orthonormality conditions $(u_k, u_j)_{L^2(\Omega)} = \delta_{kj}$ (the Kronecker symbol) to show that

$$u(\mathbf{x}) = -\sum_{k=1}^{\infty} f_k g_k(\mathbf{x}).$$

Compare with formula (8.84) and deduce that $g_k(\mathbf{x}) = -u_k(\mathbf{x})/\lambda_k$.]

Chapter 9
Weak Formulation of Evolution Problems

9.1 Parabolic Equations

In Chap. 2 we have considered the diffusion equation and some of its generalizations, as in the reaction-diffusion model (Sect. 2.5) or in the Black-Scholes model (Sect. 2.9). This kind of equations belongs to the class of *parabolic equations,* that we have already classified in spatial dimension 1 (Sect. 5.5.1) and that we are going to define in a more general setting.

Let $\Omega \subset \mathbb{R}^n$ be a *bounded* domain, with boundary $\Gamma = \partial\Omega$, $T > 0$, and consider the space-time cylinder $Q_T = \Omega \times (0, T)$(see Fig. 9.1). Let $\mathbf{A} = \mathbf{A}(\mathbf{x}, t)$ be a square matrix of order n, $\mathbf{b} = \mathbf{b}(\mathbf{x}, t)$, $\mathbf{c} = \mathbf{c}(\mathbf{x}, t)$ vectors in $\mathbb{R}^n$, $a = a(\mathbf{x}, t)$ and $f = f(\mathbf{x}, t)$ real functions. Equations for $u = u(\mathbf{x}, t)$ *in divergence form* of the type

$$u_t - \operatorname{div}(\mathbf{A}\nabla u) + \operatorname{div}(\mathbf{b}u) + \mathbf{c} \cdot \nabla u + au = f \tag{9.1}$$

or in *non-divergence form* of the type

$$u_t - \operatorname{Tr}(\mathbf{A}D^2 u) + \mathbf{c} \cdot \nabla u + au = f \tag{9.2}$$

are called **parabolic** in Q_T if

$$A(\mathbf{x}, t)\boldsymbol{\xi} \cdot \boldsymbol{\xi} > 0, \qquad \forall (\mathbf{x}, t) \in Q_T, \forall \boldsymbol{\xi} \in \mathbb{R}^n,\ \boldsymbol{\xi} \neq \mathbf{0}.$$

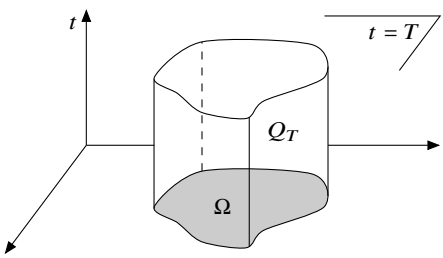

Fig. 9.1 Space-time cylinder

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
https://doi.org/10.1007/978-3-031-21853-8_9

For parabolic equations we may repeat the considerations concerning elliptic equations made in Sects. 8.1 and 8.2. Also in this case, different notions of solutions may be given, with the obvious corrections due to the evolutionary nature of (9.1) and (9.2). For identical reasons, we develop the basic theory for divergence form equations. Thus, we consider parabolic operators of the type[1]

$$\mathcal{P}u = u_t + \mathcal{E}u \equiv u_t - \operatorname{div}(\mathbf{A}(\mathbf{x},t)\nabla u) + \mathbf{c}(\mathbf{x},t)\cdot\nabla u + a(\mathbf{x},t)u.$$

The matrix $\mathbf{A} = (a_{i,j}(\mathbf{x},t))$ encodes the anisotropy of the medium with respect to diffusion. For instance (see Sect. 2.6.2) a matrix of the type

$$\begin{pmatrix} \alpha & 0 & 0 \\ 0 & \varepsilon & 0 \\ 0 & 0 & \varepsilon \end{pmatrix}$$

with $\alpha \gg \varepsilon > 0$, denotes higher propensity of the medium towards diffusion along the x_1-axis, than along the other directions. As in the stationary case, it is important to compare the effects of the drift, reaction and diffusion terms, to control the stability of numerical algorithms. We make the following assumptions:

a) $\mathcal{P}$ is *uniformly parabolic*: there are positive numbers ν and M such that:

$$\nu\,|\boldsymbol{\xi}|^2 \le \sum_{i,j=1}^{n} a_{ij}(\mathbf{x},t)\xi_i\xi_j \quad \text{and} \quad \left|a_{ij}(\mathbf{x},t)\right| \le M, \quad \forall \boldsymbol{\xi} \in \mathbb{R}^n, \text{ a.e. in } Q_T. \tag{9.3}$$

b) The coefficients $\mathbf{c}$ and a are bounded (i.e. all belong to $L^\infty(Q_T)$), with

$$|\mathbf{c}| \le \gamma_0, \ \ |a| \le \alpha_0, \text{ a.e. in } Q_T. \tag{9.4}$$

We consider initial-boundary value problems of the form:

$$\begin{cases} u_t + \mathcal{E}u = f & \text{in } Q_T \\ \mathcal{B}u = 0 & \text{on } S_T \\ u(\mathbf{x},0) = g(\mathbf{x}) & \text{in } \Omega \end{cases} \tag{9.5}$$

where $S_T = \Gamma \times (0,T)$ is the lateral part of Q_T and $\mathcal{B}u = 0$ stands for one of the usual boundary conditions. For instance, $\mathcal{B}u = \mathbf{A}\nabla u \cdot \boldsymbol{\nu} + hu$ for the Neumann/Robin condition, $\mathcal{B}u = u$ for the Dirichlet condition.

We start with the Cauchy-Dirichlet problem for the heat equation to introduce a possible *weak formulation*. Our approach makes use of integrals for functions with values in a Hilbert space and of Sobolev spaces involving time. A brief account of these notions is presented in Sect. 7.11. Then we recast into a general abstract framework all the most common initial-boundary value problems for divergence form equations. As in the elliptic case, the main tool for solving the abstract problem is provided by a sort of evolution version of the Lax Milgram Theorem (Theorem 9.6).

[1] For simplicity we assume $\mathbf{b} = \mathbf{0}$.

9.2 The Cauchy-Dirichlet Problem for the Heat Equation

Suppose we are given the problem

$$\begin{cases} u_t - \Delta u = f & \text{in } Q_T \\ u(\boldsymbol{\sigma}, t) = 0 & \text{on } S_T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega. \end{cases} \tag{9.6}$$

We recall that $\Omega \subset \mathbb{R}^n$ is a bounded domain. We want to find a weak formulation. Let us proceed formally, assuming that everything is smooth. Mimicking what we did several times in Chap. 8, we multiply the diffusion equation by a smooth function $v = v(\mathbf{x}, t)$, vanishing on S_T, and integrate over Q_T. We find

$$\int_{Q_T} u_t(\mathbf{x}, t)v(\mathbf{x}, t)d\mathbf{x}dt - \int_{Q_T} \Delta u(\mathbf{x}, t)v(\mathbf{x}, t)d\mathbf{x}dt = \int_{Q_T} f(\mathbf{x}, t)v(\mathbf{x}, t)d\mathbf{x}dt.$$

Integrating by parts the second term with respect to $\mathbf{x}$, we get, since $v = 0$ on S_T :

$$\int_{Q_T} u_t(\mathbf{x}, t)v(\mathbf{x}, t)d\mathbf{x}dt + \int_{Q_T} \nabla u(\mathbf{x}, t) \cdot \nabla v(\mathbf{x}, t)d\mathbf{x}dt = \int_{Q_T} f(\mathbf{x}, t)v(\mathbf{x}, t)d\mathbf{x}dt. \tag{9.7}$$

This looks like what we did for elliptic equations, except for the presence of u_t. Moreover, here we will have somehow to take into account the initial condition. Which could be a correct functional setting?

First of all, since we are dealing with evolution equations, it is convenient to adopt the point of view of Sect. 7.11, and consider $u = u(\mathbf{x}, t)$ as a function of t with values into a suitable Hilbert space V, that is $u : [0, T] \to V$.

When we adopt this convention, we write $u(t)$ instead of $u(\mathbf{x}, t)$ and $\dot{u}$ instead of u_t. Accordingly, we write $f(t)$ instead of $f(\mathbf{x}, t)$. Now, the homogeneous Dirichlet condition, i.e. $u(t) = 0$ on Γ for $t \in [0, T]$, suggests that the natural space for u is $L^2(0, T; V)$, where $V = H_0^1(\Omega)$ equipped with the inner product

$$(w, z)_{H_0^1(\Omega)} = (\nabla w, \nabla z)_{L^2(\Omega;\mathbb{R}^n)}$$

and corresponding norm $\|\nabla w\|_{L^2(\Omega;\mathbb{R}^n)}$. In particular, it follows that $u(t) \in V$ for a.e. $t \in (0, T)$.

To shorten the formulas, we **introduce the symbols**

$$(\cdot, \cdot)_0 \quad \text{and} \quad \|\cdot\|_0$$

for **the inner product and norm in either** $L^2(\Omega)$ **or** $L^2(\Omega; \mathbb{R}^n)$. With these notations, (9.7) becomes, separating space from time:

$$\int_0^T (\dot{u}(t), v(t))_0 \, dt + \int_0^T (\nabla u(t), \nabla v(t))_0 \, dt = \int_0^T (f(t), v(t))_0 \, dt. \tag{9.8}$$

By looking at the first integral, it would seem appropriate to require that $\dot{u} \in L^2(0,T;L^2(\Omega))$. This, however, is not coherent with the choice $u(t) \in V$, a.e. $t \in (0,T)$, since we have $\Delta u(t) \in V^* = H^{-1}(\Omega)$ and

$$\dot{u}(t) = \Delta u(t) + f(t) \tag{9.9}$$

from the diffusion equation. Thus, we deduce that the natural space for $\dot{u}$ is $L^2(0,T;V^*)$. Consequently, the first term in (9.8) has to be interpreted as

$$\langle \dot{u}(t), v(t) \rangle_* ,$$

where $\langle \cdot, \cdot \rangle_*$ denotes the duality between V^* and V.

Similarly, a reasonable assumption on f is $f \in L^2(0,T;V^*)$ and instead of $(f(t), v(t))_0$ we should write $\langle f(t), v(t) \rangle_*$. Now, from Theorem 7.100, p. 488, we know that

$$u \in C([0,T]; L^2(\Omega))$$

so that, if we choose $g \in L^2(\Omega)$, the initial condition $u(0) = g$ makes perfect sense and also

$$\|u(t) - g\|_0 \to 0 \ \text{ as } t \to 0.$$

The above arguments motivate our *first weak formulation.*

Definition 9.1 Let $f \in L^2(0,T;V^*)$ and $g \in L^2(\Omega)$. We say that $u \in L^2(0,T;V)$ is a *weak solution* of problem (9.6) if $\dot{u} \in L^2(0,T;V^*)$ and:

1. For every $v \in L^2(0,T;V)$,

$$\int_0^T \langle \dot{u}(t), v(t) \rangle_* \, dt + \int_0^T (\nabla u(t), \nabla v(t))_0 dt = \int_0^T \langle f(t), v(t) \rangle_* \, dt. \tag{9.10}$$

2. $u(0) = g$.

It is not a difficult task to check that, if u, g and f are a smooth functions and u is a weak solution, then u solves problem (9.6) in a classical sense. Thus, under regularity conditions, the weak and the classical formulation are equivalent. The Definition 9.1 is rather satisfactory and works well in many kind of applied situations, like for instance in dealing with so called *optimal control problems*. However, for the numerical treatment of problem (9.6) it is often more convenient an alternative formulation that we give below.

Definition 9.2 A function $u \in L^2(0,T;V)$ is called *weak solution* of problem (9.6) if $\dot{u} \in L^2(0,T;V^*)$ and:

1. For every $w \in V$ and a.e. $t \in (0,T)$,

$$\langle \dot{u}(t), w \rangle_* + (\nabla u(t), \nabla w)_0 = \langle f(t), w \rangle_* . \tag{9.11}$$

2. $u(0) = g$.

Remark 9.3 Equation (9.11) may be interpreted in the sense of distributions. To see this, observe that, for every $w \in V$, the real function $t \mapsto z(t) = \langle \dot{u}(t), w\rangle_*$ is a distribution in $\mathcal{D}'(0,T)$ and

$$\langle \dot{u}(t), w\rangle_* = \frac{d}{dt}(u(t), w)_0 \qquad \text{in } \mathcal{D}'(0,T). \tag{9.12}$$

This means that, for every $\varphi \in \mathcal{D}(0,T)$, we have

$$\int_0^T \langle \dot{u}(t), w\rangle_* \, \varphi(t)dt = -\int_0^T (u(t), w)_0 \dot{\varphi}(t)dt.$$

In fact, since $\left|\langle \dot{u}(t), w\rangle_{V^*}\right| \le \|\dot{u}(t)\|_* \, \|w\|_V$, it follows that $z \in L^1_{\text{loc}}(0,T) \subset \mathcal{D}'(0,T)$. Moreover, by Bochner's Theorem 7.96, p. 486, with V replaced by V^*, and the definition of $\dot{u}$, we may write

$$\int_0^T \langle \dot{u}(t), w\rangle_* \, \varphi(t)dt = \left\langle \int_0^T \dot{u}(t)\varphi(t)dt, w \right\rangle_* = \left\langle -\int_0^T u(t)\dot{\varphi}(t)dt, w \right\rangle_*.$$

On the other hand, $\int_0^T u(t)\dot{\varphi}(t)dt \in V$ so that[2]

$$\left\langle -\int_0^T u(t)\dot{\varphi}(t)dt, w \right\rangle_* = \left(-\int_0^T u(t)\dot{\varphi}(t)dt \, , w \right)_0 = -\int_0^T (u(t), w)_0 \dot{\varphi}(t)dt$$

and (9.12) is true. As a consequence, equation (9.11) may be written in the form

$$\frac{d}{dt}(u(t), w)_0 + (\nabla u(t), \nabla w)_0 = (f, w)_0, \tag{9.13}$$

in the sense of distributions in $\mathcal{D}'(0,T)$, for all $w \in V$.

We now prove the equivalence of the two Definitions 9.1 and 9.2.

Theorem 9.4 *The two Definitions 9.1 and 9.2 are equivalent.*

Proof Assume that u is a weak solution according to Definition (9.2). Let $\{w_j\}_{j\ge1}$ be a countable orthonormal basis in V. For each $w = w_j$, (9.11) holds outside a set F_j of measure zero. Set $E_0 = U_{j\ge1}F_j$. Then E_0 has measure zero and (9.11) holds for all w_j, $j \ge 1$, and all $t \in (0,T)\setminus E_0$. Take any $v \in L^2(0,T;V)$ and let $c_j(t) = (v(t), w_j)_V$,

$$v_N(t) \equiv \textstyle\sum_1^N c_j(t)w_j.$$

Take $w = w_j$ in (9.11), multiply by $c_j(t)$ and sum over j from 1 to N. We find

$$\langle \dot{u}(t), v_N(t)\rangle_* + (\nabla w(t), v_N(t))_0 = \langle f(t), v_N(t)\rangle_*, \quad \forall t \in (0,T)\setminus E_0. \tag{9.14}$$

Since $v_N(t) \to v(t)$ in V for a.e. $t \in (0,T)$, it follows that

$$\langle \dot{u}(t), v(t)\rangle_* + (\nabla w(t), v(t))_0 = \langle f(t), v(t)\rangle_*, \quad \text{a.e. in } (0,T). \tag{9.15}$$

[2] Recall from Sect. 6.8 that if $u \in H$ and $w \in V$, $\langle u, w\rangle_* = (u, w)_0$.

Integrating (9.15) with respect to t over $(0,T)$ we obtain (9.1).

Assume now that u is a weak solution according to Definition 9.1. We want to show that (9.11) holds for all $v \in V$ and a.e. $t \in (0,T)$. Suppose that this is not true. Then there exists $w \in V$ and a set $E \subset (0,T)$ of positive measure, such that

$$\langle \dot{u}(t), w\rangle_* + (\nabla u(t), \nabla w)_0 - \langle f(t), w\rangle_* > 0 \ \text{ for every } t \in E.$$

Then, (9.10) does not hold for $v(t) = \chi_E(t)w$. Contradiction. □

9.3 Abstract Parabolic Problems

9.3.1 Formulation

In this section we formulate an Abstract Parabolic Problem ($\mathcal{APP}$ in the sequel) and give a notion of weak solution. All the most common linear initial-boundary value problems for a wide class of operators can be recast as an $\mathcal{APP}$. In this general framework we provide an existence, uniqueness and stability result that plays the same role of the Lax-Milgram Theorem for the elliptic case.

Guided by the example in Sect. 9.2, we first single out the main ingredients in the abstract formulation.

• *Functional setting*. The functional setting is constituted by a Hilbert triplet $\{V, H, V^*\}$, where V, H are *separable* Hilbert spaces with V *compactly embedded* into H. As usual, we denote by $\langle\cdot,\cdot\rangle_*$ the duality between V^* and V.

The choice of V depends on the type of boundary condition we are dealing with. The familiar choices are $V = H_0^1(\Omega)$ for the homogeneous Dirichlet condition, $V = H^1(\Omega)$ for the Neumann or Robin condition, $V = H^1_{0,\Gamma_D}(\Omega)$ in the case of mixed Dirichlet/Neumann or Robin conditions, with homogeneous Dirichlet conditions.

• *Spaces for data and solution*. We assume that the source term f belongs to $L^2(0,T;V^*)$ and the initial data g to H. Our solution will belong to the Hilbert space

$$H^1(0,T;V,V^*) = \left\{u : u \in L^2(0,T;V),\ \dot{u} \in L^2(0,T;V^*)\right\}$$

with the inner product

$$(u,v)_{H^1(0,T;V,V^*)} = \int_0^T (u(t),v(t))_V dt + \int_0^T (\dot{u}(t),\dot{v}(t))_{V^*} dt$$

and the norm

$$\|u\|^2_{H^1(0,T;V,V^*)} = \int_0^T \|u(t)\|_V^2\, dt + \int_0^T \|\dot{u}(t)\|^2_{V^*}\, dt.$$

• *The bilinear form*. We are given a bilinear form, possibly time dependent,

$$B(w,z;t) : V \times V \to \mathbb{R} \ \text{ for a.e. } t \in (0,T).$$

We shall adopt the following **assumptions.**

a) *Continuity*: there exists $\mathcal{M} = \mathcal{M}(T) > 0$, such that

$$|B(w, z; t)| \leq \mathcal{M} \|w\|_V \|z\|_V , \quad \forall w, z \in V, \text{ a.e. in } (0, T). \tag{9.16}$$

b) *V-H weak coercivity*: there exist numbers $\lambda \geq 0$ and $\alpha > 0$ such that

$$B(w, w; t) + \lambda \|w\|_H^2 \geq \alpha \|w\|_V^2 , \quad \forall w \in V, \text{ a.e. in } (0, T). \tag{9.17}$$

c) *t-measurability*: the map $t \mapsto B(w, z; t)$ is measurable for all $w, z \in V$.

Remark 9.5 Note that, under the assumption a), for every $u, v \in L^2(0, T; V)$, the function $t \mapsto B(u(t), v(t); t)$ belongs to $L^1(0, T)$.

Our **Abstract Parabolic Problem** can be formulated in the following way.

Find $u \in H^1(0, T; V, V^*)$ *such that:*

$\mathcal{AP}_1$. *For all* $v \in L^2(0, T; V)$,

$$\int_0^T \langle \dot{u}(t), v(t) \rangle_* \, dt + \int_0^T B(u(t), v(t); t) dt = \int_0^T \langle f(t), v(t) \rangle_* \, dt.$$

$\mathcal{AP}_2$. $u(0) = g$.

From Theorem 7.100, p. 488 we know that

$$H^1(0, T; V, V^*) \hookrightarrow C([0, T] ; H).$$

Thus condition $\mathcal{AP}_2$ makes perfectly sense and moreover it implies

$$\|u(t) - g\|_H \to 0$$

as $t \to 0$.

Condition $\mathcal{AP}_1$ can be stated in the following equivalent form:

$\mathcal{AP}'_1$. *For all* $w \in V$ *and a.e.* $t \in (0, T)$,

$$\langle \dot{u}(t), w \rangle_* + B(u(t), w; t) = \langle f(t), w \rangle_* . \tag{9.18}$$

The equivalence of the two conditions $\mathcal{AP}_1$ and $\mathcal{AP}'_1$ follows from the separability of V and Theorem 6.18, p. 360, by repeating almost verbatim the proof of Theorem 9.4. In particular, from this proof it follows that another equivalent formulation is:

$\mathcal{AP}''_1$. *For all* $v \in L^2(0, T; V)$ *and a.e.* $t \in (0, T)$,

$$\langle \dot{u}(t), v(t) \rangle_* + B(u(t), v(t); t) = \langle f(t), v(t) \rangle_* . \tag{9.19}$$

Our purpose is to prove the following theorem.

Theorem 9.6 *The Abstract Parabolic Problem has a unique solution u. Moreover, the following energy estimates hold:*

$$\|u(t)\|_H^2 + \alpha \int_0^t \|u(s)\|_V^2 \, ds \le e^{2\lambda t} \left\{ \|g\|_H^2 + \frac{1}{\alpha} \int_0^t \|f(s)\|_{V^*}^2 \, ds \right\} \tag{9.20}$$

and

$$\int_0^t \|\dot{u}(s)\|_{V^*}^2 \, ds \le \left\{ C_0 \, \|g\|_H^2 + C_1 \int_0^t \|f(s)\|_{V^*}^2 \, ds \right\} \tag{9.21}$$

for every $t \in [0,T]$, *with* $C_0 = 2\alpha^{-1}\mathcal{M}^2 e^{2\lambda t}$, $C_1 = 2\alpha^{-2}\mathcal{M}^2 e^{2\lambda t} + 2$.

Observe how the constants in the estimates (9.20) and (9.21) deteriorates as t becomes large, unless the bilinear form is coercive. Indeed, in this case $\lambda = 0$ and no time dependence appears in those constants. This turns out very useful for studying asymptotic properties of the solution as $t \to +\infty$.

We split the proof of Theorem 9.6 into the two Lemmas 9.7 and 9.10. In the first one we prove that, assuming that u is a solution of $\mathcal{APP}$, the energy estimates (9.20) and (9.21) hold. These estimates provide a control of the relevant norms

$$\|u\|_{H^1(0,T;V,V^*)} \quad \text{and} \quad \|u\|_{C([0,T];H)}$$

of the solution, in terms of the norms $\|f\|_{L^2(0,T;V^*)}$ and $\|g\|_H$ of the data. Uniqueness and stability follow immediately.

In Lemma 9.10 we prove the existence of the solution. To this purpose, we approximate our $\mathcal{APP}$ by means of a sequence of Cauchy problems for suitable ordinary differential equations in finite dimensional spaces. Their solutions are called *Faedo-Galerkin approximations.* The task is to show that we can extract from the sequence of Faedo-Galerkin approximations a subsequence converging in some sense to a solution of our $\mathcal{APP}$. This is a typical compactness problem in Hilbert spaces. The key tool is Theorem 6.60, p. 389: *in a Hilbert space any bounded sequence has a weakly convergent subsequence*. The required bounds are once again provided by the energy estimates.

9.3.2 Energy estimates. Uniqueness and stability

We prove the following lemma.

Lemma 9.7 *Let u be a solution of the Abstract Parabolic Problem. Then the energy estimates* (9.20) *and* (9.21) *hold for u. In particular the Abstract Parabolic Problem has at most one weak solution.*

For the proof we shall use the following elementary but very useful lemma, known as *Gronwall's Lemma*.

Lemma 9.8 *Let* Ψ, G *be continuous in* $[0,T]$, *with* G *nondecreasing and* $\gamma > 0$. *If*

$$\Psi(t) \le G(t) + \gamma \int_0^t \Psi(s)ds, \qquad \textit{for all } t \in [0,T]\,, \tag{9.22}$$

then

$$\Psi(t) \le G(t)e^{\gamma t}, \qquad \textit{for all } t \in [0,T]\,.$$

Proof Let

$$R(s) = \gamma \int_0^s \Psi(r)dr.$$

Then, for all $s \in [0,T]$,

$$R'(s) = \gamma\Psi(s) \le \gamma \left[G(s) + \gamma \int_0^s \Psi(r)dr \right] = \gamma \left[G(s) + R(s) \right].$$

Multiplying both sides by $\exp(-\gamma s)$, we can write the above inequality in the form

$$\frac{d}{ds} \left[R(s) \exp(-\gamma s) \right] \le \gamma G(s) \exp(-\gamma s).$$

An integration over $(0,t)$ yields ($R(0) = 0$):

$$R(t) \le \gamma \int_0^t G(s)e^{\gamma(t-s)}ds \le G(t)(e^{\gamma t} - 1), \qquad \text{for all } t \in [0,T]\,.$$

and from (9.22) the conclusion follows easily. □

Proof (of Lemma 9.7) Let $v = u$ in (9.19). We get,

$$\langle \dot{u}(s), u(s) \rangle_* + B(u(s), u(s); s) = \langle f(s), u(s) \rangle_*\,, \quad \text{a.e. in } (0,T). \tag{9.23}$$

Now, from Remark 7.101, p. 489, we have

$$\langle \ddot{u}(s), u(s) \rangle_* = \frac{1}{2} \frac{d}{ds} \|u(s)\|_H^2\,.$$

Also,[3]

$$\left| \langle f(s), u(s) \rangle_* \right| \le \|f(s)\|_{V^*} \|u(s)\|_V \le \frac{1}{2\alpha} \|f(s)\|_{V^*}^2 + \frac{\alpha}{2} \|u(s)\|_V^2\,.$$

Thus, using assumption (9.17) on the bilinear form B, we can write, after simple rearrangements,

$$\frac{1}{2} \frac{d}{ds} \|u(s)\|_H^2 + \frac{\alpha}{2} \|u(s)\|_V^2 \le \frac{1}{2\alpha} \|f(s)\|_{V^*}^2 + \lambda \|u(s)\|_H^2\,.$$

Integrating over $(0,t)$, and using $u(0) = g$, we obtain, for every $t \in (0,T)$:

$$\|u(t)\|_H^2 + \alpha \int_0^t \|u(s)\|_V^2\, ds \le \|g\|_H^2 + \frac{1}{\alpha} \int_0^t \|f(s)\|_{V^*}^2\, ds + 2\lambda \int_0^t \|u(s)\|_H^2\, ds. \tag{9.24}$$

In particular, setting

$$\Psi(t) = \|u(t)\|_H^2\,, \quad G(t) = \|g\|_H^2 + \frac{1}{\alpha} \int_0^t \|f(s)\|_{V^*}^2\, ds, \quad \gamma = 2\lambda,$$

(9.24) implies

$$\Psi(t) \le G(t) + \gamma \int_0^t \Psi(s)ds.$$

[3] Recall the elementary inequality $2ab \le a^2/\alpha + \alpha b^2$, $\forall a, b \in \mathbb{R}$, $\forall \alpha > 0$.

Then, the Gronwall Lemma gives $\Psi(t) \le G(t)e^{\gamma t}$ in $[0,T]$, that is

$$\|u(t)\|_H^2 \le e^{2\lambda t}\left\{\|g\|_H^2 + \frac{1}{\alpha}\int_0^t \|f(s)\|_{V^*}^2\, ds\right\}. \tag{9.25}$$

Using once more (9.24) and (9.25) we get

$$\begin{aligned}\|u(t)\|_H^2 + \alpha\int_0^t \|u(s)\|_V^2\, ds &\le \left(1 + \int_0^t 2\lambda e^{2\lambda s} ds\right)\left\{\|g\|_H^2 + \frac{1}{\alpha}\int_0^t \|f(s)\|_{V^*}^2\, ds\right\}\\ &= e^{2\lambda t}\left\{\|g\|_H^2 + \frac{1}{\alpha}\int_0^t \|f(s)\|_{V^*}^2\, ds\right\}. \end{aligned}\tag{9.26}$$

We still have to estimate $\|\dot u\|^2_{L^2(0,t;V^*)} = \int_0^t \|\dot u(s)\|_{V^*}^2\, ds$. Write (9.18) in the form

$$\langle \dot u(s), w\rangle_* = -B(u(s), w; s) + \langle f(s), w\rangle_*$$

and observe that, using assumption (9.16) on B, for all $w \in V$ and a.e. $s \in (0,t)$, we can write:

$$|\langle \dot u(s), w\rangle_*| \le \left\{\mathcal{M}\,\|u(s)\|_V + \|f(s)\|_{V^*}\right\}\|w\|_V$$

and therefore, by the definition of norm in V^*, we have $\|\dot u(s)\|_{V^*} \le \mathcal{M}\,\|u(s)\|_V + \|f(s)\|_{V^*}$.

Squaring and integrating over $(0,t)$ we obtain

$$\int_0^t \|\dot u(s)\|_{V^*}^2\, ds \le 2\mathcal{M}^2\int_0^t \|u(s)\|_V^2\, ds + 2\int_0^t \|f(s)\|_{V^*}^2\, ds.$$

Finally, using (9.26) we infer

$$\int_0^t \|\dot u(s)\|_{V^*}^2\, ds \le \frac{2\mathcal{M}^2 e^{2\lambda t}}{\alpha}\left\{\|g\|_H^2 + \frac{1}{\alpha}\int_0^t \|f(s)\|_{V^*}^2\, ds\right\} + 2\int_0^t \|f(s)\|_{V^*}^2$$

from which (9.21) follows. □

9.3.3 The Faedo-Galerkin approximations

As we have mentioned at the beginning of this section, to prove the existence of a solution, we approximate our $\mathcal{APP}$ by a sequence of Cauchy problems for suitable systems of ODEs in finite dimensional spaces. To realize this program we proceed through the following steps.

• *Selection of a countable basis for V.* Since V is separable with compact embedding into H, by Proposition 6.18, p. 360, we can select a *countable orthogonal basis* $\{w_k\}_{k=1}^{\infty}$, also *orthonormal*[4] *in H*. In particular, the finite dimensional subspaces $V_m = \text{span}\,\{w_1, \dots, w_m\}$ satisfy the conditions

$$V_m \subset V_{m+1}, \quad w_{m+1} \perp V_m \ \text{ and } \ \overline{\cup V_m} = V. \tag{9.27}$$

[4] For instance, the eigenfunctions w_k, normalized in H, of the abstract variational problem $(w_k, v)_V = \lambda_k (w_k, v)_H$ for all $v \in V$.

Moreover, since V is dense in H, there exists a sequence $g_m = \sum_{j=1}^m \hat{g}_j w_j \in V_m$ such that $g_m \to g$ in H.

- *The Faedo-Galerkin approximations*. Set now

$$u_m(t) = \sum_{j=1}^m c_{jm}(t) w_j, \tag{9.28}$$

with $c_{jm} \in H^1(0,T)$, and look at the following finite dimensional problem.

Determine $u_m \in H^1(0,T;V_m)$ such that, for every $s = 1, \dots, m$,

$$\begin{cases} (\dot{u}_m(t), w_s)_H + B(u_m(t), w_s; t) = \langle f(t), w_s \rangle_* \ \text{ a.e. in } (0,T) \\ u_0(0) = g_m. \end{cases} \tag{9.29}$$

Note that, since the differential equation in (9.29) is true for each element of the basis w_s, $s = 1, \dots, m$, then it is true for every $v \in V_m$. Moreover, since, in particular, $\dot{u}_m \in L^2(0,T;V)$, we have (see (6.91), p. 394)

$$(\dot{u}_m(t), v)_H = \langle \dot{u}_m(t), v \rangle_* \quad \forall v \in V.$$

We call u_m a *Faedo-Galerkin approximation of the solution* u. Inserting (9.28) into (9.29), since the $w_k' s$ are orthonormal in H, we see that (9.29) is equivalent to the following system of ODEs for the unknowns c_{jm} : *for every* $s = 1, \dots, m$,

$$\begin{cases} \dot{c}_{sm}(t) + \sum_{j=1}^m B(w_j, w_s; t) c_{jm}(t) = \langle f(t), w_s \rangle_* \ \text{ a.e. in } (0,T) \\ c_{sm}(0) = g_m \end{cases} \tag{9.30}$$

Introduce the $m \times m$ matrix $\mathbf{B} = \mathbf{B}(t)$ whose entries are $b_{sj}(t) = B(w_j, w_s; t)$ and the vectors

$$\mathbf{C}_m^\top(t) = (c_{1m}(t), \dots, c_{mm}(t)), \ \mathbf{g}_m^\top = (\hat{g}_{1m}, \dots, \hat{g}_{mm}), \ \mathbf{F}_m^\top(t) = (f_1(t), \dots, f_m(t)),$$

where $f_s(t) = \langle f(t), w_s \rangle_*$. Then problem (9.30) can be written in the form

$$\dot{\mathbf{C}}_m(t) + \mathbf{B}(t)\mathbf{C}_m(t) = \mathbf{F}_m(t), \ \mathbf{C}_m(0) = \mathbf{g}_m. \tag{9.31}$$

By a Carathéodory Theorem (see e.g. *E. A. Coddington, N. Levinson*, [36]), the system of linear ODEs with bounded measurable coefficients (9.31) has a unique solution $\mathbf{C}_m \in H^1(0,T;\mathbb{R}^m)$. As a consequence, $u_m \in H^1(0,T;V_m)$ and satisfies the assumptions of Lemma 9.7. Since $\|g_m\|_H \le \|g\|_H$, the following lemma holds.

Lemma 9.9 *Problem (9.29) has a unique solution* $u_m \in H^1(0,T;V_m)$ *and* u_m *satisfies the energy estimates*

$$\|u_m\|_{L^2(0,T;V)} + \|\dot{u}_m\|_{L^2(0,T;V^*)} \le C_0 \left\{ \|g\|_H + \|f\|_{L^2(0,T;V^*)} \right\} \tag{9.32}$$

where $C_0 = C_0(\alpha, \lambda, \mathcal{M}, T)$.

9.3.4 Existence

We now prove the existence of the solution.

Lemma 9.10 *The sequence of Faedo-Galerkin approximations converges weakly in* $H^1(0,T;V,V^*)$ *to the solution of our Abstract Parabolic Problem.*

Proof Let $\{u_m\}$ be the sequence of Faedo-Galerkin approximations. From (9.32), the sequences $\{u_m\}$ and $\{\dot{u}_m\}$ are bounded in $L^2(0,T;V)$ and $L^2(0,T;V^*)$, respectively. Then, the weak compactness Theorem 6.60, p. 389, implies that there exists a subsequence $\{u_{m_k}\}$ such that[5]

$$u_{m_k} \rightharpoonup u \text{ in } L^2(0,T;V) \quad \text{and} \quad \dot{u}_{m_k} \rightharpoonup \dot{u} \text{ in } L^2(0,T;V^*).$$

We prove that u is the unique solution of our $\mathcal{APP}$.

First of all, to say that $u_{m_k} \rightharpoonup u$, *weakly in* $L^2(0,T;V)$, as $k \to \infty$, means that

$$\int_0^T (u_{m_k}(t), v(t))_V \, dt \to \int_0^T (u(t), v(t))_V \, dt$$

for all $v \in L^2(0,T;V)$. Similarly, $\dot{u}_{m_k} \rightharpoonup \dot{u}$ *weakly in* $L^2(0,T;V^*)$ means that

$$\int_0^T \left\langle \dot{u}_{m_k}(t), v(t) \right\rangle_* dt \to \int_0^T \langle \dot{u}(t), v(t) \rangle_* \, dt$$

for all $v \in L^2(0,T;V)$.

By the equivalence of conditions (9.18) and (9.19), we can write

$$\int_0^T \left\langle \dot{u}_{m_k}(t), v(t) \right\rangle_* dt + \int_0^T B(u_{m_k}(t), v(t); t) dt = \int_0^T \langle f(t), v(t) \rangle_* \, dt \tag{9.33}$$

for every $v \in L^2(0,T;V_{m_k})$. Let $N \le m_k$ and $w \in V_N$. Let $\varphi \in C_0^\infty(0,T)$ and insert $v(t) = \varphi(t)w$ into (9.33). Keeping N fixed and letting $m_k \to +\infty$, thanks to the weak convergence of u_{m_k} and $\dot{u}_{m_k}$ in their respective spaces, we infer

$$\int_0^T \left\{ \langle \dot{u}(t), w \rangle_* + B(u(t), w; t) - \langle f(t), w \rangle_* \right\} \varphi(t) dt = 0. \tag{9.34}$$

Letting now $N \to \infty$, recalling (9.27), we deduce that (9.34) holds for all $w \in V$. Then, the arbitrariness of φ implies

$$\langle \dot{u}(t), w \rangle_* + B(u(t), w; t) = \langle f(t), w \rangle_* \tag{9.35}$$

for all $w \in V$ and for almost every $t \in (0,T)$.

It remains to check that u satisfies the initial condition $u(0) = g$. In (9.34), choose $\varphi \in C^1([0,T])$, $\varphi(0) = 1$, $\varphi(T) = 0$. Integrating by parts the first term (see Theorem 7.100, p. 488, c)) we find, since $\langle u(t), w \rangle_* = (u(t), w)_H$,

$$\int_0^T \left\{ -(u(t), w)_H \dot{\varphi}(t) + B(u(t), w; t)\varphi(t) - \langle f(t), w \rangle_* \, \varphi(t) \right\} dt = (u(0), w)_H. \tag{9.36}$$

Similarly, inserting $v(t) = \varphi(t)w$ with $w \in V_{m_k}$ into (9.33), and integrating by parts, we get

[5] More rigorously, $\dot{u}_{m_k} \rightharpoonup z$ in $L^2(0,T;V^*)$ and one shows that $z = \dot{u}$ (see Problem (9.2)).

$$\int_0^T \left\{-(u_{m_k}(t), w)_H \dot{\varphi}(t) + B(u_{m_k}(t), w; t)\varphi(t) - \langle f(t), w\rangle_* \varphi(t)\right\} dt = (g_{m_k}, w)_H.$$

If $m_k \to +\infty$, the left hand side of the last equation converges to the left hand side of (9.36), while

$$(g_{m_k}, w)_H \to (g, w)_H.$$

Thus, we deduce that $(u(0), w)_H = (g, w)_H$ for every $w \in V$. The density of V in H yields $u(0) = g$.

Therefore u is a solution of our $\mathcal{APP}$. Finally, note that, due to the uniqueness of the solution of $\mathcal{APP}$, the whole sequence $\{u_m\}$ converges to u, not only a subsequence. Thus the proof is complete. □

9.4 Parabolic PDEs

9.4.1 Problems for the heat equation

In this section we examine some examples of applications of the results in Sect. 9.3 to the initial-boundary value problem (9.5). All the results hold in spatial dimension $n \geq 1$. In dimension $n = 1, \Omega$ is always a bounded interval $(a, b) \subset \mathbb{R}$ and, as usual, $\partial_\nu u$ at the boundary means $-u_x(a, t)$ and $u_x(b, t)$. We start with the diffusion equation.

• *The Cauchy-Dirichlet problem.* We have already introduced the weak formulation of the Cauchy-Dirichlet problem (9.6) (see Definitions 9.1 or 9.2). For this problem the Hilbert triplet is $V = H_0^1(\Omega), H = L^2(\Omega), V^* = H^{-1}(\Omega)$, while the bilinear form is $B(w, z) = (\nabla w, \nabla z)_0$; Ω is a bounded domain. Referring to the assumptions on B on page 547, a) holds with $\mathcal{M} = 1$ and b) holds with $\alpha = 1, \lambda = 0$, since B is coercive. Assumption c) is empty since B is independent of t. Theorem 9.6 yields:

Problem (9.6) *has a unique weak solution* $u \in H^1(0, T; H_0^1(\Omega), H^{-1}(\Omega))$. *Moreover, the following estimates hold, for every* $t \in [0, T]$*:*

$$\|u(t)\|_0^2 + \int_0^t \|\nabla u(s)\|_0^2 \, ds \leq \|g\|_0^2 + \int_0^t \|f(s)\|_{H^{-1}(\Omega)}^2 \, ds$$

and

$$\int_0^t \|\dot{w}(s)\|_{H^{-1}(\Omega)}^2 \, ds \leq 2\|g\|_0^2 + 4\int_0^t \|f(s)\|_{H^{-1}(\Omega)}^2 \, ds.$$

• *The Cauchy-Neumann/Robin problem.* Let Ω be a bounded *Lipschitz* domain. Consider the following problem:

$$\begin{cases} u_t - \Delta u = f & \text{in } Q_T \\ \partial_\nu u + hu = q & \text{on } S_T \\ u(0) = g & \text{in } \Omega. \end{cases} \tag{9.37}$$

Assume that $f \in L^2(0,T;L^2(\Omega)) = L^2(Q_T)$, $g \in L^2(\Omega)$, $h \in L^\infty(S_T)$, $h \geq 0$ and $q \in L^2(0,T;L^2(\Gamma)) = L^2(S_T)$. Choosing the Hilbert triplet $V = H^1(\Omega)$, $H = L^2(\Omega)$, $V^* = H^1(\Omega)^*$ and the bilinear form

$$B(w,z;t) = (\nabla w, \nabla z)_0 + \int_\Gamma h(\boldsymbol{\sigma},t) w(\boldsymbol{\sigma}) z(\boldsymbol{\sigma})\, d\sigma dt. \tag{9.38}$$

a weak formulation of problem (9.37) reads:

Find $u \in H^1(0,T;H^1(\Omega),H^1(\Omega)^*)$ such that $u(0) = g$ and

$$\langle \dot{u}(t), v\rangle_* + B(u(t),v;t) = (f(t),v)_0 + (q(t),v)_{L^2(\Gamma)}, \quad \forall v \in H^1(\Omega), \text{ a.e. in} (0,T).$$

We check assumptions (i), (ii), p. 547. Recalling the Trace Theorem 7.77, p. 471, we can write

$$|B(w,z;t)| \leq (1 + c_{tr}^2 \, \|h\|_{L^\infty(S_T)}) \, \|w\|_{H^1(\Omega)} \, \|z\|_{H^1(\Omega)} \,,$$

so that B is continuous with $\mathcal{M} = 1 + c_{tr}^2 \, \|h\|_{L^\infty(S_T)}$. Furthermore, since $h \geq 0$ a.e. on $\partial\Omega$,

$$B(w,w;t) \geq \|\nabla w\|_0^2 = \|w\|_{H^1(\Omega)}^2 - \|w\|_0^2 \,.$$

Thus B is weakly coercive with $\alpha = \lambda = 1$. Finally, since $\|f(s)\|_* \leq \|f(s)\|_0$ and $\|q(s)\|_* \leq c_{tr} \, \|q(s)\|_{L^2(\Gamma)}$ a.e. in $(0,T)$, we conclude that:

Problem (9.37) has a unique weak solution $u \in H^1(0,T;H^1(\Omega),H^1(\Omega)^*)$ and, for every $t \in [0,T]$,

$$\|u(t)\|_0^2 + \int_0^t \|u(s)\|_{H^1(\Omega)}^2 \, ds \leq e^{2t} \left\{ \|g\|_0^2 + \int_0^t \|f(s)\|_0^2 \, ds + c_{tr}^2 \int_0^t \|q(s)\|_{L^2(\Gamma)}^2 \, ds \right\}$$

and

$$\int_0^t \|\dot{u}(s)\|_*^2 \, ds \leq C \, \|g\|_0^2 + (C+2) \left\{ \int_0^t \|f(s)\|_0^2 \, ds + c_{tr}^2 \int_0^t \|q(s)\|_{L^2(\Gamma)}^2 \, ds \right\}$$

with $C = 2\mathcal{M}^2 e^{2t}$.

• *Mixed Dirichlet-Neumann boundary conditions*. Let Ω be a bounded *Lipschitz* domain. We consider the problem

$$\begin{cases} u_t - \Delta u = f & \text{in } Q_T \\ u(0) = g & \text{in } \Omega \\ \partial_\nu u = q & \text{on } \Gamma_N \times (0,T) \\ u = 0 & \text{on } \Gamma_D \times (0,T), \end{cases} \tag{9.39}$$

where Γ_D is a relatively open subset of $\partial\Omega$ and $\Gamma_N = \partial\Omega \setminus \Gamma_D$. For the weak formulation we choose $H = L^2(\Omega)$ and $V = H^1_{0,\Gamma_D}(\Omega)$, with inner product $(w,z)_V = (\nabla w, \nabla z)_0$. Recall that in $H^1_{0,\Gamma_D}(\Omega)$ a Poincaré's inequality holds:

$$\|w\|_0 \le C_P \|\nabla w\|_0 . \tag{9.40}$$

A weak formulation of problem (9.39) reads:
Find $u \in H^1(0,T;V,V^*)$ *such that* $u(0) = g$ *and*

$$\langle \dot{u}(t), v\rangle_* + B(u(t),v) = (f(t),v)_0 + (q(t),v)_{L^2(\Gamma_N)}, \quad \forall v \in V, \text{ a.e. in}(0,T),$$

where $B(u,v) = (\nabla u, \nabla v)_0$.

The bilinear form $B(w,v) = (\nabla w, \nabla v)_0$ is continuous, with $\mathcal{M} = 1$, and coercive, with $\alpha = 1$. Moreover

$$\|f(t)\|_{V^*} \le C_P \|f(t)\|_0 \text{ and } \|q(t)\|_{V^*} \le c_{tr} \|q(t)\|_{L^2(\Gamma_N)}$$

for almost every $t \in (0,T)$. We conclude that:

If $f \in L^2(0,T;L^2(\Omega))$, $g \in L^2(\Omega)$, $q \in L^2(0,T;L^2(\Gamma_N))$, *the problem (9.39) has a unique weak solution* $u \in H^1(0,T;V,V^*)$. *Moreover, the following inequalities hold for all* $t \in [0,T]$:

$$\|u(t)\|_0^2 + \int_0^t \|\nabla u(s)\|_0^2 \, ds \le \|g\|_0^2 + C_P^2 \int_0^t \|f(s)\|_0^2 \, ds + c_{tr}^2 \int_0^t \|q(s)\|_{L^2(\Gamma_N)}^2 \, ds$$

and

$$\int_0^t \|\dot{u}(s)\|_*^2 \, ds \le 2\|g\|_0^2 + 4\left\{C_P^2 \int_0^t \|f(s)\|_0^2 \, ds + c_{tr}^2 \int_0^t \|q(s)\|_{L^2(\Gamma_N)}^2 \, ds\right\}.$$

9.4.2 Regularity

As in the elliptic case, the regularity of the solution improves with the regularity of the data. In particular, for the interior regularity, if f and all the coefficients are smooth, then $u \in C^\infty(Q_T)$, regardless the regularity of the initial and boundary conditions. Regarding the global regularity, we limit ourselves to give the details in the case of the Cauchy-Dirichlet problem (9.6) for the heat equation. Precisely, we have:

Theorem 9.11 *Let* Ω *be a bounded, Lipschitz domain and* u *be the weak solution of problem (9.6). If* $g \in H_0^1(\Omega)$ *and* $f \in L^2(0,T;L^2(\Omega))$, *then* $u \in L^\infty(0,T;H_0^1(\Omega))$, $\dot{u} \in L^2(0,T;L^2(\Omega))$ *and*

$$\operatorname*{ess\,sup}_{s\in[0,T]} \|\nabla u(s)\|_0^2 + \int_0^T \|\dot{u}(s)\|_0^2 \, ds \le \|\nabla g\|_0^2 + \int_0^T \|f(s)\|_0^2 \, ds. \tag{9.41}$$

If in addition, Ω *is a* C^2*-domain, then* $u \in L^2(0,T;H^2(\Omega))$ *and*

$$\int_0^T \|u(s)\|^2_{H^2(\Omega)}\, ds \le C(n,\Omega)\left\{\|\nabla g\|_0^2 + \int_0^T \|f(s)\|_0^2\, ds\right\}. \tag{9.42}$$

Proof Let us go back to the weak formulation for the Faedo-Galerkin approximations, that in this case reads

$$\begin{cases} (\dot u_m(t), w_s)_0 + (\nabla u_m(t), \nabla w_s)_0 = (f(t), w_s)_0, & \text{a.e. } t \in [0,T] \\ u_m(0) = g_m, \end{cases} \tag{9.43}$$

for $s = 1, \dots, m$. Take as a basis $\{w_s\}$ in $H^1_0(\Omega)$ a sequence of Dirichlet eigenfunctions for the Laplace operators in Ω. In particular (see Theorem 8.8, p. 507), we may choose the w_s to be orthonormal in $L^2(\Omega)$ and orthogonal in $H^1_0(\Omega)$. Write $u(t) = \sum_{j=1}^\infty c_j(t) w_j$ with convergence in V for a.e. t, and $u_m(t) = \sum_{j=1}^m c_j(t) w_j$. Note that $\dot u_m(t) \in L^2(\Omega)$ for almost every $t \in (0,T)$. Multiplying the differential equation in (9.43) by $\dot c_j(t)$ and summing for $j = 1, \dots, m$, we get

$$\|\dot u_m(t)\|_0^2 + (\nabla u_m(t), \nabla \dot u_m(t))_0 = (f(t), \dot u_m(t))_0, \tag{9.44}$$

for a.e. $t \in [0,T]$. Now, observe that

$$(\nabla u_m(t), \nabla \dot u_m(t))_0 = \frac{1}{2}\frac{d}{dt}\|\nabla u_m(t)\|_0^2, \qquad \text{a.e. in } (0,T)$$

and that, by the Schwarz inequality,

$$(f(t), \dot u_m(t))_0 \le \|f(t)\|_0 \|\dot u_m(t)\|_0 \le \frac{1}{2}\|f(t)\|_0^2 + \frac{1}{2}\|\dot u_m(t)\|_0^2.$$

From this inequality and (9.44), we infer

$$\frac{d}{dt}\|\nabla u_m(t)\|_0^2 + \|\dot u_m(t)\|_0^2 \le \|f(t)\|_0^2, \qquad \text{a.e. in } (0,T).$$

An integration over $(0,t)$ yields, since $\|\nabla g_m\|_0^2 \le \|\nabla g\|_0^2$,

$$\|\nabla u_m(t)\|_0^2 + \int_0^t \|\dot u_m(s)\|_0^2\, ds \le \int_0^t \|f(s)\|_0^2\, ds + \|\nabla g\|_0^2. \tag{9.45}$$

Passing to the limit as $m \to \infty$, (see Proposition 7.98) we deduce that the same estimate holds for u and therefore, that $u \in L^\infty(0,T; H^1_0(\Omega))$, $\dot u \in L^2(0,T; L^2(\Omega))$ and (9.41) holds. In particular, since $\dot u(t) \in L^2(\Omega)$ for almost all $t \in (0,T)$, for every $v \in V$ we may write

$$(\nabla u(t), \nabla v)_0 = (f(t) - \dot u(t), v)_0, \qquad \text{a.e. in} (0,T).$$

Now, if Ω is a C^2-domain, the regularity theory for elliptic equations (Theorem 8.30, p. 530) implies that $u(t) \in H^2(\Omega)$ for almost all $t \in (0,T)$ and that

$$\|u(t)\|^2_{H^2(\Omega)} \le C(n,\Omega)\left\{\|f(t)\|_0^2 + \|\dot u(t)\|_0^2\right\}.$$

Integrating over $(0,T)$ and using (9.45) to estimate $\int_0^t \|\dot u(s)\|_0^2\, ds$, we obtain both $u \in L^2(0,T; H^2(\Omega))$ and the estimate (9.42). □

Remark 9.12 A result similar to Theorem 9.11 holds for the homogeneous Neumann problem ($q = 0$). Also the proof is the same, using the Neumann eigenvectors for the Laplace operators instead of the Dirichlet ones (see Problem 9.4)

Further global regularity requires that the data f and g satisfy suitable compatibility conditions on $\Gamma \times \{t = 0\}$. Assume that Ω is smooth, $f \in C^\infty(\overline{Q}_T)$ and $g \in C^\infty(\overline{\Omega})$. Suppose $u \in C^\infty(\overline{Q}_T)$. Since $u = 0$ on the lateral side, we have

$$u = \partial_t u = \cdots = \partial_t^j u = \cdots = 0, \quad \forall j \geq 0, \text{ on } S_T \tag{9.46}$$

which hold, by continuity, also for $t = 0$ on $\partial\Omega$. On the other hand, the heat equation gives

$$\partial_t u = \Delta u + f, \qquad \partial_t^2 u = \Delta(\partial_t u) + \partial_t f = \Delta^2 u + \Delta f + \partial_t f$$

and, in general ($\Delta^0 = \text{Identity}$)

$$\partial_t^j u = \Delta^j u + \textstyle\sum_{k=0}^{j-1} \Delta^k \partial_t^{j-1-k} f, \quad \forall j \geq 1, \text{ in } Q_T.$$

For $t = 0$ we have $\Delta^j u(0) = \Delta^j g$ and from (9.46) we get the conditions

$$g = 0 \quad \text{and} \quad \Delta^j g + \textstyle\sum_{k=0}^{j-1} \Delta^k \partial_t^{j-1-k} f(0) = 0, \quad \forall j \geq 1, \text{ on } \Gamma. \tag{9.47}$$

Thus, conditions (9.47) are necessary in order to have $u \in C^\infty(\overline{Q}_T)$. It turns out that they are sufficient as well (for the proof, see e.g. *H. Brezis*, [35]):

Theorem 9.13 *Let Ω be a bounded smooth domain and u be the weak solution of problem (9.6). If $f \in C^\infty(\overline{Q}_T)$, $g \in C^\infty(\overline{\Omega})$ and the compatibility conditions (9.47) hold, then $u \in C^\infty(\overline{Q}_T)$.*

9.4.3 General Equations

The weak formulation of the general problem (9.5) follows the pattern of the previous sections. Choose a Hilbert triplet $\{V, H, V^*\}$, where $H = L^2(\Omega)$ and V is a Sobolev space, $H_0^1(\Omega) \subseteq V \subseteq H^1(\Omega)$. In general Ω is a bounded Lipschitz domain. A weak formulation of problem (9.5) reads:

Find $u \in H^1(0, T; V, V^*)$ such that $u(0) = g$ and

$$\langle \dot{u}(t), v \rangle_* + B(u(t), v; t) = \langle f(t), v \rangle_*, \quad \forall v \in V, \text{ a.e. in } (0, T)$$

where $(w, z \in V)$

$$B(w, z; t) = \int_\Omega \{\mathbf{A}(\mathbf{x}, t)\nabla w \cdot \nabla z + (\mathbf{c}(\mathbf{x}, t) \cdot \nabla w)z + a(\mathbf{x}, t)wz\}\, d\mathbf{x},$$

or, in the case of Robin conditions,

$$= \int_\Omega \{\mathbf{A}(\mathbf{x}, t)\nabla w \cdot \nabla z + (\mathbf{c}(\mathbf{x}, t) \cdot \nabla w)z + a(\mathbf{x}, t)wz\}\, d\mathbf{x} + \int_\Gamma h(\boldsymbol{\sigma}, t)wz\, d\sigma$$

with $h \in L^\infty(S_T)$, $h \geq 0$ a.e. on S_T.

Notice that B is time dependent, in general. Under the hypotheses (9.3) and (9.4), p. 542, by following what we did in the elliptic case in Sects. 8.4.2, 8.4.3 and 8.4.4, it is not difficult to show that

$$|B(w,z;t)| \le \mathcal{M}\,\|w\|_V\,\|z\|_V$$

so that B is continuous in V. The constant $\mathcal{M}$ depends only on n, T and on M, γ_0, α_0, that is on the size of the coefficients a_{ij}, c_j, a (also on the trace constant c_{tr} in Thm. 7.77 and $\|h\|_{L^\infty(S_T)}$ in the case of the Robin condition).

Also, B is weakly coercive. In fact from (9.4), we have, for every $\varepsilon > 0$:

$$\int_\Omega (\mathbf{c}\cdot\nabla u)u\,d\mathbf{x} \ge -\gamma_0\,\|\nabla u\|_0\,\|u\|_0 \ge -\frac{\gamma_0}{2}\left[\varepsilon\,\|\nabla u\|_0^2 + \frac{1}{\varepsilon}\,\|u\|_0^2\right]$$

and

$$\int_\Omega a u^2 d\mathbf{x} \ge -\alpha_0\,\|u\|_0^2\,,$$

whence, as $h \ge 0$ a.e. on $\partial\Omega$,

$$B(u,u;t) \ge \left(\nu - \frac{\gamma_0\varepsilon}{2}\right)\|\nabla u\|_0^2 - \left(\frac{\gamma_0}{2\varepsilon} + \alpha_0\right)\|u\|_0^2\,. \tag{9.48}$$

Thus, if $\gamma_0 = 0$, i.e. $c = 0$, assumption (ii), p. 547, holds with any $\lambda > \alpha_0$. If $\gamma_0 > 0$, choose in (9.48)

$$\varepsilon = \frac{\nu}{\gamma_0} \quad\text{and}\quad \lambda = 2\left(\frac{\gamma_0}{2\varepsilon} + \alpha_0\right) = 2\left(\frac{\gamma_0^2}{2\nu} + \alpha_0\right).$$

Then

$$B(u,u;t) + \lambda\,\|u\|_0^2 \ge \frac{\nu}{2}\,\|\nabla u\|_0^2 + \frac{\lambda}{2}\,\|u\|_0^2 \ge \min\left\{\frac{\nu}{2}, \frac{\lambda}{2}\right\}\|u\|_{H^1(\Omega)}^2$$

so that B is weakly coercive independently of the choice of V, with $\alpha = \min\left\{\frac{\nu}{2}, \frac{\lambda}{2}\right\}$. Moreover, for fixed u, v in V, the function

$$t \longmapsto B(u,v;t)$$

is measurable by Fubini's Theorem.

Thus, from Theorem 9.6, we can draw the following conclusion.

Theorem 9.14 *If $f \in L^2(0,T;V^*)$ and $g \in L^2(\Omega)$, there exists a unique weak solution u of problem (9.5). Moreover*

$$\max_{t\in[0,T]}\|u(t)\|_0^2 + \int_0^T \|u(t)\|_V^2\,dt + \int_0^T \|\dot u(t)\|_*^2\,dt \le C\left\{\int_0^T \|f(t)\|_*^2\,dt + \|g\|_0^2\right\}$$

where, in general, C depends only on α, λ and $\mathcal{M}$.

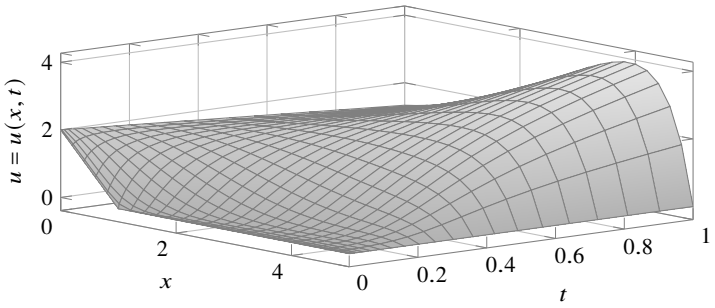

Fig. 9.2 The solution of problem (9.49)

Remark 9.15 The method works with nonhomogeneous boundary conditions as well. For instance, for the initial-Dirichlet problem, if the boundary data is the trace on S_T of a function $\varphi \in L^2(0,T;H^1(\Omega))$ with $\dot{\varphi} \in L^2(0,T;L^2(\Omega))$, the change of variable $w = u - \varphi$ reduces the problem to homogeneous boundary conditions. Also, the other nonhomogeneous initial-boundary value problems can be treated as in Section 9.4.1.

Example 9.16 Figure 9.2 shows the graph of the solution of the following Cauchy-Dirichlet problem:

$$\begin{cases} u_t - u_{xx} + 2u_x = 0.2tx & 0 < x < 5, t > 0 \\ u(x,0) = \max(2-2x,0) & 0 < x < 5 \\ u(0,t) = 2 - t/6, u(5,t) = 0 & t > 0. \end{cases} \tag{9.49}$$

Note the tendency of the drift term $2u_x$, to "transport to the right" the initial data and the effect of the source term $0.2tx$ to increase the solution near $x = 5$, more and more with time.

Remark 9.17 The regularity Theorem 9.11 can be extended to general divergence form equations. For instance, assume that $\mathbf{A}$ is symmetric and the coefficients a_{ij}, c_j and a (also h in case of Robin conditions) *do not depend on* t. Then, if $f \in L^2(0,T;H)$ and $g \in V$, the weak solution u of the homogeneous Cauchy-Dirichlet or Cauchy-Robin/Neumann belongs to $L^\infty(0,T;V)$ while $\dot{u} \in L^2(0,T;H)$.

Moreover, if Ω is of class C^2 and the coefficients a_{ij} are Lipschitz continuous in $\overline{\Omega}$, then $u \in L^2(0,T;H^2(\Omega))$. The proof goes along the lines of that of Theorem 9.11 (see Problem 9.10). For further global regularity, we refer to the more specialized books in the references.

9.5 Weak Maximum Principles

Weak solutions satisfy maximum principles that generalize those in Chap. 2. Consider the operator

$$\mathcal{P}u = u_t - \operatorname{div}(\mathbf{A}(\mathbf{x},t)\nabla u) + \mathbf{c}(\mathbf{x},t)\cdot\nabla u + a(\mathbf{x},t)u,$$

where $\mathbf{A}$, $\mathbf{c}$ and a satisfies the hypotheses (9.3) and (9.4), and the associated bilinear form

$$B(w,v;t) = \int_\Omega \{\mathbf{A}(\mathbf{x},t)\nabla w\cdot\nabla v + (\mathbf{c}(\mathbf{x},t)\cdot\nabla w)v + a(\mathbf{x},t)wv\}\,d\mathbf{x}.$$

Let Ω be a bounded Lipschitz domain, $f \in L^2(0,T;H^{-1}(\Omega))$ and $g \in L^2(\Omega)$.

Under this assumptions, we know that there exists a unique weak solution $u \in H^1(0,T;H_0^1(\Omega),H^{-1}(\Omega))$ of the problem

$$\langle \dot u(t), v\rangle_* + B(u(t),v;t) = \langle f(t), v\rangle_* \tag{9.50}$$

for almost all $t \in (0,T)$ and all $v \in H_0^1(\Omega)$, with $u(0) = g$.

Definition 9.18 We say that $f(t) \geq 0$ in $\mathcal{D}'(\Omega)$ for almost every $t \in (0,T)$ if $\langle f(t),\varphi\rangle \geq 0$ a.e. in $(0,T)$, $\forall\varphi \in \mathcal{D}(\Omega)$, $\varphi \geq 0$ in Ω.

The following result holds.

Theorem 9.19 *Let u be a weak solution of problem (9.50). If $g \geq 0$ a.e. in Ω and $f(t) \geq 0$ in $\mathcal{D}'(\Omega)$ a.e. in $(0,T)$, then, for every $t \in [0,T]$,*

$$u(t) \geq 0 \text{ a.e. in } \Omega.$$

Proof For simplicity, we give the proof under the additional assumptions that the coefficients a_{ij}, a are independent of time, $f \in L^2(\Omega)$ and $g \in H_0^1(\Omega)$. Then, from Remark 9.17, we know that $\dot u \in L^2(0,T;L^2(\Omega))$ and u satisfies the equation

$$(\dot u(t), v)_0 + B(u(t),v) = (f(t),v)_0 \tag{9.51}$$

for all $v \in H_0^1(\Omega)$ and a.e. in $(0,T)$.

Since $u = 0$ on S_T, then $u^-(t) = \max\{-u(t),0\} \in H_0^1(\Omega)$ for almost every $t \in (0,T)$. Thus, we may choose $v(t) = -u^-(t) \leq 0$ as a test function in (9.51). Recalling that $u(t) = u^+(t) - u^-(t)$ and $u^+u^- = 0$ a.e, we get, from (9.48),

$$\begin{aligned} B(u(t),-u^-(t);t) &= \int_\Omega \left\{\mathbf{A}(\mathbf{x})\nabla u^-(t)\cdot\nabla u^-(t) + (\mathbf{c}(\mathbf{x},t)\cdot\nabla u^-)u^- + a(\mathbf{x})(u^-(t))^2\right\} d\mathbf{x} \\ &\geq \left(\nu - \frac{\gamma_0\varepsilon}{2}\right)\|\nabla u^-\|_0^2 - \left(\frac{\gamma_0}{2\varepsilon} + \alpha_0\right)\|u^-\|_0^2 \qquad (\varepsilon > 0). \end{aligned}$$

Choosing e.g. $\varepsilon = 2\nu/\gamma_0$ if $\gamma_0 > 0$, in any case we get

$$B(u(t),-u^-(t)) \geq -\left(\frac{\gamma_0^2}{4\nu} + \alpha_0\right)\|u^-(t)\|_0^2$$

and, since $f \geq 0$ a.e. in Ω,

$$(f(t),-u^-(t))_0 \leq 0, \quad \text{a.e. } t \in (0,T).$$

Thus, from (9.51) and $(\dot u(t),-u^-(t))_0 = \frac{1}{2}\frac{d}{dt}\|u^-(t)\|_0^2$, we infer that

$$\frac{1}{2}\frac{d}{dt}\|u^-(t)\|_0^2 \le \left(\frac{\gamma_0^2}{4\nu}+\alpha_0\right)\|u^-(t)\|_0^2 \tag{9.52}$$

for almost all $t \in (0,T)$. Now, $g = u(0) \ge 0$ a.e. in Ω entails $\|u^-(0)\|_0^2 = 0$. From (9.52) it follows that $u^-(t) = 0$ and therefore $u(t) = u^+(t) \ge 0$, a.e. in Ω, for all $t \in [0,T]$ since $u^- \in C([0,T]; L^2(\Omega))$. □

Roughly speaking, Theorem 9.19 says that *if u is a weak supersolution for the operator $\mathcal{P}$, that is $\mathcal{P}u \ge 0$ in Q_T, which is nonnegative on the parabolic boundary $\partial_p Q_T$, then u is nonnegative in all Q_T.*

Remark 9.20 Maximum principles for other boundary value problems are also available (see Problem 9.8). For instance, let $u \in H^1(0,T; H^1(\Omega), H^1(\Omega)^*)$ be a weak solution of the Cauchy-Neumann problem

$$\langle \dot{u}(t), v\rangle_* + B(u(t), v; t) = (f(t), v)_0 + (h(t), v)_{L^2(\Gamma)} \tag{9.53}$$

a.e. in $(0,T)$ and for all $v \in H^1(\Omega)$, with $u(0) = g$.

Assume that $f \in L^2(Q_T)$, $g \in L^2(\Omega)$ and $h \in L^2(S_T)$. In this case, the maximum principle states that, *if $g \ge 0$ a.e. in Ω, $f \ge 0$ a.e. in Q_T and $h \ge 0$ a.e. on S_T, then $u(t) \ge 0$ a.e in Ω, for all $t \in [0,T]$.*

In other words, if

$$\begin{cases} \mathcal{P}u \ge 0 & \text{in } Q_T \\ \mathbf{A}\nabla u \cdot \boldsymbol{\nu} \ge 0 & \text{on } S_T \\ u(\mathbf{x},0) \ge 0 & \text{in } \Omega, \end{cases}$$

in a weak sense, then $u \ge 0$ a.e. in Q_T.

9.6 The Wave Equation

9.6.1 Hyperbolic Equations

The wave propagation in a nonhomogeneous and anisotropic medium leads to second order hyperbolic equations, generalization of the classical wave equation $u_{tt} - c^2\Delta u = f$. With the same notations of Section 9.1, an equation in divergence form of the type

$$u_{tt} - \operatorname{div}(\mathbf{A}(\mathbf{x},t)\nabla u) + \mathbf{c}(\mathbf{x},t)\cdot\nabla u + a(\mathbf{x},t)u = f(\mathbf{x},t) \tag{9.54}$$

or in non-divergence form of the type

$$u_{tt} - \operatorname{tr}(\mathbf{A}(\mathbf{x},t)D^2u) + \mathbf{c}(\mathbf{x},t)\cdot\nabla u + a(\mathbf{x},t)u = f(\mathbf{x},t) \tag{9.55}$$

is called *hyperbolic* in $Q_T = \Omega \times (0,T)$ if

$$\mathbf{A}(\mathbf{x},t)\boldsymbol{\xi}\cdot\boldsymbol{\xi} > 0, \qquad \text{a.e. in } Q_T, \forall \boldsymbol{\xi} \in \mathbb{R}^n, \boldsymbol{\xi} \ne \mathbf{0}.$$

The typical problems for hyperbolic equations are those already considered for the wave equation. Given f in Q_T, we want to determine a solution u of (9.54) or (9.55) satisfying the initial conditions

$$u(\mathbf{x},0) = g(\mathbf{x}), \quad u_t(\mathbf{x},0) = h(\mathbf{x}) \qquad \text{in } \Omega$$

and one of the usual boundary conditions (Dirichlet, Neumann, Robin or mixed) on the lateral boundary $S_T = \partial\Omega \times (0,T]$.

Even if the hyperbolic equations display substantial differences from the parabolic ones from the phenomenological point of view, for divergence form equations it is possible to give a similar weak formulation, which can be analyzed by means of Faedo-Galerkin method. Since for general equations the theory is rather technical, we will limit ourselves to the Cauchy-Dirichlet problem for the wave equation.

9.6.2 The Cauchy-Dirichlet problem

Consider the problem

$$\begin{cases} u_{tt} - c^2\Delta u = f & \text{in } Q_T \\ u(\mathbf{x},0) = g(\mathbf{x}),\ u_t(\mathbf{x},0) = h(\mathbf{x}) & \text{in } \Omega \\ u(\boldsymbol{\sigma},t) = 0 & \text{on } S_T. \end{cases} \tag{9.56}$$

As usual, to find a weak formulation, we proceed formally and multiply the wave equation by a smooth function $v = v(\mathbf{x},t)$, vanishing on S_T. Integrating over Q_T, we find

$$\int_{Q_T} u_{tt}(\mathbf{x},t)v(\mathbf{x},t)d\mathbf{x}dt - c^2\int_{Q_T}\Delta u(\mathbf{x},t)v(\mathbf{x},t)d\mathbf{x}dt = \int_{Q_T} f(\mathbf{x},t)v(\mathbf{x},t)d\mathbf{x}dt.$$

Integrating by parts the second term with respect to $\mathbf{x}$, we get, since $v = 0$ on S_T,

$$\int_{Q_T} u_{tt}(\mathbf{x},t)v(\mathbf{x},t)d\mathbf{x}dt + c^2\int_{Q_T}\nabla u(\mathbf{x},t)\cdot\nabla v(\mathbf{x},t)d\mathbf{x}dt = \int_{Q_T} f(\mathbf{x},t)v(\mathbf{x},t)d\mathbf{x}dt$$

which becomes, in the notations of the previous sections, separating space from time,

$$\int_0^T (\ddot{u}(t),v(t))_0\,dt + c^2\int_0^T (\nabla u(t),\nabla v(t))_0\,dt = \int_0^T (f(t),v(t))_0\,dt,$$

where $\ddot{u}$ stays for u_{tt}. Again, the natural space for u is $L^2(0,T;H_0^1(\Omega))$. Thus, for almost all $t > 0$, $u(t) \in V = H_0^1(\Omega)$, and $\Delta u(t) \in V^* = H^{-1}(\Omega)$. On the other hand, from the wave equation we have

$$u_{tt} = c^2\Delta u + f$$

and it is natural to require $\ddot{u} \in L^2(0,T;V^*)$. Accordingly, a reasonable assumption for $\dot{u}$ is $\dot{u} \in L^2(0,T;H)$, $H = L^2(\Omega)$, an intermediate space between $L^2(0,T;V)$ and $L^2(0,T;V^*)$. Thus, we look for solutions u such that

$$u \in L^2(0,T;V), \quad \dot{u} \in L^2(0,T;H), \quad \ddot{u} \in L^2(0,T;V^*). \tag{9.57}$$

For simplicity, we also assume $f \in L^2(0,T;H)$ and

$$u(0) = g \in V, \ \dot{u}(0) = h \in H. \tag{9.58}$$

Since u satisfies (9.57) and $H \hookrightarrow V^*$, by Theorem 7.99, p. 488 it follows that

$$u \in C([0,T];H) \quad \text{and} \quad \dot{u} \in C([0,T];V^*)$$

and therefore the initial conditions (9.58) makes sense.

The above considerations lead to the following definition, analogous to Definition 9.1, p. 544, for the heat equation.

Definition 9.21 Let $f \in L^2(0,T;H)$ and $g \in V$, $h \in H$. We say that $u \in L^2(0,T;V)$ is a *weak solution* to problem (9.56) if

$$\dot{u} \in L^2(0,T;H), \qquad \ddot{u} \in L^2(0,T;V^*)$$

and:

1. For all $v \in L^2(0,T;V)$,

$$\int_0^T \langle \ddot{u}(t), v(t) \rangle_* \, dt + c^2 \int_0^T (\nabla u(t), \nabla v(t))_0 dt = \int_0^T (f(t), v(t))_0 dt. \tag{9.59}$$

2. $u(0) = g$, $\dot{u}(0) = h$.

As in the case of the heat equation, condition 1 can be stated in the following equivalent form:

1′. For all $v \in V$ and a.e. in $(0,T)$,

$$\langle \ddot{u}(t), v \rangle_* + c^2 (\nabla u(t), \nabla v)_0 = (f(t), v)_0. \tag{9.60}$$

Remark 9.22 The reader can easily check that if f, g, h are smooth functions and u is a smooth weak solution of problem (9.56), then u is actually a classical solution.

Remark 9.23 As in Remark 9.3, p. 545, the equation (9.60) may be interpreted in the sense of distributions in $\mathcal{D}'(0,T)$. Indeed (see Problem 9.15) for all $v \in V$, the real function $t \mapsto w(t) = \langle \ddot{u}(t), v \rangle_*$ is a distribution in $\mathcal{D}'(0,T)$ and

$$w(t) = \frac{d^2}{dt^2}(u(t), v)_0, \qquad \text{in } \mathcal{D}'(0,T). \tag{9.61}$$

As a consequence, the equation (9.60) may be written in the form

$$\frac{d^2}{dt^2}(u(t),v)_0 + c^2(\nabla u(t),\nabla v)_0 = (f(t),v)_0$$

for all $v \in V$ and in the sense of distributions in $\mathcal{D}'(0,T)$.

9.6.3 The method of Faedo-Galerkin

We want to show that problem (9.56) has a unique solution, which continuously depends on the data, in the appropriate norms. Once more, we are going to use the method of Faedo-Galerkin, but we use it somehow more directly, without reference to an abstract result. Here are the main steps, emphasizing the differences with the parabolic case.

1. Let $\{w_k\}_{k=1}^{\infty}$ be the sequence of normalized Dirichlet eigenfunctions for the Laplace operator in Ω. Recall that they constitute

an *orthogonal basis in V* and an *orthonormal basis in H*.

In particular, we can write

$$g = \sum_{k=1}^{\infty} \hat{g}_k w_k, \qquad h = \sum_{k=1}^{\infty} \hat{h}_k w_k,$$

where $\hat{g}_k = (g, w_k)_0$, $\hat{h}_k = (h, w_k)_0$, with the series converging in V and H, respectively.

2. Let $V_m = \text{span}\,\{w_1, w_2, \ldots, w_m\}$ and define

$$u_m(t) = \sum_{k=1}^{m} r_k(t) w_k, \qquad g_m = \sum_{k=1}^{m} \hat{g}_k w_k, \qquad h_m = \sum_{k=1}^{m} \hat{h}_k w_k.$$

We construct the sequence of Faedo-Galerkin approximations u_m, by solving the following projected problem:

Determine $u_m \in H^2(0,T;V)$ such that, for all $s = 1, \ldots, m$,

$$\begin{cases} (\ddot{u}_m(t), w_s)_0 + c^2(\nabla u_m(t), \nabla w_s)_0 = (f(t), w_s)_0, & 0 \le t \le T \\ u_m(0) = g_m,\ \dot{u}_m(0) = h_m. \end{cases} \tag{9.62}$$

Since the differential equation in (9.62) must hold for each element of the basis w_s, $s = 1, \ldots, m$, it must hold for every $v \in V_m$. Moreover, since $u_m \in H^2(0,T;V)$ we have $\ddot{u}_m \in L^2(0,T;V)$, so that

$$\langle \ddot{u}_m(t), v \rangle_* = (\ddot{u}_m(t), v)_0.$$

3. We show that the sequences $\{u_m\}$, $\{\dot{u}_m\}$ and $\{\ddot{u}_m\}$ are bounded in $L^2(0,T;V)$, $L^2(0,T;H)$ and $L^2(0,T;V^*)$, respectively (energy estimates). Then, once again, the weak compactness Theorem 6.60, p. 389, implies that a subsequence $\{u_{m_k}\}$ converges weakly in $L^2(0,T;V)$ to u, while $\{\dot{u}_{m_k}\}$ and $\{\ddot{u}_{m_k}\}$ converge weakly in $L^2(0,T;H)$ and $L^2(0,T;V^*)$ to $\dot{u}$ and $\ddot{u}$, respectively.

4. We prove that the function u in step **3** is the unique weak solution of problem (9.56), so that the whole sequences $\{u_m\}$, $\{\dot{u}_m\}$ and $\{\ddot{u}_m\}$ converges weakly to u, $\dot{u}$, $\ddot{u}$ in their respective spaces.

9.6.4 Solution of the approximate problem

The following lemma holds.

Lemma 9.24 *For all $m \geq 1$, there exists a unique solution to problem (9.62). In particular, since $u_m \in H^2(0,T;V)$, we have $u_m \in C^1([0,T];V)$.*

Proof Observe that, since $w_1, w_2, \ldots, w_m$ are orthonormal in H,

$$(\ddot{u}_m(t), w_s)_0 = \sum_{k=1}^{m} (w_k, w_s)_0 \ddot{r}_k(t) = \ddot{r}_s(t)$$

and since they are orthogonal in V,

$$c^2 \sum_{k=1}^{m} (\nabla w_k, \nabla w_s)_0 r_k(t) = c^2 (\nabla w_s, \nabla w_s)_0 r_s(t) = c^2 \|\nabla w_s\|_0^2 r_s(t).$$

Set: $F_s(t) = (f(t), w_s)_0$, $\mathbf{F}(t) = (F_1(t), \ldots, F_m(t))^\top$
and $\mathbf{R}_m(t) = (r_1(t), \ldots, r_m(t))^\top$, $\mathbf{g}_m = (\hat{g}_1, \ldots, \hat{g}_m)^\top$, $\mathbf{h}_m = (\hat{h}_1, \ldots, \hat{h}_m)^\top$.
If we introduce the diagonal matrix $\mathbf{W} = \text{diag}\left\{\|\nabla w_1\|_0^2, \|\nabla w_2\|_0^2, \ldots, \|\nabla w_m\|_0^2\right\}$ of order m, problem (9.62) is equivalent to the following system of m *uncoupled* linear ordinary differential equations, with constant coefficients and right hand side in $L^2(0,T;\mathbb{R}^m)$:

$$\ddot{\mathbf{R}}_m(t) + c^2 \mathbf{W} \mathbf{R}_m(t) = \mathbf{F}_m(t), \qquad \text{a.e. in } (0,T) \tag{9.63}$$

with initial conditions $\mathbf{R}_m(0) = \mathbf{g}_m$, $\dot{\mathbf{R}}_m(0) = \mathbf{h}_m$. Since $\mathbf{F}_m \in L^2(0,T;\mathbb{R}^m)$, system (9.63) has a unique solution $\mathbf{R}_m(t) \in H^2(0,T;\mathbb{R}^m)$. From

$$u_m(t) = \sum_{k=1}^{m} r_k(t) w_k,$$

we deduce $u_m \in H^2(0,T;V)$. □

9.6.5 Energy estimates

We want to show that, from the sequence of Galerkin approximations $\{u_m\}$, it is possible to extract a subsequence converging to the weak solution of the original

problem. We are going to prove that the relevant Sobolev norms of u_m can be controlled by the norms of the data, in a way that does not depend on m. Moreover, the estimates must be powerful enough in order to pass to the limit as $m \to +\infty$ in the approximating equation

$$(\ddot{u}_m(t), v)_0 + c^2(\nabla u_m(t), \nabla v)_0 = (f(t), v)_0.$$

In this case we can estimate the norms of u_m in $L^\infty(0,T;V)$, of $\dot{u}_m$ in $L^\infty(0,T;H)$ and of $\ddot{u}_m$ in $L^2(0,T;V^*)$, that is the norms

$$\max_{t\in[0,T]} \|\nabla u_m(t)\|_0\,, \qquad \max_{t\in[0,T]} \|\dot{u}_m(t)\|_0 \quad \text{and} \quad \int_0^T \|\ddot{u}_m(t)\|_*^2\, dt.$$

Theorem 9.25 *Let u_m be the solution of problem (9.62). Then, for all $t \in [0,T]$,*

$$\|\dot{u}_m(t)\|_0^2 + c^2\,\|\nabla u_m(t)\|_0^2 \le e^t \left\{ c^2\,\|\nabla g\|_0^2 + \|h\|_0^2 + \int_0^t \|f(s)\|_0^2\, ds \right\}. \tag{9.64}$$

Proof Since $u_m \in H^2(0,T;V)$, we may choose $v = \dot{u}_m(t)$ as a test function in (9.62). We find

$$(\ddot{u}_m(t), \dot{u}_m(t))_0 + c^2(\nabla u_m(t), \nabla \dot{u}_m(t))_0 = (f(t), \dot{u}_m(t))_0 \tag{9.65}$$

for a.e. $t \in [0,T]$. Observe that, for almost every $t \in (0,T)$,

$$(\ddot{u}_m(t), \dot{u}_m(t))_0 = \frac{1}{2}\frac{d}{dt}\,\|\dot{u}_m(t)\|_0^2 \qquad \text{and} \qquad (\nabla u_m(t), \nabla \dot{u}_m(t))_0 = \frac{c^2}{2}\frac{d}{dt}\,\|\nabla u_m(t)\|_0^2\,.$$

Moreover, by the Schwarz inequality,

$$(f(t), \dot{u}_m(t))_0 \le \|f(t)\|_0\,\|\dot{u}_m(t)\|_0 \le \frac{1}{2}\,\|f(t)\|_0^2 + \frac{1}{2}\,\|\dot{u}_m(t)\|_0^2\,,$$

so that, from (9.65), we deduce

$$\frac{d}{dt}\left\{\|\dot{u}_m(t)\|_0^2 + c^2\,\|\nabla u_m(t)\|_0^2\right\} \le \|f(t)\|_0^2 + \|\dot{u}_m(t)\|_0^2\,.$$

We now integrate over $(0,t)$, recalling that $u_m(0) = g_m$, $\dot{u}_m(0) = h_m$ and observing that

$$\|\nabla g_m\|_0^2 \le \|\nabla g\|_0^2\,, \quad \|h_m\|_0^2 \le \|h\|_0^2\,,$$

by the orthogonality of the basis. We find:

$$\begin{aligned}\|\dot{u}_m(t)\|_0^2 + c^2\,\|\nabla u_m(t)\|_0^2 &\le \|h_m\|_0^2 + c^2\,\|\nabla g_m\|_0^2 + \int_0^t \|f(s)\|_0^2\, ds + \int_0^t \|\dot{u}_m(s)\|_0^2\, ds \\ &\le \|h\|_0^2 + c^2\,\|\nabla g\|_0^2 + \int_0^t \|f(s)\|_0^2\, ds + \int_0^t \|\dot{u}_m(s)\|_0^2\, ds.\end{aligned}$$

Let

$$\Psi(t) = \|\dot{u}_m(t)\|_0^2 + c^2\,\|\nabla u_m(t)\|_0^2\,, \quad G(t) = \|h\|_0^2 + c^2\,\|\nabla g\|_0^2 + \int_0^t \|f(s)\|_0^2\, ds.$$

Note that both Ψ and G are continuous in $[0,T]$ and G is nondecreasing. Then we have

$$\Psi(t) \le G(t) + \int_0^t \Psi(s)\,ds$$

and Gronwall's Lemma 9.8, p. 548, yields, for every $t \in [0,T]$,

$$\|\dot{u}_m(t)\|_0^2 + c^2 \|\nabla u_m(t)\|_0^2 \le e^t \left\{ \|h\|_0^2 + c^2 \|\nabla g\|_0^2 + \int_0^t \|f(s)\|_0^2\,ds \right\}. \qquad \square$$

We now give a control of the norm of $\ddot{u}_m$ in $L^2(0,T;V^*)$.

Theorem 9.26 *Let u_m be the solution of problem (9.62). Then, for all $t \in [0,T]$,*

$$\int_0^t \|\ddot{u}_m(s)\|_*^2\,ds \le C_0 \left\{ \|h\|_0^2 + c^2 \|\nabla g\|_0^2 \right\} + C_1 \int_0^t \|f(s)\|_0^2\,ds, \tag{9.66}$$

where $C_0 = 2c^2(e^t - 1)$ and $C_1 = C_0 + 2C_P^2$, where C_P is the Poincaré constant.

Proof Let $v \in V$ and write

$$v = w + z$$

with $w \in V_m = \text{span}\{w_1, w_2, \ldots, w_m\}$ and $z \in V_m^{\perp}$. Since $w_1, \ldots, w_k$ are orthogonal in V, we have

$$\|\nabla w\|_0 \le \|\nabla v\|_0 .$$

Choosing w as a test function in problem (9.62), we obtain

$$(\ddot{u}_m(t), v)_0 = (\ddot{u}_m(t), w)_0 = -c^2(\nabla u_m(t), \nabla w)_0 + (f(t), w)_0.$$

Since

$$|(\nabla u_m(t), \nabla w)_0| \le \|\nabla u_m(t)\|_0 \|\nabla w\|_0 , \quad |(f(t), w)_0| \le C_P \|f(t)\|_0 \|\nabla w\|_0$$

we may write

$$\begin{aligned} |(\ddot{u}_m(t), v)_0| &\le \left\{ c^2 \|\nabla u_m(t)\|_0 + C_P \|f(t)\|_0 \right\} \|\nabla w\|_0 \\ &\le \left\{ c^2 \|\nabla u_m(t)\|_0 + C_P \|f(t)\|_0 \right\} \|\nabla v\|_0 . \end{aligned}$$

Thus, by the definition of norm in V^*, we infer

$$\|\ddot{u}_m(t)\|_* \le c^2 \|\nabla u_m(t)\|_0 + C_P \|f(t)\|_0 .$$

Squaring and integrating over $(0,t)$ we obtain

$$\int_0^t \|\ddot{u}_m(s)\|_*^2\,ds \le 2c^4 \int_0^t \|\nabla u_m(s)\|_0^2\,ds + 2C_P^2 \int_0^t \|f(s)\|_0^2\,ds$$

and (9.64) gives (9.66). $\square$

9.6.6 Existence, uniqueness and stability

Theorems 9.25 and 9.26 imply that the sequences $\{u_m\}$ and $\{\dot{u}_m\}$ are bounded in $L^\infty(0,T;V)$ and $L^\infty(0,T;H)$ respectively, hence, in particular, in $L^2(0,T;V)$ and $L^2(0,T;H)$, while the sequence $\{\ddot{u}_m\}$ is bounded in $L^2(0,T;V^*)$.

The weak compactness Theorem 6.60, p. 389, implies that there exists a subsequence, which for simplicity we still denote by $\{u_m\}$, such that, as $m \to \infty$,[6]

$$\begin{aligned} u_m &\rightharpoonup u \quad \textit{weakly in } L^2(0,T;V) \\ \dot{u}_m &\rightharpoonup \dot{u} \quad \textit{weakly in } L^2(0,T;H) \\ \ddot{u}_m &\rightharpoonup \ddot{u} \quad \textit{weakly in } L^2(0,T;V^*). \end{aligned}$$

The following theorem holds:

Theorem 9.27 *Let $f \in L^2(0,T;H)$, $g \in V$, $h \in H$. Then u is the unique weak solution of problem (9.56). Moreover,*

$$\|u\|^2_{L^\infty(0,T;V)}\,,\ \|\dot{u}\|^2_{L^\infty(0,T;H)}\,,\ \|\ddot{u}\|^2_{L^2(0,T;V^*)} \le C\left\{\|f\|^2_{L^2(0,T;H)} + \|\nabla g\|^2_0 + \|h\|^2_0\right\}$$

with $C = C(c,T,\Omega)$.

Proof **Existence**. We know that:

$$\int_0^T (\nabla u_m(t), \nabla v(t))_0\, dt \to \int_0^T (\nabla u(t), \nabla v(t))_0\, dt$$

for all $v \in L^2(0,T;V)$,

$$\int_0^T (\dot{u}_m(t), w(t))_0 dt \to \int_0^T (\dot{u}(t), w(t))_0 dt$$

for all $w \in L^2(0,T;H)$, and

$$\int_0^T (\ddot{u}_m(t), v(t))_0 = \int_0^T \langle \ddot{u}_m(t), v(t)\rangle_*\, dt \to \int_0^T \langle \ddot{u}(t), v(t)\rangle_*\, dt$$

for all $v \in L^2(0,T;V)$.

We want to use these properties to pass to the limit as $m \to +\infty$ in problem (9.62), keeping in mind that the test functions have to be chosen in V_m. Let $N \le m$ and $w \in V_N$. Let $\varphi \in C_0^\infty(0,T)$ and insert $v(t) = \varphi(t)w$ as a test function into (9.62). Integrating over $(0,T)$, we get

$$\int_0^T \left\{(\ddot{u}_m(t), w)_0 + c^2(\nabla u_m, \nabla w)_0 - (f(t), w)_0\right\} \varphi(t) dt = 0. \tag{9.67}$$

Keeping N fixed and Letting $m \to +\infty$, thanks to the weak convergence of u_m and $\ddot{u}_m$ in their respective spaces, we obtain

$$\int_0^T \left\{(\ddot{u}(t), w)_0 + c^2(\nabla u(t), \nabla w)_0 - (f(t), w)_0\right\} \varphi(t) dt = 0. \tag{9.68}$$

Letting now $N \to \infty$, we infer that (9.68) holds for all $w \in V$. Then, the arbitrariness of φ entails

$$\langle \ddot{u}(t), w\rangle_* + c^2(\nabla u(t), \nabla w)_0 = (f(t), w)_0$$

for all $w \in V$ and almost all $t \in (0,T)$. Therefore u satisfies (9.60) and we know that $u \in C([0,T];H)$, $\dot{u} \in C([0,T];V^*)$.

[6] Rigorously, $u_m \rightharpoonup u$ *in* $L^2(0,T;V)$, $\dot{u}_m \rightharpoonup w$ *in* $L^2(0,T;H)$, $\ddot{u}_m \rightharpoonup z$ *in* $L^2(0,T;V^*)$ and one shows that, $w = \dot{u}$, $z = \ddot{u}$.

To check the initial conditions, we proceed as in Lemma 9.10, p. 552. Let $v(t) = \varphi(t)w + \psi(t)z$ with $w, z \in V$ and $\varphi, \psi \in C^2([0,T])$ such that

$$\varphi(0) = 1,\ \dot\varphi(0) = \varphi(T) = \dot\varphi(T) = 0,\quad \dot\psi(0) = 1,\ \psi(0) = \psi(T) = \dot\psi(T) = 0.$$

With this v, integrating by parts twice, using Theorem 7.100, p. 488, c), and the density of H into V^*, we can write, since $\dot u(0) \in V^*$,

$$\begin{aligned}\int_0^T \langle \ddot u(t), v(t)\rangle_*\, dt &= -\int_0^T (\dot u(t), \dot v(t))_0 dt - (\dot u(0), w)_0\\ &= \int_0^T (u(t), \ddot v(t))_0 dt + \langle \dot u(0), z\rangle_* - (u(0), w)_0.\end{aligned}$$

Thus, inserting this v into (9.59) we obtain

$$\int_0^T \left\{(u(t), \ddot v(t))_0 + c^2(\nabla u(t), \nabla v(t))_0 - (f(t), v(t))_0\right\} = -\langle \dot u(0), z\rangle_* + (u(0), w)_0. \tag{9.69}$$

On the other hand, let $v_N(t) = \varphi(t)w_N + \psi(t)z_N$, where w_N, z_N are the projections of w, z, respectively, into V_N. Insert this v_N as a test function into (9.62), integrate twice by parts and let first $m \to +\infty$ and then $N \to \infty$. We deduce

$$\int_0^T \left\{\langle u(t), \ddot v(t)\rangle_* + c^2(\nabla u(t), \nabla v(t))_0 - (f(t), v(t))_0\right\} = -(h, z)_0 + (g, w)_0. \tag{9.70}$$

From (9.69) and (9.70) we get

$$\langle \dot u(0), z\rangle_* - (u(0), w)_0 = (h, z)_0 - (g, w)_0 = \langle h, z\rangle_* - (g, w)_0,$$

for every $w, z \in V$. Since V is dense in H, the arbitrariness of w and z gives

$$\dot u(0) = h \quad \text{and} \quad u(0) = g.$$

Uniqueness. Assume $g = h \equiv 0$ and $f \equiv 0$. We want to show that $u \equiv 0$. The proof would be easy if we could choose $\dot u$ as a test function in (9.59), but $\dot u(t)$ does not belong to V. For fixed s, set (we follow *J. Smoller*, [20])

$$v(t) = \begin{cases} \int_t^s u(r)\,dr & \text{if } 0 \le t \le s\\ 0 & \text{if } s \le t \le T.\end{cases}$$

We have $v(t) \in V$ for all $t \in [0,T]$, so that we may insert it into (9.59). We deduce

$$\int_0^s \left\{\langle \ddot u(t), v(t)\rangle_* + c^2(\nabla u(t), \nabla v(t))_0\right\} dt = 0. \tag{9.71}$$

An integration by parts yields,

$$\int_0^s \langle \ddot u(t), v(t)\rangle_*\, dt = -\int_0^s (\dot u(t), \dot v(t))_0 dt = \int_0^s (\dot u(t), u(t))_0 dt = \frac{1}{2}\int_0^s \frac{d}{dt}\,\|u(t)\|_0^2\, dt,$$

since $v(s) = \dot u(0) = 0$ and $\dot v(t) = -u(t)$, if $0 < t < s$. On the other hand,

$$\int_0^s (\nabla u(t), \nabla v(t))_0 dt = -\int_0^s (\nabla \dot v(t), \nabla v(t))_0 dt = -\frac{1}{2}\int_0^s \frac{d}{dt}\,\|\nabla v(t)\|_0^2\, dt.$$

Hence, from (9.71),

$$\int_0^s \frac{d}{dt}\left\{\|u(t)\|_0^2 - c^2\,\|\nabla v(t)\|_0^2\right\} dt = 0$$

or, since $u(0) = 0$ and $\nabla v(s) = \mathbf{0}$,

$$\|u(s)\|_0^2 + c^2 \|\nabla v(0)\|_0^2 = 0$$

which entails $u(s) \equiv 0$.

Stability. To prove the estimates for u and $\dot{u}$, use Proposition 7.98, p. 487, to pass to the limit as $m \to \infty$ in (9.64). The estimate for $\ddot{u}$ follows from the weak lower semicontinuity of the norm in $L^2(0,T;V^*)$. □

Problems

9.1 Consider the Abstract Parabolic Problem of Sect. 9.3, under the assumptions on p. 547. Show that $w(t) = e^{-\lambda_0 t} u(t)$, with $\lambda_0 > \lambda$, satisfies a similar problem with a coercive bilinear form.

9.2 Let $\{V, H, V^*\}$ be a Hilbert triplet. Let $\{u_m\}_{m\geq 1} \subset L^2(0,T;V)$ satisfy the following conditions:

a) $u_m \rightharpoonup u$ in $L^2(0,T;V)$.
b) $\dot{u}_m \rightharpoonup z$ in $L^2(0,T;V^*)$.

Prove that $z = \dot{u}$.

[*Hint.* Note that $\int_0^T \langle \dot{u}_m(t), v\rangle_* \varphi(t)dt = -\int_0^T (u_m(t), v)_H \dot{\varphi}(t)dt$ for all $\varphi \in \mathcal{D}(0,T)$ and all $v \in V$.]

9.3 Consider the problem

$$\begin{cases} u_t - (a(x)u_x)_x + b(x)u_x + c(x)u = f(x,t) & 0 < x < 1, 0 < t < T \\ u(x,0) = g(x), & 0 \leq x \leq 1 \\ u(0,t) = 0, u(1,t) = k(t). & 0 \leq t \leq T. \end{cases}$$

1. Reduce the problem to homogeneous Dirichlet conditions.
2. Write a weak formulation for the new problem.
3. Prove the well-posedness of the problem, under suitable hypotheses on the coefficients a, b, c and the data f, g, k. Write a stability estimate for the original u.

9.4 Consider the Cauchy-Neumann problem

$$\begin{cases} u_t - \Delta u = f & \text{in } \Omega \times (0,T) \\ \partial_{\boldsymbol{\nu}} u = 0, & \text{on } S_T \\ u(0) = g. & \text{in } \Omega \end{cases}$$

Assume that $f \in L^2(0,T;L^2(\Omega))$, $g \in H^1(\Omega)$.

a) Show that $u \in L^\infty(0,T;H^1(\Omega))$ and $\dot{u} \in L^2(0,T;L^2(\Omega))$.
b) If, in addition, Ω is a C^2-domain, show that $u \in L^2(0,T;H^2(\Omega))$.

9.5 Let

$$\Omega = \left\{\mathbf{x} \in \mathbb{R}^2 : x_1^2 + 4x_2^2 < 4\right\}, \Gamma_D = \partial\Omega \cap \{x_1 \geq 0\}, \Gamma_N = \partial\Omega \setminus \Gamma_D.$$

Consider the mixed problem

$$\begin{cases} u_t - \operatorname{div}(A_\alpha(\mathbf{x})\nabla u) + \mathbf{b}(\mathbf{x}) \cdot \nabla u - \alpha u = x_2 & \text{in } \Omega \times (0,T) \\ u(\mathbf{x},0) = \mathcal{H}(x_1) & \text{in } \Omega \\ u(\boldsymbol{\sigma},t) = 0 & \text{on } \Gamma_D \times [0,T] \\ A_\alpha(\boldsymbol{\sigma})\nabla u(\boldsymbol{\sigma},t) \cdot \boldsymbol{\nu}(\boldsymbol{\sigma}) = -\sigma_1 & \text{on } \Gamma_N \times [0,T] \end{cases}$$

where $\mathcal{H}$ is the Heaviside function and

$$A_\alpha(\mathbf{x}) = \begin{pmatrix} 1 & 0 \\ 0 & \alpha e^{|\mathbf{x}|^2} \end{pmatrix}, \quad \mathbf{b}(\mathbf{x}) = \begin{pmatrix} x_2 \\ x_2/|x_2| \end{pmatrix}.$$

Determine for which values of the real parameter α, the problem is uniformly parabolic. Give a weak formulation and analyze the well-posedness.

9.6 The potassium concentration $c = c(\mathbf{x},t)$, $\mathbf{x} = (x,y,z)$, in a cell of spherical shape Ω and radius R, satisfies the following evolution problem:

$$\begin{cases} c_t - \operatorname{div}(\mu \nabla c) - \sigma c = 0 & \text{in } \Omega \times (0,T) \\ \mu \nabla c \cdot \boldsymbol{\nu} + \chi c = c_{\text{ext}}, & \text{on } S_T \\ c(\mathbf{x},0) = c_0(\mathbf{x},0). & \text{in } \Omega \end{cases}$$

where: c_{ext} is a given external concentration, σ and χ are positive scalars and μ is strictly positive. Write a weak formulation and analyze the well-posedness, providing suitable assumptions on μ, c_{ext} and c_0.

9.7 State and prove an analogue of Theorem 9.11, p. 555, for the heat equation and homogeneous Robin/Neumann boundary conditions.

9.8 State and prove an analogue of the weak maximum principle in Theorem 9.19, p. 560, for the initial Robin/Neumann boundary value problem.

[*Hint.* Mimic the proof of Theorem 9.19. Observe that if $v \in H^1(\Omega)$ and $v \geq 0$ a.e. in Ω, then the trace of v on $\partial\Omega$ is nonnegative on $\partial\Omega$.]

9.9 Let Ω be a bounded Lipschitz domain and u be the weak solution of

$$\begin{cases} u_t - \Delta u = 0 & \text{in } Q_T \\ u = 0 & \text{on } S_T \\ u(\mathbf{x},0) = g(\mathbf{x}) & \text{in } \Omega. \end{cases}$$

a) Prove that, if $g \in L^\infty(\Omega)$, $\min\{0, \operatorname{ess\,inf}_\Omega g\} = u(\mathbf{x},t) \leq \max\{0, \operatorname{ess\,sup}_\Omega g\}$ in Q_T.
b) Prove that, if $g \in C(\overline{\Omega})$ with $g = 0$ on $\partial\Omega$, then $u \in C^\infty(Q_T) \cap C(\overline{Q}_T)$.

[*Hint.* b) Let $\{g_m\} \subset C_0^\infty(\Omega)$ such that $\|g_m - g\|_{L^\infty(\Omega)} \to 0$ as $m \to +\infty$. Denote by u_m the weak solution corresponding to the initial data g_m. Check that $u_m \in C^\infty(\overline{Q}_T)$. Show that

$$\|u_m(t) - u(t)\|_{L^\infty(\Omega)} \leq \|g_m - g\|_{L^\infty(\Omega)}$$

for all $t \in [0,T]$.]

9.10 Let Ω be a bounded, Lipschitz domain and u be the weak solution of

$$\begin{cases} u_t - \operatorname{div}(\mathbf{A}(\mathbf{x})\nabla u) + \mathbf{c}(\mathbf{x}) \cdot \nabla u + a(\mathbf{x})u = f(\mathbf{x},t) & \text{in } Q_T \\ u(\mathbf{x},0) = g(\mathbf{x}) & \text{in } \Omega \\ u = 0 & \text{on } S_T. \end{cases}$$

Assume that $f \in L^2(0,T;L^2(\Omega))$, $g \in H_0^1(\Omega)$ and that $\mathbf{A}$ is *symmetric*.

a) Show that $u \in L^\infty(0,T;H_0^1(\Omega))$, $\dot{u} \in L^2(0,T;L^2(\Omega))$ and prove a stability estimate similar to (9.41).
b) If Ω is of class C^2 and the coefficients a_{ij} are Lipschitz continuous in $\overline{\Omega}$, show that $u \in L^2(0,T;H^2(\Omega))$ and prove a stability estimate similar to (9.42).

9.11 Let $\Omega \subset \mathbb{R}^n$ be a bounded, Lipschitz domain, $Q_T = \Omega \times (0,T)$. For $k \geq 1$, let u_k be the unique weak solution of the following Cauchy-Dirichlet problem:

$$\begin{cases} \mathcal{P}_k u_k = \partial_t u_k - \operatorname{div}(\mathbf{A}_k(\mathbf{x},t)\nabla u_k) + \mathbf{c}_k(\mathbf{x},t)\cdot\nabla u_k + a_k(\mathbf{x},t)u_k = f & \text{in } Q_T \\ u_k = 0 & \text{on } S_T \\ u_k(\mathbf{x},0) = g(\mathbf{x}) & \text{in } \Omega \end{cases}$$

where $f \in L^2(0,T;H^{-1}(\Omega))$, $g \in L^2(\Omega)$ and $\mathcal{P}_k$ is a uniformly parabolic operator satisfying conditions (9.3) and (9.4), p. 542. Assume that, as $k \to +\infty$,

$$\mathbf{A}_k \to \mathbf{A}_0 \text{ in } L^\infty(Q_T;\mathbb{R}^{n^2}),\ \mathbf{c}_k \to \mathbf{c}_0 \text{ in } L^\infty(Q_T;\mathbb{R}^n),\ a_k \to a_0 \text{ in } L^\infty(Q_T).$$

Denote by u_0 the unique weak solution of the same problem for the limit operator $\mathcal{P}_0$. Show that $u_0 \to u$ in $L^2(0,T;H_0^1(\Omega))$ and in $C([0,T];L^2(\Omega))$, and $\dot{u}^k \to \dot{u}_0$ in $L^2(0,T;H^{-1}(\Omega))$.

9.12 Consider the Cauchy-Neumann problem

$$\begin{cases} u_{tt} - c^2\Delta u = f & \text{in } Q_T \\ \partial_\nu u(x) = 0 & \text{on } S_T \\ u(\mathbf{x},0) = g(\mathbf{x}),\ u_t(\mathbf{x},0) = h(\mathbf{x}) & \text{in } \Omega. \end{cases}$$

Write a weak formulation of the problem and prove the analogues of Theorems 9.25, p. 566, 9.26, p. 567, and 9.27, p. 568.

9.13 (*Concentrated reaction*) Consider the problem

$$\begin{cases} u_{tt} - u_{xx} + u(x,t)\,\delta(x) = 0 & -1 < x < 1,\ 0 < t < T \\ u(x,0) = g(x),\ u_t(x,0) = h(x) & -1 \leq x \leq 1 \\ u(-1,t) = u(1,t) = 0 & 0 \leq t \leq T, \end{cases}$$

where $\delta(x)$ denotes the Dirac δ at the origin.

a) Write a weak formulation for the problem.
b) Prove the well-posedness of the problem, under suitable hypotheses on g and h.

[*Hint.* a) Let $V = H_0^1(-1,1)$ and $H = L^2(-1,1)$. A weak formulation is: *find* $u \in L^2(-1,1;V)$, *with* $\dot{u} \in L^2(-1,1;H)$ *and* $\ddot{u} \in L^2(-1,1;V^*)$, *such that, for every* $v \in V$,

$$\langle \ddot{u}(t), v\rangle_* + (u_x(t), v_x) + u(0,t)v(0) = 0 \qquad \text{a.e. in } (0,T)$$

and $\|u(t) - g\|_H \to 0$, $\|\dot{u}(t) - h\|_{V^*} \to 0$ *as* $t \to 0$.]

9.14 Show that the two conditions (9.60) and (9.59), p. 563, are equivalent.

9.15 Let $u \in L^2(0,T;V)$, *with* $\dot{u} \in L^2(0,T;H)$ *and* $\ddot{u} \in L^2(0,T;V^*)$. Show that the function $t \mapsto w(t) = \langle \ddot{u}(t), v\rangle_*$ is a distribution in $\mathcal{D}'(0,T)$ and

$$w(t) = \frac{d^2}{dt^2}(u(t),v)_0 \quad \text{in } \mathcal{D}'(0,T).$$

Chapter 10
More Advanced Topics

10.1 The Linear Elastostatic System

The system of equations of linear elastostatics models the *small* deformations and displacements of a solid body, which at rest occupies a bounded domain $\Omega \subset \mathbb{R}^3$. We assume that Ω is regular (or polyhedral) and write $\partial\Omega = \Gamma_D \cup \Gamma_N$, $\Gamma_D \cap \Gamma_N = \emptyset$, with Γ_D of *positive surface measure*. Under the action of a body force $\mathbf{f}$ in Ω and of a traction force $\mathbf{h}$, acting on Γ_N, the body undergoes a deformation and each point moves from position $\mathbf{x}$ to $\mathbf{x} + \mathbf{u}(\mathbf{x})$. Here $\mathbf{f}$ and $\mathbf{h}$ are vector fields in $\mathbb{R}^3$. Given $\mathbf{f}$ and $\mathbf{h}$, our goal is to determine the unknown displacement $\mathbf{u}$, in an equilibrium situation. As usual we need general and constitutive laws for $\mathbf{u}$.

To this purpose, we introduce the *deformation tensor* given by

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2}(\nabla\mathbf{u} + (\nabla\mathbf{u})^\top) = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)_{i,j=1,2,3}$$

and the *stress tensor* $\mathbf{T}$, which takes values on the set of the symmetric 3×3 matrices. $\mathbf{T}$ describes the internal forces trying to return the body to equilibrium. These two tensors are related through the *Hooke law* of linear elasticity

$$\mathbf{T} = 2\mu\boldsymbol{\varepsilon}(\mathbf{u}) + \lambda(\operatorname{div}\mathbf{u})\mathbf{I}_3, \tag{10.1}$$

where μ, λ are the so called *Lamé coefficients* and $\mathbf{I}_3$ is the identity matrix in $\mathbb{R}^3$. Assuming that our body is homogeneous and isotropic with respect to the deformations, μ, λ are constants verifying the conditions

$$\mu > 0,\ 2\mu + 3\lambda > 0. \tag{10.2}$$

We add to the two constitutive laws (10.1), (10.2), the law of conservation of the linear momentum and the equation expressing the balance of the traction forces on the part Γ_N of the body surface. Moreover we assume that some part Γ_D of $\partial\Omega$, of *positive measure*, is kept fixed.

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Thus we are led to the mixed problem:

$$\begin{cases} \operatorname{div} \mathbf{T} + \mathbf{f} = \mathbf{0} & \text{in } \Omega \\ \mathbf{u} = \mathbf{0} & \text{on } \Gamma_D \\ \mathbf{T} \cdot \boldsymbol{\nu} = \mathbf{h} & \text{on } \Gamma_N . \end{cases} \tag{10.3}$$

Since $\operatorname{div} \mathbf{T} = \mu \Delta \mathbf{u} + (\mu + \lambda) \operatorname{grad} \operatorname{div} \mathbf{u}$, the first equation in (10.3) can be written in the equivalent form

$$-\mu \Delta \mathbf{u} - (\mu + \lambda) \operatorname{grad} \operatorname{div} \mathbf{u} = \mathbf{f},$$

known as the stationary *Navier equation*.

To solve Problem (10.3), we derive a variational formulation and analyze its well posedness. The boundary conditions suggest that the natural functional setting is given by the Sobolev space

$$V = H^1_{0,\Gamma_D}(\Omega; \mathbb{R}^3) = \left\{ \mathbf{v} \in H^1(\Omega; \mathbb{R}^3) : \mathbf{v} = \mathbf{0} \text{ on } \Gamma_D \right\}.$$

Since the Poincarè inequality holds in V, V is a Hilbert space with the norm

$$\|\mathbf{v}\|_V^2 = \int_\Omega |\nabla \mathbf{v}|^2 \, d\mathbf{x} = \sum_{i,j=1}^3 \int_\Omega \left(\frac{\partial v_i}{\partial x_j} \right)^2 d\mathbf{x}. \tag{10.4}$$

We now multiply the differential equation in (10.3) by $\mathbf{v} \in V$ (a "*virtual displacement*") and integrate by parts the first integral, using the boundary conditions. Recalling that $(\operatorname{div} \mathbf{T})_i = \sum_{j=1}^3 \frac{\partial T_{ij}}{\partial x_j}$, we find

$$\begin{aligned} 0 &= \int_\Omega (\operatorname{div} \mathbf{T} + \mathbf{f}) \cdot \mathbf{v} \, d\mathbf{x} = \int_\Omega \sum_{i,j=1}^3 \frac{\partial T_{ij}}{\partial x_j} v_i \, d\mathbf{x} + \int_\Omega \mathbf{f} \cdot \mathbf{v} \, d\mathbf{x} \\ &= - \int_\Omega \sum_{i,j=1}^3 T_{ij} \frac{\partial v_i}{\partial x_j} \, d\mathbf{x} + \int_{\Gamma_N} \mathbf{h} \cdot \mathbf{v} \, d\sigma + \int_\Omega \mathbf{f} \cdot \mathbf{v} \, d\mathbf{x}. \end{aligned}$$

From Hooke's law (10.1), we deduce

$$\sum_{i,j=1}^3 T_{ij} \frac{\partial v_i}{\partial x_j} = 2\mu \sum_{i,j=1}^3 \varepsilon_{ij}(\mathbf{u}) \frac{\partial v_i}{\partial x_j} + \lambda \operatorname{div} \mathbf{u} \operatorname{div} \mathbf{v}.$$

Moreover, given the symmetry of the deformation tensor, it follows that

$$\sum_{i,j=1}^3 \varepsilon_{ij}(\mathbf{u}) \frac{\partial v_i}{\partial x_j} = \sum_{i,j=1}^3 \varepsilon_{ij}(\mathbf{u}) \varepsilon_{ij}(\mathbf{v})$$

and therefore we are lead to the following variational formulation of the *linear elastostatics problem*:

Determine $\mathbf{u} \in V$ *such that*

$$\int_\Omega [2\mu \sum_{i,j=1}^{3} \varepsilon_{ij}(\mathbf{u})\varepsilon_{ij}(\mathbf{v}) + \lambda \operatorname{div} \mathbf{u} \operatorname{div} \mathbf{v}]\, d\mathbf{x} = \int_{\Gamma_N} \mathbf{h} \cdot \mathbf{v}\, d\sigma + \int_\Omega \mathbf{f} \cdot \mathbf{v}\, d\mathbf{x} \quad (10.5)$$

for every $\mathbf{v} \in V$.

It is not difficult to check that, under regularity assumptions, the weak formulation is equivalent to (10.3).

We will need the following important inequality, which gives a control of the full L^2-norm of $\mathbf{v}$ by the L^2-norm of its symmetric part $\boldsymbol{\varepsilon}(\mathbf{v})$.

Lemma 10.1 (Korn's inequality) *Let* Ω *be a regular domain or a polyhedron. There exists a constant* $\gamma > 0$ *depending on* Ω*, such that*

$$\int_\Omega \{ \sum_{i,j=1}^{3} \varepsilon_{ij}^2(\mathbf{v}) + |\mathbf{v}|^2 \} d\mathbf{x} \geq \gamma \, \|\mathbf{v}\|^2_{H^1(\Omega;\mathbb{R}^3)}, \qquad \forall \mathbf{v} \in H^1(\Omega;\mathbb{R}^3).$$

For the proof, see e.g. *R. Dautray, J.-L. Lions*, [27], vol. 2.

The following theorem holds.

Theorem 10.2 *Let* $\mathbf{f} \in L^2(\Omega;\mathbb{R}^3)$ *and* $\mathbf{h} \in L^2(\Gamma_N;\mathbb{R}^3)$. *If* Γ_D *has positive surface measure, then the variational elastostatic problem has a unique solution* $\mathbf{u} \in V$. *Moreover*

$$\|\mathbf{u}\|_V \leq \frac{C}{\mu} \left\{ \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^3)} + \|\mathbf{h}\|_{L^2(\Gamma_N;\mathbb{R}^3)} \right\}.$$

Proof We use Lax-Milgram Theorem 6.42, p. 376. Define

$$B(\mathbf{u},\mathbf{v}) = \int_\Omega [2\mu \sum_{i,j=1}^{3} \varepsilon_{ij}(\mathbf{u})\, \varepsilon_{ij}(\mathbf{v}) + \lambda \operatorname{div} \mathbf{u} \operatorname{div} \mathbf{v}]\, d\mathbf{x}$$

and

$$F\mathbf{v} = \int_{\Gamma_N} \mathbf{h} \cdot \mathbf{v}\, d\sigma + \int_\Omega \mathbf{f} \cdot \mathbf{v}\, d\mathbf{x}.$$

Then we want to determine $\mathbf{u} \in V$ such that

$$B(\mathbf{u},\mathbf{v}) = F\mathbf{v}, \qquad \text{for every } \mathbf{v} \in V.$$

Using the Schwarz and Poincaré inequalities and the trace Theorem 7.80, p. 473, it is easy to check that $F \in V^*$.

Also the continuity of the bilinear form B in $V \times V$ follows from a simple application of the Schwarz and Poincaré inequalities. We leave the details to the reader. The difficulty is to show the coercivity of B. Since

$$B(\mathbf{v},\mathbf{v}) = \int_\Omega [2\mu \sum_{i,j=1}^{3} \varepsilon_{ij}(\mathbf{v})\, \varepsilon_{ij}(\mathbf{v}) + \lambda (\operatorname{div} \mathbf{v})^2] d\mathbf{x} \geq \int_\Omega [2\mu \sum_{i,j=1}^{3} \varepsilon_{ij}(\mathbf{v})\, \varepsilon_{ij}(\mathbf{v})] d\mathbf{x},$$

the coercivity of B follows if we prove that there exists $\theta > 0$ such that

$$\sum_{i,j=1}^{3} \int_\Omega \varepsilon_{ij}^2(\mathbf{v})\, d\mathbf{x} \geq \theta \|\mathbf{v}\|_V^2, \quad \forall \mathbf{v} \in V. \tag{10.6}$$

We use a contradiction argument. Define

$$\|\boldsymbol{\varepsilon}(\mathbf{v})\|_d^2 = \sum_{i,j=1}^{3} \int_\Omega \varepsilon_{ij}^2(\mathbf{v})\, d\mathbf{x}$$

and suppose (10.6) is not true. Then, for every integer $n > 0$, we can find $\mathbf{v}_n$ such that

$$\|\boldsymbol{\varepsilon}(\mathbf{v}_n)\|_d < \frac{1}{n} \|\mathbf{v}_n\|_V .$$

Normalize the sequence $\{\mathbf{v}_n\}$ by setting

$$\mathbf{w}_n = \frac{\mathbf{v}_n}{\|\mathbf{v}_n\|_{L^2(\Omega;\mathbb{R}^3)}}.$$

Still we have

$$\|\boldsymbol{\varepsilon}(\mathbf{w}_n)\|_d < \frac{1}{n} \|\mathbf{w}_n\|_V , \quad \text{for all } n \geq 1.$$

By Lemma 10.1, we deduce

$$\|\mathbf{w}_n\|_V^2 \leq \gamma^{-1}(\|\boldsymbol{\varepsilon}(\mathbf{w}_n)\|_d^2 + \|\mathbf{w}_n\|_{L^2(\Omega;\mathbb{R}^3)}^2) \leq \gamma^{-1}\left(\frac{1}{n^2}\|\mathbf{w}_n\|_V^2 + 1\right)$$

and, for $n^2 > 2/\gamma$, we get

$$\frac{1}{2}\|\mathbf{w}_n\|_V^2 \leq \gamma^{-1}.$$

Thus the sequence $\{\mathbf{w}_n\}$ is bounded in V. Moreover, as $n \to +\infty$,

$$\|\mathbf{w}_n\|_{L^2(\Omega;\mathbb{R}^3)} = 1 \quad \text{and} \quad \|\boldsymbol{\varepsilon}(\mathbf{w}_n)\|_d \to 0.$$

By Rellich Theorem 7.86, p. 478, there exists a subsequence, that we still call $\{\mathbf{w}_n\}$, such that

$$\mathbf{w}_n \rightharpoonup \mathbf{w}, \quad \text{in } V,$$

and

$$\mathbf{w}_n \to \mathbf{w}, \quad \text{in } L^2(\Omega;\mathbb{R}^3).$$

By the weak lower semicontinuity of a norm (see Theorem 6.59, p. 388) we can write

$$\|\boldsymbol{\varepsilon}(\mathbf{w})\|_d \leq \lim \inf_{n\to+\infty} \|\boldsymbol{\varepsilon}(\mathbf{w}_n)\|_d = 0,$$

from which $\boldsymbol{\varepsilon}(\mathbf{w}) = \mathbf{0}$ a.e. in Ω. By a classical result[1] in the theory of rigid bodies, $\boldsymbol{\varepsilon}(\mathbf{w}) = \mathbf{0}$ if and only if there exist $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$ such that

$$\mathbf{w}(\mathbf{x}) = \mathbf{a} + \mathbf{b} \times \mathbf{x} \qquad \text{for all } \mathbf{x} \in \Omega.$$

Thus $\mathbf{w}$ is an infinitesimal rigid body displacement. Now $|\Gamma_D| > 0$, and since $\mathbf{w} = \mathbf{0}$ on Γ_D, we infer that $\mathbf{a} = \mathbf{b} = \mathbf{0}$, contradicting

$$1 = \|\mathbf{w}_n\|_{L^2(\Omega;\mathbb{R}^3)} \to \|\mathbf{w}\|_{L^2(\Omega;\mathbb{R}^3)} . \qquad \square$$

[1] See e.g. *J. Nečas, I. Hlaváček*, Mathematical Theory of Elastic and Elasto-Plastic Bodies: an Introduction, Vol. 3. Studies in Applied Mechanics, Elsevier 1981.

Remark 10.3 As in Remark 8.2, p. 501, the variational formulation (10.5) corresponds to the *principle of virtual work* in Mechanics. In fact it expresses the balance between the works done by the internal elastic forces (given by $B(\mathbf{u}, \mathbf{v})$) and by the external forces (given by $F\mathbf{v}$) due to the *virtual displacement* $\mathbf{v}$. Moreover, given the symmetry of the bilinear form B, the solution $\mathbf{u}$ minimizes the energy

$$E(\mathbf{v}) = \frac{1}{2}\int_\Omega [2\mu \sum_{i,j=1}^{3} \varepsilon_{ij}^2(\mathbf{v}) + \lambda(\operatorname{div}\mathbf{v})^2]d\mathbf{x} - \int_{\Gamma_N} \mathbf{h}\cdot\mathbf{v}\, d\sigma + \int_\Omega \mathbf{f}\cdot\mathbf{v}\, d\mathbf{x}$$

among all the admissible virtual displacements. Accordingly, the variational formulation coincides with the Euler equation of the energy functional E.

Remark 10.4 If $\Gamma_D = \partial\Omega$, that is when $\mathbf{v} \in H_0^1(\Omega;\mathbb{R}^3)$, the proof of Korn's inequality is not difficult. In fact, first observe that we can write

$$\sum_{i,j=1}^{3} \varepsilon_{ij}^2(\mathbf{v}) = \sum_{i,j=1}^{3} \frac{1}{4}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)^2 = \frac{1}{2}\sum_{i,j=1}^{3}\left[\left(\frac{\partial v_i}{\partial x_j}\right)^2 + \frac{\partial v_i}{\partial x_j}\frac{\partial v_j}{\partial x_i}\right]. \tag{10.7}$$

Integrating by parts twice (recalling that $\mathbf{v}$ vanishes on $\partial\Omega$), we find

$$\int_\Omega \frac{\partial v_i}{\partial x_j}\frac{\partial v_j}{\partial x_i}d\mathbf{x} = -\int_\Omega \frac{\partial^2 v_i}{\partial x_i \partial x_j} v_j\, d\mathbf{x} = \int_\Omega \frac{\partial v_i}{\partial x_i}\frac{\partial v_j}{\partial x_j}\, d\mathbf{x}.$$

Thus

$$\sum_{i,j=1}^{3}\int_\Omega \frac{\partial v_i}{\partial x_j}\frac{\partial v_j}{\partial x_i}d\mathbf{x} = \int_\Omega (\operatorname{div}\mathbf{v})^2 d\mathbf{x}$$

whence, recalling (10.4), Korn's inequality easily follows.

Incidentally, observe that

$$|\operatorname{curl}\mathbf{v}|^2 = \sum_{i,j=1, i\neq j}^{3}\left[\left(\frac{\partial v_i}{\partial x_j}\right)^2 - \frac{\partial v_i}{\partial x_j}\frac{\partial v_j}{\partial x_i}\right].$$

Therefore, from (10.7), by adding and subtracting

$$\frac{1}{2}\sum_{i,j=1,\ i\neq j}^{3}\frac{\partial v_i}{\partial x_j}\frac{\partial v_j}{\partial x_i},$$

we infer

$$\int_\Omega \sum_{i,j=1}^{3}\varepsilon_{ij}^2(\mathbf{v})d\mathbf{x} = \frac{1}{2}\int_\Omega |\operatorname{curl}\mathbf{v}|^2\, d\mathbf{x} + \int_\Omega (\operatorname{div}\mathbf{v})^2 d\mathbf{x}$$

from which the remarkable formula

$$\int_\Omega |\nabla\mathbf{v}|^2\, d\mathbf{x} = \int_\Omega |\operatorname{curl}\mathbf{v}|^2\, d\mathbf{x} + \int_\Omega (\operatorname{div}\mathbf{v})^2 d\mathbf{x}.$$

10.2 The Stokes System

We have seen, in Sect. 3.6.3, that the Navier-Stokes equations

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla)\mathbf{u} = \nu \Delta \mathbf{u} - \frac{1}{\rho}\nabla p + \frac{1}{\rho}\mathbf{f} \tag{10.8}$$

$$\operatorname{div} \mathbf{u} = 0 \tag{10.9}$$

express, respectively, the balance of linear momentum and the incompressibility for the motion of a homogeneous, viscous fluid. Here $\mathbf{u}$ represents the fluid velocity, p its pressure, ρ its density (constant), while $\mathbf{f}$ is a body force per unit volume and $\nu = \mu/\rho$ is the *kinematic viscosity* (constant). Note that the pressure appears in (10.8) only through its gradient so that it is determined up to an additive constant, as it should be.

We add to the equations (10.8), (10.9) an *initial condition*

$$\mathbf{u}(0) = \mathbf{g}$$

and conditions on the boundary of a (bounded) domain $\Omega \subset \mathbb{R}^n$ ($n = 2, 3$), where the fluid is confined. Typically, the observation of real fluids reveals that the normal and tangential components of their velocity vector on a rigid wall must agree with those of the wall itself. Thus if the wall is at rest we have $\mathbf{u} = \mathbf{0}$ (*no slip condition*).

In order to emphasize the role of the convective term $(\mathbf{u} \cdot \nabla)\mathbf{u}$, we pass to dimensionless variables by choosing a reference length L related to Ω, for instance $L =$ diameter of Ω or $L = |\Omega|^{1/n}$, and a reference velocity $U = \|\mathbf{g}\|_{L^\infty(\Omega;\mathbb{R}^n)}$.

Then, we rescale the quantities in our problem in the following way[2]:

$$\mathbf{y} = \frac{\mathbf{x}}{L}, \quad \tau = \frac{\nu t}{L^2}, \quad \mathbf{v} = \frac{\mathbf{u}}{U}, \quad P = \frac{pL}{\rho \nu U}, \quad \mathbf{F} = \frac{L^2}{\nu U}\mathbf{f}.$$

After elementary calculations, equation (10.9) remains unaltered while (10.8) takes the following dimensionless form:

$$\frac{\partial \mathbf{v}}{\partial \tau} + \mathcal{R}(\mathbf{v} \cdot \nabla)\mathbf{v} = \Delta \mathbf{v} - \nabla P + \mathbf{F} \tag{10.10}$$

in the rescaled domain $\Omega' = \{\mathbf{y} : L\mathbf{y} \in \Omega\}$, where the only parameter left is

$$\mathcal{R} = \frac{UL}{\nu},$$

known as *Reynolds number*. Low (high) Reynolds number means high (low) viscosity or motion weakly (strongly) affected by the initial data. As $\mathcal{R} \to 0$ the equation (10.10) linearizes into the diffusion equation for the velocity field. In a stationary regime we are led to the following *Stokes system*:

[2] Note that the physical dimensions of ν are $[\text{length}]^2 \times [\text{time}]^{-1}$.

$$\begin{cases} -\Delta \mathbf{v} = \mathbf{F} - \nabla P & \text{in } \Omega' \\ \operatorname{div} \mathbf{v} = 0 & \text{in } \Omega' \\ \mathbf{v} = \mathbf{0} & \text{on } \partial\Omega'. \end{cases} \tag{10.11}$$

We want to analyze the well posedness of problem (10.11) by first deriving a variational formulation. Thus, take a test function $\boldsymbol{\varphi} \in C_0^\infty(\Omega'; \mathbb{R}^n)$, multiply by $\boldsymbol{\varphi}$ the first equation in (10.11) and integrate over Ω'. We find:

$$\int_{\Omega'} -\Delta \mathbf{v} \cdot \boldsymbol{\varphi} \, d\mathbf{x} = \int_{\Omega'} \mathbf{F} \cdot \boldsymbol{\varphi} \, d\mathbf{x} - \int_{\Omega'} \nabla P \cdot \boldsymbol{\varphi} \, d\mathbf{x}.$$

After an integration by parts in the first and in the third term we get:

$$\int_{\Omega'} \nabla \mathbf{v} : \nabla \boldsymbol{\varphi} \, d\mathbf{x} = \int_{\Omega'} \mathbf{F} \cdot \boldsymbol{\varphi} \, d\mathbf{x} + \int_{\Omega'} P \operatorname{div} \boldsymbol{\varphi} \, d\mathbf{x}, \qquad \forall \boldsymbol{\varphi} \in C_0^\infty(\Omega'; \mathbb{R}^n) \tag{10.12}$$

where

$$\nabla \mathbf{v} : \nabla \boldsymbol{\varphi} = \sum_{i,j=1}^{n} \frac{\partial u_i}{\partial x_j} \frac{\partial v_i}{\partial x_j}.$$

Now, multiply the equation $\operatorname{div} \mathbf{v} = 0$ by $q \in L^2(\Omega')$ and integrate over Ω'; we get:

$$\int_{\Omega'} q \operatorname{div} \mathbf{v} \, d\mathbf{x} = 0, \qquad \forall q \in L^2(\Omega'). \tag{10.13}$$

Viceversa, under regularity assumptions, if $\mathbf{v} = 0$ on $\partial\Omega'$ and it satisfies (10.12), going back with the integration by parts we find

$$\int_{\Omega} (-\Delta \mathbf{v} - \mathbf{F} + \nabla P) \cdot \mathbf{v} \, d\mathbf{x} = 0, \qquad \forall \mathbf{v} \in C_0^\infty(\Omega'; \mathbb{R}^n)$$

which implies, by a usual density argument,

$$-\Delta \mathbf{v} - \mathbf{F} + \nabla P = \mathbf{0} \ \text{ in } \Omega'.$$

Moreover, from (10.13) we deduce

$$\operatorname{div} \mathbf{v} \ = 0 \ \text{ in } \Omega'.$$

Thus under regularity assumptions, for functions vanishing on $\partial\Omega$, the Stokes system (10.11) is equivalent to the system of the two equations (10.12), (10.13).

These two equations and the homogeneous boundary condition suggest that the natural functional setting is the space $H_0^1(\Omega'; \mathbb{R}^n)$. Observe that, by density, the equation (10.12) holds for every $\boldsymbol{\varphi} \in H_0^1(\Omega'; \mathbb{R}^n)$. To select uniquely the pressure, introduce the Hilbert space[3]

[3] Other normalization of the pressure are possible and sometimes more convenient from a numerical point of view.

$$Q = \left\{ q \in L^2(\Omega') : \int_\Omega q = 0 \right\}.$$

Finally, let $\mathbf{F} \in L^2(\Omega'; \mathbb{R}^n)$.

Definition 10.5 A *variational solution* of the problem (10.11) is a pair $(\mathbf{v}, P)$ such that:

$$\begin{cases} \mathbf{v} \in H_0^1(\Omega'; \mathbb{R}^n),\ P \in Q \\ \int_{\Omega'} \nabla \mathbf{v} : \nabla \boldsymbol{\varphi}\, d\mathbf{x} = \int_{\Omega'} \mathbf{F} \cdot \boldsymbol{\varphi}\, d\mathbf{x} + \int_{\Omega'} P \operatorname{div} \boldsymbol{\varphi}\, d\mathbf{x} & \forall \boldsymbol{\varphi} \in H_0^1(\Omega'; \mathbb{R}^n) \\ \int_{\Omega'} q \operatorname{div} \mathbf{v}\, d\mathbf{x} = 0 & \forall q \in Q. \end{cases} \tag{10.14}$$

We want to show existence, uniqueness and stability of a weak solution. However, a direct use of the Lax-Milgram Theorem is prevented by the presence of the unknown pressure P. A way to overcome this difficulty is to choose, instead of $H_0^1(\Omega'; \mathbb{R}^n)$, its closed subspace V_{div}, whose elements are the vectors with *vanishing divergence*. With the norm

$$\|\mathbf{v}\|_{V_{\text{div}}}^2 = \int_{\Omega'} |\nabla \mathbf{v}|^2\, d\mathbf{x} = \sum_{i,j=1}^{n} \int_{\Omega'} \nabla \mathbf{v} : \nabla \mathbf{v},$$

V_{div} is a Hilbert space and if we choose $\boldsymbol{\varphi} \in V_{\text{div}}$, the second equation in (10.14) becomes

$$\int_{\Omega'} \nabla \mathbf{v} : \nabla \boldsymbol{\varphi}\, d\mathbf{x} = \int_{\Omega'} \mathbf{F} \cdot \boldsymbol{\varphi}\, d\mathbf{x}, \qquad \forall \boldsymbol{\varphi} \in V_{\text{div}}. \tag{10.15}$$

The bilinear form

$$B(\mathbf{v}, \boldsymbol{\varphi}) = \int_{\Omega'} \nabla \mathbf{v} : \nabla \boldsymbol{\varphi}\, d\mathbf{x}$$

is clearly bounded and coercive in V_{div}. Moreover, $\mathbf{F} \in (V_{\text{div}})^*$ since $\mathbf{F} \in L^2(\Omega'; \mathbb{R}^n)$. The Lax-Milgram Theorem gives a unique solution $\overline{\mathbf{v}} \in V_{\text{div}}$, that also incorporates the incompressibility condition $\operatorname{div} \mathbf{v} = 0$. Moreover, using also Poincaré's inequality,

$$\|\overline{\mathbf{v}}\|_{\text{div}} \le C_P \|\mathbf{F}\|_{L^2(\Omega'; \mathbb{R}^n)}.$$

Still we have to determine the pressure. To this purpose, note that the equation (10.15) asserts that the vector

$$\mathbf{g} = \Delta \overline{\mathbf{v}} + \mathbf{F},$$

which belongs to $H^{-1}(\Omega'; \mathbb{R}^n)$, satisfies the equation

$$\langle \mathbf{g}, \boldsymbol{\varphi} \rangle_{H^{-1}, H_0^1} = 0, \quad \forall \boldsymbol{\varphi} \in V_{\text{div}}.$$

In other words, the functional $\mathbf{g}$ vanishes over V_{div}. It turns out that the elements of $H^{-1}(\Omega'; \mathbb{R}^n)$ that vanish over V_{div} have a special form, as indicated in the following lemma.

Lemma 10.6 *Let $\Omega' \subset \mathbb{R}^n$ be a bounded Lipschitz domain. A functional $\mathbf{g} \in H^{-1}(\Omega'; \mathbb{R}^n)$ satisfies the condition*

$$\langle \mathbf{g}, \boldsymbol{\varphi} \rangle_{H^{-1}, H_0^1} = 0, \qquad \forall \boldsymbol{\varphi} \in V_{\text{div}} \tag{10.16}$$

if and only if there exists $P \in L^2(\Omega')$ *such that*

$$-\nabla P = \mathbf{g}.$$

The function P *is unique up to an additive constant.*

Justification The proof is rather complex (see *G. P. Galdi*, [4]). We only give a formal argument to support the plausibility of the result. Let us work in dimension $n = 3$, assuming that $\mathbf{g}$ is smooth and that Ω is, say, a cube. First note that if $\mathbf{V} \in \mathcal{D}(\Omega'; \mathbb{R}^3)$,

$$\operatorname{div} \operatorname{curl} \mathbf{V} = \mathbf{0},$$

and therefore (10.16) holds for

$$\boldsymbol{\varphi} = \operatorname{curl} \mathbf{V}.$$

From Gauss formula 8 in Appendix B we have

$$0 = \int_{\Omega'} \mathbf{g} \cdot \operatorname{curl} \mathbf{V} \, d\mathbf{x} = \int_{\Omega'} \operatorname{curl} \mathbf{g} \cdot \mathbf{V} \, d\mathbf{x} - \int_{\partial\Omega'} (\mathbf{g} \times \mathbf{V}) \cdot \boldsymbol{\nu} \, d\sigma = \int_{\Omega'} \operatorname{curl} \mathbf{g} \cdot \mathbf{V} \, d\mathbf{x}.$$

Since $\mathbf{V}$ is arbitrary, we infer that

$$\operatorname{curl} \mathbf{g} = \mathbf{0}.$$

Thus, there exists a scalar potential P such that $-\nabla P = \mathbf{g}$. □

Thanks to Lemma 10.6, there exists a unique $\overline{P} \in Q$ such that

$$\mathbf{g} = \Delta \overline{\mathbf{v}} + \mathbf{F} = -\nabla \overline{P}.$$

Hence, $(\overline{\mathbf{v}}, \overline{P})$ is the unique solution of problem (10.14). We have proved the following result:

Theorem 10.7 *Let* $\Omega' \subset \mathbb{R}^3$ *be a bounded, Lipschitz domain and* $\mathbf{F} \in L^2(\Omega'; \mathbb{R}^n)$. *The problem* (10.11) *has a unique weak solution* $(\overline{\mathbf{v}}, \overline{P})$, *with* $\overline{\mathbf{v}} \in H_0^1(\Omega'; \mathbb{R}^n)$ *and* $\overline{P} \in Q$.

Remark 10.8 (the pressure as a multiplier) Given the symmetry of the bilinear form in (10.15), the solution $\overline{\mathbf{v}}$ is a minimizer in V_{div} of the energy functional

$$E(\mathbf{v}) = \frac{1}{2} \int_{\Omega'} \left\{ |\nabla \mathbf{v}|^2 - 2\mathbf{F} \cdot \mathbf{v} \right\} d\mathbf{x}.$$

Equivalently, $\overline{\mathbf{v}}$ minimizes $E(\mathbf{v})$ in all $H_0^1(\Omega'; \mathbb{R}^n)$ under the constraint $\operatorname{div} \mathbf{v} = 0$. Introducing a multiplier $q \in Q$ and the Lagrangian

$$\mathcal{L}(\mathbf{v}, q) = \int_{\Omega'} \left\{ \frac{1}{2} |\nabla \mathbf{v}|^2 - \mathbf{F} \cdot \mathbf{v} - q \operatorname{div} \mathbf{v} \right\} d\mathbf{x},$$

the system (10.14) expresses a necessary optimality condition (Euler-Lagrange equation). Thus the pressure P appears as a Lagrange multiplier associated to the solenoidality constraint $\operatorname{div} \mathbf{v} = 0$.

10.3 The Stationary Navier Stokes Equations

In this section we examine a Dirichlet problem for the stationary Navier-Stokes equations. We use the Schauder Theorem 6.88, p. 413, to show existence and the contraction mapping Theorem 6.83, p. 410, to show uniqueness for small data.

10.3.1 Weak formulation

Let $\Omega \subset \mathbb{R}^n$ $(n = 2, 3)$ be a bounded domain and consider the Navier-Stokes system, (we assume $\rho = 1$):

$$\begin{cases} -\nu\Delta\mathbf{u} + (\mathbf{u}\cdot\nabla)\mathbf{u} = -\nabla p + \mathbf{f} & \text{in } \Omega \\ \operatorname{div}\mathbf{u} = 0 & \text{in } \Omega \\ \mathbf{u} = \mathbf{0} & \text{on } \partial\Omega. \end{cases} \tag{10.17}$$

Keeping the same notations of the last section, a natural weak formulation of problem (10.17) is the following.

Definition 10.9 A *weak* (or *variational*) *solution* of problem (10.17) is a pair $(\mathbf{u}, p)$ such that $\mathbf{u} \in H_0^1(\Omega;\mathbb{R}^n)$, $p \in Q$ and

$$\int_\Omega \{\nu\nabla\mathbf{u} : \nabla\mathbf{v} + (\mathbf{u}\cdot\nabla)\mathbf{u}\cdot\mathbf{v}\}\, d\mathbf{x} = \int_\Omega \{\mathbf{f}\cdot\mathbf{v} + p\operatorname{div}\mathbf{v}\}\, d\mathbf{x}, \quad \forall\mathbf{v} \in H_0^1(\Omega;\mathbb{R}^n),$$

$$\int_\Omega q\operatorname{div}\mathbf{u}\, d\mathbf{x} = 0, \qquad \forall q \in Q.$$

The presence of the nonlinear term introduces nontrivial difficulties and, in general, we cannot expect uniqueness, unless $\|\mathbf{f}\|_{L^2(\Omega:\mathbb{R}^n)}$ is small or ν is large enough (see *G. P. Galdi*, [4]).

10.3.2 Existence, and uniqueness for small data

We first examine the trilinear form

$$b(\mathbf{u},\mathbf{w},\mathbf{v}) = \int_\Omega (\mathbf{u}\cdot\nabla)\mathbf{w}\cdot\mathbf{v}\, d\mathbf{x}, \quad \text{for } \mathbf{u},\mathbf{w},\mathbf{v} \in V_{\operatorname{div}}.$$

Since $\operatorname{div}\mathbf{u} = 0$ and $\mathbf{w}, \mathbf{v}$ vanish on $\partial\Omega$, we have:

$$b(\mathbf{u},\mathbf{w},\mathbf{v}) + b(\mathbf{u},\mathbf{v},\mathbf{w}) = \int_\Omega \sum_{i,j=1}^n \left[u_i \frac{\partial w_j}{\partial x_i} v_j + u_i \frac{\partial v_j}{\partial x_i} w_j\right] d\mathbf{x}$$

$$= \int_\Omega \mathbf{u} \cdot \nabla(\mathbf{w} \cdot \mathbf{v})\, d\mathbf{x} = -\int_\Omega (\mathbf{w} \cdot \mathbf{v}) \operatorname{div} \mathbf{u}\, d\mathbf{x} = 0$$

so that

$$b(\mathbf{u}, \mathbf{w}, \mathbf{v}) = -b(\mathbf{u}, \mathbf{v}, \mathbf{w}). \tag{10.18}$$

In particular

$$b(\mathbf{u}, \mathbf{w}, \mathbf{w}) = 0. \tag{10.19}$$

Now

$$|(\mathbf{u} \cdot \nabla)\mathbf{w} \cdot \mathbf{v}| \le \sum_{i,j=1}^{n} \left| u_i \frac{\partial w_j}{\partial x_i} v_j \right| \le |\mathbf{u}|\, |\nabla \mathbf{w}|\, |\mathbf{v}|$$

and, since $\frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$, we can write[4]

$$\int_\Omega |\mathbf{u}|\, |\nabla \mathbf{w}|\, |\mathbf{v}|\ d\mathbf{x} \le \|\mathbf{u}\|_{L^4(\Omega;\mathbb{R}^n)}\, \|\mathbf{w}\|_{V_{\text{div}}}\, \|\mathbf{v}\|_{L^4(\Omega;\mathbb{R}^n)}\,.$$

Thus

$$|b(\mathbf{u}, \mathbf{w}, \mathbf{v})| \le \|\mathbf{u}\|_{L^4(\Omega;\mathbb{R}^n)}\, \|\mathbf{w}\|_{V_{\text{div}}}\, \|\mathbf{v}\|_{L^4(\Omega;\mathbb{R}^n)}. \tag{10.20}$$

From the embedding Theorem 7.92, p. 483, we have, if $n = 2$, $H_0^1(\Omega) \subset L^s(\Omega)$ for every $s \ge 2$, with compact embedding. If $n = 3$, $H_0^1(\Omega) \subset L^s(\Omega)$ for every $s \in [2, 6]$, with compact embedding whenever $s < 6$. Moreover the following estimate

$$\|\mathbf{u}\|_{L^s(\Omega;\mathbb{R}^n)} \le \tilde{C}_s\, \|\mathbf{u}\|_{V_{\text{div}}} \tag{10.21}$$

holds in all cases, with $\tilde{C}_s = \tilde{C}(s, \Omega)$. Then we may also write:

$$|b(\mathbf{u}, \mathbf{w}, \mathbf{v})| \le \tilde{C}_4^2\, \|\mathbf{u}\|_{V_{\text{div}}}\, \|\mathbf{w}\|_{V_{\text{div}}}\, \|\mathbf{v}\|_{V_{\text{div}}}\,. \tag{10.22}$$

We can now prove the following result.

Theorem 10.10 *Let $\Omega \subset \mathbb{R}^n$ ($n = 2, 3$) be a bounded, Lipschitz domain and $\mathbf{f} \in L^2(\Omega; \mathbb{R}^n)$. Then there exists a weak solution $(\mathbf{u}, p)$ of problem (10.17).*

Moreover, if $\tilde{C}_4$ is the constant in the embedding inequality (10.21) for $s = 4$ and

$$\frac{C_P \tilde{C}_4^2}{\nu^2}\, \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)} < 1, \tag{10.23}$$

then, the solution is unique.

Proof We split it in several steps.

1. *Pressure elimination*. As in the case of Stokes equation, we incorporate the incompressibility condition $\operatorname{div} \mathbf{u} = 0$, by solving in V_{div} the problem

$$\nu \int_\Omega \nabla \mathbf{u} : \nabla \mathbf{v}\, d\mathbf{x} = \int_\Omega \mathbf{f} \cdot \mathbf{v}\, d\mathbf{x} - \int_\Omega (\mathbf{u} \cdot \nabla)\mathbf{u} \cdot \mathbf{v}\, d\mathbf{x}, \qquad \forall \mathbf{v} \in V_{\text{div}}. \tag{10.24}$$

[4] We use the generalized Hölder inequality: $\left| \int_\Omega fgh\, d\mathbf{x} \right| \le \|f\|_{L^p}\, \|g\|_{L^q}\, \|h\|_{L^r}$ with $p, q, r > 1$, $\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = 1$.

Once $\mathbf{u}$ is determined, we recover the pressure $p \in L^2(\Omega)$ using Lemma 10.6.

2. *Reduction to a fixed point problem.* Write (10.24) in the form

$$\nu \int_\Omega \nabla \mathbf{u} : \nabla \mathbf{v} \, d\mathbf{x} + b(\mathbf{u}, \mathbf{u}, \mathbf{v}) = \int_\Omega \mathbf{f} \cdot \mathbf{v} \, d\mathbf{x}.$$

Now fix $\mathbf{w} \in V_{\text{div}}$ and consider the *linear* problem

$$a(\mathbf{u}, \mathbf{v}) = F\mathbf{v}, \qquad \forall \mathbf{v} \in V_{\text{div}}, \tag{10.25}$$

where

$$a(\mathbf{u}, \mathbf{v}) = \nu \int_\Omega \nabla \mathbf{u} : \nabla \mathbf{v} \, d\mathbf{x} + b(\mathbf{w}, \mathbf{u}, \mathbf{v}) \qquad \text{and} \qquad F\mathbf{v} = \int_\Omega \mathbf{f} \cdot \mathbf{v} \, d\mathbf{x}.$$

We shall use the Lax-Milgram Theorem to prove that, for every $\mathbf{w} \in V_{\text{div}}$, the problem (10.25) has a unique solution

$$\mathbf{u} = T(\mathbf{w}) \in V_{\text{div}}.$$

Now, any function $\mathbf{u}^* \in V_{\text{div}}$ is a solution of the equation (10.24) if and only if $\mathbf{u}^*$ is a *fixed point* of the map $T : V_{\text{div}} \to V_{\text{div}}$, that is such that $T(\mathbf{u}^*) = \mathbf{u}^*$. Thus to show existence, we must show that T has a fixed point and to show uniqueness we must show that such fixed point is unique.

3. *Solution of the linear problem (10.25).* The bilinear form $a(\mathbf{u}, \mathbf{v})$ is continuous in V_{div} (from (10.22)) and coercive, since $b(\mathbf{w}, \mathbf{v}, \mathbf{v}) = 0$, and therefore

$$a(\mathbf{v}, \mathbf{v}) = \nu \int_\Omega \nabla \mathbf{v} : \nabla \mathbf{v} \, d\mathbf{x} + b(\mathbf{w}, \mathbf{v}, \mathbf{v}) = \nu \, \|\mathbf{v}\|^2_{V_{\text{div}}} .$$

Since $\mathbf{f} \in L^2(\Omega; \mathbb{R}^n)$ the functional $F\mathbf{v} = \int_\Omega \mathbf{f} \cdot \mathbf{v} \, d\mathbf{x}$ clearly belongs to $(V_{\text{div}})^*$.

By the Lax-Milgram Theorem, there exists a unique solution $\mathbf{u} = T(\mathbf{w})$ of (10.25) and

$$\|T(\mathbf{w})\|_{V_{\text{div}}} \leq \frac{C_P}{\nu} \, \|\mathbf{f}\|_{L^2_{\text{div}}(\Omega;\mathbb{R}^n)} \tag{10.26}$$

where C_P is a Poincaré constant.

4. *T has a fixed point.* Set $M = \frac{C_P}{\nu} \, \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)}$ and define

$$B_M = \{\mathbf{w} \in V_{\text{div}} : \|\mathbf{w}\|_{V_{\text{div}}} \leq M\}.$$

Then B_M is a closed, bounded, convex subset of V_{div} and $T : B_M \to B_M$. To apply Schauder Theorem 6.88, p. 413, we must check that

$T : B_M \to B_M$ is compact and continuous.

a) T is **compact**. Let $\{\mathbf{w}_k\}$ be a bounded sequence in B_M, that is

$$\|\mathbf{w}_k\|_{V_{\text{div}}} \leq M. \tag{10.27}$$

We want to prove that there exists a convergent subsequence $\{T(\mathbf{w}_{k_j})\}$.

Let $\mathbf{w}, \mathbf{z} \in B_M$. By definition, using (10.18),

$$\nu \int_\Omega \nabla [T(\mathbf{w}) - T(\mathbf{z})] : \nabla \mathbf{v} \, d\mathbf{x} = -b(\mathbf{w}, T(\mathbf{w}), \mathbf{v}) + b(\mathbf{z}, T(\mathbf{z}), \mathbf{v}) = b(\mathbf{w}, \mathbf{v}, T(\mathbf{w})) - b(\mathbf{z}, \mathbf{v}, T(\mathbf{z})).$$

By adding an subtracting $b(\mathbf{w}, \mathbf{v}, T(\mathbf{z}))$, we find:

$$\nu \int_\Omega \nabla [T(\mathbf{w}) - T(\mathbf{z})] : \nabla \mathbf{v} \, d\mathbf{x} = b(\mathbf{w}, \mathbf{v}, T(\mathbf{w}) - T(\mathbf{z})) + b(\mathbf{w} - \mathbf{z}, \mathbf{v}, T(\mathbf{z})).$$

Choosing $\mathbf{v} = T(\mathbf{w}) - T(\mathbf{z})$, from (10.20), (10.21) with $s = 4$, and (10.27), we obtain, after simplifying by $\|T(\mathbf{w}) - T(\mathbf{z})\|_{V_{\text{div}}}$,

$$\|T(\mathbf{w}) - T(\mathbf{z})\|_{V_{\text{div}}} \leq \frac{\tilde{C}_4 M}{\nu} \|\mathbf{w} - \mathbf{z}\|_{L^4(\Omega;\mathbb{R}^n)} . \tag{10.28}$$

By the compactness of the embedding $H_0^1(\Omega;\mathbb{R}^n) \hookrightarrow L^4(\Omega;\mathbb{R}^n)$, there exists a subsequence $\{\mathbf{w}_{k_j}\}$ convergent in $L^4(\Omega;\mathbb{R}^n)$. In particular

$$\|\mathbf{w}_{k_j} - \mathbf{w}_{k_m}\|_{L^4(\Omega;\mathbb{R}^n)} \to 0 \tag{10.29}$$

as $j, m \to \infty$. Apply (10.28) to $\mathbf{w} = \mathbf{w}_{k_j}$, $\mathbf{z} = \mathbf{w}_{k_m}$. We infer that $\{T(\mathbf{w}_{k_j})\}$ is a Cauchy sequence, and therefore convergent in V_{div}.

b) T is **continuous**. From (10.28) and (10.21) we get

$$\|T(\mathbf{w}_k) - T(\mathbf{w})\|_{V_{\text{div}}} \leq \frac{\tilde{C}_4^2 M}{\nu} \|\mathbf{w}_k - \mathbf{w}\|_{V_{\text{div}}} \tag{10.30}$$

from which the (Lipschitz) continuity of T follows.

5. *Uniqueness for small data.* Since B_M is a closed subset of V_{div}, it is, in particular, a complete space with respect to the metric induced by the V_{div}-norm. Now, by (10.30), if

$$\frac{\tilde{C}_4^2 M}{\nu} = \frac{C_P \tilde{C}_4^2}{\nu^2} \|\mathbf{f}\|_{L^2(\Omega;\mathbb{R}^n)} < 1,$$

T is a strict contraction. By the Contraction Theorem 6.83, p. 410, T has a unique fixed point $\mathbf{u} \in B_M$, which is also a solution of problem (10.17). On the other hand, if $\mathbf{u}$ is a solution of problem (10.17), then $\mathbf{u} \in B_M$ and therefore $\mathbf{u}$ is the unique solution. □

10.4 A Control Problem

Control problems are more and more important in modern technology. We give here an application of the variational theory we have developed so far, to a fairly simple temperature control problem.

10.4.1 Structure of the problem

Suppose that the temperature u of a homogeneous body, occupying a smooth bounded domain $\Omega \subset \mathbb{R}^3$, satisfies the following stationary conditions:

$$\begin{cases} \mathcal{E}u \equiv -\Delta u + \operatorname{div}(\mathbf{b}u) = z & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega, \end{cases} \tag{10.31}$$

where $\mathbf{b} \in C^1(\overline{\Omega};\mathbb{R}^3)$ is given, with $\operatorname{div} \mathbf{b} \geq 0$ in Ω.

In (10.31) we distinguish two types of dependent variables: the **control** variable z, that we take in $H = L^2(\Omega)$, and the **state** variable u. Coherently, (10.31) is called the **state system**. Given a control z, from Corollary 8.24, (10.31) has a unique weak solution $u[z] \in V = H_0^1(\Omega)$.

Thus, setting

$$a(u,v) = \int_\Omega (\nabla u \cdot \nabla v - u\mathbf{b} \cdot \nabla v)\, d\mathbf{x},$$

$u[z]$ satisfies the **state** equation

$$a(u[z], v) = (z, v)_H \qquad \forall v \in V \tag{10.32}$$

and

$$\|u[z]\|_V \le C_P \|z\|_H . \tag{10.33}$$

From elliptic regularity (Theorem 8.30, p. 530) it follows that $u \in H^2(\Omega) \cap H_0^1(\Omega)$, so that u is a *strong solution* of the state equation and satisfies it in the a.e. pointwise sense as well.

Our problem is to choose the source term z in order to minimize the "distance" of u from a given target state u_d.

Of course there are many ways to measure the distance of u from u_d. If we are interested in a distance which involves u and u_d over an open subset $\Omega_0 \subseteq \Omega$, a reasonable choice may be

$$J(u,z) = \frac{1}{2}\int_{\Omega_0} (u - u_d)^2 d\mathbf{x} + \frac{\beta}{2}\int_\Omega z^2 d\mathbf{x} \tag{10.34}$$

where $\beta > 0$.

$J(u,z)$ is called **cost functional** or **performance index**. The second term in (10.34) is called *penalization term*; its role is, on one hand, to avoid using "too large" controls in the minimization of J, on the other hand, to assure coercivity for J, as we shall see later on. In any case, the penalization term can be fine-tuned by choosing a suitable size for β.

Summarizing, we may write our control problem in the following way:

Find $(u^*, z^*) \in H \times V$, such that

$$\begin{cases} J(u^*, z^*) = \min\limits_{(u,z)\in V\times H} J(u,z) \\ \text{under the conditions} \\ \mathcal{E}u = z \ \text{ in } \Omega, \ \ u = 0 \ \text{ on } \partial\Omega. \end{cases} \tag{10.35}$$

If (u^*, z^*) is a minimizing pair, u^* and z^* are called **optimal state** and **optimal control**, respectively.

Remark 10.11 When the control z is defined in an open subset Ω_0 of Ω, we say that it is a *distributed control.* In some cases, z may be defined only on $\Gamma \subseteq \partial\Omega$ and then it is called *boundary control*.

Similarly, when the cost functional (10.34) involves the observation of u in $\Omega_0 \subseteq \Omega$, we say that the observation is *distributed*. On the other hand, one may observe u or $\partial_\nu u$ on $\Gamma \subseteq \partial\Omega$. These cases correspond to *boundary observations* and the cost functional has to take an appropriate form. Two examples are given in Problems 10.7, 10.8.

The main questions to face in a control problem are:

- Establish existence and/or uniqueness of an optimal pair (u^*, z^*).
- Derive necessary and/or sufficient optimality conditions.
- Construct algorithms for the numerical approximation of (u^*, z^*).

10.4.2 Existence and uniqueness of an optimal pair

Given $z \in H$, we may substitute into J the unique solution $u = u[z]$ of (10.32) to get the *reduced functional*

$$\tilde{J}(z) = J(u[z], z) = \frac{1}{2}\int_{\Omega_0} (u[z] - u_d)^2 d\mathbf{x} + \frac{\beta}{2}\int_{\Omega} z^2 d\mathbf{x},$$

depending only on z. Thus, our minimization problem (10.35) is reduced to find an optimal control $z^* \in H$ such that

$$\tilde{J}(z^*) = \min_{z \in H} \tilde{J}(z). \tag{10.36}$$

Once z^* is known, the optimal state is given by $u^* = u[z^*]$.

The strategy to prove existence and uniqueness of an optimal control is to use the relationship between minimization of quadratic functionals and abstract variational problems corresponding to symmetric bilinear forms, expressed in Theorem 6.46, p. 381. The key point is to write $\tilde{J}(z)$ in the following way:

$$\tilde{J}(z) = \frac{1}{2} b(z, z) + Lz + q \tag{10.37}$$

where $q \in \mathbb{R}$ (irrelevant in the optimization) and:

- $b(z, w)$ is a bilinear form in H, *symmetric, continuous and coercive*.
- L is a *linear, continuous functional* in H.

Then, by Theorem 6.46, there exists a unique minimizer $z^* \in H$. Moreover z^* is the minimizer if and only if z^* satisfies the Euler equation (see (6.62))

$$\tilde{J}'(z^*)w = b(z^*, w) - Lw = 0, \qquad \forall w \in H. \tag{10.38}$$

This procedure yields the following result.

Theorem 10.12 *There exists a unique optimal control $z^* \in H$. Moreover, z^* is optimal if and only if the following Euler equation holds $(u^* = u[z^*])$:*

$$\tilde{J}'(z^*)w = \int_{\Omega_0} (u^* - u_d) u[w] \, d\mathbf{x} + \beta \int_{\Omega} z^* w = 0, \qquad \forall w \in H. \tag{10.39}$$

Proof By direct calculations we infer

$$\tilde{J}(z) = \frac{1}{2}\int_{\Omega_0} u[z]^2 d\mathbf{x} + \frac{\beta}{2}\int_{\Omega} z^2 d\mathbf{x} - \int_{\Omega_0} u[z]u_d \, d\mathbf{x} + \frac{1}{2}\int_{\Omega_0} u_d^2 \, d\mathbf{x}.$$

According to the above strategy, we see that $\tilde{J}(z)$ is written in the form (10.37) where

$$b(z,w) = \int_{\Omega_0} u[z]u[w] \, d\mathbf{x} + \beta \int_{\Omega} zw \, d\mathbf{x} \tag{10.40}$$

is a bilinear form,

$$Lw = \int_{\Omega_0} u[w]u_d \, d\mathbf{x} \tag{10.41}$$

is a linear functional in H, and $q = \int_{\Omega_0} u_d^2 d\mathbf{x}$.

Moreover, b is symmetric (obvious), continuous and coercive. In fact, from (10.33) and the Schwarz and Poincaré inequalities, we have, since $\Omega_0 \subseteq \Omega$,

$$\begin{aligned} |b(z,w)| &\le \|u[z]\|_{L^2(\Omega_0)} \|u[w]\|_{L^2(\Omega_0)} + \beta \|z\|_H \|w\|_H \\ &\le (C_P^2 + \beta) \|z\|_H \|w\|_H \end{aligned}$$

which gives the continuity of b. The coerciveness of b follows from

$$b(z,z) = \int_{\Omega_0} u^2[z] d\mathbf{x} + \beta \int_{\Omega} z^2 \ge \beta \|z\|_H^2 .$$

Finally, from (10.33) and Poincarè's inequality,

$$|Lw| \le \|u_d\|_{L^2(\Omega_0)} \|u[w]\|_{L^2(\Omega_0)} \le C_P \|u_d\|_H \|w\|_H ,$$

and we deduce that L is continuous in H.

Then, Theorem 6.46 yields existence and uniqueness of the optimal control and Euler equation (10.38) translates into (10.39) after simple computations. □

10.4.3 Lagrange multipliers and optimality conditions

The Euler equation (10.39) gives a characterization of the optimal control z^* but it is not suitable for its computation. To obtain more manageable optimality conditions, let us change point of view by regarding the state equation $\mathcal{E}u[z] = -\Delta u + \operatorname{div}(\mathbf{b}u) = z$, with $u = 0$ on $\partial\Omega$, as a *constraint* for our minimization problem. Then, the key idea is to introduce a *Lagrange multiplier* $p \in V$, to be chosen suitably later on, and write $\tilde{J}(z)$ in the augmented form

$$\frac{1}{2}\int_{\Omega_0} (u[z] - u_d)^2 d\mathbf{x} + \frac{\beta}{2}\int_{\Omega} z^2 d\mathbf{x} + \int_{\Omega} p[z - \mathcal{E}u[z]] d\mathbf{x}. \tag{10.42}$$

In fact, we have just added zero. Since clearly,

$$\tilde{L}z = \int_{\Omega} p\mathcal{E}u[z] d\mathbf{x}$$

is a continuous linear functional in H, Theorem 10.12 yields the Euler equation:

$$\tilde{J}'(z^*)w = \int_{\Omega_0} (u^* - u_d)u[w]d\mathbf{x} + \int_\Omega (p + \beta z^*)w \, d\mathbf{x} - \int_\Omega p \, \mathcal{E}u[w]d\mathbf{x} = 0 \quad (10.43)$$

for every $w \in H$. Now we integrate twice by parts the last term, recalling that $u[w] = 0$ on $\partial\Omega$. We find:

$$\begin{aligned}\int_\Omega p\mathcal{E}u[w]d\mathbf{x} &= \int_{\partial\Omega} p(-\partial_\nu u[w] + (\mathbf{b}\cdot\nu)u[w])d\sigma + \int_\Omega (-\Delta p - \mathbf{b}\cdot\nabla p)u[w]d\mathbf{x} \\ &= -\int_{\partial\Omega} p \, \partial_\nu u[w]d\sigma + \int_\Omega \mathcal{E}^* p \, u[w]d\mathbf{x},\end{aligned}$$

where the operator $\mathcal{E}^* = -\Delta - b\cdot\nabla$ is the *formal adjoint* of $\mathcal{E}$.

Now we choose the multiplier: let p^* be the solution of the following **adjoint** problem:

$$\begin{cases} \mathcal{E}^* p = (u^* - u_d)\chi_{\Omega_0} & \text{in } \Omega \\ p = 0 & \text{on } \partial\Omega. \end{cases} \quad (10.44)$$

Using (10.44), the Euler equation (10.43) becomes

$$\tilde{J}'(z^*)w = \int_\Omega (p^* + \beta z^*)w \, d\mathbf{x} = 0 \qquad \forall w \in H, \quad (10.45)$$

equivalent to $p^* + \beta z^* = 0$.

Summarizing, we have proved the following result:

Theorem 10.13 *The control z^* and the state $u^* = u(z^*)$ are optimal if and only if there exists a multiplier $p^* \in V$ such that z^*, u^* and p^* satisfy the following optimality conditions:*

$$\begin{cases} -\Delta u^* + \operatorname{div}(\mathbf{b}u^*) = z^* & \text{in } \Omega, \quad u^* = 0 \text{ on } \partial\Omega \\ -\Delta p^* - \mathbf{b}\cdot\nabla p^* = (u^* - u_d)\chi_{\Omega_0} & \text{in } \Omega, \quad p^* = 0 \text{ on } \partial\Omega \\ p^* + \beta u^* = 0 & \text{a.e. in } \Omega, \text{ (Euler equation).} \end{cases}$$

Remark 10.14 The optimal multiplier p^* is also called **adjoint state**. Also note that $\tilde{J}'(z^*) \in H^*$

10.4.4 An iterative algorithm

From Euler equation (10.45) and the Riesz Representation Theorem, we infer that

$$p^* + \beta z^* \text{ is the Riesz element associated with } \tilde{J}'(z^*),$$

called the **gradient of** J **at** z^* and denoted by the usual symbol $\nabla\tilde{J}(z^*)$ or by $\delta z(z^*, p^*)$. Thus, we have

$$\nabla\tilde{J}(z^*) = p^* + \beta z^*.$$

It turns out that $-\nabla\tilde{J}(z^*)$ plays the role of the *steepest descent* direction for $\tilde{J}$, as in the finite-dimensional case. This suggests an iterative procedure to compute a sequence of controls $\{z_k\}_{k\geq 0}$, convergent to the optimal one.

Select an initial control z_0. If z_k is known ($k \geq 0$), then z_{k+1} is computed according to the following scheme.

1. Solve the state equation $a(u_k, v) = (z_k, v)_{L^2(\Omega)}$, $\forall v \in V$.
2. Knowing u_k, solve the adjoint equation

$$a^*(p_k, \varphi) = (u_k - u_d, \varphi)_{L^2(\Omega_0)}, \qquad \forall \varphi \in V.$$

3. Set

$$z_{k+1} = z_k - \tau_k \nabla\tilde{J}(z_k) \tag{10.46}$$

and select the *relaxation parameter* τ_k in order to assure that

$$\tilde{J}(z_{k+1}) < \tilde{J}(z_k). \tag{10.47}$$

Clearly, (10.47) implies the convergence of the sequence $\{\tilde{J}(z_k)\}$, though in general not to zero. Concerning the choice of the relaxation parameter, there are several possibilities. For instance, if $\beta \ll 1$, we may chose

$$\tau_k = \tilde{J}(z_k) \left\|\nabla\tilde{J}(z_k)\right\|^{-2}.$$

With this choice, (10.46) is a Newton type method:

$$z_{k+1} = z_k - \frac{\nabla\tilde{J}(z_k)}{\left\|\nabla\tilde{J}(z_k)\right\|^2}\tilde{J}(z_k).$$

10.5 Reaction-Diffusion Models

We have seen in Sect. 2.10.2 that Fisher's equation

$$v_\tau = Dv_{yy} + rv\left(1 - \frac{v}{M}\right) \qquad \tau > 0, y \in \mathbb{R},$$

models the spatial spread of an advantageous gene in a population, over an infinitely long one-dimensional habitat. Here v denotes the gene concentration, and D, r, and M are positive parameters. As we noticed, Fisher's equation is an example of *semilinear reaction-diffusion equation*. This kind of equations can be used in a wide variety of models, whenever some density (of mass, population, probability...) experiences the opposing effects of *diffusion*, i.e. random dispersal, and *reaction*, i.e. the production or absorption of matter. The tools we have developed in Chapters 8 and 9 allow a quite general description of the main features of reaction-diffusion models. To illustrate this, here we deal with the remarkable case of a biological

species, growing and dispersing in an heterogeneous habitat. This model was first introduced by Skellam in a seminal paper[5].

10.5.1 Dynamics of a population in a heterogeneous environment

Let $u = u(\mathbf{x}, t)$ denote the (nonnegative) density of a population at time t in the position $\mathbf{x} \in \Omega$, where $\Omega \subset \mathbb{R}^n$ is a smooth, bounded domain ($n = 2, 3$). We assume that the random dispersal is modeled by a Brownian motion. From Ch. 2 we infer that u satisfies the equation

$$u_t - D\Delta u = f(\mathbf{x}, u) \qquad \text{in } Q_T = \Omega \times (0, T), \tag{10.48}$$

for some diffusion coefficient $D > 0$, the *motility* of the population, and a nonlinear reaction term f which encodes the *law of population growth*.

Roughly, f results from the balance between births and deaths, and its main feature can be clarified considering the model in the absence of dispersal. In this case, (10.48) reduces to the family of ODEs

$$u_t = f(\mathbf{x}, u) \qquad \text{in } Q_T,$$

indexed by $\mathbf{x}$, and f positive (resp. negative) corresponds to a local increase (decrease) of the population. Accordingly, natural basic assumptions on f are:

$$f(\mathbf{x}, 0) = 0; \tag{10.49}$$

$$\text{for some } M > 0,\ f(\mathbf{x}, u) < 0 \text{ whenever } u > M. \tag{10.50}$$

These two conditions have easy interpretation in terms of the model: (10.49) asserts that, in the absence of population, no birth or death occurs; according to (10.50), the habitat has a *maximal carrying capacity* (less than or equal to M), and when the population density is larger than this threshold it must decrease. A typical example of reaction f is the Verhulst logistic law, homogeneous in space, which is the one considered by Fisher in his model:

$$f(\mathbf{x}, u) = ru\left(1 - \frac{u}{M}\right) = ru - \frac{r}{M}u^2. \tag{10.51}$$

In this context, r is the intrinsic growth coefficient of the population, while the term $-rM^{-1}u^2$ accounts for the competition for resources within the population. When the habitat is heterogeneous, the vital coefficients may depend on position[6], resulting in a reaction of the form

$$f(\mathbf{x}, u) = m(\mathbf{x})u - \beta(\mathbf{x})u^2. \tag{10.52}$$

[5] *Skellam, J.G.*, Random Dispersal in Theoretical Populations, Biometrika **38**, 196–218, 1951.

[6] In general, the coefficients may depend also on time, to encode e.g. seasonal effects.

In particular, while in general $\beta > 0$, m may be positive or negative in different subsets of Ω, representing the *favorable* and *hostile* regions of the environment, respectively.

Once the most fitting reaction has been identified, (10.48) must be complemented with a nonnegative, nontrivial initial condition, and with suitable boundary conditions on $\partial\Omega$. For instance, homogeneous Dirichlet boundary conditions are used in case of a *lethal boundary*, while homogeneous Neumann ones encode a boundary acting as a *reflecting barrier*, resulting in an isolated habitat. Analogously, in the case only a fraction of the population meeting the boundary exits Ω, while the other part remains inside, then homogeneous Robin boundary conditions are appropriate. Nonhomogeneous and/or mixed boundary conditions can be easily interpreted as well.

This kind of models, as well as many variants, has been discussed intensively in the literature, see e.g. *R. S. Cantrell, C. Cosner*, [25]. For concreteness, in the next sections we analyze the prototypical case of an isolated habitat, and we focus on the Cauchy-Neumann problem

$$\begin{cases} u_t - D\Delta u = f(\mathbf{x}, u) & \text{in } Q_T = \Omega \times (0,T) \\ \partial_\nu u = 0 & \text{on } S_T = \partial\Omega \times (0,T) \\ u(0) = g & \text{in } \Omega, \end{cases} \tag{10.53}$$

with $g \geq 0$ in $\overline{\Omega}$.

Beyond existence, uniqueness and regularity, the main question that we are interested in is the long-time behavior of the solution: *does the population survive for all times, or can it undergo extinction?* It turns out that this issue is deeply connected with the existence of nontrivial steady states $U = U(\mathbf{x})$, satisfying

$$\begin{cases} -D\Delta U = f(\mathbf{x}, U) & \text{in } \Omega \\ \partial_\nu U = 0 & \text{on } \partial\Omega. \end{cases} \tag{10.54}$$

In the next sections we discuss the well-posedness of (10.53), both in weak and in classical sense, together with some general relations with (10.54). As we will see, this can be done for a much more general class of semilinear parabolic problems. We will get back to our population dynamics model in Sect. 10.5.5, to draw the consequences of the general theory in terms of persistence or extinction of the species.

10.5.2 A monotone iteration scheme

In this section we use the linear variational theory developed in Chapter 9 to deal with semilinear problems such as (10.53). In particular, maximum and comparison principles allow to construct an iterative procedure to solve these problems. Actually, the procedure is quite flexible and, by means of the results of Chapter 8, it can be adapted also to handle stationary problems like (10.54).

We consider the Cauchy-Neumann problem

$$\begin{cases} u_t - \Delta u = f(\mathbf{x}, u) & \text{in } Q_T \\ \partial_\nu u = 0 & \text{on } S_T \\ u(0) = g & \text{in } \Omega, \end{cases} \tag{10.55}$$

where $\Omega \subset \mathbb{R}^n$, $n \geq 1$, is a bounded Lipschitz domain and $f \in C^1(\overline{\Omega} \times \mathbb{R})$, $g \in L^\infty(\Omega) \cap H^1(\Omega)$.

The simple example of the ODE $\dot{u} = u^2$ shows that we cannot in general expect existence of a solution in the whole interval $[0, T]$. In our case, we are able to prove existence and uniqueness through the construction of a suitable monotone iterative sequence. To control the behavior of such sequence we need the notions of upper and lower solution, or super/subsolution, of problem (10.55). Let $V = H^1(\Omega)$, and let the symbols $(\cdot, \cdot)_0$ and $\|\cdot\|_0$ denote the inner product and norm in either $L^2(\Omega)$ or $L^2(\Omega; \mathbb{R}^n)$.

Definition 10.15 We say that a function $u \in L^2(0, T; V)$ with $\dot{u} \in L^2(0, T; L^2(\Omega))$, is a *weak supersolution (resp. subsolution)* of problem (10.55) if[7] $u(0) \geq g$ (resp. $\leq$) and

$$(\dot{u}(t), v)_0 + (\nabla u(t), \nabla v)_0 \geq (f(\mathbf{x}, u(t)), v)_0 \qquad (\text{resp } \leq) \tag{10.56}$$

for a.e. $t \in (0, T)$ and all $v \in V$, $v \geq 0$ a.e. in Ω.

It is easy to check that any regular function u satisfying

$$\begin{cases} u_t - \Delta u \geq f(\mathbf{x}, u) & \text{in } Q_T \\ \partial_\nu u \geq 0 & \text{on } S_T \\ u(0) \geq g & \text{in } \Omega \end{cases}$$

in a classical sense is a weak supersolution of (10.55). Clearly, u is a weak solution of (10.55) if u is both super and subsolution. The following lemma gives a comparison between bounded super and subsolutions.

Lemma 10.16 *Let v and w be, respectively, bounded weak sub and supersolution of problem (10.55). Then*

$$v \leq w \quad \text{a.e. in } Q_T.$$

If moreover $\partial\Omega$ is C^2, $v, w \in C^{2,1}(Q_T) \cap C^1(\overline{Q}_T)$ and $v(\mathbf{x}_1, t_1) = w(\mathbf{x}_1, t_1)$, with $\mathbf{x}_1 \in \overline{\Omega}$ and $t_1 > 0$, then

$$v = w \quad \text{in } \overline{Q}_{t_1}.$$

Proof For the first part, let $z = w - v$. The function z satisfies, in the weak sense,

$$\begin{cases} z_t - \Delta z \geq f(\mathbf{x}, w) - f(\mathbf{x}, v) & \text{in } Q_T \\ \partial_\nu z \geq 0 & \text{on } S_T \\ z(\mathbf{x}, 0) \geq 0 & \text{in } \Omega. \end{cases}$$

[7] Recall that $u \in C([0, T]; L^2(\Omega))$ by Thm. 7.100, p. 488.

Now, for a.e. $(\mathbf{x}, t) \in Q_T$ we have

$$f(\mathbf{x}, w) - f(\mathbf{x}, v) = \int_0^1 \frac{d}{ds} f(\mathbf{x}, s(w-v)+v)\, ds = z \int_0^1 \partial_u f(\mathbf{x}, s(w-v)+v)\, ds.$$

Moreover, since w and v are bounded, we have $a \le v, w \le b$ a.e. in Q_T, for some $a, b \in \mathbb{R}$. Thus, setting

$$r(\mathbf{x}, t) = -\int_0^1 \partial_u f(\mathbf{x}, s(w-v)+v)\, ds,$$

we infer

$$|r| \le L \equiv \max_{\overline{\Omega}\times[a,b]} |\partial_u f| \qquad \text{and} \qquad z_t - \Delta z + r(\mathbf{x}, t) z \ge 0.$$

The weak maximum principle (see Remark 9.20) yields $z \ge 0$ a.e. in Q_T, and the first part follows.

Now let us assume that v, w are classical sub/supersolutions, and write

$$\tilde{z} = e^{Lt} z = e^{Lt}(w - v).$$

Then $\tilde{z} \ge 0$ in Q_T and

$$\tilde{z}_t - \Delta \tilde{z} = e^{Lt}\left[z_t + Lz - \Delta z\right] = e^{Lt}(L - r(\mathbf{x}, t))z \ge 0.$$

Then, in case $\mathbf{x}_1 \in \Omega$, the strong maximum principle, Theorem 2.6, p. 41, applies and we obtain that $\tilde{z}$ identically vanish on $\overline{Q}_{t_1}$. On the other hand, if $\mathbf{x}_1 \in \partial\Omega$ then, being $\partial\Omega$ regular, the interior sphere condition holds true at $\mathbf{x}_1$ and we can conclude by the parabolic Hopf's principle, stated in Theorem 10.17 below. □

Theorem 10.17 (Hopf's principle) *Let $u \in C^{2,1}(Q_T) \cap C(\overline{Q}_T)$ satisfy*

$$u_t - \Delta u \ge 0 \qquad \text{in } Q_T.$$

Assume that:

1. *at $\mathbf{x}_1 \in \partial\Omega$ an interior sphere condition (Definition 3.11, p. 125) holds, and denote by $\boldsymbol{\nu}$ the outward unit normal to the sphere at $\mathbf{x}_1$;*
2. *$u(\mathbf{x}_1, t_1) = m = \min_{\overline{Q}_{t_1}} u$, for some $t_1 > 0$.*

If the normal derivative of u at $(\mathbf{x}_1, t_1)$ exists, then

$$\partial_{\boldsymbol{\nu}} u(\mathbf{x}_1, t_1) < 0$$

unless $u \equiv m$ in $\overline{Q}_{t_1}$.

For a proof of this result, we refer the reader e.g. to *M. H. Protter, H. F. Weinberger*, [16].

We are now ready to produce a monotone iterative scheme leading to both existence and uniqueness of the solution to problem (10.55).

Theorem 10.18 *Let $\overline{u}$ and $\underline{u}$ be bounded weak super and subsolution of problem (10.55), respectively. Then there exists a unique weak solution $u = u_g$ such that*

$$\underline{u} \le u_g \le \overline{u} \quad \text{a.e. in } Q_T. \tag{10.57}$$

Proof We have $a \le \overline{u}, \underline{u} \le b$ a.e. in Q_T, for suitable $a, b \in \mathbb{R}$. Set $F(\mathbf{x}, s) = f(\mathbf{x}, s) + Ls$, where $L = \max_{\overline{\Omega}\times[a,b]} |\partial_u f|$, so that

$$\partial_u F(\mathbf{x}, s) = \partial_u f(\mathbf{x}, s) + L \ge -L + L = 0,$$

and therefore F is increasing in u in $[a, b]$ for every $\mathbf{x} \in \overline{\Omega}$. Rewrite the original equation in the form

$$u_t - \Delta u + Lu = f(\mathbf{x}, u) + Lu \equiv F(\mathbf{x}, u).$$

Define $u_0 = \underline{u}$ and let u_1 be the weak solution of the linear problem

$$\begin{cases} \partial_t u_1 - \Delta u_1 + Lu_1 = F(\mathbf{x}, u_0) & \text{in } Q_T \\ \partial_\nu u_1 = 0 & \text{on } S_T \\ u_1(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega. \end{cases} \tag{10.58}$$

Since $F(\mathbf{x}, u_0)$ is bounded in Q_T and $g \in H^1(\Omega)$, by Remark 9.12, p. 556, there exists a unique weak solution $u_1 \in C([0, T]; L^2(\Omega)) \cap L^2(0, T; V)$, with $\dot{u}_1 \in L^2(0, T; L^2(\Omega))$.

The function $w_1 = u_1 - u_0$ is a weak solution of the problem

$$\begin{cases} \partial_t w_1 - \Delta w_1 + Lw_1 \ge F(\mathbf{x}, u_0) - F(\mathbf{x}, u_0) = 0 & \text{in } Q_T \\ \partial_\nu w_1 \ge 0 & \text{on } S_T \\ w_1(\mathbf{x}, 0) \ge 0 & \text{in } \Omega. \end{cases}$$

From the maximum principle and Lemma 10.16, we obtain $a \le u_0 \le u_1$, a.e. in Q_T. A similar argument gives $u_1 \le \overline{u} \le b$, a.e. in Q_T. Thus we have:

$$a \le u_0 \le u_1 \le \overline{u} \le b, \quad \text{a.e. in } Q_T.$$

We now proceed by induction. Assume that $k > 1$ and that we are given $u_{k-1} \in C([0, T]; L^2(\Omega)) \cap L^2(0, T; V)$, with $\dot{u}_{k-1} \in L^2(0, T; L^2(\Omega))$, such that

$$a \le u_0 \le u_1 \le \cdots \le u_{k-2} \le u_{k-1} \le \overline{u} \le b \quad \text{a.e. in } Q_T. \tag{10.59}$$

Define u_k as the solution of the problem

$$\begin{cases} \partial_t u_k - \Delta u_k + Lu_k = F(\mathbf{x}, u_{k-1}) & \text{in } Q_T \\ \partial_\nu u_k = 0 & \text{on } S_T \\ u_k(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega. \end{cases} \tag{10.60}$$

Then also $u_k \in C([0, T]; L^2(\Omega)) \cap L^2(0, T; V)$, with $\dot{u}_k \in L^2(0, T; L^2(\Omega))$. Moreover, the function $w_k = u_k - u_{k-1}$ is a weak solution of the problem

$$\begin{cases} \partial_t w_k - \Delta w_k + Lw_k \ge F(\mathbf{x}, u_{k-1}) - F(\mathbf{x}, u_{k-2}) & \text{in } Q_T \\ \partial_\nu w_k \ge 0 & \text{on } S_T \\ w_k(\mathbf{x}, 0) \ge 0 & \text{in } \Omega. \end{cases}$$

Since F is increasing, from (10.59) we get $F(\mathbf{x}, u_{k-1}) - F(\mathbf{x}, u_{k-2}) \ge 0$. Then, once more, the maximum principle and Lemma 10.16 yield

$$\underline{u} \le u_0 \le u_1 \le \cdots \le u_{k-1} \le u_k \le \overline{u} \le b \quad \text{a.e. in } Q_T.$$

Being monotone increasing and bounded, $\{u_k\}$ converges a.e. and in $L^2(Q_T)$ to a bounded function u. Moreover, recalling Remark 9.12, by the estimates (9.41) and (9.42), we may extract a subsequence $\{u_{k_m}\}$ converging weakly to u in $L^2(0, T; V)$, with $\{\dot{u}_{k_m}\}$ converging weakly to $\dot{u}$ in $L^2(0, T; L^2(\Omega))$. Since the limit of any convergent subsequence is u, the whole sequences converge weakly to u and $\dot{u}$, respectively. Moreover, since $F(\mathbf{x}, a) \le F(\mathbf{x}, u_k) \le F(\mathbf{x}, b)$, by the

Dominated Convergence Theorem we infer that $F(\mathbf{x}, u_k) \to F(\mathbf{x}, u)$ in $L^2(Q_T)$. This is enough to pass to the limit as $k \to +\infty$ into the equation

$$(\dot{u}_k, v)_0 + (\nabla u_k, \nabla v)_0 + L(u_k, v)_0 = (F(u_{k-1}), v)_0$$

and to deduce that u is a weak solution of problem (10.55). This gives existence.

The uniqueness follows from Lemma 10.16. Indeed, if u_1 and u_2 are two solutions to problem (10.55), being both bounded super and subsolutions, that lemma yields $u_1 \geq u_2$ and $u_2 \geq u_1$. □

Remark 10.19 Starting the iteration in the proof of Theorem 10.18 from $u_0 = \overline{u}$, we obtain a *decreasing* sequence $\{u_k\}$, converging to the solution u_g.

Remark 10.20 In case $\underline{u}(0) \leq g_1 \leq g_2 \leq \overline{u}(0)$, then $u_{g_1} \leq u_{g_2}$ on Q_T. This follows by Lemma 10.16, since u_{g_1} is a subsolution of the problem solved by u_{g_2}.

10.5.3 Long time behavior and steady states

In many interesting cases, it is possible to find *time independent* super/subsolutions. In this case, the solution obtained in Theorem 10.18 is *global in time*, i.e. it exists for all times $t > 0$. Moreover, it is possible to infer the existence of steady states, i.e. weak solutions of the stationary problem

$$\begin{cases} -\Delta U = f(\mathbf{x}, U) & \text{in } \Omega \\ \partial_\nu U = 0 & \text{on } \partial\Omega. \end{cases} \tag{10.61}$$

Finally, this provides a description of the dynamics of the equation. All these results are intertwined, and they rely again on direct consequences of the maximum and comparison principles, contained in the next lemma.

Lemma 10.21 *Let $\varphi, \psi \in L^\infty(\Omega) \cap H^1(\Omega)$ be time independent subsolution and supersolution of problem (10.55), respectively, and let u_φ, u_ψ denote the corresponding solutions of (10.55), with $u_\varphi(0) = \varphi$, $u_\psi(0) = \psi$. Then*

1. *u_φ, u_ψ and u_g exist for all $t \geq 0$, and*

$$\varphi(\mathbf{x}) \leq u_\varphi(\mathbf{x}, t) \leq u_g(\mathbf{x}, t) \leq u_\psi(\mathbf{x}, t) \leq \psi(\mathbf{x}) \qquad \text{a.e. in } \Omega \times [0, \infty), \tag{10.62}$$

   for all $\varphi \leq g \leq \psi$;
2. *$u_\varphi(\mathbf{x}, t_2) \geq u_\varphi(\mathbf{x}, t_1)$ a.e. in Ω, if $t_2 > t_1$;*
3. *$u_\psi(\mathbf{x}, t_2) \leq u_\psi(\mathbf{x}, t_1)$ a.e. in Ω, if $t_2 > t_1$.*

Proof 1 is immediate, since we can apply Theorem 10.18 in Q_T, for every $T > 0$, and (10.62) follows by construction and Remark 10.20. As far as 2 is concerned, let $\delta > 0$ and consider problem (10.55) with $g(\mathbf{x}) = u_\varphi(\mathbf{x}, \delta)$. By (10.62), $\varphi \leq g \leq \psi$ and thus $u_\varphi \leq u_g$. Since $u_g(\mathbf{x}, t) = u_\varphi(\mathbf{x}, t + \delta)$ by uniqueness, this yields

$$u_\varphi(\mathbf{x}, t) \leq u_\varphi(\mathbf{x}, t + \delta) \qquad \text{in } \Omega \times [0, \infty),$$

and 2 follows. The proof of 3 is analogous. □

The monotonicity of u_φ and u_ψ is crucial in proving their convergence to steady states, which capture the asymptotic dynamic of the problem.

Theorem 10.22 *Let φ, ψ be as in Lemma 10.21, and define*

$$\Phi(\mathbf{x}) = \lim_{t\to\infty} u_\varphi(\mathbf{x}, t), \qquad \Psi(\mathbf{x}) = \lim_{t\to\infty} u_\psi(\mathbf{x}, t),$$

a.e. in Ω. Then Φ, Ψ solve problem (10.61), and any other steady state U, with $\varphi \le U \le \psi$, satisfies $\Phi \le U \le \Psi$.

In particular, if $\Phi = \Psi$ a.e. in Ω then such a steady state is unique, and

$$u_g(t) \to \Phi \qquad \text{as } t \to \infty,$$

a.e. and in $L^2(\Omega)$, for every initial datum $\varphi \le g \le \psi$.

Proof First notice that Φ an Ψ are well defined by the monotonicity of u_φ, u_ψ, respectively. We show that the convergence is true also in $L^2(\Omega)$, and that Φ solves (10.61); the proof for Ψ is analogous and all the other assertions are consequences of (10.62), possibly taking $g = U$.

For $k = 0, 1, 2, \dots$ let us consider the functions $u_k \in L^2(0, 1; V) \cap H^1(0, 1; L^2(\Omega))$ defined as $u_k(t) = u_\varphi(t + k)$, $0 < t < 1$. Then

$$\begin{cases} \partial_t u_k - \Delta u_k = F(\mathbf{x}, u_k) & \text{in } Q_1 \\ \partial_\nu u_k = 0 & \text{on } S_1 \\ u_k(\mathbf{x}, 0) = u_\varphi(\mathbf{x}, k) & \text{in } \Omega. \end{cases}$$

Using again Lemma 10.21 we have that

$$\varphi \le u_0 \le u_1 \le \cdots \le u_{k-1} \le u_k \le \psi \qquad \text{a.e. in } Q_1.$$

We deduce that $\{u_k\}$ converges a.e. and in $L^2(Q_1)$ to the time independent function Φ. As in the proof of Theorem 10.18, we can show that $\{u_k\}$ weakly converges to Φ in $L^2(0, 1; V)$, and that $\{\dot{u}_k\}$ weakly converges to $\dot{\Phi} = 0$ in $L^2(Q_1)$. Moreover, by Aubin's compactness lemma[8] the convergence is also strong in $L^2(Q_1)$. Passing to the limit into the equation, we deduce that Φ is a weak solution of problem (10.55), constant in time, and thus it solves (10.61). □

Remark 10.23 As a byproduct of Theorem 10.22 we have that the existence of ordered stationary super/subsolutions implies the existence of (at least) a stationary solution. This can be proved directly, using only the elliptic theory developed in Chapter 8, see Problem 10.9.

Remark 10.24 In case $\Phi \not\equiv \Psi$, then u_g is bounded, but it does not necessarily converge to a steady state.

10.5.4 Regularity

Since the weak solution u_g is bounded, then $f(\mathbf{x}, u_g)$ is bounded too. In the case Ω is a C^2-domain, Remark 9.12 implies that $u_g \in L^2(0, T; H^2(\Omega)) \cap H^1(0, T; L^2(\Omega))$.

[8] J.P. Aubin, *Un théorème de compacité*, C. R. Acad. Sci. Paris 256 (1963), 5042–5044.

Recalling the regularization properties of the diffusion equation, we expect that if Ω and f are regular enough then u_g is also a classical solution, at least for $t > 0$. The standard argument to obtain this result is called *bootstrap*. It consists in transferring regularity from $f(\mathbf{x}, u_g)$ to u_g, in an inductive way. We illustrate this general idea in the following lemma. We restrict to smooth reactions.

Lemma 10.25 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain and $g \in L^2(\Omega)$. Let $u \in L^2(0,T;H^2(\Omega)) \cap H^1(0,T;L^2(\Omega))$ be the weak solution of the following problem:*

$$\begin{cases} u_t - \Delta u = f(\mathbf{x}, u) & \text{in } Q_T \\ \partial_\nu u = 0 & \text{on } S_T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega. \end{cases} \tag{10.63}$$

Assume that $a \leq u \leq b$ a.e. in Q_T.

If $f \in C^\infty(\overline{\Omega} \times \mathbb{R})$, then

$$\frac{\partial^k}{\partial t^k} u \in L^2(t_0, T; H^{2m+2-2k}(\Omega)) \tag{10.64}$$

for every $t_0 \in (0,T)$, every $m \geq 0$ and every $k = 0, \ldots, m+1$. In particular, u is a classical solution in $C^\infty(\overline{\Omega} \times (t_0, T])$.

Proof For simplicity we restrict to dimension $n \leq 3$. Let $\tau \in (0,T)$ and $r(t) = e^{-\frac{1}{t-\tau}} \chi_{(\tau,+\infty)}(t)$. Note that $r \in C^\infty(\mathbb{R})$ and its support is $[\tau, +\infty)$ while $r \geq e^{-\frac{1}{\tau}} > 0$ for $t \geq 2\tau$. Then the function $w = ru$ belongs to $L^2(\tau, T; H^2(\Omega)) \cap H^1(\tau, T; L^2(\Omega))$ and it is a (weak) solution of the problem

$$\begin{cases} w_t - \Delta w = f_1(\mathbf{x}, t) & \text{in } \Omega \times (\tau, T) \\ \partial_\nu w = 0 & \text{on } \partial\Omega \times (\tau, T) \\ w(\mathbf{x}, \tau) = 0 & \text{in } \Omega. \end{cases} \tag{10.65}$$

where

$$f_1(\mathbf{x}, t) = r(t) f(\mathbf{x}, u(\mathbf{x}, t)) + r'(t) u(\mathbf{x}, t).$$

Note that $f_1 \in L^2(\tau, T; H^1(\Omega))$. Differentiating with respect to t, we get that $z_1 = w_t = r'u + ru_t$ is a solution of the problem

$$\begin{cases} \partial_t z_1 - \Delta z_1 = \tilde{f}_1(\mathbf{x}, u) & \text{in } \Omega \times (\tau, T) \\ \partial_\nu z_1 = 0 & \text{on } \partial\Omega \times (\tau, T) \\ z_1(\mathbf{x}, \tau) = 0 & \text{in } \Omega. \end{cases} \tag{10.66}$$

with $\tilde{f}_1(\mathbf{x}, u) = r'(t) f(\mathbf{x}, u) + [r(t) f_u(\mathbf{x}, u) + r'(t)] u_t + r''(t) u$.

Since u is bounded and in $H^1(\tau, T; L^2(\Omega))$, then $\tilde{f}_1 \in L^2(\tau, T; L^2(\Omega))$. Therefore (see Remark 9.12) $z_1 = w_t \in L^2(\tau, T; H^2(\Omega)) \cap H^1(\tau, T; L^2(\Omega))$ which implies, using the elliptic regularity in problem (10.65),

$$w \in L^2(\tau, T; H^3(\Omega)).$$

In turn, by Sobolev embedding, $H^3(\Omega) \subset C^1(\overline{\Omega})$, so that $f_1 \in L^2(\tau, T; H^2(\Omega))$ and, by elliptic regularity, $w \in L^2(\tau, T; H^4(\Omega))$. Thus,

$$u \in L^2(2\tau, T; H^3(\Omega)) \cap H^1(2\tau, T; H^2(\Omega)) \cap H^2(2\tau, T; L^2(\Omega)).$$

Since τ is arbitrary, it follows that

$$u \in L^2(t_0, T; H^4(\Omega)) \cap H^1(t_0, T; H^2(\Omega)) \cap H^2(t_0, T; L^2(\Omega)) \quad \forall t_0 \in (0, T). \tag{10.67}$$

Now proceed by induction, assuming that (10.64) is true for some integer $m \geq 1$. The step $m = 1$ corresponds to (10.67). Let τ and $r = r(t)$ be as before and consider problem (10.66). By the inductive hypothesis we deduce that

$$\frac{\partial^k}{\partial t^k} z_1 \in L^2(\tau, T; H^{2m+2-2k}(\Omega)), \quad k = 0, 1, \ldots, m+1 \tag{10.68}$$

so that

$$\frac{\partial^k}{\partial t^k} u \in L^2(2\tau, T; H^{2m+4-2k}(\Omega)), \quad k = 1, \ldots, m+2.$$

By Sobolev embedding, $H^{2m+2}(\Omega) \subset C^{2m}(\overline{\Omega})$, so that $f_1 \in L^2(2\tau, T; H^{2m+1}(\Omega))$ and, via elliptic regularity, $u \in L^2(2\tau, T; H^{2m+3}(\Omega))$. This in turn implies $f_1 \in L^2(2\tau, T; H^{2m+2}(\Omega))$ and $u \in L^2(2\tau, T; H^{2m+4}(\Omega))$.

Since τ is arbitrary, we finally deduce, for every $t_0 \in (0, T)$,

$$\frac{\partial^k}{\partial t^k} u \in L^2(t_0, T; H^{2m+4-2k}(\Omega)), \quad 0 = 1, \ldots, m+2. \qquad \square$$

One relevant consequence of further regularity is that it allows to apply strong comparison principles. As a byproduct, we obtain that *ordered, partially coinciding initial data evolve in strictly different solutions*. This is true even if the initial data are satisfied only in weak sense.

Corollary 10.26 *Under the assumptions of Lemma 10.25, assume that $g_1, g_2 \in L^2(\Omega)$, $g_1 \leq g_2$ a.e. in Ω and $g_1 < g_2$ in a set of positive measure. Then*

$$u_{g_1} < u_{g_2} \qquad \text{in } \overline{\Omega} \times (0, \infty).$$

Proof By Lemma 10.25, both u_{g_1} and u_{g_2} are classical solutions in $\overline{\Omega} \times [t_0, t_1]$, for every $0 < t_0 < t_1$. Furthermore, by Remark 10.20 we already know that $u_{g_1} \leq u_{g_2}$ for $t \geq 0$. Assume by contradiction that $u_{g_1}(\mathbf{x}_1, t_1) = u_{g_2}(\mathbf{x}_1, t_1)$, with $\mathbf{x}_1 \in \overline{\Omega}$ and $t_1 > 0$. Then Lemma 10.16 yields $u_{g_1} \equiv u_{g_2}$ in $\overline{\Omega} \times [t_0, t_1]$. Since t_0 is arbitrary, and recalling that both $u_{g_i} \in H^1(0, T; L^2(\Omega))$, we infer that $g_1 = g_2$ a.e. in Ω, contradiction. $\square$

Finally, consider a steady state $U = U(\mathbf{x})$. Then $u_U = U$, and the previous results can be applied. We obtain:

Lemma 10.27 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain and $f \in C^\infty(\overline{\Omega} \times \mathbb{R})$. If $U \in H^1(\Omega) \cap L^\infty(\Omega)$ solves (10.61) then $U \in C^\infty(\overline{\Omega})$. Moreover, if U_1, U_2 are such solutions and $U_1 \leq U_2$, then*

$$\text{either } U_1 = U_2 \qquad \text{or } U_1 < U_2 \text{ in } \overline{\Omega}.$$

10.5.5 Persistence versus extinction

The results obtained so far readily apply to the case of reaction-diffusion problems, with a reaction f that satisfies the basic assumptions (10.49), (10.50), namely

$$f(\mathbf{x}, 0) = 0 \qquad \text{and} \qquad f(\mathbf{x}, u) < 0 \text{ for } u > M.$$

Indeed, on the one hand, $\underline{u}(\mathbf{x},t) \equiv 0$ solves the reaction-diffusion equation[9]; consequently, if $g \geq 0$ then $\underline{u}(\mathbf{x},t)$ is a subsolution of (10.53). On the other hand, taking $\overline{u}(\mathbf{x},t) \equiv M' = \max\{M, \|g\|_\infty\}$ we obtain

$$(\partial_t - D\Delta)\overline{u} = 0 \geq f(\mathbf{x},\overline{u})$$

by (10.50), and $\overline{u}$ is a supersolution of (10.53). Theorems 10.18, 10.22 and Corollary 10.26 yield the following result.

Theorem 10.28 *Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain, $g \in L^\infty(\Omega) \cap H^1(\Omega)$ be nonnegative and nontrivial, and $f \in C^1(\overline{\Omega} \times [0,\infty))$ satisfy (10.49), (10.50). Then (10.53) admits a unique weak solution u_g, globally defined in time, satisfying*

$$0 \leq u_g \leq u_{M'} \leq M' \qquad \text{in } \Omega \times [0,\infty),$$

where $M' = \max\{M, \|g\|_\infty\}$ and $u_{M'}(\mathbf{x},t) \to \Psi(\mathbf{x})$ as $t \to \infty$, with $\Psi \geq 0$ solution of (10.54).

If furthermore f is of class C^∞, then $u_g > 0$ for all $\mathbf{x} \in \overline{\Omega}$, $t > 0$.

By Theorem 10.22 it is clear that, if $\Phi(\mathbf{x}) \equiv 0$ is the only nonnegative steady state, then $\Psi = \Phi$ and the population extinguishes as $t \to \infty$, regardless of the initial datum g. On the other hand, if $\Psi > 0$ on Ω then at least some initial data provide persistence of the population. From this point of view assumptions (10.49), (10.50) are too generic, and the situation may greatly vary, depending on the specific form of the reaction f, see e.g. Problem 10.12.

A precise picture can be provided under the further assumption that the local relative growth rate is strictly decreasing:

$$\frac{f(\mathbf{x},u_2)}{u_2} < \frac{f(\mathbf{x},u_1)}{u_1}, \qquad \text{whenever } u_2 > u_1 > 0. \tag{10.69}$$

In particular, since f is at least of class C^1, taking the limit as $u_1 \to 0$ we infer

$$\frac{f(\mathbf{x},u_2)}{u_2} < \lim_{u_1 \to 0} \frac{f(\mathbf{x},u_1)}{u_1} = \lim_{u_1 \to 0} \frac{f(\mathbf{x},u_1) - f(\mathbf{x},0)}{u_1} = \partial_u f(\mathbf{x},0).$$

Thus we can define

$$m(\mathbf{x}) = \partial_u f(\mathbf{x},0) \in C(\overline{\Omega}) \tag{10.70}$$

obtaining

$$f(\mathbf{x},u) < m(\mathbf{x})u \text{ for all } u > 0. \tag{10.71}$$

Remark 10.29 Notice that the class of reactions satisfying (10.69) includes the heterogeneous logistic law (10.52), indeed if

$$f(\mathbf{x},u) = m(\mathbf{x})u - \beta(\mathbf{x})u^2 \qquad \text{then} \qquad \frac{f(\mathbf{x},u)}{u} = m(\mathbf{x}) - \beta(\mathbf{x})u$$

[9] Accordingly, $\Phi(\mathbf{x}) \equiv 0$ is a steady state.

is strictly decreasing as long as $\beta > 0$, and the definition of m is consistent with (10.70).

To investigate persistence and extinction a key role is played by the *linearized stationary operator at* $u = 0$, given by

$$w \mapsto -D\Delta w - m(\mathbf{x})w,$$

and by the associated *principal Neumann eigenvalue*. By arguing as Section 8.3.3, p. 506 (see also Problem 8.19, p. 538) one can easily prove the following result.

Proposition 10.30 *Let*

$$\lambda_1[m] = \min\left\{\frac{\int_\Omega \left\{D\,|\nabla u|^2 - m(\mathbf{x})u^2\right\} d\mathbf{x}}{\int_\Omega u^2\, d\mathbf{x}} : u \in H^1(\Omega),\, u \not\equiv 0\right\}. \tag{10.72}$$

There exists a nonnegative $\varphi_1 \in H^1(\Omega)$ *such that*

$$\begin{cases} -D\Delta\varphi_1 - m(\mathbf{x})\varphi_1 = \lambda_1[m]\varphi_1 & \text{in } \Omega \\ \partial_\nu \varphi_1 = 0 & \text{on } \partial\Omega. \end{cases}$$

It turns out that, under (10.69), **persistence and extinction of the population are completely characterized by the sign of** $\lambda_1[m]$. Notice that, since we avoided adimensionalization, $\lambda_1[m]$ is not a pure number and it has dimension $[\text{time}]^{-1}$. Anyway, its sign is independent of scaling, so that the following results are dimensionally consistent.

Case $\lambda_1[m] \geq 0$. In this case we have extinction, for every initial datum.

Theorem 10.31 (Extinction) *Let* $f \in C^1(\overline{\Omega} \times [0, \infty))$ *satisfy* (10.49), (10.50) *and* (10.69).

If $\lambda_1[m] \geq 0$ *then the only nonnegative solution of problem* (10.54) *is the null function. Consequently,* $u_g(t) \to 0$ *as* $t \to \infty$, *a.e. and in* $L^2(\Omega)$, *for every nonnegative* $g \in L^\infty(\Omega) \cap H^1(\Omega)$.

Proof Assume that $U \in H^1(\Omega) \cap L^\infty(\Omega)$, $U \geq 0$ is a weak solution of (10.54):

$$\int_\Omega \nabla U \cdot \nabla v\, d\mathbf{x} = \int_\Omega f(\mathbf{x}, U)v\, d\mathbf{x}, \qquad \forall v \in H^1(\Omega). \tag{10.73}$$

Choosing $v = U$ and using (10.71) and the definition of $\lambda_1[m]$ we obtain

$$0 = \int_\Omega \left\{|\nabla U|^2 - f(\mathbf{x}, U)U\right\} d\mathbf{x} \geq \int_\Omega \left\{|\nabla U|^2 - m(\mathbf{x})U^2\right\} d\mathbf{x} \geq \lambda_1[m] \int_\Omega U^2\, d\mathbf{x} \geq 0.$$

This forces

$$\int_\Omega f(\mathbf{x}, U)U\, d\mathbf{x} = \int_\Omega m(\mathbf{x})U^2\, d\mathbf{x}$$

and finally $U = 0$, again by (10.71). □

Case $\lambda_1[m] < 0$. In this case we can exclude extinction by constructing a positive subsolution. For simplicity, we now restrict to the case when $\Omega \subset \mathbb{R}^n$, $n \leq 3$, is a

smooth bounded domain and $f \in C^\infty(\overline{\Omega} \times [0,\infty))$, so that all solutions are smooth and the strong maximum principle holds.

Lemma 10.32 *Let $\lambda_1[m] < 0$, and assume that $g > 0$ on $\overline{\Omega}$. There exists $\bar\varepsilon > 0$ such that, for every $0 < \varepsilon \le \bar\varepsilon$, $v_\varepsilon(\mathbf{x}) = \varepsilon\varphi_1(\mathbf{x})$ is a strictly positive, time-independent subsolution of problem* (10.53).

Proof By assumption $m \in C^\infty(\overline{\Omega})$. Lemma 10.27, with $f(\mathbf{x},u) = (m(\mathbf{x}) - \lambda_1[m])u$, implies that $\varphi_1 \in C^\infty(\overline{\Omega})$ and $\varphi_1 > 0$ in $\overline{\Omega}$ (actually, the regularity already follows by Theorem 8.31, p. 530). We can choose its normalization in such a way that $\|\varphi_1\|_{L^\infty(\Omega)} = 1$. On the one hand it is clear that if $\varepsilon \le \varepsilon_1 \equiv \min_{\overline{\Omega}} g$ then $v_\varepsilon \le g$. On the other hand, since f is smooth and $\lambda_1[m] < 0$, we can find $\varepsilon_2 \le 1$ such that

$$f(\mathbf{x},u) \ge (\partial_u f(\mathbf{x},0) + \lambda_1[m])u = (m(\mathbf{x}) + \lambda_1[m])u \qquad \text{for all } 0 \le u \le \varepsilon_2 \tag{10.74}$$

(it suffices to choose $\varepsilon_2 = -2\lambda_1[m]/\|\partial_{uu} f\|_{L^\infty(\Omega\times(0,1))}$). Taking $\varepsilon \le \varepsilon_2$ (10.74) implies

$$-D\Delta v_\varepsilon = (m(\mathbf{x}) + \lambda_1[m])v_\varepsilon \le f(\mathbf{x}, v_\varepsilon).$$

The lemma follows taking $\bar\varepsilon = \min\{\varepsilon_1, \varepsilon_2\}$. □

As a byproduct of the previous lemma we obtain that, *in case $\lambda_1[m] < 0$, there exists a nontrivial steady state U* satisfying

$$0 < \bar\varepsilon\varphi_1 < U \le M \qquad \text{in } \overline{\Omega}.$$

The last step consists in showing that U is unique.

Lemma 10.33 *Let U_1, U_2 be solutions of* (10.54)*, positive in $\overline{\Omega}$. Then $U_1 = U_2$.*

Proof By Lemma 10.27 we have that both U_1 and U_2 are smooth and strictly positive. This allows to test the equation for U_1 with $-U_2^2/U_1$, obtaining

$$-\int_\Omega f(\mathbf{x},U_1)\frac{U_2^2}{U_1} = -\int_\Omega \nabla U_1 \cdot \nabla \frac{U_2^2}{U_1} = \int_\Omega \left\{-2\frac{U_2}{U_1}\nabla U_1 \cdot \nabla U_2 + \frac{U_2^2}{U_1^2}|\nabla U_1|^2\right\}. \tag{10.75}$$

On the other hand, testing the equation for U_2 with U_2 itself,

$$\int_\Omega f(\mathbf{x},U_2)U_2 = \int_\Omega |\nabla U_2|^2. \tag{10.76}$$

Summing up (10.75) and (10.76) we get

$$\int_\Omega \left\{\frac{f(\mathbf{x},U_2)}{U_2} - \frac{f(\mathbf{x},U_1)}{U_1}\right\} U_2^2 = \int_\Omega \left|\nabla U_2 - \frac{U_2}{U_1}\nabla U_2\right|^2 \ge 0, \tag{10.77}$$

and exchanging the role of U_1 and U_2,

$$\int_\Omega \left\{\frac{f(\mathbf{x},U_1)}{U_1} - \frac{f(\mathbf{x},U_2)}{U_2}\right\} U_1^2 \ge 0. \tag{10.78}$$

Summing (10.77) and (10.78) we finally obtain

$$\int_\Omega \left\{\frac{f(\mathbf{x},U_1)}{U_1} - \frac{f(\mathbf{x},U_2)}{U_2}\right\} (U_1^2 - U_2^2) \ge 0,$$

and (10.69) forces $U_1 = U_2$ on $\overline{\Omega}$. □

By Lemma 10.32 and Theorem 10.22 we obtain that, in case $\lambda_1[m] < 0$, for every strictly positive initial datum g the solution u_g converges to the unique positive steady state U as $t \to \infty$. Actually, by regularity, this is true for any nonnegative, nontrivial g: indeed by Corollary 10.26 we have that, for any such g, $u_g(\mathbf{x}, 1) > 0$ in $\overline{\Omega}$. Then we can apply Lemma 10.32 to $v(\mathbf{x}, t) = u_g(\mathbf{x}, t + 1)$ and obtain

$$\lim_{t\to\infty} u_g(t) = \lim_{t\to\infty} v(t) = U,$$

for every $g \geq 0$, $g \not\equiv 0$. Resuming, we have shown the following counterpart to Theorem 10.31:

Theorem 10.34 (Persistence) *Let $\Omega \subset \mathbb{R}^n$ be a smooth bounded domain and $f \in C^\infty(\overline{\Omega} \times [0, \infty))$ satisfy (10.49), (10.50) and (10.69).*

If $\lambda_1[m] < 0$ then there exists a unique nonnegative, nontrivial solution U of problem (10.54). Moreover $U \in C^\infty(\overline{\Omega})$, $U > 0$ in $\overline{\Omega}$, and $u_g(t) \to U$ as $t \to \infty$, a.e. and in $L^2(\Omega)$, for every nonnegative, nontrivial $g \in L^\infty(\Omega) \cap H^1(\Omega)$.

Remark 10.35 (Linear stability of the null solution) The above results can be interpreted in terms of linear stability/instability of the trivial solution. The null function $U \equiv 0$, as solution of the nonlinear problem (10.54), is said *linearly stable* if the solution of the linearized problem

$$\begin{cases} v_t - D\Delta v = \partial_u f(\mathbf{x}, 0)v = m(\mathbf{x})v & \text{in } Q_T \\ \partial_\nu v = 0 & \text{on } S_T \\ v(0) = g & \text{in } \Omega \end{cases}$$

goes to 0, as $t \to \infty$, for every initial datum g. Arguing as in Sect. 8.3.4, p. 509, we infer that this is the case when $\lambda_1[m] > 0$.

• *The role of habitat and motility.* The principal eigenvalue $\lambda_1[m]$ is sometimes referred to as the *survival threshold* for the population. It is clear from the definition (10.72) that such threshold does not depend only on m, but also on D and Ω, and sometimes it is convenient to write $\lambda_1[m, D]$, or $\lambda_1[m, D, \Omega]$, to stress such dependence. In particular, if Ω is fixed, *to decrease $\lambda_1[m, D]$ (and thus favor survival) one has either to lower D, or to increase m*. Actually, we can sharpen the analysis in order to clarify the role of the habitat and of the motility.

Proposition 10.36 *Let $m \in C^\infty(\overline{\Omega})$. Then:*

1. *(pointwise hostile habitat) if $m \leq 0$ in $\overline{\Omega}$ then $\lambda_1[m, D] \geq 0$, for every $D > 0$;*
2. *(favorable on average habitat) if*

$$\int_\Omega m(\mathbf{x})\, d\mathbf{x} \geq 0 \qquad \text{and} \qquad m \text{ is not identically } 0$$

then $\lambda_1[m, D] < 0$, for every $D > 0$;

3. if

$$\max_{\overline{\Omega}} m > 0 \qquad \textit{and} \qquad \int_\Omega m(\mathbf{x})\, d\mathbf{x} < 0 \tag{10.79}$$

then there exists $D^* > 0$, depending on m and Ω, such that

a. if $0 < D < D^*$ then $\lambda_1[m, D] < 0$;
b. if $D \geq D^*$ then $\lambda_1[m, D] \geq 0$.

Proof First notice that, for every m, D,

$$\lambda_1[m, D] = D\frac{\int_\Omega |\nabla\varphi_1|^2\, d\mathbf{x}}{\int_\Omega \varphi_1^2\, d\mathbf{x}} - \frac{\int_\Omega m(\mathbf{x})\varphi_1^2 d\mathbf{x}}{\int_\Omega \varphi_1^2\, d\mathbf{x}} \geq -\max_{\overline{\Omega}} m, \tag{10.80}$$

so that 1 follows by the definition of λ_1.

On the other hand, using a constant function u in (10.72) we infer, for every m, D,

$$\lambda_1[m, D] \leq -\frac{1}{|\Omega|}\int_\Omega m(\mathbf{x})d\mathbf{x} \tag{10.81}$$

and 2 follows as well, as long as m has strictly positive average. To complete the proof of 2 we have to discuss the case $\lambda_1[m, D] = \int_\Omega m(\mathbf{x})d\mathbf{x} = 0$. Using the equation for the corresponding eigenfunction φ_1, which is strictly positive in $\overline{\Omega}$, we get

$$0 = \int_\Omega m(\mathbf{x})d\mathbf{x} = -D\int_\Omega \frac{\Delta\varphi_1}{\varphi_1}d\mathbf{x} = D\int_\Omega \frac{|\nabla\varphi_1|^2}{\varphi_1^2}d\mathbf{x}.$$

Thus φ_1 is constant and, using again the equation, m is constant too. We deduce that m is identically zero and 2 follows.

The proof of 3. is more elaborated. Let m be fixed, satisfying (10.79) and denote with $\lambda_1[D]$, φ_D the corresponding eigenpair, with normalization $\|\varphi_D\|_{L^2(\Omega)} = 1$. Then (10.80), (10.81) imply

$$D\int_\Omega |\nabla\varphi_1|^2\, d\mathbf{x} \leq \lambda_1[m, D] + \max_{\overline{\Omega}} m \leq \max_{\overline{\Omega}} m - \frac{1}{|\Omega|}\int_\Omega m(\mathbf{x})d\mathbf{x}. \tag{10.82}$$

We are going to show that:

i) the map $D \mapsto \lambda_1[D]$ is nondecreasing and continuous for $D > 0$;
ii) $\lim_{D\to 0^+} \lambda_1[D] < 0$;
iii) $\lim_{D\to\infty} \lambda_1[D] > 0$.

Then 3 will follow by defining $D^* = \sup\{D : \lambda_1[D] < 0\} = \inf\{D : \lambda_1[D] > 0\}$.

To prove i), notice that, for every D_1, D_2,

$$\lambda_1[D_1] \leq \int_\Omega \left\{D_1\left|\nabla\varphi_{D_2}\right|^2 - m(\mathbf{x})\varphi_{D_2}^2\right\} d\mathbf{x} = \lambda_1[D_2] - (D_2 - D_1)\int_\Omega \left|\nabla\varphi_{D_2}\right|^2 d\mathbf{x}. \tag{10.83}$$

Thus if $D_1 > D_2$ then $\lambda_1[D_1] < \lambda_1[D_2]$. On the other hand, (10.83) and (10.82) imply that

$$|\lambda_1[D_1] - \lambda_1[D_2]| \leq \frac{1}{D_2}\left[\max_{\overline{\Omega}} m - \frac{1}{|\Omega|}\int_\Omega m(\mathbf{x})d\mathbf{x}\right] |D_1 - D_2|,$$

so that the map $D \mapsto \lambda_1[D]$ is locally Lipschitz.

To prove ii) we notice that by (10.79) we can find a domain $\omega \subset \Omega$ and a constant $\delta > 0$ such that $m \geq \delta > 0$ in ω. Then taking $z \in \mathcal{D}(\omega)$ with $\|z\|_{L^2(\omega)} = 1$,

$$\lambda_1[D] \le \int_\omega \left\{D\,|\nabla z|^2 - m(\mathbf{x})z^2\right\} d\mathbf{x} \le D\int_\omega |\nabla z|^2\,d\mathbf{x} - \delta < 0$$

for D sufficiently small.

Finally, we prove iii). We already know that $\|\varphi_D\|_{L^2(\Omega)} = 1$ and, by (10.82), $\|\nabla\varphi_D\|_{L^2(\Omega)} \to 0$ as $D \to \infty$. Then we can argue e.g. as in the proof of Thm. 7.87, p. 479 to show that φ_D converges to the constant $|\Omega|^{-1/2}$, strongly in $H^1(\Omega)$, as $D \to \infty$. This yields

$$\lambda_1[D] \le -\int_\Omega m(\mathbf{x})\varphi_D^2 d\mathbf{x} \to \frac{1}{|\Omega|}\int_\Omega m(\mathbf{x})d\mathbf{x} < 0$$

by (10.79). This concludes the proof of the proposition. □

Corollary 10.37 *Consider the reaction-diffusion problem*

$$\begin{cases} u_t - D\Delta u = m(\mathbf{x})u - \beta(\mathbf{x})u^2 & \text{in } Q_T = \Omega\times(0,T) \\ \partial_\nu u = 0 & \text{on } S_T = \partial\Omega\times(0,T) \\ u(0) = g & \text{in } \Omega, \end{cases}$$

where $m, \beta \in C^\infty(\overline{\Omega})$ and $\beta > 0$ in the closure of the bounded smooth domain $\Omega \subset \mathbb{R}^n$. Then

1. *if m is pointwise hostile then u undergoes extinction, for every $g \ge 0$;*
2. *if m is favorable on average then u persists, for every nontrivial $g \ge 0$;*
3. *in the remaining cases, persistence holds true if and only if D is sufficiently small.*

Remark 10.38 (Effect of spatial dilation on the principal eigenvalue) Let $d > 0$. Given Ω, m as above one may consider an eigenvalue problem in the dilated domain $\tilde{\Omega} = d\Omega$, with coefficient $\tilde{m}(\mathbf{x}) = m(\mathbf{x}/d)$. Notice that m and $\tilde{m}$ have the same maximum and the same average. Thus if either case 1 or 2 of Proposition 10.36 applies to the original domain, the same is true for the scaled one. This is no longer true for case 3. Indeed, it is straightforward to check that if φ_1 achieves the minimum in (10.72) then $\tilde{\varphi}_1(\mathbf{x}) = \varphi_1(\mathbf{x}/d)$ minimizes an analogous Rayleigh quotient, and finally

$$\lambda_1[\Omega, m, D] = \lambda_1\left[\tilde{\Omega}, \tilde{m}, \frac{D}{d^2}\right].$$

In terms of the model, this roughly says that increasing the size of the domain has the same (positive) effect, on the survival threshold, than decreasing the motility. We already noticed this effect in the simple one-dimensional, linear reaction-diffusion model discussed in Sec. 2.5.5, p. 66.

Remark 10.39 The arguments we developed can be rephrased to deal with Dirichlet or Robin boundary conditions. In this cases, the *survival threshold* is encoded by a Dirichlet or Robin principal eigenvalue, respectively. From this point of view one can see that, for the same habitat, **the Neumann conditions are the most favorable while the Dirichlet ones are the most hostile**, see Problem 10.13.

Problems

10.1 Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain and $g : \mathbb{R} \to \mathbb{R}$ be bounded and Lipschitz continuous, with Lipschitz constant K and $g(0) = 0$. Consider the following problem:

$$\begin{cases} -\Delta u + u = 1 & \text{in } \Omega \\ \partial_\nu u = g(u) & \text{on } \partial\Omega. \end{cases} \tag{10.84}$$

Give a weak formulation and show that there exists a weak solution $\overline{u} \in H^2(\Omega)$.

[*Hint.* Fix $w \in H^1(\Omega)$. Check that $g(w)$ has a trace in $H^{1/2}(\partial\Omega)$ and

$$\|g(w)\|_{H^{1/2}(\partial\Omega)} \leq C_0(n, \Omega) K \, \|(w)\|_{H^1(\Omega)} .$$

Let $u = T(w)$ be the unique solution of the problem

$$\begin{cases} -\Delta u + u = 1 & \text{in } \Omega \\ \partial_\nu u = g(w) & \text{on } \partial\Omega. \end{cases}$$

Use the regularity theory for elliptic equations and Rellich Theorem 7.86, p. 478, to show that $T : H^1(\Omega) \to H^1(\Omega)$ is compact and continuous. Use Schauder Theorem 6.88, p. 413, to conclude.]

10.2 Let $\Omega \subset \mathbb{R}^3$ be a bounded, smooth domain and $f \in L^2(\Omega)$. Consider the following semilinear problem:

$$\begin{cases} -\Delta u + u^3 = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{10.85}$$

a) Give a weak formulation of the problem (recall that $H_0^1(\Omega) \hookrightarrow L^6(\Omega)$, for $n = 3$).
b) Show that there exists a unique solution $z \in H^2(\Omega) \cap H_0^1(\Omega)$ of the problem

$$\begin{cases} -\Delta z + w^3 = f & \text{in } \Omega \\ z = 0 & \text{on } \partial\Omega, \end{cases}$$

for every fixed $w \in H_0^1(\Omega)$, and derive a stability estimate.
c) Use the Leray-Schauder Theorem 6.89, p. 413, to show that there exists a solution $u \in H_0^1(\Omega)$ of problem (10.85).
d) Show that the solution is unique.

10.3 Let $\Omega \subset \mathbb{R}^3$ be a bounded, smooth domain and $f \in L^2(\Omega)$. Consider the following semilinear problem:

$$\begin{cases} -\Delta u - u^3 = f & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega. \end{cases} \tag{10.86}$$

a) As in Problem 10.2, give a weak formulation of the problem and show that there exists a unique solution $z = T(w)$ of the problem

$$\begin{cases} -\Delta z - w^3 = f & \text{in } \Omega \\ z = 0 & \text{on } \partial\Omega, \end{cases}$$

for every fixed $w \in H_0^1(\Omega)$, where $T : H_0^1(\Omega) \to H_0^1(\Omega)$ is continuous and compact.
b) Prove that there exist positive constants A, B such that, if $u = sT(u)$, $0 \leq s \leq 1$, then

$$\|u\|^2_{H_0^1(\Omega)} \leq A\|u\|^4_{H_0^1(\Omega)} + B\|f\|_{L^2(\Omega)}\|u\|_{H_0^1(\Omega)}.$$

c) Use the Leray-Schauder Theorem 6.89, p. 413, to show that there exists a solution $u \in H_0^1(\Omega)$ of problem (10.86)[10].

[*Hint.* c) Show the following elementary property: if a, b, t are positive numbers, b is sufficiently small and $t - at^3 \le b$ then there exists $M > 0$, depending only on a and b, such that either $t < M$ or $t > M$; it is possible to choose $M = (3a)^{-1/2}$ and $b < 2M/3$.]

10.4 Let $B = \{\mathbf{x} \in \mathbb{R}^n : |\mathbf{x}| < 1\}$ e $a : \mathbb{R}^n \to \mathbb{R}$, $a \in C^\infty(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$. Consider the following problem:

$$\begin{cases} -\Delta u + a(\nabla u) = 1 & \text{in } B \\ \partial_\nu u + u = 0 & \text{on } \partial B. \end{cases} \tag{10.87}$$

a) Show that there exists a solution $\overline{u} \in H^2(B)$.
b) Let $w \in C^\infty(\overline{B})$ be a solution of the *linear problem*

$$\begin{cases} -\Delta z + \mathbf{b}(\mathbf{x}) \cdot \nabla z = 0 & \text{in } B \\ \partial_\nu z + z = 0 & \text{on } \partial B. \end{cases} \tag{10.88}$$

where $b(\mathbf{x}) \in C^\infty(\overline{B}; \mathbb{R}^n)$. Show that $z = 0$. Deduce that the nonlinear problem (10.89) has the only solution $\overline{u} \in H^2(B)$.

10.5 Let $\Omega \subset \mathbb{R}^3$ be a bounded, smooth domain and $f, g \in C^\infty(\overline{\Omega})$. Consider the following Dirichlet problem.

$$\begin{cases} -\Delta u + u = \dfrac{|v|}{1 + |u| + |v|} + f & \text{in } \Omega \\ -\Delta v + v = \dfrac{|u|}{1 + |u| + |v|} + g & \text{in } \Omega \\ u = v = 0 & \text{on } \partial\Omega. \end{cases} \tag{10.89}$$

Give a weak formulation of the problem and prove that:

a) (10.89) has a weak solution $(u, v) \in H_0^1(\Omega) \times H_0^1(\Omega)$.
b) If (u, v) is a weak solution of (10.89) then $(u, v) \in H^3(\Omega) \times H^3(\Omega)$.
c) If (u, v) is a weak solution of (10.89) and $f \ge 0$, $g \ge 0$ in Ω then $u \ge 0$, $v \ge 0$ in Ω. Moreover $(u, v) \in C^\infty(\overline{\Omega}) \times C^\infty(\overline{\Omega})$, that is (u, v) is a classical solution of (10.89).
d) If (u, v) is a weak solution of (10.89) and $0 \le f \le g$ then $u \le v$ in Ω. In particular, if $f \equiv g \ge 0$ then $u = v$ and the solution of problem (10.89) is unique.

[*Hint.* a) Let $X = L^2(\Omega) \times L^2(\Omega)$ with the norm $\|(u, v)\|_X^2 = \|u\|_{L^2(\Omega)}^2 + \|v\|_{L^2(\Omega)}^2$.
For fixed $(u_0, v_0) \in X$, solve the linear uncoupled problem

$$\begin{cases} -\Delta u + u = \dfrac{|v_0|}{1 + |u_0| + |v_0|} + f & \text{in } \Omega \\ -\Delta v + v = \dfrac{|u_0|}{1 + |u_0| + |v_0|} + g & \text{in } \Omega \\ u = v = 0 & \text{on } \partial\Omega. \end{cases} \tag{10.90}$$

Let $T : X \to X$ the solution map so defined. Show that T is compact and continuous. Use Schauder Theorem 6.88, p. 413, to prove that T has a fixed point.
b) Use the regularity theory for elliptic equations.
c) Use maximum principle, get rid of the absolute value signs and use bootstrapping.]

[10] Referring to Problem 10.2,d), in this case uniqueness is not feasible: for instance, if $f \equiv 0$ then (10.86) admits infinitely many solutions, see: A. Ambrosetti, P. Rabinowitz, "Dual variational methods in critical point theory and applications", *J. Functional Analysis* 14 (1973).

10.6 Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain. Consider the system:

$$\begin{cases} -\Delta\varphi + u\varphi = 1 & \text{in } \Omega \\ -\operatorname{div}(e^{\varphi}\nabla u) = 0 & \text{in } \Omega \\ \varphi = 0,\ u = g & \text{on } \partial\Omega \end{cases} \tag{10.91}$$

where $g \in H^{1/2}(\partial\Omega)$ is the trace of $\tilde{g} \in H^1(\Omega)$. Assume that $0 < \alpha \le g \le \beta < \infty$. on $\partial\Omega$. Define

$$X = \left\{u \in L^2(\Omega) : \alpha \le u \le \beta \text{ a.e. in } \Omega\right\}.$$

a) Give a weak formulation of problem (10.91).
b) Prove the existence of a weak solution $(u, \varphi) \in H^1(\Omega) \times H_0^1(\Omega)$, by following the steps below.

1. Fix $\tilde{u} \in X$ and construct a map $T_1 : X \to H_0^1(\Omega)$, where $\varphi = T_1(\tilde{u})$ is the unique solution of the problem
$$-\Delta\varphi + \tilde{u}\varphi = 1 \text{ in } \Omega,\ \varphi = 0 \text{ on } \partial\Omega. \tag{10.92}$$
In particular, derive the estimate $\|\varphi\|_{H_0^1(\Omega)} \le C_1$, with C_1 *independent of* $\tilde{u}$.
2. Using that
$$v = \varphi^- = \max\{-\varphi, 0\}$$
is an admissible test function in the weak formulation of (10.92), show that $\varphi \ge 0$ a.e. in Ω. Similarly, using
$$v = (\varphi - 1/\alpha)^+ = \max\{\varphi - 1/\alpha, 0\},$$
show that $\varphi \le 1/\alpha$ a.e. in Ω.
3. Let $\varphi = T_1(\tilde{u})$ be the the unique solution of problem (10.92). Construct a map $T_2 : H_0^1(\Omega) \to X$, defining $u = T_2(\varphi)$ as the solution of the problem.
$$-\operatorname{div}(e^{\varphi}\nabla u) = 0 \text{ in } \Omega,\ \ u = g \text{ on } \partial\Omega. \tag{10.93}$$
In particular, prove the following estimates: $\|\nabla u\|_{L^2(\Omega)} \le C_2$, with C_2 independent of u and $\alpha \le u \le \beta$ a.e. in Ω.
4. Prove that the operator
$$T = T_2 \circ T_1 : X \to X$$
is *continuous in* $L^2(\Omega)$, i.e. if $\{\tilde{u}_m\} \subset X$ and $\tilde{u}_m \to \tilde{u}$ in $L^2(\Omega)$, then $T(\tilde{u}_m) \to T(\tilde{u})$ in $L^2(\Omega)$.
5. Show that $\overline{T(X)}$ is compact in X. Deduce that T has a fixed point u_f, that is
$$u_f = T_2(T_1(u_f)).$$
Check that if $\varphi_f = T_1(u_f)$, then (u_f, φ_f) is a solution in $H^1(\Omega) \times H_0^1(\Omega)$ of (10.91).

10.7 (*Distributed observation and control, Neumann conditions*) Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain and Ω_0 an *open* (nonempty) subset of Ω. Set $V = H^1(\Omega)$, $H = L^2(\Omega)$, and consider the following control problem:

Minimize the cost functional

$$J(u, z) = \frac{1}{2}\int_{\Omega_0}(u - u_d)^2 d\mathbf{x} + \frac{1}{2}\int_{\Omega} z^2 d\mathbf{x}$$

over $(u, z) \in V \times H$, with state system

$$\begin{cases} \mathcal{E}u = -\Delta u + a_0 u = z & \text{in } \Omega \\ \partial_{\nu} u = g & \text{on } \partial\Omega \end{cases} \tag{10.94}$$

where a_0 is a positive constant, $g \in L^2(\partial\Omega)$ and $z \in H$.

a) Show that there exists a unique minimizer.
b) Write the optimality conditions: adjoint problem and Euler equations.

[*Hint.* a) Follow the proof of Theorem 10.12, observing that, if $u\,[z]$ is the solution of (10.94) the map $z \mapsto u\,[z] - u\,[0]$ is linear. Then write

$$\tilde{J}(z) = \tfrac{1}{2}\int_{\Omega_0} (u\,[z] - u\,[0] + u[0] - u_d)^2 d\mathbf{x} + \tfrac{1}{2}\int_{\Omega} z^2 d\mathbf{x}$$

and adjust the bilinear form (10.40) accordingly.]

[*Answer.* b) The adjoint problem is ($\mathcal{E} = \mathcal{E}^*$)

$$\begin{cases} -\Delta p + a_0 p = (u - z_d)\chi_{\Omega_0} & \text{in } \Omega \\ \partial_{\boldsymbol{\nu}} p = 0 & \text{on } \partial\Omega. \end{cases}$$

Where χ_{Ω_0} is the characteristic function of Ω_0. The Euler equation is: $p + z = 0$ in $L^2(\Omega)$.]

10.8 (*Boundary observation and distributed control, Dirichlet conditions*) Let $\Omega \subset \mathbb{R}^n$ be a bounded, smooth domain. Consider the following control problem:

Minimize the cost functional

$$J(u, z) = \frac{1}{2}\int_{\partial\Omega} (\partial_{\boldsymbol{\nu}} u - u_d)^2 d\sigma + \frac{\beta}{2}\int_{\Omega} z^2 d\mathbf{x}$$

over $(u, z) \in (H_0^1(\Omega) \cap H^2(\Omega)) \times L^2(\Omega)$, with state system

$$\begin{cases} -\Delta u + \mathbf{c}\cdot\nabla u = z, & \text{in } \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

where $u_d \in H^{1/2}(\partial\Omega)$, $\mathbf{c}$ is a constant vector and $f \in L^2(\Omega)$.

a) Show that, by elliptic regularity, $\tilde{J}(z) = J(u[z], z)$ is well defined and that

$$\tilde{J}'(z)w = \int_{\partial\Omega} (\partial_{\boldsymbol{\nu}} u[z] - u_d)\partial_{\boldsymbol{\nu}} u[w] d\sigma + \beta\int_{\Omega} zw d\mathbf{x} \qquad \forall w \in L^2(\Omega).$$

b) Mimicking the proof of Theorem 10.12, show that there exists a unique optimal control.
c) Show that the optimality conditions are

$$\begin{cases} -\Delta u^* + \mathbf{c}\cdot\nabla u^* = z^* \ \text{ in } \Omega & u^* = 0 \qquad\qquad \text{on } \partial\Omega \\ -\Delta p^* - \mathbf{c}\cdot\nabla p^* = 0 \quad \text{in } \Omega & p^* = \partial_{\boldsymbol{\nu}} u^* - u_d \quad \text{on } \partial\Omega \\ p^* + \beta z^* = 0 \quad \text{a.e. in } \Omega. \end{cases}$$

10.9 Consider the problem

$$\begin{cases} -\Delta u = f(\mathbf{x}, u) & \text{in } \Omega \\ u = g & \text{on } \partial\Omega, \end{cases} \tag{10.95}$$

where $\Omega \subset \mathbb{R}^n$ is a bounded Lipschitz domain, $f \in C^1(\overline{\Omega} \times \mathbb{R})$ and $g \in H^{1/2}(\partial\Omega)$. A *weak subsolution* of (10.95) is a function $u_* \in H^1(\Omega)$ such that $u_* \le g$ on $\partial\Omega$ and

$$\int_{\Omega} \nabla u_* \cdot \nabla u \, d\mathbf{x} \le \int_{\Omega} f(\mathbf{x}, u_*) v \, d\mathbf{x} \qquad \forall v \in H_0^1(\Omega),\ v \ge 0 \text{ a.e. in } \Omega,$$

and the notion of weak supersolution can be defined analogously.

Assume that there exist a weak subsolution u_* and a weak supersolution u^* of (10.95) satisfying

$$a \le u_* \le u^* \le b, \qquad \text{a.e. in } \Omega,\ a, b \in \mathbb{R}$$

(in particular $a \le g \le b$ a.e. on $\partial\Omega$).

a) construct a monotone sequence of functions $u_k \in H^1(\Omega)$ weakly solving

$$\begin{cases} -\Delta u_{k+1} + L u_{k+1} = f(\mathbf{x}, u_k) & \text{in } \Omega \\ u_{k+1} = g & \text{on } \partial\Omega, \end{cases}$$

for a suitable choice of u_0 and of the constant L;

b) Show that the sequence converges, in a suitable sense, to a solution u of problem (10.95) satisfying $u_* \le u \le u^*$ a.e. in Ω.

10.10 Consider the problem

$$\begin{cases} u_t - \Delta u = f(\mathbf{x}, u) & \text{in } Q_T \\ \partial_\nu u = 0 & \text{on } S_T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega, \end{cases}$$

where $g \in L^\infty(\Omega) \cap H^1(\Omega)$ and $f \in C^1(\overline{\Omega} \times \mathbb{R})$ satisfies $|f(\mathbf{x}, u)| \le C_1 + C_2|u|$, $C_1, C_2 \in \mathbb{R}$.

a) Find upper and lower solutions, homogeneous in space, solving suitable linear problems.

b) Deduce existence and uniqueness of a solution, defined for all positive times.

10.11 Use the method of sub/supersolutions to solve the problem

$$\begin{cases} u_t - \Delta u = u^2 + h(\mathbf{x}) & \text{in } Q_T \\ \partial_\nu u = 0 & \text{on } S_T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega, \end{cases}$$

where $g \in L^\infty(\Omega) \cap H^1(\Omega)$, $h \in L^\infty(\Omega)$, and both g and h are nonnegative and smaller than 1, a.e. in Ω. For which times is it possible to guarantee the existence of a solution?

[*Answer.* For every $T < \pi/4$.]

10.12 *(Bistable reaction)* Let $\vartheta \in (0, 1)$ and consider the problem

$$\begin{cases} u_t - D\Delta u = u(1-u)(\vartheta - u) & \text{in } Q_T \\ \partial_\nu u = 0 & \text{on } S_T \\ u(\mathbf{x}, 0) = g(\mathbf{x}) & \text{in } \Omega, \end{cases}$$

where $g \in L^\infty(\Omega) \cap H^1(\Omega)$. Check that $f(u) = u(1-u)(\vartheta - u)$ satisfies the assumptions of Theorem 10.28. Deduce that

a) if $0 \le g \le \vartheta$ a.e. in Ω and $g < \vartheta$ on a set of positive measure then $u_g(\cdot, t) \to 0$, as $t \to \infty$, uniformly on $\overline{\Omega}$;

b) if $g \ge \vartheta$ a.e. in Ω and $g > \vartheta$ on a set of positive measure then $u_g(\cdot, t) \to 1$, as $t \to \infty$, uniformly on $\overline{\Omega}$.

[*Hint.* Find suitable sub/supersolutions homogeneous in space.]

10.13 Let $m \in C(\overline{\Omega})$, $a \in C(\partial\Omega)$, $a \ge 0$, and consider the Robin eigenvalue problem

$$\begin{cases} -D\Delta u - m(\mathbf{x})u = \lambda u & \text{in } \Omega \\ \partial_\nu u + a(\mathbf{x})u = 0 & \text{on } \partial\Omega, \end{cases}$$

a) Write a formula for the principal eigenvalue $\lambda^{\mathrm{Rob}}[m, a]$ as the minimization of a suitable Rayleigh quotient. Deduce that the Neumann principal eigenvalue for the same equation satisfies $\lambda^{\mathrm{Neu}}[m] = \lambda^{\mathrm{Rob}}[m, 0]$ and that the Dirichlet one satisfies, formally, $\lambda^{\mathrm{Dir}}[m] = \lambda^{\mathrm{Rob}}[m, +\infty]$.

b) Show that $a_1 \le a_2$ on $\partial\Omega$ implies $\lambda^{\mathrm{Rob}}[m, a_1] \le \lambda^{\mathrm{Rob}}[m, a_2]$, with strict inequality as long as a_1 and a_2 do not coincide.

c) Show that $\lambda^{\mathrm{Dir}}[m] < \lambda^{\mathrm{Rob}}[m, a] < \lambda^{\mathrm{Neu}}[m]$, for every nontrivial a.

Chapter 11
Systems of Conservation Laws

11.1 Introduction

The motion of a compressible fluid with negligible viscosity (gas dynamics) or the propagation of waves in shallow waters are typical phenomena leading to systems of first order conservation laws. This chapter constitutes an introduction to the basic concepts in this important area of nonlinear PDEs. Let us introduce the following notations:

$$\mathbf{u} : \mathbb{R}^n \times [0,T] \to \mathbb{R}^m, \qquad \mathbf{F} : U \subseteq \mathbb{R}^m \to \mathcal{M}_{m,n}$$

where $\mathcal{M}_{m,n}$ is the set of $m \times n$ matrices. In this context, $\mathbf{u}$ is a *state variable,* and we write it as a column vector or as a point in $\mathbb{R}^m$. If $\Omega \subset \mathbb{R}^n$ is a bounded domain, the equation

$$\frac{d}{dt}\int_\Omega \mathbf{u}\, d\mathbf{x} = -\int_{\partial\Omega} \mathbf{F}(\mathbf{u}) \cdot \boldsymbol{\nu}\, d\sigma \tag{11.1}$$

expresses the balance between the rate of variation of $\int_\Omega \mathbf{u}\, d\mathbf{x}$ and the *inward* flux of $\mathbf{u}$ through $\partial\Omega$, governed by the *flux function* $\mathbf{F}$. As usual, $\boldsymbol{\nu}$ denotes the outward unit normal to $\partial\Omega$. Using Gauss formula, we rewrite (11.1) in the form

$$\int_\Omega [\mathbf{u}_t + \operatorname{div} \mathbf{F}(\mathbf{u})]\, d\mathbf{x} = \mathbf{0}. \tag{11.2}$$

If (11.2) holds in an arbitrary region Ω, we deduce the *conservation law*

$$\mathbf{u}_t + \operatorname{div} \mathbf{F}(\mathbf{u}) = \mathbf{0}. \tag{11.3}$$

We shall limit ourselves to present the very basic concepts and results in the case $n = 1$. For the sake of simplicity, we develop the theory for $U = \mathbb{R}^m$. If $n = 1$, (11.3) becomes

$$\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = \mathbf{0}. \tag{11.4}$$

Under smoothness condition for $\mathbf{F}$, we can take the derivative with respect to x in the second term and write (11.4) into the *non-conservative form*

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$$\mathbf{u}_t + D\mathbf{F}(\mathbf{u})\mathbf{u}_x = \mathbf{0}, \tag{11.5}$$

where $D\mathbf{F}$ denotes the Jacobian matrix of $\mathbf{F}$. System (11.5) is a particular case of *first order quasilinear system,* of the form

$$\mathbf{u}_t + \mathbf{A}(x,t,\mathbf{u})\mathbf{u}_x = \mathbf{f}(x,t,\mathbf{u}), \tag{11.6}$$

where $\mathbf{A}$ is a $m \times m$ matrix and

$$\mathbf{f} : \mathbb{R} \times \mathbb{R} \times \mathbb{R}^m \to \mathbb{R}^m.$$

If $\mathbf{A}$ does not depend on $\mathbf{u}$ the system is *semilinear*; if, moreover,

$$\mathbf{f}(x,t,\mathbf{u}) = \mathbf{B}(x,t)\mathbf{u} + \mathbf{c}(x,t),$$

the system is *linear.*

For example, the change of variables

$$u_x = w_1 \quad \text{and} \quad u_t = w_2$$

transforms the wave equation $u_{tt} - c^2 u_{xx} = f$ into the linear system

$$\mathbf{w}_t + \mathbf{A}\mathbf{w}_x = \mathbf{f}, \tag{11.7}$$

where $\mathbf{w} = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix}$, $\mathbf{f} = \begin{pmatrix} 0 \\ f \end{pmatrix}$ and

$$\mathbf{A} = \begin{pmatrix} 0 & -1 \\ -c^2 & 0 \end{pmatrix}.$$

Note that the matrix $\mathbf{A}$ has the two real distinct eigenvalues $\lambda_\pm = \pm c$, with eigenvectors

$$\mathbf{v}_+ = \begin{pmatrix} 1 \\ -c \end{pmatrix} \quad \text{and} \quad \mathbf{v}_- = \begin{pmatrix} 1 \\ c \end{pmatrix}$$

normal to the two families of characteristics $x - ct = k$, $x + ct = k$, reflecting the *strict hyperbolic* nature of the wave equation. More precisely, we give the following definition[1].

Definition 11.1 We say that the system (11.6), or simply the matrix $\mathbf{A}$, is *strictly hyperbolic* if for every $(x,t,\mathbf{u}) \in \mathbb{R} \times \mathbb{R} \times \mathbb{R}^m$, the matrix $\mathbf{A}$ has m real and distinct eigenvalues:

$$\lambda_1(x,t,\mathbf{u}) < \lambda_2(x,t,\mathbf{u}) < \ \ldots < \lambda_m(x,t,\mathbf{u}).$$

If $\mathbf{A}$ is strictly hyperbolic, there exist m linearly independent corresponding eigenvectors

$$\mathbf{r}_1(x,t,\mathbf{u}),\ \mathbf{r}_2(x,t,\mathbf{u}), \ldots, \mathbf{r}_m(x,t,\mathbf{u})$$

[1] The notion of hyperbolicity is deeply analyzed e.g. in *M. Renardy, R. C. Rogers*, [18].

and if $\mathbf{R} = (\mathbf{r}_1 \mid \mathbf{r}_2 \mid \ldots \mid \mathbf{r}_m)$ is the matrix of the **column** vectors $\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_m$, we can diagonalize $\mathbf{A}$, obtaining

$$\mathbf{R}^{-1}\mathbf{A}\mathbf{R} = \mathbf{\Lambda} = \mathrm{diag}(\lambda_1,\ \lambda_2, \ldots, \lambda_m). \tag{11.8}$$

The **rows** of $\mathbf{R}^{-1}$, denoted by

$$\mathbf{l}_1^\top(x,t,\mathbf{u}),\ \mathbf{l}_2^\top(x,t,\mathbf{u}), \ldots, \mathbf{l}_m^\top(x,t,\mathbf{u}),$$

are *left eigenvectors of* $\mathbf{A}$, since $\mathbf{R}^{-1}\mathbf{A} = \mathbf{\Lambda}\mathbf{R}^{-1}$. Clearly

$$\mathbf{l}_j^\top \mathbf{r}_k = \delta_{jk} \tag{11.9}$$

where δ_{jk} denotes the Kronecker symbol.

The following two important examples will help us to illustrate the various concepts during the development of the theory, throughout the chapter.

Example 11.2 (Gas dynamics) Consider an inviscid gas in motion along a one-dimensional pipe. We assume that the effect of gravity is negligible. Using Eulerian variables, denote by x the coordinate along the axis of the pipe and by $\rho(x,t)$, $v(x,t)$, $p(x,t)$, $e(x,t)$, density, velocity, pressure and *specific internal* energy[2] of the gas, at the point x and time t, respectively. Setting

$$E = e + \frac{v^2}{2} \quad (\textit{total specific energy}),$$

for smooth flows, the motion is governed by the laws of conservation of mass, momentum and energy that give for the above four thermodynamic variables the following system of three equations:

$$\begin{cases} \dfrac{D\rho}{Dt} + \rho v_x = \rho_t + v\rho_x + \rho v_x = 0 & \text{(mass)} \\ \dfrac{Dv}{Dt} + \rho^{-1} p_x = v_t + vv_x + \rho^{-1} p_x = 0 & \text{(momentum)} \\ \dfrac{DE}{Dt} + \rho^{-1}(pv)_x = E_t + vE_x + \rho^{-1}(pv_x + vp_x) = 0 & \text{(energy).} \end{cases} \tag{11.10}$$

To close the model, we need a constitutive (or state) equation, for instance of the form $p = p(\rho, e)$. A special case is the *ideal gas law* $p = R\rho e/c_v$, where R is the universal gas constant and c_v is the specific heat at constant volume.
The third equation in (11.10) can be simplified by introducing the *specific entropy* s and taking into account the first law of Thermodynamics:

$$T\frac{Ds}{Dt} = \frac{De}{Dt} - \frac{p}{\rho^2}\frac{D\rho}{Dt}, \tag{11.11}$$

[2] Energy per unit mass.

where T is the absolute temperature. In fact, from the first two equations in (11.10) we get

$$v_x = -\frac{1}{\rho}\frac{D\rho}{Dt}, \quad p_x = -\rho\frac{Dv}{Dt}.$$

Substituting into the third equations we find

$$0 = \frac{De}{Dt} + v\frac{Dv}{Dt} + \frac{1}{\rho}pv_x + \frac{1}{\rho}vp_x = \frac{De}{Dt} - \frac{p}{\rho^2}\frac{D\rho}{Dt}$$

and from (11.11) we deduce

$$\frac{Ds}{Dt} = s_t + vs_x = 0.$$

Thus the system (11.10) can be written in the form :

$$\begin{cases} \rho_t + v\rho_x + \rho v_x = 0 \\ v_t + vv_x + \rho^{-1}p_x = 0 \\ s_t + vs_x = 0. \end{cases} \tag{11.12}$$

In this case, the equation of state is given in the form $p = p(\rho, s)$ with the properties $p_s > 0$ and

$$\begin{cases} p_\rho \equiv c^2 > 0 \\ p_{\rho\rho} + \frac{2}{\rho}p_\rho = \frac{2c}{\rho}(\rho c_\rho + c) > 0 \end{cases} \tag{11.13}$$

(see *E. Godlewski, P.-A. Raviart*, [46]). Note that c (> 0) has the dimension of a velocity. For an ideal polytropic gas, $p(\rho, s) = (\gamma - 1)e^{s/c_v}\rho^\gamma$, with $\gamma = c_p/c_v$, where c_p is the specific heat at constant pressure. Observing that

$$p_x = p_\rho\rho_x + p_s s_x$$

and setting

$$\mathbf{z} = \begin{pmatrix} \rho \\ v \\ s \end{pmatrix}, \qquad \mathbf{A}(\mathbf{z}) = \mathbf{A}(\rho, v, s) = \begin{pmatrix} v & \rho & 0 \\ p_\rho/\rho & v & p_s/\rho \\ 0 & 0 & v \end{pmatrix},$$

(11.12) can be written in the form

$$\mathbf{z}_t + \mathbf{A}(\mathbf{z})\mathbf{z}_x = \mathbf{0}.$$

The eigenvalues of $\mathbf{A}$ are

$$\lambda_1(\rho, v, s) = v - c, \quad \lambda_2(\rho, v, s) = v, \quad \lambda_3(\rho, v, s) = v + c, \tag{11.14}$$

with corresponding eigenvectors

$$\mathbf{r}_1 = \begin{pmatrix} \rho \\ -c \\ 0 \end{pmatrix}, \mathbf{r}_2 = \begin{pmatrix} p_s \\ 0 \\ -c^2 \end{pmatrix}, \mathbf{r}_3 = \begin{pmatrix} \rho \\ c \\ 0 \end{pmatrix}. \tag{11.15}$$

Thus, the system of gas dynamics (11.12) is *strictly hyperbolic*.

To deal with nonsmooth flows (e.g. shock waves) it is convenient to write the gas dynamic equations in *conservative form.* To do it we write the conservation laws for the three variables[3] ρ, ρv and ρE. We find the system

$$\begin{cases} \rho_t + (\rho v)_x = 0 \\ (\rho v)_t + (\rho v^2)_x + p_x = 0 \\ (\rho E)_t + (\rho E v)_x + (pv)_x = 0 \end{cases} \tag{11.16}$$

which is equivalent to (11.10) for smooth flows, as it can be checked by routine calculations. Setting

$$\mathbf{q} = \begin{pmatrix} \rho \\ \rho v \\ \rho E \end{pmatrix} \quad \text{and} \quad \mathbf{F}(\mathbf{q}) = \begin{pmatrix} \rho v \\ \rho v^2 + p \\ (\rho E + p)v \end{pmatrix},$$

we may rewrite (11.16) in the form

$$\mathbf{q}_t + \mathbf{F}(\mathbf{q})_x = \mathbf{0}.$$

Example 11.3 (the p-system) Another model for the motion of a gas along a pipe can be derived using Lagrangian coordinates. Let $w = \rho^{-1}$ be the specific volume (thus $w > 0$) and let x denote a spatial coordinate which, in the Lagrangian description, represents a given gas particle in motion. Under the hypothesis of isoentropic flow[4], the equation of state is of the form $p = p(w)$ and the equations of motion can be reduced to the system

$$\begin{cases} w_t - v_x = 0 \\ v_t + p(w)_x = 0, \end{cases} \tag{11.17}$$

also known as *p-system.* Natural hypotheses on p are:

$$a(w) \equiv -p'(w) > 0 \quad \text{and} \quad a'(w) = -p''(w) < 0$$

for all $w > 0$. Moreover, we assume that

$$\lim_{w \to 0^+} p(w) = +\infty.$$

The above assumptions are satisfied for instance by an ideal polytropic gas, for which $p(w) = kw^{-\gamma}$, where $k > 0$ and $\gamma \geq 1$.

The p-system can be written in the form (11.4) with

[3] A conservation law for the quantity z, with flux zv, is given by: $z_t + (zv)_x =$ external contribution.

[4] No heat exchange occurs among the fluid particles and entropy is at thermodynamical equilibrium.

$$\mathbf{u} = \begin{pmatrix} w \\ v \end{pmatrix} \quad \text{and} \quad \mathbf{F}(\mathbf{u}) = \begin{pmatrix} -v \\ p(w) \end{pmatrix}.$$

Since

$$\mathbf{A}(\mathbf{u}) = D\mathbf{F}(\mathbf{u}) = \begin{pmatrix} 0 & -1 \\ -a(w) & 0 \end{pmatrix}$$

we see that $\mathbf{A}$ is strictly hyperbolic with eigenvalues

$$\lambda_1 = -\sqrt{a(w)} \quad \text{and} \quad \lambda_2 = \sqrt{a(w)} \tag{11.18}$$

and corresponding eigenvectors

$$\mathbf{r}_1(w, v) = \begin{pmatrix} 1 \\ \sqrt{a(w)} \end{pmatrix} \quad \text{and} \quad \mathbf{r}_2(w, v) = \begin{pmatrix} 1 \\ -\sqrt{a(w)} \end{pmatrix}. \tag{11.19}$$

Our main goal is the analysis of the Riemann problem (see Sect. 11.4) for the system (11.4), due to its importance as a model problem and as a key tool in the numerical approximation methods. After some basic facts about linear hyperbolic system, we first solve the Riemann problem in the case of a linear homogeneous system with constant coefficients. In Sect. 11.3 we start the analysis of the quasilinear system (11.4) under the strict hyperbolicity assumption, introducing the notions of characteristic and of Riemann invariant. Then, as in the scalar case $m = 1$, treated in Chap. 4, we construct special solutions under the form of rarefaction waves, contact discontinuities and shocks, introducing an entropy condition to select admissible discontinuities. The last section is devoted to a complete analysis of the Riemann problem for the p-system.

11.2 Linear Hyperbolic Systems

In this section we extend the method of characteristics to linear **strictly** hyperbolic systems.

11.2.1 Characteristics

We have seen in Chap. 4 that the method of *characteristics* plays a key role in the analysis of scalar first order equations ($m = 1$). The method can be extended to strictly hyperbolic systems. We start by examining the following Cauchy problem in the linear case:

$$\begin{cases} \mathbf{u}_t + \mathbf{A}(x,t)\mathbf{u}_x = \mathbf{B}(x,t)\mathbf{u} + \mathbf{c}(x,t) & x \in \mathbb{R},\ t > 0 \\ \mathbf{u}(x,0) = \mathbf{g}(x) & x \in \mathbb{R}. \end{cases} \tag{11.20}$$

Let us briefly review the scalar case

$$u_t + a(x,t)u_x = b(x,t)u + c(x,t) \tag{11.21}$$

with initial data

$$u(x,0) = g(x), \qquad x \in \mathbb{R}.$$

To evaluate u at a point $(\bar{x},\bar{t})$, we find the backward characteristic curve Γ through $(\bar{x},\bar{t})$, of equation $x = x(t)$, obtained as a solution of the ODE

$$\dot{x} = a(x,t)$$

with final condition $x(\bar{t}) = \bar{x}$. Let $x(0) = \xi$ the point at which Γ hits the x-axes. Evaluating u along this characteristic, that is setting $z(t) = u(x(t),t)$, and differentiating this expression, we get

$$\dot{z} = u_t + \dot{x}u_x = u_t + a(x,t)u_x.$$

Using (11.21) and the initial condition, we end up with the following Cauchy problem:

$$\begin{cases} \dot{z} = b(x(t),t)z + c(x(t),t) \\ z(0) = u(\xi,0) = g(\xi). \end{cases} \tag{11.22}$$

Solving (11.22), we obtain the values of u along our characteristic and, in particular, $u(\bar{x},\bar{t}) = z(\bar{t})$.

Note that actually, $\xi = \xi(\bar{x},\bar{t})$, $x(t) = x(t;\bar{x},\bar{t})$ and that u satisfies the integral equation

$$u(\bar{x},\bar{t}) = g(\xi(\bar{x},\bar{t})) + \int_0^{\bar{t}} [b(x(s;\bar{x},\bar{t}),s)u(x(s;\bar{x},\bar{t}),s) + c(x(s;\bar{x},\bar{t}),s)]ds,$$

obtained by integrating over $(0,\bar{t})$ the differential equation in (11.22) and using the initial condition.

For strictly hyperbolic systems we can reproduce something similar. Indeed, let us choose m linearly independent eigenvectors

$$\mathbf{r}_1(x,t),\ \mathbf{r}_2(x,t),\dots,\mathbf{r}_m(x,t)$$

and form the matrix $\mathbf{R} = (\mathbf{r}_1 \mid \mathbf{r}_2 \mid \dots \mid \mathbf{r}_m)$. Set $\mathbf{v} = \mathbf{R}^{-1}\mathbf{u}$. Using (11.8), the system (11.20) for $\mathbf{v}$ can be written in the *canonical form*

$$\begin{cases} \mathbf{v}_t + \mathbf{\Lambda}\mathbf{v}_x = \mathbf{B}^*\mathbf{v} + \mathbf{c}^* & x \in \mathbb{R}, t > 0 \\ \mathbf{v}(x,0) = \mathbf{R}^{-1}(x,0)\mathbf{g}(x) = \mathbf{g}^*(x) & x \in \mathbb{R}, \end{cases} \tag{11.23}$$

where $\mathbf{B}^* = \mathbf{R}^{-1}\mathbf{B}\mathbf{R} - \mathbf{R}^{-1}(\mathbf{A}\mathbf{R}_x + \mathbf{R}_t)$ and $\mathbf{c}^* = \mathbf{R}^{-1}\mathbf{c}$.

In the left hand side of system (11.23), the unknowns are *uncoupled* and the equation for the component v_k of $\mathbf{v}$ takes the following form:

$$\frac{\partial v_k}{\partial t} + \lambda_k \frac{\partial v_k}{\partial x} = \sum_{j=1}^{m} b^*_{kj} v_j + c^*_k. \tag{11.24}$$

Note that if $b^*_{kj} = 0$ for $j \neq k$, then the right hand side is uncoupled as well and each equation is of the form (11.21). Thus, it is natural to call *characteristics* the solutions of the equations

$$\frac{dx}{dt} = \lambda_k(x,t), \qquad k = 1, \dots, m.$$

11.2.2 Classical solutions of the Cauchy problem

We know examine problem (11.20). Under suitable hypotheses on the matrices $\mathbf{A}, \mathbf{B}$ and on the vectors $\mathbf{c}$ and $\mathbf{g}$, there exists a unique classical solution, i.e. of class C^1 in the strip $\mathbb{R} \times [0,T]$. To prove it we use the Contraction Mapping Theorem 6.83, p. 410. Precisely, we prove the following theorem.

Theorem 11.4 *Let $S = \mathbb{R} \times [0,T]$. Assume that $\mathbf{A}$ is strictly hyperbolic and moreover:*

a) $\mathbf{A}$ and $\mathbf{B}$ have entries of class $C^2(S)$, bounded with bounded derivatives;
b) $\mathbf{c}$ and $\mathbf{g}$ are of class $C^1(S)$ and $C^1(\mathbb{R})$, respectively, bounded with bounded derivatives.

Then problem (11.20) has a unique solution $\mathbf{u} \in C^1(S)$.

Proof It is enough to solve problem (11.23). First note that, since each eigenvalue $\lambda_k = \lambda_k(x,t)$ of $\mathbf{A}$ is simple, then $\lambda_k \in C^1(S)$ and it is bounded with bounded derivatives[5]. In particular, there exists a number $L > 0$ such that

$$|\lambda_k(x,t)| \leq L \qquad x \in \mathbb{R}, 0 \leq t \leq T$$

for all $k = 1, \dots, m$. Moreover, from these bounds and the hypotheses a) and b) we can find β, γ, η such that:

$$\sup_S \left| b^*_{ij}(x,t) \right| \leq \beta, \quad \sup_S \left| \partial_x b^*_{ij}(x,t) \right| \leq \beta_1, \quad i,j = 1, \dots, m$$
$$\sup_S |\mathbf{c}^*(x,t)| \leq \gamma, \quad \sup_{\mathbb{R}} |\mathbf{g}^*(x)| \leq \eta.$$

Consider now a point $(\bar{x}, \bar{t})$, $\bar{t} \leq T_1 \leq T$, with T_1 to be suitably chosen later, and the m backward characteristics Γ_i issued from $(\bar{x}, \bar{t})$, of equation

$$x_i(t) = x_i(t; \bar{x}, \bar{t}),$$

where $x_i = x_i(t)$ solves the ODE

$$\frac{dx_i}{dt} = \lambda_i(x_i, t).$$

Note that x_i is continuously differentiable[6] with respect to $\bar{x}, \bar{t}$. If $\bar{t}$ is small, each characteristic hits the x-axes at a point $x_i(0; \bar{x}, \bar{t}) = \xi_i(\bar{x}, \bar{t})$ so that each Γ_i is well defined in the interval $0 \leq t \leq \bar{t}$.

[5] It is an application of the Implicit Function Theorem.

[6] We use here the theorems on dependence from initial data for solutions of systems of ODEs. See e.g. *E. A. Coddington, N. Levinson*, [36].

Define $w_i(t;\bar{x},\bar{t}) = v_i(x_i(t;\bar{x},\bar{t}),t)$. Then

$$\frac{dw}{dt} = \lambda_i \partial_x v_i + \partial_t v_i$$

and from (11.23) we deduce

$$\frac{dw}{dt} = c_i^*(x_i(t;\bar{x},\bar{t}),t) + \sum_{j=1}^{m} b_{ij}^*(x_i(t;\bar{x},\bar{t}),t))w_j(t).$$

Integrating over $(0,\bar{t})$ recalling that $w_i(\bar{t}) = v_i(\bar{x},\bar{t})$ and $w_i(0) = g_i^*(\xi_i(\bar{x},\bar{t}))$, we find the system of integral equations

$$v_i(\bar{x},\bar{t}) = g_i^*(\xi_i(\bar{x},\bar{t})) + \int_0^{\bar{t}} [c_i^*(x_i(s;\bar{x},\bar{t}),s) + \sum_{j=1}^{m} b_{ij}^*(x_i(s;\bar{x},\bar{t}),s)v_j(x_i(s;\bar{x},\bar{t}),s)]ds \tag{11.25}$$

for every $i = 1,\dots,m$. Let $\mathbf{G}(\bar{x},\bar{t})$ denote the vector whose components are

$$g_i^*(\xi_i(\bar{x},\bar{t})) + \int_0^{\bar{t}} c_i^*(x_i(s;\bar{x},\bar{t}),s)ds, \quad i = 1,\dots,m.$$

Then (11.25) can be written in the fixed point form

$$\mathbf{v} = \mathbf{G} + \mathcal{P}(\mathbf{v}), \tag{11.26}$$

where $\mathbf{z} = \mathcal{P}(\mathbf{v})$ has components

$$z_i(\bar{x},\bar{t}) = \int_0^{\bar{t}} \sum_{j=1}^{m} b_{ij}^*(x_i(s;\bar{x},\bar{t}),s)v_j(x_i(s;\bar{x},\bar{t}),s)ds.$$

We now introduce the Banach space

$$X = C_b(\mathbb{R}\times[0,T_1];\mathbb{R}^m)$$

of the bounded and continuous vector-valued functions in $S_1 = \mathbb{R}\times[0,T_1]$, equipped with the norm

$$\|\mathbf{v}\|_X = \sup_{i=1,\dots,m} \{|v_i(x,t)|;\ x\in\mathbb{R}, 0\le t\le T_1\}.$$

We have $\mathcal{P}: X\to X$ and moreover

$$\|\mathcal{P}(\mathbf{v})\|_X \le T_1 m\beta \|\mathbf{v}\|_X.$$

Then, if $T_1 m\beta < 1$, $\mathcal{P}$ is a strict contraction and therefore, by Theorem 6.83, equation (11.26) has a unique solution $\mathbf{v}\in X$. Moreover, the sequence

$$\mathbf{v}^{n+1} = \mathbf{G} + \mathcal{P}(\mathbf{v}^n), \quad \mathbf{v}_0 = \mathbf{0} \tag{11.27}$$

converges to $\mathbf{v}$ in X. Thus, $\mathbf{v}$ solves the integral system (11.25). Still, to infer that $\mathbf{v}$ solves the problem (11.23), we need to show that $\mathbf{v}$ possesses continuous partial derivatives. To this purpose, we introduce the narrower Banach space

$$X^1 = \{\mathbf{w}\in X : \mathbf{w}_x \in X\},$$

equipped with the norm $\|\mathbf{w}\|_{X^1} = \max\{\|\mathbf{w}\|_X, \|\mathbf{w}_x\|_X\}$. Let $\mathcal{P}^1$ be the restriction of $\mathcal{P}$ to X^1. Then $\mathcal{P}^1: X^1\to X^1$ and

$$\left\|\mathcal{P}^1(\mathbf{w})\right\|_{X^1} \le \rho \left\|\mathbf{w}\right\|_{X^1},$$

where

$$\rho = T_1 m^2 \max\left\{\left(\beta + \sup_S \left|\partial_x b^*_{ij}(x,t)\right|\right) \sup_{t\in(0,\bar t);(\bar x,\bar t)\in S_1} \left|\partial_{\bar x} x_i(t;\bar x,\bar t)\right|, \beta\right\}.$$

If $\rho < 1$, i.e. if T_1 is sufficiently small, depending *only* on the bounds of the coefficients of the original equation, then $\mathcal{P}^1$ is a strict contraction. Therefore also $\partial_{\bar x}\mathbf{v}^n$ converges in X and the limit is $\partial_{\bar x}\mathbf{v}$. Now, since $\mathbf{v}^n \in X^1$, we can differentiate $\mathcal{P}(\mathbf{v}^n)$ with respect to $\bar t$. We deduce that also $\partial_{\bar t}\mathbf{v}^n$ converges in X and the limit is $\partial_{\bar t}\mathbf{v}$. Thus $\mathbf{v}$ solves problem (11.23) and it is clearly unique, due to the way it has been constructed.

The above procedure can be iterated, by solving the Cauchy problem with initial time $t = T_1$ and initial data $\mathbf{g}(x) = \mathbf{v}(x, T_1)$. In this way we obtain an extension of the solution to the interval $0 \le t \le 2T_1$. After a finite number of steps we get the solution in the whole strip $\mathbb{R} \times [0, T]$. □

Example 11.5 (The telegrapher system) The following system

$$\begin{cases} LI_t + V_x + RI = 0, \\ CV_t + I_x + GV = 0 \end{cases} \qquad (x \in \mathbb{R}, t > 0) \tag{11.28}$$

describes the flow of electricity in a line, such as a coaxial cable, that we assume of infinite length. The variable x is a coordinate along the cable. $I = I(x,t)$ and $V(x,t)$ represent the current intensity in the inner wire and the voltage across the cable, respectively. The electrical properties of the line are encoded into the positive constants C, capacitance to ground, R, resistance, L, inductance and G, conductance to ground, all per unit length. We assign initial conditions $I(x,0) = I_0(x)$, $V(x,0) = V_0(x)$.

Letting

$$\mathbf{u} = \begin{pmatrix} I \\ V \end{pmatrix}, \ \mathbf{g} = \begin{pmatrix} I_0 \\ V_0 \end{pmatrix}, \ \mathbf{A} = \begin{pmatrix} 0 & 1/L \\ 1/C & 0 \end{pmatrix}, \ \mathbf{B} = \begin{pmatrix} R/L & 0 \\ 0 & G/C \end{pmatrix},$$

we can write (11.28) in the form

$$\mathbf{u}_t + \mathbf{A}\mathbf{u}_x = -\mathbf{B}\mathbf{u}$$

with the initial condition $\mathbf{u}(x,0) = \mathbf{g}(x)$.

The system (11.28) is strictly hyperbolic with eigenvalues $\lambda_1 = 1/\sqrt{LC}, \lambda_2 = -1/\sqrt{LC}$ and associated eigenvectors

$$\mathbf{r}_1 = \begin{pmatrix} \sqrt{C} \\ \sqrt{L} \end{pmatrix}, \ \mathbf{r}_2 = \begin{pmatrix} \sqrt{C} \\ -\sqrt{L} \end{pmatrix}.$$

Letting

$$\mathbf{R} = \begin{pmatrix} \sqrt{C} & \sqrt{C} \\ \sqrt{L} & -\sqrt{L} \end{pmatrix}, \ \mathbf{v} = \mathbf{R}^{-1}\mathbf{u} = \frac{1}{2\sqrt{CL}} \begin{pmatrix} \sqrt{L} & \sqrt{C} \\ \sqrt{L} & -\sqrt{C} \end{pmatrix} \mathbf{u}$$

we write (11.28) in the canonical form

$$\begin{cases} \mathbf{v}_t + \mathbf{\Lambda}\mathbf{v}_x = -\mathbf{B}^*\mathbf{v} & x \in \mathbb{R}, t > 0 \\ \mathbf{v}(x,0) = \mathbf{R}^{-1}\mathbf{g}(x) = \mathbf{g}^*(x) & x \in \mathbb{R}, \end{cases} \tag{11.29}$$

where

$$\mathbf{B}^* = \mathbf{R}^{-1}\mathbf{B}\mathbf{R} = \frac{1}{2CL}\begin{pmatrix} RC+LG & RC-LG \\ RC-LG & RC+LG \end{pmatrix}$$

and

$$\mathbf{g}^*(x) = \frac{1}{2\sqrt{CL}}\begin{pmatrix} \sqrt{L}I_0(x) + \sqrt{C}V_0(x) \\ \sqrt{L}I_0(x) - \sqrt{C}V_0(x) \end{pmatrix}.$$

There are 2 characteristics Γ_1, Γ_2 issued from an arbitrary point $(\bar{x}, \bar{t})$, of equation

$$x_1(t) = x_1(t;\bar{x},\bar{t}) = \bar{x} + \frac{1}{\sqrt{CL}}(t-\bar{t}), \quad x_2(t) = x_2(t;\bar{x},\bar{t}) = \bar{x} - \frac{1}{\sqrt{CL}}(t-\bar{t}),$$

corresponding to the eigenvalues λ_1, λ_2. These characteristics hit the x-axes at the points

$$\xi_1 = \xi_1(\bar{x},\bar{t}) = \bar{x} - \frac{1}{\sqrt{CL}}\bar{t}, \quad \xi_2 = \xi_2(\bar{x},\bar{t}) = \bar{x} + \frac{1}{\sqrt{CL}}\bar{t}.$$

Thus, in our case, the system (11.25) is

$$v_1(\bar{x},\bar{t}) = g_1^*(\xi_1) - \frac{1}{2CL}\int_0^{\bar{t}} [(RC+LG)v_1(x_1(s),s) + (RC-LG)v_2(x_1(s),s)]ds$$

$$v_2(\bar{x},\bar{t}) = g_2^*(\xi_2) - \frac{1}{2CL}\int_0^{\bar{t}} [(RC-LG)v_1(x_2(s),s) + (RC+LG)v_2(x_1(s),s)]ds.$$

In general there are not closed formulas for the solution to this system and one has to resort to numerical methods. However, if $RC = LG$ (*distorsionless case*), the two equations are uncoupled and the solution is given, dropping the bars, by the following damped travelling wave (see Problem 11.1)

$$v_1(x,t) = g_1^*\left(x - \frac{1}{\sqrt{CL}}t\right)e^{-\frac{R}{L}t}, \quad v_2(x,t) = g_2^*\left(x + \frac{1}{\sqrt{CL}}t\right)e^{-\frac{R}{L}t}.$$

11.2.3 Homogeneous systems with constant coefficients. The Riemann problem

In the particular case of *homogeneous systems*, with constant coefficients, that is when

$$\mathbf{u}_t + \mathbf{A}\mathbf{u}_x = \mathbf{0}, \qquad x \in \mathbb{R},\ t > 0, \tag{11.30}$$

equation (11.24) becomes

$$\frac{\partial v_k}{\partial t} + \lambda_k \frac{\partial v_k}{\partial x} = 0, \qquad x \in \mathbb{R},\ t > 0 \tag{11.31}$$

and its general solution is the travelling wave $v_k(x,t) = w_k(x - \lambda_k t)$, where w_k is an arbitrary and differentiable function. Then, since $\mathbf{u} = \mathbf{R}\mathbf{v}$, the general solution of (11.30) is given by the following linear combination of travelling waves:

$$\mathbf{u}(x,t) = \sum_{k=1}^{m} v_k(x,t)\mathbf{r}_k = \sum_{k=1}^{m} w_k(x - \lambda_k t)\mathbf{r}_k. \tag{11.32}$$

Choosing $w_k = g_k^*$, we obtain the unique solution satisfying $\mathbf{u}(x,0) = \mathbf{g}(x)$.

We can explicitly compute (11.32) in the particularly important case (Riemann problem):

$$\mathbf{g}(x) = \begin{cases} \mathbf{u}_l & x < 0 \\ \mathbf{u}_r & x > 0 \end{cases} \qquad (\mathbf{u}_l \neq \mathbf{u}_r).$$

Since the eigenvectors $\mathbf{r}_1, \ldots, \mathbf{r}_k$ constitute a basis in $\mathbb{R}^m$, we can write:

$$\mathbf{u}_l = \sum_{k=1}^{m} \alpha_k \mathbf{r}_k \quad \text{and} \quad \mathbf{u}_r = \sum_{k=1}^{m} \beta_k \mathbf{r}_k. \tag{11.33}$$

Then, from (11.32), each scalar v_k solves the equation (11.31) with initial data

$$v_k(x,0) = \begin{cases} \alpha_k & x < 0 \\ \beta_k & x > 0. \end{cases}$$

Thus

$$v_k(x,t) = \begin{cases} \alpha_k & x < \lambda_k t \\ \beta_k & x > \lambda_k t. \end{cases}$$

To write the analytic expression of $\mathbf{u}$, we divide the half plane x, t into $m+1$ sectors, delimited by the characteristics as in Fig. 11.1. Recall that

$$\lambda_1 < \lambda_2 < \cdots < \lambda_{m-1} < \lambda_m.$$

Inside the sector $S_0 = \{x < \lambda_1 t,\ t > 0\}$ we have $x < \lambda_k t$ for every $k = 2, \ldots, m$. Therefore, in S_0, we have $v_k(x,t) = \alpha_k$ for every $k = 1, \ldots, m$. From (11.32), (11.33) we deduce:

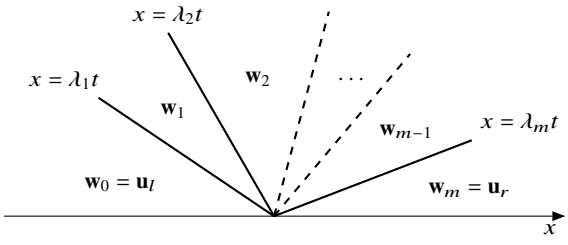

Fig. 11.1 Characteristics and solution of the Riemann problem

$$\mathbf{u}(x,t) = \sum_{k=1}^{m} v_k(x,t)\mathbf{r}_k = \sum_{k=1}^{m} \alpha_k \mathbf{r}_k = \mathbf{u}_l.$$

Similarly, inside the sector $S_m = \{x > \lambda_m t,\ t > 0\}$ we have $v_k(x,t) = \beta_k$ for every $k = 1, \dots, m$ and therefore $\mathbf{u} = \mathbf{u}_r$.

Let now $1 \le j \le m-1$. Inside the sector $S_j = \{\lambda_j t < x < \lambda_{j+1} t,\ t > 0\}$ we have

$$\cdots < \lambda_{j-1} < \lambda_j < \frac{x}{t} < \lambda_{j+1} < \lambda_{j+2} < \cdots, \qquad (t > 0)$$

and $\mathbf{u}$ assumes the constant value

$$\mathbf{w}_j = \sum_{k=1}^{j} \beta_k \mathbf{r}_k + \sum_{k=j+1}^{m} \alpha_k \mathbf{r}_k \qquad 1 \le j \le m-1. \tag{11.34}$$

Summarizing, the solution of the Riemann problem is a self-similar solution of the form (Fig. 11.1):

$$\mathbf{u}(x,t) = \mathbf{w}\left(\frac{x}{t}; \mathbf{u}_l, \mathbf{u}_r\right) = \begin{cases} \mathbf{w}_0 = \mathbf{u}_l & \frac{x}{t} < \lambda_1 \\ \mathbf{w}_1 & \lambda_1 < \frac{x}{t} < \lambda_2 \\ \vdots & \vdots \\ \mathbf{w}_{m-1} & \lambda_{m-1} < \frac{x}{t} < \lambda_m \\ \mathbf{w}_m = \mathbf{u}_r & \lambda_m < \frac{x}{t}. \end{cases} \tag{11.35}$$

Example 11.6 Let $m = 3$, with initial states

$$\mathbf{u}_L = \sum_{k=1}^{3} \alpha_k \mathbf{r}_k \ \text{ and } \ \mathbf{u}_R = \sum_{k=1}^{3} \beta_k \mathbf{r}_k.$$

The construction of the solution is described in Fig. 11.2: the contribution of the initial data to the value of $\mathbf{u}$ at (x,t) is carried along the characteristics through (x,t).

Going back to the solution (11.35), if we define

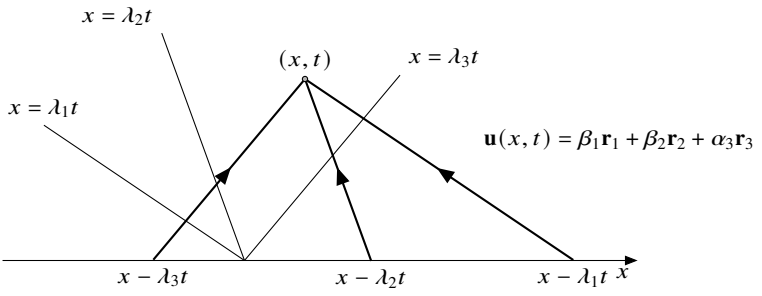

Fig. 11.2 Computation of the solution of the Riemann problem of Example 11.6, at the point (x,t)

$$J^-(x,t) = \{k : x < \lambda_k t\} \quad \text{and} \quad J^+(x,t) = \{k : x > \lambda_k t\},$$

we can write the analytic expression of $\mathbf{u}$ in the following way:

$$\mathbf{u}(x,t) = \mathbf{u}_l + \sum_{k \in J^+(x,t)} (\beta_k - \alpha_k)\mathbf{r}_k = \mathbf{u}_r - \sum_{k \in J^-(x,t)} (\beta_k - \alpha_k)\mathbf{r}_k. \tag{11.36}$$

Now, for every $k = 1, \dots, m$, along the characteristic $x = \lambda_k t$, the solution $\mathbf{u}$ undergoes a jump discontinuity from left to right given by

$$[\mathbf{u}(\cdot,t)]_k = \mathbf{w}_k - \mathbf{w}_{k-1} = (\beta_k - \alpha_k)\mathbf{r}_k, \qquad (t > 0) \tag{11.37}$$

and hence, (11.36) expresses the value of $\mathbf{u}$ as a superposition of these jumps. Thus, the initial discontinuity breaks into *m discontinuities*, each one propagating with its own speed λ_k, $k = 1, 2, \dots, m$. In this case we say that the states $\mathbf{u}_l$ and $\mathbf{u}_r$ are connected by *m contact discontinuities* (according to the terminology of Sect. 4.6.3).

Since $\mathbf{A}\mathbf{r}_k = \lambda_k \mathbf{r}_k$, we infer that, for $t > 0$,

$$A\,[\mathbf{u}(\cdot,t)]_k = \lambda_k(\beta_k - \alpha_k)r_k = \lambda_k\,[\mathbf{u}(\cdot,t)]_k\,. \tag{11.38}$$

We shall see that the equation (11.38) corresponds to the Rankine-Hugoniot condition for the *k-contact discontinuity*.

• *Hugoniot straight lines.* The relation (11.38) holds in general only for $t > 0$, unless the initial jump $\mathbf{u}_r - \mathbf{u}_l$ is an *eigenvector* of $\mathbf{A}$. When this happens, that is if

$$\mathbf{u}_r - \mathbf{u}_l = (\beta_j - \alpha_j)\mathbf{r}_j$$

for some index j, then it must be $\beta_k = \alpha_k$ for $k \neq j$, and from (11.36) we infer that

$$\mathbf{u}(x,t) = \begin{cases} \mathbf{u}_l & \text{for } x < \lambda_j t \\ \mathbf{u}_r & \text{for } x > \lambda_j t. \end{cases}$$

Thus the solution takes the initial discontinuity and propagates it with speed λ_j. The other characteristics do not carry any jump. In this situation the states $\mathbf{u}_l$ and $\mathbf{u}_r$ are connected by the single *j-contact discontinuity*.

Changing perspective, let us fix $\mathbf{u}_l$ and ask which states $\mathbf{u}$ can be connected to $\mathbf{u}_l$ (on the right of $\mathbf{u}_l$) by a single j-contact discontinuity, for some $j = 1, \dots, m$. From what we have just seen, this is possible if (and only if) the vectors $\mathbf{u} - \mathbf{u}_l$ are parallel to $\mathbf{r}_j$, that is if

$$\mathbf{A}(\mathbf{u} - \mathbf{u}_l) = \lambda_j(\mathbf{u} - \mathbf{u}_l). \tag{11.39}$$

The set of the states $\mathbf{u}$ satisfying (11.39) coincides with the straight line

$$\mathbf{u}(s) = \mathbf{u}_l + s\mathbf{r}_j, \qquad s \in \mathbb{R}$$

called the *j^{th}-Hugoniot line issued from* $\mathbf{u}_l$.

Summarizing: a state $\mathbf{u}$ can be connected to $\mathbf{u}_l$ (*on the right of* $\mathbf{u}_l$) by a single contact discontinuity if and only if it belongs to one of the m Hugoniot lines issued from $\mathbf{u}_l$.

Remark 11.7 When the relevant domain is a quadrant, say $x > 0, t > 0$, or a half-strip $(a, b) \times (0+\infty)$, some caution is necessary to get a well posed problem. For instance, consider the problem

$$\mathbf{u}_t + \mathbf{A}\mathbf{u}_x = \mathbf{0}, \qquad x \in (0, R), t > 0 \tag{11.40}$$

with the initial condition

$$\mathbf{u}(x, 0) = \mathbf{g}(x), \qquad x \in [0, R] .$$

Which other data (and where) should be assigned to uniquely determine $\mathbf{u}$? Look at the k-th equation of the uncoupled problem, that is

$$\frac{\partial v_k}{\partial t} + \lambda_k \frac{\partial v_k}{\partial x} = 0.$$

Suppose that $\lambda_k > 0$, so that the corresponding characteristic γ_k is *inflow on* $x = 0$ and *outflow on* $x = R$. Guided by the scalar case (Sect. 4.2.4), we must assign the value of v_k on $x = 0$. On the contrary, if $\lambda_k < 0$, the value of v_k has to be assigned on $x = R$. Thus, we can draw the following conclusion: suppose that r eigenvalues (say $\lambda_1, \lambda_2, \ldots, \lambda_r$) are positive and the other $m - r$ eigenvalues are negative. *Then the values of* $v_1, \ldots, v_r$ *have to be assigned on* $x = 0$ *and the values of* $v_{r+1}, \ldots, v_m$ *on* $x = R$. In terms of the original unknown $\mathbf{u}$, this amounts to assign, on $x = 0$, r independent linear combinations of the $\mathbf{u}$ components:

$$(\mathbf{R}^{-1}\mathbf{u})_k = \sum_{j=1}^{m} c^{jk} u_j \qquad k = 1, 2, \ldots, r,$$

while other $m - r$ have to be assigned on $x = R$.

11.3 Quasilinear Conservation Laws

11.3.1 Characteristics and Riemann invariants

In this section we consider quasilinear systems of the form

$$\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = \mathbf{0} \tag{11.41}$$

in $\mathbb{R} \times (0, +\infty)$, with $\mathbf{F} : \mathbb{R}^m \to \mathbb{R}^m$, smooth. We assume that the jacobian matrix $\mathbf{A} = D\mathbf{F}$ is strictly hyperbolic with eigenvalues

$$\lambda_1(\mathbf{u}) < \lambda_2(\mathbf{u}) < \cdots < \lambda_m(\mathbf{u})$$

and corresponding eigenvectors $\mathbf{r}_1(\mathbf{u}), \ldots, \mathbf{r}_m(\mathbf{u})$. Under these hypotheses, the eigenvalues and the eigenvectors are smooth functions of $\mathbf{u}$ (see e.g. *F. John*, [9]).

To each eigenvalue $\lambda_k(\mathbf{u})$, $k = 1, \ldots, m$, we can associate a characteristic line, defined by the ODE

$$\dot{x}(t) = \lambda_k(\mathbf{u}(x,t)).$$

In the scalar case, the solution u is constant along a characteristic so that this line carries the initial value of u. If $m > 1$ this is not true anymore and we may ask if there are scalar functions $R(\mathbf{u})$ that remain constant along a characteristic. We distinguish the cases $m = 2$ and $m > 2$.

• **Case** $m = 2$. Write $\mathbf{u} = (w, v)^\top$ and let Γ_1 be a characteristic of equation $\dot{x} = \lambda_1(\mathbf{u}(x,t))$. If $R(\mathbf{u}(x,t))$ remains constant along Γ_1 it must be

$$\frac{d}{dt} R(\mathbf{u}(x(t),t)) = \nabla R(\mathbf{u}) \cdot (\dot{x}\mathbf{u}_x + \mathbf{u}_t) = \nabla R(\mathbf{u}) \cdot (\lambda_1(\mathbf{u})\mathbf{u}_x + \mathbf{u}_t) \equiv 0. \tag{11.42}$$

Since $\mathbf{u}_t = -\mathbf{A}(\mathbf{u})\mathbf{u}_x$, (11.42) is equivalent to

$$[\nabla R(\mathbf{u})\lambda_1(\mathbf{u}) - \nabla R(\mathbf{u})\mathbf{A}(\mathbf{u})] \cdot \mathbf{u}_x \equiv 0. \tag{11.43}$$

Now, (11.43) is satisfied if $\nabla R(\mathbf{u})$ is a *left* (row) eigenvector of $\mathbf{A}(\mathbf{u})$ corresponding to the eigenvalue $\lambda_1(\mathbf{u})$. Since we are in dimension $m = 2$, recalling (11.9), this implies that

$$\nabla R(\mathbf{u}) \cdot \mathbf{r}_2(\mathbf{u}) \equiv 0. \tag{11.44}$$

Since (11.44) involves the eigenvector $\mathbf{r}_2$, R is called a 2-*Riemann invariant*. Letting

$$\mathbf{r}_2(w,v) = \begin{pmatrix} r_{21}(w,v) \\ r_{22}(w,v) \end{pmatrix}$$

and $R = R(w,v)$, (11.44) may be written in the more explicit form

$$r_{21}R_w + r_{22}R_v = 0,$$

which is a first order scalar equation for R.

Similarly, a 1-*Riemann invariant* is a scalar function S that remains constant along a characteristic of equation $\dot{x} = \lambda_2(\mathbf{u}(x,t))$ and hence it satisfies the condition

$$\nabla S(\mathbf{u}) \cdot \mathbf{r}_1(\mathbf{u}) \equiv 0. \tag{11.45}$$

Letting

$$\mathbf{r}_1(w,v) = \begin{pmatrix} r_{11}(w,v) \\ r_{12}(w,v) \end{pmatrix}$$

and $S = (w,v)$, we can write (11.45) in the more explicit form

$$r_{11}S_w + r_{12}S_v = 0.$$

• **Case** $m > 2$. Changing slightly point of view, the equations (11.44) and (11.45) express the fact that the Riemann invariants S and R *remain constant along the integral curves of the eigenvectors* $\mathbf{r}_1$ *and* $\mathbf{r}_2$, respectively. In this form, Riemann invariants can be defined also for $m > 2$. Thus, in general, we can give the following definition.

Definition 11.8 A smooth function $w_k : \mathbb{R}^m \to \mathbb{R}$ is called a *k-Riemann invariant* if it satisfies the following first order PDE:

$$\nabla w_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) \equiv 0. \tag{11.46}$$

If $\xi \longmapsto \mathbf{v}(\xi)$ is an integral curve of $\mathbf{r}_k$, that is

$$\frac{d}{d\xi}\mathbf{v}(\xi) = \mathbf{r}_k(\mathbf{v}(\xi)),$$

then

$$\frac{d}{d\xi}w_k(\mathbf{v}(\xi)) = \nabla w_k(\mathbf{v}(\xi)) \cdot \frac{d}{d\xi}\mathbf{v}(\xi) = \nabla w_k(\mathbf{v}(\xi)) \cdot \mathbf{r}_k(\mathbf{v}(\xi)) \equiv 0$$

and therefore a k-Riemann invariant is constant along the trajectories of $\mathbf{r}_k$.

In general, the Riemann invariants exist only locally. In fact, we have:

Proposition 11.9 *For every* $k = 1, \dots, m$*, there exist, locally,* $m - 1$ *independent k-Riemann invariants, that is with linearly independent gradients.*

For the proof see e.g. *J. Smoller*, [20].

Example 11.10 (the p-system) Referring to Example 11.3, the equations for the two Riemann invariants are:

$$R_w - \sqrt{a(w)}R_v = 0 \text{ and } S_w + \sqrt{a(w)}S_v = 0.$$

It is easy to check that

$$R(w,v) = v + \int_{w_0}^{w} \sqrt{a(s)}ds, \qquad S(w,v) = v - \int_{w_0}^{w} \sqrt{a(s)}ds$$

($w_0 > 0$, arbitrary).

Example 11.11 (Gas dynamics) Turning to system (11.12) in Example 11.2, since $m = 3$, we have 3 pairs of invariants.

The pair of 1-invariants $w_{1j} = w_{1j}(\rho, v, s)$, $j = 1, 2$, can be found solving the first order PDE

$$\nabla w \cdot \mathbf{r}_1 = \rho w_\rho - c w_v = 0.$$

The characteristic system for this equation can be written, formally, as (see Chap. 4, Remark 4.25, p. 232)

$$\frac{d\rho}{\rho} = -\frac{dv}{c} = \frac{ds}{0}. \tag{11.47}$$

From the last equation we have $ds = 0$. Hence $w_{11}(\rho, v, s) = s$ is a first integral of the system (11.47) and therefore $w_{11} = s$ is a 1-Riemann invariant. From the first equation we get, recalling that $c^2 = p_\rho$,

$$-dv = \frac{\sqrt{p_\rho}}{\rho} d\rho.$$

Then, another first integral of (11.47) is

$$w_{12}(\rho, v, s) = v + l(\rho, s)$$

where $l(\rho, s) = \int \frac{\sqrt{p_\rho(\rho,s)}}{\rho} d\rho$. Thus $w_{12} = v + l(\rho, s)$ is the second 1-Riemann invariant. Similar computations give $w_{31}(\rho, v, s) = s$ and $w_{32}(\rho, v, s) = v - l(\rho, s)$ for the pair of 3-invariants.

To compute the 2-invariants $w_{2j} = w_{2j}(\rho, v, s)$, $j = 1, 2$, we solve the PDE

$$\nabla w \cdot \mathbf{r}_2 = p_s w_\rho - c^2 w_s = 0$$

whose characteristic system reads

$$\frac{d\rho}{p_s} = \frac{dv}{0} = -\frac{ds}{p_\rho}.$$

Thus $dv = 0$ and $p_\rho d\rho + p_s ds = dp = 0$ from which we find the two 2-invariants

$$w_{21}(\rho, v, s) = v \quad \text{and} \quad w_{22}(\rho, v, s) = p.$$

Summarizing, the three pairs of invariants are:

$$\{s, v + l\}, \quad \{v, p\}, \quad \{s, v - l\}.$$

The Riemann invariants can be employed to find the explicit solution in problems of practical interest in the context of classical fluid dynamics, like for instance the Riemann problem in gas dynamics, for which we refer e.g. to *E. Godlewski, P.-A. Raviart*, [46].

11.3.2 Weak (or integral) solutions and the Rankine-Hugoniot condition

The definition of weak or integral solution of the Cauchy problem

$$\begin{cases} \mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = \mathbf{0} & \text{in } \mathbb{R} \times (0, +\infty) \\ \mathbf{u}(x, 0) = \mathbf{g}(x) & \text{in } \mathbb{R} \end{cases} \tag{11.48}$$

parallels the analogous definition for the scalar case (Definition 4.7, p. 206).

Definition 11.12 We say that a locally bounded function $\mathbf{u} : \mathbb{R} \times [0, +\infty) \to \mathbb{R}^m$ is a *weak solution* of problem (11.48) if the equation

$$\int_0^\infty dt \int_{\mathbb{R}} [\mathbf{v}_t \cdot \mathbf{u} + \mathbf{v}_x \cdot \mathbf{F}(\mathbf{u})]\, dx + \int_{\mathbb{R}} \mathbf{v}(x, 0) \cdot \mathbf{g}(x)\, dx = 0 \tag{11.49}$$

holds for every smooth $\mathbf{v} : \mathbb{R} \times [0, +\infty) \to \mathbb{R}^m$, vanishing outside a compact subset of the half plane $\mathbb{R} \times [0, +\infty)$.

As in the scalar case, a weak solution of class $C^1(\mathbb{R} \times [0, +\infty))$ satisfies the Cauchy problem in classical sense. Moreover, in the class of weak piecewise C^1 solutions (see Sect. 4.4.3), the only admissible discontinuities in the half plane $t > 0$ are those prescribed by the following vector Rankine-Hugoniot condition valid along a discontinuity line Γ of equation $x = s(t)$:

$$\sigma\, [\mathbf{u}_+ - \mathbf{u}_-] = [\mathbf{F}(\mathbf{u}_+) - \mathbf{F}(\mathbf{u}_-)]\,. \tag{11.50}$$

The condition (11.50) consists of m scalar equations, where $\mathbf{u}_+$ and $\mathbf{u}_-$ are the values that $\mathbf{u}$ attains on the right and left sides of Γ and $\sigma = \dot{s}(t)$ is the speed of the discontinuity. The proof is identical to the proof of Theorem 4.9, p. 207.

11.4 The Riemann Problem

Our purpose is to construct a solution of the system

$$\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = \mathbf{0} \tag{11.51}$$

in $\mathbb{R} \times (0, +\infty)$, satisfying the initial condition

$$\mathbf{g}(x, 0) = \begin{cases} \mathbf{u}_l & \text{for } x < 0 \\ \mathbf{u}_r & \text{for } x > 0, \end{cases} \tag{11.52}$$

where $\mathbf{u}_l$ and $\mathbf{u}_r$ are constant states. We always assume that $\mathbf{F}$ is a smooth function in $\mathbb{R}^m$. Let us briefly review the scalar case $m = 1$,

$$u_t + q(u)_x = 0, \qquad x \in \mathbb{R},\, t > 0. \tag{11.53}$$

We have seen in Sect. 4.5 that the Riemann problem for equation (11.53) has a unique entropic solution, for which we can write an explicit formula. Suppose

$$q \in C^2(\mathbb{R}), \quad q''(u) > 0,$$

so that the equation is *genuinely nonlinear*. The case q concave is similar.

If $u_l < u_r$, the solution u is constructed connecting the two states u_l and u_r through a rarefaction wave, that is:

$$u(x,t) = \begin{cases} u_l & x < q'(u_l)t \\ f\left(\frac{x}{t}\right) & q'(u_l)t < x < q'(u_r)t \\ u_r & x > q'(u_r)t \end{cases}$$

where $f = (q')^{-1}$, i.e. the inverse of q'.

If $u_l > u_r$, then u is constructed connecting the two states u_l and u_r through a shock wave, that is

$$u(x,t) = \begin{cases} u_l & x < \sigma(u_l, u_r)t \\ u_r & x > \sigma(u_l, u_r)t \end{cases} \tag{11.54}$$

where $\sigma = \sigma(u_l, u_r)$ is the shock speed, given by the *Rankine-Hugoniot* condition

$$\sigma(u_l, u_r) = \frac{q(u_r) - q(u_l)}{u_r - u_l}.$$

Recall that the *entropy* condition, imposed to get rid of non-physically acceptable solutions, implies the inequalities

$$q'(u_r) < \sigma(u_l, u_r) < q'(u_l). \tag{11.55}$$

Geometrically, (11.55) means that the characteristics "impinge" into the shock curve from both sides.

On the other hand, in the linear case $u_t + au_x = 0$, the solution is given by

$$u(x,t) = \begin{cases} u_l & x < at \\ u_r & x > at \end{cases}$$

exhibiting a discontinuity line parallel to the characteristics (*contact discontinuity,* see Fig. 11.3).

Consider now the m-dimensional case, $m > 1$. Guided by the case of linear systems with constant coefficients, we expect that the initial discontinuity breaks down, in general, into m waves, each one propagating with its own speed. In the nonlinear case, however, these waves are of different kinds, namely rarefaction waves, shocks and contact discontinuities. Thus, we first introduce and construct

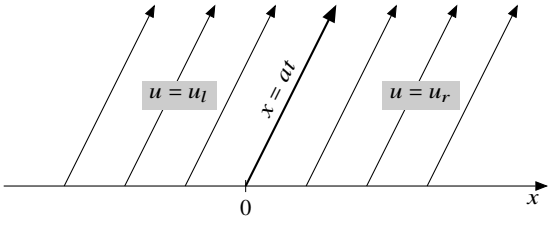

Fig. 11.3 Contact discontinuity

these special solutions. Then, to solve the Riemann problem, we first examine which pairs of constant states $\mathbf{u}_l$ and $\mathbf{u}_r$ can be connected by a single rarefaction wave or by a single shock wave or by a single contact discontinuity. Finally, we obtain a solution in the general case by a mix of those waves.

11.4.1 Rarefaction waves. Genuinely nonlinear systems

We start by examining the existence of solutions to (11.51) of the type

$$\mathbf{u}(x,t) = \mathbf{v}(\theta(x,t)), \tag{11.56}$$

where θ is a scalar function. These solutions are called *simple waves*. Inserting (11.56) into (11.51), we find the equation

$$\mathbf{v}'(\theta)\theta_t + \mathbf{A}(\mathbf{v}(\theta))\mathbf{v}'(\theta)\theta_x = \mathbf{0}$$

that, assuming $\theta_x \neq 0$, we rewrite as

$$\mathbf{A}(\mathbf{v}(\theta))\mathbf{v}'(\theta) = -\frac{\theta_t}{\theta_x}\mathbf{v}'(\theta). \tag{11.57}$$

Disregarding the trivial case $\mathbf{v}'(\theta) = \mathbf{0}$, it is apparent that (11.57) is satisfied if

$$\mathbf{A}(\mathbf{v}(\theta))\mathbf{v}'(\theta) = \lambda(\mathbf{v}(\theta))\mathbf{v}'(\theta),$$

where $\theta = \theta(x,t)$ is a solution of the scalar equation

$$\theta_t + \lambda(\mathbf{v}(\theta))\theta_x = 0. \tag{11.58}$$

In other terms, (11.57) is satisfied if $\mathbf{v}'(\theta)$ coincides with an *eigenvector* $\mathbf{r}_k(\mathbf{v}(\theta))$ of $\mathbf{A}(\mathbf{v}(\theta))$, with corresponding *eigenvalue* $\lambda_k(\mathbf{v}(\theta))$. Note that if $\mathbf{v}'(\theta) \neq \mathbf{0}$ in a given interval (θ_0, θ_1), the index k does not depend on θ, since the eigenvalues are distinct and $\theta \mapsto \lambda_k(\mathbf{v}(\theta))$ is a smooth function. Thus we can use the notation $\mathbf{v}_k(\theta)$ and write

$$\mathbf{v}_k'(\theta) = \mathbf{r}_k(\mathbf{v}_k(\theta)). \tag{11.59}$$

The equation (11.59) means that $\mathbf{v}_k = \mathbf{v}_k(\theta)$ is an integral curve of the vector field $\mathbf{r}_k$.

Due to the smoothness of $\mathbf{r}_k$, $\mathbf{v}_k(\theta)$ is uniquely defined, at least in a neighborhood of θ_0, and we can solve for $\theta = \theta(x,t)$ the equation (11.58), which becomes

$$\theta_t + \lambda_k(\mathbf{v}_k(\theta))\theta_x = 0. \tag{11.60}$$

Setting

$$q_k(\theta) = \int_{\theta_0}^{\theta} \lambda_k(\mathbf{v}_k(s))ds,$$

the equation (11.60) can be written in the form

$$\theta_t + q_k(\theta)_x = 0 \tag{11.61}$$

which is a scalar conservation law. Since

$$q_k'(\theta) = \lambda_k(\mathbf{v}_k(\theta))$$

and

$$q_k''(\theta) = \nabla\lambda_k(\mathbf{v}_k(\theta)) \cdot \mathbf{v}_k'(\theta) = \nabla\lambda_k(\mathbf{v}_k(\theta)) \cdot \mathbf{r}_k(\mathbf{v}(\theta)), \tag{11.62}$$

we distinguish some cases according to the following definition, where by k^{th}-*characteristic field* we mean the eigenvalue-eigenvector pair $\lambda_k(\mathbf{u}), r_k(\mathbf{u})$.

Definition 11.13 We say that:

a) The k^{th}-characteristic field $\lambda_k(\mathbf{u}), \mathbf{r}_k(\mathbf{u})$ is *genuinely nonlinear* if

$$\nabla\lambda_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) \neq 0, \qquad \text{for all } \mathbf{u} \in \mathbb{R}^m. \tag{11.63}$$

b) *The system is genuinely nonlinear* if

$$\nabla\lambda_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) \neq 0, \qquad \text{for all } \mathbf{u} \in \mathbb{R}^m \text{ and } k = 1, \ldots, m.$$

c) The k^{th}-characteristic field $\lambda_k(\mathbf{u}), \mathbf{r}_k(\mathbf{u})$ is *linearly degenerate* if

$$\nabla\lambda_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) = 0, \qquad \text{for all } \mathbf{u} \in \mathbb{R}^m. \tag{11.64}$$

Remark 11.14 Let $\lambda_k(\mathbf{u}), \mathbf{r}_k(\mathbf{u})$ be linearly degenerate. Then, $w = \lambda_k(\mathbf{u})$ is a k-Riemann invariant by definition.

Going back to the construction of a simple wave, assume that the pair $\lambda_k(\mathbf{u})$, $r_k(\mathbf{u})$ is *genuinely nonlinear*. Then the function

$$\theta \mapsto q_k'(\theta) = \lambda_k(\mathbf{v}_k(\theta)) \tag{11.65}$$

is invertible by (11.62). Setting $f_k = (q_k')^{-1}$, a solution of (11.61) is given by

$$\theta_k(x,t) = f_k\left(\frac{x}{t}\right).$$

Thus, from (11.56) we obtain *the simple wave*

$$\mathbf{u}_k(x,t) = \mathbf{v}_k\left(f_k\left(\frac{x}{t}\right)\right). \tag{11.66}$$

Note that if $\mathbf{v}_k(\theta) = \mathbf{u}$ then $\mathbf{u}_k = \mathbf{u}$ on the half straight line $x = \lambda_k(\mathbf{u})t$, $t > 0$.

Definition 11.15 The simple wave $\mathbf{u}_k$ defined in (11.66) is called k-*rarefaction wave* centered at $(0,0)$.

Summarizing, given a genuinely nonlinear characteristic field $\lambda_k(\mathbf{v})$, $\mathbf{r}_k(\mathbf{v})$, we may construct a *k-rarefaction wave centered at the origin and equal to* $\mathbf{u}_0$ on the half straight line $x = \lambda_k(\mathbf{u}_0)t$, $t > 0$, by first solving the Cauchy problem

$$\begin{cases} \mathbf{v}_k'(\theta) = \mathbf{r}_k(\mathbf{v}_k(\theta)) \\ \mathbf{v}_k(\theta_0) = \mathbf{u}_0. \end{cases} \tag{11.67}$$

Once $\mathbf{v}_k = \mathbf{v}_k(\theta)$ is known, we compute f_k, i.e. the inverse function of

$$\theta \mapsto \lambda_k(\mathbf{v}_k(\theta)),$$

and then we use (11.66), obtaining

$$\mathbf{u}_k(x,t) = \mathbf{v}_k\left(f_k\left(\frac{x}{t}\right)\right).$$

Example 11.16 The *p-system*. From (11.18) and (11.19), we have:

$$\nabla\lambda_1(\mathbf{u})\cdot\mathbf{r}_1(\mathbf{u}) = \frac{-a'(w)}{2\sqrt{a(w)}} > 0, \qquad \nabla\lambda_2(\mathbf{u})\cdot\mathbf{r}_2(\mathbf{u}) = \frac{a'(w)}{2\sqrt{a(w)}} < 0$$

for all (w, v), $w > 0$. Hence the *p-system* is genuinely nonlinear.

Example 11.17 (Gas dynamics) From (11.14) and (11.15), we have:

$$\nabla\lambda_1 = \begin{pmatrix} -c_\rho \\ 1 \\ -c_s \end{pmatrix}, \quad \nabla\lambda_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \nabla\lambda_3 = \begin{pmatrix} c_\rho \\ 1 \\ c_s \end{pmatrix}$$

whence, by (11.13),

$$\begin{aligned} &\nabla\lambda_1\cdot\mathbf{r}_1 = -(c + \rho c_\rho) < 0 \\ &\nabla\lambda_2\cdot\mathbf{r}_2 \equiv 0 \\ &\nabla\lambda_3\cdot\mathbf{r}_3 = c + \rho c_\rho > 0. \end{aligned}$$

The pairs $\lambda_1, \mathbf{r}_1$ and $\lambda_3, \mathbf{r}_3$ are *genuinely nonlinear*, while $\lambda_2, \mathbf{r}_2$ is *linearly degenerate*.

11.4.2 Solution of the Riemann problem by a single rarefaction wave

We are now ready to explore the possibility to solve problem (11.51), (11.52) by means of a single rarefaction wave. In other words, we look for the pairs of states $\mathbf{u}_l$, $\mathbf{u}_r$ that can be connected by a rarefaction wave. To this purpose, given a state $\mathbf{u}_0 \in \mathbb{R}^m$, we denote by $R_k(\mathbf{u}_0)$ the integral curve of $\mathbf{r}_k$ satisfying the initial condition $\mathbf{v}_k(\theta_0) = \mathbf{u}_0$, for some θ_0. We now examine the structure of the integral curves $R_k(\mathbf{u}_l)$, $k = 1, \ldots, m$. If $\lambda_k(\mathbf{u}_l), \mathbf{r}_k(\mathbf{u}_l)$ is genuinely nonlinear, $R_k(\mathbf{u}_l)$ takes the name of *k-rarefaction curve, issued from* $\mathbf{u}_l$.

Theorem 11.18 *Let $\lambda_k(\mathbf{u}), r_k(\mathbf{u})$ be genuinely nonlinear, $1 \le k \le m$. Near a given state $\mathbf{u}_0$, there exists a parametrization of the curve $R_k(\mathbf{u}_0)$ given by*

$$\varepsilon \longmapsto \boldsymbol{\varphi}_k(\varepsilon; \mathbf{u}_0), \qquad |\varepsilon| \le \varepsilon_0,$$

for some $\varepsilon_0 > 0$, such that

$$\boldsymbol{\varphi}_k(\varepsilon; \mathbf{u}_0) = \mathbf{u}_0 + \varepsilon \mathbf{r}_k(\mathbf{u}_0) + \frac{\varepsilon^2}{2} D\mathbf{r}_k(\mathbf{u}_0)\mathbf{r}_k(\mathbf{u}_0) + o(\varepsilon^2). \tag{11.68}$$

Moreover, we can decompose $R_k(\mathbf{u}_0)$ into the following three disjoint sets:

$$R_k(\mathbf{u}_0) = R_k^+(\mathbf{u}_0) \cup \{\mathbf{u}_0\} \cup R_k^-(\mathbf{u}_0) \tag{11.69}$$

where

$$R_k^+(\mathbf{u}_0) = \{\mathbf{u} \in R_k(\mathbf{u}_0) : \lambda_k(\mathbf{u}) > \lambda_k(\mathbf{u}_0)\} \tag{11.70}$$

and

$$R_k^-(\mathbf{u}_0) = \{\mathbf{u} \in R_k(\mathbf{u}_0) : \lambda_k(\mathbf{u}) < \lambda_k(\mathbf{u}_0)\}. \tag{11.71}$$

Proof Normalize $\mathbf{r}_k(\mathbf{u})$ to have

$$\nabla \lambda_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) \equiv 1. \tag{11.72}$$

Let $\theta \longmapsto \mathbf{v}_k(\theta)$ be the solution of the Cauchy problem (11.67), with $\theta_0 = \lambda_k(\mathbf{u}_0)$:

$$\begin{cases} \mathbf{v}_k'(\theta) = \mathbf{r}_k(\mathbf{v}_k(\theta)) \\ \mathbf{v}_k(\lambda_k(\mathbf{u}_0)) = \mathbf{u}_0. \end{cases}$$

The function $\mathbf{v}_k = \mathbf{v}_k(\theta)$ exists at least in some interval

$$\lambda_k(\mathbf{u}_0) - \varepsilon_0 \le \theta \le \lambda_k(\mathbf{u}_0) + \varepsilon_0, \qquad \varepsilon_0 > 0.$$

Using (11.72), we write:

$$\frac{d}{d\theta}\lambda_k(\mathbf{v}_k(\theta)) = \nabla \lambda_k(\mathbf{v}_k(\theta)) \cdot \mathbf{r}_k(\mathbf{v}_k(\theta)) \equiv 1.$$

In particular, note that $\theta \mapsto \lambda_k(\mathbf{v}_k(\theta))$ is strictly increasing. Integrating from $\lambda_k(\mathbf{u}_0)$ to θ and using $\mathbf{v}_k(\lambda_k(\mathbf{u}_0)) = \mathbf{u}_0$, we find

$$\lambda_k(\mathbf{v}_k(\theta)) = \theta.$$

Therefore, near $\mathbf{u}_0$, we may write

$$R_k(\mathbf{u}_0) = \{\mathbf{v}_k(\theta) : \lambda_k(\mathbf{u}_0) - \varepsilon_0 \le \lambda_k(\mathbf{v}_k(\theta)) \le \lambda_k(\mathbf{u}_0) + \varepsilon_0\}. \tag{11.73}$$

Put $\theta = \lambda_k(\mathbf{u}_0) + \varepsilon$ and

$$\boldsymbol{\varphi}_k(\varepsilon; \mathbf{u}_0) = \mathbf{v}_k(\lambda_k(\mathbf{u}_0) + \varepsilon), \quad |\varepsilon| \le \varepsilon_0.$$

We have $\boldsymbol{\varphi}_k(0; \mathbf{u}_0) = \mathbf{v}_k(\lambda_k(\mathbf{u}_0)) = \mathbf{u}_0$ and

$$\frac{d}{d\varepsilon}\boldsymbol{\varphi}_k(\varepsilon; \mathbf{u}_0)_{|\varepsilon=0} = \mathbf{r}_k(\mathbf{v}_k(\lambda_k(\mathbf{u}_0))) = \mathbf{r}_k(\mathbf{u}_0). \tag{11.74}$$

Moreover, since

$$\frac{d^2}{d\theta^2}\mathbf{v}_k(\theta) = D\mathbf{r}_k(\mathbf{v}_k(\theta))\frac{d}{d\theta}\mathbf{v}_k(\theta) = D\mathbf{r}_k(\mathbf{v}_k(\theta))\mathbf{r}_k(\mathbf{v}_k(\theta)),$$

we deduce that

$$\frac{d^2}{d\varepsilon^2}\varphi_k(\varepsilon;\mathbf{u}_0)_{|\varepsilon=0} = D\mathbf{r}_k(\mathbf{u}_0)\mathbf{r}_k(\mathbf{u}_0). \tag{11.75}$$

From (11.74) and (11.75), (11.68) follows.

The decomposition (11.69) follows from (11.73), by putting

$$R_k^-(\mathbf{u}_0) = \{\mathbf{v}(\theta)\colon \lambda_k(\mathbf{u}_0) - \varepsilon_0 \le \lambda_k(\mathbf{v}_k(\theta)) < \lambda_k(\mathbf{u}_0)\}$$

and

$$R_k^+(\mathbf{u}_0)\ \{\mathbf{v}(\theta)\colon \lambda_k(\mathbf{u}_0) < \lambda_k(\mathbf{v}_k(\theta)) \le \lambda_k(\mathbf{u}_0) + \varepsilon_0\}\,. \qquad \square$$

Remark 11.19 We emphasize that, under the normalization condition (11.72), the part $R_k^+(\mathbf{u}_0)$ of $R_k(\mathbf{u}_0)$ corresponds to the positive values of ε, in the parametrization (11.68).

Let now $\mathbf{u}_0 = \mathbf{u}_l$ be the left initial state in the Riemann problem, i.e. assigned on $x < 0$. By the structure Theorem 11.18 we can write

$$R_k(\mathbf{u}_l) = R_k^+(\mathbf{u}_l) \cup \{\mathbf{u}_l\} \cup R_k^-(\mathbf{u}_l)$$

with

$$R_k^+(\mathbf{u}_l) = \{\mathbf{u} \in R_k(\mathbf{u}_l) : \lambda_k(\mathbf{u}) > \lambda_k(\mathbf{u}_l)\}\,. \tag{11.76}$$

The next theorem shows that if $\mathbf{u}_r \in R_k^+(\mathbf{u}_l)$, then the Riemann problem can be solved by a single k-rarefaction wave.

Theorem 11.20 *Assume that for some k, $1 \le k \le m$:*

a) The pair $\lambda_k(\mathbf{u}), \mathbf{r}_k(\mathbf{u})$ is genuinely nonlinear.
b) $\mathbf{u}_r \in R_k^+(\mathbf{u}_l)$.

Then there exists a weak solution of the Riemann problem with initial states $\mathbf{u}_l$, $\mathbf{u}_r$, given by a rarefaction wave.

Proof Since $\mathbf{u}_r \in R_k^+(\mathbf{u}_l)$, there exist two numbers θ_r and θ_l such that

$$\mathbf{u}_l = \mathbf{v}_k(\theta_l) \quad \text{and} \quad \mathbf{u}_r = \mathbf{v}_k(\theta_r).$$

Assume that $\theta_l < \theta_r$. The opposite case is similar. From b) we deduce

$$\lambda_k(\mathbf{u}_r) > \lambda_k(\mathbf{u}_l).$$

From (11.65), p. 632, we get

$$q_k'(\theta_r) = \lambda_k(\mathbf{u}_r) \quad \text{and} \quad q_k'(\theta_l) = \lambda_k(\mathbf{u}_l),$$

so that

$$q_k'(\theta_l) < q_k'(\theta_r).$$

Since $q''(\theta) \ne 0$, q_k' is strictly increasing in the interval $[\theta_l, \theta_r]$ and therefore q_k is strictly convex. Then we can solve the scalar equation (11.61), that is

$$\theta_t + (q_k(\theta))_x = 0,$$

with initial data

$$g(x,0) = \begin{cases} \theta_l & x < 0 \\ \theta_r & x > 0. \end{cases}$$

The solution is given by the formula

$$\theta_k(x,t) = \begin{cases} \theta_l & x < \lambda_k(\mathbf{u}_l)t \\ f_k\left(\frac{x}{t}\right) & \lambda_k(\mathbf{u}_l)t < x < \lambda_k(\mathbf{u}_r)t \\ \theta_r & x > \lambda_k(\mathbf{u}_r)t \end{cases}$$

where $f_k = (q_k')^{-1}$. Then, the k-rarefaction wave

$$\mathbf{u}_k(x,t) = \mathbf{v}_k(\theta_k(x,t)) = \begin{cases} \mathbf{u}_l & x < \lambda_k(\mathbf{u}_l)t \\ \mathbf{v}_k\left(f_k\left(\frac{x}{t}\right)\right) & \lambda_k(\mathbf{u}_l)t < x < \lambda_k(\mathbf{u}_r)t \\ \mathbf{u}_r & x > \lambda_k(\mathbf{u}_r)t \end{cases} \tag{11.77}$$

is a solution of the Riemann problem with initial states $\mathbf{u}_l, \mathbf{u}_r$. □

11.4.3 Lax entropy condition. Shock waves and contact discontinuities

We have seen in Sect. 11.3.2 that, along a jump discontinuity line Γ, the conservation law for a piecewise continuous solution translates into the Rankine-Hugoniot condition

$$\sigma\,[\mathbf{u}_r - \mathbf{u}_l] = [\mathbf{F}(\mathbf{u}_r) - \mathbf{F}(\mathbf{u}_l)] \qquad \text{on } \Gamma, \tag{11.78}$$

where σ is the speed along Γ and $\mathbf{u}_l$ and $\mathbf{u}_r$ denote the values that $\mathbf{u}$ attains on the shock line Γ from the left and the right sides, respectively. We also know from the scalar case, that these conditions are not enough to select a physically meaningful integral solution. Here an entropy criterion comes into play. We will limit ourselves to the so called *Lax entropy condition*, direct generalization of (4.66), p. 212. We will say that a shock wave is *admissible* or *entropic* if it satisfies this condition.

To introduce it, let us consider the situation in the *scalar* case from a different (heuristic) perspective. In order to uniquely determine a shock, we need to know the values u_l, u_r and σ. The Rankine-Hugoniot condition gives an equation for these three unknowns. Now, in the case of strictly convex/concave flux function q, the entropy condition requires that

$$q'(u_r) < \sigma(u_l, u_r) < q'(u_l),$$

prescribing that the characteristics *impinge into* Γ, both from the left and the right sides. In other words, the characteristics carry into Γ the values of u_l and u_r from the initial data, respectively. Thus, in principle, we have the correct number of information to select in a unique way u_l, u_r, σ on Γ.

In the vector case, the Rankine-Hugoniot conditions (11.78) provide m scalar equations for the speed σ and the $2m$ components of $\mathbf{u}_l$ and $\mathbf{u}_r$, that is, $2m + 1$

scalars. Somehow we are missing $m+1$ information. Where do we find them? Assume that, for some k, $1 \le k \le m$, the k^{th}-characteristic field $\lambda_k(\mathbf{u}_r)$, $\mathbf{r}_k(\mathbf{u})$ is genuinely nonlinear and that,

$$\lambda_k(\mathbf{u}_r) < \sigma < \lambda_{k+1}(\mathbf{u}_r).$$

This means that k characteristics impinge Γ from the *right*, thus providing k information on the values of $\mathbf{u}_r$.

Similarly, assume that, for some j, $1 \le j \le m$,

$$\lambda_j(\mathbf{u}_l) < \sigma < \lambda_{j+1}(\mathbf{u}_l).$$

This means that $m-j$ characteristics impinge Γ from the *left*, thus providing $m-j$ information on the values of $\mathbf{u}_l$. We need that

$$k + (m-j) = m+1$$

from which

$$j = k-1.$$

In conclusion, a possible generalization of the entropy condition is the following: **for some index k it holds**

$$\lambda_k(\mathbf{u}_r) < \sigma < \lambda_{k+1}(\mathbf{u}_r) \tag{11.79}$$

and

$$\lambda_{k-1}(\mathbf{u}_l) < \sigma < \lambda_k(\mathbf{u}_l). \tag{11.80}$$

We may rewrite the last two relations in the following form:

$$\lambda_k(\mathbf{u}_r) < \sigma < \lambda_k(\mathbf{u}_l) \tag{11.81}$$

and

$$\lambda_{k-1}(\mathbf{u}_l) < \sigma < \lambda_{k+1}(\mathbf{u}_r). \tag{11.82}$$

Condition (11.81) implies that the shock speed σ is intermediate between the speeds of the k^{th} characteristics coming from the right and the left sides of Γ. From (11.82) we infer that this happens for the index k **only**.

Observe that if $m=1$, (11.82) is empty and (11.81) reduces to the (11.55). If $k=1$ or $k=m$, (11.82) becomes, respectively

$$\sigma < \lambda_2(\mathbf{u}_r) \quad \text{or} \quad \lambda_{m-1}(\mathbf{u}_l) < \sigma.$$

Definition 11.21 If the k^{th} characteristic field $\lambda_k(\mathbf{u})$, $r_k(\mathbf{u})$ is genuinely nonlinear and the entropy conditions (11.81), (11.82) hold, we say that the discontinuity is a *k-shock*.

We will see that, if the pair $\lambda_k(\mathbf{u})$, $r_k(\mathbf{u})$ is linearly degenerate, then

$$\lambda_k(\mathbf{u}_r) = \sigma = \lambda_k(\mathbf{u}_l)$$

and the entropy inequality is satisfied in a weaker sense. We shall call this type of discontinuity a *k-contact discontinuity.*

11.4.4 Solution of the Riemann problem by a single *k*-shock

We now explore the possibility to solve problem (11.51), (11.52) by means of a single shock wave. In other words, we look for the pairs of states $\mathbf{u}_l$, $\mathbf{u}_r$ that can be connected by a k-shock. Guided by the linear case in Sect. 11.2.3, we introduce the following set.

Definition 11.22 Given $\mathbf{u}_0 \in \mathbb{R}^m$, the *Rankine-Hugoniot locus* of $\mathbf{u}_0$, denoted by $S(\mathbf{u}_0)$, is the set of the states $\mathbf{u} \in \mathbb{R}^m$ for which there exists a scalar $\sigma = \sigma(\mathbf{u}, \mathbf{u}_0)$ such that

$$\mathbf{F}(\mathbf{u}) - \mathbf{F}(\mathbf{u}_0) = \sigma(\mathbf{u}, \mathbf{u}_0)(\mathbf{u} - \mathbf{u}_0). \tag{11.83}$$

For $\mathbf{u}$ close to $\mathbf{u}_0$, we can write

$$\mathbf{F}(\mathbf{u}) - \mathbf{F}(\mathbf{u}_0) \sim D\mathbf{F}(\mathbf{u}_0)(\mathbf{u} - \mathbf{u}_0),$$

so that $\sigma(\mathbf{u}, \mathbf{u}_0) \sim \lambda(\mathbf{u}_0)$, with $\lambda(\mathbf{u}_0)$ eigenvalue of $D\mathbf{F}(\mathbf{u}_0)$. On the other hand, in the linear case, $S(\mathbf{u}_0)$ coincides with the union of m Hugoniot straight lines issued from $\mathbf{u}_0$, each one directed as one of the eigenvectors $\mathbf{r}_k$. Thus, in the nonlinear case, we expect that $S(\mathbf{u}_0)$ is given by the union of m regular curves, each one tangent at $\mathbf{u}_0$ to an eigenvector of $D\mathbf{F}(\mathbf{u}_0)$. Precisely, the following theorem holds[7].

Theorem 11.23 *Let* $\mathbf{u}_0 \in \mathbb{R}^m$. *Near* $\mathbf{u}_0$, $S(\mathbf{u}_0)$ *consists of the union of m curves* $S_k(\mathbf{u}_0)$ *of class* C^2. *Moreover, for each* $k = 1, 2, \ldots, m$, *there exists a parametrization of* $S_k(\mathbf{u}_0)$ *given by*

$$\varepsilon \longmapsto \boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_0), \qquad |\varepsilon| \le \varepsilon_0$$

for some $\varepsilon_0 > 0$, *with the following properties:*

a) $\boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_0) = \mathbf{u}_0 + \varepsilon \mathbf{r}_k(\mathbf{u}_0) + \dfrac{\varepsilon^2}{2} D\mathbf{r}_k(\mathbf{u}_0)\mathbf{r}_k(\mathbf{u}_0) + o(\varepsilon^2)$.

b) $\sigma(\boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_0), \mathbf{u}_0) = \lambda_k(\mathbf{u}_0) + \dfrac{\varepsilon}{2} \nabla\lambda_k(\mathbf{u}_0) \cdot \mathbf{r}_k(\mathbf{u}_0) + o(\varepsilon)$.

Remark 11.24 Theorems 11.18 and 11.23 imply that *the curves* $S_k(\mathbf{u}_0)$ *and* $R_k(\mathbf{u}_0)$ *have a second order contact at* $\mathbf{u}_0$.

When the k^{th}-characteristic field $\lambda_k(\mathbf{u})$, $r_k(\mathbf{u})$ is genuinely nonlinear, $S_k(\mathbf{u}_0)$ *takes the name of k-shock curve issued from* $\mathbf{u}_0$.

[7] For the proof, see e.g. *E. Godlewski, P.-A. Raviart*, [46].

Claim In this case, near a given a *left* state $\mathbf{u}_l$, we can decompose $S_k(\mathbf{u}_l)$ into the following three disjoint sets:

$$S_k(\mathbf{u}_l) = S_k^+(\mathbf{u}_l) \cup \{\mathbf{u}_l\} \cup S_k^-(\mathbf{u}_l)$$

where

$$S_k^-(\mathbf{u}_l) = \{\mathbf{u} \in S_k(\mathbf{u}_l) : \ \lambda_k(\mathbf{u}) < \sigma(\mathbf{u}, \mathbf{u}_l) < \lambda_k(\mathbf{u}_l)\} \tag{11.84}$$

and

$$S_k^+(\mathbf{u}_l) = \{\mathbf{u} \in S_k(\mathbf{u}_l) : \ \lambda_k(\mathbf{u}_l) < \sigma(\mathbf{u}, \mathbf{u}_l) < \lambda_k(\mathbf{u})\}\,. \tag{11.85}$$

Proof (of the claim) Let us use the normalization

$$\nabla\lambda_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) \equiv 1. \tag{11.86}$$

Then, by Theorem 11.23, we can write

$$\begin{aligned}\sigma_k(\varepsilon) &\equiv \sigma(\boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_l), \mathbf{u}_l) = \lambda_k(\mathbf{u}_l) + \frac{\varepsilon}{2} + o(\varepsilon)\\ \mathbf{u}_k(\varepsilon) &\equiv \boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_l) = \mathbf{u}_l + \varepsilon\mathbf{r}_k(\mathbf{u}_l) + o(\varepsilon).\end{aligned}$$

Hence

$$\begin{aligned}\lambda_k(\mathbf{u}_k(\varepsilon)) &= \lambda_k(\mathbf{u}_l) + \varepsilon\nabla\lambda_k(\mathbf{u}_l) \cdot \mathbf{r}_k(\mathbf{u}_l) + o(\varepsilon) = \lambda_k(\mathbf{u}_l) + \varepsilon + o(\varepsilon)\\ &= \sigma_k(\varepsilon) + \frac{\varepsilon}{2} + o(\varepsilon).\end{aligned}$$

Thinking of $\mathbf{u}_k(\varepsilon)$ as a possible *initial* state for $x > 0$, in the Riemann problem, we select which states satisfy the entropy conditions (11.79), (11.80), that is:

$$\begin{aligned}\lambda_k(\mathbf{u}_k(\varepsilon)) < \sigma_k(\varepsilon) < \lambda_{k+1}(\mathbf{u}_k(\varepsilon))\\ \lambda_{k-1}(\mathbf{u}_l) < \sigma_k(\varepsilon) < \lambda_k(\mathbf{u}_l).\end{aligned}$$

First of all, we have $\lambda_k(\mathbf{u}_k(\varepsilon)) < \sigma_k(\varepsilon)$ if and only if $\varepsilon < 0$ and $|\varepsilon|$ small enough. Also, since $\sigma_k(\varepsilon) \to \lambda_k(\mathbf{u}_l)$ and $\lambda_{k+1}(\mathbf{u}_k(\varepsilon)) \to \lambda_{k+1}(\mathbf{u}_l)$ as $\varepsilon \to 0$, we get

$$\sigma_k(\varepsilon) < \lambda_{k+1}(\mathbf{u}_k(\varepsilon)),$$

for $|\varepsilon|$ small enough. On the other hand, $\sigma_k(\varepsilon) < \lambda_k(\mathbf{u}_l)$ if and only if $\varepsilon < 0$ and $|\varepsilon|$ is small enough. Since $\lambda_{k-1}(\mathbf{u}_l) < \lambda_k(\mathbf{u}_l)$ we deduce

$$\lambda_{k-1}(\mathbf{u}_l) < \sigma_k(\varepsilon)$$

for $|\varepsilon|$ small enough.

Thus we can distinguish on the curve $S_k(\mathbf{u}_l)$, the *admissible* (*entropic*) part given by

$$S_k^-(\mathbf{u}_l) = \{\mathbf{u}_k(\varepsilon) : \ \lambda_k(\mathbf{u}_k(\varepsilon)) < \sigma_k(\varepsilon) < \lambda_k(\mathbf{u}_l)\}$$

corresponding to $\varepsilon < 0$, and the other part, given by

$$S_k^+(\mathbf{u}_l) = \{\mathbf{u}_k(\varepsilon) : \ \lambda_k(\mathbf{u}_l) < \sigma_k(\varepsilon) < \lambda_k(\mathbf{u}_k(\varepsilon))\}\,.$$

These two sets correspond precisely to (11.84) and (11.85), respectively. □

At this point, the proof of the following theorem is immediate.

Theorem 11.25 *Assume that, for some k, $1 \le k \le m$:*

a) The pair $\lambda_k(\mathbf{u}), r_k(\mathbf{u})$ *is genuinely nonlinear.*
b) $\mathbf{u}_r \in S_k^-(\mathbf{u}_l)$.

Then there exists a k-shock solution of the Riemann problem with initial data $\mathbf{u}_l$, $\mathbf{u}_r$, *given by*

$$\mathbf{u}_k(x,t) = \begin{cases} \mathbf{u}_l & \text{for } x < \sigma(\mathbf{u}_l, \mathbf{u}_r)t \\ \mathbf{u}_r & \text{for } x > \sigma(\mathbf{u}_l, \mathbf{u}_r)t. \end{cases} \tag{11.87}$$

11.4.5 The linearly degenerate case

Let us now examine the linearly degenerate case. Once more we first prove a structure lemma.

Lemma 11.26 *Let the pair* $\lambda_k(\mathbf{u}), r_k(\mathbf{u})$ *be linearly degenerate. Then, for every* $\mathbf{u}_0 \in \mathbb{R}^m$*:*

a) $S_k(\mathbf{u}_0) = R_k(\mathbf{u}_0)$.
b) $\sigma(\boldsymbol{\psi}_k(\varepsilon;\mathbf{u}_0), \mathbf{u}_0) = \lambda_k(\boldsymbol{\psi}_k(\varepsilon;\mathbf{u}_0)) = \lambda_k(\mathbf{u}_0)$.

Proof Let $\mathbf{v}_k = \mathbf{v}_k(\theta)$ be the equation for $R_k(\mathbf{u}_0)$, with $\mathbf{v}_k(0) = \mathbf{u}_0$. Thus $\mathbf{v}_k'(\theta) = \mathbf{r}_k(\mathbf{v}(\theta))$. Since

$$\nabla\lambda_k(\mathbf{u}) \cdot \mathbf{r}_k(\mathbf{u}) \equiv 0,$$

the function $\theta \longmapsto \lambda_k(\mathbf{v}_k(\theta))$ is constant and equal to $\lambda_k(\mathbf{u}_0)$. Thus, we can write:

$$\begin{aligned} \mathbf{F}(\mathbf{v}_k(\theta)) - \mathbf{F}(\mathbf{u}_0) &= \int_0^\theta \frac{d}{ds}\mathbf{F}(\mathbf{v}_k(s))ds \\ &= \int_0^\theta D\mathbf{F}(\mathbf{v}_k(s))\frac{d}{ds}\mathbf{v}_k(s)ds = \int_0^\theta D\mathbf{F}(\mathbf{v}_k(s))\mathbf{r}_k(\mathbf{v}_k(s))ds \\ &= \int_0^\theta \lambda_k(\mathbf{v}_k(s))\mathbf{r}_k(\mathbf{v}_k(s))ds = \lambda_k(\mathbf{u}_0)\int_0^\theta \frac{d}{ds}\mathbf{v}_k(s)ds \\ &= \lambda_k(\mathbf{u}_0)(\mathbf{v}_k(\theta) - \mathbf{u}_0). \end{aligned}$$

It follows that $\mathbf{v}_k = \mathbf{v}_k(\theta)$ defines both $S_k(\mathbf{u}_0)$ and $R_k(\mathbf{u}_0)$ and that b) holds. □

As an immediate consequence we have the following theorem.

Theorem 11.27 *Let the pair* $\lambda_k(\mathbf{u}), r_k(\mathbf{u})$ *be linearly degenerate and* $\mathbf{u}_r \in S_k(\mathbf{u}_l)$. *Then a solution of the Riemann problem with initial data* $\mathbf{u}_l$, $\mathbf{u}_r$ *is given by*

$$\mathbf{u}(x,t) = \begin{cases} \mathbf{u}_l & \text{for } x < \sigma t \\ \mathbf{u}_r & \text{for } x > \sigma t \end{cases} \tag{11.88}$$

with

$$\sigma = \sigma(\mathbf{u}_r, \mathbf{u}_l) = \lambda_k(\mathbf{u}_l) = \lambda_k(\mathbf{u}_r).$$

The solution (11.88) is called a *k-contact discontinuity*. Indeed, the characteristics on both sides are parallel to the discontinuity line (see Fig. 11.3, p. 630).

11.4.6 Local solution of the Riemann problem

Let us summarize the consequences of the theory we have developed so far. Select $k \in \{1, \dots, m\}$. If the pair $\lambda_k(\mathbf{u}), \mathbf{r}_k(\mathbf{u})$ is genuinely nonlinear, define

$$W_k(\mathbf{u}_l) = S_k^-(\mathbf{u}_l) \cup \{\mathbf{u}_l\} \cup R_k^+(\mathbf{u}_l).$$

By Theorems 11.18 and 11.23, $W_k(\mathbf{u}_l)$ is a curve of class C^2 in a neighborhood of $\mathbf{u}_l$ and we can use for $W_k(\mathbf{u}_l)$ the following parametrization:

$$\mathbf{u}_k(\varepsilon; \mathbf{u}_l) = \begin{cases} \boldsymbol{\varphi}_k(\varepsilon; \mathbf{u}_l) & \varepsilon > 0 \\ \boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_l) & \varepsilon < 0 \end{cases}$$

with $|\varepsilon| < \varepsilon_0$. We have seen that, if $\mathbf{u}_r$ coincides with one of the states on $W_k(\mathbf{u}_l)$, for some $k \in \{1, \dots, m\}$, then the Riemann problem can be solved by a single k-rarefaction wave or by a k-shock.

If $\lambda_k(\mathbf{u}), \mathbf{r}_k(\mathbf{u})$ is linearly degenerate, define

$$W_k(\mathbf{u}_l) = S_k(\mathbf{u}_l) = R_k(\mathbf{u}_l)$$

and we can choose the following parametrization:

$$\mathbf{u}_k(\varepsilon; \mathbf{u}_l) = \boldsymbol{\psi}_k(\varepsilon; \mathbf{u}_l), \qquad \text{for } |\varepsilon| < \varepsilon_0.$$

In this case, if $\mathbf{u}_r$ coincides with one of the states on $W_k(\mathbf{u}_l)$, then the Riemann problem can be solved by a single k-contact discontinuity.

When $\mathbf{u}_r$ does not belong to any $W_k(\mathbf{u}_l)$, still we can solve the Riemann problem if $\mathbf{u}_l$ and $\mathbf{u}_r$ are close enough. Precisely, the following result holds[8]. *We denote by $\mathcal{S}$ the class of integral solutions, consisting of at most $m+1$ constant states, connected by admissible shocks, rarefaction waves or contact discontinuities.*

Theorem 11.28 *Assume that for all k, $1 \le k \le m$, the k^{th}-characteristic field $\lambda_k(\mathbf{u}), r_k(\mathbf{u})$ is genuinely nonlinear or linearly degenerate. Given $\mathbf{u}_l \in \mathbb{R}^m$, there exists a neighborhood $N(\mathbf{u}_l)$ of $\mathbf{u}_l$ such that, if $\mathbf{u}_r \in N(\mathbf{u}_l)$, the Riemann problem has a unique solution belonging to $\mathcal{S}$.*

11.5 The Riemann Problem for the p-system

In this section we provide a complete analysis of the Riemann problem for the p-system

$$\begin{cases} w_t - v_x = 0 \\ v_t + p(w)_x = 0. \end{cases}$$

[8] For the proof, see e.g. *E. Godlewski, P.-A. Raviart*, [46].

We already know that this system is genuinely nonlinear, with eigenvalues

$$\lambda_1(w,v) = -\sqrt{a(w)} \quad \text{and} \quad \lambda_2(v,u) = \sqrt{a(w)}$$

and corresponding eigenvectors given, respectively, by

$$\mathbf{r}_1(w,v) = \begin{pmatrix} 1 \\ \sqrt{a(w)} \end{pmatrix} \qquad \mathbf{r}_2(w,v) = \begin{pmatrix} 1 \\ -\sqrt{a(w)} \end{pmatrix}.$$

We recall that p is decreasing and convex, that is

$$a(w) = -p'(w) > 0 \quad \text{and} \quad a'(w) = -p''(w) < 0 \tag{11.89}$$

and moreover

$$\lim_{w \to 0^+} p(w) = +\infty. \tag{11.90}$$

Given the two states $\mathbf{u}_l = (w_l, v_l)$ and $\mathbf{u}_r = (w_r, v_r)$ with $\mathbf{u}_l \neq \mathbf{u}_r$ and $w_l, w_r > 0$, we first look for shock solutions.

11.5.1 Shock waves

Since $m = 2$, we have two types of admissible shock waves. To determine them we first write the Rankine-Hugoniot locus of $\mathbf{u}_l$. Since

$$F(\mathbf{u}) = \begin{pmatrix} -v \\ p(w) \end{pmatrix},$$

$S(\mathbf{u}_l)$ is the set of the states $\mathbf{u} = (w,v)^\top$ satisfying the system

$$\begin{cases} (w - w_l)\sigma = (v_l - v) \\ (v - v_l)\sigma = (p(w) - p(w_l)), \end{cases} \tag{11.91}$$

with $\sigma = \sigma(u, u_l)$. Solving for v, we find

$$v - v_l = \pm\sqrt{(p(w) - p(w_l))(w_l - w)}. \tag{11.92}$$

• **1-shock**. Consider $\lambda_1, \mathbf{r}_1$. The states on the admissible part $S_1^-(\mathbf{u}_l)$, defined in (11.84) for $k = 1$, satisfy the entropy condition

$$\lambda_1(w,v) < \sigma < \lambda_1(w_l, v_l)$$

(see (11.81)), that is $-\sqrt{a(w)} < \sigma < -\sqrt{a(w_l)}$, which implies $\sigma < 0$ (thus it is *back shock*) and $w_l > w$, since $a' < 0$. Then, from the first equation in (11.91) it follows that $v_l > v$. Thus, for $S_1^-(\mathbf{u}_l)$ we must pick the minus sign in the right hand side of (11.92), i.e.

$$v = v_l - \sqrt{(p(w) - p(w_l))(w_l - w)}, \qquad w_l > w. \tag{11.93}$$

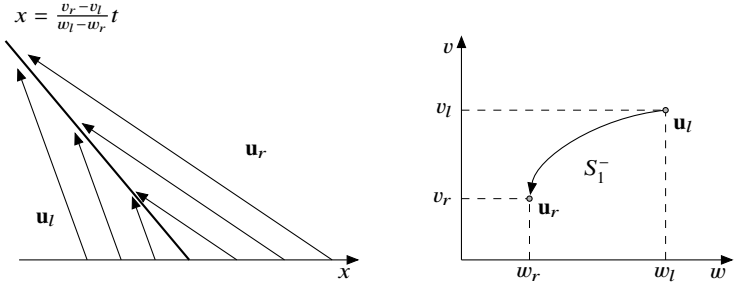

Fig. 11.4 States connected by a 1-shock

Since $p'(w) < 0$, $p(w) > p(w_l)$ and

$$\frac{dv}{dw} = -\frac{p'(w)(w_l - w) - (p(w) - p(w_l))}{2\sqrt{(p(w) - p(w_l))(w_l - w)}} > 0, \tag{11.94}$$

v is increasing. Moreover, one can check that $d^2v/dw^2 < 0$.

Summarizing, we have: *the initial states* $\mathbf{u}_r$ *that can be connected by a* 1*-shock to the state* $\mathbf{u}_l$ *belong to a curve* $S_1^- = S_1^-(\mathbf{u}_l)$ *increasing and concave (see Fig. 11.4). The solution of the corresponding Riemann problem is given by*

$$\mathbf{u}(x,t) = \begin{cases} \mathbf{u}_l & \text{for } x < \sigma t \\ \mathbf{u}_r & \text{for } x > \sigma t, \end{cases} \tag{11.95}$$

where $\sigma = \frac{v_r - v_l}{w_l - w_r}$.

• **2-shock**. Consider now $\lambda_2, \mathbf{r}_2$. Let $S_2^-(\mathbf{u}_l)$ be defined by (11.84), for $k = 2$. The states on $S_2^-(\mathbf{u}_l)$ satisfy the entropy condition

$$\lambda_2(w, v) < \sigma < \lambda_2(w_l, v_l)$$

that is $\sqrt{a(w)} < \sigma < \sqrt{a(w_l)}$.

In particular, we infer that $\sigma > 0$ (thus it is a *front shock*) and that $w_l < w$. Then, from the first equation in (11.91) it follows that $v_l > v$. Thus the curve $S_2^-(\mathbf{u}_l)$ is again defined by the equation with the minus sign in (11.92), i.e.

$$v = v_l - \sqrt{(p(w) - p(w_l))(w_l - w)}, \qquad \text{with } w_l < w. \tag{11.96}$$

From (11.94) it follows that $dv/dw < 0$ and one can check that $d^2v/dw^2 > 0$.

Summarizing, we have: *the initial states* $\mathbf{u}_r$ *that can be connected by a* 2*-shock to the state* $\mathbf{u}_l$ *belong to a curve* $S_2^- = S_2^-(\mathbf{u}_l)$ *decreasing and convex (see Fig. 11.5). The solution of the corresponding Riemann problem is still given by (11.95).*

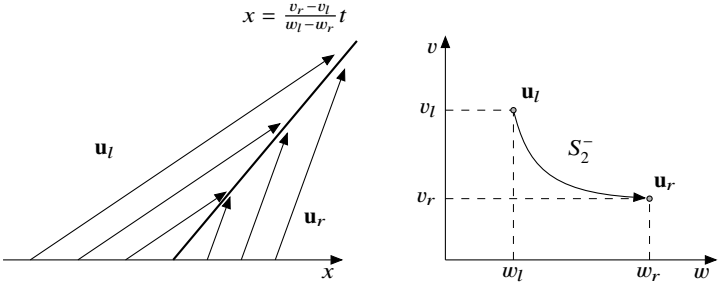

Fig. 11.5 States connected by a 2-shock

11.5.2 Rarefaction waves

Since the system is genuinely nonlinear we have two types of rarefaction waves. We first construct the rarefaction curves $R_1(\mathbf{u}_l)$ and $R_2(\mathbf{u}_l)$.

- **1-rarefaction wave**. Consider the 1st-characteristic field

$$\lambda_1(w, v) = -\sqrt{a(w)}, \quad \mathbf{r}_1(w, v) = \begin{pmatrix} 1 \\ \sqrt{a(w)} \end{pmatrix}.$$

Since $\lambda_1 < 0$, the corresponding solution is called a *back rarefaction wave*. To construct the **curve** $R_1(\mathbf{u}_l)$ we solve system (11.59), p. 631. Writing $\mathbf{v}(\theta) = (w(\theta), v(\theta))$, we find

$$\frac{dw}{d\theta} = 1, \qquad \frac{dv}{d\theta} = \sqrt{a(w(\theta))}. \tag{11.97}$$

The choice of $\theta = w$ as a parameter yields the equation

$$\frac{dv}{dw} = \sqrt{a(w)}. \tag{11.98}$$

The solution of (11.98) with initial data $\mathbf{u}_l$ is given by

$$v(w) = v_l + \int_{w_l}^{w} \sqrt{a(s)}ds. \tag{11.99}$$

The states on the admissible part $R_1^+(\mathbf{u}_l)$ of $R_1(\mathbf{u}_l)$ must satisfy the relation $\lambda_1(w, v) > \lambda_1(w_l, v_l)$, i.e. $\sqrt{a(w)} < \sqrt{a(w_l)}$, which implies $w > w_l$. Thus, the curve $R_1^+(\mathbf{u}_l)$ has equation

$$v(w) = v_l + \int_{w_l}^{w} \sqrt{a(s)}ds, \qquad \text{with } w > w_l.$$

Observe that $w \mapsto v(w)$ is increasing and $d^2v/dw^2 = a'(w)/2\sqrt{a(w)} < 0$. In particular, imposing $v_r = v(w_r)$, $w_r > w_l$, we get $v_l < v_r$.

Finally we solve for $w = w(x, t)$ the scalar conservation law

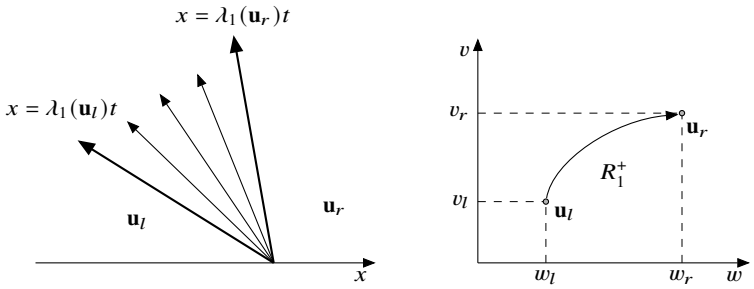

Fig. 11.6 States connected by a 1-rarefaction wave

$$w_t + \lambda_1(w)w_x = w_t - \sqrt{a(w)}w_x = 0.$$

From Thm. 4.13, b), p. 213, we find $w(x,t) = f(-\frac{x}{t})$, where f is the inverse of the function $w \longmapsto \sqrt{a(w)}$.

Summarizing, we have: *the initial states* $\mathbf{u}_r$ *that can be connected by a* 1*-rarefaction wave to the state* $\mathbf{u}_l$ *belong a curve* $R_1^+ = R_1^+(\mathbf{u}_l)$ *increasing and concave (see Fig. 11.6). The solution of the corresponding Riemann problem is given by*

$$\mathbf{u}_1(x,t) = \begin{cases} \mathbf{u}_l & x \le \sqrt{a(w_l)}t \\ \mathbf{f}_1\left(-\frac{x}{t}\right) & \sqrt{a(w_l)}t < x < \sqrt{a(w_r)}t \\ \mathbf{u}_r & x \ge \sqrt{a(w_r)}t, \end{cases}$$

where

$$\mathbf{f}_1(r) = \left(f(r), v_l + \int_{w_l}^{f(r)} \sqrt{a(s)}ds\right)^\top.$$

- **2-rarefaction wave**. Consider now the 2^{nd}-characteristic field

$$\lambda_2(w,v) = \sqrt{a(w)}, \quad \mathbf{r}_2(w,v) = \begin{pmatrix} 1 \\ -\sqrt{a(w)} \end{pmatrix}.$$

Since $\lambda_2 > 0$, the corresponding solution is called a *front rarefaction wave*. To construct the curve $R_2(\mathbf{u}_l)$ we can proceed as before. The system (11.59) is given by

$$\frac{dw}{d\theta} = 1, \qquad \frac{dv}{d\theta} = -\sqrt{a(w(\theta))}$$

which yields the equation

$$\frac{dv}{dw} = -\sqrt{a(w)}. \tag{11.100}$$

The solution of (11.100) with initial data $\mathbf{u}_l$ is given by

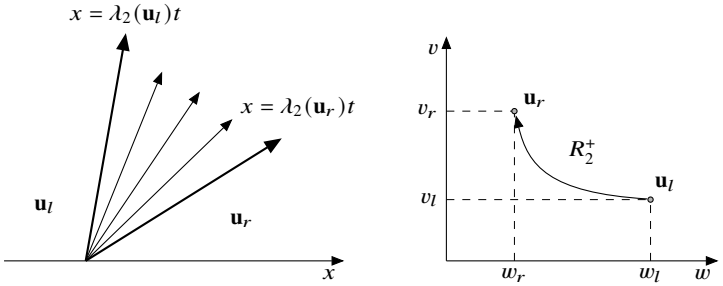

Fig. 11.7 States connected by a 2-rarefaction wave

$$v(w) = v_l - \int_{w_l}^{w} \sqrt{a(s)}ds.$$

The states on the admissible part $R_2^+(\mathbf{u}_l)$ of $R_2(\mathbf{u}_l)$ must satisfy the relation $\lambda_2(w, v) > \lambda_2(w_l, v_l)$, i.e. $\sqrt{a(w)} > \sqrt{a(w_l)}$, which implies $w < w_l$. Thus, the curve $R_2^+(\mathbf{u}_l)$ has equation

$$v(w) = v_l - \int_{w_l}^{w} \sqrt{a(s)}ds \qquad \text{with } w < w_l.$$

Observe that $w \mapsto v(w)$ is decreasing and

$$\frac{d^2v}{dw^2} = -a'(w)/2\sqrt{a(w)} > 0.$$

In particular, imposing $v_r = v(w_r)$, with $w_r < w_l$, we infer $v_l < v_r$. Finally solving

$$w_t + \lambda_2(w)w_x = w_t + \sqrt{a(w)}w_x = 0,$$

we find $w(x,t) = f(\frac{x}{t})$, where f is the inverse function of $w \longmapsto \sqrt{a(w)}$.

Summarizing, we have: *the initial states* $\mathbf{u}_r$ *that can be connected by a 2-rarefaction wave to the state* $\mathbf{u}_l$ *belong to a curve* $R_2^+ = R_2^-(\mathbf{u}_l)$ *decreasing and convex (Fig. 11.7). The solution of the corresponding Riemann problem is given by*

$$\mathbf{u}_2(x,t) = \begin{cases} \mathbf{u}_l & x \le \sqrt{a(w_l)}t \\ \mathbf{f}_2\left(\frac{x}{t}\right) & \sqrt{a(w_l)}t < x < \sqrt{a(w_r)}t \\ \mathbf{u}_r & x \ge \sqrt{a(w_r)}t, \end{cases}$$

where

$$\mathbf{f}_2(r) = \left(f(r), v_l - \int_{w_l}^{f(r)} \sqrt{a(s)}ds\right)^\top.$$

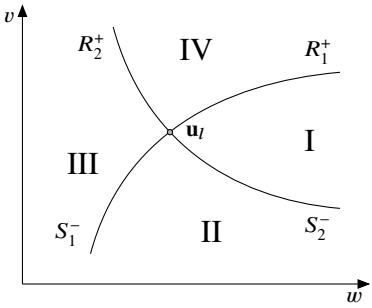

Fig. 11.8 Partition of the plane v, w induced by the curves $W_1(\mathbf{u}_l)$ and $W_2(\mathbf{u}_l)$

11.5.3 The solution in the general case

Based on the results in the previous subsections, given a left initial datum $\mathbf{u}_l = (v_l, w_l)$, the two C^2-curves

$$W_1(\mathbf{u}_l) = R_1^+(\mathbf{u}_l) \cup \{\mathbf{u}_l\} \cup S_1^-(\mathbf{u}_l)$$

and

$$W_2(\mathbf{u}_l) = R_2^+(\mathbf{u}_l) \cup \{\mathbf{u}_l\} \cup S_2^-(\mathbf{u}_l)$$

partition the plane v, w in four regions I, II, III, IV, as it is shown in Fig. 11.8.

If $\mathbf{u}_r$ belongs to one of these two curves, the solution of the Riemann problem is given by a single wave, rarefaction or shock. In general, by Theorem 11.28, p. 641, there exists a neighborhood $N(\mathbf{u}_l)$ such that, if

$$\mathbf{u}_r \in N(\mathbf{u}_l),$$

the Riemann problem has a unique solution belonging to the class $\mathcal{S}$.

For the p-system we have precise information on $N(\mathbf{u}_l)$. Indeed, we show that if $\mathbf{u}_r$ belongs to one of the regions I, II, III the Riemann problem is always solvable, while $\mathbf{u}_r$ must be sufficiently close to $\mathbf{u}_l$ if $\mathbf{u}_r$ belongs to the region IV. We start with the following result.

Theorem 11.29 *If $\mathbf{u}_r$ belongs to one of the regions I, II, III, there exists a unique weak solution of the Riemann problem, belonging to $\mathcal{S}$.*

Proof Let $\mathcal{F}$ be the family of curves W_2 starting from points on $W_1(\mathbf{u}_l)$. Precisely, let us define

$$\mathcal{F} = \{W_2(\mathbf{u}_0) : \mathbf{u}_0 \in W_1(\mathbf{u}_l)\}.$$

To prove the theorem, it is enough to show that every point $\mathbf{u}_r$ in one of the regions I, II, III belongs to one and only one curve $W_2(\mathbf{u}_0)$ of the family $\mathcal{F}$. In fact, if this is true, we can construct the unique entropic solution by first connecting $\mathbf{u}_l$ to $\mathbf{u}_0$ by a 1-rarefaction wave or a 1-shock and then by connecting $\mathbf{u}_0$ to $\mathbf{u}_r$ by a 2-rarefaction wave or a 2-shock. Clearly, the location of $\mathbf{u}_r$ determine the type of waves to be chosen.

To be specific, assume that $\mathbf{u}_r$ belongs to region I. With reference to Fig. 11.9, consider the points

$$\mathbf{p} \in W_1(\mathbf{u}_l)$$

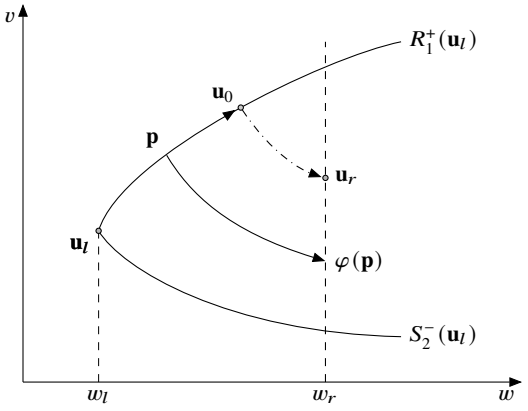

Fig. 11.9 The map φ in the proof of Theorem 11.29

within the vertical lines $w = w_l$ and $w = w_r$. In particular, being $\mathbf{p} \in R_1^+(\mathbf{u}_l)$, the coordinates of $\mathbf{p}$ are $(w, v(w))$, where (see (11.99))

$$v(w) = v_l + \int_{w_l}^{w} \sqrt{a(s)}\, ds.$$

Since the slope of $S_2^-(\mathbf{p})$ is negative and bounded, all these curves intersect the straight line $w = w_r$ at some point $\varphi(\mathbf{p}(w))$ of coordinates $(w_r, V(w))$ where, since $\varphi(\mathbf{p}) \in S_2^-(\mathbf{p})$, (see (11.96)):

$$\begin{aligned} V(w) &= v(w) - \sqrt{(p(w_r) - p(w))(w - w_r)} \\ &= v_l + \int_{w_l}^{w} \sqrt{a(s)}\, ds - \sqrt{(p(w_r) - p(w))(w - w_r)}. \end{aligned}$$

Hence the map

$$w \longmapsto \varphi(\mathbf{p}(w))$$

is continuous and if w is close to w_r, the point $\varphi(\mathbf{p}(w))$ is above $\mathbf{u}_r$. Since $\varphi(\mathbf{u}_l)$ is below $\mathbf{u}_r$, by continuity, there exists a point $\mathbf{u}_0$ such that $\varphi(\mathbf{u}_0) = \mathbf{u}_r$.

This shows that every point $\mathbf{u}_r$ in the region I belongs to one of the curves of the family $\mathcal{F}$. To prove uniqueness of this curve, it is enough to check that dV/dw is strictly positive. In fact, we have

$$\frac{dV}{dw} = \sqrt{a(w)} - \frac{p(w_r) - p(w) + p'(w)(w_r - w)}{2\sqrt{(p(w_r) - p(w))(w - w_r)}} > 0,$$

since $p'(w) < 0$ and $w_r > w$. Thus, when $\mathbf{u}_r$ belongs to the region I the proof is complete.

If $\mathbf{u}_r$ belongs to the region III the argument is similar. For $\mathbf{u}_r$ in the region II, we show that the horizontal line $v = v_r$ intersects $S_1^-(\mathbf{u}_l)$ and $S_2^-(\mathbf{u}_l)$ at two points

$$\mathbf{p}_1 = (w_1, v_r) \qquad \text{and} \qquad \mathbf{p}_2 = (w_2, v_r),$$

uniquely determined. To find w_2, we solve the equation (see (11.96))

$$v_r = v_l - \sqrt{(p(w_l) - p(w))(w - w_l)}, \quad w > w_l. \tag{11.101}$$

Now, the function

$$w \mapsto v_l - \sqrt{(p(w_l) - p(w))(w - w_l)}$$

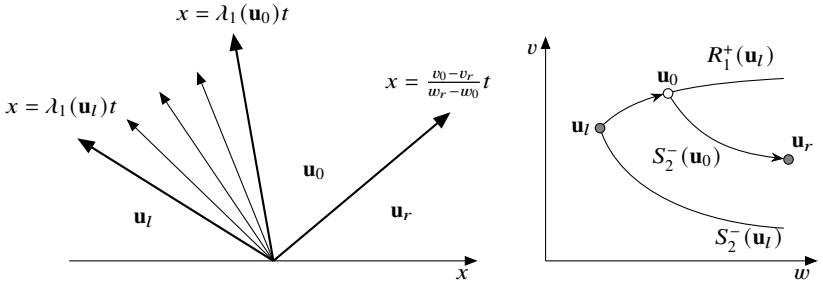

Fig. 11.10 Solution of the Riemann problem when $\mathbf{u}_r$ belongs to region I

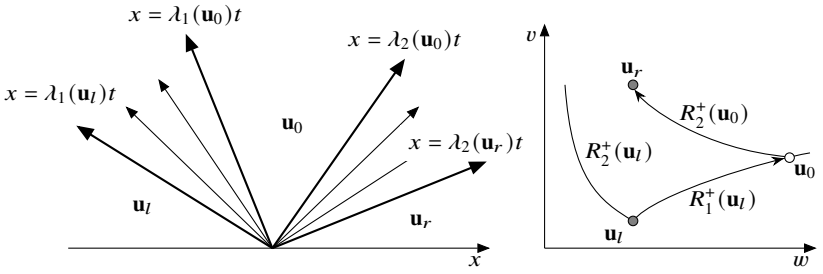

Fig. 11.11 The solution of the Riemann problem when $\mathbf{u}_r$ belongs to region IV

is one-to-one from $[w_l, +\infty)$ onto $[-\infty, v_l)$. Since $v_r < v_l$, there exists exactly one solution w_2 of (11.101).

Similarly, using the fact that

$$p(w) \to +\infty \text{ as } w \to 0^+,$$

it follows that there exists a unique solution w_1 of the equation (see (11.93))

$$v_r = v_l - \sqrt{(p(w_l) - p(w))(w - w_l)}, \quad w < w_l.$$

The rest of the proof follows closely the argument we used for the region I. We leave the details to the reader. □

The construction of the solution when $\mathbf{u}_r$ is in the region I is described in Fig. 11.10: $\mathbf{u}_l$ is connected to $\mathbf{u}_0$ by a *back rarefaction wave* and then $\mathbf{u}_0$ is connected to $\mathbf{u}_r$ by a *front-shock*.

When $\mathbf{u}_r$ is in the region IV, *sufficiently close to* $\mathbf{u}_l$, the solution can be constructed as in the other cases, as shown in Fig. 11.11: $\mathbf{u}_l$ is connected to $\mathbf{u}_0$ by a *back rarefaction wave* and then $\mathbf{u}_0$ is connected to $\mathbf{u}_r$ by a *front rarefaction wave*.

We now show that, when $\mathbf{u}_r$ belongs to the region IV, it is not always possible to construct a solution in the class we are considering.

Proposition 11.30 *Let* $\mathbf{u}_r$ *belong to the region IV and assume that*

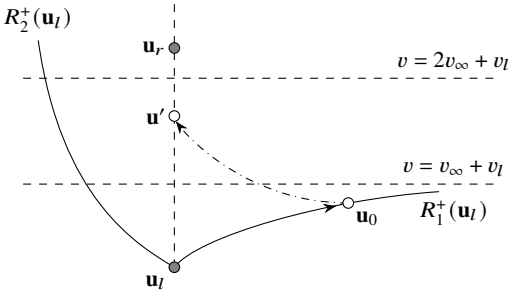

Fig. 11.12 Non-existence for the Riemann problem

$$v_\infty \equiv \int_{w_l}^{+\infty} \sqrt{a(w)}dw < \infty. \tag{11.102}$$

If

$$v_r > v_l + 2v_\infty,$$

the Riemann problem cannot have a solution belonging to the class $\mathcal{S}$.

Proof If (11.102) holds, the rarefaction curve $R_1^+(\mathbf{u}_l)$, of equation

$$v = v_l + \int_{w_l}^{w} \sqrt{a(s)}ds$$

has a horizontal asymptote, given by the straight line $v = v_\infty + v_l$. With reference to Fig. 11.12, consider the state $\mathbf{u}_r = (w_l, v_r)$ with

$$v_r > 2v_\infty + v_l.$$

We show that no 2-rarefaction curve issued from a point on $R_1^+(\mathbf{u}_l)$ can reach $\mathbf{u}_r$.

In fact, for every point $\mathbf{u}_0 = (w_0, v_0) \in R_1^+(\mathbf{u}_l)$ we have

$$v_0 = v_l + \int_{w_l}^{w_0} \sqrt{a(s)}ds < v_l + v_\infty.$$

Therefore, if

$$\mathbf{u}' = (w_l, v') \in R_2^+(\mathbf{u}_0) \cap \{w = w_l\},$$

then

$$v' = v_0 - \int_{w_0}^{w_l} \sqrt{a(s)}ds = v_0 + \int_{w_l}^{w_0} \sqrt{a(s)}ds < v_l + 2v_\infty < v_r.$$

Thus, if $v_r > v_l + 2v_\infty$, there is no way to connect $\mathbf{u}_l$ with $\mathbf{u}_r$ through an intermediate state $\mathbf{u}_0 \in R_1^+(\mathbf{u}_l)$, and this is clearly the only way to construct a solution in the class $\mathcal{S}$, with initial data $\mathbf{u}_l, \mathbf{u}_r$. □

Remark 11.31 Note that (11.102) is true in the important case of a polytropic gas, for which

$$a(w) = \gamma k w^{-\gamma-1} \qquad \gamma > 1.$$

To get a physical interpretation of what happens, consider the extreme situation

$$v_r = 2v_\infty + v_l.$$

Then $\mathbf{u}_r = (w_r, v_r)$ formally belongs to a 2- rarefaction curve coming from $\mathbf{u}_0 = (+\infty, v_\infty + v_l)$. Since

$$\lambda_1(\mathbf{u}_0) = \lambda_2(\mathbf{u}_0) = 0,$$

the solution of the Riemann problem is described in Fig. 11.11, but with the two rarefaction waves separated by the vertical line $x = 0$. On this line $w = 1/\rho = +\infty$ or $\rho = 0$, which, in the case of the motion of a gas along a tube, corresponds to a formation of a vacuum zone.

Problems

11.1 (*The telegrapher system, distorsionless case*) Consider the system (see Sect. 11.2.2)

$$\begin{cases} LI_t + V_x + RI = 0, \\ CV_t + I_x + GV = 0 \end{cases} \qquad (x \in \mathbb{R}, t > 0)$$

a) Show that in the case $RC = GL$, the full system is uncoupled and reduces to the *equations*

$$w_t^\pm \pm \frac{1}{\sqrt{LC}} w_x^\pm = -\frac{R}{L} w^\pm$$

with initial conditions

$$w^\pm(x, 0) = \frac{1}{2}\left[\frac{I_0(x)}{\sqrt{C}} \pm \frac{V_0(x)}{\sqrt{L}}\right] \equiv w_0^\pm(x).$$

b) Find an explicit formula for the original solution $(U(x,t), V(x,t))$.

[*Answer.* b) The solution is given by:

$$U(x,t) = \left\{w_0^+(x + t/\sqrt{LC}) + w_0^-(x - t/\sqrt{LC})\right\}\sqrt{C}e^{-\frac{R}{L}t}$$
$$V(x,t) = \left\{w_0^+(x + t/\sqrt{LC}) - w_0^-(x - t/\sqrt{LC})\right\}\sqrt{L}e^{-\frac{R}{L}t},$$

which is a superposition of damped travelling waves.]

11.2 Show that a k-Riemann invariant is constant along a k-rarefaction wave.

11.3 Consider the p-system. Is it possible to have a shock with only one discontinuous component?

11.4 Consider the solution of the Riemann problem for the p-system when $\mathbf{u}_r$ is in the region I. Instead of proceeding as in Fig. 11.10, can we alternatively connect $\mathbf{u}_l$ to an intermediate state $\mathbf{u}_0$ on $S_2^-(\mathbf{u}_l)$ by a *front shock* and then connect $\mathbf{u}_0$ to $\mathbf{u}_r$ by a *back rarefaction wave?*

11.5 Describe as in Figs. 11.10 and 11.11 the structure of the solution of the Riemann problem for the p-system, when $\mathbf{u}_r$ belongs to one of the regions II and III.

11.6 In the Riemann problem for the gas dynamic system one assigns the two states

$$(\rho_l, v_l, p_l) \qquad \text{and} \qquad (\rho_r, v_r, p_r).$$

Assume that we are in presence of a discontinuity, propagating at speed σ. Let

$$U = v - \sigma,$$

which represents the flow velocity relative to the discontinuity.

a) Using the system in the conservation form (11.16), show that the Rankine-Hugoniot jump conditions can be written in the following form:

$$\begin{cases} [\rho U] = 0 \\ \left[\rho U^2 + p\right] = 0 \\ \left[(\rho(E + \frac{1}{2}U^2) + p)U\right] = 0. \end{cases}$$

b) Set $M = \rho U$. What does M represent? Show that if $M = 0$ the discontinuity must be a contact discontinuity.

[*Answer.* b) M represent the mass flux across the discontinuity curve. If $M = 0$, no mass goes through the discontinuity and we necessarily have $U_l = U_r = 0$. Hence $v_r = v_l = \sigma$ and, from the second Rankine-Hugoniot condition, we infer $p_r = p_l$. Since we have a discontinuity, it must be $\rho_l \neq \rho_r$ and therefore we have a 2-contact discontinuity with $\sigma = \lambda_2 = v$. It can be shown that, if $M \neq 0$, we get a 1 or 2-shock.]

11.7 (*Shallow water waves – I*) The so called *shallow water system* governs the motion of water, with constant density, under the condition that the ratio between the water depth and the typical wave length of the free surface is small. Other simplifying assumptions are:

1. The motion is essentially bidimensional and we use a reference frame where x and y are the horizontal and vertical coordinates, respectively. In this coordinate system, the free surface is described by a graph
$$y = h(x, t),$$
while the bottom is flat, at level $y = 0$ (see Fig. 11.13).
2. The vertical acceleration is small compared to g, the gravitational acceleration, and the horizontal component u of the fluid velocity has no relevant variations along the vertical direction. Thus,
$$u = u(x, t).$$

Under the above conditions, the governing system reduces to mass conservation and momentum balance along the x direction, and takes the following form:

$$\begin{cases} h_t + hu_x + uh_x = 0 \\ u_t + uu_x + gh_x = 0. \end{cases} \tag{11.103}$$

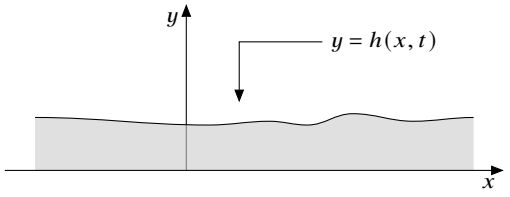

Fig. 11.13 Reference frame for the shallow water system

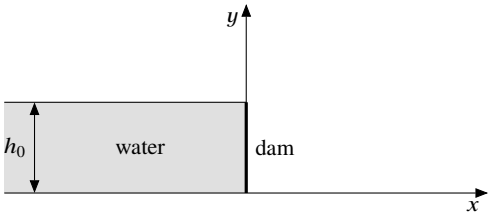

Fig. 11.14 Schematic representation for the dam break problem

a) Show that (11.103) is strictly hyperbolic and genuinely nonlinear.
b) Compute the Riemann invariants.

[*Answer.* a) The eigenvalues are

$$\lambda_1(h,u) = u - \sqrt{gh}, \qquad \lambda_2(h,u) = u + \sqrt{gh},$$

with eigenvectors

$$\mathbf{r}_1(h,u) = \begin{pmatrix} -\sqrt{h} \\ \sqrt{g} \end{pmatrix}, \quad \mathbf{r}_2(h,u) = \begin{pmatrix} \sqrt{h} \\ \sqrt{g} \end{pmatrix}.$$

Moreover

$$\nabla\lambda_1 \cdot \mathbf{r}_1 = \nabla\lambda_2 \cdot \mathbf{r}_2 = \frac{3}{2}\sqrt{g}.$$

b) The functions

$$R(h,u) = u - 2\sqrt{hg}, \qquad S(h,u) = u + 2\sqrt{hg}$$

are the Riemann invariants]

11.8 (*Shallow water waves – II: the* dam break *problem*) Assume that a quantity of shallow water of height h_0 is held motionless in the domain $x < 0$ by a dam located at $x = 0$ (see Fig. 11.14). Ahead of the dam there is no water. Suppose that the dam suddenly breaks at time $t = 0$ and that we want to determine the water flow u and the profile h of the free surface for $t > 0$. Note that $0 \le h \le h_0$ and $u \ge 0$. According to the shallow water model in Problem 11.7, we have to solve the Riemann problem for system (11.103), with data

$$(h_l, u_l) = (h_0, 0), \qquad (h_r, u_r) = (0, 0).$$

We ask the reader to provide an explicit analytic solution by filling in the details in the following steps, along the guideline of the analysis of the p-system[9]. We set

$$c_0 = \sqrt{gh_0}.$$

a) Examine which waves can connect (h,u) to the right of $(h_0,0)$. Note that $h < h_0$ for $x > 0$. Using the Rankine-Hugoniot conditions and the entropy condition, show that it cannot be a 1-shock. Show instead that these states belong to the rarefaction curve $R_1^+(h_0,0)$ given by the parametric equations

$$h(\theta) = \left(\sqrt{h_0} - \frac{\theta}{2}\right)^2, \quad u(\theta) = \sqrt{g}\theta, \qquad \text{with } \theta \ge 0, \tag{11.104}$$

or, eliminating θ, by the equation

[9] The solution can be also found by a direct use of the Riemann invariants. See e.g. *J. Billingham, A. C. King*, [24].

$$u = 2(c_0 - \sqrt{gh}), \tag{11.105}$$

in the first quadrant of the state plane $h \geq 0, u \geq 0$. In particular, check that, if

$$(h, u) \in R_1^+(h_0, 0),$$

then

$$\lambda_1(h, u) = u - c_0 = 2c_0 - 3\sqrt{gh} > -c_0 = \lambda_1(h_0, 0) \tag{11.106}$$

since $h < h_0$.

b) Deduce from (11.105) and (11.106), that the states $(h_0, 0)$ and $(0, 2c_0)$, the right end point of $R_1^+(h_0, 0)$, are connected by a 1-rarefaction wave defined in the sector

$$-c_0 t = \lambda_1(h_0, 0)t < x < \lambda_1(0, 2c_0) = 2c_0 t$$

and that the state $(0, 2c_0)$ is connected to $(0, 0)$ by a 2-shock, for

$$x > 2c_0 t.$$

c) To derive the following analytical expression for the 1-rarefaction wave, solve for θ the equation

$$\lambda_1(h(\theta), u(\theta)) = \frac{3}{2}\sqrt{g}\theta - c_0 = \frac{x}{t},$$

and substitute into (11.104):

$$h(x, t) = \frac{1}{9g}\left(2c_0 - \frac{x}{t}\right)^2, \quad u(x, t) = \frac{2}{3}\left(c_0 + \frac{x}{t}\right).$$

In Fig. 11.15 the profiles of the characteristics configuration and the profiles of h and u.

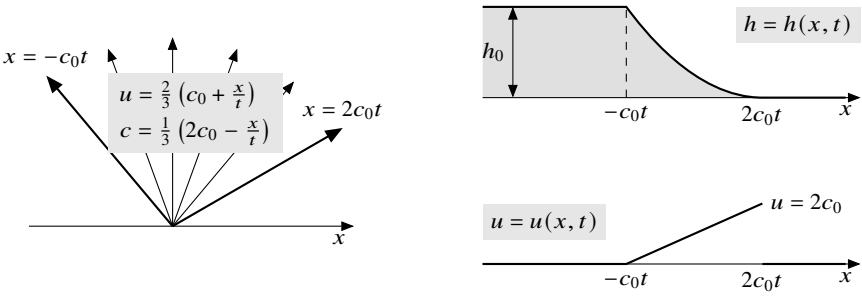

Fig. 11.15 Characteristics and the h and u profiles in the dam break problem at time t

Appendix A
Measures and Integrals

A.1 A Counting Problem

Two persons, that we denote by $\mathcal{R}$ and $\mathcal{L}$, must compute the total value of M coins, ranging from 1 to 50 cents. $\mathcal{R}$ decides to group the coins arbitrarily in piles of, say, 10 coins each, then to compute the value of each pile and finally to sum all these values. $\mathcal{L}$, instead, decides to partition the coins according to their value, forming piles of 1-cent coins, of 5-cents coins and so on. Then he computes the value of each pile and finally sums all their values.

In more analytical terms, let

$$V : M \to \mathbb{N}$$

be a *value function* that associates to each element of M (i.e. each coin) its value. $\mathcal{R}$ partitions the **domain** of V in disjoint subsets, sums the values of V in such subsets and then sums everything. $\mathcal{L}$ considers each point p in the **image** of V (the value of a single coin), considers the inverse image $V^{-1}(p)$ (the pile of coins with the same value p), computes the corresponding value and finally sums over every p.

These two ways of counting correspond to the strategy behind the definitions of the integrals of Riemann and Lebesgue, respectively. Since V is defined on a discrete set and is integer valued, in both cases there is no problem in summing its values and the choice is determined by an efficiency criterion. Usually, the method of $\mathcal{L}$ is more efficient.

In the case of a real (or complex) function f, the "sums of its values" corresponds to an integration of f. While the construction of $\mathcal{R}$ remains rather elementary, the one of $\mathcal{L}$ requires new tools.

Let us examine the particular case of a *bounded* and *positive* function, defined on an interval $[a, b] \subset \mathbb{R}$. Thus, let

$$f : [a, b] \to [\inf f, \sup f] .$$

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S. Salsa, G. Verzini, *Partial Differential Equations in Action*, UNITEXT 147,
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To construct the Riemann integral, we partition $[a, b]$ in subintervals $I_1, \ldots, I_N$ (the piles of $\mathcal{R}$), then we choose in each interval I_k a point ξ_k and we compute $f(\xi_k)l(I_k)$, where $l(I_k)$ is the length of I_k, (i.e. the value of the k-th pile). Now we sum the values $f(\xi_k)l(I_k)$ and set

$$(\mathcal{R})\int_a^b f = \lim_{\delta\to 0}\sum_{k=1}^{N} f(\xi_k)l(I_k),$$

where $\delta = \max\{l(I_1), \ldots, l(I_N)\}$. If the limit is finite and moreover is independent of the choice of the points ξ_k, then this limit defines the Riemann integral of f in $[a, b]$.

Now, let us examine the Lebesgue strategy. This time we partition the interval $[\inf f, \sup f]$ in subintervals $[y_{k-1}, y_k]$ (the values of each coin for $\mathcal{L}$) with

$$\inf f = y_0 < y_1 < \ldots < y_{N-1} < y_N = \sup f.$$

Then we consider the inverse images $E_k = f^{-1}([y_{k-1}, y_k])$ (the piles of homogeneous coins) and we would like to compute their ... *length*. However, in general E_k is **not** an interval or a union of intervals and, in principle, it could be a very irregular set so that it is not clear what is the "length" of E_k.

Thus, the need arises to associate with every E_k a *measure,* which replaces the length when E_k is an irregular set. This leads to the introduction of the *Lebesgue measure* of (practically every) set $E \subseteq R$, denoted by $|E|$.

Once we know how to measure E_k (the number of coins in the $k - th$ pile), we choose an arbitrary point $\overline{\alpha}_k \in [y_{k-1}, y_k]$ and we compute $\overline{\alpha}_k\,|E_k|$ (the value of the $k - th$ pile). Then, we sum all the values $\overline{\alpha}_k\,|E_k|$ and set

$$(\mathcal{L})\int_a^b f = \lim_{\rho\to 0}\sum_{k=1}^{N} \overline{\alpha}_k\,|E_k|$$

where ρ is the maximum among the lengths of the intervals $[y_{k-1}, y_k]$. It can be seen that under our hypotheses, *the limit exists, is finite and is independent of the choice of* $\overline{\alpha}_k$. Thus, we may always choose $\overline{\alpha}_k = y_{k-1}$. This remark leads to the definition of the Lebesgue integral in Sect. A.3: the number $\sum_{k=1}^{N} y_{k-1}\,|E_k|$ is nothing else that the integral of a *simple function*, which approximates f from below and whose range is the finite set $y_0 < \ldots < y_{N-1}$. The integral of f is the supremum of these numbers.

The resulting theory has several advantages with respect to that of Riemann. For instance, the class of integrable functions is much wider and there is no need to distinguish among bounded or unbounded functions or integration domains.

Especially important are the convergence theorems presented in Sect. A.4, which allow the possibility of interchanging the operation of limit and integration, under rather mild conditions.

Finally, the construction of the Lebesgue measure and integral can be greatly generalized as we will mention in Sect. A.5.

For the proofs of the theorems stated in this Appendix, the interested reader can consult e.g. *W. Rudin*, [39], or *R. L. Wheeden, A. Zygmund*, [42].

A.2 Measures and Measurable Functions

A measure in a set Ω is a *set function*, defined on a particular class of subsets of Ω called *measurable sets*, which "behaves well" with respect to union, intersection and complementation. Precisely:

Definition A.1 A collection $\mathcal{F}$ of subsets of Ω is called σ*-algebra* if:

1. $\emptyset, \Omega \in \mathcal{F}$.
2. $A \in \mathcal{F}$ implies $\Omega \setminus A \in \mathcal{F}$.
3. If $\{A_k\}_{k\in\mathbb{N}} \subset \mathcal{F}$ then also $\bigcup A_k$ and $\bigcap A_k$ belong to $\mathcal{F}$.

Example A.2 If $\Omega = \mathbb{R}^n$, we can define the smallest σ-algebra containing all the open subsets of $\mathbb{R}^n$, called the *Borel* σ*-algebra*. Its elements are called *Borel sets*, typically obtained by countable unions and/or intersections of open sets.

Definition A.3 Given a σ-algebra $\mathcal{F}$ in a set Ω, a *measure* on $\mathcal{F}$ is a function $\mu : \mathcal{F} \to \mathbb{R}$ such that:

1. $\mu(A) \geq 0$ for every $A \in \mathcal{F}$.
2. If $A_1, A_2, \ldots$ are pairwise disjoint sets in $\mathcal{F}$, then

$$\mu(\cup_{k\geq 1} A_k) = \sum_{k\geq 1} \mu(A_k) \qquad (\sigma\text{-additivity}).$$

The elements of $\mathcal{F}$ are called *measurable sets*.

The Lebesgue measure in $\mathbb{R}^n$ is defined on a σ*-algebra* $\mathcal{M}$ containing the Borel σ-algebra, through the following theorem.

Theorem A.4 *There exists in* $\mathbb{R}^n$ *a* σ*-algebra* $\mathcal{M}$ *and a measure*

$$|\cdot|_n : \mathcal{M} \to [0, +\infty]$$

with the following properties:

1. *Each open and closed set belongs to* $\mathcal{M}$.
2. *If* $A \in \mathcal{M}$ *and* A *has measure zero, every subset of* A *belongs to* $\mathcal{M}$ *and has measure zero.*
3. *If*

$$A = \left\{\mathbf{x} \in \mathbb{R}^n : a_j < x_j < b_j;\ j = 1, \ldots, n\right\}$$

then $|A| = \prod_{j=1}^{n} (b_j - a_j)$.

The elements of $\mathcal{M}$ are called *Lebesgue measurable sets* and $|\cdot|_n$ (or simply $|\cdot|$ if no confusion arises) is called the *n-dimensional Lebesgue measure.* Unless explicitly said, from now on, *measurable* means *Lebesgue measurable* and the measure is the Lebesgue measure.

Not every subset of $\mathbb{R}^n$ is measurable. However, the nonmeasurable ones are quite . . . pathological[1]!

The *sets of measure zero* are quite important. Here are some examples: all countable sets, e.g. the set $\mathbb{Q}$ of rational numbers; straight lines or smooth curves in $\mathbb{R}^2$; straight lines, smooth curves, hyperplanes and surfaces in $\mathbb{R}^3$.

Notice that a straight line segment has measure zero in $\mathbb{R}^2$ but, of course not in $\mathbb{R}$.

We say that a *property holds almost everywhere in* $A \in \mathcal{M}$ (in short, a.e. in A) *if it holds at every point of A except that in a subset of measure zero.*

For instance, the sequence $f_k(x) = \exp(-n \sin^2 x)$ converges to zero a.e. in $\mathbb{R}$, a Lipschitz function is differentiable a.e. in its domain (Rademacher's Theorem 1.9, p. 18).

The Lebesgue integral is defined for *measurable* functions, characterized by the fact that the inverse image of every closed set is measurable.

Definition A.5 Let $A \subseteq \mathbb{R}^n$ be measurable, and $f : A \to \mathbb{R}$. We say that f is *measurable* if

$$f^{-1}(C) \in \mathcal{F}$$

for any closed set $C \subseteq \mathbb{R}$.

If f is continuous, is measurable. The sum and the product of a finite number of measurable functions is measurable. The pointwise limit of a sequence of measurable functions is measurable.

If $f : A \to \mathbb{R}$, is measurable, we define its *essential supremum* or *least upper bound* by the formula:

$$\operatorname{ess\,sup} f = \inf \{K : f \leq K \text{ a.e. in } A\}.$$

Note that, if $f = \chi_{\mathbb{Q}}$, the characteristic functions of the rational numbers, we have $\sup f = 1$, but $\operatorname{ess\,sup} f = 0$, since $|\mathbb{Q}| = 0$.

Every measurable function may be approximated by **simple functions**. A function $s : A \subseteq \mathbb{R}^n \to \mathbb{R}$ is said to be **simple** if its range is constituted by a *finite number* of values $s_1, \ldots, s_N$, attained respectively on measurable sets $A_1, \ldots, A_N$, contained in A. Introducing the characteristic functions χ_{A_j}, we may write

$$s = \sum_{j=1}^{N} s_j \chi_{A_j}.$$

We have:

[1] See e.g. *W. Rudin*, [39].

Theorem A.6 *Let $f : A \to \mathbb{R}$, be measurable. There exists a sequence $\{s_k\}$ of simple functions converging pointwise to f in A. Moreover, if $f \geq 0$, we may choose $\{s_k\}$ increasing.*

A.3 The Lebesgue Integral

We define the Lebesgue integral of a measurable function on a measurable set A. For a simple function

$$s = \sum_{j=1}^{N} s_j \chi_{A_j}$$

we set:

$$\int_A s = \sum_{j=1}^{N} s_j \left|A_j\right|,$$

with the agreement that, *if $s_j = 0$ and $\left|A_j\right| = +\infty$, then $s_j \left|A_j\right| = 0$.*

If $f \geq 0$ is measurable, we define

$$\int_A f = \sup \int_A s,$$

where the supremum is computed over the set of all simple functions s such that $s \leq f$ in A.

In general, if f is measurable, we write $f = f^+ - f^-$, where $f^+ = \max\{f, 0\}$ and $f^- = \max\{-f, 0\}$ are the positive and negative parts of f, respectively. Then we set:

$$\int_A f = \int_A f^+ - \int_A f^-,$$

under the condition that at least one of the two integrals in the right hand side is finite.

If both these integrals are finite, the function f is said to be **integrable** or **summable** in A. From the definition, it follows immediately that a measurable functions *f is integrable in A if and only if $|f|$ is integrable in A.*

All the functions Riemann integrable in a set A are Lebesgue integrable as well. An interesting example of non-Lebesgue integrable function in $(0, +\infty)$ is given by $h(x) = \sin x/x$. In fact[2]

$$\int_0^{+\infty} \frac{|\sin x|}{x} dx = +\infty.$$

[2] We may write

$$\int_0^{+\infty} \frac{|\sin x|}{x} dx = \sum_{k=1}^{\infty} \int_{(k-1)\pi}^{k\pi} \frac{|\sin x|}{x} dx \geq \sum_{k=1}^{\infty} \frac{1}{k\pi} \int_{(k-1)\pi}^{k\pi} |\sin x|\, dx = \sum_{k=1}^{\infty} \frac{2}{k\pi} = +\infty.$$

On the contrary, it may be proved that

$$\lim_{N\to+\infty}\int_0^N \frac{\sin x}{x}dx = \frac{\pi}{2}$$

and therefore the *improper* Riemann integral of h is finite.

The set of the integrable functions in A is denoted by $L^1(A)$. If we identify two functions when they agree a.e. in A, $L^1(A)$ becomes a Banach space with the norm

$$\|f\|_{L^1(A)} = \int_A |f|$$

(see Chap. 6). We denote by $L^1_{\mathrm{loc}}(A)$ the set of *locally summable* functions, i.e. of the functions which are summable in every compact subset of A.

A.4 Some Fundamental Theorems

The following theorems are among the most important and useful in the theory of integration.

Theorem A.7 (Dominated Convergence Theorem) *Let $\{f_k\}$ be a sequence of summable functions in A such that $f_k \to f$ a.e. in A. If there exists $g \geq 0$, summable in A and such that $|f_k| \leq g$ a.e. in A, then f is summable and*

$$\|f_k - f\|_{L^1(A)} \to 0 \quad \text{as } k \to +\infty.$$

In particular

$$\lim_{k\to\infty}\int_A f_k = \int_A f.$$

Theorem A.8 *Let $\{f_k\}$ be a sequence of summable functions in A such that*

$$\|f_k - f\|_{L^1(A)} \to 0 \quad \text{as} \quad k \to +\infty.$$

Then there exists a subsequence $\{f_{k_j}\}$ such that $f_{k_j} \to f$ a.e. as $j \to +\infty$.

Theorem A.9 (Monotone Convergence Theorem) *Let $\{f_k\}$ be a sequence of non-negative, measurable functions in A such that*

$$f_1 \leq f_2 \leq \ldots \leq f_k \leq f_{k+1} \leq \ldots .$$

Then

$$\lim_{k\to\infty}\int_A f_k = \int_A \lim_{k\to\infty} f_k.$$

Example A.10 A typical situation we often encounter in this book is the following. Let $f \in L^1(A)$ and, for $\varepsilon > 0$, set $A_\varepsilon = \{|f| > \varepsilon\}$. Then, we have

$$\int_{A_\varepsilon} f \to \int_A f \quad \text{as } \varepsilon \to 0.$$

This follows from Theorem A.7 since, for every sequence $\varepsilon_j \to 0$, we have $|f|\,\chi_{A_{\varepsilon_j}} \le |f|$ and therefore

$$\int_{A_{\varepsilon_j}} f = \int_A f\chi_{A_{\varepsilon_j}} \to \int_A f \quad \text{as } \varepsilon \to 0.$$

Let $C_0(A)$ be the set of continuous functions in A, compactly supported in A. An important fact is that any summable function may be approximated by a function in $C_0(A)$.

Theorem A.11 *Let $f \in L^1(A)$. Then, for every $\delta > 0$, there exists a continuous function $g \in C_0(A)$ such that*

$$\|f - g\|_{L^1(A)} < \delta.$$

The fundamental theorem of calculus extends to the Lebesgue integral in the following form:

Theorem A.12 (Differentiation) *Let $f \in L^1_{loc}(\mathbb{R})$. Then*

$$\frac{d}{dx}\int_a^x f(t)dt = f(x) \qquad a.e.\ x \in \mathbb{R}.$$

Differentiation under the integral sign can be performed according to the next theorem (compare with Sect. 1.4).

Theorem A.13 (Differentiation under the integral sign) *Let $\Omega \subseteq \mathbb{R}^n, T \subseteq \mathbb{R}^m$ be open sets, and $f : \Omega \times T \to \mathbb{R}$. Assume that:*

a) $\mathbf{x} \mapsto f(\mathbf{x}, \mathbf{t}) \in L^1(\Omega)$ for all $\mathbf{t} \in \mathbb{R}^m$ and $\mathbf{t} \mapsto f(\mathbf{x}, \mathbf{t}) \in C^1(T)$ for a.e. $\mathbf{x} \in \mathbb{R}^n$;
b) there exist $m + 1$ functions $g_0(\mathbf{x}), g_1(\mathbf{x}), ..., g_m(\mathbf{x})$ integrable in Ω such that, a.e. in Ω,

$$|f(\mathbf{x}, \mathbf{t})| \le g_0(\mathbf{x}), \ \left|f_{t_j}(\mathbf{x}, \mathbf{t})\right| \le g_j(\mathbf{x}), \ j = 1, ..., m.$$

Then, $\mathbf{t} \mapsto \int_\Omega f(\mathbf{x}, \mathbf{t})d\mathbf{x} \in C^1(T)$ and

$$\frac{\partial}{\partial t_j}\int_\Omega f(\mathbf{x}, \mathbf{t})d\mathbf{x} = \int_\Omega f_{t_j}(\mathbf{x}, \mathbf{t})d\mathbf{x}.$$

Finally, the integral of a summable function can be computed via iterated integrals in any order. Precisely, let

$$I_1 = \{\mathbf{x} \in \mathbb{R}^n : -\infty \le a_i < x_i < b_i \le \infty;\ i = 1, \dots, n\}$$

and

$$I_2 = \left\{\mathbf{y} \in \mathbb{R}^m : -\infty \le a_j < y_j < b_j \le \infty;\ j = 1, \dots, m\right\}.$$

Theorem A.14 (Fubini) *Let f be summable in $I = I_1 \times I_2 \subset \mathbb{R}^n \times \mathbb{R}^m$. Then*

1. *$f(\mathbf{x}, \cdot) \in L^1(I_2)$ for a.e. $\mathbf{x} \in I_1$, and $f(\cdot, \mathbf{y}) \in L^1(I_1)$ for a.e. $\mathbf{y} \in I_2$.*
2. *$\int_{I_2} f(\cdot, \mathbf{y}) d\mathbf{y} \in L^1(I_1)$ and $\int_{I_1} f(\mathbf{x}, \cdot) d\mathbf{x} \in L^1(I_2)$.*
3. *The following formulas hold:*

$$\int_I f(\mathbf{x}, \mathbf{y})\ d\mathbf{x}d\mathbf{y} = \int_{I_1} d\mathbf{x} \int_{I_2} f(\mathbf{x}, \mathbf{y})\ d\mathbf{y} = \int_{I_2} d\mathbf{x} \int_{I_1} f(\mathbf{x}, \mathbf{y})\ d\mathbf{y}.$$

Let $f \in L^1(B_R(\mathbf{p}))$. The following formula can be considered a version of Fubini's Theorem in spherical coordinates:

$$\int_{B_R(\mathbf{p})} f(\mathbf{x}) d\mathbf{x} = \int_0^R ds \int_{\partial B_s(\mathbf{p})} f(\boldsymbol{\sigma}) d\sigma. \tag{A.1}$$

A.5 Probability Spaces, Random Variables and Their Integrals

Let $\mathcal{F}$ be a σ-*algebra* in a set Ω. A *probability measure* P on $\mathcal{F}$ is a measure in the sense of definition A.2, such that $P(\Omega) = 1$ and

$$P : \mathcal{F} \to [0, 1]\,.$$

The triplet $(\Omega, \mathcal{F}, P)$ is called a *probability space*. In this setting, the elements ω of Ω are *sample points*, while a set $A \in \mathcal{F}$ has to be interpreted as an *event*. $P(A)$ is the probability of (occurrence of) A.

A typical example is given by the triplet

$$\Omega = [0, 1]\,, \qquad \mathcal{F} = \mathcal{M} \cap [0, 1]\,, \qquad P(A) = |A|$$

which models a *uniform random choice* of a point in $[0, 1]$.

The probability of an event A, under the supplementary condition that another event B, having positive probability, has occurred, is called the *conditional probability of A given B* and it is denoted by $P(A\,|B)$. The following formula holds:

$$P(A\,|B) = \frac{P(A \cap B)}{P(B)}.$$

If $P(A\,|B) = P(A)$ or, equivalently, $P(A \cap B) = P(A)P(B)$, the two events A and B are *independent.* More generally, *the events* $A_1, A_2, \cdots, A_N$ are:

- *pairwise independent* if, for every pairs A_j, A_k, $j \neq k$,

$$P(A_j \cap A_k) = P(A_j)P(A_k);$$

- *independent* if for every subfamily $A_{j_1}, A_{j_2}, \cdots, A_{j_k}$ of distinct events

$$P(A_{j_1} \cap A_{j_2} \cap \cdots \cap A_{j_k}) = P(A_{j_1})P(A_{j_2}) \cdots P(A_{j_k}).$$

A 1-*dimensional random variable* in $(\Omega, \mathcal{F}, P)$ is a function

$$X : \Omega \to \mathbb{R}$$

such that X is $\mathcal{F}$*-measurable*, that is

$$X^{-1}(C) \in \mathcal{F}$$

for each closed set $C \subseteq \mathbb{R}$.

Example A.15 The *number k of steps to the right after N steps* in the random walk of Sect. 2.4 is a (discrete) random variable. Here Ω is the set of walks of N steps.

By the same procedure used to define the Lebesgue integral we can define the integral of a random variable with respect to a probability measure. We sketch the main steps.

If X is *simple*, i.e. $X = \sum_{j=1}^{N} s_j \chi_{A_j}$, we define

$$\int_\Omega X \, dP = \sum_{j=1}^{N} s_j P(A_j).$$

If $X \geq 0$ we set

$$\int_\Omega X \, dP = \sup \left\{ \int_\Omega Y \, dP : Y \leq X, Y \text{ simple} \right\}.$$

Finally, if $X = X^+ - X^-$ we define

$$\int_\Omega X \, dP = \int_\Omega X^+ \, dP - \int_\Omega X^- \, dP$$

provided at least one of the integral on the right hand side is finite.

In particular, if

$$\int_\Omega |X| \, dP < \infty,$$

then

$$E(X) = \langle X \rangle = \int_\Omega X \, dP$$

is called the *expected value* (or *mean value* or *expectation*) of X, while

$$\mathrm{Var}(X) = \left\langle X^2 \right\rangle - \langle X \rangle^2 = \int_\Omega (X - E(X))^2 \, dP$$

is called the *variance* of X.

If $X, Y : \Omega \to \mathbb{R}$ are random variables with *finite* variance, the quantity

$$\operatorname{cov}(X,Y) = \langle XY\rangle - \langle X\rangle \langle Y\rangle = \int_\Omega (X - E(X))(Y - E(Y))\, dP$$

is called the *covariance of* X and Y.

Let $\mathbf{X}$ be an n-dimensional random variable,

$$\mathbf{X} = (X_1, ..., X_n) : \Omega \to \mathbb{R}^n.$$

If

$$\int_\Omega \left|X_j\right| \, dP < \infty \qquad j = 1, ..., n$$

the *expected value* and the *variance* of $\mathbf{X}$ are the vectors of components

$$E(\mathbf{X})_j = \langle X\rangle_j, \quad \operatorname{Var}(\mathbf{X})_j = \left\langle X_j^2 \right\rangle - \left\langle X_j \right\rangle^2$$

while the *covariance of* $\mathbf{X}$ is the matrix $\operatorname{cov}(\mathbf{X})$ whose entries are $\operatorname{cov}(X_i, X_j)$, $i, j = 1, ...n$, provided each X_j has finite variance.

Appendix B
Identities and Formulas

B.1 Gradient, Divergence, Curl, Laplacian

Let $\mathbf{F}$ be a smooth vector field and f a smooth real function, in $\mathbb{R}^3$.

Orthogonal cartesian coordinates

1. *Gradient*:
$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}.$$
2. *Divergence* ($\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$):
$$\operatorname{div}\mathbf{F} = \frac{\partial}{\partial x}F_1 + \frac{\partial}{\partial y}F_2 + \frac{\partial}{\partial z}F_3.$$
3. *Laplacian*:
$$\Delta f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2}.$$
4. *Curl*:
$$\operatorname{curl}\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ F_1 & F_2 & F_3 \end{vmatrix}.$$

Cylindrical coordinates

$$x = r\cos\theta,\ y = r\sin\theta,\ z = z \qquad (r > 0,\ 0 \le \theta \le 2\pi)$$
$$\mathbf{e}_r = \cos\theta\mathbf{i} + \sin\theta\mathbf{j}, \quad \mathbf{e}_\theta = -\sin\theta\mathbf{i} + \cos\theta\mathbf{j}, \quad \mathbf{e}_z = \mathbf{k}.$$

1. *Gradient*:

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$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{\partial f}{\partial z}\mathbf{e}_z.$$

2. *Divergence* ($\mathbf{F} = F_r\mathbf{e}_r + F_\theta\mathbf{e}_\theta + F_z\mathbf{k}$):

$$\operatorname{div}\mathbf{F} = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{1}{r}\frac{\partial}{\partial \theta}F_\theta + \frac{\partial}{\partial z}F_z.$$

3. *Laplacian*:

$$\Delta f = \frac{1}{r}\frac{\partial}{\partial r}(r\frac{\partial f}{\partial r}) + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2} = \frac{\partial^2 f}{\partial r^2} + \frac{1}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial z^2}.$$

4. *Curl*:

$$\operatorname{curl}\mathbf{F} = \frac{1}{r}\begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\theta & \mathbf{e}_z \\ \partial_r & \partial_\theta & \partial_z \\ F_r & rF_\theta & F_z \end{vmatrix}.$$

Spherical coordinates

$$x = r\cos\theta\sin\psi,\ y = r\sin\theta\sin\psi,\ z = r\cos\psi \quad (r > 0, 0 \le \theta \le 2\pi, 0 \le \psi \le \pi)$$

$$\begin{aligned} \mathbf{e}_r &= \cos\theta\sin\psi\mathbf{i} + \sin\theta\sin\psi\mathbf{j} + \cos\psi\mathbf{k} \\ \mathbf{e}_\theta &= -\sin\theta\mathbf{i} + \cos\theta\mathbf{j} \\ \mathbf{e}_\psi &= \cos\theta\cos\psi\mathbf{i} + \sin\theta\cos\psi\mathbf{j} - \sin\psi\mathbf{k}. \end{aligned}$$

1. *Gradient*:

$$\nabla f = \frac{\partial f}{\partial r}\mathbf{e}_r + \frac{1}{r\sin\psi}\frac{\partial f}{\partial \theta}\mathbf{e}_\theta + \frac{1}{r}\frac{\partial f}{\partial \psi}\mathbf{e}_\psi.$$

2. *Divergence* ($\mathbf{F} = F_r\mathbf{e}_r + F_\theta\mathbf{e}_\theta + F_\psi\mathbf{e}_\psi$):

$$\operatorname{div}\mathbf{F} = \underbrace{\frac{\partial}{\partial r}F_r + \frac{2}{r}F_r}_{\text{radial part}} + \frac{1}{r}\underbrace{\left[\frac{1}{\sin\psi}\frac{\partial}{\partial \theta}F_\theta + \frac{\partial}{\partial \psi}F_\psi + \cot\psi F_\psi\right]}_{\text{spherical part}}.$$

3. *Laplacian*:

$$\Delta f = \underbrace{\frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r}}_{\text{radial part}} + \frac{1}{r^2}\underbrace{\left\{\frac{1}{(\sin\psi)^2}\frac{\partial^2 f}{\partial \theta^2} + \frac{\partial^2 f}{\partial \psi^2} + \cot\psi\frac{\partial f}{\partial \psi}\right\}}_{\text{spherical part (Laplace-Beltrami operator)}}.$$

4. *Curl*:

$$\operatorname{rot}\mathbf{F} = \frac{1}{r^2\sin\psi}\begin{vmatrix} \mathbf{e}_r & r\mathbf{e}_\psi & r\sin\psi\mathbf{e}_\theta \\ \partial_r & \partial_\psi & \partial_\theta \\ F_r & rF_\psi & r\sin\psi F_z \end{vmatrix}.$$

B.2 Formulas

Gauss formulas

In $\mathbb{R}^n$, $n \geq 2$, let:

- Ω be a bounded smooth domain and $\boldsymbol{\nu}$ the outward unit normal on $\partial\Omega$.
- $\mathbf{u}, \mathbf{v}$ be vector fields of class $C^1(\overline{\Omega})$.
- φ, ψ be real functions of class $C^1(\overline{\Omega})$.
- $d\sigma$ be the area element on $\partial\Omega$.

1. $\displaystyle\int_\Omega \operatorname{div} \mathbf{u}\, d\mathbf{x} = \int_{\partial\Omega} \mathbf{u} \cdot \boldsymbol{\nu}\, d\sigma$ (Divergence Theorem).
2. $\displaystyle\int_\Omega \nabla\varphi\, d\mathbf{x} = \int_{\partial\Omega} \varphi\boldsymbol{\nu}\, d\sigma.$
3. $\displaystyle\int_\Omega \Delta\varphi\, d\mathbf{x} = \int_{\partial\Omega} \nabla\varphi \cdot \boldsymbol{\nu}\, d\sigma = \int_{\partial\Omega} \partial_{\boldsymbol{\nu}}\varphi\, d\sigma.$
4. $\displaystyle\int_\Omega \psi \operatorname{div} \mathbf{F}\, d\mathbf{x} = \int_{\partial\Omega} \psi\mathbf{F} \cdot \boldsymbol{\nu}\, d\sigma - \int_\Omega \nabla\psi \cdot \mathbf{F}\, d\mathbf{x}$ (Integration by parts).
5. $\displaystyle\int_\Omega \psi\Delta\varphi\, d\mathbf{x} = \int_{\partial\Omega} \psi\partial_{\boldsymbol{\nu}}\varphi\, d\sigma - \int_\Omega \nabla\varphi \cdot \nabla\psi\, d\mathbf{x}$ (Green's identity I).
6. $\displaystyle\int_\Omega (\psi\Delta\varphi - \varphi\Delta\psi)\, d\mathbf{x} = \int_{\partial\Omega} (\psi\partial_{\boldsymbol{\nu}}\varphi - \varphi\partial_{\boldsymbol{\nu}}\psi)\, d\sigma$ (Green's identity II).
7. $\displaystyle\int_\Omega \operatorname{curl} \mathbf{u}\, d\mathbf{x} = -\int_{\partial\Omega} \mathbf{u} \times \boldsymbol{\nu}\, d\sigma.$
8. $\displaystyle\int_\Omega \mathbf{u} \cdot \operatorname{curl} \mathbf{v}\, d\mathbf{x} = \int_\Omega \mathbf{v} \cdot \operatorname{curl} \mathbf{u}\, d\mathbf{x} - \int_{\partial\Omega} (\mathbf{u} \times \mathbf{v}) \cdot \boldsymbol{\nu}\, d\sigma.$

Identities

1. $\operatorname{div}\operatorname{curl} \mathbf{u} = 0.$
2. $\operatorname{curl} \nabla\varphi = \mathbf{0}.$
3. $\operatorname{div}(\varphi\mathbf{u}) = \varphi \operatorname{div} \mathbf{u} + \nabla\varphi \cdot \mathbf{u}.$
4. $\operatorname{curl}(\varphi\mathbf{u}) = \varphi \operatorname{curl} \mathbf{u} + \nabla\varphi \times \mathbf{u}.$
5. $\operatorname{curl}(\mathbf{u} \times \mathbf{v}) = (\mathbf{v} \cdot \nabla)\mathbf{u} - (\mathbf{u} \cdot \nabla)\mathbf{v} + (\operatorname{div} \mathbf{v})\mathbf{u} - (\operatorname{div} \mathbf{u})\mathbf{v}.$
6. $\operatorname{div}(\mathbf{u} \times \mathbf{v}) = \operatorname{curl} \mathbf{u} \cdot \mathbf{v} - \operatorname{curl} \mathbf{v} \cdot \mathbf{u}.$
7. $\nabla(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \times \operatorname{curl} \mathbf{v} + \mathbf{v} \times \operatorname{curl} \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{v} + (\mathbf{v} \cdot \nabla)\mathbf{u}.$
8. $(\mathbf{u} \cdot \nabla)\mathbf{u} = \operatorname{curl} \mathbf{u} \times \mathbf{u} + \frac{1}{2}\nabla |\mathbf{u}|^2.$
9. $\operatorname{curl}\operatorname{curl} \mathbf{u} = \nabla(\operatorname{div} \mathbf{u}) - \Delta\mathbf{u}.$

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Index

A

Absorbing barriers 112
Adjoint of a bilinear form 395
Adjoint problem 589
Alternative
  for the Dirichlet problem 516
  for the Neumann problem 518
Angular frequency 254
Arbitrage 93

B

Barenblatt solutions 104
Barrier 138
Bessel function 74, 291
Bilinear form 375
Bond number 327
Bootstrap 598
Boundary conditions 25
  Dirichlet 25, 36
  mixed 26, 36
  Neumann 25, 36
  Robin 25, 36
Breaking time 202
Brownian motion 56
Brownian path 56

C

Canonical form 287, 288
Canonical isometry 370
Characteristic 183, 223, 616, 618
  parallelogram 271
  strip 238
  system 238
Chebyshev polynomials 362
Closure 8
Comparison 40
Compatibility conditions 396, 398
Complete integral 244
Condition
  compatibility 118
  E 219
  entropy 636
  Rankine-Hugoniot 629
Conjugate exponent 11, 350
Conormal derivative 517
Contact discontinuity 624, 630, 638
Continuous isomorphism 368, 378
Convection 63, 182
Convergence
  least squares 14
  uniform 12
  weak 387
Convolution 422, 443
Cost functional 586
Critical mass 77
Critical survival value 67
Curve
  rarefaction 631

D

d'Alembert formula 270
d-harmonic function 120
Darcy's law 103
Diffusion 22
Diffusion coefficient 55
Dirac comb 427
Dirac measure 47
Direct product 441
Direct sum 356
Dirichlet eigenfunctions 507
Dirichlet Principle 536
Dispersion 280
  relation 255, 281, 328
Dissipation
  external/internal 279
Distribution 425
  composition 436
  division 439
Distributional derivative 430
Domain 8
  C^1, C^k 17
  Lipschitz 18
  of dependence 272, 309
  smooth 17
Drift 62, 91
Duhamel method 80, 185, 278

E

Eigenfunction 361
  of a bilinear form 405
Eigenspace 400, 402, 405
Eigenvalue 361, 402
  of a bilinear form 404
Eigenvector 402

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Elastic restoring force 112
Elliptic equation 495
Entropy condition 211
Envelope 203, 248
Equal area rule 203
Equation
  backward 96
  backward heat 42
  Bernoulli 323
  Bessel 74, 363
  biharmonic 521
  Black-Scholes 3, 95
  Bukley-Leverett 235
  Burgers 4
  Chebyshev 361
  Clairaut 244
  diffusion 2, 21
  eikonal 4, 241
  elastostatics 574
  elliptic 282
  Fisher 4
  fully nonlinear 2
  Hermite 363
  hyperbolic 282
  Klein-Gordon 280
  Laplace 3
  Legendre 362
  linear elasticity 5
  linear, nonlinear 2
  Maxwell 5
  minimal surface 4
  Navier 574
  Navier-Stokes 5, 153, 578
  Navier-Stokes, stationary 582
  parabolic 282, 541
  parametric Bessel (of order p) 363
  partial differential 2
  Poisson 3, 115
  porous medium 4, 104
  quasilinear 2
  reduced wave 180
  Schrödinger 3
  semilinear 2
  stochastic differential 91
  Sturm-Liouville 361
  transport 2
  Tricomi 282
  uniformly parabolic 542
  vibrating plate 3
  vibrating string 262
  viscous Burgers 217
  wave 3
Equicontinuity 385
Equipartition of energy 335
Escape probability 143
Essential support 421
Essential supremum 349
Euler equation 382
European options 90
Expected value 71
Expiry date 90
Extension operator 468
Exterior Dirichlet problem 163
Exterior domain 163
Exterior Robin problem 164, 179

F

Fick's law 63
Final payoff 96
First exit time 142
First integral 230, 232
First variation 382
Flux function 181
Focusing effect 337
Forward cone 306
Fourier coefficients 359
Fourier law 24
Fourier series 31
Fourier transform 445, 463
Fourier-Bessel series 76, 364
Frequency 254
Froude number 327
Function
  Bessel's of first kind and order p 364
  characteristic 10
  compactly supported 10
  continuous 10
  d-harmonic 119
  essentially bounded 349
  Green's 156
  Hölder continuous 348
  harmonic 22, 115
  Heaviside 47
  piecewise continuous 206
  summable 349
  test 51, 421
  weight 361
Functional 368
Fundamental solution 46, 50, 148, 276, 305, 433

G

Gas dynamics 613
Gaussian law 57, 70
General solution 231, 245
Genuinely nonlinear 632
Global Cauchy problem 27, 37, 78
  nonhomogeneous 81
Gram-Schmidt process 360
Greatest lower bound 9
Group velocity 255

H

Harmonic lifting 136
Harmonic measure 144
Harnack's inequality 131
Heisenberg Uncertainty Principle
  for the first eigenvalue 414
Helmholtz decomposition formula 151
Hermite polynomials 363
Hilbert triplet 394
Homeomorphism 411
Hooke's law 573
Hopf's principle 126, 594
Hugoniot line 624

I

Identity
  Green's (first and second) 20
  strong Parseval's 450
  weak Parseval's 448
Inequality
  Hölder 349
Infimum 9
Inflow/outflow boundary 231
Inflow/outflow characteristics 187
Inner product space 350
Integral surface 223
Integration by parts 20
Interior sphere condition 125
Invasion problem 113

Inward heat flux 36
Isometry 352
Ito's formula 91

K

Kernel 365
Kinematic condition 324
Kinetic energy 260

L

Lagrange multiplier 581
Lattice 68, 118
Least squares 32
Least upper bound 9
Lebesgue spine 140
Legendre polynomials 362
Light cone 242
Limit point 9
Linearly degenerate 640
Liouville Theorem 131
Local chart 17
Local diffeomorphism 16
Local wave speed 192
Localization 468
Logarithmic potential 150
Logistic growth 106
Lognormal density 93

M

Mach number 301
Markov properties 58, 71
Mass conservation 62
Material derivative 153
Maximum principle 84, 120
  weak 39, 523, 560
Mean value property 122
Method 27
  Duhamel 82
  electrostatic images 157
  Galerkin's 378
  of characteristics 191
  of descent 309
  of Faedo-Galerkin 550, 564
  of stationary phase 257
  reflection 467
  separation of variables 27, 29, 264, 297, 400, 509
  time reversal 312
  vanishing viscosity 215
Metric space 345
Minimax property (of the eigenvalues)
  for the first eigenvalue 409, 418
Mollifier 422
Monotone scheme 594
Multi-index 133, 425
Multidimensional symmetric random walk 68
Multiplicity (of an eigenvalue) 402

N

Neumann eigenfunctions 508
Neumann function 162
Norm
  Integral of order p 349
  least squares 347
  maximum 347
  maximum of order k 348
Normal probability density 46
Normed space 345
Numerical sets 7

O

Open covering 468
Operator
  adjoint 371
  bounded, continuous 365
  compact 390
  discrete Laplace 119
  linear 365
  mean value 118
Optimal control 586
Optimal state 586
Orthonormal basis 359

P

Parabolic
  boundary 26, 37
Parabolic dilations 42
Parallelogram law 351
Partition of unity 468
Perron method 137
Phase speed 254
Poincaré's inequality 457, 479
Point 7
  boundary 7
  interior 7
  limit 7
Point source solution
  two dimensional 338
Poisson formula 130
Potential 115
  double layer 165
  energy 261
  Newtonian 149
  retarded 315, 338
  single layer 169
Principal Dirichlet eigenvalue 507
Principle of virtual work 577
Probability
  measure 662
  space 662
Problem
  abstract parabolic 546
  abstract variational 375
  characteristic Cauchy 336
  eigenvalue 30
  Goursat 336
  ill posed (heat equation) 337
  inverse 312
  Riemann 622
  well posed 6, 24
Projected characteristics 230
Projection
  on closed convex sets 416
Put-call parity 99

Q

Quantum mechanics
  harmonic oscillator 415

R

Random variable 56
Random walk 51
  with drift 59
Range 365
  of influence 272, 306
Rankine-Hugoniot condition 200, 206

Rarefaction/simple waves 196
Rayleigh quotient 407, 507
Reaction 65
Reflecting barriers 111
Regular point 138
Resolvent 400
  of a bilinear form 404
  of a bounded operator 402
Retarded potential 316
retrograde cone 294
Reynolds number 578
Riemann invariant 626
Riemann problem 213
Rodrigues' formula 362, 363

S

Schwarz inequality 351
Schwarz reflection principle 176
Self-financing portfolio 94, 101
Selfadjoint operator 372
Sequence
  Cauchy 346
  fundamental 346
Set
  bounded 8
  closed 8
  compact 8
  compactly contained 8
  connected 8
  convex 8
  dense 8
  open 8
  precompact 385
  sequentially closed 8
  sequentially compact 8, 385
Shallow water waves 652
Shock
  curve 200
  speed 200
  wave 200
Shock curve 638
Sobolev exponent 482
Solution
  classical 497
  distributional 497
  integral 206
  self-similar 44, 105
  steady state 28
  strong 497
  unit source 49
  variational 497
  viscosity 497
  weak 206
Sommerfeld condition 180
Space
  separable 359
Space-like curve 243
Spectral decomposition
  of a matrix 400
  of an operator 404
Spectrum 400
  continuous 402
  of a bilinear form 404
  of a bounded operator 402
  point 402
  residual 403
Spherical waves 255
Stability estimate 378
Standing wave 265
Stationary phase (method of) 333
Steepest descent 590
Stiffness matrix 379
Stochastic process 56, 70
Stokes System
  equazione biarmonica 578
Stopping time 59, 142
Strike price 90
Strip condition 239
Strong Huygens' principle 307, 309
Sub/superharmonic function 135
Sub/supersolution 39, 593
Superposition principle 21, 78, 262
Support 10
  of a distribution 428
Supremum 9
Surface
  of the unit sphere (ω_n) 7
  tension 322, 324
Symbol
  "big O" 65
  "little o" 11, 54
  Kronecker 359
System
  p-system 615
  hyperbolic 612
  telegrapher 620

T

Tempered distribution 446
Tensor
  deformation 573
  stress 153, 573
Tensor product 441
Term by term
  differentiation 12
  integration 12
Theorem
  Ascoli-Arzelà 385
  Contraction mapping 409
  Dominated Convergence 660
  Fubini 662
  Lax-MIlgram 376
  Leray-Schauder 413
  Monotone Convergence 660
  projection 356
  Riesz's representation 369
  Riesz-Fréchet-Kolmogorov 386
  Schauder 411
Theorem Rellich 478
Time-like curve 243
Topology 7, 345
  euclidean 8
  relative 8
Trace 470
Trace inequality 476
Traffic in a tunnel 248
Transition function 70
Transition layer 217
Transition probability 58
Transmission conditions 536
Travelling wave 184, 192, 215
Trivial extension 467
Tychonov class 84

U

Uniform ellipticity 511
Unit impulse 47
Upper, lower limit 9

V

Value function 90
Variational formulation
  Dirichlet problem 500, 512
  Mixed problem 506, 519
  Neumann problem 503, 517
  Robin problem 505
Variational inequality
  on closed convex sets 416
Variational principle
  for the k-th eigenvalue 408
  for the first eigenvalue 407
Volatility 91

W

Wave
  capillarity 330
  cylindrical 290
  gravity 330
  harmonic 253
  incoming/outgoing 292
  linear 322
  linear gravity 338
  monochromatic/harmonic 290
  number 254
  packet 256
  plane 255, 290
  rarefaction, p-system 644, 645
  shock 637
  shock, p-system 642, 643
  simple 631
  spherical 291
  standing 254
  travelling 253
Weak coerciveness 515
Weak formulation
  Cauchy-Dirichlet problem 544
  Cauchy-Robin/Neumann problem 554, 555, 557
  Initial-Dirichlet problem (wave eq.) 563
Weakly coercive (bilinear form) 395, 558
Weierstrass test 12, 33

Y

Young modulus 336